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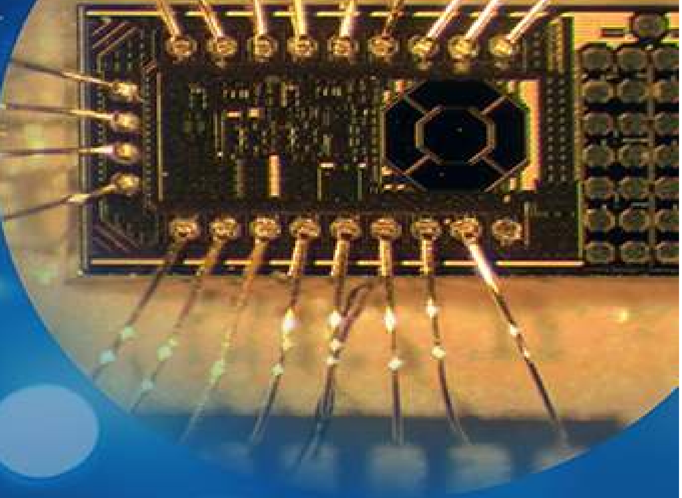

Optoelectronic Devices and Optical Fiber Communication

P Chakrabarti

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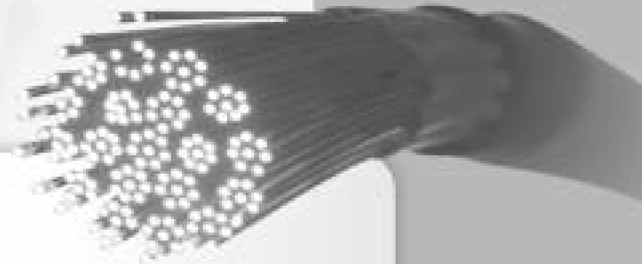

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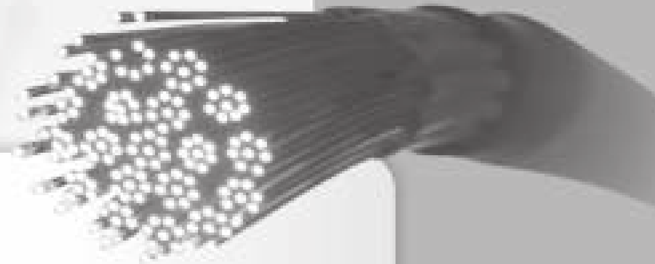

Optoelectronic Devices and Optical Fiber Communication

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Preface

The CBS edition of the present book entitled *Optoelectronic Devices and Optical Fiber Communication* is an updated and revised version of the original textbook entitled "Optical Fiber Communication" published by McGraw Hill Education, (India), which is currently available in the e-book form.

The proposal from the current publisher for bringing out an updated version of the original text is based on the demand for hardcopies of the text. A thorough revision of the text has been carried out to take care of errors, omissions, and typos. The sections and subsections in different chapters in this book are presented in a structured manner for ease of understanding. Numerous solved examples, unsolved problems and extensive list of references to relevant literatures are included at the end of each chapter.

The book can be used as a primary text on a basic course on the subject at the undergraduate and graduate levels of electrical/electronics engineering and electronics science. The book has been planned to become a self-sufficient text for understanding the subject. Interested students can explore further details on various topics dealt within the book by consulting the relevant references listed at the end of each chapter. The solved problems are illustrated to give the readers an idea about the practical values of different parameters and their implications on practical system design. The book can be used as a supporting text for courses on semiconductor optoelectronic devices and systems.

I am thankful to McGraw-Hill Education, (India), for reverting the copyright of the original book and thereby allowing me to bring out the revised book with CBS Publishers. I gratefully acknowledge the interest shown by CBS Publishers in the book which motivated me to work on this title. I am thankful to the entire team of CBS Publishers, for their untiring effort and meticulous workmanship to bring this book to its present form. Despite our sincere efforts, there may be a few slips, omissions, or typos. The suggestions and feedback for improvement of the text will be highly appreciated. Finally, on a personal note, I would like to thank my wife Runa, and our daughter Ishita, for their support during the revision of this text.

P Chakrabarti

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List of Symbols

Symbol	**Meaning**
a	Fiber core radius
A_S	Emission area, area of cross-section
A_c	Carrier amplitude
A_m	Amplitude of the modulating signal
A_{common}	Common area of overlapping of two fibers
A_{21}	Einstein's coefficient
B_0	Brightness of the source along the direction normal to the plane of emission
$B_\perp$	Brightness of the source in the direction perpendicular to the emission plane
$B_\parallel$	Brightness of the source in the direction parallel to the emitting surface
B	Bandwidth
B_{12}, B_{21}	Einstein's coefficient
B_F	Modal birefringence
B_{max}	Maximum 3-dB bandwidth
B_{mod}	Bandwidth of a fiber limited by modal dispersion
B_r	Recombination coefficient
B_T	Transmission bandwidth of FM
b	Normalized propagation constant
C	Capacitance, constant
C_a	Input capacitance of the receiver amplifier
C_d	Capacitance of the photodetector
C_f	Capacitance associated with the feedback resistor
C_j	Junction capacitance of the photodetector
C_T	Total capacitance
c	Velocity of light in free space
C_t	Transmitter coupling loss
C_r	Receiver coupling loss
D	Total intramodal/chromatic dispersion of a fiber in ps nm^{-1}
D_{mat}	Material dispersion of a fiber in ps nm^{-1} km^{-1}
D_{tot}	Total material dispersion in ps nm^{-1} km^{-1}
D_{wg}	Waveguide dispersion of a fiber in ps nm^{-1} per km^{-1}
D_{pro}	Profile dispersion
D_{PMD}	Average value of polarization mode dispersion measured in ps/$\sqrt{}$km
D_f	Frequency deviation ratio of FM

Symbol	Meaning
D_L	Dispersion equalization penalty
d	Diameter of the fiber core
E	Electronic energy
E_F	Fermi energy
E_g	Bandgap energy
F_n	Noise figure
$F(M)$	Excess noise factor
f	Frequency
f_m	Frequency of the modulating signal
f_{LO}	Local oscillator frequency
f_{IF}	Intermediate frequency
f_{3dB}	3-dB bandwidth of a photodetector
G	Open loop gain
G_s	Single-pass gain
g	Degeneracy factor; laser cavity gain
g_m	Transconductance of FET
g_{th}	Threshold gain for lasing
$H(f)$	Transfer function
$H_{OL}(f)$	Open loop transfer function
$H_{CL}(f)$	Closed loop transfer function
$H_{out}(f)$	Transfer function of optical receiver output
h	Planck constant
h_{eff}	Effective height of the planar waveguide
$h_p(t)$	Input pulse shape to an optical receiver
$h_{out}(t)$	Output pulse shape
I	Electrical current/optical intensity
I_B	Current produced in a photodetector due to background radiation
I_D	Dark current in a photodetector
I_{eq}	Equivalent photodetector current
I_P	Average photocurrent
I_{th}	Threshold current for lasing of ILD
I_0	Reverse saturation current
i_{det}	Photodetector current
$i_p(t)$	Time-varying photogenerated current in a photodetector
i_{det}	Output current of a photodetector
$\langle i_s^2 \rangle$	Mean square value of the shot-noise current
$\langle i_T^2 \rangle$	Mean square value of the thermal noise current

Symbol	Meaning
$J_n(x)$	Bessel function of the first kind of order n and argument x
J_{th}	Threshold current density of an ILD
j	$\sqrt{(-1)}$
k	Boltzmann constant/free space propagation constant/ratio of the ionization coefficient
k_1	Propagation constant in the core region
k_2	Propagation constant in core region
L	Fiber length
L_B	Beat length in a single- mode fiber
L_{RS}	Loss due to Rayleigh scattering
S_l	Splice loss in dB
D_L	Dispersion power penalty
P_T	Transmitted power
P	Total power carried by the fiber
P_{core}	Power carried by fiber core
P_{clad}	Power carried by fiber cladding
P_R	Received power
t_r	Rise-time of an RC circuit
T_R	Receiver rise-time in nano-second
T_C	Fiber rise-time due to chromatic dispersion
t_{FWHM}	Time corresponding to full-width half-maximum
$p(t)$	Gaussian pulse in the time domain
$P_{op}(t)$	Time-varying optical power from of an intensity-modulated optical transmitter
$P(f)$	Fourier transform of the pulse $p(t)$
R_T	Effective resistance value of bias and amplifier resistances
T_{sys}	System rise-time in nano-second
q	Electronic charge
i^2_{signal}	Mean square value of the signal current
F_n	Noise figure of an amplifier
S	Signal power
N	Noise power; Group refractive index
S/N	Signal-to-noise power ratio
m^2	Mean square gain of an APD
M	Average gain of an APD; total number of modes in a fiber
M_1	Mean value (first temporal moment)
M_2	Mean-square value (second temporal moment)
M_{eff}	Effective number of modes in a curved graded-index fiber
p	Average photoelectric constant for Rayleigh scattering

Symbol	**Meaning**
P_B	Threshold optical power for stimulated Brillouin scattering
P_R	Threshold optical power required for stimulated Raman scattering
R_a	Amplifier resistance
R_b	Photodetector bias resistance
R_L	Load resistance
R_c	Critical radius of curvature for macro-bending of a multimode fiber
R_{cs}	Critical radius of curvature for macro-bending of a single mode fiber
r	Electro-optic coefficient
T_F	Fictive temperature
T_{max}	Time taken by the most oblique ray to travel a length L along the fiber
T_{min}	Time taken by the axial ray to travel a length L along the fiber
N_1	Group refractive index in the core of a fiber
n	Refractive index of any material
n_1	Core refractive index
n_2	Cladding refractive index
$n(r)$	Refractive index of the fiber material as a function of radial distance
$n(0)$	Refractive index of a graded-index fiber at the centre or axis of the core
$n(a)$	Refractive index of a graded-index fiber at the core-cladding interface and inside cladding
$n(E)$	Electric field dependent refractive index
V	V-number or normalized frequency of a fiber
$K_n(x)$	Modified Bessel function of the second kind of order n and argument x
V_π	Voltage required to create a phase difference of π in an electro-optic modulator
P	Total optical power carried by a fiber
P_{core}	Power carried by the core of the fiber
P_{clad}	Power carried by the cladding of the fiber
T	Absolute temperature
T_b	Bit period
B_T	Bit rate
B_e	Bit-error rate
n_0	Electron concentration in thermal equilibrium
p_0	Hole concentration in thermal equilibrium
N_c	Effective density of state in the conduction band
N_v	Effective density of state in the valence band
m^+n	Effective mass of electrons
m^+p	Effective mass of holes
E_c	Conduction band energy
E_v	Valence band energy

Symbol	**Meaning**
E_F	Fermi level energy
n_i	Intrinsic carrier concentration
N_D	Donor concentration
N_A	Acceptor concentration
V_0	Built-in potential
J	Bias current density
d	Thickness of the active layer
r_r	Radiative recombination rate
r_{nr}	Non-radiative recombination rate
R_r	Total radiative recombination rate
R_{nr}	Total non-radiative recombination rate
P_{int}	Optical power generated within LED
N_T	Trap density
v_{th}	Thermal velocity
R_{AU}	Rate of Auger recombination
C	Auger recombination coefficient
$T(0)$	Fresnel transmission coefficient for normal incidence
$P_{emitted}$	Emitted optical power
S	Surface recombination velocity
D_n	Diffusion coefficient of electrons
L_n	Electron diffusion length
f_{3dB}	3-dB bandwidth
$f_{(3dB\text{-}op)}$	3-dB optical bandwidth
$f_{(3dB\text{-}el)}$	3-dB electrical bandwidth
E_1	Energy corresponding to state 1
E_2	Energy corresponding to state 2
N_1	Population in the energy state E_1
N_2	Population in energy state E_2
R_{12}	Rate of absorption in presence of a radiation field
$(R_{21})_{sp}$	Spontaneous emission rate
$(R_{21})_{st}$	Spontaneous emission rate
D	Normalized waveguide thickness
$I(z)$	Optical field intensity at any point z
$g(0)$	Gain of the cavity at the peak wavelength
R_1	Reflection coefficient of the front mirror of the FP laser
R_2	Reflection coefficient of the rear mirror of the FP laser
t_d	Time delay between the application of the current pulse and attaining lasing threshold
f_r	Relaxation oscillation frequency

Symbol	**Meaning**
$\langle P_e^2 \rangle$	Square of the mean optical power
$S_{RIN}(f)$	Power spectral density of relative intensity noise
P_s	Power emitted by the source
P_F	Power coupled to the fiber
$P_{coupled}$	Power coupled to the fiber after Fresnel reflection
A_f	Fiber area
n_{th}	Threshold carrier density
r_s	Radius of the circular emitting area of the source
r_L	Radius of the micro-spherical lens
v_{in}	Input voltage
v_{out}	Output voltage
v_g	Group velocity
f_p	Fraction of the bundle area covered by the fiber core
L_F	Loss encountered at the fiber-to-fiber joint
L_{lat}	Loss due to lateral misalignment of the fibers
R_f	Feedback resistance of the TZ amplifier
$\boldsymbol{\mathcal{B}}$	Magnetic induction
$B(\theta,\Phi)$	Brightness or radiance of an optical source
$\langle \delta P_e^2 \rangle$	Mean square value of power fluctuation
$\boldsymbol{\mathcal{D}}$	Electric displacement
$\mathcal{E}$	Electric field vector
$\mathcal{H}$	Magnetic field
$\mathcal{L}$	Total loss/attenuation of a fiber in dB
α_f	Average loss in the fiber per km
τ_e	(1/e) –full width pulse broadening due to dispersion
τ_{wg}	Group delay due to waveguide dispersion
τ	Mean lifetime of carriers
α	Absorption coefficient per km
σ	RMS pulse spreading due to dispersion; capture cross-section
$\sigma_{intramodal}$	RMS pulse spreading due to intramodal dispersion
σ_{modal}	RMS pulse spreading due to intermodal dispersion
τ_{mat}	Group delay due to material dispersion
σ_{mat}	RMS pulse spreading due to material dispersion
σ_λ	RMS spectral width of the source
σ_0	RMS pulse broadening in absence of mode coupling
σ_c	RMS pulse broadening in presence of mode coupling
$\mathcal{R}$	Responsivity of a photodetector; Fresnel reflection coefficient

Symbol	**Meaning**
$\delta\tau_{pol}$	Delay difference because of polarization mode dispersion
τ_{gx}	Group delay in *x*-direction (H-polarization)
τ_{gy}	Group delay in *y*-direction (V-polarization)
v_{gx}	Group velocity in *x*-direction
v_{gy}	Group velocity in *y*-direction
μ	Index of modulation in the case of intensity modulation
ν	Frequency of the optical signal
η	Quantum efficiency
η_{int}	Internal quantum efficiency
η_{ext}	External quantum efficiency
δ_{pol}	Differential group delay due to birefringence
δT_{mod}	Delay difference between the highest and lowest order mode
α	Attenuation (dB/km) in a fiber; absorption coefficient; electron ionization coefficient; profile index factor of a graded-index fiber
α_n	Attenuation coefficient of a fiber in nepers
$\alpha_I(l,m)$	Attenuation coefficient for a mode of order (l, m)
$\alpha(r)$	Attenuation coefficient at any radial distance r from the centre of a graded-index fiber
α_r	Bending loss
α_{IR}	Attenuation due to infrared absorption
α_{SR}	Attenuation in dB due to Rayleigh scattering
α_{uv}	Attenuation due to ultraviolet absorption
α	Average loss in a laser cavity
$\alpha(\lambda)$	Absorption coefficient of a material as a function of wavelength
Γ	Confinement factor
β	Spontaneous emission coefficient; hole ionization coefficient; z-component of the propagation constant
β_c	Isothermal compressibility
γ_R	Rayleigh scattering coefficient
ϕ_c	Critical angle measured with respect to the normal drawn on the plane of incidence
Φ_s	Total phase-shift associated with a single-pass optical amplifier
Δ	Refractive index deviation
λ	Wavelength of light
θ_c	Critical angle measured with respect to the axis of the fiber
$\theta_{0\,max}$	Maximum acceptance angle for rays to enter into the fiber
Ω	Solid acceptance angle of a fiber

Symbol	**Meaning**
ω	Angular frequency
ζ	Electro-optic coefficient
τ_{wg}	Time delay due to waveguide dispersion
ΔE_g	Energy bandgap difference in a heterojunction
ΔE_c	Energy corresponding to conduction band-edge discontinuity
ΔE_v	Energy corresponding to valence band-edge discontinuity
δ_p	Energy separation between the Fermi level and the valence band edge
δ_n	Energy separation between the Fermi level and the conduction band edge
τ_{SRH}	Mean lifetime of the carriers due to Shockley-Read-Hall recombination
τ_{AU}	Mean lifetime of carriers due to Auger recombination
Δn	Excess photogenerated carriers (electrons)
τ	Mean lifetime of the carriers
τ_r	Mean lifetime of the carriers for radiative recombination
τ_{nr}	Mean lifetime of the carriers for non-radiative recombination
τ_{eff}	Effective mean lifetime of the carriers
ω_{3dB}	3-dB angular frequency bandwidth
$\rho(\nu)$	Radiation field density
$\Delta\nu$	Frequency separation between two adjacent modes of a FP laser diode
$\Delta\lambda$	Wavelength separation between two adjacent modes of a FP laser diode
τ_{sp}	Spontaneous emission lifetime of the carriers
τ_{st}	Stimulated emission lifetime of carriers
τ_{ph}	Photon lifetime of the carriers
Φ	Photon flux density
Φ_s	Steady-state photon flux density
$T(\Phi)$	Fresnel transmission coefficient for an angle of incidence of Φ
$P(\omega)$	Power spectrum of LED output
$g(\lambda)$	Gain of the laser cavity as a function of wavelength
$m(\beta)$	Number of modes in a graded-index fiber for a given propagation constant
$\theta_{\parallel}$	Angular width of the beam in direction parallel to the plane of the *p*-*n* junction
$\theta_{\perp}$	Angular width of the beam in the direction perpendicular to the plane of the *p*-*n* junction
η_F	Fiber-to-fiber coupling efficiency
η_{lat}	Coupling efficiency due to lateral misalignment

List of Abbreviations

Abbreviation	Meaning
A/D	Analog to digital
AC	Alternating current
AGC	Automatic gain control
AM	Amplitude modulation
APD	Avalanche photodiode
ASK	Amplitude shift keying
ATM	Asynchronous transmission mode
BER	Bit error rate
CATV	Common antenna television
CCITT	International Telephone and Telegraph Consultative Committee
CCTV	Closed circuit television
CNR	Carrier-to-noise ratio
CW	Continuous wave
CWDM	Coarse wavelength division multiplexing
D/A	Digital to analog
dB	Decibel
DBR	Distributed Bragg reflector
dBm	Decibel with reference to 1 mW power
D-IM	Direct intensity modulation
DC	Depressed cladding
DC	Direct current
DF	Dispersion flattened
DFB	Distributed feedback
DH	Double heterostructure
DPSK	Differential phase shift keying
DS	Dispersion shifted
DSB	Double sideband
DWDM	Dense wavelength division multiplexing
EDFA	Erbium doped fiber amplifier
EIA	Electronics Industries Association
ELED	Edge-emitter light emitting diode
EMI	Electromagnetic interference
erf	Error function
erfc	Complementary error function
FA	Fiber amplifier

Abbreviation	Meaning
FBT	Fiber biconical taper
FDM	Frequency division multiplexing
FET	Field effect transistor
FM	Frequency modulation
FOTP	Fiber optic test procedure
FPA	Fabry-Pérot amplifier
FSK	Frequency-shift keying
FWHP	Full width half power
FWHM	Full-width half maximum
GI	Graded index (fiber)
GRIN	Graded-index (rod lens)
HB	High birefringence
HBT	Heterojunction bipolar transistor
HEMT	High electron mobility transistor
He-Ne	Helium-Neon (LASER)
NF	High frequency
IF	Intermediate frequency
ILD	Injection laser diode
IM-DD	Intensity modulation–direct detection
IO	Integrated optics
I–O	Input-output
ISI	Intersymbol interference
ISDN	Integrated services digital network
LAN	Local area network
LB	Low birefringence
LED	Light-emitting diode
LP	Linearly polarized
LPE	Liquid phase epitaxy
MAN	Metropolitan area network
MBE	Molecular beam epitaxy
MC	Matched cladding
MCVD	Modified chemical vapor deposition
MESFET	Metal semiconductor field effect transistor
MFD	Mode field diameter
MISFET	Metal insulator field effect transistor
MMF	Multimode fiber
MOSFET	Metal-oxide semiconductor field effect transistor
MOVPE	Metal organic vapor phase epitaxy
MQW	Multiquantum well
MSM	Metal-semiconductor-metal

Abbreviation	Meaning
NRZ	Non-return to zero
OEIC	Optoelectronic integrated circuit
OFDM	Optical frequency division multiplexing
OOK	On-off keying
OTDR	Optical time domain reflectometer
OVPO	Outside vapor phase oxidation
PCM	Pulse code modulation
PCS	Plastic clad silica (fiber)
PCVD	Plasma-activated chemical vapour deposition
PD	Photodiode
PDF	Probability distribution function
p-i-n-PET	p-i-n-photodetector followed by PET
PLL	Phase locked loop
PM	Phase modulation
PMF	Polarization maintaining fiber
PoLSK	Polarization shift keying
PON	Passive optical network
PSK	Phase-shift keying
RAPD	Reach-through avalanche photodiode
RIN	Relative intensity noise
RMS (rms)	Root mean square
RO	Relaxation oscillator
RZ	Return to zero
SAM	Separate absorption and multiplication (APD)
SAW	Surface acoustic wave
SBS	Stimulated Brillouin scattering
SC	Subcarrier connector
SCM	Subcarrier multiplexing
SDH	Synchronous digital hierarchy
SDM	Space division multiplexing
SHF	Super high frequency
SI	Step-index (fiber)
SL	Superlattice
SLA	Semiconductor laser amplifier
SLD	Semiconductor laser diode
SLED	Surface emitting LED
SMF	Single mode fiber
SNR	Signal-to-noise ratio
SONET	Synchronous optical network
SOP	State of polarization

Abbreviation	Meaning
SQW	Single quantum well
SRS	Stimulated Raman scattering
ST	Straight tip
TDM	Time-division multiplexing
TDMA	Time division multiple access
TE	Transverse electric
TEM	Transverse electromagnetic
TM	Transverse magnetic
TWA	Traveling wave amplifier
UHF	Ultra high frequency
VAD	Vapour axial deposition
VCO	Voltage controlled oscillator
VHF	Very high frequency
VPE	Vapour phase epitaxy
VSB	Vestigial sideband
WDM	Wavelength-division multiplexing
WKB	Wenzel-Kramer-Brillouin (method)
ZMD	Zero material dispersion
ZTD	Zero total dispersion

Introduction

CHAPTER 1

Optoelectronic devices are electronic devices that convert an optical signal (light) to an electrical signal (O–E conversion) or an electrical signal to an optical signal (E–O conversion). In the electrical domain, a signal can be processed with the help of electrical technology using electronic components and circuits. The electrically processed signal can be subsequently converted to an optical signal (E–O) with the help of an appropriate optoelectronic device (optical emitter). Similarly, an electrical signal once converted to an optical signal can be processed with the help of several optical (photonic) devices and components. The processed optical signal can be taken back to the electrical domain with the help of a suitable optoelectronic device (optical detector). The frequent conversion of the signal from one domain to the other (E–O and O–E) is an integral part of a conventional optical fiber communication system. However, optoelectronic devices can be deployed in both telecommunication and non-telecommunication applications, these devices are primarily fabricated using semiconductor materials and are mostly available in the form of solid-state devices even though photomultiplier devices are also available in the form of vacuum tubes. The major optoelectronic devices include photodiodes (photodetectors and solar cells), optical sources or emitters (light-emitting diode, injection laser diode), phototransistors, photoconductors, photoresistors, semiconductor laser amplifiers. The present text focuses on optoelectronic devices and other photonic components like optical modulators, switches, attenuators, fiber amplifiers, and electro-optical switches which are key components of optical fiber communication systems. For applications in optical fiber communication systems, optoelectronic devices are designed to operate in the near-infrared (NIR) region of the optical spectrum. Optoelectronic devices for several non-telecommunication applications are also available for operation in the ultraviolet (UV), mid-infrared (MIR), and long wavelength infrared (LWIR) regions of the optical spectrum (Krier, 2004; Ali, 2010). Long-wavelength infrared sources and detectors are also used for free-space optical communication systems.

The purpose of telecommunication or simply communication is to establish a link (wired or wireless) between two distant points to transfer information (or message) from one point to the other (either one way or both ways). This transfer of information from the source to the destination point can be efficiently carried out with the help of electrical technology. The arrangement required for establishing an electrical communication link between two distant points is called a communication system. The fundamental elements of a typical communication system are shown in Fig. 1.1.

In practice, information at the source point may appear in a variety of forms like voice, speech, music, picture (static or video), or data. To make the information available in a non-electrical form suitable for electrical communication we need to convert it into an electrical form with the help of a suitable transducer as shown in Fig. 1.1.

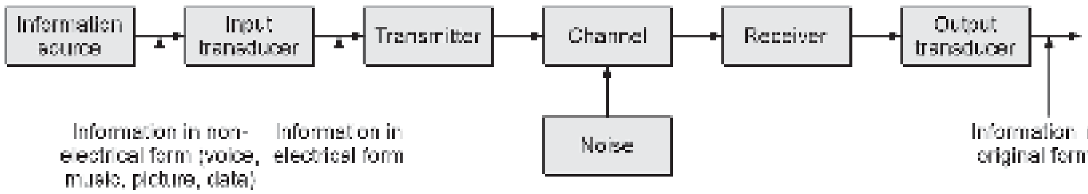

Fig. 1.1 Block diagram of an electrical communication system

The electrical form of the information is subsequently used for transmission. The transfer of information in the electrical form is generally achieved by modulating the information signal with the help of a high-frequency electromagnetic wave which acts as a carrier. This modulation process is done by the transmitter unit. The transmitter then couples the modulated signal onto the transmission channel. Before transmission one needs to ensure that the characteristics of the channel support the modulated information signal and do not cause unacceptable degradation of the signal during propagation of the signal through the channel. The channel is considered as a medium that links the transmitter and the receiver. The channel may be a guided transmission link such as a twin-wire line, coaxial cable, waveguide, or even an unguided atmospheric or space channel. When the transmitted signal travels towards the destination point it is subjected to two major constraints, e.g. noise added in the channel and the attenuation of the signal caused by the absorption of the signal by the channel. While the attenuation can be easily compensated by boosting the signal with the help of signal boosters, the management of noise added by the channel can be quite challenging in some cases (Chakrabarti, 2010; 2018). The transmitted signal to the receiver via the channel then undergoes a reverse process by which the original message signal is extracted from the modulated signal. The process is known as demodulation. The *demodulated* electrical information is finally converted in the original form with the help of a proper transducer.

The information is finally delivered to the message destination. The complexity of this simple arrangement arises from the fact that the transmitted signal gets progressively weakened (attenuated) and mutilated by noise during propagation through the channel. As a result, the receiver has to process a weak and mutilated signal to extract the original message signal. This makes the design of a communication receiver much more complex as compared to the transmitter. Ultra-sophisticated systems have been developed over the past decades for the successful implementation of communication links using electromagnetic waves at radio frequencies, microwave, and millimeter wave frequencies. The successful demonstration of a laser source by Maiman (Maiman, 1960) motivated researchers to use a laser beam as a carrier for the information paving a new avenue for another mode of communication called *optical communication*. In general, optical communication links can be implemented in a variety of forms, for example, it can be in an unguided form (free space optical communication) or a guided form (optical fiber or fiber optic communication). Moreover, light can be used either for intensity modulation by the modulating signal followed by direct detection (IM/DD) or as a coherent carrier in a manner analogous to electrical communication (coherent optical communication). A communication system that uses light as a carrier for the transportation of information from the source to the destination is called an optical communication system.

1.1 GENESIS OF ELECTRICAL AND OPTICAL COMMUNICATION

The era of electrical communication began with the invention of the telegraph by Samuel FB Morse in 1838 (Agrawal, 1995). The first commercial telegraph service could be implemented in 1844. In 1874, Alexander Graham Bell successfully demonstrated the conversion of sound waves into electric current through a small magnet and patented his primitive telephone set in 1876. These two major discoveries paved the way for electrical communication. On the other hand, the feasibility of optical communication could be envisioned nearly after a century with the invention of the LASER (light amplification by stimulated emission of radiation) by Maiman in 1960 (Maiman, 1960). Ironically the earliest mode of communication used by human beings happened to be some form of optical link. In ancient times (the eighth century BC) fire signals were used by the Greeks for sending alarms, distress calls, or making public announcements of events. Later on around 150 BC alphabets were encoded using different optical signals to exchange messages based on some prearranged understanding (protocol). In this mode of optical communication, the human eye was used to work as a receiver. As a result, the speed of this mode of communication was extremely poor and the practical application of this mode was further constrained by the requirement of line-of-sight transmission and the presence of obstacles in the path because of rain, fog, and other atmospheric disturbances. This mode of communication could not be pursued afterward because of technological limitations.

1.1.1 Emergence of Optical Communication

Following the invention of telegraphy in 1838 and the successful implementation of commercial telegraph service in 1844, the first telephone exchange was established in New Haven in 1878. At this time wire cable was the only medium used as the transmission channel. In 1880, Alexander Graham Bell, reported the transmission of speech using a light beam as the carrier and the atmosphere as the transmission medium (Bell, 1876). However, as mentioned earlier, the transmission of light beams through the atmosphere was restricted to line-of-sight paths and affected atmospheric disturbances such as rain, fog, etc. Most importantly, the non-availability of a proper optical source then severely impaired the emergence of optical communication in the early part of the twentieth century (Allard, 1989). James Maxwell theorized a mathematical interpretation of electromagnetic waves in 1864. Later on, in 1884, Heinrich Hertz discovered long-wavelength radiowaves and demonstrated the applicability of Maxwell's theory. Hertz's demonstration revolutionized the concept and scope of electrical communication. In 1895, Marconi G, demonstrated the transmission of radiowaves through free space. In the ensuing years lower frequency (longer wavelength) electromagnetic waves (radio and microwaves) turned out to be suitable carriers for message transmission via the atmospheric channel. Such transmissions are less sensitive to variations in the atmospheric conditions. The era of wireless electrical communication continued to flourish in the ensuing decades. The electromagnetic carrier waves could be transmitted over considerably large distances without having significant attenuation or distortion. However, the information-carrying capacity of these high-frequency electromagnetic waves is directly related to their frequency[1].

In principle, the higher the frequency of the carrier, the larger the transmission efficiency and consequently the higher the information-carrying capacity of the communication system. This fact has been the driving force behind the development of subsequent wireless electrical communication systems that used progressively higher frequencies starting from VHF (very high frequency), and UHF (ultra high frequency) to microwaves and finally to millimeter wave for transmission. The relative frequency and wavelength of various types

[1] The information carrying capacity is directly related to the bandwidth of the modulated carrier which is a fixed fraction of the carrier frequency.

of electromagnetic waves are illustrated in Fig. 1.2. It can be seen that the transmission media used in different ranges of frequency spectrum include metallic wires coaxial cables,

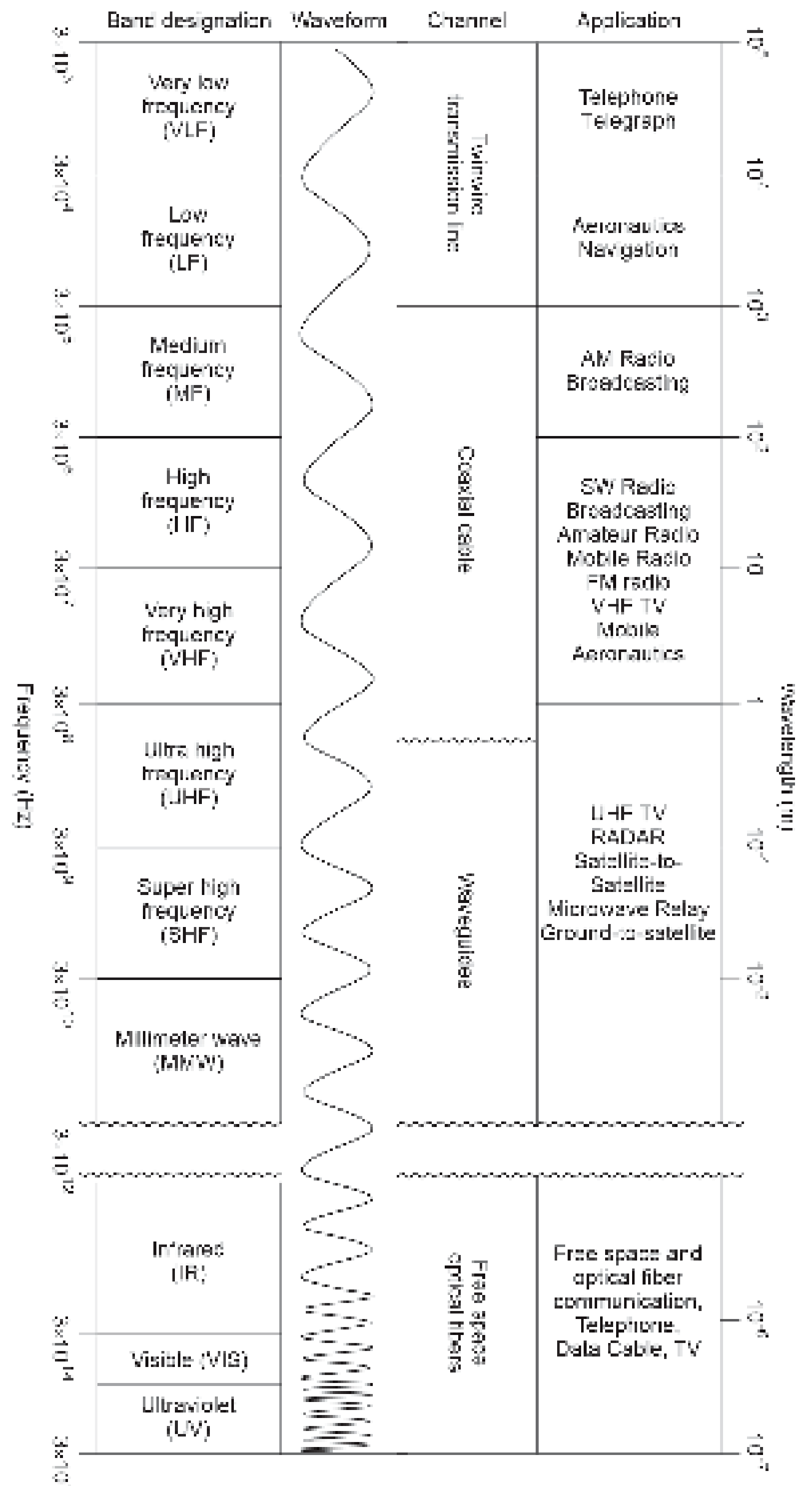

Fig. 1.2 EM spectrum

microwave, and millimeter-wave waveguides, and radiowaves that utilize atmosphere as the channel. The electrical communication systems that utilize various ranges of the electromagnetic spectrum include telephone (landline as well as mobile), AM and FM radio, transmission, radar, satellite-to-satellite links, etc. This frequency range utilized in various commercial applications extends from about 300 Hz in the audio band to 90 GHz in the millimeter wave band.

It is interesting to note here that if the frequency of the electromagnetic carrier is further pushed upwards to encompass the optical region of the electromagnetic spectrum, the bandwidth of the existing microwave transmission can be increased by a factor of 10^4. This would lead to an enormous information-carrying capacity for the new communication system. The system that uses an optical signal as the carrier for transporting the message signal from the source to the destination is called the *optical communication system.* As it stands today this mode of communication has several distinct advantages over conventional electrical communication. For optical communication, it is customary to specify the range of the electromagnetic spectrum in terms of wavelength rather than frequency. However, similar to conventional communication electrical systems, both waveguides and atmospheric channels can be used for the transmission of optical signals in optical communication. The waveguide used in optical communication is generally an optical fiber. Optical communication is thus classified as free space optical communication and optical fiber communication depending on whether a free space channel or a fiber waveguide is used as the transmitting medium.

1.1.2 Evolution of Optical Fiber Communication

This section reviews the most important discoveries that laid the foundation for modern optical communication. The first significant contribution may be traced back to 1917 when Albert Einstein mathematically formulated the conditions for stimulated emission. In 1955 Towns observed stimulated emission leading to microwave amplification. In the subsequent year, Bloembergen demonstrated MASER (microwave amplification by stimulated emission of radiation). The first solid-state MASER was developed by Bell Labs in 1957. Maiman T, demonstrated LASER operation in ruby rods at Hughes research labs in 1960 (Maiman, 1960). By 1962, many of the research labs including IBM and GE succeeded in developing semiconductor laser sources. The invention of the laser source by Maiman in 1960 led to the availability of a coherent optical source operating at a frequency of the order of 5×10^{14} Hz. The invention created interest among researchers in exploring the potential of an optical signal being used as a carrier of message signals in the optical communication system. The early 1960s witnessed several interesting experiments carried out by researchers using the atmosphere as the optical channel (Davis, 1996). It was demonstrated that a coherent optical carrier can be modulated at very high frequencies. Although the low beam divergence of the laser beam extended the free space transmission distance, the high installation cost together with limitations imposed by atmospheric obstructions such as rain, fog, etc. finally made this high-speed system unattractive for practical applications. Today, the application of unguided optical communication (one that uses free space atmosphere as the channel) is restricted to linking a TV camera to a base vehicle, for baseband data linking over a few hundred meters between buildings, and also to long distance earth-to-space and satellite-to-satellite linking (Davis, 1996; De Cusatis *et al* 1998; Fowles, 1975; Gower, 1984; Keiser, 1991; Green, 1993; Hoss, 1993; Senior, 1992; Snyder & Love 1983; Wilson & Hawkes, 1989).

The limitations of free space optical communication using laser as an optical source motivated the researchers to explore the guided transmission of the optical signal via a dielectric waveguide or optical fiber made from glass as a channel. In 1966, Kao[2] and Hockham of proposed dielectric fiber waveguides for the transmission of signals at optical frequencies (Kao *et al* 1966) to avoid the degradation of optical signals by the atmosphere. In principle, this type of guided optical communication should be much more reliable and versatile as compared to free-space optical communication. However, early fibers measured high transmission losses to the tune of 1000 dB/km or so. This extremely high loss associated with the early glass fibers fabricated using traditional glass-making methods was considered impractical for system implementation. Kao and Hockham, attributed this high loss associated with the fiber to impurities present in the glass. It was not until 1970 that Felix Kapron *et al* at Corning Glass Works successfully fabricated a silica fiber having a loss of 20 dB/km at 850 nm. This improvement made the optical fibers look like a viable transmission medium. In subsequent years further improvement in optical fiber fabrication technology reduced the loss from 20 dB/km to 1 dB/km at 1300 nm by 1976. This led to the commercial implementation of the first optical fiber communication link in 1978. By 1982, the fiber loss was brought down to 0.5 dB/km at 1550 nm. Thousands of kilometers of optical fiber lines were installed worldwide by the early 1980s. The other problem associated with guided optical communication in those days included difficulties in satisfactorily jointing the fiber cables. The joint loss used to be significantly high. In the following two decades (1970–1990) extensive research work was carried out worldwide to improve the quality of fiber, reduce the loss, and devise new techniques for jointing fibers. At present, fiber fabrication technology is mature enough to yield silica-based optical fibers that provide an attenuation as low as 0.16 dB/km which is very close to the theoretical limit of 0.14 dB/km.

The growth of optical fiber communication has been sustained by a parallel development in the area of semiconductor devices and technology which provided the necessary light sources, photodetectors, and associated electronic circuits for the optical communication system. Design and development of semiconductor optical sources in the form of semiconductor injection laser diode (ILD), light-emitting diode (LED), and semiconductor photodetectors such as *p-i-n* photodetector, avalanche photodiodes, MSM photodiodes, phototransistors which are compatible in size with the optical fibers (of the order of 100 μm diameter) have been instrumental in successful implementation of practical optical fiber communication link. The semiconductor laser sources of the early 1960s were unsuitable for practical application because of their inability to operate continuously for long hours at room temperature. The early and mid-1970s witnessed dramatic development in the field of semiconductor laser sources and made the fabrication of laser sources with a lifetime of several thousand hours possible. These sources were made of AlGaAs (aluminum-gallium-arsenide), a ternary III-V alloy that emits light in the range of 800–900 nm.

Ever since the successful breakthrough in the development of low-loss optical fibers in 1970, the area of optical fiber communication progressed steadily toward its present maturity. Hundreds of thousands of kilometers of optical fiber cables were installed worldwide in three decades after 1970. Over the past three decades optical communication systems have gone through several different generations of technology which is primarily distinguished based on operating wavelength. The fibers available in the early 1970s

[2] Kao has been awarded 2009, Nobel Prize, along with two others in Physics for his contribution in the area of optical fiber communication.

exhibited low attention loss (~ 5 dB/km) window near 0.8 μm (800 nm). The first generation (1G) optical communication links operated at 800 nm using GaAs-based optical sources and silicon photodetectors. Optical fibers used in this generation were multimode silica fibers. The 1G optical links were used primarily in telephone systems in the United States, parts of Europe, and Japan. The operating bit rate in the 1G system ranged between 30–140 Mbps. The repeater spacing in this generation was nearly 10 km. The availability of good quality optical fibers with extremely low attenuation (~0.5 dB/km) at 1300 nm in the early 1980s motivated the researchers and design engineers to shift the operating wavelength from 800 nm (in the 1G system) to 1300 nm in the second generation (2G) system. This shift resulted in a substantial increase in the repeater spacing thus making optical communication links quite beneficial for long-haul communication, especially telephone trunks. The 2G optical fiber communication system also found applications in intercity links and local area networks (LANs) using multimode silica fiber. The quaternary III-V alloy InGaAsP was available by that time in semiconductor industries to provide laser diodes and light-emitting diodes at this operating wavelength with lifetimes of 25 years and 100 years respectively. As single-mode fibers were found to exhibit lower loss and significantly large bandwidth (because of extremely low dispersion), the 2G system soon switched over from the use of multimode to single-mode fiber particularly in long-haul communication. Bit rates over 500 Mbps and some cases upto 4 Gbps have been reported with a typical repeater spacing of 40 km. The first transatlantic system (TAT-8) operating at 1300 nm was installed in 1988.

Silica-based optical fibers were found to offer the lowest attenuation of 1550 nm. The variation of attenuation of silica-based optical fiber with wavelength is shown in Fig. 1.3, the three windows are shown in the figure as shaded regions.

The fact that silica fiber offers the least attenuation at 1550 nm motivated the design engineers to consider shifting the operating wavelength to 1550 nm. However, a major constraint on optical communication at this wavelength was initially found to be higher signal dispersion in silica fiber as compared to that at 1300 nm. However, this

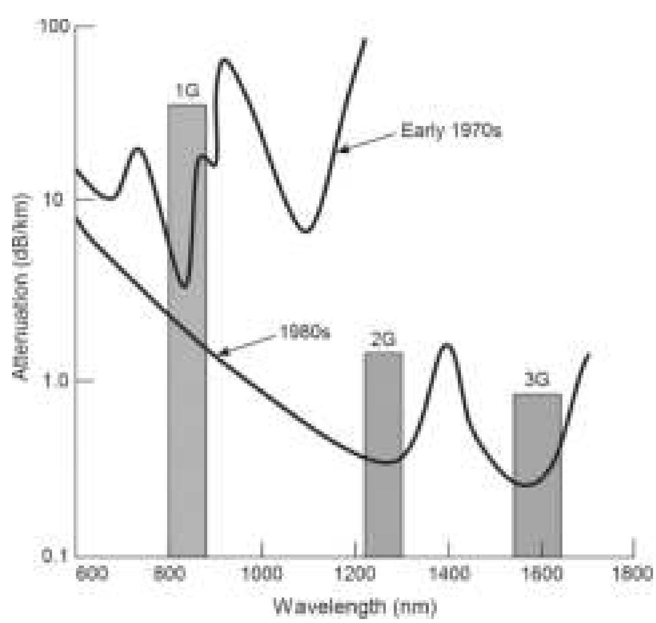

Fig. 1.3 Attenuation of the optical fibers of the 1970s and 1980s

dispersion problem could be tackled easily by making use of artificial techniques for shifting or flattening the minimum dispersion wavelength to the desired value. The third generation (3G) fiber optic system finally adopted 1550 nm as the operating wavelength for high capacity, long-haul, and under-sea optical fiber links. This change of operating wavelength from 1300 nm to 1550 nm in 3G system was possible because of the availability of reliable sources at photodetector based on InP/InGaAs technology. The 3G optical fiber communication system makes use of both an intensity modulation-direct detection (IM/DD) scheme as well as a coherent optical communication scheme. The former scheme involves linear modulation of the intensity of the light source by the input electrical signal (message) at the transmitter and subsequent reconversion of the intensity-modulated signal to the corresponding electrical signal (message) with the help of a photodetector acting as a photon counter at the receiver. In this mode, no attention is paid to the frequency or phase of the optical carrier. This simple mode of optical fiber communication is most popular even though it suffers from limited sensitivity and its inability to take full advantage of the enormous bandwidth of the optical fibers. Coherent optical communication on the other hand involves modulation of amplitude, frequency, phase, or polarization of the optical signal from the light source following the modulating electrical signal (message). At the receiver end the modulated signal is demodulated using a coherent detection technique which is very similar to a conventional electrical communication receiver. A coherent optical fiber communication system offers a significant improvement in the receiver sensitivity over the IM/DD system and enables one to use electrical equalization techniques to compensate for the effect of optical dispersion in fibers. However, there are some practical difficulties with a coherent optical system that has restricted the widespread adoption of this type of optical link in commercial applications.

Theoretically, optical fiber communication systems have unlimited information carrying capacity (~50 THz in practice). However, standard electrical interface schemes such as time division multiplexing (TDM) in the form of synchronous optical network (SONET) or synchronous digital hierarchy (SDH) impose limits on the overall data transmission rate of the optical network. This is due to the limited data-handling capabilities of associated electrical circuits involving amplifiers, multiplexers, demultiplexers, regenerators, etc. The major bottleneck of the present system arises from the frequent conversion of the signal from optical to electrical domain.

The field of optical fiber communication is undergoing dramatic developments even today. There has been a continuous improvement in transmission technology with respect to speed, reliability, and cost. The immense bandwidth potential of optical fibers (~50 THz) has constantly motivated researchers to explore various possibilities for further upgradation of state-of-the-art technology. The information-carrying capacity of optical fibers is doubling almost every two years. It appears that this field of technology would need many more decades to attain the projected goal of information superhighway (optical link). Research carried out in the recent past indicates that there is a possibility of developing systems that would make use of all processing in the optical domain by making use of all-optical switches, couplers, repeaters, etc. This would drastically improve the speed of the existing systems that are constrained by the frequent use of optical-to-electrical (O–E) and electrical-to-optical (E–O) conversions. Further research in the area of soliton transmission opened up a new avenue for improving the information-carrying capacity of optical fibers. A soliton is a special type of non-dispersive pulse that makes judicious use of non-linear effects in the fiber to compensate for the chromatic dispersion of the fiber. It is projected that by making use of dispersion-shifted fiber it would be possible to use soliton pulses to transmit data virtually error-free at 50 Gbps over a distance of 20000 km without any repeater.

1.1.3 Classification of Optical Communication Systems

Optical communication transmission systems can be classified the basis of a variety of criteria such as spectral wavelength range of operation, number of channels (single or multiple), bit rates, type of WDM, channel spacing, nature of application, interface characteristics, etc. From the discussion in the foregoing section, it can be seen that there has been tremendous development in the design and implementation of optical telecommunication links in different countries. Unfortunately, there was a lack of uniformity and compatibility of equipment manufactured by different companies. The International Telecommunication Union–Telecommunication Standardization Sector (ITU-T) plays an important role in achieving interworking among the equipment of different make. Based on the ITU-T recommendations optical communication systems are classified as follows (ITU-T Manual 2009).

In this section, we shall discuss the classification as per ITU-T standard based on operating wavelength bands. It is always desirable to have a large spectral band to achieve a high bit rate. However, the spectral range in each band is constrained by the characteristics of the particular type of fiber, the nature of the source, attenuation, and dispersion in the channel. The prescribed bands and the recommended applications are listed below:

1. The "Original" O-band (1260 nm to 1360 nm)

Historically, this is the first wavelength band used for optical communication because the optical fibers of the 1970s used to exhibit minimum loss in this spectral band. While the lower wavelength (1260 nm) limit is decided by the cut-off wavelength of the fiber cable, the upper limit (1360 nm) is decided by the rising edge of the attenuation peak (occurring at 1383 nm) in glass fiber caused by hydroxyl ions.

2. The "Extended" E-band (1360 nm to 1460 nm)

With the advances in the production of high-quality glass fiber preforms and the invention of dehydrated production of glass fibers, it was possible to achieve fibers exhibiting attenuation less than that exhibited by normal fibers in the "O" band (ITU-T G.652.D). However, the use of the "E" band is very limited because fibers installed before the beginning of this millennium exhibit a high attenuation in this band.

3. The "Conventional" C-band (1530 nm to 1565 nm)

The new-generation fibers exhibit the lowest attenuation in this band. This band is feasible for use in long-haul, submarine, and trans-oceanic optical transmission. This band enables the use of WDM and EDFA (erbium-doped fiber amplifier) technologies.

4. The "Short wavelength" S-band (1460 nm to 1530 nm)

This band is assigned by taking the lower limit to be equal to the upper limit of the E-band and the upper limit to be the lower limit of the C-band. With the availability of EDFAs having wider gain characteristics, the S-band is becoming attractive for future DWDM applications. This band can also be used for pumping fiber amplifiers including the Raman type amplifiers. The S-band is also used for some PON (passive optical network) systems as downstream wavelength.

5. *The "Long wavelength" L-band (1565 nm to 1625 nm)*

After the C-band, this is the second lowest attenuation wavelength band and is used when the C-band alone is not sufficient for a particular application. This band also supports EDFA and WDM technologies. The combined C plus L band is now suitable for the development of submarine transoceanic transmission using WDM technology.

6. *The "Ultra-long wavelength" U-band (1625 nm to 1675 nm)*

Unlike the other five bands (O, E, S, C, and L), U- band is not used as a transmission channel in optical communication systems. This band is reserved for testing and maintenance of the optical fiber links after installation. Optical time domain refractometer (OTDR) is used in this band for fault detection and loss testing purposes.

The single-mode spectral bands are listed in Table 1.1.

Table 1.1 Single-mode spectral bands

Band designation	*Full-form*	*Wavelength range (nm)*
O-band	Original	1260 to 1360
E-band	Extended	1360 to 1460
S-band	Short wavelength	1460 to 1530
C-band	Conventional	1530 to 1565
L-band	Long wavelength	1565 to 1625
U-band	Ultra-long wavelength	1625 to 1675

1.2 GENERAL OPTICAL FIBER COMMUNICATION SYSTEM

A generalized optical fiber communication system is shown in Fig. 1.4. The major elements of the system are shown by blocks.

The information source provides an electrical signal usually derived from a message signal which is not generally an electrical signal (e.g. sound, picture). The electrical signal from the information source is fed to a transmitter comprising an electrical stage that drives an optical source to produce modulation of light wave carriers. It is important to note that in the IM/DD system, the modulating signal is used to modulate the intensity of the light source only. It may be emphasized that, unlike conventional electrical modulation schemes where the amplitude, frequency, or phase of the carrier is altered following the modulating signal, the frequency or phase of the optical signal from the optical source remains unaltered. From this viewpoint, the function of the optical source in a way is to provide electrical to-optical (E–O) conversion. This is usually achieved with the help of a light-emitting diode (LED) or a semiconductor laser diode which is generally known as

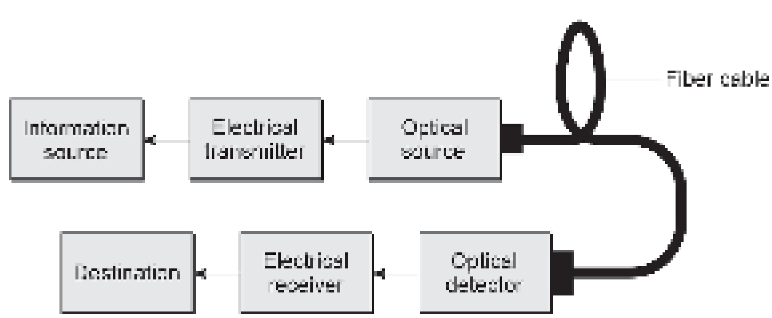

Fig. 1.4 Block diagram of an optical fiber communication system

an injection laser diode (ILD). These optical sources are lightweight, compact, and most importantly compatible in size with the optical fiber which is used as a waveguide for subsequent transmission of the signal. Moreover, both LED and ILD sources consume low electrical power and can generate light waves at different wavelength regions of the optical spectrum where the silica fibers offer less attenuation. The modulated lightwave output from the optical source is coupled to the transmission medium consisting of an optical fiber cable. The fiber cable contains a group of optical fibers which are generally long thin strands (typically of the order of 100 μm diameter) of ultrapure glass that provide low loss at the transmitting wavelength.

An optical fiber consists of two coaxial solid cylinders having slightly different refractive indices. The inner solid cylinder called the core has a higher refractive index as compared to the outer cylinder known as *cladding*. The optical signal propagates through the fiber by total internal reflection. As the optical signal propagates down the fiber length, it gets attenuated due to absorption, scattering, etc. within the fiber and at the same time gets distorted and broadened because of various dispersion mechanisms. The weak distorted optical signal is received by the receiver at the destination. The key component of the receiver is an optical detector which converts the weak and distorted information-bearing optical signal to an electrical signal that is a replica of the modulating signal. The signal is then processed by an electrical receiver and the output is finally sent to the destination.

A practical optical communication system is much more complex than the simple block diagram shown in Fig. 1.4. Like electrical communication, optical communication can be either analog or digital type. A practical digital optical communication link looks more like one shown in Fig. 1.5. The basic components of a practical optical communication link consist of an optical transmitter, optical repeater, optical receiver, optical fiber waveguide, connector, splice, splitter, optical amplifier, etc. The message signal may be in a continuous analog form or the form of digital pulses representing bits 1s and 0s. The message signal is used to modulate the intensity of the optical source with the help of an electrical drive circuit (modulator). The manufacturers generally provide optical sources with a small portion of an optical fiber (1–2 m length) attached to it in an optimum fashion. This is called *fiber pigtail or flylead* which can be easily plugged in for connection with the line fiber by using a demountable connector. The optical signal propagates down the fiber toward the receiver end. While the signal propagates along the fiber it gets attenuated due to the absorption of the optical signal by the fiber material for several reasons to be discussed afterward. In addition to this attenuation, the signal also gets distorted due to the dispersion phenomenon to be discussed afterward. The weak and distorted optical signal is subsequently allowed to travel along the fiber and before the signal gets distorted beyond recognition, it is necessary to have an arrangement to regenerate the signal and retransmit the reconstructed and boosted to travel further over the transmission link. This is achieved with the help of regenerative repeaters. In the present case (Fig. 1.5) only one repeater unit is shown for illustration. The actual number of repeaters needed along a transmission line depends on the transmission characteristics of the channel (optical fiber) and the total distance to be covered.

At the end of the link, the received optical signal which is attenuated and distorted during transmission down the optical fiber is reconverted from optical-to-electrical (O–E) for further electrical processing and extraction of the original electrical message signal. The key element at the receiving end is an optical detector (*p-i-n* or avalanche photodiode) which converts the intensity variation in the recovered optical signal into a corresponding electrical signal. The size of the optical detector should be compatible with

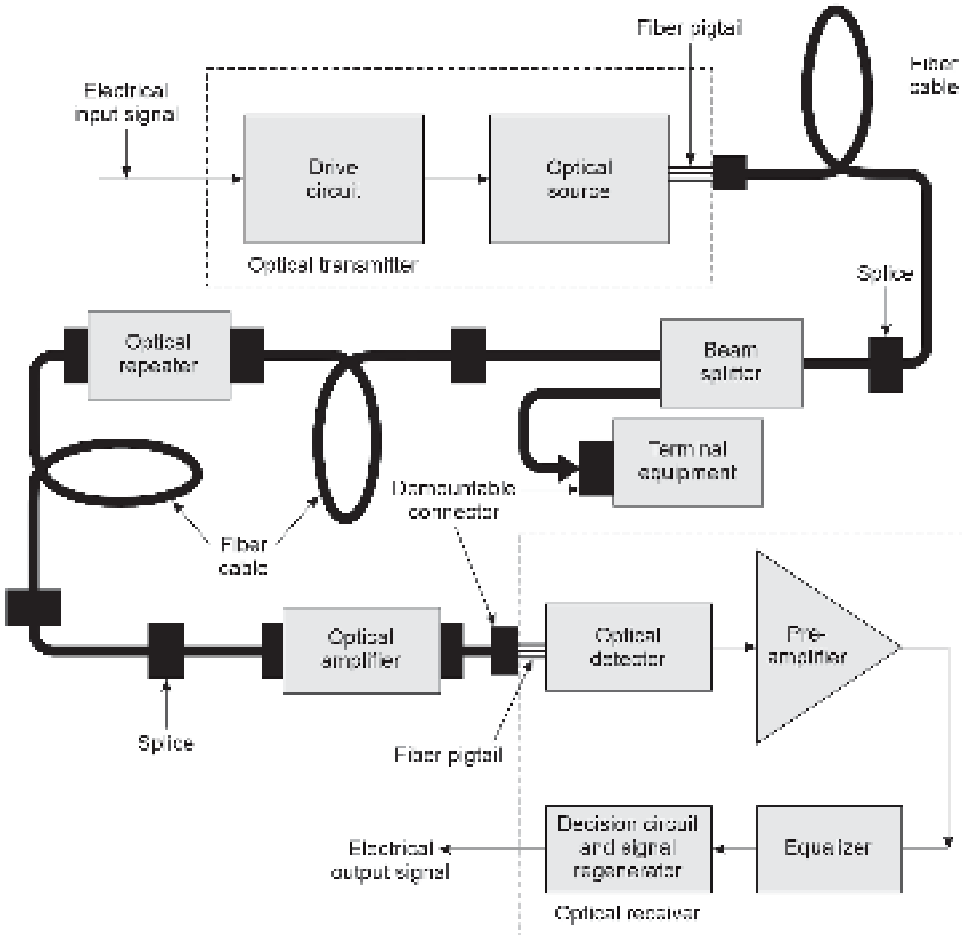

Fig. 1.5 Schematic of an optical communication system

the optical fiber size. The important requirements of photodetector characteristics include linearity, high-speed response, high responsivity, and low-noise behavior. In a practical optical communication system additional components such as optical connectors, splices, couplers, and optical amplifiers are used. The connectors and splices are used for joining two fibers. The connectors are generally demountable while the splices provide permanent joints. The couplers are in-line buses that are used at terminal points to remove a portion of the optical signal from the trunk line at intermediate points or inject additional optical signals onto the trunk. Optical amplifiers provide online amplification to the propagating optical signal. Such amplifiers are useful for compensating for the attenuation caused by the optical fiber during the propagation of the signal. Both semiconductor laser amplifier (SLA) and erbium-doped fiber amplifier (EDFA) are used for providing amplification of the signal in the optical domain.

The generalized optical communication system shown in Fig. 1.5 may be an analog or a digital link, depending on whether the lightwave carrier is modulated using an analog or digital information signal. In an analog system, the modulation involves the variation of the light emitted from the optical source continuously. On the other hand in digital systems, discrete changes in the light intensity (on-off optical pulses) are generally used for transmission of lightwave carriers. In general, analog–optical communication systems are easier to implement but they are less efficient and need a larger signal-to-noise ratio at the receiver end than the digital counterpart. Further, the semiconductor sources do not provide good linearity characteristics (particularly at high modulation frequency) which is essential for the implementation of analog optical links. As a result, digital optical communication is generally preferred for long-haul and high-speed optical links while analog optical links are restricted to use for short distance and low bandwidth operation.

1.3 ADVANTAGES OF OPTICAL FIBER COMMUNICATION

An optical communication system that uses lightwave as the carrier and optical fiber as the waveguide has many attractive features over the conventional electrical communication system that uses copper cable as the waveguide. Some of the features were apparent when the technique was first conceived. The additional features became apparent with the technological development in related areas. Some of the distinct advantages of optical communication include the following.

1. **Large potential bandwidth:** The frequency of lightwave carriers in the infrared region is of the order of 10^{14} Hz (10^5 GHz) which yields a far greater transmission bandwidth as compared to conventional metallic cable systems. For example, coaxial cables provide a bandwidth of the order of 500 MHz. The information carrying capacity of an optical fiber is far superior to the best copper cable system. It may be pointed out that the full potential bandwidth of optical fibers (~50 THz) is not being utilized at present because of technological constraints. With the advent of the wavelength division multiplexing (WDM) technique, it would be possible to enhance bandwidth utilization significantly in the future.
2. **Small size and lightweight:** Optical fibers have a very small diameter (of the order of human hair diameter ~100 μm). The optical fibers coated with protecting layers also turn out to be smaller and lighter as compared to conventional metallic cables. The small size and light weight make them especially attractive for use in aircraft, satellites, and ships.
3. **Electrical isolation:** The optical fibers are made of dielectric materials such as glass or plastic which are electrical insulators. As a result, these waveguides do not exhibit earth loop and interface problems. The optical fiber transmission system is convenient for use in electrical hazardous environments as it does not create sparks. This feature also enables easy interfacing of equipment.
4. **Immunity to interference and cross-talk:** Optical fibers are made of dielectric material and therefore are free from electromagnetic interference (EMI) and radiofrequency interference (RFI). Unlike metal cables, optical fiber cables are free from inductive pick-up from other electrical signal-carrying wires or lightning. In other words, the function of fiber optic communication system remains unaffected even in an electrically noisy environment. The optical inference between individual fibers in an optical fiber cable is also absent and as a result, there is no cross-talk effect which is quite common in conventional electrical communication that uses metal cables.
5. **Signal security:** The optical signals are well confined in optical fiber waveguides and as a result, there is practically no leakage of optical power from the fibers. Emanations, if any get absorbed in opaque jackets surrounding the fibers. This feature provides a high degree of signal security in optical fiber-based communication systems. Unlike the situations with copper cables, signals cannot be tapped from optical fibers during transmission in a non-invasive manner. This feature makes optical communication systems very feasible for military, banking, and other secure data transmission applications.

6. **Low transmission loss:** Optical fibers have much lower loss as compared to conventional copper cables. The development in the field of optical fiber fabrication over the past two decades has resulted in the production of high-quality optical fibers that provide extremely low loss (less than 1 dB/km). Fibers have been fabricated with loss to the tune of 0.2 dB/km. The low-loss fibers have considerably enhanced the repeater spacing and significantly cut down the system cost.
7. **Ruggedness and flexibility:** With proper protecting layers and cabling structures, optical fiber cables remain flexible yet rugged enough to bear stresses during installation. As compared to its metal counterpart, optical fiber cable is superior for transportation, storage, handling, and installation. The optical fiber cables can be used for under-sea installation and other abusive environments without causing any damage.
8. **Reliability and easy maintenance:** The availability of extremely low-loss and low-dispersion single-mode fibers has improved the reliability of long-haul optical links with a lesser number of repeaters as compared to conventional metal cable systems. The average lifetime of the state-of-the-art optical fiber system is nearly 20 years. Moreover, optical communication subsystems require minimum maintenance in the long run.
9. **Potential low cost:** The overall system cost of an optical communication link for long-haul applications is considerably less than its electrical counterpart using metal cables. Extremely low loss and large bandwidth of optical fiber are primarily responsible for low-cost system development using lightwave technology. Although high-quality optical sources (such as ILD), optical fiber connectors, and couplers are still very expensive, the raw material used in making the silica fibers is abundantly available in nature and found in ordinary sand. This significantly reduces the cost of this waveguide used in optical communication systems in comparison to metal cables.

The advantages of optical fiber communication discussed above have made this technology almost indispensable for long-haul optical links and are preferred over conventional electrical communication using electrical transmission lines and even microwave and millimeter wave systems. Initially, optical fiber communication link was intended to be used only in intercity and intercontinental trunk lines. However, with the increasing demand for larger bandwidth and the advent of ISDN (Integrated Service Digital Network) involving transmission of voice, video, facsimile, computer data, etc. optical fibers have finally entered into subscribers loop.

The subject of optical fiber communication has been discussed in several excellent textbooks (Keiser, 1991; Senior, 1992; Gower, 1984).

1.4 SCOPE OF THE BOOK

The primary objective of the book is to provide an understanding of the basic principles of an optical fiber communication system. An attempt has been made throughout the text to focus attention on fundamental concepts underlying various techniques used for the transmission and reception of lightwave carriers through optical fibers. As the field of lightwave technology as a whole is undergoing dramatic changes even today and the older technology is being replaced by newer ones, the intricacies of practical optical fiber communication systems have been deliberately avoided in the text.

The material presented in the subsequent portion of the text has been distributed among 12 chapters including the current chapter to cover the introductory fundamentals and various issues related to the generation, transmission, distribution, and reception of lightwave signals in a variety of optical communication systems. The chapters are organized as follows.

Chapter 2 deals with the concept of optical fiber as a transmission medium. The classifications of optical fibers and their structural variations are also discussed. This is followed by simple analyses based on the principles of geometrical optics. Some important parameters of the fibers are estimated based on simple ray analysis. This chapter also deals with the descriptions of materials used for making fibers, and various techniques for fabrication of optical fibers. The chapter ends with a discussion of issues concerning the strength of optical fibers and different techniques of fiber cabling for strengthening the fibers for field installation. Vanious issues associated withe the installation of optical fiber cables are also dissussed in this chapter.

Ray analysis which is used to explain the propagation of light through an optical fiber discussed in the previous chapter is approximated and is applicable in the case of fibers of large size (such as multimode fibers). An accurate analysis of the propagation of light through an optical fiber can only be done with the help of Maxwell's electromagnetic field theory and by treating optical fibers as a dielectric medium. In Chapter 3, the propagation of light through planar waveguide and also through dielectric waveguide has been analyzed based on electromagnetic wave theory. The chapter begins with a brief discussion of Maxwell's equations and their applications for finding modal equations for the transmission of light as electromagnetic waves through fibers considered cylindrical waveguides of dielectric media. The mathematical treatment is rigorous and may be skipped without affecting the basic understanding of the later part of the book. Nevertheless, the analysis leads to some fundamental formulas related to various propagating modes, cut-off conditions, power flow, etc. which are essential for a thorough understanding of the subject.

Optical signals get progressively degraded and distorted as they propagate down the fiber. The primary reasons for this degradation are attenuation caused by absorption and scattering of the signal by the fiber medium and dispersion of the optical signal. Various mechanisms responsible for the attenuation and distortion of the signal during propagation through optical fibers are discussed in Chapter 4. The methods available for reducing signal degradation in practical fibers are also mentioned in this chapter.

Optical fiber constitutes the transmission medium of an optical fiber communication system. Apart from the transmission medium, the other two major units of an optical communication system are the optical transmitter and the optical receiver. The key component of an optical transmitter is an optical source. A variety of optical sources are commercially available. However, sources that are compatible with the size and transmission characteristics of the optical fibers can only be used for optical fiber communication. Chapter 5 discusses the light sources employed in optical fiber communication. The discussion is primarily confined to semiconductor sources which are predominantly used in optical transmitters because of the stringent requirements stated above. The basic physical principles underlying the operation of semiconductor optical sources (light-emitting diode and injection laser diode) are discussed in this chapter. A thorough understanding of the operation of the sources requires basic concepts of semiconductor devices. It is presumed that the readers have done a basic course on semiconductor devices. A few sections have, however, been devoted to the discussion of basic principles of semiconductor devices which are essential for a proper understanding of the mechanism of operations of semiconductor optical sources and detectors. Further, compound semiconductors and alloys are mostly used for making these optoelectronic devices. Keeping this fact in mind, adequate material about III-V materials which are mostly used for making semiconductor sources and optical detectors has been added in this chapter.

An optical communication system often called an optical link is an interconnection of optical transmitters, receivers, and other optical components along the route for transmission, distribution, and reception of optical signal. The light signal generated by an optical source is generally modulated with the help of a driver circuit. The intensity-modulated light is subsequently launched into the optical fiber for transmission. Chapter 6 deals with various issues related to the launching of power from optical sources to optical fibers. Different techniques used for coupling power from one fiber to the other are also discussed in this chapter. The major factors that affect the coupling efficiency are also discussed in this chapter

At the destination point, the transmitted signal is processed to reproduce the original message/information signal. This is done by an optical receiver. The key component of an optical receiver is an optical detector, also known as a photodetector. It converts the optical signal to an electrical signal (E–O conversion). Chapter 7 is devoted to the study of semiconductor optical detectors and photovoltaic devices. It discusses the principles of photo-detection mechanisms and different non-multiplying and multiplying photodetector structures used in optical communication systems. The noise characteristics of different photodetectors have also been discussed since the noise characteristics of the photodetector play an important role in deciding the sensitivity of an optical receiver. A brief discussion of solar cells is also included at the end of this chapter the solar cells one ophelectric cells that find application as an alternative source of electrical energy by making use solar energy.

After optical detection, the electrical signal undergoes various processing by the subsequent stages of the optical receiver. The theory and design of optical receivers are discussed in Chapter 8. The design of an optical receiver is much more complex than that of an optical transmitter. This is because the signal is generally weak and mutilated at the receiving end. The processing of this weak and distorted signal in the presence of various electrical noise components arising from the processing units is challenging and makes the design of the receiver very complex. A variety of receiver configurations including the state-of-the-art monolithic optoelectronic receiver have been discussed in this chapter.

In optical communication, information, transmission and reception can be either in analog or in digital form. The primary purpose behind digital optical communication is to link telephone exchanges with digital integrated circuits which offer reliable transmission and reception of both voice and data signals at a substantially low cost. In analog optical communication, the message signals are superimposed on radio-frequency (RF) subcarriers. The multiplexed electrical signal is subsequently used to modulate the optical carrier. Chapter 9 deals with various aspects and system requirements of both analog and digital forms of optical communication.

A major bottleneck of a traditional optical communication system lies in the frequent conversion of the signal from the optical to the electrical domain and vice-versa. This conversion drastically affects the speed of the system. The signal after conversion to the electrical domain gets constrained by the limited bandwidth available in the electrical system. On the contrary, if the entire processing can be done in the optical domain the speed of the existing optical communication system can be greatly improved. Because of the above fact, several active optical devices and components have been developed to enable the processing of optical signals without changing them to the electrical domain. Chapter 10 discussed various active optical devices and components that are used in manipulating the signal in the optical domain itself. Optical amplifiers, optical modulators, optical switches, and other elements of integrated optics (IO) such as beam-splitters, directional couplers, etc. are also discussed in this chapter.

Chapter 11 is devoted to the study of the essentials of advanced optical communication systems and optical networks. The enormous bandwidth of optical fibers (particularly single-mode fibers) can be efficiently utilized by making use of Wavelength-Division-Multiplexing (WDM) techniques. This chapter discusses WDM systems and various components used in the design. Traditionally intensity–intensity modulated optical signals are detected by an optical detector which essentially acts as a photon counter and converts the received intensity modulated light to the corresponding electrical signal. In the process, neither the phase nor-polarization of the light comes into the picture. The performance of this type of Intensity Modulation/Direct Detection (IM/DD) optical communication system is limited by the poor sensitivity of the receiver whose sensitivity can never exceed the quantum limit. With the advent of technology, it is possible to add the received light signal with a locally generated optical signal and then detect the resultant signal. This process enables one to achieve an improvement in the sensitivity by –20 dB over the IM/DD system. This is the essence of coherent optical communication. The principles of coherent optical fiber communication have been discussed in this chapter. The Chapter also covers various principles and architectures of optical network which is the backbone of today's communication network. It covers SONET/SDH systems, wavelength-routed, and WDM-based networks. The chapter concludes with the latest development in the area of soliton pulses and soliton-based optical communication systems which is envisaged as the future fourth-generation (4G) optical communication system.

In the concluding chapter (Chapter 12) various standards, tools, and measurement procedures followed in fiber optics are discussed. Special types of measuring equipment are needed to test and characterize optical and optoelectronic components and optical fibers. This chapter discusses all essential measuring equipment and standard practices for making measurements and checking the performances of optical communication systems.

A large number of solved numerical examples are given throughout the text. These examples are intended for a better understanding of the subject and also for having some idea about various parameters and standards of practical systems. Some unsolved problems (qualitative and quantitative) are also included at the end of each chapter to enable the readers to make a self-assessment of their understanding of the subject topics dealt with in various chapters of the book.

At the end of every chapter, a separate list of references has been included. These references would help the readers to find additional materials on relevant topics. Further supplementary study materials can be found in other textbooks listed in the reference sections.

1.5 MAJOR MILESTONES

The breakthrough inventions and developmental work in the field of optoelectronic devices, components, and systems that paved the way for the widespread successful implementation of modern-day optical communication systems and several other non-telecommunication applications are listed below (Hecht, 1999).

1609	Galileo (Italy)	Galilean telescope
1668	Newton (UK)	Reflection telescope
1870	Tyndall (UK)	The light guided in a thin water jet
1873	Maxwell (UK)	Electromagnetic theory
1897	Rayleigh (UK)	Waveguide analysis
1930	Lamb (Germany)	Experiments with silica fibers
1951	Heel, Hopkins, Kapany (UK)	Fiber optic endoscopy

(*Contd.*)

1953	Charles Hard Townes (USA)	Invention of MASER
1954	Bell Lab, USA	First Si solar cell
1958	Goubau (USA)	Experiments with lens guide
1959	Kapany (UK)	Optical fiber with cladding
1960	Maiman (USA) Javan (USA)	First LASER (Ruby) Operation of He-Ne laser
1961	Robert Biard and Gary Pittman for Texas Instruments	First infrared LED
1962	Nick Holonyack (USA)	Visible LED Semiconductor LASER
1966	Kao and Hockham (UK)	Optical fibers for long-distance transmission
1969	Uchida (Japan) Miller at Bell Lab	Graded-index fiber Integrated optics
1970	Kapron and Keck (USA) Delange	Fiber transmission loss <20 dB/km Concept of WDM
1970	Fairchild Optoelectronics	Low-cost LED
1972	Gambling *et al* (UK)	GHz bandwidth over 1 km optical link
1975	Payne and Gambling (UK)	Prediction of zero material dispersion at 1.3 μm
1976	Thomas P Pearsal	High-efficiency and high-radiance LED for fiber optic communication
1977	Kenichi Iga of the Tokyo Institute of Technology, USA	Vertical cavity surface emitting laser
1978	NTT Ibaraki Lab	Commercial optical fiber link Single-mode fiber with record 0.2 dB/km loss at 1.55 μm
1980	Bell Labs	Single-mode 1.3 μm technology for the first transatlantic-fiber-optic cable, TAT-8
1981	Moungi Bawendi of MIT, Louis Brus of Columbia University, and Alexei Ekimov of Nanocrystals Inc.	Semiconductor quantum dots
1982	British Telecom	Single-mode fiber to replace GI fibers
1985	Bell Labs	Single-mode fiber across the USA to carry long-haul telephone signals at 400 Mbps and more
1986	Rome Air Development Centre	Silicon Photonics
1987	Dave Payne, University of Southampton	First erbium-doped fiber amplifier operating at 1.55 μm
	Yablonovitch and John	Photonic crystal (PC)
1987	Ching Wan Tang and Steven van Slyke at Eastman Kodak	First organic light emitting diode (OLED)
1988	Linn Mollenauer	First soliton transmission through 4000 km of single-mode fiber
Dec. 1988	Bell Labs	First transatlantic fiber-optic cable, using 1.3 μm lasers and single-mode fiber
1989	Yablonovitch	Birth of photonic crystals

(*Contd.*)

1993	TAT-8	Transmission of the first soliton signals over 180 million kilometers
1994	Federico Capasso and Alfred Y Cho at Bell Lab, USA	Quantum cascade laser
1996	Ciena Corporation	Commercial deployment of DWDM
	Nakazawa Fujitsu, NTT labs and Bell Labs	Successful transmission of one trillion bits per second through single optical fibers
2012	MetroPCS at Dallas, Texas, USA	Voice over long-term evolution (VoLTE) service began
2015	Sumimoto electric	Multi-core fiberoptic cable

Summary

- The chapter discusses the evolution of today's optical communication and major milestones that paved the way for the incredible expansion of optical fiber communication network all over the globe.
- Major breakthrough works in this area include the successful demonstration of laser by Maiman in 1960 followed by the path breaking work of Kao in 1966 related to the possibility of transmission of light using dielectric waveguide.
- The principle of operation of an IM/DD scheme for realizing a practical optical transmitter and receiver system has been discussed.
- The essence of coherent optical communication and its advantage in terms of increased sensitivity and the disadvantage in terms of increased complexities and practical difficulties with implementation are discussed.
- The major bottleneck of today's optical communication is the speed which is greatly hampered by frequent E–O and O–E conversions.
- The major advantages of an optical communication system include increased bandwidth, reduced size and less weight, security of the signal, ruggedness of the system, low cost, etc.
- The three generations of optical communication centered on 850 nm, 1330 nm, and 1550 nm and the rationale behind the choice of these attenuation windows are discussed in this chapter.
- It outlines the importance of WDM system in terms of enhancing the capacity of existing optical fiber link.
- The scope of the book is outlined in this chapter by highlighting the sequence in which the rest of the chapters have been organized along with a brief discussion of various topics covered.
- The chapter ends with a list of the major milestones that shaped the modern optical communication systems.

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Optical Fiber Fundamentals and Fiber Cables

CHAPTER 2

The most important role in optical fiber communication is played by the optical fiber. The performance of an optical fiber link largely depends on the characteristics of this fiber through which the lightwave propagates. This chapter addresses several fundamental issues about optical fibers such as the structures of an optical fiber, classification of fibers, propagation of light through optical fibers, fiber composition, fabrication techniques, etc.

Over several decades in the past different dielectric waveguide structures have been proposed and studied for the transmission of optical signals (lightwave) (Keiser, 1991). The study of the propagation of light through dielectric waveguide structure was first reported at the beginning of the twentieth century (Hondros, 1910; Schriever, 1920). In the simplest form, a dielectric waveguide for the transmission of light may be viewed essentially as a transparent silica rod surrounded by air. However, this type of unsupported waveguide proves to be impractical because of high loss and associated aberrations. Furthermore, this type of waveguide needs mechanical support when the core region is made very thin to limit the large number of modes responsible for undesired aberrations. Later on, it was proved that the light transmission through the long strands of silica fiber can be greatly improved by incorporating an over-layered material of a slightly lower refractive index. This central region through which the light is intended to propagate is called the core region and the supporting outer layer is called *cladding*. In 1950 long human hair size (~100 μm diameter) clad dielectric waveguides were proposed (Snyder *et al* 1983; Marcatili, 1979; Marcuse, 1991; Midwinter, 1979; Buck, 1995). The applications of these dielectric waveguides were initially restricted to several non-telecommunication systems. Historically, the term fiber optics was introduced by Hopkins and Kapany (Hopkins *et al* 1954) and a flexible fiberscope was developed by them for the transmission of images from inaccessible points. This flexible fiberscope later found applications for inspecting jet engines, and reactor vessels and examining interior parts of the human body. An advanced form of flexible fiberscope is being used today in the medical field as an optical fiber endoscope. It may be emphasized that optical fibers were in use in several non-telecommunication field appliances before it was successfully brought into application in the field of guided optical communication.

In 1966 Kao[1] and Hockham (Kao *et al* 1966) proposed the use of clad dielectric fiber waveguide as a medium for establishing an optical communication link. The performance of a viable transmission through optical fiber was severely restricted in the beginning by the huge loss (~1000 dB/km) provided by the optical fibers of the late 1960s. They, however,

[1] Kao shared the Nobel Prize in Physics, in the year 2009 along with two other scientists for his historic invention in the field of optical communication.

rightly predicted that clad optical fiber waveguides might be used as an efficient channel for transporting optical signals if the loss in the fibers could be reduced significantly. The observations made by Kao *et al* stimulated serious attempts by different companies to fabricate fibers with lower loss by improving the fabrication techniques. The first success came in 1970 when Corning Glass Works reported a silica fiber with a low value of loss (~20 dB/km) which is acceptable for communication purposes (Gambling, 2000; Hecht, 1999). This was followed by a series of works reporting fibers with lower and lower losses. The rest is history and is discussed in detail in Chapter 1.

A typical cladded dielectric fiber waveguide structure is shown in Fig. 2.1. It consists of a cylindrical transparent case with a refractive index of n_1 surrounded by another coaxial cylindrical cladding of slightly lower refractive index n_2. The surrounding cladding region is usually thicker and provides mechanical support to the inner core structure and also helps in reducing the radiation loss of the propagating optical signal through the inner core region. In actual practice, light is not strictly confined to the central core region only but also propagates partially through the supporting cladding layer.

To appreciate the transmission of light through optical fibers whose diameter is extremely small, it is necessary to understand the nature of light. It is understood that propagation of light through optical fibers can be explained by using the principles of geometrical optics which is based on rectilinear propagation of light. This type of analysis is known as ray theory or ray analysis and is the subject of discussion in the present chapter. However, we shall see later in this chapter that the analysis is an approximate one and is valid under certain conditions. The other explanation which is more generalized in nature, is based on electromagnetic theory and is known as *mode analysis*. Before discussing ray analysis it would be worthwhile to examine the existing theories dealing with the nature of light. Historically, the theory of optics (light) has undergone various transformations and evolved over the years right from the primitive corpuscular theory put forward by Sir Isaac Newton, in the seventeenth century. The existing theory had to be modified multiple times to explain multiple phenomena invented to be exhibited by light. It may be pointed out that even today no single theory can explain all the phenomena exhibited by light.

2.1 NATURE OF LIGHT AND RELATED THEORIES

Light is everywhere and is considered as one form of energy. However, it has deluded scientists the most over centuries by exhibiting different kinds of phenomena depending on the situation. As a result, several theories of light have been developed by scientists to explain the nature of light. As per the existing theories light is considered to exhibit both wave and particle nature, different theories of light are discussed afterwards. The branch of physics that deals with the study of light is known as *optics*.

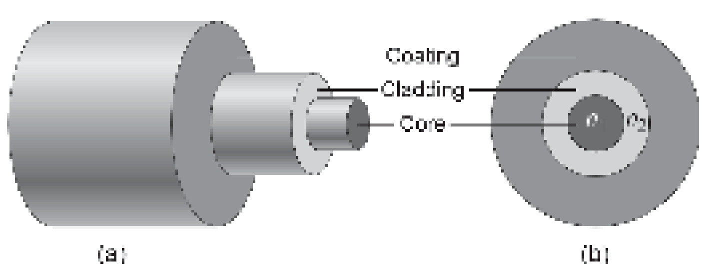

Fig. 2.1 (a) A typical optical fiber (b) Cross-sectional view

2.1.1 Corpuscular Theory

The earliest theory concerning the nature of light was the corpuscular theory proposed by Newton (Newton, 1704). According to this theory, light constitutes a stream of minute particles that are emitted by luminous optical sources. These particles are considered to be tiny with zero or negligible mass and perfectly elastic and can travel in all directions in a homogeneous medium with the speed of light. The mass of the luminous body does not change much even if it emits light for a long time due to the negligible mass of the corpuscles. According to corpuscular theory, the emitted corpuscles travel in a straight line without getting affected by gravity and are either attracted when they approach a surface between two media. The attraction of the corpuscles at the interface causes the light to exhibit the phenomenon of refraction while the repulsion of the corpuscles by the interface is responsible for the reflection of light. The early theory of light was good enough to explain large-scale optical phenomena such as reflection and refraction. The corpuscular theory of Newton suggested that the velocity of light in an optically denser medium (higher refractive index) is greater than the velocity of light in a rarer medium (lower refractive index). Later on, the experimental results of Foucault and Michelson demonstrated that the velocity of light in a denser medium is lesser than that in a rarer medium. The corpuscular theory also failed to explain two other important phenomena exhibited by light, e.g. interference, diffraction, and polarization.

2.1.2 Wave Theory

Christiaan Huygens developed his wave theory of light in 1678. His theory was later published in his work–*Treatise on Light* (Huygens, 1690). According to his wave theory light is emitted in all directions as a series of waves in a medium which he termed as luminiferous ether. Contrary to what Newton's corpuscular theory suggested, Huygens assumed that light slows down upon entering a denser medium. The wave theory predicted that lightwaves could interfere with each other like sound waves as reported by Thomas Young. Young's well-known double-slit diffraction experiment demonstrated that light behaves as waves. He also suggested that different colours of light are caused by different wavelengths of light. Later on in 1817, Fresnel AJ, put forward his wave theory of light. Fresnel's wave theory could overcome the shortcomings of the previous wave theory. Later on, Poisson advanced Fresnel's mathematical work to further strengthen the wave theory. According to Fresnel's theory diffraction phenomenon could be explained by considering light to be a wave motion and an approximate rectilinear propagation character of light is valid under certain conditions. At the early stage of propagation of wave theory of light, the nature of wave motion (longitudinal or transverse) was not clearly understood. The polarization phenomenon exhibited by light confirmed that light is a transverse wave. In 1821, Fresnel showed mathematically that polarization could be explained only by the wave theory of light and only if the light is considered entirely transverse, with no longitudinal vibration whatsoever (Crew, 1900). The major weakness of the wave theory lies in the presumption that like sound waves, light also needs a medium for transmission. The existence of the hypothetical ether was later proved to be wrong by the famous Michelson-Morley experiment. Another contradiction between Newton's corpuscular theory and the wave theory of Huygens and others was related to the speed of light in a denser medium. Newton's corpuscular theory implied that light would travel faster in a denser medium, while the wave theory of Huygens and others predicted just the opposite. An accurate measurement of the speed of light by Foucault, in 1850 supported the wave theory. Newton's corpuscular theory was finally overturned by the wave theory of light.

2.1.3 Maxwell's Electromagnetic Theory

The wave theory of Fresnel faced another challenge when Michael Faraday in 1845 demonstrated experimentally that the plane of polarization of linearly polarized light is rotated when the light rays travel along the direction of the magnetic field in the presence of a transparent dielectric. The effect is now known as Faraday rotation which established for the first time that light was some form of energy related to electromagnetism. The evidence of the rotation of light in the presence of a magnetic field prompted Faraday to propose in 1847 that light was a high-frequency electromagnetic vibration. He also pointed out that light could propagate even in the absence of a medium such as the ether.

James Clerk Maxwell was inspired by the work of Faraday and started working on electromagnetic radiation and light. Maxwell discovered that self-propagating electromagnetic waves can travel through space at a constant speed. Interestingly, the speed of these electromagnetic waves turned out to be equal to the previously measured speed of light. Maxwell immediately concluded that light is nothing but a form of electromagnetic radiation. The full mathematical description of the behaviour of electric and magnetic fields was reported by Maxwell in two famous volumes entitled *A Treatise on Electricity and Magnetism*, published in 1873 (Maxwell, 1873). His published work later became popular as Maxwell's equations in electromagnetic theory. Soon after, Maxwell's electromagnetic theory was experimentally demonstrated by Heinrich Hertz. Hertz successfully generated and detected electromagnetic waves which he termed radiowaves in the laboratory. He also demonstrated that these waves behave exactly like visible light, exhibiting properties such as reflection, refraction, diffraction, and interference. Maxwell's theory and Hertz's experiments were instrumental to the development of modern radio (wireless) communication including optical communication. The only difference between normal radio communication and optical communication is that the frequency of the electromagnetic wave in the latter case is in the tune of 10^{14} Hz.

2.1.4 Quantum Theory

The wave theory successfully explained almost all optical and electromagnetic phenomena but it remained a matter of controversy over the constant speed of light predicted by Maxwell's equations and later confirmed by the Michelson-Morley experiment (Michelson *et al* 1887). This is because the assumption of constant speed for light contradicts classical laws of motion that states that all speeds are relative to the speed of the observer. Albert Einstein resolved this ambiguity in 1905 by proposing that space and time are interchangeable which accounted for constant speed of light. He also proposed in his well-known theory of relativity that energy and mass are equivalent. The wave theory of light suffered from a major setback when it failed to explain the photoelectric effect reported by Einstein. It was demonstrated that light striking a metal surface ejects electrons from the surface, causing an electric current to flow across an applied voltage. It was also observed that the energy of each ejected electron was proportional to the frequency, rather than the intensity of the light. Furthermore, photoelectric emission is not possible below a certain minimum frequency depending on the particular metal. These observations were in direct conflict with the wave theory. The supporters of wave theory continued to search for an explanation for the photoelectric effect in vain and finally, Einstein once again came to the rescue. In 1905, Einstein resolved this contradiction by supporting the particle theory of light to explain the observed effect. But this time the particle theory was different from the corpuscular theory proposed earlier by Newton. The photoelectric theory was successfully explained with Max Planck's theory of blackbody radiation. According to Planck's theory blackbodies emit light (and also other electromagnetic radiation) in the form of discrete bundles or packets of energy. These packets were called *quanta*, and the particle of light was termed *photon* in compliance with other particle names such as electron and proton. Mathematically, the energy E of a photon is related to the frequency ν of the emitted light as

$$E = h\nu = \frac{hc}{\lambda} \tag{2.1}$$

where h is the constant of proportionality known as Planck's constant which has a value of 6.623×10^{-34} Js (c is the velocity of light in free space and λ is the wavelength of the emitted light). The new form of the particle theory of light supported by Einstein for an explanation of the photoelectric effect initially faced stiff opposition from the firm believers of wave theory. The new particle theory became popular as the *quantum theory of light*. Einstein's explanation of the photoelectric effect formed the basis for wave-particle duality and also to a great extent the wave mechanics form of quantum mechanics formulated by Schrödinger. Generally speaking, everything has both a particle nature and a wave nature, and different experiments demonstrate their co-existence and manifestation in one form or the other. In 1924, Louis de Broglie startled the scientific community by proposing that even electrons exhibit wave–particle duality which was later experimentally demonstrated by Davisson and Germer in 1927[2].

It is interesting to note that various theories on light such as rectilinear propagation of light which is only an approximation of the generalized wave theory of light proposed by Fresnel, Fresnel's wave theory, Maxwell's electromagnetic theory, and quantum theory of light coexist. The apparent contradictions are largely overcome with the application of wave-particle duality. Each of these theories has some limitations when it is applied to explain a particular optical phenomenon. The most appropriate theory is largely dependent on the type of the phenomenon. For example, to explain a large-scale optical phenomena it is sufficient to consider that light is emitted from the source in the form of rays that travel in a straight line. The rectilinear propagation of light is the basis of geometrical optics. The finer optical phenomena such as interference and diffraction can only be explained with the help of the wave theory of light. Fresnel's wave theory forms the basis of physical optics. Further for an explanation of the polarization of light, one has to take the help of the electromagnetic theory of light proposed by Maxwell. Again for explaining the interaction of light with matter one has to apply the quantum theory of light. For example, important optical phenomena such as emission, and absorption of light can only be understood by considering light as a stream of photons. In this subject, one has the opportunity to see the application of all the existing theories of light in the context of explaining different optical phenomena involved in the generation, transmission, and detection of optical signals. It is also worthwhile to note that even though Newton's corpuscular theory was finally discarded, the concept of the particle nature of light was later accepted in a different form (quantum theory). The particle nature of light envisioned by Newton in the seventeenth century simply suggests why Newton is regarded as one of the greatest scientists of all time.

[2] Einstein received the Nobel Prize in 1921 for explaining photoelectric effect with wave-particle theory and de Broglie followed in 1929 for extending the theory to other particles.

2.2 FUNDAMENTALS OF GEOMETRICAL (OR RAY) OPTICS

In this section, the fundamental laws of optics are reviewed. This discussion will help the reader to understand the mechanism of propagation of light through different types of optical fibers.

According to the wave theory of light or principle of physical optics, a point optical source radiates electromagnetic waves in the form of a train of spherical wave fronts with the source at the center as shown in Fig. 2.2. A wavefront is the locus of all points in the wave train which have the same phase. In general, wavefronts are obtained by joining the maxima (crests) or minima (troughs) of the wave. As all points in a given wavefront have the same phase, a wavefront is often referred to as a phase front. The wavefronts are separated by one wavelength (λ). When the wavelength of light is very much smaller than the size of the object it encounters, the wavefronts appear as straight lines to the object. Under this condition, a lightwave can be considered a plane wave and its direction of travel can be represented by a light ray drawn perpendicular to the wavefront. Figure 2.2(b) indicates a set of plane wavefronts with light rays emerging perpendicular to the wave fronts. The rays originating from a particular set of wavefronts are identical and constitute a ray congruence. Each ray is representative of the ray congruence or the set of original wave fronts. The large-scale optical phenomena like reflection and refraction can be explained by geometrical methods of the ray tracing approach. This is the basis of geometrical optics. This picture of ray tracing makes it easy to visualize and appreciate the propagation of light through optical fibers because the direction of the ray indicates the direction of flow of the energy. It may be pointed out that the concept of ray tracing or geometric optics does not apply to all types of optical fibers. For fibers with a core diameter only of the order of a few wavelengths of light, the ray analysis becomes invalid. The propagation of light through such fibers is best understood with the help of electromagnetic mode theory based on the solution of Maxwell's equations under appropriate boundary conditions.

2.2.1 Ray Theory

This approach is based on the principles of geometrical optics allowing one to assume rectilinear propagation of light and thereby representing the path of light through a medium by straightlight rays. Some basic concepts of ray optics relevant to optical fiber are discussed below.

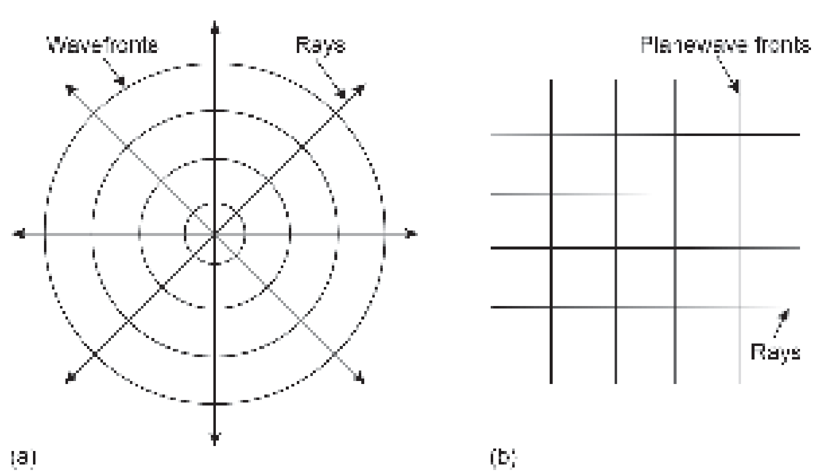

Fig. 2.2 (a) Spherical wavefronts from a luminous point source (b) Plane wavefronts

2.2.2 Total Internal Reflection

An important optical parameter of a material is its refractive index (or index of refraction). In free space, light travels with a velocity $c = 3 \times 10^8$ m/s. When light enters a dielectric medium, the wave travels with a velocity v which is less than c. The speed with which light travels in the medium is a characteristic of the medium and is representative of the optical density of the medium. The ratio of the velocity of light in vacuum to that of the material is defined as the *refractive index* of the material given by

$$n = \frac{\text{Velocity of light in free space}}{\text{Velocity of light in the medium}} = \frac{c}{v} \tag{2.2}$$

where $c = \nu\lambda$, λ being the wavelength and ν the frequency of the lightwave in free space. The value of the refractive index for free space or air is 1.00, water is 1.33 and pure silica glass is 1.5. It may be pointed out here that the refractive index of a particular medium depends on the wavelength of the light. For example, the refractive index of pure silica glass is 1.458 at $\lambda = 850$ nm.

When a light ray encounters a boundary separating two different media a portion of the incident light is reflected into the first medium and the rest is bent and transmitted into the second medium. The latter phenomenon is known as *refraction*. The refraction (or bending) of light rays at the interface results from the difference in the speed of light between two media which have different refractive indices. The reflection and refraction of a light ray at the interface of two media are illustrated in Fig. 2.3. In this case light is assumed to travel from a denser to a rarer medium ($n_2 < n_1$). This type of reflection is referred to as *internal reflection*. On the other hand, when light traveling in the medium is reflected off an optically denser medium, the reflection is called *external reflection*. According to the laws of reflection, the incident ray, the reflected ray, and the normal drawn at the point of incidence on the interface lie in one plane. Further, the angle of incidence is equal to the angle of reflection.

In the case of refraction, the ray traveling in the denser medium (n_1) makes an angle of incidence ϕ_1 with the normal drawn on the plane of incidence at the point of incidence. The refracted ray traveling in the other dielectric would make an angle of refractive value ϕ_2 with the normal, where ϕ_2 is greater than ϕ_1. The angle of incidence and angle of refraction are governed by Snell's law, given by

$$n_1 \sin \phi_1 = n_2 \sin \phi_2$$

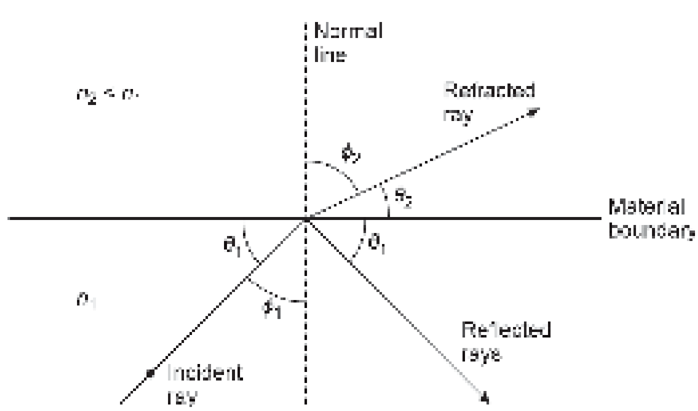

Fig. 2.3 Illustrating refraction for light travelling from a denser to a rarer medium

that is,

$$\frac{\sin\phi_1}{\sin\phi_2} = \frac{n_2}{n_1} \tag{2.3}$$

As the angle of incidence ϕ_1 is increased, the refracted ray bends more and more away from the normal *W* increasing the value of ϕ_2. At a certain value of ϕ_1, the angle of refraction ϕ_2 becomes 90^o and the refracted ray emerges parallel to the interface between the dielectric (Fig. 2.4(a)). If the angle of incidence is more the critical value ϕ_c, the entire light energy is reflected in the denser medium as illustrated in Fig. 2.4(b). This is known as *total internal reflection*. The limiting value of the angle of incidence ϕ_1 (= ϕ_c) in the denser medium for which the refracted ray grazes the interface between the two media is called the *critical angle*, given by

$$n_1 \sin \phi_c = n_2$$

or,

$$\phi_c = \sin^{-1}\left(\frac{n_2}{n_1}\right) \tag{2.4}$$

We shall see later in the subsequent sections that it is often convenient to measure the angle of incidence and reflection with respect to the interface of the two media rather than the normal drawn on the plane at the point of incidence. These angles made by the incident and the refracted rays with respect to the interface of the two media are represented by ϕ_1 and ϕ_2 respectively as shown in Fig. 2.3. If the angles are measured with respect to the interface plane, the critical condition of refraction would be reached when θ_1 is decreased. Snell's law can be written in terms of θ_1 and θ_2 as

$$n_1 \sin\left(\frac{\pi}{2} - \theta_1\right) = n_2 \sin\left(\frac{\pi}{2} - \theta_2\right)$$

that is,

$$n_1 \cos \theta_1 = n_2 \cos \theta_2 \tag{2.5}$$

The critical angle of incidence θ_c to the boundary of these two media can be obtained by putting $\theta_2 = 0$ in Eq. (2.5). Thus, the measured critical angle to the interface plane of the two media can be expressed as

$$\theta_c = \cos^{-1}\left(\frac{n_2}{n_1}\right) \tag{2.6}$$

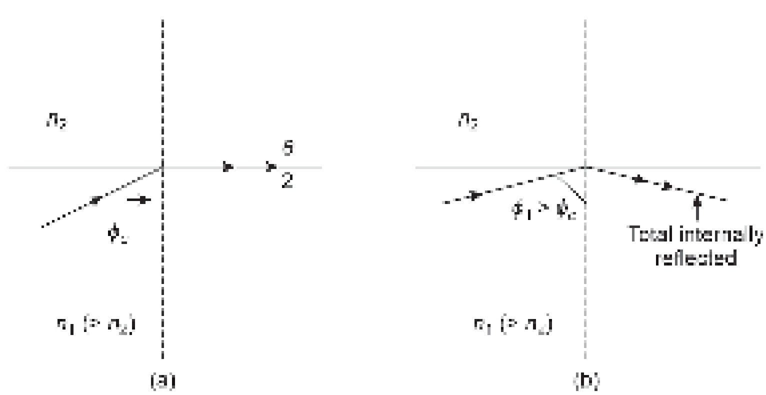

Fig. 2.4 Illustrating total internal reflection for light travelling from a denser to a rarer medium

When the angle of incidence (to the interface) is decreased below the critical angle θ_c, there would be no refraction, and the light would be total internally reflected. This is illustrated with the help of the ray diagram shown in Fig. 2.5.

When the angle of incidence θ_1 (measured to the interface boundary plane of the two media) is equal to the critical angle θ_c the refracted ray grazes the interface (becomes parallel to the interface plane). In other words, θ_c is the minimum value of the angle of incidence for refraction to occur at the interface. When the angle of incidence is less than the critical angle, the total energy is reflected into the original medium, and no part of it is refracted in the rarer medium. This phenomenon is known as total internal reflection. The total internally reflected ray is shown with the help of a double arrow on the reflected ray. In actual practice, nearly 99.9% of the incident light is reflected in the original medium when total internal reflection occurs. The small loss at the interface cannot be explained with the help of ray analysis. The wave theory can account for this loss at the interface. The above simplistic approach underlying the concept of total internal reflection can be easily extended to understand the propagation of light through an optical fiber. It may be pointed out that this simplistic approach does not apply to all types of optical fibers, particularly single-mode fibers or few-mode fibers. Figure 2.5 illustrates the propagation of a light ray through multiple total internal reflections at different points along the interface of the silica core and the cladding with a slightly lower refractive index of a large core-size optical fiber. The dotted line corresponds to the axis of the fiber. The incident ray strikes the core-cladding interface of the fiber at an angle $\theta_1 < \theta_c$. As a result, the ray gets total internally reflected before it strikes the core-cladding interface below the axis of the fiber at the same angle of incidence. Further, the axis of the fiber being parallel to the core-cladding interface plane, the angle of incidence is also equal to the angle the ray makes with the axis. The path of the ray which can be obtained with the help of simple principles of geometrical optics is depicted in Fig. 2.6. It has been assumed that the path of the ray lies in the plane containing the axis of the fiber. Such a ray that lies in the plane containing the axis of the fiber is called a *meridional ray*. The simplistic picture of the transmission of light rays through an optical fiber also presumes that the fiber is perfect and there are no imperfections and/or discontinuities at the core-cladding interface. In actual practice, the propagation of light through an optical fiber is much more complex than the simplistic mechanism discussed above.

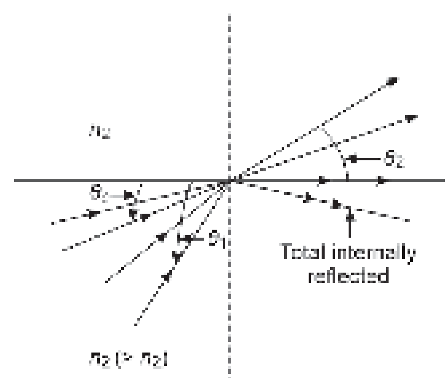

Fig. 2.5 Illustration of total internal reflection

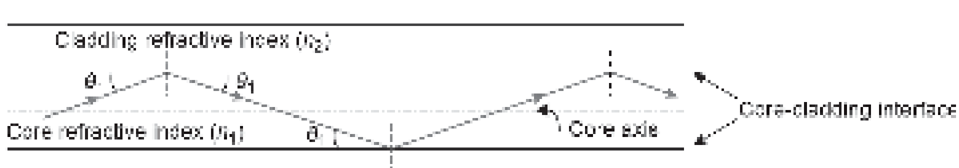

Fig. 2.6 A longitudinal section of an ideal fiber showing the propagation of a light ray along the meridional plane

Example 2.1 Calculate the value of critical angle ϕ_c, at the interface of glass (n_1 = 1.458) and air.

Solution The critical angle (measured to the interface) ϕ_c is given by [Eq. (2.6)]

$$\theta_c = \cos^{-1}\left(\frac{n_2}{n_1}\right) = \cos^{-1}\left(\frac{1}{1.458}\right) = 46°.69$$

For silica glass, the refractive index is taken as n_1 = 1.5. In that case, θ_c turns out to be nearly 48°. Therefore, any ray that makes an angle less the 48°, with the interface of glass and air would be total internally reflected.

Example 2.2 Calculate the value of the critical angle (to the interface) when light travels from glass (n_1 = 1.5) into water (n_2 = 1.33). What is the value of the critical angle with respect to the normal drawn on the interface plane at the point of incidence?

Solution The critical angle measured to the interface can be estimated in this case as

$$\theta_c = \cos^{-1}\left(\frac{n_2}{n_1}\right) = \cos^{-1}\left(\frac{1.33}{1.5}\right) = 27°.54$$

Therefore, the critical angle measured with respect to the normal drawn on the interface at the point of incidence

$$\theta_c = 90° - 27°.54 = 62°.46$$

2.2.2.1 Phase shift in total internal reflection

The simple phenomenon of total internal reflection discussed above based on ray theory exhibits some important features that can be best understood with the help of wave theory. One such feature of total internal reflection is the phaseshift associated with reflected light. The phaseshift between the incident and the reflected light is dependent on the angle of incidence (Fig. 2.7). It may be recalled that in ray representation each ray originates from a set of wave fronts which undergo a phase shift after total internal reflection. It is further found that the phaseshift depends on the polarization of light and increases as the angle of incidence approaches the critical angle. This phaseshift amounts to the Fresnel reflection coefficient becoming a complex number rather than a real one. The mathematical derivations of polarization-dependent phase-shift can be found in the literature (Jenkins *et al* 1957; Born *et al* 1975; Ghatak, 1977). In the case of total internal reflection, the phase change of the reflected wave for the normal and parallel components with respect to the plane of incidence can be expressed as

$$\delta_\perp = 2\tan^{-1}\left(\frac{\sqrt{\dfrac{n_1^2}{n_2^2}\cos^2\theta_1 - 1}}{\dfrac{n_1}{n_2}\sin\theta_1}\right) \quad (2.7)$$

$$\delta_\parallel = 2\tan^{-1}\left(\frac{\sqrt{\dfrac{n_1^2}{n_2^2}\cos^2\theta_1 - 1}}{\dfrac{n_2}{n_1}\sin\theta_1}\right) \quad (2.8)$$

where θ_1 ($<\theta_c$) is the angle of incidence measured with respect to the interface plane of the two media.

Example 2.3 Estimate the maximum and minimum values of the phaseshift encountered by the normal and parallel components of the reflected waves in total internal reflection. Also, plot the variations of these components with the angle of incidence assuming that the light travels from glass ($n_1 = 1.5$) to air ($n_2 = 1.0$).

Solution It can be easily seen from Eqs (2.7) and (2.8) that the phase change for both components decreases with an increase in the angle of incidence to the interface upto the critical angle θ_c.

The minimum values of the phase shift for the two components can be obtained by putting $\theta_1 = \theta_c = \cos^{-1}(n_2/n_1)$ in the Eqs (2.7) and (2.8) as $\delta_\perp = 0$ and $\delta_{||} = 0$.

The maximum values of the phase shift of the reflected wave for the two components can be obtained when the angle of incidence $\theta_1 \to 0$. The maximum values of the phase shift of the reflected wave for the two components can be obtained as $\delta_\perp = 180^\circ$ and $\delta_{||}$ equal to 180°.

The value of the critical angle (θ_c) can be estimated as

$$\theta_c = \cos^{-1}\left(\frac{1}{1.5}\right) = 48^\circ.18$$

The values of phase-shifts of the normal and parallel components of the reflected wave correspond to different values of the angle of incidence θ_1 ($<\theta_c$) can be estimated using Eqs (2.7) and (2.8). The variations of the phase shift with the angle of incidence are shown in Fig. 2.7.

Example 2.4 Estimate the values of the phase-shift encountered by the normal and parallel components of the reflected waves in total internal reflection when the angle of incidence (to the interface) is 20°. Assume that the light travels from the glass ($n_1 = 1.458$) to water ($n_2 = 1.33$). Estimate the value of the angle of incidence for which the phase-shift encountered by each component is zero.

Solution The value of phase shift for the normal component can be estimated as

$$\delta_\perp = 2\tan^{-1}\left(\frac{\sqrt{\frac{(1.458)^2}{(1.33)^2}\cos^2 20^\circ - 1}}{\frac{1.458}{1.33}\sin 20^\circ}\right) = 66^\circ.81$$

and the value of parallel component can be estimated as

$$\delta_{||} = 2\tan^{-1}\left(\frac{\sqrt{\frac{(1.458)^2}{(1.33)^2}\cos^2 20^\circ - 1}}{\frac{1.33}{1.458}\sin 20^\circ}\right) = 76^\circ.80$$

The phase-shift undergone by each component is zero when

$$\theta_1 = \theta_c = \cos^{-1}\left(\frac{1.33}{1.458}\right) = 24^\circ.19$$

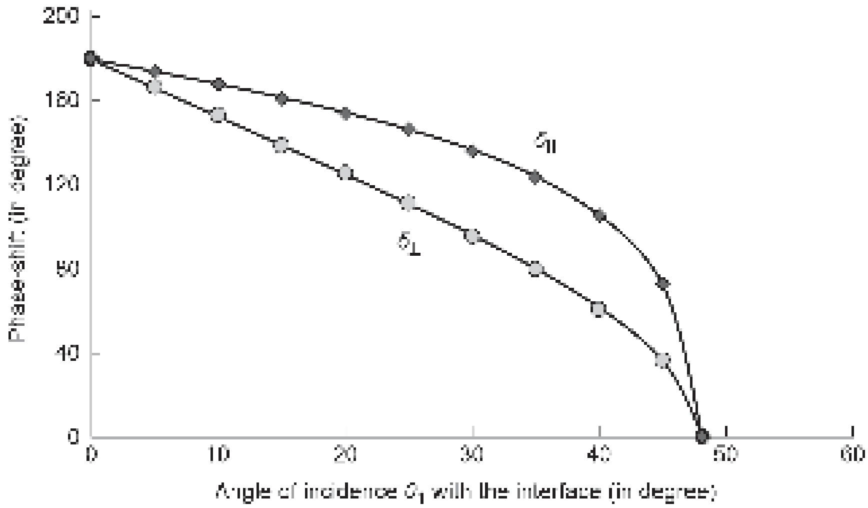

Fig. 2.7 Variation of phase-shift in the reflected wave for parallel and normal components encountering total internal reflection at the interface of glass and air

2.2.2.2 Goos-Hänchen shift

Another important effect associated with total internal reflection arises from the phase shift encountered by the reflected wave. It is observed that the reflected beam is laterally shifted from the expected path predicted by the simple ray analysis. This lateral shift is caused by the propagation of the evanescent-wave[3] across the interface towards the lighter medium. This lateral shift is known as Goos-Hänchen shift and is illustrated with the help of Fig. 2.8. It may be stressed here that even though it is assumed that the entire energy of the incident wave is reflected into the originating medium in total internal reflection, there is some penetration into the lighter medium at the boundary. The evanescent wave is believed to travel along the interface of the two media.

A consequence of the Goos-Hänchen shift in ray analysis is that the geometric reflection appears to occur from a virtual plane that is parallel to the interface and situated in the lighter medium close to the interface boundary. The lateral shift of the reflected beam from the path (d) predicted by ray theory can be estimated using wave theory (Midwinter, 1979). The shift which ranges generally from a few tens of nanometers to 100 nm goes unnoticed

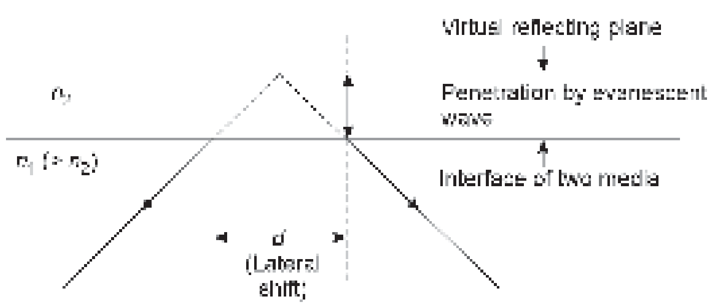

Fig. 2.8 Lateral shift of the reflected beam in total internal reflection (Goos-Hänchen shift)

[3] A near-field standing wave decaying exponentially with distance from the boundary at which the wave is formed.

in a macro-scale. Nevertheless, the principle underlying the Goos-Hänchen shift provides useful insight into the mechanism of propagation of light through dielectric waveguides including optical fibers.

2.3 CLASSIFICATION OF OPTICAL FIBERS

Before applying ray analysis further for exploring the propagation of light through an optical fiber it is worthwhile to have a look at the various types of fibers used in optical communication. Even though ray analysis is applicable for describing the propagation of light through optical fibers with large-size cores, an accurate analysis is possible only with mode analysis based on Maxwell's electromagnetic equations. The modes may be viewed as electromagnetic field patterns which are solutions of Maxwell's equations under the given boundary conditions. In this section, we present a brief description of the fiber configurations and the basis of their classification. This will be followed by the quantitative formulation of some useful parameters of the optical fiber by applying ray analysis.

An optical fiber is a cylindrical dielectric waveguide that confines electromagnetic energy in the form of light and guides it in a direction parallel to its axis. The transmission characteristics of the fiber depend on the structure of the fiber. An optical fiber, in general, consists of a solid dielectric cylinder of radius, a and a refraction index of n_1. This cylinder is known as the core of the fiber and is generally surrounded by another solid coaxial dielectric cylinder of refractive index n_2 ($<n_1$) forming the cladding. In principle, cladding is not essential for light to propagate along the core alone. However, as discussed earlier cladding reduces scattering loss occurring otherwise from the bare core surface. It also provides mechanical support to the fiber and protects the core from undesirable containments. In the simplest form, an optical fiber generally has an elastic jacket on the top of the cladding. A schematic of a single fiber generally used in laboratories is illustrated in Fig. 2.9. In practice, a variety of materials are used for making the core and cladding of an optical fiber. Principle materials used for making optical are usually glass and plastic. Plastic fiber uses polymer materials of different refractive indices for making the core and cladding regions. Glass or silica fibers on the other hand use glass as the principle material for making both the core and the cladding. The refractive index variation in glass is usually achieved by adding a tiny amount of an appropriate dopant. Both plastic and glass fiber are generally encapsulated in elastic jackets to add further strength and protection.

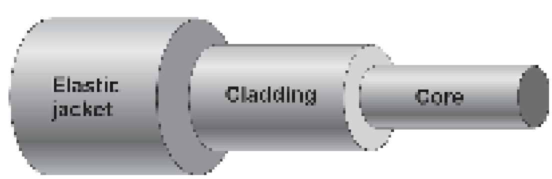

Fig. 2.9 Schematic of a single fiber strand

2.3.1 Step-index (SI) and Graded Index (GI) Fibers

The variation in the refractive index in the core and the cladding region is caused by the compositional variation of the materials used for making them. Based on the nature of variation of the refractive index in the core and cladding regions, optical fibers are classified under two categories, e.g. step-index (SI) fiber and graded-index (GI) fiber. In the core of step index fiber, the refractive index (RI) of the core is constant (n_1) and uniform throughout. The core refractive index undergoes an abrupt change to n_2 at the core-cladding interface and remains the same in the entire cladding region.

Mathematically, the refractive index of a step-index (SI) fiber can be expressed as

$$n(r) = \begin{cases} n_1 & \text{for } r < a \\ n_2 & \text{for } r \geq a \end{cases} \tag{2.9}$$

where r is the distance measured from the center of the core along the radius, a is the core radius and n_1 (core RI) $> n_2$ (cladding RI) so that light may travel along the core of the fiber by total internal reflection at the core-cladding interface. The core-cladding index difference or deviation Δ can be expressed as

$$\Delta = \frac{n_1 - n_2}{n_1} \tag{2.10}$$

Alternatively, the cladding refractive index can be expressed in terms of the core refractive index and the index deviation in a step-index fiber as

$$n_2 = n_1(1 - \Delta) \tag{2.11}$$

In practical optical fibers n_2 is chosen close to n_1 for reasons to be discussed later. The value of Δ generally ranges between 0.2 to 3. Typically the value of Δ is chosen as 0.01.

On the other hand, in a GI fiber the core refractive index is made to vary as a function of the radial distance from the center of the fiber core. The nature of the shape of the variation of the refractive index of a GI fiber may range from triangular, or parabolic to any higher-order profile. The nature of this variation can be adjusted during the fabrication of the fiber. Mathematically, the variation of refractive index along the radial distance of a graded-index fiber can be expressed as

$$n(r) = \begin{cases} n_1\left[1 - 2\Delta\left(\dfrac{r}{a}\right)^{\alpha}\right]^{1/2} & r < a \\ n_1(1-2\Delta)^{1/2} \approx n_1(1-\Delta) = n_2 & r \geq a \end{cases} \tag{2.12}$$

where r is the distance measured from the center of the core along the radius, a is the core radius, n_1 is the refractive index at the center of the core, i.e. at the axis of the core and n_2 is the refractive index at the core-cladding interface which remains constant throughout the cladding region and α is a dimensionless parameter that defines the shape of the refractive index profile. For example, if $\alpha = 1$, the index profile becomes triangular, and for $\alpha = 2$ the index profile becomes parabolic. When $\alpha = \infty$, the index profile becomes abrupt and the GI fiber reduces to an SI fiber. In practical GI fibers, α is set close to 2 to limit one form of dispersion (intermodal dispersion). The index difference of a GI fiber can be obtained from Eq. (2.12) as

$$\Delta = \frac{n_1^2 - n_2^2}{2n_1^2} = \frac{(n_1 + n_2)(n_1 - n_2)}{2n_1^2} \approx \frac{n_1 - n_2}{n_1} \tag{2.13}$$

In the above derivation, it has been assumed that $n_1 \approx n_2$ thereby, $n_1 + n_2 \approx 2n_1$.

It is interesting to note that the index difference of a GI fiber can be approximated by the same expression as that of a SI fiber (Eq. (2.10)). For the sake of comparison of an SI fiber with a corresponding GI fiber, the core refractive index of the GI fiber at the center or axis of the core and that at the core-cladding interface must be related to the corresponding core and cladding refractive indices of the SI fiber as

$$n(r = 0) = n_1$$
$$n(r = a) = n_2$$

The schematics of step-index and graded-index fibers are shown in Fig. 2.10 along with their refractive index profiles.

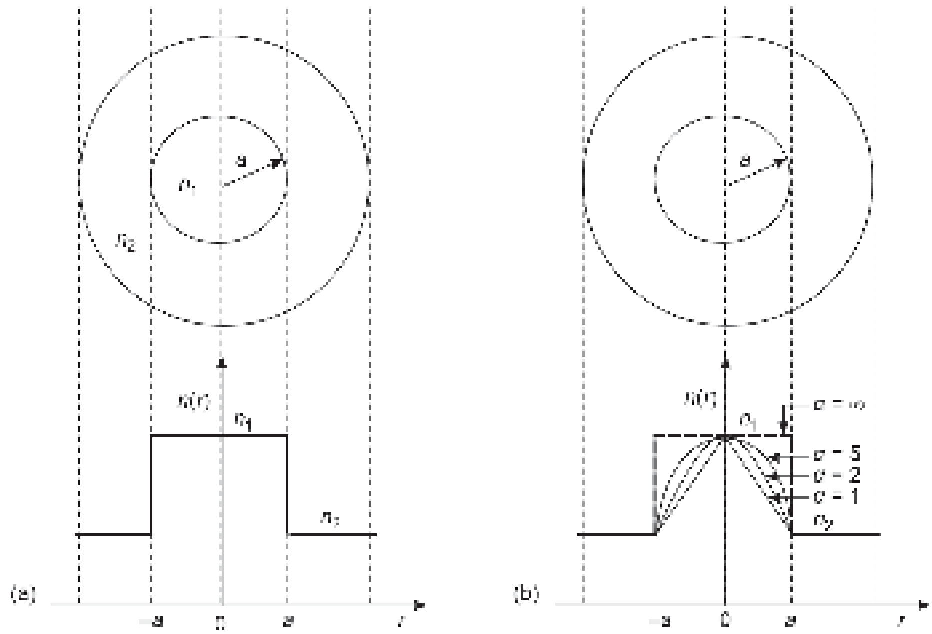

Fig. 2.10 Refractive index profiles along radial distance measured from the center of the core taken as reference in (a) SI fiber (b) GI fiber showing the transverse sections of the fibers

2.3.2 Single-mode and Multimode Fibers

Although ray analysis enables one to visualize the transmission of lightwaves through optical fiber via total internal reflection, it should be noted that ray analysis is an approximate one and cannot be applied in all situations. An alternative method of understanding the propagation of lightwave through an optical fiber is based on the electromagnetic theory of Maxwell. The propagation of light through a fiber waveguide is described in this method in terms of a set of guided electromagnetic field patterns called modes in the waveguide. A detailed analysis of these modes will be discussed afterward. As will be seen later in this chapter, each fiber supports a discrete number of modes determined by the parameters of the fiber and the operating wavelength. Based on the modes propagating in an optical fiber, both SI and GI fibers can be further classified as single-mode fiber and multimode fiber. As the names imply, a single mode supports only one mode (fundamental mode) of propagation for wave guiding. On the other hand, multimode fibers may contain a few modes to hundreds of modes for propagation along the fiber. The core diameter of a single-mode fiber is extremely small (8–12 μm) as compared to that of a multimode fiber which varies between 50–200 μm for SI fiber and 50–100 μm for GI fiber. Because of the larger dimension, it is easier to launch power from an optical source to a multimode fiber. On the other hand, single-mode fibers require a highly directive source for launching power. For example, a diffuse source like a light-emitting diode (LED) can be used to launch optical power to a multimode fiber while a directive source like an injection laser diode (ILD) is generally used for launching power to a single-mode fiber.

A major disadvantage associated with a multimode fiber arises from intermodal dispersion. This will be discussed in detail in Chapter 4. To have a qualitative understanding of this effect, let us assume that a light pulse is launched into a multimode fiber from an optical source. The light energy gets distributed within the fiber among all the modes that the fiber can support. Each of these modes propagates through the fiber but each with a slightly different velocity. This difference in velocity of the individual modes causes the modes in a given pulse to arrive at the end of the fiber end at different instants of time. This causes the pulse to spread out in time. This effect is known as intermodal dispersion. The width of the received pulse becomes larger than the time allocated to the pulse at the time of transmission. The spreading of the pulse causes interference with adjacent bits causing inter symbol interference (ISI). In Chapter 4, we will see that ISI caused by intermodal dispersion highly restricts the rate at which we can transmit optical pulse through a multimode fiber. The intermodal dispersion can be significantly reduced by making use of a graded-index fiber. The grading in the refractive index of the fiber core helps the oblique rays corresponding to the higher order modes to travel faster and make up for the delay caused by various modes. The data rate transmission capacity and so also the bandwidth of multimode fibers can be significantly improved by making use of a graded index fiber in place of a step-index fiber. Further, in a single-mode fiber, only one mode is present and as a result, it is free from intermodal dispersion. Thus, single-mode fiber has a much larger bandwidth and is attractive for use in long-haul optical communication.

The propagation of light rays through different types of fibers is illustrated with the help of ray diagrams in Fig. 2.11. Based on ray analysis, it is easier to visualize the transmission of light through a multimode step-index fiber by considering multiple total internal reflection episodes at different points along the core-cladding interface for all rays that make angles less than the critical angle θ_c with the interface. As the ray picture does not hold for a small geometry fiber, such as a single-mode fiber the propagation through a single-mode fiber is indicated in Fig. 2.11(b) by using an axial beam. Further, in a graded index fiber the refractive index decreases progressively along the radius of the fiber as

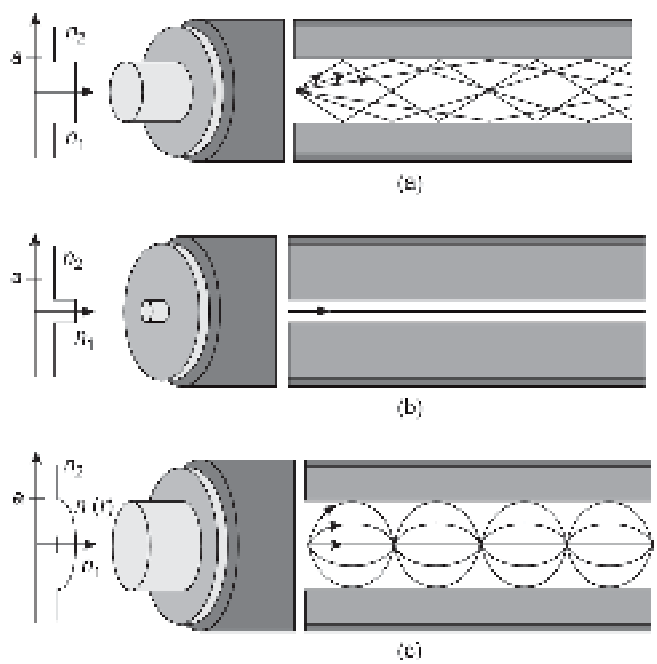

Fig. 2.11 Illustrating the path of light rays through (a) Multimode step-index fiber (b) Single mode step-index fiber (c) Multimode graded-index fiber

one moves from the center of the core towards the core-cladding interface. As a result, an incident ray making a certain angle with the core axis gets bent continuously as it moves toward the cladding. The rays propagating through GI fibers are therefore indicated with the help of curved ray paths as indicated in Fig. 2.11(c). To appreciate the propagation of light rays through a GI fiber let us consider that core to consist of several coaxial cylindrical layers of slightly different refractive indices as shown in Fig. 2.12. The refractive index of the central cylinder is the highest and it decreases progressively in the outer cylindrical layers. For the sake of simplicity, each cylindrical layer may be assumed to have a constant refractive index. As a meridional ray[4] travels from the central region towards the core-cladding interface, it suffers multiple refractions at successive interfaces of high to low refractive indices of the imaginary cylindrical layers. As the ray travels from a denser to rarer medium at each interface, the angle of incidence at successive interfaces goes on increasing (because of the bending of the ray away from the normal at each refraction) until at a certain layer the condition of total internal reflection is satisfied. When the angle of incidence at a particular interface becomes less than the critical angle, the ray gets total internally reflected and starts traveling back towards the core axis as shown in Fig. 2.12. The incident ray making different angles with the axis of the GI fiber is thus reflected from different radial distances from the axis as shown in Fig. 2.13.

It is interesting to note that a multimode fiber is generally made to have a graded-index profile so that intermodal dispersion can be minimized. From this view point, a single-mode

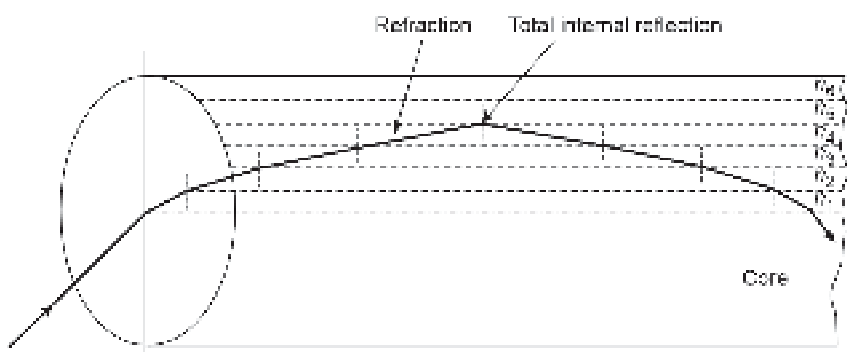

Fig. 2.12 Illustrating the trajectory of a meridional ray through the core of a graded-index fiber approximated by several coaxial solid cylinders of different refractive indices

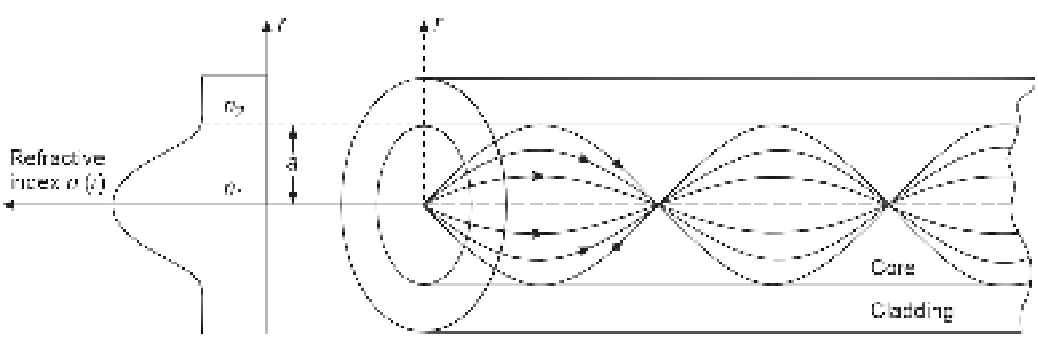

Fig. 2.13 The path of light rays in a graded-index fiber looking like sinusoids because of total internal reflections for different rays occurring at different radial distances

[4] A ray which lies in the plane containing the axis of symmetry of the fiber is called a *meridional ray*.

fiber that does not suffer from intermodal dispersion should have a step-index profile. However, single-mode fibers are sometimes deliberately designed to have graded-index profiles, not to combat intermodal dispersion but to have additional flexibility in the design in terms of larger allowable diameters at a given wavelength of operation.

2.4 RAY OPTICS REPRESENTATION

The ray optics is valid when the wavelength of light is very small as compared to the size of the object (in this case core radius). In the case of multimode fiber, the core radius is much larger than the wavelength of light (typically of the order of 1 μm) used in optical fiber communication. The propagation mechanism of light in an ideal multimode step-index fiber can be visualized by simple ray (geometrical) optics representation.

2.4.1 Meridional and Skew Rays

Two types of rays can propagate in a fiber, e.g. meridional rays and skew rays. Meridional rays are those rays that are confined to the meridional planes of the fiber. Meridional planes are those which certain the axis of symmetry that is, the fiber core axis. As a particular meridional ray lies in a single plane, it is easy to track its path as it propagates through the fiber. Meridional rays are further divided into two categories, e.g. bound rays or trapped rays and unbound rays. The bound rays propagate along the fiber core following the laws of total internal reflection and unbound rays are refracted out of the core into the cladding region. The paths of both ray categories of meridional rays in a step-index fiber are illustrated in Fig. 2.14.

The other type of rays that propagate through the core without passing through the fiber axis is called *skew rays*. These rays are larger in number as compared to meridional rays and follow a helical path along the fiber. As these rays do not lie in the same plane it is very difficult to track them. Nevertheless, the skew rays constitute a large section of the total bound rays propagating through the fiber. These skew rays circulate along the core following a zigzag path and cannot be easily visualized in two dimensions. The helical path followed by skew rays is illustrated in Fig. 2.15 showing the path down the fiber as well as its appearance at the cross-section of the fiber. It is clear from Fig. 2.15(b) that the skew rays change direction by 2γ at each reflection where γ is the angle between the projection of the ray in two dimensions and the radius of the fiber core at the point of reflection.

2.4.2 Numerical Aperture (NA) and Acceptance Angle for Meridional Rays

The path of meridional rays in an ideal step-index fiber is shown in Fig. 2.16. Consider a light ray OA entering the fiber core from a medium of refractive index n_0. Let the ray

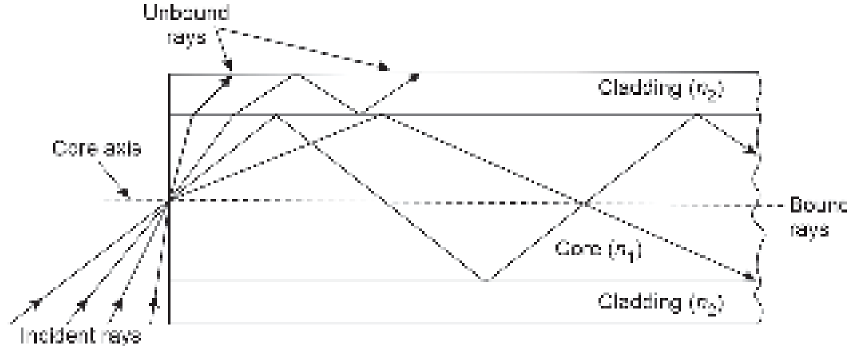

Fig. 2.14 Bound and unbound rays in the SI optical fiber in ray representation

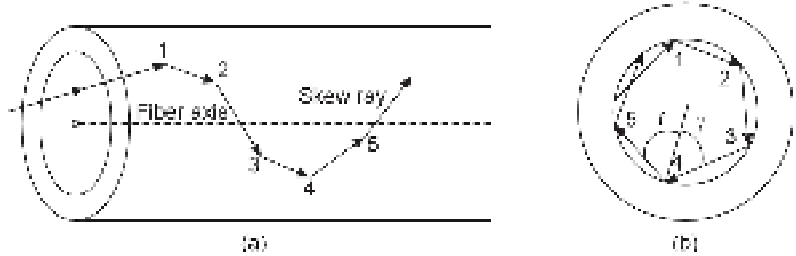

Fig. 2.15 The helical trajectory of a skew ray (a) Down the fiber (b) Cross-sectional view of the fiber

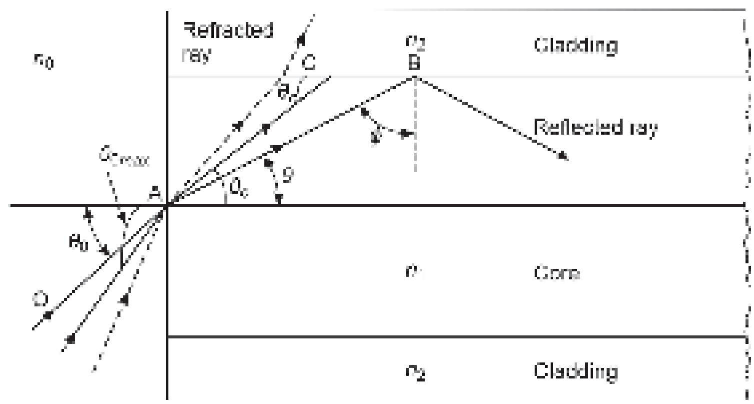

Fig. 2.16 Representation of meridional ray path through a step-index fiber for computation of numerical aperture

make an angle θ_0 with the axis of the core. The incident ray is refracted into the core region having a refractive index of n_1 and strikes the core-cladding interface at a point B where it makes an angle ϕ with the normal drawn on the interface at B that is, $\theta\left(=\frac{\pi}{2}-\phi\right)$ with the interface or with the axis of the core which is parallel to the interface.

If the angle θ is less than the critical angle θ_c, the ray would be total internally reflected as shown in Fig. 2.16, and follow a zigzag path down the fiber length and after each reflection, it will pass through the axis of the fiber. If the angle of incidence θ_0 is increased gradually, a stage will be reached when the refracted ray in the core region will strike the core-cladding interface at an angle $\theta(=\theta_c)$ with the interface or $\phi(=f_c)$ with the normal. Under this condition, the refracted ray would just graze the interface (as shown by the double arrow). If θ_0 is increased further, the ray entering the core region will strike the core-cladding interface at an angle $\theta > \theta_c$ and the ray would be refracted out of the core as shown by the dashed ray path.

Therefore, the maximum angle (to the core-cladding interface) that would support total internal reflection can be obtained by applying Snell's law at point B. That is,

$$n_1 \sin\phi_c = n_2 \sin\frac{\pi}{2} = n_2$$

$$\Rightarrow \qquad n_1 \sin\left(\frac{\pi}{2}-\theta_c\right) = n_2 \qquad (2.14)$$

$$\Rightarrow \qquad \cos\theta_c = \frac{n_2}{n_1} \qquad (2.15)$$

The maximum entrance angle $\theta_{0\max}$ in the incident medium of refractive index n_0 that would support total internal reflection of the ray entering into the core region can be obtained by applying Snell's law at the point of entrance, A. Thus

$$n_0 \sin\theta_{0\max} = n_1 \sin\theta_c = n_1\sqrt{1-\cos^2\theta_c} \tag{2.16}$$

The parameter $n_0 \sin\theta_{0\max}$ is called the numerical aperture (NA) of the step-index fiber and can be written using equations as

$$\mathrm{NA} = n_0 \sin\theta_{0\max} = n_1\sqrt{1-\frac{n_2^2}{n_1^2}}$$

$$= \sqrt{n_1^2 - n_2^2} \tag{2.17}$$

The NA of the step-index fiber can also be expressed in terms of the index difference parameter, Δ. Equation (2.17) can be written as

$$NA = n_1\sqrt{\frac{n_1^2-n_2^2}{n_1^2}} \approx n_1\sqrt{\frac{(n_1-n_2)(n_1+n_2)}{n_1^2}}$$

$$\approx n_1\sqrt{\frac{2n_1(n_1-n_2)}{n_1^2}} \approx n_1\sqrt{2\Delta} \tag{2.18}$$

The above derivation is based on the approximation, $n_1 \approx n_2$ which is valid when Δ is very small. This approximation is generally true for all practical fibers because Δ is generally 0.01 or less.

It may be noted that all rays that have incident angles less than $\theta_{0\max}$ will be total internally reflected at the core-cladding interface and guided along the fiber. On the other hand, the rays that make an angle larger than $\theta_{0\max}$ at the entrance would be refracted out into the cladding region and lost. Therefore, $\theta_{0\max}$ is the maximum angle to the fiber axis at which light may enter the fiber and propagate subsequently. This is illustrated in Fig. 2.17. This conical half-angle $\theta_{0\max}$ is referred to as the maximum acceptance angle or simply the acceptance angle of the fiber. The acceptance angle for a step-index fiber can be written as

$$\theta_{0\max} = \sin^{-1}\left(\frac{\mathrm{NA}}{n_0}\right) = \sin^{-1}\left(\frac{\sqrt{n_1^2-n_2^2}}{n_0}\right) \tag{2.19}$$

In case light enters the fiber from the air, i.e. $n_0 = 1$ and we may obtain the acceptance angle as

$$(\theta_{0\max})_{\mathrm{air}} = \sin^{-1}(\mathrm{NA}) = \sin^{-1}\left(\sqrt{n_1^2-n_2^2}\right) \tag{2.20}$$

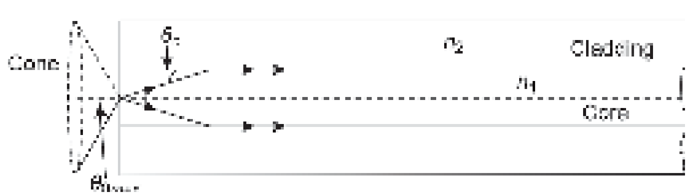

Fig. 2.17 Illustrating the maximum acceptance angle of the step-index fiber and the solid acceptance cone

It may be stressed here that in actual practice, more often than not the light enters the fiber core from another medium (called index matching fluid) having nearly the same refractive index. This reduces the Fresnel's reflection loss at the entrance.

Further if $\theta_{0\max}$ is very small, we may approximate

$$\sin\theta_{0\max} \approx \theta_{0\max} = \sqrt{n_1^2 - n_2^2} \tag{2.21}$$

For a small value of $\theta_{0\max}$ the maximum solid acceptance angle of the fiber can be approximated as

$$\Omega = \int_0^{\theta_{0\max}} \int_0^{2\pi} \sin\theta_0 d\theta_0 d\phi \approx 2\pi \int_0^{\theta_{0\max}} \theta_0 d\theta_0$$

$$\pi\theta_{0\max}^2 = \pi\left(n_1^2 - n_2^2\right) \tag{2.22}$$

The numerical aperture of a fiber is a dimensionless quantity and ranges between 0.15 and 0.50 depending on the index difference between the core and the cladding. The plastic fibers generally have a large difference between the refractive index of the core and that of the cladding. As a result, plastic fibers have a large value of numerical aperture. It may be pointed out that the importance of numerical aperture is that it measures the light-gathering power of the optical fiber from a source. For example, the light coupled from an LED source to a step-index fiber is proportional to the square of the numerical aperture. This means that a larger value of numerical aperture will mean a larger light-gathering power of the fiber. One should prefer to have optical fibers with a larger numerical aperture. However, from an optical fiber communication point of view, there is an adverse consequence of using fibers with a large numerical aperture. A large numerical aperture calls for a large difference between the refractive indices of the core and the cladding. This large difference in refractive index value results in large intermodal dispersion which severely restricts the bandwidth of transmission.

Example 2.5 A step-index silica fiber has a core refractive index 1.48 and a cladding refractive index 1.47. Calculate (i) the critical angle (wrt axis) at the core-cladding interface (ii) the NA of the fiber and (iii) the acceptance angle assuming the entrance medium to be air.

Solution (i) The critical angle θ_c at the core-cladding interface (wrt the axis of the fiber) is

$$\theta_c = \cos^{-1}\left(\frac{n_2}{n_1}\right) = \cos^{-1}\left(\frac{1.47}{1.48}\right) = 6°.67$$

Note that the critical angle to normal on the interface at the point of incidence

$$\phi_c = \frac{\pi}{2} - 6°.67 = 83°.33$$

(ii) The numerical aperture of the fiber [Eq. (2.17)] is

$$\text{NA} = \sqrt{n_1^2 - n_2^2} = \sqrt{(1.48)^2 - (1.47)^2} = 0.17$$

(iii) Using Eq. (2.20), the maximum acceptance angle can be obtained as

$$\theta_{0\max} = \sin^{-1}(\text{NA}) = \sin^{-1}(0.17) = 9°.8$$

Example 2.6 The relative refractive index difference of a silica fiber is 1%. Calculate the solid acceptance angle of the fiber in air assuming the core-refractive index to be 1.458.

Solution The NA of the fiber is given by [Eq. (2.14)]

$$\mathrm{NA} = n_1\sqrt{2\Delta} = 1.458\sqrt{0.02} = 0.21$$

The solid acceptance angle of the fiber can be obtained for Eq. (2.22) as

$$\Omega_{\max} = \pi(\mathrm{NA})^2 = 3.14 \times (0.21)^2 = 0.14 \text{ sr (steradian}^5)$$

Example 2.7 A step-index silica fiber with a core radius much longer than the operating wavelength of light has a core refractive index of 1.50 and a cladding refractive index of 1.48. Estimate the values of (i) the numerical aperture of the fiber (ii) the maximum acceptance angle in air and (iii) the maximum acceptance angle in water having a refractive index of 1.33.

Solution The large dimension of the core in comparison with the operating wavelength allows us to apply the ray analysis approach for the computation of numerical aperture and acceptance angle.

1. The numerical aperture of the fiber is

$$\mathrm{NA} = \sqrt{n_1^2 - n_2^2} = \sqrt{(1.50)^2 - (1.48)^2} = 0.24$$

2. The acceptance angle of the fiber in the air is [Eq. (2.20)]

$$(\theta_{0\max})_{\text{air}} = \sin^{-1}(\mathrm{NA}) = \sin^{-1}(0.24) = 14°.13$$

3. The acceptance angle of the fiber in water (see Eq. (2.19))

$$(\theta_{0\max})_{\text{water}} = \sin^{-1}\left(\frac{\mathrm{NA}}{n_0}\right) = \sin^{-1}\left(\frac{0.24}{1.33}\right) = 10°.39$$

2.4.3 Acceptance Angle for Skew Rays

As the propagation of skew rays is not confined to a single plane that contains the axis of symmetry of the fiber, it is necessary to define the direction of the ray in two perpendicular planes. The maximum acceptance angle for the skew rays in a step-index fiber can be estimated (Senior, 1992)

$$\sin(\theta_{0\max})_{\text{skew}} = \frac{\mathrm{NA}}{\cos\gamma} = \frac{\sqrt{n_1^2 - n_2^2}}{\cos\gamma} \tag{2.23}$$

That is,

$$(\theta_{0\max})_{\text{skew}} = \sin^{-1}\left(\frac{\mathrm{NA}}{\cos\gamma}\right) \tag{2.24}$$

where γ is the angle between the core radius and projection of the skew ray in two dimensions. The helical path followed by a skew ray through a fiber when viewed in two dimensions appears to give a change in direction by 2γ. A comparison of Eq. (2.20) and Eq. (2.24) reveals that the skew rays are accepted by the fiber at a larger acceptance angle than the meridional rays depending on the value of $\cos\gamma$ (≤ 1). It may be noted that a particular skew ray is characterized by the angle γ which may vary between 0° for

[5] Solid angles are measured in steradian (sr).

meridional rays and 90° for skew rays entering at the core-cladding interface giving a maximum acceptance angle of $\pi/2$. For meridional rays $\cos\gamma = 1$ and the acceptance angle of skew rays reduces to that for meridional rays.

Example 2.8 A step-index fiber is air has a numerical aperture of 0.22. Calculate the acceptance angle in the air for skew rays that change direction by 110° at each reflection.

Solution The skew rays under consideration change direction by 110° at each reflection.
Therefore, $2\gamma = 110°$
That is $\gamma = 55°$
Using Eq. (2.24), we get

$$(\theta_{0\,\max})_{\text{skew}} = \sin^{-1}\left(\frac{\text{NA}}{\cos\gamma}\right) = \sin^{-1}\left(\frac{0.22}{\cos 55°}\right) = 22°.55$$

Example 2.9 A step-index fiber has core and cladding refractive indices of 1.45 and 1.43 respectively. Calculate the maximum acceptance angle of the fiber in the air for meridional rays. Compare this value with that for skew rays that change direction within the core by 100° at each reflection.

Solution The numerical aperture of the fiber is

$$\text{NA} = \sqrt{n_1^2 - n_2^2} = \sqrt{(1.45)^2 - (1.43)^2} = 0.24$$

The maximum acceptance angle of the fiber in the air ($n_0 = 1$) for the meridional rays is

$$\theta_{0\,\max} = \sin^{-1}(0.24) = 13°.88$$

As the skew rays change direction by 100° at each reflection, we have

$$2\gamma = 100°$$

That is, $\gamma = 50°$

Using Eq. (2.24), we get

$$(\theta_{0\,\max})_{\text{skew}} = \sin^{-1}\left(\frac{\text{NA}}{\cos\gamma}\right) = \sin^{-1}\left(\frac{0.24}{\cos 50°}\right) = 21°.92$$

Thus the acceptance angle for the skew rays is nearly 8° greater than that for meridional rays.

Example 2.10 A step-index fiber in air accepts skew rays at a maximum axial angle of 45°. If the skew rays change direction by 90° at each reflection within the fiber core, estimate the acceptance angle for the meridional rays for the fiber in the air.

Solution In the case

$$2\gamma = 90°$$

That is, $\gamma = 45°$

Using Eq. (2.23), we get

$$\sin 45° = \frac{\text{NA}}{\cos 45°}$$

That is,

$$\text{NA} = \sin 45° \cdot \cos 45° = \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}} = 0.5$$

The maximum acceptance angle of the fiber in the air ($n_0 = 1$) for the meridional rays is

$$\theta_{0\max} = \sin^{-1}(0.5) = 30°$$

2.4.4 Numerical Aperture of Graded-index Fibers

The refractive index of a graded-index fiber is a function of radial distance r from the center of the core and is constant in the cladding. As a result, it is expected that the numerical aperture of GI fiber will also be a function of radial distance. Recalling the expression of the numerical aperture of a SI fiber, the local numerical aperture of a GI fiber can be defined in an analogous way (Gloge *et al* 1973).

$$\mathrm{NA}(r) = \sqrt{n^2(r) - n_2^2} \tag{2.25}$$

Using the expression of $n(r)$ from Eq. (2.12), we get

$$\mathrm{NA}(r) = \sqrt{n_1^2\left[1 - 2\Delta\left(\frac{r}{a}\right)^{\alpha}\right] - n_2^2}$$

$$= \sqrt{n_1^2 - n_2^2}\left[1 - \frac{2n_1^2\Delta}{n_1^2 - n_2^2}\left(\frac{r}{a}\right)^{\alpha}\right]^{1/2}$$

$$= \mathrm{NA}(0)\sqrt{1 - \left(\frac{r}{a}\right)^{\alpha}} \tag{2.26}$$

where $NA(0)$ is the numerical aperture at the center of the core, also called the axial numerical aperture of the GI fiber, given by

$$\mathrm{NA}(0) = \sqrt{n^2(0) - n_2^2} = \sqrt{n_1^2 - n_2^2} \approx n_1\sqrt{2\Delta} \tag{2.27}$$

The numerical aperture of a GI fiber has a maximum value NA(0) at the center of the core and reduces to zero at the core-cladding interface ($r = a$). The variation of the numerical aperture of a GI fiber with radial distance is shown in Fig. 2.18. It is interesting to note that the local numerical aperture of a GI fiber at a particular point depends on the value of α which defines the shape of the index profile. For an SI fiber ($\alpha = \infty$) the numerical aperture is constant and equal to NA(0) of a corresponding GI fiber and drops abruptly to zero at the core-cladding interface as shown in Fig. 2.18.

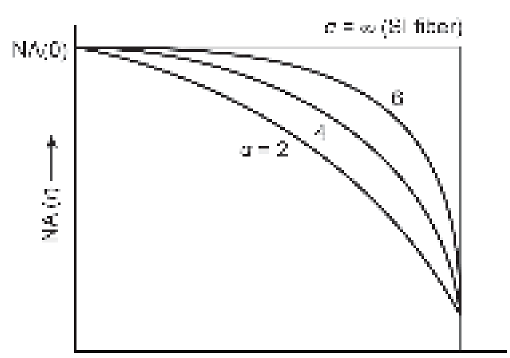

Fig. 2.18 Variation of numerical aperture of a GI fiber with radial distance from the center of the core

Example 2.11 A GI fiber with a parabolic profile has a refractive index of 1.458 at the center of the core. Estimate the value of the numerical aperture of the fiber at a point exactly midway between the axis of the core and the core-cladding interface assuming $\Delta = 0.01$.

Solution The axial numerical aperture of the GI fiber is

$$\mathrm{NA}(0) \approx n_1\sqrt{2\Delta} = 1.458\sqrt{2 \times 0.01} = 0.206$$

For a GI fiber with a parabolic-index profile, $\alpha = 2$. Therefore, the local numerical aperture at $r = a/2$ is

$$\mathrm{NA}(r = a/2) = \mathrm{NA}(0)\sqrt{1-\left(\frac{r}{a}\right)^2} = 0.206\sqrt{1-\left(\frac{1}{2}\right)^2} = 0.178$$

Example 2.12 A step-index fiber has an acceptance angle of 18° in air. The fiber has a relative refractive index of 2.5%. Estimate the value of the critical angle at the core-cladding interface of the fiber and also the numerical aperture of the fiber.

Solution The acceptance angle $\theta_{0\max}$ can be expressed as

$$\sin\theta_{0\max} = \sqrt{n_1^2 - n_2^2} = n_1\sqrt{2\Delta}$$

That is,

$$\sin 20° = n_1\sqrt{2\times 0.025} = 0.223 n_1$$

Therefore, $n_1 = 1.533$

The refractive index of the cladding can be estimated as

$$n_2 = 1.533\ (1 - 0.025) = 1.494$$

The value of the critical angle at the core-cladding interface can be estimated as

$$\theta_c = \cos^{-1}\left(\frac{n_2}{n_1}\right) = \cos^{-1}\left(\frac{1.494}{1.533}\right) = 12°.95$$

The value of the numerical aperture is

$$\mathrm{NA} = n_1\sqrt{2\Delta} = 1.533\sqrt{2\times 0.025} = 0.34$$

2.5 FIBER MATERIALS

Optical fibers are long, thin, and flexible strands of optically transparent materials that work as optical waveguides. The principal materials used for making optical fibers are generally based on some form of glass or plastic material or a combination of both. Fused silica (amorphous silicon dioxide, SiO_2) is the most widely used material for the fabrication of high-quality optical fibers. Silica glass exhibits the following properties which make it especially attractive for making optical fibers.

1. Silica has good optical transparency in the NIR wavelength region ranging from 0.85 μm to 1.65 μm. High-quality silica glass exhibits the lowest attenuation of 0.2 dB/km around 1.5 μm wavelength.
2. Long strands of fibers can be drawn from molten silica at reasonably high temperatures.
3. Silica-based fibers can be spliced and cleaved without much practical difficulties.
4. A silica fiber has an extremely high mechanical strength against pulling and even bending, provided that the fiber is not too thick and that the surfaces are well prepared. The mechanical strength of a fiber can be further improved with a suitable polymer jacket. Even simple cleaving (breaking) of silica fiber ends can provide nicely flat surfaces with sufficient optical quality.
5. Silica is chemically very stable and does not react with most of the chemicals. Silica fibers can thus be used even in abusive environments.

6. Silica glass can be doped with various materials to increase or decrease the refractive index precisely. This property of glass enables one to achieve compatible materials with a slight difference in the values of refractive indices needed for creating core and cladding regions.
7. Silica has a particularly low Kerr nonlinearity, which makes them suitable for optical communication since non-linear effects are often detrimental for such applications.

Many of the above properties are also exhibited by many polymers, generally referred to as plastics. These polymers are also used in making plastic optical fiber (POF). However, plastic fibers have limited application because of the substantially higher attenuation as compared to silica fibers. Further, silica fibers exhibit larger dispersion because of the large numerical aperture. The application of POF is restricted to short-distance optical communication systems designed for low-bit-rate purposes.

2.5.1 Glass Fibers

Glass is a non-crystalline solid (NCS) (or amorphous) solid. It is in general, is a hard substance, usually brittle and transparent, composed chiefly of silicates and an alkali fused at high temperature. Glass is obtained by fusing mixtures of elements, metal oxides, halides, sulfides, tellurides, or selenides. Most of the commercially available glasses are prepared by melting and quenching. Alternatively, glass can be obtained by deposition from a vapor or a liquid solution. Technically, glass formation is a property of any material but in practice limited to a relatively small number of substances. A majority of commercial glasses, available in various shapes and sizes are essentially silicates of one type or another, i.e. materials based on silica (SiO_2). The major glass-forming substances are listed in Table 2.1.

Glass does not have a well-defined structure like crystalline materials and instead has a randomly oriented molecular network. As a result, glass does have a fixed melting point. At room temperature glass is generally hard and it continues to stay in that state when heated to several hundred degrees celsius depending on the constituents. As the temperature is increased beyond 1000°C, silica glass generally softens and further in the temperature around 1400°C–1600°C glass comes into a viscous state. This extended range of temperature is referred to as the melting temperature of glass rather than the melting point. The melting temperature of silica glass can be reduced by adding soda-lime. The most widely used glass for making optical fiber is generally oxide glass (Nobukazu, 1981; Quinn, 1990). Among the oxide glasses most commonly used glass for making optical fiber is silica (SiO_2). It has a refractive index of 1.458 at a wavelength of 850 nm.

Table 2.1 Glass forming substances (Dormus, 1973)

Constituents	*Particulars*
Elements	S, Se, P, etc.
Oxides	SiO_2, GeO_2, P_2O_5, B_2O_3, PbO_3, etc.
Halides	BeF_2, AlF_3, NaF, $ZnCl_2$, etc.
Sulfides	As_2S_3, CS_2, Sb_2S_3, etc.
Selenides	SnSe, PbSe, As_2Se_3, etc.
Tellurides	SnTe, PbTe, Sb_2Te_3, As_2Te_3, etc.
Nitrides	KNO_3, $Ca(NO_3)_2$, etc.
Sulfates	$KHSO_4$
Carbonates	K_2CO_3, $MgCO_3$, etc.
Polymers	Polystyrene, Polycarbonate, Nylon, etc.

One of the major advantages of glass is that its properties can be changed by changing the composition of glass. For example, by adding a tiny amount of dopant such as fluorine or other metal oxides the refractive index of pure silica at a given wavelength can be varied by controlling the amount of dopant added. This enables one to achieve similar materials that are compatible despite having different values of refractive index. In other words, in the case of glass fiber, the core and the cladding can be created by controlling the dopant properly to maintain a small difference in refractive index between the two. The refractive index of pure silica glass (SiO_2) at a given wavelength can be increased by introducing another oxide component such as GeO_2, P_2O_5 to form a binary oxide glass. Likewise, by doping silica with fluorine (F) or by adding B_2O_3 in silica, the refractive index can be decreased suitably by adjusting the content of the dopant. The dopants are generally introduced during the formation of preforms which are subsequently used for drawing fibers. The variation of the refractive index of silica at 850 nm with the percentage of the added dopant is illustrated in Fig. 2.19 (Keiser G).

The refractive index of the core in an optical fiber is higher than the cladding refractive index and silica-based fibers can be designed by suitably controlling the dopant. A few possible options are listed in Table 2.2 wherein SiO_2:GeO_2 indicates GeO_2 doped silica glass and so on.

Table 2.2 Composition of silica-based fibers

Core	*Cladding*
SiO_2	SiO_2:B_2O_3
SiO_2	SiO_2:F
SiO_2:GeO_2	SiO_2:GeO_2: B_2O_3
SiO_2:B_2O_3:P_2O_5	SiO_2:B_2O_3

The principal raw material used for making silica fiber is sand which is abundantly available in nature. Silica has high transparency in both visible and NIR regions which are extensively used for both short- and long-haul optical fiber communication systems. However, the attenuation characteristics and some other optical properties such as non-linear effects in glass can be changed by changing the composition of glass. Further, new properties such as lasing can be induced by incorporating rare earth elements in normal silica glass. This type of special glass exhibits new optical and magnetic properties which can be exploited for making a variety of active optical components. A few special glass fibers which are useful in optical fiber communication are discussed here.

2.5.2 Fluoride Fibers

Heavy metal fluoride (HMF) glasses were first reported by Lucas *et al* in 1974, at the University of Rennes, France (Lucas, 1989; Tran *et al* 1984). Fluoride optical fibers are based on fluoride glasses, e.g. fluoroaluminate or fluorozirconate glasses. The cations of

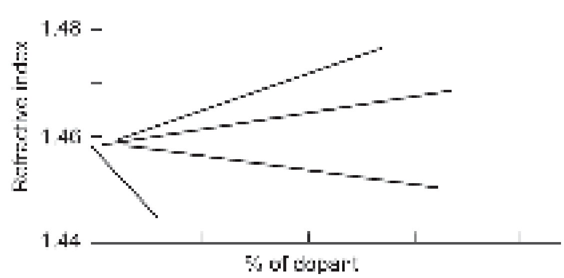

Fig. 2.19 Variation of refractive index of silica glass with percentage of dopant

such glasses are usually from heavy metals such as zirconium or lead. Fluorozirconate glass (where ZrF_4 is the major component) is a typical member of the halide glass family. It consists primarily of ZrF_4 as the principal constituent. Other constituents are added to provide moderate resistance to crystallization. Among the various halide glass ZBLAN glass (ZrF_4-BaF_2-LaF_3-AlF_3-NaF) is most extensively investigated for making optical fibers. The detailed composition of ZBLAN glass is listed in Table 2.3. ZBLAN glass is used for making the core of the optical fiber. To create the cladding of a lower refractive index ZBLAN is usually doped with HaF_4. HaF_4 replaces partially ZrF_4 in ZBLAN and results in a reduction of refractive index. A halide fiber thus constitutes a ZBLAN core with ZBLAN:HaF_4 cladding. The heavy metal fluoride constituents lead to low phonon energies. As a result, fluoride fibers exhibit high optical transparency, i.e. extremely low optical absorption in the mid-infrared (MIR) wavelength region (2–8 μm) unlike common silica fiber which absorbs light significantly beyond 2 μm wavelength.

Table 2.3 Constituents of ZBLAN glass and their molecular percentage

Constituent	*Molecular percentage*
ZrF_4	54
BaF_2	20
LaF_3	4.5
AlF_3	3.5
NaF	18

Fluorozirconate (ZrF_4) and fluorohafnate (HfF_4) material combination has been extensively used for making low-loss fluoride glass. A host of fluoride fibers using various constituent components are listed in Fig. 2.4 (Sakaguchi *et al* 1987). Initially, fluoride fibers were envisaged to be used for optical fiber communication beyond 2 μm wavelength because of the low intrinsic losses (0.01–0.001 dB/km) of these fibers in the mid-IR region to replace silica fibers, which are transparent only up to 2 μm. However, the commercial application of these fibers did not materialize because of the high brittleness and extremely high cost associated with the fabrication of halide fibers. A major problem associated with pure halide fiber is that fluoride glass tends to form micro-crystallites which increases the scattering of light resulting in increased attenuation (Table 2.4). However, other possible applications of fluoride fibers were explored subsequently. The mid-infrared transparency of fluoride glasses was exploited to develop mid-infrared spectroscopy, fiber-optic sensors, MIR imaging, etc. Fluoride fibers have been used to transport light over a short distance from Er:YAG lasers operating at 2.9 μm wavelength in many medical appliances.

Table 2.4 Fluoride glass-based optical fibers (Sakaguchi *et al* 1987)

Glass	*Composition (mol %)*							
	ZrF_4	*BaF_2*	*GdF_3*	*LaF_3*	*YF_3*	*AlF_3*	*LiF*	*NaF*
ZBG	63	33	4	–	–	–	–	–
ZBGA	60	32	4	–	–	4	–	–
ZBLAL	52	20		5		3	20	–
ZBLYAL	49	22	–	3	3	3	20	–
ZBLYAN	47.5	23.5	–	2.5	2	4.5	–	20

Further, halide glass is especially attractive for the realization of various kinds of fiber lasers and fiber amplifiers because of suppressed multi-phonon transitions in fluoride glasses (Tran *et al* 1984; Lucas, 1989; France *et al* 1990 and Sakaguchi, 1987; Wetenkamp *et al* 1992; Poulain, 1998). Heavy metal halide glasses in the family of ZrF_4-BaF_2-LaF_3-NaF-AlF_3 systems offer extremely low loss in the longer wavelength and are the most suited for operation in mid-infrared region (Ohsawa *et al* 1984). The ZBLNA system is reasonably stable enough to be drawn into fibers continuously without devitrification. The value of the Rayleigh scattering coefficient for ZBLNA glass is reported to be considerably smaller than that for other fluorozirconate glasses. The values of transmission loss for the fluoride glass fiber are 0.001 dB/km at 3.2 μm and 0.005 dB/km at 3.5 μm (Ohsawa *et al* 1984).

2.5.3 Active Glass Fibers

Optical fibers used as a channel in guided optical communication are generally viewed as a passive component in the sense that the output power available at the receiver end is always less than the power launched at the input end (transmitter) of the fiber. However, by incorporating rare-earth elements into a normally passive fiber it is possible to induce new optical and magnetic properties in the fiber. These properties can be subsequently exploited to obtain amplification, phase retardation, and other non-linear behaviors of light propagating through such fibers. This kind of fiber is referred to as active fiber. The first rare-earth doped fiber amplifier was reported in the early 1960s using Nd^{3+} (Snitzer, 1961; Koester *et al* 1964). The possibility of using Er^{3+} as a rare-earth element in silica-based fiber for obtaining amplification was demonstrated by Payne *et al* (Payne *et al* 1987). A variety of glasses can be suitably doped to act as active fibers for making fiber lasers and fiber amplifiers (Ainslie, 1991; Miniscalco, 1991; Simpson, 1993; Poole *et al* 1985; Poole *et al* 1986; Townsend, 1987; Fermann, 1988).

Fiber lasers and fiber amplifiers are nearly generally based on active glass fibers which are doped with a trace amount (0.005–0.05 mole percent) of laser-active rare earth elements (in the fiber core). The rare-earth elements include erbium, neodymium, ytterbium, etc. These rare-earth dopant ions create metastable states in the energy gap of the principal glass material to create a situation for stimulated emission. Such active fibers absorb pump light at a shorter wavelength than that of the lasing wavelength. The active fibers in fiber laser or amplifier act as gain media to provide high gain efficiency, resulting from strong optical confinement in the waveguide structure. Some commonly used rare-earth elements and host glasses with emission wavelength from the corresponding active fibers are listed in Table 2.5. For rare-earth-doped active fibers, the material composition of the core is normally modified by adding additional dopants. In place of using pure silica

Table 2.5 Laser-active rare-earth dopants and host glass with the desired wavelength of emission

Rare-earth ions	*Host glass*	*Laser wavelength*
Neodymium (Nd^{3+})	Silica glass and phosphate glass	1.03–1.1 μm, 0.9–0.95 μm, 1.32–1.35 μm
Erbium (Er^{3+})	Silica glass, fluoride glass and phosphate glass	1.5–1.6 μm, 2.7 μm, 0.55 μm
Ytterbium (Yb^{3+})	Silica glass	1.0–1.1 μm
Thulium (Tm^{3+})	Silica glass, fluoride glass	1.7–2.1 μm, 1.45–1.53 μm, 0.48 μm, 0.8 μm
Neodymium (Nd^{3+})	Chalcogenide glass	1.08 μm
Praseodymium (Pr^{3+})	Chalcogenide glass	1.3, 1.6, 2.9, 3.4, 4.5, 4.8, 4.9, and 7.2 μm
Holmium (Ho^{3+})	YAG, YLF, silica	2.1 μm, 2.8–2.9 μm

core some form of aluminosilicate, germanosilicate, or phosphosilicate glass is used to facilitate the lasing action. Halide fibers are also extensively used for making fiber lasers and fiber amplifiers as already discussed. Out of the rare-earth elements ytterbium- and neodymium-doped gain media are generally used for making lasers while erbium-doped fibers are mostly used for making erbium-doped fiber amplifiers (EDFA) (Judd, 1962; Dieke *et al* 1963; Kenyon, 2002).

2.5.4 Chalcogenide Glass Fibers

Chalcogenide glasses contain at least one of the chalcogen elements such as S, Se, or Te. The potential of sulfur-based fiber was proposed in 1967. Since then chalcogenide glass optical fibers have been extensively studied. Chalcogenide glasses exhibit high optical non-linearity and as a result, have found applications in non-linear optics ranging from optical amplifiers to all-optical switches and the glass containing one of the chalcogens mentioned above is generally co-doped with some other elements such as P, I, Cl, Br, Cd, Si, etc. for tailoring the refractive index and other optical properties and *improving thermal and mechanical stability*. High-quality fiber has been successfully achieved using alloys containing a variety of chalcogen elements (Sanghera *et al* 1994; Sanghera, 1998; Brady, 1998; Harrington, 2004). The most widely investigated material from the Chalcogenide glass family is As_2S_3. Chalcogenide glass-based single-mode fiber with $As_{40}S_{58}Se_2$ (core) and As_2S_3 (cladding) is reported to exhibit a loss as low as 1 dB/km. However, the delicate nature, complicated fabrication methodology, and excessive cost of chalcogenide glasses have restricted their widespread application and commercialization. The chalcogenide glass fibers are also an excellent host for rare-earth ions, not only for the near-infrared region (1.3–1.55 μm) telecommunication systems but also for applications at longer wavelengths. In recent years special chalcogenide glasses have been developed for doping with rare-earth elements for laser emission in the mid-infrared wavelength region and beyond (2–12 μm).

2.5.5 Plastic Optical Fiber

Plastic optical fibers are manufactured from a variety of polymers commonly referred to as plastic materials such as polystyrene, polycarbonates, and polymethyl methacrylate (PMMA) (Dutton, 1998; Kioke *et al* 1995; Ishigure *et al* 2000; White, 2005). These materials exhibit transmission windows mostly in the visible wavelength region (500–800 nm). However, the attenuation of optical signal in such fibers at these wavelengths is very high typically in the range from 150 dB/km for PMMA to 1000 dB/km for polystyrene. The attenuation of POF is much higher than that of high-quality single-mode glass fibers, which have a loss only to the tune of 0.2 dB/km. Further, dispersion in POF is also very high because of a larger numerical aperture compared to glass fibers arising out of large differences between the refractive indices of core and cladding. The large dispersion in POF severely restricts the transmission speed of data through such fibers. The plastic fibers are limited to short-distance applications at low data rates. Nevertheless, plastic fibers have several advantages that make them suitable for applications such as industrial controls, automobiles, sensors for detecting high-energy particles, and general illumination purposes including short-haul data links. Extensive reviews of plastic optical fibers are available in the literature (Jubia *et al* 2001; Koike *et al* 2011). A special type of plastic optical fiber made from an amorphous fluorinated polymer called CYTOP has been reported to exhibit even lower attenuation (~50 db/km) (Naritomi, 1996).

POF offers several advantages over glass fibers. A large value of numerical aperture enables light to be coupled easily even from inexpensive diffused optical sources such as light-emitting diodes. The low cost of POF and other associated optical components makes POF-based optical communication systems cheaper than glass fiber-based optical fiber communication systems. Other advantages of POF include lighter weight, operation in the visible region, greater flexibility, and resiliency to bending, shock, and vibration, ease in handling and connecting (POF diameters are 1 mm compared with 8–100 μm for glass), needs simple and inexpensive test equipment, splices and connectors. However, the lack of industrial standards and high loss make POF less attractive for sophisticated applications. Historically, plastic optical fibers using polymethacrylates (PMMA) were reported way back in the early 1960s. The loss of the early PMMA-based plastic fiber was as high as 1000 dB/km. Mitsubishi Rayon was later developed PMMA-based POF to offer loss close to the theoretical limit of 150 dB/km at 650 nm. This was a step-index fiber with a bandwidth of 50 Mbps over 100 meters. Koike *et al* subsequently developed a process to manufacture graded-index POF using PMMA and reported a GI- POF having a bandwidth of 3 GHz-km with losses of 150 dB/km. In 1995 the same group developed a graded index POF based on perfluorinated polymer (PPF) with losses of less than 50 dB/km over a range of 650 nm to 1300 nm. The attenuation in POF is caused by the molecular vibrational absorption of the groups C-H, N-H, and O-H and also by the absorption due to electronic transitions between different energy levels within molecular bonds. Additional factors include scattering arising from composition, orientation, and density fluctuations (Zubia *et al* 2001). The attenuation profiles of the step-index and graded-index of POFs are shown in Fig. 2.20. The fibers exhibit attenuation windows primarily in the visible region. The attenuation characteristics of a typical PFP fiber are shown in Fig. 2.21. The PPF fiber exhibits a low value of attenuation in the near-infrared region.

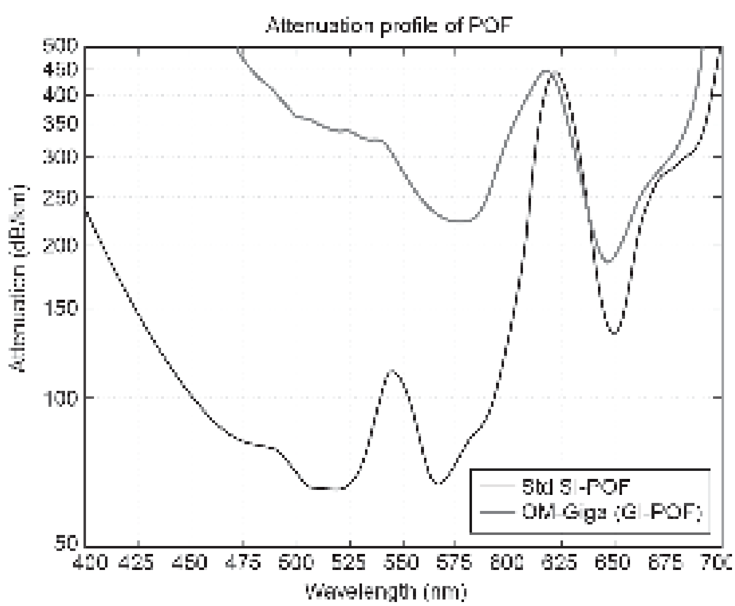

Fig. 2.20 Attenuation profiles of a standard SI-POF and a GI-POF

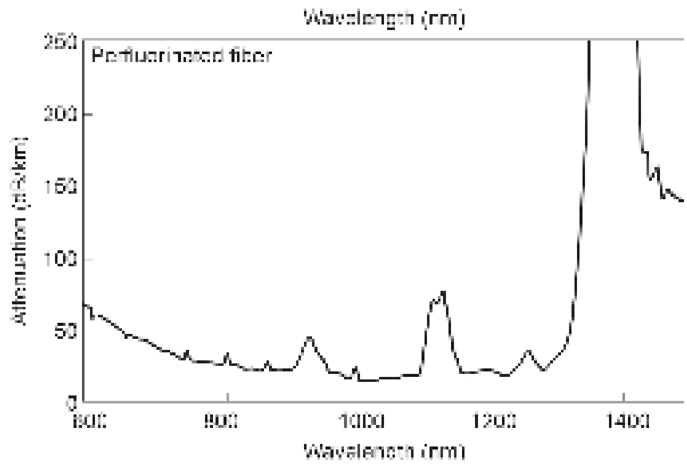

Fig. 2.21 Variation of attenuation of PFP fiber with wavelength

2.5.6 Plastic Clad Silica Fiber

Plastic-clad silica (PCS) is a compromise between high-performance silica fiber and less efficient plastic fibers. It consists of a core made of silica glass and cladding made of a compatible polymer of a lower refractive index. Commercial PCS fibers consist of a pure silica core, a soft silicone cladding, and a protective jacket. PCS fibers are an economical alternative to all-silica fibers. The advantages of PCS fiber include high light collection efficiency, insensitivity to bending, and excellent transmission. The parameters of a typical commercial PCS fiber are listed in Table 2.6.

Table 2.6 Specifications of a typical commercial PCS fiber

Parameter	*Values*
Core diameter	200 μm to 1500 μm
Numerical aperture	0.40
Core-to-clad ratio	1.1, 1.2, 1.4
Cladding material	Silicone
Jacket material	Nylon, Tefzel

2.6 FIBER FABRICATION TECHNIQUES

These are different for glass and plastic fibers. This section deals with various fabrication techniques available for making all glass fibers. Two basic techniques are there for the fabrication of glass fiber. The first is based on the traditional glass-making approach where fibers are directly drawn from molten glass. The other method involves a so-called preform which is a prefabricated clear solid glass rod or tube. The preforms are typically of the order of 1–10 cm diameter and 1 m length. Like optical fibers, the central region has a higher refractive index as compared to the outer region. Preform-based methods are most suitable for the fabrication of high-quality glass fibers.

2.6.1 Fiber Pulling from a Preform

High-quality glass fibers are generally fabricated by pulling long strands from a so-called preform using a fiber-drawing tower. The fiber drawing machine is an apparatus that is typically several meters high. Along its axis, the preform contains a region with an increased refractive index, which forms the fiber core after drawing fiber from the preform (Jaeger, 1979; Paek, 1986; Brehm *et al* 1988; Chu *et al* 1989). The preform is heated close to the melting point using a drawing furnace fixed at the top of the drawing tower. A thin fiber is generally pulled out of the bottom of the preform. The fiber from a single preform in this process can have lengths of several kilometers. During the pulling process, the fiber diameter is held constant by automatically adjusting the pulling speed of the take-up drum fixed at the bottom of the tower. The speed of the take-up drum determines how fast the fibers are drawn. This in turn also determines the thickness of the fiber produced in the process. To maintain uniform thickness the motor of the take-up drum must rotate at a pre-decided uniform speed. An optical fiber thickness monitor is used in a feedback loop to monitor the speed of the drum. The furnace temperature is also maintained with the help of an automatic feedback system. A typical fiber drawing tower is shown in Fig. 2.22.

Before the fiber is wound up, it usually receives a primary buffer polymer coating followed by a secondary coating for mechanical and chemical protection of the fiber. Ultraviolet-based coating curing system usually follows the coating applicators as indicated in the figure. Multiple coatings help in the suppression of microbends. Typical coating materials include acrylate, silicone, and polyimide. The fiber is finally wound up in the take-up drum whose speed is precisely controlled by the feedback mechanism used for monitoring the thickness of the bare fiber. Additional PVC or similar protective jackets are usually provided by extrusion after the drawing process.

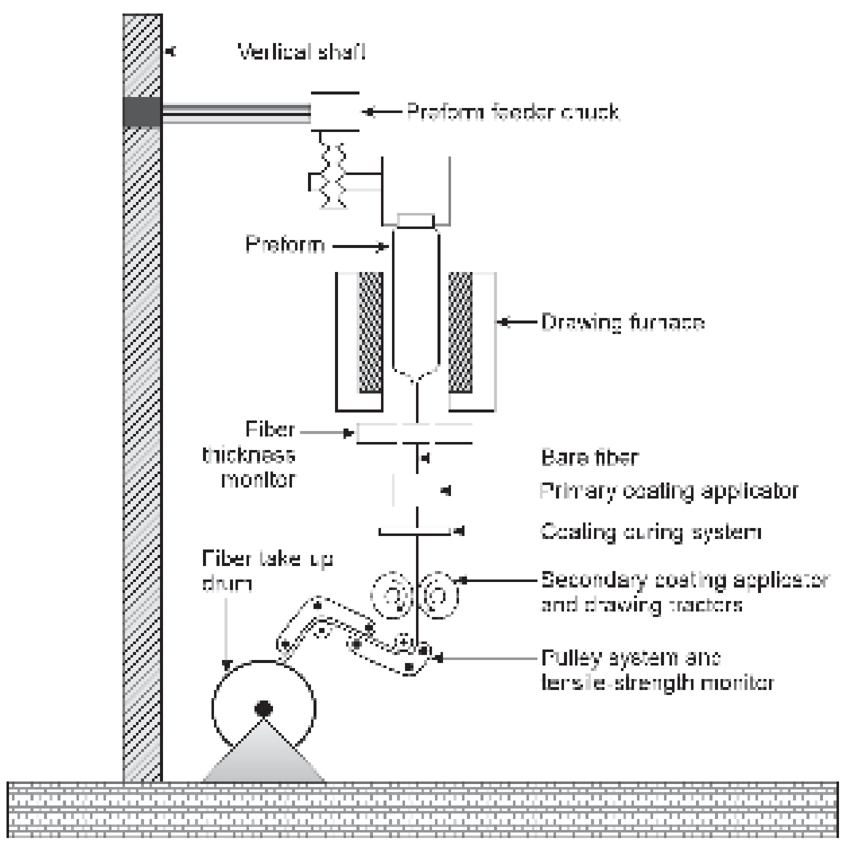

Fig. 2.22 Schematic of a fiber-drawing tower

2.6.2 Fabrication of Fiber Preforms

Optical fiber preforms are generally fabricated with a process called *chemical vapor deposition* or *vapor phase oxidation technique*. The basic method and many variations of this method have been extensively used for making silica-based optical fibers since the 1970 (pioneering contributions in this field have been from the University of Southampton (UK), Bell Telephone Laboratories (Bell Labs), and Corning Glass Works, later on, a more sophisticated technique based on *plasma deposition* was developed by the scientists at Philips Co. In general, a mixture of oxygen, and metal halides such as silicon tetrachloride ($SiCl_4$), and germanium tetrachloride ($GeCl_4$) are made to react to produce a white powder of SiO_2 containing GeO_2. The later oxide in trace amounts is used to vary the refractive index depending on the requirement. The white dust powder is subsequently collected on a substrate in the form of soot which looks opaque. The soot is subsequently transformed into a homogeneous glass by heating without melting. This process is known as *sintering*. The following chemical reactions take place during the oxidation of metal halides.

$$SiCl_4 + O_2 = SiO_2 + 2Cl_2 \text{ (dry oxidation)} \tag{2.28}$$

$$GeCl_4 + O_2 = GeO_2 + 2Cl_2 \text{ (dry oxidation)} \tag{2.29}$$

$$SiCl_4 + 2H_2O \text{ (steam)} = SiO_2 + 4HCl \text{ (wet oxidation)} \tag{2.30}$$

$$GeCl_4 + 2H_2O \text{ (steam)} = GeO_2 + 4HCl \text{ (wet oxidation)} \tag{2.31}$$

Other forms of impurity such as P_2O_5 can be added in the formation of soot during the reaction by incorporating $POCl_3$ among the reactants. For making active optical fibers rare-earth dopants are also passed with the reacting gases. Four basic methods used for making optical fibers are outside vapor phase oxidation (OVPO) or outside vapor deposition (OVD), modified chemical vapor deposition (MCVD), vapor axial deposition (VAD), and plasma-activated chemical vapor deposition (PCVD).

2.6.2.1 Outside vapor deposition

The breakthrough technique involving the vapor deposition of silica for making high-quality optical fiber preforms was proposed and patented by Keck and Schultz (Keck and Schultz, 1973). This invention paved a new avenue for making optical fibers in place of using a traditional glass-making approach involving molten glass. The process of outside vapor deposition for fabricating high-quality optical fiber preform has been reviewed in the literature with a detailed discussion of various steps involved in the process starting from metal halides, formation of porous soot, sintering and finally drawing fiber from the preform (Schultz, 1980; Blankenship *et al* 1982).

In this process, the silica soot is deposited on the surface of a rotating target rod (e.g. a graphite or ceramic mandrel) from a burner moving along the target rod. The method is illustrated in Fig. 2.23. The material precursors such as $SiCl_4$, $GeCl_4$, etc. along with a fuel gas such as hydrogen or methane is supplied to the burner which is moved back and forth along the rotating rod to deposit soot layer by layer on the target rod. The carrier gas along with the chemical vapor is supplied through a standard "bubbler" delivery system shown in Fig. 2.24. The bubbler is fitted with a mass flow controller (MFC) to monitor the flow of carrier gas. This carrier gas is made to bubble through the liquid in the container for vaporizing some of the chemicals. The mixture of the carrier gas and the desired chemical vapors is then allowed to flow to the deposition burner as shown in the Fig. 2.24 (Blankenship *et al* 1982). The soot adheres to the target rod to form porous glass preform. The supply of constituent metal halides is controlled during the deposition process to create a pre-decided thickness of the preform with the desired refractive index

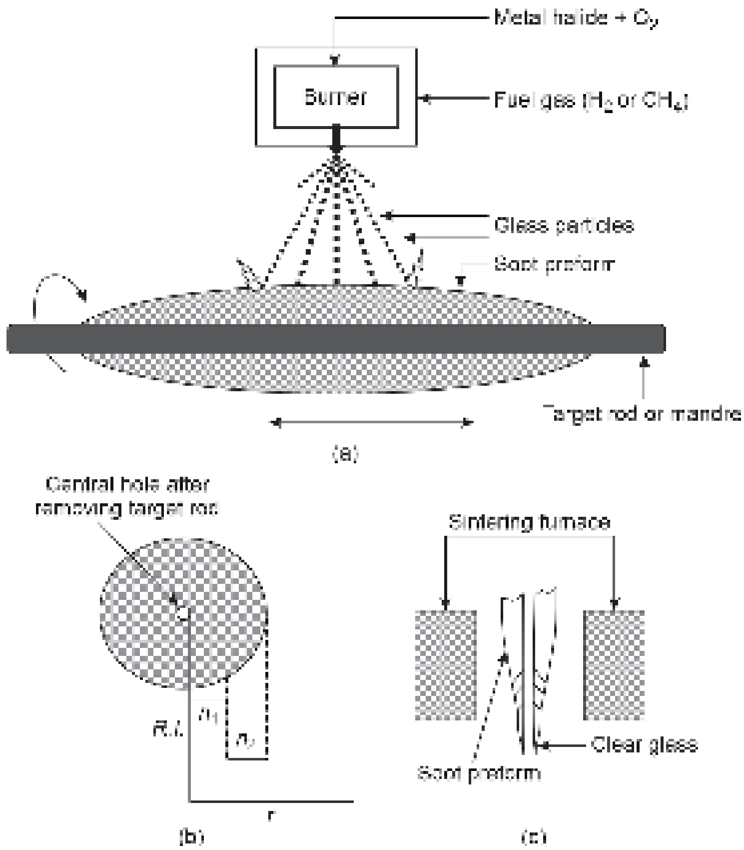

Fig. 2.23 Illustrating the schematic of the OVD technique (a) Basic arrangement for soot deposition and collection on the target rod (b) Cross-sectional view of the porous preform showing variation of refractive index for a SI preform (c) Sintering of porous preform to produce clear glass preform

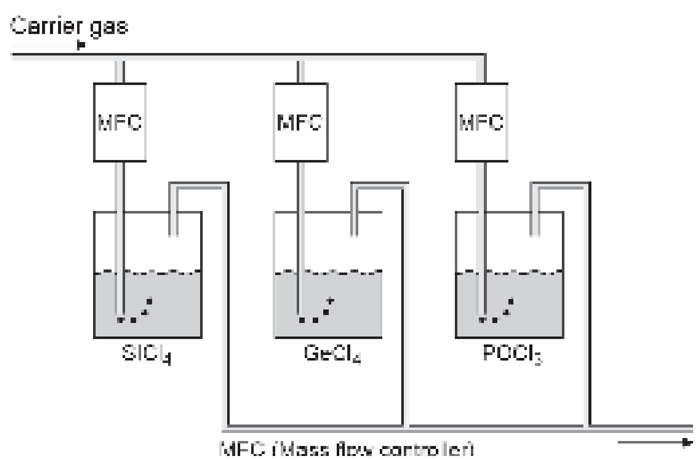

Fig. 2.24 The 'bubbler' delivery system for feeding the burner

difference in different layers. Both step-index (SI) and graded-index (GI) silica fibers can be fabricated by this method. For making GI fibers, the supply of dopant metal halide is made to vary during the growth of shoot layers. The OVD technique was developed by Corning Glass Works for making optical fibers with a loss of less than 20 dB/km. After the deposition process is completed, the target rod is removed, and the porous preform is vitrified in a furnace at a high temperature (usually above 1400°C) and purged with a drying gas to lower the hydroxyl content. The vitrified preform is transformed into a clear glass preform which is subsequently mounted in a fiber drawing tower to draw fibers from the preform melt. When the sintered preform is heated to melting point the central hole collapses during the fiber drawing process. However, the central hole sometimes leads to the trapping of air bubbles during the fiber drawing process, and as a result scattering of light takes place in such fibers. Moreover, the OVD is prone to contamination as the oxidation is done in an open atmosphere.

2.6.2.2 Vapor phase axial deposition

The OVD process is suitable for producing good quality fiber preform. However, this method is not suitable for the continuous production of fiber for large-scale application. Moreover, as the process involves oxidation in the outside atmosphere there is a possibility of contamination. The basic vapor phase deposition method was suitably modified by Izawa *et al* for the continuous production of low-loss fiber in a controlled environment. This process is known as the vapor axial deposition (VAD) technique. This process is very similar to the OVD process except for the fact that VAD is done in a closed environment and it uses a modified geometry, where the deposition occurs at the end of the target rod (Sudo *et al* 1978). The rod is continuously pulled away from the burner, and very long preforms can be drawn in the process. It is most suitable for the production of preform on large scales. At the bottom of the VAD chamber (Fig. 2.25) flame hydrolysis (same process as used in OVD) torches produce porous silica soot (including appropriate dopants) which is deposited across the base of a silica seed rod. A porous preform is grown in the axial direction by rotating the seed rod and pulling it upward by a precision mechanism. The rotation of the seed rod is needed for cylindrical symmetry of the porous preform. As the porous preform is drawn further upwards it crosses a localized heating zone where the porous preform gets sintered into a clear glass preform. Very long-length fiber preforms can be manufactured by this technique. The solid preforms can be cut into smaller pieces and loaded in a fiber drawing tower for drawing fibers. Another important difference between VAD and OVD is that the doping profile is determined in the former case only by the torch (burner) geometry, rather than by a variation of the gas mixture over time. The advantages of the VAD method over the OVD technique are that there is no central hole in the preform, the preforms can be grown with continuous length just by replenishing the reactants in the torches and the entire process of making preform is carried out in a closed and clean environment. As a result, the quality of VAD fibers is better than those obtained by using the VAD method. Some further developments in VAD technology have been reported in the literature (Murata, 1986).

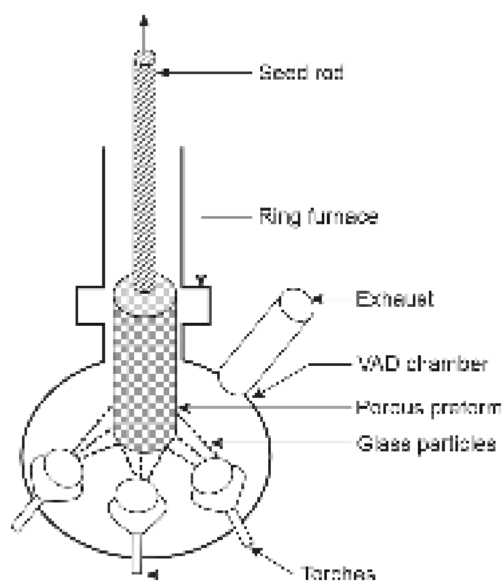

Fig. 2.25 Schematic of a vapor axial deposition (VAD) system

2.6.2.3 Modified chemical vapor deposition (MCVD)

This process was developed by Bell laboratories (French *et al* 1944) and by Southampton University (Payne *et al* 1974) primarily for making a high-quality step- and graded-index silica fiber.

Unlike OVD, the glass particles are created by burning metal halides in oxygen inside a silica tube. The technique is illustrated schematically in Fig. 2.26. The soot is collected on the inner wall of the rotating silica tube. The rotation of the tube is necessary for cylindrical symmetry in the deposition of the soot. The opaque soot collected inside the tube is sintered to clear glass by using an oxyhydrogen torch which moves back and forth along the length of the silica tube. After completion of the deposition of the desired thickness of soot, the flow of the reactant gases is stopped. The silica tube with the freshly deposited glass on the inner wall is then heated to a high temperature so that the tube collapses to form a solid preform. The preform is finally loaded in a fiber drawing tower to draw fibers. The vapor-deposited glass on the inner wall of the tube provides the core and the cladding consists of the material used in the original silica tube. One of the major disadvantages of this technique is that OH^- ions get into the fiber core because of the use of an oxyhydrogen burner for sintering.

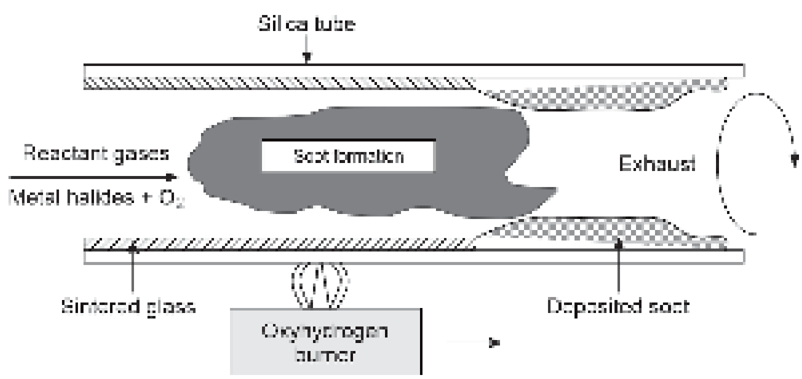

Fig. 2.26 Schematic of the modified chemical vapor deposition (MCVD) setup

2.6.2.4 Plasma chemical vapor deposition (PCVD)

Plasma chemical vapor deposition is similar to MCVD deposition inside a silica tube discussed in the previous section. The method uses a non-isothermal low-pressure plasma to initiate a gas phase reaction for the deposition of doped and undoped silica without involving the sintering process separately. The method was invented and brought to production by a group of scientists at Philips Research Laboratory (Jaeger *et al* 1978; Lydtin, 1986). In this case, the reactants containing metal halides and oxygen are passed through a silica tube. Unlike the MCVD technique, the reaction is initiated in the PCVD system by a non-isothermal microwave plasma operating at low pressure. Microwaves are used in place of burner for heating the deposition region (Fig. 2.27). The silica tube is pre-heated at temperatures in the range of 1000–1200°C to reduce mechanical stress during the growth of glass film inside the tube. A moving microwave resonator operating at 2.45 GHz generates plasma inside the tube and activates the chemical reaction. Following the reaction clear glass material is directly deposited on the inner wall of the pre-heated tube without the formation of any soot. As a result, no separate sintering is required in this case. After deposition of the desired thickness of the glass on the inner wall of the tube, the tube is heated to a high temperature to collapse and form a solid preform. The

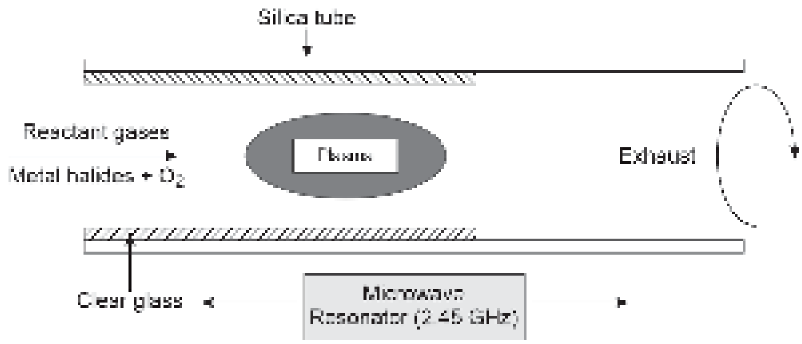

Fig. 2.27 Schematic of the plasma chemical vapor deposition (PCVD) setup

preform is subsequently loaded on a fiber drawing tower for drawing fibers. A modified method with particularly high precision is plasma impulse chemical vapor deposition (PICVD), where short microwave pulses are used in place of a CW microwave source. Another modified form of PCVD is called plasma-enhanced chemical vapor deposition (PECVD), which operates at atmospheric *pressure with a fairly high deposition* rate. The preforms for multimode fibers, particularly for large core fibers, are often fabricated using plasma outside deposition (POD), where an outer fluorine-doped layer with a depressed refractive index, later forming the fiber cladding, is made with a plasma torch. The core can then be made of pure silica, without any dopant.

In general vapor deposition methods can produce extremely low propagation losses very close to intrinsic loss of silica glass of 0.2 dB/km. This is because highly pure preforms can be obtained by using high-purity metal halides such as $SiCl_4$ and $GeCl_4$ which can be easily purified by distillation, as they are liquid at room temperature. Further contamination can be avoided when no hydrogen is present as fuel gas for the burners. Therefore, using a process like PCVD the water content in the preforms can be greatly reduced and the strong loss peak at 1400 nm can be greatly reduced (1 ppb of OH^- ion can give rise to a loss of 1 dB/km). The different vapor deposition techniques discussed above differ in terms of material purity, precision and flexibility of refractive index control, and the mechanical strength of the fabricated fibers.

2.6.3 Fiber Fabrication Method Without Involving Preforms

There are at least two more common methods of fabricating fibers that do not require the formation of fiber preforms. These methods are the rod-in-tube method and the double crucible method.

2.6.3.1 Rod-in-tube technique

In this method, a solid rod of a glass (say, SiO_2:GeO_2) with a higher refractive index is inserted into a glass tube (say, SiO_2) with a lower refractive index. When the outer tube is heated to a high temperature both the rod and the tube get well connected. The combination is then strongly heated so that the combination of the rod and the tube melts and the bottom of the tube collapses due to surface tension. Long fibers can be drawn from the molten material. The rod material constitutes the core while the cladding material comes from the tube. However, utmost care is required to avoid the trapping of air bubbles during fiber drawing. Casting methods where the molten core glass is poured into the cladding tube or sucked into the tube using a vacuum pump are also used for making fibers without using a preform.

2.6.3.2 Double crucible method

Soft glass fibers are generally fabricated by using the double crucible method (Midwinter, 1979; Beales *et al* 1976). The double crucible method is based on a traditional glass-making approach wherein fibers are directly drawn from molten glass. The double crucible setup consists of two concentric crucibles. The inner crucible is meant for holding the melt of core glass and the outer is for holding the melt of cladding glass. The two crucibles terminate in a common orifice as shown in Fig. 2.28. Solid rods of different refractive indices are used as feedstock for the two crucibles. Feedstock is generally obtained by first melting ultrapure glass powder and then forming small pieces of rods. The rods are heated by a furnace to a very high temperature for melting the rods. The fiber is drawn from the molten state through the common orifice at the bottom of the two concentric crucibles. The method can be easily adapted for a variety of glasses without much difficulty as compared to preform-based fabrication methods. However, the double crucible method is not suitable for making ultrapure fibers with very low losses as required in long-haul optical communication. The primary reason for this is that there are sources of contamination in this method. The contaminants may come from the environment as well as from the crucibles themselves. The glass rods are generally obtained by fusing glass powder in a silica crucible to avoid other metal contaminations. However, in the double crucible method, crucibles are generally made of metals that contaminate the molten glass. However, the use of platinum crucibles reduces the contamination from crucibles. Plastic optical fibers are generally produced in a simple extrusion process, which is very similar to the double crucible method. A schematic of the double crucible system for drawing halide fiber is shown in Fig. 2.28.

For practical applications, optical fibers need to be incorporated into some kind of cable structure depending on the applications. Before discussing various cabling techniques

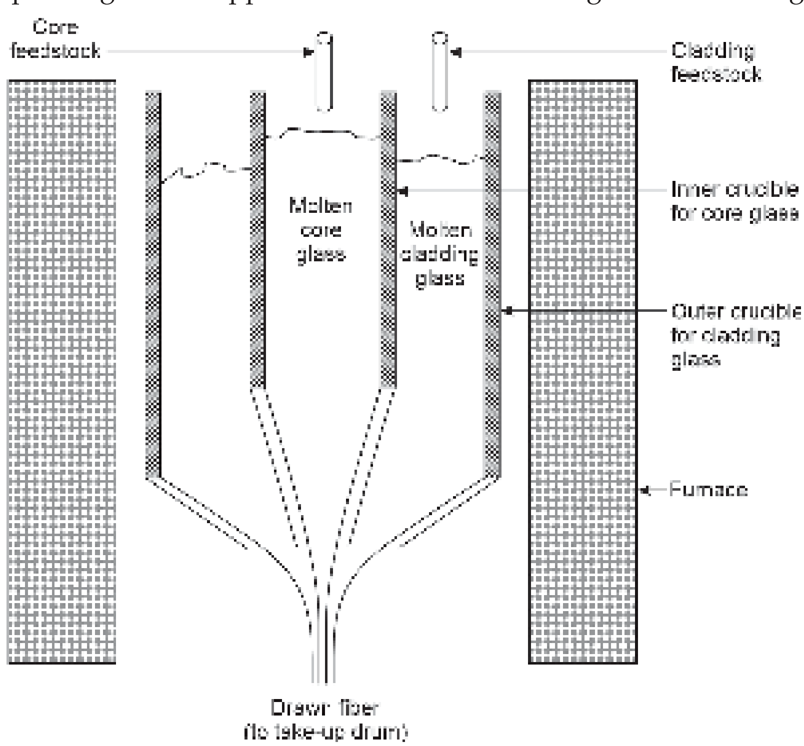

Fig. 2.28 Schematic of a double-crucible arrangement

adopted in optical fiber it is necessary to study the mechanical properties of optical fibers which are very different from metal cables.

2.7 MECHANICAL PROPERTIES OF OPTICAL FIBERS

In addition to having optical transparency in the desired wavelength regions, the optical fiber must have high mechanical strength for field applications. Optical fibers are generally incorporated in some form of cable structure. During the cabling process as well as during field installation the fibers are subjected to a variety of stresses. Fibers must therefore be designed in a way such that they can withstand erratic stress and strain during cabling, installation, and servicing. Inadequate mechanical strength may lead to the rupture of fibers leading to cable failure. The fibers generally encounter loads that can be either impulsive or slowly varying in nature also the slowly varying loads may arise out of a variation of temperature or during initial settling following the installation of cables. The mechanical behaviors of optical fibers are generally determined by two characteristic parameters, e.g. strength and static fatigue (Kurkjian *et al* 1989). Glass is generally viewed as a fragile material that is less strong as compared to conducting metal wires. The intrinsic strength of a material is usually determined by the strength of the cohesive bonds between the constituent atoms. Therefore, the strength of glass varies with the composition. For short-length fiber, the tensile strength of 14 GPa has been reported to be achieved. This value is close to 20 GPa tensile strength exhibited by steel wires. Good quality magnesium aluminosilicate glasses used for reinforcement in composite structural applications exhibit Young's Modulus of ~90 GPa; shear modulus of ~30 GPa and poisson's ratio of ~0.2 at room temperature. One of the major problems with glass fiber is that it cannot be elongated like metal wires beyond the elastic limit. This is because, unlike glass, metal can elongate plastically beyond the elastic limit whereas glass tends to break or develop cracks beyond the elastic limit.

The mechanical strength of optical fibers is often tested by static fatigue. In this testing, a constant load higher than the threshold load is applied to the fiber, and the rupture is observed after some time. The fracture depends on several factors for a given load such as quality of coating, temperature, and presence of contaminants (do Nascimento *et al* 2006). One of the major contaminants that severely affects the mechanical properties of fiber is water. The OH^- weakens the silica bond and causes the fiber to rupture much below the threshold load. The facture of optical fibers is also caused by initial flaws or micro-cracks already present in the fiber due to manufacturing defects. When the fiber is subjected to loads the micro-cracks start propagating causing large size cracks that finally lead to the rupture of the fiber. For testing the dynamic fatigue the fibers are subjected to a time-varying load to observe rupture. Fiber cable failure may also occur with aging without any applied load when the cables are used in an abusive environment for which they are not designed. The quality of an optical fiber can be described by fatigue susceptibility (n) given by (Mrotek *et al* 2002)

$$\log(\sigma_f) = \frac{\log(\delta)}{(1+n)} + c \tag{2.32}$$

where σ_f is the facture stress, δ is the stress rate and c is a constant.

To provide additional mechanical strength and protection from an abusive environment, buffered optical fibers are generally sealed in the form of cable. The type of cabling largely depends on the nature of field applications.

2.8 OPTICAL FIBER CABLES

To protect optical fibers it is necessary to incorporate them in some form of cable structure. The structure of the cable will largely depend on the type of installation (aerial, underground duct, mini-trench, buried underground or submarine, etc.) and other environmental conditions. These factors must be given due consideration at the time of cable construction. The environmental conditions can be of two types, e.g. natural external factors such as temperature variations, wind pressure, water contamination, earthquakes, etc. and man-made factors such as smoke, air pollution, fire, etc. The above factors will have a direct impact on the performance of optical fibers. The purpose of cabling is to protect fibers incorporated in the cable in a manner to ensure the proper functioning of the optical fibers, under all the kinds of conditions to which the cables may be exposed during manufacture, installation, and operation. One of the major factors that need attention in the cable design is the residual fiber stress caused by tension, torsion, and micro- or macro-bending during manufacture, installation, or operation. This kind of strain is likely to shorten my cause the growth of cracks leading to permanent rupture which is often accelerated by the presence of environmental contaminants including water. As discussed in the previous section the mechanical strength of optical fibers differs significantly from the conducting wires in conventional metal cables. In a metal cable, the main stress-bearing components are the conducting wires themselves. Metals can elongate plastically beyond the elastic limit while fibers tend to break when the stress exceeds the elastic limit. As a result, additional strength materials (steel wires or organic yarn strength materials) are generally incorporated in optical fibers so that they act as primary load-bearing elements of the fiber cables. The fiber cables may be designed strong enough to bear load during installation comparable to metal cables. This enables the fiber installation crew members to use equipment that is similar to those used for the installation of metal cables. Every fiber contains some flaws or micro-cracks that are created during the fabrication process. The strength of fibers is mainly governed by the size of flaws already present in the fiber. Under the influence of stress the fiber weakens. The effect of stress may not be apparent immediately after installation but the performance degrades over some time depending on the load to which the cable is subjected and may result in permanent failure of the cable. For designing cable suitable for a particular type of fiber, it is important to know the minimum strength of the fibers. The optical fibers are proof-tested to a certain stress level during manufacture. A good cable design protects the fiber from external stress and enhances the life of the fibers. Another factor that may cause the rupture of fiber is impulsive strain arising during manufacture (fiber drawing from preform), and installation (pulling fiber through ducts). The number and magnitude of these strains can allow a crack to reach a critical size, causing instantaneous breakage of the optical fiber. For protecting the optical fibers from the lateral force caused by impact, a buffer coating layer is generally used. Additional protective layers are provided in the form of strands of synthetic materials and a special type of sheath in the fiber cables.

A variety of optical fiber cables are commercially available. The external factors and type of installation for the field application of optical fibers determine the structure and the materials needed for designing the fiber cable. The major components of an optical cable depending on the desired application may include primary and secondary buffer coating of the fiber, a suitable strength material for the core of the cable, additional strength members in the form of steel wires or organic strength materials, water blocking materials (for underwater cables) and sheath materials. Even though silica fibers are intrinsically very strong, they become weak because of the presence of flaws. Secondary protection of

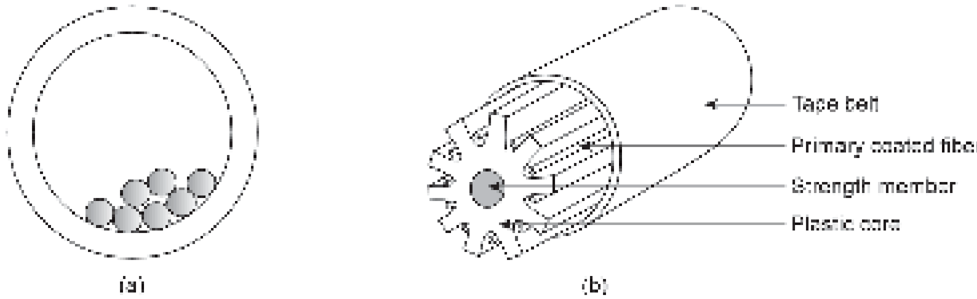

Fig. 2.29 Secondary coated fibers (a) Loosely placed primary coated fiber (b) Primary coated fiber placed in V-grooved cylindrical core surrounded by outer tube

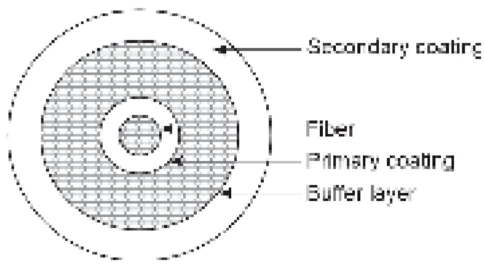

Fig. 2.30 Tight secondary coating

primary coated fibers is therefore provided in different forms of cables. While the primary coating protects the fiber from mechanical shocks during manufacturing, the secondary coatings protect the fiber from installation and operational hazards. Secondary protection of primary coated fibers is provided in different ways depending on the nature of the application and the environment in which the fiber is intended to be installed.

In loose packaging within a tube or groove the primary coated fibers are loosely placed in a hard outer tube reinforced with a composite wall (Fig. 2.29(a)). In certain cable configurations, the primary coated fibers are laid in a V-grooved cylindrical core surrounded by the outer tube (Fig. 2.29(b)). Sometimes tight polymer coating with a composite primary layer, an optional buffer layer, and a polymer secondary coating are used. A secondary coated fiber stranded around a central strength member is shown in Fig. 2.30. Single primary or secondary coated fibers are generally used for plant application. For field applications, a large number of primary or secondary coated fibers are generally placed around a central strength material fastened by plastic tape surrounded by an outer jacket. An assembly of seven buffer-coated (primary/secondary coated). This is further strengthened by incorporating organic strength material followed by an outer jacket. The cross-section view of such a fiber is illustrated in Fig. 2.31. The fiber cable also shows additional insulated copper wires which add strength to the cable. Further, these wires may be used to supply power to unmanned repeaters. The outer sheath protects the fiber from certain abrasions. However, two or more plastic jackets are not adequate to protect against penetration of water into the fiber. For water protection, additional measures such as axially laid aluminum foil or polyethylene laminated film on the inner side of the sheath are incorporated into the cable. For underwater cables special water-resistant fillers are used in the cables to protect the fibers. The fiber design of optical fiber cables varies widely depending on the application and field environment. Some specialized applications of optical fiber cables include undersea, dispensed links, data buses for avionics, and radiation hard cables for military use. The first experimental installation of

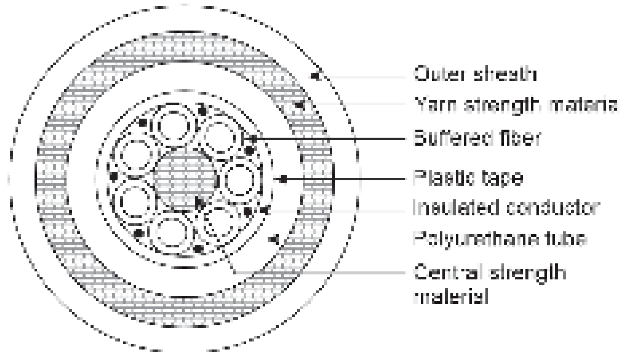

Fig. 2.31 Fiber cable with seven primary/secondary coated fiber around a central strength material

undersea cables was done in 1980 (Worthington, 1980). The cable design technology has evolved dramatically. The first international link was established between the United Kingdom and Belgium (UK-B5) using the STC NL 1 system (Black, 1986). The undersea cable configurations are illustrated in Fig. 2.32.

A special type of cable design is needed for application in radiation environments such as in nuclear reactors or in space vehicles that pass through Van Allen belts of the earth. A specially designed optical fiber for such applications is illustrated in Fig. 2.33.

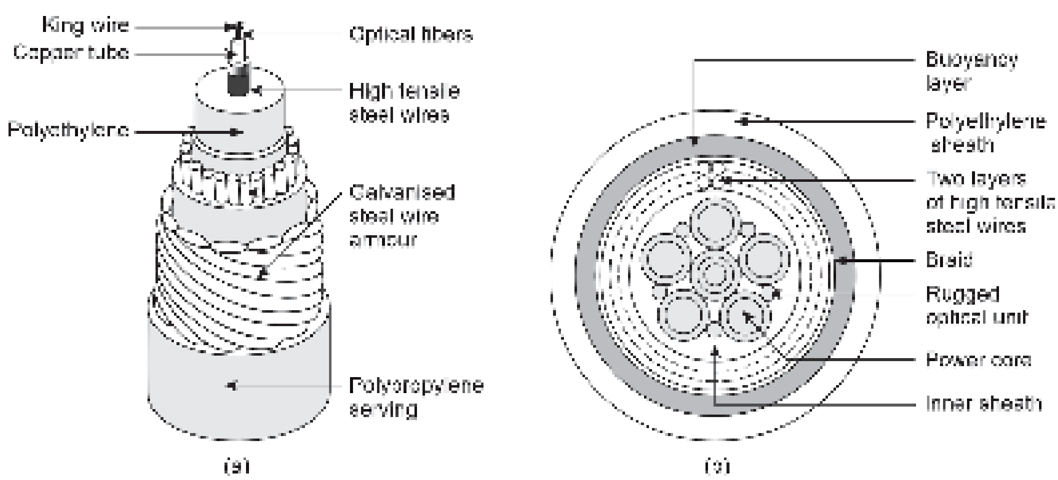

Fig. 2.32 (a) Submarine cables for unburied routes (b) Combined power and optical cable for undersea application (Black, 1986)

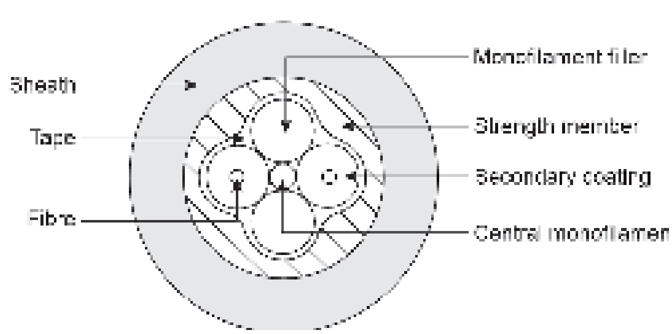

Fig. 2.33 Cross-section of a rugged metal free radiation hard cable

More technical details can be found in the handbooks and manuals of optical fiber cables (ITU Fiber Cable Manual, 2009).

2.9 OPTICAL FIBER CABLE INSTALLATION

A particular type of optical fiber cable has a specified strain limit. Therefore, it is necessary to take special care and make appropriate arrangements during installation so that no damage is caused to the fibers placed tightly inside the cable. Often it is not possible to ascertain the damage immediately but the effect may be apparent even years after installation. The aim of the installation technician and crew should be a strain-free laying of the cables as far as practicable. A variety of cable installation methods are available and their deployment depends on the nature of the installation requirement. Different types of cable installation situations are discussed in this section.

2.9.1 Installation of Cables in Underground Ducts

One of the most difficult jobs for the cable installation crew involves the installation of optical fiber cables in underground ducts. The complexity of the installation operation depends on the condition and the geometry of the duct caused by the presence of existing cables and excessive curvatures within the duct. The existing ducts through which cable installation is to be executed must be clean and clear without any obstruction. The cable installation crew must take utmost care to protect the cable from excessive strain axially or in bending at the time of installation. The major concern in such installation involves the prediction of maximum cable tension expected during installation vis-a-vis the strain-bearing capability of the cable. A typical cable installation situation through a duct is illustrated in Fig. 2.34.

2.9.2 Cable Installation Tension Calculation

An estimation of expected cable tension likely to be created at the time of installation of the cable through a near-perfect, clear, and sound duct can be obtained using the following equations depending on the actual level of placement of the cable inside the duct (ITU-T Manual, 2009). In this calculation, it is customary to consider the route between deviations and inclinations as a straight cabling portion (for example, the portions RS, ST, and UV of the cable as shown in Fig. 2.34). In the figure, the straight portions are indicated by "Level" whereas the inclined sections are indicated in terms of up/down depending on the nature and magnitude of the inclination. The points shown as empty circles are used in the figure to represent the duct piece-wise depending on the geometry of the duct for calculating the cable installation tension when pulled through the duct.

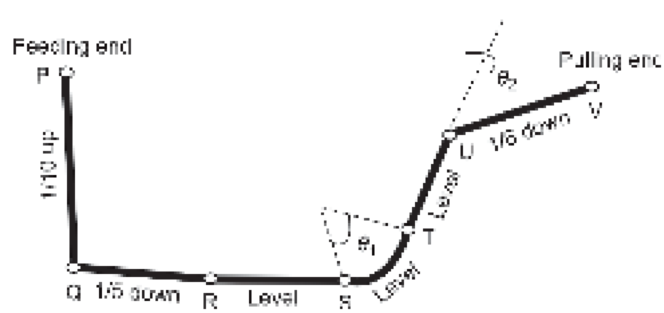

Fig. 2.34 Illustrating tension prediction for an optical fiber cable pulled through a duct with an arbitrary topography

For the straight portion of cabling, the cable tension can be obtained as

$$T = T_i + \mu lwg \times 10^{-3} \tag{2.33}$$

For the inclined section, the cable tension is estimated as

$$T = T_i + lwg(\mu \cos\theta + \sin\theta) \times 10^{-3} \tag{2.34}$$

For the curved section, the cable tension can be obtained as

$$T = T_i e^{\mu\beta} \tag{2.35}$$

where,

T_i = Tension (in kN) at the starting end of the cable portion under consideration

T = Tension (in kN) at the end of the section

l = Length (in m) of the section of the cable

μ = Friction coefficient between the cable and the duct

w = Specific mass (kg/m) of the cable

θ = Inclination in radians (up = positive (+)/down = negative (–))

β = Deviation (in radians) and

γ = Acceleration due to gravity (m/s^2)

For the successful and hassle-free installation of optical fibers, several special techniques and accessories are used. The accessories include winching devices, rope, guiding, and bending devices. The special techniques include the following.

a. **Cable overload protection method:** This is intended to protect the cable and the fibers from excessive load requirements for installation. The method involves the use of additional devices for protection such as primary and/or intermediate winch and at the cable-rope interface. Depending on the type of the winch mechanical clutches, stalling motors, and hydraulic valves are used. These devices can be preset at a predetermined load. Mechanical fuses and sensing devices are generally used at the cable-rope interface (ITU-T Manual, 2009).
b. **Cable friction reduction:** For reduction of cable friction during installation proper lubrication of the cable is undertaken depending on the nature of the installation.
c. **Air-assisted installation:** This method involves the use of continuous high-speed airflow along the length of the cable by using a suitable source for forcing air at a high speed.
d. **Water pumping or floating technique:** This method is effective for the installation of fiber cables through ducts. It involves forcing water into the duct with the help of a suitable water pumping system to minimize the friction between the cable and the duct.
e. **Trenchless technique:** This method allows the installation crew to lay the fiber underground without the need for excavation. This technique involves the creation of a horizontal bore below the existing underground service infrastructure lines (gas pipelines, electrical cables, water pipelines, etc.) with the help of specialized equipment.

Summary

- The chapter addresses several fundamental issues related to optical fibers in respect of structure, classification, composition and fabrication techniques, and propagation mechanism.
- The perception of light among the scientific community starting from Newton's corpuscular theory to quantum theory is outlined.

- The fundamental principles of geometrical optics and especially total internal reflection (TIR) have been discussed for understanding the mechanism of propagation of light through a large-size optical fiber.
- TIR is associated with lateral shift of the reflected light ray called Goos-Hänchen shift.
- Optical fibers are classified on the structural basis under step index (SI) and graded index (GI) fibers. For an SI fiber,

$$n(r)\begin{cases} n_1 & \text{for} \quad r < a \\ n_2(< n_1) & \text{for} \quad r \geq a \end{cases}$$

  where, r is the distance measured from the centre of the core along the radius, a is the core radius. For a GI fiber

$$n(r)\begin{cases} n_1\left[1-2\Delta\left(\dfrac{r}{a}\right)^{\propto}\right]^{1/2} & r < a \\ n_1(1-2\Delta)^{1/2} \approx n_1(1-\Delta) = n_2 & r \geq a \end{cases}$$

  where, n_1 is the refractive index at the centre of the core, i.e. at the axis of the core and n_2 is the refractive index at the core-cladding interface and $\propto$ is a dimensionless constant.
- The numerical aperture which is a measure of the light gathering power of an optical fiber is expressed for an SI fiber as

$$\text{NA} = \sqrt{n_1^2 - n_2^2} \approx n_1\sqrt{2\Delta}$$

  For a GI fiber NA is a function of position given by,

$$\text{NA}(r) = \left(\sqrt{n_1^2 - n_2^2}\right)\sqrt{1-\left(\frac{r}{a}\right)^{\propto}}$$

- Optical fibers are generally made of glass or plastic. The principal material used in glass fiber is silica (SiO_2) with GeO_2, B_2O_3, P_2O_5 as dopants. Plastic fibers use a host of transparent optical polymers. Erbium doped fibers are used for making fiber lasers.
- Fibers are drawn from preforms using fiber drawing machines. Preforms are small size rods of ultra-pure glass with similar type of index variations as the fibers.
- Optical fiber preforms can be made using various techniques such as outside vapour phase oxidation (OVPO), modified chemical vapour deposition (MCVD), vapour axial deposition (VAD), plasma chemical vapour deposition (PCVD).
- Fibers can be directly drawn from the molten state using double crucible technique.

Problems

2.1 Derive the expressions for phase shifts encountered by the normal and parallel components following total internal reflection as given by Eqs (2.7) and (2.8).

2.2 Explain the limitation of ray analysis. Illustrate two cases in the context of the optical fiber when ray analysis is invalid.

2.3 What is the difference between external reflection and total internal reflection?

2.4 Calculate the frequency of the light emitted by a monochromatic light source operating at 1300 nm. How much energy (in eV) does it correspond to?

2.5 A 60/125 μm step-index fiber operating at 850 nm has a core refractive index of 1.5 and a relative index deviation of 1%. What is the numerical aperture of the fiber? What is the radius of the core? Is it a single or a multimode fiber?

2.6 What is the purpose of cladding in an optical fiber?

2.7 Is it possible to transmit light through a glass fiber without having a cladding?

2.8 What is the typical value of the diameter of an optical fiber?

2.9 Which wavelength range is most suited for the transmission of light through silica fiber?

2.10 Distinguish between meridional rays and skew rays in an optical fiber.
2.11 What is the significance of the numerical aperture (NA) of a fiber?
2.12 What are the advantages of having a large NA?
2.13 What are the disadvantages of having a large NA?
2.14 How would you enhance the value of NA at a given wavelength?
2.15 Explain why the value of NA of an optical fiber is kept low.
2.16 Which fiber has a larger NA: A glass fiber or a plastic fiber?
2.17 Which fiber has a larger bandwidth: A single-mode fiber or a multimode fiber?
2.18 What is the advantage of a GI fiber over an SI fiber?
2.19 How does GI fiber help in enhancing the bandwidth?
2.20 Which phenomenon exhibited by light confirms that light is a transverse wave?
2.21 What is the raw material for glass fiber?
2.22 What is POF? What are the advantages and disadvantages of POF over silica fiber?
2.23 What is a PCS fiber? What is the advantage of this fiber over plastic fiber?
2.24 The attenuation of early fibers was extremely high. What are the primary reasons behind this?
2.25 Explain why it is not possible even theoretically to make the attenuation of glass zero at any wavelength.
2.26 List the composition of ZBLAN fiber.
2.27 Which wavelength range is best suited for transmission through ZBLAN fiber?
2.28 What are the major disadvantages of halide fibers?
2.29 Define mathematically, the refractive index profile in the core and the cladding region of triangular and parabolic indexed GI fibers.
2.30 Obtain an expression for the numerical aperture of a triangular indexed GI fiber.
2.31 Derive an expression for the numerical aperture (NA) of a parabolic indexed GI fiber and plot the variation of the NA with the distance measured from the center of the core.
2.32 For what value of α [Eq. (2.12)], does the index profile of a GI fiber become the same as that of an SI fiber?
2.33 What is the condition for total internal reflection to occur at the core-cladding interface?
2.34 Does every ray that gets total internally reflected manage to propagate throughout the fiber?
2.35 A multimode step-index glass fiber has a core refractive index of 1.458 and a cladding refractive index of 1.433. What is the value of NA? Calculate the value of the maximum acceptance angle when the fiber is (i) placed in the air (RI = 1); and (ii) placed in water (RI = 1.33).
2.36 A step-index fiber has a core refractive index of 1.51 and a cladding refractive index is 1.49. Estimate the value of NA by using the exact formula. Compare and contrast the value with that obtained with the help of the approximate formula, e.g. $\mathrm{NA} = n_1\sqrt{2\Delta}$.
2.37 A triangular index GI fiber has a refractive index of 1.458 at the axis of the core and a refractive index of 1.433 at the core-cladding interface. Estimate the value of the numerical aperture of the fiber at a point midway between the center of the core and the core-cladding interface.
2.38 A step-index fiber has an NA of 0.18 and a core refractive index of 1.458. What is the value of the relative index deviation ratio?
2.39 The speed of light in the core of a step-index fiber is 2×10^8 m/s. The NA of the fiber is 0.22. What is the refractive index of the cladding region?
2.40 Show that the numerical aperture of a GI fiber with a parabolic index profile can be expressed as

$$\mathrm{NA}(r) = n_1\left[2\Delta\left(1 - \frac{r^2}{a^2}\right)\right]^{1/2}$$

2.41 A step-index fiber has a numerical aperture 0.2 and a core refractive index 1.458. What is the value of the acceptance angle of the fiber when the light enters the fiber from an index-matching fluid with a refractive index of 1.40?

2.42 A step-index fiber has an acceptance angle of 22° in air and has a relative index deviation of 2.5%. Estimate the values of the refractive index of the core and the cladding region.

2.43 What is an active fiber?

2.44 What is a preform? What are the typical dimensions of a preform?

2.45 What are the common contaminants in a fiber drawn using the double crucible method?

2.46 What are the disadvantages of outside vapor-phase oxidation?

2.47 Which impurity causes maximum absorption in optical fiber?

2.48 Why is sintering not needed in plasma chemical vapor deposition?

2.49 Estimate the value of the total cable tension cumulatively by working through the following sections/points of the cable installed in a duct as illustrated in Fig. 2.24.

(a) P-Q (length 200 m)
(b) at Q
(c) Q-R (length 100 m)
(d) R-S (length 100 m)
(e) S-T (length 50 m)
(f) at T
(g) T-U (length 50 m)
(h) at U
(i) U-V (length 75 m)

The following values may be used in the computation assuming a suitable value of the tension at the feeding end.

$w = 0.9$ kg/m $\mu = 0.55$ $g = 9.81$ m/s^2

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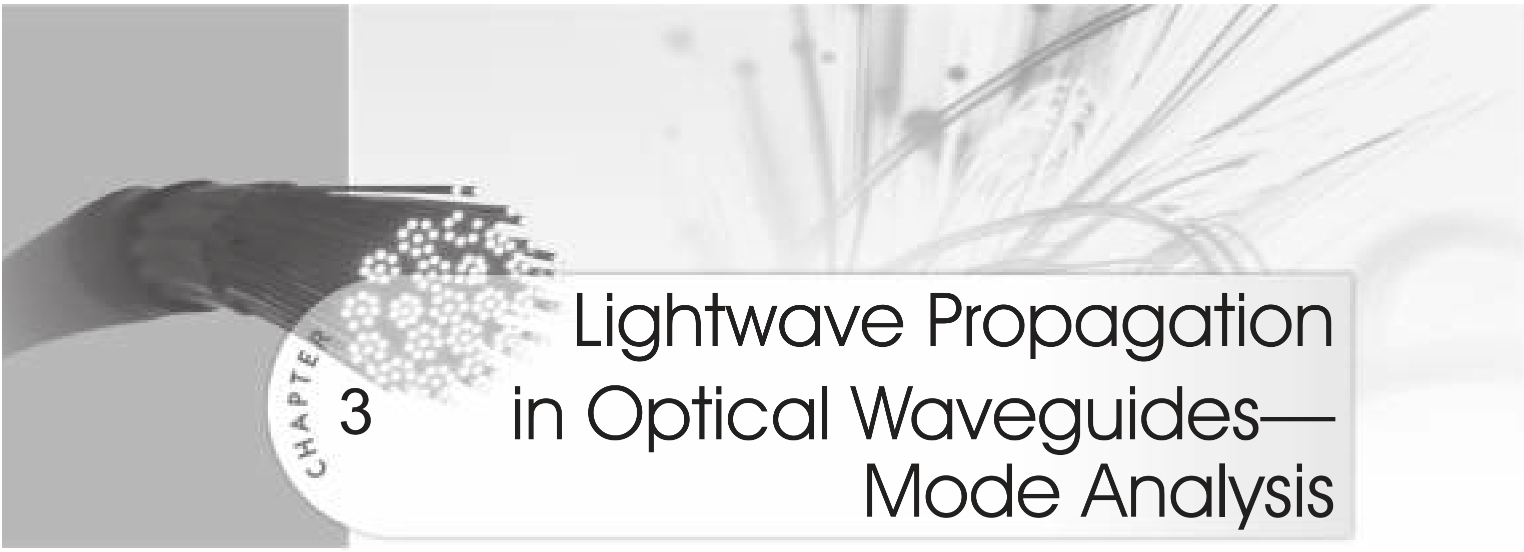


CHAPTER 3

Lightwave Propagation in Optical Waveguides—Mode Analysis

In the previous chapter, we have seen that light can be considered as an electromagnetic wave having frequency in the optical range. The mode analysis presented in this chapter for understanding the propagation of lightwave through planar waveguides as well as optical fibers is based on this principle. The propagation of light through optical fibers can be best analyzed with the help of mode analysis by solving Maxwell's equations under appropriate boundary conditions. Ray analysis is only an approximated analysis and is applicable under zero wavelength limits. In the case of optical fibers the ratio of the size of fiber core radius to wavelength should be ideally infinite or should be very large so that ray analysis becomes valid. For multimode fibers, the condition is satisfied to some extent in the sense that the core radius is generally very large as compared to the typical wavelength of operation (~1 μm). This is, however, not true for single-mode fibers used in optical fiber communication through which the lightwave propagates. This chapter addresses several fundamental issues about optical fibers such as the structures of an optical fiber, classification of fibers, propagation of light through optical fibers, fiber composition, fabrication techniques, etc.

Over several decades in the past different dielectric waveguide structures have been proposed and studied for the transmission of the optical signal (lightwave) (Marcatilli, 1979; Hondros *et al* 1910; van Heel *et al* 1954; Midwinter, 1979). The investigation of the propagation of light through dielectric waveguide structure was first reported at the beginning of the twentieth century. In the simplest form, a dielectric waveguide for the transmission of light may be viewed essentially as a transparent silica rod surrounded by air. However, this type of unsupported waveguide proves to be impractical because of high loss and associated aberrations. Furthermore, this type of waveguide needs mechanical support when the core region is made very thin to limit the large number of modes responsible for undesired aberrations. Later on, it was proved that the light transmission through the long strands of silica fiber can be greatly improved by incorporating an over-layered material of a slightly lower refractive index. This central region through which the light is intended to propagate is called the core region and the supporting outer layer is called cladding. In the 1950's long human hair size (~100 μm diameter) clad dielectric waveguides were proposed (Midwinter, 1979). The applications of these dielectric waveguides were initially restricted to several non-telecommunication systems including medical appliances like fiberoptic endoscopes.

3.1 GENERIC MODE EQUATIONS

In Chapter 2, we have described the propagation of light in a multimode fiber by using the principles of geometrical optics which are based on the assumption that light travels in a straight line (rectilinear propagation of light). This simple picture though helps one to visualise the propagation of light through a relatively large diameter light pipe but the same does not apply to the case of single-mode fibers. In addition, simple ray analysis cannot explain several other phenomena associated with the propagation of light through optical fibers. This chapter is devoted to mode analysis based on Maxwell's equations. The analysis helps one to understand various modes that are generated in the transverse direction while light propagates along the axis of the fiber. In mode analysis to follow, light is considered as an electromagnetic wave propagating through an optical fiber which is made of dielectric material (Felsen, 1974; Snyder *et al* 1974; Marcuse, 1991).

3.1.1 Electromagnetic Waves

The analysis is based on the particular nature of light that treats light as an electromagnetic wave and allows one to apply Maxwell's equations to explain its propagation through dielectric waveguides. An electromagnetic wave comprises two fields, e.g. an electric field and a magnetic field. Both the fields are vectors having a direction and a magnitude (amplitude). The two fields are orthogonal to each other and move with the speed of light. The electric and magnetic field distribution of a train of plane or linearly polarised electromagnetic wave is shown in Fig. 3.1 by arbitrarily assuming that the electric field is oriented along the x-axis and the magnetic field along the y-axis. Under this condition, the direction of propagation of light will be along the z-direction. Thus the electric field vector always oscillates along the x-direction (vertically polarised) and thus the plane wave corresponding to the electric field is linearly polarised with the polarisation vector $\hat{\boldsymbol{e}}_x$. Likewise, the magnetic field is linearly polarised with

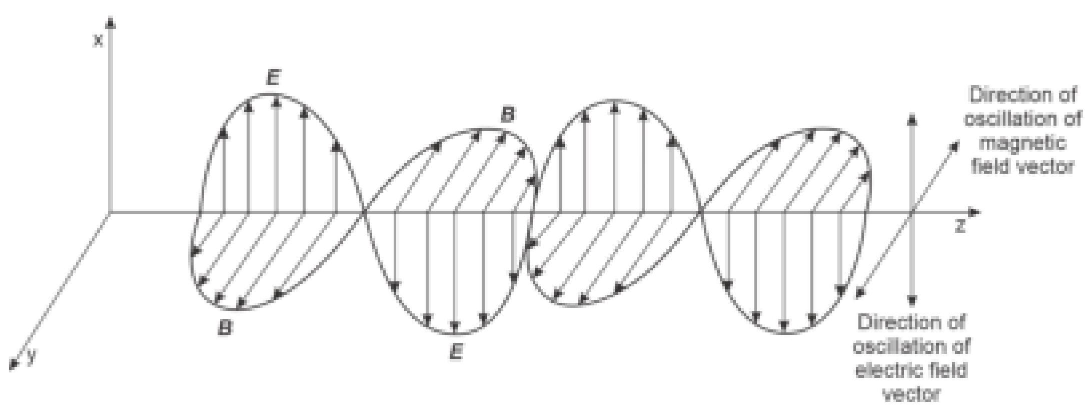


Fig. 3.1 Electromagnetic wave propagating in the z-direction where the electric field vector oscillates in the x-z plane (vertical polarisation) while the magnetic field vector is confined in x-y plane (horizontal polarisation)

polarisation vector $\hat{\boldsymbol{e}}_y$ (horizontally polarised). It may be pointed out that we could have drawn the magnetic field in the vertical plane and the electric field in the horizontal plane. Polarisation refers to the orientation of the electromagnetic field about some plane or boundary towards which the wave advances. At any instant in time, the field vectors may be oriented vertically, horizontally, or somewhere in between the two extremes. The general state of polarisation can be determined by expressing the electric field of a linearly polarised wave mathematically as

$$\boldsymbol{E}_x(z, t) = Re(\boldsymbol{E}) = \hat{\boldsymbol{e}}_x E_{0x} \cos(\omega t - \beta z) \tag{3.1}$$

where the electric field vector has a generalised form

$$\boldsymbol{\varepsilon}(z, t) \sim \exp[j(\omega t - \beta z)] \tag{3.2}$$

and $\hat{e}_x$ is the unit vector along the x-direction, ω is the angular frequency, β is the z-component of the propagation constant and E_{0x} is the amplitude of the electric vector along the x-direction.

To describe the general state of polarisation let us consider another linearly polarised wave that is orthogonal and independent of $\boldsymbol{E}_x(z, t)$. We assume this linearly polarised wave to be

$$\boldsymbol{E}_y(z, t) = \hat{\boldsymbol{e}}_y E_{0y} \cos(\omega t - \beta z + \delta) \tag{3.3}$$

where $\hat{\boldsymbol{e}}_y$ is the unit vector along y-direction, E_{0y} is the amplitude of the electric field vector and δ is the phase difference between the two orthogonal waves.

The resultant of the two waves can be expressed as

$$\boldsymbol{E}(\boldsymbol{z}, \boldsymbol{t}) = \boldsymbol{E}_x(z, t) + \boldsymbol{E}_y(z, t) \tag{3.4}$$

When $\delta = 0$ or $2\pi m$, (m being an integer) the two orthogonal waves are in phase and the resultant wave is also linearly polarised. The magnitude (amplitude) of the resultant electric vector is given by

$$|\bar{\boldsymbol{E}}| = \sqrt{E_{0x}^2 + E_{0y}^2} \tag{3.5}$$

The polarisation vector makes an angle θ with the x-axis given by

$$\theta = \tan^{-1}\left(\frac{E_{0y}}{E_{0x}}\right) \tag{3.6}$$

The resultant linearly polarised wave is shown in Fig. 3.2. Thus, two orthogonal linearly polarised waves having the same phase combine to form another linearly polarised wave. Conversely, an arbitrary linearly polarised wave can be decomposed into two independent orthogonal linearly polarised components having the same phase.

For arbitrary values of δ, Eq. (3.4) can be represented as

$$\left(\frac{E_x}{E_{0x}}\right)^2 + \left(\frac{E_y}{E_{0y}}\right)^2 - 2\left(\frac{E_x}{E_{0x}}\right)\left(\frac{E_y}{E_{0y}}\right)\cos\delta = \sin^2\delta \tag{3.7}$$

which is the generalised equation of an ellipse and the wave is elliptically polarised.

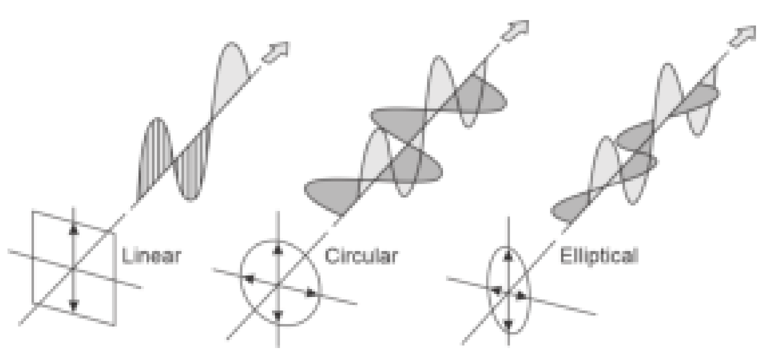


Fig. 3.2 Illustrating the formation of linearly polarised, circularly polarised and elliptically polarised waves by combining two independent orthogonally plane polarised waves with different phase relationships

Further, when

$$E_{0x} = E_{0y} = E_0 \text{ (say)}$$

and

$$\delta = 2\pi m \pm \frac{\pi}{2} \quad (m = 0, \pm 1, \pm 2,...)$$

Eq. (3.7) becomes

$$E_x^2 + E_y^2 = E_0^2 \tag{3.8}$$

which is the generalised equation of a circle and the wave is said to be circularly polarised. The formation of elliptically and circularly polarised waves is illustrated in Fig. 3.2. In actual practice, circular and elliptical polarisations occur when the propagation speed of the two independent linearly polarised with orthogonal plane of polarisations are slightly different. This is usually caused by the materials that exhibit slightly different refractive index values in the two orthogonal directions. This gives rise to a phenomenon known as birefringence.

3.1.2 Maxwell's Electromagnetic Equations in Non-homogeneous Medium

The mode analysis is based on electromagnetic wave equations derived based on Maxwell's equations (Maxwell, 1873). Maxwell's curl and the divergence equations involving the electric field ***E*** and magnetic field ***H***, the electric flux density (electric displacement), and the magnetic flux density (magnetic induction) ***B*** are given by

$$\nabla \times \mathcal{E} = -\frac{\partial \mathcal{B}}{\partial t} \tag{3.9}$$

$$\nabla \times \mathcal{H} = \mathcal{J} + \frac{\partial \mathcal{D}}{\partial t} \tag{3.10}$$

$$\nabla \cdot \mathcal{D} = \rho \tag{3.11}$$

$$\nabla \cdot \mathcal{B} = 0 \tag{3.12}$$

where $\mathcal{J} = \sigma\mathcal{E}$ is the conduction current density, σ is the conductivity of the medium, ρ is the volume density of electric charge. The electric and the magnetic flux density vectors are related to their corresponding field vectors as

$$\mathcal{D} = \in \mathcal{E} = \in_0 n^2 \mathcal{E} \tag{3.13}$$

and

$$\mathcal{B} = \mu \mathcal{H} \tag{3.14}$$

where $\in$ is the dielectric permittivity and μ is the magnetic permeability of the medium. For a pure dielectric medium, the conductivity(σ) is zero.

Transverse Electric (TE) and Transverse Magnetic (TM) Modes in a Planar Waveguide

Assuming that no electric charge is enclosed, Maxwell's Eqs (3.9) to (3.12) for an isotropic, linear, non-conducting, and nonmagnetic medium can be expressed as (Adam, 1981; Ghatak *et al* 1991)

$$\nabla \times \mathcal{E} = -\mu_0 \frac{\partial \mathcal{H}}{\partial t} \tag{3.15}$$

$$\nabla \times \mathcal{H} = \in \frac{\partial \mathcal{E}}{\partial t} = \in_0 n^2 \frac{\partial \mathcal{E}}{dt} \tag{3.16}$$

$$\nabla \cdot \mathcal{D} = 0 \tag{3.17}$$

$$\nabla \cdot \mathcal{B} = 0 \tag{3.18}$$

where the relative permeability has been assumed to be unity ($\mu_r = 1$) and ($n = \sqrt{\in_r}$) is the refractive index of the propagating medium.

Taking the curl on both sides of Eq. (3.15), we get

$$\nabla \times (\nabla \times \boldsymbol{\mathcal{E}}) = -\mu_0 \frac{\partial(\nabla \times \boldsymbol{\mathcal{H}})}{\partial t} = -\mu_0 n^2 \frac{\partial^2 \boldsymbol{\mathcal{E}}}{\partial t^2} \tag{3.19}$$

Using Eq. (3.16), we may write Eq. (3.19) as

$$\nabla(\nabla \cdot \boldsymbol{\mathcal{E}}) - \nabla^2 \boldsymbol{\mathcal{E}} = -\mu_0 \in_0 n^2 \frac{\partial^2 \boldsymbol{\mathcal{E}}}{\partial t^2} \tag{3.20}$$

where ∇^2 is the Laplacian operator.

Further using Eq. (3.17), we may write

$$\nabla \cdot \boldsymbol{\mathcal{D}} = \in_0 \nabla \cdot (n^2 \boldsymbol{\mathcal{E}}) = \in_0 [\nabla n^2 \cdot \boldsymbol{\mathcal{E}} + n^2 \nabla \cdot \boldsymbol{\mathcal{E}}] = 0 \tag{3.21}$$

That is,

$$\nabla \cdot \boldsymbol{\mathcal{E}} = -\frac{1}{n^2} \nabla n^2 \cdot \boldsymbol{\mathcal{E}} \tag{3.22}$$

Substituting the value of $\nabla \cdot \boldsymbol{E}$ from Eq. (3.22) into Eq. (3.21), we get

$$\nabla^2 \cdot \boldsymbol{\mathcal{E}} + \nabla\left(\frac{1}{n^2} \nabla n^2 \cdot \boldsymbol{\mathcal{E}}\right) - \mu_0 \in_0 n^2 \frac{\partial^2 \boldsymbol{\mathcal{E}}}{\partial t^2} = 0 \tag{3.23}$$

The above equation reveals that in a non-homogeneous medium, the equations for the components $\boldsymbol{\mathcal{E}}_x$, $\boldsymbol{\mathcal{E}}_y$ and $\boldsymbol{\mathcal{E}}_z$ are coupled. For a homogeneous medium, the second term of Eq. (3.23) vanishes and takes the following scalar form

$$\nabla^2 \cdot \boldsymbol{\mathcal{E}} = \mu_0 \in_0 n^2 \frac{\partial^2 \boldsymbol{\mathcal{E}}}{\partial t^2} \tag{3.24}$$

Equation (3.24) suggests that each cartesian component of the electric field satisfies the scalar equation independently and the components are no longer coupled. This means Eq. (3.24) can also be expressed as

$$\nabla^2 \psi = \mu_0 \in_0 n^2 \frac{\partial^2 \psi}{\partial t^2} \tag{3.25}$$

That is,

$$\nabla^2 \psi = \frac{1}{v_p^2} \frac{\partial^2 \psi}{\partial t^2} \tag{3.26}$$

where v_p is the phase velocity which corresponds to the velocity of propagation of a point of constant phase on the wave in the medium, given by

$$v_p = \frac{1}{\sqrt{\mu_0 \in_0 \in_r}} \tag{3.27}$$

For the free space, $\in_r = 1$ and the phase velocity given by Eq. (3.27) corresponds to the velocity of light in the free space given by

$$c = \frac{1}{\sqrt{\mu_0 \in_0}} \tag{3.28}$$

Similarly, by taking the curl on both sides of Eq. (3.16) and using Eqs (3.18) and (3.15), we get

$$\nabla^2 \boldsymbol{\mathcal{H}} + \frac{1}{n^2} \nabla n^2 \times (\nabla \times \boldsymbol{\mathcal{H}}) - \in_0 \mu_0 n^2 \frac{\partial^2 \boldsymbol{\mathcal{H}}}{\partial t^2} = 0 \tag{3.29}$$

If we now assume that the refractive index varies only in the directions transverse (x and y) to the direction of propagation (z-axis), we may write

$$n^2 = n^2(x, y) \tag{3.30}$$

The solution of Eqs (3.23) and (3.29) can be obtained by writing the cartesian components of the equations and separating the z- and the time components as

$$\boldsymbol{\mathcal{E}} = \boldsymbol{E}(x, y)e^{j(\omega t - \beta z)} \tag{3.31}$$

and

$$\boldsymbol{\mathcal{H}} = \boldsymbol{H}(x, y)\, e^{j(\omega t - \beta z)} \tag{3.32}$$

where β is the propagation constant in the z-direction.

If we now assume that the refractive index depends on the x-coordinate only, then we may write

$$n^2 = n^2(x) \tag{3.33}$$

As the wave is propagating in the z-direction, we may neglect the propagation of the wave in the y-direction by assuming the propagation constant in the y-direction to be zero. Therefore, without any loss of generality, we may write the solutions following Eqs (3.31) and (3.32) as

$$\boldsymbol{\mathcal{E}}_i = \boldsymbol{E}_i(x)e^{j(\omega t - \beta z)}\ ;\ i = x, y\ z \tag{3.34}$$

and

$$\mathcal{H}_i = H_i\,(x)e^{j(\omega t - \beta z)}\ ;\ i = x, y\ z \tag{3.35}$$

Substituting the expression for the electric field given by Eq. (3.34) in Eq. (3.15), we get

$$\begin{vmatrix} \hat{\boldsymbol{i}} & \hat{\boldsymbol{j}} & \hat{\boldsymbol{k}} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial x} \\ \boldsymbol{\varepsilon}_x & \boldsymbol{\varepsilon}_y & \boldsymbol{\varepsilon}_z \end{vmatrix} = -j\omega\mu_0\left[\,\hat{\boldsymbol{i}}H_x + \hat{\boldsymbol{j}}H_y + \hat{\boldsymbol{k}}H_z\,\right] \tag{3.36}$$

where $\hat{\boldsymbol{i}}, \hat{\boldsymbol{j}}$ and $\hat{\boldsymbol{k}}$ are unit vectors along x-, y-, and z-axes respectively.

Equation (3.36) can be expanded as

$$\hat{\boldsymbol{i}}(j\beta E_y) + \hat{\boldsymbol{j}}\left(-j\beta E_x - \frac{\partial E_z}{\partial x}\right) + \hat{\boldsymbol{k}}\frac{\partial E_y}{\partial x} = -j\omega\mu_0\left[\hat{\boldsymbol{i}}H_x + \hat{\boldsymbol{j}}H_y + \hat{\boldsymbol{k}}H_z\right] \tag{3.37}$$

Equating the terms component-wise (x, y, and z) on both sides of Eq. (3.37), we may write

$$j\beta E_y = -j\omega\mu_0 H_x \tag{3.38}$$

$$-j\beta E_x - \frac{\partial E_z}{\partial x} = -j\omega\mu_0 H_y \tag{3.39}$$

$$\frac{\partial E_y}{\partial x} = -j\omega\mu_0 H_z \tag{3.40}$$

Similarly, substituting the expression for the magnetic field given by Eq. (3.35) in Eq. (3.16), we may write

$$\begin{vmatrix} \hat{\boldsymbol{i}} & \hat{\boldsymbol{j}} & \hat{\boldsymbol{k}} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix} = j\omega\epsilon_0 n_2\left[\hat{\boldsymbol{i}}\varepsilon_x + \hat{\boldsymbol{j}}\varepsilon_y + \hat{\boldsymbol{k}}\varepsilon_z\right] \tag{3.41}$$

That is,

$$\hat{\boldsymbol{i}}(j\beta H_y) + \hat{\boldsymbol{j}}\left(-j\beta H_x - \frac{\partial H_z}{\partial x}\right) + \hat{\boldsymbol{k}}\left(\frac{\partial H_y}{\partial x}\right) = j\omega\epsilon_0 n^2\left[\hat{\boldsymbol{i}}E_x + \hat{\boldsymbol{j}}E_y + \hat{\boldsymbol{k}}E_z\right] \tag{3.42}$$

Equating the terms componentwise (x, y, and z) on both sides of Eq. (3.42), we may write

$$j\beta H_y = j\omega\epsilon_0 n^2 E_x \tag{3.43}$$

$$-j\beta H_x - \frac{\partial H_z}{\partial x} = j\omega\epsilon_0 n^2 E_y \tag{3.44}$$

$$\frac{\partial H_y}{\partial x} = j\omega\epsilon_0 n^2 E_z \tag{3.45}$$

The two sets of Eqs (3.38)–(3.40) and (3.43)–(3.45) can be rearranged in the following two groups with specific characteristics

$$\left.\begin{aligned} H_x &= -\frac{\beta}{\omega\mu_0}E_y && (3.46)\\ H_z &= \frac{j}{\omega\mu_0}\frac{\partial E_y}{\partial x} && (3.47)\\ -j\beta H_x - \frac{\partial H_z}{\partial x} &= j\omega\epsilon_0 n^2 E_y && (3.48) \end{aligned}\right\}\text{TE mode}$$

and

$$\left.\begin{aligned} E_x &= \frac{\beta}{\omega\epsilon_0 n^2}H_y && (3.49)\\ E_z &= \frac{1}{j\omega\epsilon_0 n^2}\frac{\partial H_y}{\partial x} && (3.50)\\ j\beta E_x + \frac{\partial E_z}{\partial x} &= j\omega\mu_0 H_y && (3.51) \end{aligned}\right\}\text{TM mode}$$

It can be seen from the rearranged sets of Eqs (3.46)–(3.48) and (3.49)–(3.51) that the first set of equations involve only E_y, H_x and H_z while the second set of equations involves only E_x, E_z and H_y. Thus, for a planar waveguide under consideration, Maxwell's equations essentially boil down to two independent sets of equations. A closer examination of the two sets of equations reveals that the first set of Eqs (3.46)–(3.48) correspond to non-vanishing values of E_y, H_x and H_z have vanishing E_x, E_z and H_y components. This set of equations gives rise to an electric field that has only a transverse component and gives rise to transverse electric (TE) modes. On the other hand, the second set of Eqs (3.49)–(3.51) correspond to non-vanishing E_x, E_z and H_y with E_y, H_x and H_z vanishing. In this case, as the magnetic field has only a transverse component, the equations give rise to transverse magnetic (TM) modes. In the following section, we will consider a step-index planar waveguide to explore the propagation characteristics. The planar optical waveguides find applications in integrated optics (IO).

3.2 MODES IN A STEP-INDEX PLANAR WAVEGUIDE

In the preceding analysis, we have considered the refractive index to be an arbitrary function of x coordinate. In this section, we shall analyse the simplest planar waveguide which consists of a thin dielectric film having a refractive index n_1 sandwiched between materials of slightly lower refractive indices. The refractive index of the top layer and the bottom layer may have different values to form an asymmetric waveguide structure. For a symmetric waveguide structure, the values of the refractive index of the top and the bottom layers are the same (say, $n_2 < n_1$). In the analysis, we shall consider a symmetrical structure where the central region (core) has a refractive index n_1 slightly higher than the refractive index, n_2 of the top and the bottom layers.

The schematic of the planar waveguide under consideration along with the rectangular cartesian coordinate axes is shown in Fig. 3.3.

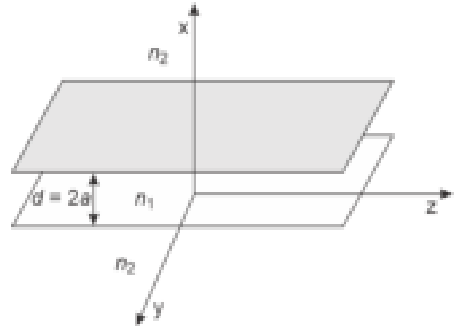


Fig. 3.3 Schematic of a symmetrical planar waveguide ($n_1 > n_2$) with infinite extensions in the *y*- and *z*-directions. The propagation of the electromagnetic wave is considered in the z-direction

For a symmetrical step-index planar waveguide, the variation of refractive index along the x-axis is shown in Fig. 3.4. The longitudinal axis of symmetry of the sandwiched layer is considered as the reference.

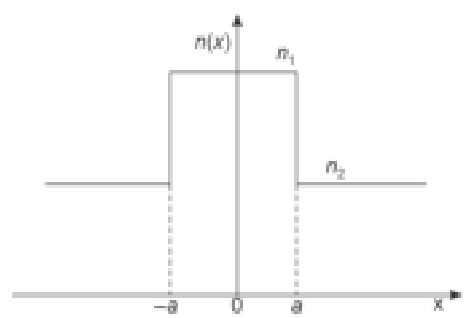


Fig. 3.4 Refractive index profile of a symmetric step-index planar waveguide along x-axis

3.2.1 Transverse Electric (TE) Modes in a Symmetric Planar Waveguide

For analysing the transverse electric modes characterised by the set of Eqs (3.46)–(3.48), we substitute the values of H_x and H_z from Eqs (3.46) and (3.47) into Eq. (3.49) and recall the fact that the refractive index is a function of x only (Eq. (3.33)) to obtain the following ordinary differential equation, given by

$$\frac{d^2E_y}{dx^2} + \left[k^2n^2(x) - \beta^2\right]E_y = 0 \tag{3.52}$$

where k is the free-space wavenumber given by

$$k = \omega\sqrt{\mu_0\epsilon_0} = \frac{\omega}{c}$$

c being the velocity of light in free space.

For the symmetric step-index planar waveguide under consideration

$$n(x) = \begin{cases} n_1 & \text{for } |x| < a \\ n_2 & \text{for } |x| > a \end{cases} \quad (3.53) \; (3.54)$$

Let us examine the boundary conditions governing Eq. (3.52). It may be noted that E_y and H_z correspond to the tangential components and must be continuous at the interface at $x = \pm a$. Further, Eq. (3.47) suggests that H_z is proportional to the derivative of E_y with x, and therefore, the boundary conditions can be expressed as

$$E_y \text{ and } \frac{dE_y}{dx} \text{ continuous at } x = \pm a \quad (3.55)$$

Using Eqs (3.53) and (3.54), the electric field equation given by Eq. (3.52) can be resolved into the following two equations

$$\frac{d^2E_y}{dx^2} + [k^2n_1^2 - \beta^2]E_y = 0 \quad \text{for } |x| < a \quad (3.56)$$

$$\frac{d^2E_y}{dx^2} + [k^2n_2^2 - \beta^2]E_y = 0 \quad \text{for } |x| > a \quad (3.57)$$

The above equations are subjected to the boundary conditions given by Eq. (3.55).

For the modes to be guided through the central region of the planar waveguide, the field must vary harmonically within the central region and decay exponentially in the surrounding region having a refractive index, $n_2(< n_1)$. To have the solution of the field Eqs (3.56) and (3.57) that would satisfy the above requirements, we must have

$$\beta^2 > k^2n_2^2 \quad (3.58)$$

$$\beta^2 < k^2n_1^2 \quad (3.59)$$

In other words, most of the energy associated with the modes will remain confined in the central region provided that

$$k^2n_2^2 < \beta^2 < k^2n_1^2 \quad (3.60)$$

Based on the above condition, Eqs (3.56) and (3.57) can be written as

$$\frac{d^2E_y}{dx^2} + u^2E_y = 0 \quad \text{for } |x| < a \quad (3.61)$$

$$\frac{d^2E_y}{dx^2} - w^2E_y = 0 \quad \text{for } |x| > a \quad (3.62)$$

where

$$u^2 = k^2\,n_1^2 - \beta^2 \quad (3.63)$$

and

$$w^2 = \beta^2 - k^2n_2^2 \quad (3.64)$$

The solution of Eq. (3.61) can be written in the form

$$E_y(x) = A_1\cos(ux) + A_2\sin(ux) \qquad \text{for } |x| < a \quad (3.65)$$

where A and B are arbitrary constants and can be determined using appropriate boundary conditions.

For the surrounding confining regions $x > a$ and $x < -a$ the solutions of the Eq. (3.62) for the two regions can be written as

$$E_y = \begin{cases} B_1 e^{wx} + B_2 e^{-wx} & \text{for } x < -a \quad (3.66) \\ C_1 e^{wx} + C_2 e^{-wx} & \text{for } x > a \quad (3.67) \end{cases}$$

where B_1, B_2, C_1 and C_2 are arbitrary constants.

Since the electric fields have to decrease exponentially in the surrounding confining regions, the exponentially increasing terms in the solutions given by Eqs (3.66) and (3.67) must vanish. Therefore, neglecting the exponentially increasing terms from the above solutions, we may write the solutions for the field in the surrounding regions as

$$E_y = \begin{cases} B_1 e^{wx} & \text{for } x < -a \quad (3.68) \\ C_2 e^{-wx} & \text{for } x > a \quad (3.69) \end{cases}$$

When we apply the boundary conditions given by the Eq. (3.55) which ensures the continuity of E_y and dE_y/dx at the interface, i.e. $x = \pm d/2$, we will get four equations arising out of Eq. (3.65) and the set comprising Eqs (3.68) and (3.69). Application of boundary conditions will lead to a transcendental equation that can be solved to obtain the allowed values of β for which the solution of the four equations obtained by applying the boundary conditions will exist. The discrete values of the propagation constant β in the range $kn_2 < \beta < kn_1$ for which the solution of the solution of the four equations will exist to sustain the corresponding modes can be obtained numerically.

For a textbook analysis, we may adopt a simpler approach when the refractive index in the central region is assumed to be symmetric about $x = 0$, that is

$$n^2(-x) = n^2(x) \tag{3.70}$$

Under this condition, the solutions of the electric field equations which correspond to the modes may be symmetric or anti-symmetric. Mathematically, we may write

$$E_y(-x) = E_y(x) \qquad \text{for symmetric modes} \tag{3.71}$$

$$E_y(-x) = -E_y(x) \qquad \text{for anti-symmetric modes} \tag{3.72}$$

3.2.1.1 Symmetric modes

For the symmetric mode, we may express the field equation as

$$E_y(x) = \begin{cases} A\cos(ux) & \text{for } |x| < a \quad (3.73) \\ Be^{-w|x|} & \text{for } |x| > a \quad (3.74) \end{cases}$$

where A and B are arbitrary constants of the solution for the symmetric case.

Applying the boundary conditions concerning the continuity of $E_y(x)$ and dE_y/dx at the interface $x = \pm a$, we get

$$A\cos(ua) = Be^{-wa} \tag{3.75}$$

and

$$-uA\sin(ua) = -wBe^{-wa} \tag{3.76}$$

Dividing Eq. (3.76) by Eq. (3.75), we may write the combined equation in the form

$$ua\tan(ua) = wa \tag{3.77}$$

We may recall Eqs (3.63) and (3.64) to find the following

$$w^2 = \beta^2 - k^2 n_2^2 = k^2 n_1^2 - u^2 - k^2 n_2^2 \tag{3.78}$$

That is,

$$w^2 = k^2(n_1^2 - n_2^2) - u^2 \tag{3.79}$$

We may rearrange Eq. (3.79) in the form

$$w^2a^2 = a^2k^2(n_1^2 - n_2^2) - u^2a^2 \tag{3.80}$$

That is,

$$wa = \left[a^2k^2\left(n_1^2 - n_2^2\right) - u^2a^2\right]^{\frac{1}{2}} \tag{3.81}$$

Let us substitute the following notations

$$V = ak\left(n_1^2 - n_2^2\right)^{\frac{1}{2}} \tag{3.82}$$

$$\chi = ua = a\left[k^2n_1^2 - \beta^2)\right]^{\frac{1}{2}} \tag{3.83}$$

with the above substitution, Eq. (3.81) can be written as

$$wa = \left[V^2 - \chi^2\right]^{\frac{1}{2}} \tag{3.84}$$

Substituting Eq. (3.84) into Eq. (3.77), we may write

$$\chi \tan \chi = \left[V^2 - \chi^2\right]^{\frac{1}{2}} \tag{3.85}$$

3.2.1.2 Anti-symmetric modes

Similarly, for the anti-symmetric mode, we may write the solution of the electric field equation as

$$E_y(x) = \begin{cases} C \sin(ux) & \text{for } |x| < a \qquad (3.86) \\ \dfrac{x}{|x|} De^{-w|x|} & \text{for } |x| > a \qquad (3.87) \end{cases}$$

where C and D are arbitrary constants.

3.2.1.3 Solution of modal equations

Following the procedure followed in the symmetric case, we may get the following relationship for the anti-symmetric case

$$-\chi \cot \chi = \left[V^2 - \chi^2\right]^{\frac{1}{2}} \tag{3.88}$$

The following two transcendental equations can be solved numerically to find the allowed values of χ for the existence of the solution of the equations.

$$\chi \tan \chi = \left[V^2 - \chi^2\right]^{\frac{1}{2}} \quad \text{(symmetric case)} \tag{3.89}$$

$$-\chi \cot \chi = \left[V^2 - \chi^2\right]^{\frac{1}{2}} \quad \text{(anti-symmetric case)} \tag{3.90}$$

The discrete values of χ so obtained can be subsequently used to calculate the corresponding discrete values of β in the range $kn_2 < \beta < kn_1$ for which the solution of the field equation representing different modes will exist.

The above two equations can be solved graphically as follows. We may write the right side as

$$\gamma = \left[V^2 - \chi^2\right]^{\frac{1}{2}} \tag{3.91}$$

That is,

$$\gamma^2 + \chi^2 = V^2 \tag{3.92}$$

It can be easily seen that Eq. (3.92) is the equation of a circle in the coordinates (γ, χ) having a radius of V. In Fig. 3.5, the portion of circles for two different values of V are shown with the help of dashed curves in the first quadrant when γ along the y-axis is plotted against $\chi(= ua)$ along the x-axis. If we now plot the two functions $\chi \tan \chi$ and $-\chi \cot \chi$ along y-axis against $\chi(= ua)$ along x-axis in the same figure, the points of intersections of these curves with the circles will give the allowed values of χ for the existence of the solutions of Eqs (3.89) and (3.90). These discrete values of χ can be subsequently used to determine the corresponding values of β using Eq. (3.83). It can be easily appreciated that the number of intersections (roots of the equations) will depend on the value of the parameter V. This signifies that for a given value of V, there will be a finite number of solutions of the electric field in the planar waveguide representing the corresponding modes (TE modes). From the above graphical analysis, we may summarise the following observations about the modes.

(i) Single-mode waveguide

We assume that the value of V is such that

$$0 < V < \pi/2 \tag{3.93}$$

In this case, it can be easily seen from Fig. 3.7 that in this range there is only one point of interaction of the curves or only one discrete TE mode of the waveguide and the mode is symmetric in x. The waveguide under this condition supports only one mode and is called a single-mode waveguide.

Example 3.1 A planar waveguide shown in Fig. 3.5 has a refractive index of the central region, n_1 = 1.485 and that of the surrounding regions, n_2 = 1.475. The waveguide is designed to support the propagation of light having λ = 1.55 μm. Calculate the thickness of the central region so that the waveguide ensures single-mode operation.

Solution We have seen in the preceding analysis that for single-mode operation, the value of V must satisfy the following condition

$$V < \frac{\pi}{2} \tag{E3.1}$$

Using Eq. (3.82), we may write

$$V = ak\left(n_1^2 - n_2^2\right)^{\frac{1}{2}} < \frac{\pi}{2} \tag{E3.2}$$

That is,

$$a \cdot \frac{2\pi}{1.55 \times 10^{-6}}[(1.485)^2 - (1.475)^2]^{(1/2)} < \frac{\pi}{2}$$

Therefore,

$$a < 2.25 \text{ μm}$$

That is, the thickness of the central film region should be

$$d = 2a < 4.50 \text{ μm}$$

This means that if the thickness of the central film is less than 4.50 μm, it will support only one mode and the waveguide will behave as a single-mode waveguide at the operating wavelength of λ = 1.55 μm.

It may be emphasised here that the same waveguide may support more than one mode if the operating wavelength is smaller. Further, since the film thickness cannot be zero for a physical waveguide, the waveguide will always support at least one mode.

(ii) Number of modes for a given V

We can easily see from Fig. 3.7 that

when $\pi/2 < V < \pi$

we have one symmetric and one anti-symmetric mode that is, a total of two TE modes; when $\pi < V < 3\pi/2$

we have two symmetric and one anti-symmetric mode or a total of 3 TE modes.

Therefore, in general considering m as an integer including 0 that is, m = 0,1,2,3... we may write

when

$$2m\frac{\pi}{2} < V < (2m+1)\frac{\pi}{2} \tag{3.94}$$

or,

$$m*pi < V < (m + 1/2)*pi \tag{3.95}$$

$$m < \frac{V}{\pi} < m + \frac{1}{2} \tag{3.96}$$

we have (m + 1) number of symmetric modes and m number of anti-symmetric modes, that is, a total of (2m + 1) number of TE modes.

Further, when

$$(2m+1)\frac{\pi}{2} < V < (2m+2)\frac{\pi}{2} \tag{3.97}$$

or,

$$m + \frac{1}{2} < \frac{V}{\pi} < m+1 \tag{3.98}$$

we have (m + 1) number of symmetric modes and (m + 1) number of anti-symmetric modes, that is, a total number of 2(m + 1) number of TE modes.

Example 3.2 A symmetrical planar waveguide (Fig. 3.5) having n_1 = 1.50 and n_2 = 1.48 has a 9 μm thick central film. Calculate the number of TE modes supported by the waveguide at an operating wavelength of λ = 650 nm.

Solution The value of V in this case can be computed using Eq. (3.82) as

$$\frac{V}{\pi} = \frac{9/2 \times 10^{-6} \times 2}{650 \times 10^{-9}}\left[(1.5)^2 - (1.48)^2\right]^{\frac{1}{2}} = 1.38$$

From the generic formula for calculation of the number of TE modes given by Eq. (3.96), in particular, we get two symmetric and one anti-symmetric mode or a total of three TE modes.

(iii) Calculation of propagation constants for $V \gg 1$

It can be easily seen from Fig. 3.5 that for a very large value of V ($\gg 1$) the points of intersection of the V-circle given by Eq. (3.92) with $\chi \tan \chi$ (symmetric modes) and $-\chi \cot \chi$ (anti-symmetric modes) come very close to

$$\chi = \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi \cdots \tag{3.99}$$

The values of the propagation constant for $V \gg 1$ can be accordingly expressed as

$$\chi_m = ua = a\left[k^2 n_1^2 - \beta_m^2\right]^{\frac{1}{2}} \approx (m+1)\frac{\pi}{2} \tag{3.100}$$

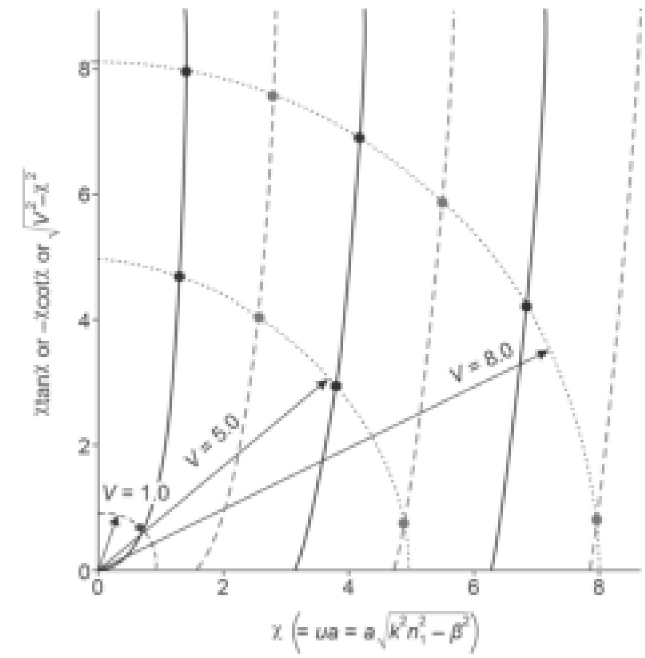


Fig. 3.5 The variations of $\chi\tan\chi$ (shown by solid lines), $-\chi$ cot χ (shown by dashed line) with χ along with the quadrants of the circles (shown by dotted lines) with different values of *V* as determined by Eq. (3.92). The points of intersections of the circles with the tangent and cotangent functions for a given value of *V* correspond to the values of propagation constant, β for symmetric and anti-symmetric modes

where the symmetric modes correspond to $m = 0, 2, 4,\cdots$ and the anti-symmetric modes $m = 1, 3, 5,\cdots$

(iv) Designation of mode numbers

The variation of the modal field and the normalised $|E_y|^2$ for a symmetrical step-index planar waveguide with x is illustrated in Fig. 3.6 for the first three TE modes. We have seen that for $0 < V < \pi/2$, the value of $\chi(= ua)$ always lies between 0 and $\pi/2$ and there is only one mode (symmetric). Therefore, the corresponding electric

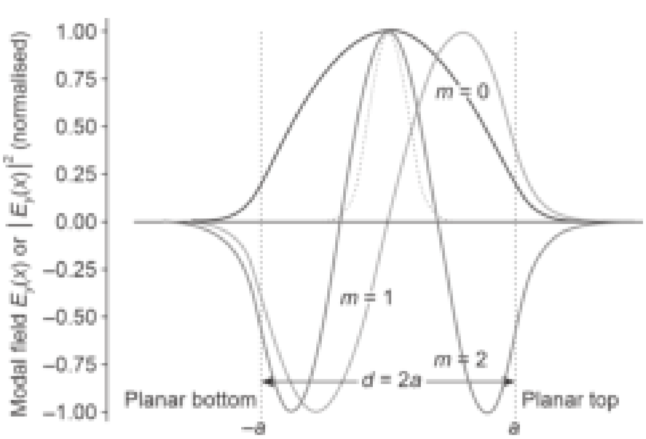


Fig. 3.6 Illustrating the modal fields (symmetric and anti-symmetric) profile for a step-index planar waveguide for $V = 2.2\pi$ wherein the even values of *m* corresponds to symmetric and the odd values of *m* correspond to anti-symmetric modes along with the normalized energy.

field variation $E_y(x)$ will have no zeroes for $|x| < a$, i.e. within the central film region. This mode is referred to as the fundamental mode or the zero-order mode designated by TE_0 mode. The next mode (anti-symmetric) occurs for $\chi(= ua)$ lying between $\pi/2$ and π. As a result, the corresponding electric field constituting the mode, $E_y(x)$ will have one zero at the center of the film ($x = 0$). As the mode contains one field zero in the central film region, the mode is designated as TE_1 mode. Likewise, the electric field $E_y(x)$ having two zeroes in the central film will be designated as TE_2 mode (Fig. 3.6). In general, the m^{th} order mode is designated as TE_m where m represents the number of field zeroes within the central region.

(v) Normalised propagation constant and cut-off condition

It is often convenient to determine the modes supported by a waveguide in terms of the parameter called normalised propagation constant b. The parameter is defined as

$$b = \frac{w^2a^2}{V^2} = 1 - \frac{u^2a^2}{V^2} = 1 - \frac{\chi^2}{V^2} \tag{3.101}$$

Substituting the values of w from Eq. (3.78) and V from Eq. (3.82) into Eq. (3.101), we may write

$$b = \frac{\beta^2 - k^2n_2^2}{k^2(n_1^2 - n_2^2)} \tag{3.102}$$

That is,

$$b = \frac{\left(\dfrac{\beta}{k}\right)^2 - n_2^2}{n_1^2 - n_2^2} = 1 - \frac{\chi^2}{V^2} \tag{3.103}$$

From the foregoing discussion, it can be easily understood that the allowed values of χ for different values of V can be calculated by solving Eqs (3.89) and (3.90) under appropriate boundary conditions. These values of χ for different values of V can be subsequently used for calculation of the normalised propagation constant, b using Eq. (3.103). The variation of b with the parameter, V is shown in Fig. 3.7 for the first few modes which include TE (shown by solid lines) and TM (shown by dashed line) modes. The TM modes will be discussed in the next section. This graph (Fig. 3.7) is very useful for determining the modes supported by a particular waveguide at a specific wavelength. For example, in the case of a particular waveguide operating at a given wavelength, we can first determine the value of V for the waveguide. The value of V so obtained can be used to read the value of b from the universal b *vs* V graph in Fig. 3.7. From the value of b obtained from the graph we can determine the value of β by writing Eq. (3.102) in the following form

$$\beta^2 = k^2\left[n_2^2 + b\left(n_1^2 - n_2^2\right)\right] \tag{3.104}$$

We may recall that for guided modes we must have

$$n_2 < \frac{\beta}{k} < n_1 \tag{3.105}$$

This means that for guided modes the normalised propagation constant, b will satisfy the following condition

$$0 < b < 1 \tag{3.106}$$

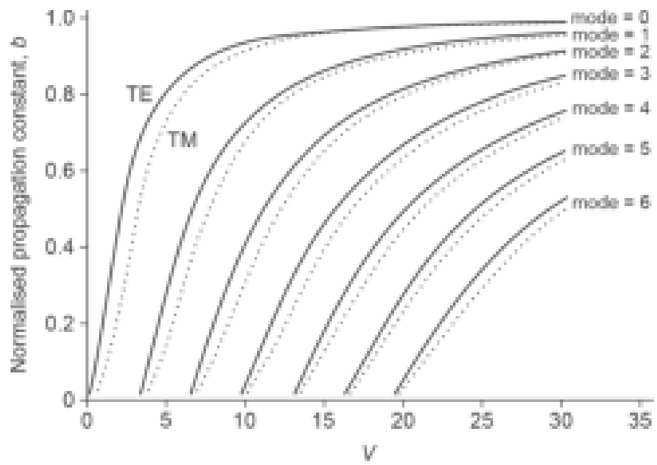


Fig. 3.7 Variation of the normalised propagation constant, *b* as a function of *V* for a step-index planar waveguide showing both TE (solid line) and TM (dashed line) modes. The cut-off of a mode corresponds to the value of *V* at which $b = 0$.

It is understood that for the guided mode β cannot be less than kn_2. When for a particular mode β becomes equal to kn_2, that is b becomes equal to zero, the mode is considered to be reached "cut-off". Therefore, at cut off we may write

$$\beta = kn_2, \qquad w = 0 \qquad \text{and} \qquad b = 0 \tag{3.107}$$

For the symmetric planar waveguide discussed above, at the cut-off condition, we find from Eq. (3.84) that

$$\chi = V \tag{3.108}$$

Therefore, under cut-off conditions, the TE modes can be determined by

$$V \tan V = 0 \tag{3.109}$$

$$-V \cot V = 0 \tag{3.110}$$

Based on the above cut-off equations, the cut-off values of different TE modes can be determined from the following equation

$$V\,(\text{cut-off}) = V_c = m\frac{\pi}{2}, \quad m = 0, 1, 2, 3,\cdots \tag{3.111}$$

In the above cut-off equation, even and odd values of m correspond to symmetric and anti-symmetric TE modes in the step-index planar waveguide under consideration. As mentioned earlier, the fundamental mode does not have any cut-off because the parameter V can have a zero value only when $a = 0$. Therefore, any waveguide having a finite thickness at the central region will have at least one guided mode.

(vi) Radiation modes

So far we have considered only the guided modes in the step-index planar waveguide for which it is necessary that

$$kn_2 < \beta < kn_1 \tag{3.112}$$

However, if there is a situation such that

$$\beta > kn_2 \tag{3.113}$$

Under the above condition the solution of the field equation for the region $|x| > a$ will have a solution (say for TE modes) that will vary harmonically in that region. These modes are called radiation modes. It can be easily verified that under the condition stated by Eq. (3.113), the electric field, i.e. Eq. (3.62) can be written in the form

$$\frac{d^2E_y}{dx^2} + \delta^2 E_y = 0 \quad for\,|x| > a \tag{3.114}$$

where,

$$\delta^2 = k^2 n_2^2 - \beta^2 > 0 \tag{3.115}$$

Under this condition, the solution of Eq. (3.114) will have a solution of the form

$$E_y(x) \sim e^{\pm\delta x} \tag{3.116}$$

This means that the electric field in the regions surrounding the central region $|x| > a$ will be oscillatory and constitute radiation modes unlike in the case of guided modes where the electric fields outside the central region decay exponentially. It may be noted that β can never become equal to or exceed the value kn_1 because the electric field under that situation will be unbounded when $x \to \infty$ or $x \to -\infty$.

3.2.2 Transverse Magnetic (TM) Modes in a Step-index Planar Waveguide

So far, we have considered TE modes in a symmetrical step-index planar waveguide. A similar analysis can be carried out for the TM modes which involve the three field components, e.g. E_x, E_z and H_y given by the Eqs (3.49)–(3.51). Substituting E_x and E_z from Eqs (3.49) and (3.50) respectively into Eq. (3.51) we get

$$n^2(x)\frac{d}{dx}\left[\frac{1}{n^2(x)}\frac{dH_y(x)}{dx}\right] + \left(k^2n^2(x) - \beta^2\right)H_y(x) = 0 \tag{3.117}$$

The above equation can be rearranged in the form

$$\frac{d^2H_y}{dx^2} - \left[\frac{1}{n^2(x)}\frac{dn^2(x)}{dx}\right]\frac{dH_y}{dx} + \left(k^2n^2(x) - \beta^2\right)H_y(x) = 0 \tag{3.118}$$

For a symmetrical step-index planar waveguide structure under consideration, Eq. (3.118) can be split into the following equations depending on the value of x

$$\frac{d^2H_y}{dx^2} + \left(k^2n_1^2 - \beta^2\right)H_y = 0, \quad |x| < a \tag{3.119}$$

$$\frac{d^2H_y}{dx^2} - \left(\beta^2 - k^2n_2^2\right)H_y = 0, \quad |x| > a \tag{3.120}$$

Applying the boundary conditions involving the continuity of tangential components of H_y and E_z at the interface $x = a$ and proceeding similarly as in the case of TE modes, we arrive at the following equations for the TM modes

$$\chi \tan \chi = \left(\frac{n_1^2}{n_2^2}\right)\left(V^2 - \chi^2\right)^{\frac{1}{2}} \text{ (symmetric TM modes)} \tag{3.121}$$

$$-\chi \cot \chi = \left(\frac{n_1^2}{n_2^2}\right)\left(V^2 - \chi^2\right)^{\frac{1}{2}} \text{ (anti-symmetric TM modes)} \tag{3.122}$$

It may be pointed out here that in the previous section, we have discussed about single-mode symmetric planar waveguide to support a single TE mode when $0 < V < \pi/2$. The waveguide supports two modes under this condition, e.g. one TE mode and one TM mode with slightly different propagation constants. In practice, the incident field is generally plane-polarised and therefore if **E** is along y-axis it will excite TE mode and if the **E** is along x-axis it will excite TM mode. If the incident field has a polarisation that makes an angle with the x-axis, in that case, both TE and TM modes will be excited. As these two modes have slightly different propagation constants they will superimpose on one another with different phases as the wave propagates along the z-direction. As a result, the resultant polarisation will change the state continuously depending on the position along the z-axis.

3.3 POWER ASSOCIATED WITH MODES IN A PLANAR WAVEGUIDE

The power flow through the waveguide can be calculated using the Poynting vector given by

$$\langle S\rangle = \frac{1}{2}\text{Re}(\boldsymbol{\mathcal{E}} \times \boldsymbol{\mathcal{H}}^*) \tag{3.123}$$

In this section, we will consider power associated with TE modes only. From Eqs (3.34) and (3.35), we find

$$\mathcal{E}_y = E_y(x)e^{j(\omega t - \beta z)} \tag{3.124}$$

and

$$\mathcal{H}_x = H_z(x)e^{j(\omega t - \beta z)} \tag{3.125}$$

For TE mode, we may use Eq. (3.46) and substitute for $\mathcal{E}_y$ from Eq. (3.124) to write

$$\mathcal{H}_x = -\frac{\beta}{\omega\mu_0}\mathcal{E}_y = -\frac{\beta}{\omega\mu_0}E_y(x)e^{j(\omega t-\beta z)} \tag{3.126}$$

Similarly, using Eq. (3.47) for TE mode and taking $\mathcal{E}_y$ from Eq. (3.124), we may write

$$\mathcal{H}_z = \frac{j}{\omega\mu_0}\frac{\partial \mathcal{E}_y}{\partial x} = \frac{j}{\omega\mu_0}\frac{dE_y}{dx}e^{j(\omega t-\beta z)} \tag{3.127}$$

Using Eqs (3.124), (3.126) and (3.127), we can easily find the following

$$\langle S_y\rangle = 0 \tag{3.128}$$

$$\langle S_x\rangle = \frac{1}{2}\text{Re}(\mathcal{E}_y\mathcal{H}_z^*) = 0 \tag{3.129}$$

and

$$\langle S_z\rangle = -\frac{1}{2}\text{Re}(\mathcal{E}_y\mathcal{H}_x^*) \tag{3.130}$$

Substituting for $\mathcal{E}_y$ and $\mathcal{H}_x$ Eqs (3.124) and (3.126) into Eq. (3.130) and remembering that E_y is real, we get

$$\langle S_z\rangle = \frac{\beta}{2\omega\mu_0}|E_y|^2 \tag{3.131}$$

The power associated with the mode per unit length in the y-direction can be calculated by integrating $\langle S_z\rangle$ in the x direction as

$$P = \frac{1}{2}\frac{\beta}{\omega\mu_0}\int_{-\infty}^{\infty}|E_y|^2 dx \tag{3.132}$$

Let us consider the symmetric modes to begin with. Using Eqs (3.73) and (3.74), we may expand the above integration for the symmetrical step-index planar waveguides having infinite thickness for the layers surrounding the central waveguide region as follows

$$P = \frac{1}{2}\frac{\beta}{\omega\mu_0}2\left[\int_0^a A^2\cos^2 ux\, dx + B^2\int_a^{\infty} e^{-2wx}dx\right] \tag{3.133}$$

That is,

$$P = \frac{\beta A^2}{\omega\mu_0}\left[\int_0^a \cos^2 ux\, dx + \frac{B^2}{A^2}\int_a^\infty e^{-2wx} dx\right] \tag{3.134}$$

Substituting $d = 2a$ and carrying out the above integrations, we get

$$P = \frac{\beta A^2}{2\omega\mu_0}\left[\frac{d}{2} + \frac{1}{2u}\sin(ud) + \frac{B^2}{A^2}\frac{1}{w}e^{-wd}\right] \tag{3.135}$$

Using Eq. (3.75), the above equation can be written as

$$P = \frac{\beta A^2}{4\omega\mu_0}\left[d + \frac{1}{u}\sin(ud) + \frac{2}{w}\cos^2(ud/2)\right] \tag{3.136}$$

That is,

$$P = \frac{\beta A^2}{4\omega\mu_0}\left[d + \frac{2\sin\left(\frac{ud}{2}\right)\cos\left(\frac{ud}{2}\right)}{u} + \frac{2}{w}\left\{1 - \sin^2\left(\frac{ud}{2}\right)\right\}\right] \tag{3.137}$$

Rearranging the above equation and after simplification, we get

$$P = \frac{\beta A^2}{4\omega\mu_0}\left[d + \frac{2}{w} + \frac{2\sin\left(\frac{ud}{2}\right)\cos\left(\frac{ud}{2}\right)}{uw}\left\{w - u\tan\left(\frac{ud}{2}\right)\right\}\right] \tag{3.138}$$

Using Eq. (3.77), we find that the term within the parenthesis { } in the above equation is zero. Therefore, the power associated with the symmetric mode per unit length in the y-direction can be obtained as

$$P = \frac{\beta A^2}{4\omega\mu_0}\left[d + \frac{2}{w}\right] = \frac{\beta A^2}{2\omega\mu_0}\left[a + \frac{1}{w}\right] \tag{3.139}$$

In a similar manner, we can calculate the power associated with the anti-symmetric TE modes. The power associated with the TM modes can also be calculated following a similar procedure. The detailed calculations of both are left to the readers as problems.

3.4 PHYSICAL INTERPRETAION OF MODES IN A PLANAR WAVEGUIDE

Before we carry out the mode analysis in a cylindrical optical fiber structure it is imperative to understand the formation of modes in a planar waveguide structure. In the previous section, we considered an idealistic situation involving a symmetrical step-index planar waveguide in which the central region having a thickness $d(= 2a)$ and refractive index n_1 is sandwiched between two surrounding layers on the top and the bottom having a refractive index of $n_2(< n_1)$. The thickness of these layers is assumed to be infinite. The planar waveguide structure in the following discussion consists of a dielectric slab of refractive index n_1 sandwiched between two regions of refractive index $n_2(< n_1)$ as shown in Fig. 3.3. This structure closely resembles the longitudinal section of an optical fiber along the axis of symmetry except for the circular symmetry of later. Detailed analysis of optical waveguide theory can be found in the literature (Snyder, 1983; Adams, 1988; Marcuse, 1991; Okamoto, 1992; Ghatak *et al* 1998; Hunsperger, 2009).

The ray analysis for guided waves and modes in optical waveguides has been studied extensively in the past (Maurer *et al* 1967; 1970; Felsen, 1974; Unger, 1977). Conceptually the relationship between the ray analysis with the wave theory can be appreciated by considering a plane monochromatic wave propagating along the planar waveguide in the direction of the ray path traced in Fig. 3.8. Let us consider a ray that makes an angle $\theta(< \theta_e)$ with the axis of the fiber and continues to propagate along the fiber by total internal reflections at the core-cladding interface as shown in the figure. The plane wave associated with the ray (considered as the wave propagation vector) can be resolved into two orthogonal component plane waves propagating along the z-(horizontal) and x-(vertical) directions as shown in the figure. The components of the propagation constant (β) in the two directions can be expressed as

$$\beta_z = n_1 k \cos\theta \tag{3.140}$$

$$\beta_x = n_1 k \sin\theta \tag{3.141}$$

where $k = 2\pi/\lambda$ is the free space propagation constant which increases in an optically denser medium with a refractive index $n_1(> 1)$ as the free space wavelength is reduced to λ/n_1.

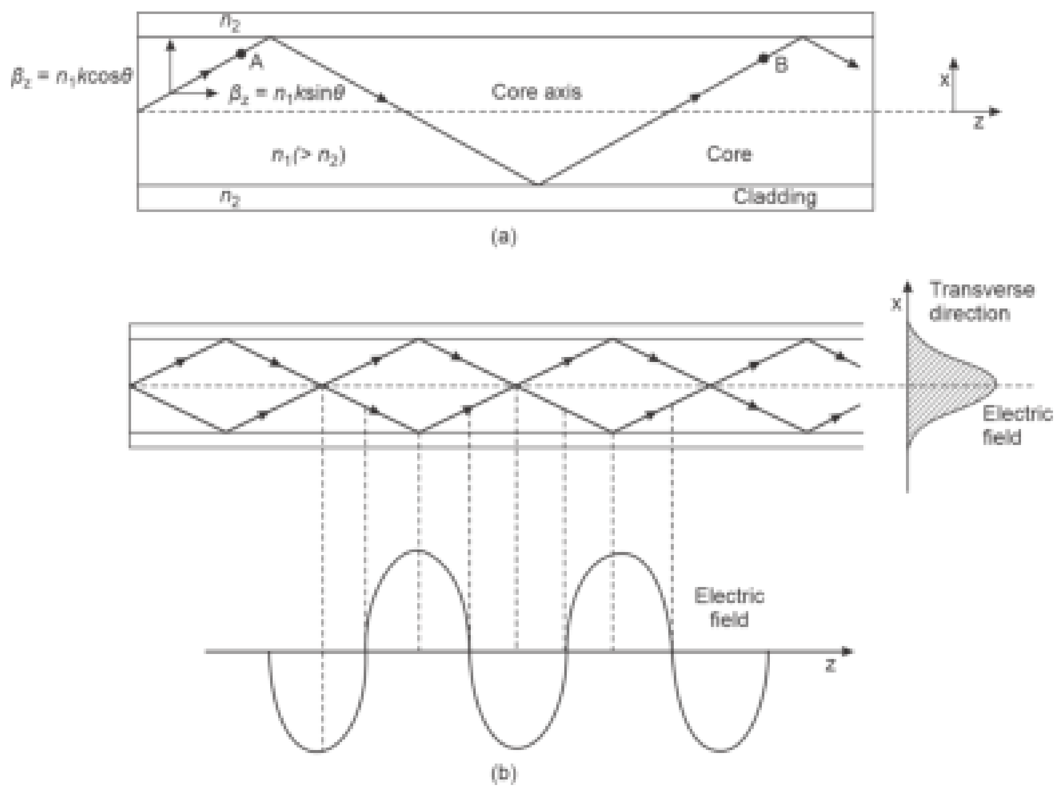


Fig. 3.8 Illustrating the formation of a mode in a planar waveguide (a) Plane wave propagating through the guide shown in the form of equivalent ray which corresponds to the wave vector (b) Formation of the lowest order mode through superposition of plane waves in the transverse direction

The x-component of the plane wave is total internally reflected at the interface of the central core region and the outer cladding region of lower refractive index. If after two successive reflections from the upper and lower interfaces (between points A and B) the total phase change is equal to 2πm, (m being an integer) then constructive interference of the waves occurs and a standing wave pattern is created in the transverse x-direction. This is illustrated in Fig. 3.8. In this illustration, it is assumed that the constructive interference of the waves forms the lowest-order standing wave (m = 0) in which the electric field is maximum at the center. The electric field is effectively confined in the central region but decays exponentially towards zero in the cladding region beyond the interfaces on both sides. The variation of the electric field in the transverse x-direction for the lowest-order mode is shown in Fig. 3.8(a). While the wave advances in the z-direction the electric field distribution does not change in the transverse x-direction. This stable field distribution in the x-direction with periodic z-dependence is called a *mode.* The variation of the electric field in the z-direction is shown in Fig. 3.8(b). It is interesting to note that a particular mode originates from constructive interference of the plane waves corresponding to a ray congruence (for which a representative ray is shown in the figure) making a specific angle with the core-cladding interface or the axis of the fiber. In other words, each representative ray propagating through the fiber corresponds to a mode. It may be emphasised that any ray making an angle less than the critical angle with the core-cladding interface does not necessarily propagate through the fiber. The propagation is sustained only for those rays

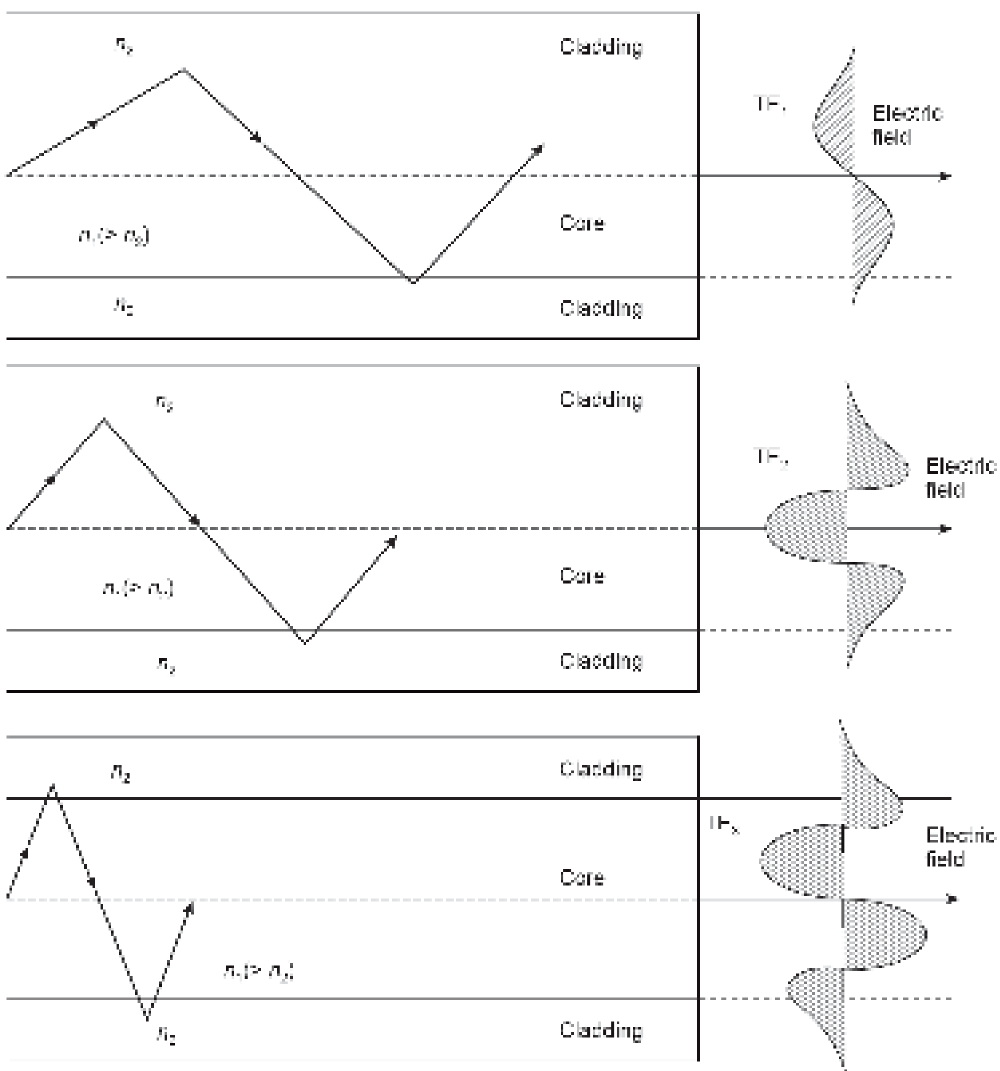


Fig. 3.9 Illustrating transverse electric modes TE_m (m = 1,2,3) along with ray paths showing penetration in the cladding region. The penetration of the field in the cladding region increases with the order of the mode.

which make angles less than the critical angle and the plane waves associated with the rays interfere constructively during propagation. The lowest-order mode corresponds to the least oblique ray while the highest-order mode corresponds to the most oblique ray to the axis of the fiber. The light propagating through a waveguide thus forms a discrete number of modes corresponding to discrete values of $\theta(< \theta_c)$ which the equivalent rays make with the core-cladding interface. A few higher-order modes are shown in Fig. 3.9. The electric field distributions in the transverse direction for different modes corresponding to $m = 1,2,3$ are shown in the figure. It should be noted that m denotes the number of field zeros in the transverse direction and signifies the order of the mode. In mode analysis, light is considered as electromagnetic waves with $\boldsymbol{E}$ and $\boldsymbol{H}$ fields varying periodically in orthogonal directions. In Fig. 3.9, the electric field is considered to be perpendicular to the direction of propagation that is, $E_z = 0$ while the magnetic field is non-zero in the z-direction, i.e. ($H_z \neq 0$). These modes are designated as TE_m modes. Similarly when the electric field is in the direction of propagation that is, in the z-direction and $H_z = 0$, the modes are called transverse magnetic mode TM_m modes. When $E_z = 0$ and $H_z = 0$, then the total field lies in the transverse plane and the mode is called transverse electromagnetic mode (TEM). However, TEM modes are rarely found in optical waveguides.

The field pattern of a particular mode is invariant in the transverse direction while it has a periodic z-dependence of the form $\exp(-j\beta_z z)$, where β_z is the z-component of the propagation constant. The direction of propagation of light is considered conventionally to be along the z-axis, it is customary to represent β_z as β. Considering the time dependence of the monochromatic electromagnetic field in the form $\exp(j\omega t)$, ω being the angular frequency, the propagating mode can be expressed as $\sim\exp[j(\omega t - \beta_z)]$.

3.5 MODES IN CYLINDRICAL OPTICAL FIBERS

In the previous section, we discussed qualitatively the formation of different modes in a planar waveguide. The relevance of the above discussion is apparent from the fact that a longitudinal section of an optical fiber along the axis of symmetry closely resembles a planar waveguide except for the circular symmetry of the former. Moreover, planar waveguides are also used in integrated optics as well as in optoelectronic integration for guiding light from one device to the other. However, in guided optical communication the channel is invariably the cylindrical waveguide structures based on dielectric materials popularly known as *optical fibers*. Analysis of the formation of different kinds of modes is of utmost importance to study various mechanisms of power flow, attenuation, and dispersion in optical fibers. Rigorous analysis of propagation of light through optical fibers has been extensively studied in the past and is available in the literature (Snyder, 1969; Snitzer, 1971; Gloge, 1971; 1972; Olshansky, 1979). Just like planar waveguides transverse electric (TE), transverse magnetic (TM) modes are created in optical fibers. In addition, hybrid modes (EH or HE) modes having both E_z, $H_z \neq 0$ are also created in an optical fiber. Hybrid modes are the characteristics of an optical fiber that are not generally found when electromagnetic wave propagates through the hollow metallic waveguides. We, discuss the mode analysis of a step-index optical fiber. An optical fiber consists of a solid cylindrical core surrounded by a solid coaxial cylindrical cladding and is axially symmetric. The refractive index of the core (n_1) is slightly higher than that of the cladding $n_2(< n_1)$.

3.5.1 Bound or Guided Modes

When light propagates through an optical fiber along the axis (considered as the z-direction), the plane-polarised electromagnetic waves get total internally reflected repeatedly from the core-cladding boundary. The superposition of these plane-polarised waves creates standing wave patterns in the transverse direction which remain stationary while the wave advances in the z-direction. These modes are solutions of Maxwell's equations under given boundary conditions. The actual number of modes created in a waveguide depends on the parameters of the waveguide as well as the wavelength of the monochromatic electromagnetic wave propagating through the fiber. Like in a planar waveguide, the modes are largely confined to the core and partly extending in the cladding. These modes vary harmonically in the core region and decay exponentially in the cladding region. These modes are referred to as core modes or bound modes. Lower-order modes are tightly concentrated near the axis of the core while the higher-order modes are less tightly bound to the axis of the core and tend to spread towards the boundary of the inner core and penetrate deeper in the cladding region (Snitzer, 1961; Snyder, 1969; Olshansky, 1979; Marcuse, 1979; Yeh, 1979; Gloge, 1979; Senior, 2008; Keiser, 2000; Ramsey *et al* 1980).

3.5.2 Cladding Modes

When light is launched in an optical fiber it is almost inevitable that a fraction of the light also enters the cladding. This happens because some light enters beyond the fiber acceptance angle which is finally refracted out in the cladding region. In addition, there exist several micro-bends in the fiber and other imperfections in the core-cladding interface which may cause the light to leave the core and enter the cladding. The cladding being a dielectric medium supports the formation of modes by the light entering into the cladding region. These modes which are not bound in the core region but are still solutions to the boundary-value problem are called *cladding or radiation modes*. As the core and cladding modes propagate through the fiber the coupling of modes occurs between higher-order core modes and the cladding modes. This is because higher-order core modes are less bound to the core and extend more in the cladding region. The mode coupling generally causes the transfer of power between the core and the cladding modes. The net result is a loss of power from the core mode to the cladding modes. In the process, higher-order core modes get eliminated. This leads to a reduction of overall intermodal dispersion which is a desirable effect. However, in normal fibers, the cladding is surrounded by a lossy coating of a higher refractive index. This minimises the reflection between the cladding and coating and minimises guidance at the cladding-coating interface. As a result, these modes cannot propagate over a long distance although a few may manage to propagate over a considerable distance. The cladding modes are also responsible for significant dispersion and therefore it is desirable to get rid of the undesired cladding modes.

3.5.3 Leaky Modes

When light is launched from a source into the core of a multimode fiber there may be a few modes that do not strictly satisfy the conditions for being guided inside the core. These modes continuously leak power from the core by a quantum mechanical tunneling process. The modes are called *leaky modes* and they occur when the upper and lower bounds on propagation constant β are not satisfied. For a mode to remain guided it is necessary that the propagation constant β satisfies the condition

$$(n_2 k =)k_2 < \beta < k_1(= n_1 k) \tag{3.142}$$

where n_1 and n_2 are the refractive indices of the core and the cladding respectively and $k(=2\pi/\lambda)$ is the free space propagation constant. The transition from a guided mode to a leaky mode occurs when $\beta = k_2(=n_2 k)$. This is known as the cut-off condition of a guided mode. When β falls below n_2k for a particular guided mode, the mode gets eliminated by leaking its power to the cladding. These leaky modes nevertheless travel there for some considerable distance and may carry a significant amount of power in short-length fibers.

3.6 FORMULATION OF WAVEGUIDE EQUATIONS IN CYLINDRICAL COORDINATES

The propagation of light along the axis of the fiber considered as z-direction is shown in Fig. 3.10. As the fiber is a cylindrical structure it is convenient to consider a cylindrical coordinate system $\{r, \phi, z\}$, where the z-axis is considered along the axis of the fiber which is also the direction of propagation of light. In the cylindrical coordinates, the electric and magnetic field vectors of an electromagnetic wave propagating along the z-direction can be expressed as (Keiser, 2000; Ghatak *et al* 1991)

$$\mathcal{E}(r, \phi, z, t) = \boldsymbol{E}(r, \phi) \exp [j(\omega t - \beta z)] \tag{3.143}$$

$$\mathcal{H}(r, \phi, z, t) = \boldsymbol{H}(r, \phi) \exp [j(\omega t - \beta z)] \tag{3.144}$$

where,

$$\boldsymbol{\mathcal{E}} = \hat{\boldsymbol{r}} E_r + \hat{\boldsymbol{\phi}} E_\phi + \hat{\boldsymbol{z}} \ E_z \tag{3.145}$$

$$\boldsymbol{\mathcal{H}} = \hat{\boldsymbol{r}} H_r + \hat{\boldsymbol{\phi}} H_\phi + \hat{\boldsymbol{z}} \ E_z \tag{3.146}$$

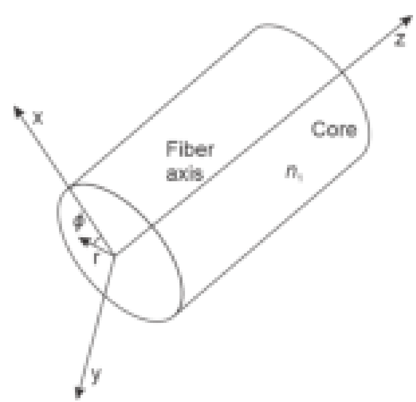


Fig. 3.10 Optical fiber showing the chosen cylindrical coordinates for mode analysis

Here, $\hat{\boldsymbol{r}}, \hat{\boldsymbol{\phi}}, \hat{\boldsymbol{z}}$ are respectively unit vectors in the radial, azimuthal, and longitudinal directions.

Writing Maxwell's curl Eq. (3.15) in cylindrical coordinates, we get

$$\begin{vmatrix} \frac{1}{r}\hat{\boldsymbol{r}} & \hat{\phi} & \frac{1}{r}\hat{\boldsymbol{z}} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ E_r & rE_\phi & E_z \end{vmatrix} = -j\omega\mu[\hat{\boldsymbol{r}} H_r + \hat{\phi} H_\phi + \hat{\boldsymbol{z}} H_z]$$

That is,

$$\frac{1}{r}\hat{\boldsymbol{r}}\left[\frac{\partial E_z}{\partial \phi} + j\beta r E_\phi\right] + \hat{\phi}\left[-j\omega\beta E_r - \frac{\partial E_z}{\partial r}\right] + \frac{1}{r}\hat{\boldsymbol{z}}\left[\frac{\partial (rE_\phi)}{\partial r} - \frac{\partial E_r}{\partial \phi}\right]$$

$$= j\omega\mu[\hat{\boldsymbol{r}} H_r + \hat{\phi} H_\phi + \hat{\boldsymbol{z}} H_z] \tag{3.147}$$

Equating the components on both sides of the Eq. (3.147), we get

$$\frac{1}{r}\left[\frac{\partial E_z}{\partial \phi} + j\beta r E_\phi\right] = -j\omega\mu H_r \tag{3.148}$$

$$j\omega\beta E_r + \frac{\partial E_z}{\partial r} = j\omega\mu H_\phi \tag{3.149}$$

$$\frac{1}{r}\left[\frac{\partial (rE_\phi)}{\partial r} - \frac{\partial E_r}{\partial \phi}\right] = -j\omega\mu H_z \tag{3.150}$$

Similarly writing the other curl equation, Eq. (3.16) in cylindrical coordinates and equating the components of the radial, azimuthal, and longitudinal directions, we get

$$\frac{1}{r}\left[\frac{\partial H_z}{\partial \phi} + j\beta r H_\phi\right] = j\omega\in E_r \tag{3.151}$$

$$j\omega\beta H_r + \frac{\partial H_z}{\partial r} = -j\omega\in E_\phi \tag{3.152}$$

$$\frac{1}{r}\left[\frac{\partial (rH_\phi)}{\partial r} - \frac{\partial H_r}{\partial \phi}\right] = j\omega\in E_z \tag{3.153}$$

Expressing the tangential components of electric and magnetic fields (E_r, E_ϕ, H_r, H_ϕ) in terms of longitudinal components of electric and magnetic field (E_z, H_z), we get

$$j\beta E_\phi - j\omega\mu H_r = \frac{1}{r}\frac{\partial E_z}{\partial \phi} \tag{3.154}$$

$$-j\omega\beta E_r + j\omega\mu H_\phi = \frac{\partial E_z}{\partial r} \tag{3.155}$$

$$j\omega\in E_r - j\beta H_\phi = \frac{1}{r}\frac{\partial H_z}{\partial \phi} \tag{3.156}$$

$$j\omega\in E_\phi + j\omega\beta H_r = -\frac{\partial H_z}{\partial r} \tag{3.157}$$

The radial and the azimuthal field components of the electric and magnetic fields can be obtained in terms of the longitudinal components (E_z and H_z) by making use of the above four equations. For example, the radial components of the electric field can be obtained by eliminating H_ϕ from Eqs (3.155) and (3.156). This can be achieved by multiplying Eq. (3.155) by β and Eq. (3.156) by $\omega\mu$ and adding the resultant equations. After algebraic manipulation, we obtain

$$E_r = -\frac{j}{\omega^2\mu\in - \beta^2}\left[\beta\frac{\partial E_z}{\partial r} + \frac{\omega\mu}{}\frac{\partial H_z}{\partial \phi}\right] \tag{3.158}$$

Similarly eliminating H_r from Eqs (3.154) and (3.157), we get the azimuthal component of the electric field as

$$E_\phi = -\frac{j}{\omega^2\mu\in - \beta^2}\left[\frac{\beta}{r}\frac{\partial E_z}{\partial \phi} - \omega\mu\frac{\partial H_z}{\partial \phi}\right] \tag{3.159}$$

Likewise eliminating E_ϕ from Eqs (3.154) and (3.157) and E_r from Eqs (3.155) and (3.156), we may express the radial and azimuthal components of the magnetic field (H_r and H_ϕ) as

$$H_r = -\frac{j}{\omega^2\mu\in - \beta^2}\left[\beta\frac{\partial H_z}{\partial r} - \frac{\omega\in}{r}\frac{\partial E_z}{\partial \phi}\right] \tag{3.160}$$

$$H_\phi = -\frac{j}{\omega^2\mu\epsilon-\beta^2}\left[\frac{\beta}{r}\frac{\partial H_z}{\partial\phi}+\omega\epsilon\frac{\partial E_z}{\partial\phi}\right] \tag{3.161}$$

Substituting (H_r and H_ϕ) from Eqs (3.160) and (3.161) into Eq. (3.153), we get

$$\frac{\partial}{\partial r}\left[\beta\frac{\partial H_z}{\partial\phi}+\omega\epsilon r\frac{\partial E_z}{\partial r}\right]-\frac{\partial}{\partial\phi}\left[\beta\frac{\partial H_z}{\partial r}-\frac{\omega\epsilon}{r}\frac{\partial E_z}{\partial\phi}\right]=j\omega\epsilon rE_z \tag{3.162}$$

That is,

$$\omega\epsilon r\frac{\partial^2 E_z}{\partial r^2}+\omega\epsilon\frac{\partial E_z}{\partial r}+\frac{\omega\epsilon}{r}\frac{\partial^2 E_z}{\partial\phi^2}+(\omega^2\mu\epsilon-\beta^2)\,\omega\,\epsilon rE_z=0 \tag{3.163}$$

The above equation can be rearranged as

$$\frac{\partial^2 E_z}{\partial r^2}+\frac{1}{r}\frac{\partial E_z}{\partial r}+\frac{1}{r^2}\frac{\partial^2 E_z}{\partial\phi^2}+(\omega^2\mu\,\epsilon-\beta^2)E_z=0 \tag{3.164}$$

Similarly substituting the values of E_r and E_ϕ from Eqs (3.158) and (3.159) into Eq. (3.150), we get

$$\frac{\partial^2 H_z}{\partial r^2}+\frac{1}{r}\frac{\partial H_z}{\partial r}+\frac{1}{r^2}\frac{\partial^2 H_z}{\partial\phi^2}+(\omega^2\mu\epsilon-\beta^2)H_z=0 \tag{3.165}$$

It can be seen that in Eqs (3.164) and (3.165) it is possible to obtain independent differential equations in terms of either E_z or H_z. This suggests that E_z and H_z are uncoupled and each equation can be solved independently. However, coupling between E_z and H_z may occur as per the requirement of the boundary conditions. In case the boundary conditions do not call for such coupling, Eq. (3.164) can be solved by assuming $H_z = 0$. The corresponding modes resulting from the solution of the Eq. (3.50) are called transverse magnetic (TM) modes. Likewise, when $E_z = 0$, Eq. (3.165) can be solved to obtain transverse electric modes. However, when both E_z and H_z are non-zero we get hybrid modes designated as EH or HE modes depending on whether E_z or H_z contribute respectively more towards the transverse field. Hybrid mode is a special feature of optical fibers and these modes are not found in hollow metallic waveguides. Analysis of the propagation of light through an optical fiber is thus much more complex.

We now proceed to find the guided modes in a step-index fiber by solving the equations under appropriate boundary conditions. For this purpose, we adopt the standard method of separation of variables. We note that both E_z and H_z are functions of four parameters, e.g. (r, ϕ, z, t). Assuming the solution to be of the form

$$E_z(r, \phi, z, t) \text{ or } H_z(r, \phi, z, t) = C\,R(r)\,\Phi(\phi)Z(z)T(t) \tag{3.166}$$

where C is an arbitrary constant.

Further, we note that the dependence of E_z and H_z on the parameters z and t are harmonic. Therefore, we may write

$$Z(z)T(t) = \exp\,[j(\omega t - \beta z] \tag{3.167}$$

It may be pointed out that we are interested in the solutions of Eqs (3.164) and (3.165) should be obtained in a way that the field values must return to the same value when the coordinate ϕ is increased by 2π. This means that the function $\Phi(\phi)$ must be periodic with a period 2π. Therefore, we may assume

$$\Phi(\phi) = \exp\,[(jl\phi) \tag{3.168}$$

where l is an integer (positive or negative) so that

$$\Phi(\phi + 2\pi) = \exp\,[jl(\phi + 2\pi)] = \exp(jl\phi) \tag{3.169}$$

Finally, the solutions take the form

$$E_z(r, \phi, z, t) \text{ or } H_z(r, \phi, z, t) = C\, R(r) \exp(jl\phi) \exp[j(\omega t - \beta z)] \tag{3.170}$$

Using the above forms of E_z and H_z Eqs (3.164) and (3.165) take the following form (Keiser, 2000; Saleh *et al* 1991; Ghatak *et al* 1991)

$$\frac{d^2R(r)}{dr^2} + \frac{1}{r}\frac{dR(r)}{dr} + \left(\omega^2 \mu \in -\beta^2 - \frac{l^2}{r^2}\right)R(r) = 0 \tag{3.171}$$

Equation (3.171) is a well-known form of differential equations and the solution is in the form of Bessel's functions. It may be pointed out that both E_z and H_z take the same form as a result final forms of solutions will be similar except for different constant values which are determined based on given boundary conditions.

It may be recalled that the modes (solutions of Eq. (3.171)) will remain bound or guided so long as the propagation constant satisfies the condition $(n_2k =)k_2 < \beta < k_1$ $(= n_1k)$,, where k is the free space propagation constant and n_1 and n_2 are the values of the refractive index in the core and the cladding region. It may be pointed out that the refractive index (n) of a dielectric medium is related to the relative permittivity($\in_r$) as $n = \sqrt{\in_r}$. In the analysis of propagation of light through an optical fiber, we are concerned with the optical property of the dielectric medium we need to consider different values of permittivity for the core and the cladding region. Assuming that the permeability of the core and the cladding region is the same we introduce the following new parameters for the core and the cladding regions for a step-index fiber

$$u^2 = \omega^2\mu\in_1 - \beta^2 \qquad (r < a\text{: core region}) \tag{3.172}$$

$$w^2 = \beta^2 - \omega^2\mu\in_2 \qquad (r > a\text{: cladding region}) \tag{3.173}$$

in such a way that both u^2 and w^2 are positive so that u and w are both real. After the above substitution Eq. (3.171) takes the following forms for the core and the cladding regions (Saleh *et al* 1991)

$$\frac{d^2R(r)}{dr^2} + \frac{1}{r}\frac{dR(r)}{dr} + \left(u^2 - \frac{l^2}{r^2}\right)R(r) = 0 \qquad (r < a\text{: core}) \tag{3.174}$$

$$\frac{d^2R(r)}{dr^2} + \frac{1}{r}\frac{dR(r)}{dr} - \left(w^2 + \frac{l^2}{r^2}\right)R(r) = 0 \qquad (r > a\text{: cladding}) \tag{3.175}$$

The solution of Eq. (3.174) is in the form of $J_l(ur)$, Bessel's function of the first kind of order l and argument (ur) while that of Eq. (3.175) is in the form of $K_l(wr)$, modified Bessel's function of the second kind of order l and argument (wr). The solutions of Eqs (3.174) and (3.175) are of the form (Fig. 3.12)

$$R(r) \sim J_l(ur) \qquad (\text{for } r < a\text{: core}) \tag{3.176a}$$

$$R(r) \sim K_l(wr) \qquad (\text{for } r > a\text{: cladding}) \tag{3.176b}$$

Accordingly the longitudinal components of electric and magnetic fields (E_z and H_z) for the core region can be expressed as

$$E_z(r < a) = AJ_l(ur) \exp(jl\phi) \exp[j(\omega t - \beta z)] \tag{3.177}$$

$$H_z(r < a) = BJ_l(ur) \exp(jl\phi) \exp[j(\omega t - \beta z)] \tag{3.178}$$

where A and B are arbitrary constants.

The longitudinal components of electric and magnetic fields in the cladding region can be expressed as

$$E_z(r > a) = CK_l(wr) \exp(jl\phi) \exp[j(\omega t - \beta z)] \tag{3.179}$$

$$H_z(r > a) = DK_l(wr) \exp(jl\phi) \exp[j(\omega t - \beta z)] \tag{3.180}$$

where C and D are arbitrary constants.

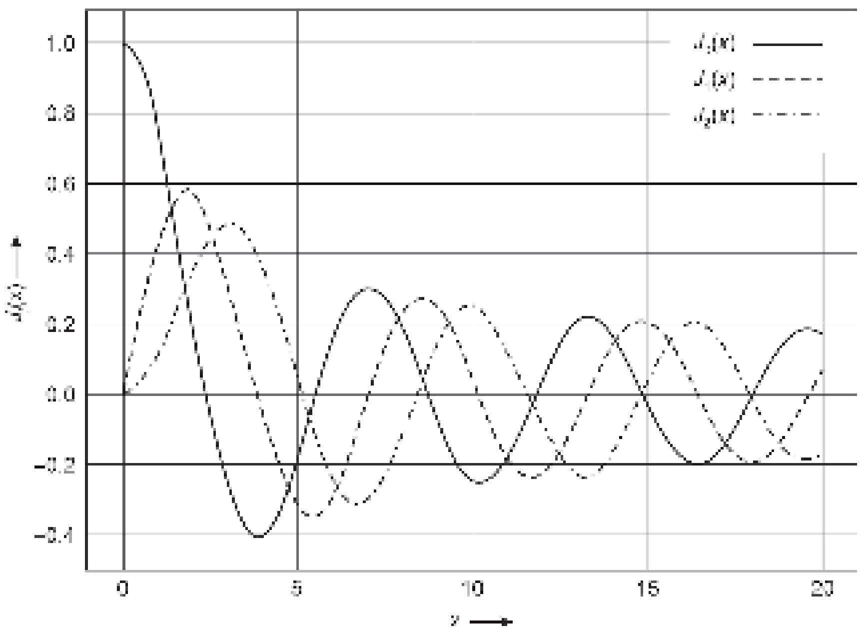


Fig. 3.11 The variation of the Bessel's function of the first kind for different orders with argument *x*

Before we proceed further let us examine the nature of Bessel's function of the first kind $J_l(x)$ (Fig. 3.11) and the modified Bessel's function of the second kind, $K_l(x)$. The function $J_l(x)$ oscillates like a sinusoidal signal with decaying amplitude given by

$$J_l(x) = \frac{1}{2\pi}\exp[j(x\sin\theta - n\theta] \tag{3.181}$$

The Bessel's function can be expanded in power series like any other trigonometric function. For a large value of the argument ($x \gg 1$) the Bessel's function of the first kind can be approximated in the limit as (Saleh *et al* 1991)

$$J_l(x) \approx \left(\frac{2}{\pi x}\right)^{\frac{1}{2}} \cos\left[x - \left(l + \frac{1}{2}\right)\frac{\pi}{2}\right] \qquad (x \gg 1) \tag{3.182}$$

The modified Bessel's function of the second kind in the same limit on the other hand decays exponentially with increasing x

$$\mathrm{K}_l(x) \approx \left(\frac{\pi}{2x}\right)^{\frac{1}{2}} \left(1 + \frac{4l^2 - 1}{8x}\right) \exp(-x) \;\; (x \gg 1) \tag{3.183}$$

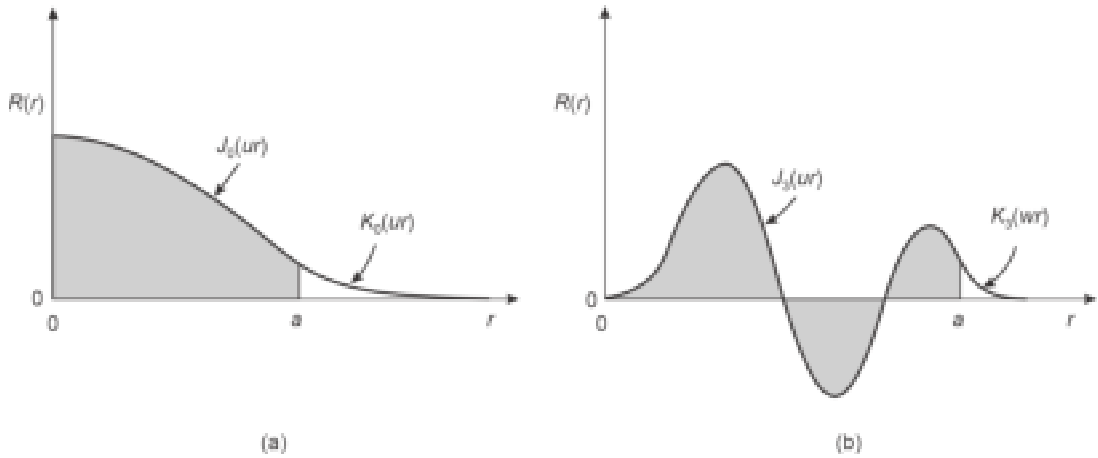
As $u^2 > 0$, u is real and so also the function $R(r) \sim J_l(ur)$ and the modes are bound to the core region (Fig. 3.12).

Further, in this case as $w > 0$ and therefore the modified Bessel function

$$K_l(wr) \;\rightarrow \exp(-wr) \;\rightarrow 0 \text{ as } r \rightarrow \infty \tag{3.184}$$

The cut-off condition is the critical point where the modes are no longer bound to the core region and is attained when $\beta = k_2$ $(= n_2 k)$. For a mode to vary harmonically in the core region and decay exponentially in the cladding region it is necessary that

$$(n_2 k =)k_2 \geq \beta \leq k_1 \; (= n_1 k) \tag{3.185}$$



3.12 Illustrating radial distribution of $R(r)$ for (a) for l = 0 (b) for l = 3. The unshaded portion shows the extension in the cladding region

It is interesting to note that the values of the parameters u and w determine the fashion by which the radial distribution varies. For example, a large value of u suggests that the field in the core region will have faster oscillation while a large value of w would lead to a rapid decay of the radial field in the cladding region resulting in less penetration there. It may be further observed that

$$\begin{aligned} u^2 + w^2 &= k_1^2 - \beta^2 + \beta^2 - k_2^2 \\ &= k_1^2 - k_2^2 = k^2(n_1^2 - n_2^2) = k^2(\mathrm{NA})^2 = \text{constant} \end{aligned} \tag{3.186}$$

This means that when u increases w must decrease and the radial field oscillates faster in the core region and decays slowly in the cladding region. Further, when $\beta > k_2$, w becomes imaginary and u^2 exceeds the value of $(\mathrm{NA})^2$ and the radial field ceases to be bound in the core region signifying the extinction of the corresponding mode.

3.6.1 The Normalised Frequency or V-number

It is often convenient to use the normalised values of u and w by normalising them with the core radius. We introduce the two parameters (Saleh *et al* 1991)

$$U = ua \tag{3.187a}$$

$$W = wa \tag{3.187b}$$

The normalised frequency or the V-number, often referred to as the V-parameter of the fiber is defined as

$$V^2 = U^2 + W^2 \tag{3.188}$$

That is, $V^2 = a^2(u^2 + w^2) = a^2k^2(\mathrm{NA})^2$ (3.189)

Therefore, the normalised frequency or the V-number can be expressed as (Saleh *et al* 1991; Keiser, 2000)

$$V = \frac{2\pi a}{\lambda}(\mathrm{NA}) = \frac{2\pi a}{\lambda}\sqrt{n_1^2 - n_2^2} \approx \frac{2\pi a}{\lambda} n_1 \sqrt{2\Delta} \tag{3.190}$$

V-number is an important parameter that determines the number of modes that are supported by a particular type of fiber. From the foregoing discussion, it is clear that for a mode to be bounded to the core the parameter U must be less than the value of V.

3.6.2 Boundary Conditions and Eigenvalue Equation

To find the longitudinal components of the electric and magnetic fields we need to determine the four constants A, B, C and D with the help of the boundary conditions. The boundary conditions require the tangential components of the electric field vector $\boldsymbol{E}$ that is, E_ϕ and E_z should be continuous inside and outside the dielectric at the core-cladding interface. Likewise, the magnetic field vector $\boldsymbol{H}$ that is, H_ϕ and H_z should also be continuous inside and outside the dielectric at the core-cladding interface. Mathematically, we may write

$$E_z|_{r\to a-0} = E_z|_{r\to a+0} \tag{3.191a}$$
$$E_\phi|_{r\to a-0} = E_\phi|_{r\to a+0} \tag{3.191b}$$
$$H_z|_{r\to a-0} = H_z|_{r\to a+0} \tag{3.191c}$$
$$H_\phi|_{r\to a-0} = H_\phi|_{r\to a+0} \tag{3.191d}$$

Substituting the values of E_z, H_z, E_ϕ and H_ϕ from Eqs (3.177)–(3.180) and (3.157) and (3.159) into the above boundary conditions, Eqs (3.191a)–(3.191d), we get

$$AJ_l(ua) - CK_l(wa) = 0 \tag{3.192}$$

$$-\frac{j}{u^2}\left[A\frac{j\beta l}{a}J_l(ua) - B\omega\mu u J_l'(ua)\right] - \frac{j}{w^2}\left[C\frac{j\beta l}{a}K_l(wa) - B\omega\mu w K_l'(wa)\right] = 0 \tag{3.193}$$

$$BJ_l(ua) - DK_l(wa) = 0 \tag{3.194}$$

$$-\frac{j}{u^2}\left[B\frac{j\beta l}{a}J_l(ua) + A\omega\epsilon_1 u J_l'(ua)\right] - \frac{j}{w^2}\left[D\frac{j\beta l}{a}K_l(wa) + C\omega\epsilon_2 w K_l'(wa)\right] = 0 \tag{3.195}$$

The four equations above can be rearranged in the matrix form as

$$\begin{pmatrix} J_l(ua) & 0 & -K_l(wa) & 0 \\ \frac{\beta l}{au^2}J_l(ua) & \frac{j\omega\mu}{u}J_l'(ua) & \frac{\beta l}{aw^2}K_l(wa) & \frac{j\omega\mu}{w}K_l'(wa) \\ 0 & J_l(ua) & 0 & -K_l(wa) \\ -\frac{j\omega\epsilon_1}{u}J_l'(ua) & \frac{\beta l}{au^2}J_l(ua) & -\frac{j\omega\epsilon_2}{w}K_l'(wa) & \frac{\beta l}{aw^2}K_l(wa) \end{pmatrix}\begin{pmatrix} A \\ B \\ C \\ D \end{pmatrix} = 0 \tag{3.196}$$

To obtain non-trivial solutions for the coefficients A, B, C, and D, the determinants of the coefficients of Eq. (3.196) must be zero. That is,

$$\begin{vmatrix} J_l(ua) & 0 & -K_l(wa) & 0 \\ \frac{\beta l}{au^2}J_l(ua) & \frac{j\omega\mu}{u}J_l'(ua) & \frac{\beta l}{aw^2}K_l(wa) & \frac{j\omega\mu}{w}K_l'(wa) \\ 0 & J_l(ua) & 0 & -K_l(wa) \\ -\frac{j\omega\epsilon_1}{u}J_l'(ua) & \frac{\beta l}{au^2}J_l(ua) & -\frac{j\omega\epsilon_2}{w}K_l'(wa) & \frac{\beta l}{aw^2}K_l(wa) \end{vmatrix} = 0 \tag{3.197}$$

Since the right side is zero with little mathematical manipulation, the determinant reduces to

$$J_l^2(ua)K_l^2(wa)\begin{vmatrix} 1 & 0 & -1 & 0 \\ \dfrac{\beta l}{au^2} & \dfrac{j\omega\mu J_l'(ua)}{uJ_l(ua)} & \dfrac{\beta l}{aw^2} & \dfrac{j\omega\mu}{w}\dfrac{K_l'(wa)}{\mathrm{K}_l(wa)} \\ 0 & 1 & 0 & -1 \\ -\dfrac{j\omega\epsilon_1}{u}\dfrac{J_l'(ua)}{J_l(ua)} & \dfrac{\beta l}{au^2} & \dfrac{j\omega\epsilon_2}{w}\dfrac{K_l'(wa)}{K_l(wa)} & -\dfrac{\beta l}{aw^2} \end{vmatrix} = 0 \tag{3.198}$$

Assuming that $J_l(ua), K_l(wa) \neq 0$ and adding column 1 and column 3 and interchanging rows 1 and 3 we finally get the characteristic equation for the existence of the solution of Eq. (3.196) can be obtained as

$$\begin{vmatrix} \dfrac{\beta l}{a}\left(\dfrac{1}{u^2}+\dfrac{1}{w^2}\right) & \dfrac{j\omega\mu}{u}\dfrac{J_i'(ua)}{J_l(ua)} + \dfrac{j\omega\mu}{w}\dfrac{K_l'(wa)}{K_l(wa)} \\ -\dfrac{j\omega\epsilon_1}{u}\dfrac{J_l'(ua)}{J_l(ua)} + \dfrac{j\omega\epsilon_2}{w}\dfrac{K_l'(wa)}{K_l(wa)} \quad \dfrac{\beta l}{au^2} & \dfrac{\beta l}{a}\left(\dfrac{1}{u^2}+\dfrac{1}{w^2}\right) \end{vmatrix} = 0$$

After simplification, the eigenvalue equation for β takes the form

$$\left(\frac{J_l'(ua)}{uJ_l(ua)} + \frac{K_l'(wa)}{wK_l(wa)}\right)\left(k_1^2\frac{J_l'(ua)}{uJ_l(ua)} + k_2^2\frac{K_l'(wa)}{wK_l(wa)}\right) = \left(\frac{\beta l}{a}\right)^2\left(\frac{1}{u^2}+\frac{1}{w^2}\right)^2 \tag{3.199}$$

Equation (3.199) is a transcendental equation that must be satisfied by the propagation constant β for the existence of modes. It is interesting to note β can only have discrete values within the range $k_2 \leq \beta \leq k_1$. In general, Eq. (3.199) is a complicated implicit equation that can only be solved with the help of numerical techniques and represents the characteristic equation for hybrid modes ($E_z \neq 0, H_z \neq 0$). It can be further seen that Bessel's function of the first kind $J_l(ua)$ has oscillatory nature (Fig. 3.11) like sinusoidal functions while the modified Bessel's function of the second kind has an exponentially decaying nature. As a result, a given value of l, Eq. (3.199) will have an integer number (say, m) of roots. The corresponding roots of Eq. (3.199), will be values of β designated as β_{lm}. A mode is therefore described by the indices l and m characterising its azimuthal and radial distributions, respectively. The function $R(r)$ depends on both l and m. It may be pointed out that $l = 0$ corresponds to meridional rays. Moreover, there exist two independent distributions of the $\boldsymbol{E}$ and $\boldsymbol{H}$ vectors for each mode, corresponding to two states of polarisation. The hybrid modes for $l \neq 0$ are accordingly designated either as EH_{lm} or as HE_{lm} modes. However, when $l = 0$, the modes are either transverse electric (TE_{0m}) modes or transverse magnetic (TM_{0m}) modes.

3.6.3 Transverse Electric (TE) and Transverse Magnetic (TM) modes

In an optical fiber the modes in general, are hybrid ($E_z \neq 0, H_z \neq 0$). In the special case when $l = 0$, the modes become either transverse electric (TE_{0m}) or transverse magnetic (TM_{0m}) depending on the condition satisfied by the characteristic equation. It can be easily seen that when $l = 0$, the right side of the characteristic equation vanishes and we obtain

$$\left(\frac{J_0'(ua)}{uJ_0(ua)} + \frac{K_0'(wa)}{wK_0(wa)}\right)\left(k_1^2\frac{J_0'(ua)}{uJ_0(ua)} + k_2^2\frac{K_0'(wa)}{wK_0(wa)}\right) = 0 \tag{3.200}$$

Equation (3.86) leads to two different eigenvalue equations given by either,

$$\left(\frac{J_0'(ua)}{uJ_0(ua)} + \frac{K_0'(wa)}{uK_0(wa)}\right) = 0 \tag{3.201a}$$

or else,

$$\left(k_1^2\frac{J_0'(ua)}{uJ_0(ua)} + k_2^2\frac{K_0'(wa)}{wK_0(wa)}\right) = 0 \tag{3.201b}$$

Because when both the factors are zero, the case would be trivial only.

It can be easily verified that when Eq. (3.201a) is satisfied the corresponding modes are (TE_{0m}) modes. On the other hand, when Eq. (3.201b) is satisfied, the modes correspond to (TM_{0m}) modes.

Substituting $l = 0$, Eqs (3.192) and (3.195) can be written as

$$AJ_0\,(ua) - CK_0\,(wa) = 0 \tag{3.202}$$

$$\frac{1}{u^2}[\mathrm{A}\omega \in_1 uJ_0'(ua)] + \frac{1}{w^2}[\mathrm{C}\omega\in_2 wK_0'\,(wa)] = 0 \tag{3.203}$$

Eliminating C from the above equations by substituting C in terms of A in Eq. (3.203), we get

$$A\frac{\omega\in_1}{u}J_0'(ua) + AJ_0(ua)\frac{\omega\in_2}{w}\frac{K_0'(wa)}{K_0(wa)} = 0$$

That is,

$$AJ_0(ua)\left[\omega\in_1\frac{J_0'(ua)}{uJ_0(ua)} + \omega\in_2\frac{K_0'(wa)}{wK_0(wa)}\right] = 0 \tag{3.204}$$

Multiplying both sides of the Eq. (3.204) by the factor ($\omega\mu$), we get

$$AJ_0(ua)\left[k_1^2\frac{J_0'(ua)}{uJ_0(ua)} + k_2^2\frac{K_0'(wa)}{wK_0(wa)}\right] = 0 \tag{3.205}$$

From Eq. (3.205), we find

either

$$AJ_0(ua) = 0 \tag{3.206a}$$

or else

$$\left[k_1^2\frac{J_0'(ua)}{uJ_0(ua)} + k_2^2\frac{K_0'(wa)}{wK_0(wa)}\right] = 0 \tag{3.206b}$$

Further, from our previous discussion we find (see Eqs (3.201a) and (3.201b) for non-trivial solution

when

$$\left(\frac{J_0'(ua)}{uJ_0(ua)} + \frac{K_0'(wa)}{wK_0(wa)}\right) = 0$$

$$\left[k_1^2\frac{J_0'(ua)}{uJ_0(ua)} + k_2^2\frac{K_0'(wa)}{wK_0(wa)}\right] \neq 0$$

From Eqs (3.206a) and (3.206b) we also note that if

$$\left[k_1^2\frac{J_0'(ua)}{uJ_0(ua)} + k_2^2\frac{K_0'(wa)}{wK_0(wa)}\right] \neq 0$$

then

$$AJ_0(ua) = 0$$

Equivalently, we may also write when

$$\left(\frac{J_0'(ua)}{uJ_0(ua)} + \frac{K_0'(wa)}{wK_0(wa)}\right) = 0$$

then

$$AJ_0(ua) = 0$$

This means that when $AJ_0(ua) = 0$, the electric field (see Eq. (3.177))

$$E_z(r < a) = 0$$

As the z-component of the electric field is zero the corresponding mode is purely transverse electric (TE).

In a similar way by using Eqs (3.194) and (3.195) and substituting D in terms of B we may write

either,

$$\left(\frac{J_0'(ua)}{uJ_0(ua)} + \frac{K_0'(wa)}{wK_0(wa)}\right) = 0 \tag{3.207}$$

or else,

$$BJ_0(ua) = 0 \tag{3.208}$$

For non-trivial cases, Eq. (3.208) will be valid when

$$\left(\frac{J_0'(ua)}{uJ_0(ua)} + \frac{K_0'(wa)}{wK_0(wa)}\right) \neq 0 \tag{3.209}$$

However, when (3.209) is true then

$$\left[k_1^2 \frac{J_0'(ua)}{uJ_0(ua)} + k_2^2 \frac{K_0'(wa)}{wK_0(wa)}\right] = 0 \tag{3.210}$$

Therefore, it is clear that when Eq. (3.210) is true the z-component of the magnetic field

$$H_z(r < a) = 0 \tag{3.211}$$

since, $BJ_0(ua)$ in that case is zero.

Using the following identities for $J_l'(x)$ and $K_l'(x)$

$$J_l'(x) = \pm J_{l\mp1}(x) \mp l\frac{J_l(x)}{x} \tag{3.212a}$$

$$K_l'(x) = -K_{l\mp1}(x) \mp l\frac{K_l(x)}{x} \tag{3.212b}$$

we obtain

$$\frac{J_0'(ua)}{uJ_0(ua)} + \frac{K_0'(wa)}{wK_0(wa)} = \frac{J_1(ua)}{uJ_0(ua)} + \frac{K_1(wa)}{wK_0(wa)} \tag{3.213a}$$

$$k_2^1 \frac{J_0'(ua)}{uJ_0(ua)} + k_2^2 \frac{K_0'(wa)}{wK_0(wa)} = k_1^2 \frac{J_1(ua)}{uJ_0(ua)} + k_2^2 \frac{K_1(wa)}{wK_0(wa)} \tag{3.213b}$$

Therefore, we may summarize the following

$$\frac{J_1(ua)}{uJ_0(ua)} + \frac{K_1(wa)}{wK_0(wa)} = 0 \tag{3.214}$$

corresponds to transverse electric modes that is, TE_{0m} modes, and

$$k_1^2 \frac{J_1(ua)}{uJ_0(ua)} + k_2^2 \frac{K_1(wa)}{wK_0(wa)} = 0 \tag{3.215}$$

corresponds to transverse magnetic modes that is, TM_{0m} modes.

The transverse electric field distributions of the lowest-order transverse electric (TE_{01}) and lowest-order transverse magnetic (TM_{01}) are shown in Fig. 3.13.

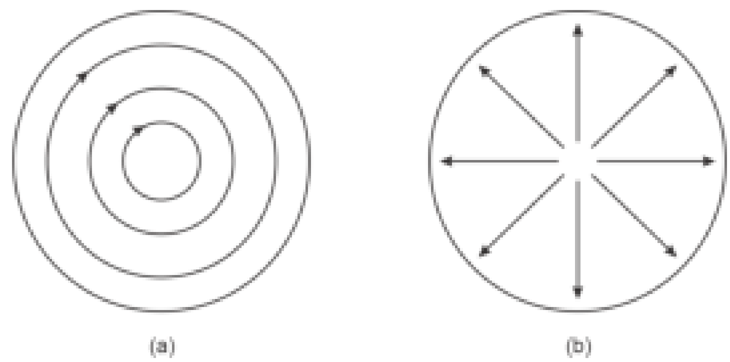


Fig. 3.13 Cross-sectional view of the transverse electric field (a) TE_{01} mode (b) TM_{01} mode

The solution of the eigenvalue Eq. (3.199) becomes very complex when $l \neq 0$. Numerical methods can be used to solve the equation. However, it is possible to obtain an approximate yet fairly accurate solution to the equation by assuming that the refractive index values of the core and the cladding are nearly equal $n_1 \approx n_2$. Such fibers are called weakly guiding fibers and the modes are often referred to as weakly guided modes (Marcuse, 1991; Snyder, 1969; Gloge, 1971).

3.6.4 Mode Cut-offs

In the generalised case, a mode ceases to exist when $\beta = k_2 = n_2 k$. In that case, the mode is no longer bound to the core nor does the field decay exponentially in the cladding and the cut-off is said to occur. The cut-offs of various order modes can be obtained by solving the eigenvalue equation under the condition $w^2 \to 0$. An exact solution of the Eq. (3.199) under the above condition is beyond the scope of the book. Nevertheless, it is important to note the cut-off conditions of different order modes listed in Table 3.1.

Table 3.1 Cut-off conditions of different modes

Azimuthal index, l	*Designated mode*	*Cut-off condition*
0	TE_{0m}, TM_{0m}	$J_0(ua) = 0$
1	HE_{1m}, EH_{1m}	$J_1(ua) = 0$
≥ 2	EH_{lm}	$J_l(ua) = 0$
	HE_{lm}	$\left(\frac{n_1^2}{n_2^2} + 1\right) J_l(ua) = \frac{ua}{l-1} J_{l-1}(ua)$

Alternatively, the cut-off conditions for various modes are expressed in terms of the V-number or normalised frequency of an optical fiber defined by Eq. (3.190). The dimensionless V-number is directly related to the number of modes supported by a particular fiber. It is often convenient to represent the cut-off conditions in terms of normalised propagation constant defined as

$$b = \frac{a^2 w^2}{V^2} = \frac{a^2(\beta^2 - k_2^2)}{a^2(k_1^2 - \beta^2 + \beta^2 - k_2^2)}$$

That is,

$$b = \frac{(\beta / k)^2 - n_2^2}{n_1^2 - n_2^2} \tag{3.216}$$

It can be easily seen that the normalised propagation constant, b varies between 0 and 1, when β varies from k_2 to k_1. The variation of b with the V-number or normalised frequency of the fiber is shown in Fig. 3.14 for a few lower-order modes. It can be easily seen from the figure that b can have a non-zero value only when $\beta > k_2$. In other words, the modes are cut off when $\beta/k = n_2$. Therefore, for a given value of V-number, there can exist only a finite number of modes. It can be further seen from the figure that the hybrid HE_{11} mode does not have any cut-off. The mode ceases to exist only when the core radius is zero. This means that for a practical fiber, there would be at least one mode that cannot be eliminated by reducing the V-number. This is the essence of the principle of a single-mode fiber. It can be verified that when $V \leq 2.405$ all modes except the HE_{11} mode gets eliminated and the fiber is left with only one mode and the fiber is called a single mode fiber. Mathematically, the design formula for obtaining a single-mode fiber can be expressed as

$$V = \frac{2\pi a}{\lambda}(n_1^2 - n_2^2)^{\frac{1}{2}} \leq 2.405 \tag{3.217}$$

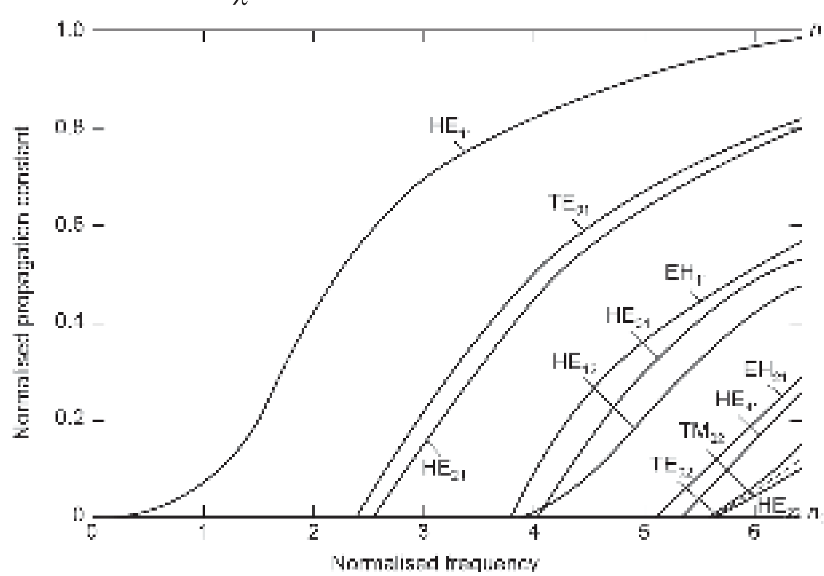


Fig. 3.14 Variation of normalised propagation constant with normalised frequency

For given values of a, n_1 and n_2 the wavelength of the light can be so adjusted as to satisfy Eq. (3.217). Under this condition, the optical fiber would behave as a single-mode fiber. Alternatively, for given values of n_1, n_2 and λ the radius of the core a can be so adjusted as to satisfy Eq. (3.217) allowing the fiber to support only one mode. It is interesting to note that at this value, $J_0 = 0$ and therefore all the lowest modes including the TE_{01} and TM_{01} and excepting HE_{11} are cut-off (see Table 3.1).

The cut-off value of the normalised frequency for a single-mode step-index fiber can be written as

$$V_c = \frac{2\pi a}{\lambda_c}(n_1^2 - n_2^2)^{\frac{1}{2}} = 2.405 \tag{3.218}$$

where λ_c is the cut-off wavelength. For a given fiber, the cut-off wavelength for single-mode operation can be expressed by using Eqs (1.103) and (1.104) as

$$\lambda_c = \frac{V\lambda}{2.405} \tag{3.219}$$

Problem 3.3 A step-index optical fiber with a core radius of 4.6 μm has a core refractive index of n_1 = 1.465 and index deviation of Δ = 0.2%. Estimate the cut-off wavelength for the fiber to exhibit single-mode operation.

Solution The product of normalised frequency or *V*-number of the fiber and the wavelength of operation, i.e. $V\lambda$ can be obtained from Eq. (3.103) as

$$V\lambda = 2\pi a(n_1^2 - n_2^2)^{\frac{1}{2}} \approx 2\pi a n_1 \sqrt{2\Delta}$$

$$= 2 \times 3.14 \times 4.6 \times 10^{-6} \times 1.465 \times \sqrt{2 \times 0.002} = 2.677 \times 10^{-6}$$

Using Eq. (3.219) the cut-off wavelength can be estimated as

$$\lambda_c = \frac{2.677 \times 10^{-6}}{2.405} = 1113 \text{ nm}$$

Problem 3.4 Find the core radius necessary for single-mode operation at 1300 nm of a step-index fiber with core and cladding refractive indices equal to 1.450 and 1.447 respectively.

Solution For single-mode operation

$$V = V_c = \frac{2\pi a}{\lambda}(n_1^2 - n_2^2)^{\frac{1}{2}} = 2.405$$

That is,

$$a = \frac{2.405 \times 1300 \times 10^{-9}}{6.28 \times \sqrt{(1.450)^2 - (1.447)^2}} \text{ m} = 5.24 \text{ μm}$$

3.6.5 Relationship between Number of Modes and *V*-number

It is apparent from the foregoing discussion that the number of modes supported by a particular type of fiber depends on the *V*-number. However, it is not convenient to compute the number of modes following the procedure described above. On the other hand, it is possible to relate the *V*-number with the number of modes (M) supported by a particular fiber when M is very large. The ray theory can be used to derive the relationship between the *V*-number and the number of modes for multimode step-index fiber.

The number of modes emanating from a waveguide at a wavelength λ can be expressed as

$$M \cong \frac{2A}{\lambda^2}\Omega \tag{3.220}$$

where A is the area of cross-section through which the modes are entering or leaving the waveguide and Ω is the solid angle through which the modes are emanated or accepted by the fiber.

From the ray analysis, we found that the ray congruence incident at the fiber end will be accepted by the fiber only when it lies within the fiber acceptance angle given by

$$\Omega = \pi\theta_{0\max}^2 \approx \pi(\mathrm{NA})^2 \tag{3.221}$$

Further, the numerical aperture of a step-index fiber can be expressed as

$$\mathrm{NA} = n_0 \sin\theta_{0\max} \approx \theta_{0\max} = \left(n_1^2 - n_2^2\right)^{\frac{1}{2}} \tag{3.222}$$

The total number of modes M entering a step-index fiber can be expressed as

$$M \approx \frac{2A}{\lambda^2}(n_1^2 - n_2^2)^{\frac{1}{2}} \tag{3.223}$$

Assuming the area through which the modes are entering the fiber is equal to the cross-sectional area of the fiber core, i.e. πa^2, we may express the total number of modes entering the step-index fiber as

$$M \approx \frac{2A}{\lambda^2}(n_1^2 - n_2^2)^{\frac{1}{2}} = \frac{2\pi^2 a^2}{\lambda^2}(n_1^2 - n_2^2) = \frac{V^2}{2} \tag{3.224}$$

Problem 3.5 Calculate the number of modes in a step-index multimode fiber at 1550 nm if the core radius of the fiber is 25 μm the refractive index of the core is 1.46 and the relative refractive index difference is 0.2%.

Solution The total number of modes in the fiber can be obtained as

$$M \approx \frac{2\pi^2 a^2}{\lambda^2}(n_1^2 - n_2^2) \approx \frac{2\pi^2 a^2}{\lambda^2}(2n_1^2\Delta)$$

That is,

$$M = \frac{2(3.14)^2(25\times10^{-6})^2}{(1550\times10^{-9})^2}[2(1.46)^2 \times 0.002] \approx 48$$

Example 3.6 The V-number of a step-index fiber operating at 1330 nm is 50. Estimate the number of modes that propagate through the fiber.

Solution The number of modes in the step-index fiber can be obtained as

$$M \approx \frac{V_2}{2} = \frac{(50)^2}{2} = 1250$$

3.7 LINEARLY POLARISED MODES

In a practical fiber, the difference between the refractive indices of the core and the cladding region in a step-index fiber is usually very small (Saleh *et al* 1991; Gloge 1974; Keiser 2000). The analysis of the modes in such fibers can be greatly simplified by assuming, $\Delta \ll 1$ or $n_1 = n_2$, that is

$$k_1^2 = k_2^2 \approx \beta^2 \tag{3.225}$$

A highly simplified yet fairly accurate solution can be obtained under this assumption. This approximation is known as a *weakly guided approximation*. Under this approximation modes with similar propagation characteristics can be clubbed together to form a single linearly polarised (LP) mode. Further, a close look at Fig. 3.14 reveals that the propagation constants at a given value of V or their variations with V-number for the mode pairs $\mathrm{EH}_{l-1,m}$ and $\mathrm{HE}_{l+1,m}$ are very similar (Marcuse *et al* 1979; Gloge, 1971).

Using the approximation given by Eq. (3.225) under weakly guiding fiber approximation, the eigenvalue Eq. (3.199) takes the following simplified form

$$\left(\frac{J_l'(ua)}{uJ_l(ua)}+\frac{K_l'(wa)}{wK_l(wa)}\right)=\pm\frac{l}{a}\left(\frac{1}{u^2}+\frac{1}{w^2}\right) \tag{3.226}$$

For $l = 0$, we get

$$\frac{J_0'(ua)}{uJ_0(ua)}+\frac{K_0'(wa)}{wK_0(wa)}=0 \tag{3.227}$$

The corresponding modes are TE_{0m} and TM_{0m} modes. It is interesting to note that the condition for the existence of TE_{0m} and TM_{0m} is the same under weakly guiding approximation. This can also be verified by applying the approximation given by Eq. (3.225) to Eqs (3.201a) and (3.201b).

The Equation (3.112) can be resolved into two sets of equations for the positive and negative signs, such as

$$\left(\frac{J_l'(ua)}{uJ_l(ua)}+\frac{K_l'(wa)}{wK_l(wa)}\right)=+\frac{l}{a}\left(\frac{1}{u^2}+\frac{1}{w^2}\right) \tag{3.228a}$$

the solution of which gives a set of EH modes and

$$\left(\frac{J_l'(ua)}{uJ_l(ua)}+\frac{K_l'(wa)}{wK_l(wa)}\right)=-\frac{l}{a}\left(\frac{1}{u^2}+\frac{1}{w^2}\right) \tag{3.228b}$$

the solution of which gives a set of HE modes.

Using the recurrence relations for J_l' and K_l' (see Appendix) Eqs (3.228a) and (3.228b) can be expressed as

$$\frac{J_{l+1}(ua)}{uJ_l(ua)}+\frac{K_{l+1}(ua)}{wK_l(ua)}=0 \tag{3.229a}$$

for EH modes and

$$\frac{J_{l-1}(ua)}{uJ_l(ua)}-\frac{K_{l-1}(ua)}{wK_1(ua)}=0 \tag{3.229b}$$

for HE modes.

Further, using recurrence relations of the Bessel function of the first and second kinds Eq. (3.229*b*) can be rearranged as

$$\frac{uJ_{l-2}(ua)}{J_{l-1}(ua)}=-\frac{wK_{l-2}(ua)}{K_{l-1}(ua)} \tag{3.230}$$

for HE modes.

Equation (3.229a) can be simply rearranged as

$$\frac{uJ_1(ua)}{J_{l+1}(ua)}=-\frac{wK_l(ua)}{K_{l+1}(ua)} \tag{3.231}$$

for EH modes.

Equations (3.230) and (3.231) can be combined to form a unified relation by using a new parameter, *i* as

$$\frac{uJ_{i-1}(ua)}{J_i(ua)}=-\frac{wK_{i-1}(ua)}{K_i(ua)} \tag{3.232}$$

where the parameter i is related to azimuthal parameter l for different modes as

$$i = \begin{cases} 1 \text{ for TE and TM modes} \\ l+1 \qquad \text{for EH modes} \\ l-1 \qquad \text{for HE modes} \end{cases} \tag{3.233}$$

It is interesting to note that under weakly guiding approximation all modes can be characterised by a common set of parameters (i, m) and one characteristic Eq. (3.232) that corresponds to linearly polarised (LP) modes. In other words, similar types of modes degenerate to form a single linearly polarised (LP)mode characterised by a set of values of (i, m). For example, it can be easily verified that for the same radial order m, an $HE_{l+1,m}$ mode degenerates with an $EH_{l-1,m}$ mode to constitute a single LP_{im} mode (Gloge, 1971; Ghatak *et al* 1998) demonstrated that under weakly guiding approximation each of the TE, TM, EH or HE modes loses their identity and degenerates into linearly polarised mode denoted by LP_{im}.

The cut-off condition of a particular order LP_{im} mode can be obtained by setting $w = 0$. Thus the cut-off condition of LP_{im} mode can be expressed as

$$J_{i-1}(ua) = 0 \tag{3.234}$$

For $l = 0$, this includes the roots of the Bessel function of

$$J_{-1}(ua) = 0 \tag{3.235}$$

Recalling that $J_{-1}(ua) = -J_1(ua)$, the cut-off condition of LP_{0m} modes should include $J_1(0) = 0$ as the first root. The cut-off conditions of LP_{0m} and LP_{1m} modes are shown in Fig. 3.15. It can be easily verified with the help of Eq. (3.233) that for $i = 0$, we can have LP_{0m} modes which comprise only HE_{1m} modes. For $i = 1$, the LP_{1m} modes on the other hand comprise a combination of TE, TM and HE modes.

In general, we obtain the following LP modes that result from the degeneracy of different forms of transverse electric, transverse magnetic, and hybrid modes.

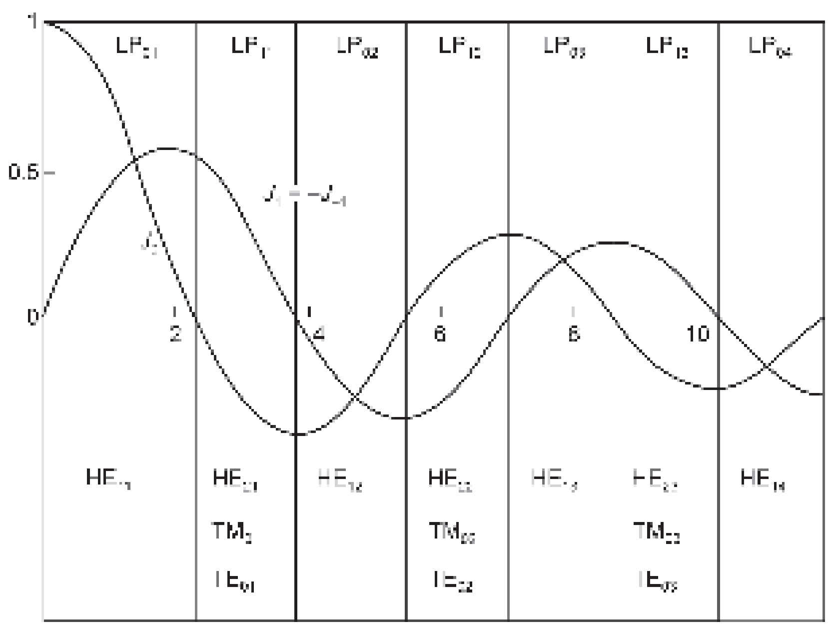


Fig. 3.15 The cut-off regions of LP_{0m} and LP_{1m} modes (after Gloge, 1971)

i. LP_{0m} modes result from the degeneration of HE_{1m} modes
ii. LP_{1m} modes result from the degeneration of TE_{0m}, TM_{0m} and HE_{2m} modes.
iii. LP_{im} $(i > 1)$ modes result from the degeneration of $HE_{l+1,m}$ and $EH_{l-1,m}$ modes.

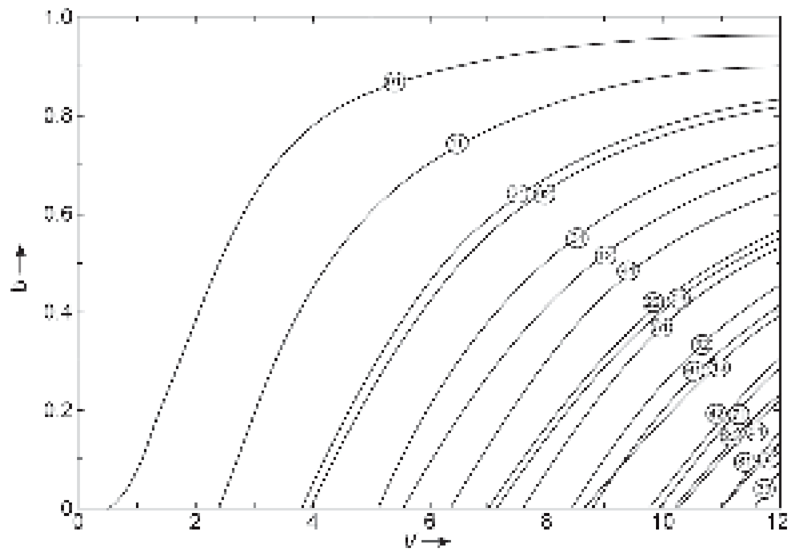


Fig. 3.16 Variation of propagation constant with *V*-number for different order LP modes (after Gloge, 1975)

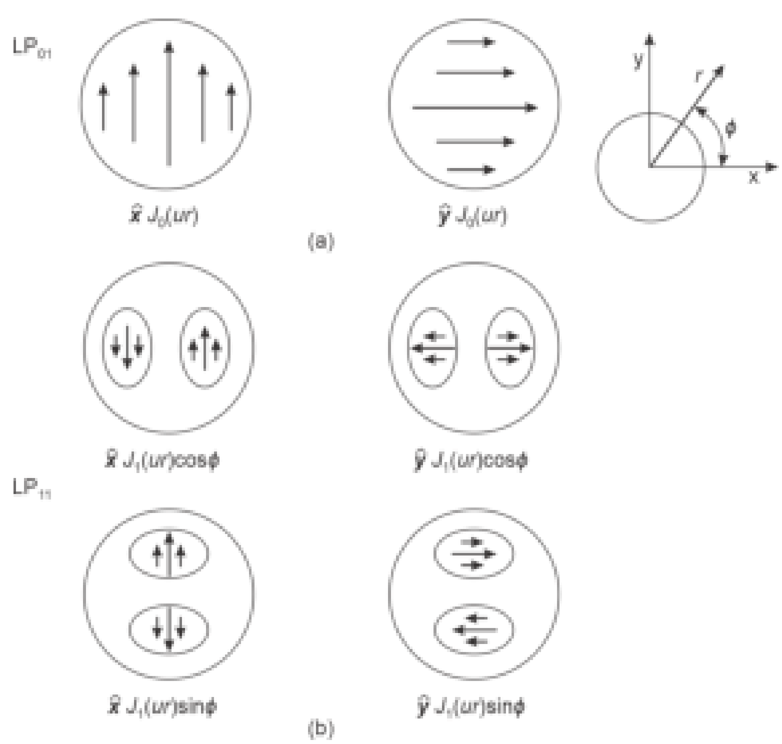


Fig. 3.17 Schematic of the modal field pattern for LP_{01} and LP_{11} modes showing intensity distribution with arrows indicating the electric field vectors

The variation of normalised propagation constant b with the V-number of a step-index fiber under weakly guiding approximation for various orders LP_{im} modes are shown in Fig. 3.16. In addition to the fact that an LP_{im} mode breaks up into modes with terms $(l + 1)\phi$ which are recognised as $HE_{l+1,m}$ modes and those with terms $(l-1)\phi$ recognised as $EH_{l-1,m}$ or TE_{0m} and TM_{0m} modes, it is often convenient to visualise a mode in this representation. Unlike in a generalised representation involving TE, TM, EH and HE modes, LP mode representation enables one to choose the electric field vector, $\boldsymbol{E}$ in any arbitrary direction with the magnetic field vector $\boldsymbol{H}$ perpendicular to it (Gloge, 1971). Further, there exist equivalent solutions with reverse field polarities because each of the two possible polarisation directions can be associated with a non-zero value of the azimuthal parameter ($i \neq 0$), with either a $\cos(i\phi)$ or a $\sin(i\phi)$ dependence. As a result a single LP_{im} ($i \neq 0$) corresponds to four discrete mode patterns. This is illustrated for LP_{01} and LP_{11} modes in Fig. 3.17.

The correspondence between the LP modes under weakly guiding approximations and the generalised TE, TM, EH and HE modes as discussed earlier is listed in Table 3.2 along with the number of degenerate modes in a given LP mode.

Table 3.2 Constitution of a few lower-order linearly polarised (LP) modes

Azimuthal parameter (i)	*Radial parameter (m)*	*Designated LP mode*	*Constituent modes with polarisations*	*Total number of degenerate modes*
0	1	LP_{01}	$HE_{11} \times 2$	2
1	1	LP_{11}	TE_{01}, TM_{01}, $HE_{21} \times 2$	4
2	1	LP_{21}	$HE_{11} \times 2$, $HE_{31} \times 2$	4
0	2	LP_{02}	$HE_{12} \times 2$	2
3	1	LP_{31}	$EH_{21} \times 2$, $HE_{41} \times 2$	4
1	2	LP_{12}	TE_{02}, TM_{02}, $HE_{22} \times 2$	4
4	1	LP_{41}	$EH_{31} \times 2$, $HE_{51} \times 2$	4
2	2	LP_{22}	$EH_{12} \times 2$, $HE_{32} \times 2$	4
0	3	LP_{03}	$HE_{13} \times 2$	2
5	1	LP_{51}	$EH_{41} \times 2$, $HE_{61} \times 2$	4

3.8 POWER FLOW IN A STEP-INDEX OPTICAL FIBER

The modal analysis also helps one to find expressions for fractional power carried by the core and the cladding region of a step-index fiber. From the foregoing analysis, we note the following.

(i) Each mode varies harmonically in the core region and decays exponentially in the cladding region

(ii) The lower order modes (those far away from cut-off frequency) in a multimode fiber are more concentrated near the core while the higher order modes extend more in the cladding region

(iii) The modes no longer decay exponentially in the cladding region but become radiative at cut-off.

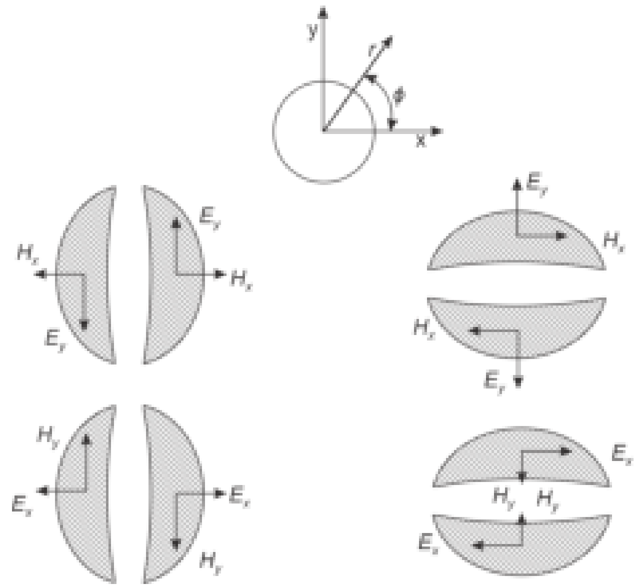


Fig. 3.18 LP_{11} mode showing four possible distributions of electric and magnetic fields

Under weakly guiding approximation we can represent modes whose transverse field is polarised in one direction as shown in Fig. 3.18 for LP_{11} mode. For an electromagnetic wave oscillating at a frequency, ω the time-averaged Poynting vector $\langle \boldsymbol{S} \rangle$ is given by

$$\langle \boldsymbol{S} \rangle = \frac{1}{2}\mathrm{Re}(\boldsymbol{\mathcal{E}} \times \boldsymbol{\mathcal{H}}^*) \tag{3.236}$$

where $\boldsymbol{\mathcal{E}}$ is the complex electric field and $\boldsymbol{\mathcal{H}}^*$ is the complex conjugate of the magnetic field which is orthogonal to the electric field. The time-averaged Poynting vector gives the power flowing along the axial direction which is perpendicular to the plane containing the orthogonal electric and magnetic field vectors.

The power flowing through the core and the cladding regions can be obtained by integrating the Poynting vectors. The power flowing through the core and the cladding regions can be obtained (Gloge, 1971) as follow:

$$P_{\text{core}} = \frac{1}{2}\int_0^a \int_0^{2\pi} r(E_x H_y^* - E_y H_x^*) d\phi\, dr \tag{3.237}$$

$$P_{\text{clad}} = \frac{1}{2}\int_a^\infty \int_0^{2\pi} r(E_x H_y^* - E_y H_x^*) d\phi\, dr \tag{3.238}$$

where E_x, E_y, H_x and H_y correspond to horizontal (cos ϕ part) and vertical (sin ϕ part) components of the electric and magnetic field vectors and the asterisk symbol is used to indicate the complex conjugate of the corresponding component.

Under weakly guiding approximation ($\Delta \ll 1$) the total power P in a given mode is distributed between the core and the cladding (Gloge, 1975)

$$\frac{P_{\text{core}}}{P} = \frac{a^2 w^2}{V^2}\left[1 - \frac{J_l^2(ua)}{J_{l+1}(ua)J_{l-1}(ua)}\right] \tag{3.239}$$

and

$$\frac{P_{\text{clad}}}{P} = 1 - \frac{P_{\text{core}}}{P} \tag{3.240}$$

The variation of the fractional powers flowing through the core and the cladding regions of a step-index fiber (under weakly guiding approximation) with V-number of the fiber is shown in Fig. 3.19 for different orders of LP_{jm} modes.

The power density at the core–cladding interface averaged over the circumference can be obtained (Gloge 1972; Gloge 1975)

$$p(a) = \frac{P}{\pi a^2}\left(1 - \frac{a^2 u^2}{V^2}\right)\frac{J_l^2(ua)}{J_{l+1}(ua)J_{l-1}(ua)} \tag{3.241}$$

When light is launched into a fiber from an incoherent source (such as an LED or an incandescent lamp), it tends to excite all possible modes in the fiber with the same amount of power. The average power can be calculated by taking the sum of the power carried by each mode and dividing it by the total number of modes. Assuming the number of modes to be very large the average power carried by the core and the cladding can be computed by integrating the power carried by a single mode over all possible modes. The total average power density at the core–cladding interface under this condition can be obtained as

$$\left[\frac{P_{\text{clad}}}{P}\right]_{\text{total}} = \frac{1}{2\pi a^2} \tag{3.242}$$

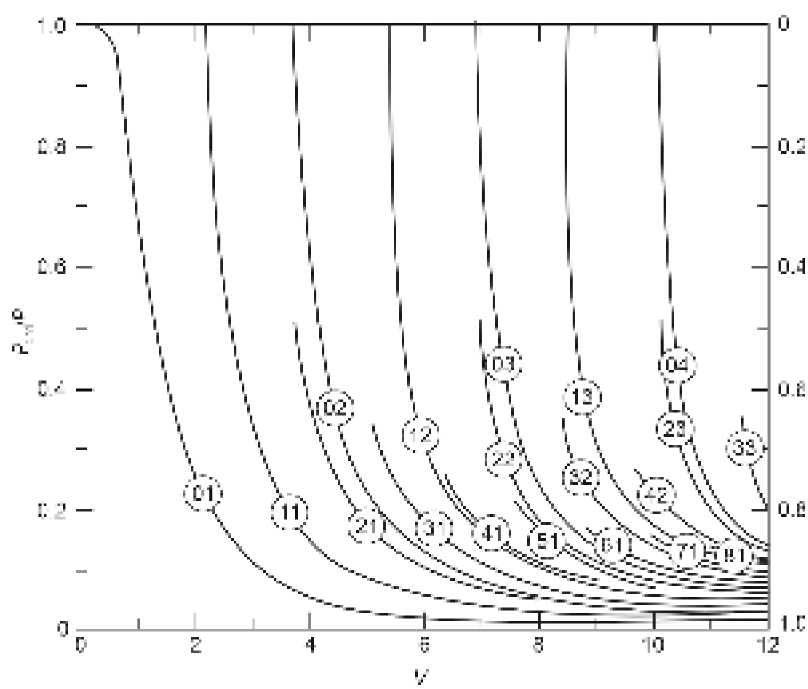


Fig. 3.19 Fractional power flowing through the core and cladding of a step-index fiber for different orders of LP_{jm} modes (after Gloge, 1971)

It can be seen from Eq. (3.240) that the total mean power at the core–cladding interface averaged over all possible modes is independent of the total number of modes or V-number of the fiber. The total average cladding power under this condition can be obtained as

$$\left[\frac{P_{\text{clad}}}{P}\right]_{\text{total}} = \frac{4}{3}\frac{1}{\sqrt{M}} \tag{3.243}$$

where M is the total number of modes. For a step-index fiber, the number of modes is related to the V-number as

$$M = \frac{V^2}{2} \tag{3.244}$$

Therefore, the power carried by the cladding in a multimode step-index fiber decreases inversely with the V-number.

Example 3.7 A 50/125 μm step-index multimode fiber has a core refractive index of 1.458 and an index deviation of $\Delta = 0.01$. Estimate the percentage of power carried by the cladding of the fiber when power is launched into the fiber from a source operating at 850 nm.

Solution The given fiber has the following parameters

$$a = 25\ \mu\text{m} \qquad n_1 = 1.458$$
$$\Delta = 0.01 \qquad \lambda = 0.85\ \mu\text{m}$$

The V-number of the fiber is given by

$$V = \frac{2\pi a}{\lambda}(n_1\sqrt{2\Delta}) = \frac{6.28\times 25\times 10^{-6}}{0.85\times 10^{-6}}(1.485\times\sqrt{0.02}) = 38$$

The number of modes can be obtained as

$$M \cong \frac{V^2}{2} = \frac{(38)^2}{2} = 722$$

Therefore the percentage of the total power carried by the cladding can be estimated as

$$\left(\frac{\mathrm{P_{clad}}}{P}\right)_{\text{total}} (\text{in percent}) = \frac{4}{3}\frac{1}{\sqrt{722}}\times 100 \approx 5\%$$

3.9 SINGLE MODE FIBERS

In the previous chapter, it is discussed that two distinctly different types of optical fibers are available in the form of single-mode fiber and multimode fiber. The names imply that the former supports only one mode for propagation through the fiber while the later allows more than one mode to propagate through it. In the absence of any other mode, the former also avoids distortion arising out of delay differences among the modes and thereby provides the ultimate achievable bandwidth (Snyder, 1981; Neumann, 1988; Gloge, 1995). From the modal analysis, it is clear that single-mode operation becomes possible when $V \leq 2.405$. Single-mode operation generally occurs when the size of the core radius is set to a few times the operating wavelength (4–6) with a small value of index difference Δ (ranging between 0.2 to 1.0 %). It is seen that all modes except HE_{11} are eliminated when the V-number is conveniently set to 2.4. At this value, fairly large variations in core size and index deviation are possible by maintaining single-mode operation. Theoretically, there is no cut-off of the fundamental mode when V-number decreases and the mode ceases to exist only when $V = 0$ which is not possible for any practical fiber. In principle, the V-number can assume any value greater than 0 but less than 2.405. However, theoretical calculations reveal that the power carried by the core decreases very fast as the V-number is decreased below 2. It is reported that only 0.1 pecent of the total power is confined in the core when $V = 0.6$ and propagation below this is virtually impossible (Gloge, 1975). Single-mode fibers offer very large bandwidth but their operation is limited by the core size and the index difference for a given wavelength of operation. A typical single-mode fiber operating at a wavelength of 0.85 μm with a core radius of $a = 3$ μm and $V = 2.4$ would have a NA = 0.1.

3.9.1 Mode-Field Diameter (MFD)

It is understood that in a single-mode optical fiber the optical power flows in both the core and the cladding. In many situations, it is not just enough to consider the physical size of the core. For example, in the situation of joining two single-mode fibers, it is necessary to know the extent of the region beyond the core of the emitting fiber[1] that carries significant optical power. As nearly 20% of the total power is carried by the cladding of a single-mode fiber, it is not sufficient to consider the core diameter for such applications. This can be best understood by considering the geometric distribution of optical power in the propagating mode of the single-mode fiber as illustrated in Fig. 3.20. It can be easily seen from the figure that the electric field has a significant value even beyond the core radius on either side from the center of the core. To consider the region of the fiber in which the electric field of the fundamental LP_{01} mode is dominant we introduce an effective value of the core diameter which is referred to as the mode-field diameter (MFD) of the fiber. It can be easily appreciated that the physical diameter of the core is always less than the mode-field diameter. The actual value of the mode-field diameter depends on the distribution of the electric field for the fundamental mode. In a multimode fiber, the power mainly flows through the core region and very little fraction flows through the cladding region. From this viewpoint,

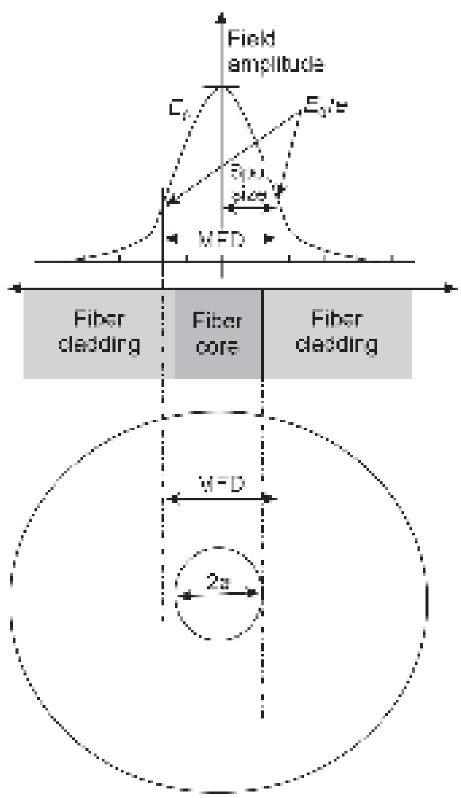


Fig. 3.20 Distribution of the electric field of the fundamental mode in a single mode fiber illustrating the core size and mode-field diameter

[1] The fiber that carries optical power which is to be coupled to a subsequent fiber (referred to as a receiving fiber)

mode-field diameter may be considered analogous to the core diameter of a multimode fiber in the sense that beyond mode-field diameter the power flowing in the remaining cladding region is negligibly small.

There are different criteria for defining and measuring the mode-field diameter of a single mode fiber (Artiglia *et al* 1989; Jeunhomme, 1989; Franzen *et al* 1985). One method of computing the mode-field diameter is to define the field distribution mathematically as a function of the radial distance and subsequently locate the points on both sides of the symmetrical field distribution up to which the field value is considered to be significant as illustrated in Fig. 3.20. For example, let the electric field distribution for the fundamental mode in a single-mode fiber be described as a Gaussian distribution given by

$$E(r) = E_0 \exp(-r^2/W_0^2) \tag{3.245}$$

where r is the distance measured from the center of the core along the radius and W_0 is a measure of the width of the electric field distribution. The mode-field diameter has been calculated by different methods depending on the criterion used to define the significant value of the electric field as compared to the peak value. The simplest method is to consider the field to be significant until it falls to $(1/e)$ times the peak value E_0. It can be easily verified for $r = \pm W_0$, the electric field becomes

$$E(r = \pm W_0) = e^{-1} E_0 \tag{3.246}$$

The width $2W_0$ in this case, is considered as the mode-field diameter. This value is larger than the core diameter, $2a$ of the fiber, and accounts for a major portion of the power flowing through the cladding. It may be noted that the optical power falls to e^{-2} of the peak optical power beyond $\pm W_0$ on the two sides. The mode-field diameter (MFD) of the LP_{01} the mode can be defined as

$$MFD = 2W_0 = 2\left[\frac{2\int_0^\infty r^3 E^2(r)dr}{\int_0^\infty rE^2(r)dr}\right]^{\frac{1}{2}} \tag{3.247}$$

where $E(r)$ is the electric field distribution of LP_{01} mode.

There are other methods of defining the mode-field diameter (Anderson *et al* 1983; 1987). Further, the mode-field dimension depends on the refractive index profile that results in the non-Gaussian distribution of the electric field.

3.9.2 Birefringence in a Single-mode Fiber

In practical optical fibers, the state of polarisation (SoP) of light is not generally maintained during its propagation beyond a few meters. Intensity modulation/direct detection (IM/DD) based optical fiber communication is not generally affected by the change of state of polarisation because the detector in this case behaves as a photon counter which is insensitive to the polarisation state of the received light. A coherent optical communication system on the other hand may be adversely affected by the change in polarisation state of light. Therefore, there is a need for a fiber that would maintain the state of polarisation over a significant length of a fiber. Single-mode fibers can be designed to meet this requirement and such fibers are referred to as polarisation-maintaining fibers (PMFs). The phenomenon of birefringence in a single-mode fiber can be exploited for maintaining the polarisation state of light during its propagation through the fiber. In the absence of maintenance of SoP, the single-mode fibers give rise to a kind of dispersion known as polarisation mode dispersion (PMD).

It is understood from the previous discussion that in a single-mode fiber the lowest-order mode LP_{01} mode which consists of a pair of HE_{11} modes (HE_{11}^x and HE_{11}^y) polarised in orthogonal planes propagate through the fiber. These modes are very similar except for the fact that their planes of polarisation are orthogonal. The components can be chosen arbitrarily along the principal axes (x and y) determined by the symmetry elements of the fiber cross-section and as shown in Fig. 3.21. Under ideal conditions, the two components propagate with the same phase velocity. The practical fibers, however, are not perfectly circular, and as a result, the two orthogonally polarised modes propagate with slightly different phase and group velocities. Additional factors such as bending and anisotropic stress aggravate the situation causing the fiber to behave as a birefringent medium leading to different refractive indices (n_x and n_y) in the two orthogonal directions. As a result, the two orthogonal modes exhibit different propagation constants β_x and β_y. The modal birefringence of the fiber is defined as the difference between the effective refractive indices given by

$$B_F = n_y - n_x \tag{3.248}$$

Equivalently, we may write (Kaminow, 1981)

$$\beta_F = k(n_y - n_x) = \beta_y - \beta_x \tag{3.249}$$

where k is the free space propagation constant and λ is the wavelength of the light.

When light is so injected in the fiber that both the modes with orthogonal polarisation are excited, then the two modes develop a relative phase difference as they propagate through the fiber. When the phase difference between the two components becomes an integral multiple of 2π, the modes beat with each other to return to the same state of polarisation as at the input. The length over which the reproduction of the original state of polarisation is achieved is called the *fiber beat length*, given by

$$L_p = \frac{2\pi}{\beta_F} = \frac{2\pi}{\beta_y - \beta_x} = \frac{\lambda}{n_y - n_x} \tag{3.250}$$

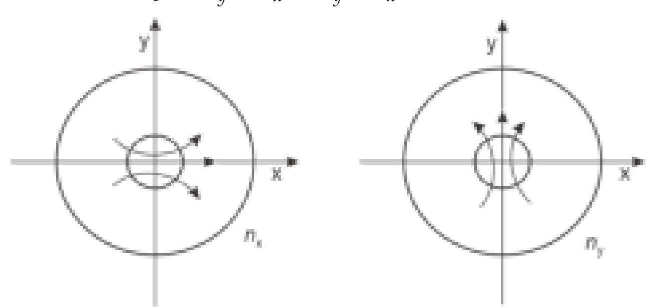


Fig. 3.21 Fundamental HE_{11} mode in a single mode fiber decomposed into two orthogonal components

It is interesting to note that the state of polarisation of light propagating through a single-mode fiber varies periodically along the length. The linear phase retardation along the direction of propagation of light (z-direction) depends on the fiber length. Assuming the phase coherence to be maintained between the two orthogonal components of the fundamental HE_{11} mode, the phase retardation, ϕ can be written as

$$\phi = (\beta_y - \beta_x)L \tag{3.251}$$

where L is the length of the fiber traversed by the light.

Figure 3.17 illustrates the state of polarisation which is elliptical in general but changes over the coherence length (Rashleigh *et al* 1978). For example, if the incident linear polarisation is 45° about the x-axis (Fig. 3.22), then becomes circular when $\phi = \pi/2$ and linear again when $\phi = \pi$. The change of state of polarisation continues in the same way and becomes circularly polarised again when $\phi = 3\pi/2$ and finally returns to the same initial state of polarisation when $\phi = 2\pi$ (Kaminow, 1981). The length over which the total phase changes by 2π is the beat length, L_P defined by Eq. (3.250).

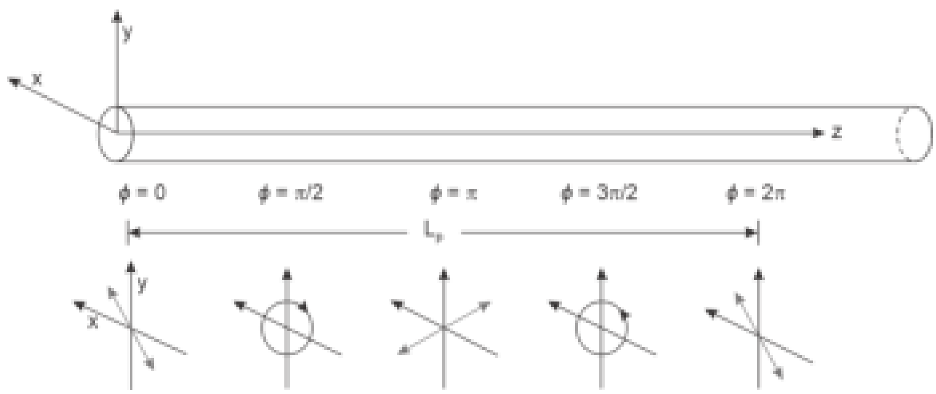


Fig. 3.22 Illustration of change of polarization state of the light propagating through a single mode fiber over coherence length showing the beat length

Example 3.8 A single mode fiber operating at 1330 nm has a modal birefringence of 1.5×10^{-5}. Estimate the fiber beat length.

Solution The beat length can be calculated as

$$L_p = \frac{\lambda}{B_F} = \frac{1330 \times 10^{-9}}{1.5 \times 10^{-5}} \approx 8.87 \text{ cm}$$

This is an intermediate type birefringence fiber.

The birefringence, B_F of a single-mode fiber may vary in the range from 10^{-3} (high birefringence) to 10^{-8} (low birefringence). Accordingly, the beat length may vary from a few millimeters to several tens of meters. High-birefringence (Hi-Bi) fibers that maintain the state of polarisation of the incident light over large distances are called polarisation-maintaining fibers (PMFs).

3.10 MODE ANALYSIS FOR A GRADED-INDEX (GI) FIBER

The mode analysis for a graded-index fiber is much more complex than that for a step-index fiber because the refractive index of a graded-index fiber is a function of position. Rigorous analysis of the propagation of light through a graded-index fiber has been reported in the past (Okamoto *et al* 1976; Gloge *et al* 1973; Gloge, 1975). For a graded-index fiber, the parameter q in Eq. (3.57) becomes a function of the position given by (Gloge *et al* 1973; Gloge, 1975)

$$q(r) = \left[k^2 n^2(r) - \beta^2 - \frac{l^2}{r^2} \right]^{\frac{1}{2}} \tag{3.252}$$

Accordingly, Eq. (3.171) takes the following form (Gloge, 1975; Gloge *et al* 1973)

$$\frac{d^2 R(r)}{dr^2} + \frac{1}{r}\frac{dR(r)}{dr} + \left(k^2 n^2(r) - \beta^2 - \frac{l^2}{r^2} \right) R(r) = 0 \tag{3.253}$$

The solution of this equation by using boundary conditions and calculation of the spatial distribution of field components cannot be obtained in a straightforward way as in the case of a step-index fiber. In this section, we assume the field distribution as a quasi-plane wave traveling within the core. It can be easily seen that the differential Eq. (3.253) contains a slowly varying parameter $n(r)$ which varies over distances of the order of an optical wavelength. The solution of this type of differential equation is generally obtained in an asymptotic form that also contains a slowly varying parameter changing over the desired range of the equation. This type of approximate solution can be obtained with the help of the WKB (Wentzel-Kramers-Brillouin) approach (Gloge *et al* 1973).

Using the WKB approach the solution of Eq. (3.253) may be assumed to be of the form

$$R(r) = A\exp(jkS(r)) \tag{3.254}$$

where A is a constant independent of r and $S(r)$ is a function of r. Substituting the trial solution in the original differential Eq. (3.253), we get

$$jk\left(\frac{d^2S(r)}{dr^2}\right) - \left(k\frac{dS(r)}{dr}\right)^2 + \frac{jk}{r}\frac{dS(r)}{dr} + \left(k^2n^2(r) - \beta^2 - \frac{l^2}{r^2}\right) = 0 \tag{3.255}$$

It may be noted that the parameter $n(r)$ varies slowly over a distance on the order of an optical wavelength, λ and therefore expansion of $S(r)$ in the powers of λ would enable the solution to converge rapidly. Further, we note that $k^{-1} = \lambda/2\pi$. We may therefore express $S(r)$ alternatively in powers of k^{-1} as (Gloge, 1975; Keiser, 2000)

$$S(r) = S_0 + k^{-1}S_1 + \cdots \tag{3.256}$$

where S_0, S_1, ⋯ are some functions of r.

Substituting Eq. (3.256) in Eq. (3.255) and rearranging the terms containing the same powers of k we get

$$\left[-k^2\left(\frac{dS_0}{dr}\right)^2 + \left(k^2n^2(r) - \beta^2 - \frac{l^2}{r^2}\right)\right] + \left[jk\frac{d^2S_0}{dr^2} - 2k\left(\frac{dS_0}{dr}\right)\left(\frac{dS_1}{dr}\right) + \frac{jk}{r}\left(\frac{dS_0}{dr}\right)\right]$$

$$+ \text{ terms containing powers of } (k^0, k^{-1}, k^{-2}, \cdots) = 0 \tag{3.257}$$

Equation (3.257) is true only when each of the terms containing equal powers of k is individually equal to zero. A sequence of relations defining the functions S_0, S_1,⋯, etc. can thus be obtained by equating each term containing equal powers of k to zero. The first two terms thus yield

$$-k^2\left(\frac{dS_0}{dr}\right)^2 + \left(k^2n^2(r) - \beta^2 - \frac{l^2}{r^2}\right) = 0 \tag{3.258}$$

and

$$jk\frac{d^2S_0}{dr^2} - 2k\left(\frac{dS_0}{dr}\right)\left(\frac{dS_1}{dr}\right) + \frac{jk}{r}\left(\frac{dS_0}{dr}\right) = 0 \tag{3.259}$$

Integrations of Eqs (3.258) yields

$$kS_0 = \int_r \left(k^2n^2(r) - \beta^2 - \frac{l^2}{r^2}\right)^{\frac{1}{2}} dr \tag{3.260}$$

where the negative sign and constant of integration are omitted for the sake of simplicity. Further, we note that a mode will be bound in the core of the fiber when it varies harmonically in the core which is possible only when S_0 is real. For S_0 to be real it is required that the radical of the integrand must be greater than zero, that is

$$\left(k^2n^2(r) - \beta^2 > \frac{l^2}{r^2}\right) \tag{3.261}$$

It can be seen that for a given azimuthal mode number l there exist two values of r (say, r_1 and r_2) for which the radical of the integrand of Eq. (3.256) is zero. This is depicted in Fig. 3.23 along with the cross-sectional projection of a skew ray in a graded index fiber. These two values r_1 and r_2 correspond to the lower and upper bounds of the integral in the Eq. (3.260). It should be noted that both r_1 and r_2 are functions of the mode number l and the guided modes exist for values of r lying between these two extreme values. For other values of r the function S_0 is imaginary and results in a decaying field and should be discarded for bound modes. In other words, the path followed by a typical skew ray is bounded between two coaxial cylindrical surfaces with inner and outer radii of r_1 and $r_2$respectively. These coaxial cylindrical surfaces are called *caustic surfaces*. The solution in terms of function S_0 can be written in the form

$$kS_0 = \int_{r_1(l)}^{r_2(l)} \left(k^2 n^2(r) - \beta^2 - \frac{l^2}{r^2} \right)^{\frac{1}{2}} dr \tag{3.262}$$

For a given value of l the corresponding values of r_1 and r_2 represent the turning points at which the ray turns back and forth. It can be easily seen that for a given value of l if the function (l^2/r^2) falls below $k^2n^2(r) - \beta^2$ over a certain range of r (Fig. 3.23), there exist two intersection points corresponding to r_1 and r_2 within which the in Eq. (3.261) remains valid and bound modes exist in the region. On the other hand, when (l^2/r^2) function lies above $k^2n^2(r) - \beta^2$ term for a given value of l over the entire core region, then the function S_0 becomes imaginary leading to unbound decaying mode corresponding to evanescent field.

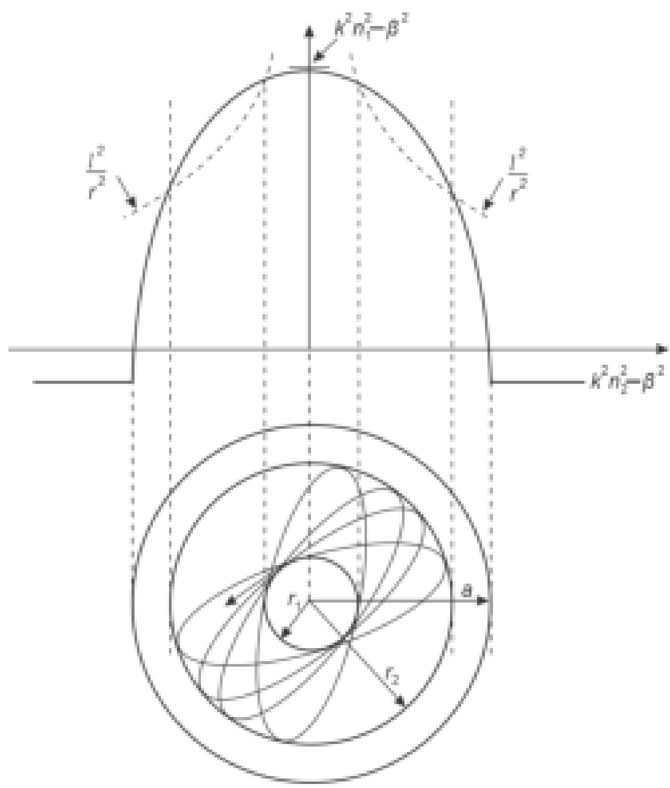


Fig. 3.23 Bound modes in a GI fiber illustrating graphical representation of WKB solution and oscillatory field between the turning points projected across a transverse section of the fiber

It may be recalled that to constitute a bound mode the waves associated with a particular ray congruence corresponding to the mode must interfere constructively to form a standing wave pattern in the transverse direction. The complete analysis reported elsewhere in the literature (Gloge, 1975) is beyond the scope of this book. The requirement of constructive interference of the waves imposes the condition that the phase function S_0 between the turning points r_1 and r_2 should be an integral multiple of half-periods that is, π in terms of phase angle. Therefore, we may write (Keiser, 2000)

$$kS_0 = \int_{r_1(l)}^{r_2(l)} \left(k^2 n^2(r) - \beta^2 - \frac{l^2}{r^2} \right)^{\frac{1}{2}} dr = m\pi \tag{3.263}$$

where $m = 0, 1, 2,\cdots$ corresponds to the radial mode number signifying the number of half-periods between the turning points.

The total number of bound modes as a function of propagation constant β can be obtained by taking the sum of Eq. (3.263) over all possible values of l from 0 to l_m where l_m is the maximum value of the azimuthal mode number for a given value of β. Assuming that lm is very large, the discrete summation can be replaced by definite integration within limits $(0, l_m)$. Further, for a given set of (l, m) there exists a degenerate group of four modes with different polarisations (Gloge, 1975). Therefore, the total number of modes as a function of the propagation constant can be expressed as

$$m(\beta) = \frac{4}{\pi} \int_0^{l_m} \int_{r_1(l)}^{r_2(l)} \left(k^2 n^2(r) - \beta^2 - \frac{l^2}{r^2} \right)^{\frac{1}{2}} dr \cdot dl \tag{3.264}$$

Interchanging the order of the integrals and suitably changing the lower limit of r to consider all the modes, we may write

$$m(\beta) = \frac{4}{\pi} \int_0^{r_2} \int_0^{l_m} \left(k^2 n^2(r) - \beta^2 - \frac{l^2}{r^2} \right)^{\frac{1}{2}} dl \cdot dr \tag{3.265}$$

The upper limit of l can be obtained from the following condition

$$k^2 n^2(r) - \beta^2 - \frac{l_m^2}{r^2} = 0 \tag{3.266}$$

That is,

$$l_m = r(k^2 n^2(r) - \beta^2)^{\frac{1}{2}} \tag{3.267}$$

The right side of the Eq. (3.266) is a standard integral (see Appendix) and substitution of the limits of integration finally yields

$$m(\beta) = \frac{4}{\pi} \int_0^{r_2} [k^2 n^2(r) - \beta^2] r\, dr \tag{3.268}$$

The maximum value of the radius r_2 can be obtained from the condition

$$k^2 n^2(r_2) - \beta^2 = 0 \tag{3.269}$$

The generalised refractive index profile of a graded index within the core ($r < a$) can be expressed as

$$n(r < a) = n_1 \left[1 - 2\Delta \left(\frac{r}{a} \right)^{\alpha} \right]^{\frac{1}{2}} \tag{3.270}$$

Combining Eqs (3.268) and (3.270), we find

$$k^2 n_1^2 \left[1 - 2\Delta \left(\frac{r_2}{a} \right)^{\alpha} \right] - \beta^2 = 0 \tag{3.271}$$

That is,

$$r_2 = a\left[\frac{1}{2\Delta}\left(\frac{k^2n_1^2 - \beta^2}{k^2n_1^2}\right)\right]^{\frac{1}{\alpha}} \tag{3.272}$$

Substituting $n(r)$ from Eq. (3.270) into Eq. (3.268), we get

$$m(\beta) = \frac{4}{\pi}\int_0^{r_2}\left[k^2n_1^2\left[1 - 2\Delta\left(\frac{r}{a}\right)^{\alpha}\right] - \beta^2\right]r \cdot dr \tag{3.273}$$

Carrying out the above integration and substituting the upper limit r_2 from Eq. (3.272), we get

$$m(\beta) = a^2k^2n_1^2\Delta\left(\frac{\alpha}{\alpha+2}\right)\left(\frac{k^2n_1^2 - \beta^2}{2\Delta k^2n_1^2}\right)^{\frac{2+\alpha}{\alpha}} \tag{3.274}$$

It is understood that for the existence of a bound mode, the following condition has to be satisfied

$$\beta \geq k_2 (= kn_2) \tag{3.275}$$

The maximum number of modes in a graded-index fiber can be obtained under the following condition

$$\beta = kn_2 = kn_1(1 - \Delta) \tag{3.276}$$

The maximum number of modes in a graded-index fiber can therefore be obtained from Eq. (3.274) under the condition given by Eq. (3.276) can be obtained as

$$M_{\text{GI}} = m(\beta = kn_2) = a^2k^2n_1^2\Delta\left(\frac{\alpha}{\alpha+2}\right) \tag{3.277}$$

For a corresponding step-index fiber we have already seen that

$$M_{\text{SI}} = \frac{V^2}{2} = a^2k^2n_1^2\Delta \tag{3.278}$$

Therefore, the number of modes in a GI fiber is reduced by a factor, $\alpha/(\alpha + 2)$, where α is the grade index of the refractive index profile of the GI fiber. For a parabolic index GI fiber, $\alpha = 2$ and consequently

$$(M_{\text{GI}})_{\text{parabolic}} = \frac{M_{SI}}{2} \tag{3.279}$$

This means that a parabolic index GI fiber carries just half the number of modes carried by a corresponding SI fiber at a given wavelength.

Example 3.9 Calculate the number of modes in a graded-index fiber with a parabolic index profile operating at 1330 nm. The axial refractive index of the fiber is $n_1 = 1.49$ and that the core–cladding interface is $n_2 = 1.47$. The core radius of the fiber is 25 μm. How many modes will be supported by a corresponding SI fiber at this wavelength?

Solution The value of the index deviation, Δ of the fiber can be estimated as

$$\Delta = \frac{n_1 - n_2}{n_1} = \frac{1.49 - 1.47}{1.49} = 0.0136$$

Therefore, the number of modes supported by the parabolic index GI fiber can be estimated as

$$(M_{\text{GI}})_{\text{parabolic}} = (25 \times 10^{-6})^2\left(\frac{2 \times 3.14}{1330 \times 10^{-9}}\right)^2(1.49)^2(0.013)\left(\frac{2}{4}\right) \approx 201$$

The number of modes supported by a corresponding step-index fiber would be

$$M_{SI} \approx 2 \times 201 = 402.$$

Example 3.10 A step-index fiber has the following specifications

$$n_1 = 1.460,\ n_2 = 1.457 \text{ and } a = 6\ \mu\text{m}$$

What is the maximum value of the wavelength of operation for which the fiber will behave as a single-mode fiber? Calculate the number of modes supported by the fiber when operated at 0.65 μm wavelength. How many modes will be supported at this wavelength by an equivalent parabolic index fiber?

Solution For the given fiber to work as a single-mode fiber, it is necessary that

$$V = V_c = \frac{2\pi a}{\lambda_{max}}(n_1^2 - n_2^2)^{\frac{1}{2}} = 2.405$$

That is,

$$\lambda_{max} = \frac{2 \times 3.14 \times 6 \times 10^{-6} \times \left((1.46)^2 - (1.457)^2\right)^{\frac{1}{2}}}{2.405} = 1.465\ \mu\text{m}$$

The V-number of the fiber at λ = 0.65 μm for the fiber is

$$V = \frac{2\pi \times 6 \times 10^{-6} \times \left((1.46)^2 - (1.457)^2\right)^{\frac{1}{2}}}{0.65 \times 10^{-6}} = 5.4$$

The number of modes supported by the fiber at 0.65 μm wavelength is

$$M_{SI} = 14$$

The corresponding parabolic index fiber at this wavelength will support half the number of modes that is, 7 modes only.

Summary

- The propagation of light through optical fibers can be best analysed with the help of mode analysis by solving Maxwell's equations under appropriate boundary conditions.
- The analysis of propagation of light in the form of electromagnetic wave in a simple planar waveguide forms the basis of mode analysis in optical fiber and other optical waveguides used in integrated optics.
- A symmetrical step-index planar waveguide supports TE and TM modes.
- For $0 < V < \pi/2$ the step-index planar waveguide supports one mode only.
- Just like planar waveguides, transverse electric (TE) ($E_z = 0$) and transverse magnetic (TM) modes ($H_z = 0$) are created in optical fibers. In addition, hybrid modes (EH or HE) for which both $E_z, H_z \neq 0$ are also created in an optical fiber. Hybrid modes are a speciality of an optical fiber.
- The modes vary harmonically in the core region and decay exponentially in the cladding region. These modes are referred to as core modes or bound modes.
- Lower order modes are tightly concentrated near the axis of the core while the higher order modes are less tightly bound to the axis of the core and tend to spread towards the boundary of the inner core and penetrate deeper in the cladding region.
- The cladding being a dielectric medium supports the formation of modes by the light entering into the cladding region. These modes which are not bound in the core region but are still solutions of the boundary-value problem are called cladding or radiation modes.

- For a mode to remain guided it is necessary that the propagation constant β satisfies the condition,

$$(n_2 k =)\, k_2 < \beta < k_1 (= n_1 k)$$

where, n_1 and n_2 are the refractive indices of the core and the cladding, respectively and $k\,(= 2\pi/\lambda)$ is the free space propagation constant.
- The transition of a guided mode to a leaky mode occurs when $\beta = k_2 (= n_2 k)$. This is known as the cut-off condition of a guided mode.
- The normalised frequency or the V-number of an SI fiber is,

$$V = \frac{2\pi a}{\lambda}(\text{NA}) = \frac{2\pi a}{\lambda}\sqrt{n_1^2 - n_2^2} \approx \frac{2\pi a}{\lambda} n_1 \sqrt{2\Delta}$$

- V-number is an important parameter that determines the number of modes that is supported by a particular type of fiber.
- The eigenvalue equation that β has to satisfy for the existence of a mode in the fiber is

$$\left(\frac{J_l'(ua)}{uJ_l(ua)} + \frac{K_l'(wa)}{wK_l(wa)}\right)\left(k_1^2 \frac{J_l'(ua)}{uJ_l(ua)} + k_2^2 \frac{K_l'(wa)}{wK_l(wa)}\right) = \left(\frac{\beta l}{a}\right)^2 \left(\frac{1}{u^2} + \frac{1}{w^2}\right)^2$$

This is a transcendental equation in which β can only have discrete values within the range $k_2 \le b \le k_1$.
- For TE mode

$$\left(\frac{J_0'(ua)}{uJ_0(ua)} + \frac{K_0'(wa)}{uK_0(wa)}\right) = 0$$

- For TM mode

$$\left(k_1^2 \frac{J_0'(ua)}{uJ_0(ua)} + k_2^2 \frac{K_0'(wa)}{wK_0(wa)}\right) = 0$$

- In the generalised case, a mode ceases to exist when $\beta = k_2 = n_2 k$. In that case, the mode is no longer bound to the core nor does the field decays exponentially in the cladding and the cut-off is said to occur.
- The cut-off value of the normalised frequency for a single-mode operation in the case of a step-index fiber is

$$V_c = \frac{2\pi a}{\lambda_c}\left(n_1^2 - n_2^2\right)^{\frac{1}{2}} = 2.405$$

where, λ_c is the cut-off wavelength.
- For a given fiber, the cut-off wavelength for single mode operation can be expressed by using equations (3.103) and (3.104) as,

$$\lambda_c = \frac{V\lambda}{2.405}$$

- Number of modes emanating from a waveguide at a wavelength, λ, can be expressed as,

$$M \approx \frac{2A}{\lambda^2}\left(n_1^2 - n_2^2\right)^{\frac{1}{2}} = \frac{2\pi^2 a^2}{\lambda^2}\left(n_1^2 - n_2^2\right) = \frac{V^2}{2}$$

- Under weakly guiding approximation each of the TE, TM, EH, or HE modes looses their identity and degenerates into linearly polarised mode denoted by LP_{im}.
- The total average cladding power carried by the cladding is,

$$\left[\frac{P_{\text{clad}}}{P}\right]_{\text{total}} = \frac{4}{3}\frac{1}{\sqrt{M}}$$

where, M is the total number of modes. For a step-index fiber the number of modes is related to the V-number as,

$$M = \frac{V^2}{2}$$

- The power carried by the cladding in a multimode step-index fiber decreases inversely with the V-number.
- The total number of modes supported by a GI fiber is given by

$$M_{\mathrm{GI}} = a^2 k^2 n_1^2 \Delta \left(\frac{\alpha}{\alpha + 2} \right)$$

- A parabolic index GI fiber carries just half the number of modes carried by a corresponding SI fiber at a given wavelength.

Problems

3.1 Verify the vector identity

$$\nabla \times (\nabla \times \boldsymbol{\varepsilon}) = \nabla(\nabla \cdot \boldsymbol{\varepsilon}) - \nabla^2 \boldsymbol{\varepsilon}$$

3.2 Verify the generalised representation of an elliptically polarised light in terms of two orthogonally plane-polarised electromagnetic waves with an arbitrary phase difference, δ as given in Eq. (3.7).

3.3 Consider a symmetrical step-index planar waveguide (Fig. 3.3). Obtain the expressions for E_x, E_z and H_y for transverse magnetic (TM) modes.

3.4 Starting from Maxwell's equations derive the following magnetic field equations for a symmetrical step-index planar waveguide (Fig. 3.3) using appropriate boundary conditions.

$$\chi \tan \chi = \left(\frac{n_1^2}{n_2^2} \right) \left[V^2 - \chi^2 \right]^{\frac{1}{2}} \quad \text{(symmetric TM modes)}$$

$$-\chi \cot \chi = \left(\frac{n_1^2}{n_2^2} \right) \left[V^2 - \chi^2 \right]^{\frac{1}{2}} \quad \text{(anti-symmetric TM modes)}$$

where

$$V = ak\left(n_1^2 - n_2^2\right)^{\frac{1}{2}}$$

and

$$\chi = ua = a\left[k^2 n_1^2 - \beta^2\right]^{\frac{1}{2}}$$

assuming $2a = d$ as the thickness of the central film.

[**Hint:** As H_y and E_z are tangential components on the planes at $x = \pm a$ (see Eq. (3.50)), the boundary condition at the interface will be as follows

$$H_y \text{ and } \frac{1}{n^2}\frac{dH_y}{dx} \text{ shall be continuous at } x = \pm a]$$

3.5 A linearly polarised wave is incident in a symmetrical step-index planar waveguide (Fig. 3.3) with the electric vector making an angle 45° with the x-and y-axes. Determine the field distributions at $z = 0$ and $z > 0$ for the TE and TM modes assuming the difference between the propagation constants between the two modes to be $\Delta\beta$.

[**Hint:** The electric field distributions at $z = 0$ can be obtained as

$$\varepsilon_x = E_0 \cos 45° \cos \omega t$$

$$\varepsilon_y = E_0 \sin 45° \cos \omega t]$$

3.6 In the above problem (P3.5) determine the value of z at which the incident wave will be circularly polarised.

3.7 Determine the value of z at which the incident wave will restore its state of polarisation. What is this length called?

[**Hint:** For the incident wave to return to its original state of polarisation it is necessary that

$$\Delta\beta \cdot z = 2\pi$$

The length is called beat length,

$$L_b = \frac{2\pi}{\Delta\beta}\ ]$$

3.8 Show that for a symmetrical step-index planar waveguide structure (Fig. 3.3) in the case of TE modes

$$\left|\frac{H_z}{H_x}\right| \le \left(\frac{n_1^2 - n_2^2}{n_2^2}\right)^{\frac{1}{2}}$$

3.9 Repeat the problem (P3.8) for the TM mode case.

3.10 Show that for a symmetrical step-index planar waveguide with $n_1 = 1.46$ and $n_2 = 1.45$, the z-component of the magnetic field is less than or equal to 10% of the strength of the transverse component, H_x.

3.11 Derive the expression for the power associated with the asymmetric TE mode per unit length in the y-direction for a step-index planar waveguide (Fig. 3.3).

3.12 Show that for a symmetrical step-index planar waveguide, the power associated with the TM mode per unit length in the y-direction for both the symmetric and anti-symmetric case is given by

$$P = \frac{\beta A^2}{2\omega \in_0 n_1^2}\left[\frac{d}{2} + \frac{(n_1 n_2)^2}{w}\frac{k^2(n_1^2 - n_2^2)}{n_2^4 u^2 + n_1^4 w^2}\right]$$

3.13 Obtain the cut-off value of the V-parameter to support the fundamental TE mode.

3.14 A step-index planar waveguide has the following parameters

$$n_1 = 3.62 \text{ and } n_2 = 3.53$$

Estimate the maximum value of the thickness of the guide slab to support only the fundamental TE mode at 1300 nm.

3.15 Show that the expression for the power confinement factor can be expressed as

$$\Gamma = \frac{wa\left[1 + \frac{\sin(ua)\cos(ua)}{ua}\right]}{\cos^2(ua) + wa\left[1 + \frac{\sin(ua)\cos(ua)}{ua}\right]}$$

where $2a$ is the thickness of the planar waveguide.

3.16 Derive Eq. (3.22) for a homogeneous isotopic dielectric medium reproduced below

$$\nabla^2 \boldsymbol{H} = \mu\in\frac{\partial^2 \boldsymbol{H}}{\partial t^2}$$

3.17 Verify Eqs (3.34) to (3.36) and Eqs (3.37) to (3.39).

3.18 Derive the expressions for E_r, E_Φ, H_r, H_Φ described in Eqs (3.44)-(3.47).

3.19 Discuss the limitations of the ray analysis approach over mode analysis in the context of propagation of light through an optical fiber.

3.20 How many modes would you expect to propagate through a 75/250 μm GI fiber with a parabolic index profile at an operating wavelength of 1.5 μm? Given that the refractive index values of the fiber at the center of the core and the core–cladding interface are 1.458 and 1.443 respectively.

3.21 Discuss the cut-off conditions of various modes an optical fiber. Derive the condition for the existence of LP_{01}mode in an SI fiber under weakly guiding approximation.

3.22 What is the mode field diameter of a single-mode fiber? Explain with necessary mathematical steps the method of computation of mode field diameter of a single mode SI fiber assuming the power distribution to be Gaussian.

3.23 For a graded index fiber having a triangular index profile, show that

$$\Delta = \frac{[\mathrm{NA}(r)]^2}{2n_1^2\left(1-\frac{r}{a}\right)}$$

3.24 Explain clearly the meaning of a graded-index (GI) fiber, giving an expression for the possible refractive index profile. Using the simple ray theory concept, discuss the transmission of light through GI fiber. What are the major limitations of the ray theory?

3.25 A 40/125 micron GI fiber with a core axis refractive index of 1.5 has a characteristic index profile (α) of 1.92, a relative refractive index difference of 1%. Estimate the number of modes propagating in the fiber when the transmitted light has a wavelength of 1550 nm. Determine the cut-off value of the normalised frequency for single-mode transmission in the fiber.

3.26 What is V-number?

Consider a fiber with a 25 μm core radius, a core index $n_1 = 1.48$, and $\Delta = 0.01$.

(i) If $\lambda = 1330$ nm so, what is the value of V and how many modes propagate in the fiber?

(ii) What percent of power flows in the cladding?

(iii) If the relative core–cladding index deviation is reduced $\Delta = 0.003$, how many modes does the fiber support, and what fraction of the optical power flows in the cladding?

3.27 A 50/400 μm GI silica fiber having a parabolic index profile has $n_1 = 1.458$ and $\Delta = 0.01$. Calculate the maximum value of the numerical aperture of the fiber. How many modes propagate through this fiber at a wavelength of 0.85 μm? How does this value compare with that in a corresponding SI fiber?

3.28 Distinguish between local NA and rms value of NA in a graded index fiber. Obtain the expressions for the above parameters for an a-profile GI fiber. Estimate the rms value of a 50/125 GI fiber with triangular index profile having $n(0) = 1.465$ and $n(a) = 1.445$, a being the radius of the core.

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CHAPTER 4

Transmission Characteristics of Optical Fibers

In this chapter, the basic transmission characteristics of different types of fibers have been discussed. These characteristics largely determine the degradation of optical signals as light propagates along the fiber. In Chapter 2, we have discussed the fundamentals of optical fibers and introduced different types of optical fiber, here we explore the suitability of different types of optical fibers for optical communication system. From this perspective, the two most important transmission characteristics of an optical fiber are attenuation (or loss) and the dispersion. Attenuation limits the optical power transmitted through the fiber while dispersion restricts the bandwidth or rate at which data can be transmitted through the fiber. Both these factors play a significant role in the design of optical fiber communication link.

The feasibility of transmission of a signal through the dielectric waveguide was envisaged several decades ago (Kao *et al* 1966). However, early generation fibers exhibited high attenuation. Subsequent researches revealed that the pure silica glass has a very low intrinsic attenuation in the near infrared (NIR) region and the loss observed in early generation fibers was primarily due to the presence of a host of impurities such as Fe, Cu, Mn including hydroxyl ions. Out of these impurities, hydroxyl ions turned out to be a major contributor to the attenuation caused in glass fibers. The first breakthrough with a reasonably low attenuation (~20 dB/km) optical fiber was reported in 1970 (Kapron *et al* 1970). Extensive improvement in the fiber fabrication process enabled the design engineers to achieve fibers with a very low attenuation close to the theoretically predicted intrinsic attenuation value.

For the transmission of optical signal over a long distance requires low attenuation as well as low dispersion characteristics of the fiber. After achieving low attenuation of optical fibers attention of the researchers turned towards dispersion properties. The dispersion of light in the fiber causes temporal spreading of optical pulses and subsequently restricts the rate at which data in the form of optical pulses can be transmitted through the fiber. Extensive research on dispersion characteristics of optical fibers revealed that dispersion properties can be manipulated and tailored to a great extent unlike attenuation. Special types of optical fibers such as graded index multimode fibers can be designed appropriately to enhance the bandwidth (or bit rate) as compared to conventional step-index fibers. Further, single mode fibers provide extremely low dispersion making them very attractive for long-haul optical communication. This chapter discusses all the major issues related to attenuation and dispersion characteristics of a fiber that tends to degrade the propagating signal.

4.1 ATTENUATION OR LOSS IN AN OPTICAL FIBER

Attenuation or loss in an optical fiber primarily decides the maximum transmission distance (distance between the optical transmitter and the receiver) without using any repeater which generally restores the signal at intermediate points in a long haul communication system. Extremely low loss of optical fibers (~1 dB/km) made fiber-based optical communication more attractive as compared to conventional electrical communication systems based on metal cables which generally offer attenuation in the range of 3–5 dB/km.

The attenuation or loss in an optical fiber is measured in terms of decibel (dB) in a way similar to that measured for any other communication channel. Ideally, when light travels through an optical fiber, the power decreases exponentially with the distance traversed by the light. Assume an optical fiber through which the light propagates along the length (z-direction). If $P(0)$ is the optical power launched in a fiber at $z = 0$ of the optical power available at a point z away from the input end would be given by

$$P(z) = P(0) \exp(-\alpha_n z) \tag{4.1}$$

where α_n is the attenuation coefficient of the fiber which is a function of wavelength, given by

$$\alpha_n = \frac{1}{z}\ln\left[\frac{P(0)}{P(z)}\right] \tag{4.2}$$

As the power ratio in neper (NP) is expressed as

$$= \frac{1}{2}\ln\left[\frac{P(0)}{P(z)}\right] \tag{4.3}$$

Accordingly, the product $2za_n$ can be expressed in terms of neper.

The most convenient way of expressing the attenuation of an optical signal in an optical fiber is to use decibles. In this form, the attenuation or loss in a fiber can be expressed as

$$\alpha(\text{dB / km}) = \frac{10}{L}\log_{10}\left[\frac{P(0)}{P(L)}\right] \tag{4.4}$$

where L is the length along the fiber traversed by the light. Alternatively, in terms of power ratio, we may write

$$\left[\frac{P(0)}{P(L)}\right] = 10^{\alpha L(\text{dB})/10} \tag{4.5}$$

An ideal fiber has no attenuation and therefore, $P(z) = P(0)$. It may be pointed out here that practical optical fibers are generally passive components (excepting active fibers) in the sense that optical power decreases as it propagates through the fiber that is, $P(z) < P(0)$. To obtain the attenuation (in dB) as a positive quantity, it is customary to express the ratio in terms of the input to output power. Equation (4.5) is convenient to convert the attenuation or loss from dB to a simple power ratio. The logarithmic representation of attenuation or loss has the advantage that the operation of multiplication and division can be translated in terms of addition and subtraction and similarly, the powers and roots are reduced to multiplication and division. This kind of representation is very convenient in power budgeting of optical fiber links. The optical power level used in optical communication system is generally very low. Therefore, it is often convenient to express optical power in terms of dBm which corresponds to decibel power with respect to 1 mW reference power. The power in dBm can be expressed as

$$\text{Power (in dBm)} = 10\ \log\left(\frac{P}{1\text{ mW}}\right) \tag{4.6}$$

It can be easily verified that when $P = 1$ mW, the power in dBm corresponds to zero. That is,

$$1\text{ mW} = 0\text{ dBm} \tag{4.7}$$

Example 4.1 A 50 km long optical fiber link operating at 850 nm offers an average attenuation of 0.5 dB/km. An optical power of 100 μW is launched into the fiber at the input. What is the value of optical power at a distance of 30 km from the input? Also express the power in μW and in dBm What is the output power at the end of the link?

Solution Using Eq. (4.5), we get

$$\frac{P(0)}{P(30\text{ km})} = 10^{\frac{0.5\times 30}{10}}$$

That is, $P(30\text{ km}) = 10^{-1.5} \times 100\ \mu\text{W} = 3.16\ \mu W$

The power in dBm can be obtained using Eq. (4.6) as

$$P(30\text{ km})\ (\text{in dBm}) = 10\log\left(\frac{3.16\times 10^{-6}}{10^{-3}}\right) = -25\text{ dBm}$$

Alternatively, we may express the input power in dBm as

$$P(0)\ (\text{in dBm}) = 10\log\left(\frac{100\times 10^{-6}}{10^{-3}}\right) = -10\text{ dBm}$$

Therefore, the power in dBm at a distance of 30 km can be obtained as

$$P(30\text{ km})\text{ in dBm} = -10\text{ dBm} - (0.5\text{ dB/ km})\,(30\text{ km}) = -25\text{ dBm}$$

The output power at the end of the link can be obtained as

$$\frac{P(0)}{P(50\text{ km})} = 10^{\frac{0.5\times 50}{10}}$$

That is, $P(50\text{ km}) = 10^{-2.5} \times 100\ \mu\text{W} = 0.316\ \mu\text{W}$
The output power in dBm can be obtained as
$P(50\text{ km})$ in dBm = –10 dBm – (0.5 dB/km) (50 km) = –35 dBm

Example 4.2 150 μW optical power is launched at the input of a 10 km long optical fiber link operating at 850 nm. The output power available is 5 μW. Estimate the total attenuation in dB over the link length neglecting all connector and splice losses. What is the average attenuation per km?

Solution The average attenuation of the fiber can be obtained as

$$\alpha = \frac{10}{10}\log\left(\frac{150\times 10^{-6}}{5\times 10^{-6}}\right) = 1.48\text{ dB / km}$$

The total loss over the link length is

$$\alpha \times L = 1.48 \times 10 = 14.8\text{ dB}$$

Example 4.3 In the previous problem (Example 4.2), assume that the 10 km long link is made by splicing 5 pieces of fibers each of 2 km length. Each splice joint gives a loss of 0.8 dB. Calculate the total signal loss over the link. What is the output power in presence of splice loss in μW? In dBm?

Solution The 5 pieces will have four splice joints. The additional loss over the link would be

$$\alpha_{\text{splice}} = 0.8 \times 4\text{ dB} = 3.2\text{ dB}$$

The total loss over the link length in presence of the splice loss would be

$$\alpha' = 14.8 + 3.2 = 18.0\text{ dB}$$

The output power in presence of splice loss can be estimated as

$$\frac{P(0)}{P'(10\text{ km})} = 10^{18/10} = 63$$

That is,

$$P'(10\text{ km}) = 2.38\ \mu\text{W}$$

The corresponding power in dBm can be obtained as

$$P'(10\text{ km})\ (\text{in dBm}) = 10\log\left(\frac{2.38\times10^{-6}}{10^{-3}}\right) = -26.23\text{ dBm}$$

Alternatively,

$$P(0\text{ km})\ (\text{in dBm}) = 10\ \log\left(\frac{150\times10^{-6}}{10^{-3}}\right) = -8.23\text{ dBm}$$

Therefore,

$$P'(10\text{ km})\ (\text{in dBm}) = -8.23\text{ dBm} - 18\text{ dB} = -26.23\text{ dBm}$$

4.2 CAUSES OF ATTENUATION IN FIBERS

Attenuation in optical fibers is viewed as transmission loss and causes a reduction in the intensity of the optical signal as it propagates along the length. Attenuation in optical fibers is caused primarily by absorption and scattering. Additional factors such as bending (both micro- and macrobending) and compositional variations in core and cladding also affect the overall attenuation in optical fibers. The principal material used for making optical fiber is glass. The following discussion is therefore primarily focused on glass fiber. A typical plot of attenuation *versus* wavelength for a standard glass fiber is shown in Fig. 4.1. A good quality single mode fiber exhibits an attenuation of 0.5 dB/km at 1300 nm and an attenuation as low as 0.3 dB/km at 1550 nm. All practical fibers generally exhibit an attenuation peak corresponding to OH^- ion absorption around 1400 nm.

4.2.1 Absorption Loss

It is a mechanism by which the light energy is lost in the propagating medium through a variety of processes. Typically, light in the form of photons transfers their energy to electrons or constituent atoms of the material. Absorption of light in optical fibers is

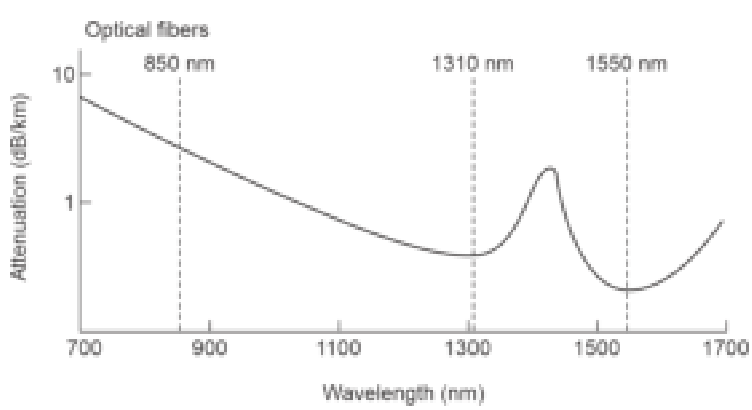

Fig. 4.1 Attenuation *vs* wavelength of a typical glass fiber showing major attenuation windows and absorption peak caused by water molecules

generally classified as intrinsic or extrinsic absorption. Absorption may also be caused by defects already present or created in the constituent material of the fiber.

4.2.2 Intrinsic Absorption

Intrinsic absorption refers to the absorption caused by basic fiber-material (for example: SiO_2) when it is in the purest form and does not contain any impurities or imperfections. This is the fundamental transmission limit of the material and no practical fiber made of this material can exhibit lower attenuation than that caused by intrinsic absorption. There are two major intrinsic absorption mechansims, e.g.

(i) Electronic absorption in the ultraviolet region
(ii) Atomic absorption in the infrared region

4.2.2.1 Electronic absorption

This involves absorption of photons that results in excitations of electrons from the valence band of glass to the conduction band. The amorphous glass is viewed as an insulator having a large bandgap. Electronic absorption takes place when a photon associated with the propagating light interacts with an electron in the valence band and transfers its energy to the electron to excite it to a higher energy state in the conduction band. This type of absorption needs a relatively high energy photon because of the large bandgap. This absorption is significant in the ultraviolet region (high frequency or small wavelength) for glass. The ultraviolet absorption near the absorption edge is governed by the standard Urbach's rule and given by the empirical relation (Olshansky, 1979)

$$\alpha_{uv} = C \exp\left(\frac{h\nu}{E_0}\right) \tag{4.8}$$

where C and E_0 are empirical constants and $h\nu$ is the energy of the photon, h is the Planck's constant and ν the frequency of the photon given by

$$\nu = \frac{c}{\lambda} \tag{4.9}$$

As the wavelength of the light increases, the frequency as well as the energy decreases and as a result absorption decreases exponentially. This is illustrated in Fig. 4.2.

4.2.2.2 Atomic absorption

This type of absorption is associated with the characteristic vibrational frequency of the chemical bond involving the constituent atoms of the material. At a particular temperature, the molecular bonds vibrate with a certain characteristic frequency. When light in the form of an electromagnetic wave propagates through the material, at some frequency it so happens that the former loses energy by transferring its energy to the vibrating bonds. The loss of energy by this mechanism is generally dominant in the infrared region and is manifested in the form of attenuation caused by absorption at the atomic level. The infrared absorption loss is given by (Ohashi, 1992)

$$\alpha_{IR} = C \exp\left(-\frac{D}{\lambda}\right) \tag{4.10}$$

where C and D are empirical constants. With increase in wavelength of the light, the loss due to atomic absorption increases very fast. The absorption wavelength depends on the constituent bonds. The fundamental absorption wavelengths in high quality silica glass for B-O, P-O, Si-O and Ge-O bonds are reported to be 7.3, 8.0, 9.0 and 11.0 μm respectively

(Osanai, 1976). It can be seen from Fig. 4.2 that the loss due to atomic absorption is very low in the near infrared (NIR) region (below 1.7 μm). This is the primary reason behind the use of NIR (0.7–1.6 μm) band for silica-based optical fiber communication. The absorption of optical signal by electronic and atomic absorption can be reduced significantly by changing the composition of glass. Optical fibers based on heavy metal halides exhibit very low loss even far beyond the mid-infrared region.

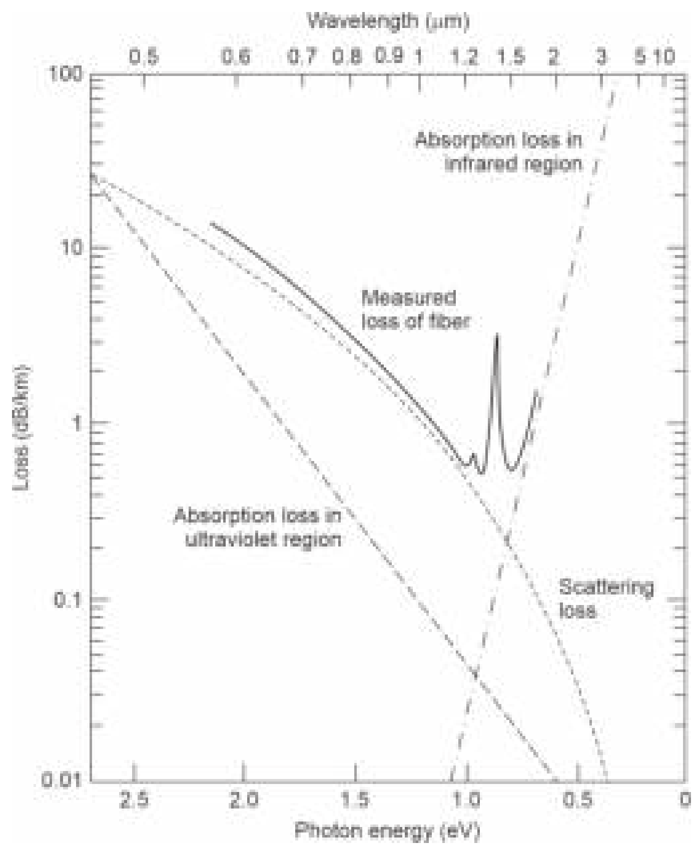

Fig. 4.2 Optical attenuation characteristics illustrating the contributions of intrinsic absorption loss in the UV and infrared regions alongwith scattering loss due to Rayleigh scattering vis-à-vis measured loss in a practical fiber

4.2.3 Extrinsic Absorption

An optical signal can also be absorbed in fibers by impurities present in the fiber material. These impurities may come from the raw material used for making the fiber or from contamination arising out of improper processing. This type of absorption is called extrinsic because the absorption is caused by external elements which are not the constituents of the intrinsic glass. Fibers fabricated by traditional melting technique generally contain trace amounts of transition elements (chromium, copper, iron, nickel etc) which give rise to significant attenuation. Extrinsic absorption caused by commonly present metal impurities in glass fiber is listed in Table 4.1 (Schultz, 1973). The impurity-induced extrinsic absorption may occur because of electronic transition between the energy levels associated with partially filled subshells or because of charge transition from one impurity ion to the other.

Table 4.1 Absorption caused by common metal impurities

Metal ions	*Peak wavelength (nm)*	*Loss (dB/km) (one part per billion)*
Fe^{2+}	1100	0.68
Cu^{2+}	850	1.1
V^{4+}	725	2.7
C^{2+}	685	0.1
Ni^{2+}	650	0.1
Cr^{3+}	625	1.6
Mn^{3+}	460	0.2
Fe^{3+}	400	0.15

It can be seen from the table that impurities such as Fe^{2+}, Cu^{2+} affects the transmission in the desired NIR region used in optical fiber communication. Extrinsic absorption by other metal ions affects the transmission in the visible region.

The other major component that causes significant extrinsic absorption in optical fibers is hydroxyl ion (OH^-). Hydroxyl ion contamination may result from the use of oxyhydrogen flame for hydrolysis reactions of $SiCl_4$ and $GeCl_4$. The excessive loss exhibited by early fibers was found to be due to the presence of a large amount of hydroxyl ions. The number of hydroxyl must be reduced to the order of only a few parts per billion to keep the attenuation of the fiber to an acceptable limit. The hydroxyl ions get bonded in the glass structure and cause fundamental absorption peaks at 1380 nm, 950 nm and 720 nm (Keck *et al* 1973). By reducing the hydroxyl ion content to the level of 1 ppb (part per billion) it is possible to fabricate high quality single mode silica fiber to offer a loss to the tune of 0.5 dB/km in the window near 1330 nm and about 0.2 dB/km in the window near 1550 nm which is very close to the intrinsic attenuation of 0.18 dB/km for silica fiber (Beales *et al* 1980).

4.2.4 Defect Loss

In addition to intrinsic and extrinsic loss, a fiber may suffer from additional loss induced by atomic defects arising out of imperfections in the atomic structure which may include a missing atom or a molecule, high density cluster of atoms or oxygen deficiencies, etc. The loss due to these factors is generally low in good quality fibers. However, if the fibers are subjected to high energy ionizing radiations such as cosmic radiation and nuclear radiation, this component of the loss becomes significant. In many practical applications, the fibers are actually subjected to this type of ionizing radiation. For example, optical fibers are sometimes used in nuclear reactors where they are exposed to numerous ionizing radiations. Similarly, fibers used in satellites are often subjected to cosmic radiation in the Van Allen belt, etc. A high radiation dose may cause a significant amount of loss by creating defect centres in the fiber. The dose of ionizing radiation received by a material is expressed in terms of the unit of rad. Thus, the dose corresponding to 1 rad (Si) refers to the radiation energy absorbed in bulk silicon, defined as

$$1 \text{ rad (Si)} = 100 \text{ erg}\cdot\text{g}^{-1} = 0.01 \text{ J}\cdot\text{kg}^{-1} \tag{4.11}$$

It is reported that the attenuation caused by ionizing radiation increases with the increase in the total radiation dose received by the fiber and the attenuation may be as high as 5 dB/km when the total radiation dose is 10^4 rad (Si) (West *et al* 1994).

4.2.5 Scattering Loss

Unlike loss due to absorption, scattering loss generally occurs when the propagating light wave interacts with a particle the fiber material in a manner that the energy is transferred in a different direction. In an optical fiber this is viewed as the transfer of optical power from one mode to the other. In many cases, the transfer of power may take place from a propagating mode to a leaky or radiating mode which do not survive over a long distance and are radiated out of the fiber. Scattering thus results in a loss of optical power as the light propagates along the fiber and occurs because of microscopic variation in material density, structural non-homogeneity or compositional variations over distance of the order of wavelength of the propagating light. Scattering is generally classified under two categories: Linear scattering and non-linear scattering.

In linear scattering, the optical power transferred to a different mode is proportional to the power contained in the propagating mode. Linear scattering is characterized by the fact that there is no change in the frequency of the scattered wave because of the transfer of power from the propagating mode. Linear scattering is further classified in two categories: Rayleigh scattering and Mie scattering.

4.2.5.1 Rayleigh scattering

Rayleigh scattering, named after Physicist Lord Rayleigh, is caused by inhomogeneities that occur on a small scale compared with the wavelength of light. These microscopic variations arise from density and compositional variations and result in fluctuation in the refractive index over distances which are much less than the value of the wavelength.

Rayleigh scattering generally accounts for more than 95% of the attenuation in the optical attenuation in a fiber. When light travels in the core, it interacts with the silica molecules in the core and the elastic collisions lead to Rayleigh scattering. If the scattered light does not fall within the angle accepted by the fiber, it deviates from the direction of propagation leading to loss of optical power. It may so happen sometimes that the scattered light is reflected back towards the source. The scattered light in such cases can be used to detect the presence of defects in an optical fiber and is the underlying principle of operation of an optical time domain reflectometer (OTDR). Rayleigh scattering in glass is similar to scattering of sunlight that makes the sky look blue. The scattering due to density fluctuation occurs in all directions and results in an attenuation that is proportional to λ^{-4}. For a single component glass, the Rayleigh scattering coefficient, γ_R is given by (Olshansky, 1979)

$$\gamma_R = \frac{8\pi^3}{3\lambda^4} n^8 p^2 \beta_c k T_F \tag{4.12}$$

where λ is the wavelength of the propagating light, n is the refractive index of the medium, p is the average photoelectric constant, β_c is the isothermal compressibility at a fictive temperature[1] T_F and k is the Boltzmann's constant. The transmission loss factor can be calculated by using the Rayleigh scattering coefficient as (Gagliardi *et al* 1976)

$$\mathcal{L} = \exp(-\gamma_R L) \tag{4.13}$$

where L is the length of the fiber. It can be easily seen from Eq. (4.12) that the effect of Rayleigh scattering is strongly influenced by wavelength of operation. The effect reduces at longer wavelength. Therefore, to minimise the effect of Rayleigh scattering it is always

[1] The temperature at which the glass can reach a state of thermal equilibrium.

desirable to operate at the longest wavelength in the permissible wavelength band. The corresponding attenuation in decibels per unit length due to Rayleigh scattering can be calculated from

$$\alpha_{RS} = 10 \log \left(\frac{1}{\mathcal{L}} \right) \tag{4.14}$$

For silica glass, the loss due to Rayleigh scattering at $\lambda = 0.6328$ μm is reported to be 3.9 dB/km (Schroeder *et al* 1973). The loss due to Rayleigh scattering usually dominates the overall loss in an optical fiber below $\lambda = 1$ μm. The infrared absorption loss on the other hand dominates over loss due to Rayleigh scattering beyond 1μm. This is illustrated in Fig. 4.3.

Example 4.4 A silica fiber operating at 650 nm has a core refractive index of 1.46. The photoelastic coefficient and isothermal compressibility of the silica glass are 0.3 and 7×10^{-11} m^2/N respectively. Estimate the loss due to Rayleigh scattering in the fiber assuming the fictive temperature of glass to be 1400 K.

Solution The Rayleigh scattering coefficient can be estimated as

$$\gamma_R = \frac{8 \times (3.14)^3 \times (1.46)^8 \times (0.3)^2 \times 7 \times 10^{-11} \times 1.38 \times 10^{-23} \times 1400}{3 \times (650 \times 10^{-9})^4}$$

$$= 1.161 \times 10^{-3} \text{ m}^{-1}$$

The transmission loss factor over 1 km length of the fiber is

$$\mathcal{L} = \exp(-1.161 \times 10^{-3} \times 10^3) = 0.313$$

The attenuation due Rayleigh scattering can be accordingly calculated with the help of Eq. (4.14) as

$$\alpha_{RS} = 10 \log \left(\frac{1}{0.313} \right) = 5.04 \text{ dB/km}$$

Example 4.5 Compare and contrast the values of attenuation due to Rayleigh scattering at optical wavelengths of 1 μm and 1.33 μm using the data given in Example 4.4 and assuming that the refractive index is independent of wavelength.

Solution It can be easily verified that at $\lambda = 1$ μm = 1000 nm, the Rayleigh scattering coefficient will be scaled as

$$\gamma_R = 1.161 \times 10^{-3} \times \left(\frac{650}{1000} \right)^4 = 0.207 \times 10^{-3} \text{ m}^{-1}$$

Therefore, transmission loss factor at 1 μm is

$$\mathcal{L}(\lambda = 1 \text{ μm}) = \exp(-0.207 \times 10^{-3} \times 10^3) = 0.813$$

The attenuation due to Rayleigh scattering at this wavelength would be

$$\alpha_{RS}(\lambda = 1 \text{ μm}) = 10 \log \left(\frac{1}{0.813} \right) = 0.89 \text{ dB/km}$$

For operation at $\lambda = 1.33$ μm = 1330 nm, the corresponding parameters translate as

$$\gamma_R = 1.161 \times 10^{-3} \times \left(\frac{650}{1330} \right)^4 = 0.066 \times 10^{-3} \text{ m}^{-1}$$

$$\mathcal{L}(\lambda = 1.33\ \mu\text{m}) = \exp(-0.066 \times 10^{-3} \times 10^{3}) = 0.936$$

$$\alpha_{RS}(\lambda = 1.33\ \mu\text{m}) = 10 \log\left(\frac{1}{0.936}\right) = 0.29\ \text{dB/km}$$

From the above example, it can be seen that the loss due to Rayleigh scattering can be significantly reduced by operating at the longest possible wavelength.

4.2.5.2 Mie scattering

Mie scattering named after German physicist Gustav Mie, is the other form of linear scattering which is less common in high quality optical fibers. Mie scattering occurs due to inhomogeneities which are comparable in size to the guided wavelength. For optical fibers, such inhomogeneities may arise due to imperfections caused by the manufacturing process and may include irregularities at core–cladding interface, index difference between core and cladding, presence of bubbles, irregular size of the ore, etc. Mie scattering becomes significant when the size of the irregularities exceeds $\lambda/10$. Mie scattering can be controlled significantly by controlling the irregularities (Senior, 2008).

4.2.5.3 Non-linear scattering loss

It is generally believed that optical fibers behave as linear waveguides in the sense that the output power increases proportionately with the increase in input optical power. This is not always true. Several non-linear effects such as non-linear scattering become dominant at high optical power levels. The non-linear scattering results in the transfer of power from one mode to at a different frequency. The optical power may also be transferred from a mode in either forward or backward direction. Two types of non-linear scattering, e.g. stimulated Brillouin scattering (SBS) and stimulated Raman scattering (SRS) are generally observed in long single mode optical fibers at high power levels. Non-linear scattering results in transfer of power from one mode to another at a different frequency. Therefore, non-linear scattering mechanism can be exploited to give optical gain but at a different frequency at the expense of attenuation of light transmission at a particular wavelength. From this view point, non-linear scattering is undesirable so far as conventional optical communication is concerned. However, the power level required for non-linear scattering to dominate is generally much above the level of power used in practical optical communication systems. As a result, the contribution of non-linear scattering in the total attenuation (or loss) in an optical fiber remains unnoticed. It is interesting to note that non-linear scattering can be used to provide optical amplifiers which find extensive application in long distance optical communication systems.

4.2.5.4 Stimulated Brillouin scattering (SBS)

This type of scattering (SBS) occurs from the scattering of the propagating light by thermal molecular vibrations of the material. The interaction of the photon with the vibrating molecules of the material results in a phonon of acoustic frequency as well as a scattered photon of a different energy (or wavelength). The spectrum of the scattered light thus appears in the form of upper and lower sidebands which are separated from the incident light by the modulation frequency. For SBS the frequency shift is a maximum in the backward direction and zero in the forward direction. Therefore, SBS is viewed as a backward process.

The threshold power required for stimulated Brillouin scattering to occur depends on the wavelength of the operating wavelength and the line width of the optical source.

Assuming that the polarization of the scattered light is not maintained, the threshold power, P_B required for SBS can be obtained as (Stolen, 1979)

$$P_B = 4.4 \times 10^{-3}\, d^2 \lambda^2\, \alpha_{\text{dB}}\, \Delta v \text{ (watt)} \tag{4.15}$$

where d is the core diameter in micrometer, λ is the operating wavelength in micrometer, α_{dB} is the fiber attenuation in decibel per kilometer and Δv is the line-width (in GHz) of the injection laser source.

4.2.5.5 Stimulated Raman scattering (SRS)

This scattering is associated with the generation of a high frequency optical phonon, unlike stimulated Brillouin scattering which is associated with the generation of an acoustic phonon. In contrast with SBS, SRS may occur both in the forward as well as in the backward direction and requires threshold optical power which is several orders higher than that required for SBS to occur. The threshold optical power needed for SRS can be obtained as (Stolen, 1980)

$$P_R = 5.9 \times 10^{-2}\, d^2 \lambda \alpha_{\text{dB}} \text{ (watt)} \tag{4.16}$$

where the symbols are same as specified following Eq. (4.15).

Example 4.6 Light is launched from an injection laser diode operating at 1.55 μm to a 8/125 μm single mode fiber. The bandwidth of the laser source is 500 MHz. The single mode fiber offers an average loss of 0.3 dB/km. Estimate the values of threshold optical power for the cases of stimulated Brillouin scattering and stimulated Raman scattering.

Solution The threshold power required for SBS can be obtained as

$$P_B = 4.4 \times 10^{-3} \times 8^2 \times (1.55)^2 \times 0.4 \times 0.5 \text{ W}$$
$$= 135.3 \text{ mW}$$

The threshold power required for SRS can be obtained as

$$P_R = 5.9 \times 10^{-2} \times 8^2 \times 1.55 \times 0.4 \text{ W}$$
$$= 2.34 \text{ W}$$

From the above example, it can be seen that the threshold optical power level required to be launched into the fiber for SRS to occur is much higher than that required for SBS. It is interesting to note that both the threshold values are much above the power generally used in optical communication. Therefore, SBS and SRS do not contribute to attenuation in optical fiber communication.

From the above discussion, it is apparent that both the intrinsic loss as well as loss due to Rayleigh scattering are important factors that determine the overall attenuation of the fiber in the near-infrared region used for optical communication. Further Rayleigh scattering depends not on the type of material and also on the relative size of the particles with respect to the wavelength of operation. The loss due to Rayleigh scattering decreases rapidly with increase in wavelength because of its dependence in the form of λ^{-4}. As a result shorter wavelengths are scattered more as compared to longer wavelengths. It can be seen from Fig. 4.3, that light signal with wavelength below 800 nm is unusable for optical communication because attenuation due to Rayleigh scattering is too high. Figure 4.3 shows that the overall attenuation of a practical fiber matches closely with the predicted value of attenuation. The attenuation peak near 1400 nm is due to absorption by residual water molecules in the fiber.

4.3 BENDING LOSS

Additional loss in optical fibers may occur from bends in optical fibers. The bends in optical fibers can be classified into two categories: (i) microscopic bends, which have small radii of curvature and comparable to fiber diameter and (ii) macroscopic bends which have radii of curvature much longer than the core diameter. Both micro- and macrobending can cause significant attenuation in optical fibers.

4.3.1 Microbending Loss

Microbend loss is caused by small-scale variations in the radius of curvature of the fiber. These variations are created by non-uniform lateral forces often encountered by the fiber during manufacturing and/or cabling processes. These irregularities may also be caused by non-uniform speed of the fiber drawing machines during fabrication. This kind of microbends may act as a facilitator for coupling power from a guided mode to a leaky or unbound mode causing significant attenuation of optical power (Fig. 4.3). Fibers containing microbends can be made relatively flat by using a compressible plastic jacket and applying appropriate external forces (Gloge, 1975). This method can significantly reduce the loss in optical fiber caused by microbends. The loss due to microbends may be as high as 1–2 dB/km.

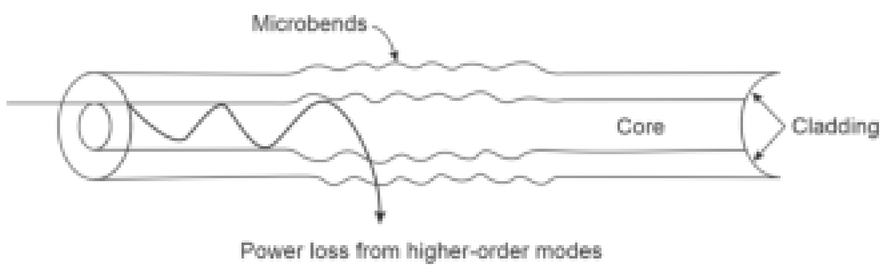

Fig. 4.3 Microbending loss from higher order mode (Keiser)

4.3.2 Macrobending Loss

Macrobend (large bend) occurs when a fiber is bent into a relatively large radius of curvature with respect to the fiber diameter. These bends can cause a significant power loss when the radius of curvature falls below a certain critical value. Macrobends are formed when the fibers are wound in the form of a spool or a fiber cable roll. The power losses in these cases do not cause significant radiation loss if the radius of curvature is large enough. The macrobends can also be caused when a fiber cable is bent uniformly to take a turn. The bending loss is primarily due to radiation of energy from the fiber when the evanescent field fails to keep up pace with the part of the mode varying harmonically in the core. This can be qualitatively understood with the help of Fig. 4.4 illustrating the propagation of a mode. A mode is considered as an electromagnetic field pattern created in the transverse direction which varies harmonically in the core region and decay exponentially in the cladding region. A mode is considered to be bound when the evanescent field tail in the cladding region moves alongwith the part moving within the core. Consider a fiber is uniformly bent as illustrated in Fig. 4.4. The field tail on the other side of the centre of curvature is required to move faster relative to the part on the inner side in order to keep up with the part moving through the core region. This is only

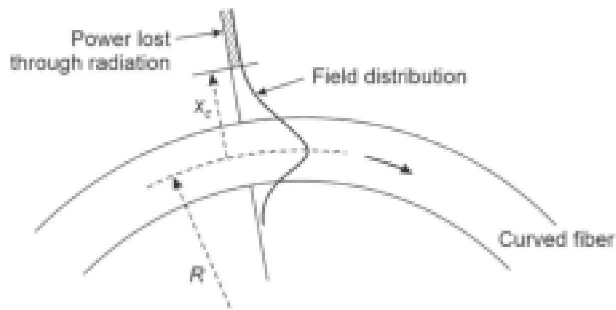

Fig. 4.4 Illustrating macrobending loss with the help of the fundamental mode field

possible upto a certain critical value of bending decided by the radius of curvature of the bending. When the radius of curvature is smaller than a certain critical value decided by the refractive indices of the core and the cladding as well as the wavelength of operation, a situation may so arise that the field tail needs to move with a speed larger than the speed of light in order to catch up the part moving in the core region. Since this is not possible, the field tail under such a condition is radiated out of the fiber causing a loss of optical power propagating through the fiber.

The bending loss depends on the radius of curvature and can be empirically expressed as (Ramsay *et al* 1980)

$$\alpha_r = c_1 \exp(-c_2 R) \tag{4.17}$$

where c_1 and c_2 are empirical constants and R is the radius of curvature of the bending.

For a multimode fiber, the critical value of the radius of curvature of the macrobending of a fiber is given by (Wolf, 1979)

$$R_c = \frac{3n_1^2 \lambda}{4\pi\sqrt{n_1^2 - n_2^2}} \tag{4.18}$$

where n_1 and n_2 are the values of the refractive index of the core and the cladding and λ is the wavelength of the light propagating through the fiber. When the bending is so large that the radius of curvature falls below this critical value, the bending loss of the fiber tends to become very large.

For a single mode fiber the critical radius of curvature can be obtained as (Gambling, 1979)

$$R_{cs} = \frac{20\lambda}{\sqrt{n_1^2 - n_2^2}}\left(2.748 - 0.996\frac{\lambda}{\lambda_c}\right)^{-3} \tag{4.19}$$

where λ_c is the critical wavelength.

The effective number of modes guided by a curved graded index fiber has been derived by Gloge (Gloge, 1972) as

$$M_{\text{eff}} = M_\infty\left[1 - \frac{\alpha+2}{2\alpha\Delta}\left\{\frac{2a}{R} + \left(\frac{3}{2n_2 kR}\right)^{\frac{2}{3}}\right\}\right] \tag{4.20}$$

where α is the grade index of the GI fiber, Δ is the index deviation and $k(=2\pi/\lambda)$ and R is the radius of curvature of the bending. M_∞ corresponds to the number of modes through a graded-index straight fiber given by

$$M_\infty = a^2 k^2 n_1^2 \Delta\left(\frac{\alpha}{\alpha+2}\right) \tag{4.21}$$

Example 4.7 A 50/125 μm GI fiber with a parabolic index profile has a core refractive index of 1.458 at the centre of the core and a relative index deviation of $\Delta = 0.01$. Estimate the number of modes supported by the fiber at 850 nm. The fiber is now uniformly bent with a radius of curvature of 2 cm. Calculate the expected number of modes to be radiated out due to bending of the fiber.

Solution The number of modes in the GI fiber with parabolic index profile ($\alpha = 2$) under straight conditions can be estimated using Eq. (4.21) as

$$M_\infty = (25\times10^{-6})^2 \times\left(\frac{2\times3.14}{850\times10^{-9}}\right)^2 \times 0.01\times\frac{2}{2+2}$$
$$\approx 170$$

The number of modes when the fiber is bent with a radius of curvature $R = 2$ can be calculated from Eq. (4.20) as

$$M_{\text{eff}} = 170\times\left[1-\frac{2+2}{2\times2\times0.01}\left\{\frac{2\times25\times10^{-6}}{2\times10^{-2}}+\left(\frac{3\times850\times10^{-9}}{2\times1.458\times(1-0.01)\times2\times3.14\times2\times10^{-2}}\right)^{\frac{2}{3}}\right\}\right]$$
$$\approx 126$$

This means that nearly 25% modes will be radiated out of the fiber because of bending.

Example 4.8 A 62.5/125 μm step-index fiber has core and cladding refractive index values of 1.50 and 1.48 respectively at a wavelength of operation of 1330 nm. Estimate the value of the critical radius of curvature from the viewpoint of macrobending loss.

Solution The critical radius of curvature beyond which bending loss becomes very high, can be estimated by using Eq. (4.18) as

$$R_c = \frac{3\times(1.5)^2\times1330\times10^{-9}}{4\times3.14\times\sqrt{(1.5)^2-(1.48)^2}}$$
$$= 2.92\ \mu\text{m}$$

4.4 CORE-CLADDING LOSS

In a practical fiber, the total loss is contributed by all kinds of dissipative and scattering mechanisms involving the core and the cladding regions of the fiber. The core and the cladding carry different amounts of optical power. Further, the refractive indices of the core and so their attenuation coefficient are also different. As a result, the attenuation coefficients of the two regions must be considered to be different for calculation of overall attenuation of a particular mode propagating through the fiber. In the absence of mode coupling, the attenuation coefficient for a mode of order (l, m) can be expressed as (Gloge, 1975)

$$\alpha_I(l, m) = \alpha_1\left(\frac{P_{\text{core}}}{P}\right) + \alpha_2\left(\frac{P_{\text{clad}}}{P}\right) \tag{4.22}$$

where α_1 and α_2 are the attenuation coefficients in dB/km in the core and the cladding regions, P_{core}/P and P_{clad}/P correspond to the fractional power carried by the core and the cladding.

Equation (4.22) can also be expressed using Eq. (3.240) as

$$\alpha_I(l, m) = \alpha_1 + (\alpha_2 - \alpha_1)\left(\frac{P_{\text{clad}}}{P}\right) \tag{4.23}$$

For a graded-index fiber the loss is expected to follow the variation of the refractive index along the radius. Accordingly, the loss at any distance r from the axis of the core can be written as (Gloge, 1975)

$$\alpha(r) = \alpha_1 + (\alpha_2 - \alpha_1)\frac{n^2(0) - n^2(r)}{n^2(0) - n_2^2} \tag{4.24}$$

where α_1 and α_2 are the axial and cladding attenuation coefficients of the GI fiber respectively.

The overall attenuation exhibited by a given mode can be obtained as (Gloge, 1975)

$$\alpha_{GI} = \frac{\int_0^\infty \alpha(r)p(r)rdr}{\int_0^\infty p(r)rdr} \tag{4.25}$$

where $p(r)$ is the radial distribution of power of that particular mode.

4.5 DISPERSION

In addition to the attenuation, the transmission of optical signal through an optical fiber is adversely, by dispersion of the signal by the dielectric medium. For example, when optical signal in the form of a pulse of optical power of a certain duration travels through an optical fiber, its power gets dispersed resulting in a spreading of the pulse into a wider time interval. Dispersion is essentially broadening of light pulses and is a critical factor that limits the quality of signal transmission through an optical link. The physical properties and the geometry of the transmission medium are responsible for the dispersion that causes degradation in the quality of the signal as it propagates along the fiber. Depending on the origin, dispersion is broadly classified under two categories: intramodal dispersion and intermodal dispersion. The term intramodal dispersion refers to dispersion or spreading of the pulse that occurs within a particular mode and is generally found in all types of fibers. On the other hand, intermodal dispersion is caused by the time delay between various modes of travel to the destination point. Thus, intermodal dispersion is found to be present only in a multimode fiber which supports more than one mode to carry the optical power and the delay is caused by the time difference between the lowest and highest order modes. Intermodal dispersion is not found in a single mode fiber because it supports only one mode. Single mode fibers generally suffer from a special type of dispersion called polarization mode dispersion (PMD) arising out of the birefringence phenomenon discussed in Chapter 3.

4.5.1 Intersymbol Interference and Bandwidth

Optical communication can be either in analog or in digital form of transmission of optical signal. In the analog form of transmission, the intensity of the light is allowed to vary in a continuous fashion analogous to the information signal. On the other hand digital optical communication is achieved by transmitting the coded signal (1's and 0's) in the form of optical pulses of finite duration (bit period). Dispersion (intramodal and intermodal) affects the transmission of optical signal in case of both analog and digital transmission of optical signal. For an analog optical communication system, dispersion and attenuation affect the signal-to-noise ratio (SNR) available at the optical receiver and restricts the bandwidth to maintain the desired SNR. The spreading of pulses caused by dispersion in the optical fiber results in overlapping of the flattened pulses as they propagate through the fiber. As the undesired overstepping increases with the distance, a stage is eventually reached when the pulses become unrecognizable due to unacceptable amount of overlapping. As we will see shortly that this phenomenon leads to inter symbol interference (ISI). This overlapping finally leads to errors in making decisions regarding 1's and 0's. A large ISI may lead to the increasing number of errors. The error in digital optical communication is measured in terms of bit-error-rate (BER) which is measured in terms of the number of errors incurred in a given bit stream. For example, a BER = 10^{-9} corresponds to 1 error in a bit stream of 10^9. It can be easily appreciated that to maintain a given BER, one must restrict the rate at which the pulses are transmitted through the fiber to avoid excessive intersymbol interference. This parameter is called *bitrate* (B_T). Bitrate may be viewed as the analogue of bandwidth in the case of analog optical communication. Similarly, the bit-error rate and signal-to-noise ratio are also related to each other. The exact relationship depends on the characteristics of the channel (fiber). The effect of intersymbol interference arising from dispersion phenomenon in optical fiber is illustrated in Fig. 4.5 for the case of digital optical transmission. It can be easily seen from the figure that the rectangular pulses originally launched into the fiber get dispersed more and more as they propagate along the fiber. Further, up to a certain distance along the fiber the dispersed signals remain distinguishable. However, beyond this the overlapping of the broadened (dispersed) pulses is so large that they barely remain distinguishable. The overlapping of pulses to this extent may lead to a wrong decision regarding 1's and 0's which is represented by the presence or absence of a rectangular pulse. In the above example, the original coded signal is in the form of a bit stream 1 0 1 in the form of rectangular pulses that get so much dispersed that at an instant of time, t_4 one would experience the presence of signal in the bit slot allocated for the '0', because of overlapping of the pulses from the adjacent bits (1's). This overlapping will force one to

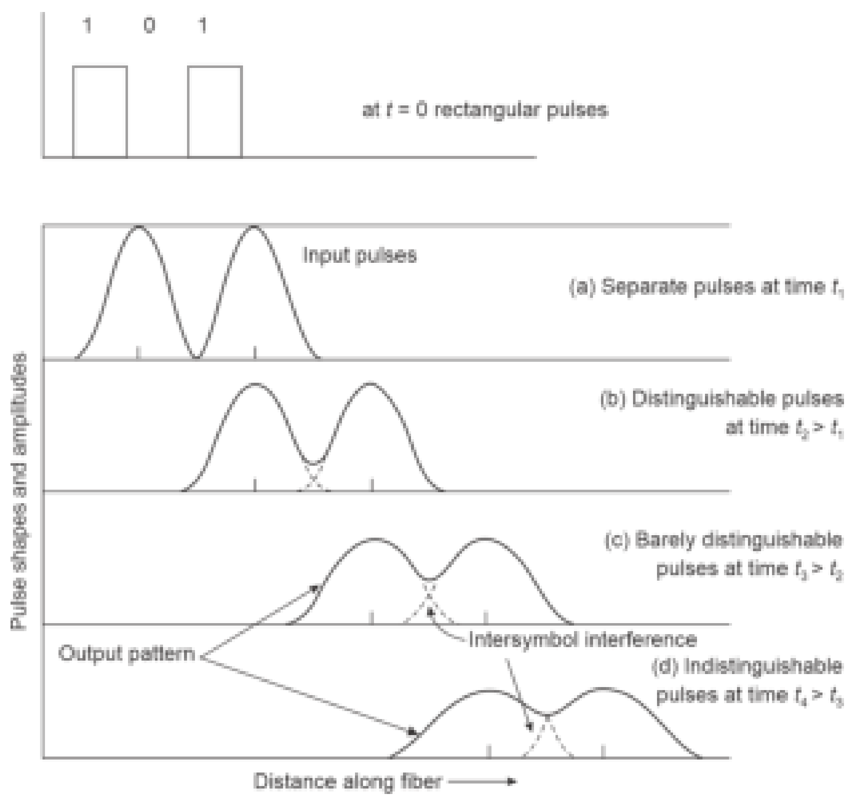

Fig. 4.5 Illustrating the dispersion of the rectangular optical pulses as they propagate along the fiber

interpret the received signal as 1 1 1 in place of 1 0 1 as was transmitted originally. This undesirable effect results in interference among the symbols and is termed as intersymbol interference (ISI). Thus, to avoid the effect of intersymbol interference one must control the rate at which the bits are being transmitted. A slower bit rate of transmission will make the separation between the pulses large and the effect of ISI will remain unnoticed. Alternatively, one may use a high quality fiber which exhibits low dispersion. Currently, high quality single mode fibers with a small value of dispersion are available. These fibers are generally used for long-haul optical communication link.

If T_b is the duration of a single pulse, a conservative estimate of the maximum bit rate that can be obtained on an optical channel without overlapping of bits as

$$B_T \le \frac{1}{2T_b} \tag{4.26}$$

The above relation is based on the assumption that the spreading of the pulses due to dispersion in the channel is also T_b. In actual practice, a realistic approximation of the maximum bit rate that would allow a tolerable amount of overlapping within the limits of the desired SNR or bit error rate can be obtained by assuming the light pulse output to be Gaussian in nature.

Consider an optical channel that disperses the optical pulses to a Gaussian shape with a variance of σ^2, which is equivalent to an rms width of σ. The Gaussian pulse shown as a function of time in Fig. 4.6 can be mathematically expressed as

$$p(t) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2\sigma^2}\right) \tag{4.27}$$

A convenient way of defining the width of the pulse in the time domain can be obtained by finding the time when the power falls to $(1/e)$ times the peak power. It can be easily seen that at

$$t = \pm\sqrt{2}\sigma, \frac{p(t = \pm\sqrt{2}\sigma)}{p(0)} = \frac{1}{e} \tag{4.28}$$

Therefore, the width of the pulse (also called $1/e$ pulse width) can be obtained as

$$\tau_e = 2\sqrt{2}\sigma \tag{4.29}$$

It may be noted that $1/e$ pulse width is different from the rms pulse width σ which corresponds to the standard deviation.

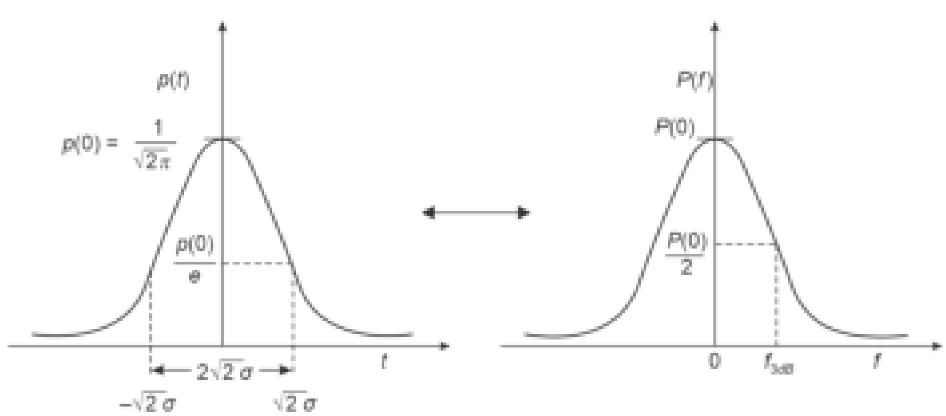

Fig. 4.6 Gaussian pulse in the time domain and its Fourier transform (also Gaussian) showing the calculation of 3-dB frequency

To find the 3-dB bandwidth of the pulse we need to find the Fourier transform of the Gaussian pulse given in the time domain. The Fourier transform of the Gaussian pulse given by Eq. (4.27) can be expressed as

$$P(f) = \frac{1}{\sqrt{2\pi}} \exp(-2\pi^2 f^2 \sigma^2) = P(0) \exp(-2\pi^2 f^2 \sigma^2) \tag{4.30}$$

The 3-dB frequency can be obtained as

$$\frac{P(f = f_{3\text{dB}})}{P(0)} = \frac{1}{2} = \exp(-2\pi^2 f_{3\text{-dB}}^2 \sigma^2)$$

That is,

$$f_{3\text{dB}} = \frac{\sqrt{\ln(2)}}{\sqrt{2}\pi\sigma} = \frac{0.187}{\sigma} \tag{4.31}$$

Thus, the optical bandwidth can be written as

$$B = f_{3\text{dB}} = \frac{0.187}{\sigma} \approx \frac{0.2}{\sigma} \text{Hz} \tag{4.32}$$

In digital optical communication, it is convenient to describe the transmission of signal over a channel in terms of bit rate rather than bandwidth. The exact relationship between the bit rate and the bandwidth depends on the pulse format (return-to-zero or non-return-to-zero) of the digital coding. For example, in a non-return to zero (NRZ) code formatting the relationship between the bandwidth (B) and bit rate (B_T) can be expressed as

$$B_T = B \quad \text{(NRZ)} \tag{4.33}$$

On the other hand, for a return to zero scheme of coding the maximum value of the bit rate that can be achieved is given by (Senior, 2008)

$$B_T(\text{max}) = 2B \quad \text{(RZ)} \tag{4.34}$$

This can be easily appreciated by examining the relationship between the wavelength to the bits in the digital coding for the NRZ and RZ schemes as illustrated in Fig. 4.7.

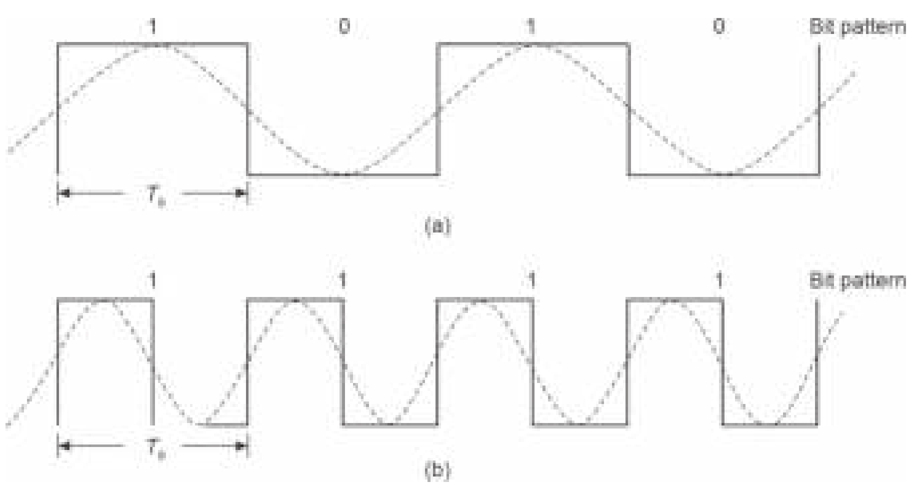

Fig. 4.7 Illustrating the relationship between digital codes and corresponding wavelength for the cases (a) Non-return to zero (NRZ) (b) Return-to-zero (RZ) schemes (after Senior)

4.5.2 Intramodal Dispersion

Intramodal dispersion also known as chromatic dispersion (CD) or group velocity dispersion (GVD) refers to pulse broadening that occurs within a mode because of the finite spectral width of the source. None of the optical sources used in optical communication systems is a strictly monochromatic source. Therefore, there will be propagation delay differences between different spectral components in the optical signal launched into the fiber from an optical source. For example, consider a multimode fiber in which light is launched from a light emitting diode operating at 850 nm with a spectral width of 40 nm. This means that the LED has a peak optical emission power at 850 nm and the power emitted by the source is essentially confined within a spectral wavelength band of 40 nm ranging from 830 nm to 870 nm. In the multimode fiber, the total optical power launched into the fiber is distributed among various modes supported by the fiber. All the modes jointly carry the power and deliver it at the destination point. Each of these modes contains all the spectral components of the light present in the source. The delay difference may be caused within a mode by the dispersive properties of the waveguide material arising from the dependence of refractive index on the wavelength of light (material dispersion) and also by guidance effects within the fiber structure (waveguide dispersion). Since both types of dispersion occur within a particular mode, they are referred to as *intramodal dispersion*. Out of these two components waveguide dispersion is generally dominant in a single mode fiber because nearly 20% of the total power flows through the cladding region and only 80% is confined in the core region. On the other hand, in a multimode fiber the power flow is negligibly small and as a result the waveguide dispersion component remains unnoticed in a multimode fiber. The waveguide dispersion can be tailored by waveguide design. Both the material and waveguide components of intramodal dispersion can be significantly reduced by using a laser source with a smaller spectral width. A good quality laser diode provides a spectral width as low as 1–2 nm for a multimode laser source and 10^{-4} nm for a single mode laser source.

In general, the dispersion mechanism in an optical fiber is quite complex and both material and waveguide dispersions are intricately related to each other and one cannot be isolated from the other. However, the analysis becomes quite involved when all the mechanisms responsible for the total spreading of the pulse are considered simultaneously. On the other hand, a simpler approach which is also fairly accurate is often adopted to compute the overall dispersion of a fiber due to intramodal effects. In this approach, each component is computed independently in the absence of the other. Excessive spreading of pulses may result in overflow from their allotted time slots. This in turn results in overlapping among the adjacent bits. The undesirable overlapping effect leads to ISI and restricts the rate at which data can be sent over the fiber.

4.5.2.1 Material dispersion

Material dispersion arises from the dependence of the refractive index of the fiber material on the wavelength of light which is not strictly monochromatic for optical sources which emit power in a finite spectral width. As a result, the spectral components propagate with different group velocities and cause broadening of pulse due to material dispersion. To estimate the effect of material dispersion on pulse broadening, consider a plane wave propagating through a homogeneous dielectric medium of refractive index n_1. The propagation constant (β) can be expressed as (Ghatak *et al* 1998)

$$\beta = \frac{2\pi}{\lambda} n_1(\lambda) \tag{4.35}$$

Therefore,

$$\frac{dB}{d\lambda} = 2\pi\left[-\frac{n_1}{\lambda^2} + \frac{1}{\lambda}\frac{dn_1}{d\lambda}\right] = -\frac{2\pi}{\lambda^2}\left[n_1 - \lambda\frac{dn_1}{d\lambda}\right] = -\frac{2\pi}{\lambda^2}N_1 \tag{4.36}$$

The parameter

$$N_1 = n_1 - \lambda\frac{dn_1}{d\lambda} \tag{4.37}$$

is called the group index because the group velocity of the wave can be expressed in terms of this parameter. For example, the group velocity can be obtained as

$$v_g = \left(\frac{d\beta}{d\omega}\right)^{-1} = \left(\frac{d\beta}{d\lambda}\frac{d\lambda}{d\omega}\right)^{-1} \tag{4.38}$$

Further,

$$\omega = 2\pi\left(\frac{c}{\lambda}\right) \tag{4.39}$$

That is,

$$\frac{d\omega}{d\lambda} = -\frac{2\pi c}{\lambda^2} \tag{4.40}$$

Substituting Eq. (4.40) into Eq. (4.38), we get

$$v_g = -\frac{2\pi c}{\lambda^2}\left(\frac{d\beta}{d\lambda}\right)^{-1} \tag{4.41}$$

Using Eqs (4.36) and (4.41), we may express the group velocity as

$$v_g = \frac{c}{N_1} \tag{4.42}$$

The group delay in the optical fiber over the length L arising from the group velocity of the spectral components can be expressed as

$$\tau_g = \frac{L}{v_g} = \frac{L}{c}N_1 = \frac{L}{c}\left(n_1 - \lambda\frac{dn_1}{d\lambda}\right) \tag{4.43}$$

For material dispersion, the group delay is denoted by τ_{mat} and therefore

$$\tau_{\text{mat}} = \frac{L}{c}\left(n_1 - \lambda\frac{dn_1}{d\lambda}\right) \tag{4.44}$$

Consider that the source used for launching power into the optical fiber operates at a wavelength λ and has a rms spectral width of σ_λ. The rms pulse broadening due to material dispersion, σ_λ can be obtained from Taylor series expansion of Eq. (4.44) about the operating wavelength, λ as

$$\sigma_{\text{mat}} = \left(\frac{d\tau_{\text{mat}}}{d\lambda}\right)\sigma_\lambda + 2\left(\frac{d^2\tau_{\text{mat}}}{d\lambda^2}\right)\sigma_\lambda + \cdots \tag{4.45}$$

Neglecting the effect of higher order derivatives which are generally small as compared to the first order derivative, particularly for sources used in the near infrared (NIR) region, we may approximate the material dispersion as

$$\sigma_{\text{mat}} \approx \left(\frac{d\tau_{\text{max}}}{d\lambda}\right)\sigma_\lambda \tag{4.46}$$

Using Eq. (4.44) we may derive the dependence of group delay due to material dispersion on wavelength as

$$\frac{d\tau_{\text{mat}}}{d\lambda} = \frac{L\lambda}{c}\left[\frac{dn_1}{d\lambda} - \frac{d^2n_1}{d\lambda^2} - \frac{dn_1}{d\lambda}\right] = -\frac{L\lambda}{c}\frac{d^2n_1}{d\lambda^2} \tag{4.47}$$

Therefore, the rms pulse broadening due to material dispersion can be obtained from Eq. (4.46) with the help of Eq. (4.47) as

$$\sigma_{\text{mat}} = \frac{L\sigma_\lambda}{c}\left|\lambda \frac{d^2 n_1}{d\lambda^2}\right| \tag{4.48}$$

The material dispersion of an optical fiber is often designated by dispersion (D) parameter defined for the material dispersion as

$$D_{\text{mat}} = \frac{1}{L}\left(\frac{d\tau_{\max}}{d\lambda}\right) = \frac{\lambda}{c}\left|\frac{d^2 n_1}{d\lambda^2}\right| \tag{4.49}$$

It should be noted that the dispersion parameter, D_{mat} is a function of wavelength and is measured as the spreading of the pulse as a function of wavelength in pico-seconds per unit length in kilometer and per unit wavelength in nanometer (ps km^{-1} nm^{-1}).

The variation of the material dispersion parameter D_{mat} with the wavelength of light is shown in Fig. 4.8 for pure silica glass (Payne *et al* 1975). It is observed that the material dispersion characteristic curve *versus* wavelength goes through zero at a wavelength of 1.27 μm. This point is called zero material dispersion (ZMD) point. It can be further seen that the material dispersion remains very small to maintain values close to zero for pure silica glass at a wavelength around 1300 nm. The second-generation optical communication was focused at this wavelength also because the attenuation at this wavelength in pure silica glass is also very low. It is apparent that the material dispersion can be reduced either by operating at a longer wavelength or reducing the spectral width of the source. Use of a laser source in place of an LED source will thereby reduce the material dispersion effect in optical fibers because of a smaller spectral width of the former.

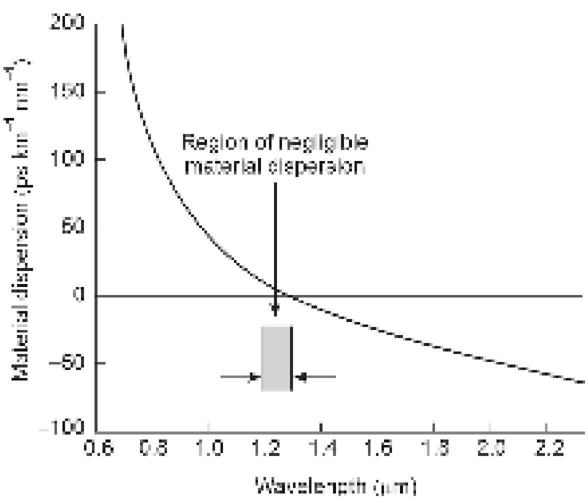

Fig. 4.8 Variation of material dispersion with wavelength for silica (after Payne *et al* 1975)

Example 4.9 A 20 km long optical fiber exhibits an rms pulse broadening of 20 ns due to material dispersion alone when the power is launched from an LED operating at 850 nm with a spectral width of 30 nm. Estimate the material dispersion parameter of the fiber.

Solution The rms pulse broadening in terms of dispersion parameter can be expressed as

$$\sigma_{\text{mat}} = \sigma_\lambda L D_{\text{mat}}$$

Therefore, the material dispersion can be estimated as

$$D_{\text{mat}} = \frac{20 \times 10^3}{20 \times 30} = 100 \text{ ps km}^{-1}\text{nm}^{-1}$$

4.5.2.2 Waveguide dispersion

The waveguide dispersion originates from the variation in group velocity with wavelength for a particular mode. Each mode can be identified with a corresponding ray which makes a particular angle with the fiber axis. When the source has a finite spectral width each of these rays will contain all the spectral components. As a result, the angle made with the fiber axis by a particular ray corresponding to a mode will also vary with the wavelength. Subsequently there will be differences between the time taken by different components leading to pulse broadening. This is known as *waveguide dispersion*. In terms of mode analysis waveguide dispersion arises because of the dependence of the propagation constant with wavelength and is found in fibers when $d^2\beta/d\lambda^2 \neq 0$. In a multimode fiber, waveguide dispersion is generally negligible as compared to material dispersion because majority modes propagate far from the cut-off. On the other hand, waveguide dispersion is significant in the case of a single mode fiber. Further, waveguide dispersion cannot be separated from material dispersion because of their interrelations. In the analysis of waveguide dispersion therefore, the dependence of refractive index on wavelength will be ignored. For evaluating the group delay due to waveguide dispersion it would be convenient to express the group delay in terms of propagation constant by using Eqs (4.41) and (4.43) as

$$\tau_g = \frac{L}{v_g} = -\frac{\lambda^2 L}{2\pi c}\left(\frac{d\beta}{d\lambda}\right) \tag{4.50}$$

Further,

$$\frac{d\beta}{d\lambda} = \left(\frac{d\beta}{dk}\right)\left(\frac{dk}{d\lambda}\right) = \frac{d}{d\lambda}\left[\frac{2\pi}{\lambda}\right]\left(\frac{d\beta}{dk}\right) = -\frac{2\pi}{\lambda^2}\left(\frac{d\beta}{dk}\right) \tag{4.51}$$

Combining Eqs. (4.50) and Eq. (4.51), the group delay for waveguide dispersion can be expressed as

$$\tau_{wg} = \frac{L}{c}\left(\frac{d\beta}{dk}\right) \tag{4.52}$$

The rms pulse spreading due to waveguide dispersion can be obtained as

$$\sigma_{wg} = \left(\frac{d\tau_{wg}}{d\lambda}\right)\sigma_\lambda \tag{4.53}$$

The normalized propagation constant for a step-index fiber can be expressed as

$$b = \frac{a^2 w^2}{V^2} = \frac{a^2(\beta^2 - k_2^2)}{a^2(k_1^2 - \beta^2 + \beta^2 - k_2^2)} = \frac{\beta^2 - k_2^2}{k_1^2 - k_2^2} \tag{4.54}$$

When the index deviation is very small that is, $n_1 \approx n_2$, the normalized propagation constant can be approximated as (Gloge, 1971)

$$b = \frac{\left(\frac{\beta}{k}\right)^2 - n_2^2}{n_1^2 - n_2^2} \approx \frac{(\beta / k) - n_2}{n_1 - n_2} \tag{4.55}$$

Rearranging Eq. (4.55), we may write

$$\beta = kn_2 + kb(n_1 - n_2) = kn_2\left[1 + b\left(\frac{n_1 - n_2}{n_2}\right)\right] \tag{4.56}$$

Further, since $n_1 \approx n_2$, the propagation constant, β can be approximated as

$$\beta \approx kn_2(1 + b\Delta) \tag{4.57}$$

Using the above expression for β and Eq. (4.52) the group delay due to waveguide dispersion can be obtained by assuming n_2 to be independent of wavelength (that is, in absence of material dispersion) as

$$\tau_{wg} = \frac{L}{c}\frac{d\beta}{dk} = \frac{L}{c}\left[n_2 + n_2\Delta\frac{d(kb)}{dk}\right] \tag{4.58}$$

Further, the normalized propagation constant b or the propagation constant β is generally expressed in terms of V-number of the fiber, given by

$$V = ak(n_1^2 - n_2^2)^{\frac{1}{2}} \approx akn_1\sqrt{2\Delta} \approx akn_2\sqrt{2\Delta} \tag{4.59}$$

Taking the derivative of V-number with respect to k and remembering the fact that n_2 is independent of wavelength, we may write

$$\frac{dV}{dk} \approx an_2\sqrt{2\Delta} \tag{4.60}$$

Equation (4.58) can be rewritten to express the group delay due to waveguide dispersion in terms of V rather than in terms of k, we may write

$$\tau_{wg} = \frac{L}{c}\left[n_2 + n_2\Delta\frac{d(kb)}{dV}\frac{dV}{dk}\right] = \frac{L}{c}\left[n_2 + n_2\Delta\frac{d(Vb)}{dV}\right] \tag{4.61}$$

The second term on the right hand side corresponds to group delay caused by waveguide dispersion. Further, the derivative term can be approximated as (Gloge[b], 1971)

$$\frac{d(Vb)}{dV} = b\left[1 - \frac{2J_l^2(ua)}{J_{l+1}(ua)J_{l-1}(ua)}\right] \tag{4.62}$$

The variation of $d(Vb)/dV$ with the V-number for various LP modes is shown in Fig. 4.9. It can be seen that for a given value of V-number the group delay is different for different modes.

The rms pulse spreading due to waveguide dispersion can be expressed as

$$\sigma_{wg} \approx \left|\frac{d\tau_{wg}}{d\lambda}\right|\sigma_\lambda \tag{4.63}$$

Taking the derivative of Eq. (4.61) and using Eq. (4.63), we get

$$\sigma_{wg} = \frac{Ln_2\Delta\sigma_\lambda}{c}\left|\frac{d}{d\lambda}\left[\frac{d(Vb)}{dV}\right]\right| = \frac{Ln_2\Delta\sigma_\lambda}{c}\left|\frac{d}{dV}\left[\frac{d(Vb)}{dV}\right]\left(\frac{dV}{d\lambda}\right)\right| \tag{4.64}$$

Further, taking derivative on both sides of Eq. (4.59) with respect to λ we get

$$\frac{dV}{d\lambda} = \frac{d}{d\lambda}\left[a\left(\frac{2\pi}{\lambda}\right)n_2\sqrt{2\Delta}\right] = a\left(-\frac{2\pi}{\lambda^2}\right)n_2\sqrt{2\Delta} = -\frac{V}{\lambda} \tag{4.65}$$

Substituting the value of $dV/d\lambda$ in Eq. (4.64) we obtain

$$\sigma_{wg} = \frac{Ln_2\Delta\sigma_\lambda}{c\lambda}V\frac{d^2(Vb)}{dV^2} \tag{4.66}$$

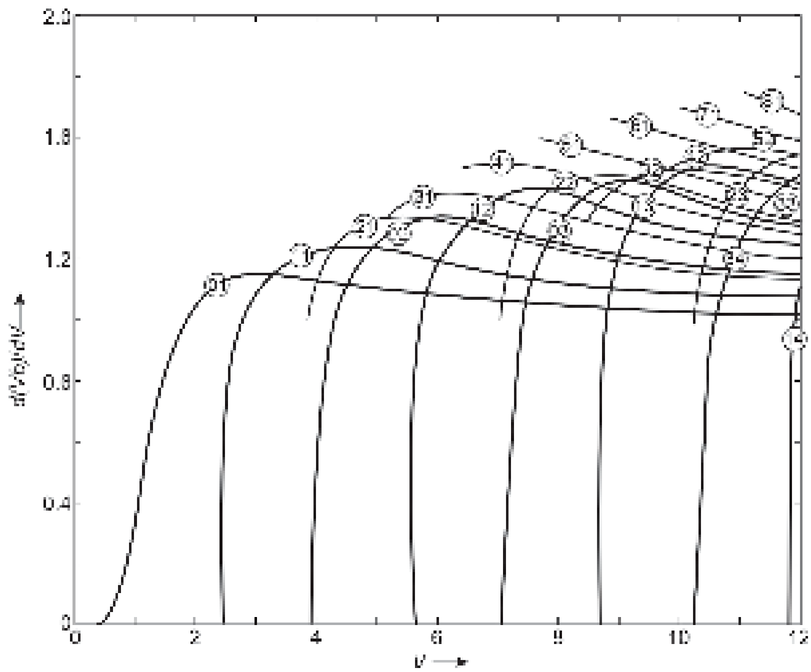

Fig. 4.9 Variation of $d(Vb)/dV$ with V-number of the fiber (after Gloge[b] 1971)

Consequently, the waveguide dispersion parameter can be expressed as

$$D_{wg} = \frac{1}{L}\frac{d\tau_{wg}}{d\lambda} = -\frac{n_2\Delta}{c\lambda}V\frac{d^2(Vb)}{dV^2} \tag{4.67}$$

The variation of b, $d(Vb)/dV$ and $V(d^2(Vb)/dV^2)$ with V number of the fiber for the lowest order LP_{01} mode is shown in Fig. 4.10. For a single-mode fiber, the waveguide dispersion increases significantly when the V-number is set near 1.5. For V-number above 2, the waveguide dispersion for the lowest mode is very small as can be seen from Fig. 4.10. For this reason the V number of a single mode fiber is kept close to and less than the cut-off value of 2.405.

4.5.2.3 Waveguide dispersion in a single-mode fiber

It can be seen from Eq. (4.67) that the waveguide dispersion factor depends on the second derivative of the product of V-number and the normalized propagation constant, b with respect to the V-number. For a multimode fiber, this factor is negligibly small and as a result the waveguide dispersion is insignificant in the case of multimode fiber as compared to material dispersion. On the other hand, for a single mode fiber waveguide dispersion may become comparable to material dispersion. To obtain a quantitative value of waveguide dispersion, the parameter ua can be expressed for the lowest order mode (LP_{01}) as (Gloge[a], 1971)

$$ua = \frac{(1+\sqrt{2})V}{1+(4+V^4)^{\frac{1}{4}}} \tag{4.68}$$

Substituting the value of ua from Eq. (4.68) into Eq. (4.54) we get

$$Vb(V) = V\left\{1 - \frac{(1+\sqrt{2})^2}{\left[1+(4+V^4)^{\frac{1}{2}}\right]^2}\right\} \tag{4.69}$$

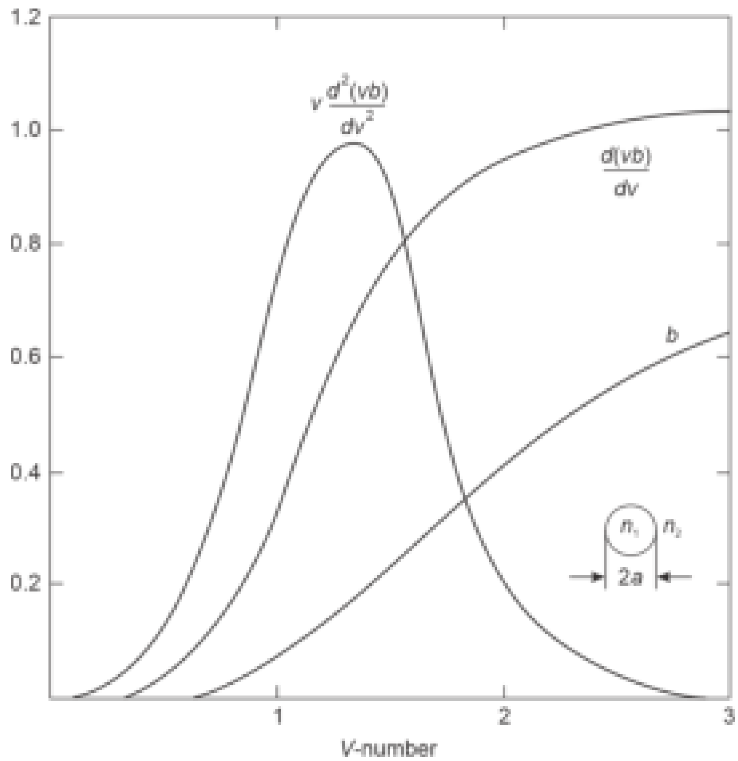

Fig. 4.10 Variation of $b, d(Vb)/dV$ and $Vd^2(Vb)/dV^2$ with V-number for the lowest order mode (LP_{01}) (after Gloge[b] 1971)

The second derivative of the product (Vb) with respect to V can be obtained by using Eq. (4.69). The variations of the parameters b, $d(Vb)/dV$ and $Vd^2(Vb)/dV^2$ with the V-number for LP_{01} (HE_{11}) is shown in Fig. 4.10. It can be seen that the quantity ($V\,d^2(Vb)/dV^2$) has a positive value over the entire region of single mode operation that is, for $0 < V \leq 2.405$ with a maximum value at $V = 1.15$. As a result, the waveguide dispersion, D_{wg} is negative in the entire region of single mode operation. The quantity ($V\,d^2(Vb)/dV^2$) goes to zero at $V = 3.0$ beyond the region of true single mode operation and attains a negative value beyond this point (not shown in Fig. 4.10). For a single mode fiber with $V = 2.4$ operating at 1320 nm having $n^2 = 1.5$ and $\Delta = 0.2\%$, the value of $d^2(Vb)/dV^2 \approx 0.1$. The waveguide dispersion parameter of the fiber can be estimated as

$$D_{wg} = -\frac{1}{L}\frac{n_2\Delta}{c\lambda}V\frac{d^2(Vb)}{dV^2} = -\frac{1}{10^{-3}} \times \frac{1.5 \times 0.002}{3 \times 10^8 \times 1320} \times 2.4 \times 0.1$$
$$= -1.8 \text{ ps nm}^{-1}\text{ km}^{-1}$$

Example 4.10 Consider a single mode fiber with the following parameters.

Core refractive index=1.48; $\Delta = 0.1\%$ and $\lambda = 1330$ nm. Assume the core radius of the fiber to be 4 μm. Compare and contrast the value of waveguide dispersion of the fiber with that obtained in the previous example.

Solution The cladding refractive index of the fiber can be estimated as

$$n_2 = 1.48(1 - 0.001) = 1.4785$$

The V-number of the fiber can be obtained as

$$V = \frac{2 \times 3.14 \times 4 \times 10^{-6}}{1330 \times 10^{-9}} \times 1.48 \times \sqrt{2 \times 0.001} = 1.25$$

It may be noted from Fig. 4.11, that at this V-number value (1.25), the quantity

$$V \frac{d^2(Vb)}{dV^2} \approx 1$$

Therefore, the waveguide dispersion value of the single mode fiber can be estimated as

$$D_{wg} = -\frac{1}{L}\frac{n_2 \Delta}{c\lambda} V \frac{d^2(Vb)}{dV^2} = -\frac{1}{10^{-3}} \times \frac{1.4785 \times 0.001}{3 \times 10^8 \times 1330} \times 1.0 = -3.7 \text{ ps km}^{-1} \text{ nm}^{-1}$$

It can be seen that the waveguide dispersion is nearly double the value obtained in Example 4.9. It is interesting to note that the increase in the value of waveguide dispersion in this case is because of a lower value of V-number of the fiber which results in a larger value of the product ($V\, d^2(Vb)/dV^2$) causing the waveguide dispersion to be more. This is another reason behind choosing the V-number of the fiber close to 2.405 so that the value of the above factor is low and consequently the dispersion caused by waveguide dispersion becomes negligible.

In addition to material dispersion and waveguide dispersion, single mode fibers suffer from other two forms of dispersion: Profile dispersion and polarization mode dispersion.

4.5.2.4 Profile dispersion

The profile dispersion arises from the dependence of the index deviation, Δ on the operating wavelength of the light. The profile dispersion parameter is proportional to $d\Delta/d\lambda$. The value of this dispersion is generally very small (< 0.5 ps nm^{-1} km^{-1}) and usually goes unnoticed. For a multimode fiber, the profile dispersion is insignificant because the majority of the modes that carry the light through the fiber propagate far away from the cut-off. In a multimode fiber the intramodal dispersion is thus dominated by material dispersion and waveguide dispersion only (Fig. 4.11). Further, the V-number of a multimode fiber is generally high and as a result the waveguide dispersion is very small as compared to material dispersion. The total dispersion in a standard single mode fiber is generally dominated by both material dispersion as well as waveguide dispersion. Considering the effect of profile dispersion, D_{pro} the total dispersion, D_{tot} of a single mode fiber can be expressed as (Neumann, 1988)

$$D_{\text{tot}} = D_{\text{mat}} + D_{\text{wg}} + D_{\text{pro}} \tag{4.70}$$

It has already been seen (Fig. 4.9) that the material dispersion attends a value zero at the zero material dispersion (ZMD) point corresponding to the wavelength of 1.27 μm. This means that the pulse broadening due to material dispersion can be made zero at this wavelength. Further, it has been demonstrated (Fleming, 1978) that the ZMD point can be shifted conveniently to a suitable wavelength by changing the constituents of the glass. For example, by changing the concentration of GeO_2 in a pure silica glass from 0–15% it is possible to shift the ZMD point from 1.27 to 1.37 μm (Fleming, 1978). However, the overall dispersion is affected by other components such as waveguide and profile dispersion components as well. Therefore, the zero pulse broadening point does not actually correspond to ZMD where the material dispersion component is zero. It may be pointed out that for wavelengths longer

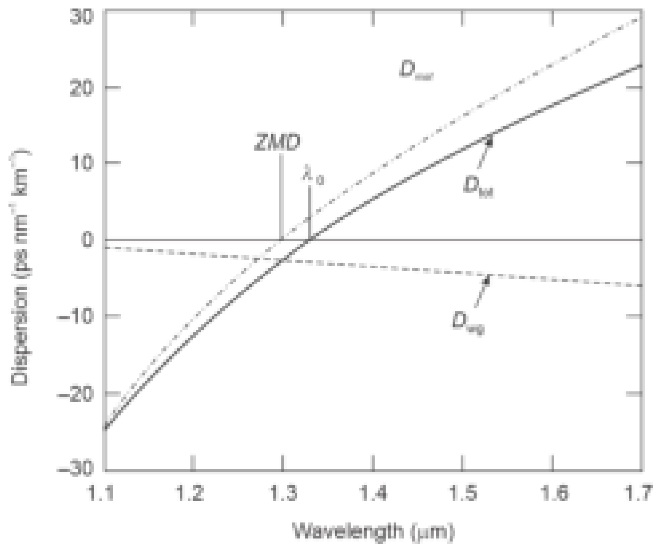

Fig. 4.11 Variations of material dispersion and waveguide dispersion in a single mode fiber with wavelength illustrating ZMD point as well as the wavelength, λ_0 at which the total dispersion is zero (after Senior)

than the ZMD point, the material dispersion is positive whereas the waveguide dispersion is negative for conventional single mode operation region. The variation of material dispersion and waveguide dispersion with wavelength of operation of a conventional single mode fiber is illustrated in Fig. 4.12. The total dispersion in a single mode fiber is approximately equal to the sum of material and waveguide dispersion because the profile dispersion is negligible and intermodal dispersion is not present in a single mode fiber. It can be easily appreciated that there has to be

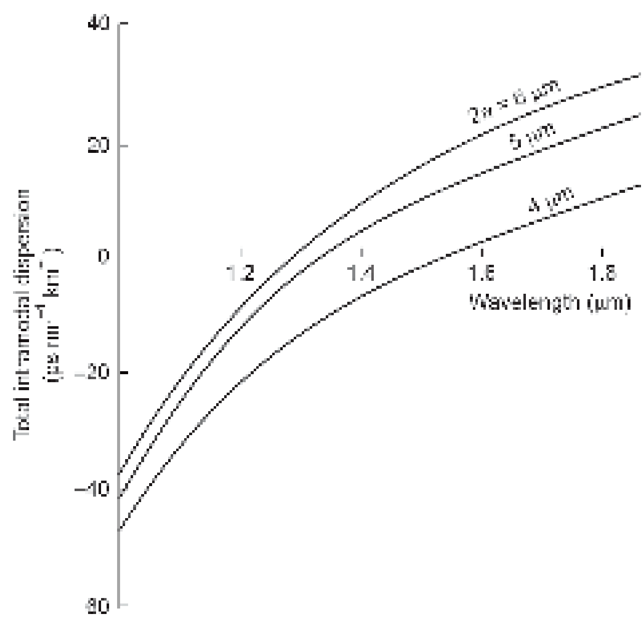

Fig. 4.12 Variation of total intramodal dispersion in a single mode fiber with wavelength for different values of core radius (after Gambling *et al* 1981)

a certain wavelength at which the waveguide dispersion exactly compensates the material dispersion and the total dispersion becomes zero. The wavelength, λ_0 at which the total first order dispersion in a single mode fiber is zero, is slightly larger than the wavelength corresponding to ZMD point. It can be seen that ZMD occurs at 1.27 μm and the waveguide dispersion component shifts this minimum dispersion point to a longer wavelength λ_0 = 1.32 μm for minimum total dispersion. A very low value of total intramodal dispersion in a single mode fiber which is again free from intramodal dispersion effects enables them to offer a very high value of bandwidth-length product (~100 GHz km^{-1}) (Yamada *et al* 1978) around this wavelength. This is one of the major reasons behind shifting the wavelength to this region in the second-generation (2G) optical fiber communication. It may, however, be pointed out that silica fibers exhibit a moderately high value of attenuation resulting primarily from Rayleigh scattering. It may be pointed out that the material dispersion and waveguide dispersion components can be tailored by adjusting the material composition and the geometry of the fiber. As a result, it is possible to alter the value of the wavelength, λ_0 corresponding to the first order zero total dispersion point. It has been demonstrated that λ_0 can be selected in the range of 1.3 μm to 1.6 μm by adjusting the core diameter and the index profile (Gambling *et al* 1979). The variation of total intramodal dispersion with wavelength for a single mode fiber with different values of core radius is illustrated in Fig. 4.12.

4.5.2.5 Polarization mode dispersion (PMD)

The birefringence phenomenon discussed earlier affects the polarization state of the light propagating through cylindrical optical fibers. The birefringence manifests itself in the form of an additional pulse broadening component generally termed polarization mode dispersion (PMD). A small departure from circular symmetry of the core (less than even 1% of the circularity of the core) in a single mode fiber may cause the fundamental HE_{11} mode to decompose into two orthogonal components, e.g. HE_{11}^x and HE_{11}^y to support bimodal transmission. Other factors such as bending, twisting, etc. of the fiber may also be responsible for birefringence. In all practical systems fibers generally have non-perfect geometry because of one or more factors and as a result the polarization state of the fiber can only be maintained over a few meters of length of the fiber. The polarization state as such does not otherwise affect optical transmission systems that involve some kind of intensity modulation of light. For example, in IM/DD based optical communication system the light is intensity modulated at the transmitter and it is subsequently detected with the help of a photodetector which is primarily a kind of photon counter and is insensitive to the polarization state or phase of the light in the fiber. However, there are more sophisticated applications (Kaminow, 1981) including coherent optical communication systems where polarization state of the light is important. Nevertheless, polarization mode dispersion can be very critical for high bit-rate long-haul transmission link operating over 100 Gbps. km. Therefore, in certain situations it is necessary to maintain the polarization state of the fiber over a significant length.

Consider a uniformly birefringent fiber. The light energy in the fundamental mode now travels in the bimodal form supporting two components with orthogonal polarization states. The birefringence of the medium will cause the two orthogonal components to travel at a slightly different velocity. As a result, the polarization orientation of the propagating light will rotate with distance. The two modes exhibit different group delays of τ_{gx} and τ_{gy}. The resulting delay difference, $\delta\tau_{pol}$ occurring between the two orthogonally polarized components thus gives rise to pulse spreading. The delay difference can be expressed as (Rashleigh *et al* 1978)

$$\delta\tau_{\text{pol}} = \left|\tau_{gx} - \tau_{gy}\right| = \left|\frac{L}{v_{gx}} - \frac{L}{v_{gy}}\right| \tag{4.71}$$

where v_{gx} and v_{gy} are the group velocities of the corresponding orthogonal components and L is the length of the fiber. The parameter $\delta\tau_{\text{pol}}$ is the polarization mode dispersion of the fiber.

The group delay $\delta\tau_{\text{pol}}$ caused by the two orthogonal components leads to a pulse spreading of ($\delta\tau_{\text{pol}} L$) over a length L of the fiber. The product ($\delta\tau_{\text{pol}}.L$) can be used as a good approximation for calculation of 3 dB bandwidth given by (Kitayama *et al* 1988)

$$B = \frac{0.9}{(\delta\tau_{\text{pol}} L)} \text{ Hz/km} \tag{4.72}$$

It is further interesting to note that polarization mode dispersion varies randomly along the fiber length whereas chromatic dispersion remains more or less stable along the length of the fiber. The randomness of PMD is attributed to the fact the perturbations responsible for birefringence are dependent on temperature. Therefore, PMD is manifested in the form of time varying fluctuation about the mean value of the group delay, $\delta\tau_{\text{pol}}$. Equation (4.71) is strictly valid under ideal conditions of a fiber with uniform stable birefringence property. For practical applications, the PMD is often expressed in terms of mean value of the differential group delay as

$$\delta_{\text{pol}} \approx D_{\text{PMD}} \sqrt{L} \tag{4.73}$$

where D_{PMD} is the average value of the PMD measured in $\text{ps}/\sqrt{\text{km}}$. The value of D_{PMD} usually ranges between $0.1 \text{ to } 1 \text{ ps}/\sqrt{\text{km}}$ and largely depends on the environmental conditions and type of installation (Cameron *et al* 1998). For example, the value of PMD is generally large for aerial optical fiber cables as compared to buried cables. This is attributed to sudden changes in temperature and/or movements caused by the wind in the former case.

4.6 INTERMODAL DISPERSION

So far we have considered intramodal dispersion which involves various dispersion mechanisms operative within a particular mode. All types of fibers (single and multimode) are affected by intramodal dispersion. In this section pulse broadening caused by intermodal dispersion (also called modal dispersion) is discussed. Intermodal dispersion arises from the propagation delay difference between different modes in a multimode fiber. It may be pointed out that an optical pulse launched into a multimode fiber propagates in the form of various modes which jointly carry the total power. These modes travel along the fiber with different group velocities causing different modes to have different transmission times to reach the destination. This results in the broadening of the pulse at the output of the fiber. The pulse broadening is apparently decided by the differences in transmission times of the slowest and the fastest mode. It is often easier to appreciate and estimate the pulse broadening due to intermodal dispersion by visualizing the propagation of light with the help of ray tracing approach. The ray tracing approach works well because the size of the core of a multimode fiber is much larger than the wavelength of light propagating through it. We have seen previously that steeper the angle of propagation (with respect to the core-axis) of the ray congruence, the higher is the order of the mode and larger is the value of the mode number and slower is the axial group velocity of the mode. Therefore, the fastest mode in ray approach corresponds to the axial ray which travels with the maximum group velocity and the slowest mode corresponds to the ray that is most oblique corresponding to critical angle. The intermodal dispersion which causes pulse broadening essentially arises from the difference in time, T_{max} taken by the longest ray congruence path (most oblique ray) corresponding to the highest order mode and the time, T_{min} taken by the shortest ray congruence path (axial ray) corresponding to the lowest order mode. This dispersion vanishes in the case of single mode operation. Pulse broadening due to intermodal dispersion is most significant in the case of a step-index multimode fiber. This can be controlled to a great extent by using a graded-index profile. When an optical fiber is designed to have a near parabolic refractive index profile, the pulse broadening due to intermodal dispersion can be minimized. As a consequence, the bandwidth of a graded-index multimode fiber is much larger than that of a corresponding step-index multimode fiber. Single mode fibers on the other hand do not suffer from pulse broadening arising from intermodal dispersion and therefore they offer largest possible bandwidths which are limited only by intramodal dispersion effects.

4.6.1 Pulse Broadening in a Multimode Step-index Fiber

Consider t-mode multimode step-index fiber. According to ray theory, the fastest and the slowest modes can be represented by the axial and the most oblique ray respectively. The most oblique ray is one that is incident at the core-cladding interface at the critical angle, θ_c with the core-cladding interface or ϕ_c with the normal drawn on the core-cladding interface at the point of incidence as shown in Fig. 4.13. It is interesting to note that both the rays travel with the same velocity through the core region having a constant refractive index. The delay difference actually arises from the path difference between the two rays. The time taken by the axial ray to travel the length, L along the fiber is T_{min} given by

$$T_{min} = \frac{L}{\text{Velocity of light in the medium}}$$

$$= \frac{L}{(c/n_1)} = \frac{Ln_1}{c} \quad (4.74)$$

where n_1 is the core refractive index and c is the velocity of light in free space.

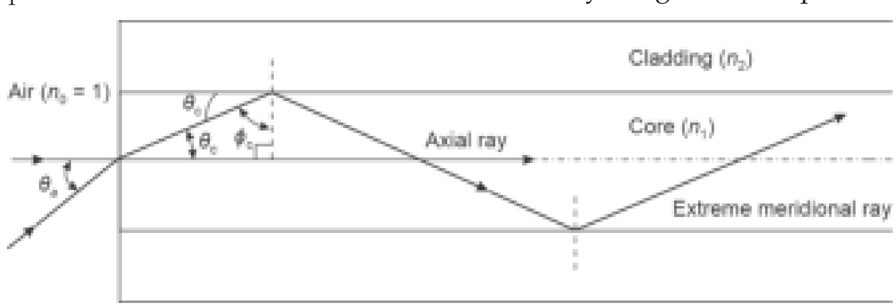

Fig. 4.13 Ray diagram in a multimode step-index fiber showing the paths of the axial ray and the extreme meridional ray

The maximum delay time exhibited by the most oblique ray is T_{max} given by

$$T_{max} = \frac{L/\cos\theta_c}{c/n_1} = \frac{Ln_1}{c\cos\theta_c} \quad (4.75)$$

Applying Snell's law at the core-cladding interface, we get

$$n_1 \cos\theta_c = n_2$$

That is,

$$\cos\theta_c = \frac{n_2}{n_1} \tag{4.76}$$

where n_2 is the refractive index of the cladding.

Substituting the value of $\cos\theta_c$ from Eq. (4.76) into Eq. (4.75) we get

$$T_{\max} = \frac{Ln_1^2}{cn_2} \tag{4.77}$$

The delay difference between the two extreme rays corresponding to the highest and the lowest order mode can be obtained by using Eqs (4.74) and (4.77) as

$$\begin{aligned}\delta T_{\text{mod}} &= T_{\max} - T_{\min} = \frac{Ln_1^2}{cn_2} - \frac{Ln_1}{c} \\ &= \frac{Ln_1}{c}\left(\frac{n_1 - n_2}{n_2}\right) \approx \frac{Ln_1\Delta}{c}\end{aligned} \tag{4.78}$$

The above equation is derived with the assumption that $\Delta \ll 1$ so that $n_1 \approx n_2$. Alternatively, Eq. (4.78) can be expressed in the following form

$$\delta T_{\text{mod}} = \frac{Ln_1^2}{cn_2}\left(\frac{n_1 - n_2}{n_1}\right) = \frac{Ln_1^2\Delta}{cn_2} \tag{4.79}$$

Further, the numerical aperture of a step-index fiber with $\Delta \ll 1$ can be approximated as

$$NA \approx n_1\sqrt{2\Delta} \tag{4.80}$$

Using the approximation, the delay difference responsible for intermodal dispersion can be expressed as

$$\delta T_{\text{mod}} = \frac{L(NA)^2}{2n_1 c} \tag{4.81}$$

It may be stressed that Eq. (4.81) takes into account only the meridional rays and gives a fairly good approximation of the delay difference caused by the two extreme rays in a step-index fiber only. The delay difference is proportional to the square of the numerical aperture. In order to keep the delay difference low it is necessary to have a small difference in the values of the refractive index of the core and the cladding, that is $n_1 \approx n_2$ or equivalently $\Delta \ll 1$.

4.6.2 RMS Pulse Broadening

The delay difference derived above can be used to estimate the broadening of the pulse caused by intermodal dispersion. For this purpose, it is necessary to assume the shape of the pulse initially launched in the fiber. For computation of rms pulse broadening due to intermodal dispersion it is assumed that there is no intramodal dispersion. Consider a monochromatic rectangular light pulse, $p(t)$ of unit area shown in Fig. 4.14 such that (Stremler, 1982; Senior, 2008)

$$\int_{-\infty}^{\infty} p(t)dt = 1 \tag{4.82}$$

The rms broadening due to intermodal dispersion can be obtained by considering the rectangular pulse of unit area shown in Fig. 4.14. The rectangular pulse has been assumed to have a height of $1/\delta T_{\text{mod}}$ and a width of δT_{mod} ranging from $-\delta T_{\text{mod}}/2$ to $\delta T_{\text{mod}}/2$ so that the total area of the pulse remains unity. It can be seen from Fig. 4.14 that

$$\int_{-\delta T_{\text{mod}}/2}^{\delta T_{\text{mod}}/2} p(t)dt = 1 \tag{4.83}$$

Mathematically, the broadened pulse can be expressed as

$$p(t) = \frac{1}{\delta T_{\text{mod}}} \quad \text{for } -\frac{\delta T_{\text{mod}}}{2} \le t \le \frac{\delta T_{\text{mod}}}{2}$$
$$= 0 \qquad \text{otherwise} \tag{4.84}$$

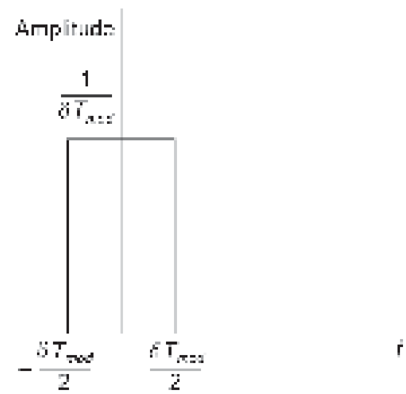

Fig. 4.14 The broadened rectangular pulse due to intermodal dispersion

The mean-square value (variance) of the pulse broadening due to intermodal dispersion can be obtained as

$$\sigma_{\text{mod}}^2 = M_2 - M_1^2 \tag{4.85}$$

where M_1 is the mean value (first temporal moment) and M_2 is the mean square value (second temporal moment) of the pulse. That is,

$$M_1 = \int_{-\infty}^{\infty} tp(t)dt \tag{4.86}$$

and

$$M_2 = \int_{-\infty}^{\infty} t^2 p(t)dt \tag{4.87}$$

For the broadened pulse under consideration

$$M_1 = \frac{1}{\delta T_{\text{mod}}} \int_{-\frac{\delta T_{\text{mod}}}{2}}^{\frac{\delta T_{\text{mod}}}{2}} tdt = \frac{1}{\delta T_{\text{mod}}} \frac{t^2}{2} \Bigg|_{-\frac{\delta T_{\text{mod}}}{2}}^{\frac{\delta T_{\text{mod}}}{2}} = 0 \tag{4.88}$$

and

$$M_2 = \frac{1}{\delta t_{\text{mod}}} \int_{-\frac{\delta T_{\text{mod}}}{2}}^{\frac{\delta T_{\text{mod}}}{2}} t^2 dt = \frac{1}{\delta T_{\text{mod}}} \frac{t^3}{3} \Bigg|_{-\frac{\delta T_{\text{mod}}}{2}}^{\frac{\delta T_{\text{mod}}}{2}} = \frac{1}{3}\left(\frac{\delta T_{\text{mod}}}{2}\right)^2 \tag{4.89}$$

The rms value of the pulse broadening due to intermodal dispersion can be

$$\sigma_{\text{mod}} = \frac{1}{2\sqrt{3}} \delta T_{\text{mod}}$$

Substituting the value of δT_{mod} from Eq. (4.81) into Eq. (4.89) we get

$$\sigma_{\text{mod}} = \frac{L(\text{NA})^2}{4\sqrt{3}n_1 c} \approx \frac{L n_1 \Delta}{2\sqrt{3}c} \tag{4.90}$$

From Eq. (4.90) it is clearly seen that the rms pulse broadening due to intermodal dispersion in a multimode step-index fiber depends on the relative index difference,

Δ and so also on the square of the numerical aperture. It is imperative to conclude, that to minimize the intermodal dispersion in a multimode fiber it is necessary to have a small value of relative index difference, Δ. In other words, a weakly guided multimode step-index fiber causes less spreading due to intermodal dispersion. However, a smaller value of Δ reduces the numerical aperture of the fiber. We shall see later in Chapter 6 that a smaller value of numerical aperture reduces the acceptance angle of the fiber and lowers the capability of the fiber to gather power from the source. This means that there is a trade-off between the power launched into a multimode step-index fiber and the intermodal dispersion caused by the fiber. Further, the rms pulse broadening also depends on the length of the fiber. This means that there exists another trade-off between the bandwidth and length of the fiber. The multimode step-index fibers generally suffer from excessive intermodal dispersion and are therefore not suitable for long-haul optical communication. It has been pointed out earlier that the rms pulse broadening due to intermodal dispersion can be greatly reduced by making use of the graded-index profile in multimode fibers.

Example 4.11 Consider a 10 km optical fiber link using a multimode step-index fiber with the following parameters.

Core refractive index, n_1 = 1.458; Relative index deviation, Δ = 0.002.

Estimate the delay time difference between the axial ray and the most oblique ray. What is the value of rms pulse broadening due to intermodal dispersion? Estimate the bandwidth and the maximum bit-rate of transmission assuming RZ formatting and neglecting intramodal dispersion.

Solution The time delay difference between the axial ray and the most oblique ray can be estimated by using Eq. (4.78) as

$$\delta T_{\text{mod}} = \frac{Ln_1\Delta}{c} = \frac{10\times 10^3 \times 1.458\times 0.002}{3\times 10^8} = 97.2 \text{ ns}$$

The rms pulse broadening due to intermodal dispersion can be accordingly calculated by using Eq. (4.90) as

$$\sigma_{\text{mod}} = \frac{Ln_{1\Delta}}{2\sqrt{3}c} = \frac{10\times 10^3 \times 1.458\times 0.002}{2\sqrt{3}\times 3\times 10^8} \approx 28 \text{ ns}$$

The bandwidth of transmission can be obtained using Eq. (4.32) as

$$B \approx \frac{0.2}{\sigma_{\text{mod}}} = \frac{0.2}{28\times 10^{-9}} = 7.14 \text{ MHz}$$

The maximum bit-rate can be obtained for NRZ mode by using Eq. (4.33) as

$$B_T = 2\times B = 2\times 7.14 = 14.28 \text{ Mbps}$$

4.6.3 Intermodal Dispersion in a Multimode Graded-index Fiber

In Chapter 2, we have seen that the intermodal dispersion in a multimode fiber can be reduced by using a graded index profile in which the refractive index has a maximum value at the centre of the core and it progressively decreases along the radius of the core to attain the minimum value equal to the cladding refractive index at the core-cladding interface. This type of profile slows down the axial ray (because of maximum optical density, i.e. highest refractive index along the axis) and makes the propagation of the oblique rays progressively easier and thereby compensates the delay difference between the extreme rays to a large extent. As a result, multimode graded index fibers exhibit a significant improvement in bandwidth over multimode step-index fibers. It can be mathematically shown that there exists an optimum value of the grade parameter (α) for which the intermodal dispersion has a minimum value. For example, it can be theoretically shown that for a square-law index profile the group velocity is almost independent of the order of the mode that is, mode number.

4.6.4 Optimum Refractive-index Profile of a Graded-index Fiber

Consider a graded-index fiber with the refractive-index profile given by

$$n(r) = \begin{cases} n_1\left[1-2\Delta\left(\dfrac{r}{a}\right)^{\alpha}\right]^{\frac{1}{2}} & r < a \\ n_1(1-2\Delta)^{\frac{1}{2}} \approx n_1(1-\Delta) = n_2 & r \geq a \end{cases} \tag{4.91}$$

It has been shown in Chapter 3, that the number of modes for a given value of the propagation constant, β can be expressed by using WKB approximation in the form (Gloge, 1973)

$$m = a^2k^2n_1^2\Delta\left(\frac{\alpha}{\alpha+2}\right)\left(\frac{k^2n_1^2-\beta^2}{2\Delta k^2n_1^2}\right)^{\frac{2+\alpha}{\alpha}} \tag{4.92}$$

The Eq. (4.92) can be rearranged as

$$\beta = \left[k^2n_1^2 - 2\left(\frac{\alpha+2}{\alpha}\frac{m}{\alpha^2}\right)^{\frac{\alpha}{\alpha+2}}(k^2n_1^2\Delta)^{\frac{2}{\alpha+2}}\right]^{\frac{1}{2}} \tag{4.93}$$

The propagation constant can be alternatively expressed as

$$\beta = kn_1\left[1-2\Delta\left(\frac{m}{M}\right)^{\frac{\alpha}{\alpha+2}}\right]^{\frac{1}{2}} \tag{4.94}$$

where M is the total number of possible guided modes through the graded-index fiber given by (see Eq. (3.277))

$$M = a^2k^2n_1^2\Delta\left(\frac{\alpha}{\alpha+2}\right) \tag{4.95}$$

The group delay associated with the modes propagating through a graded-index fiber can be estimated using Eq. (4.58) and remembering the fact that n_1 and Δ are also functions of k, as

$$\tau_g = \frac{L}{c}\frac{\partial\beta}{\partial k} \tag{4.96}$$

Taking the first derivative of the propagation constant β with respect to k using Eq. (4.93), we may finally express the group delay τ_g as

$$\tau_g = \left[\frac{L}{c}\left(\frac{kn_1}{\beta}\right)\left(n_1 + k\frac{\partial n_1}{\partial k}\right) - \frac{4\Delta}{\alpha+2}\left(\frac{m}{M}\right)^{\frac{\alpha}{\alpha+2}}\left(\left(n_1 + k\frac{\partial n_1}{\partial k}\right) + \frac{n_1k}{2\Delta}\frac{\partial\Delta}{\partial k}\right)\right] \tag{4.97}$$

Making the following substitutions

$$N_1 = n_1 + k\frac{\partial n_1}{\partial k} \tag{4.98a}$$

and

$$\in = \frac{2n_1 k}{\left(n_1 + k\dfrac{\partial n_1}{\partial k}\right)\Delta}\frac{\partial \Delta}{\partial k} = \frac{2n_1 k}{N_1 \Delta}\frac{\partial \Delta}{\partial k} \tag{4.98b}$$

Equation (4.97) can be expressed as

$$\tau_g = \frac{LN_1}{c}\left(\frac{kn_1}{\beta}\right)\left[1 - \frac{\Delta}{\alpha+2}\left(\frac{m}{M}\right)^{\frac{\alpha}{\alpha+2}}(4+\in)\right] \tag{4.99}$$

Using Eq. (4.94), we may write

$$\frac{kn_1}{\beta} = \left[1 - 2\Delta\left(\frac{m}{M}\right)^{\frac{\alpha}{\alpha+2}}\right]^{-\frac{1}{2}} \tag{4.100}$$

Further, it may be noted that the core-cladding index difference $\Delta \ll 1$ and $m/M < 1$ for a given value of β lying between kn_1 and kn_2. Substituting

$$x = \Delta\left(\frac{m}{M}\right)^{\frac{\alpha}{\alpha+2}} \ll 1 \tag{4.101}$$

Eq. (4.100) can be expanded binomially to approximate as

$$\frac{kn_1}{\beta} = [1-2x]^{-\frac{1}{2}} \approx 1 + x + \frac{3x^2}{2} \tag{4.102}$$

That is,

$$\frac{kn_1}{\beta} \approx 1 + \Delta\left(\frac{m}{M}\right)^{\frac{\alpha}{\alpha+2}} + \frac{3}{2}\Delta^2\left(\frac{m}{M}\right)^{\frac{2\alpha}{\alpha+2}} \tag{4.103}$$

Substituting the value of (kn_1/β) from Eq. (4.103) into Eq. (4.99), the group delay can be obtained as (Olshansky *et al* 1976)

$$\tau_g = \frac{LN_1}{c}\left[1 + \Delta\left(\frac{m}{M}\right)^{\frac{\alpha}{\alpha+2}} + \frac{3}{2}\Delta^2\left(\frac{m}{M}\right)^{\frac{2\alpha}{\alpha+2}}\right]\left[1 - \frac{\Delta}{a+2}\left(\frac{m}{M}\right)^{\frac{\alpha}{\alpha+2}}(4+\in)\right] \tag{4.104}$$

That is,

$$\tau_g = \frac{N_1 L}{c}\left[1 + \frac{\alpha - 2 - \in}{\alpha+2}\Delta\left(\frac{m}{M}\right)^{\frac{\alpha}{\alpha+2}} + \frac{3\alpha - 2 - 2\in}{2(\alpha+2)}\Delta^2\left(\frac{m}{M}\right)^{\frac{2\alpha}{\alpha+2}} + O(\Delta^3)\right] \tag{4.105}$$

where $O(\Delta^3)$ is the term containing the term Δ^3 and can be ignored since $\Delta \ll 1$.

It can be seen from Eq. (4.105) that the first order term in Δ of the delay difference between the modes is zero when

$$\alpha = 2 + \in \tag{4.106}$$

It is interesting to note that the value of $\in$ is generally very small. Therefore it may be concluded that minimum intermodal dispersion will result from the group delay between the various modes of a multimode fiber when $\alpha \approx 2$, that is, the refractive index profile of the GI fiber is nearly parabolic.

The rms pulse broadening due to intermodal dispersion can be calculated only when the power carried by individual mode is known.

The rms value of pulse spreading due to intermodal dispersion can be expressed as (Olshansky *et al* 1976)

$$\sigma_{\text{intermodal}} = \left(\left\langle \tau_g^2 \right\rangle - \left\langle \tau_g \right\rangle^2\right)^{\frac{1}{2}} \tag{4.107}$$

where

$$\left\langle \tau_g^2 \right\rangle = \sum_{l,m} \frac{P_{lm}\tau_g^2(l,m)}{M} \tag{4.108}$$

and

$$\left\langle \tau_g \right\rangle = \sum_{l,m} \frac{P_{lm}\tau_g(l,m)}{M} \tag{4.109}$$

Here P_{lm} is the optical power contained in the mode of order (l, m) and M is the total number of modes.

Assume that all modes are equally excited meaning thereby that each mode carries the same amount of power. Mathematically, this amounts to $P_{lm} = P$ for all modes irrespective of the mode order designated by (l, m). Further, if we assume that the total number of modes is very large then the summations of Eqs. (4.108) and (4.109) can be replaced by corresponding integrals. On the basis of the above assumptions and subsequent substitution of Eq. (4.105) into Eq. (4.107) yields (Olshansky *et al* 1976)

$$\sigma_{\text{intermodal}} = \frac{Ln_1\Delta}{2c}\frac{\alpha}{\alpha+1}\left(\frac{\alpha+2}{3\alpha+2}\right)^{\frac{1}{2}} \times \left[c_1^2 + \frac{4c_1c_2(\alpha+1)\Delta}{2\alpha+1} + \frac{16\Delta^2c_2^2(\alpha+1)^2}{(5\alpha+2)(3\alpha+2)}\right]^{\frac{1}{2}} \tag{4.110}$$

where

$$c_1 = \frac{\alpha - 2 - \epsilon}{\alpha + 2} \tag{4.111a}$$

and

$$c_2 = \frac{3\alpha - 2 - 2\epsilon}{2(\alpha+2)} \tag{4.111b}$$

The foregoing analysis clearly suggests that the intermodal dispersion can be greatly reduced by making the refractive index profile nearly parabolic (i.e. $\alpha \approx 2$). Further, when $\alpha - 2 - \epsilon$ is of the order of Δ there is a partial cancellation of the two mode dependent terms in Eq. (4.105) and the optimal value of α for minimum rms pulse broadening due to intermodal dispersion is (Olshansky, 1975)

$$\alpha_c = 2 + \epsilon - \Delta\frac{(4+\epsilon)(3+\epsilon)}{5+2\epsilon} \tag{4.112}$$

However, the overall pulse broadening also depends on the chromatic or intramodal dispersion which has been ignored in the analysis so far. In a practical measurement, both intramodal and intermodal dispersion phenomena manifest in the form of the overall pulse broadening caused by the fiber. To obtain the optimum index profile of a GI fiber it is therefore necessary to take into account the effect of intramodal dispersion which not only depends on the transmission characteristics of the core and the cladding but also the spectral property of the source.

The pulse broadening due to intramodal dispersion can be expressed as (Olshansky *et al* 1976; Einarsson, 1986)

$$\alpha_{\text{intermodal}}^2 = \left(\frac{\sigma_\lambda}{\lambda}\right)^2 \left\langle \left(\lambda\frac{d\tau_g}{d\lambda}\right)^2 \right\rangle \tag{4.113}$$

where σ_λ is the rms spectral width of the source.

Using Eq. (4.105) and neglecting all terms containing second and higher orders of the index deviation term $\Delta(\ll 1)$ and terms containing negligibly small factors such as $d\Delta/d\lambda$ and $\Delta\, dn_1/d\lambda$, we obtain

$$\lambda \frac{d\tau_g}{d\lambda} \approx -\frac{L}{c}\lambda^2 \frac{d^2 n_1}{d\lambda^2} + \frac{N_1 L\Delta}{c} \frac{\alpha - 2 - \epsilon}{\alpha+2} \frac{2\alpha}{\alpha+2}\left(\frac{m}{M}\right)^{\frac{\alpha}{\alpha+2}} \tag{4.114}$$

It may be pointed out that for a GI fiber both the terms on the right hand side of the Eq. (4.113) are generally comparable when α is very different from 2. On the other hand for a near parabolic index fiber $\alpha \approx 2$ and the second term becomes negligible.

To evaluate the rms spreading due to intramodal dispersion we again assume that all modes are equally excited to carry the same amount of power by each mode. Further assuming that number of modes is very large in number, the summations used for computation of the average value can be conveniently replaced by integration. Substituting the value of $\lambda d\tau_g/d\lambda$ from Eq. (4.114) into Eq. (4.113), we finally arrive at (Olshansky *et al* 1976)

$$\sigma_{\text{intramodal}} = \left[\frac{L}{c}\frac{\sigma_\lambda}{\lambda}\left(-\lambda^2 \frac{d^2 n_1}{d\lambda^2}\right)^2 - N_1 c_1 \Delta\left(2\lambda^2 \frac{d^2 n_1}{d\lambda^2}\frac{\alpha}{\alpha+1} - N_1 c_1 \Delta \frac{4\alpha^2}{(\alpha+2)(3\alpha+2)}\right)\right]^{\frac{1}{2}} \tag{4.115}$$

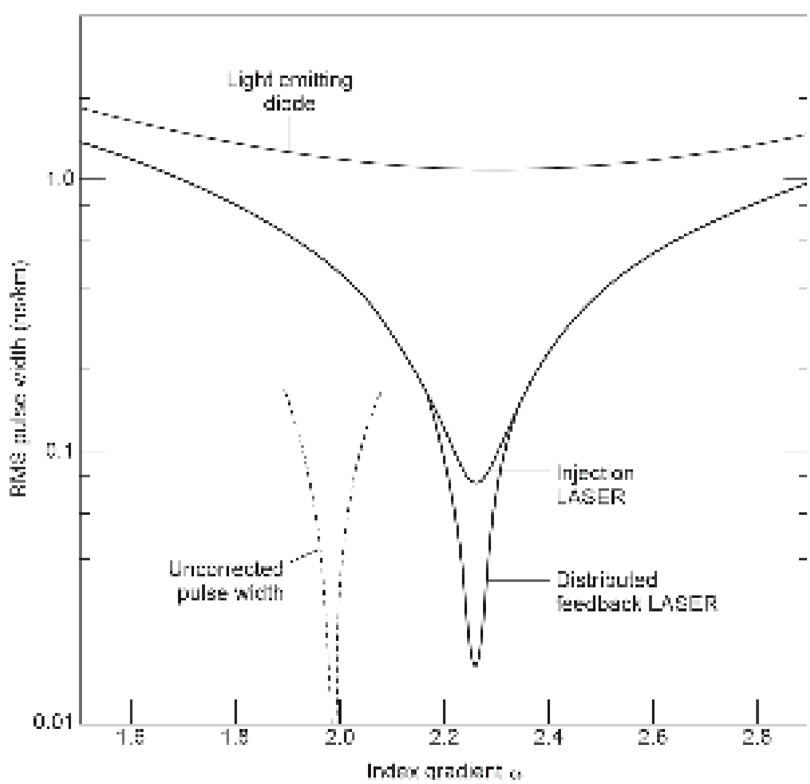

Fig. 4.15 Variation of rms pulse broadening with index gradient α at 900 nm (after Olshansky *et al* 1976)

The overall rms pulse broadening, σ caused by intermodal as well as intramodal dispersion in a graded-index fiber can be finally estimated by using the following relation (Personick, 1973)

$$\sigma = \left[\sigma_{\text{intermodal}}^2 + \sigma_{\text{intramodal}}^2\right]^{\frac{1}{2}} \tag{4.116}$$

The total rms pulse broadening of an α-profile GI fiber can be obtained by substituting the values of $\sigma_{\text{intermodal}}$ and $\sigma_{\text{intramodal}}$ from Eqs. (4.110) and (4.115) respectively. It can be easily appreciated that the overall pulse broadening depends on the wavelength of operation as well as the rms spectral width of the source when intramodal dispersion is taken into consideration. In other words the optimum value of α as predicted by considering intermodal dispersion only may not be the same for minimizing the overall pulse broadening. Theoretical and experimental studies with titania doped silica fiber (Olshansky *et al* 1976) revealed that the optimum value of α indeed depends on the value of the wavelength of light as well as the rms spectral width of the source. Figure 4.15 shows the variation of the rms pulse width as a function of α for three different types of GaAs source, e.g. an LED, an injection laser diode and a Distributed Feedback (DFB) laser all operating at 900 nm with different values of rms spectral width of 15 nm, 1 nm and 0.2 nm respectively. In Fig. 4.15, the dashed curve represents the rms pulse broadening by considering intermodal dispersion only and completely neglecting intramodal dispersion. It is further seen from the figure that the overall rms pulse broadening attains the minimum value for the DFB laser source which has the lowest rms spectral width. It has been understood that the optimal value of α that minimizes the overall rms pulse spreading depends strongly on the operating wavelength. This is accounted for the different dispersive properties of the core and the cladding regions. To achieve minimum pulse broadening that is maximum information carrying capacity the radial index profile must be modified accordingly to compensate for this (Olshansky *et al* 1976).

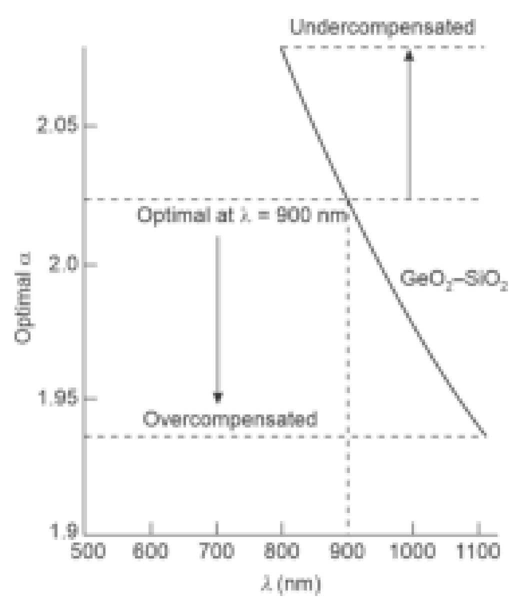

Fig. 4.16 Variation of optimal value of index profile parameter α of a GI fiber with wavelength (after Cochen, 1978)

It is interesting to note that a fiber with a given refractive index profile α will exhibit different values of rms pulse spreading depending on the wavelength of operation. This is attributed to profile dispersion effect which arises from the compositional variation of the glass material in a graded-index fiber and results in different variations of refractive index with wavelength in the different layers. The wavelength dependence of the optimal value of α that minimizes pulse dispersion in a GeO_2–SiO_2 fiber is shown in Fig. 4.16 (Cohen *et al* 1978). It can be seen from the figure that the optimal value of the profile index α decreases with increase in the operating wavelength. For example, an optical fiber having an optimal profile index parameter α_c at 900 nm offers a minimum pulse broadening that is, peak bandwidth at this wavelength. For longer wavelength of operation, the index profile becomes overcompensated while at shorter wavelength the index becomes under-compensated. The multimode fibers with index profile parameter $\alpha > \alpha_c$ (optimal value corresponding to 900 nm) are considered to have undercompensated profile and these fibers tend to exhibit minimum pulse broadening or peak bandwidth at a shorter wavelength. Similarly, multimode fibers having $\alpha < \alpha_c$ (900 nm) are considered to have an overcompensated profile and the profile index of these fibers becomes optimal at a longer wavelength where they exhibit minimum pulse broadening or peak bandwidth.

Example 4.12 A GI fiber with an optimal index profile has an axial refractive index n_1 = 1.458 and an index deviation of Δ of 1%. Estimate the rms pulse broadening by assuming

$$\frac{dn_1}{d\lambda} = 0 \text{ and } \frac{d\Delta}{d\lambda} = 0$$

Compare and contrast the value with that of a corresponding step-index fiber. Comment on the bandwidth improvement in the case of GI fiber over the SI fiber. Assume that the pulse broadening due to intramodal dispersion is negligible.

Solution The rms pulse broadening due to intermodal dispersion in a GI fiber can be obtained from Eq. (4.110). In the present case

$$N_1 = n_1 \text{ and } \epsilon = 0$$

and using Eq. (4.111) the optimal profile index parameter can be obtained as

$$\alpha_c = 2 - \frac{12}{5}\Delta$$

Therefore the rms pulse spreading due to intermodal dispersion in the GI with optimal profile index reduces ton (4.110) as

$$(\sigma_{\text{intermodal}})_{GI} = \frac{n_1 L \Delta^2}{2\sqrt{3}c} \tag{4.117}$$

Substituting the values of the given parameters, the rms pulse broadening in the case of an optimal index GI fiber can be obtained as 14 ps km^{-1}.

For a corresponding SI fiber, the rms pulse spreading due to intermodal dispersion is given by Eq. (4.90) as

$$(\sigma_{\text{intermodal}})_{SI} = \frac{n_1 L \Delta}{2\sqrt{3}c} \tag{4.118}$$

Substituting the values of the parameters, the rms pulse broadening in the case of SI fiber can be obtained as 14 ns km^{-1}.

Therefore, the ratio of the pulse broadening in a SI fiber to that in a GI fiber is given by

$$\frac{(\sigma_{\text{intermodal}})_{SI}}{(\sigma_{\text{intermodal}})_{GI}} = \frac{10}{\Delta} = \frac{10}{0.01} = 10^3$$

It may be recalled that the bit rate or information carrying capacity of the fiber is inversely related to the rms value of the pulse spreading. Thus the result indicates that there is an improvement in the magnitude of information carrying capacity in a GI fiber by a factor of 10^3 as compared to that of a corresponding SI fiber.

Example 4.13 A multimode step-index fiber has a numerical aperture of 0.22 and a core refractive index of 1.458. The fiber exhibits an overall intramodal dispersion of 200 ps km^{-1}. Calculate overall value of the rms pulse broadening per kilometre of the fiber when the LED source operating at 850 nm has a rms spectral width of 40 nm. Estimate the bandwidth of a 10 km link based on this fiber.

Solution The rms pulse broadening per kilometre due to material dispersion can be obtained as

$$\sigma_{\text{intra}}(1\text{ km}) = 40 \times 1 \times 200 = 8 \text{ ns km}^{-1}$$

The rms pulse broadening due to intermodal dispersion can be obtained as

$$\sigma_{\text{inter}}(1\text{ km}) = \frac{L(NA)^2}{4\sqrt{3}n_1 c} = \frac{10^3 \times 0.22}{4\sqrt{3} \times 1.458 \times 3 \times 10^8} = 15.97 \text{ ns km}^{-1}$$

The overall rms pulse broadening can be obtained as

$$\sigma = \left[\sigma_{\text{intra}}^2 + \sigma_{\text{inter}}^2\right]^{\frac{1}{2}} = \sqrt{8^2 + (15.97)^2} = 17.86 \text{ ns km}^{-1}$$

For a fiber link of 10 km length, the total rms pulse broadening would be

$$\sigma_T = 17.86 \times 10^{-9} \times 10 = 178.6 \text{ ns}$$

The bandwidth of the link can be estimated as

$$B \approx \frac{0.2}{178.6 \times 10^{-9}} = 1.11 \text{ MHz}$$

4.6.5 Mode Coupling

Theoretical calculation generally overestimates the pulse dispersion in an optical fiber. The analysis is based on a number of drastic assumptions which are not really valid for a practical optical fiber. For example, in the theoretical analysis the fiber is assumed to be ideal without any structural imperfection or undesirable index variation or any other kind of deformities including macro- or micro-bends. Furthermore, it is assumed that all modes are equally excited in the fiber. This means that every mode in the fiber carries the same amount of power which is not true for a practical fiber. In actual practice, the pulse dispersion is much less than the theoretically predicted value after a certain initial length of the fibers when the modes attain equilibrium. After traversing some initial distance along the fiber the modes become stable and the pulse dispersion increases less rapidly because of mode coupling and differential mode loss (Olshansky, 1975). In the initial length of the fiber immediately after launching coupling of energy from one mode to the other takes place. This kind of exchange of power from one mode to the other is initiated by random perturbations in the structure, refractive index, and cabling induced microbends. The mode coupling tends to compensate the propagation delays associated with various modes and reduce the pulse broadening due to intermodal dispersion. The reduction in the pulse spreading due to intermodal dispersion arises from the fact that some power originally travelling in the fast mode is eventually transferred to a slow mode and conversely a portion of the power from the slow mode gets transferred partially in the fast mode in a way that the extreme of the group velocity spread is partly equalized (Marcuse, 1975). In the presence of mode coupling with steady-state transfer of power the rms pulse broadening due to intermodal dispersion is no longer proportional to the length L of the fiber rather it varies as the square root of the product of the length L and the coupling length, L_c that is, with $\sqrt{LL_c}$. The rms value of the pulse spreading due to intermodal dispersion in multimode fiber is thus reduced by a factor of $\sqrt{L_c / L}$. In the case of strong coupling ($L_c \ll L$) the effect may be significant (Marcuse, 1975).

Mode coupling is generally associated with an additional loss. The improvement in the pulse spreading caused by mode coupling is related to the loss over a distance $Z < L_c$ is related to the excess loss hZ, h being the loss in dB/km by the equation

$$hZ\left(\frac{\sigma_c}{\sigma_0}\right)^2 = C \tag{4.119}$$

where σ_0 and σ_c are the values of rms pulse broadening in the absence of mode coupling and presence of strong mode coupling and hZ is the excess attenuation arising out of mode coupling and C is a constant.

It may be further pointed out that splices, connectors and other passive components in an optical fiber link may also cause additional mode coupling and affect the overall performance of the system.

4.7 DISPERSION OPTIMIZATION OF SINGLE MODE FIBERS

From the foregoing discussion, it is clear that multimode fibers are generally affected by both intramodal dispersion as well as intermodal dispersion. The intermodal dispersion, however, can be minimized in a multimode fiber by making the refractive index profile graded. The graded index fiber with an optimal index profile can significantly increase the bandwidth of transmission of this fiber over its step-index counterpart. On the other hand, in a single mode fiber the intermodal dispersion is totally absent because the fiber supports only one mode and the question of group delay arising from various modes does not arise in this case. Therefore, single mode fibers are affected by intramodal dispersion which comprises material dispersion and waveguide dispersion. It may be recalled here that waveguide dispersion in a single mode fiber is relatively large as compared to that in a multimode fiber because the cladding carries a significant amount of power (nearly 20%) of the total power in the case of the former while very little power is carried by the cladding in the case of the later. However, the overall dispersion of single mode fibers is much less than that of multimode fibers. Therefore, single mode fibers are widely used for high-speed long-haul optical communication systems. It may be pointed out that the dispersion characteristics of single mode fibers can be tailored by changing the geometry and/or index profile of the fiber core and cladding. This section deals with various design techniques for tailoring the dispersion characteristics of single mode fibers.

4.7.1 Dispersion-shifted Fiber (DSF)

The single mode fibers do not suffer from intermodal dispersion. The overall dispersion of the fiber is thus determined by the intramodal dispersion which has two components, e.g. material dispersion and waveguide dispersion. Out of these two components, the material dispersion of the fiber cannot be changed much. On the other hand, the waveguide dispersion component can be significantly by changing the refractive index profile from the conventional step-index profile to a more complex index profiles (Ainsle *et al* 1986). It has been seen earlier (Fig. 4.11) that at wavelengths longer than the ZMD (zero-material dispersion) point the material dispersion and waveguide dispersion components are of opposite sign. Therefore, it is possible that these two components cancel each other at some longer wavelength. In other words, the wavelength corresponding to zero material dispersion can be shifted to a longer wavelength to cause lowest intermodal dispersion. Further, since the waveguide dispersion can be tailored by using different designs, the overall dispersion can be made minimum at a desired wavelength. In general, the fibers are designed in such a way that the minimum dispersion point is shifted to 1.55 μm. This is done because silica fibers offer minimum loss at this wavelength and therefore, one may get the benefit of both the lowest attenuation and lowest dispersion at this wavelength. The third-generation optical communication uses single mode fibers operating at 1.55 μm wavelength with optical sources and detectors based on matured InP/InGaAs technology. The single mode fibers in which the minimum dispersion point is shifted to the desired wavelength are called dispersion-shifted fibers (DSFs). The dispersion characteristics of single mode fibers are often modified in a way so as to exhibit a low dispersion window over the entire low-loss region ranging typically from 1.3 to 1.6 μm for silica based fibers. This type of single mode fibers allows the spectral requirements of the source to be less stringent and finds application in flexible wavelength-division multiplexing (WDM) scheme. Such single mode fibers are known as Dispersion-Flattened Fibers (DFFs).

It can be easily seen from Eq. (4.67) that the waveguide dispersion component D_{wg} depends on the core radius a (because V-number at a given wavelength depends on the core radius), the index deviation, Δ and also on the shape of the refractive index profile (Ainsle *et al* 1986). In principle, a variety of refractive index profiles are capable of tailoring the waveguide dispersion component so as to adjust the overall minimum dispersion point to the desired wavelength above the ZMD point.

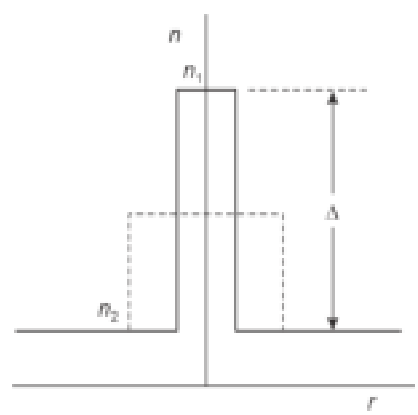

Fig. 4.17 The refractive index profile of a dispersion shifted fiber (solid line) shown against the profile of a non-shifted standard fiber (dotted line) showing reduced radius of the core and increased Δn in the former case

The simplest technique involves reduction of core radius and a concomitant increase in the value of index deviation ratio, Δ of a conventional step-index profile of a fiber as illustrated in Fig. 4.17 so as to maintain the cut-off of LP_{11} mode in the wavelength range of

1–1.3 μm. It has been reported that by reducing the core radius of a single mode step-index fiber from 5.5 μm to 1.8 μm it is possible to shift the wavelength for zero overall intramodal dispersion from 1.3 μm to 1.75 μm, assuming a constant value of the material dispersion (White *et al* 1979). Figure 4.18 illustrates typical material and waveguide dispersion curves for step-index fibers with different sizes and material composition of the core (Ainsle *et al* 1986). The resultant total dispersion curves are also shown in the figure. The V-number of the fibers has been maintained in the range ($1.5 < V < 2.4$) so as to ensure the confinement of the fundamental mode. In order to achieve this it is necessary to reduce the core radius linearly while increasing Δ as a square function. The latter is achieved by increasing the concentration of Germania in the core.

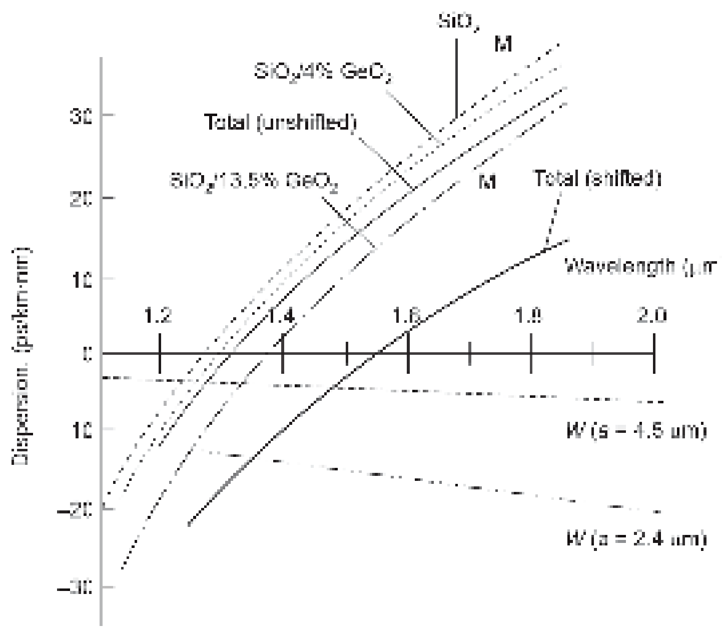

Fig. 4.18 Variation of material dispersion, waveguide dispersion and total dispersion for a dispersion–shifted fiber as well as a standard fiber with wavelength for the single-mode step-index case (after Ainslie *et al* 1986)

The first dispersion shifted fiber was demonstrated by Cohen *et al* (Cohen *et al* 1979) by drawing fibers with core radii of 3.4 and 2.5 μm from a fiber pre-form with a step-index profile having a high value of index deviation, Δ. These two fibers exhibited zero total intramodal dispersion at 1.375 and 1.54 μm respectively. However, these fibers exhibited high value of loss (> 2 dB/km). Following the step-index design approach other researchers also reported dispersion-shifted single mode fibers exhibiting zero total dispersion at the desired 1.55 μm (Kawana *et al* 1980; Miya *et al* 1983). However, all these fibers exhibited rather high loss when operated in the regime of 1.55 μm. The excessive loss of the dispersion-shifted fibers derived from the step-index design approach is generally attributed to stress-induced defects at the core-cladding interface, inhomogeneities associated with waveguide variation at core-cladding interface (Ainslie *et al* 1986). In order to reduce the stress induced across the core-cladding interface caused by abrupt change in the material composition graded-index single mode fiber approach was adopted by different researchers.

Several graded-index structures tried for making dispersion-shifted fibers are shown in Fig. 4.19. These include the triangular, trapezoidal, depressed cladding triangular and Gaussian refractive index profile structures.

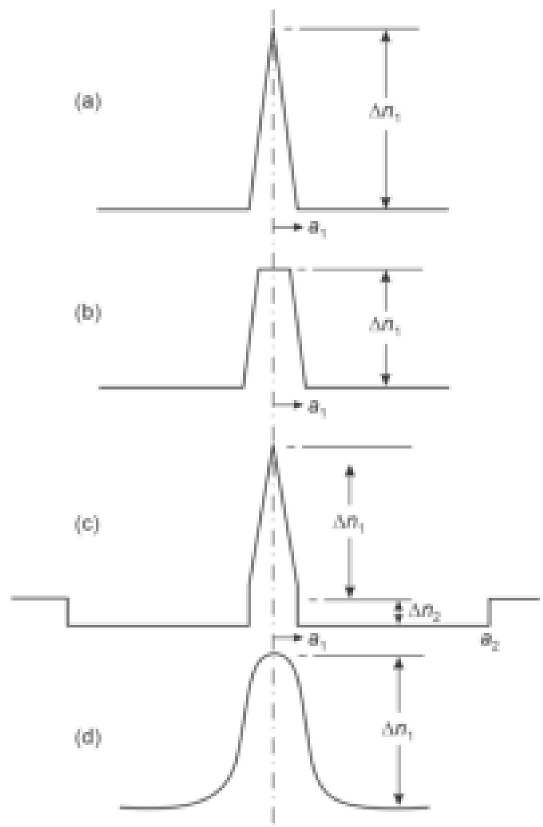

Fig. 4.19 Refractive-index profiles of dispersion-shifted graded-index fibers (a) Triangular profile (b) Trapezoidal profile (c) Depressed-cladding triangular profile (d) Gaussian profile (after Ainslie *et al* 1986)

The first single-mode fiber with triangular profile (Fig. 4.19(a) reported by Saifi *et al* exhibited a loss of 0.3 dB/km (almost the same low-loss as a non-shifted fiber) at 1.55 μm and zero dispersion at a wavelength 1.40 μm. The triangular index profile fibers can reduce the loss significantly as compared with step profiles. This has been discussed extensively in the literature (Ainslie, 1983; Saifi, 1983). It is understood that the loss and dispersion characteristics of the fiber largely depend on the drawing tension during the fabrication of the fiber. The trapezoidal profile fibers and the depressed-cladding triangular profile designs shown in Figs 4.19(b) and (c), respectively, have reduced the sensitivity to microbending caused by shifting of the LP_{11} mode cut-off to a longer wavelength. In the simple triangular index profile fiber the optimization of dispersion and loss parameters at 1.55 μm causes the cut-off of LP_{11} mode to occur in the wavelength range 0.85–0.9 μm resulting in an increased sensitivity to microbending loss at 1.55 μm. In an attempt to reduce the microbending loss several other graded-index profiles have been tried in the past. These include a trapezoidal index profile, triangular profile with depressed cladding and Gaussian profile as shown in Fig. 4.19. The first single-mode fiber with a Gaussian profile made by the VAD technique (Miyamoto *et al* 1985) exhibited a loss of 0.21 dB/km at 1.55 μm. It has been reported that the sensitivity to microbending loss can be greatly reduced by making use of a triangular index profile incorporated in a depressed cladding index configuration (Sang *et al* 1985), shown in Fig. 4.19(c). The core-cladding diameter ratio was maintained above 8.5 to ensure minimum leakage loss of the LP_{01} mode at 1.55 μm. The depressed cladding triangular profile single mode fiber shifts the LP_{11} mode cut-off to 1.1 μm. These fibers are however susceptible to increased splice loss.

Many other complex index profile structures including a dual-shaped core DSF have also been investigated in an attempt to provide an improvement in the microbending loss performance at the 1.55 μm wavelength region (Chung *et al* 1985; Bernard *et al* 1984). These structures include triangular-profile multiple index, segmented-core triangular profile and dual-shaped core as shown in Fig. 4.20. The dual-shaped core DSF exhibits an attenuation in the tune of 0.22–0.24 dB km^{-1} at 1.55 μm with a dispersion of 0–2 $psnm^{-1}km^{-1}$. Dispersion-shifted fibers have been commercially deployed for high-speed single-channel transmission at 1.55 μm. These fibers are not suitable for wavelength-division multiplex (WDM) operation. These fibers suffer from cross-talk which occurs when multiple signals are grouped around 1.55 μm to reduce dispersion. This is the reason that dispersion-shifted fibers are no longer recommended for commercial deployment.

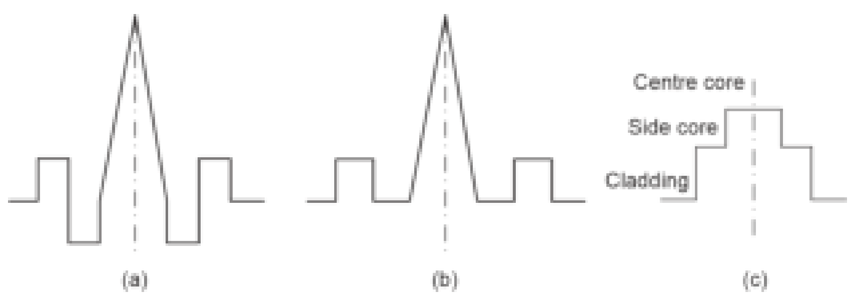

Fig. 4.20 Complex refractive index profiles for dispersion-shifted fibers (a) Triangular profile multiple index (b) Segmented-core triangular profile (c) Dual-shaped core

4.7.2 Dispersion-flattened Fiber (DFF)

An alternative method of achieving low-dispersion over a range of wavelength was first proposed in 1974 (Kawakami *et al* 1974). The original *"W"* fiber structure (Fig. 4.21(a)) was proposed by them. The relatively narrow depressed cladding region helps in modifying the waveguide dispersion to give an overall intramodal dispersion curve which turned

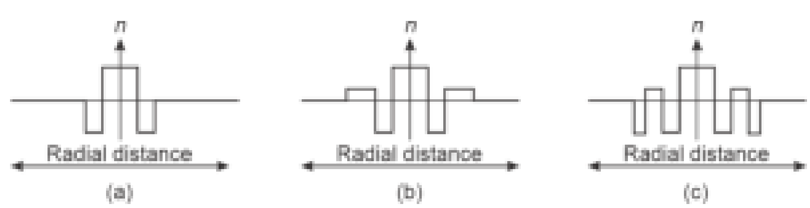

Fig. 4.21 Refractive-index profiles for dispersion-flattened fibers (a) "W" fiber (double clad) (b) Triple clad fiber and (c) Quadruple fiber

out to give two wavelengths for zero dispersion as shown in Fig. 4.22 (Ainsle *et al* 1986). Specific parameters of "W" fiber structure for confining the dispersion to ± 1 pskm^{-1}nm^{-1} in the wavelength range of 1.35–1.67 μm was later proposed in 1979 (Okamoto *et al* 1979; Okamoto, 1979) (Fig. 4.22).

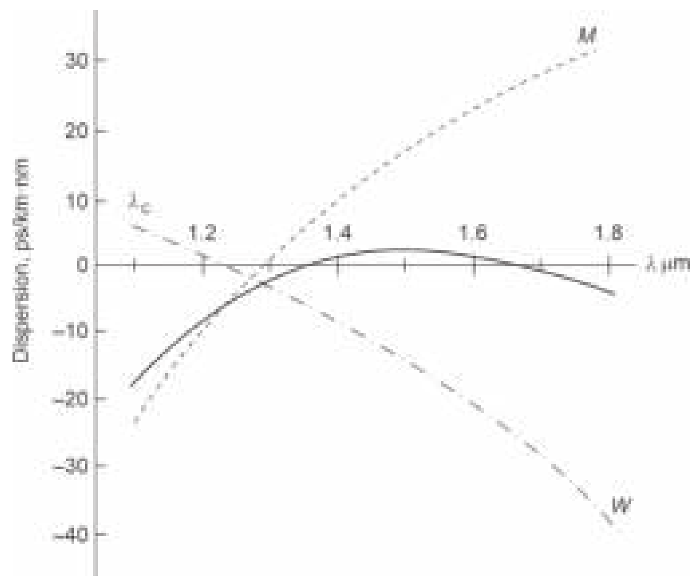

Fig. 4.22 Variation of material dispersion, waveguide dispersion, and total dispersion for a dispersion-flattened W-profile fiber with wavelength (after Okamoto 1979)

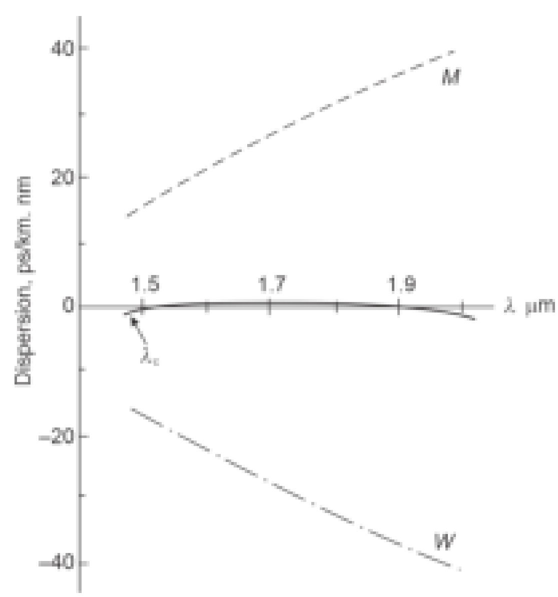

Fig. 4.23 Variation of material dispersion, waveguide dispersion and overall dispersion of a dispersion flattened fiber (after Okamoto *et al* 1979)

A step-index fiber with a very large value of index difference and small core diameter was also studied by Okamoto *et al* (Okamoto *et al* 1979) in an attempt to create waveguide dispersion that mirrors the material dispersion so as to give almost zero dispersion in 1.5–2 μm region (Fig. 4.23). The major drawback of this fiber is that it suffers from excessive loss arising out of large index difference. The first practical dispersion-flattened fiber with a "W" structure was reported in 1981 (Miya *et al* 1981). A major limitation of this type of fiber arises from the requirement of high degree of dimensional control for making reproducible DFF in addition to relatively high overall fiber attenuation and sensitivity of such fibers to bending losses. In an attempt to reduce the sensitivity of "W" structure dispersion-flattened fiber to bending losses the basic "W" structure is modified to form triple clad (TC) and quadruple clad (QC) structures shown in Figs 4.21(b) and (c) respectively (Cohen *et al* 1982; Jang *et al* 1983). Several other structures including one involving segmented-core for single-mode dispersion-flattened operation have been proposed (Bhagavatula *et al* 1983). These complex DF fiber structures are reported to have low bending loss yet maintain a very low dispersion in the desired wavelength region. However, successful commercial deployment of these fibers is largely dependent on the reproducibility, cost and compatibility vis-à-vis conventional single mode fibers.

4.7.3 Polarization Maintaining Fibers

Practical single-mode fibers generally suffer from the birefringence phenomenon, discussed earlier in this chapter. Birefringence in turn causes the polarization state of the light to change as it propagates down the fiber. Conventional photodetectors used in IM/DD systems are insensitive to the polarization state of the incoming light. As a result birefringence phenomenon does not affect conventional optical communication system. On the other hand, it may be a cause of major concern in coherent optical communication systems wherein the incoming light signal is superimposed on locally generated optical signal. Further, the orthogonally polarized modes in a birefringent single-mode fiber may manifest in the form of polarization modal noise resulting from the interference of the two components. Likewise, the delay difference between the two orthogonal birefringent components may lead to polarization mode dispersion (PMD). In addition, a change in polarization state of the light may be of major concern when a single-mode fiber is used in conjunction with optical components such as optical modulators, couplers and various other forms of optical waveguides. Therefore, in several situations it is necessary for a single mode fiber to maintain the state of polarization of light propagating through the fiber. A single mode fiber that maintains the state of polarization of the light propagating through it is referred to as polarization maintaining (PM) fiber. Polarization maintained fibers can be classified into two major groups: High-birefringence (HB) and low-birefringence (LB) fibers. Conventional single mode fibers generally exhibit birefringence in the range of $B_F = 10^{-6}$ to 10^{-5} (Stolen *et al* 1987). A single mode needs to have B_F better than atleast 10^{-4} for polarization maintenance (Noda *et al* 1986). High-birefringence (HB) fibers can be further classified under two categories, e.g. single-polarization and two-polarization fibers. A single-polarization fiber can be designed to allow only one mode to propagate by imposing a cut-off condition on the other mode by exploiting the difference in bending loss between the two modes with orthogonal polarization.

Summary

- The basic transmission characteristics of different types of fibers that degrade the quality of the propagating optical signal are discussed.
- Two major factors that degrade the signal are, attenuation (or loss) and dispersion.

- Attenuation determines the maximum repeaterless distance between the transmitter and the receiver.
- Dispersion determines the maximum rate at which the transmission is possible over the fiber.
- Attenuation is measured in decibels given by,

$$\alpha(\text{dB/km}) = \frac{10}{L}\log_{10}\left(\frac{P_{\text{in}}}{P_{\text{out}}}\right)$$

- The major factors that contribute to attenuation are as follows:
  – Intrinsic absorption by fiber material
  – Extrinsic absorption caused by foreign materials present in fiber
  – Atomic defect caused by high energy ionizing radiation
  – Scattering loss (primarily Rayleigh type)
  – Bending loss
  – Core–cladding loss due to different attenuation in the two regions.
- Dispersion causes intersymbol interference in digital optical communication.
- There are two types of dispersion, e.g. intramodal dispersion occurring within a particular mode and intermodal or simply modal dispersion occurring among various modes.
- Intramodal dispersion has two forms, e.g. material dispersion due to dependence of the refractive index on the wavelength and waveguide dispersion.
- Intermodal or modal dispersion arises out of the difference in velocity of individual modes in a multimode fiber. Different modes travel with different velocity causing a spreading of pulse.
- The dispersion is measured in terms of RMS spreading of the pulse.
- Overall spreading of pulse due to both intramodal and intermodal can be expressed as the following:

$$\sigma = \left(\alpha_{\text{intra}}^2 + \sigma_{\text{mod}}^2\right)^{\frac{1}{2}}$$

- The bandwidth can be estimated assuming the pulse shape to be Gaussian in nature as:

$$B \approx \frac{0.186}{\sigma}$$

- RMS pulse spreading due to material dispersion is,

$$\sigma_{\text{mat}} = \frac{L\sigma_\lambda}{c}\left|\lambda\frac{d^2 n_1}{d\lambda^2}\right|$$

- RMS pulse spreading due to waveguide dispersion is,

$$D_{wg} = \frac{1}{L}\frac{d\tau_{wg}}{d\lambda} = -\frac{n_2\Delta}{c\lambda}V\frac{d_2(Vb)}{dV^2}$$

- The overall RMS pulse spreading can be obtained as:

$$\sigma_{\text{inter}} = \left(\sigma_{\text{mat}}^2 + \sigma_{wg}^2\right)^{\frac{1}{2}}$$

- RMS pulse spreading due to intermodal dispersion is,

$$\sigma_{\text{mod}} = \frac{L(\text{NA})^2}{4\sqrt{3}n_1 c} \approx \frac{Ln_1\Delta}{2\sqrt{3}\,c}$$

- The modal dispersion can be reduced by making use of graded-index fiber.
- The first order modal dispersion can be eliminated by making use of a parabolic index fiber with $\alpha \approx 2$.
- In a practical fiber, modal dispersion is automatically reduced because of natural mode coupling where higher order modes transfer their energy to cladding modes and get eliminated.

- In addition to the above dispersion components, single mode fibers exhibit Polarization Mode Dispersion (PMD) arising out of birefringence property of the fiber.
- Single mode fibers do not suffer from intermodal dispersion and can therefore support much higher bit rates as compared to multimode fibers.
- The wavelength corresponding to intramodal dispersion minimum, can be adjusted by suitably adjusting waveguide dispersion.
- The tailoring of intramodal dispersion can be used to realize dispersion-shifted and dispersion-flattened fiber so that the minimum attenuation window around 1550 nm can be used simultaneously with low dispersion of the fiber in this region.
- Further, single mode fibers can be designed suitably to maintain the desired state of polarization. This type of single mode fiber is called Polarization Maintaining (PM) fiber.

Problems

4.1 An optical power of 0 dBm is launched into an optical fiber having an average loss of 0.8 dB/km. Estimate the value of optical power (in mW) available at the end of the fiber, if the length of the fiber is 5 km.

4.2 A repeater-less optical link has a length of 2 km. Estimate the value of minimum detectable power (in dBm) needed for the detector at the receiver end when an optical power of 0.5 mW is launched into the fiber at the transmitter end. The fiber has an average loss of 0.5 dB/km and total connector loss of 1.2 dB.

4.3 A 5 km long optical fiber link uses a photodetector at the receiver end with a minimum detectable power of 2 μW. The fiber has an average loss of 0.7 dB/km and a total joint loss over the link of 2.5 dB. Calculate the minimum value of power needed to be launched into the fiber at the transmitter end for reliable operation of the link.

4.4 Repeat problem 4.3 by considering an additional safety margin of 3 dB.

4.5 Distinguish between linear and non-linear scattering in an optical fiber. Why is non-linear fiber is not a cause of major concern in conventional optical fiber communication?

4.6 Discuss some of the major applications of non-linear scattering.

4.7 A 50/200 mm step-index fiber has a core refractive index of 1.458 at a wavelength of operation of 850 nm. The critical radius of curvature from the view point of macro-bending lossis 3 mm. Estimate the value of the cladding refractive index.

4.8 A 10 km long optical fiber causes an rms pulse broadening of 25 ns due to material dispersion alone when the power is launched from an LED operating at 1300 nm with a spectral width of 40 nm. Estimate the material dispersion parameter of the fiber.

4.9 For a single mode step-index fiber with $V = 2.4$ operating at 1330 nm has a core refractive index of $n_1 = 1.458$ and $\Delta = 0.1\%$. Estimate the value of the waveguide dispersion parameter assuming the value of $d^2(Vb)/dV^2 \approx 0.1$ at an operating wavelength of 850 nm.

4.10 A multimode step-index fiber has a numerical aperture of 0.18 and a core refractive index of 1.5. The fiber exhibits an overall intramodal dispersion of 225 ps km^{-1}. Estimate the value of the rms pulse broadening when light is launched into the fiber from an LED operating at 1330 nm having an rms spectral width of 30 nm. Calculate the value of the bandwidth of a 20 km link based on this fiber.

4.11 Compare and contrast the rms values of pulse broadening of the following fibers:
  (i) A multimode SI fiber with core refractive index $n_1 = 1.49$ and relative index difference $\Delta = 1\%$.
  (ii) A GI fiber having an optimum parabolic index profile with $n(0) = 1.49$ and relative index difference $\Delta = 0.5\%$.

4.12 What is birefringence? Explain how this phenomenon gives rise to PMD in a single mode fiber. How would you design a single mode fiber to combat dispersion and attenuation simultaneously at a given operating wavelength?

4.13 What is material dispersion? How does this parameter affect the bit rate of transmission? The material dispersion parameter for a glass fiber is 25 ps nm^{-1} km^{-1} at an operating wavelength

of 1.55 mm. Compare and contrast the values pulse broadening due to material dispersion within the fiber for the following cases.
(i) Light is launched from a DH-LED having a peak wavelength of 1.55 mm and a spectral width of 30 nm.
(ii) Light is launched from an ILD operating at the same peak wavelength with a spectral width of 2 nm.

4.14 What are the two factors that limit intermodal dispersion in practical fibers?
4.15 What is PMD? How does it affect an IM-DD system and coherent OC system?
4.16 What are the major components of dispersion in a single mode fiber? Compare and contrast the relative contribution of these components with suitable mathematical formulations. What is ZMD? How would you shift the ZMD point?
4.17 A multimode SI fiber has a relative refractive index difference of 1% and a core refractive index of 1.46. The rms pulse broadening of the fiber due to material dispersion is 1.96 ns km^{-1} and that due to waveguide dispersion is 0.2 ns km^{-1}. Estimate the bandwidth of a 5 km long link assuming RZ code when there is no mode coupling.
4.18 Repeat Problem 4.18 assuming that the mode coupling gives a characteristic length equivalent to 60% of the actual fiber length.

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CHAPTER 5

Optical Sources

The transmitter of an optical fiber communication system consists of an optical source, optical interconnects, and associated electronics necessary for the modulation of the light output following the information or intelligence signal. An optical source is the key component of the optical transmitter unit. The purpose of an optical source is to convert an electrical signal reliably into optical radiation (E–O conversion).

5.1 SOURCES FOR OPTICAL FIBER COMMUNICATION

There are a variety of optical sources that convert electrical energy to an optical signal (light). However, because of the compatibility with the dimensions of an optical fiber, semiconductor optical sources are generally used in fiber optic communication systems. There are two types of semiconductor optical sources, e.g. light-emitting diode (LED) and injection laser diode (ILD) often simply referred to as laser diode (LD). An optical source should ideally meet the following requirements for use in the transmitter unit of an optical fiber communication system (Senior, 1992; Keiser, 2000).

- The size of the emitting optical source must be compatible with the size of the optical fiber.
- The emitted light should be preferably directive for easy launching of light from the source to the fiber.
- The light output (optical power) must vary linearly with the electrical input for faithful E–O conversion.
- The source must have a reasonably high E–O conversion efficiency.
- The emission wavelength should match the attenuation window of the fiber (the wavelength at which the fiber offers low attenuation).
- The spectral width of the source should be small to reduce chromatic dispersion during propagation through the fiber.
- The source should have a high modulation capability, i.e. a large bandwidth to meet the large information-carrying capacity of the fiber.
- The source must be able to couple sufficient optical power to the fiber so that it can travel a long distance and still deliver the required power to the detector for faithful conversion of the optical signal into an electrical signal (O–E conversion).
- The source should have a moderately long life.

Both LED and ILD meet the basic requirements of an optical source for use in optical fiber communication systems. Wideband continuous spectra sources such as incandescent lamps cannot be used for optical fiber communication systems. Structurally both LED and ILD consist of *p-n* junction made of direct bandgap semiconductor materials (most commonly III-V materials). When forward-biased electrons and holes are injected into *p* and *n* regions respectively where they recombine with the majority carriers. In direct bandgap material, the recombination is normally radiative giving rise to the emission of photons (light). However, all recombination is not radiative. Many of them are non-radiative and the energy is released in the form of heat which is absorbed by the lattice. The principle of operation of a laser diode differs significantly from that of an LED. The light from an LED results from spontaneous emission following the random radiative recombination of the carriers. On the other hand, a laser diode works on the principle of stimulated emission which dominates only under special circumstances. The characteristics of an LED significantly differ from those of an ILD. Some of the major differences between the two sources in the context of their application in optical fiber communication systems are discussed below.

One of the major differences between an LED and an ILD is that the optical output from the former is incoherent whereas that from the latter is coherent. This is because light in an ILD is produced in an optical resonator that ensures both spatial and temporal coherence of the light emanating from the cavity. The spatial coherence ensures that the output light is highly monochromatic whereas temporal coherence means that the output beam is highly directional. Since no cavity resonator is used in the case of LED, the light output generally exhibits a relatively large spectral width. The half-power beam width is also quite large because light is emitted in a hemisphere with a cosine power distribution. In Chapter 3, we have seen that total dispersion of an optical fiber depends on the rms spectral width of the source. From this viewpoint, laser diodes are superior to light-emitting diodes. Further, a relatively large beam width in the case of LED makes it extremely difficult to launch power from this kind of source to a single-mode fiber is extremely difficult. LEDs are generally used with multimode fibers. Incoherent light from LED can couple power to a large number of modes supported by the fiber. The bandwidth of an LED is much lower than that of an ILD. The transmission rate in LED-based optical communication systems is generally restricted to 100–200 Mbps. The output power of an LED is much lower than that of an ILD. The output power from an LED can be significantly improved by modifying the structure. However, LEDs have several advantages over ILDs. The average life of an LED is nearly 20 years. The cost of an LED is much lower than that of an ILD. LEDs can be used to launch power efficiently in low-cost plastic fibers. In short-distance optical communication use of LED with plastic fibers can substantially cut down the overall cost of the system. The maximum transmission rate (bps) is, however, severely restricted by the limited modulation capability of the LED and relatively large dispersion in plastic fibers. The performance of an injection laser diode degrades with aging. The cost of an ILD is also very high. ILDs require complex driver circuits as compared to those required in the case of LEDs. A thorough understanding of the principle of operation of light-emitting diodes and injection laser diodes requires some fundamental concepts of semiconductor device physics. In the next few sections, we will discuss some selected topics on semiconductor devices and materials that are essential for understanding various optoelectronic devices discussed in the text. These include direct and indirect bandgap semiconductors, intrinsic and extrinsic semiconductors, various generation and recombination mechanisms, *p-n* junction fundamentals, semiconductor heterojunctions, etc.

5.2 SELECTED TOPICS FROM SEMICONDUCTOR DEVICES

Optical sources and photodetectors are key components of an optical transmitter and receiver respectively of an optical fiber communication system. Both the source and the detector are based on semiconductor devices. To understand the mechanism of operation of these devices it is necessary to know the basics of the semiconductor devices. In the following subsections, some essential concepts are outlined with the presumption that the readers have an elementary understanding of semiconductor device fundamentals.

5.2.1 Energy Band Diagram: Direct and Indirect Bandgap Semiconductor

Discrete allowed electron energies of an isolated atom split into a band of allowed energies as the atoms are brought together closely to form a crystal. The energy band theory of semiconductors is quite complex and is beyond the scope of this book. A rigorous analysis of allowed and forbidden energy bands of a semiconductor can be found in advanced textbooks on semiconductor devices (Kittel, 1992, McKelvey, 1992; Neamen, 2007). The energy band theory of the semiconductor can be derived with the help of quantum mechanics that make use of Schrodinger's wave equation and the one-dimensional Kronig–Penney model of the periodic potential function representing a one-dimensional single crystal lattice. This simple model can explain many important features of the quantum behavior of electrons moving in a crystal. One major result of the analysis reveals that electrons in a crystal can occupy certain allowed energy bands and are prohibited from occupying certain forbidden energy bands. The upper energy band is called the *conduction band* (CB) while the lower energy band is termed as *valence band* (VB). A simplistic energy band diagram is shown in Fig. 5.1(a) illustrating the allowed energy bands and the forbidden energy band. The energy corresponding to the forbidden band is measured in terms of energy (usually in eV) and is called the energy bandgap or simply the bandgap of the semiconductor. The energy bandgap of a semiconductor is the property of the materials and depends on the temperature. In the simple energy band diagram, the energy of the allowed and forbidden bands are shown along the ordinate but the abscissa does not represent anything. A more accurate description of the allowed and forbidden energy bands of a semiconductor is represented by energy (E) versus wave-vector ($\boldsymbol{k}$) diagram, often called as E–$\boldsymbol{k}$ diagram illustrated in Fig. 5.1(b). The E–$\boldsymbol{k}$ diagram can be obtained for a semiconductor by applying quantum mechanics. However, it is sometimes more convenient to describe the operations of semiconductor devices by using the simplistic energy band diagram shown in Fig. 5.1(a). At absolute zero ($T = 0$ K), the electrons are in the lowest energy state so that all states at the lower valence band are filled up and all states at the upper conduction band are empty. The bandgap energy is measured from the top of the valence band to the bottom of the conduction band.

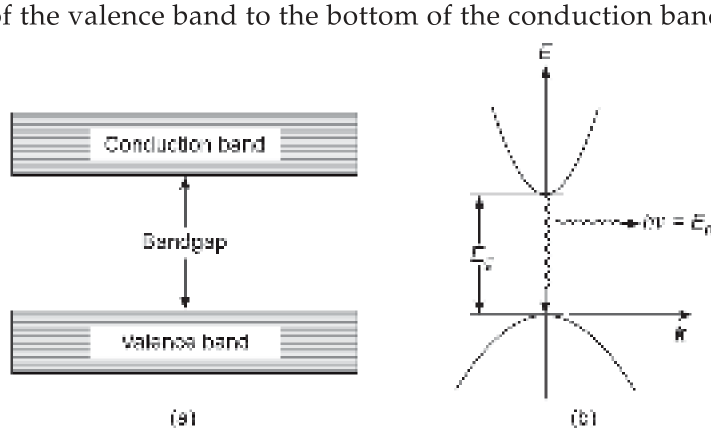


Fig. 5.1 (a) Simplistic energy band diagram (b) E–$\boldsymbol{k}$ diagram

In a Si crystal, the silicon atoms form covalent bonds. Each Si atom at the center shares 4 electrons in the outermost orbit shares one electron each from the four Si atoms surrounding it. Thus, each Si atom is surrounded by eight valence electrons at $T = 0$ K. Fig. 5.2(a) illustrates a two-dimensional representation of the covalent bonding in a single-crystal Si lattice. All the electrons shown in Fig. 5.2(a) occupy and fill the valence band at 0 K making the covalent bonds completed. On the other hand, the upper allowed energy band that is, the conduction band is empty at 0 K. When the temperature increases above 0 K, several electrons may get enough thermal energy to break the covalent bond and jump to the conduction band creating one electron in the conduction band and leaving a vacant position (a missing electron) in the valence band. This missing electron turns the particular bond incomplete. The vacancy in the bond is subsequently filled by another electron from an adjacent bond which in turn becomes incomplete creating a vacant position. The vacant position thus shifts from one covalent bond to the other in the valence band. The vacancy created in the valence band is called a hole which is viewed as a positive charge in the macroscopic description of the transport of charge carriers in semiconductors. The transition of an electron from the valence band to the conduction band therefore results in a generation of an electron in the conduction band a hole in the valence band that is an electron-hole pair (EHP) in total. This is illustrated schematically in Fig. 5.2(b) for a single EHP generation in the lattice. The energy band diagrams of a semiconductor at 0 K and $T > 0$ are illustrated in Fig. 5.3. As the temperature increases above 0 K, many electrons in the valence band get enough thermal energy to break the covalent bond and jump to the conduction band creating more electron-hole pairs. It is interesting to note that the number of electrons in the conduction band created by thermal generation is the same as the number of holes in the valence band (Fig. 5.3(b)). Both the free electrons in the conduction band and holes are mobile in the semiconductor material and both contribute to the conduction of current. This means that electrons in the valence band can move into a vacant hole. In other words, a hole moves in the direction opposite to the direction in which an electron flows.

Fig. 5.2 Silicon crystal structure in two-dimension showing the formation of covalent bond by sharing four outermost electrons from adjacent atoms (a) Structure at 0 K (b) Formation of electron hole pair at $T > 0$ K

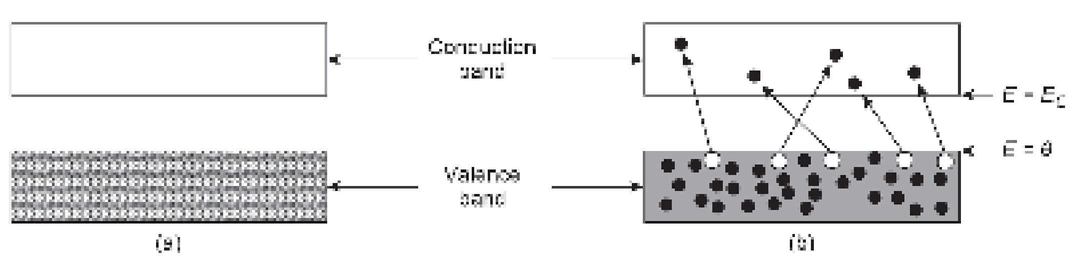


Fig. 5.3 Simplistic energy band diagram of a semiconductor (a) At 0 K showing completely filled valence band and empty conduction band (b) At $T > 0$ K showing partially filled conduction band due to transfer of electrons from the valence band

The electrons excited from the valence band to the conduction band do not stay permanently in the conduction band. The excited electrons look for an opportunity to return to the valence band by releasing the balance energy. This process in which an electron from the conduction band jumps to occupy a vacant electron position (hole) in the valence band is called recombination. In the process, we lose one electron from the conduction band and a hole from the valence band that is, an electron-hole pair in total. From this view point recombination process is just the opposite of the generation process. However, there are a variety of ways by which this recombination may take place in a semiconductor. One way is the characteristics of the band structure of a particular material that decides the type of recombination. Depending on the nature of the variation of the conduction band and valence band energy with the momentum vector ($\boldsymbol{k}$), the semiconductors are classified either as direct-bandgap or indirect bandgap materials. For the transition of an electron from the conduction band to the valence band to take place both energy and momentum must be conserved. Macroscopically, the electrons and holes move in opposite directions. When an electron collides with a hole, the recombination is successful only when the momentum and energy are conserved and both the colliding electron and the hole vanish in the process.

If the recombination is not successful it would result in a scattering process. The difference in the recombination mechanisms in direct and indirect bandgap semiconductors can be best understood with the help of E–$\boldsymbol{k}$ diagrams. The actual E–$\boldsymbol{k}$ diagrams are fairly complex and are illustrated in Fig. 5.4 for three semiconductors, e.g. Ge, Si, and GaAs. shown in Fig. 5.4. It can be easily seen that the conduction band energy of each material has multiple valleys (e.g. L, Γ, and X valley) occurring at different energy minima values depending on the crystal directions. The valence band consists of three parabolas corresponding to heavy holes, light holes, and a spin-split-off band. The difference in the energy between the bottom of the conduction band and the top of the valence band is defined as the energy bandgap of the material. The three energy band diagrams shown in Fig. 5.4 broadly fall under two categories. The basic difference between the two categories considering E–$\boldsymbol{k}$ diagram is that the maximum of the valence band energy (top of the valence band) and the minimum of the conduction band energy (bottom of the conduction band) occur at the same k value (i.e. $k = 0$) in the case of GaAs (Fig. 5.4(c)) and at different $\boldsymbol{k}$ values in the case of Ge (Fig. 5.4(a)) and Si (Fig. 5.4(b)). The simplistic E–$\boldsymbol{k}$ diagrams for the two categories are depicted in Fig. 5.5.

Let us consider the recombination of an electron in the conduction band with a hole in the valence band in the two cases. The most probable recombination will be that where the electron and the recombining hole have the same momentum as shown in Fig. 5.5(a). This type of recombination is accompanied by the release of balance energy ($\sim E_g$) in the form of light that is, in the form of a photon. This type of transition is called direct transition or band-to-band (BTB) transition and also as a photon-assisted transition. The materials having an E–$\boldsymbol{k}$ diagram similar to the one shown in Fig. 5.5(a) and exhibiting recombination associated with the emission of photons are referred to as direct bandgap semiconductors. This form of transition in a direct bandgap material provides an efficient mechanism for photon emission which is exploited in developing semiconductor optical source. In direct bandgap semiconductors the minority carrier lifetime that is, the average time during which the minority carriers remain in a free state before recombination is generally short (~0.1 ns to 10 ns) causing the recombination to occur very fast. GaAs and many other III-V materials (discussed in Section 5.2) fall in the category of direct bandgap semiconductors.

However, there are certain semiconductors in which the maximum of the valence band and the minimum of the conduction band occur at different values of momentum

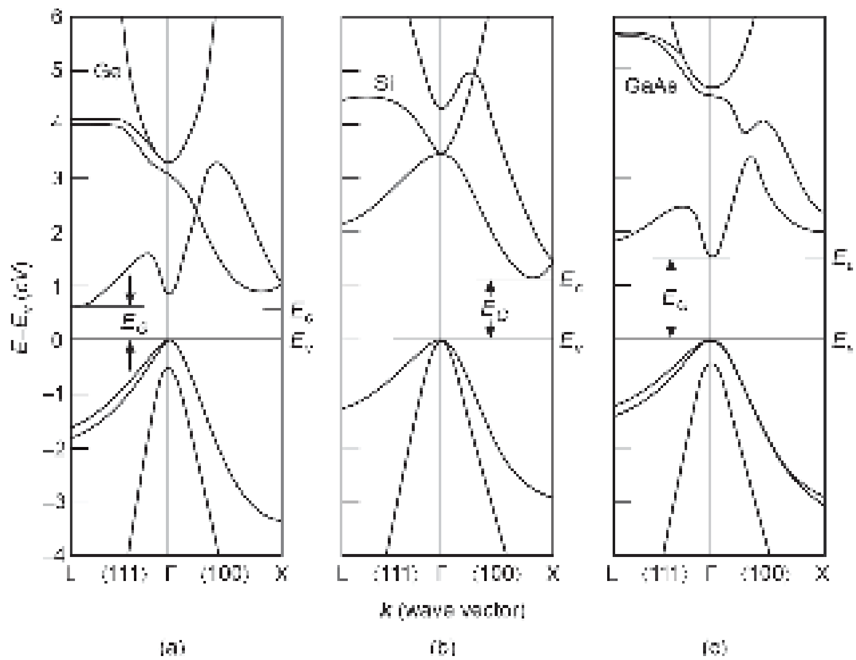


Fig. 5.4 *E*–***k*** diagram showing the variations of conduction and valence band energy with wave vector, ***k*** for (a) Ge (b) Si (c) GaAs (after Sze, 2003)

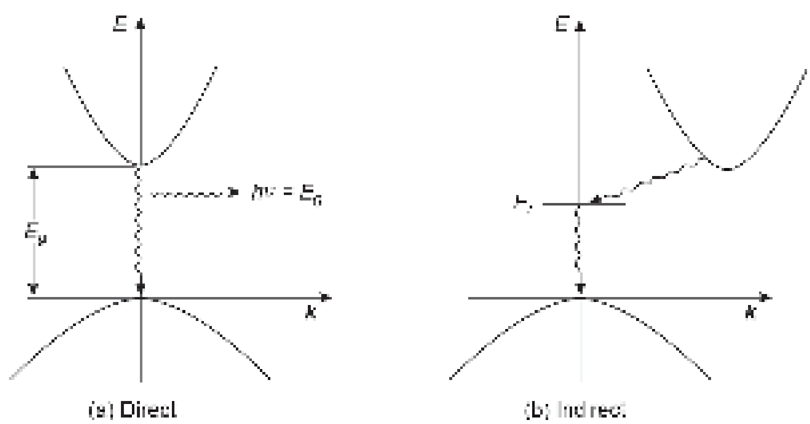


Fig. 5.5 Simplistic *E*–***k*** diagram illustrating direct and indirect recombination

as illustrated in Fig. 5.5(b). In this case, an electron-hole recombination can only take place when the electron in the conduction band loses momentum to have a momentum corresponding to the top of the valence band. The conservation of momentum thus needs the involvement of a third particle, called phonon (quantized lattice vibration). This type of three-particle recombination is generally less probable and referred to as indirect transition or phonon-assisted transition. Semiconductor materials exhibiting *E*–***k*** diagrams similar to the one depicted in Fig. 5.5(b) are called indirect bandgap semiconductors. Both Si and Ge fall in the category of indirect bandgap semiconductors along with some III-V materials. The indirect recombination process is generally slow as compared to direct recombination. The indirect recombination lifetime is generally large (~10 ms to 0.1 ms). This type of transition is less probable and is more likely to be non-radiative. Non-radiative recombination processes may involve lattice defects and other impurities present in the semiconductor (either added intentionally or incorporated unintentionally). These kinds of non-radiative recombination are relatively faster. As the recombination in an indirect bandgap semiconductor is dominated by non-radiative transitions, both Si and Ge exhibit poor levels of electroluminescence. In general, both direct and indirect recombination is possible in any semiconductor. In a direct bandgap semiconductor radiative recombination is generally dominant over non-radiative recombination.

5.2.2 Compound Semiconductors

From the foregoing discussion, it is apparent that only direct bandgap semiconductor materials can be used for making optical sources because electrons and holes recombine radiatively and directly across the bandgap without requiring the involvement of a third particle to conserve momentum. This radiative recombination is quite high enough to produce a significant level of optical emission. None of the elemental semiconductors (e.g. Si and Ge) are direct bandgap materials and therefore they cannot be used for making efficient optical sources. On the other hand, there are a host of compound semiconductors and their alloys which are direct bandgap materials and can produce a significant level of light output in the desired wavelength region for optical fiber communication. In addition, these materials are more flexible and offer several advantages over elemental semiconductors which make them almost indispensable for making optoelectronic devices. In this section, we discuss briefly these semiconductors (particularly III-V materials) for a better understanding of the mechanism of operation of semiconductor sources and photodetectors discussed in this text.

Compound semiconductors, as the name suggests are produced by combining elements from different groups of the periodic table. These materials include III-V, II-VI, IV-VI, and IV-IV compounds. Among these materials, III-V compounds and their alloys are most extensively used in making optoelectronic devices suitable for optical fiber communication. In this section, we will confine our discussion mainly to III-V compounds and their alloys. A binary III-V semiconductor is a compound formed by the chemical combination of one element (say A) from Group-III with another (say B) from Group-V of the periodic table. The resultant compound is often denoted as A_{III} $\mathrm{B_V}$. It is interesting to note that a III-V compound semiconductor has the same number of valence electrons (a total of 8 outer electrons) per atom as Si. GaAs, InP, GaP, AlAs, and InSb are some examples of binary III-V compound semiconductors. It may be pointed out that Si crystallizes in the diamond structure forming pure covalent bonds. On the other hand, compound semiconductors such as GaAs crystallize in the zincblende structure by forming bonds that are predominantly covalent and partially ionic (Bhattacharya, 2007). Historically, InSb was the first binary III-V semiconductor reported in 1950. The material can be synthesized easily and has interesting electronic and optical properties. The direct bandgap of the InSb is 0.17 eV at 300 K and it has an electron mobility of 8 $\mathrm{m^2\ V^{-1}\ m^{-1}}$ (Sze, 2003). The material has been extensively used for the development of optoelectronic devices for far infrared applications. The most important members of the binary III-V compound semiconductor family are GaAs and InP. GaAs has a direct bandgap energy of 1.42 eV at 300 K and an electron mobility of 0.8 $\mathrm{m^2\ V^{-1}\ m^{-1}}$ (Sze, 2003). These materials have been most extensively used for making semiconductor laser sources, Gunn diode, and many other semiconductor devices. The other binary material that has drawn considerable attention from the researchers for development of optical sources is GaP. The material has a bandgap of 2.1 eV which corresponds to the wavelength in the visible region of the spectrum and has found application in the development of visible light-emitting diodes. It may be pointed out that GaP is an indirect bandgap material but can be doped suitably to improve the radiative transition efficiency. Some important physical parameters of a few selected binary III-V semiconductors are listed in Table 5.1. The lattice constant values (in Å) of some important materials are plotted against their bandgap values (in eV) in Fig. 5.6. It is seen that there exists a large number of binary III-V semiconductors that fall in the category of direct bandgap material and can therefore, be used for efficient emission of light at different wavelengths. The fundamental quantum mechanical relationship between the energy bandgap, E, and the frequency, ν can be expressed as

$$E = h\nu$$

where h is Planck's constant and $\nu (= c/\lambda)$; c being the velocity of light and λ the wavelength of the light emitted.

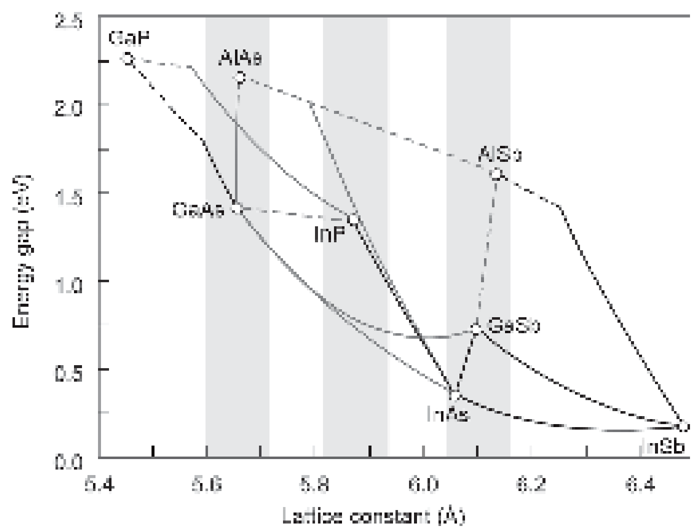


Fig. 5.6 Bandgap-lattice constant diagram for some important binary III-V semiconductors

As the most probable recombination in a direct bandgap material occurs across the energy bandgap in eV, the peak wavelength of emission in micrometers can be obtained as

$$\lambda = \frac{hc}{E_g} \tag{5.1}$$

Table 5.1 Physical parameters of a few selected binary III-V semiconductors

Material	*Bandgap energy (eV) at 300 K (Bhattacharya, 2007)*	*Lattice constant (Å) at 298 K (Sze, 2003)*
GaAs	1.42	5.6532
InP	1.35	5.8697
AlAs	2.16	5.6611
GaP	2.26	5.4495
InAs	0.36	6.0584
InSb	0.17	6.479
GaSb	0.72	6.095
AlSb	1.58	5.136

After substituting the values of the physical parameters, the emission wavelength can be expressed as

$$\lambda = \frac{1.24}{E_g(\text{eV})}(\mu\text{m}) \tag{5.2}$$

Another important feature of binary compound semiconductors is that they can easily form ternary and quaternary alloys. These alloys are solid solutions of corresponding binary compound semiconductors. A ternary III-V alloy consists of three elements from Group-III and Group-V (two from one Group and one from the other). For example, a ternary III-V alloy can be formed by mixing GaAs and AlAs. The resultant alloy is represented as $Al_xGa_{1-x}As$, x being the mole fraction. This means that in the $Al_xGa_{1-x}As$ crystal lattice x atoms of Al and (1–x) atoms of Ga are randomly mixed in the Group-III sublattice and all the Group-V lattice sites are occupied by As. Similarly, GaAs may form a solid solution with GaP to form a new alloy $Al_xGa_{1-x}P$ in which the Group-V sublattice sites are shared by x atoms of P and (1–x) atoms of As while all Group-III lattice are filled with Ga atoms. The value of x can vary from 0 to 1. The ternary material $Al_xGa_{1-x}As$ has emerged as a potential material for application in several novel semiconductor devices while $Ga_x As_{1-x} P$ has been extensively used for developing visible LED. For LEDs operating in the near-infrared region around 1.55 μm in the 3G optical fiber communication is based on $In_xGa_{1-x}As$. In addition to binary compound semiconductors and their ternary alloys, there exist quaternary alloys. A quaternary III-V alloy contains a total of four elements from Group-III and Group-V.

For example, a quaternary alloy such as $Ga_xIn_{1-x}As_{1-y}P_y$ contains two elements from Group-III (e.g. In and Ga) and two elements from Group-V (e.g. As and P) and can be obtained by dissolving P in the ternary alloy $In_xGa_{1-x}As$ or making solid solutions of corresponding binary crystals. Another distinct advantage of alloying binary III-V compounds is that it is possible to vary the bandgap of the resultant alloy continuously and monotonically by changing the composition. Further, the band structure and other optical and electronic properties can be tailored by varying the mole fraction x in the case of ternary and x and y both in the case of quaternary alloys. This particular feature of III-V materials provides enormous flexibility in the design of optoelectronic devices including optical sources and photodetectors operating in the different wavelength regions.

Many physical parameters of ternary alloy semiconductors can be derived from the parameters of the corresponding binaries by considering linear dependence. For example, the lattice constant, a of a ternary alloy $A_x B_{1-x} C$ in terms of the lattice constants of the constituent binaries (AC and BC) can be expressed following Vegard's law as

$$a(A_xB_{1-x}C) = x\cdot a(AC) + (1-x)\cdot a(BC) \tag{5.3}$$

However, the linear relationship does not hold strictly for all properties of mixed crystal alloys. For example, the bandgap of a ternary alloy is a non-linear function of the mole fraction and can be expressed as

$$E_g(A_xB_{1-x}C) = x\cdot E_g(AC) + (1-x)\cdot E_g(BC) - b\cdot x(1-x) \tag{5.4}$$

where b is called the bowing parameter which depends on the combination (AC–BC) of the constituent binary semiconductors.

5.2.3 Intrinsic and Extrinsic Semiconductors

A pure semiconductor containing no impurity is called an *intrinsic semiconductor*. The electrons and holes in an intrinsic semiconductor are generated by thermal vibrations in the atom. The thermal generation process creates electron–hole pairs as discussed earlier. This process may be viewed as the opposite of the recombination process. The conductivity of an intrinsic semiconductor is generally low and can be greatly varied by adding a tiny and controlled amount of chemical impurity in the semiconductor. For example, if a Group-V element (say P) is added to silicon, the P atom containing five electrons in the outermost orbit will substitute a Si atom. Four of its five electrons in the outermost orbit would form covalent bonds with four Si atoms surrounding the P atom and the remaining one electron (called the fifth electron) will remain loosely bound to the nucleus of the P atom. With a small amount of thermal energy, this electron can be easily elevated to the conduction band of the semiconductor. This type of impurity donates electrons and is therefore known as *donor impurity*. Note that the donor impurity atom (P here) becomes a positively charged ion after donating the electron. Further, the electron created in this process does not have an accompanied hole created in the valence band. In other words, by adding a controlled amount of donor impurity it is possible to increase the number of electrons so that the total number of electrons will be much larger than the holes (created by thermal generation). This type of semiconductor in which the electrons are the majority carriers and holes are the minority carriers is called an *n*-type semiconductor. On the other hand, if we add a Group-III atom (say Al) in pure silicon, the three outermost electrons will form three complete covalent bonds with adjacent Si atoms when the Al atom occupies one of the Si sites. The covalent bond with the fourth Si atom would remain incomplete with a missing electron. This means that the addition of Group-III atoms in Si would result in the generation of holes which would not be accompanied by the creation of additional electrons. In this process, the total number of holes in a Si sample can be increased by increasing the concentration of Group-III atoms. The empty position in the incomplete bond is occupied by electrons and the impurity atom becomes negatively charged ions. As the Al dopant accepts electrons this type of impurity is called *acceptor impurity*. Therefore, by adding a controlled amount of acceptor impurity in an intrinsic semiconductor, it is possible to make the total number of holes exceed the number of electrons. This type of semiconductor is called *p*-type semiconductor in which holes are the majority carriers. The donor and acceptor concentrations are expressed per unit volume as N_D and N_A respectively. The donor and acceptor atoms introduce discrete donor and acceptor states inside the forbidden bandgap. The donor level which corresponds to the energy of the fifth electron is close to the conduction band edge in an *n*-type semiconductor while the acceptor level is close to the valence band edge of a *p*-type semiconductor. Qualitative sketches showing the position of donor and acceptor levels and illustrating the formation of electrons and holes are shown in Fig. 5.7.

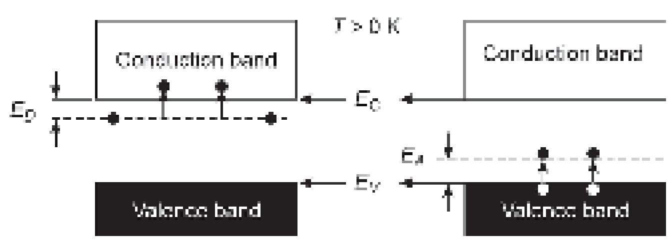


Fig. 5.7 Illustrating the positions of donor level (DL) and acceptor level in the semiconductor energy band diagram

The doping of III-V semiconductors such as GaAs and related materials is more complex than that of elemental semiconductors such as Si and Ge. For example, GaAs can be made *p*-type by introducing a Group-II element (such as beryllium, zinc, and cadmium). The Group-II atom replaces the Ga atom belonging to Group-III and becomes a substitutional acceptor in GaAs creating a hole. Likewise, a Group-VI element such as (selenium, or tellurium) occupies a Group-V site (As in this case). For each Group-VI atom introduced in GaAs the sixth electron is donated for conduction. Group-VI elements thus act as a donor impurity in GaAs.

Group-IV elements such as Si and Ge can also be used for doping GaAs and related materials. In III-V materials, a Group-IV atom (such as Si) may either replace a Group-III (Ga in this case) or a Group-V (As in this case). If a Si atom replaces the Group-III atom (Ga in this case), the Si impurity will work as a donor but if a Si atom replaces a Group-V atom (As in this case) then it will act as an acceptor impurity. Such impurities are called amphoteric. Experimental studies reveal that Ge predominantly works as a donor impurity while Si acts primarily as an acceptor impurity in GaAs (Neamen, 2007).

The introduction of donor or acceptor impurity in a semiconductor changes the distribution of electrons and holes in the semiconductor. The distribution of electrons in the conduction band and holes in the valence band follow Fermi-Dirac statistics. The Fermi energy level lies in the middle of the energy bandgap of the semiconductor when it is intrinsic. However, the Fermi energy changes when a dopant atom is added. For example, when donor impurities are added to a semiconductor, the Fermi energy level moves from near the middle of the bandgap toward the conduction band edge. In this case, the electron concentration is larger than the hole concentration and the semiconductor is *n*-type. Likewise, when the acceptor is added the Fermi energy level moves from near the middle of the bandgap towards the valence band edge. The concentration of holes in the valence band in this case is larger than the electron concentration in the conduction band and the semiconductor becomes *p*-type. The density of state functions in the conduction and valence bands, the Fermi-Dirac distribution, and its variation with energy across the bandgap of the semiconductor are illustrated in Fig. 5.8 along with the distribution of electrons and holes in the conduction and valence band respectively for *p*- and *n*-types semiconductors.

The electron and hole concentrations under thermal equilibrium can be expressed as (Neamen, 2007)

$$n_0 = N_c \exp\big((E_F - E_c)/kT\big) \tag{5.5a}$$

$$p_0 = N_v \exp\big((E_v - E_F)\big)/kT) \tag{5.5b}$$

where k is the Boltzmann constant and T is the absolute temperature, E_c, E_v and E_F correspond to conduction band edge, valence band edge, and Fermi level energies respectively and N_c and N_v are the effective density of states in the conduction and the valence bands respectively, of the semiconductor given by

$$N_c = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2} \tag{5.6a}$$

$$N_v = 2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2} \tag{5.6b}$$

where h is the Planck's constant and m_n^* and m_p^* are the electron and the hole effective mass respectively.

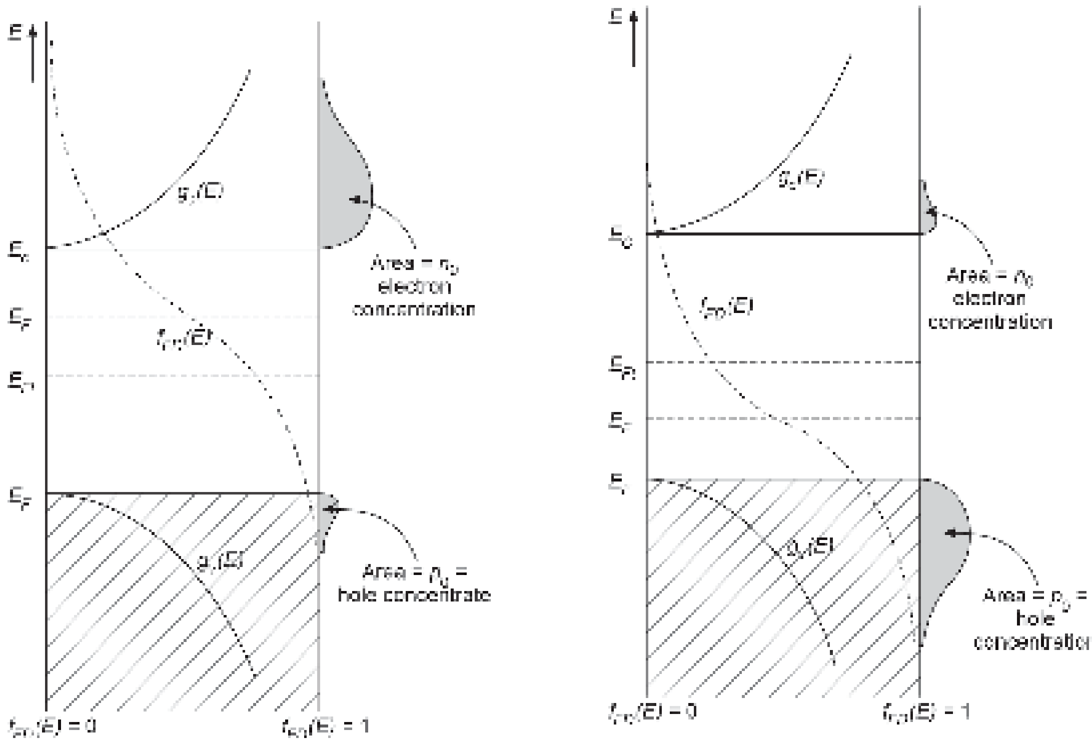


Fig. 5.8 Variations of density of state functions in the conduction and valence bands, (g_c (*E*) and g_v (*E*)), the Fermi-Dirac distribution, f_F (*E*) with energy across the bandgap of the semiconductor along with the distribution of electrons and holes in the conduction and valence bands respectively, for *p*- and *n*-type semiconductors

In thermal equilibrium, the product of n_0 and p_0 is a constant and is given by (Neamen, 2007) as

$$n_0p_0 = N_cN_v \exp(-(E_c - E_v)/kT) = N_cN_v \exp(-E_g/kT) \tag{5.7}$$

where E_g corresponds to the bandgap energy of the semiconductor.

Further, for an intrinsic semiconductor, we may write

$$n_i = N_c \exp((E_i - E_c)/kT) \tag{5.8a}$$

$$p_i = N_v \exp((E_v - E_i)/kT) \tag{5.8b}$$

and

$$n_i = p_i \tag{5.9}$$

Using Eqs (5.8a) and (5.8b), we may write

$$n_i^2 = N_cN_v \exp(-E_g/kT) \tag{5.10}$$

Combining Eqs (5.7) and (5.10), we may write for any semiconductor under thermal equilibrium

$$n_i^2 = n_0\,p_0 = N_cN_v \exp(-E_g/kT) \tag{5.11}$$

That is,

$$n_i = \sqrt{N_cN_v}\exp\left(-\frac{E_g}{2kT}\right)$$

$$= 2\left(\frac{2\pi kT}{h^2}\right)^{3/2}(m_n^* m_p^*)^{3/4}\exp(-E_g/2kT) \tag{5.12}$$

where the parameter n_i is called the intrinsic carrier concentration of the semiconductor. The intrinsic carrier concentration of a semiconductor is a characteristic of the material and depends on temperature.

Consider a semiconductor doped with a donor impurity concentration of N_D. Assuming that all donor atoms are ionized at room temperature, we may write

$$n_0 \approx N_D \quad \text{and} \quad p_0 = \frac{n_i^2}{n_0} \approx \frac{n_i^2}{N_D} \tag{5.13}$$

Similarly, for a p-type semiconductor doped with an acceptor impurity concentration of N_A, we may write

$$p_0 \approx N_A \quad \text{and} \quad n_0 = \frac{n_i^2}{p_0} \approx \frac{n_i^2}{N_A} \tag{5.14}$$

Example 5.1 Calculate the intrinsic carrier concentration of GaAs at 300 K using the following parameters.

$$E_g = 1.43 \text{ eV};\ N_c = 4.4 \times 10^{23} \text{ m}^{-3} \text{ and } N_v = 8.2 \times 10^{24} \text{ m}^{-3}$$

Solution At 300 K, the value of the product (kT) in eV becomes

$$kT = 1.38\times10^{-23}\times300 \text{ J} = \frac{4.14\times10^{-21}}{1.6\times10^{-19}} = 0.0259 \text{ eV}$$

The intrinsic carrier concentration of GaAs at 300 K can be estimated using Eq. (5.12) as

$$n_i = \sqrt{4.4\times10^{23}\times8.2\times10^{24}}\exp\left(-\frac{1.43}{2\times0.0259}\right) = 1.96\times10^{12} \text{ m}^{-3}$$

5.2.4 Compensated Semiconductors

A compensated semiconductor is one in which both acceptor and donor impurities are simultaneously present. For example, a compensated n-type semiconductor can be obtained by diffusing donor impurities in a p-type semiconductor such that, $N_D > N_A$. Similarly, a p-type compensated semiconductor can be obtained by diffusing acceptor impurities in an n-type semiconductor such that, $N_A > N_D$. A semiconductor is completely compensated when $N_D = N_A$ and the semiconductor behaves like an intrinsic one.

Assuming that all donors and acceptor impurity atoms are ionized at room temperature, a semiconductor doped with a donor concentration of N_D and acceptor concentration of N_A in the same region, we may write the charge neutrality condition as

$$N_D + p_0 = N_A + n_0 \tag{5.15}$$

That is,

$$N_D + \frac{n_i^2}{n_0} = N_A + n_0 \tag{5.16}$$

which can be rearranged as a quadratic equation in terms of n_0 as

$$n_0^2 + (N_A - N_D)n_0 - n_i^2 \tag{5.17}$$

Solving Eq. (5.17), we may write

$$n_0 = \frac{N_D - N_A}{2} + \sqrt{\left(\frac{N_D - N_A}{2}\right)^2 + n_i^2} \tag{5.18}$$

It can be easily seen that when $N_D = N_A$, we find

$$n_0 = n_i \tag{5.19}$$

It is important to note that both the electron and hole concentration of a semiconductor doped equally with donor and acceptor impurity are equal to the intrinsic carrier concentration similar to the case of an intrinsic semiconductor. However, the difference lies in the fact that unlike intrinsic semiconductors there exists a large number of donor and acceptor ions in the former case.

Further, when $N_D \gg N_A$, Eq. (5.18) can be approximated by neglecting n_i as

$$n_0 \approx N_D$$

5.2.5 The *p-n* Junction Basics

A *p-n* junction is formed by bringing a *p*-type semiconductor into intimate contact with an *n*-type semiconductor. In such a contact the entire semiconductor is a single-crystal material in which one region is doped with acceptor impurity to form a *p*-type region and the adjacent region is doped with donor impurity to form an *n*-type region. For example, a *p-n* junction can be formed by using an *n*-type material as the starting substrate and compensating the donor impurity of a selected region by the desired acceptor impurity to change the conductivity to *p*-type. The interface between the *n*-type and *p*-type regions is called the *metallurgical junction*.

When a *p-n* junction is created the holes from the *p*-region migrate to the *n* region and electrons from the *n*-region to the *p*-region by diffusion process because the *p*-side is rich in holes and *n*-side rich in electrons. This flow continues unless and until the Fermi level is aligned on both sides. The migration of electrons and holes from *n*-side and *p*-side leaves uncompensated donor and acceptor ions on the *n*- and *p*-side respectively. Thus, a space charge region spreading on both sides of the metallurgical junction is created across the *p-n* junction (Streetman *et al* 2008; Pierret 1988). This region is depleted or empty of free electrons and holes and is, therefore, often referred to as a depletion region. The remaining portions on both sides continue to behave as neutral regions with their original conductivity type. The formation of the space charge region calls for a built-in potential and so also a built-in electric field (extending from *n* to *p* side and confined only in the depletion region) to be developed across the junction and also results in the creation of a barrier that prevents further diffusion of carriers across the junction. It is interesting to note that the built-in electric field exists only in the depletion region. This is illustrated in Fig. 5.9(a). The electric potential in the neutral region being constant the corresponding electric field is zero. In other words, the electric field goes to zero on both sides of the depletion region. The built-in potential is created under equilibrium when no current flows across the junction. The potential cannot therefore be measured by connecting a voltmeter across the junction.

However, when the *p*-side of the junction is connected to the positive terminal of an external battery and the *n*-side to its negative terminal, the barrier height is reduced. This is shown in Fig. 5.9(b). It is interesting to note that the external electric field acts in the direction opposite to the built-in electric field and as a result, the net electric field directed from *n*- to *p*-region is reduced. The reduced built-in electric field forces the depletion region to constrict. Under this condition, the diffusion of electrons and holes takes place again by diffusion because of the reduction in barrier height which was earlier preventing the carriers from crossing the junction under equilibrium. This type of biasing of a *p-n* junction is called *forward biasing*. Under forward bias, the holes from the *p*-side diffuse to the *n*-side, and electrons from the *n*-side diffuse to the *p*-side. The flow is sustained by the difference in the Fermi energy level on the two sides. The flow continues unless and until the external supply is removed so that the Fermi level on both sides is again aligned and the current flow stops. The holes diffusing from the *p*-region to the *n*-region become minority carriers in the *n*-region and they recombine with the majority carriers on the side. Similarly, the electrons entering from the *n*-side to the *p*-side become minority carriers and they recombine with majority carriers (holes) in the *p*-region. From the previous discussion, it is understood that the recombination may be either radiative or non-radiative. For a *p-n* junction made of a direct bandgap material, this type of recombination is predominantly radiative. Thus, a forward-biased *p-n* junction can be used for the emission of light. This mechanism is known as injection electroluminescence and is the basis of the operation of a light-emitting diode (LED) (Streetman *et al* 2008).

When the *n*-side of a *p-n* junction is connected to the positive terminal of an external battery and the *p*-side to its negative terminal the diode is said to be reversed biased. Under reverse-biased conditions, the external electric field acts in the direction of the built-in electric field and the resultant field is increased. This in turn enhances the barrier height and enlarges the width of the depletion region making the diffusion of majority carriers (holes from the *p*-side and electrons from the *n*-side) across the junction extremely difficult. This is illustrated in Fig. 5.9(c). Ideally, no current should flow across a reverse-biased junction. However, minority carriers (electrons from the *p*-side and holes from the *n*-side) may flow under the field existing in the depletion region by the drift mechanism. For example, if an electron from the *p*-side somehow manages to enter the depletion region it would be immediately swept out to the *n*-region by the electric field existing in the depletion region. Similarly, a hole entering into the depletion region from the *n*-side would be swept out to the *p*-region by the same field. The flow of these carriers constitutes the reverse saturation

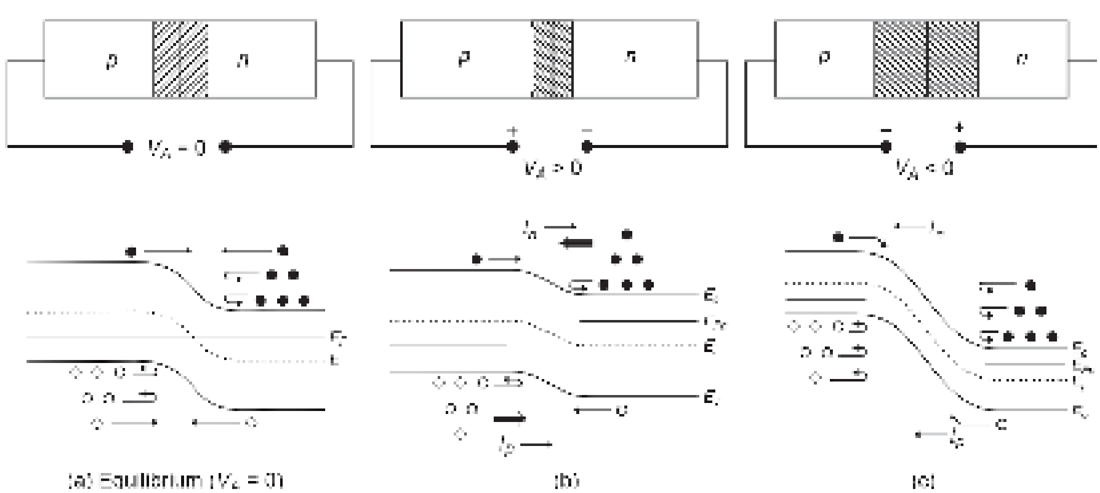


Fig. 5.9 *p-n* junction diode under various bias conditions illustrating the carrier transport with the help of energy band diagram for (a) Unbiased condition (b) Under forward bias condition (c) Under reverse bias condition

current which is generally small in magnitude as compared to the forward current as the minority carriers participate in the former case. However, the minority carriers can be increased by injecting or creating excess carriers in the regions by electrical or optical means to cause a significant change in the reverse current. The first mechanism is exploited in the case of a bipolar junction transistor (BJT) while the second mechanism is the basis of operation of a photodetector. The change in the reverse situation current in the latter case is directly related to the excess carrier generated by incident light. This mechanism has been discussed in detail in Chapter 7.

In conclusion, a *p-n* junction allows a large current to flow across the junction under forward-biased conditions and very little current under reverse-biased conditions making it ideal for rectification purposes (converting an AC to DC) or a switch. The two-terminal device is also called a *p-n* junction diode. Moreover, when a *p-n* junction is made of a direct bandgap material it is possible to obtain emission of radiation (light of wavelength matching with the energy bandgap of the material) when the junction is forward biased. This mechanism has been exploited in the realization of an LED. Further, by operating the *p-n* junction under reverse-biased conditions it is possible to use this device as a detector of optical radiation (light). Normally, a *p-n* junction is asymmetric in the sense that the donor impurity concentration (ND) on the n-side is not the same as the acceptor impurity concentration (NA). Generally, one side is heavily doped as compared to the other side. The heavily doped side is indicated by a superscript '+' so that the junctions are indicated as $n^+ - p$ or $p^+ - n$. In an asymmetrical *p-n* junction, the widths of the depletion regions on the two sides of the metallurgical junction are unequal. The depletion region spreads more on the lightly doped side than on the heavily doped side. Further, when a p^+-n junction is forward biased the injection is predominantly from p^+ to n-region and the injection of electrons from n to p^+ side is negligible. In several practical *p-n* junction devices (including LED and photodetector) asymmetric *p-n* junctions are used for improving the performance.

5.2.6 Semiconductor Heterojunctions

In the previous section, we have only considered *p-n* junctions in which the semiconductor material was homogeneous throughout the structure. Such a *p-n* junction made of the same material with different conductivities (*p* and *n*) on the two sides is referred to as a *homojunction*. When two different semiconductor materials are used to form a junction, the junction is called a *heterojunction*.

A complete treatment of heterojunction is quite involved and requires a good knowledge of quantum mechanics. This is beyond the scope of this text. The use of heterojunction can significantly improve the performance of a device compared to its homojunction counterpart. For example, heterojunctions provide optical and electrical confinement in injection laser diodes and double-heterostructure LEDs. In a bipolar transistor, a heterojunction at the base-emitter interface can significantly improve the emitter injection efficiency and reduce the base resistance. This is exploited in the realization of a heterojunction bipolar transistor (HBT). In a heterojunction field-effect transistor (HFET) the two-dimensional electron gas at the heterointerface can be utilized to design an ultra-fast transistor known as a high-electron-mobility-transistor (HEMT) also called a modulation-doped field-effect transistor (MODFET) or also known as two-dimensional electron-gas field-effect transistor (TEGFET). These heterojunction devices are used for designing the pre-amplifier of an optical receiver discussed in Chapter 8. In a photodetector, heterojunctions are used to improve the quantum efficiency and speed of the detectors. Heterojunction devices are extensively used for making optical sources and photodetectors for optical fiber communication. In this section, an overview of semiconductor heterojunctions is presented so that the principle of heterojunction devices discussed in the text can be appreciated without referring to other books. An extensive discussion on semiconductor heterojunctions can be found in many textbooks (Sharma *et al* 1974; Bhattacharya, 2007).

When two different materials having different values of energy bandgap and other parameters form a heterojunction, the energy band diagram experiences a discontinuity at the junction interface because of the misalignment of the conduction band and valence band on the two sides of the heterointerface. From the technological point of view, a heterojunction consists of two distinct materials having different values of lattice constant. A heterojunction is formed by chemical bonding at the heterointerface. If the values of the lattice constant of the two materials used for forming the heterojunction are very different, a large amount of misfit at the heterointerface introduces dislocations resulting in interface states. In some cases, the misfit may become so large that the device turns out to be virtually useless. We are more interested in near-perfect heterostructures where the difference between the lattice constants of the two materials is very small. Such material combinations are called lattice-matched materials and are only used for making heterojunctions by epitaxial process. The percentage lattice mismatch can be obtained from the difference in the values of the lattice constants of the two materials. For Ge and GaAs the percentage lattice mismatch at the heterointerface is of the order of 0.13% which is an acceptable figure. Ge–GaAs heterojunctions have been studied by several workers for making useful heterojunction devices. In a heterojunction, one side will have a bandgap lower than the other side. The conductivity of the narrow bandgap material is conventionally indicated by using a lowercase symbol (*n* or *p* as the case may be). On the other hand, the conductivity of the wider bandgap material is denoted by using the upper case symbol (N or P as the case may be). In a heterojunction, the conductivity of the two materials used for making a heterojunction does not have to be different always. In a heterojunction when the type of conductivity of the two materials is the same (both *n*-type or both *p*-type), the heterojunction is called *isotype heterojunction*. For example, an *n*-Ge/N-GaAs forms an isotype heterojunction. However, when the materials on the two sides have different types of conductivity, the heterojunction is called an *anisotype heterojunction*. Both types of heterojunctions are extensively used in optical sources and optical detectors used for optical fiber communication. The values of electron affinity (measured in terms of the energy difference between the conduction band edge and the vacuum level) of the two semiconductors (with different values of bandgap) forming the heterojunction are different. As a result, the energy band diagrams of the two semiconductors get differently oriented on the two sides of the heterointerface depending on the electron affinity values of these materials. Based on the relative orientations, the heterojunctions are generally classified under three categories (Type-I, -II and -III) as illustrated in Fig. 5.10. The energy band diagram of a heterojunction between a narrow bandgap material and a wide bandgap material with a different type of conductivity is illustrated in Fig. 5.11. The figure shows the energy band diagrams of a typical *p-n* anisotype heterojunction before and after the formation of the junction. The energy band diagram of the ideal heterojunction is based on Anderson's model (Sharma *et al* 1974) which neglects the presence of any interface states at the heterointerface. Practical heterojunctions do have significant interface state charges which may cause deviation in the energy band diagram from the idealized picture depicted in Fig. 5.11.

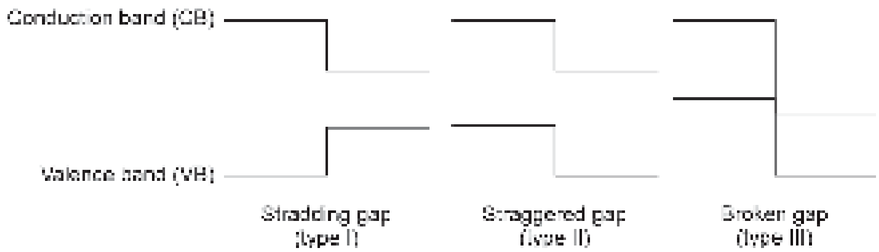
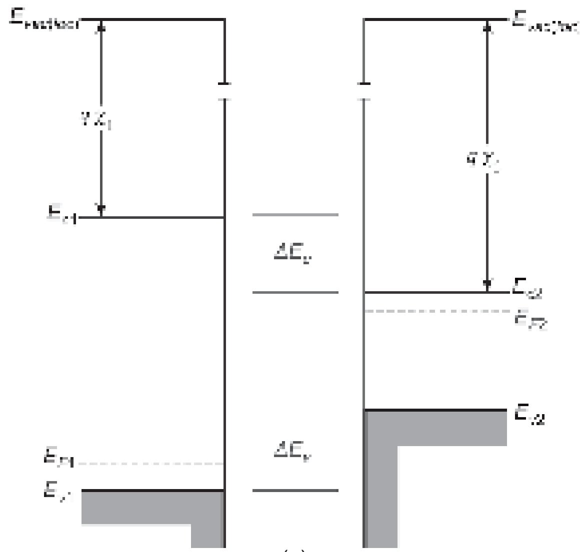
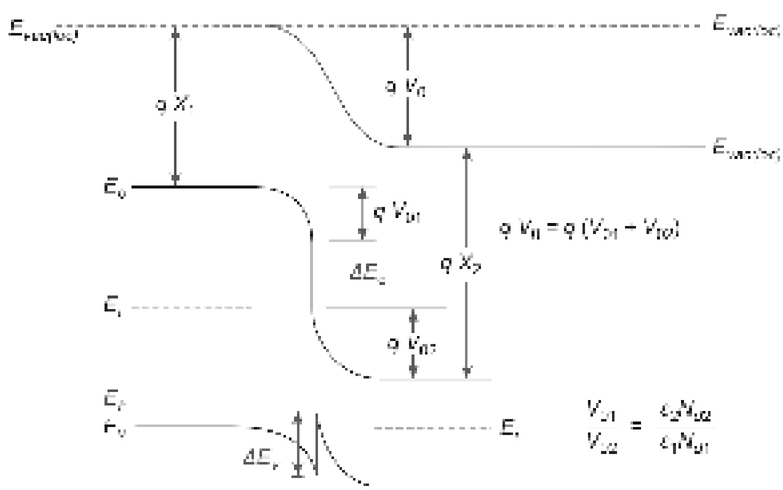


Fig. 5.10 Type-I, II and III heterojunctions

Heterojunction can be analyzed by using Anderson's model (Anderson, 1962). For the heterojunction under consideration, the inter-relations between various parameters can be derived.



(a)



(b)

Fig. 5.11 Energy band diagram of a *p-n* heterojunction (a) Before formation of junction (b) After formation of junction [Note that a local vacuum level is required to maintain the fact that the electron affinity is the characteristic of the material and does not change after formation of the heterojunction]

The energy bandgap difference, ΔE_g is given by

$$\Delta E_g = E_{g2} - E_{g1} \tag{5.20}$$

The energy bandgap difference is asymmetrically distributed between the conduction and valence band edges on the two sides, given by

$$\Delta E_g = \Delta E_c + \Delta E_v \tag{5.21}$$

where ΔE_c corresponds to conduction band edge discontinuity and ΔE_v corresponds to valence band edge discontinuity at the heterointerface. The conduction band edge discontinuity, ΔE_c at the heterointerface can be estimated from the electron affinity values of the two semiconductors for an ideal heterojunction as

$$\Delta E_c = q(\chi_2 - \chi_1) \tag{5.22}$$

Further, writing

$$\delta_P = E_F - E_{v1} \tag{5.23}$$

$$\delta_n = E_{c2} - E_F \tag{5.24}$$

the energy bandgap of the wider bandgap material can be expressed as

$$\begin{aligned} E_{g1} &= \delta_P + \delta_n + qV_{02} + \Delta E_c + qV_{01} \\ &= \delta_P + \delta_n + \Delta E_c + qV_0 \end{aligned} \tag{5.25}$$

where V_0 is the total built-in potential across the heterointerface which is the sum of the built-in potentials V_{01} and V_{02} created on the p-side and n-side of the heterojunction respectively, given by

$$V_0 = V_{01} + V_{02} \tag{5.26}$$

The band edge discontinuities or band-offsets calculated based on Anderson's model assume that the interface is defect-free. According to Anderson's model, the conduction band edge discontinuity of a heterojunction can be obtained by taking the difference between the electron affinity values of the two materials. Experimentally measured values for practical heterojunctions reveal that the band offset values are very different from those predicted based on Anderson's model.

One of the major advantages of binary III-V materials and their ternary and quaternary alloys is that many of these materials can form lattice-matched pairs for the fabrication of useful heterojunction devices. For example, GaAs and AlAs have a lattice mismatch of only 0.04%. This means that the ternary material $Al_xGa_{1-x}As$ for all compositions (different values of x) remains lattice-matched almost perfectly. The GaAs/$Al_xGa_{1-x}As$ heterojunctions have been widely used for the design and fabrication of near-infrared (NIR) sources and photodetectors during the first generation (1G) optical fiber communication systems. Similarly, $In_{0.53}Ga_{0.47}As$ ($E_g = 0.74$ eV) is lattice matched to InP ($E_g = 1.35$ eV). InGaAs/InP have been extensively used for fabrication of optical sources and photodetectors operating at 1.55 μm which is the standard for the 3G optical communication system. The ideal energy band diagrams of typical N – $Al_{0.3}Ga_{0.7}As$/n-GaAs and P – $Al_{0.3}Ga_{0.7}As$/n – GaAs heterojunctions are illustrated in Fig. 5.12.

Figure 5.13 provides useful information for determining the lattice matching between the binary, ternary, and quaternary III-V alloys. The shaded regions help one to determine the lattice-matched substrates when various binary compounds are used to make ternary and quaternary alloys. For example, it can be seen that for a certain composition of $Ga_xIn_{1-x}As$ determined by the mole fraction, x the ternary material will be lattice matched to InP substrate.

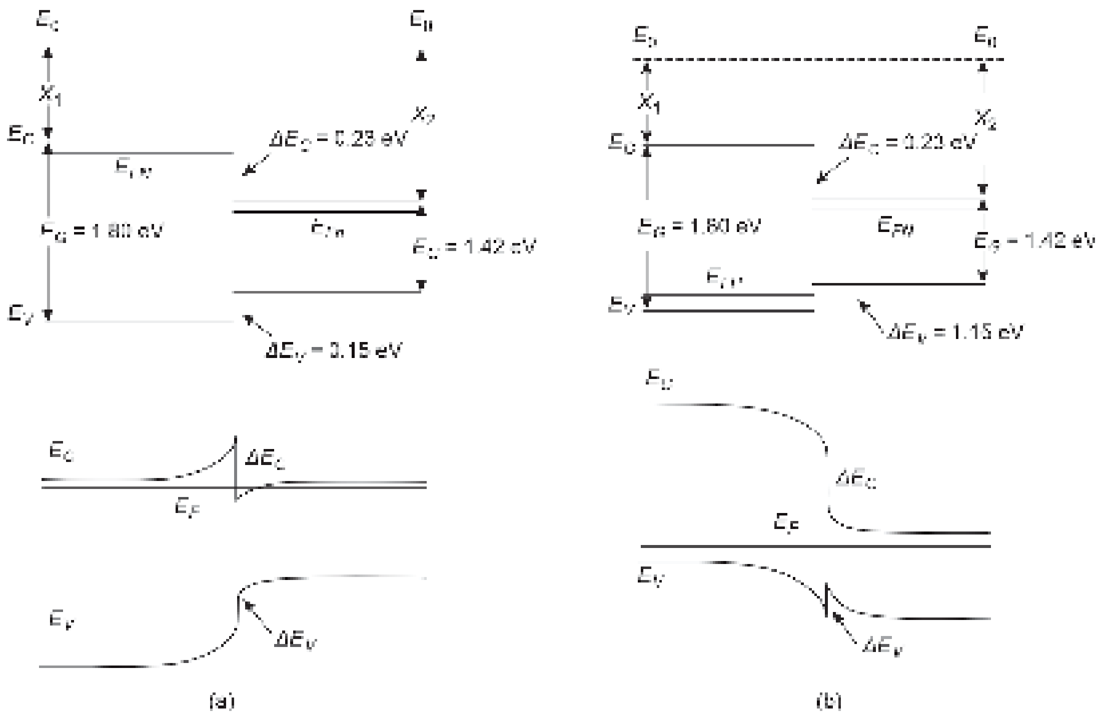


Fig. 5.12 Energy band diagrams of $Al_{0.3}Ga_{0.7}As$/GaAs heterojunction (a) *n-n* heterojunction (b) *p-n* heterojunction

5.2.7 Optical Source Materials

The near-infrared (NIR) region of the optical spectrum ranging approximately from 0.8 μm to 1.65 μm is used in optical fiber-based communication systems. The reason behind this reflects that optical fibers are primarily made of silica glass and silica fibers offer minimum loss in this wavelength region. The intrinsic loss of silica fiber is very high both in the ultraviolet region and infrared region (beyond 2 μm). Further, we have learned that the emission wavelength of a semiconductor material depends on the direct energy bandgap of the material. Many ternary and quaternary III-V semiconductor alloys of binary compounds are direct bandgap materials and can therefore be used as prospective candidates for making optical sources for use in the transmitter of an optical fiber communication system.

For operation, in 800 to 900 nm (0.8 to 0.9 μm) wavelength region of the optical spectrum, the principal material used for making optical sources is $Al_xGa_{1-x}As$. The mole fraction x can be used to vary the bandgap of the alloy to match the desired wavelength. In the ternary material $Al_xGa_{1-x}As$, the parameter x can be varied from 0 to 1. For $x = 0$, it corresponds to GaAs and for $x = 1$ the material corresponds to AlAs. It is interesting to note that GaAs is a direct bandgap material with $E_g = 1.43$ eV while AlAs is an indirect bandgap material with $E_g = 2.16$ eV. Therefore, when x is varied from 0 to 1 the alloy may not be a direct bandgap in nature for all values of x. In other words, there is a critical value of x (0.45 in this example) below which the material is a direct bandgap in nature and above which it behaves like an indirect bandgap material (Sze, 2003). The energy bandgap and the lattice constant of the ternary alloy Al_xGa_{1-x} As can be obtained from Fig. 5.13 following the line joining GaAs ($E_g = 1.43$ eV and $a = 5.64$Å) and AlAs ($E_g = 2.16$ eV and $a = 5.66$Å). The variation of energy bandgap of $Al_xGa_{1-x}As$ with mole

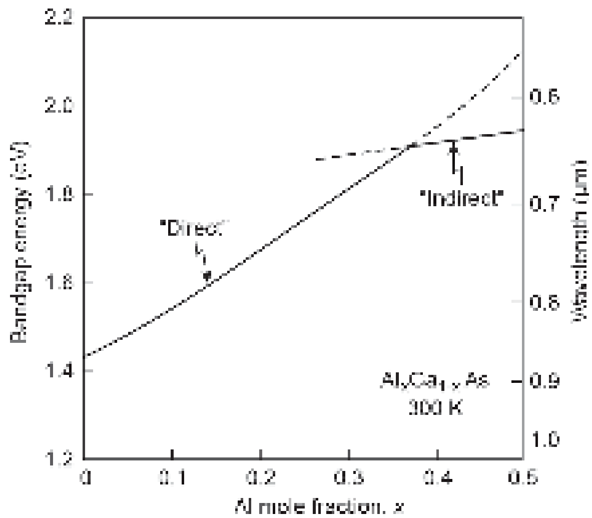


Fig. 5.13 Variation of bandgap of $Al_xGa_{1-x}As$ with mole fraction indicating corresponding emission wavelength (after Miller *et al* 1973)

fraction is depicted in Fig. 5.13 (Miller *et al* 1973). The bandgap energy of the ternary alloy $Al_xGa_{1-x}As$ with the mole fraction x between 0 to 0.37 (in the direct bandgap region) can be obtained as (Kressel *et al* 1977)

$$E_g = 1.424 + 1.266x + 0.266x^2 \qquad (0 < x < 0.37) \tag{5.27}$$

The peak wavelength of emission can be obtained by substituting the value of E_g in eV from Eq. (5.27) into Eq. (5.3).

The principal material used for making optical sources operating in 800–900 nm is based on $Al_xGa_{1-x}As$. The parameter x can be suitably adjusted to achieve the appropriate bandgap necessary for peak emission at the desired wavelength. A typical emission spectrum of AlGaAs LED is shown in Fig. 5.14. The peak wavelength of emission corresponds to the wavelength at which the relative intensity of the emitted light is

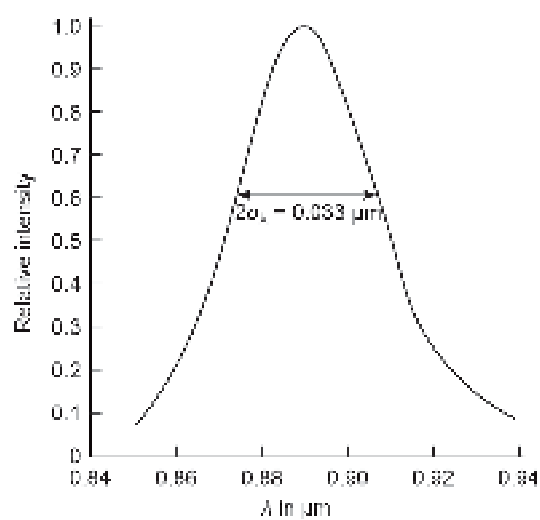


Fig. 5.14 Variation of the relative intensity of the light emitted by the AlGaAs LED source with the wavelength of light (after Miller *et al* 1973)

maximum. It is important to note that the LED emits light over a region of the spectrum centered at the wavelength of peak emission (or peak power). The spectral width of the source is generally expressed in terms of *rms* spectral width, σ_λ and varies in the range of 30–100 nm depending on the quality of the LED. In this example, the spectral width of the LED is shown as 33 nm (Miller[1] *et al* 1973).

Example 5.2 Estimate the peak wavelength of emission of an LED that uses $Al_{0.11}Ga_{0.89}As$ as the active region.

Solution The energy bandgap $Al_{0.11}Ga_{0.89}As$ can be estimated by putting $x = 0.11$ in Eq. (5.27) as

$$E_g = 1.424 + 1.266 \times 0.11 + 0.266 \times (0.11)^2 = 1.566 \text{ eV}$$

The peak emission wavelength can be estimated as

$$\lambda = 1.24/1.566 \ \mu\text{m} = 792 \text{ nm}$$

Example 5.3 The active region of an LED uses $Al_xGa_{1-x}As$. Find the value of x that would result in peak emission at 820 nm wavelength.

Solution: The energy bandgap required for emission at 820 nm that is 0.82 µm can be estimated as

$$E_g = \frac{1.24}{0.82} \text{ eV} = 1.512 \text{ eV}$$

The mole fraction x can be calculated from Fig. 5.14 as $x = 0.07$. The desired material is thus $Al_{0.07}Ga_{0.93}As$.

For application in the longer wavelength region, ternary InGaAs and quaternary InGaAsP grown on matured InP-based technology are most promising in the wavelength range of 0.92–1.65 µm. As discussed already the quaternary material such as $Ga_xIn_{1-x}As_{1-y}P_y$ is a more versatile material because the lattice constant and energy bandgap can be varied over wider ranges by varying both the the mole fractions x and y. The material is lattice-matched to InP. In particular, $In_{0.53}Ga_{0.47}As$ has a bandgap energy of 0.74 eV and is very useful for making optical sources and photodetectors for operation at 1.55 µm using InP as the lattice-matched substrate. The appropriate composition of $In_{1-x}Ga_xAs_yP_{1-y}$ for lattice matching can be obtained by selecting the bandgap of the quaternary along the upper dashed line in Fig. 5.13 passing through the InP point. The energy bandgap of $Ga_xIn_{1-x}As_{1-y}P_y$ compositions that are lattice-matched to InP can be obtained using the following empirical relation (Nahory *et al* 1978)

$$E_g\,(In_{1-x}Ga_xAs_yP_{1-y}) = 1.35 + 0.668x - 1.17y + 0.758x^2 + 0.18y^2 - 0.069xy - 0.322x^2y + 0.33xy^2 \text{ eV} \tag{5.28}$$

5.3 LIGHT-EMITTING DIODE (LED) FOR OPTICAL FIBER COMMUNICATION

Light emitting diodes are best suited for optical fiber communication systems when the power requirement of the source is not more than a few tens of microwatt and the desired bit rate is in the range of 100–200 Mbps. The structure of an LED largely depends on the desired application. For example, the structure of an LED intended for use as a display device may vary widely from that of an LED intended to be used in optical communication. For application in optical fiber transmission systems, an LED must have a high value of radiance (or brightness), a fast emission response time, and a high quantum efficiency.

The radiance or brightness of the source is measured in terms of the power emitted by the source per unit area per unit solid angle (measured in steradian). A high value of radiance ensures that sufficient power can be coupled from the source to the fiber. The emission response time is a measure of the delay in time between the application of a current pulse and the emission of light. A fast emission response time enables the source to be modulated directly by a fast-changing time-varying injected current. The quantum efficiency is a measure of the fraction of the total injected carriers that recombine radiatively to produce the emission of photons. A high value of quantum efficiency would lead to a high value of LED output power. In this section, we focus our attention on those LED structures which are used for the design of optical sources in fiber optic communication systems.

5.3.1 LED Power and Efficiency

Light-emitting diodes are primarily used for display purposes. However, specially designed LED sources operating in the near-infrared region do find applications in optical fiber communication systems where the desired bit rate is less than 200 Mbps. The operation of an LED mainly relies on spontaneous emission. The absence of optical amplification through stimulated emission restricts the quantum efficiency of a normal LED. Spontaneous emission allows a significant number of non-radiative recombination to take place via crystalline impurity and imperfection. This restricts the internal quantum efficiency (fraction of the total injected carriers that recombine radiatively) to 50% or less. However, advanced LED structures such as double-heterostructure (LED) to be discussed later in this chapter can enhance the quantum efficiency in the range of 60–80% (Lee *et al* 1978).

It is understood from the foregoing discussion that under forward-biased conditions, minority carriers are injected across the junction and the injection can be made predominant in one direction by increasing the doping concentration on one side significantly larger than that on the other side. Thus, for a forward biased $n^+ - p$ junction, the excess electrons injected from the n^+ into the p-region will decay exponentially through recombination with the holes (majority carriers in the p-region) following the relation (Senior, 1992; Keiser, 1999)

$$\Delta n(t) = \Delta n(0)\exp\left(-\frac{t}{\tau}\right) \tag{5.29}$$

where $\Delta n(0)$ is the value of the excess electron concentration injected initially at $t = 0$ and τ is the overall carrier recombination lifetime. For a low-level injection, the concentration of the injected minority carriers is only a small fraction of the total majority carriers present in the p-region. As a result, the lifetime of the carriers here refers to the minority carrier lifetime.

Under equilibrium, a constant bias current is established in the diode. The carrier recombination rate of the LED can be derived from the continuity equation and can be expressed as (Keiser, 1999)

$$\frac{d(\Delta n)}{dt} = \frac{J}{qd} - \frac{\Delta n}{\tau} \tag{5.30}$$

where q is the electronic charge, J is the bias current density flowing through the diode, and d is the thickness of the recombination region also called the active region of the LED.

Under steady-state conditions the time-rate variation of the excess carriers to zero, that is,

$$\frac{d(\Delta n)}{dt} = 0 \tag{5.31}$$

Therefore, the steady-state electron density when a constant bias current is flowing through the diode can be obtained from Eqs (5.30) and (5.31) as

$$\Delta n = \frac{J\tau}{qd} \tag{5.32}$$

When the bias current is switched off, the carrier injection stops, and Eq. (5.30) becomes

$$\frac{d(\Delta n)}{dt} = -\frac{\Delta n}{\tau} \tag{5.33}$$

The injected carriers under this condition decay exponentially following Eq. (5.29) which is the solution of Eq. (5.33).

Under steady-state conditions with a constant bias current flowing through the LED, the recombination rate which corresponds to the number of carriers recombining per unit volume per second can be obtained from Eq. (5.32) as

$$r = \frac{\Delta n}{\tau} = \frac{J}{qd} \tag{5.34}$$

It may be recalled that the recombination may be either radiative or non-radiative. Therefore, the total recombination rate given by Eq. (5.34) comprises the contribution of both the radiative as well as the non-radiative recombination rates. Thus, Eq. (5.34) can be written as

$$r = r_r + r_{nr} = \frac{J}{qd} \tag{5.35}$$

where r_r and r_{nr} are the radiative and non-radiative recombination rates respectively, given by

$$r_r = \frac{\Delta n}{\tau_r} \tag{5.36}$$

$$r_{nr} = \frac{\Delta n}{\tau_{nr}} \tag{5.37}$$

Here, τ_r and τ_{nr} correspond to the mean lifetime of carriers for radiative and non-radiative recombination respectively. It may be pointed out that each radiative recombination is associated with the emission of a photon with energy equal to the bandgap energy of the material and each no radiative recombination is responsible for a loss of an electron-hole pair and subsequent release of heat which is absorbed by the lattice leading the quantized lattice vibration or emission of a phonon. So far as the electroluminescence is concerned, any non-radiative recombination is viewed as a loss responsible for lowering the efficiency of the LED. The internal quantum efficiency of an LED is a measure of the ability of the device to convert electron-hole pairs to photons. In other words, it is a measure of the ability of the LED to convert electrical energy to optical energy (E–O conversion). The internal quantum efficiency of the LED may, therefore, be viewed as the fraction of the total recombination that occurs radiatively. Mathematically, it can be expressed as

$$\eta_{\text{int}} = \frac{r_r}{r} = \frac{r_r}{r_r + r_{nr}} \tag{5.38}$$

Substituting the values of r_r and r_{nr} from Eqs (5.36) and (5.37) into Eq. (5.38), we get

$$\eta_{\text{int}} = \frac{\Delta n / \tau_r}{\Delta n / \tau_r + \Delta n / \tau_{nr}} \tag{5.39}$$

Equation (5.39) can be simplified to express the internal quantum efficiency as

$$\eta_{\text{int}} = \frac{1/\tau_r}{1/\tau_r + 1/\tau_{nr}} = \frac{\tau}{\tau_r} \tag{5.40}$$

where τ is the effective lifetime of the carrier given by

$$\frac{1}{\tau} = \frac{1}{\tau_r} + \frac{1}{\tau_{nr}} \tag{5.41}$$

The direct bandgap semiconductors such as AlGaAs and InGaAsP which are widely used for making light-emitting diodes in the near-infrared (NIR) region for application in optical fiber communication systems have nearly equal values of lifetime of carriers for radiative and non-radiative recombination. As a result, the effective lifetime of the carriers becomes half the value of the lifetime for radiative recombination (τ_r). Therefore, LEDs based on these materials exhibit values of internal quantum efficiency of about 50% only. The internal quantum efficiency of a simple LED can be enhanced in the range of 60–80% by making use of double heterostructures discussed later in this chapter. The specially designed structure helps to reduce the non-radiative recombination through self-absorption. However, such structures under certain conditions may be dominated by surface recombinations at the heterointerfaces which are non-radiative and may lead to a reduction in the quantum efficiency of the LED. It may be pointed out that internal quantum efficiency accounts for the photons that are generated internally to the number of carriers recombining internally. The actual number of photons emitted out of the device is always less than the number of internally generated photons. The external quantum efficiency is the ratio of photons emitted out of the LED to the total number of internally generated photons. It is also often defined as the number of photons emitted out of the LED to the total number of carrier recombinations (Senior, 1992).

If an LED is forward biased so that a drive current I flows through the device under steady-state, the total recombination rate (/s) can be obtained as

$$R = R_r + R_{nr} = \frac{I}{q} \tag{5.42}$$

where R_r and R_{nr} correspond to the total radiative recombination rate and total non-radiative recombination rate (/s). The internal quantum efficiency can be alternatively defined in terms of recombination rate ratio as

$$\eta_{\text{int}} = \frac{R_r}{R_r + R_{nr}} = \frac{R_r}{R} \tag{5.43}$$

Use Eqs (5.42) and (5.43) yields

$$R_r = \eta_{\text{int}} \left(\frac{I}{q} \right) \tag{5.44}$$

It may be noted that R_r which corresponds to the total radiative recombination rate which also corresponds to the total number of photons emitted internally per unit of time (/*s*). Further, recalling that the energy associated with each emitted photon is $h\nu$, the optical power generated internally to the LED can be obtained as

$$P_{\text{int}} = R_r(h\nu) = \eta_{\text{int}}\left(\frac{I}{q}\right)(h\nu) = \eta_{\text{int}}\frac{hc}{q\lambda}It \tag{5.45}$$

It can be easily verified from Eq. (5.45) that the power generated internally by the LED is directly proportional to the drive current flowing through the device. It should be noted that the power emitted by the LED is less than the power generated internally because of several factors that lower the value of the available power from an LED. However, the LED output power also follows a similar relationship except for the fact that the internal quantum efficiency is replaced by a new factor called the *external quantum efficiency* which is the overall quantum efficiency for the device.

Example 5.4 An InGaAsP/InP LED emits peak power at 1330 nm at a drive current of 50 mA. The values of radiative and non-radiative recombination lifetime are 25 ns and 100 ns respectively. Estimate the internal quantum efficiency of the LED. Also, calculate the optical power yielded internally in the LED.

Solution The effective lifetime of the carriers is

$$\tau = \frac{\tau_r\tau_{nr}}{\tau_r + \tau_{nr}} = \frac{25\times100}{25+100}\text{ ns} = 20\text{ ns}$$

The internal quantum efficiency can be obtained as

$$\eta_{\text{int}} = \frac{20}{25} = 0.8$$

The internal power yield can be estimated as (Eq. (5.45))

$$P_{\text{int}} = \eta_{\text{int}}\frac{hc}{q\lambda}I = 0.8\times\frac{6.6254\times10^{-34}\times3\times10^{8}\times50\times10^{-3}}{1.6\times10^{-19}\times1330\times10^{-9}} = 37.36\text{ mW}$$

The power emitted by an LED depends on the radiative recombination. To improve the quantum efficiency of an LED it is necessary to make the radiative recombination lifetime much smaller than the non-radiative recombination lifetime so that radiative recombinations dominate over the non-radiative recombination. The value of non-radiative recombination lifetime depends on the properties of the defects that create energy levels in the forbidden bandgap of the material. The energy released out of the recombination of carriers at these levels is dissipated by phonons. Further, for heavily doped regions and narrow bandgap semiconductors another form of non-radiative recombination called *Auger recombination* may become significant. The other form of non-radiative recombination in LED arises from surface recombination at the heterointerfaces in double heterostructures discussed later in this chapter. In the subsequent sections, different forms of recombination processes are discussed briefly.

5.3.1.1 Shockley-Read-Hall (SRH) recombination

Non-radiative recombination is dominated by Shockley-Read-Hall (SRH) occurring via recombination centers created by traps/defects. The non-radiative SRH lifetime of the carriers is expressed as (Hall, 1952; Shockley *et al* 1952).

$$\tau_{SRH} = \frac{1}{N_T \sigma v_{th}} \tag{5.46}$$

where σ is the capture cross-section of the particular type of carrier (electron or hole), N_T is the trap density of the corresponding type of carriers and v_{th} is the carrier thermal velocity given by

$$v_{th} = \sqrt{\frac{3kT}{m^*_{n,p}}} \tag{5.47}$$

where k is Boltzmann's constant, T is the absolute temperature and $m^*_{n,p}$ is the effective mass of the carrier (electron/hole).

5.3.1.2 Auger recombination

The other non-radiative bulk recombination process which is generally dominant in heavily doped semiconductors and narrow-band semiconductors is known as Auger recombination. This process involves three carriers for completing the recombination. In this process, the excess energy released by the recombination of an electron-hole pair is transferred by Columbic collisions as kinetic energy of a third carrier, which is raised in energy deep into the respective band. The carrier finally returns to the bottom of the band by releasing the energy in the form of thermal vibration. There are different forms of Auger recombination processes. For example, Auger recombination in one case may involve two conduction band electrons and one heavy hole. In this recombination process, one conduction band electron transfers its energy to another electron in the conduction band which moves up in the band and then the first electron drops down to the valence band to recombine with a hole. This way the other Auger recombination processes may involve one conduction band electron and two holes in the valence band (any one of the heavy-hole, light-hole, or a hole from the spin-split-off band). The Auger recombination processes have been dealt with in detail elsewhere (Sze, 2003; Bhattacharya, 2007). The carrier lifetime for the Auger process can be expressed as

$$\tau_{AU} = \frac{n}{R_{AU}} = \frac{1}{Cn^2} \tag{5.48}$$

where C is the Auger recombination coefficient which is a fundamental characteristic of the material, n is the majority carrier concentration and R_{AU} is the rate of Auger recombination given by

$$R_{AU} = Cn^3 \tag{5.49}$$

When both the non-radiative recombinations are present, the overall non-radiative lifetime of the carrier can be expressed as

$$\frac{1}{\tau_{nr}} = \frac{1}{\tau_{SRH}} + \frac{1}{\tau_{AU}} \tag{5.50}$$

5.3.1.3 Radiative recombination

This process involves the direct transition of an electron in the conduction band to the valence band where it recombines with a hole. In this process, the balance energy following each recombination is released in the form of a photon light and the momentum is conserved.

The radiative recombination lifetime can be expressed as

$$\tau_r = \frac{1}{B_r(n_0 + p_0)} \tag{5.51}$$

where B_r is the coefficient for band-to-band recombination, n_0 and p_0 are the equilibrium electron and hole concentrations respectively. In the presence of excess carrier injection, the radiative recombination lifetime can be obtained as

$$\tau_r = \frac{1}{B_r(n_0 + p_0 + \Delta n)} \tag{5.52}$$

where Δn is the excess injected carrier density.

Example 5.5 The recombination processes in an LED working at 1650 nm are governed by radiative recombination with a mean lifetime of 10 ns while the non-radiative recombination is dominated by SRH and Auger processes with mean lifetime values of 20 ns and 100 ns respectively. Estimate the internal quantum efficiency of the LED.

Solution The effective lifetime of the carrier can be estimated as

$$\frac{1}{\tau} = \frac{1}{\tau_r} + \frac{1}{\tau_{nr}} = \frac{1}{\tau_r} + \frac{1}{\tau_{SRH}} + \frac{1}{\tau_{AU}}$$

$$= \left(\frac{1}{10} + \frac{1}{20} + \frac{1}{100}\right) \text{ns}^{-1}$$

That is,

$$\tau = \frac{100}{16} = 6.25 \text{ ns}$$

The internal quantum efficiency can be estimated as

$$\eta_{\text{int}} = \frac{6.25}{10} = 62.5\%$$

5.3.1.4 External Quantum Efficiency

It may be pointed out that all photons generated internally in an LED will not be able to exit the device and finally contribute to the power emitted by the device. This can be appreciated by considering the schematic of LED. It can be seen that the light generated within the LED strikes the LED surface from inside. The surface of the LED is the interface between the emitting region of the LED and the surrounding medium in which the light is finally emitted. The former has a larger refractive than that of the surrounding medium. Therefore, at the interface, there would be a critical angle ϕ_c (with the normal drawn on the interface at the point of incidence) such that the light incident at the interface at an angle larger than ϕ_c would be reflected back into the emitting region. The light falling at the interface makes an angle less than ϕ_c as shown in Fig. 5.15 would be refracted out of the LED to contribute to the emitted power. It can be easily seen from Fig. 5.15 that only the fraction of light falling within a cone subtending an angle at the vertex equal to the critical angle ϕ_c (= $\pi/2 - \theta_c$) will manage to escape out of the LED to contribute to the external power. Applying Snell's law at the interface between the LED material emitting light having a refractive index of n_1 and the outside medium has a refractive index, n_2 ($<n_1$), we have

$$\phi_c = \sin^{-1}\left(\frac{n_2}{n_1}\right) \tag{5.53}$$

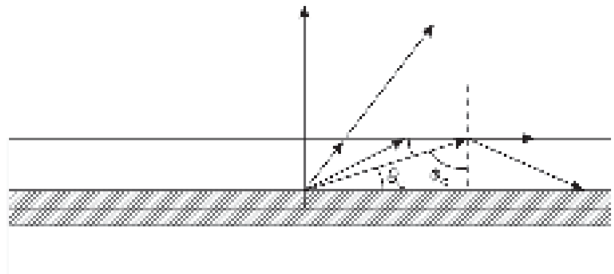


Fig. 5.15 Light emitted out of an LED through refraction showing that light falling beyond the critical angle ϕ_c with the normal is reflected back

The outside medium is generally air ($n_2 = 1$). However, in optical fiber communication applications, the source is generally cemented to a small portion of a fiber called pigtail or fly-lead with the help of some transparent adhesive made up of some index matching fluid to reduce the Fresnel loss. The external quantum efficiency of the LED can be obtained as

$$\eta_{\text{ext}} = \frac{1}{4\pi}\int_0^{\Phi_c} T(\phi)(2\pi \sin\phi)\, d\phi \tag{5.54}$$

where $T(\phi)$ is the Fresnel transmission coefficient which is a function of the angle of incidence.

For normal incidence, the Fresnel transmission coefficient can be expressed as (Snyder *et al* 1983)

$$T(0) = \frac{4n_1 n_2}{(n_1 + n_2)^2} \tag{5.55}$$

Considering the surrounding medium to be air ($n_2 = 1$) and assuming $n_1 = 1$ we may write

$$T(0) = \frac{4n}{(n+1)^2} \tag{5.56}$$

The external quantum efficiency under this condition can be approximated by using Eq. (5.54) as

$$\eta_{\text{ext}} \approx \frac{1}{n(n+1)^2} \tag{5.57}$$

The optical power available outside the LED can be expressed as

$$P_{\text{emitted}} = \eta_{\text{ext}} P_{\text{int}} \tag{5.58}$$

Example 5.6 Estimate the external quantum efficiency of a GaAs LED operating at 0.86 μm and emitting light in the surrounding medium of air. Assume the relative permittivity of GaAs to be 12.9.

Solution The refractive index of GaAs can be obtained as

$$n = \sqrt{12.9} \approx 3.6$$

The external quantum efficiency of the LED can be estimated by using Eq. (5.57) as

$$\eta_{\text{ext}} = \frac{1}{3.6 \times (1+3.6)^2} \approx 1.31\%$$

5.4 LED STRUCTURES: HOMOJUNCTIONS AND HETEROJUNCTIONS

An LED in the simplest form is essentially a *p-n* junction formed by using one of the direct bandgap semiconductors of suitable bandgap that ensures emission in the desired wavelength. Such an LED that uses the same material (say, GaAs) on both sides with different conductivity is called a *homojunction LED*. These are not useful for application in fiber optic communication systems because of their poor radiance and low quantum efficiency. The radiance and quantum efficiency of an LED can be greatly improved by making use of multiple heterojunctions. A heterostructure LED generally consists of a combination of multiple heterojunctions that secure the carriers and subsequently the emitted photon in such a way that the overall quantum efficiency and radiance of the LED increase. To provide a high value of radiance and high quantum efficiency it is necessary to make use of suitable lattice-matched materials of different bandgap values in the LED structure to achieve the confinement of the carriers for recombination and also confinement of photons emitted due to radiative recombination. The confinement of carriers is done in such a way that they are forced to recombine radiatively in the active region of the LED. This improves the quantum efficiency of the device. Once the carriers recombine radiatively and generate photons, the structure of the LED must support their confinement such that these photons do not get reabsorbed in the material. Both single heterojunction and double-heterojunction (or heterostructure) LED structures have been proposed and studied for use in the transmission system of optical fiber communication. Historically the concept of heterostructure in LED was brought from the study of laser diode structures for improving the performance of the former. A variety of LED structures ranging from simple planar LED, and dome LED to more complex double heterostructure surface-emitting LED (SLED) (Burrus type), double heterostructure edge-emitting LED (ELED), and more advanced super-luminescent diode (SLD). A planar LED is the simplest one from the fabrication point of view. It can be easily fabricated by liquid phase epitaxy (LPE) or vapor phase epitaxy method. Diffusion of *p*-type impurity on an *n*-type substrate can produce a planar LED as illustrated in Fig. 5.16(a). The first light-emitting diode suitable for optical fiber communications was fabricated by Keyes and Quist in 1962 using an *n*-GaAs with a junction formed by *Zn* doping (Keyes *et al* 1962). The LED exhibited a peak emission wavelength of 0.93 μm and a spectral width of 35 nm. The LED was reported to have a modulation bandwidth over 100 MHz. A GaAs LED with a hemispherical dome to enhance coupling was fabricated and measured by Carr (Carr, 1965). The LED had a room temperature spectral width of 25 nm centered at 900 nm.

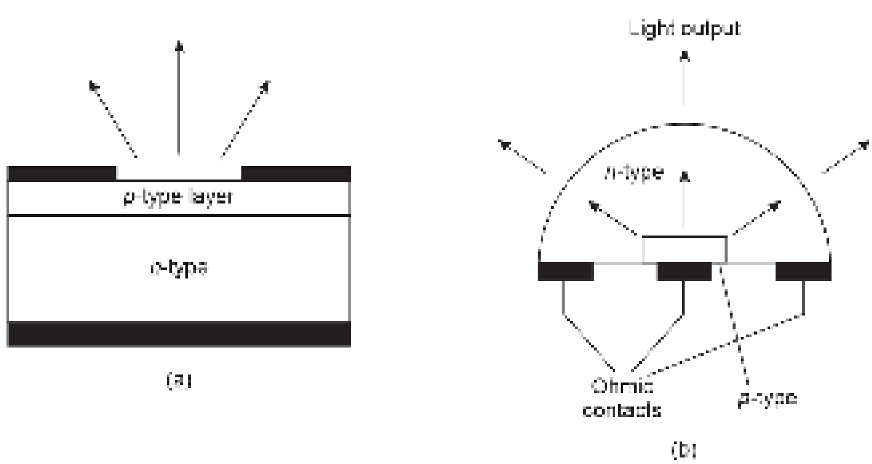


Fig. 5.16 Simple LED structure (a) Planar (b) Dome

The LED exhibited an external quantum efficiency of 7.3% and an emission rise-time of 1.6 ns at room temperature (Carr, 1965).

5.4.1 Surface-emitting LED (SLED)

A high radiance surface-emitting LED particularly suitable for optical fiber communication systems was fabricated by Burrus (Burrus *et al* 1970). In a surface-emitting LED the light is emitted in the direction perpendicular to the plane of the *p-n* junction. The surface-emitting LEDs are often referred to as Burrus type after the name of the inventor. Later on, a more sophisticated structure of the same design but based on double-heterostructure (DH) $Al_xGa_{1-x}As$ grown on GaAs by liquid-phase epitaxy (LPE) with improved efficiency and radiance was reported (Burrus *et al* 1971). The bandgap energy of the ternary material can be tailored to adjust the peak wavelength emission of the surface-emitting DH-LED anywhere within 0.75 to 0.9 μm which is where silica fiber exhibits low loss. It was further demonstrated that the DH-LED can emit power as high as 10^6 W/m^2 sr for a drive current of 150 mA. Approximately 2 mW power can be easily launched from this type of source to a multimode fiber with a relative index difference between the core and the cladding of 10%. The half-power spectral width of the source is in the range of 35–40 nm, a response time of 1–2 ns, and an external quantum efficiency of 2–3% (Burrus *et al* 1971). The double heterostructure was extensively studied by Burrus and others concerning different performance parameters for possible application as an optical source in optical fiber communication (Burrus *et al* 1971; Dawson *et al* 1971; Burrus, 1972; Lee *et al* 1972). It may be pointed out that the double heterostructure concept had been bought in the fabrication of injection laser diode before its application in LED (Panish *et al* 1970; Hayashi, 1970).

The schematic of a double-heterostructure p–$Al_x Ga_{1-x} As$/n–GaAs/n–$Al_x Ga_{1-x}$ As light emitting diode (DH-LED) is shown in Fig. 5.17 along with the energy-band diagram. The structure consists of a lightly doped active layer made up of a narrower bandgap material (shown as n-GaAs) sandwiched between the layers made up of a larger bandgap ternary material (shown as $Al_x Ga_{1-x} As$. The binary GaAs forms heterojunction with the ternary on both sides. The lightly doped n-GaAs region acts as the active region of the LED while the surrounding layers are made of $Al_xGa_{1-x} As$ acts as confining regions which help in the confinement of carriers injected in the active region as well as the photons subsequently emitted following the recombination of injected carriers in the active region. The confinement of the carriers is evident from the energy band diagram showing the electron and hole energy barriers for the injected carriers. Further, the optical confinement is ensured by the difference in the refractive index between the active region and that in the confining layers as illustrated in Fig. 5.17. The dual confinement of carriers and photons enables double-heterostructure LEDs to have high values of radiance and high quantum efficiency. Another important performance of an LED for application in optical fiber communication is the bandwidth. The bandwidth of the LED determines the modulation capability which means that the maximum frequency of the signal that can be used to modulate the intensity of the light emitted by the LED. The main performance parameters of the LED such as quantum efficiency, radiance, and bandwidth of the device are affected by the geometry as well as other device parameters such as self-absorption, heterointerface recombination velocity, doping concentration, active layer thickness, injection current density, and carrier and optical confinement factors (Lee *et al* 1978). Unlike in a conventional homojunction LED, it is possible to achieve a higher power with the help of double

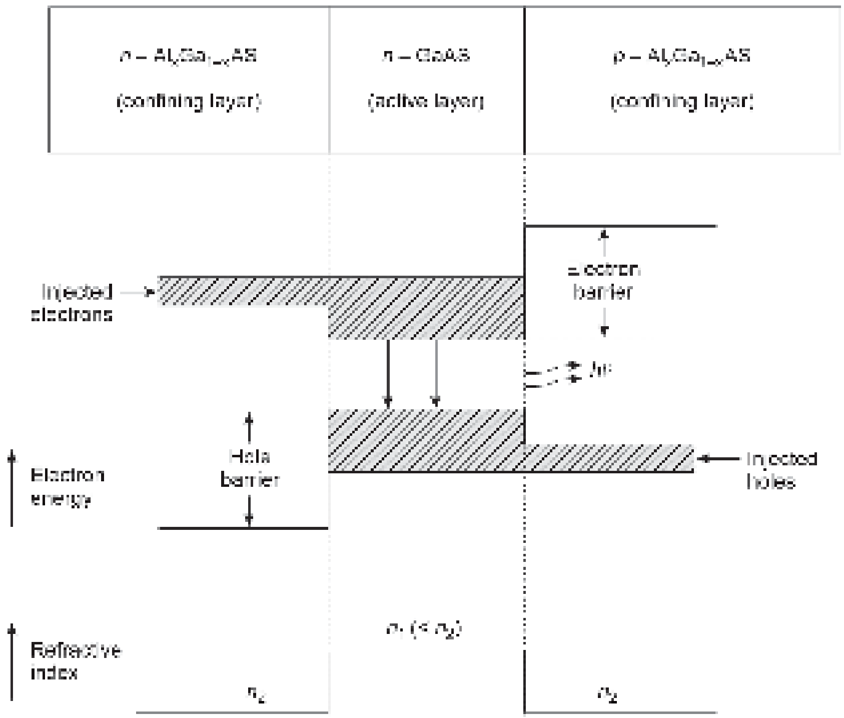


Fig. 5.17 Schematic of a double heterostructure n–$Al_xGa_{1-x}As$/n–GaAs/p–$Al_x Ga_{1-x}As$ LED showing the energy band diagram and illustrating the injection of carriers across the barriers along with the refractive index profile in the active and confining regions

heterostructure even by using a lightly doped active region in the latter case. This is because the non-radiative recombination at the surface can be greatly reduced by making use of a good quality heterointerface. The confinement of carriers enhances the injection efficiency resulting in enhanced radiative recombination caused by the reduction of recombination time by increased injected carrier density.

The performance of DH-LED however, depends on the quality of the heterojunctions on both sides of the active region. The quality of a heterojunction is often characterized in terms of interfacial recombination velocity or surface recombination velocity (S) at the heterointerface. For an ideal heterointerface (a perfectly reflecting boundary) $S = 0$. The interface between the bulk material and surface material vanishes when $S = D/L_D$, where D is the diffusion coefficient and L_D is the diffusion length in the bulk material. The ratio $D/(L_D)$ corresponds to bulk recombination velocity. A small value of surface recombination velocity is desirable for a good-quality heterojunction. The smaller the value of surface recombination velocity at the heterointerface better the quality of the heterojunction. A Burrus-type DH-LED was studied analytically by Lee *et al* (Lee *et al* 1978). The model involves the solution of the diffusion equation of charge carriers in the active region under the boundary conditions at the heterointerfaces. The analysis revealed that the surface recombination velocity at the heterointerfaces (assumed to be the same and equal to S) modifies the bulk recombination velocity to an average effective carrier lifetime given by

$$\frac{1}{\tau_{\text{eff}}} = \frac{1}{\tau} + \frac{2S}{d} \tag{5.59}$$

where τ_{eff} is the effective lifetime of the carriers in the presence of surface recombination, τ is the bulk carrier lifetime, S is the surface recombination velocity at the interface and d is the thickness of the active layer. The simplified relation given by Eq. (5.29) is obtained under the assumption that the surface recombination velocity is much less than the bulk recombination velocity that is, $(SL_D)/D \ll 1$. It can be easily verified from Eq. (5.29) that the effective lifetime reduces when S has a relatively high value. A reduction in the value of carrier lifetime results in a reduction of quantum efficiency and consequently a reduction of radiance and power output of the source. Another, factor that reduces the power output of an LED is the self-absorption of the photons emitted due to the recombination of the carriers. To couple power from a surface-emitting LED it is necessary to put the fiber perpendicular to the plane of the *p-n* junction which is parallel to the plane of the emitting surface. A surface-emitting Burrus type LED emits power with an isotropic emission pattern known as the *Lambertian pattern*. Such a source looks equally bright when viewed from any direction. The power, however, diminishes by a factor equal to the cosine of the viewing angle measured with the normal drawn on the emitting surface. This means that the power emitted in a direction reduces to 50% when the viewing angle becomes 60º (cos 60º = 0.5). The half-power beam width of a surface-emitting LED is thus 120º. The coupling schematic cross-section of a Burrus-type DH-LED with a piece of optical fiber bonded on the emitting surface is shown in Fig. 5.18.

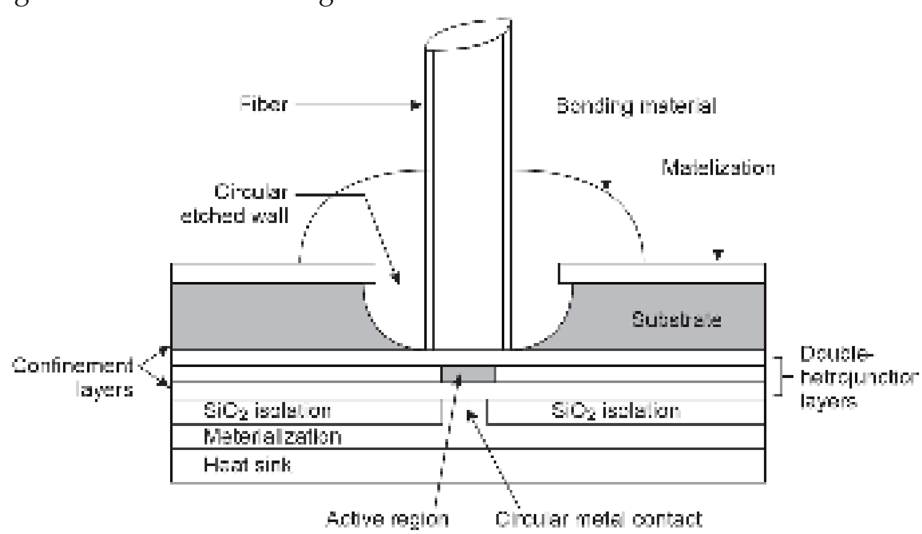


Fig. 5.18 Schematic cross-sectional view of a surface emitting DH-LED showing a section of the fiber connected perpendicular to the emitting surface for collecting light

The coupling of light from the non-coherent source to the optical fiber depends on the numerical aperture of the light-receiving fiber. SLEDs generally suffer from the problem of lateral current spreading when the contact area is less than 25 μm. In such cases, the effective emission area is much less than the contact area which results in coupling loss. The lateral current spreading can be reduced by making use of a mesa structure SLED (Uji *et al* 1985). The schematic of a mesa etched double heterostructure LED is illustrated in Fig. 5.19. The coupling can be increased by making use of a multimode fiber with a relatively large value of numerical aperture. The coupling efficiency can also be improved by making use of a micro-lensing arrangement.

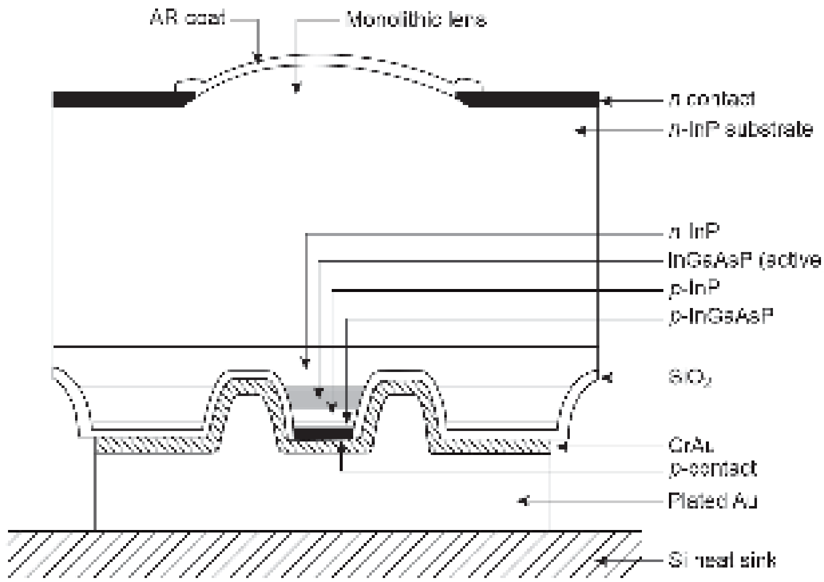


Fig. 5.19 The schematic of a mesa etched InP/InGaAsP/InP double heterostructure surface emitting LED (after *Uji et al* 1985)

5.4.2 Edge-Emitting LED (ELED)

A double-heterostructure edge emitting light emitting diode ELED uses stripe geometry similar to that used in an injection laser diode to restrict the current spreading in the lateral direction. It consists of an active region which is made of a suitable material to emit light in the desired wavelength region and is sandwiched between two guiding layers. Both the guiding regions have the same refractive index value which is lower than the refractive index of the active region but higher than the refractive index of the surrounding material. The stripe geometry forms a complex waveguide that channelizes the emitted optical power toward the core of the receiving fiber whose axis is parallel to the plane of the *p-n* junction as shown in Fig. 5.20. The light is usually collected from one end by making the rear facet reflective. The width of the contact stripe of ELED varies in the range of 50–70 μm to match the diameter of standard multimode fibers of the order of 50–100 μm. The radiation pattern of an ELED source is comparatively better than that of an SLED source. The typical radiation pattern of a GaAs/AlGaAs ELED is shown in Fig. 5.20 (Senior, 1992). The waveguiding effect causes the radiation pattern of the emitted light to be more directive in the direction perpendicular to the plane of the *p-n* junction. A half-power beam width of 30° in the transverse direction can be easily achieved in ELED. However, the half-power beam width remains 120° in the direction parallel to the plane of the *p-n* junction as there is no waveguiding in the lateral direction. A variety of modified ELED structures have been proposed, fabricated, and studied by various researchers (Newman, 1986).

5.4.3 Super-luminescent LED (SLD)

The output power of an LED can be significantly improved by making spontaneous and stimulated emissions occur simultaneously in an LED in a controlled manner. The spectral width of such a source is narrower than that of a conventional LED. An LED of this kind was first reported by Kurbatov *et al* (Kurbatov *et al*) and the device was named as superluminescent diode (SLD). Structurally an SLD is similar to a stripe contact laser

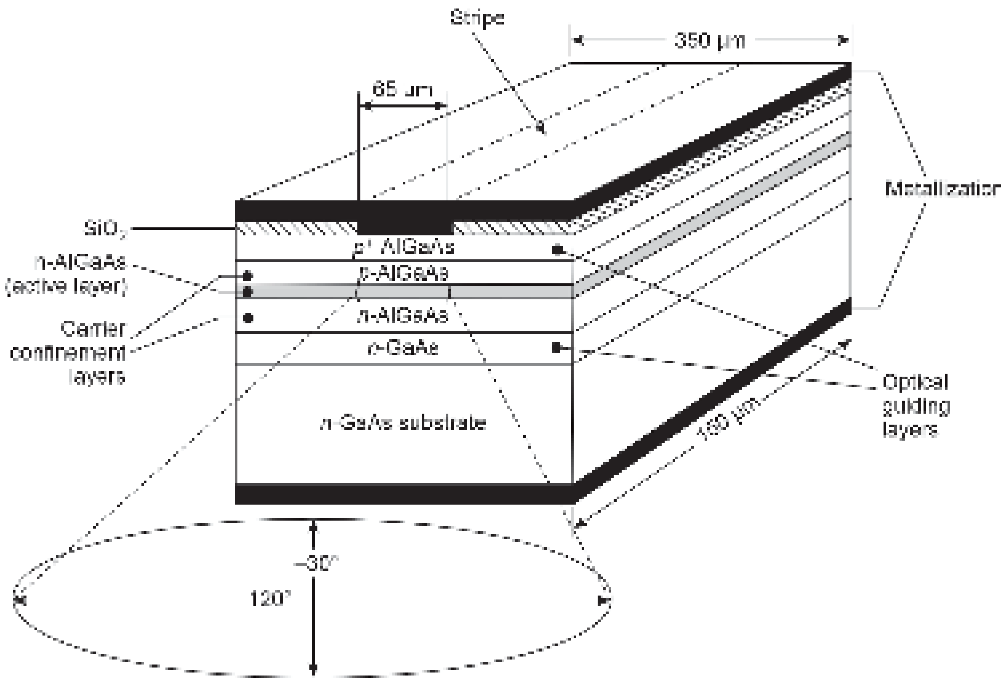


Fig. 5.20 Schematic of an edge emitting LED using stripe geometry

diode except for the fact that the stripe in the case of the former device is inclined at an angle of 10° with the normal to the emitting surface to eliminate optical feedback. As a result, the output light consists of a single-pass amplified spontaneous emission which is incoherent but 90% polarized. The SLD exhibited a spectral width of 2 nm under pulsed operation mode with a very high driving current. A double heterostructure SLD based on $GaAs/Al_xGa_{1-x}As$ was later reported (Lee *et al* 1973). The schematic of the SLD configuration is reproduced in Fig. 5.21. The SLD structure is very similar to a stripe geometry double heterostructure injection laser diode except for the fact that the optical feedback is suppressed in the former case by eliminating one of the mirrors (rear mirror)

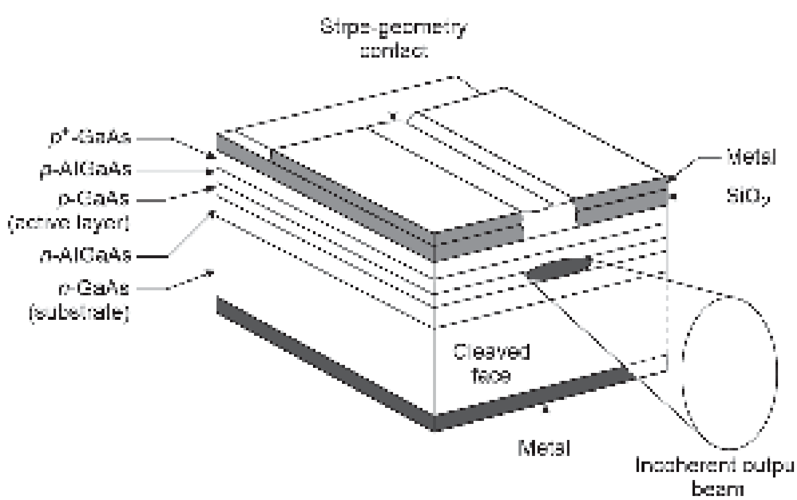


Fig. 5.21 Schematic of a superluminescent light emitting diode (after Lee *et al* 1973)

and providing absorption of the backward waves in the cavity thereby allowing single-pass gain only. The absence of optical feedback prevents the device from lasing at high drive currents by suppressing oscillations in many modes. The single-pass gain provides sufficient output power. The SLD is reported to be capable of coupling peak power as high as 50 mW under the pulsed mode of operation into a multimode fiber with a NA = 0.63. The SLD exhibited a spectral width in the tune of 5–8 nm with a preferential TE-mode polarization. Some novel SLD structures have been developed for operation in different ranges of wavelength spectrum (Senior, 1992). From the viewpoint of performance, SLDs are much inferior to injection laser diodes and require high drive current. Nevertheless, the incorporation of superluminescence in conventional LED can significantly enhance the radiance and reduce the spectral width of incoherent light emitters (Miller *et al* 1973). SLDs are commercially available for operation in different ranges of wavelength such as 1.16–1.33 µm (second attenuation window) and 1.52–1.57 µm (third attenuation window) (Senior, 1992). The SLDs generally exhibit power around 4–5 times higher than that of ELEDs.

5.5 LED CHARACTERISTICS

A light-emitting diode converts an electrical signal into light output power depending on the value of the drive current. In general, the optical power *versus* drive current characteristic of an LED is fairly linear for low and moderate values of injection current. The optical power *versus* drive current characteristics of an LED are shown in Fig. 5.22. This linearity feature makes LED an attractive optical source over injection laser diode for analog optical communication without severe impairment. At a very high value of injection drive current, the optical power output tends to saturate for a variety of reasons. The nonlinearity also depends on the configuration of the device and is generally predominant in high-radiance LED. The non-linearity in the output optical power versus the drive current of an LED leads to harmonic and intermodulation distortion in analog optical communication systems. As a result, it is often necessary to use some form of linearization circuit (Straus, 1978) to minimize the distortion.

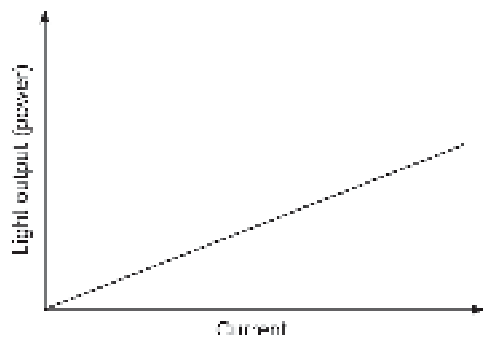


Fig.5.22 Optical power output *vs* drive current characteristics

The variation of the output power with the drive current for a Burrus-type LED is shown in Fig. 5.23 (Miller, 1973). The device exhibits reasonably linear characteristics making it suitable for direct modulation by either analog or digital signals. The modulation bandwidths are reported to be several hundreds of MHz (Burrus *et al* 1971; Dawson *et al* 1971).

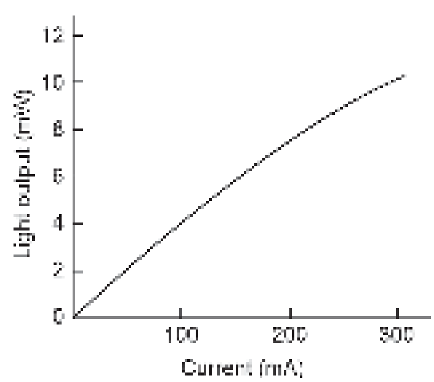


Fig. 5.23 Optical power *vs* drive current characteristic of a typical AlGaAs surface-emitting LED (after Botez *et al* 1979)

The output optical power of an LED is found to decrease with an increase in temperature. This is accounted for the reduction in internal quantum efficiency of an LED at higher operating temperature. The temperature of the *p-n* junction increases when a large drive current results in high radiance of the LED. The increased junction temperature causes the output power of the LED to drop down at a given drive current level. The variations of the output power of LEDs with an increase in junction temperature for the three basic configurations, e.g. SLED, ELED, and SLD are shown in Fig. 5.24. It is interesting to note that the output of the superluminescent LED is most sensitive to the variation of junction temperature as compared to surface and edge-emitting LEDs. This is attributed to the fact that the stimulated emission responsible for the operation of an SLD is adversely affected by the increase in temperature. The severity of non-linearity in the optical power output *versus* drive current of the SLD with an increase in ambient temperature is apparent from Fig. 5.25. The characteristics shown in Fig. 5.26 correspond to an InGaAsP ridge waveguide superluminescent LED (Kaminow *et al* 1983). The high power output of the SLD can be exploited by operating the device at a lower temperature with the help of thermoelectric coolers. At higher operating temperatures the output power is reduced and the power output *versus* drive current characteristics become highly non-linear (Kaminow *et al* 1983) associated with high non-linearity.

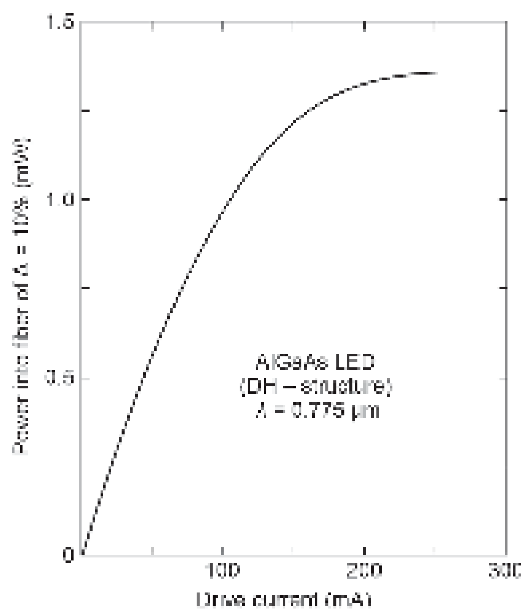


Fig. 5.24 Variation of output power of a Burrus type DH-LED with the drive current (after miller, 1973)

5.5.1 Frequency Response

The speed of an optical communication system depends not only on the information or data carrying capacity of the fiber but also on the rate at which the information or data can be modulated at the transmitter and the speed of response of the photodetector at the receiver. The rate at which the intensity of the light source can be modulated by the information or data is known as the modulation bandwidth of the source. The bandwidth is an important parameter of an LED for its application as a source in an optical communication system. However, the parameter is not that important for LEDs

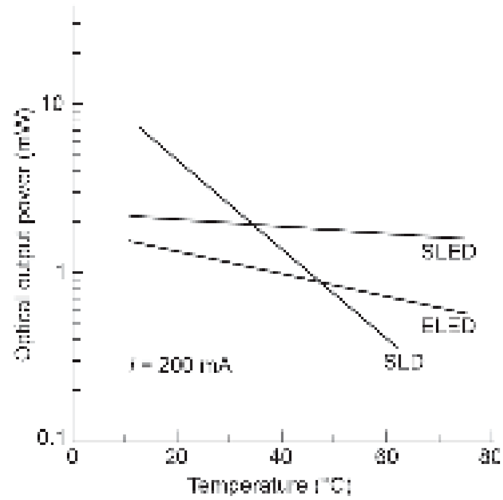


Fig. 5.25 Variation of the optical power output of three different LED configurations operating at 1.3 μm with temperature at a constant bias current (after Lee *et al* 1988)

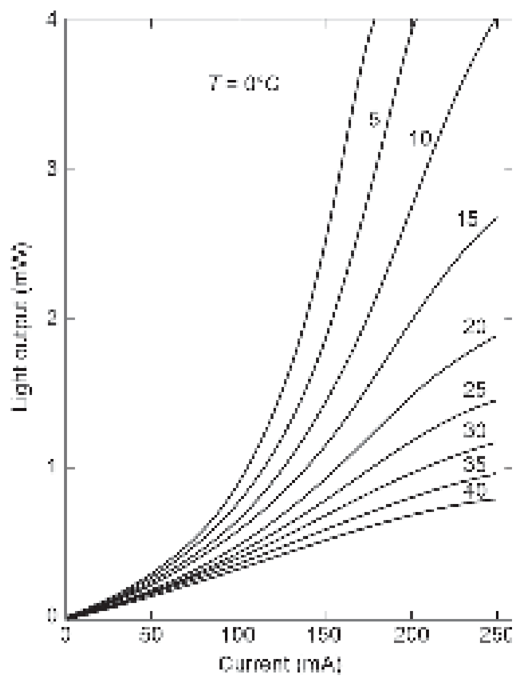


Fig. 5.26 Variation of light output of an InGaAsP ridge waveguide SLD with drive current for different ambient temperatures (after Kaminow *et al* 1983)

used as display devices. The modulation bandwidth of LEDs for application in optical fiber communication systems has been studied by several researchers for a variety of LED structures (Namizaki *et al* 1974; Liu *et al* 1975; Lee *et al* 1978). A first-hand approximation of the bandwidth of LED is often taken as the reciprocal of the lifetime of minority carriers because the radiance is caused by the radiative recombination of the injected minority carriers in the active region of the LED. Under forward conditions, minority carriers are injected across the junction. When the modulating signal changes at a high speed the stored minority charges cannot follow it. This is because the injected carriers need a finite time to vanish through recombination before they can respond to the fast-changing modulating signal. This is, however, the intrinsic factor that decides the ultimate bandwidth of an LED. The extrinsic factor that affects the bandwidth of an LED is the RC time constant arising out of the junction capacitance of the LED and the associated series resistance of the circuit. If the RC time constant is very large, then the overall bandwidth is decided by the external factor rather than the intrinsic factor.

Consider the 1-D HJ LED structure shown in Fig. 5.27. The bandwidth of an LED can be estimated by assuming a small signal AC voltage (~exp ($j\omega t$)) to be applied across the junction already biased by a DC voltage.

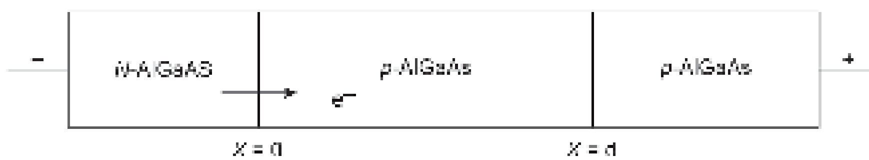


Fig. 5.27 Schematic of a heterojunction LED

The time-dependent continuity equation in this case can be expressed as

$$\frac{\partial(\Delta n(x,t))}{\partial t} = D_n \frac{\partial^2 \Delta n(x,t)}{\partial x^2} - \frac{\Delta n(x,t)}{\tau} \tag{5.60}$$

where D_n is the diffusion coefficient for electrons and τ is the mean carrier lifetime. In the presence of a small signal AC voltage superimposed on the DC bias voltage, the excess carriers (Δn) will also vary with the applied AC signal. Mathematically, the injected carriers can be expressed as

$$\Delta n(x, t) = \Delta n(x) \exp(j\omega t) \tag{5.61}$$

Substituting Eq. (5.61) in Eq. (5.60), we may write

$$\frac{\partial^2 \Delta n(x)}{\partial x^2} = \frac{\Delta n(x)}{L_n^2(\omega)} \tag{5.62}$$

where,

$$L_n(\omega) = \frac{\sqrt{D_n \tau}}{1 + j\omega\tau} = \frac{L_n}{\sqrt{1 + j\omega\tau}} \tag{5.63}$$

The excess carrier decreases exponentially with an increase in x as

$$\Delta n(x) = \Delta n(0) \exp\left(-\frac{x}{L_n(\omega)}\right) \tag{5.64}$$

The excess carriers vanish at $x = \infty$.

Therefore, total modulated light output from the LED per unit area can be obtained as

$$I(t) = \eta \int_0^\infty \Delta n(0) \exp\left(-\frac{x}{L_n(\omega)}\right) \exp(j\omega t)\, dx$$

$$= \frac{\eta \Delta n(0) L_n}{\sqrt{1 + j\omega\tau}} \exp(j\omega t) \tag{5.65}$$

where η is the quantum efficiency of the LED.

5.5.2 Optical Bandwidth

The optical output power $P(\omega)$ of an LED biased by a drive current modulated at an angular frequency ω can be expressed (Kressel *et al* 1977; Namizaki *et al* 1974; Liu *et al* 1975) as

$$P(\omega) = \frac{P(0)}{\sqrt{1+(\omega\tau)^2}} \tag{5.66}$$

where $P(0)$ corresponds to the optical power output of the LED under DC conditions (without modulation).

The variation of the optical power output of an LED modulated by the drive current at frequency ω normalized to the output of the LED in the absence of modulation (DC bias current only) with the modulation frequency is shown in Fig. 5.28.

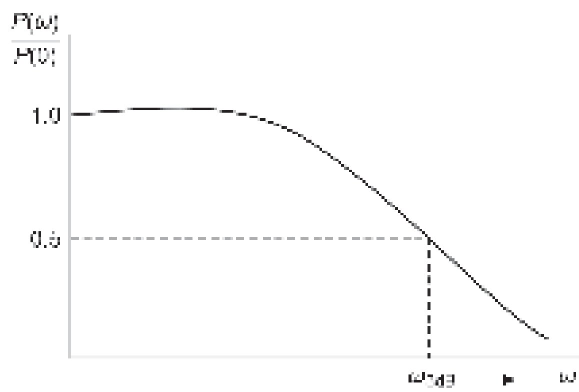


Fig. 5.28 Frequency response of an optical source showing the 3 dB bandwidth

It can be easily seen from Eq. (5.66) that the optical output power of the LED decreases with an increase in the value of the angular frequency of the modulating signal. The half-power point which corresponds to a drop of power by 3-dB from its constant value can be obtained by locating the point on the normalized frequency response curve corresponding to $P(\omega)/P(0) = 0.5$. The frequency corresponding to this point is the 3-dB bandwidth of the LED (in terms of angular frequency) as shown in Fig. 5.28. From Eq. (5.66), it can be easily seen that for the half-power (3-dB) point

$$\frac{P(\omega)}{P(0)} = \frac{1}{2} \quad \text{when } (\omega\tau)^2 = 3 \tag{5.67}$$

Designating the corresponding value of ω by ω_{3dB}, we may write

$$\omega_{3dB} = \frac{\sqrt{3}}{\tau} \tag{5.68}$$

Alternatively, the 3 dB bandwidth can be expressed as

$$f_{3dB} = \frac{\sqrt{3}}{2\pi\tau} \tag{5.69}$$

The bandwidth calculated by the above method by considering the optical power directly gives the bandwidth of the LED in the optical domain. Therefore, this value of bandwidth is often referred to as the optical bandwidth of the source.

Example 5.7 The carrier recombination lifetime for an LED is 10 ns. Estimate the optical bandwidth of the LED.

Solution The optical bandwidth of the LED can be obtained as

$$f_{3dB-op} = \frac{\sqrt{3}}{2\pi \times 10 \times 10^{-9}} = 275.8 \text{ MHz}$$

Example 5.8 The carrier recombination lifetime for an LED is 20 ns. The LED emits an optical power of 200 μW when biased with a constant DC. Estimate the optical output power when the LED is modulated by an rms drive current corresponding to the DC drive current at a frequency of 100 MHz.

Solution The optical power output of the LED when modulated by the drive current at 100 MHz can be obtained as

$$P(100\,\text{MHz}) = \frac{200 \times 10^{-6}}{\sqrt{1 + (2\pi \times 100 \times 10^{6} \times 20 \times 10^{-9})^2}} = 15.88\,\mu\text{W}$$

Example 5.9 The values of radiative and non-radiative carrier lifetime for an LED are 10 ns and 100 ns respectively. The LED emits an optical power of 100 μW at a constant bias current. Estimate the value of the optical power output of the LED when it is modulated with an rms drive current corresponding to the DC drive current at a frequency of 10 MHz. What is the bandwidth of the LED?

Solution The mean carrier lifetime in this case can be obtained as

$$\frac{1}{\tau} = \frac{1}{\tau_r} + \frac{1}{\tau_{nr}} = \frac{1}{10 \times 10^{-9}} + \frac{1}{100 \times 10^{-9}}$$

That is,

$$\tau = \frac{10 \times 100 \times 10^{-18}}{110 \times 10^{-9}} = 9.09\,\text{ns}$$

The optical output power from the LED when biased by a drive current at 10 MHz can be obtained as

$$P(10\,\text{MHz}) = \frac{100 \times 10^{-6}}{\sqrt{1 + (2\pi \times 10 \times 10^{6} \times 9.09 \times 10^{-9})^2}} = 75.75\,\mu\text{W}$$

Example 5.10 The carrier recombination lifetime for a double heterostructure LED (DH-LED) at room temperature is 20 ns. The surface recombination velocity of the carriers at each heterointerface is S =100 m/s and the thickness of the active region is 1.0 μm. Estimate the optical bandwidth of the DH-LED.

Solution The effective lifetime of the carriers for the DH-LED can be estimated using Eq. (5.59) as

$$\frac{1}{\tau_{\text{eff}}} = \frac{1}{\tau} + \frac{2S}{d} = \frac{1}{20 \times 10^{-9}} + \frac{2 \times 100}{1.0 \times 10^{-6}} = 0.5 \times 10^{8} + 2 \times 10^{8} = 2.5 \times 10^{8}$$

The effective carrier lifetime for the DH-LED is thus,

$$\tau_{\text{eff}} = \frac{1}{2.5\times 10^{8}} = 4\,\text{ns}$$

The 3-dB optical bandwidth of the DH-LED can be estimated as

$$f_{3dB-op} = \frac{1}{2\pi\times 4\times 10^{-9}} = 39.8\ \text{MHz}$$

5.5.3 Electrical Bandwidth

It has already been pointed out that the bandwidth calculated above is based on the frequency response of the optical power output of the LED when the drive current is modulated. This bandwidth corresponds to the 3-dB point of the emitted power of the LED in the optical domain. However, the bandwidth of the LED is often calculated in the electrical domain. The optical power of an LED is usually measured with the help of an optical power meter which uses an optical detector at the input port. The detector is a square-law device that converts the incident optical power to an electric current. The electrical current produced by the detector is proportional to the incident optical power. If $I(\omega)$ is the electrical current produced by the detector in response to an intensity-modulated optical power $P(\omega)$ then

$$I(\omega) \propto P(\omega) \tag{5.70}$$

Further, the electrical power $P_e(\omega)$ corresponding to the electrical current $I(\omega)$ in the detector circuit is given by

$$P_e(\omega) = \frac{I^2(\omega)}{R} \tag{5.71}$$

where R is the equivalent resistance of the detector circuit.

The ratio of the output electrical power of the LED corresponding to the optical power produced by the modulated drive current to the electrical output power produced by the LED in the absence of modulation can be obtained using Eqs (5.66), (5.70) and (5.71) as

$$\frac{P_e(\omega)}{P_e(0)} = \frac{I^2(\omega)}{I^2(0)} = \frac{P^2(\omega)}{P^2(0)} = \frac{1}{1+(\omega\tau)^2} \tag{5.72}$$

The electrical 3-dB point corresponds to the frequency where

$$P_e(\omega_{3dB}) = \frac{1}{2}P(0) \tag{5.73}$$

It can be easily seen from Eq. (5.72) that Eq. (5.73) is satisfied when $\omega\tau = 1$, that is, the 3-dB electrical bandwidth becomes

$$f_{3dB-el} = \frac{1}{2\pi\tau} \tag{5.74}$$

It is interesting to note that at the 3-dB frequency, the electrical current ratio becomes

$$\frac{I(\omega)}{I(0)} = \frac{1}{\sqrt{2}} = 0.707 \tag{5.75}$$

The bandwidth of an LED can thus be expressed by considering the half-power (3-dB) point of the optical power output of the LED or by estimating the half-power point in the electrical domain by converting the optical power output to the corresponding electrical power in the detector circuit. It can be easily seen that

$$f_{3dB-op} = \sqrt{3} f_{3dB-el} \tag{5.76}$$

It can be further verified that the optical 3-dB point occurs at a point where the ratio of detected modulated current to unmodulated DC current equals to 1/2. Figure 5.29 shows the comparison of electrical bandwidth and the optical bandwidth which corresponds to effectively 6 dB attenuation in terms of detected electrical bandwidth.

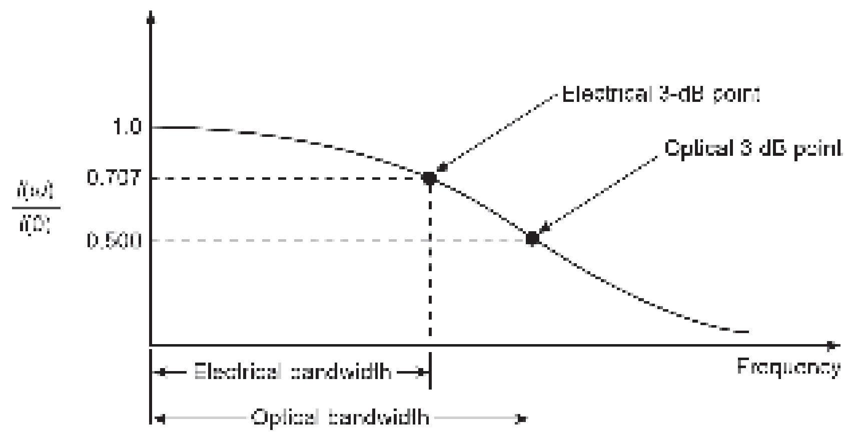


Fig. 5.29 Frequency response of an LED illustrating the electrical bandwidth and optical bandwidth

Example 5.11 The carrier recombination lifetime for an LED operating at 50 mA DC drive current is 1 ns. Estimate the values of electrical and optical bandwidth of the LED.

Solution The electrical bandwidth of the LED can be calculated as

$$f_{3dB-el} = \frac{1}{2\pi \times 1 \times 10^{-9}} = \frac{10^8}{6.28} = 159.23 \text{ MHz}$$

The optical bandwidth of the LED can be equivalently written as,

$$f_{3dB-op} = \sqrt{3} f_{3dB-el} = 275.8 \text{ MHz}$$

5.5.4 Power-bandwidth Product

It can be seen from Eq. (5.45) that the power emitted internally by an LED is

$$P_{\text{int}} = \eta_{\text{int}} \left(\frac{I}{q}\right)\left(\frac{hc}{\lambda}\right) = \left(\frac{\tau}{\tau_r}\right)\left(\frac{I}{q}\right)\left(\frac{hc}{\lambda}\right) \tag{5.77}$$

That is, the power internally generated by the LED is directly proportional to the overall recombination lifetime of the minority carriers. On the other hand, the bandwidth of the LED (see Eq. (5.69)) is inversely proportional to the overall recombination lifetime of the minority carriers.

Therefore, the product of the power internally emitted by the LED and the optical bandwidth can be expressed as

$$\text{Power–bandwidth product} = P_{\text{int}} \times \omega_{3\text{-}dB-op} = \left(\frac{\sqrt{3}}{\tau_r}\right)\left(\frac{I}{q}\right)\left(\frac{hc}{\lambda}\right) \tag{5.78}$$

The power–bandwidth product of the LED is independent of the overall recombination lifetime of the carriers and depends on the radiative recombination lifetime of the carriers.

The radiative recombination lifetime of minority carriers in the presence of injection of minority carriers under applied forward bias can be expressed as

$$\tau_r = \frac{1}{B_r(n_0 + p_0 + \Delta n)} \tag{5.79}$$

If the injection current density under forward bias is $J(=I/A)$ such that the current is entirely due to the radiative recombination of minority carriers, then the excess carriers injected in the active region of the LED can be expressed as

$$\Delta n = \frac{J\tau_r}{qd} \tag{5.80}$$

where q is the electronic charge and d is the thickness of the active region. Substituting the value of Δn from Eq. (5.80) into Eq. (5.79), we obtain the following quadratic equation of τ_r.

$$\frac{J}{qd} B_r \tau_r^2 - B_r(n_0 + p_0)\tau_r - 1 = 0 \tag{5.81}$$

Solving Eq. (5.81) for τ_r, we get

$$\tau_r = \frac{B_r(n_0 + p_0) + \left[B_r^2(n_0 + p_0)^2 + 4\left(\frac{J}{qd}B_r\right)\right]^{\frac{1}{2}}}{2\left(\frac{J}{qd}B_r\right)} \tag{5.82}$$

For a high-level injection $\Delta n \gg (n_0 + p_0)$; the radiative recombination lifetime of the carrier can be approximated using Eq. (5.82) as

$$\tau_r = \left(\frac{qd}{JB_r}\right)^{\frac{1}{2}} \tag{5.83}$$

The radiative recombination lifetime of the carrier is thus inversely proportional to $\sqrt{J}$ and directly to $\sqrt{d}$. This means that the radiative recombination lifetime of the carriers can be decreased by reducing the thickness of the active region or increasing the bias current density. An excessive increase in the bias current density may cause undesired heating of the junction leading to a heat sinking problem. This may lead to distortion of the modulated signal. It may be further noted that a thin active layer thickness of a double heterostructure LED will increase the surface recombination rate (see Eq. (5.59)) and lead to a reduction in the quantum efficiency of the LED. It may be stressed that various parameters of an LED are interrelated and must be optimized properly for the desired application. Recent investigations have led to the successful fabrication of high-radiance LED sources with surface-emitting as well as edge-emitting configurations. The bandwidth of an LED depends on the structure. An edge-emitting LED exhibits a larger bandwidth than a surface-emitting LED for the same drive current. Figure 5.30 shows the frequency response for SLED and ELED for the same bias current of 150 mA. Further, LEDs with bandwidth as high as 1 GHz have been reported for application in high-speed optical communication systems (Bhattacharya, 2007; Grothe *et al* 1979).

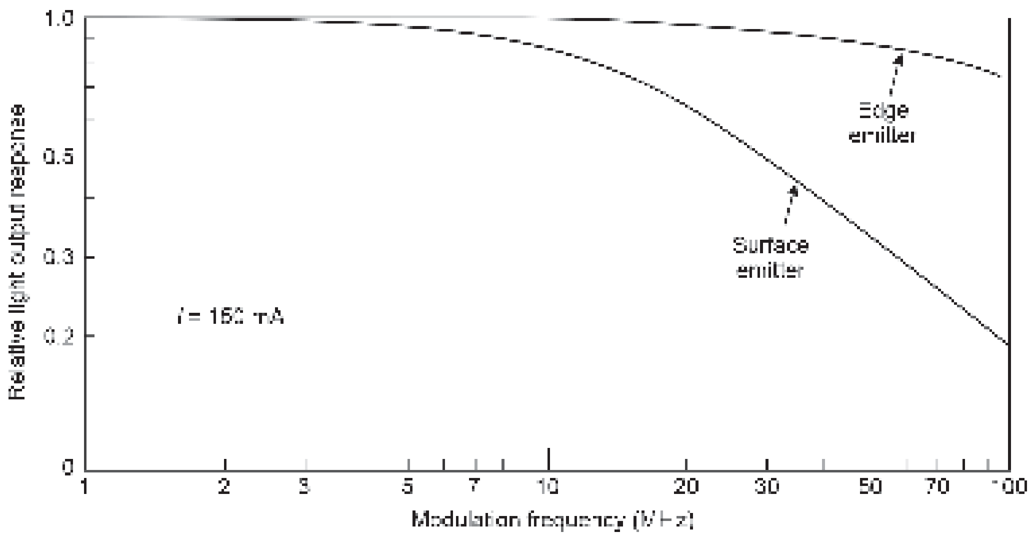


Fig. 5.30 Frequency responses of LED for the surface emitter and edge emitter configurations for a DC drive current of 150 mA

Neglecting the non-radiative recombination the 3-dB electrical bandwidth of the LED can be approximated (Saul, 1983)

$$f_{3\text{-}dB-el} \cong \frac{1}{2\pi}\left(\frac{JB_r}{qd}\right)^{\frac{1}{2}} \tag{5.84}$$

Example 5.12 A double heterojunction LED with negligible surface recombination at the heterointerfaces has an active region of thickness 0.3 μm operating at 0.85 μm wavelength region. Estimate the optical bandwidth of the LED when the drive current is neglecting the non-radiative recombination. Assume the radiative recombination coefficient $B_r = 10^{-12}$ m^3/s. The drive current of the LED without modulation is 0.5×10^5 A/m^2.

Solution The radiative recombination lifetime of the carriers for the LED can be estimated as

$$\tau_r = \left(\frac{qd}{JB_r}\right)^{\frac{1}{2}} = \left(\frac{1.6\times10^{-19}\times0.3\times10^{-6}}{0.5\times10^{5}\times10^{-14}}\right)^{\frac{1}{2}} = 9.7 \text{ ns}$$

Neglecting the non-radiative and surface recombination, the overall recombination lifetime of the carriers for the LED can be approximated as $\tau \approx \tau_r$

The optical bandwidth of the LED can be approximated as

$$f_{3\text{-}dB-op} = \frac{\sqrt{3}}{2\pi\times9.7\times10^{-9}} = 28.43 \text{ MHz}$$

5.5.5 Spectral Response

The spectral response of a light-emitting diode is essentially the variation of emitted optical power from an LED with wavelength. It may be pointed out that an LED is not a monochromatic source of light. The light is generally emitted over a finite spectral width around the peak wavelength of emission. The spectral width is measured in terms of wavelength corresponding to half maximum intensity points, i.e. full width at half power (FWHP) points as discussed earlier. The 1G optical communications was primarily focused in the wavelength range of 0.8 to 0.9 μm. The LEDs for operation in this wavelength region were based on GaAs/AlGaAs technology. These LEDs usually exhibit spectral width in the range of 25–40 nm (Fig. 5.31) (Wittke *et al* 1976). On the other hand, light-emitting diodes based on InP/InGaAsP and InP/InGaAs developed for operation in the longer wavelength region (1.3–1.65 μm) for the 2G and 3G optical communication systems exhibit larger spectral widths ranging from 50–150 nm. The spectral width of the source plays an important role in determining the intramodal dispersion associated with the optical fiber.

The spectral width and peak power of emission from an LED also depend on the doping concentration in the active region, the drive current, and the temperature of operation. Figure 5.32 shows the spectra of InGaAsP surface-emitting LED for the cases of lightly and heavily doped active regions (Carter, 1981). The spectral width also depends on the structure of the LED. For example, the spectral width of an edge-emitting LED (ELED) is almost 1.5 times more than the spectral width of a surface-emitting LED (SLED) based on the same material combination.

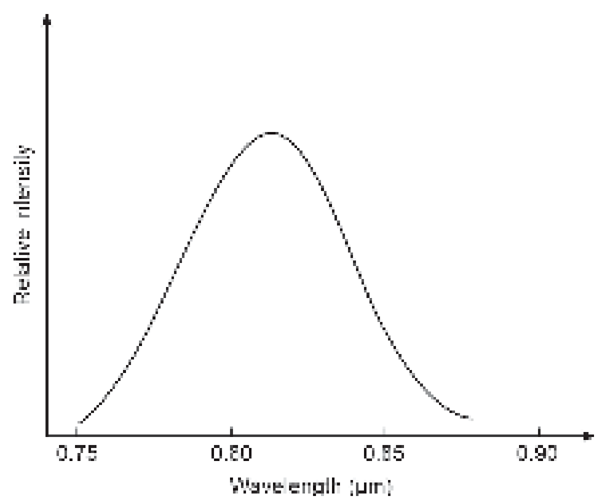


Fig. 5.31 Spectral response of an GaAs/AlGaAs surface emitting LED (after Wittke *et al* 1976)

Figure 5.33 shows the spectral characteristics of an InP/InGaAsP surface-emitting and edge-emitting LEDs (Lee *et al* 1988). It has been understood that the peak wavelength of emission and the spectral response of an LED also depend on the drive current. Figure 5.34 shows the emission spectra of an $In_{1-x}Ga_xP_zAs_{1-z}$ LED (Wright *et al* 1979). The LED exhibits a peak at 1.075 μm with a spectral width of the order of 56 nm for a bias current density of 1.2×10^7 A/m^2. The wavelength corresponding to the peak power output of the LED shifts to 1.23 μm and the spectral width increases approximately to 77 nm when the bias current density is increased to 2.5×10^7 A/m^2. InGaAsP/InP material system has been extensively used for the development of a variety of LED structures for application in the 1.3 μm window region. An exhaustive review of the advances in the performance

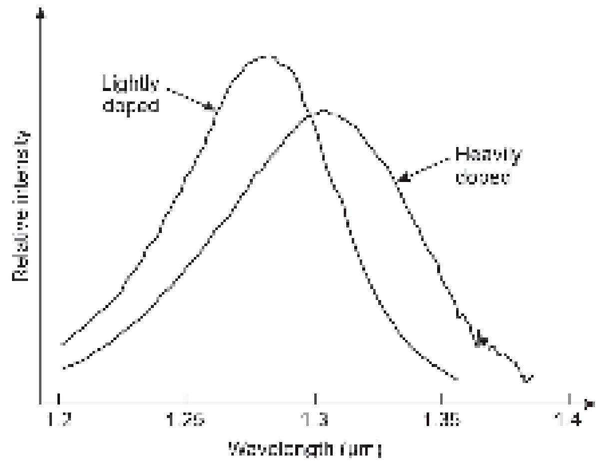


Fig. 5.32 Spectral responses of an InGaAsP surface emitting LED for the cases of lightly and heavily doped active regions (after Carter, 1981)

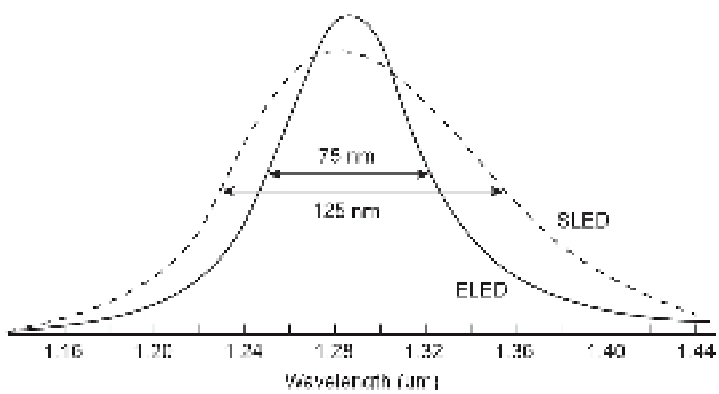


Fig. 5.33 A comparison of the relative spectral widths of a surface-emitting LED and an edge-emitting LED (after Lee *et al* 1988)

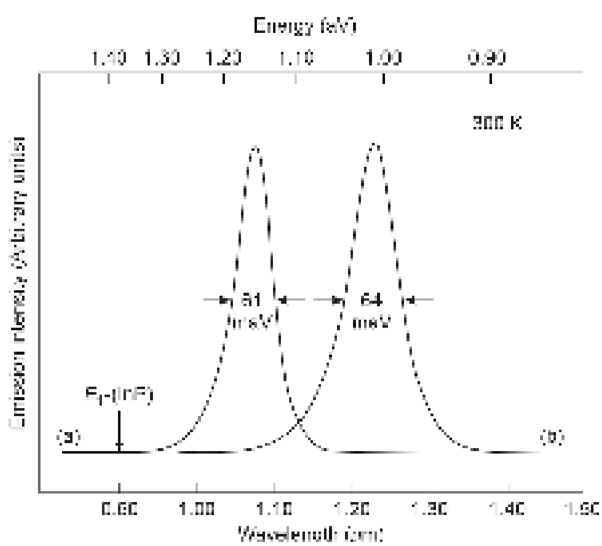


Fig. 5.34 Dependence of the spectral response of an InP/InGaPAs LED on the bias current density (after Wright *et al* 1979)

and reliability of InGaAsP-based LEDs has been reported in the literature (Saul, 1983). The reported values for power and bandwidth of 1.3 μm LEDs summarized by Saul are reproduced in Fig. 5.35 (Saul, 1983). An emitted output power of 5 mW in the air was reported to be achieved by Goodfellow *et al* (Goodfellow *et al* 1988). for LEDs with 30 MHz bandwidth. The highest bandwidth of 1.2 GHz is also reported to be exhibited by InGaAsP LED with an optical power output of 100 pW. It is further seen that for a fixed bandwidth, LEDs based on AlGaAs exhibit higher output powers than those exhibited by their counterparts based on InGaAsP.

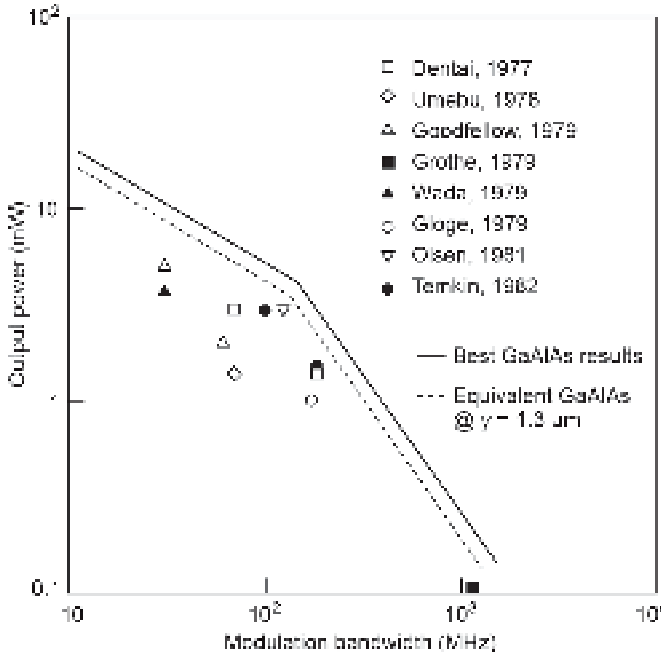


Fig. 5.35 A comparative performance of LEDs based on the reported results till early 1980's (after Saul, 1983)

5.6 ADVANCED LED STRUCTURES

With the advent of advanced technology of III-V materials, it became possible to tailor the characteristics of conventional LEDs to make them work as a more powerful source of optical power for use at the transmitter end of optical fiber communication systems. Two important advanced LED structures are resonant cavity LED and quantum dot LED discussed in the next two sections.

5.6.1 Resonant Cavity LED

A resonant cavity LED is based on a novel structure in which the active region of the device is placed in a resonant optical cavity. As a consequence, the optical power emitted from the active region is restricted to the modes of the cavity. The resonant cavity light-emitting diode (RC-LED) is based on planar technology. It consists of a Fabry–Pérot active resonant cavity between distributed Bragg reflector (DBR) mirrors. Since the cavity size is of the order of a micrometer, the RC-LED is often referred to as a microcavity light-emitting diode (Hunt *et al* 1992; Danie *et al* 2002; Jin *et al* 2003; Zhang *et al* 2003; Potfajova *et al* 2004).

The basic structure for a resonant cavity light-emitting diode (RC-LED) is shown in Fig. 5.36. The active region of RC-LED consists of InGaAsP/InP multi-quantum wells placed in the optical resonant cavity. This cavity is positioned between the top and the bottom distributed bragg's reflector (DBR) mirrors. The current confinement is achieved by selective ion implantation in the top mirror. The resonant cavity is essentially a Fabry-Pérot (FP) resonator where the optical cavity mode is in resonance and causes amplification of the spontaneous emission from the active layer. The reflectivity of the bottom DBR mirror is higher than 90% and is achieved by incorporating nearly 40 gratings. On the other hand, the surface DBR mirror is made from semi-transparent by introducing fewer gratings (nearly 15) to provide a low facet reflectivity (i.e. 40 to 60%) to allow the optical signal to exit through this mirror. Because of the incorporation of the DBR mirrors these devices are also referred to as grating-assisted RC-LEDs. This RC-LED structure is similar to that of a vertical-cavity surface-emitting laser (VCSEL) except for the fact that the light is emitted as a result of resonantly amplified spontaneous emission rather than stimulated emission in the latter case.

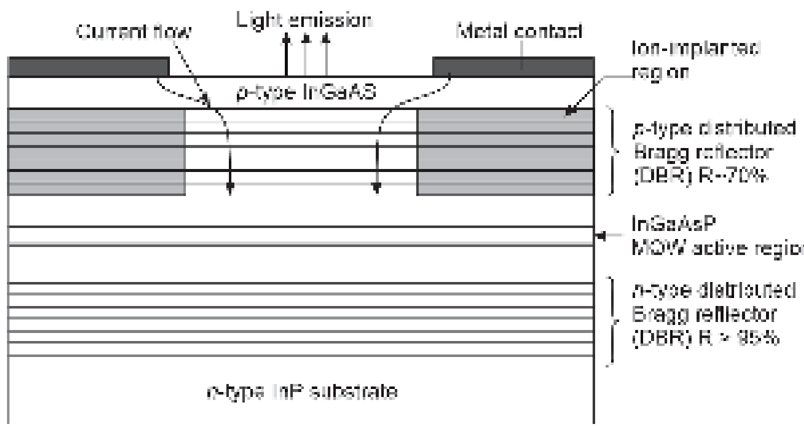


Fig. 5.36 Resonant cavity surface-emitting LED structure using DBR mirrors

Based on the cavity design, RC-LEDs can be constructed to emit light either from the bottom or from the top surface of the device structure. RC-LEDs are generally fabricated for operation over a range of wavelengths between 0.85 and 0.88 μm and also at 0.65 μm for use with plastic optical fiber. However, it is also possible to fabricate RC-LEDs for longer wavelengths (Depreter *et al* 2000; Krestnikov *et al* 2001; Jin *et al* 2003; Hunt *et al* 1992). The growth process for RC-LEDs is fairly complex compared to the fabrication of conventional devices. The RC-LEDs are attractive because of their enhanced features, such as the highly directional circular output beam and improved fiber coupling efficiency. The external quantum efficiency of the RC-LED is generally around 6 to 10% when operating at a wavelength of 1.3 μm or 1.55 μm due to the increased line-width broadening at these longer wavelengths. Nevertheless, even this value of external quantum efficiency proves sufficient to provide satisfactory data transmission at high modulation rates above 1 Gbps. In addition, device coupling limitations associated with the planar technology used for the RC-LED can be overcome by an advanced resonant cavity technique known as RC^{2}LED (Bienstman *et al* 2000). Hence a symmetric resonant cavity is created for the output coupling reflector instead of using a traditional DBR mirror. This structure produces a narrow radiation pattern and therefore it exhibits higher output signal power which is more than 50% greater than that provided by conventional RC-LED designs. An extensive review of RC-LED is available in the literature (Delbeke *et al* 2002).

The enhancement brightness and extraction efficiencies of resonant-cavity light-emitting diodes (RC-LEDs) have attracted significant interest because, compared to traditional light-emitting diodes (LEDs), RC-LEDs have spectral purity, superior emission directionality, inherent temperature stability and enhanced light extraction efficiency. RC-LEDs are benefitting from this evolution: recently, RC-LEDs emitting red light in combination with POF have been proposed as the standard for FireWire or i. Link (IEEE1394b) (Pessa *et al* 2002). This standard covers broadband service applications such as digital TV, DVD, etc. Several companies do participate in this commercialization of the RC-LED for communication applications and some extend this activity toward non-telecommunication applications. They also find application in the plastic optical fiber (POF)-based local area network for in-home multimedia and automotive industries.

5.6.2 Quantum-Dot LED (QD-LED)

Quantum-dot structures are generally used for lasers. However, it is also possible to use a quantum-dot structure in a typical RC-LED structure to enhance the emitted power output. This resultant device is referred to as a quantum-dot or QD-LED.

A single-mirror quantum-dot light-emitting diode (QD-LED) is shown in Fig. 5.37. In this structure an active layer comprising a layer of InAs quantum dots covered by InGaAs is positioned at a distance from a gold-coated mirror on the device surface. The active region consists of a single layer of quantum dots. To confine the injected carriers an additional layer of AlGaAs is incorporated between the GaAs substrate and the active region. The quantum-dot layer is placed at half the emission wavelength distance to maximize the output light power. The optical signal reflected by the mirror constructively interferes with the radiation emitted downwards from the active layer and thereby causes the optical signal power emerging from the bottom (substrate side) to increase by fourfold (Fiore *et al* 2000).

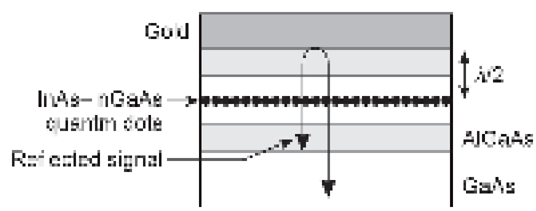


Fig. 5.37 Schematic of a quantum-dot LED (QD-LED)

QD-LEDs with output optical power as high as 10 mW have been reported for operation in the wavelength regions of 1.30 μm and 1.55 μm (Krestnikov *et al* 2001; Kicherer *et al* 2002).

5.6.3 Organic Light Emitting Diode (OLED)

An organic light emitting diode (OLED) uses a layer of an organic compound film having emissive electroluminescence properties. A host of organic polymers exhibit this kind of emissive properties. The organic semiconductor layer is placed between two electrodes. One of the electrodes is made transparent (ITO–Indium tin oxide) to deliver the emitted light output. The organic LEDs are generally used in digital display devices such as television screens, and computer monitors and in other portable systems such as mobile phones, handheld games consoles, etc. The speed of OLEDs is generally very poor and therefore does not find application in optical communication systems (Shinar, 2004).

5.7 LASER SOURCES

LASER is the acronym for light amplification by stimulated emission of radiation. It can be obtained in a variety of media such as gas, liquid, plasma, and solid (such as Ruby rod). It can also be obtained from a specially designed semiconductor *p-n* junction. Semiconductor laser sources are generally known as a laser diode (LD) or more frequently as an injection laser diode (ILD) to be more specific. The size of the laser source depends on the type of medium used for the design. Among various laser sources, semiconductor laser sources have the smallest size that is compatible with the size of an optical fiber. As a result, semiconductor laser sources are generally used in optical fiber communication. In this chapter, we shall primarily discuss various injection laser diodes. However, for a thorough understanding of the mechanism of LASER operation, it is necessary to understand the phenomenon of stimulated emission. It may be pointed out here that a laser that produces light by itself should be viewed as an optical oscillator rather than an optical amplifier. From this viewpoint, the acronym is quite misleading and as a result, actual optical amplifiers had to be named "LASER amplifiers" allowing redundancy in the nomenclature. Laser operation is generally explained with the help of energy states of atomic, molecular, and ionic systems. However, with semiconductor laser diodes the energy states of electrons are used for explaining various processes. Consider an electron within an atom occupying a low energy stable state (generally referred to as "ground" state. When external energy is supplied to the medium containing the atom, the energy is absorbed by the atom, and the electron is excited from the ground state to an "excited state". Suppose a photon with frequency, ν and energy $E = h\nu$ equal to the same amount of energy as required by the electron to give up to reach the stable state interacts with the excited atoms to trigger a resonance with the atom. The interaction forces the excited electron to leave the excited state and occupy the initial stable state releasing the energy in the form of another photon which has the same energy as the triggering photon. The resonance ensures that the emitted photon has identical wavelength, phase, and directional characteristics as the exciting photon. This feature makes a laser source a coherent source of light with spatial and temporal coherence. It is important to note that the photon that triggers or stimulates the emission process is not absorbed. The triggering electron accompanies the emitted photon to travel along the resonant medium to trigger further emission.

The basic mechanism of laser operation is discussed in the following sections.

5.7.1 Absorption, Spontaneous Emission, and Stimulated Emission

Laser involves the interaction of matter with light. It has been discussed earlier that in such cases light is considered as a stream of photons which are discrete packets of energy called "quanta". The energy of the photon can be expressed as

$$E = h\nu = \frac{hc}{\lambda} \tag{5.85}$$

where $h = 6.62 \times 10^{-34}$ Js is Planck's constant, c is the velocity of light in the free space, ν is the frequency of the light and λ is the corresponding wavelength.

According to quantum theory, atoms can only have a set of discrete energy states such that an atom can move to an excited state by absorbing energy from the external source or even move from an excited state to another lower energy state by releasing the energy. It may be stressed here that different energy states of atoms are caused by different electron configurations within the atom. An atom is excited when an electron transition within the atom takes place from the ground to a higher excited state. Absorption and emission of photons are caused by transitions of electrons from a lower to higher energy state and a higher to lower energy state respectively. The three basic processes, e.g. absorption, spontaneous emission, and stimulated emission can be clearly understood with the help of an energy state diagram associated with the atom when interacting with light. Figure 5.38

illustrates the three forms of transitions involving an atomic system with two discrete energy states or levels designated by E_1 and E_2 ($E_2 > E_1$). Consider the atom to be initially in the stable lower state, E_1 (ground state, say). When a photon with energy $h\nu_{12} = (E_2 - E_1)$ is incident on the atom then the atom may absorb the photon and excited to the energy state E_2. This process is referred to as absorption and is illustrated with the energy state diagram in Fig. 5.38(a). Consider the other situation in which the atom is already in the excited state, E_2. The atom may make a transition to the lower energy state E_1 by its own and releasing the difference in energy in the form of a photon such that $(E_2 - E_1) = h\nu_{12}$, $h\nu_{12}$ being the frequency of the emitted photon. This process is known as spontaneous emission and is illustrated with the help of the energy state diagram in Fig. 5.38(b). It may be stressed here that excited atoms in a system generally stay in the excited state for an average period called relaxation time after which they return to the stable ground state. This transition is natural and does not require the involvement of any external agent.

There may be another situation when a photon with energy exactly equal to $(E_2 - E_1)$ is made to interact with the excited atom before the relaxation period is over. The atom in the excited state is provoked by the photon to return to the lower energy state with the emission of a second photon which possesses the same energy and phase as the incident photon. This process is known as stimulated emission and is illustrated in Fig. 5.38(c) with the help of an energy state diagram.

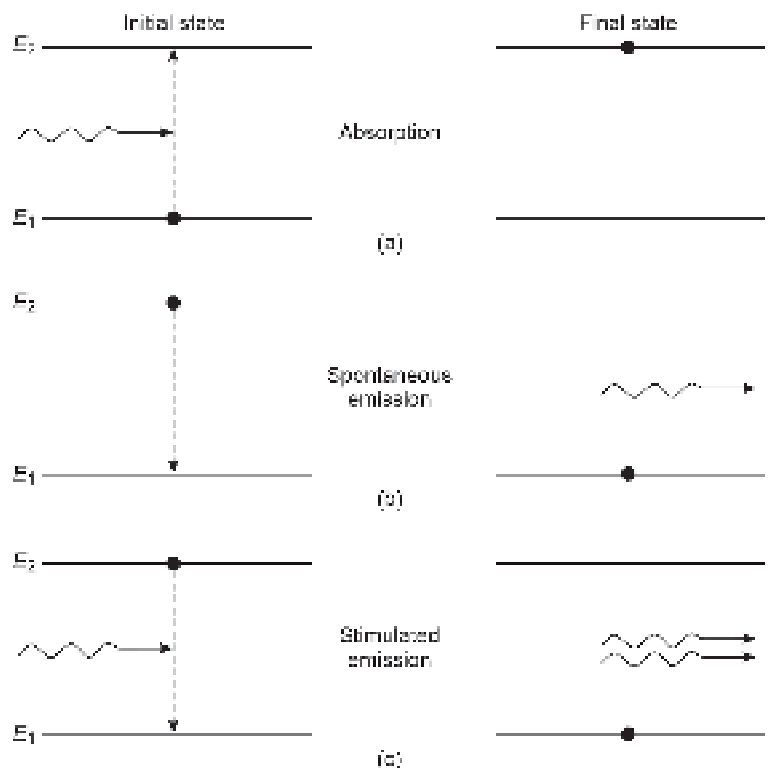


Fig. 5.38 Energy-state diagram illustrating (a) Absorption (b) Spontaneous emission (c) Stimulated emission

The basic difference between spontaneous emission and stimulated emission is that the former is a natural process whereas the latter is a forced process caused by an external photon. Because of the very origin of the spontaneous emission process, it is random, and therefore light emitted by the process following many electronic transitions from a large number of atoms results in incoherent radiation. This means that the photons emitted in the process have random phases. This kind of emission property is exploited to design semiconductor light-emitting diodes which are recognized as incoherent sources of light. On the other hand, photons produced by stimulated emission have energy usually identical to that of the incident photon, and also the phase and polarization of each of the emitted photons are the same. As a result, stimulated emission produces coherent light ideally of a single frequency[1] (or wavelength). Further, the liberated photons interact with the medium again and again to produce more such photons which constructively[2] interfere with each other to provide amplification and thereby produce an intense light beam. The phenomenon is therefore called LASER.

5.7.2 The Einstein Relations

Long before LASER was demonstrated, Einstein in 1917 (Einstein, 1917), mathematically derived the rates of two-state transitions involving absorption, spontaneous emission, and stimulated emission by considering an atomic system in thermal equilibrium. Under thermal equilibrium, the total downward–upward transitions (from energy level E_1 to energy level E_2) are equal to the total downward transitions involving both spontaneous and stimulated emission. Assume that the energy state E_1 has a population density of atoms of N_1 and that of the state E_2 has a population density of N_2. Assuming that Boltzmann's statistics are valid for the system, we may write

$$N_1 = g_1 \exp\left(-\frac{E_1}{kT}\right) \tag{5.86a}$$

$$N_2 = g_2 \exp\left(-\frac{E_2}{kT}\right) \tag{5.86b}$$

where k is Boltzmann's constant and g_1 and g_2 are the degeneracies[3] of the energy levels E_1 and E_2 respectively and T is the absolute temperature. For the non-degenerate case, $g_1 = g_2$ it can be easily verified that $N_2 < N_1$ since $E_2 > E_1$. In other words, the lower energy state is more thickly populated than the higher energy state under thermal equilibrium.

Consider that the two-state atomic system is in thermodynamic equilibrium with a radiation field comprising photons of energy $h\nu$ each and having a spectral density of $\rho(\nu)$. The atoms in the lower (ground) energy state will go to the excited state by absorbing the energy from the photons present in the radiation field. The upward transition of atoms from the lower to higher energy state is viewed as the absorption and the rate of absorption depends on the number of atoms in the ground state, i.e. population density N_1 in the ground state and the spectral density of the radiation field and is given by

$$R_{12} = B_{12}\rho(\nu)N_1 \tag{5.87}$$

where the proportionality constant B_{12} is called the Einstein constant of absorption.

[1] The incident photon with energy $h\nu$ actually stimulates atoms to emit photons with energy not always exactly equal to $h\nu$ but around this value (reasons discussed in the later sections) causing a finite spectral width of the emitted light.

[2] For understanding constructive interference the wave associated with the stream of photons should be considered.

[3] The degeneracy parameters indicate the number of sublevels within a particular energy state. If $g_1 = g_2 = 1$, the system is said to be non-degenerate.

On the other hand, atoms in the excited state have two options to transit from the higher energy state to the lower energy state (ground state) either by spontaneous emission or by stimulated emission under the radiation field. The spontaneous emission rate depends on the number of atoms in the excited state, i.e. N_2. The spontaneous emission rate can be expressed as

$$(R_{21})_{sp} = A_{21}N_2 \tag{5.88}$$

It may be pointed out that the excited atoms remain in the higher energy state on average for a period called the spontaneous lifetime or relaxation time and then come back to the ground energy state. If τ_{12} is the spontaneous lifetime of the atoms, then the spontaneous emission rate will be N_2/τ_{12} and therefore, $A_{21} = 1/\tau_{12}$. The stimulated emission rate depends on both the number of atoms in the excited state (that is, N_2) as well as the spectral density of the radiation field $\rho(\nu)$. The stimulated emission rate can be expressed as

$$(R_{21})_{st} = B_{21}N_2\rho(\nu) \tag{5.89}$$

where B_{21} is the Einstein coefficient for stimulated emission.

For a system in thermal equilibrium the rate of upward transition must be equal to the total rate of downward transition, that is

$$R_{12} = (R_{21})_{sp} + (R_{21})_{st} \tag{5.90}$$

Substituting the values of corresponding rates from rate equations, Eqs (5.87)–(5.89), we obtain

$$B_{12}\,\rho(\nu)N_1 = A_{21}N_2 + B_{21}N_2\rho(\nu) \tag{5.91}$$

The spectral density of the radiation field can be obtained from Eq. (5.91) as

$$\rho(\nu) = \frac{A_{21}N_2}{B_{12}N_1 - B_{21}N_2} \tag{5.92}$$

Equation (5.92) can be rearranged as

$$\rho(\nu) = \frac{A_{21}/B_{21}}{(B_{12}/B_{21})(N_1/N_2) - 1} \tag{5.93}$$

Using Eqs (5.86a) and (5.86b), we may write

$$\frac{N_1}{N_2} = \frac{g_1}{g_2}\exp\left[(E_2 - E_1)/kT\right] = \frac{g_1}{g_2}\exp(h\nu/kT) \tag{5.94}$$

Substituting Eq. (5.94) into Eq. (5.93), we finally obtain

$$\rho(\nu) = \frac{A_{21}/B_{21}}{(B_{12}g_1/B_{21}g_2)\exp(h\nu/kT) - 1} \tag{5.95}$$

Further, the atomic system is in thermal equilibrium and therefore the radiation density produced by the system is identical to that of the black body radiation. According to Planck's black body radiation theory, the radiation spectral density can be expressed as

$$\rho(\nu) = \frac{8\pi h\nu^3}{c^3}\frac{1}{\exp(h\nu/kT) - 1} \tag{5.96}$$

Comparing Eqs (5.95) and (5.96), we may write

$$\frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3} \tag{5.97}$$

and

$$B_{12} = \frac{g_2}{g_1} B_{21} \tag{5.98}$$

When the degeneracies of the two states are the same that is, $g_1 = g_2$, then

$$B_{12} = B_{21} \tag{5.99}$$

That is, the probability of absorption is the same as that of the stimulated emission. It is interesting to compare the rates of absorption, the spontaneous emission, and the stimulated emission. For example, the ratio of the stimulated emission rate to the spontaneous emission rate is given by

$$\frac{\text{Stimulated emission rate}}{\text{Spontaneous emission rate}} = \frac{B_{21}}{A_{21}}\rho(\nu) = \frac{1}{\exp(h\nu / kT) - 1} \tag{5.100}$$

In general, the spontaneous emission rate is far more dominant than the stimulated emission when the atomic system is in thermal equilibrium. Spontaneous emission results in photons with random phase relationship. Therefore, sources such as LEDs which exploit spontaneous emission to produce light turn out to be incoherent sources. To make stimulated emission dominant it is necessary to ensure the presence of an intense radiation field. This is generally achieved by making use of optical cavity resonators to provide optical feedback.

Further, the ratio of the stimulated emission rate to the absorption rate under thermal equilibrium can be obtained by using Eqs (5.87) and (5.89) as

$$\frac{\text{Stimulated emission rate}}{\text{Absorption rate}} = \frac{B_{21}}{B_{12}}\frac{N_2}{N_1} \tag{5.101}$$

According to Boltzmann's distribution under thermal equilibrium, the lower energy state E_1 of the two-level atomic system contains more atoms than the upper energy level E_2. That is, if $E_2 > E_1$ then $N_2 < N_1$. Therefore, the rate of stimulated emission under thermal equilibrium is less than the absorption. So that the stimulated emission may dominate over absorption it is necessary to create a situation such that the population in the higher energy level is more than that at the lower (ground) stable energy level. This situation is called *population inversion*. The populations in the two states for a two-energy level atomic system as obtained based on Boltzmann distribution are illustrated in Fig. 5.39(a). The situation required to be created under population inversion for stimulated emission to dominate is shown in Fig. 5.39(b).

From the above discussion, it is clear that absorption, spontaneous emission, and stimulated emission are three competitive processes. Under equilibrium, the three processes balance each other and both absorption and spontaneous emission rates dominate over the stimulated emission rate. Further, to make the stimulated emission rate dominate over the spontaneous emission rate, we need to create an intense radiation field which can be obtained by using an optical cavity resonator and thereby providing positive optical feedback. Moreover, it is possible to make the stimulated emission rate dominate over

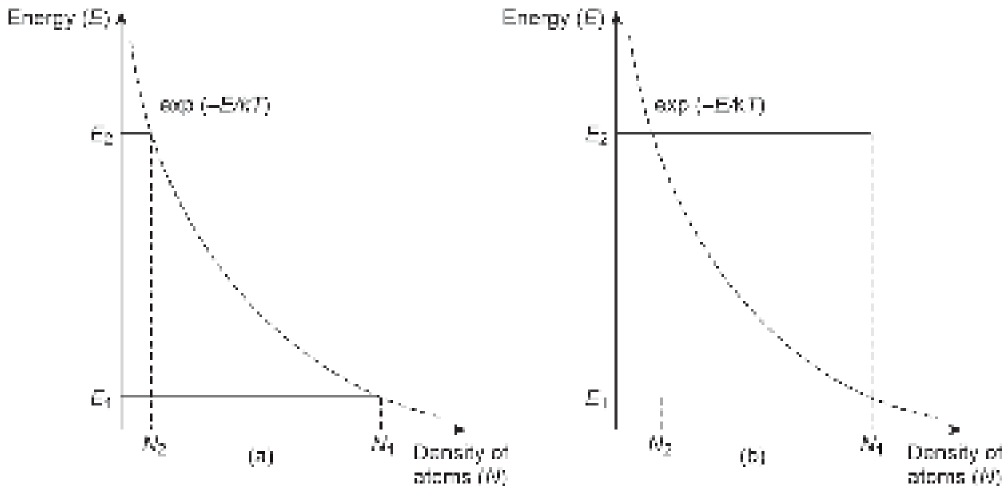


Fig. 5.39 (a) Population in the two levels under thermal equilibrium as predicted by Boltzmann distribution (b) Population inversion required to be created under non-equilibrium condition for stimulated emission to dominate over absorption.

the absorption rate by creating the situation of population inversion. This can be achieved by a technique called pumping which involves supplying instantaneous energy of short duration (such as flashlight) for exciting the atoms from the ground to the semi-stable excited state where they interact with the photons to cause stimulated emission. Therefore, population inversion and optical feedback are two primary requirements that the medium containing the atomic system must fulfill so that the stimulated emission rate dominates over both the absorption rate and the spontaneous emission rate for the laser action to occur resulting in a coherent intense beam of light. The positive feedback provided by the cavity turns a laser source into an oscillator rather than the amplifier.

Example 5.13 A tungsten lamp operating at a temperature of 1200 K emits an average frequency of 6×10^{14} Hz. Estimate the ratio of the rate of stimulated emission to the rate of spontaneous emission.

Solution The ratio of the stimulated emission rate to the spontaneous emission rate can be obtained by using Eq. (5.100) as

$$\frac{\text{Stimulated emission rate}}{\text{Spontaneous emission rate}} = \frac{1}{\exp\left(\frac{h\nu}{kT}\right) - 1}$$

$$\frac{1}{\exp\left(\frac{6.626 \times 10^{-4} \times 6 \times 10^{14}}{1.38 \times 10^{23} \times 1200}\right)} = 3.77 \times 10^{-11}$$

This means that the stimulated emission rate is negligibly small in the case of a tungsten lamp as compared to the spontaneous emission rate.

Example 5.14 The energy levels in a two-level atomic system have a separation of 1.4 eV at 300 K. Estimate the ratio of the stimulated emission rate to the rate of absorption under thermal equilibrium. Assume that Einstein's constants for the absorption and stimulated emission are the same.

Solution The ratio of the stimulated emission rate to the absorption rate can be calculated by using Eq. (5.101) and putting $B_{12} = B_{21}$ as

$$\frac{\text{Stimulated emission rate}}{\text{Absorption rate}} = \frac{N_2}{N_1} = \exp\left[-(E_2 - E_1)/kT\right]$$

$$= \exp\left(-\frac{1.4}{0.0259}\right) = 3.35 \times 10^{-24}$$

This means that under thermal equilibrium stimulated emission rate is negligible as compared to the absorption rate.

5.7.3 Population Inversion

In the previous section, it has been concluded that the stimulated emission rate is negligible in comparison with the absorption rate when the atomic system is in thermal equilibrium (also see Example 5.13). However, it is possible to create a non-equilibrium situation in which an inversion in the population in the energy states can be created so that the stimulated emission rate may dominate over the absorption rate. This so-called population inversion can be created in laser media by a variety of techniques. Before we discuss how population inversion is achieved in semiconductor laser sources let us see how it is achieved in non-semiconductor laser systems involving discrete energy states. In practice, the two-level atomic system does not work well for creating population inversion. Therefore, our discussion shall be focused on 3-state (Ruby laser) and four-state (He–Ne) atomic systems for laser action.

As already discussed atoms are needed to be excited in the higher energy state (or level) to create population inversion through non-equilibrium distribution as shown in Fig. 5.39(b). The process through which non-equilibrium distribution is attained to create population inversion is called "*Pumping*" involves the use of intense radiation of short duration such as an optical flash lamp (similar to a camera flashlight) or a radio frequency field so that the atoms can gain sufficient energy from the external source to be excited to the higher energy state. It may be noted here that in a two-level atomic system, the population inversion is not very efficient as this system can at best provide an equal population in the two energy states making the stimulated emission rate just equal to the absorption rate. For example, in a two-level system with non-degenerate or equally degenerate cases, $B_{12} = B_{21}$. This means that the probabilities of absorption and stimulated emission can be at best equal.

Consider the practical 3-level (state) and 4-level (state) atomic systems shown in Fig. 5.40 for understanding population inversion. Figure 5.40(a) corresponds to the energy levels of atoms in a Ruby laser medium. It consists of ground-level E_0 and an excited unstable level E_2 and an intermediate metastable state E_1 such that $E_2 > E_1 > E_0$. Under equilibrium conditions the energy levels will be occupied by atoms following Boltzmann distribution. If N_0, N_1 and N_2 correspond to the initial populations in the energy levels corresponding to E_0, E_1, and E_2 respectively, then under thermal equilibrium $N_0 > N_1 > N_2$. When the laser medium is subjected to suitable pumping energy then the electrons in some of the atoms[4] may be excited to the energy level E_3. Since this energy level is rather unstable, the excited electrons shall return to either E_1 or to the ground state E_0 through a non-radiative process. Because of the rapid depletion of electrons from this state the energy level E_2 remains empty always for continuous pumping. As the pumping continues for a while a situation is reached when the population of atoms in the energy state E_1 becomes more than that in E_0 where the atoms are pumped to the excited state and population inversion sets in. This is because E_1 is a metastable state where the excited atoms spend a longer time on average before returning to the ground state. While the atoms are in the excited metastable state E_1 if they are made to interact with photons of energy equal to $(E_1 - E_0)$ then stimulated emission can result following radiative electron transition from E_1 to E_0. One of the major drawbacks of the 3-level transition system of the Ruby laser is that it requires high pump power. This is because the system involves population inversion between the most populated ground state and an intermediate metastable state and lifting more than fifty percent of the ground state atoms to the excited state consumes huge pump power. On the other hand, if the simple 3-level system is slightly modified to include another intermediate state in between the ground state and the metastable state so that the final transition takes place to the new state then the power required for pumping can be reduced significantly. This concept is used in the He-Ne laser system which is viewed as a 4-level transition system as shown in Fig. 5.40(b).

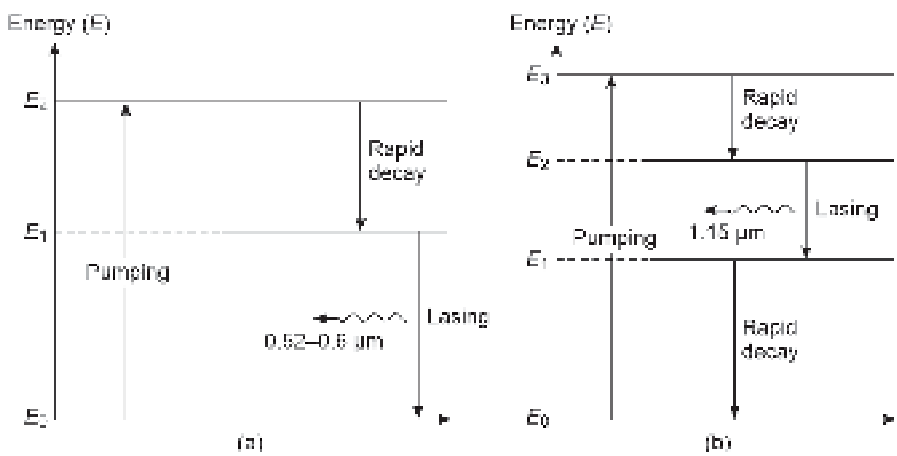


Fig. 5.40 Illustrating energy level diagram of multi-level atomic non-semiconductor laser system (a) 3-level ruby laser (b) 4-level He-Ne laser system

In the case of the He-Ne laser involving a four-level atomic system shown in Fig. 5.40(b), pumping excites the atoms from the ground state E_0 to an unstable excited state E_3. The excited atoms soon rapidly decay and drop down to the metastable energy state E_2. Further, since E_1 is an excited state above the ground level, the population in the level is small, and as a result even a small change in population in the energy state E_2 the following pumping can cause a population inversion between the states E_2 and E_1. If the excited atoms in the

[4] The excitation of atom from one energy level to the other is caused by the excitation of an electron in the particular atom between the corresponding levels. In non-semiconductor laser, excitation of atoms or electrons are interchangeably used. However, in case of semiconductor laser diode the transitions will be considered exclusively with reference to electrons.

metastable state interact with photons with energy ($E_2 - E_1$) then significant stimulated emission can occur between E_2 and E_1 to cause lasing[5].

5.7.4 Optical Feedback

Apart from population inversion, the other requirement for laser operation is optical feedback. Light amplification in a laser occurs when a photon interacts with an excited atom in such a way as to cause stimulated emission. The photon generated in the process joins the primary photon to cause further stimulated emission by repeated interaction with excited atoms in the lasing medium. The photons emitted in the process of stimulated emission are in the same phase and as a result, a kind of avalanche multiplication of photons with the same phase takes place. This leads to an intense coherent beam of light. The repeated interaction of photons with the excited atom in the lasing medium is facilitated by using an optical cavity or an optical resonator. In the simplest form, an optical resonator is a pair of parallel mirrors (plane or curved) placed at the two ends (front-end and rear-end) of the lasing medium, also called gain medium. The mirrors are generally layered with optical coatings to adjust the reflectivity to provide optical feedback through multiple reflections at the end mirrors. In general, the rear mirror is designed in a way to have high reflectivity while the front mirror is made partially transmitting so that the laser beam output can be taken from the frontend.

In actual practice, the initial photons that trigger stimulated emission in the medium are produced by spontaneous emission. The photons emitted in the process are reflected back and forth by the end mirrors to interact with the excited atoms in the medium[6] and cause more and more stimulated emission of photons with the same phase. In a way the medium provides gain and the optical cavity provides a kind of positive feedback that makes the laser source an oscillator rather than an amplifier as suggested by the acronym LASER. In general, the gain produced by the medium due to a single pass of the photons through the cavity is very small. The gain can be significantly high if the photons are allowed to have multiple passes through the medium following reflections from the end mirrors. On average, each photon passes through the gain medium several hundred times before it emerges through the front mirror as one of the constituent photons in the laser beam output. The optical cavity acts like a Fabry-Perot resonator. A stable output from the laser medium is obtained in the form of an intense light beam when the gain of the medium exceeds the total loss in the medium arising from absorption, scattering, and undesirable exit of photons through any one of the mirrors.

The gain of the lasing medium is a function of the wavelength of the light produced by stimulated emission. Oscillation in the laser cavity occurs in a small range of frequencies where the gain exceeds the total loss in the cavity. This is the reason behind the fact that a LASER source is not strictly a monochromatic source. Nevertheless, it can emit light in a very narrow spectral band (unlike an LED which emits over a relatively broad spectral band) about a central wavelength determined by the mean energy level difference of the stimulated emission transitions. It may be pointed out here that some lasers do not make use of an optical cavity. These structures are designed to produce very high optical gain

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[5] A device is said to "lase" when an intense laser beam is emitted from the device following stimulated emission. In this sense "to lase" is often used as a verb to account for the laser action.

[6] The medium is already set in a non-equilibrium condition by creating population inversion so that the stimulated emission is dominant over absorption during multiple transit of the photons in the medium.

through a single pass of the photons in the medium to produce significant amplified spontaneous emission (ASE) without needing feedback. This type of structure is viewed as a superluminescent light-emitting diode (SLD) discussed earlier in this chapter. This is because it does not involve optical feedback to cause oscillation and the emitted light has a low coherence and a relatively large spectral width.

5.8 SEMICONDUCTOR LASER DIODE

Semiconductor laser diode also known as injection laser diode finds extensive application as an optical source in the transmitter module of optical fiber communication systems requiring bandwidth over 200 MHz. The size compatibility of the semiconductor laser diode with the optical fiber makes them attractive over other forms of laser sources. The principle of operation of all kinds of laser sources is the same. However, the structure and size vary largely depending on the nature of the lasing medium (solid, liquid, gas, or semiconductor). Semiconductor laser diodes are specially designed *p-n* junctions generally realized in double heterojunction form. Configuration-wise it largely resembles a DH-LED, discussed earlier where the confining layers surrounding the active layer provide carrier confinement as well as optical confinement. It is worth mentioning here that the concept of double heterostructure was brought into LED design only after the successful demonstration of heterojunction semiconductor laser diode. From the structural design point of view, laser diodes are inherently much for complex. The biggest challenge in this respect is to confine the current in a small region in the lateral direction of the cavity. From the operational point of view, laser diodes require complicated drive circuits with the provision of an automatic thermal stabilization circuit because of the dependence of laser output on temperature. In addition, laser sources are very expensive and prone to catastrophic degradation. However, a very large bandwidth, a low spectral width, coherent light output, and large output power are some of the features that make laser diode far more superior to LEDs in optical fiber communication systems particularly for high-speed and long-haul applications.

5.8.1 Population Inversion and Optical Feedback in a Laser Diode

In a semiconductor laser diode stimulated emission results from radiative transitions between the distributed energy states in the conduction band to those in the valence band unlike in a gas or solid laser where stimulated emission is caused by radiative transitions between discrete atomic or molecular levels. From the previous discussion, it is understood that for laser action, it is necessary to create a situation of population inversion in the lasing medium and use an optical resonator to provide necessary optical feedback for the sustenance of oscillation. In a semiconductor laser diode, both population inversion and optical feedback can be achieved with the help of special design techniques as discussed below.

A semiconductor laser diode popularly known as ILD is essentially a *p-n* junction diode in which both the *p*- and *n*-region are heavily doped. Both *p* and *n* regions are so heavily doped that they become degenerate. As a result, Fermi level on the *p*-side enters into the valence band and that on the *n*-side enters into the conduction band. The situation is shown in Fig. 5.41 with the help of an energy band diagram before and after the formation of the junction. Since the Fermi level is a reference level below which all states are filled up and above which all states are empty. The filled-in states on the p-side and n-side before forming the junction are shown by shaded lines in Figs 5.41(a) and 5.41(b) shows the equilibrium energy-band diagram after the formation of the junction. Under equilibrium, the Fermi level is aligned on both sides. The filled-in states are indicated by shaded lines as before. When a forward bias is applied across the *p-n* junction the barrier height is reduced and the energy band diagram looks like one shown in Fig. 5.41(c). It can be easily seen that under forward bias a large number of electrons and holes are injected into a narrow region near the metallurgical junction as shown with the help of vertical dashed lines. It can be easily verified that in this narrow region, there are a large number of filled-in states in the conduction band (higher energy level) just opposite to a large number of empty states in the valence band (lower energy level). This means that a population inversion is created in the narrow region. This region forms the active region where stimulated emission takes place (as shown with the help of a downward arrow).

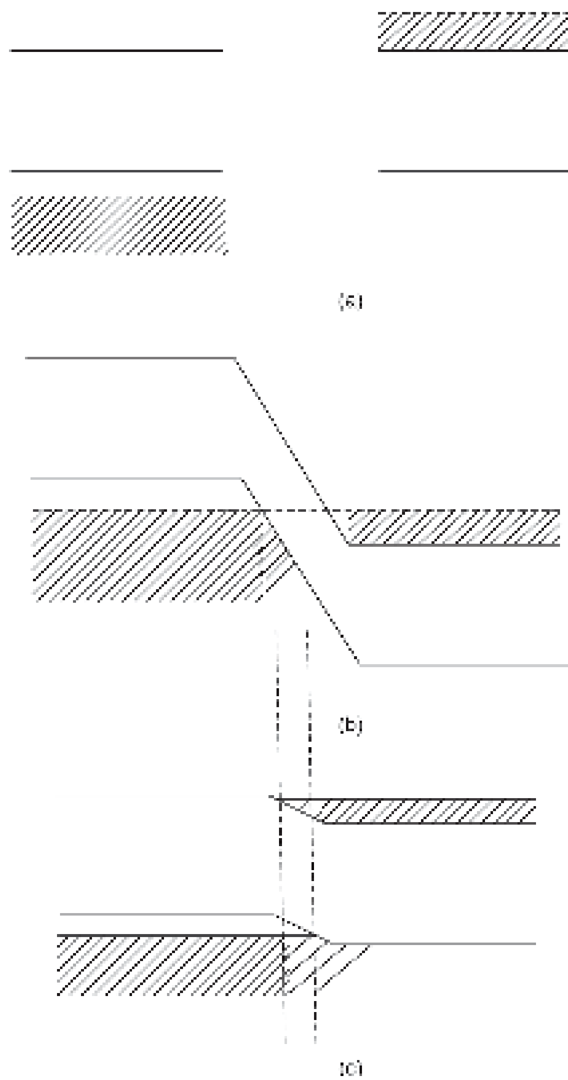

Fig. 5.41 Illustrating population inversion in an injection laser diode (a) Energy band diagrams of *p*- and *n*-type degenerate semiconductors before formation of junction (b) Energy-band diagram in equilibrium (c) Energy band diagram under forward bias

To provide strong optical feedback it is necessary to create an optical resonator structure surrounding the region in which the population inversion is created by making use of a *p-n* junction which is degeneratively doped on both sides. This can be easily achieved in a semiconductor laser diode because of the crystalline nature. Unlike in other laser sources where the Fabry-Perot resonator comprises a pair of flat and partially reflecting external mirrors, the mirror facets are constructed in a laser diode just by making two parallel clefts along natural cleavage planes of the semiconductor crystal as shown in Fig. 5.42 for a laser diode. These two mirrors (front and rear) provide strong optical feedback along the longitudinal direction to provide gain through repeated interaction of the emitted photon with the lasing medium such that the total gain compensates for the loss. This feedback turns the device into an Oscillator. It may be stressed that a laser cavity may have several resonant frequencies for each of which the gain exceeds the corresponding loss. To reduce undesirable emission from the side walls of the cavity these sides are usually abraded.

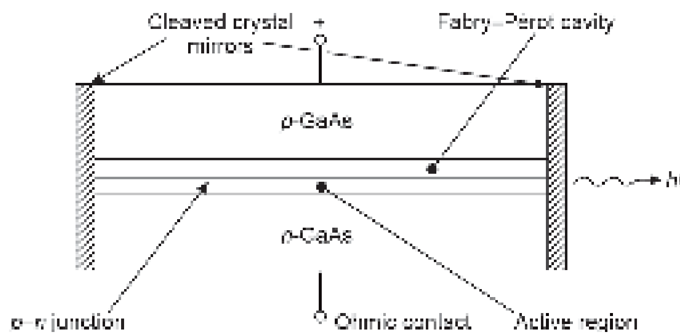

Fig. 5.42 Fabry-Pérot laser diode

There is another form of laser diode that does not make use of a Fabry-Perot (FP) cavity resonator to provide optical feedback. In place of an FP cavity, the feedback is provided by Bragg reflectors (gratings) or periodic variations in refractive index called *distributed feedback corrugations*. The corrugations are incorporated along the length of the active region as indicated in Fig. 5.43(a). The diffraction grating is generally etched close to the *p-n* junction (active region) of the diode. This grating acts more like an optical filter to select a particular wavelength which is fed back to the gain medium for lasing. In this case, the grating provides the requisite feedback for lasing and as a result, no separate mirror is required. Such laser diodes are called distributed-feedback (DFB) laser diodes. There is another form of distributed feedback configuration where the grating is incorporated only in the passive region unlike in the entire pumped region as in the case of DFB laser. This type of laser diode is termed a *distributed Bragg reflector* (DBR) laser diode and is shown in Fig. 5.43(b) (Dutta *et al* 1993). It may be pointed out here that DFB and DBR laser diodes oscillate at a single longitudinal mode, unlike FP laser diodes which generally oscillate at multiple longitudinal modes. Laser diodes are generally obtained in double heterostructure configurations discussed in the latter part of the chapter.

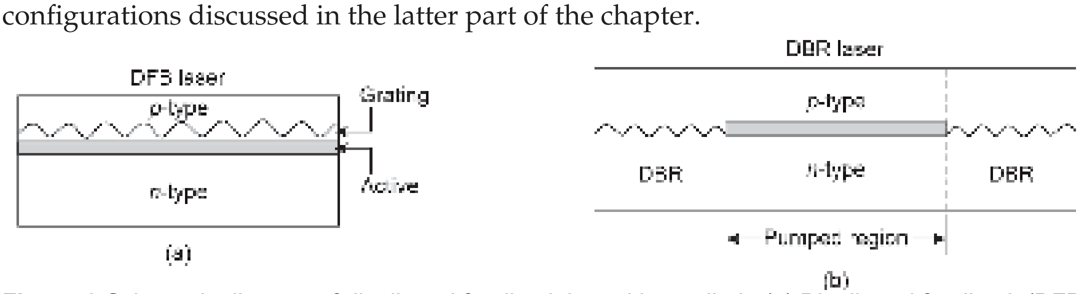

Fig. 5.43 Schematic diagram of distributed feedback based laser diode (a) Distributed feedback (DFB) laser diode (b) Distributed Bragg reflector (DBR) laser diode

5.8.2 Double Heterostructure (DH) Laser Diode

A simple homojunction laser diode generally requires a high threshold current[7] for lasing to occur. The injection laser diodes are generally fabricated with a double heterostructure configuration to have a better carrier and photon confinements that lead to a smaller threshold current for lasing. In the double heterostructure configuration, a narrow bandgap material forming the active region is generally sandwiched between two layers of wider bandgap materials. A typical GaAs/AlGaAs double heterostructure injection laser diode is shown in Fig. 5.44. In this laser diode, a thin layer of *n*-GaAs (~0.2 μm) is sandwiched between two thicker (~1 μm) layers of *p*- and *n*-type $Al_xGa_{1-x}As$. The schematic structure of the N-AlGaAs/p-GaAs/P-AlGaAs injection laser diode structure is shown in Fig. 5.45 along with the energy band diagram, refractive index profile, and photon density distribution (Yariv, 1989; Kasap, 1999). It can be easily seen that the bottom of the conduction band of AlGaAs lies above that of GaAs and as a result, a potential well is created in the GaAs region as indicated with the help of the energy band diagram

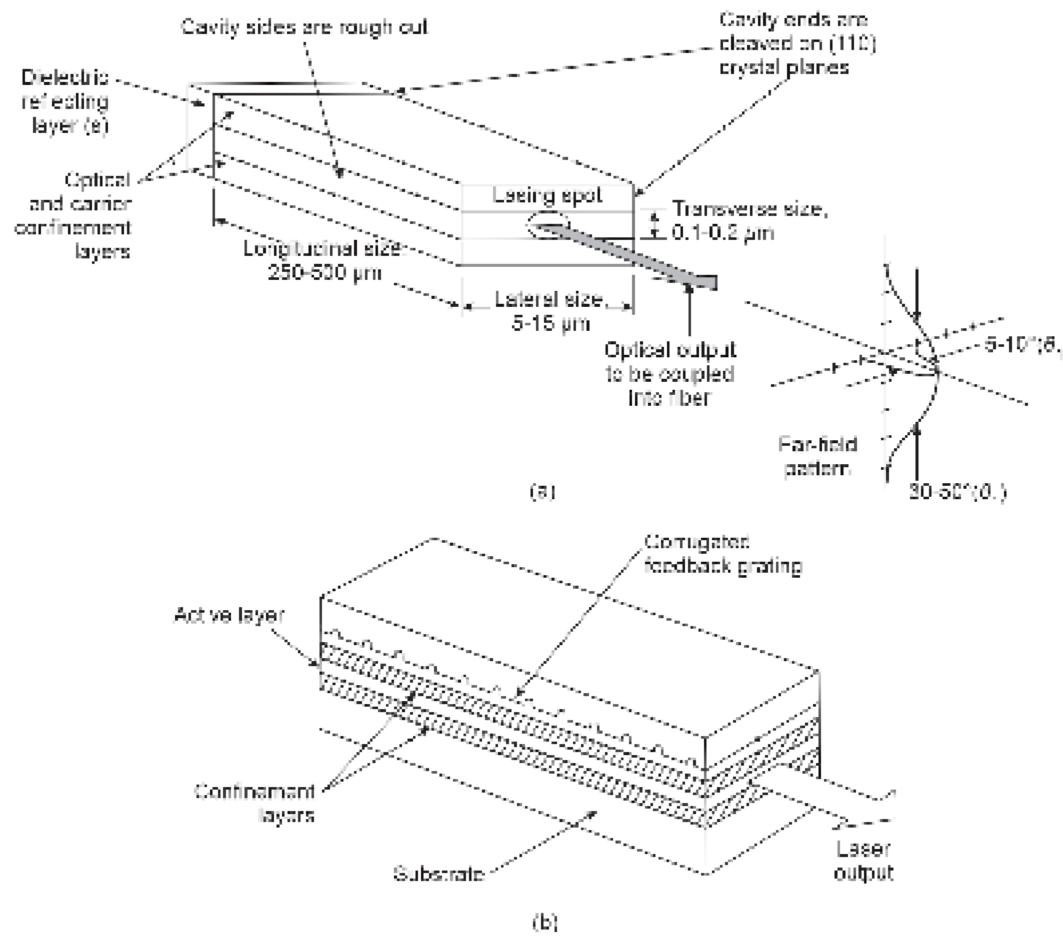

Fig. 5.44 Double heterostructure injection laser diode (a) Fabry-Perot cavity resonator laser diode (b) Distributed feedback laser diode

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[7] Threshold current is a laser diode corresponds to the bias current value which is just sufficient to provide a gain of the cavity that is just equal to the total loss in the cavity.

shown in Fig. 5.45(b). When the structure is forward biased electrons are injected from the *n*-AlGaAs region into the GaAs region. The energy barriers on two sides of the GaAs active region prevent the carriers from diffusing away from this region. Therefore, the carriers are forced to be confined in this region where a population inversion has already been created as seen from the energy band diagram. The stimulated emission occurs in the active region and the emitted photons enter into sustained oscillation with the help of a cavity resonator or distributed feedback provided in the structure. Further, it is interesting to note that the double heterostructure provides confinement of photons in addition to

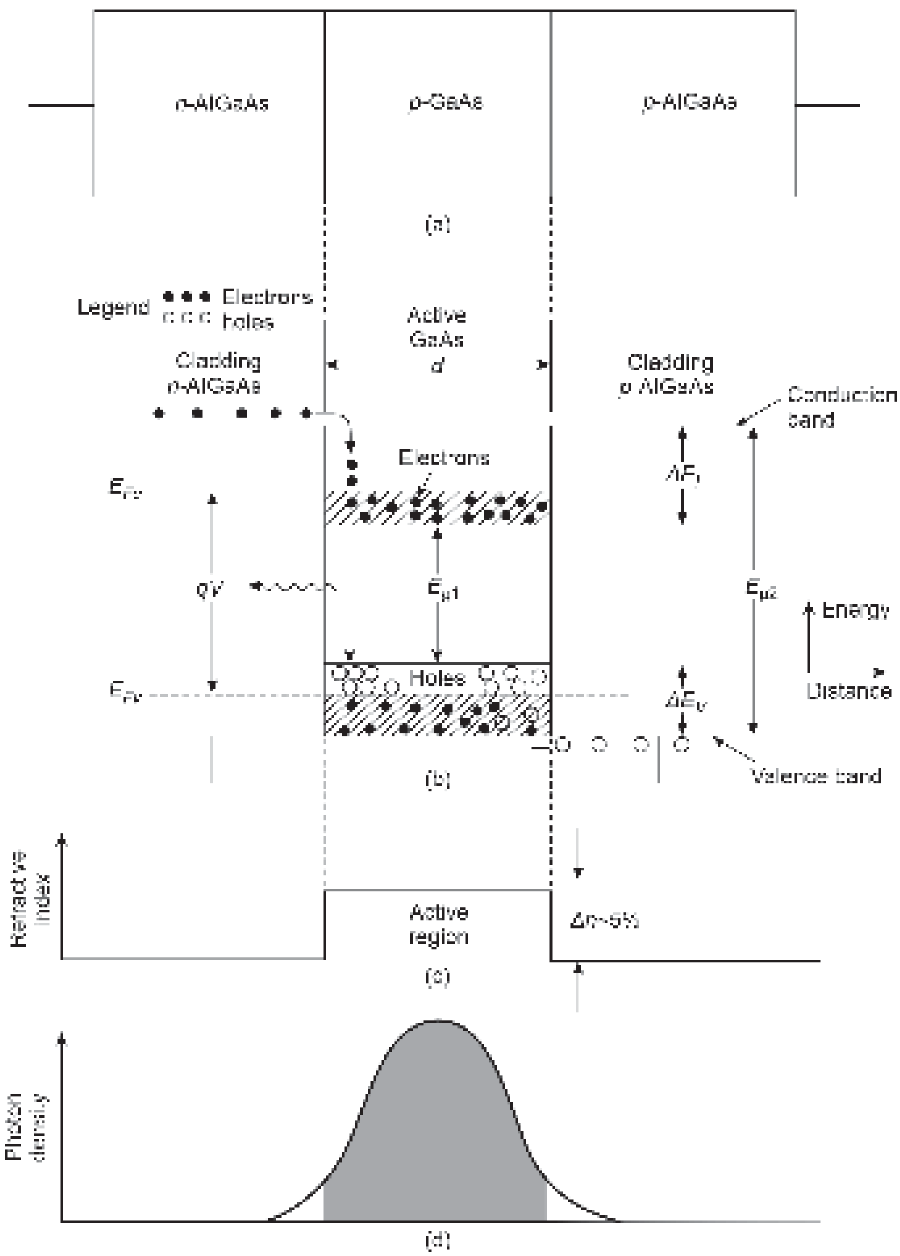

Fig.5.45 Double heterostructure AlGaAs/GaAs/AlGaAs double heterostructure laser diode (a) Schematic (b) Energy band diagram under forward bias (c) Refractive index profile (d) Photon density distribution

carrier confinement discussed above. This is possible in this particular structure because the refractive index of GaAs is more than that of AlGaAs. Thus the double heterostructure forms a three-layer waveguide structure in the form of AlGa/GaAs/AlGaAs. The refractive index profile for the waveguide structure is shown in Fig. 5.45(c). The photon generated in the process is essentially confined in the GaAs active region in the transverse direction[8] as shown in Fig. 5.45(d) and is guided along the longitudinal direction. In the lateral direction, the confinement is achieved with the help of a special structural design (gain-guided or index-guided–to be discussed afterward).

It can be easily seen from Fig. 5.45(d) that even though the photons are essentially confined to the active region they also spread into the surrounding confining regions. This can be analyzed by considering the propagation of the emitted light in the form of electromagnetic waves through the three-layer waveguide structure. The normalized waveguide thickness of the three-layer slab waveguide can be expressed as

$$D = \left(\frac{2\pi d}{\lambda}\right)\sqrt{n_a^2 - n_c^2} \tag{5.102}$$

where d is the thickness of the active region and n_a and n_c are the values of the refractive index for the active and the cladding region respectively.

The optical confinement factor, Γ is defined as the fraction of the electromagnetic energy of the guided mode that is confined in the active region. It is an important parameter that represents the effective width of the active region and the extent to which the optical confinement is provided. The optical confinement factor, Γ for the fundamental mode can be approximated (Ogasawara *et al* 2007)

$$\Gamma \approx \frac{D^2}{2 + D^2} \tag{5.103}$$

Example 5.15 A double heterostructure laser diode operating at 0.87 μm has an active layer thickness of 0.2 μm. The refractive index of the active region is 3.59 and that of the confining regions is 3.25. Estimate the value of the optical confining factor in the transverse direction.

Solution The normalized waveguide thickness can be obtained using Eq. (5.102) as

$$D = \left(\frac{2 \times 3.14 \times 0.2 \times 10^{-6}}{0.87 \times 10^{-6}}\right)\sqrt{(3.59)^2 - (3.25)^2} = 2.2$$

Therefore, the confinement factor can be calculated using the approximate relation given by Eq. (5.103) as

$$\Gamma \approx \frac{(2.2)^2}{2 + (2.2)^2} \approx 0.707$$

This means that the DH-laser diode confines approximately 70.7% of the emitted optical power in the active region along the transverse direction.

5.8.3 Lasing Conditions

Consider a Fabry-Perot laser diode cavity. A typical FP cavity resonator is generally 250–500 μm long, 5–20 μm wide. The thickness of the cavity is usually very small

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[8] The 3D structure of the double-heterostructure laser diode needs to be considered to understand the direction of confinement

(~0.1–0.2 μm). The light within the cavity may be viewed as an electromagnetic wave that sets up electromagnetic field patterns called modes within the cavity. These modes are either transverse electric (TE) or transverse magnetic type (TM). These modes are created along all three dimensions. The modes created along the length of the cavity are called longitudinal modes. The modes created along the lateral direction lie in the plane of the *p-n* junction and are called lateral modes. The modes that are created in the direction of the thickness, i.e. perpendicular to the plane of the *p-n* junction constitute the transverse modes. The longitudinal modes are related to the length of the cavity and determine the frequency spectrum of the emitted radiation. The lateral modes on the other hand decide the lateral profile of the laser beam. The transverse modes depend on the guiding properties of the three-layer waveguide structure comprising the active region and the surrounding cladding regions. These modes determine the radiation pattern and the threshold current density of the laser diode (Keiser, 2000).

To determine the lasing condition and the resonant frequencies, consider the simplistic schematic of the cavity resonator shown in Fig. 5.46. The FP cavity is made by cleaving both sides of the crystal. The cleaved edges essentially behave like mirrors because of a large refractive index difference between the semiconductor and air. Out of these two mirrors (front and rear), the reflectivity of the rear mirror is enhanced by putting additional dielectric reflecting layers while the front mirror is left semi-transmitting to allow the laser beam to be emitted out of the cavity. If L is the length of the cavity and λ corresponds to the wavelength near the peak of the spontaneous emission spectrum, then for the longitudinal modes (Bhattacharya, 2007)

$$L = m\frac{\lambda}{2} \tag{5.104}$$

where m is an integer.

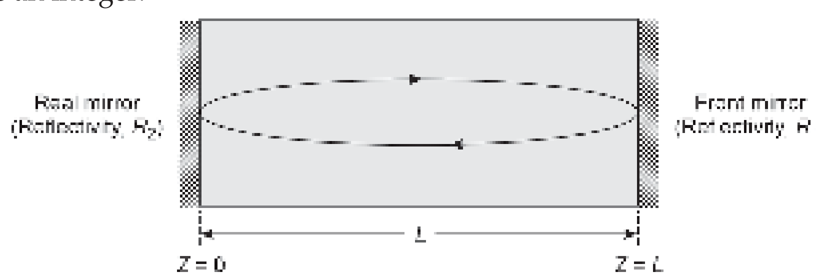

Fig. 5.46 Schematic of a Fabry-Pérot resonator

Let the reflectivity of the front and rear mirror be R_1 and R_2 respectively. Consider the electromagnetic wave propagating through the cavity along the longitudinal direction (say, z-axis). The electric field can be expressed as

$$E(z,t) = I(z)\exp\left[j(\omega t - \beta z)\right] \tag{5.105}$$

where $I(z)$ is the optical field intensity, ω is the angular frequency of the radiation field and β is the propagation constant.

The electromagnetic wave traveling along the axis of the cavity is reflected back and forth by the front and rear mirrors multiple times. In the process, the optical field intensity undergoes gain as well as loss during the transit. The gain in the cavity arises from the fact that the medium inside the cavity is preset for population inversion. Multiple interactions of the photons associated with the radiation field propagating through the cavity results in more and more stimulated emission causing an amplification of the radiation field. However, the cavity also introduces some loss of photons that tends to reduce the radiation field intensity. The loss in the cavity arises from multiple factors such as (i) emission of photons through any of the facets (ii) absorption and scattering caused by the lasing medium and (iii) absorption in the cladding regions. The lasing occurs when the gain of the cavity for a particular mode exceeds the total loss encountered by it. Both the gain and loss of the cavity depend on the energy associated with the photons that constitute the radiation field. As the radiation field travels along the cavity, the intensity of the radiation field increases exponentially due to gain and decreases exponentially due to the loss with the distance, z traveled along the length of the cavity. The optical field intensity at any point can be expressed as

$$I(z) = I(0)\exp\left[\left(\Gamma g(h\nu) - \bar{\alpha}(h\nu)\right)z\right] \tag{5.106}$$

where $\bar{\alpha}$ is absorption coefficient accounting for the average loss in the cavity per unit length, g is the gain coefficient of the cavity accounting for the gain per unit length of the cavity and Γ is the optical confinement factor which depends on the confinement of the radiation field in the transverse and lateral directions. The confinement in the transverse direction has been discussed in the previous section. The confinement in the lateral direction depends on the preparation of the side walls.

The lasing occurs for a particular mode when the gain is sufficient to exceed the total loss during one complete round trip through the cavity corresponding to $z = 2L$. During the round trip, the optical radiation is reflected by the front and the rear mirrors which have a reflectivity of R_1 and R_2 respectively. Taking into account the effect of mirror reflectivities, the intensity of the radiation field $I(0)$ after a complete round trip becomes

$$I(2L) = I(0)R_1R_2\exp\left[\left(\Gamma g(h\nu) - \bar{\alpha}(h\nu)\right)2L\right] \tag{5.107}$$

The lasing occurs when sustained oscillation takes place in the cavity under a steady state. The condition for sustained oscillation demands that the amplitude and phase of the incident wave must be the same as those of the returned wave after a round trip. This means that for lasing to occur, the following two conditions must be satisfied

$$I(2L) = I(0) \tag{5.108}$$

for the amplitude and

$$\exp[-j2\beta L] = 1 \tag{5.109}$$

for the phase.

Equation (5.108) can be used to obtain the threshold value of the cavity gain, g_{th} which is just sufficient to overcome the cavity loss. Using Eqs (5.107) and (5.108), we obtain

$$\Gamma g_{th} = \bar{\alpha} + \frac{1}{2L}\ln\left(\frac{1}{R_1R_2}\right) = \bar{\alpha} + \alpha_{end} \tag{5.110}$$

where α_{end} corresponds to the end loss of the cavity and is determined by the reflectivities of the mirrors. For 100% confinement, $\Gamma = 1$, and Eq. (5.110) can be expressed as

$$g_{th} = \bar{\alpha} + \frac{1}{2L}\ln\left(\frac{1}{R_1R_2}\right) \tag{5.111}$$

For lasing to occur one must ensure that the gain of the cavity exceeds the threshold gain, that is,

$$g \geq g_{th} \tag{5.112}$$

In the beginning, the gain of the cavity must exceed the threshold gain for lasing to occur. This is achieved through strong pumping that ensures enough population inversion to provide a gain that compensates for the overall loss in the cavity.

For an FP laser diode, the cleaved edges of the semiconductor crystal serve as mirrors. The reflectivities R_1 and R_2 correspond to the Fresnel reflection coefficient which is decided by the refractive index of the cavity and that of the surrounding medium in which the laser is emitted. The Fresnel reflection coefficient is given by

$$R = \left(\frac{n_1 - n_2}{n_1 + n_2}\right)^2 \tag{5.113}$$

where n_1 and n_2 are the refractive indices of the materials on the two sides of the reflecting boundary.

Example 5.16 A Fabry-Perot cavity resonator has uncoated facets working as mirrors. The cavity has a refractive index of 3.7 and the surrounding medium is air. Estimate the reflectivities of the mirrors.

Solution Since the uncoated facets act as mirrors for the Fabry-Perot cavity, the reflectivities of the facets are decided by the refractive indices of the media on the two sides of the reflecting boundaries. Therefore, in this case

$$R_1 = R_2 = \left(\frac{3.7-1}{3.7+1}\right)^2 = 0.33$$

Example 5.17 A Fabry-Perot laser diode has a 400 μm long cavity that uses GaAs as the material in the active region with uncoated facets. The cavity offers an average loss of 1000 m^{-1} at the operating wavelength. Estimate the value of the threshold gain assuming the refractive index of GaAs to be 3.6.

Solution The reflectivities of the uncoated facets can be estimated as

$$R_1 = R_2 = \left(\frac{3.6-1}{3.6+1}\right)^2 = 0.32$$

The threshold gain can be estimated as

$$\Gamma g_{th} = \bar{\alpha} + \frac{1}{2L}\ln\left(\frac{1}{R_1 R_2}\right)$$

$$= 1000 + \frac{1}{400\times 10^{-6}}\ln\left(\frac{1}{0.32\times 0.32}\right) = 6697.17\ \text{m}^{-1}$$

Example 5.18 A double heterostructure FP laser diode with uncoated facets has a cavity loss of 1000 m^{-1}. The threshold gain of the 400 μm long cavity is 8000 m^{-1}. Calculate the reflectivities of the facets assuming the confinement factor to be 0.9.

Solution For the uncoated facets

$$R_1 = R_2 = R \text{ (say)}$$

Using Eq. (5.110) we find

$$0.9 \times 8000 = 1000 + \frac{1}{400 \times 10^{-6}} \ln\left(\frac{1}{R^2}\right)$$

On solving, we get

$$R = 0.29$$

Example 5.19 A double-heterostructure injection laser diode with a cavity length of 400 μm offers a loss of 10^3 m^{-1}. The cavity uses uncoated facets as mirrors with a reflectivity of 0.33. Calculate the reduction in gain threshold when the reflectivity of the rear mirror is increased to 0.8. Assume the confinement factor to be 0.9.

Solution For the uncoated facets $R_1 = R_2 = 0.33$

The threshold gain for the uncoated facets is

$$\Gamma g_{\text{th}} = \frac{1}{\Gamma}\left[\bar{\alpha} + \frac{1}{2L}\ln\left(\frac{1}{R_1 R_2}\right)\right]$$

$$= \frac{1}{0.9}\left[10^3 + \frac{1}{400 \times 10^{-6}}\ln\left(\frac{1}{0.33 \times 0.33}\right)\right] = 7270 \text{ m}^{-1}$$

When the reflectivity of the rear mirror is increased, then $R_2 = 0.8$ and $R_1 = 0.33$
Therefore, the new threshold gain requirement will be

$$g'_{\text{th}} = \frac{1}{0.9}\left[10^3 + \frac{1}{400 \times 10^{-6}}\ln\left(\frac{1}{0.8 \times 0.33}\right)\right] = 4329 \text{ m}^{-1}$$

The reduction in the threshold gain becomes

$$g_{tt} - g'_{\text{th}} = 7270 - 4329 = 2941 \text{ m}^{-1}$$

5.8.4 Resonant Frequencies

It is already seen that the lasing sets alternature in when the amplitude and phase of the incident wave are the same as those of the returned wave after a round trip in the cavity. Equation (5.109) describes the phase requirement to be satisfied for sustained oscillation. The equation is valid when

$$\exp[-j2\beta L] = 1 = \exp(-j2\pi m) \tag{5.114}$$

where m is an integer[9].

That is, $$2\beta L = 2\pi m \tag{5.115}$$

The propagation constant β can be expressed as

$$\beta = nk = \frac{2\pi n}{\lambda} \tag{5.116}$$

where n is the refractive index of the cavity and λ is the operating wavelength.

[9] When the cavity resonates a standing wave pattern is formed in the cavity and the integer m stands for the number of half-wavelength spans between the two facets of the cavity.

Substituting Eq. (5.116) into Eq. (5.115), we may write

$$\beta = \frac{2nL}{\lambda} \tag{5.117}$$

Further $c = \nu\lambda$, c being the velocity of light. We may therefore alternatively write Eq. (5.117) as

$$m = \frac{2nL}{c}\nu \tag{5.118}$$

At this stage it is interesting to note that the gain of the cavity is a function of wavelength and so also a function of frequency (since $c = \nu\lambda$). A typical gain *versus* wavelength characteristics of the cavity are shown in Fig. 5.47. The gain of the cavity can be expressed as a Gaussian variation with wavelength and can be mathematically expressed as

$$g(\lambda) = g(0)\exp\left[-\frac{(\lambda-\lambda_0)^2}{2\sigma^2}\right] \tag{5.119}$$

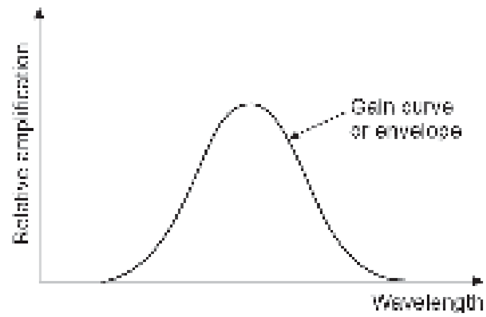

Fig. 5.47 Gain *vs* frequency characteristics of the cavity of a laser diode

where λ_0 is the wavelength corresponding to the peak gain at the center, $g(0)$ and σ are the rms spectral width of the gain curve.

It may be seen from the figure that there will be a range of wavelength (and so also frequencies) for which gain may exceed the loss and these frequencies satisfy Eq. (5.118). In other words, several frequencies can satisfy Eqs (5.108) and (5.109) to produce sustained oscillation in the cavity depending on the structure of the cavity. The gain curve of the cavity can be so designed that only one frequency can satisfy the Eqs (5.108) and (5.109). Under this condition, the laser diode generates only one mode and it is called a single-mode laser diode. When the diode generates several oscillating frequencies, it is interesting to find the separation between the adjacent oscillating modes in terms of frequency (or wavelength). Consider two adjacent longitudinal modes. Let the frequency of the mode be ν_m for which the corresponding integer that satisfies Eq. (5.118) is m. If the adjacent mode has a frequency of ν_{m-1} then Eq. (5.118) can be written for the two successive modes as

$$m = \frac{2nL}{c}\nu_m \tag{5.120}$$

and

$$m-1 = \frac{2nL}{c}\nu_{m-1} \tag{5.121}$$

Subtracting Eq. (5.120) from Eq. (5.119), we obtain

$$\frac{2nL}{c}(\nu_m - \nu_{m-1}) = 1 \tag{5.122}$$

That is,

$$\Delta\nu = \frac{c}{2nL} \tag{5.123}$$

where $\Delta\nu = \nu_m - \nu_{m-1}$ corresponds to the frequency separation between two adjacent modes.

Further, the above separation in the frequency between two adjacent modes can be translated into wavelength separation by using the relation

$$\frac{\Delta \nu}{\nu} = \frac{\Delta \lambda}{\lambda} \tag{5.124}$$

Using Eqs (5.123) and (5.124) the wavelength separation between two adjacent modes can be obtained as

$$\Delta \lambda = \frac{\lambda}{\nu} \Delta \nu = \frac{\lambda^2}{2nL} \tag{5.125}$$

The output spectrum of a typical multimode laser diode is shown in Fig. 5.48 as a function of wavelength (Peterman *et al* 1982). It can be easily seen that the spectrum follows the gain characteristics of the cavity. The number of modes and their spacing depend on the length of the cavity and the refractive index of the lasing material.

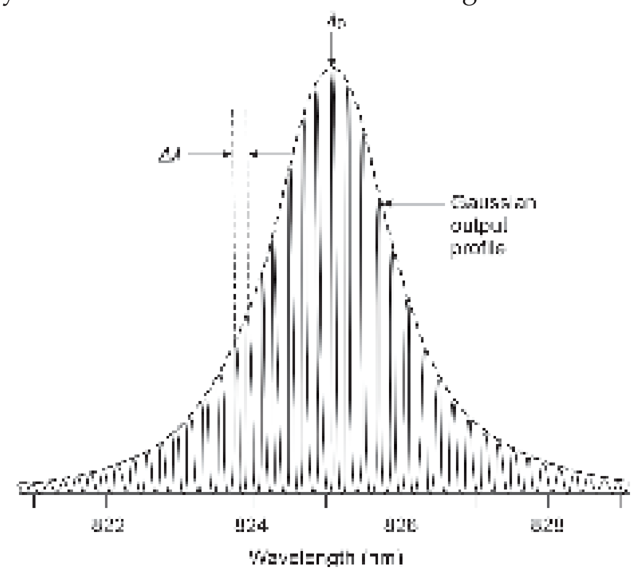

Fig. 5.48 Spectrum of a typical GaAs/AlGaAs laser diode (after Peterman *et al* 1982)

Example 5.20 The gain of an active cavity resonator of an injection laser diode follows Gaussian distribution. If the half-power point, $\lambda - \lambda_0 = 3$ nm, estimate the rms spectral width, σ of the source.

Solution The gain characteristics of the cavity are (refer to Eq. (5.119))

$$g(\lambda) = g(0) \exp\left[-\frac{(\lambda - \lambda_0)^2}{2\sigma^2} \right]$$

Given that

$$g(\lambda) = \frac{1}{2} g(0) \quad \text{for} \quad \lambda - \lambda_0 = 3 \text{ nm}$$

Therefore,

$$\frac{1}{2} = \exp\left(-\frac{(3 \times 10^{-9})^2}{2\sigma^2} \right)$$

or

$$\sigma = 1.76 \text{ nm}$$

Example 5.21 A Fabry-Perot injection laser diode with an active cavity of length 500 μm operating at 850 nm. Calculate the wavelength separation between the successive modes in the cavity assuming the refractive index of the cavity to be 3.6.

Solution The wavelength separation between the successive modes of the injection laser diode can be calculated using Eq. (5.125) as

$$\Delta\lambda = \frac{\lambda^2}{2nL} = \frac{(850\times10^{-9})^2}{2\times3.6\times500\times10^{-6}} = 0.2\,\text{nm}$$

5.8.5 Laser Diode Rate Equations

A laser diode essentially converts an electrical signal to an optical signal. The relation between the optical output power of a laser diode and the drive current can be obtained by considering the rate equations for electron concentration and photon density in the active region of the laser diode. A simplified relationship between the drive-current and optical power can be derived by assuming that the modulating frequency is much less than the cut-off frequency decided by the transit time of the optical wave in the cavity (Lau *et al* 1985).

The two rate equations that govern the electron concentration and photon density are given by

$$\frac{d\Phi}{dt} = Cn\Phi + \delta\frac{n}{\tau_{\text{sp}}} - \frac{\Phi}{\tau_{\text{ph}}} \tag{5.126}$$

$$\frac{dn}{dt} = \frac{J}{qd} - \frac{n}{\tau_{\text{sp}}} - Cn\Phi \tag{5.127}$$

where Φ is the photon flux density, n is the electron concentration (per unit volume), τ_{sp} is the spontaneous lifetime of the carriers and τ_{ph} is the photon lifetime, J is the conduction, d is the thickness of the active region, q is the electronic charge and δ is a small fractional value. The coefficient C takes into account Einstein's B coefficients and accounts for the interaction between optical absorption and emission.

The rate equations stated above are taken into consideration all the mechanisms by which carriers and photons are generated in the laser cavity. Equation (5.126) corresponds to the rate of change of photon flux density. The first term on the right-hand side of Eq. (5.126) corresponds to the photons generated through stimulated emission while the second term accounts for the photons generated through spontaneous emission. The third term on the right-hand side of the Eq. (5.126) is attributed to the loss of photons caused by the laser cavity because of various loss mechanisms discussed earlier. The Eq. (5.127) on the other hand corresponds to the rate of change of electron concentration in the active region. The first term on the right-hand side of Eq. (5.127) corresponds to the injected electrons in the region by the applied bias current. The second and third terms on the right-hand side of the Eq. (5.127) correspond to the loss of electrons through spontaneous and stimulated emission respectively.

Under steady-state conditions, the photon flux density and electron concentration values do not change with time and as such the left-hand sides of both the Eqs (5.126) and (5.127) become zero. Thus in the steady state, we may write

$$Cn\Phi + \delta\frac{n}{\tau_{\text{sp}}} - \frac{\Phi}{\tau_{\text{ph}}} = 0 \tag{5.128}$$

$$\frac{J}{qd} - \frac{n}{\tau_{sp}} - Cn\Phi = 0 \tag{5.129}$$

From Eq. (5.126) we find that the contribution of the second term on the left-hand side arising from spontaneous emission is very small and can be neglected. Further, the photon flux density can build up in the cavity, $d\Phi/dt$ must be positive initially when Φ is small. Therefore, we must have

$$Cn - \frac{1}{\tau_{ph}} \geq 0 \tag{5.130}$$

From Eq. (5.130) it can be easily interpreted that there should be a threshold value of n that would satisfy the equality condition. The photon flux density Φ can only increase when n is larger than the threshold value. The threshold value of the electron concentration, n_{th} can be obtained from Eq. (5.130) as

$$n_{\text{th}} = \frac{1}{C\tau_{ph}} \tag{5.131}$$

The threshold current J_{th} corresponding to the threshold value of electron concentration, n_{th} in the steady state when $\Phi = 0$, can be obtained from Eq. (5.131) as

$$\frac{J_{\text{th}}}{qd} = \frac{n_{\text{th}}}{\tau_{sp}} \tag{5.132}$$

The threshold current defined by Eq. (5.132) accounts for the bias current density required to sustain the decay of carriers through spontaneous emission in the absence of photon flux density. In the steady state if the photon flux density attains the value, = Φ_s, then using Eqs (5.129) and (5.132), we may write

$$\frac{J - J_{\text{th}}}{qd} - Cn_{\text{th}}\Phi_s = 0 \tag{5.133}$$

The steady-state photon flux density can be obtained by rearranging Eq. (5.133) as

$$\Phi_s = \frac{1}{Cn_{\text{th}}} \frac{(J - J_{\text{th}})}{qd} \tag{5.134}$$

Substituting the value of Cn_{th} from Eq. (5.131), we finally obtain

$$\Phi_s = \frac{\tau_{ph}}{qd}(J - J_{\text{th}}) \tag{5.135}$$

Further, since Φ_s cannot be negative, therefore in the steady state the current must exceed the threshold value, that is

$$J > J_{\text{th}} \tag{5.136}$$

The variation of the optical power output of an injection laser diode with the applied drive current is depicted in Fig. 5.49. It is seen that at a low bias current, the output power is low. However, the output power increases significantly when the bias current exceeds the threshold current. The threshold current is obtained by extrapolating the linear region of the characteristics corresponding to the lasing region. Under this condition stimulated

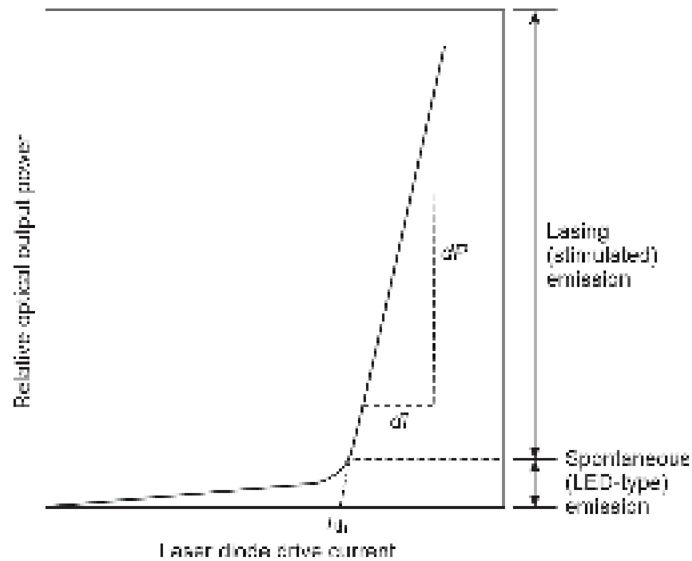

Fig. 5.49 Optical power *vs* bias current characteristics of an injection laser diode

emission takes place and a directed laser beam is emitted. In the low-bias current region, the output light power is low with poor directivity and is primarily caused by the spontaneous emission of photons. The threshold current density is related to the threshold gain coefficient of the laser diode under very strong confining conditions ($\Gamma \approx 1$) as

$$g_{th} = \beta J_{th} \tag{5.137}$$

where the proportionality constant, β (called gain factor) depends on the structure of the laser diode (Kressel *et al* 1977).

Further substituting the value of g_{th} from Eq. (5.111) into Eq. (5.137), we get

$$J_{th} = \frac{1}{\beta}\left[\alpha + \frac{1}{2L}\ln\left(\frac{1}{R_1 R_2}\right)\right] \tag{5.138}$$

Example 5.22 A Fabry-Perot cavity laser diode has a length of 500 μm and a lateral width of 100 μm. Assuming the front mirror to be an uncoated facet and the rear mirror to be a perfectly reflecting one, estimate the value of the threshold current of the laser diode when the average loss in the cavity is 1000 m^{-1}. Given that the gain factor at the operating temperature is 20 mA/cm^3 and the refractive index of the cavity is 3.6.

Solution The reflectivity of the front mirror can be estimated as

$$R_1 = \left(\frac{3.6-1}{3.6+1}\right)^2 = 0.32$$

The reflectivity of the rear mirror is $R_2 = 1$.

The value of the threshold current density can be calculated using Eq. (5.138) as

$$J_{th} = \frac{1}{20\times10^{-3}}\left[10 + \frac{1}{500\times10^{-4}}\ln\left(\frac{1}{0.32}\right)\right] = 1.63\times10^3\ \text{A/cm}^2$$

Therefore, the threshold current can be estimated as

$$I_{th} = J_{th} \times \text{area} = 1.63 \times 10^3 \times 500 \times 100 \times 10^{-8} = 0.82\,\text{A}$$

5.8.6 External Quantum Efficiency

The external quantum efficiency of a laser diode is defined as the number of photons emitted per radiative recombination of electron-hole pair above the threshold. It may be noted here that above the threshold the radiative recombination takes place through stimulated emission because the lifetime of the carriers is much shorter for stimulated emission (~10 ps) as compared to that for spontaneous emission (typically ~1 ns). Assuming that above the threshold region, the gain coefficient remains the same as g_{th} the external quantum efficiency of the laser diode can be expressed as (Kressel *et al* 1977)

$$\eta_{ext} = \frac{\eta_i(g_{\text{th}} - \bar{\alpha})}{g_{\text{th}}} \tag{5.139}$$

That is,

$$\eta_{ext} = \eta_i \left[\frac{1}{1 + \dfrac{2\bar{\alpha}L}{\ln\left(\dfrac{1}{R_1 R_2}\right)}} \right] \tag{5.140}$$

where η_i is the internal quantum efficiency. The internal quantum efficiency is defined as

$$\eta_i = \frac{\text{Number of photons created in the laser cavity}}{\text{Number of injected carriers}}$$

The value of internal quantum efficiency is very high ranging between 50 to 100%.

The external quantum efficiency of the laser diode can also be estimated from the slope of the optical power *versus* bias current characteristic in the lasing region and is often called differential quantum efficiency. The external differential quantum efficiency can be written as

$$\eta_{ext} = \frac{q}{E_g}\frac{dP}{dI} = \frac{q\lambda}{hc}\left(\frac{dP}{dI}\right) = 0.8065\lambda\left(\frac{dP}{dI}\right) \tag{5.141}$$

where E_g $(= hc/\lambda)$ is the energy bandgap of the material, h being the Planck's constant, λ is the emission wavelength and c is the velocity of light. The external quantum efficiency is usually less than 50%.

The electrical-to-optical conversion efficiency can be obtained as (Sze *et al* 2007)

$$\eta_{o/e} = \frac{P}{P_e} \times 100\% = \frac{(I - I_{\text{th}})h\nu\eta_i}{VIq}\left[\frac{\ln\left(\dfrac{1}{R_1R_2}\right)}{2\bar{\alpha}L + \ln\left(\dfrac{1}{R_1R_2}\right)}\right] \times 100\% \tag{5.142}$$

where P_e = VI is the DC input electrical power.

5.8.7 Longitudinal and Transverse Modes: Single mode Operation

In the foregoing discussion, we noted that an injection laser diode supports several modes in the longitudinal, lateral, and transverse directions. The longitudinal modes are related to the length of the cavity which is much larger than the emission wavelength and as a result, the longitudinal modes are very large in number. These modes essentially decide the frequency spectrum of the emitted light. We have also noted that the separation in frequency (or wavelength) between two successive modes is inversely proportional to the length of the cavity. This means that the frequency (or wavelength) separation, $\Delta\nu$ or $\Delta\lambda$ between the modes can be increased by reducing the length of the cavity. Therefore, it is possible to have a situation when the length of the cavity is such that the wavelength separation between the successive modes becomes larger than the spectral width of the laser source. In this situation, only one mode is supported in the longitudinal direction. This method of achieving a single longitudinal mode is not very convenient because of the fact that it is difficult to handle laser diodes with short cavity lengths and smaller dimensions usually restricting the power output to only a few milliwatts (Botez, 1987).

Example 5.23 An FP injection laser diode operating at 850 nm has a cavity length of 500 μm. Calculate the separation in the frequency between the successive longitudinal modes of the laser diode. Compare and contrast the value with that obtained when the length of the cavity is reduced to 50 μm. The refractive index of the cavity may be assumed to be 3.6.

Solution The frequency separation between the adjacent modes for the 500 μm cavity can be obtained as

$$\Delta\nu = \frac{c}{2nL} = \frac{3\times10^8}{2\times3.6\times500\times10^{-6}} = 83.33\times10^9\,\text{Hz}$$

When the cavity length is reduced to 50 μm, the frequency separation becomes

$$\Delta\nu' = \frac{c}{2nL} = \frac{3\times10^8}{2\times3.6\times50\times10^{-6}} = 833.3\times10^9\,\text{Hz}$$

Therefore, the separation in frequency increases ten times by reducing the length of the cavity.

The lateral modes are created in the plane parallel to the *p-n* junction and are usually decided by the preparation of the side walls of the cavity which are generally roughened to stop emission from the side walls. The other important modes in an injection laser diode are the transverse modes. The transverse modes are created by the standing waves formed due superposition of plane-polarized waves travelling along the length of the cavity. These modes are created in the direction perpendicular to the plane of the *p-n* junction. The transverse modes have two components: one parallel and the other perpendicular to the plane of the *p-n* junction (Yariv, 1989). The modes are confined in the transverse direction by the step-index profile of the confining layers as already discussed. When the laser beam is emitted from the cavity it exhibits a diverging field due to dispersion at the end face which is essentially the cleaved facet of the crystal. The output pattern of the beam is dominated by dispersion because the lateral width (~20 μm) and thickness (several microns) are comparable with the value of the emission frequency. The divergence angle of emission in the direction perpendicular to the plane of the *p-n* junction is measured in terms of angular width given by

$$\theta_\perp = 2\sin^{-1}\left(\frac{\lambda}{H}\right) \tag{5.143}$$

where λ is the emission wavelength and H is the thickness of the active layer.

The angular width of the beam in the direction parallel to the plane of the *p-n* junction is given by

$$\theta_{\parallel} = 2\sin^{-1}\left(\frac{\lambda}{W}\right) \tag{5.144}$$

It can be easily seen that a smaller value of the W or H will result in a larger angular beam width in the corresponding directions because of increased diffraction effects. It is possible to restrict the transverse modes to a single lowest-order TEM_{00}-mode by making W and H sufficiently small (Yariv, 1989). The beam profile showing the angular divergence in the lateral and transverse direction caused by the transverse modes is shown in Fig. 5.50.

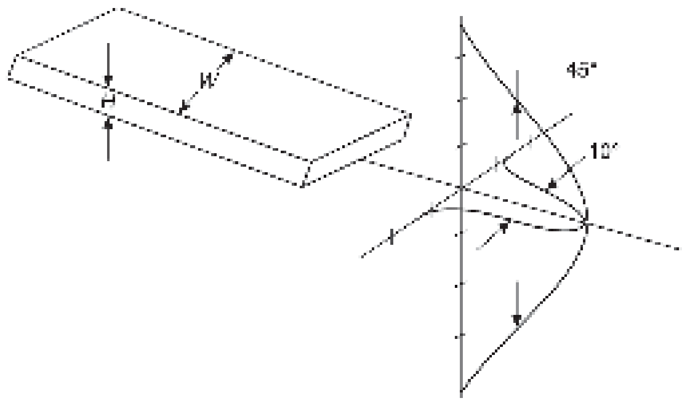

Fig. 5.50 Beam profile of laser diode showing angular divergence of the beam in the lateral and transverse direction

Example 5.24 A FP injection laser diode operating at 850 nm has a cavity width of 20 μm. Calculate the divergence angle of the emitted beam in the lateral and transverse directions of the cavity assuming the thickness of the active region to be 2 μm.

Solution The angular beam divergence of the laser diode in the transverse direction can be obtained as

$$\theta_{\perp} = 2\sin^{-1}\left(\frac{\lambda}{H}\right) = 2\sin^{-1}\left(\frac{850\times10^{-9}}{2\times10^{-6}}\right) = 50°{\cdot}3$$

The angular beam divergence in the direction transverse to the plane of the *p-n* junction is obtained as

$$\theta_{\parallel} = \sin^{-1}\left(\frac{850\times10^{-9}}{20\times10^{-6}}\right) = 4°{\cdot}8$$

The term ‘single mode operation’ of a laser source may be slightly misleading in the sense that it may either refer to a single longitudinal mode or a single transverse mode of operation. When a laser diode is operated at a single longitudinal mode it is operated in a single frequency mode by making the separation in frequency between successive modes larger than the laser transition line width. Under this condition, only one longitudinal mode falls within the gain bandwidth of the device (Keiser, 2000). The line width of such a laser diode is very narrow. As already discussed, a laser diode may also be operated with a single lowest-order transverse mode or also at a single higher-order mode by using a diffraction element in the resonator. When operated at the lowest order mode the quality of the emitted beam is affected by the diffraction effect as already discussed. It may be stressed here that even if a laser diode operates at a single transverse mode it may have multiple longitudinal modes in the axial direction of the cavity. A true single-mode operation means that the output of the laser diode must contain one longitudinal mode and one transverse mode only. In a double heterostructure laser diode configuration the thickness of the active region is kept in the range of 0.3–0.4 μm to allow only one fundamental transverse mode to be supported. The lateral modes can be confined by using various techniques discussed in subsequent sections. Thus by reducing the length of the cavity and thickness of the active region, it is possible to have only one longitudinal mode and one lowest order transverse mode respectively in an injection laser diode. In this situation, the laser diode acts as a single-mode laser. The output spectrum of a single-mode injection laser diode is shown in Fig. 5.51.

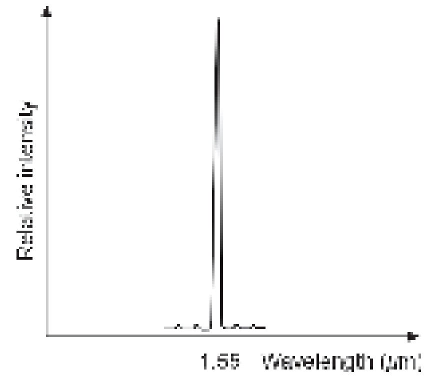

Fig. 5.51 Output spectrum of a single mode laser

5.8.8 Practical Laser Diode Structures

It is understood from the previous discussion that in a double heterostructure laser diode, optical confinement in the transverse direction is achieved by making use of two confining layers. However, there is no confinement in the lateral direction. This means that lasing takes place along the entire width of the cavity. The situation is depicted in Fig. 5.52. Even though the emission from the side walls can be prevented by roughening the side walls, the lasing over a broad region results in several undesirable effects including unacceptable geometry of the emitted beam pattern making it extremely difficult to couple power in optical fibers, large threshold current, multi-filament lasing and difficult heat sinking.

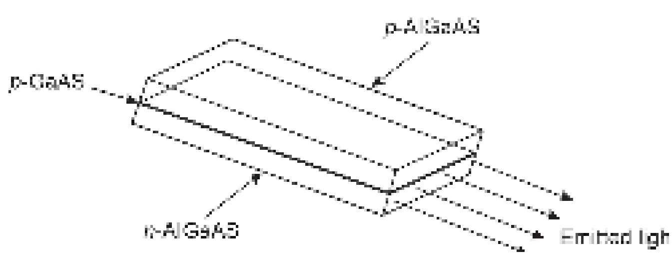

Fig. 5.52 DH-LED without lateral confinement showing emission over entire width

In an attempt to overcome these effects, a large number of laser structures have been proposed. In all these structures the active region has been kept confined in the lateral direction by some technique. The novel structures are of two basic forms, e.g. (i) gain-guided structure and (ii) index-guided structure. In the case of the gain-guided structure, the injection of current is restricted to a small region in the lateral direction along the junction plane. In the index-guided structure, the restriction of lasing in a small region in the lateral direction along the plane of the junction is achieved with the help of a built-in index profile in the lateral direction all along the length of the cavity.

5.8.8.1 Gain-guided structure

The gain-guided structure is obtained by introducing stripe geometry to the structure. In this case, the current injection is restricted to a narrow region beneath the stripe (Sheps 2002; Botez 1987). Lasing occurs in the limited region defined by the stripe electrode of a small width that runs along the length of the cavity. Injection of high-density current beneath the stripe creates population inversion resulting in a small variation in the refractive index below the stripe. A weak complex waveguide structure is thus created in the process.

The surrounding regions reflect the light back into the region beneath the stripe and thereby confine the light in the lateral direction. The schematic of a gain-guided laser structure with stripe electrode geometry is shown in Fig. 5.53. The current restriction in the lateral direction allows the diode to operate in the continuous-wave (CW) mode to deliver power exceeding 100 mW with a reasonably low value of threshold current. As a result, the requirement for heat sinking is also low. The gain-guided structure also allows the laser diode to be operated in the fundamental mode along the junction plane making it convenient to couple power to a single-mode fiber (Dutta *et al* 1993). The weak wave-guiding in the lateral direction makes the output beam highly astigmatic. The output also exhibits a doublepeak in the radiation pattern making it unsuitable for specialized applications (Botez, 1985).

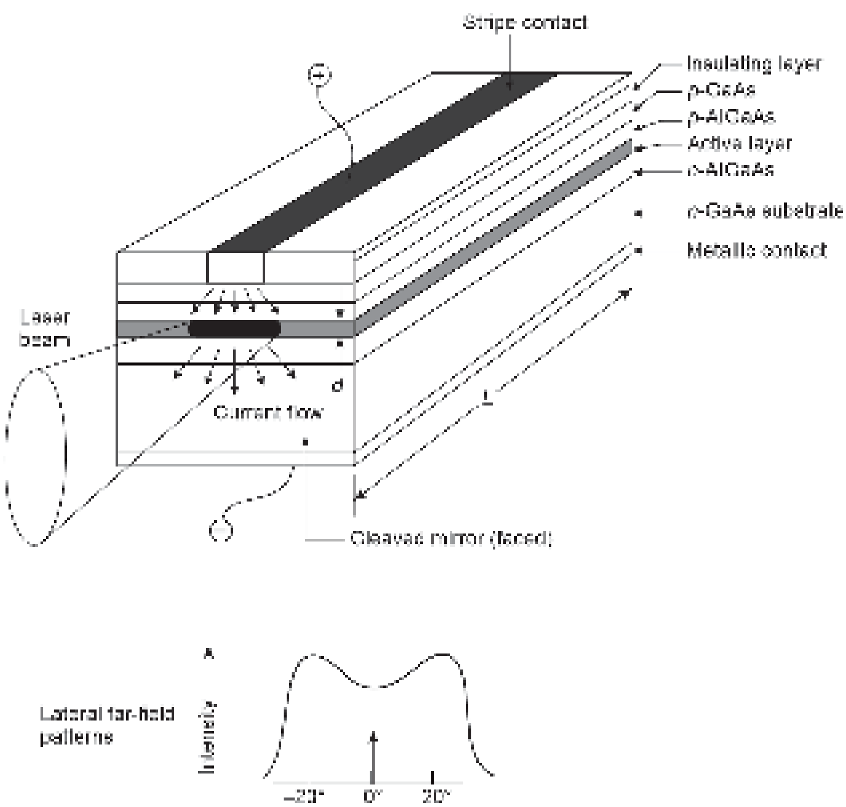

Fig. 5.53 Gain-guided laser structure with stripe electrode geometry along with intensity pattern of the emitted beam (after Scheps, 2002)

5.8.8.2 Index-guided structure

Index-guided structures make use of dielectric waveguide structures in the lateral direction running along the length of the cavity. The light in index-guided structures is

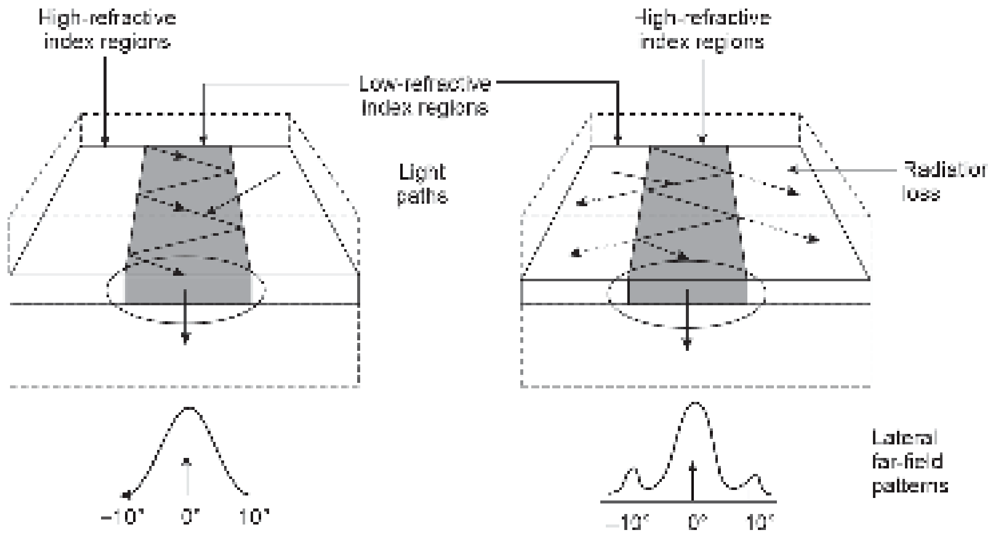

Fig. 5.54 Index-guided laser diode structures showing the intensity profile of the emitted beam: (a) Positive-index guided structure (b) Negative-index guided structure (after Botez, 1985)

guided by the variations in the refractive index of different regions. The index-guided structure provides only one fundamental transverse mode and when operated with a single longitudinal mode it emits a well-collimated beam with a bell-shaped intensity distribution as shown in Fig. 5.54. The index-guided laser structures fall into two categories: positive-index and negative-index waveguides. If the central region of the waveguide is made of a material with a higher refractive index than that of the materials forming the other region, the index-guided structure is said to be a positive-index guided structure. In this case, the light generated in the central active region is totally internally reflected at the dielectric boundaries and remains confined in the central region of higher refractive index over the restricted region in the lateral direction as shown in Fig. 5.54. The width of the high refractive index region and the difference between the refractive index values of the central region and the guiding regions are selected in such a way that only one lateral mode is supported. The positive index-guided laser gives well collimated beam with a perfect radiation pattern as shown in Fig. 5.54(a). In a negative-index guided structure, the central part of the active region is made to have a lower refractive index as compared

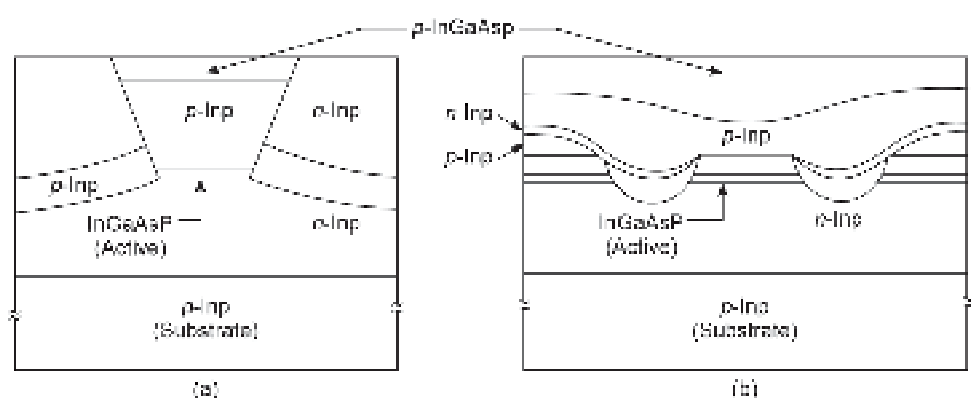

Fig. 5.55 Buried heterostructure laser diode (a) Etched-mesa buried heterostructure (EMBH) (b) Double channel planar buried heterostructure (DCPBH)

to that of the surrounding layers. In this case, light is confined in the central portion of the active region through ordinary reflection and a substantial portion of light is lost through transmission into the surrounding region. The loss appears in the form of side lobes in the radiation pattern as shown in Fig. 5.54(b).

Each of the above types of index-guided laser diode can be fabricated with the help of four fundamental structures, e.g. a buried heterostructure, a selectively diffused structure, a varying thickness structure, and a bent-layer structure (Botez, 1985). The buried heterostructure can be of two types, e.g. etched-mesa buried heterostructure (EMBH) and double channel planar buried heterostructure (DCPBH) (Dutta *et al* 1993). The cross-sectional view of the two forms of buried heterostructure laser diodes is shown in Fig. 5.55. Fabrication of buried heterostructure is technologically complex. In a EMBH structure a narrow-mesa stripe is first etched in double heterostructure material. The mesa is then embedded in a high resistivity *n*-type material (shown as *n*-InP) with a low refractive index. The high refractive index mesa region surrounded by the low index region provides strong optical confinement.

The fabrication of other forms of index-guided structures is less complex as compared to buried heterostructures (BH). In the selectively diffused structure, a Gr-II dopant such as Zn for AlGaAs laser is diffused into the active region immediately beneath the metallic contact stripe. The dopant causes a change in the refractive index in the central region of the channel creating a waveguide structure in the lateral direction. This is illustrated in the cross-sectional view of the structure shown in Fig. 5.56(a). In a varying thickness structure, a channel is first etched into the substrate. Crystal layers are subsequently regrown on the etched substrate in the channel by the epitaxial method to fill the depressions and create a variation in the thickness between the central active region and the confining regions. The varying thickness structure is shown in Fig. 5.56(b). In a bent layer structure, a mesa is etched onto the substrate. Semiconductor crystal layers are regrown on the substrate by the epitaxial method to complete the mesa structure in the bent form shown in Fig. 5.56(c). In this case, the light generated in the active region travels through the mesa region along the length of the cavity surrounded by the low-index regions on both sides of the mesa.

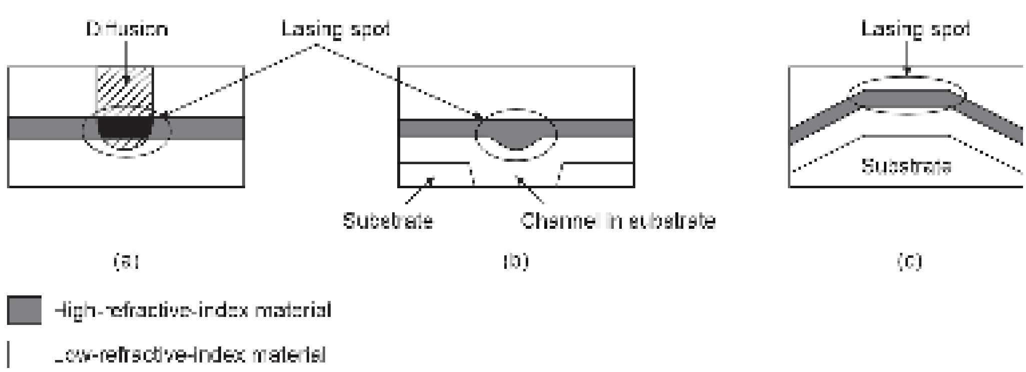

Fig. 5.56 Illustrating various index-guided laser structures (a) Selectively diffused structure (b) Varying thickness structure (c) Bent structure (after Botez, 1985)

Gain-guided structures were proposed before index-guided structures. The fabrication of gain-guided structures is less complex as compared to index-guided structures. However, gain-guided lasers generally suffer from higher threshold currents and inferior radiation patterns of the emitted beam as compared to their index-guided structures. Despite the complex fabrication steps needed for making index-guided lasers, these devices are preferred over gain-guided structures because of their superior performance in terms of large modulation bandwidth, excellent radiation pattern of the emitted beam, stable fundamental mode operation, and low threshold current (Dutta *et al* 1993).

5.8.9 DFB and DBR Laser Diodes

From the foregoing discussion, it is understood that the Fabry-Perot laser diodes are not wavelength selective and multiple modes are created in the cavity. Nevertheless, the FP laser diode can be operated as a single-mode source by reducing the size of the cavity appropriately. This is often inconvenient from the viewpoint of handling such small devices. Alternatively, a frequency-selective device may be incorporated into the laser diode to eliminate other frequencies. A grating structure can be incorporated into the laser waveguide to provide feedback that is a corrugated layer that offers periodic variation in the refractive index and acts as a passive waveguide to provide optical feedback. When the periodic grating is incorporated in the pumped region (active region where population inversion is created), the laser is termed a distributed feedback (DFB) laser. On the other hand, when the grating is incorporated in the passive region (outside the pumped region) is termed a distributed Bragg reflector (DBR) laser. The basic structures are illustrated in Fig. 5.43. Unlike the Fabry-Pérot laser diode both DFB and DBR laser diodes oscillate in a single longitudinal mode even at a very high-speed modulation (Ogasawara, 2007).

In a distributed feedback laser, the gratings or Bragg reflectors are used to replace one or both the mirrors of the conventional FP resonator. The period of the grating is chosen approximately as half of the average wavelength. This leads to constructive interference between the reflected beams. The grating is constructed by etching the substrate with a periodic pattern and then refilling the etched-out portions with suitable material of different refractive indices in the next cycle of the growth process. DBR laser can be tuned over a range of frequencies as well. This is possible because the frequency selectivity property of the grating depends on the refractive index which can be changed by changing the injected current density by the bias current. Distributed feedback (DFB) laser also uses a built-in grating structure but outside the gain region. In this structure, the reflection from the end facets is suppressed by antireflection coating. It may be pointed out here that DFB laser diodes are easy to fabricate as compared to DBR laser diodes. These devices have a low threshold current. DBR diodes on the other hand are difficult to fabricate but offer wide tunability in terms of frequency. There is a third variety that makes use of Bragg reflectors both in the active and passive regions. These structures are known as distributed reflector (DR) lasers. These structures provide improved lasing characteristics as compared to both DFB and DBR laser structures including high efficiency and high output power. The cross-sectional views of laser structures using built-in frequency selective gratings are shown in Fig. 5.57.

5.8.10 Quantum-Well (QW) Laser

Quantum-well structures have been extensively used for making laser diodes since 1990. The quantum well structure resembles the basic double-heterostructure except for the fact that the thickness of the active region in the former is made very thin (~10 nm) as compared to that of conventional DH laser diodes. The typical thickness (~1 μm) of conventional DH laser diodes is good enough for carrier and optical confinement. Other electronic and optical properties of the confined active layer remain the same as the bulk properties of the material. As a result, the key parameters of the laser

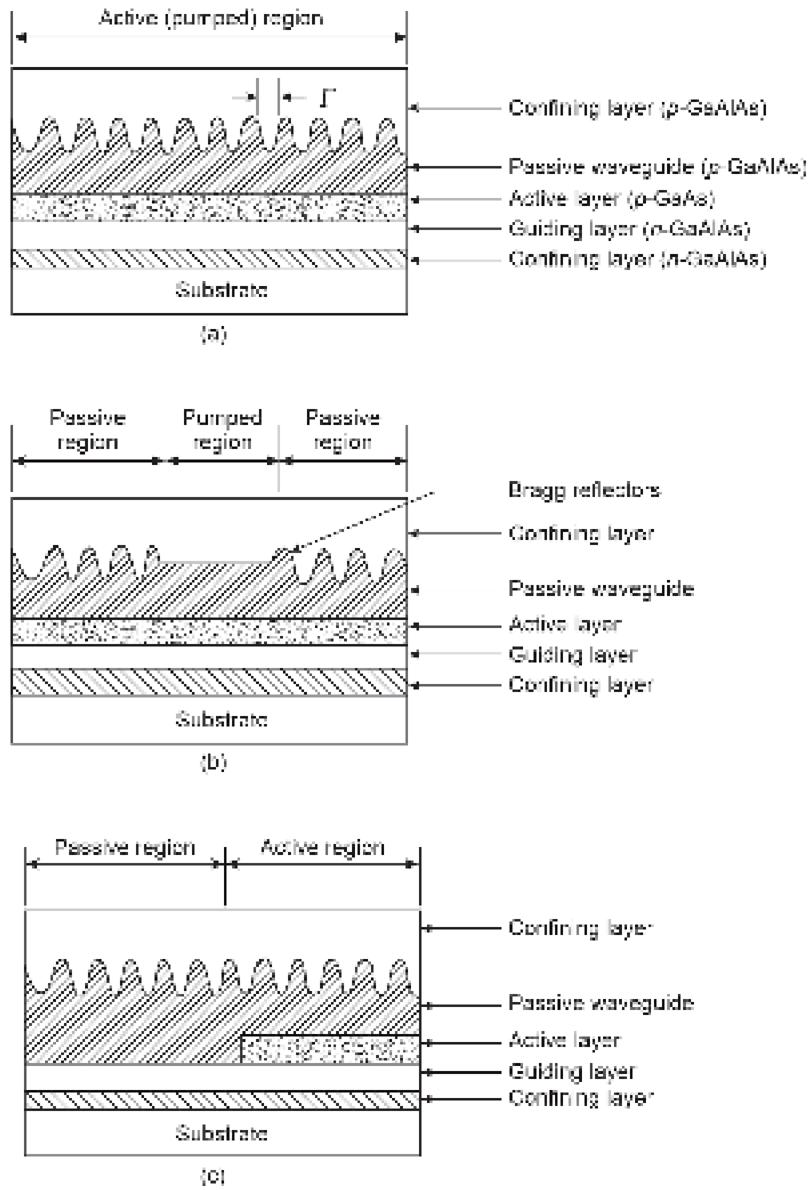

Fig. 5.57 Cross-sectional schematic of three types of laser structures using frequency selective gratings (a) Distributed feedback (DFB) laser (b) Distributed Bragg reflector (DBR) laser (c) Distributed reflector (DR) laser

diode such as threshold current, modulation capability, emission line width, etc. cannot be controlled significantly. On the other hand, if the thickness of the active region is reduced to 10 nm or so the quantum mechanical effects come into play and there is a drastic change in the property of the heterostructure. For example, under this condition, the motion of the carriers in the direction normal to the active region is restricted. This, in turn, causes quantization of kinetic energy into discrete energy levels for carriers moving in this direction. This effect is similar to one observed in the case of a one-dimensional potential well problem. The name quantum well structure originates from this fact. The net effect of the restriction imposed on the motion of the carriers in the third direction is that there is a drastic change in the electronic and optical properties of the quantum well structure as compared to bulk properties. This enables one to tailor the key performance parameters of a laser diode. Both single and multiple quantum well structures are used in making semiconductor laser diodes. The energy band diagrams of a single quantum well (SQW) and a multiple quantum well structure are shown in Fig. 5.58.

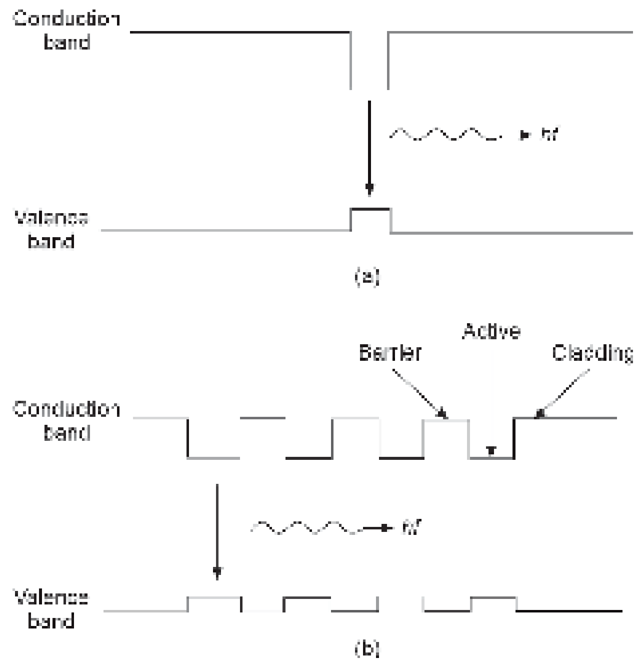

Fig. 5.58 Energy band diagram of quantum well structures (a) Single quantum well (SQW) (b) Multiple quantum well (MQW)

Both single and multi-quantum well structures have been used for the fabrication of laser diodes (Dutta, 1987; Tsang, 1987; Zory, 1993). An SQW laser uses a single active region while an MQW laser uses multiple active regions built in the form of quantum wells with barrier layers in between the successive active regions as shown in Fig. 5.58. Multi-quantum well (MQW) laser diodes offer several advantages over conventional DH laser diodes. These include lower threshold current, increased modulation bandwidth, narrower emission line width, and lower frequency chirp (Bowers *et al* 1988). One of the major drawbacks of the QW structure is that the gain volume of a QW is very small because of the small width of the active region. As a result, the interaction between the optical field and the carriers is not strong enough as in the case of conventional laser diodes. Several modified quantum well laser structures use separate waveguide structures surrounding the QW to confine the photons to near the quantum wells. Several such structures are investigated and reported in the literature. However, the discussion of advanced structures are beyond the scope of the book.

5.8.11 Vertical Cavity Surface Emitting Laser (VCSEL) Diode

Conventional laser diodes are essentially edge emitter devices in which the light is emitted in the direction parallel to the plane of the *p-n* junction. A vertical cavity surface emitting laser (VCSEL) or simply a vertical cavity laser (VCL) is a laser diode with improved efficiency and modulation bandwidth that emits light in the direction perpendicular to the plane of the *p-n* junction[10]. Unlike conventional laser diodes, VCSEL emits a nearly circular beam and operates in the single mode because of the small size of the cavity (Dutta *et al* 1993; Lee *et al* 1983; Margalit *et al* 1997; Sale, 1995; Yu, 2003).

The design of VCSEL varies widely depending on the basic structure of the active region (QW or non-QW). However, a few things are common in all types of VCSEL. The cavity length of VCSEL is very short typically a few wavelengths of the emitted light long. As a result a photon in the cavity hardly gets a chance to cause stimulated emission in a single pass. It is therefore necessary for the photons to have multiple passes before they exit through the front-end mirror. This is ensured by making use of highly reflecting mirrors. The mirrors in conventional Fabry-Perot resonator-based laser diode have uncoated facets offering reflectivity of nearly 30%. On the other hand, the mirrors used in VCSEL need reflectivity exceeding 99% to produce significant emissions in the small cavity. VCSEL generally makes use of distributed Bragg reflector (DBR) layers formed by depositing alternate layers of semiconductor and dielectric materials with different refractive indices. The multiple DBR layers used in VCSEL also carry current flowing through the device. Because of the large resistance of these layers, significant heat is generated by the device. VCSEL therefore requires proper heat sinking arrangement. A typical VCSEL is schematically shown in Fig. 5.59 for a mesa-etched and an ion-implanted structure. The mesa etched structure shown in Fig. 5.59(a) has a typical diameter of 1–2 μm and therefore a large number of VCSEL′s can be fabricated on a single substrate. This enables the integration of multiple lasers on a single chip so that 1D or 2D arrays can be created for application in wavelength-division-multiplexing (WDM) systems. In the ion-implanted VCSEL structure the active region width is restricted by creating a high resistance region surrounding the central active region with the help of ion implantation. This effectively reduces the current spreading.

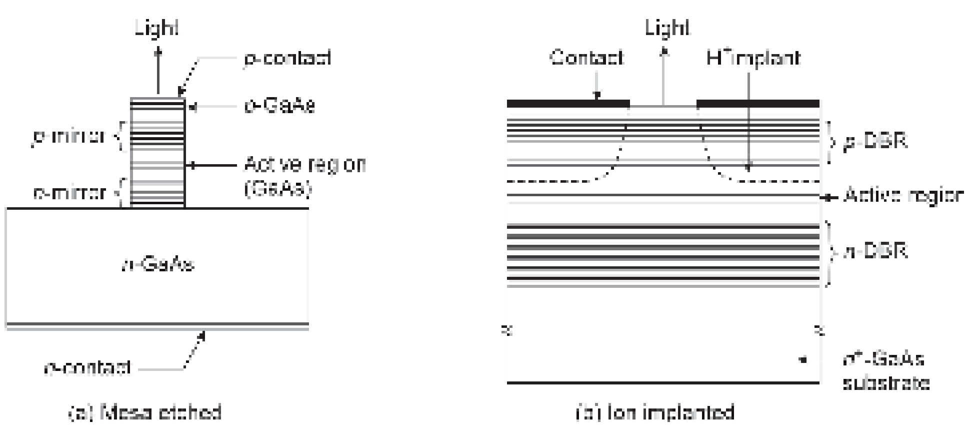

Fig. 5.59 Schematic of two different VCSEL structures (a) Mesa etched structure (b) Ion-implanted structure (after Dutta *et al* 1993)

5.8.12 Quantum-Wire (QWR) and Quantum-Dot (QD) Lasers

We have seen earlier that the charge carriers are confined in one-dimension (1D) in a quantum well (QW) semiconductor laser when the active layer is extremely thin (<20 nm). The QW lasers exhibit improved performance over conventional laser diode in terms of

[10] This feature may be compared with that of a surface emitting LED.

narrower optical gain spectra, very low threshold current, larger modulation bandwidth and reduced temperature sensitivity (Holonyak *et al* 1980; Arakawa *et al* 1986; Chen *et al* 1987; Choi *et al* 1990; Kapon *et al* 1990; Geels *et al* 1990). Further improvement in the performance of a semiconductor laser diode is possible by reducing the thickness of the active region to the de Broglie wavelength regime. This leads to 1D (wire) in quantum wires (QWR) and 0 D (island) in quantum dot (QD) or quantum box (QB) formation in semiconductor laser structures (Kapon, 1999). The QWR and QD lasers exhibit far superior characteristics as compared to their QW laser counterpart. These low-dimensional laser diodes are expected to find applications in the integration of a large number of lasers on a single chip and its integration with low power electronic circuits. Fabrication of QWR and QD semiconductor heterostructures continues to be a challenging task involving nanofabrication and crystal growth technologies. The schematic structures of quantum-wire and quantum-dot laser diodes are shown in Fig. 5.60.

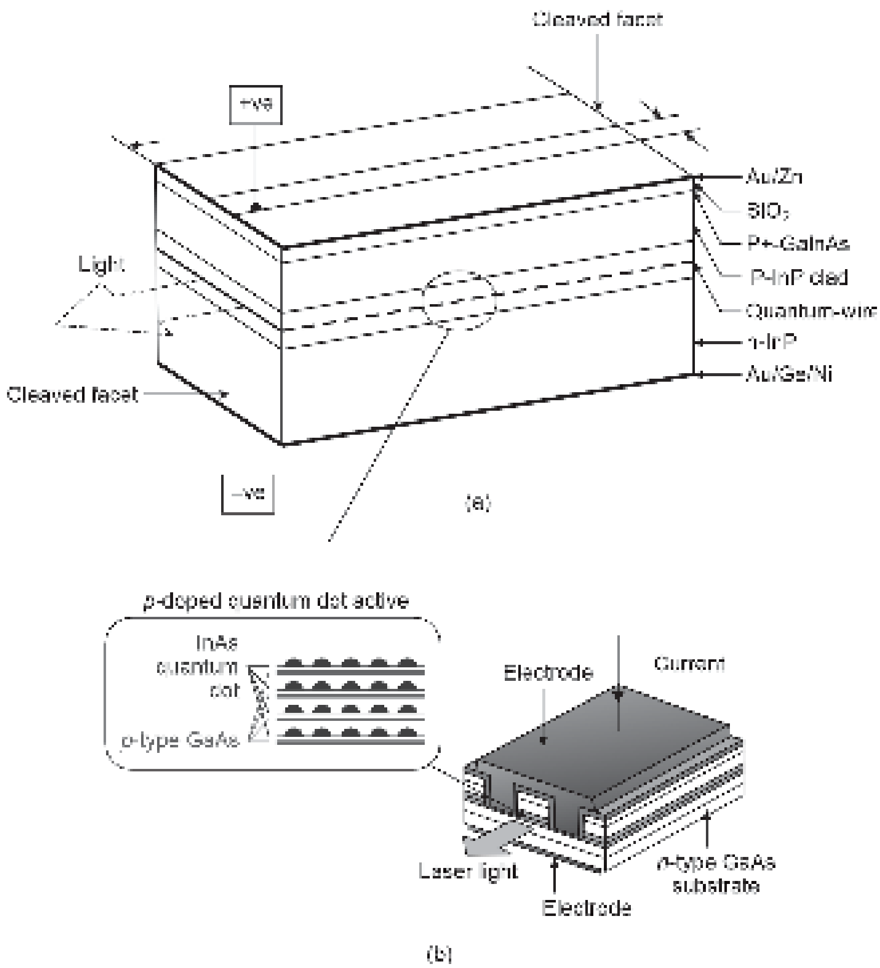

Fig. 5.60 (a) Schematic of a QWR laser (b) Schematic of a QD laser

5.9 TRANSIENT RESPONSE AND MODULATION CAPABILITY OF LASER DIODE

One of the key parameters of a laser diode source for application in optical transmitters is the maximum frequency of the modulating current signal by which the intensity of the source can be modulated. It is therefore important to study the transient or temporal behavior of the laser diode. It may be pointed out here that light output from an optical source can be either modulated by direct or indirect methods. In the direct method, the time-varying bias current is used to modulate the injected carriers and thereby control the emitted output power from the source in an analogous manner. In the indirect or external method, the light is allowed to pass through a medium whose optical properties can be controlled by the modulating signal so that some characteristics of the output light signal get modulated. For high-speed applications (>2 Gbps) indirect modulation is preferred to avoid various undesirable non-linear effects. Various indirect techniques will be discussed afterwards in Chapter 10.

To study the temporal behavior of the laser diode we need to formulate the rate equation for the carriers and photons. In laser operation, the bias current causes injection of carriers, and photons are caused by stimulated emission and repeated interaction of the emitted photons with the carriers. Let us assume that the background carrier concentration is very low before injection. This means that the carriers are supplied by the bias current and subsequently removed by both spontaneous and stimulated recombination.

The rate of change of electron population, n with time can be expressed as (Bhattacharya, 2002)

$$\frac{dn}{dt} = \frac{J}{qd} - \frac{n}{\tau} - R_{st} \tag{5.145}$$

where J is the bias current, d is the thickness of the active region, q is the electronic charge, τ is the spontaneous lifetime of the carrier given by

$$\frac{1}{\tau} = \frac{1}{\tau_r} + \frac{1}{\tau_{nr}} \tag{5.146}$$

Here, τ_r and τ_{nr} are the radiative and non-radiative lifetimes of the carriers for spontaneous emission. This means that the second term on the right-hand side accounts for the rate of removal of injected carriers by spontaneous emission which includes both radiative as well as non-radiative recombination. The last term in Eq. (5.145) corresponds to a rate of loss of injected carriers by stimulated emission. If N_p is the number of photons per unit energy interval, then we may write

$$R_{st} = h\nu \frac{dN_p}{dt} \tag{5.147}$$

where h is Planck's constant and ν is the emission frequency.

From Eq. (5.145) it is clear that the rate equation is governed by the overall lifetime (radiative and non-radiative) of the carriers. Further, the stimulated emission rate depends on the spontaneous radiative lifetime of the carriers as well as the stimulated carrier lifetime and photon lifetime (Bhattacharya, 2002).

The radiative lifetime of carriers for spontaneous emission can be estimated using Eq. (5.51). For a doping concentration of $N_D = 10^{24}$ m^{-3}, at room temperature, we may write $n_0 \approx N_D = 10^{24}$ m^{-3}. For GaAs and related materials, we may assume $B_r = 8 \times 10^{-16}$ m^{-3} s^{-1}. The lifetime for radiative recombination can be estimated as

$$\tau_r = \frac{1}{8 \times 10^{-16} \times 10^{24}} = 1.25 \text{ ns}$$

For light sources, non-radiative lifetime is usually very large.

The stimulated carrier lifetime depends on the optical density and is generally of the order of 10 ps. The photon lifetime τ_{ph} is the average lifetime for which the photons exist in the cavity before getting lost by absorption in the cavity or emission through any one of the facets. The photon lifetime in a Fabry-Perot cavity can be estimated from (Kressel *et al* 1977; Sze *et al* 2007)

$$\tau_{ph} = \frac{n}{c\left[\alpha + \frac{1}{2L}\ln\left(\frac{1}{R_1 R_2}\right)\right]} \tag{5.148}$$

Assuming the refractive index of the lasing material to be $n = 3.6$, the length of the cavity to be $L = 250$ μm and the reflectivity of the facets to be $R_1 = R_2 = 0.32$ and an average loss of the cavity to be $\overline{\alpha} = 10^3$ m^{-1}. The photon lifetime can be estimated based on the above values as

$$\tau_{ph} = \frac{3.6}{3\times 10^8}\left[\frac{1}{10^3 + \frac{1}{2\times 250\times 10^{-6}}\ln\left(\frac{1}{0.32\times 0.32}\right)}\right] = 2.16\,\text{ps}$$

The photon lifetime value sets the upper limit to the direct modulation capability of the laser diode.

As soon as a laser diode is switched on there are several transient effects that cause carrier and photon density to fluctuate. Under steady-state condition, the carrier density and photon density attain a state of dynamic equilibrium such that

$$\frac{dn}{dt} = 0 \quad \text{and} \quad \frac{dN_p}{dt} = 0$$

Using Eqs (5.145) and (5.147) we may write

$$n = \frac{J\tau}{qd} \tag{5.149}$$

5.9.1 Transient Response

As the photon lifetime is very small as compared to the carrier lifetime a laser diode can be easily pulse modulated. The upper limit to such operation can be estimated as $f_{\max} = 1/2\pi\tau_{ph}$. Assuming the photon lifetime to be approximately 2 ps, the maximum modulating frequency turns out to be approximately GHz. However, the maximum bandwidth achievable in the case of direct modulation is even less. In pulse operation, if a laser diode is completely switched off, initially there is no photon inside the cavity. Under this condition, the spontaneous carrier lifetime decides the modulation rate. The transient response of the laser diode can be studied by considering a turn-on bias current $J = J_p$ applied to an unbiased laser diode at $t = 0$ as shown in Fig. 5.61. In the absence of stimulated emission,

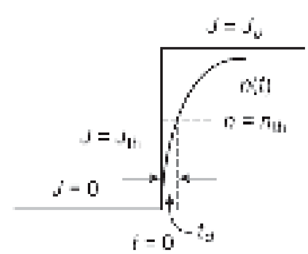

Fig. 5.61 Showing the transient response of the laser diode in terms of carrier density illustrating the delay in attaining the threshold value for lasing

the carrier rate equation in this case can be written as

$$\frac{dn}{dt} = \frac{J_p}{qd} - \frac{n}{\tau} \tag{5.150}$$

Taking Laplace transform on both sides of Eq. (5.150), we get

$$sn(s) - n(0-) = \frac{J_p}{qd}\frac{1}{s} - \frac{n(s)}{\tau} \tag{5.151}$$

As the diode is unbiased before time $t = 0$, therefore $n(0–) = 0$. Thus Eq. (5.151) can be rearranged as

$$n(s) = \frac{J_p \tau}{qd}\left[\frac{1}{s} - \frac{1}{s+\tau}\right] \tag{5.152}$$

Taking inverse Laplace transform on both sides of Eq. (1.152), we may write

$$n(t) = \frac{J_p \tau}{qd}\left[1 - \exp\left(-\frac{t}{\tau}\right)\right] \tag{5.153}$$

It can be seen that after application of the current signal, the threshold gain is not achieved. There is a time gap t_d between the application of the current pulse and reaching the threshold value so that the gain of the cavity becomes just equal to the total loss in the cavity and the laser diode just starts lasing. If n_{th} corresponds to the concentration of carriers in the cavity when the current density reaches the threshold value, J_{th} then at time $t = t_d$ then using Eq. (5.153), we may write

$$n(t = t_d) = \frac{J_p \tau}{qd}\left[1 - \exp\left(-\frac{t_d}{\tau}\right)\right] \tag{5.154}$$

Further, we note that at $t = t_d$ the carrier concentration attains threshold value n_{th} given by

$$n_{\text{th}} = \frac{J_{\text{th}} \tau}{qd} \tag{5.155}$$

Using Eqs (5.154) and (5.155) the delay time between the application of the current pulse and attaining the threshold value for lasing can be obtained as

$$t_d = \tau \ln\left(\frac{J_p}{J_p - J_{\text{th}}}\right) \tag{5.156}$$

In terms of current, $I(= J \times A)$, A being the emission active area of the laser diode, Eq. (5.156) can also be expressed as

$$t_d = \tau \ln\left(\frac{I_p}{I_p - I_{\text{th}}}\right) \tag{5.157}$$

It can be seen from Eq. (5.157) that the delay time depends on the spontaneous lifetime of the carriers. Further, every time the laser needs to attain the threshold for lasing when the diode starts from complete turn-off condition. This period can be reduced by pre-biasing the laser diode with a current I_B close to the value I_{th} so that the diode is not completely turned off in the absence of the pulse and when the pulse arrives the laser diode can reach the threshold faster. The delay in the case of a pre-biased laser can be estimated in a similar way as

$$t_d = \tau \ln\left(\frac{I_p}{I_p + I_B - I_{th}}\right) \tag{5.158}$$

Further, when the laser diode is pre-biased with a current close to the threshold the spontaneous recombination time is also reduced because of the injection of carriers even in the absence of the pulse signal. As a result, overall delay is reduced during pulse modulation of a pre-biased laser diode. This in turn increases the direct modulation capability of the laser source (Kurbator *et al* 1985).

5.9.2 Laser Modulation Bandwidth

In a laser diode, the photon population builds up very fast after the injection of carriers. As the photons build up the carrier density decreases until it falls below the steady-state carrier density. At this point the rate of change of photon density becomes negative. The photons start building up again when the carrier injection crosses the steady-state value. Damped oscillatory variations in carrier and photon density are observed when the bias current is modulated with an ac signal. The system behaves as a tuned circuit and resonance occurs at some characteristic frequency. These relaxation oscillations of the laser field set the upper limit of modulating frequency for the case of directly modulated laser diodes. The relaxation oscillation frequency depends on both the spontaneous lifetime and the photon lifetime.

The relaxation oscillation frequency can be obtained as (Bhattacharya, 2007; Sze *et al* 2007)

$$f_r = \frac{1}{2\pi}\left[\frac{1}{\tau \tau_{ph}}\left(\frac{J}{J_{th}} - 1\right)\right]^{\frac{1}{2}} \tag{5.159}$$

Under direct modulation, a laser diode cannot be modulated at a rate higher than that given by f_r in Eq. (5.159) and the rate of change of internal photon density in the cavity for a small signal change in the value of the bias current density is given by (Sze *et al* 2007)

$$\frac{\Delta N_{ph}}{\Delta J} = \frac{\tau}{qd}\left[\left(1 - \frac{f}{f_r}\right)^2 + \left(2\pi f \tau_{ph}\right)^2\right]^{-\frac{1}{2}} \tag{5.160}$$

Equation (5.160) suggests the frequency response of the laser source. It can be easily seen from the equation that the response reaches a peak value at $f = f_r$. The response is almost flat in the low frequency range and drops rapidly with frequency ($\sim f^{-2}$) above f_r, that is at a rate of 40 dB per decade. It can be seen that the resonant frequency can be increased by increasing the DC bias current of the laser diode and thereby increasing the bandwidth of the laser source.

Figure 5.62 shows the variation of the normalized modulated laser output of a typical DH laser diode (Dutta *et al* 1989) with the modulation frequency when the laser source is directly modulated with sinusoidal current superposed on DC bias current.

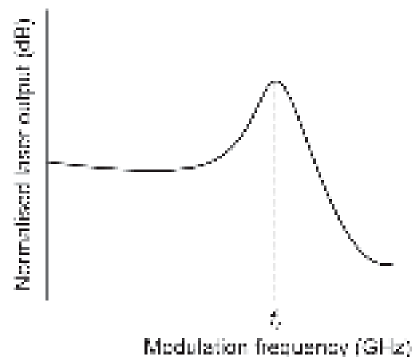

Fig. 5.62 Normalised small signal frequency response of a DH laser diode for direct modulation

Example 5.25 Estimate the upper limit of the modulation frequency of an injection laser diode driven by an injection current twice as high as the threshold current. Assume that the operation of the laser diode is limited by the relaxation oscillation frequency. The values of spontaneous recombination lifetime and photon lifetime are 1 ns and 2 ps respectively.

Solution The maximum rate at which the laser diode can be modulated can be obtained using Eq. (5.159) as

$$f_{\max} \approx f_r = \frac{1}{2 \times 3.14}\left[\frac{1}{10^{-9} \times 2 \times 10^{-12}}\right]^{\frac{1}{2}} = 3.5\text{ GHz}$$

This means that for direct modulation an injection laser diode has a modulation capability of a few gigahertz only.

5.10 SOME UNDESIRABLE EFFECTS IN LASER

Injection laser diodes exhibit some special features that need to be addressed for deploying these devices in high-speed optical communication systems. The exact behavior of these devices may vary significantly depending on the structure as well as the material used for making the laser diodes.

5.10.1 Thermal Effects

One of the major problems with the operation of a laser diode is that the threshold current depends on the temperature. It is found that the threshold current of the laser diode increases with an increase in operating temperature as depicted in Fig. 5.63. The dependence of threshold current on temperature has been reported (Kressel, 1977; Agrawal, 1993). The temperature dependence of the threshold current can be expressed as

$$I_{\text{th}}(T) \propto \exp\left(\frac{T}{T_0}\right) \tag{5.161}$$

where T is the absolute temperature of operation and T_0 is the threshold temperature coefficient. The parameter T_0 depends on the material as well as the structure of the laser diode. For example, InGaAsP-based laser diodes have threshold temperature T_0 in the range of 40–75 K whereas AlGaAs-based devices have threshold temperatures in the range of 120–190 K (Botez, 1980). This means that InGaAsP-based laser diodes are more adversely affected than their AlGaAs counterparts. This is attributed to the stronger temperature dependence of the physical parameters of InGaAsP on temperature (Casey, 1984).

The temperature dependences of the output power *versus* drive current characteristics of different gain-guided injection laser diodes are shown in Fig. 5.63 (Kirby, 1981). Both the diodes use stripe geometry with a stripe width of 20 μm. It can be easily seen that InGaAsP-based devices have a stronger dependence on temperature than AlGaAs-based lasers. For example, the threshold current increases almost 1.4 times when the temperature changes from 20– 60 °C in the case of AlGaAs laser diodes. On the other hand, the change in the threshold current value is almost twice when the temperature is varied in the same range for the case of InGaAsP devices.

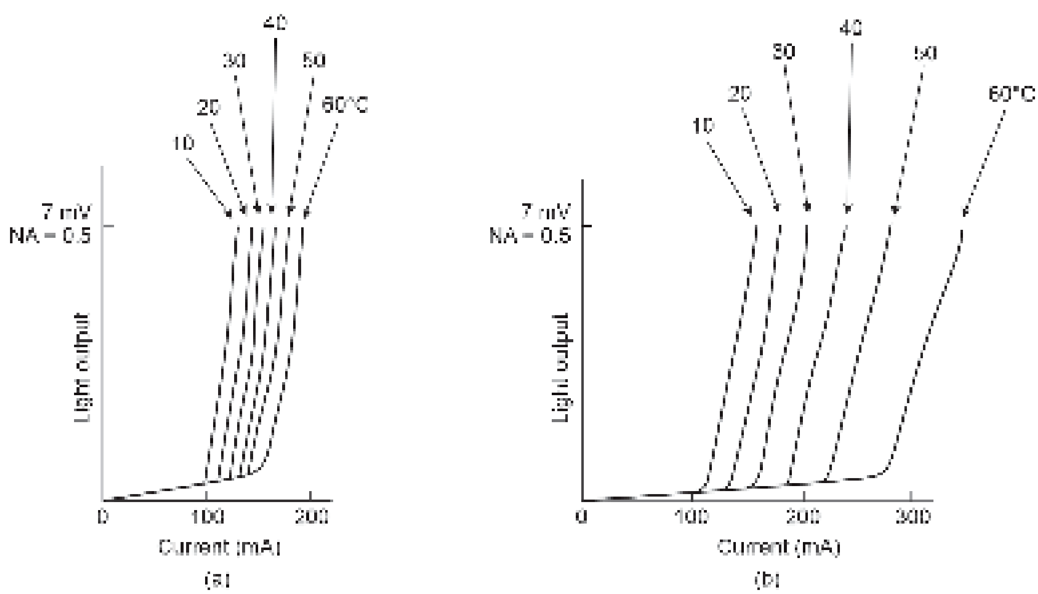

Fig. 5.63 Temperature dependence of the output power *vs* drive current characteristics of laser diodes (a) AlGaAs laser diodes (b) InGaAsP laser diodes (after Kirby, 1981)

Example 5.26 The threshold current of an AlGaAs laser diode at 20 ºC is 100 mA. The threshold temperature of the device is $T_0 = 180\,\text{K}$. Calculate the percentage change in the threshold current when the temperature of the device is increased to 60ºC.

Solution The ratio of the threshold current of the laser diode at 60 ºC to that at 20 ºC can be obtained as

$$\frac{J_{\text{th}}(60\,°\text{C})}{J_{\text{th}}(20\,°\text{C})} = \frac{\exp\left(\dfrac{333}{180}\right)}{\exp\left(\dfrac{293}{180}\right)} = \frac{6.36}{5.09} = 1.249$$

Given that J_{th} (20 ºC) = 100 mA.

Therefore, J_{th} (60 ºC) = 1.24 × 100 mA = 124.9 mA.

The percentage change in the threshold current due to variation in temperature from 20 ºC to 60 ºC is 24.9%.

Temperature control is very important for successful operation of laser diodes in optical communication systems. For general purpose, use the temperature control can be managed by using proper heat sinking arrangement. However, for high-speed long-haul optical communication systems more sophisticated laser modules are used. These modules contain an in-built thermoelectric or thermostatic cooler in a single package.

5.10.2 Aging Effects

The output light from a laser source also decreases with aging. It is understood that in FP laser diodes the mirrors get damaged with aging because of the repeated interaction of photons on the facets. As a result, the threshold current of a laser diode is also found to increase with aging. This undesirable effect causes a reduction of output power from the laser diode for the same bias current.

Both temperature changes and aging can affect the output power of a laser diode significantly. To maintain a steady output power to combat the effect of temperature variation or aging effect it is necessary to adjust the DC current suitably. Various techniques used for controlling the bias current to maintain a constant output power include the optical feedback technique, pre-distortion technique, and temperature matching transistor scheme.

The straightforward approach to counter the output power variation is to use a PIN detector inside the laser package near the rear facet of the cavity. The detector picks up a small optical signal that is proportionate with the generated light from the transmittance of the rear facet[11]. The PIN detector output is an electrical signal that is subsequently used for controlling the bias current to maintain a constant output power from the laser source (Ettenberg *et al* 1979; Chen, 1980).

5.10.3 Frequency Chirp

When a laser diode is switched-on there is an abrupt change in the carrier flux density in the cavity caused by injection of carriers. The carrier flux density changes significantly following the lasing. The change in the density of charge carriers induces a variation of the refractive index of the lasing medium. In addition, the temperature in the cavity also increases during the lasing process. This temperature variation also contributes to changing the refractive index of the material in the active region. The changes in the refractive index of the cavity due to carrier fluctuations and subsequent variations in temperature result in a phase shift of the optical field apart from producing relaxation oscillations discussed earlier. This phase change, in turn, gives rise to a change in resonant frequency of both the FP and DFB laser diodes. The shift in resonant frequency causes the wavelength of operation to shift. In the case of semiconductor laser diodes, the downward resonant frequency chirp causes the wavelength to shift to a longer wavelength than it was previously set just before switching on. Frequency chirp is not a matter of great concern for a short-distance single-channel transmission. However, in long-distance applications and particularly in WDM-based systems frequency chirp is indeed a very serious problem. This is because chirping causes broadening of the spectral width of the source and as a result the intramodal dispersion of the fiber increases (Cartledge *et al* 1989). The increase in dispersion value restricts the maximum rate of transmission of data (bit rate) even for single-mode fibers used in long-distance communication systems. The wavelength shift due to frequency chirp for direct modulation at the rate of a few Gbps has been theoretically predicted for an InGaAsP laser diode is approximately 0.05 nm which corresponds to a frequency shift of 6.4 GHz (Henry *et al* 1988). External modulators are preferred for transmission rates above 1 Gbps to avoid the effect of frequency chirp on the system performance.

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[11] The rear facet is not perfectly reflecting.

Several techniques have been proposed to reduce the frequency chirp in laser diodes including the use of modified laser structures such as quantum wells and other advanced structures. The simplest approach to combat frequency chirp is to bias the laser diode much above the threshold value so that even under the worst conditions the modulation current cannot drive it below the threshold value. This would limit the variation in the carrier density which causes the refractive index variation in the active region. This method is not, however, very attractive because the laser diode emits power even during 0 pulses causing the system to exhibit a non-zero extinction ratio (Senior, 2006)

5.10.4 Noise

An important characteristic of an ILD that affects the performance of a laser-based optical fiber communication system is the noise generated by the laser diode because of several factors. The major components of noise in a laser diode include

1. Phase or frequency noise
2. Reflection noise
3. Mode partition noise

The phase (or frequency) noise arises from the difference in phase between various randomly emitted (both spontaneous and stimulated) photons. The phase variation is a natural consequence of the very mechanism of laser operation. It is, therefore, an inherent or intrinsic property of all types of laser sources. As a result, this component cannot be eliminated. The random change in the phase of the emitted photons causes the phase of the emitted electromagnetic field to change. The spectral density of this noise component has a $1/f$ or $1/f^2$ dependence on frequency up to almost 1 MHz (Saltz, 1986). Beyond 1 MHz the noise spectral density is uniform and constant and is understood to be associated with quantum fluctuations called quantum noise. The spectral density of the noise is depicted in Fig. 5.64.

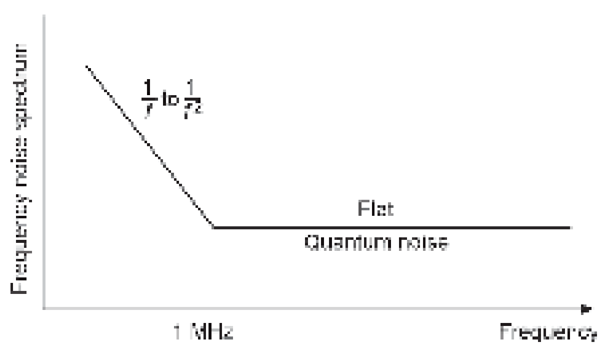

Fig. 5.64 Phase or frequency noise spectrum of an injection laser diode (after Saltz, 1986)

Reflection noise is caused by external reflection of the emitted optical signal. Unwanted reflections from couplers, joints, and splices usually return a portion of the emitted optical signal back into the laser cavity. The reflected wave has the same wavelength as that of the emitted wave. As a result, the reflected wave also gets amplified by the cavity and causes undesirable fluctuation in the light output from the cavity. The reflection may be from nearby interfaces (such as from the source-fiber coupler) or even from a distant interface within the optical link (such as from splices or other joints). The reflections from nearby interfaces can be minimized by using antireflection coatings while the effect of reflection from distant interfaces can be eliminated by making use of an isolator immediately following the laser diode. This effect is not very significant in multimode lasers because the reflected wave is distributed between various modes of the optical fiber and therefore they are weakly coupled back in the cavity (Henry, 1986; Henry *et al* 1986).

Mode partition noise is associated with multimode lasers. It arises from the intensity fluctuations of the longitudinal modes when the modes are not properly stabilized (Ogawa, 1982). The output from a multimode laser source comes from different modes which contribute to the total power. It may so happen that the total output power of the laser remains the same while the contribution from the different modes may vary due to random fluctuation of the relative intensity of various longitudinal modes in the laser output spectrum. This is illustrated in Fig. 5.65. The variation in the relative intensity amongst the various modes is manifested in the receiver output in the form of distortion. This is because different modes undergo different attenuation and delay while propagating through the fiber. Mode partition noise increases the bit error rate in optical communication systems.

Fig. 5.65 Illustrating relative intensity fluctuation among various longitudinal modes in a multimode injection laser diode

5.10.5 Relative Intensity Noise (RIN)

Relative intensity noise (RIN) is attributed to random fluctuation in amplitude or intensity of the output from an injection laser diode. It leads to optical intensity noise. The origin of this noise may be temperature variations or the random nature of spontaneous emission. Some of the photons generated through spontaneous emission may resonate with the cavity and get amplified. This may cause a fluctuation in the laser output.

The noise source created by the random intensity fluctuation is called relative intensity noise (RIN) and is an important parameter of an injection laser diode. RIN is defined as the ratio of the mean square value of the power fluctuation, $\left\langle \delta P_e^2 \right\rangle$ to the square of the mean optical power, $\left\langle P_e \right\rangle^2$. That is,

$$\text{RIN} = \frac{\left\langle \delta P_e^2 \right\rangle}{\left\langle P_e \right\rangle^2} \tag{5.162}$$

The symbol < > represents the mean value.

The mean square value of the power fluctuation can be obtained as

$$\left\langle \delta P_e^2 \right\rangle = \int_0^\infty S_{\text{RIN}}(f)df \tag{5.163}$$

where $S_{\text{RIN}}(f)$ is the power spectral density of the relative intensity noise.

Assuming the bandwidth to be 1 Hz, the RIN can be expressed using Eqs (5.162) and (5.163) as

$$\text{RIN} = \frac{S_{\text{RIN}}(f)B(=1\text{Hz})}{\langle P_e \rangle^2} \tag{5.164}$$

The RIN of a single-mode laser usually ranges between 10^{-13} to 10^{-16} per unit bandwidth. RIN is often expressed in dB Hz^{-1}. The relative intensity noise decreases with an increase in the injection current level because of the reasons already discussed following the relation (Senior, 2002)

$$\text{RIN} \propto \left(\frac{I}{I_{\text{th}}} - 1 \right)^{-3} \tag{5.165}$$

In an IM/DD system, the light is directly detected by a photodetector. If an optical power $P_{op}(t)$ is incident on the photodetector then the photocurrent generated by the photodetector will be

$$i_p(t) = \frac{q\eta}{h\nu} P_{op}(t) \tag{5.166}$$

where q is the electronic charge, h is Planck's constant, ν is the frequency of the incident optical power, and η is the quantum efficiency of the photodetector.

A fluctuation in the incident optical power by $\delta P_{op}(t)$ will produce a fluctuating current, $\delta i_p(t)$ given by

$$\delta i_p(t) = \frac{q\eta}{h\nu} \delta P_{op}(t) \tag{5.167}$$

The mean square value of the current can be obtained as

$$\langle i^2(t) \rangle = \langle \delta i_p^2(t) \rangle = \frac{q^2\eta^2}{(h\nu)^2} \langle \delta P_{op}^2(t) \rangle \tag{5.168}$$

Consider that the fluctuation in the optical power at the detector input is caused by the RIN of the laser source, then replacing P_{op} by P_e in Eq. (5.167) we can find the mean square value of the noise current in the output of the photodetector arising from RIN as (Senior, 2002)

$$\langle i_{\text{RIN}}^2 \rangle = \frac{q^2\eta^2}{(h\nu)^2} \langle \delta P_e^2(t) \rangle = \frac{q^2\eta^2}{(h\nu)^2} (\text{RIN}) \langle P_e^2 \rangle B \tag{5.169}$$

5.10.6 Mode Hopping

Laser sources generally require a good heat sinking arrangement for providing stable output power. If the proper cooling arrangement is not provided then the temperature of the device junction can increase when the laser diode is biased above the threshold. This causes a lowering of the lasing energy and consequently, the lasing output of a single-mode laser can shift to a longer wavelength longitudinal mode. This phenomenon is called mode hopping. The switching from one dominant mode to another of a longer wavelength due to hopping is shown in Fig. 5.66. The transition from one mode to the other is quite random and does not occur as a continuous function of the drive current. Mode hopping occurs in the drive current range of 1–2 mA for injection laser diodes. This mode hopping affects the optical power output *versus* drive current characteristics of a single mode laser source and often gives rise to undesirable kinks in the characteristic as illustrated in Fig. 5.67.

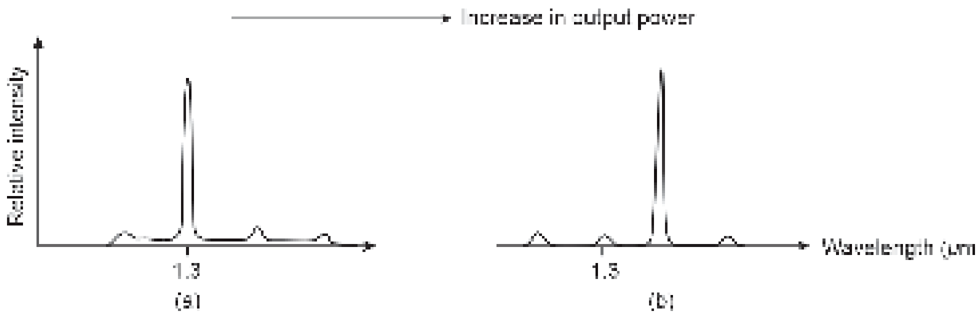

Fig. 5.66 Shifting of one mode to another at a longer wavelength in a single mode laser due to mode hopping

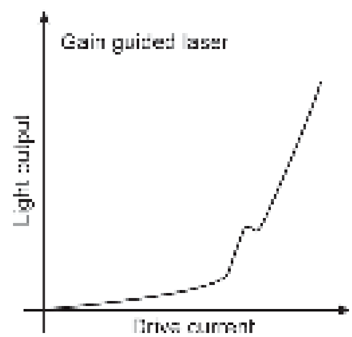

Fig. 5.67 Light output *vs* drive current characteristics showing the kink produced by mode hopping

5.10.7 Reliability Issues

One of the major concerns of injection laser diodes is the reliability. The failures of laser diodes are not very uncommon but the mechanisms of failure are not well understood (Dutta, 1987). The mechanism of failure depends on the type of structure (FP, DFB, or DBR). The failures of injection laser diodes are generally classified into two categories, e.g. catastrophic failure and gradual failure. Catastrophic degradation generally results from the mechanical damage of the mirror facets due to improper handling. In this case, the failure may be partial or total. On the other hand, gradual degradation in an injection laser diode is caused by defect formation in the active region or degradation of the current confining junctions. These factors cause an increase in threshold current with a reduction in quantum efficiency. In FP lasers the end faces get damaged with aging because of repeated interaction of the photons with the facets.

Summary

- An optical source is a key component of an optical transmitter that does the E–O conversion.
- Two types of semiconductor optical source, e.g. light emitting diode (LED) and injection laser diode (ILD) often referred to as laser diode (LD) are predominantly used as optical sources in optical fiber communication.
- Only direct bandgap semiconductors can be used for emission of light caused by Band-To-Band (BTB) radiative transition.
- Mostly III-V semiconductors are used for making sources for optical fiber communication.
- Important III-V semiconductors include AlGaAs, InGaAs, and InGaAsP materials and their heterostructures on lattice matched substrates.

- Light emitting diodes have larger spectral width, broader radiation pattern (HPBW of 120°), and lower modulation bandwidth as compared to laser diodes making the former less attractive for high-speed, long-haul applications.
- LEDs are more economic and have longer life as compared to ILDs.
- LED are of three types, e.g. surface emitting or Burrus type, edge emitting, and superluminescent type.
- Double Heterostructure (DH) LEDs have improved performance as compared to homojunction or single heterojunction LEDs.
- Double heterostructure provides optical confinement as well as electrical confinement.
- The quantum efficiency, responsivity, optical power output, and modulation bandwidth are four important parameters of an LED.
- The 3 dB optical bandwidth can be obtained as:

$$f_{3\text{dB}} = \frac{\sqrt{3}}{2\pi\tau}$$

- The 3 dB electrical bandwidth is given by,

$$f_{3\text{dB}} = \frac{1}{2\pi\tau}$$

τ being the effective lifetime of the carriers.

- The 3 dB bandwidth depends on the bias current density as:

$$f_{3\text{dB}-el} \cong \frac{1}{2\pi}\left(\frac{JB_r}{qd}\right)^{\frac{1}{2}}$$

- The gain-bandwidth product of an LED is a constant.
- Some advanced LED structures include resonant cavity LED, quantum-dot LED, organic LED, etc.
- The most attractive and expensive source used in optical fiber communication is a Laser Diode (LD). Their performance is much superior to LEDs.
- Like other forms of lasers, the condition of population inversion and optical feedback are required to be satisfied in laser diodes.
- Population inversion is created with the help of appropriately doped heterojunctions.
- Optical feedback is provided with the help of Fabry-Pérot resonator of by Distributed Feed-Back (DFB).
- The condition for laser threshold is,

$$g_{\text{th}} = \bar{\alpha} + \frac{1}{2L}\ln\left(\frac{1}{R_1R_2}\right)$$

- The laser cavity creates a number of longitudinal modes for which the gain exceeds the loss.
- The separation between the adjacent modes can be expressed as:

$$\Delta\nu = \frac{c}{2nL}$$

- In terms of wavelength it transforms into

$$\Delta\lambda = \frac{\lambda}{\nu}\Delta\nu = \frac{\lambda^2}{2nL}$$

- A single mode laser is one that supports only one longitudinal mode.
- FP laser diodes are available in various structural forms such as gain-guided structure, index-guided structure to confine light in the lateral direction.
- Laser diodes are also available in forms of Distributed Feed-Back (DFB) or Distributed Bragg Reflector (DBR). In the former case, the grating is incorporated in the active (pumped) region whereas in the latter case it is incorporated in the passive region.

- A laser structure with multiple heterojunctions is used to form Multi Quantum Well (MQW) laser diode which provides lower threshold current, increased modulation bandwidth, narrower emission line width, and lower frequency chirp.
- Laser diodes are generally edge-emitters. However, the laser cavity can be so designed that the light is emitted from the surface. This configuration is called a vertical cavity laser diode.
- The delay time between the application of a current pulse of amplitude, I_p and attaining the threshold value for lasing can be obtained as

$$t_d = \tau \ln\left(\frac{I_p}{I_p - I_{\text{th}}}\right)$$

- In order to reduce the delay time, the laser diode is generally kept prebiased just below the threshold when the signal is in "OFF" state.
- The upper limit of the modulation bandwidth of a laser diode is set by the relaxation oscillation frequency of the laser field given by

$$f_r = \frac{1}{2\pi}\left[\frac{1}{\tau\tau_{ph}}\left(\frac{J}{J_{\text{th}}} - 1\right)\right]^{\frac{1}{2}}$$

- Some of the major issues of concern with laser diodes are aging effect, frequency chirp, noise (phase or frequency noise, reflection noise, mode partition noise), mode hopping, reliability, etc.

Problems

5.1 List the advantages and drawbacks of light-emitting diodes in comparison with injection laser diodes.

5.2 With a schematic of double heterostructure LED, explain how carrier confinement and optical confinement can be achieved simultaneously. Define the quantum efficiency of an LED. Derive an expression for the internal quantum efficiency of an LED and hence discuss the effect of various recombination mechanisms on the quantum efficiency.

5.3 Explain how population inversion and optical feedback are achieved in an injection laser diode. Derive the threshold condition for lasing.

5.4 Distinguish between radiative and non-radiative recombination processes in a semiconductor. The radiative and non-radiative recombination lifetimes of the minority carriers in the active region of a DH-LED are 5 ns and 25 ns respectively. Find the internal quantum efficiency ignoring self-absorption and surface recombination. Calculate the bulk recombination lifetime if the surface recombination velocity at the heterojunction interface is 10 m/s. The active layer thickness is 2 μm.

5.5 A double heterojunction LED with negligible surface recombination at the heterointerfaces and negligible non-radiative recombination with an active region of thickness 0.3 μm operating at 0.85 μm wavelength region exhibits an electrical bandwidth of 40 MHz. Estimate the drive current density of the LED at this bandwidth by assuming the radiative recombination coefficient $B_r = 10^{-15}$ m^3/s.

5.6 A Fabry-Perot cavity resonator has uncoated facets working as mirrors. The cavity is made of GaAs which has a relative permittivity of 13.2 and the surrounding medium is air. Estimate the reflectivities of the mirrors.

5.7 A Fabry-Perot laser diode has a 400 μm long cavity made of GaAs. The rear mirror of the cavity has a reflectivity of 0.6 while the front mirror uses an uncoated facet. The cavity offers an average loss of 1000 m^{-1} at the operating wavelength. Estimate the value of the threshold gain assuming the refractive index of GaAs to be 3.6.

5.8 Repeat problem 5.7 by considering the confinement factor to be 0.85.

5.9 The active cavity of a Fabry-Pérot injection laser diode offers an average loss of 2×10^3 m^{-1}. The uncoated facets of the cavity have a reflectivity of 30% each. Determine the gain coefficient for the cavity when the length of the cavity is 500 μm.

5.10 A Fabry-Perot injection laser diode with an active cavity of length 400 μm operating at 870 nm. Calculate the frequency separation between the successive modes in the cavity assuming the refractive index of the cavity to be 3.6.

5.11 The divergence angle of the emitted beam from a laser diode in the plane of the *p-n* junction is 48° and that in the plane perpendicular to the junction is 10°. If the laser diode is working at 650 nm, estimate the values of the width and thickness of the cavity.

5.12 Estimate the value of the photon lifetime for a 250 μm long cavity FP laser diode neglecting the cavity loss. Given $n = 3.6$ and $R_1 = R_2 = 0.32$.
[Hint: In the absence of cavity loss $\tau_{ph} = 2$ nL $(1 - R_1 R_2)/c$]

5.13 Show that for a laser diode pre-biased with a current I_B just below the threshold current I_{th} the delay between the application of a turn-on current pulse of amplitude I_p and attainment of the lasing threshold can be obtained as

$$t_d = \tau \ln\left(\frac{I_p}{I_p + I_B - I_{\text{th}}}\right)$$

where τ is the spontaneous lifetime of the carriers.

5.14 The threshold current of an InGaAsP laser diode at 20ºC is 120 mA. The threshold temperature of the device is $T_0 = 60$ K. Calculate the threshold current when the temperature of the device is increased to 80ºC.

5.15 The threshold temperature of an AlGaAs laser diode is $T_0 = 160$ K. Calculate the percentage change in the threshold current of the device when the temperature is increased from 20ºC to 80ºC.

5.16 Light from a GaAS LED is to be coupled to the core of an optical fiber of refractive index 1.458. Estimate the Fresnel reflection loss at the interface if the refractive index of GaAs is 3.6 assuming that there is no air gap in between.

5.17 The 3-dB optical bandwidth of an LED is 100 MHz. What is the value of the effective lifetime of the carriers?

5.18 The carrier recombination lifetime of a DH-LED is 10 ns in the absence of surface recombination velocity. What is the electrical bandwidth of the LED?

5.19 Repeat problem 5.18 by assuming the surface recombination velocity at each interface is 10^3 m/s and the thickness of the active region is 0.2 μm.

5.20 Estimate the bandgap of each of the following materials at 300 K.
(i) $Al_{0.33}Ga_{0.67}As$
(ii) $In_{0.53}Ga_{0.47}As$

5.21 Calculate the emission wavelength in each of the above materials.

5.22 What is Auger recombination? How does it affect the bandwidth of an LED?

5.23 Calculate the external quantum efficiency of a GaAs laser diode assuming the slope of the output power–current characteristics is 0.4 mW/mA.

5.24 Consider that the refractive index of the cavity material $n(\lambda)$ is a function of wavelength. Show that under this situation the wavelength spacing between two successive longitudinal modes can be expressed as

$$\Delta\lambda = \frac{\lambda^2}{2L\left(n - \lambda \dfrac{dn}{d\lambda}\right)}$$

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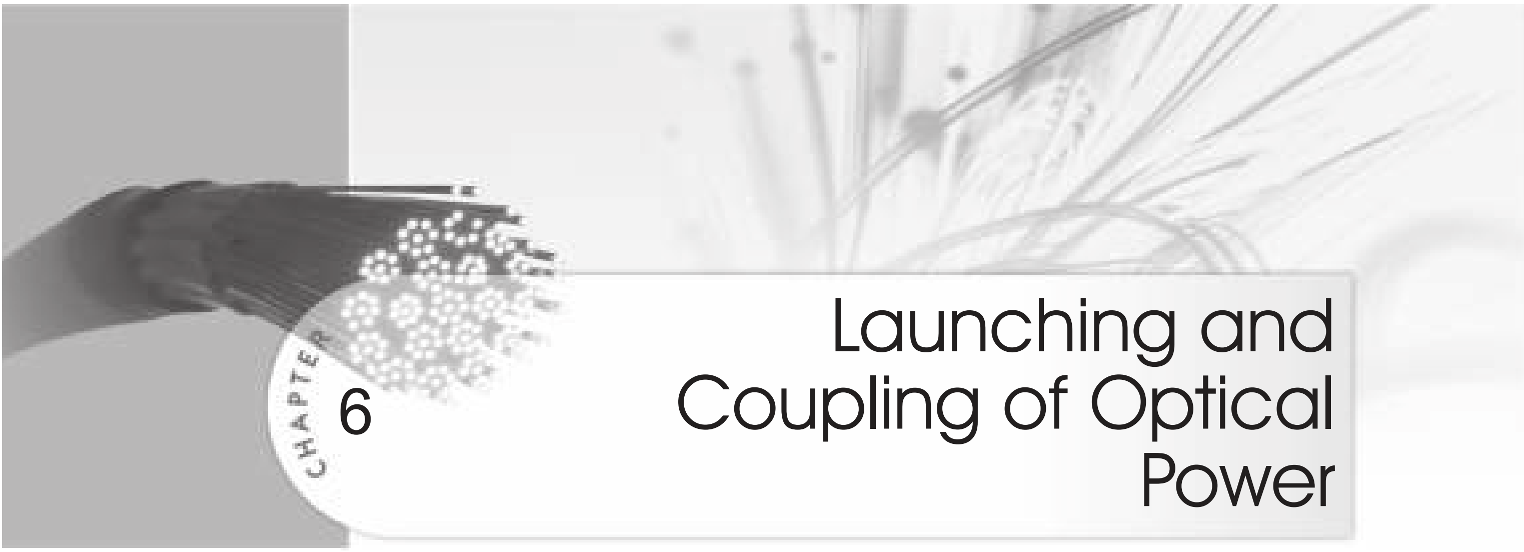

CHAPTER 6 Launching and Coupling of Optical Power

The power emitted by a source needs to be launched into the optical fiber waveguide so that the modulated light generated by the transmitter can be transported through the fiber and delivered at the receiver end. Transmission of light over thousands of kilometers through optical fibers is possible only with the help of intermediate repeaters. Between successive repeaters, we need to connect a number of sections of fiber cables as well as several other optical components in between. The number of such joints or connectors depends on the distance between successive repeaters or the distance between the transmitter and the receiver in a repeaterless optical link. Even though, technologically, it is possible to fabricate single-mode optical fibers of length around 200 km, such long fiber cables are not very convenient for transportation and installation. For field applications, fiber cables of shorter lengths are generally used. The standard separation of 40–60 km between the repeaters requires multiple connections in between fiber cables.

6.1 NEED FOR POWER COUPLING

Each fiber cable consists of a large number of fibers and each of the fibers from one cable is to be connected to the corresponding fiber of the subsequent cable. It should be borne in mind that each such connection gives rise to an additional loss in the link. The loss encountered at each joint depends on many factors including the alignment of the fibers. Therefore, it is necessary to learn about various techniques for launching power from the source to the fiber and also from one fiber to another fiber. Various practical issues need to be addressed for the successful implementation of an optical fiber link. This chapter focuses attention on some of the important aspects of coupling power from the source to a fiber and subsequently from the transmitting[1] fiber to the receiving fiber. These issues are discussed in later sections of this chapter. Further, there is a significant amount of coupling loss at the source-fiber interface. This is because there is a huge mismatch between the radiation and the pattern of the emitted light from the optical source depending on the type of source used at the transmitter side and the maximum acceptance angle of the fiber. Therefore, the larger the mismatch more will be the coupling loss at the source-fiber interface.

The coupling efficiency at the source-fiber interface can be expressed as

$$\eta = \frac{P_F}{P_S} \tag{6.1}$$

where P_S is the optical power emitted by the source and P_F is the power launched by the source into the fiber. The coupling efficiency depends on many parameters of the source and the fibers as well as the technique used for coupling.

[1] The fiber that initially carries the power which is to be launched into a second fiber is called the transmitting fiber. The second fiber is called the receiving fiber.

The acceptance angle of fibers used for practical optical communication systems is generally very small. On the other hand, sources like LEDs have broad radiation patterns. For example, a surface-emitting LED has a half-power beam width (HPBW) of 120°. Therefore it is extremely difficult to align the fiber about the source radiation pattern for maximum coupling. To resolve this tedious job, manufacturers supply optical sources[2] with a short-length optical fiber (usually less than 1 m) connected in a fashion to provide optimum power coupling. The small section of the fiber so connected is called fiber flylead or a pigtail. A commercial laser source with a pigtail fiber is shown in Fig. 6.1. This largely simplifies the job of the user because the user needs to connect the end of the pigtail to the link fiber by using standard connectors.

Fig. 6.1 Photo-image of a fiber pigtail connected to a laser source (LP-405 MF-80–Courtesy: Thorlabs Inc Newton, New Jersey, USA)

6.2 CHARACTERISATION OF RADIATION PATTERN OF OPTICAL SOURCES

To understand the mechanism of launching optical power from an optical source into an optical fiber, it is necessary to know the radiation pattern of the emitted light. The radiation pattern of the optical output from an optical source is generally characterized in terms of the parameter called 'brightness' or 'radiance' of the source. The radiance of a source is a measure of the optical power radiated by the source per unit area of the emitting surface and unit solid angle[3] ($Wm^{-2}\ sr^{-1}$).

The radiation pattern of a luminescent source depends on the type of the source. The spatial distribution of power from a source can be formulated by using spherical coordinates (R, θ, ϕ) with the polar axis considered normal to the plane of the emitting surface as shown in Fig. 6.2(a). For simplicity, we assume the emission from the source is uniform over the entire area. The brightness, B of a source is generally a function of both θ and ϕ. A lambertian source is one that obeys Lambert's cosine law that is, the intensity of the light is directly proportional to the cosine of the viewing angle measured with reference to the polar axis drawn normally to the plane of the emitting surface. The source, however, looks equally bright when viewed from any direction.

The brightness of a Lambertian source can thus be expressed as (Keiser, 2000)

$$B(\theta, \phi) = B_0 \cos\theta \tag{6.2}$$

where B_0 is the brightness of the source along the direction normal to the emitting surface.

The emitted radiation pattern of a conventional surface-emitting LED (SLED) is approximately Lambertian (Fig. 6.2(b)). The SLED has a large beam divergence and

[2] Pigtails are also used with photodetectors where similar alignment is necessary.

[3] The solid angle is measured in terms of steradian (sr).

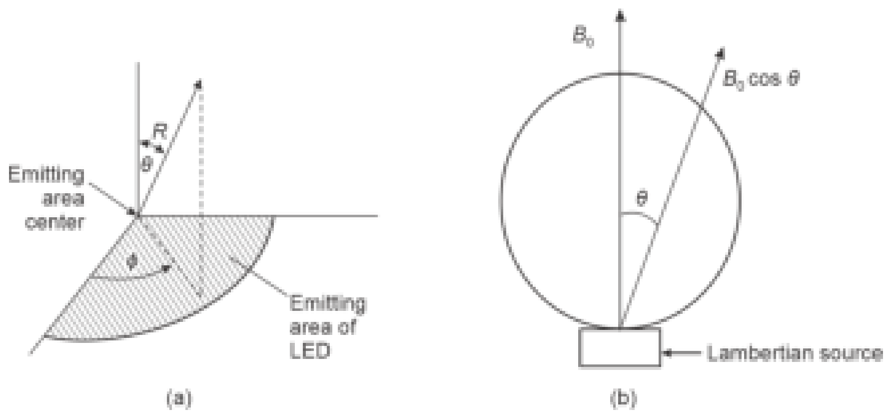

Fig. 6.2 Characterisation of optical source (a) Spherical coordinates (b) Emission pattern from a Lambertian source

therefore its radiation pattern resembles a sphere. The acceptance angle of practical fibers is generally very small. As a result, most of the total optical output emitted by such a diffuse source cannot be coupled into optical fibers.

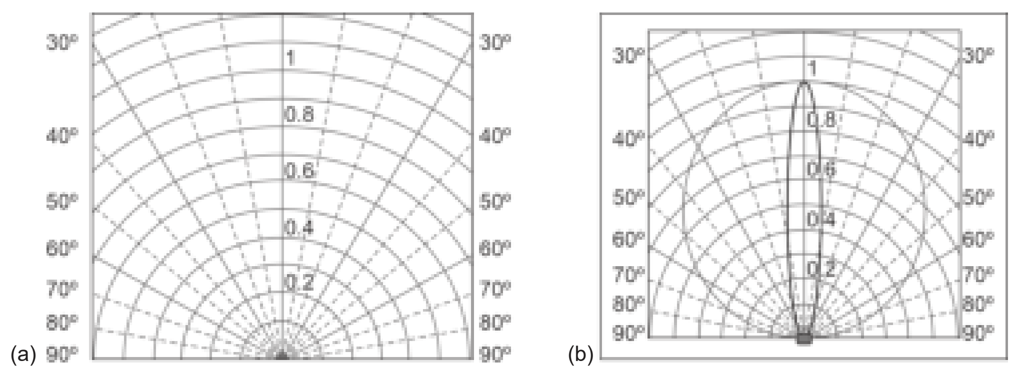

Fig. 6.3 Radiation pattern of different optical sources (a) Surface emitting LED (b) Laser source

The radiation pattern of an ideal Lambertian surface-emitting LED is shown in Fig. 6.3(a). It can be easily seen that the brightness falls to one-half of the maximum brightness when the viewing angle becomes 60º on either side of the polar axis[4]. This means that the available power in each of these directions is one-half of the power available along the normal drawn to the surface. Therefore, the half-power beam width (HPBW) of the source is 120º. The brightness or radiance of a non-ideal Lambertian source is often represented in the form

$$B(\theta, \phi) = B_0 \cos^m \theta \tag{6.3}$$

where m is an integer.

On the other hand, a laser source or even an edge-emitting LED (ELED) has a much more complex emission pattern (Fig. 6.3(b)). We have already seen in the previous chapter that the emission pattern of an ELED in the plane normal to the plane of the p-n junction has a smaller beam width as compared to that in the plane of the p-n junction. Thus, these devices have different radiances $B(\theta, 0^\circ)$ and $B(\theta, 90^\circ)$ along the planes parallel and

[4] Cosine of 60º is 0.5.

perpendicular to the plane of the emitting junction. The radiance of these devices can be expressed as (Uematsu *et al* 1979)

$$\frac{1}{B(\theta,\phi)} = \frac{\sin^2\phi}{B_\perp(\theta)} + \frac{\cos^2\phi}{B_{||}(\theta)} \tag{6.4}$$

where $B_\perp(\theta)$ and $B_{||}(\theta)$ correspond to the radiance measured in the planes perpendicular and parallel to the junction planes. These values have different order dependence on the cosine of the viewing angle.
Thus,

$$B_\perp(\theta) = \cos^n \theta \tag{6.5}$$

and

$$B_{||}(\theta) = \cos^m \theta \tag{6.6}$$

We may write Eq. (6.4) as

$$\frac{1}{B(\theta,\phi)} = \frac{\sin^2\phi}{\cos^n\theta} + \frac{\cos^2\phi}{\cos^m\theta} \tag{6.7}$$

where n and m are integers that determine the transverse and lateral power distribution and are called transverse and lateral power distribution coefficients (Keiser, 2000).

For an edge-emitting LED since the lateral distribution has the same beam width as that of SLED, we may write $m = 1$. The beam width in the transverse direction is much less for an ELED and therefore n can have a large value depending on the design. For a laser diode, both the transverse and lateral power distribution coefficients n and m can be very large.

Example 6.1 The radiance of a non-ideal Lambertian source is given by

$$B(\theta, \phi) = B_0 \cos^2\theta$$

Estimate the half-power beam width of the source.
Solution The radiance of the source is given by

$$B(\theta, \phi) = B_0 \cos^2\theta$$

The half-power beam width of the source can be found by locating the half-power points on both sides of the polar axis (axis normal to the emitting surface). This means that

$$B_0 \cos^2\theta = \frac{1}{2} B_0$$

That is,

$$\theta = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45°$$

The half-power beam width of the source is

$$\text{HPBW} = 2 \times 45° = 90°$$

Example 6.2 An edge-emitting LED has the following dependence on the radiance in the transverse and lateral directions.

$$B_\perp(\theta) = \cos^n\theta \text{ and } B_{||}(\theta) = \cos^m\theta$$

If the transverse and lateral half-power beam widths are 45° and 20° respectively, find the values of transverse and lateral power distribution coefficients.

Solution In the transverse direction ϕ = 90° and the half-power beam width is 20°. Substituting these values in Eq. (6.7), we may write

$$B(\theta = 10°, \phi = 90°) = B_0(\cos 10°)^n = \frac{1}{2}B_0$$

That is,

$$n = \frac{\log(0.5)}{\log(\cos 10°)} = 45$$

Likewise, for the half-power beam width of 45° in the transverse direction, we may write

$$B(\theta = 22°.5, \phi = 0°) = B_0(\cos 22°.5)^m = \frac{1}{2}B_0$$

That is,

$$m = \frac{\log(0.5)}{\log(\cos 22°.5)} = 9$$

6.3 COUPLING OF OPTICAL POWER FROM A LAMBERTIAN SOURCE-TO-FIBER

It has already been pointed out that the power coupling between the source and the fiber depends on the spatial distribution of the power emitted by the source rather than the total power emitted. To estimate the amount of power that can be launched into an optical fiber from a source it is necessary to know the radiance property of the source as well as the power gathering capability of the fiber. Consider a simplistic situation in which an optical fiber is perfectly and symmetrically aligned with respect to the radiation pattern of the source as shown in Fig. 6.4. We also assume that there is no gap between the emitting surface of the source and the end face of the fiber.

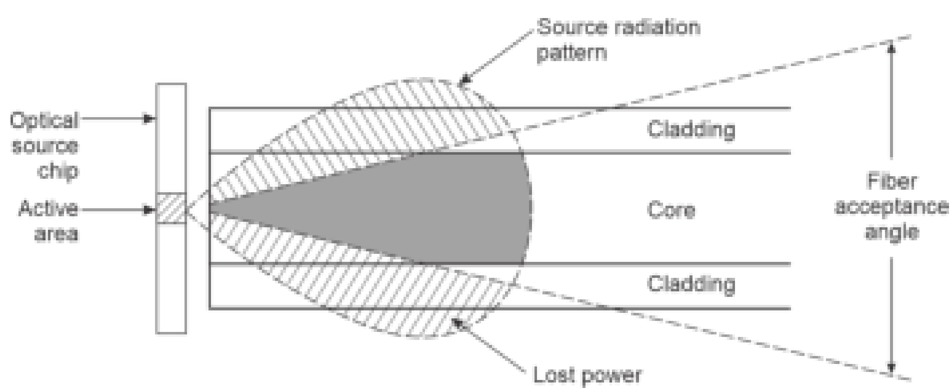

Fig. 6.4 Power coupling from source to a fiber

The radiance or brightness of the source can be expressed as a function of the emission area, A_S and the solid emission angle, Ω_S of the source, that is $B = B(A_S, \Omega_S)$. The total power emitted by the source can be obtained by integrating the radiance of the source over the entire emission area and the emission angle of the source. The entire power emitted by the source cannot be coupled to the fiber. The power coupled to the fiber will be limited by the area, A_f and the solid acceptance angle, Ω_f of the fiber. The coupled power can be expressed as

$$P_F = \int_{A_f} dA_S \int_{\Omega_f} B(A_S, \Omega_S) d\Omega_S \tag{6.8}$$

In this case, the axis is symmetrical with respect to the radiation pattern of the source. The differential solid acceptance angle of the source can be related to the viewing angle, θ by considering a spherical radiation pattern (Fig. 6.5) as

$$d\Omega_s = \frac{dA}{R^2} = \frac{(2\pi R \sin\theta)(R d\theta)}{R^2} = 2\pi \sin\theta d\theta \tag{6.9}$$

In Eq. (6.8), the contribution of each emitting point is considered by taking into account the differential incremental area on the emitting surface and integrating it over the entire area of the emitting source. The differential area on the emitting source can be expressed as

$$dA_S = d\theta_S r dr \tag{6.10}$$

The radiance of the source is assumed to be Lambertian and therefore

$$B(\theta, \phi) = B_0 \cos\theta \tag{6.11}$$

Substituting the values of dA_S, $d\Omega_S$ from Eqs (6.10) and (6.9) into Eq. (6.8) and using Eq. (6.11), we may write

$$P_F = \int_0^{r_m} \int_0^{2\pi} \left[\int_0^{\theta_{0\max}} (B_0 \cos\theta) 2\pi \sin\theta d\theta \right] d\theta_s r dr \tag{6.12}$$

Note that in Eq. (6.12), the upper limit of the inner integral is restricted by the maximum acceptance angle of the fiber. Assume that the emitting area of the source is circular with a radius of r_S. The upper limit of the outermost integral is indicated by r_m depends on the relative areas of the emitting source and the receiving fiber core, given by

$$r_m = r_S \quad \text{when} \quad r_S < a \tag{6.13}$$
$$r_m = a, \quad \text{when} \quad r_S > a \tag{6.14}$$

where a is the radius of the core.

Further assuming that the radius of the circular emitting area of the source is less than the radius of the core, the power coupled to the fiber by the source emitting over a circular area of radius r_S can be obtained from Eq. (6.12) as

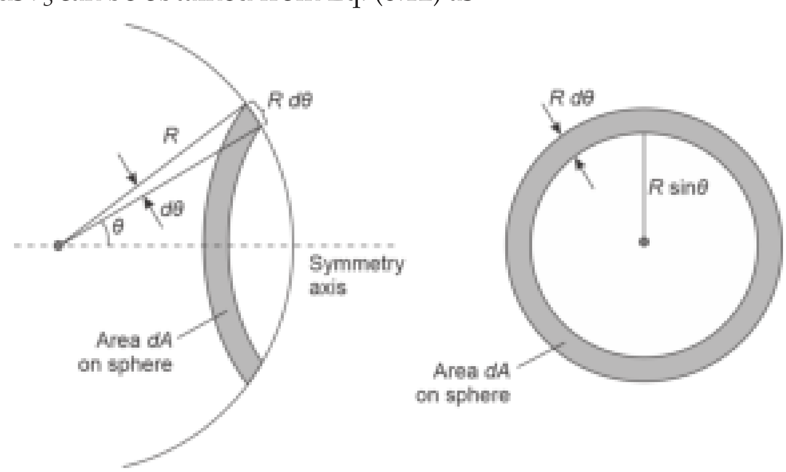

Fig. 6.5 Illustrating differential solid emission angle in terms of linear viewing angle

$$P_F = \int_0^{r_S} \int_0^{2\pi} \left[2\pi B_0 \int_0^{\theta_{0\max}} \cos\theta \sin\theta d\theta \right] d\theta_S r dr$$

That is,

$$P_F = \pi B_0 \int_0^{r_S} \int_0^{2\pi} \sin^2 \theta_{0\max} d\theta_S r dr$$

$$= \pi B_0 \int_0^{r_S} \int_0^{2\pi} (\text{NA})^2 d\theta_S r dr \tag{6.15}$$

where NA is the numerical aperture of the fiber defined in Chapter 2 in terms of the maximum acceptance angle of the fiber.

6.3.1 Power Coupling to a Step-index Fiber

If we now consider a step-index fiber for which the numerical aperture is independent of r and θ_S, the power coupled by an LED to a step-index fiber can be obtained from Eq. (6.15) as

$$(P_F)_{\text{LED-SI}} = \pi^2 r_S^2 B_0 (\text{NA})^2 \tag{6.16}$$

Further for a step-index fiber with a core refractive index $n_1 (\approx n_2)$ and the NA can be approximated as

$$\text{NA} \approx n_1 \sqrt{2\Delta} \tag{6.17}$$

Using the approximation Eq. (6.17), the power coupled from a source of circular emitting area of πr_S^2 to a step-index fiber of core area πa^2 ($> \pi r_S^2$) can be obtained from Eq. (6.16) as

$$(P_F)_{\text{LED-SI}} = 2\pi^2 r_S^2 B_0 n_1^2 \Delta \tag{6.18}$$

Example 6.3 A surface-emitting LED with a circular emitting area of radius 20 μm has an axial radiance of 10^6 Wm^{-2}sr^{-1} at a given drive current operating at 850 nm. The light emitted from the LED is to be coupled to a 60/125 μm step-index fiber with a core refractive index of 1.458 and an index deviation of 1%. Estimate the amount of power coupled to this fiber under ideal conditions and ignore any other coupling loss.

Solution In this case, the core radius of the step-index fiber is

$$a = \frac{60}{2} = 30\ \mu\text{m}$$

which is larger than the radius of the circular emitting area of the LED source of 20 μm.

The power coupled to the step-index fiber by the LED can be obtained by using Eq. (6.18) as

$$(P_F)_{\text{LED-SI}} = 2\pi^2 (20 \times 10^{-6})^2 \times 10^6 \times (1.458)^2 \times 0.01$$
$$= 0.167\ \text{mW}$$

Example 6.4 A surface-emitting LED with a circular emitting area of radius 25 μm has an axial radiance of 120×10^4 Wm^{-2}sr^{-1} at an operating wavelength of 850 m. The light emitted from the LED is to be coupled to a step-index fiber with a core radius of 30 μm. The power coupled to the fiber under ideal conditions is 250 μW. Estimate the value of the numerical aperture of the fiber.

Solution The power coupled to the fiber from the LED source can be obtained from Eq. (6.16) as

$$(P_F)_{LED-SI} = \pi^2(25 \times 10^{-6})^2 \times 120 \times 10^4(NA)^2$$

Given that

$$(P_F)_{LED-SI} = 250 \times 10^{-6}\ W$$

Therefore,

$$NA = \left(\frac{250 \times 10^{-6}}{\pi^2(25 \times 10^{-6})^2 \times 120 \times 10^4}\right)^{\frac{1}{2}} = 0.18$$

The total optical power emitted by the source over the area, $A_S(=\pi r_S^2)$ into a hemisphere with a solid acceptance angle, $\Omega_S(= 2\pi$ sr) can be obtained as

$$P_S = (\pi r_S^2)\int_0^{\frac{\pi}{2}} B_0 \cos\theta(2\pi)\sin\theta d\theta$$

That is,

$$P_S = (\pi^2 r_S^2 B_0)\int_0^{\frac{\pi}{2}} \sin 2\theta d\theta$$

$$= \pi^2 r_S^2 B_0 \qquad (6.19)$$

Using Eq. (6.19), the power coupled from an LED to a step-index fiber can be expressed using Eq. (6.16) as

$$(P_F)_{LED-SI} = P_S(NA)^2 \qquad \text{for} \qquad r_S \le a \qquad (6.20)$$

When the radius of the emitting area of the source is greater than the core radius, the power coupled to the step-index fiber from an LED source can be obtained by setting the upper limit of r_m equal to the core radius a. Following the above steps, it can be shown that

$$(P_F)_{LED-SI} = \left(\frac{a}{r}\right)^2 P_S(NA)^2 \qquad \text{for} \qquad r_S > a \qquad (6.21)$$

It can be seen from Eqs (6.20) and (6.21) that for a given LED source, the power coupled to a step-index fiber is proportional to the square of the numerical aperture of the fiber. This means larger the value of the numerical aperture of a fiber more the power coupled to it. The numerical aperture of the fiber depends on the difference between the refractive index between the core and the cladding regions of the fiber. A large value of the index difference causes an increase in the value of the numerical aperture. Plastic fibers have a large value of numerical aperture and as a result, power coupled to plastic fibers is generally large as compared to that coupled to glass fibers. It may, however, be pointed out that a large value of index difference also enhances the dispersion and lowers the transmission rate.

Example 6.5 A surface-emitting LED with a circular emission area of radius 30 μm is intended to couple power individually into the following two fibers.

(i) A step-index 50/125 μm fiber with a numerical aperture of 0.18

(ii) A step-index fiber of 50 μm core radius and a numerical aperture of 0.18

The axial radiance of the source is 10^6 $Wm^{-2}sr^{-1}$

Compare and contrast the values of power coupled in the above cases.

Solution (i) The radius of the fiber is 25 μm and is less than the radius 30 μm of the emitting area of the source.

The total power emitted by the source is

$$P_S = \pi^2 r_S^2 B_0 = (3.14)^2 \times (30 \times 10^{-6})^2 \times 10^6 = 8.87 \text{ mW}$$

The power coupled to this fiber is therefore

$$(P_F)_1 = \left(\frac{25}{30}\right)^2 \times 8.87 \times (0.18)^2 \text{mW} = 0.199 \text{ mW}$$

(ii) The radius of the fiber is 50 μm and is larger than the source radius of 30 μm.

The power coupled to this fiber is

$$(P_F)_2 = 8.87 \times (0.18)^2 \text{ mW} = 0.287 \text{ mW}$$

Therefore,

$$\frac{(P_F)_2}{(P_F)_1} = \frac{0.287}{0.199} = 1.44$$

The power coupled to the second fiber is 1.44 times that coupled to the first fiber by the same source.

6.3.2 Power Coupling to a Graded-index Fiber

In a graded-index (GI) fiber the numerical aperture depends on the distance r measured from the axis of the core given by

$$\text{NA}(r) = n^2(r) - n_2^2 \tag{6.22}$$

The power coupled to a GI fiber from an LED for $r_S < a$ can be obtained by using Eq. (6.15) as

$$(P_F)_{\text{LED-GI}} = \pi B_0 \int_0^{r_S} \int_0^{2\pi} \left[n^2(r) - n_2^2\right] d\theta_S r dr \tag{6.23}$$

$$= \pi^2 r_S^2 B_0 \int_0^{r_S} \left[n_1^2 \left\{ 1 - 2\Delta \left(\frac{r}{a}\right)^{\alpha} \right\} - n_2^2 \right]$$

$$= \pi^2 r_S^2 B_0 \left(n_1^2 - n_2^2\right) \left[1 - \frac{2}{\alpha + 2} \left(\frac{r_S}{a}\right)^{\alpha} \right] \tag{6.24}$$

Assuming that $n_1 \approx n_2$ we may write

$$n_1^2 - n_2^2 \approx n_1 (2\Delta) \tag{6.25}$$

Using the above approximation the power coupled to the GI fiber can be expressed as

$$(P_F)_{\text{LED-GI}} = 2\pi^2 r_S^2 B_0 n_1^2 \Delta \left[1 - \frac{2}{\alpha + 2} \left(\frac{r_S}{a}\right)^{\alpha} \right]$$

$$= 2 P_S n_1^2 \Delta \left[1 - \frac{2}{\alpha + 2} \left(\frac{r_S}{a}\right)^{\alpha} \right] \tag{6.26}$$

6.3.3 Fresnel Reflection Loss

In the above derivation, we have assumed that there is a perfect coupling between the source and the fiber. This is only possible when the medium between the source and the fiber has the refractive index n_1 same as that of the core of the fiber. If the medium between the source and the fiber end has a different refractive index then a fraction of the power from the source will be reflected from the interface. If the refractive index of the medium between the source emitting surface and the receiving end of the fiber is n_0, then the reflection loss at the interface for normal incidence is given by

$$R = \left(\frac{n_1 - n_0}{n_1 + n_0}\right)^2 \tag{6.27}$$

R is called the Fresnel reflection or reflectivity at the end-face of the fiber core.
The coupled power can therefore be expressed as

$$P_{\text{coupled}} = (1 - R)P_{\text{emitted}} \tag{6.28}$$

The power loss at the interface can be expressed in dB as

$$L(dB) = -10 \log\left(\frac{P_{\text{coupled}}}{P_{\text{emitted}}}\right) = -10 \log(1 - R) \tag{6.29}$$

Example 6.6 Optical power is coupled from a GaAs LED to a step-index fiber having a core refractive index of 1.5. If the refractive index of GaAs is 3.6 and the fiber is in intimate contact with the source, estimate the Fresnel reflection at the interface. Also, estimate the value of coupling loss at the interface due to Fresnel reflection loss.

Solution The Fresnel reflection at the interface can be estimated as

$$R = \left(\frac{3.6 - 1.5}{3.6 + 1.5}\right)^2 = 0.169$$

The coupling loss can be estimated using Eq. (6.29) as

$$L = -10 \log (1 - 0.169) = 0.8 \text{ dB}$$

In practice, the space between the source emitting surface and the fiber end-face is filled with an index-matching fluid to reduce the Fresnel reflection loss at the end face of the fiber core.

It may be pointed out here that the optical power that can be launched from an optical source into a fiber depends on the brightness of the source but not on the wavelength of emission. This may appear to be a little baffling in the sense that the number of modes generated in the fiber depends on the wavelength of the light and all the modes jointly carry the power launched into the fiber yet the power coupled does not depend on the wavelength. This is accounted for by the fact that the power radiated per mode by a source is directly proportional to the square of the wavelength and on the other hand the number of modes supported by the fiber is inversely proportional to the square of the wavelength. Therefore, the total power launched into a given fiber individually by two different sources with the same emission area and the same value of brightness but operating at two different wavelengths remains the same. This means that the power carried by each mode is different in the two cases (see Problem 6.9).

6.4 POWER COUPLING IMPROVEMENT SCHEMES

From the foregoing discussion, it is understood that the poor coupling of power from a Lambertian source into an optical fiber is primarily because of the huge mismatch between the large angular pattern of the light emitted by the source and the small numerical aperture or acceptance angle of the fiber. In conventional butt coupling, the fiber end-face is symmetrically placed on the emitting area of the source. The coupling can be greatly improved by placing a concentrator (microlens) between the source emitting surface and the front-end of the fiber. The concentrator transforms the wide-angular pattern of the emitted radiation to a narrower angle to match the narrow acceptance angle of the fiber. This technique enhances the *coupling efficiency* but introduces certain complexities (Ackenhusen, 1979). Firstly, the size of the lens has to be compatible with the emitting area of the source and the size of the core. This makes the handling of such small lenses extremely difficult. The introduction of additional components in the form of a lens also demands additional alignment of the microlens with the source along with the end face of the fiber. The alignment of the fiber end with the microlens becomes more critical as the radiation is concentrated at a smaller angle. Despite increased complexity concentrators in principle, also enhance the efficiency of coupling power from an optical source to a fiber (Abram *et al* 1975; Thyagarajan *et al* 1978).

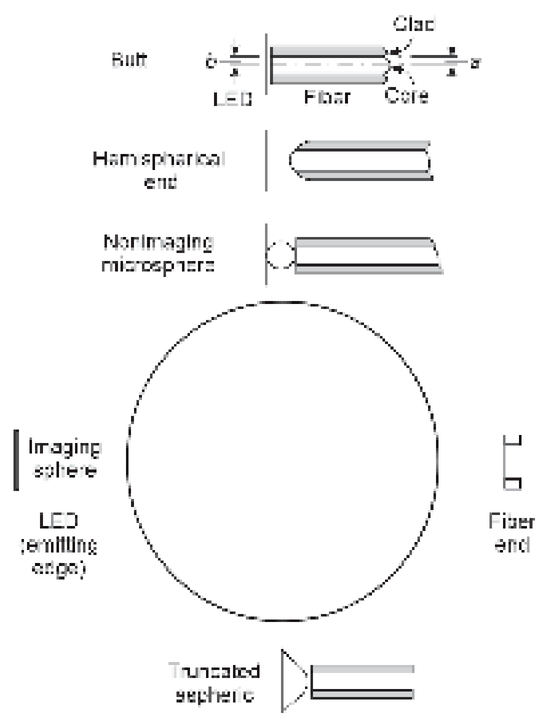

Fig. 6.6 Microlens arrangement for improving power coupling efficiency (after Ackenhusen, 1979)

A few simple schemes for improving the power coupling efficiency between the source and the fiber by using micro-lens arrangement are illustrated in Fig. 6.6. These include a butt-coupling without any concentrator, a rounded fiber front-end, a small non-imaging spherical lens placed in contact with the emitting source at one end and the fiber end-face on the other end, an imaging sphere of relatively large size to create an image of the light spot on the fiber front-end and finally a truncated aspherical lens created by using silicone on the surface of the source (Ackenhusen, 1979). When the source size is larger than the core area of the fiber, the maximum power that can be coupled into the fiber is same as given by Eq. (6.18). In this case, the coupling efficiency cannot be increased by making use of any kind of concentrator or microlens. On the other hand, if the emitting area of the source is smaller than that of the fiber core area, then the use of microlens between the source and the fiber can improve the coupling of power. If the emitting area, A_S of a Lambertian source is smaller than the core area, A_f of a step-index fiber with an effective numerical aperture of NA, then the fractional coupling efficiency that can be achieved is given by (Ackenhusen, 1979)

$$\eta = \frac{A_f}{A_S}(\mathrm{NA})^2(1-R)^n \qquad \text{for} \qquad A_S < A_f \tag{6.30}$$

Alternatively, for a circular emitting area

$$\eta = \left(\frac{a}{r_S}\right)^2 (\mathrm{NA})^2 (1-R)^n \qquad \text{for} \qquad r_S < r_f \tag{6.31}$$

where R is the angle-averaged reflective loss at each of the n interfaces with the surrounding medium considered to be air, a is the core radius and r_S is the radius of the circular emitting area of the source. It may be pointed out that the coupling efficiency given by Eq. (6.30) is the maximum value that can be achieved theoretically. It has been established that no concentrator can achieve the value prescribed by Eq. (6.30) (Welford *et al* 1978).

The most efficient microlens turns out to be the non-imaging spherical lens which collimates the light emitted by the LED source placed at the focal point. Assuming the refractive index of the spherical lens to be $n_1 = 2.0$ and that of the surrounding medium air to be $n_2 = 1.0$ and applying the Gaussian lens formula with the proper sign convention, it can be shown that for non-imaging cases[5] the focal length of the spherical lens becomes (Keiser, 2000).

$$f = 2r_L \tag{6.32}$$

where r_L is the radius of the spherical lens. This means that for collimating the light emitted from the source, the object (the emitting surface of the source) must be placed in contact with the spherical lens.

By placing the lens in contact with the source one can achieve a magnification of the emitting area by a factor M given by

$$M = \frac{\pi r_L^2}{\pi r_S^2} = \left(\frac{r_L}{r_S}\right)^2 \tag{6.33}$$

Accordingly, the power that can be coupled with the help of the microlens in the full acceptance angle can be expressed as

$$P_L = P_S \left(\frac{r_L}{r_S}\right)^2 \sin^2 \theta_{0\,\max} \tag{6.34}$$

where P_S is the power emitted by the source in the absence of the lens.

The other method of improving the coupling efficiency involves the use of fiber with a rounded end-face which can be created using flame, laser or photoresist (Kawasaki *et al* 1975; Paek *et al* 1975; Cohen *et al* 1974). The improvement in coupling efficiency that can be achieved with the help of the above techniques was studied experimentally (Ackenhusen, 1979). The measurements revealed that an improvement in the coupling efficiency of can be achieved by placing a round-end fiber at a suitable distance from the source. The variations of coupled power with the axial separation between the source and the fiber end-face are reproduced in Fig. 6.7 from the reported measurements (Ackenhusen, 1979). In this plot, the coupled power is measured relative to the power coupled by the source in the case of butt coupling normalized to unity while the distance between the source and the fiber is expressed in the units of fiber core radius.

[5] In the non-imaging case, the refracted beam out of the spherical lens is a collimated beam which means that the image is formed at infinity.

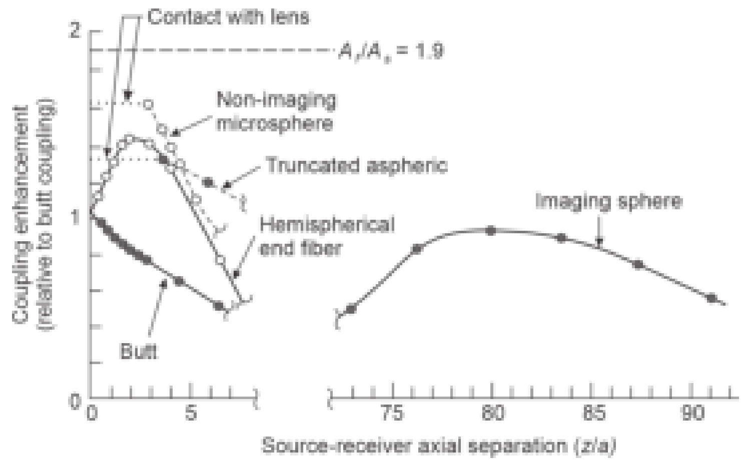

Fig. 6.7 Variation of coupling enhancement with axial separation between the LED and fiber as reported experimentally (after Ackenhusen, 1979)

Various techniques described above for improving the power coupling efficiency by using different forms of microlenses suffer from spherical aberrations encountered in fibers with large values of numerical aperture (Kawasaki *et al* 1975). Moreover, it is extremely difficult to realize the optimum spherical surface radius (Uematsu *et al* 1979). Another method of improving the coupling efficiency is to use a taper-ended fiber shown in Fig. 6.8. Oseki *et al* (Oseki *et al* 1976) have demonstrated that the power coupling efficiency between a DH-LED source and a corning fiber with a numerical aperture, NA = 0.181 can be increased to a value as high as by making use of taper-ended fiber. The taper-ended fibers can be produced by chemical etching or by heating-and-pulling method (Uematsu *et al* 1979). In a taper-ended fiber, the numerical aperture can be effectively increased (Oseki *et al* 1976) as

$$(\mathrm{NA})_{eff} = R \times (\mathrm{NA}) \tag{6.35}$$

where R is the taper ratio (Oseki *et al* 1976).

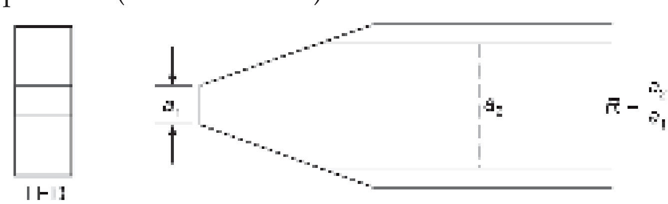

Fig. 6.8 Schematic diagram of a taper-ended fiber showing the taper ratio

Incoherent sources such as LEDs are often used to couple power to a bundle of fibers. Consider a generalized incoherent source with constant brightness in the emission plane and the angular dependence of brightness with the viewing angle, θ described by

$$B(\theta, \phi) = B_0 \cos^n\theta \tag{6.36}$$

When the source is directly butted against a bundle of fibers, the power coupling efficiency can be obtained as (Barnoski, 1976)

$$\eta_{Fb} = \frac{\text{Power coupled to the fiber bundle}}{\text{Total power emitted by the source}} \tag{6.37}$$

$$= f_p(1 - \cos^{n+1}\theta_c) \qquad \text{for} \qquad r_S < r_f \tag{6.38}$$

$$= \left(\frac{r_f}{r_S}\right)^2 f_p\left(1 - \cos^{n+1}\theta_c\right) \qquad \text{for} \qquad r_S > r_f \tag{6.39}$$

where θ_c is the acceptance angle of the fiber bundle, f_p is the fraction of the bundle area covered by fiber cores and is known as the packing fraction of the fiber and r_f and r_S correspond to the radius of the bundle and that of the source respectively (Barnoski, 1976; Thyagrajan *et al* 1976).

6.5 COUPLING OF POWER FROM A LASER SOURCE TO A FIBER

The emission pattern of conventional edge-emitting laser sources is generally much more directive as compared to LED sources. The full-width half-maximum (FWHM) of an injection laser diode is generally in the range of 5–10º in the plane parallel to the *p-n* junction edge and that in the direction perpendicular to the junction plane is in the range of 30–50º. The narrower beam width of the emitted light from a laser diode closely matches the acceptance angle of optical fibers used in practical optical communication systems. It is therefore relatively easy to launch power from a laser source into an optical fiber including a single-mode fiber. Nevertheless, the use of microlenses can further improve the coupling efficiency in the case of laser sources as well. Different forms of microlenses such as spherical lenses, cylindrical lenses, and taper-ended fibers have been tried by various researchers to improve the coupling efficiency between a laser diode and an optical fiber (Cohen *et al* 1974; Edward *et al* 1993; Bludau *et al* 1985; Presby *et al* 1989). These microlenses largely aim to match the modes of the laser and the fiber as well as the small emission area of the laser source to the area of the fiber core. Microlenses are often fabricated by tapering the fiber end down to a point and melting the end (Edward *et al* 1993). The resultant hemispherical lens generally offers coupling efficiency in the range of 50–55%.

It has been reported that optimal microlenses for laser-to-fiber coupling can be designed to achieve 90% (–0.45 dB) coupling (Presby, 1992). It has also been predicted that for an ideal laser diode, the coupling can theoretically reach 100% with the help of antireflection coating. The fabrication technique of the micro-machined fiber lens has been reported by Presby *et al* (Presby *et al* 1990). The optimal microlens profile for a typical laser diode with symmetric Gaussian mode has been demonstrated to be a hyperboloid of revolution and the uncoated lenses exhibit only 0.22 dB coupling loss (Edwards *et al* 1993). A simple technique involving spherical microlens directly attached to a laser source for a nearly self-aligned and self-supported packaging has been reported in the literature.

6.6 FIBER-TO-FIBER POWER COUPLING

So far we have discussed the launching of power from an optical source into an optical fiber and associated coupling loss in the process. Various techniques used for improving the coupling efficiency are also discussed. It is understood that the use of microlenses can greatly improve the coupling efficiency. Nevertheless, the amount of power that can be coupled from a source to a fiber is governed by the principle of conservation of brightness. In this section, attention is focused on the coupling of optical power from one fiber to the other and various sources of loss associated with such coupling.

In a long-haul optical fiber communication system it is necessary to connect an optical fiber cable to another fiber cable. The connections of two cables involve joining of each fiber of the one cable to the corresponding fiber in the other cable. It is also necessary to connect an optical fiber to the fiber pigtail of the source at the transmitter end and the pigtail of the detector at the receiver end. When the fibers are joined it is not always necessary that the two fibers have identical geometry and characteristics. The fibers are generally connected by two methods depending on the requirement. At the time of connecting two fiber cables for installation, the individual fibers are generally bonded permanently by using the splicing technique. A permanent joint is referred to as a "splice". On the other hand, when a fiber is connected to terminal equipment it may be necessary to make and break the connection at times and therefore a permanent bonding of a fiber to the terminal equipment is not desirable. In such cases, a demountable joint that allows a non-destructive make and break the connection is preferred over a permanent bonding. This type of temporary bonding is referred to as a demountable joint or simply a connector.

Whenever two fibers are connected or jointed by using any one of the above techniques there is an additional loss component arising out of the joint. The loss encountered at a joint depends on several factors including the characteristics of the two fibers, the nature of the end faces of the fibers which are jointed, and also on the alignment of the two fibers. In the context of making the joint of two fibers, the fiber which carries the light is referred to as the emitting fiber and the subsequent fiber which is jointed to this fiber to carry the light forward is called the receiving fiber. If we ignore the effects of all other factors, the amount of power that can be coupled from one fiber to the other is ultimately limited by the number of modes that can be supported by each fiber. For example, if two identical fibers each of which can support 800 modes are jointed together, then under ideal conditions all the 800 modes can be launched into the subsequent fiber. However, if the emitting fiber supports 800 modes but the subsequent receiving fiber can support only 600 modes, then only (600/800 = 3/4) the fraction that is 75% of the power from the emitted fiber can at best be coupled to the receiving fiber. For a step-index fiber, the number of modes propagating through a fiber can be approximated in terms of the V-number of the fiber given by

$$M_{\mathrm{SI}} \approx \frac{V^2}{2} = \frac{1}{2}a^2k^2(\mathrm{NA})^2 = a^2k^2n_1^2\Delta \tag{6.40}$$

where a is the core radius, k is the free space propagation constant, and $\mathrm{NA} = \sqrt{n_1^2 - n_2^2} \approx n_1\sqrt{2\Delta}$ is the numerical aperture of the step-index fiber, Δ being the index deviation.

For a graded index fiber, the total number of modes can be obtained as (see Eq. (3.159)).

$$M_{\mathrm{GI}} = a^2k^2n_1^2\Delta\left(\frac{\alpha}{\alpha+2}\right) \tag{6.41}$$

where α is the profile-index factor.

The fraction of the power that can be coupled from the emitting fiber to the receiving fiber depends on the common mode volume when a uniform distribution of energy is assumed over all modes. Under such conditions, the fiber-to-fiber coupling efficiency can be expressed as (Keiser, 2000)

$$\eta_F = \frac{M_{\text{common}}}{M_E} \tag{6.42}$$

where M_{common} refers to the common mode volume between the two fibers and M_E is the total number of modes supported by the emitting fiber.

Accordingly, the loss encountered at this fiber-to-fiber joint can be expressed as

$$L_F = -10 \log_{10} (\eta_F) \tag{6.43}$$

It is extremely difficult to theorize or even simulate the exact loss occurring at a joint because of the complex parameters involved in the process. These include non-identical geometry and characteristics of the two fibers, non-uniform distribution of power among various modes, quality of the end faces of the jointed fibers, and their alignment. Several models have been developed to estimate joint loss (DiVita *et al* 1978; 1980; 1981; Gloge, 1976). A small amount of misalignment of the two fibers may result in a huge coupling loss. When optical power is launched from an optical source to a fiber all the modes are generally excited which fills the full numerical aperture of the fiber. However, as the light propagates through the fiber many of the higher-order modes lose power to the cladding modes due to mode coupling, and only the power is primarily concentrated in the lower-order modes in the central region. In other words, the power emitted at the end of the fiber only fills the equilibrium numerical aperture which is much less than the full numerical aperture of the fiber as illustrated in Fig. 6.9. The numerical aperture of the receiving fiber is much larger than that of the equilibrium numerical aperture in which the optical power is concentrated in the emitting fiber. As a result, a small misalignment or difference in the geometries of the fiber does not usually cause a huge coupling loss (Keiser, 2000).

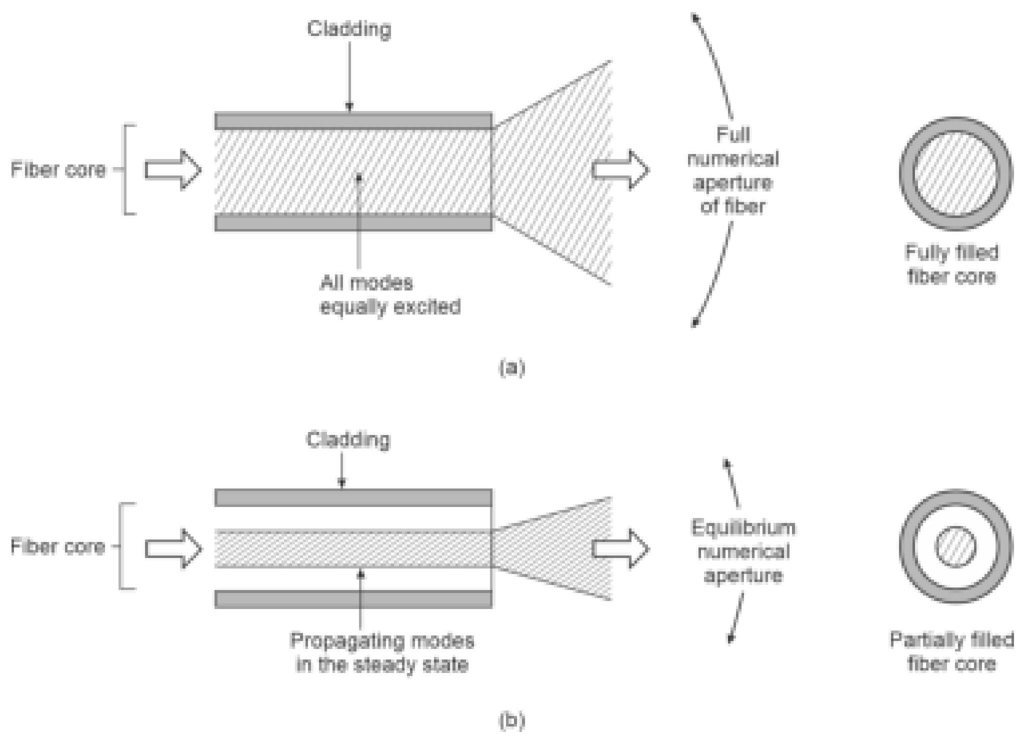

Fig. 6.9 Illustrating the initial launched state and final steady-state modal distribution of power (a) All modes equally excited to fill the full numerical aperture (b) Steady-state modal distribution when only the equilibrium numerical aperture is filled

6.6.1 Reflection Loss

When two fibers with perfect end faces are jointed even without any misalignment a small portion of the optical power is usually reflected into the emitting fiber due to

Fresnel reflection, occurs when the light from one medium encounters a step change in the refractive index at the interface. In this case, the light coming from the emitting fiber experiences a step change in the refractive index at the jointed interface (glass-air-glass). The fraction of the power reflected into the transmitting medium can be estimated with the help of the Fresnel reflection coefficient defined by Eq. (6.27) and written in the case of a small air-gap at the interface of jointed fibers as

$$R = \left(\frac{n_1 - 1}{n_1 + 1}\right)^2 \tag{6.44}$$

where n_1 is the refractive index of the core of the emitting fiber.

The loss in dB due to Fresnel reflection at a single interface between the emitting fiber and the receiving fiber can be obtained as

$$L_{\text{Fresnel}}\ (\text{dB}) = -10 \log_{10} (1 - R) \tag{6.45}$$

It may be stressed here that there are two interfaces at the joint with an air gap at the interface. The first interface is formed between the end-face of the emitting fiber and the air and the other interface is formed between the air and the end-face of the receiving fiber. The total coupling loss must, therefore, include the losses at both interfaces. If the refractive index of the core of the receiving fiber is the same as that of the emitting fiber, then by the symmetry, the loss at each interface will be the same under ideal conditions. The interface loss at the fiber-to-fiber joined can be reduced significantly by filling the air-gap with an index-matching fluid. If the refractive index of the matching fluid is the same as that of the core refractive index of the two fibers then the Fresnel reflection loss can be eliminated.

Example 6.7 Two identical step-index fibers with a core refractive index of 1.458 are butt jointed. The gap between the end-faces of the fibers is filled with air (n = 1). Estimate the total Fresnel loss at the interface under ideal conditions.

Solution The fraction of light reflected at the emitting fiber end-face/air interface can be obtained as

$$R = \left(\frac{1.458 - 1}{1.458 + 1}\right)^2 = 0.035$$

This means that about 3.5% of the emitted power is reflected back into the emitting fiber at the single interface between the emitting fiber end-face and air.

The loss of optical power in dB at a single interface can be obtained as

$$L_{\text{Fresnel}}(\text{dB}) = -10 \log_{10}(1 - 0.035) = 0.15 \text{ dB}$$

Similarly, by symmetry, the loss of optical power at the other interface between air and end-face of the receiving fiber is also 0.15 dB. Therefore, the total coupling loss at the joint due to Fresnel reflection can be obtained as

$$L_{\text{Fresnel}}(\text{dB})[\text{total}] = (0.15 + 0.15) \text{ dB} = 0.3 \text{ dB}$$

Example 6.8 In the above example (Example 6.7), assume that the gap between the end-faces of the two fibers is filled with an index-matching fluid having a refractive index of 1.3. Estimate the coupling loss at the joint.

Solution The power loss due to Fresnel at a single interface can be obtained as

$$R = \left(\frac{1.458 - 1.3}{1.458 + 1.3}\right)^2 = 0.008$$

Therefore, the Fresnel reflection loss at a single interface is

$$L_{\text{Fresnel}}(\text{dB}) = -10 \log_{10}(1 - 0.008) = 0.03 \text{ dB}$$

The total coupling loss at the joint due to Fresnel reflection is 0.06 dB.

The coupling loss due to Fresnel reflection can be substantially reduced by making use of a transparent index-matching fluid to fill the air-gap between the end-faces of the two fibers.

6.6.2 Other Intrinsic Losses

Apart from Fresnel reflection loss at the joint, there are several sources of inherent loss occurring at 'an otherwise' perfect joint. These losses depend on the matching characteristics of the two jointed fibers. A generalized fiber can be characterized in terms of three parameters, e.g. core radius, numerical aperture at the center of the core, NA(0), and the index profile constant, α. Consider the situation of jointing an emitting and a receiving fiber characterized by (a_1, NA_1 (0), α_1) and (a_2, NA_2 (0), α_2) respectively.

If $NA_1(0) = NA_2(0)$ and $\alpha_1 = \alpha_2$, the joint-loss due to unequal core-radius (in the absence of Fresnel reflection loss) can be written as (Thiel *et al* 1976)

$$L_a = \begin{cases} -10\log_{10}\left(\dfrac{a_1}{a_2}\right)^2, & a_1 < a_2 \\ 0 & , a_1 \geq a_2 \end{cases} \tag{6.46}$$

If $\alpha_1 = \alpha_2$ and $a_1 = a_2$, for the two fibers having different values of axial numerical aperture values, the joint loss in the absence of Fresnel loss can be obtained as (Thiel *et al* 1976)

$$L_{\text{NA}} = \begin{cases} -10\log_{10}\left[\dfrac{\text{NA}_1(0)}{\text{NA}_2(0)}\right]^2, & \text{NA}_1(0) < \text{NA}_2(0) \\ 0 & , \text{NA}_1(0) \geq \text{NA}_2(0) \end{cases} \tag{6.47}$$

If $NA_1(0) = NA_2(0)$ and $a_1 = a_2$, the joint loss in the absence of Fresnel reflection loss can be obtained as (Thiel *et al* 1976)

$$L_a = \begin{cases} -10\log_{10}\left[\dfrac{\alpha_1(\alpha_2 + 2)}{\alpha_2(\alpha_1 + 2)}\right], & \alpha_1 < \alpha_2 \\ 0 & , \alpha_1 \geq \alpha_2 \end{cases} \tag{6.48}$$

Example 6.9 Two graded-index fibers with the same values of core radius and axial numerical aperture are butt-jointed. The emitting fiber has an index profile parameter of 2.0 and that of the receiving fiber is 2.2. Estimate the value of joint-loss due to a mismatch of index profile parameters neglecting the Fresnel loss.

Solution The joint-loss due to mismatch of index profile parameters can be estimated using Eq. (6.48) as

$$L_a = -10\log_{10}\left(\frac{2(2.2+2)}{2.2(2+2)}\right) = 0.20 \text{ dB}$$

Example 6.10 A 50/125 μm graded-index fiber with $\alpha = 2$ is butt-jointed with a step-index fiber of the same core radius and axial numerical aperture. Estimate the joint-loss neglecting Fresnel loss.

Solution In this case, the index profile parameter of the receiving fiber is $\alpha_2 \to \infty$. The joint-loss can be estimated by using Eq. (6.48) as

$$L_\alpha = -10\log_{10}\left[\frac{\alpha_1(\alpha_2+2)}{\alpha_2(\alpha_1+2)}\right] = -10\log_{10}\left[\frac{\alpha_1\left(1+\frac{1}{\alpha_2}\right)}{\alpha_1+2}\right]$$

That is,

$$L_a = -10\log_{10}\left[\frac{2}{4}\right] = 3 \text{ dB}$$

6.6.3 Misalignment Loss

In addition to coupling loss at a fiber joint arising out of different diameters of the core/cladding, different numerical apertures, different relative index deviation, and different index profiles of the two jointed fibers, the mechanical misalignment may also result in substantial coupling loss at a fiber-to-fiber joint. These forms of losses are generally viewed as extrinsic coupling loss and can be reduced significantly by reducing different forms of misalignment. Mechanical misalignment is generally classified under three different categories, e.g. longitudinal misalignment arising out of the separation between the end-faces of the jointed fibers which are otherwise aligned axially (Fig. 6.10(a)), angular misalignment resulting from the fact that the end-faces of the two fibers are no longer parallel and the axes of the two fibers form an angle in between as illustrated in Fig. 6.10(b) and the lateral misalignment caused by the axial displacement in which the axes of the two fibers are displaced by a finite distance as illustrated in Fig. 6.10(c).

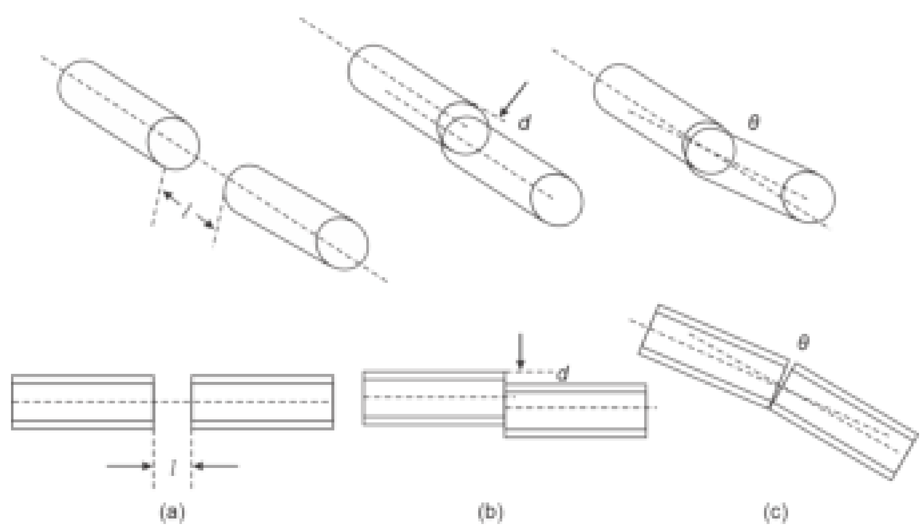

Fig. 6.10 Three possible mechanical misalignment of the jointed fibers along with longitudinal section views of the three categories (a) Lateral misalignment (b) Axial misalignment (c) Angular misalignment

Among the above three types of misalignment, lateral or axial misalignment causes larger coupling loss as compared to longitudinal misalignment. The effect of lateral misalignment can be estimated for the simplest case involving two identical step-index fibers having a core radius of a with an axial offset of d at the interface of the joint. The cross-sectional view of the joint is illustrated in Fig. 6.11. Assuming uniform modal power distribution and constant numerical aperture across the end faces of the two fibers, the optical power coupled by the emitting fiber to the receiving fiber can be considered to be proportional to the common area of overlapping of the cores of the two fibers.

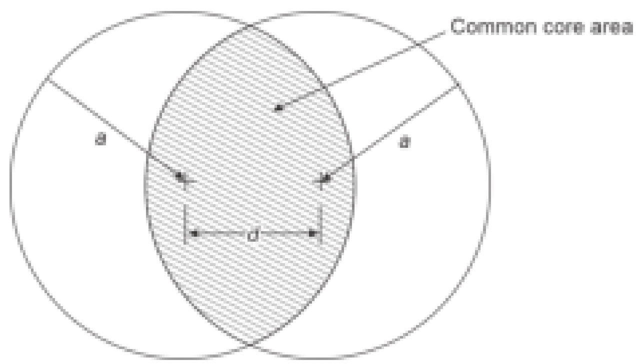

Fig. 6.11 Illustrating common area of overlapping of the fiber core areas of two jointed fibers in the case of axial misalignment at the interface

Using simple Euclidean geometry in this case, the common area of overlapping of the two cores shown by the shaded portion can be estimated as (Keiser, 2000)

$$A_{\text{common}} = 2a^2 \cos^{-1}\left(\frac{d}{2a}\right) - d\left[a^2 - \left(\frac{d}{2}\right)^2\right]^{\frac{1}{2}} \tag{6.49}$$

The coupling efficiency for two identical step-index fibers jointed with a lateral misalignment can be obtained by taking the ratio of the common core area of overlapping to the area of the core end-face and by considering the effect of Fresnel loss at the two interfaces as (Tsuchiya *et al* 1977)

$$\eta_{\text{lat}} = \frac{16(n_1/n)^2}{(1+(n_1/n))^4}\left[\frac{2}{\pi}\cos^{-1}\left(\frac{d}{2a}\right) - \frac{d}{\pi a}\left(1-\left(\frac{d}{2a}\right)^2\right)^{\frac{1}{2}}\right] \tag{6.50}$$

where n_1 is the refractive index of the core, n is the refractive index of the medium between the jointed fibers, and d is the lateral (or axial) offset between the axes of the two fibers each having a core radius of a. Consequently, the lateral misalignment loss in dB at the joint can be estimated as

$$\text{Loss}_{\text{lat}}(\text{dB}) = -10 \log_{10}\eta_{\text{lat}} \tag{6.51}$$

The formula given in Eq. (6.51) generally overestimates the predicted loss. This may be attributed to the fact that the above formulation is based on the assumption that all modes are equally excited which is not generally true in the case of practical fibers.

If the gap between the fiber end-faces is filled with an index matching fluid of refractive index same as that of the core, then $n_1/n = 1$ and there is no loss due to Fresnel reflection at the joint. In the absence of Fresnel reflection the coupling efficiency of the joint can be expressed using Eq. (6.47) as

$$\eta_{\text{lat}} = \frac{2}{\pi}\cos^{-1}\left(\frac{d}{2a}\right) - \frac{d}{\pi a}\left(1-\left(\frac{d}{2a}\right)^2\right)^{\frac{1}{2}} \tag{6.52}$$

Example 6.11 Two identical step-index fibers with a core radius of 25 μm each are jointed with a lateral misalignment of 5 μm. Estimate the coupling loss at the joint by ignoring the Fresnel reflection loss at the interface.

Solution The coupling efficiency in this case of lateral misalignment can be obtained from Eq. (6.49) as

$$\eta_{\text{lat}} = \frac{2}{3.14}\cos^{-1}\left(\frac{5}{2\times 25}\right) - \frac{5}{3.14\times 25}\left(1-\left(\frac{5}{2\times 25}\right)^2\right)^{\frac{1}{2}}$$

$$= 0.94 - 0.06 = 0.88$$

The coupling loss due to lateral misalignment can be calculated using Eq. (6.48) as

$$\text{Loss}_{\text{lat}}\ (\text{dB}) = -10\log_{10}(0.88) = 0.56\ \text{dB}$$

Example 6.12 Two identical step-index fibers with a core radius of 25 μm each are jointed with a lateral misalignment of 10 μm. Estimate the coupling loss at the joint by assuming that the gap between the end faces of the emitting and the receiving fibers at the joint is filled with air. The refractive index of the core of each fiber is 1.5.

Solution Since the gap at the interface between the two fiber end-faces at the joint is filled with air, the coupling efficiency can be estimated by using Eq. (6.50) and putting $n_1/n = 1.5$ as

$$\eta_{\text{lat}} = \frac{16\times(1.5)^2}{(1+1.5)^4}\left[\frac{2}{\pi}\cos^{-1}\left(\frac{10}{2\times 25}\right) - \frac{10}{3.14\times 25}\left(1-\left(\frac{10}{2\times 25}\right)^2\right)^{\frac{1}{2}}\right]$$

$$= 0.92(0.87 - 0.12) = 0.69$$

The coupling loss due to lateral misalignment in the presence of Fresnel loss can be obtained as

$$\text{Loss}_{\text{lat}}\ (\text{dB}) = -10\log_{10}(0.69) = 1.6\ \text{dB}$$

The calculation of power coupled from one graded index fiber to another with an axial misalignment is fairly complex because the numerical aperture varies across the end-face. Gloge (Gloge, 1976) calculated the joint-loss due to lateral misalignment in multimode-graded index fiber by assuming a uniform distribution of optical power over all guided modes. The misalignment loss at the joint of two identical graded-index fibers arising out of a small lateral off-set ($0 \le d \le 0.2a$) can be obtained as (Gloge, 1976)

$$L_t = \frac{2}{\pi}\left(\frac{d}{a}\right)\left(\frac{\alpha+2}{\alpha+1}\right) \tag{6.53}$$

where d is the lateral off-set not exceeding 20% of the core radius, a is the core radius and α is the gradient of the refractive index profile of the graded-index fiber.

The lateral coupling efficiency for the case of lateral off-set can be consequently expressed as

$$\eta_{\text{lat-GI}} = 1 - L_t \tag{6.54}$$

Example 6.13 Two identical graded-index fibers with parabolic index profiles ($\alpha = 2$) are jointed with a lateral off-set of 5 μm. Estimate the joint loss for this lateral off-set assuming uniform power distribution over all the emitted modes and neglecting the effect of Fresnel reflection at the joint. The diameter of the core of each fiber is 50 μm.

Solution The core radius of each fiber is

$$a = 50/2 = 25\ \mu\text{m}$$

and the lateral off-set is

$$d = 5\ \mu\text{m}$$

which is just 20% of the core-radius.

The joint-loss due to lateral misalignment in this case can be computed using Eq. (6.53) as

$$L_t = \frac{2}{3.14}\left(\frac{5}{25}\right)\left(\frac{2+2}{2+1}\right) = 0.169$$

The corresponding coupling loss in dB can be estimated as

$$L_t \text{ (in dB)} = -10 \log_{10} (1 - 0.169) = 0.8 \text{ dB}$$

The coupling loss associated with longitudinal misalignment arising from the separation between the end faces of the jointed fibers which are otherwise aligned axially without any angular misalignment is generally negligible.

The loss arising because of the separation of the ends of two fibers in a joint can be easily estimated using simple geometry. For example, consider the end faces of the two jointed step-index fibers with a gap z in the longitudinal direction as shown in Fig. 6.12. Assuming the fibers to be axially aligned otherwise, the loss occurring at the joint can be obtained as (Thiel *et al* 1976)

$$L_f = -10 \log_{10}\left(\frac{a}{a + z \tan\theta_c}\right) \tag{6.55}$$

where θ_c is the critical acceptance angle of the fiber and a is the radius of the core of each fiber.

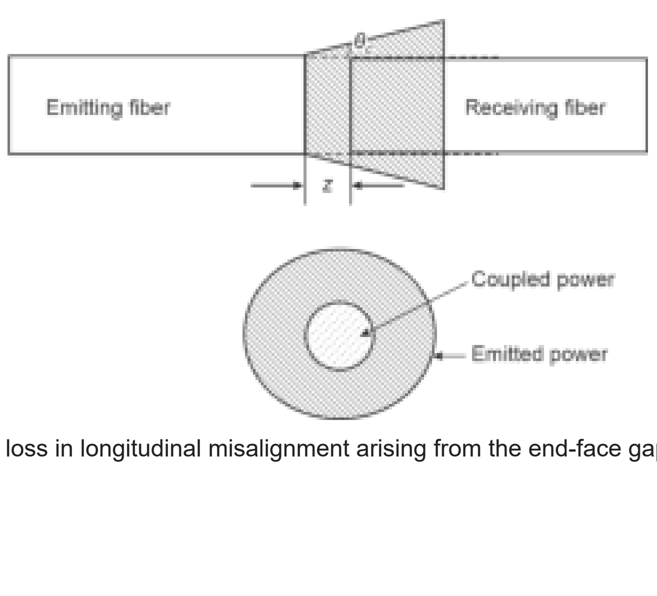

Fig. 6.12 Power loss in longitudinal misalignment arising from the end-face gap separation of *z*

6.6.3.1 Angular misalignment

Additional coupling loss occurs when the axes of the two jointed fibers are angularly misaligned as shown in Fig. 6.10(c). This is because of the power lost due to the misalignment of the solid acceptance angles of the two fibers. If the two jointed step-index fibers have an angular misalignment of θ, the optical power loss at the joint can be expressed assuming all modes to be uniformly excited, as (Thiel *et al* 1976)

$$L_f = -10\log_{10}\left[\cos\theta\left\{\frac{1}{2} - \frac{1}{\pi}p(1-p^2)^{\frac{1}{2}} - \frac{1}{\pi}\sin^{-1} - q\left(\frac{1}{\pi}r(1-r^2)^{\frac{1}{2}} + \frac{1}{\pi}\sin^{-1}r\right)\right\}\right] \tag{6.56}$$

where

$$p = \frac{\cos\theta_c(1-\cos\theta)}{\sin\theta_c\sin\theta}$$

$$q = \frac{\cos^3\theta_c}{(\cos^2\theta_c - \sin^2\theta)^{3/2}}$$

and

$$r = \frac{\cos^2\theta_c(1-\cos\theta) - \sin^2\theta}{\sin\theta_c\cos\theta_c\sin\theta}$$

wherein

$$\theta_c = \sin^{-1}\left(\frac{\text{NA}(0)}{n}\right)$$

where n is the refractive index of the material between the fibers.

Example 6.14 Two step-index fibers of radius 25 μm each are jointed with a longitudinal misalignment of 10 μm. If the gap is filled with air, calculate the loss at the joint assuming the NA of each fiber to be 0.16. Neglect Fresnel reflection loss at the interfaces.

Solution The critical acceptance angle of the fiber when the gap is filled with air ($n = 1$) can be estimated as

$$\theta_c = \sin^{-1}\left(\frac{0.16}{1}\right) = 9°.2$$

Therefore, the loss at the joint due to longitudinal misalignment can be estimated using Eq. (6.52) as

$$L_f = -10\log_{10}\left(\frac{25}{25 + 10\tan(9°.2)}\right) = 0.27\text{ dB}$$

To minimize the undesired joint-loss it is necessary to prepare the end faces of the fibers properly.

6.6.3.2 Joint loss in a single-mode fiber

Similar kinds of misalignment loss occur also in the case of jointing two single-mode fibers. The theoretical analysis concerning the loss associated with jointing two single-mode fibers has been reported (Marcuse 1977; Gambling[1] *et al* 1978; Gambling[2] *et al* 1978).

Exact calculation of joint loss in the case of single-mode fibers in terms of the true HE_{11} mode fields are quite involved and are beyond the scope of the present book. However, the assumption of Gaussian or near Gaussian shape of the propagating modes in a single-mode fiber enables one to obtain simplified expressions for joint loss in the case of lateral and angular misalignment. In the absence of angular misalignment, the joint loss due to lateral misalignment, y of two single-mode fibers can be expressed as (Gambling *et al* 1978)

$$L_l = 2.17\left(\frac{y}{w}\right)^2 \text{ dB} \tag{6.57}$$

where w is the normalized spot-size or mode-field radius of the fundamental mode, HE_{11} given by (Marcuse, 1977)

$$w = a\left[\frac{0.65 + 1.62V^{-3/2} + 2.88V^{-6}}{\sqrt{2}}\right] \tag{6.58}$$

Here, a is the core radius and V is the normalized frequency of the single-mode fiber.

Further, the insertion loss, L_a due to angular misalignment can be obtained as (Gambling *et al* 1978) as

$$L_a = 2.17\left[\frac{\theta w n_1 V}{a\,(\text{NA})}\right]^2 \text{ dB} \tag{6.59}$$

where n_1 is the core refractive index of the single-mode fiber and NA is the numerical aperture. The above derivations are based on the fact that the two single-mode jointed fibers have the same spot size as the fundamental mode.

Example 6.15 Two identical single-mode fibers are jointed with a lateral misalignment of 2 µm. The normalized frequency and core diameter of each fiber are 2.4 and 8 µm each respectively. Calculate the lateral misalignment loss at the joint neglecting Fresnel reflection.

Solution The radius of the fiber core is

$$a = 8/2 = 4 \text{ µm}$$

The normalized spot size of the single-mode fiber can be estimated by using Eq. (6.58) as

$$w = 4 \times 10^{-6}\left[\frac{0.65 + 1.62 \times (2.4)^{1.5} + 2.88 \times (2.4)^{-6}}{\sqrt{2}}\right]$$

$$= 3.11 \text{ µm}$$

The joint-loss due to lateral misalignment can be obtained using Eq. (6.57) as

$$L_l = 2.17\left(\frac{2}{3.11}\right)^2 \text{ dB}$$

$$= 0.89 \text{ dB}$$

6.7 FIBER SPLICES

A permanent joint between two fibers is called a fiber splice. Such types of permanent joints of fiber are needed is long-haul optical communication links. While splicing two fibers, it is necessary to take into consideration the difference between the geometry of the two fibers, their characteristics, and alignment that finally decide the additional loss encountered at the splice. Splices can be divided into two categories, e.g. fusion splice or mechanical splice depending on the technique used for splicing the two fibers. To join two fibers either in the form of a temporary joint or a splice it is necessary to prepare the endfaces of the two fibers properly beforehand.

6.7.1 Endface Preparation

The connector end face preparation determines the overall loss at the connector including the return loss, also known as back reflection. The return loss is measured as the ratio of the optical power reflected back into the emitting fiber (or the optical source as the case may be) to the optical power propagating through the connector in the forward direction. For a good quality joint, the endfaces of the fiber must be perfectly smooth, and circular with the endfaces perpendicular to the axis of the fiber. A good quality endface can be prepared by making use of the conventional grinding and polishing technique. The conventional grinding and polishing technique involves the use of successive polishing of the uneven endface of a freshly cleaved fiber with the help of an abrasive surface with progressively finer grain sizes. Even though for a laboratory environment the conventional technique works well, it is too time-consuming and cumbersome for field work.

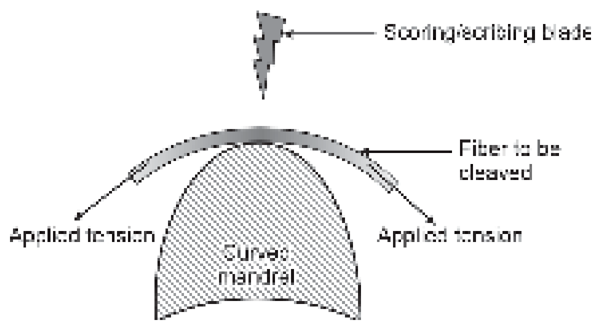

Fig. 6.13 Arrangement for cleaving fiber using controlled fracture technique

A glass fiber can break with flat endfaces perpendicular to the fiber axis by using a controlled fracture technique ([1]Bisbee, 1971; Midwinter, 1979). The enface prepared by this technique reportedly provides loss as low as 3% in the case of multimode fibers ([2]Bisbee, 1971). This technique involves creating a score or a scratch at an appropriate point to create a stress concentration. The fiber is subsequently bent over a curved mandrel as illustrated in Fig. 6.13 by applying simultaneous tension to produce a stress distribution across the fiber. This process is known as score-and-break. Many commercial tools are available in the market for cleaving fibers. The underlying theory of operation of these tools based on controlled fracture is available in the literature (Gloge *et al* 1976). If the stress across the crack initiated in the fiber by scribing is not uniformly distributed, the fractured endface of the fiber tends to produce three regions identified as mirror, mist, and hackle zones. The mirror zone is smooth and occurs in the vicinity of the crack or fracture origin. The hackle zone corresponds to the area where the fracture forks exhibit severe irregularities. The mist zone is the transition region between the mirror and the hackle zones as illustrated in Fig. 6.14(a) (Gloge *et al* 1976). It has been experimentally demonstrated that in controlled fracture, the distance from the origin of the fracture to a point on the boundary between the mirror and the mist zone, r can be expressed as (Johnson *et al* 1966)

$$Z\sqrt{r} = K \tag{6.60}$$

where K is a constant depending on the material and Z is the local stress at the point.

In order to obtain a good quality endface, the mirror zone must extend all over the endface. To achieve this the stress at all points within the fiber must be such that

$$Z\sqrt{r} < K \tag{6.61}$$

The value of Z at the origin of the crack depends on the size of the crack. The value of Z at any point within the fiber cannot be zero or negative. If Z is zero, the crack will cease to propagate. When Z is negative the crack propagates in a direction that is not perpendicular to the fiber axis. Under this condition, a lip comprising a sharp protrusion from the cleaved edge is observed. This is illustrated in Fig. 6.14(b).

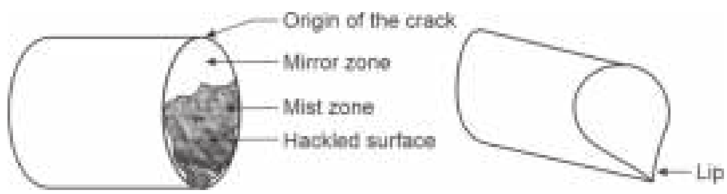

Fig. 6.14 (a) Typical fiber endface prepared by controlled fracture technique showing the formation of different zones (b) Illustrating the formation of lip

6.7.2 Fusion Splices

Fusion splices are obtained by thermal bonding of the properly prepared enfaces of two fibers to be joined (Yamada *et al* 1986; Chrin *et al* 1981; Miller *et al* 1986). This is usually achieved by heating the contact region of the two endfaces to a sufficiently high temperature to their fusing point. Before joining the fiber ends it is necessary to strip off the cabling and buffer coatings. The endfaces of the two fibers are properly positioned and aligned with the help of micromanipulators or grooved fiber holders. The butted ends of the fiber are fused with the help of an electric arc or a laser pulse so that the fiber ends momentarily melt and get bonded together permanently. A simple arrangement of the fusion splicing system is shown in Fig. 6.15. Fusion splices generally produce low splice loss (less than 0.1 dB). In general, fusion splicing takes more time than mechanical splicing. Most importantly, the quality and splice time largely depend on the skill and expertise of the fusion splice operator. Only a highly trained fusion splice operator can make consistently low-loss fusion splices. Fusion splicing often makes use of the prefusion technique which eliminates the necessity of fiber endface preparation. This technique involves the application of electric discharge before pressing the fiber endfaces together. The low-energy arc discharge helps in rounding the fiber end faces due to heating. Once the ends are rounded the fibers are pressed together and a strong electric arc is applied to cause fusion of the joint resulting in a permanent joint. The steps are illustrated with the help of schematic diagrams shown in Fig. 6.16 (Botez *et al* 1980).

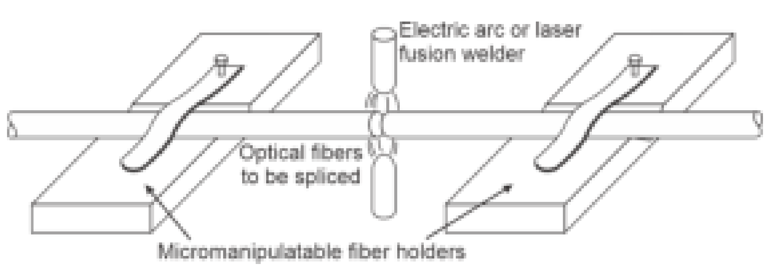

Fig. 6.15 Schematic of a typical fusion splicing arrangement

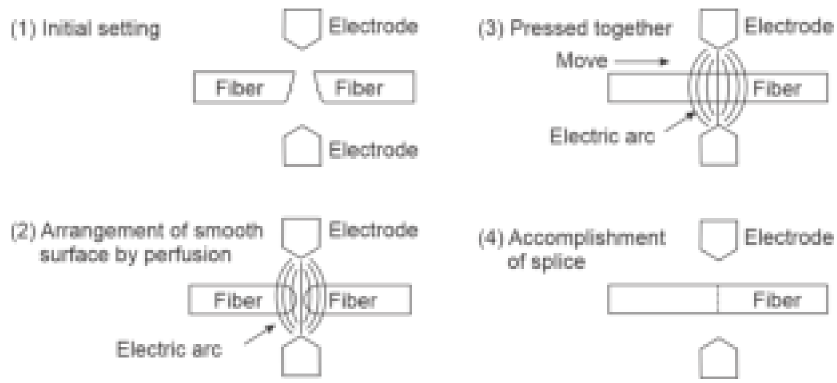

Fig.6.16 Schematic showing the steps involved in prefusion method of fusion splicing (after Botez *et al* 1980)

6.7.3 Mechanical Splices

Mechanical splices are generally of two types, e.g. V-groove splice and elastic-tube splice. The V-groove splice can be achieved by butting the duly prepared endfaces of the fibers in a V-shaped groove that is usually etched on a solid substrate made of silicon, metal, ceramic, or plastic. The endfaces are then bonded together with the help of a transparent adhesive such as epoxy. The V-groove enables self-alignment of the fibers and usually offers joint losses of the order of 0.1 dB (Midwinter, 1979). The schematic of a V-groove splice is illustrated in Fig. 6.17 (Exfiber, 2010). The epoxy holds the fiber in position. However, *V*-groove splices generally use a cover plate with a locking mechanism to protect the fiber joint. This is illustrated in Fig. 6.18 (Exfiber, 2010). The V-groove mechanical splice does not require any special tool other than a fiber stripper and cleaver. It requires little skill and much less time as compared to

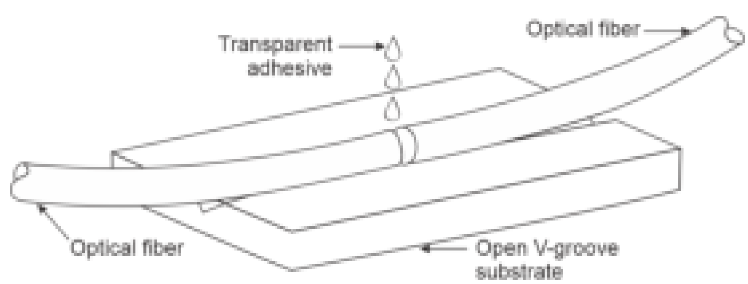

Fig. 6.17 Schematic of a V-groove optical fiber splicing system

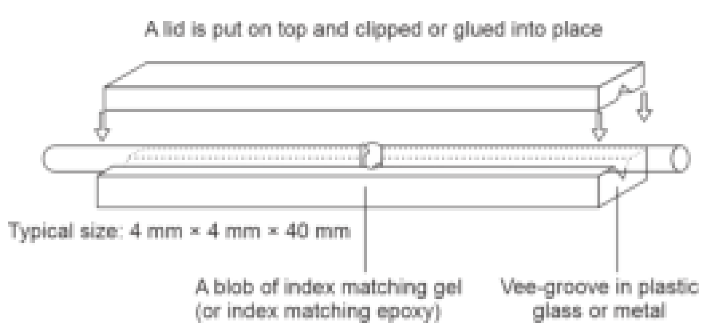

Fig. 6.18 V-groove mechanical splice with a protecting lid on the top (after Exfiber, 2010)

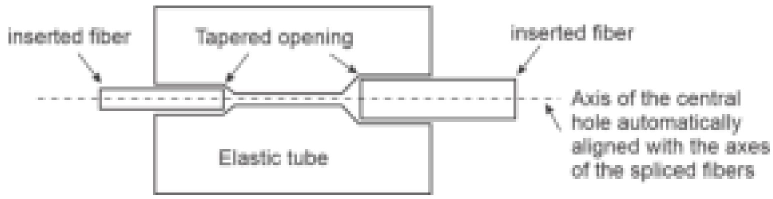

Fig. 6.19 Elastictube mechanical splice

those required in the case of fusion splices. Mechanical V-groove splices generally offer higher insertion losses as compared to their fusion counterparts. Mechanical splices are generally preferred in the case of multimode fibers. For bulk splicing, mechanical splicing turns out to be more expensive than fusion splicing. Mechanical splicing is more suitable for temporary or semi-permanent joints.

The other form of mechanical fiber is an elastictube splice shown in Fig. 6.19. It uses a single tube of an elastic material with a central hole having a diameter slightly smaller than the outer diameters of the fibers to be spliced. The central hole is tapered in either direction for the insertion of fibers to be spliced. Insertion of fiber tends to enlarge the diameter of the central hole by putting pressure on the elastic tube which in turn exerts a symmetric force on the fiber. The elastic nature of the tubes enables automatic adjustment of the fiber endfaces with respect to longitudinal, lateral, and angular alignment. One of the major advantages of elastic-tube splice is that it can be used to joint two fibers with different diameters. This technique needs no special skill or apparatus and can offer losses in the same range as those offered by fusion splices.

6.7.4 Multiple Splices

Ribbon optical fiber cables are being increasingly deployed in place of cylindrical conventional optical fiber cable in situations such as campus LAN and data center network backbones. This is because ribbon optical fiber cables provide the highest fiber density relative to cable size and easy termination. A typical ribbon optical fiber cable is illustrated in Fig. 6.20.

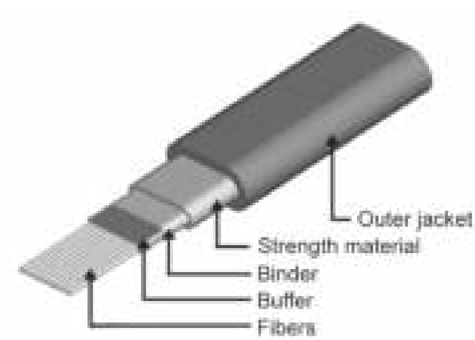

Fig. 6.20 Ribbon optical fiber cable showing 12 fiber strands (after commercial fiber optic ribbon GORE^{TM} make)

It is possible to splice simultaneously an array of optical fibers in a ribbon cable comprising multimode or single-mode fibers (Kawase *et al* 1982; Katsuyama *et al* 1985). An array of five fibers in a ribbon cable was spliced using an electric arc fusing device to provide simultaneous fusion. This type of splicing device is now commercially available. A typical multiple fusion splicing arrangement is shown in Fig. 6.21.

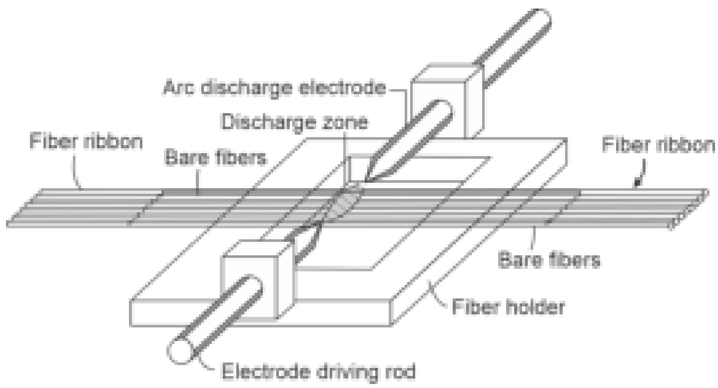

Fig. 6.21 Multiple fusion splicing arrangement

The simplest and most cost-effective multiple simultaneous splicing of a ribbon-type fiber cable comprising multimode fibers is accomplished by mechanical splicing (Miller *et al* 1986). The technique involves the stripping of ribbon and buffer coating from the fibers. The twelve fibers of a standard commercial ribbon cable are then laid into the trapezoidal grooves etched on a silicon substrate using a comb structure. The top cover silicon chip is then positioned before applying the epoxy gel to the chip-ribbon interface. After curing, the endface is prepared by grinding and polishing (Senior, 2008).

6.8 FIBER CONNECTORS

Demountable optical fiber connectors are required primarily to allow for repeated connection and disconnection of the transmission line at the joints or terminal equipment without degradation of performance. The design of such connectors needs utmost care to ensure accurate alignment every time without damaging the fiber ends during repeated make-and-break of the joint. These connectors are intended to provide low coupling loss, easy connection, and compatibility with connectors from different manufacturers. The biggest problem with the fiber optic connector is the compatibility. There is no uniform standard in the design of the connectors across the manufacturers. The type and specification depend on the manufacturer and the country of origin. A majority of the fiber connectors are like plugs comprising a male connector with a protruding ferrule that fits in a mating adapter and holds and aligns two fibers for mating. The ferrule design also helps one to connect directly to active components such as optical sources, detectors, optical amplifiers, and other optical network components.

Historically, the earliest types of demountable connectors are so-called biconic connectors. The biconic connector introduced by AT & T, Murray Hill, NJ, was molded from a glass-filled plastic (Fig. 6.22). In the original structure, the fiber used to be molded in the ferrule itself. Later on, the connector was slightly modified by inserting a 125 μm/5 mil pin in the plastic mold, and the fiber was glued into the ferrule with epoxy. This type of connector was not keyed and as a result, could rotate in the mating adapter. When mated, such connectors used to have an air gap between the mated fibers resulting in additional reflection loss.

Fig. 6.22 Biconic fiber connector

Biconic connectors for multimode fibers were used to offer losses of the order of 0.5–1 dB and those for single mode counterparts offered loss over 0.7 dB.

The fiber connectors are generally butt-type joints. Either a straight sleeve or tapered sleeve mechanism is used for the alignment of both single as well as multimode fibers in butt-joints. The schematics of these joint mechanisms are illustrated in Fig. 6.23.

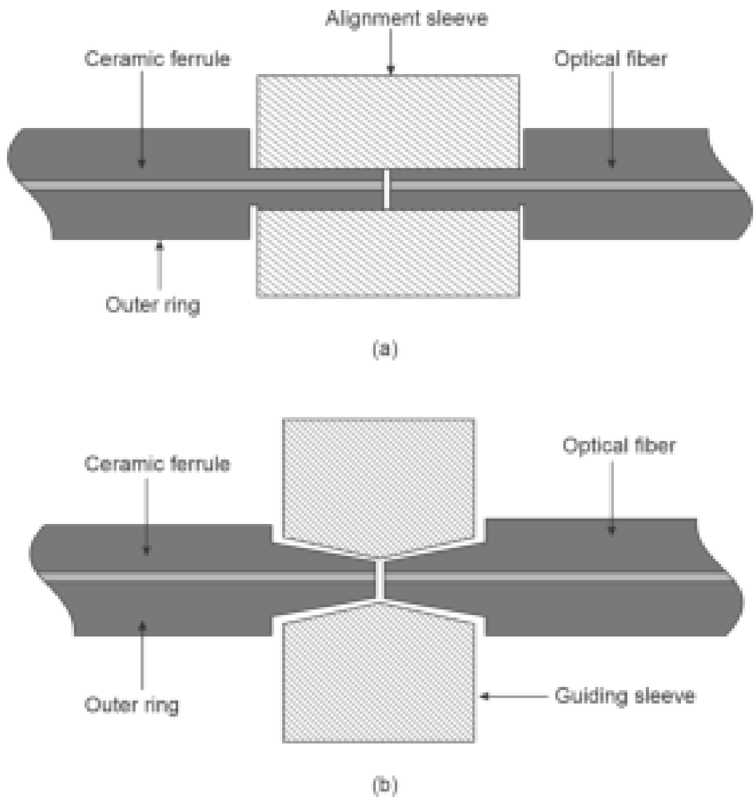

Fig. 6.23 Schematic of alignment schemes for butt-type connector (a) Straight sleeve (b) Tapered sleeve

6.8.1 ST Connector

The straight-tip ST (a trademark of AT & T) connector has been the first fiber optic connector that found widespread applications in optical networks (Fig. 6.24). ST connectors make use of a 2.5 mm ferrule to hold the fiber. The ST connectors are generally available with a

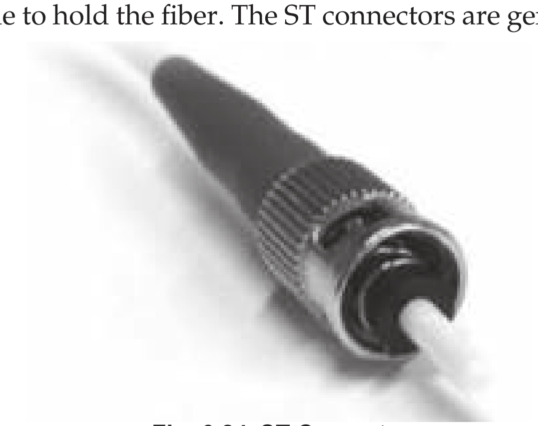

Fig. 6.24 ST Connector

round plastic or metal body. Unlike the biconic connector, the ST connector stays in place with a spring-based twist-on/twist-off bayonet-style mechanism.

6.8.2 SC Connector

The straight tip (ST) connectors subsequently got replaced by subscriber connectors (SC) developed by Nippon Telegraph & Telephone Corporation, Japan. SC connectors have an easier push-on/pull-off mechanism for mating as compared to the twist mechanism used in ST connectors (Fig. 6.25). The SC connectors are packaged in a square-shaped outfit with the same 2.5 mm ferrule for holding the fibers. In duplex form two SC connectors are housed together with a plastic clip.

(a) (b)

Fig. 6.25 SC connector (a) Single (b) Duplex

Normal SC and ST connectors are relatively large and are inconvenient for use in multiport units in a rack or closet situation. Different forms of small form factor (SFF) connectors were developed to increase the port density.

6.8.3 LC Connector

Lucent Technologies developed LC connectors which use a tab mechanism similar to one used in telephones or RJ45 connectors (Fig. 6.26). The body of the connector looks more like an SC connector with a square getup but is much smaller in size. LC connectors use 1.25 mm ferrule and are also available in a duplex pack put together with the help of a plastic clip.

Fig. 6.26 LC connector illustrating a single and a duplex package

6.8.4 FC Connector

Fixed Connector (FC) is another form of SFF connector and is most extensively used as a single-mode. FC connectors come with a threaded barrel housing and can be screwed on and fixed firmly on the adaptor and hence the name. They are suitable for application in

Fig. 6.27 FC connector

high-vibration environments. It has a 2.5 mm ferrule for holding the fiber and is the most popular connector for LAN closets. For matched FC connectors, the insertion loss is of the order of 0.25 dB. A photo-image of a typical commercial FC connector is shown in Fig. 6.27.

6.8.5 Mating Dissimilar Connectors

It is seen that there is a uniform standard in the connectors designed and developed by different manufacturers. It often creates practical problems in interfacing and terminating optical components and devices from different manufacturers. However, the ST, SC, and FC connectors share the common feature that all of them use the same 2.5 mm ferrule design. It is therefore possible that they can be mated to each other with the help of hybrid mating adapters shown in Fig. 6.28 (in the order from top-to-bottom: ST-FC; SC-FC and SC-ST) (Fiber Optic Association, 2005).

Fig. 6.28 Photo image of hybrid adaptors

6.8.6 Expanded Beam Connector

An alternative approach to butt-joint connector is an expanded beam connector which employs lenses to collimate/focus the light beam from and into the fiber. The mechanism of operation is schematically shown in Fig. 6.29. In this case, the light emerging from the emitting fiber is collimated by the lens on the left-hand side. The collimated beam is subsequently focused by the second lens to the core of the receiving fiber. The fibers should be situated at the focal point of the respective lens for the connector to work properly. The major advantages of an expanded beam connector over a butt-joint connector include less critical longitudinal and lateral alignment requirements in the former case. However, the angular alignment requirement in the case of an expanded beam connector becomes more stringent. The added advantages of an expanded beam connector include the flexibility

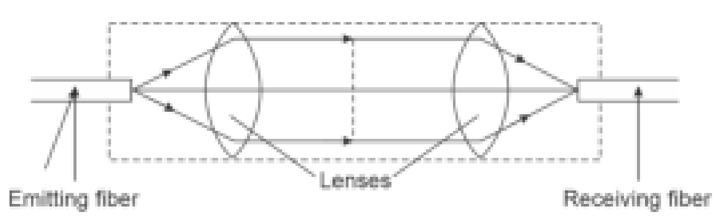

Fig. 6.29 Schematic showing the principles of expanded beam connector

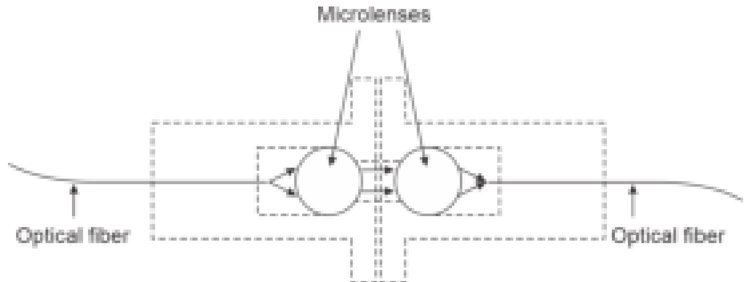

Fig. 6.30 Expanded beam connector using spherical microlenses (after Nicia, 1978)

of incorporating beam-splitting devices, switches, etc. within the connector. A variety of lens-coupled expanded beam connectors have been reported for beam collimation and focusing for multimode as well as single-mode fibers based on the principle discussed above (Nicia, 1978; Knecht *et al* 1983). An expanded-beam connector involving two spherical microlenses is shown in Fig. 6.30.

6.8.7 GRIN Lens Connector

Another popular beam expansion/collimation type of connector can be obtained with the help of a graded-index (GRIN) rod lens (Tomlinson, 1980). GRIN lenses are useful for source-to-fiber coupling as well as fiber-to-fiber coupling. The GRIN-rod lens is essentially a cylindrical glass rod with 1–2 mm diameter having a parabolic refractive index profile with a maximum at the axis. This characteristic is very similar to that of a graded index fiber. The propagation of light through a GRIN rod lens can be precisely controlled by adjusting the dimensions of the lens and exploiting the dependence of the refractive index on the wavelength of the propagating light. The lens can either be used for collimating the light beam with an adjustable divergent angle by suitably positioning the light source on or near the opposite lens face as shown in Fig. 6.31. Alternatively, the GRIN rod lens can be used to focus a collimated beam to the core of the receiving fiber.

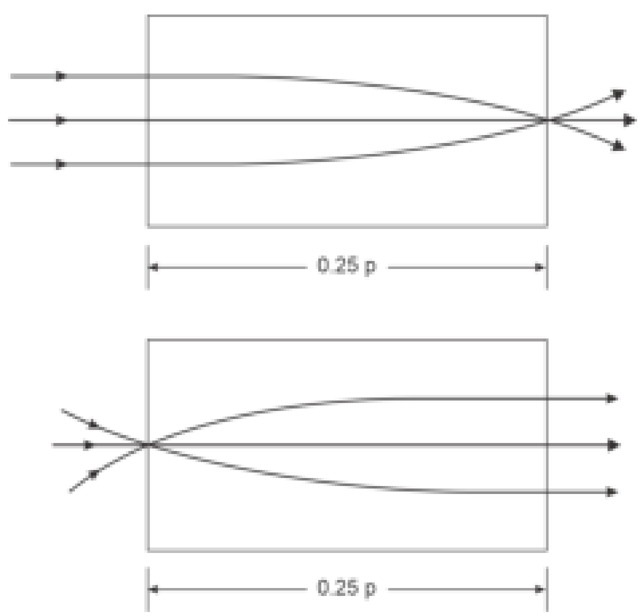

Fig. 6.31 Path of light rays through quarter-pitch GRIN rod lens

The function of a GRIN rod lens can be understood with the help of a simple mathematical analysis of the propagation of light through a graded-index medium (Senior, 2007). The refractive index variation with radius in a GRIN rod lens causes all incoming rays to follow a sinusoidal path through the lens medium. One full pitch is equivalent to the traversing of a path equal to one sinusoidal period. A GRIN rod lens with a quarter pitch corresponds to a traversed path by the light ray equivalent to a quarter of a sinusoidal period. The path of light rays through 0.25p GRIN rod lens is shown in Fig. 6.31. The pitch of the GRIN rod lens can be suitably adjusted to position the focal point outside the lens. It is also possible to adjust the pitch of the GRIN rod lens suitably in order to convert a divergent beam into a convergent beam focused outside the lens. This is illustrated in Fig. 6.32. This type of GRIN rod lens is useful for coupling a laser source to a fiber.

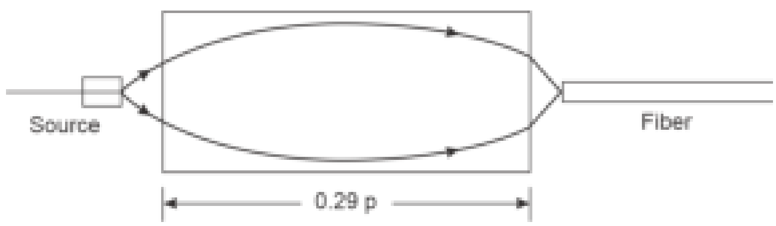

Fig. 6.32 Use of GRIN rod lens for coupling light from a laser source to an optical fiber

6.8.8 Return Loss Characteristics of Connectors

A variety of optical fiber connectors are discussed in the preceding section. The connectors generally involve two types of contacts e.g. perpendicular end face or oblique endface contacts. Either type of contact may have physical contact or have an index-matching fluid in between the endfaces to reduce the reflection loss. A fiber connection is desired to have a high return loss (low reflection level) from the joint. This is because a low reflection loss (high return loss) not only allows a larger amount of power transmitted through the joint but also reduces the unwanted power feedback to the laser cavity. The undesired returned power from the joints into the laser cavity may cause the laser source to malfunction leading to the degradation of the spectral response, linewidth, power output, and enhancing the undesired laser noise components.

Consider an ideal model of an index-matched fiber connection with perpendicular fiber end faces shown in Fig. 6.33 (Kihara *et al* 1996). The model assumes the presence of a thin sur face layer of thickness h with a high refractive index, n_2 larger than the core refractive

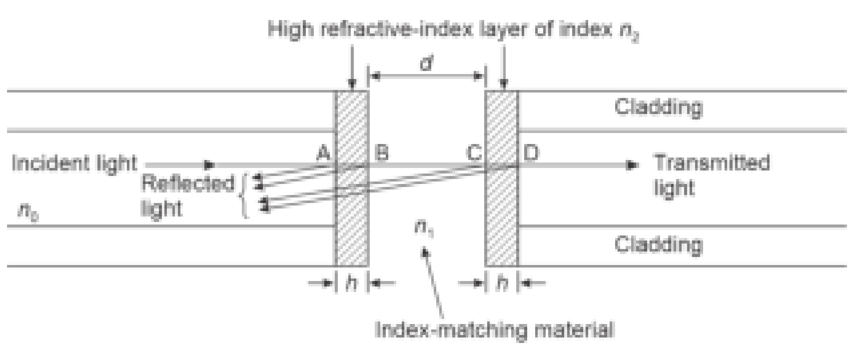

Fig. 6.33 A schematic illustrating the index-matched connection with perpendicular end faces

index, n_0. This is attributed to fiber polishing. The gap, d between the endfaces is assumed to be filled with an index matching fluid of refractive index, n_1. Assuming that there is no other misalignment at the joint, the return loss at the index-matched joint in dB can be expressed as (Kihara *et al* 1996)

$$RL_{IM}(\text{dB}) = -10\log_{10}\left[2R\left\{1-\cos\left(\frac{4\pi n_1 d}{\lambda}\right)\right\}\right] \tag{6.62}$$

where λ is the operating wavelength and R is the reflectivity of a single material-coated endface given by

$$R = \frac{r_1^2 + r_2^2 + 2r_1 r_2\cos\delta}{1 + r_1^2 r_2^2 + 2r_1 r_2\cos\delta} \tag{6.63}$$

Here,

$$r_1 = \frac{n_0 - n_2}{n_0 + n_2} \qquad \text{and} \qquad r_2 = \frac{n_2 - n_1}{n_2 + n_1} \tag{6.64}$$

correspond to the reflection coefficient[6] at the core from the high-index layer and that at the high-index layer from the index matching fluid respectively, δ being the phase difference in the high-index layer. It can be easily seen that the return loss depends on the characteristics (thickness as well as refractive index) of the high-index layers on the endfaces and also on the gap between the endfaces and the refractive index of the index matching fluid.

In the absence of any gap between the endfaces, that is, in case of physical contact of the endfaces the return loss can be expressed as

$$RL_{PC} = 10\log_{10}\left[2R_2\left\{1-\cos\frac{4\pi n_2}{\lambda}2h\right\}\right] \tag{6.65}$$

where R_2 is the reflectivity at the discontinuity between the fiber core and the high refractive index

$$R_2 = \left(\frac{n_0 - n_2}{n_0 + n_2}\right)^2 \tag{6.66}$$

In addition to the connectors discussed in this chapter, a variety of optical couplers are available for the interconnection of optical components and devices. These couplers are discussed in the later part of the text in the context of integrated optics.

Summary

- The launching of power from the source to the fiber and from one fiber to the other is discussed.
- The coupling efficiency at the source-fiber interface can be expressed as:

$$\eta = \frac{P_F}{P_S}$$

- Optical sources are characterized in terms of brightness or radiance which is equal to the amount of power radiated by the source per unit area per unit solid angle (W/m^{2}sr)
- The brightness of a Lambertian source can thus be expressed as:

$$B(\theta, \phi) = B_0\cos\theta$$

  where, B_0 is the brightness of the source along the direction normal to the emitting surface.

[6] Here the reflection coefficient correspond to the ratio of incident amplitude and reflected amplitude of the wave and is not the power ratio.

- The brightness or radiance of a non-ideal Lambertian source is often represented in the form which is as follows:

$$B(\theta, \phi) = B_0 \cos^m \theta$$

where m is an integer.

- The brightness of a laser diode is expressed as:

$$\frac{1}{B(\theta, \phi)} = \frac{\sin^2 \phi}{B_{\perp}(\theta)} + \frac{\cos^2 \phi}{B_{\parallel}(\theta)}$$

where, $B_{\perp}(\theta)$ and $B_{\parallel}(\theta)$ correspond to the radiance measured in the planes perpendicular and parallel to the junction planes.

- The power coupled by an LED to a step-index fiber is proportional to the square of the numerical aperture given by,

$$(P_F)_{\text{LED–SI}} = P_S(\text{NA})^2 \qquad \text{for} \quad r_S \le a$$

$$(P_F)_{\text{LED–SI}} = \left(\frac{a}{r_S}\right)^2 P_S(\text{NA})^2 \quad \text{for} \quad r_S > a$$

- The power coupled from an LED to a GI fiber can be expressed as

$$2P_S n_1^2 \Delta \left[1 - \frac{2}{\alpha + 2}\left(\frac{r_S}{a}\right)^{\alpha}\right]$$

- The Fresnel reflection coefficient at the interface of two media (with refractive indices of n_0 and n_1) for normal incidence is as follows:

$$R = \left(\frac{n_1 - n_0}{n_1 + n_0}\right)^2$$

This reflection occurs at the interface of the source and the fiber end at the time of launching power.

- The power loss at the interface in dB is,

$$L(\text{dB}) = -10 \log \left(\frac{P_{\text{coupled}}}{P_{\text{emitted}}}\right) = -10 \log (1 - R)$$

- Coupling between source and the fiber can be improved with the help of microlens arrangement (spherical, cylindrical, etc.).
- For long-haul communication it is necessary to have fiber joints at multiple points. In addition, joints are also required for connecting fibers to terminal equipment.
- A fiber joint can be of temporary nature and this type of joint is called demountable joint.
- Permanent fiber joints are called splices. Splices can be of mechanical type or fusion type depending on the requirement.
- The characteristics of the two fibers to be joined may not be identical. The fiber which carries the light is called emitter fiber and the other one is called the receiving fiber.
- Fiber-to-fiber coupling efficiency is,

$$\eta_F = \frac{M_{\text{common}}}{M_E}$$

where, M_{common} refers to the common mode volume between the two fibers and M_E is the total number of modes supported by the emitting fiber.

- The loss encountered at this fiber-to-fiber joint is

$$L_F = -10 \log_{10} (\eta_F)$$

- Additional coupling loss may occur due to an improper alignment of the fibers. Fiber misalignments are of three types, e.g. axial, longitudinal, and angular misalignment.
- For an axial or lateral off-set of d, between two identical SI fibers of core radius a, the coupling loss due to the misalignment is

$$\eta_{\text{lat}} = \frac{2}{\pi}\cos^{-1}\left(\frac{d}{2a}\right) - \frac{d}{\pi a}\left(1-\left(\frac{d}{2a}\right)^2\right)^{\frac{1}{2}}$$

- For a longitudinal gap of z between the end-faces of two identical fibers of core radius a, the coupling loss is,

$$L_f = -10\log_{10}\left(\frac{a}{a+z\tan\theta_c}\right)$$

  where, θ_c is the critical acceptance angle of the fiber and a is the radius of the core of each fiber.
- For single mode fibers the joint-loss due to lateral misalignment (in absence of angular misalignment) is,

$$L_l = 2.17\left(\frac{y}{w}\right)^2 \text{ dB}$$

  where, w is the normalized spot-size or mode-field radius of the fundamental mode, HE_{11}.
- Before joining two fibers it is necessary to prepare the end faces of the two fibers in such a way that the endface of each fiber is perpendicular to the axis and they are free of any scratch or discontinuity.
- A variety of optical fiber connectors of demountable type are available commercially from different manufacturers. A few of the commonly used connector types include biconical, SC, ST, LC, FC.
- Dissimilar connectors can be mated with the help of mating adaptors.
- Fibers can also be coupled optically with the help of microlenses or GRIN rod lens.
- Connector return-loss often turns out to be a major issue in optical fiber communication.

Problems

6.1 An ILD has the following dependence of the radiance in the transverse and lateral directions.

$$B_\perp(\theta) = \cos^n\theta \quad \text{and} \quad B_\parallel(\theta) = \cos^m\theta$$

If the transverse and lateral half-power beam widths are 5° each, find the values of transverse and lateral power distribution coefficients.

6.2 An edge-emitting LED has the following dependence on the radiance in the transverse and lateral directions.

$$B_\perp(\theta) = \cos^3\theta \quad \text{and} \quad B_\parallel(\theta) = \cos^{120}\theta$$

Estimate the half-power beam widths of the source in the transverse and lateral directions.

6.3 A surface-emitting LED with a circular emitting area of radius 20 μm has an axial radiance of 10^6 Wm^{-2}sr^{-1} at a given drive current operating at 850 nm. The light emitted from the LED is to be coupled to a 60/125 μm step-index fiber with a core refractive index of 1.458. If the power coupled to the fiber under ideal conditions is 170 μW, estimate the value of the index deviation of the fiber.

6.4 It is intended to couple power from a surface-emitting LED with a circular emitting area of radius 25 μm to a 60/125 μm step-index fiber. If the power needed to be coupled to the fiber is 250 μW, then estimate the value of the axial radiance of the LED needed to meet this requirement. The other coupling losses may be ignored.

6.5 Consider a 50/120 μm step-index fiber with a numerical aperture of 0.2. Power is coupled to the fiber from an LED of circular emitting area of radius 30 μm with an axial radiance of 150×10^4 Wm^{-2}sr^{-1}. Estimate the amount of power coupled by the source into the fiber. If the radius of the emitting area of the source is increased to 50 μm without changing the axial radiance estimate the amount of power coupled into the fiber.

6.6 A non-ideal Lambertian source has the following relationship between the radiance and viewing angle

$$B(\theta, \phi) = B_0 \cos^3\theta$$

Obtain an expression for the power coupled from this source to a step-index fiber for the following two cases

(i) $r_S < a$ (ii) $r_S > a$

6.7 A parabolic index fiber has a core refractive index of 1.5 and an index deviation of 1%. Plot the variation of the power coupled to the GI fiber from an LED normalized the total output power emitted by the source.

6.8 Light is coupled from a surface-emitting LED made of GaAs active region to a stepindex fiber with a core refractive index of 1.458. If GaAs has a refractive index of 3.58, estimate the value of Fresnel reflection at the endface of the fiber core by assuming the fiber endface to be placed in the closest proximity of the fiber-emitting surface. If the space between the emitting area of the source and the fiber endface is filled with an adhesive of refractive index is 1.3, find the new value of Fresnel reflection at the endface. Also, estimate the coupling loss in dB.

6.9 Compare and contrast the values of power radiated per mode by two different sources with the same emission area and same brightness but one operating at 940 nm and the other at 1300 nm. How many modes are supported by a step-index fiber of NA = 0.2 and a core radius of 25 μm at the above wavelengths?

6.10 A non-imaging spherical microlens is to be used for coupling power from an LED source into an optical fiber. The refractive index of the lens is 2.0 and the surrounding medium is air. Show that the source needs to be placed at a distance equal to twice the value of the radius of curvature of the lens measured from the vertex of the lens. Draw a neat sketch of the microlens arrangement.

6.11 Two step-index fibers with core refractive indices of 1.458 and 1.5 are butt jointed. The gap between the endfaces of the fibers is filled with air ($n = 1$). Estimate the total Fresnel loss at the interface under ideal conditions.

6.12 Estimate the reduction in the coupling loss in the above case (Problem 6.11) when the air gap is filled with an index matching fluid of refractive index 1.34.

6.13 Two identical step-index fibers are jointed with an axial misalignment of 20% of the core radius. Estimate the coupling loss at the joint by ignoring the Fresnel reflection loss at the joint.

6.14 Two identical step-index fibers with a core radius of 25 μm each are jointed with a lateral misalignment of 5 μm. Estimate the coupling loss at the joint by assuming that the gap between the end faces of the emitting and the receiving fibers at the joint is filled with an index-matching fluid with a refractive index of 1.35. The refractive index of the core of each fiber is 1.5.

6.15 Two 50/125 μm SI fibers are jointed using a transparent adhesive with a refractive index of 1.40. If the axial misalignment at the joint is 5 μm, estimate the coupling loss at the joint assuming the core refractive index of the emitting fiber to be 1.5 and that of the receiving fiber is 1.46.

6.16 Two identical 60/200 μm graded-index fibers with parabolic index profiles ($\alpha = 2$) are jointed with a lateral offset of 10% of the core radius. Estimate the joint loss in dB for this lateral offset assuming uniform power distribution over all the emitted modes and neglecting the effect of Fresnel reflection at the joint.

6.17 Compare and contrast the loss obtained in Problem 6.16 with that obtained for the case of an identical condition stated in Problem 6.16 but involving two corresponding step-index fibers.

6.18 Two step-index fibers of radius 25 μm each are jointed with an angular offset of 2° at the joint. If the gap between the jointed endfaces of the fibers is filled with air, calculate the loss at the joint assuming the NA of each fiber to be 0.16.

6.19 Two step-index fibers of radius 25 μm each are jointed with a longitudinal misalignment of 10 μm. If the gap is filled with an index matching fluid of refractive index 1.4, calculate the loss at the joint assuming the NA of each fiber to be 0.16. Neglect Fresnel reflection at the interfaces.

6.20 Two step-index fibers each with a radius of 25 μm and a numerical aperture of 0.16 are joined with an angular misalignment of 5º. If the gap is filled with air, calculate the loss at the joint due to angular misalignment neglecting Fresnel reflection loss at the interfaces.

6.21 Two graded-index fibers with identical values of core radius and refractive index profile parameter are jointed. The axial numerical aperture of the emitting fiber is 0.16 and that of the receiving fiber is 0.18. Estimate the joint loss by neglecting the Fresnel reflection loss.

6.22 Two identical single-mode fibers are joined with a misalignment of 5º. The normalized frequency and core diameter of each fiber are 2.4 and 8 μm each respectively. Calculate the lateral misalignment loss at the joint neglecting Fresnel reflection and assuming the numerical aperture of each fiber to be 0.16.

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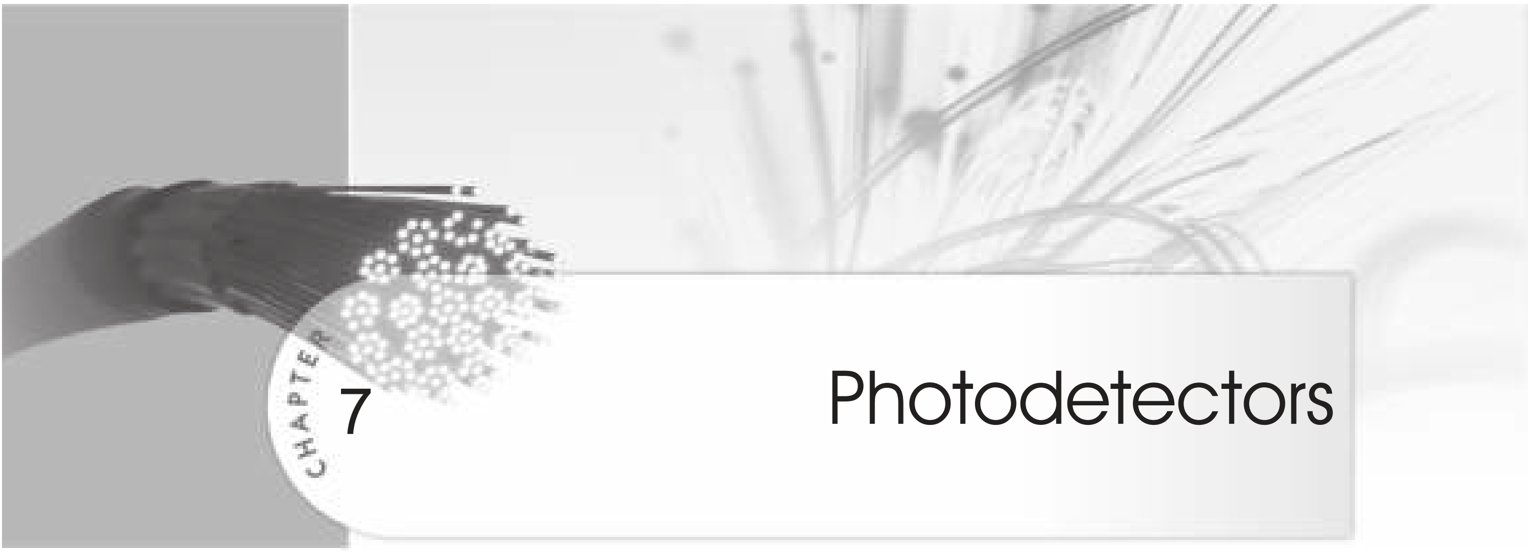


CHAPTER 7 Photodetectors

Photodetectors in general, find applications in a wide range range of systems from gas sensing instrumentation, and imaging arrays to optical receivers of both optical fiber and free-space optical communication systems. In all these applications, the function of a photodetector is to convert the received optical signal into its electrical counterpart. The performance requirement of a photodetector largely depends on the type of application.

7.1 PHOTODETECTOR REQUIREMENTS FOR OPTICAL COMMUNICATION

In this chapter, we, however, confine our discussion to those photodetectors which are used in optical fiber communication systems. A photodetector is a key component of an optical receiver where the optical signal is converted to an electrical signal and subsequently processed by associated electronic circuits. The use of a photodetector in a digital optical receiver is illustrated in Fig. 7.1. The photodetector receives the transmitted optical pulses containing information (such as voice, video, or computer data) impressed on it and converts it into an electrical signal that is supposed to be a replica of the original signal. However, in practice, the signal received by the receiver is generally weak and distorted depending on the nature of the channel (optical fiber in this case). The weak mutilated electrical signal extracted by the detector is further amplified and refined by subsequent stages of the receiver before being delivered to the output. From this application point of view, a photodetector is a transducer that converts a signal from the optical domain to the electrical domain. This process is known as optical-to-electrical (O–E) conversion. A reverse conversion from the electrical to-optical (E–O) domain is generally achieved in an optical fiber communication system by an optical source (e.g. light-emitting diode or injection laser diode) at the transmitter end.

So far as the application of a photodetector in an optical fiber communication system is concerned, the physical size of the detector must be compatible with the tiny cross-sectional size of the coupling fiber (typically human hair diameter of the order of 100 μm or so) and other associated electronic components for easy packaging and integration. This compatibility requirement makes semiconductor photodetectors as the detectors of choice over other bulky detectors such as photomultiplier tubes. Our focus in this chapter is primarily on semiconductor photodetectors used in optical fiber communication systems. It may be pointed out that in optical fiber communication systems, photodetectors are

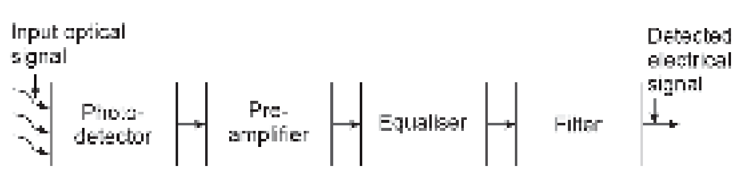


Fig. 7.1 Block diagram of a digital optical receiver

sometimes used to provide O–E feedback for controlling purposes such as maintaining a constant output of a laser source. Semiconductor photodetectors work on the principle of internal photoelectric effect. The photons of the incident optical signal having energy equal to or larger than the bandgap energy of the semiconductor material used in the detector are absorbed. This results in the excitation of electrons from the valence band to the conduction band creating an electron–hole pair. These excess carriers on and above the existing thermally generated carriers are called photo-generated carriers. The photo-generated carriers are subsequently transported over the region of their generation and are finally extracted in the external circuit with the help of a suitable biasing arrangement. The photogenerated carriers ultimately appear in the external circuit in the form of a photocurrent or a photovoltage which is subsequently processed and interpreted.

7.2 TYPES OF PHOTODETECTOR

There exists a variety of semiconductor photodetectors that are used in optical fiber communication systems. There are various ways of classifying these semiconductor photodetectors. From the structural view point, the photodetectors can be divided into two categories, e.g. (i) *bulk and (ii) junction photodetectors*. Photoconductive detectors belong to the bulk category while the host of other semiconductor photodetectors including phototransistors belong to the latter category. In general, photoconductive detectors are relatively slow and unsuitable for use in high-speed optical communication systems. Although all photodetectors work on the general principle discussed earlier, some of the photodetectors involve internal mechanisms for multiplying the photogenerated carriers. From this view point, photodetectors can be classified under two heads, viz. *non-multiplying* and *multiplying photodetectors.* A simple *p-n* junction photodiode and a *p-i-n* detector are examples of the former category while an avalanche photodiode (APD) is an example of a multiplying photodetector. Photodetectors are further classified into *intrinsic* and *extrinsic* categories depending on whether the energy of the absorbed photon is close to the bandgap or much less than that. In intrinsic photodetectors, photogenerated electron and hole pairs are created by the direct transition of electrons from the valence band to the conduction band (band-to-band transition) by consuming the energy of photons larger or equal to the bandgap energy of the semiconductor material. On the other hand, in extrinsic photodetectors, photons excite electrons from the deep-level traps to the conduction band or from the valence band to deep impurity or defect level leaving holes back in the valence band. These carriers are finally transported and extracted in the external circuit as usual. The intrinsic photodetectors are preferred over the extrinsic counterparts owing to the efficient absorption of photons and fast response speed (Bhattacharya, 1997). There is another variety of extrinsic photodetectors that make use of transitions between energy sub-bands created by a quantum well. Quantum well (QW) photodetectors belonging to this category usually absorb photons of energy (~100 meV) which is much less than the bandgap energy of the bulk materials and thereby detect optical signals in the longer wavelength such as FIR region of the optical spectrum. The performance of this detector is limited by the polarization of light that can be absorbed (Bhattacharya, 1997). The photoabsorption mechanisms involving the three basic processes discussed above are illustrated in Fig. 7.2. The principal materials used for making semiconductor photodetectors are the elemental semiconductors (like Si and Ge) as well as a host of III-V (GaAs and related materials), II-VI (HgCdTe and other related materials) alloy semiconductors. One of the important properties of alloy semiconductors is that

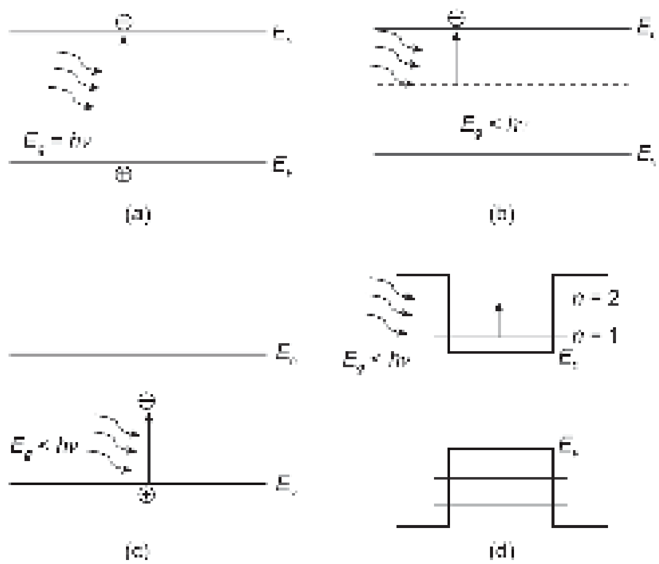


Fig. 7.2 Illustrating intrinsic and extrinsic mechanisms involving (a) Intrinsic band-to-band transition (b) Extrinsic deep level to conduction band transition (c) Extrinsic valence band to deep level transition (d) Extrinsic inter-subband transition in a quantum well

these materials can form useful heterojunctions (junctions between two dissimilar semiconductors with the same or different types of conductivity) with allied lattice-matched alloys. These heterojunction devices provide improved performance over conventional photodetectors made of the same semiconductor. From this angle, photodetectors are often classified as *homojunction* (conventional detectors involving the same materials on the *p*- and *n*-side) or *heterojunction* photodetectors. More often semiconductor photodetectors are realized in the heterojunction form particularly when they are made of alloy semiconductors. Both non-multiplying and multiplying photodetectors can be realized in the heterojunction form. The avalanche photodetectors realized in a heterojunction form are often called a heterojunction avalanche photodiode (HAPD) (Bhattacharya, 1997).

7.3 PRINCIPLE OF PHOTOGENERATION

From the foregoing discussion, it is apparent that in intrinsic photodetectors the minimum energy of the photons, correspondingly the frequency or wavelength of the photons that would be absorbed by the semiconducting material constituting the photodetector depends on the bandgap of the material. The longest wavelength corresponding to the shortest frequency (lowest energy photons) that can be absorbed by the semiconducting material can be obtained as (Keiser, 2000)

$$h\nu_{\min} = E_g \tag{7.1}$$

that is,

$$\lambda_{\max} = \frac{1.24}{E_g(\text{eV})}\,\mu\text{m} \tag{7.2}$$

The value of $\lambda_{\max}$ is identified as the long wavelength cut-off of the material. Ideally, the bandgap energy of the photodetector material is chosen slightly less than the photon energy corresponding to this longest wavelength (Bhattacharya, 1997) to ensure high absorption leading to a good photoresponse of the detector.

Optical radiation is generally characterised in terms of photon flux density or optical power and/or optical power density. The photon flux density is defined as the number of photons incident on the device per unit area per unit of time and is denoted by Φ_0. This parameter is related to the incident optical power, $P_{op}(0)$ (measured in watt) as

$$\Phi_0 = \frac{P_{op}}{Ah\nu} \tag{7.3}$$

where A is the device area exposed to optical radiation, h is Planck's constant, and ν is the frequency of light.

The incident optical radiation is absorbed in the semiconductor material and the available optical power at any distance, x from the plane of incidence ($x = 0$) decreases exponentially as

$$P_{op}(x) = P_{op}(0)\exp[-\alpha(\lambda)x] \tag{7.4}$$

where $P_{op}(0)$ is the incident optical power, $\alpha(\lambda)$ is the absorption coefficient at a wavelength λ. If we consider the effect of reflection of the incident light at the entrance, the power absorbed by the semiconductor in a distance of length W can be expressed as

$$P_{abs} = P_{op}(0)(1 - R_f)[1 - \exp(-\alpha W)] \tag{7.5}$$

where R_f is the Fresnel reflection coefficient at the entrance given by

$$R_f = \left(\frac{n_1 - n_2}{n_1 + n_2}\right)^2 \tag{7.6}$$

Here it is assumed that the light is travelling from a medium of refractive index, n_1 to a medium of refractive index, n_2.

Assuming that each photon absorbed in the semiconductor results in electron-hole pairs, the photogeneration at any point x can be expressed as (Bhattacharya, 1997)

$$\begin{aligned} G_{op}(x) &= \Phi_0\, \alpha \exp(-\alpha, x) \\ &= \frac{P_{op}(0)(1 - R_f)\,\alpha \exp(-\alpha x)}{Ah\nu} \end{aligned} \tag{7.7}$$

The average photogeneration rate (number of electron–hole pairs generated per unit volume per unit time) over an absorbing region of thickness W can be expressed as

$$G_{op} = \frac{1}{W}\int_0^w G_{op}(x)dx = \frac{P_{op}(0)(1 - R_f)[1 - \exp(-\alpha W)]}{Ah\nu W} \tag{7.8}$$

The absorption coefficient α of a semiconductor material is a function of wavelength λ. Some important semiconductor materials used in fabricating photodetectors for optical fiber communication are listed in Table 7.1 (Levinshtein *et al* 1999). The dependence of absorption coefficient of a few semiconductors on wavelength is shown in Fig. 7.3 (Bhattacharya, 1997; Senior, 2001). It is interesting to note that the photogeneration would be significant when the thickness of the absorbing region is sufficiently large (~$1/\alpha$).

Example 7.1 Calculate the long wavelength cut-off of a GaAs photodetector. The bandgap energy of GaAs is 1.424 eV at 300 K.

Solution The long wavelength cut-off of GaAs photodetector at 300 K can be obtained as

$$\lambda = \frac{1.24}{1.424}\,\mu\text{m} = 0.87\,\mu\text{m}$$

Table 7.1 Bandgap values of some important photodetector materials at room temperature	
Material	***Bandgap (eV) at 300 K***
Si (Indirect)	1.14
Ge (Indirect)	0.67
GaAs (Direct)	1.43
InP (Direct)	1.35
InAs (Direct)	0.35
AlAs (Direct)	2.16
GaSb (Direct)	0.73
$In_{0.53}Ga_{0.47}As$ (Direct)	0.75
$In_{0.70}Ga_{0.30}As_{0.64}P_{0.36}$ (Direct)	0.94

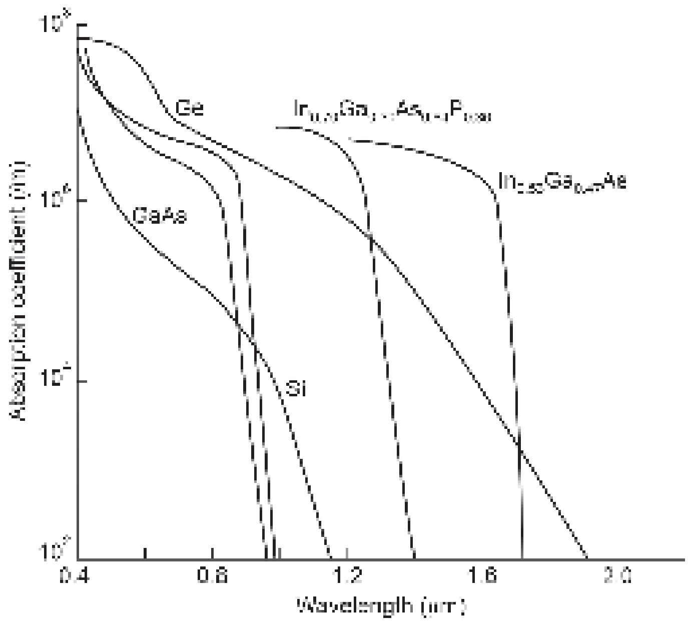


Fig. 7.3 Absorption coefficient *vs* wavelength plot for some selected semiconductors

7.4 MECHANISM OF PHOTODETECTION

The basic mechanism of operation of a photodetector can be understood by considering the operation of a *p-n* junction under reverse-biased condition illustrated in Fig. 7.3. Under reverse-biased condition, the depletion region spreads from either side increasing the barrier height. This barrier height prevents the majority of carriers from flowing across the junction by diffusion mechanism. The current that flows under reverse bias is due to the drift of minority carriers[1] (electrons from the *p*-side and holes from the *n*-side) under the action of the reverse voltage. This current is low because only minority carriers participate in the conduction of current. Further, this current cannot be increased much by increasing the reverse voltage as the carriers attain the saturation velocity after the field exceeds the critical value. The current in the reverse

[1] Note that holes flow in the direction of the electric field (which extends from *n*- to *p*-regions) while the electrons flow in the direction opposite to the field.

direction can, however, be enhanced by increasing the number of minority carriers on either side. This can be achieved either electrically by injecting minority carriers (as done in the case of a bipolar junction transistor) or optically by creating electron–hole pairs through the absorption of light of a suitable wavelength (Sze, 2003). The latter phenomenon is exploited in the operation of a photodetector.

If an optical signal comprising photons of energy greater than or equal to the bandgap of the material is incident on the device, the photons will be absorbed throughout in the neutral *p*- and *n*-regions as well as in the depletion region as shown in Fig. 7.3. The absorption of photons will result into creation of electron–hole pairs in all the three regions. The electron–hole pairs generated in the neutral region will affect the existing thermally generated carriers. However, the percentage change in the number of minority carriers would be much more than that of the majority carriers. The additional minority carriers so generated will diffuse into the depletion region and finally be swept by the field existing in the depletion region. While the carriers generated in the depletion region will be swept immediately by the high field existing in the region, the photogenerated minority carriers in the neutral *p*- and *n*-regions will take a longer time because diffusion is a slow process relative to drift. Moreover, all the minority carriers generated in the neutral *p*- and n-regions would not be able to reach the depletion region because many of these excess minority carriers will recombine with the existing majority carriers. On average, we may presume that photogenerated carriers which are created within diffusion length on either side (L_n on *p*-side and L_p on *n*-side) would be able to contribute to the photocurrent. The total photogenerated current can be expressed as

$$I_p = qA(L_n + L_p + W)\, G_{op} \tag{7.9}$$

where q is the electronic charge, A is the junction area, L_n and L_p are electron diffusion length on the *p*-side and hole diffusion length on the *n*-side respectively and G_{op} is the photogeneration rate (number of electron–hole pairs generated per unit volume per sec).

The total current under reverse bias $\left(\exp\left(-\frac{qV_r}{kT}\right) \ll 1\right)$ flowing from *p*- to *n*-region in the presence of light can be expressed as

$$I = -I_0 - qA(L_n + L_p + W)\, G_{op} \tag{7.10}$$

where $I_0 = qA\left(\frac{D_p}{L_p} p_{n0} + \frac{D_n}{L_n} n_{p0}\right)$ is the reverse saturation current in the dark condition arising out of thermally generated carriers.

Here, D_p and D_n are the electron and hole diffusion coefficients and p_{n0} and n_{p0} are the minority carriers in the *n*- and *p*-regions respectively. The current–voltage characteristics of a typical *p-n* junction photodetector corresponding to different generation rates depending on the intensity of the incident light are shown in Fig. 7.4 for illustration.

From Eq. (7.10), it is seen that the increase in current in presence of light is proportional to the photogeneration rate which in turn depends on the incident optical power. This is illustrated in Fig. 7.3. In other words, the photocurrent (difference between the current in the presence of light and that in the absence of light that is, the current under dark conditions) is a measure of the incident optical power. This property of the illuminated *p-n* junction is exploited for the detection of optical signals. A practical photodetector is, however, is much more complex as compared to a simple *p-n* junction discussed above. Before we discuss more advanced detector structures, it would be interesting to consider

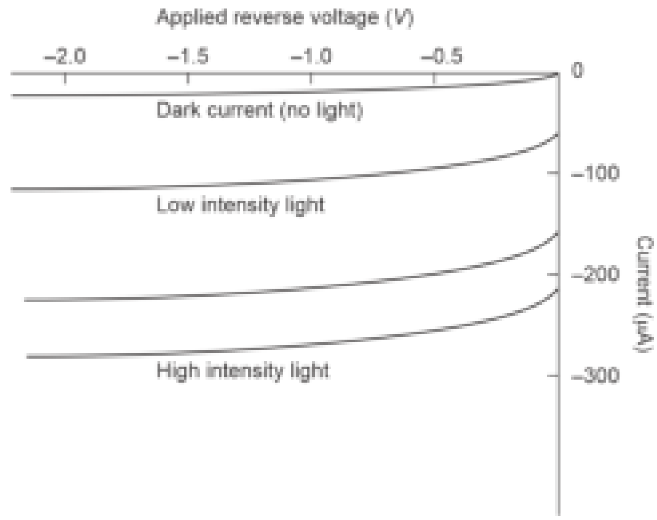


Fig. 7.4 Current–voltage characteristic of a *p-n* junction in the illuminated condition

the general characteristics of a photodetector; its quantum efficiency, responsivity, speed of response, and noise performance.

7.5 CHARACTERISTICS OF A GENERAL PHOTODETECTOR

The photodetectors used for optical fiber communication are required to meet several stringent performance standards. These include

1. **Size compatibility:** The size of the detector must be compatible with the size of the fiber carrying the light (radius of the core) so that the optical power carried by the fiber can be efficiently coupled to the photodetector.
2. **High conversion efficiency:** The photodetector must be able to produce a maximum electrical signal for a given amount of received optical signal. This efficiency is measured in terms of electron-hole pairs produced per quanta, known as *quantum efficiency*.
3. **High sensitivity at operating wavelength:** The principal material used for making fiber is silica which has several attenuation windows in the infrared region. The operating wavelength window witnessed several changes over the past decades depending on the then-available technology. The first-generation optical fiber communication based on multimode step-index fiber and centered around 0.8 to 0.9 μm wavelength region was extended to 0.8 to 1.6 μm in the second-generation based on multimode graded index fiber. In the third-generation system attention was focussed on 1.1 to 1.6 μm by making use of single-mode fiber. The present fourth-generation is primarily based on single-mode fibers specially designed to offer minimum dispersion at the operating wavelength of 1.55 μm where the attenuation in silica fiber is minimal. The fifth-generation optical fiber communication is expected to make use of the mid-infrared wavelength region where fluoride glass fiber would provide ultra-low loss for long-haul applications (Senior, 2001). The photodetectors based on suitable semiconductor materials are required in various operating wavelengths to provide a significant electrical output for a small input optical power at room temperature.
4. **High response speed:** The detector must be capable of converting fast-changing optical signals carrying millions of bits of information to the corresponding electrical signal. The present generation optical fibers can support tens of gigabits per sec of data. To

handle such a huge bit rate, the response time of the photodetectors must be very low. In other words, the response speed should be extremely high.

5. **Minimum noise.** The photodetector must have extremely low noise. The dark current, leakage current, and conductance of the photodetector must be low.

The definitions of several key parameters of a photodetector follow. The quantum efficiency of a photodetector is defined as the number of electron-hole pairs generated per quanta, i.e. per incident photons. Mathematically, the quantum efficiency can be expressed as (Sze, 2003)

$$\eta = \frac{I_p/q}{P_{op}/h\nu} \tag{7.11}$$

where I_p is the photocurrent, q is the electronic charge, P_{op} is the incident optical power, h is Planck's constant, and ν is the frequency of incident light.

The quantum efficiency depends on the absorption coefficient of the material used in the fabrication of the device. The absorption coefficient of a material is a function of wavelength and therefore, the quantum efficiency of a photodetector defined above is a function of wavelength. It is therefore customary to specify the quantum efficiency of a photodetector at a specific wavelength. The quantum efficiency of a practical photodetector is less than unity. The quantum efficiency of a photodetector is generally expressed as a percentage of quantum efficiency.

The other important parameter used for characterizing a photodetector is the current responsivity, $\mathcal{R}$ defined as

$$\mathcal{R} = \frac{I_p}{P_{op}} \tag{7.12}$$

Using Eq. (7.11), the responsivity can also be expressed as

$$\mathcal{R} = \frac{q\eta}{h\nu} \tag{7.13}$$

$$\mathcal{R} = \frac{q\eta\lambda}{hc} \tag{7.14}$$

It is thus seen that the responsivity of a photodetector varies linearly with the wavelength of operation for a given quantum efficiency. However, for a practical photodetector, the quantum efficiency is a function of wavelength and therefore the responsivity of the photodetector falls sharply at the long wavelength cut-off of the photodetector (Senior, 2001). The variations of the responsivity of an ideal photodetector and a practical photodetector with wavelength are illustrated with the help of a qualitative sketch in Fig. 7.5.

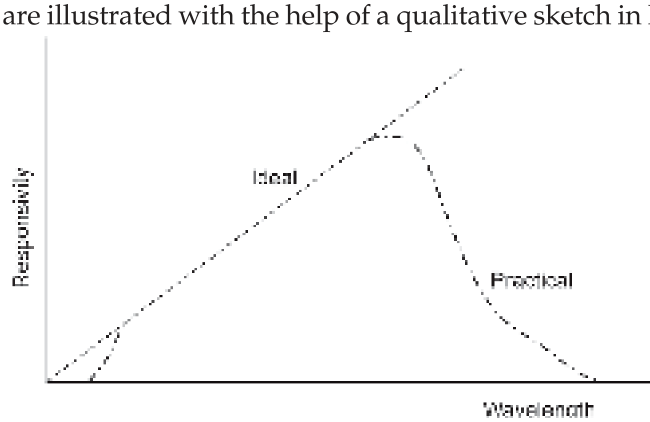


Fig. 7.5 Variation of responsivity with wavelength

It can be seen from Fig. 7.5 that for every practical photodetector the photoresponse is observed over a finite range of wavelengths. This is due to the fact that every material has a characteristic band of absorption. The responsivity characteristics of photodetector based on different materials are shown in Fig. 7.6 (Keiser, 2000).

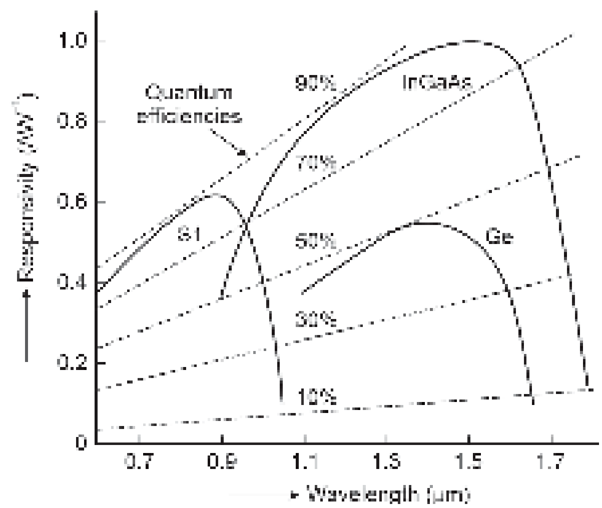


Fig. 7.6 Responsivity *vs* wavelength plots of photodetectors based on Ge, Si and InGaAs

7.6 PHOTOCONDUCTORS

A photoconductor consists of a piece of bulk semiconductor with ohmic contacts at the two opposite ends (Fig. 7.7(a)). It can also be fabricated using a thin film of semiconductor deposited on a suitable substrate with proper electrodes for electrical connection. The electrodes in a photoconductor are often realized in the form of interdigitated contacts with small gaps as illustrated in Fig. 7.7(b).

When the photoconductor is exposed to incident optical radiation of suitable wavelength as shown in Fig. 7.7, the light is absorbed in the semiconductor material resulting in the generation of excess (photogenerated) carriers. The photogenerated carriers are created by both intrinsic and extrinsic photoexcitation involving band-to-band (BTB) transition and transitions involving impurity levels respectively as shown in Fig. 7.8.

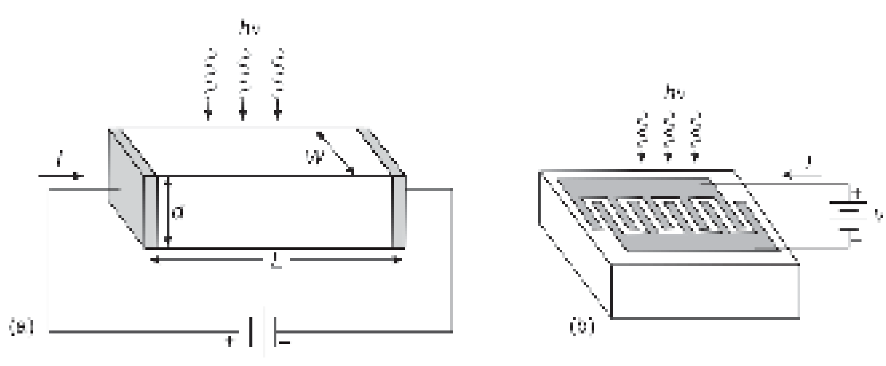


Fig. 7.7 Schematic structures of a photoconductor (a) Structure consisting of a semiconductor slab with ohmic contacts at the ends (b) Layout of a semiconductor photoconductor with interdigitated electrodes

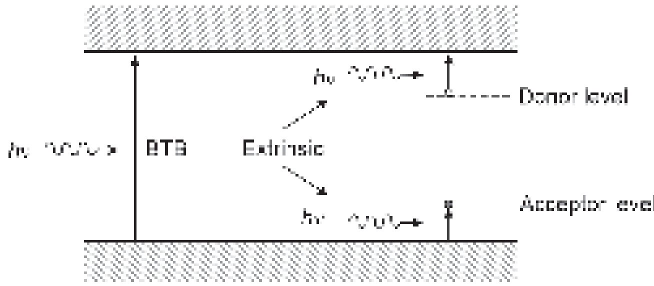


Fig. 7.8 Illustrating intrinsic photoexcitation (band-to-band) and extrinsic (between a band and an impurity level) photoexcitation caused by the incident light

The conductivity of the photoconductor in the absence of the incident light can be written as

$$\sigma = q(\mu_n n_0 + \mu_p p_0) \tag{7.15}$$

where q is the electronic charge, μ_n and μ_p correspond to electron and hole mobility respectively and n_0 and p_0 correspond to the electron and hole concentration of the semiconductor in thermal equilibrium.

In the presence of excess photogenerated carriers, the electron and hole concentrations change and as a result the conductivity of the semiconductor changes. If an external voltage is applied across the semiconductor slab, there will be a significant change in the value of the current in the illuminated condition for a given applied voltage as compared to the current in the dark condition (absence of illumination). The difference in the value of the current in dark and illuminated conditions depends on the intensity of the light. As the conductivity and so also the resistance of the semiconductor is dependent on the incident light, photoconductors are also referred to as light-dependent-resistors (LDR).

Now examine the performance of a photoconductor in terms of the photodetector parameters, e.g. quantum efficiency, gain, response time, and detectivity.

Let us assume that the front surface of the photoconductor is uniformly illuminated with steady optical power, P_{op}. The total number of photons impinging on the surface area ($A = WL$) per unit time can be written as

$$\phi_0 = \frac{P_{op}}{h\nu} \tag{7.16}$$

where $h\nu$ is the photon energy.

Assuming the thickness d of the photoconductor to be much longer than the penetration depth ($1/\alpha$), the generation and recombination rates of the carriers can be written as

$$G_n = \frac{n}{\tau} = \frac{\eta P_{op}}{h\nu(WLd)} \tag{7.17}$$

where η is the quantum efficiency that determines the number of carriers generated per photon (quanta) and τ is the carrier lifetime.

The number of carriers generated in the photoconductor by the absorption of light is much larger than the background carrier concentration (thermally generated carriers) and therefore, the steady-state carrier concentration can be written as

$$n = G_n \tau = \frac{\eta P_{op} \tau}{h\nu(WLd)} \tag{7.18}$$

For an intrinsic photoconductor ($n_0 = p_0$), the photocurrent flowing through the semiconductor across the electrodes (Ohmic contacts at the ends) can be expressed as

$$I_p = \sigma \varepsilon W d = qn(\mu_n + \mu_p)\varepsilon W d \tag{7.19}$$

where ε is the applied electric field and $n = p$ as the carriers are generated in pairs by illumination.
Using Eq. (7.18), we may express Eq. (7.19) as

$$I_p = q\frac{\eta P_{op}\tau}{h\nu L}(\mu_n + \mu_p)\varepsilon \tag{7.20}$$

The primary photocurrent caused by the illumination can be calculated from the number of electron–hole pairs generated by the incident photons and multiplying it by the electronic charge. The primary photocurrent can be written as

$$I_{ph} = q\left(\eta\frac{P_{op}}{h\nu}\right) \tag{7.21}$$

The photocurrent gain can be expressed using Eqs (7.20) and (7.21) as

$$G_a = \frac{I_p}{I_{ph}} = \frac{(\mu_n + \mu_p)\tau\varepsilon}{L} = \tau\left[\frac{1}{(L/\mu_n\varepsilon)} + \frac{1}{(L/\mu_p\varepsilon)}\right] \tag{7.22}$$

The transit time of a carrier is the time taken by the carrier from one electrode to the other, i.e. the time taken by the carrier to traverse the length of the photoconductor. Therefore, the transit time of electrons and holes can be expressed as

$$\tau_{tn} = \frac{L}{v_n} = \frac{L}{\mu_n\varepsilon} \tag{7.23}$$

$$\tau_{tp} = \frac{L}{v_p} = \frac{L}{\mu_p\varepsilon} \tag{7.24}$$

where τ_{tn} and τ_{tp} are the transit time of electrons and holes respectively and v_n and v_p correspond to the velocity of electrons and holes respectively.

The photocurrent gain can be expressed using substituting Eqs (7.23) and (7.24) into Eq. (7.22) as

$$G_a = \tau\left[\frac{1}{\tau_{tn}} + \frac{1}{\tau_{tp}}\right] \tag{7.25}$$

The gain of the photoconductor, therefore, depends on the ratio of the carrier lifetime to the transit time. To achieve a high value of gain it is necessary to make the transit time shorter as compared to the carrier lifetime. The transit time can be reduced by reducing the electrode spacing and increasing the mobility. A typical value of gain for a photoconductor is of the order of 10^3. The value of the gain can be further increased (~10^6) optimizing the device parameters and using interdigitated electrodes. It may be noted that a large value carrier lifetime reduces the speed of response which is not desirable when the photoconductor is used as a photodetector. Therefore, there is a trade-off between gain and speed of response. In general, the speed of response of a photoconductor is less than junction photodiodes.

We now consider the behavior of the photoconductor in the presence of an intensity-modulated optical power, can be written as

$$P(f) = P_{op}\,(1 + m_a e^{j2\pi ft}) \tag{7.26}$$

where P_{op} is the average optical power, m_a is the index of modulation and f is the modulation frequency.

The average value of optical current, I_p can be obtained from Eq. (7.22) can be obtained using Eq. (7.19) as

$$I_p = G_a I_{ph} = G_a q\left(\eta \frac{P_{op}}{h\nu}\right) \tag{7.27}$$

Therefore, the rms value of the photocurrent in presence in presence of intensity-modulated light can be obtained as (DiDomenico, Jr *et al* 1964)

$$i_p = \left(\frac{q\eta m_a G_a P_{op}}{\sqrt{2}\,h\nu}\right)\frac{1}{\sqrt{1+(2\pi f\tau)^2}} \tag{7.28}$$

At low frequencies, Eq. (7.28) reduces to Eq. (7.27) for 100% modulation (m_a = 1). On the other hand, at high frequencies, the signal photocurrent decreases as $1/f$.

The noise equivalent circuit of the photoconductor is shown in Fig. 7.9 (DiDomenico, Jr *et al* 1964; Sze *et al* 2007). The conductance, G accounts for the contributions arising out of the dark current, the average signal current, and the current arising out of background illumination.

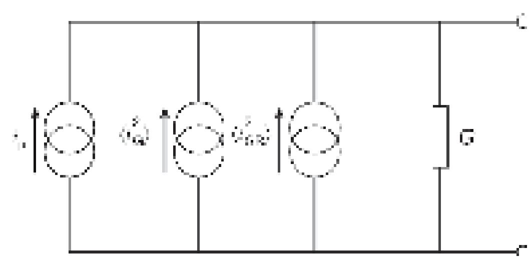


Fig. 7.9 Noise equivalent circuit of a photoconductor

The mean-square value of the thermal noise current originating from the conductance, G can be expressed as

$$\left\langle i_G^2 \right\rangle = 4kTGB \tag{7.29}$$

where B is the bandwidth.

The mean square value of the shot noise current arising out of the generation-recombination can be written as (van der Ziel, 1959)

$$\left\langle i_{GR}^2 \right\rangle = \frac{4qI_pG_aB}{1+(2\pi f\tau)^2} \tag{7.30}$$

The signal-to-noise ratio (power) can be expressed using Eqs (7.26), (7.29) and (7.30) as

$$\frac{S}{N} = \frac{i_p^2\left(\frac{1}{G}\right)}{\left(\left\langle i_G^2\right\rangle + \left\langle i_{GR}^2\right\rangle\right)\left(\frac{1}{G}\right)} = \frac{\eta m_a^2\left(\frac{P_{op}}{h\nu}\right)}{8B}\,\frac{1}{1+\dfrac{kT}{qG_a}\left[1+(2\pi f\tau)^2\dfrac{G}{I_p}\right]} \tag{7.31}$$

The noise equivalent power (NEP) of the photoconductor can be defined as the minimum value of optical power, $(P_{op})_{\min}$ needed to produce a signal power output that is equal to the total noise power per unit bandwidth. The NEP can be calculated from Eq. (7.29) by solving for P_{op} by substituting $S/N = 1$ and $B = 1$ Hz and using the following relation

$$\text{NEP} = \frac{m_a (P_{op})_{\min}}{\sqrt{2}} \tag{7.32}$$

For non-telecommunication applications of photoconductors as infrared detectors, the detectivity, D^* can be calculated using the NEP value in the following expression

$$D^* = \frac{\sqrt{AB}}{\text{NEP}}\ \text{mHz}^{1/2}\text{W}^{-1} \tag{7.33}$$

A variety of photoconductors in the form of bulk semiconductor slabs and thin films are available commercially for application in the visible, near-infrared, mid-infrared, and far-infrared and long wavelength infrared regions. For the visible region, CdS is the material of choice and for mid-infrared (MIR) and long wavelength infrared (LWIR) materials like InAs, $InAs_{1-x}Sb_x$, and MCT (mercury cadmium telluride) in the form $Hg_xCd_{1-x}Te$ are preferred materials. For the ultraviolet region GaN and several oxide semiconductor films like ZnO and TiO_2 are being used for making photoconductive detectors (Dwivedi *et al* 2007; Eisenman *et al* 1977).

7.7 *p-i-n* PHOTODETECTOR

In Section 1.3, it is stated that a simple *p-n* junction under reverse-biased conditions can be used as a photodetector. However, the response of such a detector is severely restricted by the fact that the photogenerated carriers in the neutral region take a longer time to be extracted in the external circuit as compared to carriers generated in the depletion region. This is because the carriers generated in the neutral region (zero field region) flow by diffusion mechanism which is a slow process as compared to the drift mechanism that is responsible for the transportation of the carriers generated in the drift region. To enhance the speed of response of the photodetector it is therefore necessary to modify the structure in a way that the photogenerated carriers are mostly created in the depletion region. This can be achieved by making the depletion region much wider as compared to conventional *p-n* junction. The modified structure is essentially a *p-i-n* configuration and is most widely used as a photodetector in optical fiber communication systems. In this photodetector, the width of the depletion region (*i*-region) can be tailored to optimize the quantum efficiency and the response speed (Sze, 2003).

The schematic of a *p-i-n* detector is shown in Fig 7.10 along with the energy band diagram. It consists of an intrinsic undoped *i*-region (usually lightly doped *n*-type, denoted by ν or a lightly doped *p*-region denoted by π) sandwiched between heavily doped *p*- and *n*-regions. The number of free carriers available in the *i*-region is very small and as a result, the whole region gets depleted even at a low reverse voltage. Absorption of light in the semiconductor produces electron–hole pairs in the depletion region or within a diffusion length of it contributing to the external photocurrent. To facilitate the entry of light into the semiconductor in a practical *p-i-n* detector an etched opening is either created in the top contact (front illumination) or an etched hole is created at the rear substrate end (back illumination).

Consider the *p-i-n* photodetector shown in Fig. 7.10(a). Under steady state condition, the total current density through the reverse biased depletion region is given by (Gartner, 1959)

$$J_{tot} = J_{dr} + J_{diff} \tag{7.34}$$

where J_{dr} is the drift current due to carriers generated in the depletion region and J_{diff} is the diffusion current due to carriers generated in the neutral bulk region of the semiconductor diffusing into the depletion region.

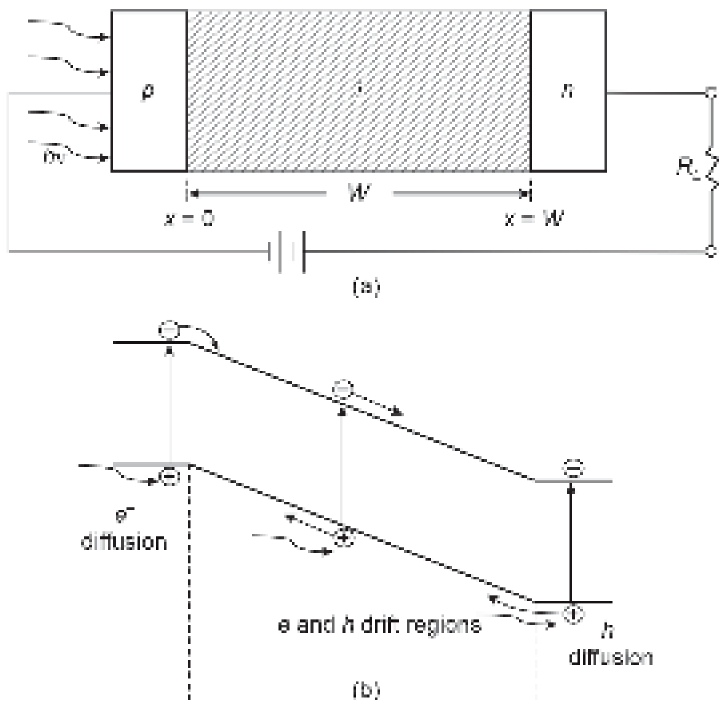


Fig. 7.10 Illustrating the operation of a *p-i-n* photodetector (a) Schematic view of a *p-i-n* detector (b) Energy band diagram showing the transportation of photogenerated carriers

7.7.1 Quantum Efficiency

It can be easily seen that the total photogeneration in the *p-i-n* structure comprises generation in the neutral *n*- and *p*-regions in addition to the generation in the fully depleted *i*-region. The photogeneration in the *p*-region can be made insignificant by thinning down the thickness of the layer ($< 1/\alpha$).

The electron–hole generation rate at any point, x (Fig. 7.5) can be expressed as

$$G_{op}(x) = \Phi_0\, \alpha \exp(-\alpha x) \tag{7.35}$$

The drift current density can be obtained as

$$J_{dr} = -q\int_0^W G_{op}(x)dx = q\Phi_0[1 - \exp(1 - \alpha W)] \tag{7.36}$$

where W is the depletion layer width (equal to the thickness of the i-region in this case). For $x > W$, the minority carrier density in the bulk of the semiconductor can be determined by the 1-D diffusion equation given by (Sze, 2003)

$$D_p \frac{\partial^2 p_n}{\partial x^2} - \frac{p_n - p_{n0}}{\tau_p} + \Phi_0 \alpha \exp(-\alpha\, x) = 0 \tag{7.37}$$

where D_p is the hole diffusion length, p_n is the hole (minority carrier) concentration at any point, x and p_{n0} is the equilibrium hole concentration in the bulk n-region.

Solving Eq. (7.37) under the following boundary conditions

$$p_n = 0 \text{ at } x = W \text{ and } p_n = p_{n0} \text{ at } x = \infty \tag{7.38}$$

Thus, the solution can be obtained in the form (Bhattacharya, 1997) as

$$p_n = p_{n0} - (p_{n0} + C\exp(-\alpha W))\exp\left(\frac{W-x}{L_p}\right) + C\exp(-\alpha x) \tag{7.39}$$

where $L_p = \sqrt{D_p \tau_p}$ and the constant C can be obtained using the boundary condition as

$$C = \frac{\Phi_0 \alpha L_p^2}{D_p(1-\alpha^2 L_p^2)} \tag{7.40}$$

The diffusion current density can be obtained as

$$J_{diff} = -qD_p \left.\frac{\partial p_n}{\partial x}\right|_{x=W} \tag{7.41}$$

The diffusion current density can be finally obtained using Eqs (7.39)–(7.41) as

$$J_{diff} = q\Phi_0 \frac{\alpha L_p}{1+\alpha L_p} exp(-\alpha W) + qp_{n0}\frac{D_p}{L_p} \tag{7.42}$$

where L_p is the hole diffusion length.
Using Eqs (7.36) and (7.42), the total current can finally be expressed as

$$J_{tot} = q\Phi_0\left(1 - \frac{\exp(-\alpha W)}{1+\alpha L_p}\right) + qp_{n0}\frac{D_p}{L_p} \tag{7.43}$$

Under usual operating conditions, the second term on the right-hand side is much smaller, and therefore the total current density is proportional to the flux density. The quantum efficiency can be expressed as

$$\eta = \frac{J_{tot}/q}{P_{op}/Ah\nu}$$
$$= (1-R_f)\left(1 - \frac{\exp(-\alpha W)}{1+\alpha L_p}\right) \tag{7.44}$$

Example 7.2 A GaAs *p-i-n* photodetector receives light from an optical source operating at 0.87 μm. The width of the *i*-region is 2 μm. The absorption coefficient of GaAs is 10^6 m^{-1}. The hole diffusion length is 10 μm. Calculate the value of quantum efficiency. The refractive index of GaAs is 3.4.

Solution The Fresnel reflection coefficient at the entrance (assuming the light comes from air into the detector)

$$R = \left(\frac{3.4-1}{3.4+1}\right)^2 = 0.29$$

The quantum efficiency of the *p-i-n* detector can be obtained as

$$\eta = (1-0.29)\left(1 - \frac{\exp\left(-10^6 \times 2 \times 10^{-6}\right)}{1+10^6 \times 10 \times 10^{-6}}\right) = 0.70$$

The quantum efficiency of the *p-i-n* detector is approximately 70%.

7.7.2 Frequency Response

The optical fiber communication system operating based on the intensity modulation-direct detection (IM-DD) principle uses intensity-modulated optical signal to transmit information. In the presence of intensity-modulated photon flux, the phase difference between photon flux and the photocurrent would be appreciable when the incident light is modulated rapidly. Assuming the incident intensity modulated photon flux to be of the form $\Phi_1 \exp(j\omega t)$, the conduction current at any point, x in the depletion region can be obtained as

$$J_{cond}(x) = q\Phi_1 \exp\left[j\omega\left(t - \frac{x}{v_s}\right)\right] \tag{7.45}$$

where ω is the frequency of the modulating signal and v_s is the saturation velocity of the carriers in the depletion region.

The total current in the presence of an intensity-modulated optical signal can be expressed as

$$J_{tot} = \frac{1}{W}\int_0^W \left(J_{cond} + \epsilon_s \frac{\partial \varepsilon}{\partial x}\right) dx \tag{7.46}$$

where ϵ_s and ε are permittivity and electric field respectively. The second term in the integrand refers to the displacement current. Assuming the electric field also varies sinusoidally following the intensity modulated flux the total current density can be obtained as (Gartner, 1959)

$$J_{tot} = \left(\frac{j\omega \epsilon_s V}{W} + q\Phi_1 \frac{1-\exp(-j\omega\tau_d)}{j\omega\tau_d}\right)\exp(j\omega t) \tag{7.47}$$

where V is the applied voltage that includes the built-in voltage and $\tau_d = W/v_s$ is the transit time of the carriers through the depletion region.

The short circuit ($V = 0$) current density is given by

$$J_{sc} = \frac{q\Phi_1[1-\exp(j\omega\tau_d)]}{j\omega\tau_d} \tag{7.48}$$

The frequency response of the *p-i-n* photodetector can be obtained by plotting the normalized current as a function of the normalized modulating frequency ($\omega\tau_d$). The 3dB bandwidth can be estimated from the value of $\omega\tau_d$ at which the normalized photocurrent falls to $1/\sqrt{2}$ of the maximum value. The variation of the magnitude and phase of the term $[1 - \exp(j\omega\tau_d)]/(j\omega\tau_d)$ with $\omega\tau_d$ is shown in Fig. 7.11. The 3dB frequency from this plot can be found to be (Gartner, 1959)

$$f_{3dB} = \frac{2.4}{2\pi\tau_d} \approx \frac{0.4 v_s}{W} \tag{7.49}$$

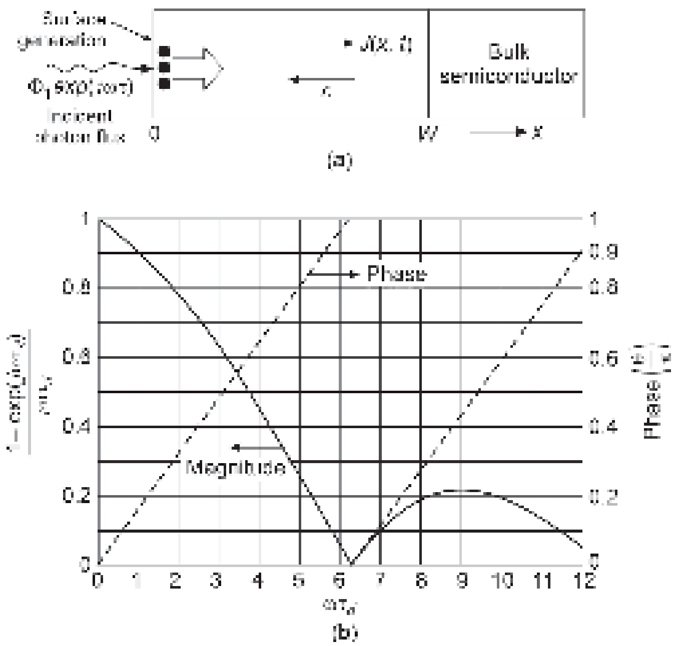


Fig. 7.11 Variation of the magnitude and phase of the ratio [1–exp($j\omega\tau_d$)]/$j\omega\tau_d$ with $\omega\tau_d$ (Gartner, 1959)

Assuming the width of the depleted *i*-region to be equal to the reciprocal of the absorption coefficient, that is $W = 1/\alpha$, the 3-dB frequency of a Si *p-i-n* photodetector turns out to be

$$f_{3\text{-dB}} \approx 0.4\alpha v_s \tag{7.50}$$

The trade-off between the quantum efficiency and the 3-dB frequency of the *p-i-n* photodetector is apparent from Eqs (7.44) and (7.49). For example, an increase in the value of the thickness, W of *i*-region enhances the quantum efficiency but reduces the 3-dB frequency. An optimization is therefore necessary to meet the specified requirement.

Example 7.3 A photodetector generates a photocurrent of 0.25 μA for an incident optical power of 0.8 μW at the operating wavelength of 0.87 μm. Estimate the quantum efficiency of the photodetector at this wavelength.

Solution The responsivity of the photodetector can be estimated by using Eq. (7.11) as

$$\eta = \frac{I_p / q}{P_{op} / h\nu} = \frac{I_p}{q} \times \frac{hc}{\lambda P_{op}}$$

$$= \frac{0.25 \times 10^{-6}}{1.6 \times 10^{-19}} \times \frac{6.62 \times 10^{-34} \times 3 \times 10^{8}}{0.8 \times 10^{-6} \times 0.87 \times 10^{-6}}$$

$$= 0.4458$$

Therefore, the quantum efficiency of the photodetector at the operating wavelength (0.87 μm) is 44.58%.

Example 7.4 A Si *p-i-n* photodetector has a 5 μm thick intrinsic region. Assuming that the field in the depleted intrinsic region is high enough to cause velocity saturation of the carriers estimate the 3-dB bandwidth of the photodetector. Assume the saturation velocity of the carrier to be 10^5 m/s.

Solution The 3-dB bandwidth of the *p-i-n* detector can be obtained using Eq. (7.31) as

$$f_{3\text{-}dB} = \frac{0.4v_s}{W} = \frac{0.4\times10^5}{5\times10^{-6}} = 8 \text{ GHz}$$

The 3-dB bandwidth of the photodetector is 8 GHz.

7.7.3 Speed of Response

The speed of response of a *p-i-n* photodetector is determined by the following three major factors.

1. Carrier drift time, τ_d in the depleted *i*-region. This is the fundamental limitation of the speed of response of a photodetector. When the electric field in the depletion region exceeds the value of the critical electric field required to cause velocity saturation of the carriers, the carriers eventually drift with the saturation velocity, v_s. For Si, the critical electric field is 2×10^6 Vm^{-1} and the scattering limited saturation velocity of the carriers is approximately 10^5 ms^{-1}. Therefore, the transit time of the carriers through a 10 μm thick *i*-region would be around 0.1 ns.
2. Diffusion time of minority carriers generated in the bulk region outside the depleted *i*-region. Carrier diffusion is a slow process as compared to the drift process. The time taken by the carriers to traverse a distance *d* by the process of diffusion is given by

$$\tau_{diff} = \frac{d^2}{2D} \tag{7.51}$$

   where *D* is the diffusion coefficient of the minority carriers. For electrons in Si, the time taken by the carriers to diffuse over a distance of 10 μm is of the order of 10 ns.
3. Input RC time constant of the detector circuit. Under reverse-biased conditions the photodetector exhibits a junction capacitance given by

$$C_j = \frac{\epsilon_s A}{W} \tag{7.52}$$

where ϵ_s is the permittivity of the semiconductor and A is the junction area. In addition to this capacitance, the photodetector has also capacitance arising out of leads and packaging. The net capacitance constitutes the RC time constant together with the effective resistance. The capacitance needs to be minimized in order to reduce the RC time constant which limits the speed of response of the photodetector.

From the foregoing discussion, it is evident that the overall response speed of the photodetector is decided by the most dominant factors of the above three components. In a *p-i-n* photodetector, the effect of diffusion is usually small because most of the carriers are generated in the depletion region and hardly any carrier is generated in the neutral region. The response speed of the photodetector can therefore be increased by reducing the thickness of the i-region. However, a reduction in the thickness of the i-region would result in a reduction of quantum efficiency [Eq. (7.44)] and an increase in the value of junction capacitance [Eq. (7.52)]. The latter in turn would increase the RC time constant and the overall response speed would be determined by the input RC time constant rather than the transit time across the depletion region. The dependence of the response speed of a photodetector on the width of the depletion region can be best understood by studying the response of the photodetector to a rectangular pulse with respect to varying thickness of the depletion region. This is illustrated in Fig. 7.12.

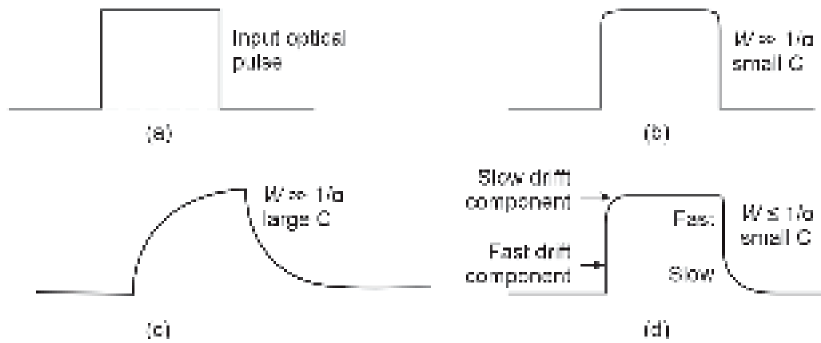


Fig. 7.12 Illustrating the effect of photodetector parameters on the response of the device to rectangular optical pulse at the input

It can be seen that for a large value of depletion layer width (much longer than the reciprocal of the absorption coefficient), the capacitance has a low value and the diffusion component is negligible. Under this condition, the response speed of the photodetector is quite fast and the rising and falling edge of the output follows the input pulse (Fig. 7.12(b)). When the detector capacitance is large, the RC time constant becomes large and the response is limited by the RC time constant rather than the drift time. This is demonstrated by the prolonged rising and falling edges of the output pulse in response to the rectangular pulse (Fig. 712(c)). When the width of the depletion region is small (of the order of the reciprocal of the absorption coefficient) and the capacitance value is small then a significant number of carriers will be created outside the depletion region. As a result, the output pulse would exhibit the effect of drift and diffusion. The steep rising edge in the beginning is caused by the collection of the fast carriers due to drift followed by a slow rising edge contributed by the collection of diffused carriers. Similarly, at the trailing edge of the pulse, the drifted carriers are collected fast followed by a long falling edge caused by the slow component arising out of diffused carriers. This is illustrated in Fig. 7.12(d).

Even though all the factors described above affect the overall response speed of a photodetector, the ultimate physical limit to the bandwidth of the photodetector is decided by the drift time of the carriers. Based on this fact, the maximum 3-dB bandwidth of the photodetector can be estimated as (Forrest, 1988)

$$B_{\max} = \frac{1}{2\pi\tau_d} = \frac{v_s}{2\pi W} \tag{7.53}$$

Further, the *p-i-n* photodetector does not provide any gain therefore the gain-bandwidth product of the photodetector is equal to the bandwidth.

In actual practice, the photodetector in an optical receiver is usually followed by a pre-amplifier. As a result, the photodetector capacitance and the input capacitance of the following stage amplifier come in parallel. The effective capacitance is thus obtained as

$$C_T = C_j + C_a \tag{7.54}$$

where C_a is the input capacitance of the amplifier.

Similarly, the load resistance comes in parallel with the input resistance of the following stage amplifier. The photodetector series resistance is usually low. The equivalent resistance R_T is given by

$$\frac{1}{R_T} = \frac{1}{R_L}\,\frac{1}{R_a} \tag{7.55}$$

where R_L is the load resistance and R_a is the input resistance of the amplifier. The photodetector in this case behaves as a simple RC low-filter and the cut-off frequency can be estimated as

$$B = \frac{1}{2\pi R_T C_T} \tag{7.56}$$

Example 7.5 A Si *p-i-n* photodetector is used in the front end of an optical receiver. The width of the *i*-region is 5 μm and the device area is 0.5×10^{-7} m^2. The load resistance and the input resistance of the amplifier are 1 kΩ and 3 kΩ respectively. The input capacitance of the amplifier is 5 pF. Estimate the bandwidth of the photodetector circuit. How does it compare with the maximum bandwidth of the photodetector in the absence of the external circuit elements? The relative permittivity of Si is 11.9. The saturation velocity of the carriers in Si is 10^5 m/s.

Solution The junction capacitance of the photodetector can be obtained as

$$C_j = \frac{\varepsilon A}{W} = \frac{8.854 \times 10^{-12} \times 11.9 \times 0.5 \times 10^{-7}}{5 \times 10^{-6}} = 1 \text{ pF}$$

The effective capacitance is given by

$$C_T = 1 \text{ pF} + 5 \text{ pF} = 6 \text{ pF}$$

The effective value of resistance can be estimated as

$$R_T = \frac{1 \times 3}{1 + 3} \text{k}\Omega = 0.75 \text{ k}\Omega$$

The bandwidth of the photodetector circuit is

$$B = \frac{1}{2 \times 3.14 \times 0.75 \times 10^3 \times 6 \times 10^{-12}} = 35.38 \text{ MHz}$$

The maximum bandwidth of the photodetector in the absence of external circuit elements can be obtained from Eq. (7.36) as

$$B_{\max} = \frac{v_s}{2\pi W} = \frac{10^5}{2 \times 3.14 \times 5 \times 10^{-6}} = 3.18 \text{ GHz}$$

The maximum bandwidth is two orders more than the circuit bandwidth of the photodetector. It may be pointed out that the photodetector is always used in a circuit and as a result, the maximum bandwidth cannot be achieved. However, by adjusting the values of various circuit parameters, the available bandwidth can be improved. In no case, the photodetector work faster than the ultimate physical limit described by Eq. (7.36).

7.7.4 Noise Performance

To study the noise performance of any junction photodiode including a *p-i-n* photodetector it is necessary to examine the various sources of noise in a generalized photodetection process illustrated in Fig. 7.13(a). The primary sources of noise include thermal noise arising out of various resistances and the shot noise arising out of the reverse saturation current of the diode under dark conditions, DC current flowing due to carriers generated by the background radiation, and that generated due to absorption of the incident light

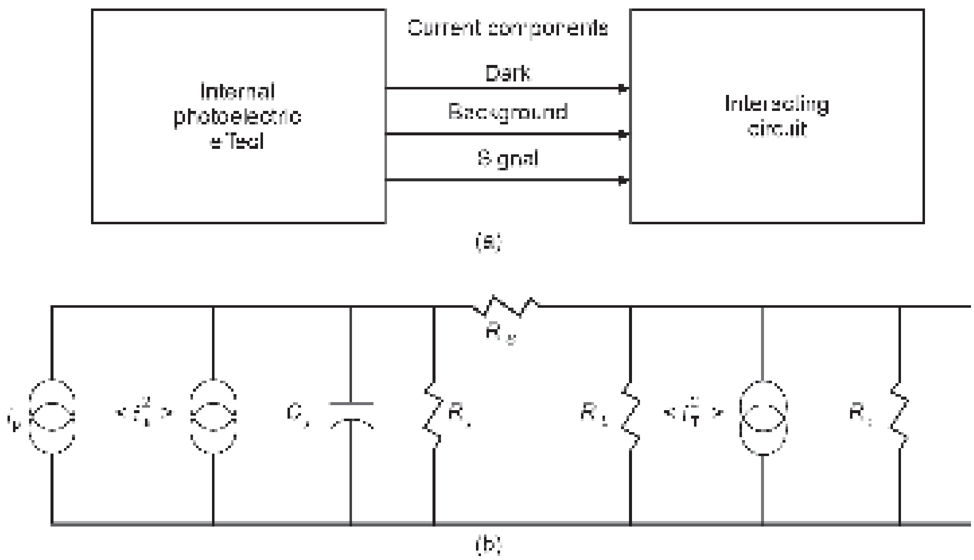


Fig. 7.13 (a) Photodetection process (b) Simplified noise equivalent circuit considering the effect of the input resistance of the following pre-amplifier stage (after Stillman *et al* 1977)

signal. A simplified noise equivalent circuit of a photodetector is shown in Fig. 7.13(b) (Stillman, 1977).

Considering the input optical signal intensity to be modulated sinusoidally with an index of modulation, μ then the rms value of the signal current can be expressed as

$$i_p = \frac{q\mu\eta P_{op}}{\sqrt{2}\,h\nu} \tag{7.57}$$

The average photocurrent due to optical signal is

$$I_P = \frac{q\eta P_{op}}{h\nu} \tag{7.58}$$

The other current components of the photodetector include the dark current I_D arising out of the thermally generated carriers and the current resulting from the carriers generated by the background radiation. The mean square value of the shot-noise current component because of the randomness of all these DC current components can be expressed as

$$\left\langle i_s^2 \right\rangle = 2q(I_D + I_B + I_P)B \tag{7.59}$$

where B is the bandwidth.

In the noise equivalent circuit of the photodetector shown in Fig. 7.8(b), C_j, R_j, R_s are the junction capacitance, junction resistance, and series resistance associated with the photodetector while R_L and R_i are the load resistance and input resistance of the following stage pre-amplifier. The resistance components contribute to additional thermal noise. Assuming the series resistance R_s to be negligibly small, the mean square value of the thermal noise current component can be approximated as

$$\left\langle i_T^2 \right\rangle = 4kT\left(\frac{1}{R_{eq}}\right)B \tag{7.60}$$

where

$$\frac{1}{R_{eq}} = \frac{1}{R_j} + \frac{1}{R_L} + \frac{1}{R_i} \tag{7.61}$$

For 100% modulated signal (μ = 1), the signal-to-noise power ratio at the output can be expressed as

$$\frac{S}{N} = \frac{i_p^2 R_{eq}}{\left(\langle i_s^2\rangle + \langle i_T^2\rangle\right) R_{eq}}$$

$$= \frac{\frac{1}{2}\left(\frac{q\eta P_{op}}{h\nu}\right)^2}{2q(I_D + I_B + I_P)B + \frac{4kTB}{R_{eq}}} \tag{7.62}$$

The minimum optical power required to obtain a given signal-to-noise ratio can be obtained from Eq. (7.62) as

$$\left(P_{op}\right)_{\min} = \frac{2h\nu B}{\eta}\left(\frac{S}{N}\right)\left\{1 + \left[1 + \frac{I_{eq}}{qB(S/N)}\right]^{\frac{1}{2}}\right\} \tag{7.63}$$

where

$$I_{eq} = I_D + I_B + \frac{2kT}{qR_{eq}} \tag{7.64}$$

It is interesting to note that when the factor $I_{eq}/qB(S/N)$ is much less than unity, the minimum optical power is determined by the quantum noise associated with the signal. On the other hand, when the quantity $I_{eq}/qB(S/N)$ is much larger than unity, the minimum power required to maintain a given S/N ratio depends on the dark current, current due to background radiation, and the equivalent resistance. Noise equivalent power (NEP) is an important figure of merit of the photodetector. The NEP is defined as the rms value of the minimum optical power at the input of the photodetector that is required to produce a unity S/N ratio at the output of the photodetector for unity bandwidth (B = 1 Hz). The NEP of the photodetector can thus be obtained from Eq. (7.64) as

$$\text{NEP} = \sqrt{2}\left(\frac{h\nu}{\eta}\right)\left(\frac{I_{eq}}{q}\right)^{\frac{1}{2}} \text{WHz}^{\frac{1}{2}} \tag{7.65}$$

To improve the NEP of the photodetector it is necessary to increase both η and R_{eq} and reduce the value of I_D and I_B.

Example 7.6 An In GaAsP *p-i-n* photodetector is connected to a load resistance of 1 MΩ followed by a pre-amplifier with a very high input resistance as compared to the load resistance. The junction resistance of the photodetector is 10 MΩ. The various components of the current of the photodetector are

$$I_D = 1 \text{ pA}; I_B = 0.2 \text{ nA}$$

The photodetector is operating at 1330 nm and has a quantum efficiency of 65%. Estimate the value of the NEP of the photodetector at room temperature (300 K).

Solution The equivalent resistance of the photodetector circuit can be obtained by using Eq. (7.48) and neglecting the input resistance of the pre-amplifier as

$$R_{eq} = \frac{1 \times 10}{11}\,\text{M}\Omega = \frac{10}{11}\,\text{M}\Omega$$

The equivalent current of the photodetector can be estimated

$$I_{eq} = 1 \times 10^{-12} + 2 \times 10^{-7} + \frac{2 \times 1.38 \times 10^{-23} \times 300 \times 11}{1.6 \times 10^{-19} \times 10 \times 10^{6}}$$
$$\approx 0.2\ \mu\text{A}$$

Thus, NEP of the photodetector

$$= \sqrt{2}\left(\frac{6.625 \times 10^{-34} \times 3 \times 10^{8}}{0.65 \times 1330 \times 10^{-9}}\right)\left(\frac{0.2 \times 10^{-6}}{1.6 \times 10^{-19}}\right)^{\frac{1}{2}} \text{WHz}^{\frac{1}{2}}$$

$$= 2.57 \times 10^{-13}\ \text{WHz}^{1/2}$$

Example 7.7 A *p-i-n* photodetector-based receiver has a bandwidth of 500 MHz. If the equivalent resistance of the front end is 100 kΩ, estimate the mean square value of the thermal noise current at 300 K.

Solution The mean square value of the thermal noise current can be obtained as

$$\langle i_T^2 \rangle = 4kT\left(\frac{1}{R_{eq}}\right)B$$

That is,

$$\langle i_T^2 \rangle = 4 \times 1.38 \times 10^{-23} \times 300 \times \frac{1}{100 \times 10^{3}} \times 500 \times 10^{6}$$
$$= 8.28 \times 10^{-17}\ \text{A}^2$$

Example 7.8 A GaAs *p-i-n* photodetector-based optical receiver operating at 0.87 μm has a bandwidth of 200 MHz. The incident optical power is 1 nW and the photodetector has a quantum efficiency of 60%. The background current is 0.02 nA. Estimate the value of the mean square shot-noise current of the photodetector. Assume the dark current to be negligibly small.

Solution The photogenerated current can be obtained as

$$I_P = \frac{q\eta\lambda}{hc} P_{op}$$

$$= \frac{1.6 \times 10^{-19} \times 0.6 \times 0.87 \times 10^{-6}}{6.625 \times 10^{-34} \times 3 \times 10^{8}} \times 10^{-9}$$

$$= 0.42\ nA$$

The mean square value of the shot-noise current in case of negligible dark current can be obtained as

$$\langle i_s^2 \rangle = 2q\left(I_B + I_P\right)B$$
$$= 2 \times 1.6 \times 10^{-19}\ (0.02 + 0.42) \times 10^{-9} \times 200 \times 10^{6}$$
$$= 2.8 \times 10^{-20}\ \text{A}^2$$

7.8 HETEROJUNCTION PHOTODETECTORS

So far, we have considered photodetectors made of a particular semiconductor material depending on the wavelength of operation matching with the bandgap of the material. A photodetector like many other semiconductor devices can also be realized in a heterojunction configuration. In a heterojunction, the junction is formed between two different semiconductors having different energy bandgaps with nearly equal lattice constant. The use of heterojunction can significantly improve the performance of conventional semiconductor junction devices using a particular semiconductor material (often called homojunction). One distinct advantage of using heterojunction in the case of a photodetector is that both the quantum efficiency and speed of response can be easily optimized. For example, in a conventional *p-i-n* detector, the quantum efficiency critically depends on the distance of the metallurgical junction from the input port through which the incoming light signal enters into the active part of the device. This problem can be largely overcome by making use of a heterojunction between a wide bandgap material and a narrow bandgap material. In such a case, the wide bandgap material used at the entrance can act as a window for transmission of the light straight into the narrow bandgap material for subsequent absorption in the region. Further, since no carrier is generated in the bulk of the large bandgap material the diffusion component of current can be eliminated. This in turn would improve the speed of response of the photodetector.

A variety of heterojunction semiconductor photodetectors have been proposed over the past decades (Bandyopadhyay *et al* 2001). To achieve a low leakage current (necessary to limit the noise), the lattice constant of the two semiconductors in a heterojunction must be closely matched (Sze, 2001). Among the III-V semiconductors, the heterojunction of GaAs and $Al_xGa_{1-x}As$ has been most widely studied. The ternary II-V semiconductor, $Al_xGa_{1-x}As$ is a direct bandgap material for $x < 0.4$ and can be epitaxially grown on GaAs to form perfect heterojunction (Casey *et al* 1978). The GaAs/$Al_xGa_{1-x}As$ heterojunctions are very attractive for optoelectronic devices operated in 0.65 to 0.85 μm wavelength regions. The photodetectors based on GaAs/$Al_xGa_{1-x}As$ have been widely used in the first-generation optical communication system. At longer wavelengths (1 to 1.6 μm) ternary alloys such as $In_xGa_{1-x}As$ and $Ga_xAl_{1-x}Sb$ grown on InP and GaSb substrates, respectively, can be used. The technology of InP/InGaAs and InP/InGaAsP has evolved significantly and the early difficulties encountered in the beginning have been largely overcome. In particular, $In_{0.53}Ga_{0.47}As$/InP heterojunction photodetector can respond up to 1.7 μm and has been extensively used in making the present generation optical fiber communication system working at 1.55 μm and 1.3 μm. Quaternary semiconductors $In_xGa_{1-x}As_{1-y}P_y$ grown on InP and $Ga_xAl_{1-x}AsSb$ grown on GaSb have been used for optical communication in longer wavelengths. InGaAs/InP is by far the most widely used heterojunction for advanced photodetector structures.

7.9 ADVANCED HETEROJUNCTION PHOTODETECTOR STRUCTURES

In principle, a simple *p-n* junction diode can be used as a photodetector. However, the performance of a simple photodetector is severely limited by the speed of response. With the advent of III-V semiconductor base heterojunction, it was possible to develop a variety of photodetector structures with improved performances (Bandyopadhyay *et al* 2001). However, in heterojunction photodetectors, the barriers created at the heterointerface cause temporary trapping of carriers resulting in a long tail in the time response of the photodetector and thereby leading to a decrease in bandwidth (Das *et al* 2001). The effect of trapping becomes significant when the thickness of the absorption layer is made small to enhance the bandwidth. Moreover, reductions of the thickness of the absorbing region cause an increase in the capacitance and so also the RC time constant and reduce the quantum efficiency. The problem associated with the increase in capacitance can be tackled by using a small area photodiode structure or a mesa structure. The problem of large barriers at the heterointerfaces can be solved by incorporating additional buffer layers. The quantum efficiency of the photodetector can be improved by making use of advanced structures (Bandyopadhyay *et al* 2001). Some of these advanced photodetector structures are discussed below.

7.9.1 Vertically Illuminated Heterojunction *p-i-n* Photodiode

The simplest heterojunction *p-i-n* photodetector with the top entry (also referred to as *front illumination*) is shown in Fig. 7.14(a). Two epitaxial layers, e.g. an *n*-InP buffer layer followed by an unintentionally doped *ν*-InGaAs absorption layer are epitaxially grown on an n^+-InP substrate (Dixon *et al* 1987). A planar *p-n* junction is formed in the *ν*-InGaAs layer by selective diffusion of Zn in the InGaAs layer. The light signal enters through the top region. The device operates in a 1.3 to 1.55 μm wavelength region which is absorbed in the InGaAs region and transmitted through the InP regions. One of the major drawbacks of the device is that the absorption of light in *p*-InGaAs results in a reduction of quantum efficiency. Moreover, it is extremely difficult to create an optical input port along with a metallic contact on the top keeping the device size small. Many variations of this basic structure have been reported in the literature. An alternative approach is to allow the light to enter through the transparent InP substrate region which would enable one to keep the device area small to have a low value of capacitance.

Further, the introduction of a p^+-InGaAsP layer provides a heterostructure that improves the quantum efficiency. This device fabricated in a mesa structure is shown in Fig. 7.14(b) (Miura *et al* 1987). The charge trapping at *n*-InGaAs/p^+-InGaAsP heterointerface may limit the speed of response (Bowers *et al* 1987).

7.9.2 Edge Illuminated Waveguide Photodetectors

The bandwidth of a *p-i-n* diode can be extended maximum up to ~50 GHz by paying the penalty in terms of reduced quantum efficiency. This is because the ultimate limit to the speed of response is decided by the transit time which can be reduced by reducing the thickness of the absorbing region. This has a direct consequence on the quantum efficiency of the device and more pronounced in indirect bandgap materials. The quantum efficiency of high-speed heterojunction *p-i-n* photodetectors can be improved significantly by allowing the light to enter parallel to the junction.

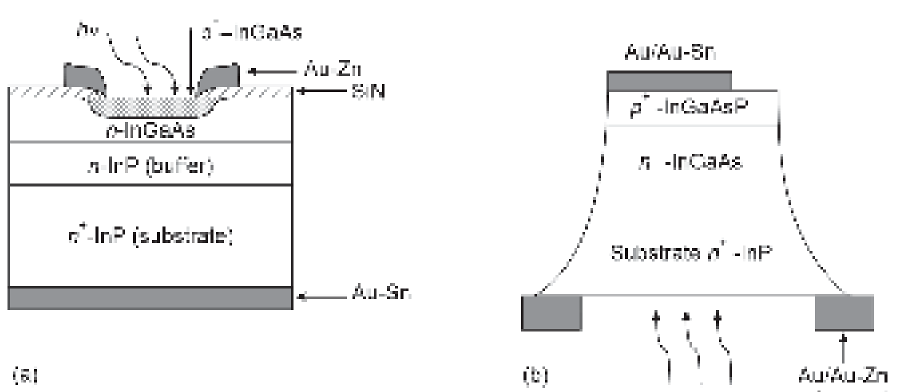


Fig. 7.14 (a) Planar InGaAs *p-i-n* photodiode (b) Backside illuminated InGaAs *p-i-n* photodiode

This type of configuration allows the light collection path to be longer keeping the current collection path shorter (Bowers & Burrus, 1987). Both high quantum efficiency and high response speed can be simultaneously achieved by using this structure. The other advantage of this type of waveguide photodetector is that the use of a very thin *i*-layer also permits one to operate this device at zero bias (Bowers Burrus, 1986). This feature of this photodetector makes it attractive for use in satellite systems. Moreover, operation at zero bias reduces the dark current significantly and thereby reduces the shot noise associated with the detector. The waveguide is designed in such a way that the light collected at the edge remains confined to the absorbing region only. The only factor that tends to reduce the speed is the diffusion component of current from the undepleted region outside the absorbing layer. A variety of waveguide photodetector structures have been proposed for operation in different wavelength ranges suitable for optical fiber communication systems. These include waveguide structures comprising GaAs with AlGaAs cladding for operation in the 0.8–0.9 μm wavelength region and InGaAs or InGaAsP with InP cladding for the 1.3–1.55 μm wavelength region. The cross-sectional view of an AlGaAs/GaAs waveguide photodetector is shown schematically in Fig. 7.15. An InGaAs/InP waveguide detector has a similar structure (Bowers & Burrus, 1987) for operation at 1.55 μm. A comprehensive theoretical analysis of a waveguide photodetector has been reported (Alping, 1989). The quantum efficiency of the photodetector depends on the optical confinement factor of the waveguide. For a typical waveguide photodetector with an intrinsic layer thickness of 0.2 μm having a 1 μm wide and 10 μm long waveguide with anti-reflection-coated facets, the transit time limited bandwidth has been estimated to be ~150 GHz with a quantum efficiency of 50% for optical confinement of 50% at an operating wavelength of 1.3 μm (Bandyopadhyay *et al* 2001). The coupling of optical power through the edge of the waveguide parallel to the plane of the semiconductor is extremely difficult for the application of the detector as a discrete component. This is also the major constraint that restricts quantum efficiency. However, in integrated circuit applications such as in an optoelectronic integrated circuit (OEIC), the edge-coupled waveguide detectors can be integrated with an optical waveguide to improve the quantum efficiency (Giboney *et al* 1992). The design and fabrication of a waveguide photodetector based on semi-insulating InP with high bandwidth has been reported recently (Nikoufard *et al* 2008). The photodetector was optimized for use as an optical amplifier or laser. The efficient and high-speed photodetector allows for easy integration of source, detector, and passive optical components on a single chip. A 3-dB bandwidth

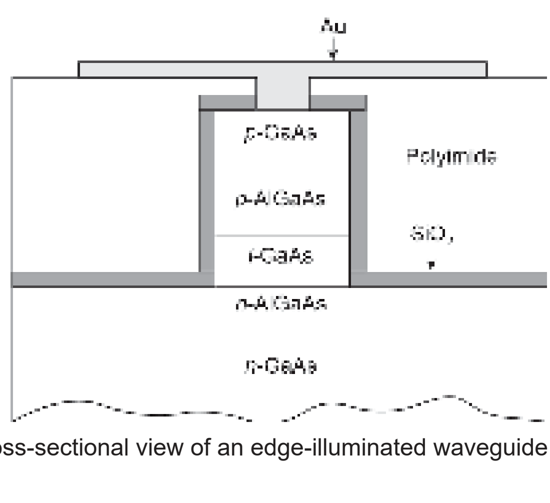


Fig. 7.15 Cross-sectional view of an edge-illuminated waveguide photodetector

of 35 GHz and an external of 0.25 A/W responsivity at 1.55 μm wavelength has been achieved at a reverse voltage of −4 V.

7.9.3 Resonant Cavity-enhanced Photodiode

A resonant-cavity-enhanced (RCE) photodetector can be realized by placing a thin absorption region of a photodetector in a resonant cavity. This enhances the quantum efficiency through multiple reflections of the optical field. Thus, a thin absorption layer is sufficient for achieving a high quantum efficiency. The thin absorption region also leads to a large BW due to the reduction in carrier-transit time through the active region. However, too thin an active layer may lead to a reduction in the BW due to RC time constant limitations. RCE photodetectors are attractive for potential applications in ultra-high-speed optical interconnects and communication systems (Unlu *et al* 2000; Kato, 1999). These photodetectors offer high-quantum efficiency and large bandwidth (BW). The two end mirrors of the cavity are generally formed by quarter-wave ($\lambda/4$) stacks of large bandgap semiconductors. The schematic representation of a generalized RCE *p-i-n* photodetector is shown in Fig. 7.16 (Unlu *et al* 1992).

The quantum efficiency of the RCE *p-i-n* photodetector for a cavity of length βL can be expressed as (Unlu *et al* 1992)

$$\eta = \left(\frac{1 + r_2 \exp(-\alpha W)}{1 - \sqrt{2r_1 r_2} \exp(-\alpha W)\cos(2\beta L) + r_1 r_2 \exp(-\alpha W)} \right)(1 - r_1)[1 - \exp(-\alpha W)] \tag{7.66}$$

where W is the thickness of the active region, r_1 and r_2 are the reflectivities of the top and the bottom mirrors of the distributed-Bragg-reflector (DBR) type, α is the optical absorption

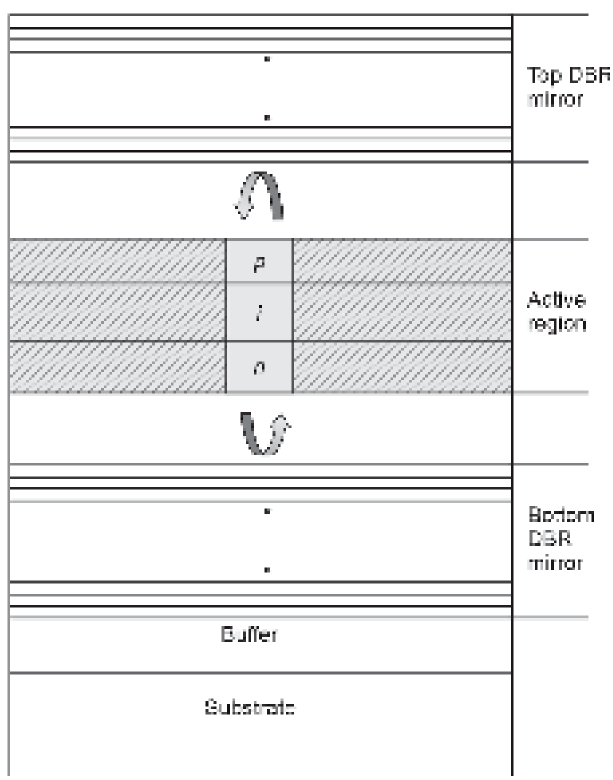


Fig . 7.16 Schematic of a generalised RCE *p-i-n* photodetector coefficient, and $\beta = 2\pi/\lambda$. The maxima of quantum efficiency occur at $\beta L = m\pi$, m being an integer.

A variety of RCE configurations have been proposed and studied over the past decades. The concept of resonant cavity has been applied to photodetector structures such as metal-semiconductor-metal (MSM) photodetector and avalanche photodiode (APD).

An RCE *p-i-n* photodetector based on InP/InGaAsP/GaAs has been reported for operation at 1.55 μm (Unlu *et al* 1992). The structure makes use of quarter-wave dielectric and InP/InGaAsP stack as top and bottom mirrors respectively. A quantum efficiency η of 82% has been reported to be achieved by using a 200 nm InGaAs absorption layer with $r_1 = 0.7$ and $r_2 = 0.95$ (Dentai, 1991). A further improvement in the quantum efficiency by 30% has been reported to be achieved by optimally placing the absorption region at cavity antinodes (Huang, 1993).

Apart from the various heterostructure *p-i-n* photodetectors discussed above, a few more structures have been proposed and studied. One such structure is the drift-enhanced dual absorption *p-i-n* photodiode (Sankaralingam, 2005). Unlike edge-coupled waveguide photodetectors this is a vertically illuminated heterojunction photodetector that manages efficiently the bandwidth-quantum efficiency trade-off of conventional *p-i-n* photodetectors. It uses a dual absorption region and a wide bandgap drift enhancement layer. Theoretically, estimated value of bandwidth-efficiency (BWE) is reported to be as high as 26 GHz.

7.10 SCHOTTKY BARRIER PHOTODETECTORS

A schottky barrier photodetector is essentially a metal-semiconductor rectifying contact (Ahlstrom, 1962) version of a *p-n* junction photodetector discussed earlier. A planar Schottky barrier photodiode is very attractive both for short-wavelength optical links, in which the maximum data rate is not limited by the inter-symbol interference caused by the dispersion of optical fibers but by the speed of the photodetectors. A schematic of the metal-semiconductor Schottky barrier photodetector is shown in Fig. 7.17(a). The diode is generally illuminated through the metal contact. To facilitate the transmission of light the metal film on the top of the semiconductor is usually made very thin (~100 Å) and accompanied by an antireflection coating. A metal-i-n photodetector similar to a *p-i-n* diode can be made by replacing the *n*-type material with an intrinsic material. A point contact metal-semiconductor photodetector is shown in Fig. 7.17(b). The small volume of the active region of the point contact photodiode results in extremely low drift time

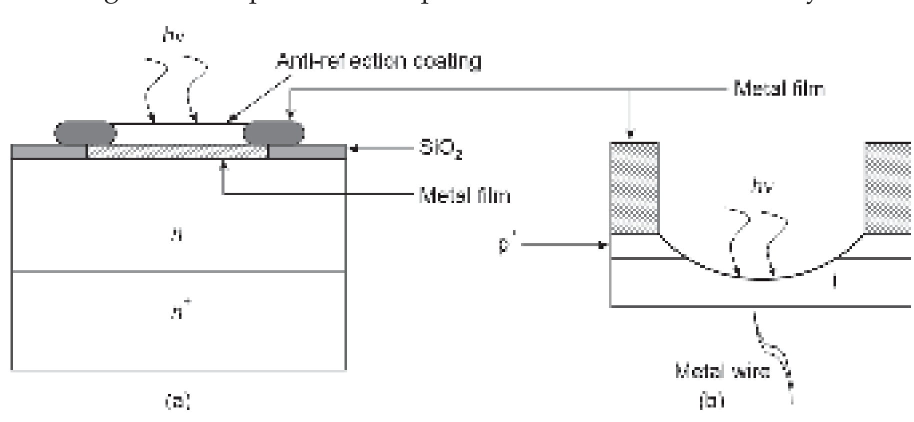


Fig. 7.17 Schematic of Schottky barrier photodiodes (a) Metal-semiconductor photodiode (b) Semiconductor point contact photodetector (after Melchior, 1972)

and capacitance enabling the device to operate at high modulation frequency (Sharpless, 1964). Lower parasitic resistance and capacitance of the Schottky barrier photodiode enable it to operate at frequencies >100 GHz (Bandyopadhyay & Deen, 2001). A narrow active region also causes lower quantum efficiency. The performance of the Schottky barrier photodetector is also affected by surface traps and recombination centers which reduces the responsivity significantly. The responsivity of Schottky barrier photodiodes can be improved by making use of direct bandgap materials with a large photo-absorption coefficient. Nearly 35% improvement in quantum efficiency over a broadband of infrared light of wavelength shorter than 1.6 μm has been reported to be developed using GaSb (Nagao *et al* 1981). A high-speed, high-sensitivity GaAs/AlGaAs heterojunction Schottky barrier photodiode capable of operating beyond 40 GHz has been developed (Lee, 1989). Fabrication and characterization of a GaAs Schottky barrier photodiode based on transparent ITO has been reported with a quantum efficiency of 60% and a bandwidth of 20 GHz at 0.83 μm (Parker, 1985). The two major difficulties associated with Schottky barrier photodiodes suitable for use in optical fiber communication systems are relatively poor quantum efficiency and non-availability of suitable metal for the formation of schottky barriers with III-V materials like InGaAs which operate in 1.3 to 1.6 μm. The first difficulty can be largely overcome by placing the active region within a resonant structure (Chin & Cheng, 1990), while the second problem associated with low schottky barrier height can be solved by artificially tailoring the energy band diagram through the incorporation of multiple layers of intermediate bandgap materials.

7.11 METAL-SEMICONDUCTOR-METAL (MSM) PHOTODETECTOR

A metal-semiconductor-metal (MSM) photodetector is essentially two schottky barrier diodes connected back to back. Unlike a schottky barrier photodiode in which there is one Schottky contact on one side and an Ohmic contact on the other side, an MSM photodetector has Schottky contact on both sides. MSM photodetectors are becoming increasingly attractive for application in optical fiber communication systems. The principal advantages of MSM photodetectors include their high responsivity, large bandwidth, integrated circuit compatibility, easy on-chip fabrication process matching perfectly with the existing planar integrated circuit technologies, relatively low voltage operation, and much lower capacitance per unit area as compared to *p-i-n* detectors (Soole *et al* 1992). The schottky electrode of an MSM photodetector is very similar to the gate metallization of a field-effect-transistor which is generally used for designing the pre-amplifier following the photodetector stage in an optical receiver. This feature enables one to develop optoelectronic integrated circuit (OEIC) receivers easily. With the advent of the technology of III-V materials and the introduction of GaAs/AlGaAs, InP/InGaAs, and other heterostructures, MSM photodetectors have become the most dominant photodetector for the realization of high-speed optical receiver systems. The basic MSM photodetector structure is shown in Fig. 7.18(a). It consists of two Schottky contacts on the top surface of an undoped semiconductor. It is designed in such a way that the region in between is completely depleted. In a planar configuration, the two Schottky contacts are made of Au, Au/Ti, or tungsten silicide (WSi_x). The contacts may be single strips or interdigitated metal fingers as shown in (Fig. 7.18). The energy band diagram of an MSM photodetector is shown in Fig. 7.18(b). The flat-band voltage V_{FB} can be expressed as

$$V_{FB} = \frac{qN_D L^2}{2\epsilon_s\epsilon_0} \tag{7.67}$$

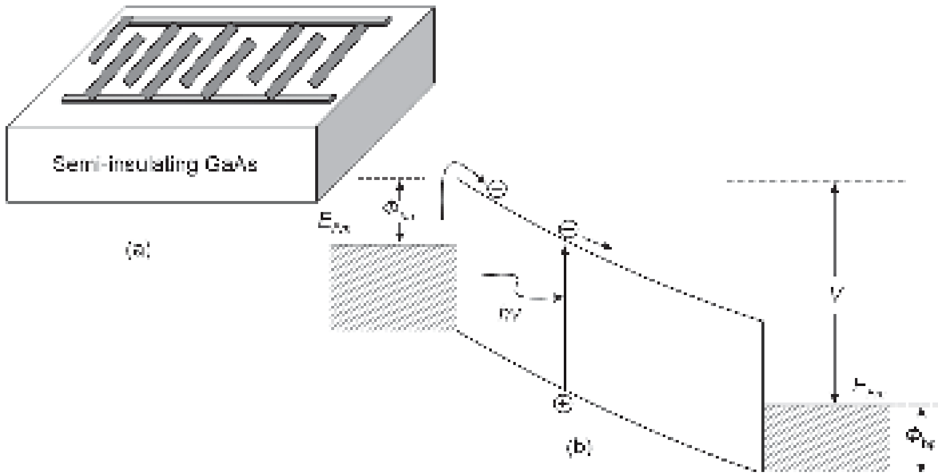


Fig. 7.18 Illustrating MSM photodetector (a) Schematic of an interdigitated MSM detector (b) Energy band diagram under bias

where L is the electrode spacing and N_D is the donor density of the underlying semiconductor. At low bias voltage electron injection at the reverse biased junction is dominant (Fig. 7.18(b)). As the bias increases, the hole injection at the forward bias condition also starts dominating. When the reach-through condition (the two depletion regions touch each other) is attained, the hole injection dominates the net current. The dark current of the MSM photodetector is determined by thermionic emission across the barrier. The total current density under these conditions is given by (Bhattacharya, 2007)

$$J = A_n^{**}T^2 \exp\left[-\frac{q\left(\Phi_{bn} - \Delta\Phi_{bn}\right)}{kT}\right] + A_p^{**}T^2 \exp\left[-\frac{q\left(\Phi_{bp} - \Delta\Phi_{bp}\right)}{kT}\right] \quad (7.68)$$

where $A_{n,p}^{**}$ are the respective Richardson constants and $\Phi_{b,n,p}$ are the respective barrier height lowering due to the image force effect (Fig. 7.18(b)).

The incident light enters the semiconductor from the top surface of the *MSM* structure and is subsequently absorbed within the underlying semiconductor resulting in the generation of electron–hole pairs (EHPs). The application of a suitable bias voltage across interdigitated electrodes creates an electric field in the underlying semiconductor. The applied electric field extracts the photogenerated carriers out of the device. The speed of response of the MSM detector is decided by how fast these carriers are collected while how many carriers manage to reach the contacts within a particular collection time determines the responsivity of the photodetector (Salem *et al* 1994). The photogenerated carriers created deep within the semiconductor have to travel a longer distance before they are collected at the contacts as compared to those generated near the surface. The magnitude of the electric field within the semiconductor determines the time needed to collect those carriers generated deep in the device. For a low applied voltage (5–10 V), this time is unacceptably large for high-speed applications. The MSM photodetector also suffers from low external quantum efficiency. This is primarily because the interdigitated electrodes shadow the light-gathering surface of the semiconductor. The trade-off between quantum efficiency and the speed of response of an MSM photodetector based on a single semiconductor material is very similar to a *p-i-n* photodetector. However, several novel MSM photodetectors based on heterostructures have been proposed in recent years for the improvement of quantum efficiency without affecting the speed. One approach to improve the speed of response of an MSM photodetector is to use a double heterostructure layer in place of a single semiconductor (Figueroa & Slayman, 1981). The double heterostructure layer blocks those photogenerated carriers created deep within the device structure. As a result, only those carriers photogenerated within the top absorption layer are collected. This in turn, leads to a fast overall speed of response. However, the responsivity of the MSM photodetector also depends upon the number of photogenerated carriers collected at the electrodes. Since many of the photogenerated carriers are produced deep within the bulk of the semiconductor layer, the insertion of a double heterostructure layer to improve the speed of the device necessarily reduces its responsivity as well (Salem *et al* 1994). Therefore, the fundamental trade-off between the speed of response and responsivity also exists in a heterostructure MSM detector. A double heterostructure MSM photodetector structure consisting of an AlGaAs layer sandwiched between the top GaAs active, absorption layer and the bottom GaAs substrate with interdigitated electrodes has been reported (Salem *et al* 1994). The effect of the depth of the AlGaAs layer on the speed and responsivity of the MSM devices has been examined. It is demonstrated that there is an optimal depth, at fixed applied bias, of the AlGaAs layer within the structure that provides maximum responsivity at minimal compromise in speed. Many more advanced MSM photodetector structures have also been reported over the past decades. The epi-layer structure of an InGaAs MSM photodetector grown on a semi-insulating InP: Fe substrate is shown in Fig. 7.19. The structure consists of an InGaAs photo-absorption region (~ 1 μm) sandwiched between two composition-graded InAlAs/InGaAs short-period superlattice (SPSL) layers.

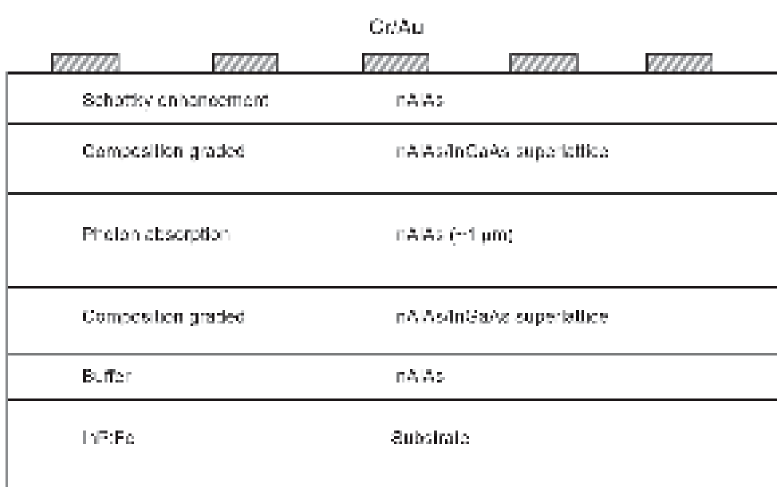


Fig. 7.19 Cross-sectional view of an epi-layer structure MSM photodiode with Cr/Au electrodes (after Kim *et al* 1997)

The structure is grown on a semi-insulating InP substrate. A buffer layer of InAlAs is used for better lattice matching and an additional layer of a thin (~30 nm) InAlAs layer on the top is used for enhancement of schottky barrier height (Kim *et al* 1997). The electrodes are in the form of interdigitated metal strips. The compositional grading layer prevents the carrier trapping effect and enhances the bandwidth of the MSM photodetector. Further, the introduction of the Schottky barrier enhancement layer helps in improving the Schottky barrier height between InGaAs and the metal which is very low usually for all metals (~0.2–0.3 eV). The enhanced schottky barrier height results in a low dark current and improves the noise performance of the MSM photodetector. This type of MSM photodetector is especially attractive for use in the OEIC receiver front-end comprising the photodetector and a FET-based pre-amplifier. An $In_{0.52}Al_{0.48}As/In_{0.53}Ga_{0.47}As$ metal-semiconductor-metal (MSM) photodetector and a high-electron-mobility transistor (HEMT) have been reported to be grown on GaAs substrate (Hong *et al* 1992) by organometallic chemical vapor deposition (MOCVD) for OEIC applications. The photodetector exhibited a responsivity of 0.45 A/W and leakage currents of 10 to 50 nA.

The reduction of quantum efficiency in an MSM photodetector because of the shadowing effect of the electrodes can be tackled by either using backside illumination (Kim *et al* 1997) or by using transparent electrodes (Seo *et al* 1992). The practical use of the backside illuminated MSM photodiode is limited because of critical and complicated processes involved in the fabrication. On the other hand, shadowing effect due to interdigitated metal electrodes in the case of front-illuminated MSM photodetectors can be significantly improved by making use of transparent contacts made of indium tin oxide (ITO) or cadmium tin oxide (CTO) on the top of Si or GaAs MSM photodiodes. A GaAs MSM photodiode with ITO electrodes has been reported to exhibit responsivity twice as high as that of a conventional MSM with Ti-Au electrodes (Seo, 1992). The main disadvantage of ITO is that it cannot be used as electrodes in InGaAs-based MSM because ITO exhibits high absorption in 1.3–1.6 μm suited for InGaAs. CTO can however be used for making transparent electrodes onto InGaAs for fabrication of MSM detectors (Berger, 1992; Khan *et al* 1994). Many alternative approaches have been adopted to improve the transparency of metal electrodes. This includes the addition of H_2-N_2 during sputtering deposition of ITO (Seo *et al* 1993), the use of semi-transparent Au schottky contact with overlaid SiN_x antireflection coating etc. (Yuang *et al* 1996). The InGaAs MSM photodiode using 10 nm thick Au electrodes is reported to exhibit a DC responsivity of 0.7 A/W at 1.55 μm.

Other significant developments in the area include metal-semiconductor-metal photodetectors (MSMPDs) with 0.87 ps response time and 510 GHz bandwidth on low-temperature GaAs (Chou, 1992); 10.7 ps and 41 GHz on bulk Si (Hsiang *et al* 1992) and a Si MSMPD with 3.7 ps response and 110 GHz bandwidth (Chou, 1993) with projected new possibilities for even faster (~400 GHz) operation; MSM traveling-wave photodetector (Shi *et al* 2001); two-dimensional gas-based vertical field MSM photodetector consisting of two Schottky contacts on top of a layer modulation-doped AlGaAs/GaAs heterostructure (Zhao *et al* 2008); InGaAs MSM photodetector on Si-substrate having performance comparable with InGaAs counterpart grown on lattice-matched InP (Wehmann *et al* 1996).

7.12 AVALANCHE PHOTODIODES

So far, we have discussed photodetectors which do not have any internal mechanism to multiply the photogenerated carriers. Avalanche photodiode is a specially designed photodetector that can internally multiply the primary signal photocurrent before it is delivered to the input circuitry of the following stage pre-amplifier. This increases the receiver sensitivity because the signal current is multiplied before encountering the thermal noise associated with the receiver circuit (Keiser, 2001). An avalanche photodiode is a more complex and sophisticated structure than a non-multiplying *p-i-n* photodetector. So for the photogenerated carriers can be multiplied they must travel through a high electric field region (2–5 × 10^7 V/m) where they may gain sufficient energy to ionize bound electrons in the valence band upon colliding with the lattice. This carrier multiplication mechanism is known as impact ionization. The primary carriers thus produce secondary carriers which again gain energy from the high field to produce tertiary carriers and the process of carrier generation builds up continuously. This phenomenon is known as the avalanche effect. A photodetector exploiting this mechanism to provide internal gain is called an avalanche photodiode (APD). The multiplication gain provided by the impact ionization is random. This randomness in the multiplication process is manifested in the form of additional noise which is often called excess noise in an APD.

The most commonly used structure for achieving carrier multiplication process with a minimal amount of excess noise is the reach-trough structure (Webb *et al* 1974). A schematic of the reach-through APD is shown in Fig. 7.20(a) and the energy band diagram of the structure is shown in Fig. 7.20(b). The structure consists of a high resistivity p-type material layer grown epitaxially on a heavily doped p^+ substrate. The lightly doped p-type (π) layer is followed by a moderately doped p-region and a heavily doped n-region. This structure is referred to as $p^+ - \pi - p - n^+$ *RAPD*. The π region is essentially an intrinsic region that unintentionally has some p-type doping because of improper purification during processing. When the reverse voltage applied across the structure increases the depletion layer widens across the p-region until it "reaches through" to the nearly intrinsic (lightly doped π region (Senior, 2001). In practice, the width of the π region is much wider than the p-region and as a consequence, the electric field in the π region is much lower than that at p-n^+ junction. It can be seen from the electric profile across the RAPD structure (Fig. 7.20(b)) that there exists a narrow region close to the metallurgical p-n^+ junction where the electric field is very high. Avalanche multiplication through impact ionization takes place in this narrow region. In this case, light enters through the p^+ region and is absorbed in the π region. The absorption of photons in this region creates electron-hole pairs which are separated by the existing electric field in this region. The photogenerated electrons drift through the π region into the high electric field at the p-n^+ junction. The carrier multiplication takes place at this high-field region. It may be pointed out that the electric field in the π region is smaller than that in the avalanche region nearly by a factor of 10, but the field in the drift region is kept high so that the carriers may be swept out by the electric field at a speed equal to the scattering limited velocity of the carriers. This is necessary to ensure the fast response of the APD. For a silicon RAPD for fiber optic communication application at

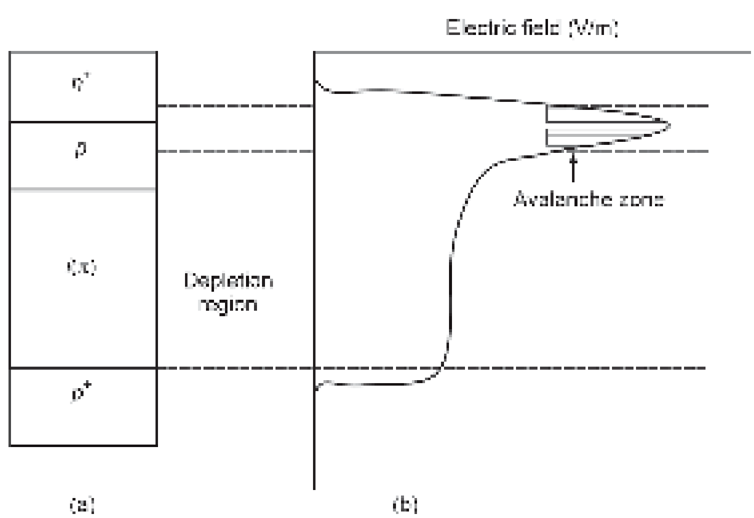


Fig. 7.20 Illustrating reach-through-avalanche-photodiode (RAPD) (a) Structure (b) Electric field profile

0.825 μm the quantum efficiency has been reported to be nearly 100%. The dark current of the APD is low and is affected by the applied reverse voltage. Ge has a photo-absorption band covering the entire optical range (0.8 to 1.6 μm) of interest in silica-based optical fiber communication systems. Silica, which is the primary constituent of an optical fiber offers low loss in this wavelength region having several attenuation windows with minimum attenuation occurring at 1.55 μm. As a result, Ge had drawn considerable interest in the beginning for the development of sensitive and fast avalanche photodiodes. However, high dark current associated with edge and surface effects resulting from difficulties in passivating Ge and excess noise made these APDs less demanding as compared to their Si counterparts. In the late 1970s and early 1980s when the attention on fiber optic communication was shifted to longer wavelength regions (1.1–1.6 μm) several APD structures based on Ge were reported (Ando *et al* 1978; Mikawa *et al* 1981, 1983; Mikami *et al* 1980; Kagawa *et al* 1981; Niwa *et al* 1984). The dark current of Ge APD was reported to be very sensitive to temperature variation (Newman *et al* 1986). The Ge APDs also exhibited a higher excess noise factor as compared to their Si counterpart. The only advantage of a Ge APD is its lower breakdown voltage.

Because of the drawbacks of Ge APDs for longer wavelength operation, III-V semiconductors have drawn considerable attention from the researchers for development of long wavelength APD for optical fiber communication. Both the ternary alloy InGaAs and the quaternary alloy InGaAsP lattice matched to InP have been most widely studied for the development of heterojunction APD (HAPD). A variety of APD structures have been proposed, fabricated and characterized over the past decades to match the stringent requirements of photodetectors for high-speed long haul optical fiber communication systems. An exhaustive review of various APDs for telecommunication applications has been recently reported (Campbell, 2007). Before we discuss the evolution of III-V-based APD it would be imperative to discuss the basic principle of operation of an avalanche photodiode.

7.12.1 Multiplication Gain

The multiplication gain in an avalanche photodiode can be best understood by quantifying the impact ionization process in terms of the electron ionization coefficient (the number of electron-hole pairs created by an electron per unit length) denoted by α and the hole ionization coefficient (number of electron-hole pairs created by holes per unit length denoted by β. These two parameters are believed to be functions of the local electric field and are characteristics of the material (McIntyre, 1966). The ionization coefficients are empirically expressed as functions of electric fields given by

$$\alpha = A_n \exp\left[-\left(\frac{B_n}{E}\right)^{m_n}\right] \tag{7.69}$$

$$\beta = A_p \exp\left[-\left(\frac{B_p}{E}\right)^{m_p}\right] \tag{7.70}$$

where A_n, B_n, A_p, B_p, m_n, m_p are empirical constants valid over a given range of electric field and E is the applied electric field. Impact ionization in most of the III–V materials used for making avalanche photodiodes for optical fiber communication requires an electric field in the range of 2–5 × 10^7 V/m. The impact ionization process has been widely investigated theoretically and experimentally by the researchers. An extensive review of the early works in this area has been reported (Stillman *et al* 1984). The concept of impact ionization and its role in determining the multiplication gain in avalanche photodiodes have changed over the past decades. Interested readers may refer to the new concept of impact ionization based on history-dependent theory (McIntyre, 1999). However, for a sufficiently large avalanche region, the local field theory works well.

The electric field required to create impact ionization depends on the energy bandgap of the material. The minimum energy required to cause impact ionization is referred to as the ionization threshold. The amount is different for electrons and holes. For the simplest case of parabolic conduction and valence bands, the ionization thresholds for electrons and holes can be expressed as (Ridley, 1982)

$$E_{ie} = E_g\left(1 + \frac{m_n^*}{2m_p^*}\right) \tag{7.71}$$

$$E_{ip} = E_g\left(1 + \frac{m_p^*}{2m_n^*}\right) \tag{7.72}$$

where E_g is the bandgap energy of the semiconductor, m_n^* and m_p^* are the effective masses of electrons and holes respectively.

To obtain the expression of gain arising out of impact ionization in an avalanche photodiode, let us consider a uniform avalanching region shown in Fig. 7.21. Assume that the reverse voltage applied across the p-n junction is sufficient to cause impact ionization of carriers in the high field region shown by the shaded area. Assume that a primary current I_{p_0} is injected into the high field region at $x = 0$ and similarly a primary electron current I_{n_0} is injected in the high field region at $x = W$. The hole current $I_p(x)$ will increase with distance x because of impact ionization and will reach a value $M_p I_{p_0}$ at $x = W$, where M_p measures the multiplication gain for holes over the avalanching region. Similarly, the primary electron current I_{n_0} injected at $x = W$ will increase to $M_n I_{n_0}$ at $x = 0$, M_n being the electron multiplication gain over the region. The total current at any point in the device ($I = I_p(x) + I_n(x)$) is constant in the steady state. The incremental hole current at any point, x over a distance dx is governed in the steady state by the following equation (Sze, 2003)

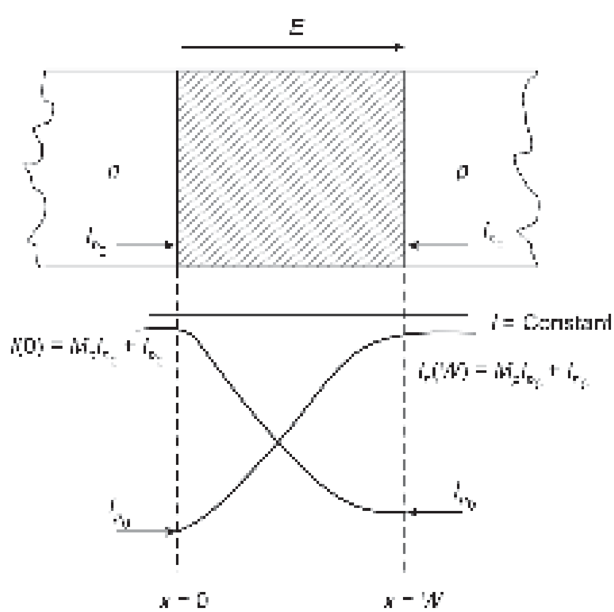


Fig. 7.21 Illustrating a uniform avalanching zone for the multiplication of electrons and holes

$$d\left(\frac{I_p(x)}{q}\right) = \left(\frac{I_p(x)}{q}\right)(\beta\, dx) + \left(\frac{I_n(x)}{q}\right)(\alpha\, dx) \tag{7.73}$$

The above equation can be rearranged in the standard differential equation form as

$$\frac{dI_p(x)}{dx} - (\beta - \alpha)I_p(x) = \alpha I \tag{7.74}$$

Equation (7.47) can be solved under the following boundary conditions

at $$x = 0,\ I_p(x = 0) = I_{p_0} \tag{7.75}$$

at $$x = W,\ I_p(x = W) = M_p I_{p_0} \tag{7.76}$$

Solving Eq. (7.74) with the help of the above boundary condition, we obtain

$$I_p(x) = \frac{I\left[\dfrac{1}{M_p} + \int_0^x \alpha \exp\left\{-\int_0^x (\beta - \alpha)dx'\right\}dx\right]}{\exp\left[-\int_0^x (\beta - \alpha)\right]dx'} \tag{7.77}$$

where M_p is the hole multiplication factor defined as

$$M_p = \frac{I_p(W)}{I_p(0)} \tag{7.78}$$

Equation (7.77) can be expressed as (Sze, 2003)

$$1 - \frac{1}{M_p} = \int_0^W \beta \exp\left\{-\int_0^x (\beta - \alpha)dx'\right\}dx \tag{7.79}$$

The avalanche breakdown occurs when $M_p \to \infty$. The condition for breakdown is given by the ionization integral

$$\int_0^W \beta \exp\left\{-\int_0^x (\beta - \alpha)dx'\right\}dx = 1 \tag{7.80}$$

When the ionization is initiated by electrons instead of holes, the ionization integral becomes

$$\int_0^W \alpha \exp\left\{-\int_x^W (\alpha - \beta)dx'\right\}dx \tag{7.81}$$

Equations (7.80) and (7.81) are equivalent in the sense that the breakdown condition depends on the ionization condition in the avalanching region and not on the type of carrier initiating the avalanche process (Sze, 2003). For semiconductors with equal ionization coefficients ($\alpha = \beta$), the breakdown condition is reduced to

$$\int_0^W \alpha\, dx = 1 \tag{7.82}$$

It may be noted that both α and β are functions of the position so long as the local field is a function of position, x. If the electric field in the entire region is uniform, the breakdown condition boils down to $\alpha W = 1$.

The breakdown occurs in a diode when the maximum field in the depletion region is equal to or greater than the critical electric field E_{CR}. For a reverse-biased abrupt p^+-n junction, the reverse breakdown voltage V_{BR} can be expressed in terms of $E_{CR} = E_m$ as (Bhattacharya, 1997)

$$V_{BR} = \frac{\epsilon_S \epsilon_0 E_{CR}^2}{2qN_D} \tag{7.83}$$

where N_D is the donor concentration on the n-side and the built-in voltage has been ignored in the deriving Eq. (7.83).

It is seen that the multiplication factor, M also called the avalanche gain is related to the impact ionization coefficients of electrons and holes. A very high value of gain ~1000 can be achieved by biasing the photodiode near the breakdown voltage. However, a very high value of avalanche gain can be achieved when sufficient time is available for avalanche build-up. For detection of high-frequency modulated light signals, this may severely restrict the bandwidth. This is the fundamental limitation of an APD that makes the gain-bandwidth product constant.

Example 7.9 The energy bandgap of InSb at 300 K is 0.17 eV. The ratio of the effective mass of the hole to that of the electron for this material is 40. Estimate the values of the ionization threshold of electrons and holes for InSb.

Solution The ionization threshold for electrons in InSb is

$$\varepsilon_{ie} = 0.17\left(1+\frac{1}{2}\left(\frac{1}{40}\right)\right) \approx 0.17 \text{ eV}$$

The ionization threshold for holes can be obtained as

$$\varepsilon_{ih} = E_g\left(1+\frac{40}{2}\right) = 3.57 \text{ eV}$$

7.12.2 Avalanche Multiplication Noise

The inherent gain in an avalanche photodiodes results from the multiplication of the carriers. However, the avalanche multiplication process is statistical. This is because every electron-hole pair generated anywhere in the high-field region does not experience the same multiplication. Thus, the avalanche gain fluctuates and the mean square value of the gain is always greater than the square of the mean value of the gain. This statistical randomness manifests in the form of an extra component of noise in addition to the existing shot and thermal noise. This extra noise is characterized in terms of excess noise in an APD defined as

$$F(M) = \frac{\langle m^2 \rangle}{\langle m \rangle^2} \tag{7.84}$$

where $\langle\ \rangle$ denotes an ensemble average and $\langle m \rangle = M$ is the average carrier gain defined by

$$M = \frac{I_M}{I_p} \tag{7.85}$$

Here, I_M is the average value of the total multiplied output and I_p is the primary (unmultiplied) photocurrent. The mean square value of the gain can be empirically expressed as

$$\langle m^2 \rangle = M^{2+x} \tag{7.86}$$

The parameter x varies between 0 and 1.0 depending on the structure of the photodiode and the constituting material.

The excess noise factor can thus be expressed as

$$F(M) = \frac{M^{2+x}}{M^2} = M^x \tag{7.87}$$

The excess noise factor thus depends on the multiplication gain which in turn is a function of the ratio of the ionization coefficients of electrons and holes and the carrier multiplication process. A full derivation of an expression for $F(M)$ is fairly complex. The excess noise factors for injected electrons and holes can be expressed as (McIntyre, 1972)

$$F_n = \frac{k_2 - k_1^2}{1-k_2} M_n + 2\left[1 - \frac{k_1(1-k_1)}{1-k_2}\right] - \frac{(1-k_1)^2}{M_n(1-k_2)} \tag{7.88}$$

$$F_p = \frac{k_2 - k_1^2}{k_1^2(1-k_2)} M_p + 2\left[\frac{k_2(1-k_1)}{k_1^2(1-k_2)} - 1\right] - \frac{(1-k_1)^2 k_2}{M_h \mathrm{k}_1^2(1-k_2)} \tag{7.89}$$

where k_1 and k_2 are weighted ionization ratios that take into account the non-uniformity of the gain and the carrier ionization coefficients in the avalanche region. These factors are given by

$$k_1 = \frac{\int_0^W \beta(x)M(x)dx}{\int_0^W \alpha(x)M(x)dx} \tag{7.90}$$

$$k_2 = \frac{\int_0^W \beta(x)M^2(x)dx}{\int_0^W \alpha(x)M^2(x)dx} \tag{7.91}$$

where the electron and hole ionization rates are considered to be functions of x.

To a first-order approximation, we ignore the changes in the values of k_1 and k_2 with variations in gain and consider these parameters to be constant and equal. Under these conditions, Eqs (7.88) and (7.89) are simplified as (Webb, 1974)

$$F_n = M_n\left[1-\left(1-k_{eff}\right)\left(1-\frac{1}{M_n}\right)^2\right] = k_{eff}M_n + \left(2-\frac{1}{M_n}\right)\left(1-k_{eff}\right) \tag{7.92}$$

for electron injection.

$$F_p = M_p\left[1-\left(1-\frac{1}{k'_{eff}}\right)\left(1-\frac{1}{M_p}\right)^2\right] = k'_{eff}M_p - \left(2-\frac{1}{M_p}\right)\left(k'_{eff}-1\right) \tag{7.93}$$

for hole injection. The effective ionization coefficient ratios are given by

$$k_{eff} = \frac{k_2 - k_1^2}{1-k_2} \approx k_2 \tag{7.94}$$

$$k'_{eff} = \frac{k_{eff}}{k_1^2} = \frac{k_2}{k_1^2} \tag{7.95}$$

Figure 7.22 shows the variation of F_n as a function of the average electron multiplication gain M_n for various values of effective ionization coefficient ratio.

If the ionization coefficients for electrons and holes are the same for the semiconductor material, the excess noise factor is maximum. It can be easily seen that in this case,

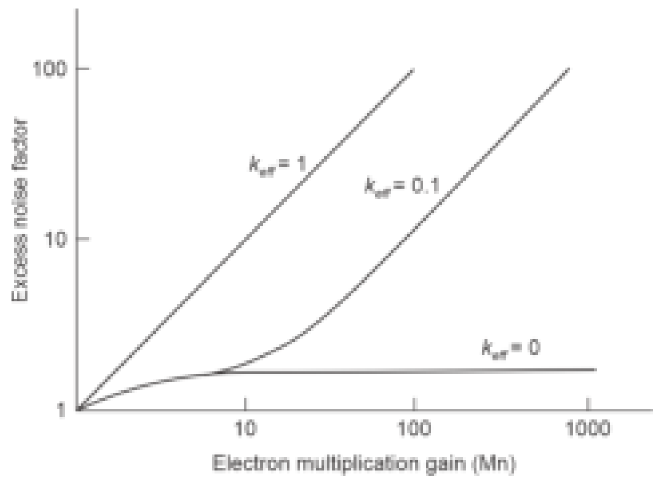


Fig. 7.22 Variation of the electron excess noise factor, F_n with the electron multiplication gain, M_n for three different values of the effective ionization coefficient ratio (after Webb *et al* 1974)

the excess noise factor becomes equal to the multiplication gain, M_n. As the ratio β/α decreases from unity, the electron ionization coefficient starts playing a dominant role in the creation of electron-hole pairs through impact ionization and the excess noise factor becomes smaller. When $\beta = 0$, only electrons cause impact ionization and the excess noise factor reaches the lowest limit of 2 (Fig. 7.22). It may therefore be concluded that to keep the excess noise factor minimum, it is necessary to have a small value of k_{eff}, which is essentially the ratio of the ionization coefficients of electrons and holes. It is interesting to compare the ratio of the ionization coefficients, k_{eff} of three important materials, e.g. Si, InGaAs, and Ge. The values of k_{eff} vary between 0.015–0.035 for Si, 0.3–0.5 for InGaAs, and 06–1.0 for Ge (Keiser, 2001). The smallest value of the ratio of ionization coefficients in the case of Si made it attractive for making APD during the first-generation optical fiber communication system. Later on, when the focus on operating wavelength shifted to longer wavelength regions it became necessary to look for newer materials (mostly the alloy semiconductors). Unfortunately, most of the III-V semiconductors suitable for operation in the longer wavelength region exhibited nearly equal values of ionization coefficients. This turned out to be a major drawback of these materials for their potential application in avalanche photodiodes. An extensive review of the work related to the study of ionization coefficients of III–V materials is available in the literature (Stillman *et al* 1983 and references listed therein).

7.12.3 Noise Equivalent Power (NEP)

The noise equivalent circuit of an avalanche photodiode can be obtained similarly as done earlier in the case of a *p-i-n* detector. The only difference is that the APD offers an internal multiplication gain which affects the current and shot noise components accordingly. If the multiplication gain of the APD is M, the signal photocurrent can be obtained as

$$i_p = \frac{q\eta\mu P_{op}}{\sqrt{2}\, h\nu} M \tag{7.96}$$

The mean square value of the shot-noise current following multiplication can be obtained as

$$\langle i_s^2 \rangle = 2q\left(I_D + I_B + I_P\right)\langle M^2 \rangle B \tag{7.97}$$

Using Eq. (7.68), (7.81) can be written as

$$\left\langle i_s^2 \right\rangle = 2q\left(I_D + I_B + I_P\right)M^2F(M)B \tag{7.98}$$

For 100% modulation (μ = 1), the signal-to-noise ratio can be expressed as

$$\frac{S}{N} = \frac{\frac{1}{2}\left(\frac{q\eta P_{op}}{h\nu}\right)^2}{2q\left(I_D + I_B + I_P\right)M^2F(M)B + \frac{4kTB}{R_{eq}M^2}} \tag{7.99}$$

That is,

$$\frac{S}{N} = \frac{\frac{1}{2}\left(\frac{q\eta P_{op}}{h\nu}\right)^2}{2qI_{eq}B + 2qM^2F(M)I_PB} \tag{7.100}$$

where R_{eq} is the equivalent resistance as defined earlier and

$$I_{eq} = I_D + I_B + \frac{2kTB}{qR_{eq}M^2} \tag{7.101}$$

It can be seen that the noise equivalent power for the APD can be expressed as

$$\mathrm{NEP} = \left(\frac{h\nu}{\eta}\right)\sqrt{\frac{2I_{eq}}{q}}\mathrm{WHz}^{\frac{1}{2}} \tag{7.102}$$

The NEP of avalanche photodiode is improved through the reduction of overall I_{eq} by the multiplication gain M (Stillman *et al* 1977).

7.12.4 III–V Semiconductor–based Avalanche Photodiodes

Si-based *p-i-n* and APD were mainly used in the first-generation optical fiber communication systemsoperating in the 0.8 to 0.9 µm wavelength range. To exploit the advantage of lower attenuation of optical fibers in 1.3 to 1.5 µm the operating wavelength was shifted to a longer wavelength side during the next generations. Among the III–V materials, the ternary alloy $In_xGa_{1-x}As$ and quaternary alloy $In_xGa_{1-x}As_{1-y}P_y$ turned out to be most attractive for use in developing photodetectors for the new generation. $In_{0.53}Ga_{0.47}As$ became the material of choice because the bandgap of the material closely matches the wavelength of operation at 1.55 µm. At this wavelength silica-based optical fiber has the lowest loss (Keiser, 2001). Moreover, the dispersion characteristics of the fibers can be tailored to have minimum dispersion at this wavelength. InGaAs-based *p-i-n* photodetectors soon became very popular for use in optical receiver systems. On the other hand, avalanche photodiodes based on InGaAs encountered a major setback during the early stage of development of excess dark current resulting from tunneling under the action of a high electric field (Forrest *et al* 1980; Ando *et al* 1980). Later on, the tunneling component of the dark current was successfully eliminated by making use of separate absorption and multiplication (SAM) APD structures (Nishida *et al* 1979; Stillman, 1987). In the SAM-APD structure, the high-field multiplication region (*p-n* junction) is located in a wide bandgap semiconductor such as InP where the tunneling is insignificant and the absorption occurs in the subsequent narrow bandgap layer such as InGaAs. A schematic of a SAM–APD along with the electric field is shown in Fig. 7.23. The structure enables one to control the charge density in the multiplication layer to maintain a high electric field sufficient

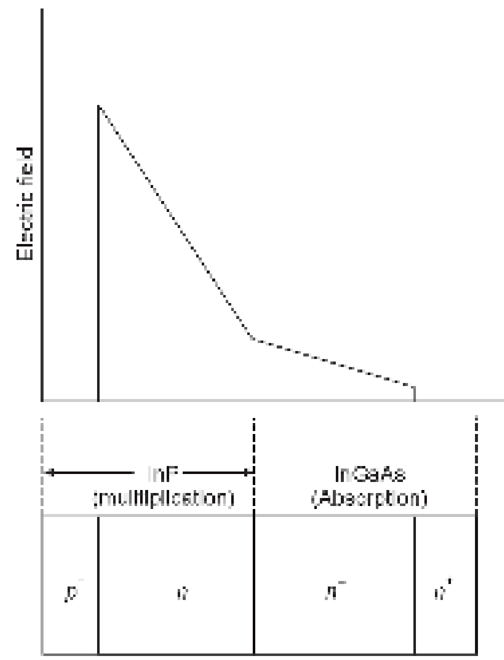


Fig. 7.23 Schematic of a separate absorption multiplication SAM–APD with electric field profile

to achieve a good avalanche gain while maintaining a low electric field in the absorbing region. This flexibility allows one to minimize the tunneling and impact ionization in the InGaAs absorber. However, precise control of doping concentration and thickness of *n*-InP is very critical for maintaining low leakage current. A well-designed SAM-APD can offer a reasonably good quantum efficiency (~80%) with a low leakage current (<10 nA) and an acceptable capacitance of 0.5 pF. The operating voltage of SAM-APD is quite high (`~100V) (Senior, 2001). The frequency response of the originally proposed SAM APDs was very poor because of the accumulation of photogenerated holes at the absorption/multiplication heterojunction interface (Forrest *et al* 1982). This trapping of holes at InP/InGaAs heterointerface can be largely overcome by incorporating a thin grading layer of InGaAsP in between having an intermediate bandgap between InP and InGaAs. The intermediate layer helps to smooth out the band-edge discontinuity at the InP/InGaAs heterointerface as illustrated in Fig. 7.24. This in turn reduces the hole trapping effect and

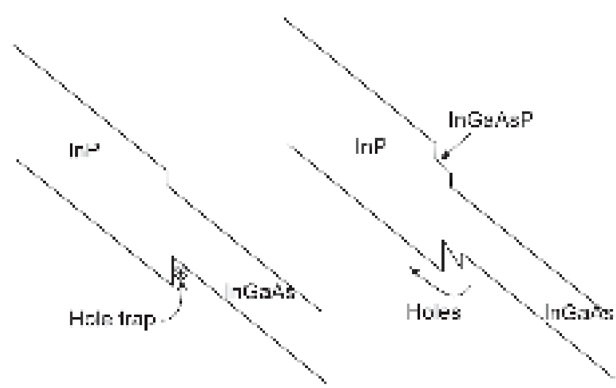


Fig. 7.24 Energy band diagram of SAM-APD for (a) Originally proposed InP/InGaAs (b) Modified structure using an intermediate bandgap layer of InGaAsP

improves the response speed of InP/InGaAs SAM-APD (Cambell *et al* 1983; Matsushima *et al* 1982; Forrest *et al* 1982). The gain-bandwidth product of the improved SAM-APD is still in the range of 10–20 GHz (Newman & Ritchie, 1986). Several approaches to eliminate the slow release of trapped holes including the (i) a chirped-period InP/InGaAs superlattice to form a pseudo-quaternary $In_xGa_{1-x}As_{1-y}P_y$ layer (Capasso, 1984); a continuous grading of the transition region (Kuwatsuka *et al* 1991); a transition region consisting of one or more latticed-matched intermediate-bandgap $In_xGa_{1-x}As_{1-y}P_y$ layers (Campbell *et al* 1983). Another approach to improve the performance of SAM-APD involved the inclusion of a high–low doping profile in the multiplication region (Capasso *et al* 1984; Tarof *et al* 1990). A wide-bandgap multiplication region consisting of a lightly doped layer where the field is high, and an adjacent doped charge layer are included in this structure. This type of APD structure is also known as separate absorption, charge, and multiplication (SCAM).

The function of the additional "Charge" layer is to decouple the thickness of the multiplication region from the charge density constraint in the conventional SAM-APD. Most of the early work on InP/InGaAsP/InGaAs SAM and SACM APDs utilized mesa structures because of their fabrication simplicity and reproducibility (Campbell, 2007). A backside illuminated mesa structure separate absorption, grading, and multiplication (SAGM) InGaAs APD is shown in Fig. 7.25 (Kasper & Campbell, 1987). The photodetector is reported to exhibit a gain-bandwidth product of 70 GHz which means that the detector can be operated at a bit rate exceeding 5 Gbps with a moderate value of gain. However, the fact that planar structures are more reliable than mesa-type photodiodes motivated the researchers to develop planar photodiodes. Some of the novel planar APD structures utilize a laterally extended guard ring (Taguchi *et al* 1988; Matsushima *et al* 1984), floating guard rings (Liu *et al* 1992; Cho *et al* 2000; Wei *et al* 2002), etched diffusion well (Tarof *et al* 1995). These approaches have been successful in suppressing an edge breakdown (Campbell, 2007).

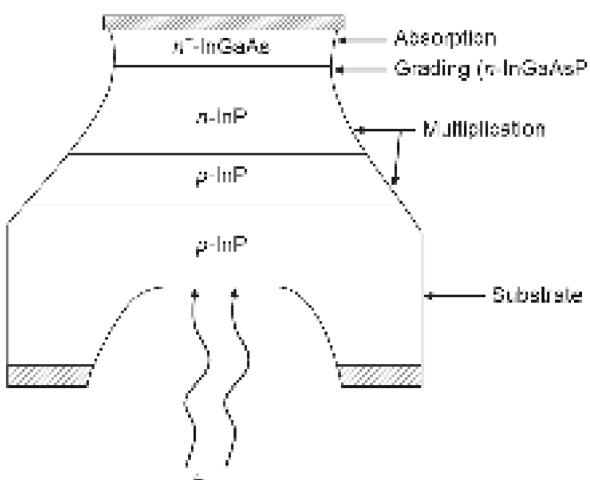
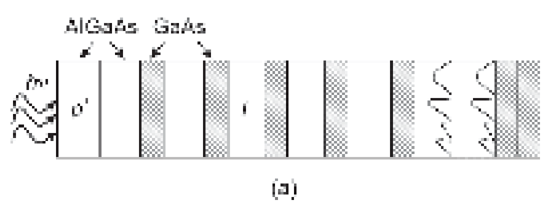


Fig. 7.25 Separate absorption, grading, and multiplication (SAGM) APD structure

Apart from the APD structures discussed above many other heterojunction APD configurations have been studied and investigated over the past decades for high-speed applications. These include photo-DOVATT (Double Velocity Avalanche Transit Time) photodiode (Chakrabarti *et al* 1987; 1990), high-speed resonant cavity SAM-APD (Nie *et al* 1987). Another novel concept to improve the sensitivity of avalanche photodiodes is based on semiconductor superlattices in the form of multi-quantum well structure (MQW) (Chin *et al* 1980; Batra *et al* 1988) and stair-case APDs (Capasso, 1983). The gain regions of these APDs consist of multi-quantum wells formed by alternately growing thin layers of wide and narrow bandgap materials such as AlGaAs and GaAs (Chin *et al* 1980; Chakrabarti & Pal, 1987) or InP and InGaAs (Batra *et al* 1988). The schematic of a multi-quantum well APD also known as superlattice APD (SL-APD) is illustrated in Fig. 7.26 along with the energy band diagram. By exploiting the asymmetry in the band-edge discontinuities in the valence and conduction bands of the wide and narrow bandgap materials, it is possible to improve the ratio of effective ionization coefficients for electrons and holes. Improvements in the ratio of α/β by a factor of 8 for an applied electric field of 2×10^7 V/m in the case of AlGaAs/GaAs superlattice APD (Chin *et al* 1980) and β/α by a factor of 16 for an applied electric field of 3×10^7 V/m for InP/InGaAs superlattice APD have been reported (Batra *et al* 1988). This artificial improvement in the ratio of the ionization coefficients (reduction of k_{eff} value) of III–V materials which otherwise have nearly equal values of electron and hole ionization coefficients (for bulk materials) are responsible for better noise performance of these APDs. Theoretical analysis of the noise behavior of superlattice APDs has been studied for estimating the noise equivalent power and output S/N ratio of these devices (Chakrabarti, 1989; Rajamani and Chakrabarti *et al* 1999). The other major advantage of SL-APD is that the bandwidth is significantly improved in this structure because of the reduction of avalanche build-in time. A step-like MQW energy band structure can be obtained by tailoring the energy gap of the wide bandgap material with the help of graded material composition. The compositional grading is done in such a way that as the carriers move through the wide bandgap material they encounter larger and larger bandgap (so also larger ionization threshold). Therefore, after gaining energy in the wide bandgap material when they enter into the narrow bandgap material they can ionize more effectively. As in conventional APD, the asymmetry of the band-edge discontinuities can be exploited to make ionization by one type of carrier dominant thereby reducing the effective ionization coefficient ratio. This concept works well with GaAs/AlGaAs but not with InP-based MQW structures because of excessive tunneling. Another SL-APD based on InGaAs/InAlAs, InGaAsP/InAlAs, and



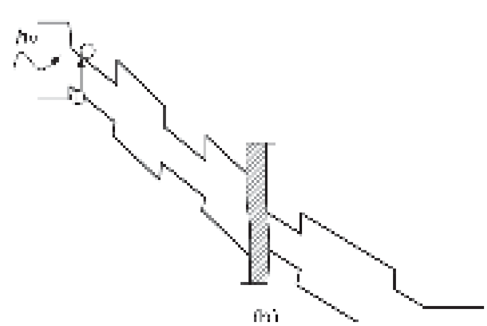


Fig. 7.26 (a) Schematic of an SL-APD (b) Energy band diagram under reverse bias

InGaAs/InAlAs have been extensively studied to overcome the limitations of conventional APDs based on InP/InGaAs. These APDs are reported to achieve gain-bandwidth products over 100 GHz (Bandyopadhyay & Deen, 2001).

It is seen from the previous discussion that the effective ratio of α and β can be increased by making use of superlattice structure consisting of multiple heterojunctions of wide bandgap and narrow bandgap materials. A more complex structure consisting of a compositionally graded narrow bandgap material can drastically improve the ionization rate ratio. The structure is known as staircase APD and is shown in Fig. 7.27. It consists of a narrow bandgap material region which is compositionally graded in such a way as to create an energy bandgap difference of at least twice over a distance of the order of 20 nm in the form of a step as shown in the figure. At the end of each step, the energy bandgap drops to the original value of the narrow bandgap material and continues to increase with compositional grading till the end of the step. Consider an electron traveling through any step. When the electron advances through the graded step it continuously experiences an increase in the ionization threshold because of the increased bandgap caused by the compositional grading and continues to gain energy from the applied electric field. As soon as the electron enters into the next step it suddenly experiences a huge reduction in the ionization threshold caused by the reduction of bandgap energy. This will enable the electrons to cause heavy ionization. However, the electrons and holes will not experience the same situation because of the asymmetry in the band structure. The net result will be a large increase in the avalanche multiplication gain caused by one type of carrier (electrons in this example). The large increase in the ionization rate ratio in turn will make the device less noisy. Therefore there will be an enormous improvement in the ratio of the effective ionization rates. Another advantage is that the transition of carriers from wide bandgap to narrow bandgap regions can cause impact ionization at a much lower electric field. The staircase APD thus requires a much lower voltage for operation than the conventional superlattice APD structures discussed earlier. Staircase APD structure based on the concept of bandgap engineering was originally proposed by Capasso (Capasso, 1984).

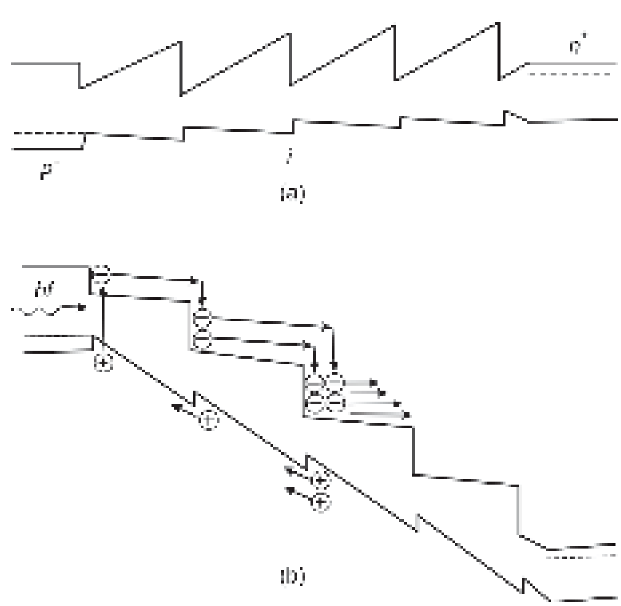


Fig. 7.27 The energy band diagram of a staircase APD (a) Unbiased (b) Under reverse bias

7.13 CHARGE COUPLED DEVICE (CCD) AS AN IMAGE SENSOR

CCD is essentially an array of Metal Insulator Semiconductor (MIS) structures that can be operated under non-equilibrium conditions with the help of a large gate pulse. Most commonly Si based MOS capacitor array is used in CCDs. The device can be deployed as an image sensor or as a shift register. For the latter application, the device is termed as a charge transfer device. As an imaging device, CCD finds applications in video recording and camera and works as a photodetector. The concept of CCD was introduced primarily as a shift register by Boyle & Smith[2] in 1970 (Boyle *et al* 1970; 1971). In the same year, an 8-bit shift register based on the charge-couple device was successfully demonstrated by Tompsett *et al* (Tompsett *et al* 1970) as a linear scanning system. In the following year, a detailed experimental report on the charge-coupled imaging devices was published by the same group (Tompsett *et al* 1971; Amelio *et al* 1971). Later on, CCD-based area scanning system was developed and reported in 1972 (Bertram *et al* 1972). However, Si-based CCDs cannot be used for imaging beyond the detectable wavelength of Si. As a result, the researchers started exploring the possibility of using compound semiconductor-based CCD structures for imaging at longer wavelengths (Tao *et al* 1973; Kim 1973). In the subsequent years, CCDs were deployed commercially as solid-state imaging devices for applications in civil and military situations.

The basic structure of a CCD-based imaging device is similar to a MOS capacitor except for the fact that the metal gate is made of semi-transparent for the light from the object to be captured as an image passes through the gate into the bulk material. To make the gate semi-transparent the thickness of the metal is reduced significantly or replaced by a polysilicon gate or a similar transparent electrode. The other alternative is to allow the light to enter from the back side (substrate side). As the substrate is very thick it is necessary to reduce the thickness of the substrate with the help of a suitable etching technique for making the back illumination an effective alternative. For a thick substrate most of the photons will be absorbed in the substrate before reaching the depletion region to cause the desired effect. The CCD imaging device in the form of MIS can be fabricated in the form of a surface channel device or a buried channel device. The longitudinal section of a typical surface-channel CCD (SCCCD) imager in the form of an array of MIS structures fabricated on a single semiconductor substrate is shown in Fig. 7.28. The figure shows the structure of a surface channel CCD imager with its energy band diagram under deep depletion.

Under the application of a large gate pulse of a suitable polarity (positive in this case) to the device shown in Fig. 7.29(a), the device is driven in the deep depletion mode and the surface potential can be written as (Sze *et al* 2007)

$$V_G - V_{FB} = \frac{qN_A W_D}{C_i} + \psi_s \tag{7.103}$$

where V_{FB} is the flat-band voltage, q is the electronic charge, N_A is the acceptor concentration in the p-type semiconductor, C_i is the insulator capacitance per unit area, and ψ_s is the surface potential under deep depletion given by

$$\psi_s = \frac{qN_A W_D^2}{2\epsilon_s} \tag{7.104}$$

[2]Willard S Boyle and George E Smith, invented the CCD memory for which they earned the 2009 Nobel Prize in Physics.

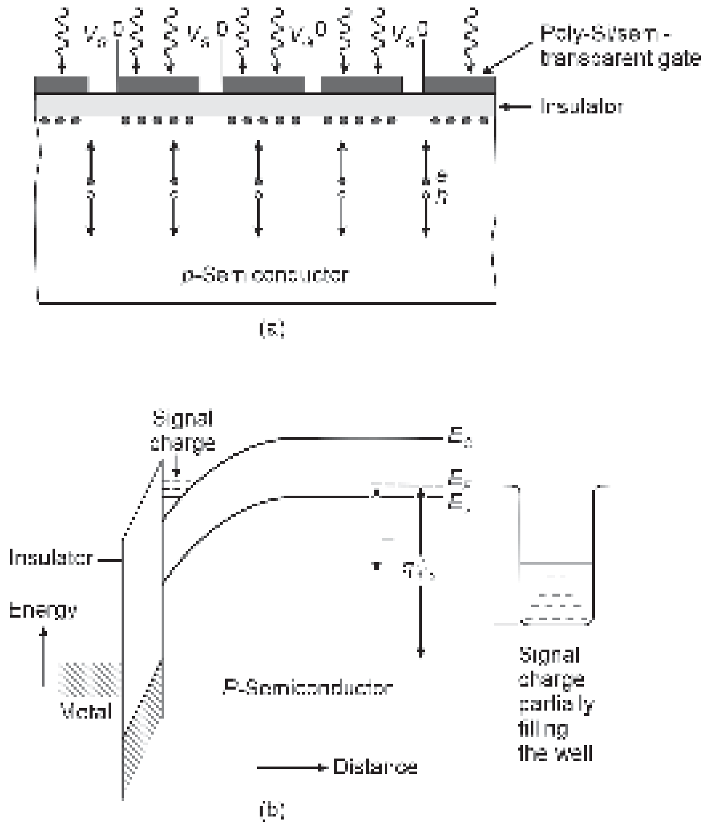


Fig. 7.28 Surface channel CCD imaging device (a) Schematic of a CCD imager (b) The energy band diagram showing band-bending under deep-depletion and partially filled well with signal charge

Here, ϵ_s is the permittivity of the semiconductor. The depletion width, W_D under deep depletion is larger than the maximum width of the depletion region under equilibrium conditions.

The large value of surface potential under deep-depletion results in the creation of a potential well as indicated in Fig. 7.28(b). The light received from the imaging object through the semi-transparent metal gate is absorbed in the *p*-semiconductor resulting in the creation of electron-hole pairs. The photogenerated electrons generated by the optical signal move toward the surface and partially fill the potential well. On the other hand, the photogenerated holes diffuse to the substrate. The internal quantum efficiency, η for the front-side illumination can be expressed as (Sze *et al* 2007)

$$\eta = 1 - \frac{e^{-\alpha W_D}}{1 + \alpha L_n} \tag{7.105}$$

where α is the optical absorption coefficient of the semiconductor at the operating wavelength and L_n is the diffusion length of electrons in *p*-semiconductor. The value of the internal quantum efficiency within W_D is very high (~100%).

The surface-channel CCD imaging device is very simple from the structural point of view but suffers from multiple associated problems that restrict its use as a reliable imaging device. The reason behind this is that in an SC-CCD, the channel is formed in the semiconductor adjacent to the semiconductor-insulator interface. The interface layer contains many crystal irregularities and defects in the crystal lattice that give rise to trap charges. This unwanted trapping of signal charges (photogenerated electrons, for example) leads to a smear image of the object to be captured.

The effect of the interface-trapping phenomenon can be greatly reduced by creating a situation where the channel can be created away from the interface. This can be achieved simply by incorporating another layer of the same semiconductor material with an opposite conductivity to separate the substrate semiconductor from the insulating layer. For example, if the SCCCD uses a *p*-type Si substrate and SiO_2 as the insulating layer, we can incorporate an *n*-type Si layer just below the SiO_2 layer. If the newly inserted *n*-type semiconductor is highly doped, the under application of a large gate voltage a complex depletion region is created with a potential minimum that appears far away from the interface. As a result, the potential well appears substantially "buried" inside the bulk semiconductor. This structure is called "Buried-Channel" CCD (BCCCD) (Walden *et al* 1972). The commercial CCD devices are based on this structure. Figure 7.29 shows the structure and energy band diagram of a buried channel CCD imager where a semiconductor of opposite conductivity is created just below the insulator.

The solid-state CCD imaging devices matured as the major consumer imaging products in the form of digital cameras and video recorders. The CCD-based camera largely replaced the conventional optical cameras because of its low cost and compact size. From the late 1990s, conventional CCD imaging devices were replaced by CMOS-based

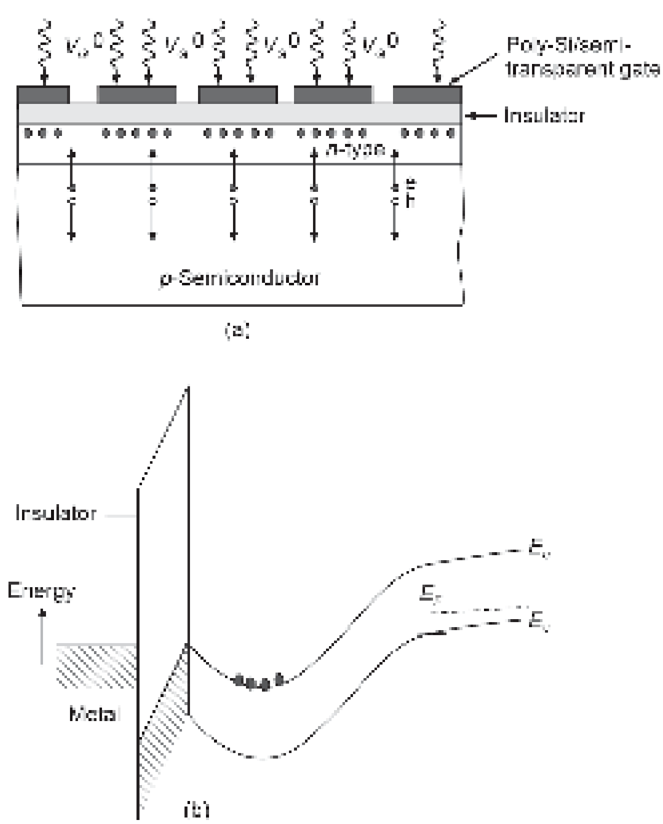


Fig. 7.29 Buried channel CCD (BCCCD) (a) Schematic structure showing the incorporation of another layer of semiconductor of opposite polarity (*n*-type) (b) Energy-band diagram showing the formation of the well away from the semiconductor-insulator interface

imaging devices (Fossum, 1994; Gamal *et al* 2005). This is because of the high integration capability of CMOS-based imaging sensors with other circuitry necessary for processing images. The digital camera on a single chip was reported for the first time in 1997 (Fossum, 1997).

7.14 QUANTUM WELL INFRARED PHOTODETECTOR (QWIP)

Quantum well-infrared photodetectors involve inter-subband electronic transition in quantum wells to absorb incident photons. In a conventional photodetector, photo absorption causes the transition of electrons between the energy bands (valence band and the conduction band) or between the impurity states and the bands. For detection of infrared radiation in a particular wavelength region it is necessary to adjust the geometry of structure in such a way that the energy difference between the inter-subband transition states matches with the wavelength of the incident radiation.

The inter-subband transition was reported by several researchers in the early 1980s (Chiu *et al* 1983; Smith *et al* 1983). The first successful infrared detector based on inter-subband absorption of radiation using resonant GaAlAs superlattice was reported in 1987, by Levine *et al* (Levine *et al* 1987). Improved versions of the photodetector were later reported by the same group (Choi *et al* 1987; Levine *et al* 1988; Yu *et al* 1991). The fundamental mechanism of operation of a QWIP involves inter-subband transitions caused by the absorption of photons resulting in the change of conductivity of the device. The following three types of inter-subband transitions may occur in QWIP-based photodetectors (Chiu *et al* 1983; Sze *et al* 2007; Rogalski, 2022).

i. The *bound-to-bound* transition involves the movement of an electron from one quantised energy state to the other. These quantised states are created by the QW structures and they are confined and situated below the energy barrier. A photon entering the QW excites an electron in the ground state to the first bound state in the QW. The excited electron subsequently tunnels out of the well. All the photo-excited electrons created in this way contribute to the change in the conductivity of the device enabling it to detect the incident radiation. This is illustrated in Fig. 7.30(a).

ii. The *bound-to-continuum* transition of electrons occurs when the first energy state in the QW structure above the ground level appears outside the barrier as illustrated in Fig. 7.30(b). It is seen that the continuum state available outside the barrier enables the photo-excited electrons from the ground state to move easily outside the barrier and contribute to conduction. This type of QWIP structure is most attractive because it offers a larger absorption of incident photons, a lower dark current, and a broader spectral response.

iii. The *bound-to-miniband* transition of electrons occurs between the ground state and a miniband (Yu *et al* 1991). A miniband is often created in the device because of the superlattice structure. When the separations of multiple quantum wells are of the same order as the well thickness, the quantum well structure behaves as a superlattice structure leading to the formation of a miniband. The miniband appears just below the conduction band edge in superlattice structures as shown in Fig. 7.30 (c).

As discussed earlier, the quantum wells are fabricated by using lattice-matched direct bandgap materials such as GaAs/AlGaAs, and InP/InGaAs. The separation between the confined energy states in the QW can be tailored by changing the thickness of the quantum well.

The main disadvantage of a QWIP structure is that the incident infrared radiation normal to the surface has zero absorption. This is because the inter-subband transition is

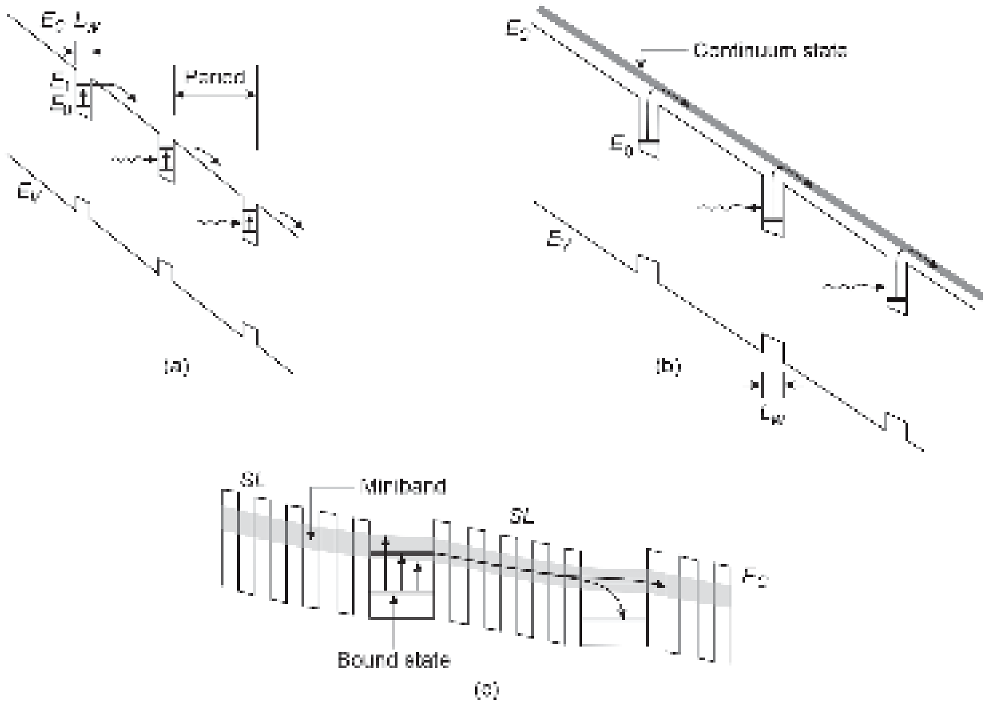


Fig. 7.30 Inter-subband transitions in quantum-wells (a) Bound-to-bound transition (b) Bound-to-continuum transition (superlattice) (c) Bound-to-miniband (formed by SL structure) transition

possible only when the electric field associated with the infrared electromagnetic radiation has components normal to the quantum-well plane. There are two approaches to fulfill this requirement so that the light is coupled properly to the light-sensitive active regions of the QWIP structure. The two popular methods are illustrated in Fig. 7.31. Figure 7.31(a) shows the schematic of a QWIP based on GaAs/AlGaAs heterostructure wherein a polished 45°-facet is made at the edge adjacent to the QW detector. The substrate acts as a window to allow the light to reach the QW region at an angle of 45° to the normal drawn on the surface of the quantum well. Alternatively, a grating can be used on the top surface of the device structure so that the light can be refracted to fall on the QW detector portion obliquely as shown in Fig. 7.31(b). It is also possible to use the grating on the backside (substrate side) of the QWIP structure so that the scattered light can reach the QW detector portion obliquely. In GaAs/AlGaAs QWIP, the thickness of the well region (GaAs region) is usually of the order of 5–10 nm while the thickness of the barrier layers (AlGaAs layers) varies in the range of 30–60 nm. A QWIP structure involves 25 or more layers of heterostructures. The well region consists of n-type GaAs having a doping concentration of $10^{23}/m^3$ while the barrier layers are undoped and made of AlGaAs, the larger bandgap material.

The QWIP essentially works like a photoconductor with the exception that it exploits the inter-subband transition of electrons caused by photons rather than the band-to-band transition that occurs in bulk photoconductors discussed earlier. The photocurrent generated in the QWIP can be expressed as

$$I_{op} = q\Phi_0 \eta G_a \tag{7.106}$$

where q is the electronic charge, Φ_0 is the total photon flux (/s), η is the quantum efficiency and G_a is the photoconductive gain.

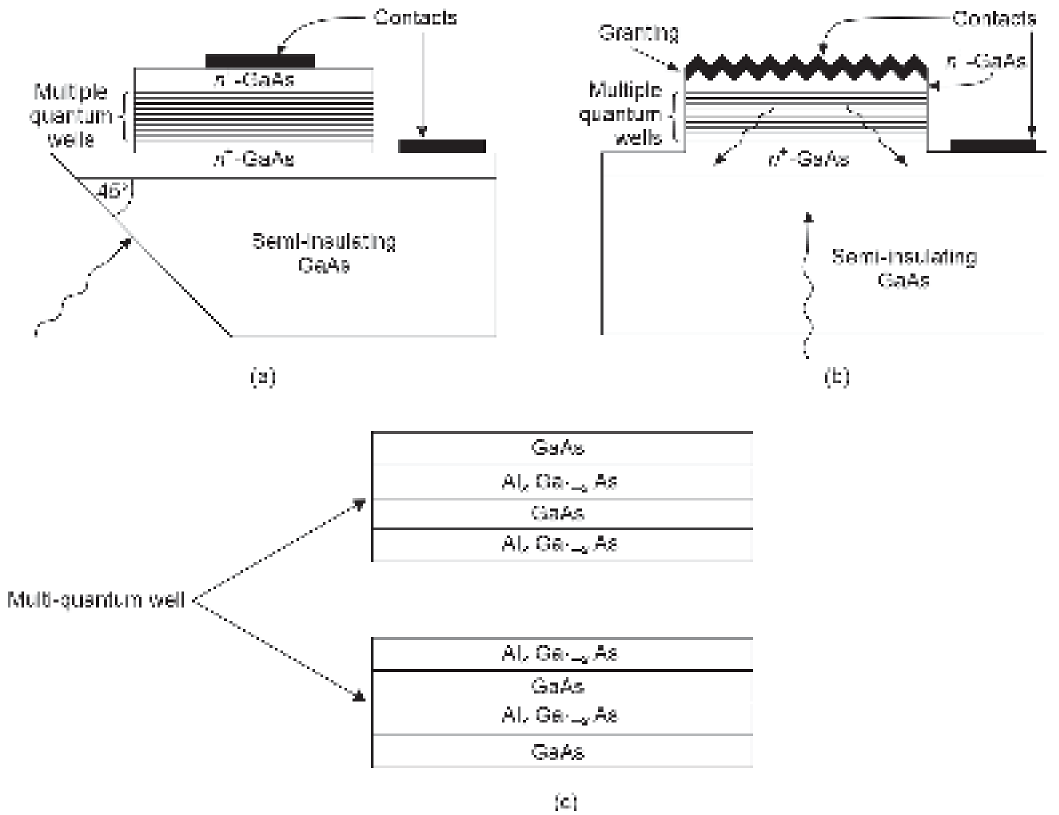


Fig. 7.31 Schematic of QWIP structures (a) Illustrating polished 45°-facet made at the edge adjacent to the QW (b) Front-end grading structure (c) Structural geometry of multi-quantum well (MQW)

The quantum efficiency of the QWIP device can be expressed by considering the fact that unlike in bulk photoconductor the carriers are only generated in the *QW* region only, as (Levine *et al* 1992; Sze *et al* 2007)

$$\eta = (1-R)\left[1-\exp\{-\alpha N_w L_w\} N_{op}\right] p_e P_c \tag{7.107}$$

where, R is the Fresnel reflection coefficient, α is the absorption coefficient of the QW structure, N_w is the number of quantum wells, L_w is the width of each quantum well, N_{op} is the number of optical passes, p_e is the escape probability of excited electrons signifying the fraction of the excited carriers that manage to escape out of the quantum wells, and P_c is the polarization correction factor (Levine *et al* 1992). Unlike in bulk devices, the absorption coefficient α in the case of a QWIP device depends on the incident angle and is proportional to $\sin^2\theta$, where θ represents the angle between the direction of incidence of the incident light and the normal drawn on the plane of quantum wells. The polarization correction factor, P_c for GaAs/AlGaAs QW structures is 0.5 for *n*-GaAs QW and 1.0 for *p*-GaAs QW. The value of E_p depends on the applied bias voltage. The escape probability depends on the inter-subband relaxation time, τ_{relax} and the average escape time τ_{esc} and is given by (Rogalski, 2022).

$$p_e = \frac{\tau_{\text{relax}}}{\tau_{\text{relax}} + \tau_{\text{esc}}} \tag{7.108}$$

The gain of the QWIP device can be expressed (Levine, 1993; Liu, 1992) under the condition that $p_e \approx 1$, as

$$G_a = \frac{1}{p_c N_w} \tag{7.109}$$

It may be pointed out here that the approximation $p_e \approx 1$ holds for QWIPs where the transition is from bound-to-continuum.

Here, p_c is the capture probability of an electron while traveling through a single QW given by

$$p_c = \frac{\tau_t}{\tau_c} \tag{7.110}$$

where, τ_t is the transit time across a single period of the QW region and τ_c is the capture time associated with the trapping of an electron into the ground state subband.

Using Eq. (7.110), the photoconductive gain can be expressed as (Liu, 1992)

$$G_a = \frac{\tau_c}{N_w \tau_t} = \frac{\tau_c}{\tau_{t\text{-}total}} \tag{7.111}$$

where, the capture time, τ_c determines the mean lifetime of the carriers and $\tau_{t\text{-}total}$ corresponds to the total transit time over the entire QWIP length L including the well and the barrier widths.

If the QWIP works in the linear region of the carrier velocity before saturation, the gain of the QWIP can be expressed using Eq. (7.111) as

$$G_a = \frac{\tau_c \mu V}{L^2} \tag{7.112}$$

where, μ is the mobility of the charge carriers in the linear regime and V is the applied voltage and L is the total length of the QWIP. Equation (7.112) is obtained by substituting

$$\tau_t = \frac{L}{v_d} = \frac{L}{\mu \varepsilon} = \frac{L}{\mu (V/L)} \tag{7.113}$$

where, ε is the electric field and v_d is the drift velocity of the charge carriers.

The current responsivity of a QWIP can be expressed as (Rogalski, 2022)

$$\mathcal{R} = q\left(\frac{\eta \lambda}{hc}\right) G_a \tag{7.114}$$

The performance of an MQW photodetector like QWIP is restricted by a relatively high value of dark current caused by thermally assisted tunneling, photon-assisted tunneling and thermionic emission out of QWs (Sze *et al* 2007; Rogalski, 2022). As the QWIPs are intended to be used in the long wavelength region (typically 3–20 μm), the energy barriers for transition generally range between a few hundred millivolts to a few tens of millivolts. A high value of dark current contributes to high detector noise. It is therefore, necessary to operate the device in low temperature region, i.e. 4–77 K.

The empirical formula for calculation of detectivity, D^* of n-type GaAs/AlGaAs QWIP at 77 K based on the best fit to experimental values is given by (Levine *et al* 1988; Rogalski, 2022)

$$D^* = 1.1 \times 10^4 \exp\left(\frac{hc}{2kT\lambda_c}\right) \text{mHz}^{1/2}\,\text{W}^{-1} \tag{7.115}$$

For p-type GaAs/AlGaAs QWIP the detectivity is given by

$$D^* = 2.0 \times 10^3 \exp\left(\frac{hc}{2kT\lambda_c}\right) \mathrm{mHz}^{1/2}\,\mathrm{W}^{-1} \tag{7.116}$$

A major competitor of QWIP is a conventional photodetector based on narrow bandgap materials such as InAsSb and HgCdTe, the latter is commonly known as Mercury-Cadmium-Telluride (MCT). A detailed comparison of the detectivity of GaAs/AlGaAs QWIPs with the theoretical performance of n^+-p HgCdTe photodiode is reported by Rogalski (Rogalski, 1997). The performance of a photodetector based on a narrow bandgap material is severely restricted by Auger recombination, at room temperature. As compared to QWIP photodetectors, the detectivity of a HgCdTe photodetectors is much higher than a GaAs/AlGaAs-based QWIP in the cut-off wavelength 8 μm ≤ λ_c ≤ 24 μm near liquid nitrogen operating temperature (77 K). For operation below 50 K, the performance of HgCdTe detectors deteriorates very fast due to inherent issues related to the material instability. The QWIP photodetectors find application in focal-plane-array (FPA) two-dimensional imaging systems. These detectors exhibit fast response due to short carrier lifetime in quantum wells. A major constraint with QWIPs based on n-GaAs wells is that the device exhibits zero absorption of photons for normal incidence making the coupling of light to the device difficult. The absorption spectra and its magnitude in a QWIP largely depend on the design of the active region.

The long wavelength infrared photodetectors based on quantum well structure and those based on conventional narrow bandgap materials do not find application in optical fiber communication systems. This is because, the range of operating wavelength of this detector is much above the near-infrared (NIR) wavelength (0.8–1.65 μm) which is supported by glass fibers. However, these photodetectors can find applications in free space optical communication systems. Specifically, two wavelengths, e.g. 9.6 μm and 10.6 μm are very important for free space optical communication as these wavelengths are atmospheric windows which allow the light signal to travel with minimum attenuation (Dwivedi *et al* 2010; 2011). The LWIP photodetectors find major applications in infrared imaging systems and night vision cameras.

7.15 PHOTOTRANSISTORS

Phototransistors belong to another class of photodetectors that can provide a high gain of the primary photocarriers through transistor action. The advantage of a phototransistor over an avalanche photodiode (APD) lies in the fact that a phototransistor does not require a very high DC voltage for operation like APD. Moreover, in APD, the multiplication gain is accompanied by an increased noise but the phototransistor gain does not lead to any significant increase in noise (DeLaMoneda *et al* 1971). A phototransistor can be realised in bipolar or even in unipolar transistor form (Adibi *et al* 1989).

7.15.1 Photo Bipolar Transistor

The first bipolar junction phototransistor was introduced by William Shockley in 1951. The Bell Laboratories first announced the invention of the phototransistor in 1950 based on the research work carried out by John Northrup Shive, and his team to use a light beam in place of a wire in a point contact transistor to act as the emitter for creating holes that subsequently flow to the collector (Bell Laboratories, 1950; Shive, 1951). An n-p-n phototransistor structure based on germanium (Ge) was subsequently reported to exhibit

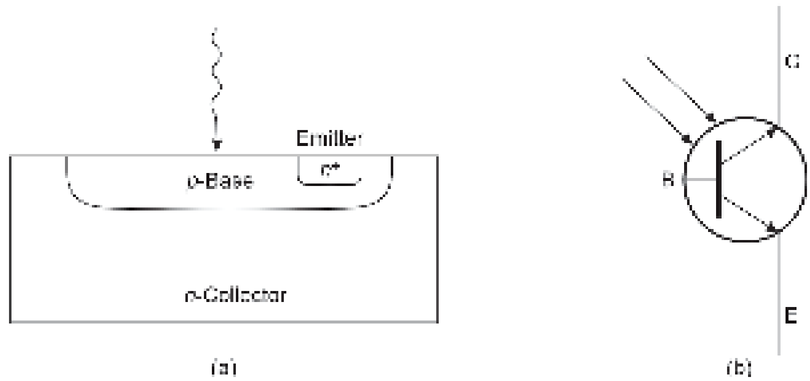


Fig. 7.32 Bipolar phototransistor (a) Schematic structure of an *n-p-n* phototransistor (b) Circuit symbol of an *n-p-n* phototransistor

an optical gain of about 100. Absorption of light in the base-collector junction creates electron–hole pairs which affect the charge neutrality condition and call for carrier injection from the base-emitter junction affecting the base current and causing multiplication of the primary photogenerated current. The schematic structure of a bipolar phototransistor is shown in Fig. 7.32 along with the circuit symbol. The equivalent circuit model and the energy band diagram showing the components of current in the presence of light are shown in Fig. 7.33.

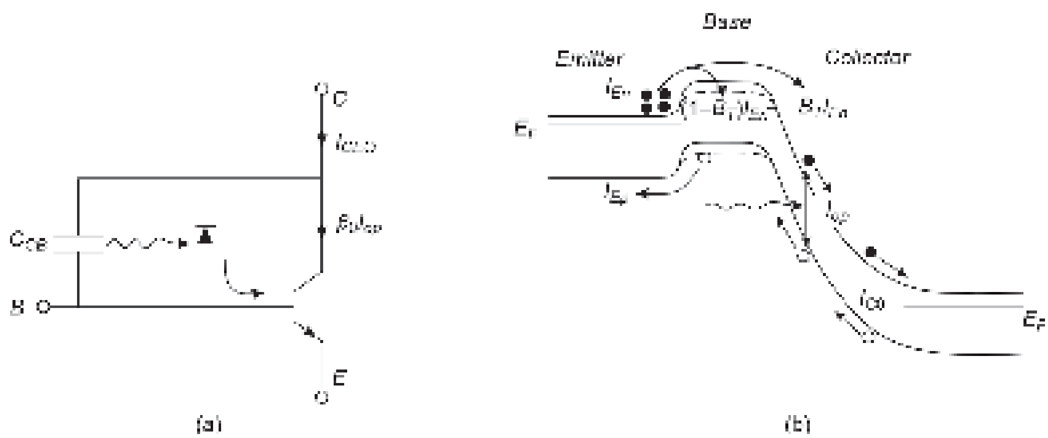


Fig. 7.33 Phototransistor operation (a) Basic equivalent circuit (b) Energy band diagram in presence of light showing different components of currents

A phototransistor can be operated in the two-terminal (2T) mode by keeping the base floating or in the three-terminal (3T) mode like a conventional transistor with appropriate biasing to work in the active region. Unlike a conventional bipolar junction transistor, a phototransistor has an extended base-collector region to facilitate the absorption of light. A phototransistor can be equivalently represented as a parallel combination of the capacitor arising from the base–collector junction in parallel with a *p-n* junction corresponding to the base-collector junction and find application in optically coupled isolators or opto-isolators in which the light from an optical source (LED or LASER) is focused on a photodiode or a phototransistor which converts the optical signal into the corresponding electrical signal which is used to modulate the optical source. These opto-isolators are useful for connecting electrical circuits having impedance mismatch issues. The use of a phototransistor in place of a conventional photodiode in an opto-isolator can significantly improve the performance in terms of increased current transfer ratio that represents the ratio of the output photocurrent to the input drive current of the LED or LASER source.

For floating base application in 2T mode, the collector of the phototransistor is connected to the positive terminal of the battery with respect to the emitter for an *n-p-n* structure. The equivalent circuit of the phototransistor and the energy band diagram of the device in presence of incident light is shown in Fig. 7.33. The incident light on the phototransistor is absorbed primarily in the base–collector junction and electrons and holes are created in the process. The holes generated in the depletion region and within the diffusion length move towards the base region while the photogenerated electrons are collected in the collector. The accumulation of excess electrons in the base region disturbs the electrical neutrality in the base region and calls for extra electrons to be injected from the emitter to the base region. These electrons finally collected by the collector provided that the transit time through the base region is much smaller than the lifetime of the electrons in the base region. Thus, a small change in the hole current caused by the photons results in a large current in the collector current due to internal gain mechanism of the transistor originating from the emitter injection efficiency.

The net collector current when the collector-base junction absorbs the incident radiation can be expressed as (Sze *et al* 2007)

$$I_c = I_{op} + I_{C0} + B_T I_{En} \tag{7.117}$$

where, I_{op} is the photocurrent generated by the incident photons, I_{C0} is the reverse saturation current of the collector–base junction, B_T is the base transport factor of the transistor and I_{En} is the electron component of base current.

The base transport factor is the fraction of the electron component of emitter current that constitutes the electron component of the collector current for an *n-p-n* transistor and is given by

$$B_T = \frac{I_{Cn}}{I_{En}} \tag{7.118}$$

The emitter injection efficiency for the *n-p-n* transistor is given by

$$\gamma = \frac{I_{En}}{I_{En} + I_{Ep}} = \frac{I_{En}}{I_E} \tag{7.119}$$

That is,

$$I_{En} = \gamma I_E \tag{7.120}$$

where, I_{Ep} is the hole component of the emitter current caused by the holes injected from the base to the emitter region, and I_E is the total emitter current.

For 2T application, the base is open and therefore, the base current is zero. We can therefore write,

$$I_C = I_E \tag{7.121}$$

That is,

$$I_{op} + I_{C0} + B_T I_{En} = I_{En} + I_{Ep} \tag{7.122}$$

We may rearrange Eq. (7.122) as

$$I_{Ep} + (1 - B_T) I_{En} = I_{op} + I_{C0} \tag{7.123}$$

When the base is open, it can be shown that

$$I_{CE_0} + (\beta_0 + 1)(I_{op} + I_{C0}) \approx \beta_0 I_{op} \tag{7.124}$$

where, I_{CE_0} is the collector-to-emitter current when the base is open and β_0 is the static common-emitter current gain given by

$$\beta_0 \equiv h_{FE} = \frac{\alpha_0}{1-\alpha_0} \tag{7.125}$$

Here α_0 is the collector-to-emitter current gain which is related to emitter injection efficiency and the base transport factor. Equation (7.124) shows that the collector-to-emitter current under open base condition is proportional to the primary photocurrent generated in the collector-base junction by the incident light.

The major drawback of a phototransistor using a conventional bipolar transistor is that the dark current arising out of the thermally generated carrier also gets amplified by the same multiplication factor due to transistor action. As a result, a very high gain cannot be achieved with the help of this simple structure. The gain of a simple phototransistor ranges between 50–100. Moreover, the phototransistor gain depends on the intensity of the incident light and therefore, the linearity is greatly compromised making the device not suitable for practical applications. Another limitation of the phototransistor arises from the charging times of the emitter, and the collector.

The phototransistors originally proposed during the early stage of optical fiber communication in the late 1970s started drawing renewed interest from the researchers as possible replacements for APDs. Several phototransistor configurations have been proposed, fabricated, and studied over the past decades. However, these devices are yet to develop for finding meaningful applications in fiber optic communication systems. Bipolar phototransistors are generally realized in heterojunction form and are known as heterojunction phototransistors (HPT). HPTs can be operated in both two-terminal (2T) or three-terminal (3T) modes. A schematic cross-sectional view of an *n-p-n* InGaAsP/InP phototransistor is shown in Fig. 7.34. In the 2T mode, the base of the transistor is kept floating.

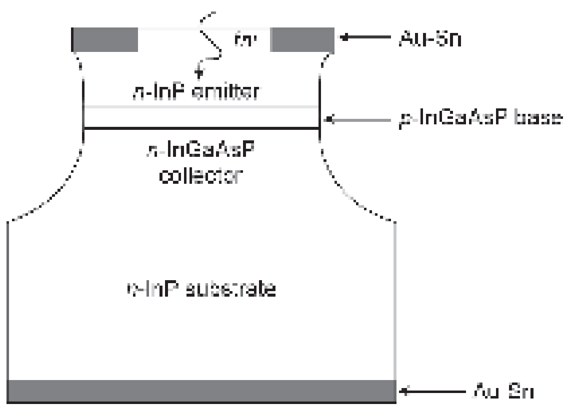


Fig. 7.34 Schematic cross-sectional view of a *n-p-n* heterojunction bipolar transistor

The base–collector junction acts as a light-gathering element. The heterojunction InP/InGaAsP phototransistor shown in Fig. 7.34, can be grown by liquid-phase epitaxy (Wright *et al* 1980). It consists of a thin lattice-matched InGaAsP base layer sandwiched between a wide bandgap n-type InGaAsP collector and an *n*-type InP emitter. The incident light is collected from the top emitter side. The light passes almost unattenuated through the wide bandgap InP layer which acts as a window and gets absorbed in the base, base-collector depletion region. A large secondary photocurrent is obtained between the emitter and the collector as the photogenerated holes are swept into the base region, increasing the forward bias. The heterojunction enables one to achieve low emitter-base and collector-base capacitance together with low base resistance. The barrier at the heterointerface enables one to dope the base region moderately without affecting the emitter injection efficiency thereby reducing the base resistance. The optical gain of the HPT is expressed as (Wright *et al* 1980).

$$G_0 = \eta h_{FE} \tag{7.126}$$

That is,

$$G_0 = \frac{h\nu}{q}\frac{I_c}{P_{0p}} \tag{7.127}$$

where, h_{FE} is the common emitter current gain, I_c is the collector current, P_{op} is the optical power incident on the emitter of the phototransistor. The HPT based on InGaAsP can be designed suitably for operation in 0.9 to 1.3 μm wavelength region giving gain of the order 100. It is expected that HPT may emerge as an alternative to APD (Tubatabaie-Alavi and Fonstad, 1981). For longer wavelength operation, InP/InGaAs HPT is more promising. Extensive theoretical investigations have been carried out on InP/InGaAs-based HPT because of their applications in optoelectronic integrated circuit receivers Chakrabarti, *et al* 2003; 2005).

7.15.2 Photo Field Effect Transistor

Phototransistors based on field-effect based transistors have been the subject of interest for quite some time because of their application in optically controlled microwave monolithic integrated circuits (MMIC) and OEIC. Almost all field-effect-transistor configurations have been explored for possible use as photodetectors. A metal-semiconductor field-effect transistor (MESFET) is one of the first typess of GaAs-based transistors investigated as phototransistors (Baack *et al* 1977; Pan, 1978; Gammel & Ballantyne, 1978; MacDonald, 1981). The MESFET-based photodetectors are also known as optically controlled field effect transistors (OPFET). The characteristics of optically controlled MESFET were extensively investigated theoretically to examine their potential as optically controlled MMIC or planar photodetector in OEIC (Chakrabarti *et al* 1992; Madjar, 1990; Paollela, 1994; Chakrabarti *et al* 1996; 1998). Other FET structures investigated for photoresponse characteristics include high-electron-mobility phototransistors (Pal *et al* 1992; Chakrabarti, 1992). The photoresponse of MIS capacitor and MISFET have also been reported (Chakrabarti, 1992; Chakrabarti *et al* 1990). The biggest advantage of a FET-based photodetector is the low value of input capacitance (~*fF*). Further planar structure and integrated circuit compatibility of transistor-based photodetectors make them attractive for the implementation of monolithic OEIC receivers (Rajamani & Chakrabarti, 1999). A major drawback of field-effect transistors is that the photoresponse is poor because light is generally allowed to enter from the gate side. To improve the photoresponse, the gate is made semi-transparent. This, however, increases the gate series resistance and affects the speed of response.

Example 7.10 An InP/InGaAs phototransistor operating at 1.55 μm produces a photocurrent of 12 mA for an input optical power of 100 μW. Calculate the gain of the phototransistor.

Solution The gain of the phototransistor can be obtained as

$$G_0 = \frac{hc}{\lambda q}\frac{I_c}{P_{op}} = \frac{6.625\times10^{-34}\times3\times10^{8}}{1.6\times10^{-19}\times1.55\times10^{-6}}\times\frac{12\times10^{-3}}{100\times10^{-6}}$$
$$= 96.16$$

7.16 STATE-OF-THE-ART TECHNOLOGY

A variety of photodetector structures have been proposed, analysed, simulated, fabricated, and tested over the past decades for application as a sensitive detector in the front-end of optical receivers in optical fiber communication systems. The focus has been on the ways and means for improving the photoresponse (quantum efficiency), and speed of response (or bandwidth). A large number of modifications of conventional photodetectors have been suggested for this purpose to achieve ever-increasing quantum efficiency and speed of response. In all the cases it was observed that there is a fundamental trade-off between the two important parameters (quantum efficiency and speed of response) and proper optimization is needed to achieve the highest sensitivity of the receiver as a whole. The use of an internal gain mechanism in avalanche photodiodes turns out to be a boon for improving the sensitivity of an optical receiver. APDs are thus found to be suitable for application in optical receivers of long-haul optical fiber communication systems. However, excess noise associated with APDs based on III–V materials for operation in 1.3–1.6 μm remains a challenge to the design engineers. Complex APD structures including MQW SL-APD have been proposed to improve the performance of conventional APDs. The technological complexities associated with the fabrication of such APDs and their incompatibility with other components of the receiver make them unacceptable for the implementation of OEIC receivers. By far Si *p-i-n* and APDs are most widely used in the shorter wavelength region of optical fiber communication. Successful integration of *p-i-n* photodetector with field-effect-transistor provides a compact integrated circuit PINFET for use in the front-end of an optical receiver. The most important factor that decides the ultimate performance of an ideal photodetector (η = 100%) is the quantum limit decided by the statistics of the photodetection process. The sensitivity of an optical receiver in a direct detection scheme can never be better than that dictated by the quantum limit. The use of optical amplifiers, e.g. semiconductor laser amplifiers (SLA) or fiber amplifiers (FA) before photodetection by any one of the non-multiplying photodetectors discussed in this chapter can significantly improve the performance of an optical receiver. SLA or EDFA (erbium-doped fiber amplifier) based optical receiver turns out to be a better choice in long haul optical communication systems than APD-based receivers. This is because the performance of the APD-based receiver deteriorates faster when the bit rate is increased beyond 10 Gbps at moderate values of gain. In the present generation optical fiber communication system operating in 1.3 – 1.55 μm, InGaAs is the material of choice for the development of photodetectors. The successful implementation of present-generation fiber optic communication has become possible only because of the parallel development in the field of III–V materials technology that has helped InP/InGaAs technology to mature. With the advent of new generation optical fibers based on heavy metal halides having extremely low loss up to 12 μm, it would be possible to develop an optical fiber communication system operating beyond 2 μm wavelength. InAsSb and HgCdTe are potential materials to be used in this wavelength region for developing semiconductor photodetectors.

Summary

- Photodetector is the key component of an optical receiver which converts optical signal into the corresponding electrical signal (O/E).
- Semiconductor photodetectors are ideal for application in optical fiber communication.
- Photodetectors can be either of bulk type (photoconductor) or junction type.
- Photodetectors can be nonmultiplying type (such as pin) or multiplying type (such as avalanche photodiode).
- Semiconductor photodetectors works on the principle of internal photoelectric effect.
- The maximum wavelength of detection is,

$$\lambda_{\max} = \frac{1.24}{E_g(\text{eV})}\,\mu\text{m}$$

- The received optical signal is characterized in terms of photon flux density or optical power.
- Important performance parameters of a photodetector are quantum efficiency, responsivity, speed of response, and noise performance often expressed in terms of noise equivalent power.
- The quantum efficiency is,

$$\eta = \frac{I_p/q}{P_{op}/h\nu}$$

  where, I_p is the photocurrent, q is the electronic charge, P_{op} is the incident optical power, h is Planck's constant and ν is the frequency of incident light.
- The responsivity is,

$$\mathcal{R} = \frac{I_p}{P_{op}}$$

- A reverse biased in principle can be used as a photodetector. This simple structure however does not satisfy all the requirements for deployment of this device in practical optical fiber communication system.
- PIN photodetector is by far the most commonly used non-multiplying photodetector for optical fiber communication system.
- There is a trade-off between the responsivity and the speed of response of a pin photodetector. The width of the i-region needs to be optimized to meet the requirement.
- The noise-equivalent power of a pin detector can be expressed in terms of equivalent current which takes into account the effects of dark current, background current, and effective load resistance, as

$$\text{NEP} = \sqrt{2}\left(\frac{h\nu}{\eta}\right)\left(\frac{I_{eq}}{q}\right)^{1/2}\text{W}$$

- The performance of a photodetector can be improved by making use of heterojunctions.
- Some important heterojunction pin structures are based on InP/InGaAs and InP/InGaAsP.
- Some advanced pin photodetectors include waveguide photodetector, resonant cavity enhanced PIN photodetector.
- The most commonly used multiplying photodetector is an Avalanche Photo Diode (APD), which provides inherent multiplication through a process of impact ionization.
- Earliest type of photodetector is called Reach-through Avalanche Photo Diode (RAPD).
- Heterojunction APD based on III-V heterostructures are most commonly used photodetectors in optical fiber communication.
- The gain of APD is accompanied by an excess noise given by

$$F(M) \approx M^x$$

  M being the multiplication gain and x is an empirical constant.

- Some advanced APD structures include SAM (Separate Absorption and Multiplication region) APD, Graded-gap APD, Superlattice APD, Stair-case APD, Q-dot photodetector, etc.
- Other photodetectors include Schottky barrier photodiode, MSM detector, phototransistors, photo-FET, etc.

Problems

7.1 The bandgap energy of $Al_xGa_{1-x}As$ in eV can be obtained by using the following empirical formula.

$$E_g = 1.424 + 1.266x + 0.266x^2 \text{ for } 0 \le x \le 0.37$$

Estimate the value of long wavelength cut-off of $Al_{0.33}Ga_{0.67}As$.

7.2 List some important III–V materials for the fabrication of semiconductor photodetectors in the near-infrared (NIR) region.

7.3 List some important III–V and II–VI semiconducting materials for the fabrication of photodetectors for operation in the mid-infrared (MIR) region. What are the drawbacks of these materials for room temperature operation?

7.4 A Si *p-i-n* photodetector is used in the front end of an optical receiver. The photodetector capacitance is 2 pF. The effective resistance of the photodetector circuit including the load resistance and the input resistance of the amplifier is 3 kΩ. Estimate the bandwidth of the photodetector circuit. How does it compare with the maximum bandwidth of the photodetector in the absence of the external circuit elements? The scattering limited velocity of the carriers in Si is 10^5 m/s.

7.5 A GaAs *p-i-n* photodetector is used in the front-end of an optical receiver. The width of the i-region is 10 μm and the device area is 0.5×10^{-7} m^2. The load resistance and the input resistance of the amplifier are 2 kΩ and 4 kΩ respectively. The input capacitance of the amplifier is 5 pF. Estimate the bandwidth of the photodetector circuit. The dielectric constant of GaAs is 12.9.

7.6 An InGaAs *p-i-n* photodetector is connected to a load resistance of 1 MΩ followed by a pre-amplifier with a very high input resistance ($\gg R_L$) as compared to the load resistance. The junction resistance of the photodetector is 20 MΩ. The various components of the current of the photodetector are

$$I_D = 1 \text{ nA};\ I_B = 0.2 \text{ nA}$$

The photodetector is operating at 1330 nm and has a quantum efficiency of 0.6. Estimate the value of the NEP of the photodetector at a room temperature of 298 K.

7.7 Calculate the NEP of a photodetector at 300 K having the following parameters.

$$R_j = 10 \text{ M}\Omega$$
$$R_L = 1 \text{ M}\Omega$$
$$R_i = 1 \text{ M}\Omega$$
$$I_D = 0.1 \text{ nA}$$
$$I_B = 0.2\ \mu\text{A}$$

The photodetector is operated at 1.55 μm with a quantum efficiency of 70%.

7.8 A Si *p-i-n* photodetector has a 5 μm wide *i*-region. The photodetector receives light at 0.87 μm where the absorption coefficient is 10^4 m^{-1}. Assuming that the product of the absorption coefficient and the hole diffusion length are much less than unity, estimate the quantum efficiency of the photodetector. Neglect the Fresnel reflection coefficient at the entrance.

7.9 Calculate the photocurrent density in Problem 7.6 assuming the incident optical power density to be 6×10^3 Wm^{-2}.

7.10 Repeat Problem 7.6, by assuming that the light enters the *p-i-n* detector from the air. The refractive index of Si is 3.42.

7.11 Repeat Problem 7.6, by assuming that the photodetector is operated at a suitable wavelength so that the absorption coefficient is 10^6 m^{-1}. The hole diffusion length is 10 μm.

7.12 Repeat Problem 7.9, by assuming that the light enters the *p-i-n* detector from the air. The refractive index of Si is 3.49.

7.13 The front-end of a photoreceiver has an equivalent resistance of 100 kΩ. If the bandwidth of the receiver is 1 GHz, estimate the mean square value of the thermal noise current.

7.14 Estimate the mean square value of the thermal noise current and the mean value of the shot-noise current (under dark conditions) using the data given in Problem 7.5. The bandwidth of the receiver is 500 MHz.

7.15 The dark current and background current of a *p-i-n*-based receiver are 1 nA and 10 nA respectively. The receiver operates with a bandwidth of 100 MHz. Estimate the mean square value of the shot noise current introduced by the photodetector.

7.16 The photodetector in Problem 7.14 is illuminated with an optical power density of 5×10^3 Wm^{-2} at 1.55 μm. Calculate the value of the mean square shot noise current introduced by the photodetector. If the equivalent resistance is 1 MΩ, estimate the mean square value of the thermal noise current. Also, calculate the NEP of the photodetector at 300 K.

7.17 The ionization coefficient of electrons and holes for GaAs can be approximated by the empirical formulas given by

$$\alpha(E) = 1.25 \times 10^7 \exp(-2.75 \times 10^7/E) \text{ m}^{-1}$$
$$\beta(E) = 5.2 \times 10^6 \exp(-3.25 \times 10^7/E) \text{ m}^{-1}$$

The above empirical relations are valid in the electric field range of 2–4 × 10^7 V/m.

Plot the variations of α and β with the inverse of the electric field in the above region.

7.18 An InGaAs APD has a quantum efficiency of 40% at 1.55 μm. When illuminated with an optical power of 0.3 μW at this wavelength, the APD produces an output photocurrent of 6 μA. The excess noise factor of the APD may be approximated as

$$F(M) \approx M^{0.6}$$

Calculate the NEP of the APD by assuming the modulation to be 100%. The following parameters may be used for computation

Dark current = 1 nA

Background illumination current = 0.2 nA

Equivalent load resistance = 1 kΩ

7.19 Explain clearly the meaning of the terminologies in the context of an optical detector:

Quantum efficiency; noise equivalent power; detectivity; speed of response and responsivity

7.20 An InP/InGaAsP heterojunction phototransistor operating at 1330 nm has a quantum efficiency of 50%. The phototransistor produces a collector current of 10 mA for an incident optical power of 120 μW. Calculate the common emitter current gain of the phototransistor.

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CHAPTER 8 Direct Detection Optical Receivers

In the previous chapter, we discussed various types of photodetectors. The key component of a direct detection optical receiver is a photodetector. In the intensity modulation/direct detection (IM/DD) system, a photodetector acts as a photon counter and produces an electrical output signal depending on the intensity of light received by it. A direct detection optical receiver usually consists of a photodetector followed by a low-noise amplifier and a few other additional signal processing circuits. The electrical signal produced by the photodetector is first amplified by the low-noise amplifier usually called a preamplifier and is subsequently processed by other processing circuits to extract the original information signal carried by the light signal.

8.1 INTENSITY MODULATION (IM)/DIRECT DETECTION (DD) SYSTEM

In an IM/DD system, the intensity of the optical signal is modulated by some characteristics of the modulating signal and is then transmitted through an optical fiber either in the analog or digital format. The signal after traveling through the optical fibers generally gets attenuated and distorted to some extent depending on the characteristics of the fiber. The weak and distorted signal is received by the photodetector which is the input port of an optical receiver. The weak mutilated signal received by the photodetector is converted into a weak signal at the output of the photodetector. The weak electrical signal produced by the photodetectors gets adversely affected by the random noise generated by the photodetection process. The weak signal is amplified subsequently with the help of an amplifier which also adds its noise components and further corrupts the signal. Sufficient care must be taken to design an optical receiver giving due consideration to various noise components that may affect the performance of the receiver otherwise the noise may override the signal making the recovery of the signal impossible. Various noise components present in a receiver generally set the lower limit of the signal that can be reliably reproduced by the receiver. The minimum optical power that can be detected by a receiver is usually known as the sensitivity of the receiver. It may be emphasised that the photodetector and the first-stage amplifier in an optical receiver are crucial so far as the noise performance of the receiver is concerned. The signal must dominate over the noise following the amplifier stage for proper detection of the signal. There is no way to improve the signal-to-noise ratio beyond this point. In other words, the noise performance measured in terms of the sensitivity of a receiver is determined by the combination of the photodetector and the preamplifier. The combination constitutes the front-end of the receiver and is considered the primary source of noise that determines the signal-to-noise ratio at the output.

In this chapter, we shall consider various sources of noise in an optical receiver of an IM/DD system. Mathematical modeling of these noise components is essential for the design of an optical receiver. It may be pointed out here that the noise performance of an analog optical communication is generally characterised in terms of signal-to-noise (S/N) ratio. On the other hand for a digital optical communication system, the noise performance is usually characterized in terms of bit-error-rate (BER). BER is obtained by dividing the number of bits in error by the total number of bits in a given bit stream. Thus,

$$\text{BER} = \frac{N_e}{Bt}$$

where, N_e corresponds to the number of errors, the product Bt corresponds to the total number of bits over the time interval, t and B is the bit-rate given by

$$B = \frac{1}{T_b}$$

T_b being the bit period.

In the previous chapter, noise characteristics of different photodetectors have been discussed. For example, the noise equivalent power (NEP) of photodetectors has been considered in the previous chapter. However, in a receiver circuit, a photodetector is interfaced with other electronic and circuit components which affect the overall noise performance and NEP no longer serves as a proper parameter for noise characterization. In this chapter, different types of optical receiver configurations are discussed. It is understood that the first-stage amplifier must be a low noise amplifier to have a good overall sensitivity of the receiver. Field-effect transistors (FETs) are generally used as amplifiers in optical receivers because of the low noise behavior of these transistors. A variety of photodetectors are used in conjunction with different forms of FETs to constitute the front-end of a direct detection optical receiver. The most commonly used front-end configuration is a PINFET which uses a *p-i-n* photodetector followed by a FET-based amplifier. The mandates of the design of an optical receiver are low noise and large bandwidth with a wide dynamic range. The front-end block of an optical receiver is shown in Fig. 8.1.

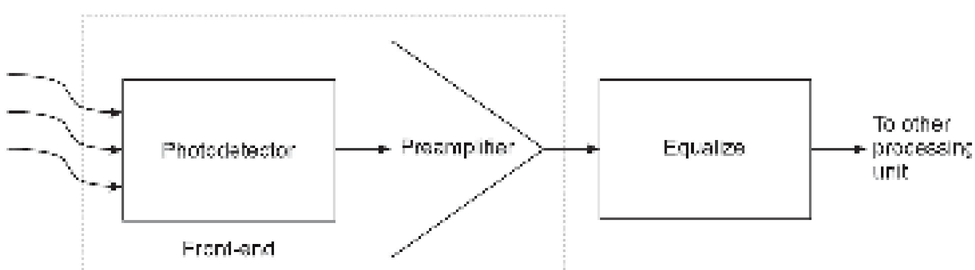

Fig. 8.1 Block diagram of a direct detection receiver showing the front-end

8.1.1 Noise Sources

In communication systems, noise is generally viewed as undesired electrical disturbance signals that tend to affect adversely the transmission and/or processing of the desired information signal. The noise may be either external (atmospheric or man-made) in nature or may be internal to the system. For example, consider the processing of a signal by an electronic circuit comprising different active and passive components. The active components consisting of transistors or other devices give rise to shot-noise[1]

[1] The term shot-noise is derived from the fact that when such signals are reproduced using loudspeakers they sound like lead shots hitting a metal plate

and other passive components give rise to thermal noise. The shot-noise is generally due to the discrete nature of current flow which for example, may arise out of random injection of carriers across the *p-n* junction in the case of bipolar junction transistor (BJT). The thermal noise also known as white or Johnson noise arises from the random motion of carriers through conductors due to the temperature effect. These two noise components get added to the signal under processing. The noise signals tend to mutilate the desired signal and need to be controlled for reliable processing of the signal. For further details, readers may refer to the standard textbooks on *Communication* (Carlson, 1996; Lathi, 1998).

In optical communication systems as the signal is transmitted in the form of light, the radiofrequency electromagnetic disturbances do not affect the signal. However, at the receiver when the signal is converted from the optical to the electrical domain all the electrical noise components start interfering with the desired signal. As the converted electrical signal in generally weak at the receiver end, it is necessary to devise ways and means to protect the signal from getting overridden by the noise signals. The interplay of signal and noise in an optical receiver largely depends on whether the optical communication is coherent or incoherent type. In an incoherent or direct detection scheme, the optical power received by the receiver is in the intensity-modulated form and the detector is essentially a photon counter that translates the signal in the electrical domain analogously. The various sources of noise present in the front-end of an optical receiver of an IM/DD system are depicted in Fig. 8.2.

8.1.2 Excess Noise Factor

In a direct detection receiver, the intensity-modulated light received by the detector experiences random arrival of photons which give rise to random generation of electron–hole pairs in the photodetector. This random generation of electrons and holes due to the absorption of photons manifests in the form of quantum (or shot) noise. The quantum or shot-noise produced in the process depends on the photogenerated current. For non-multiplying photodetectors, such as a *p-i-n* diode, the photogenerated carriers produce the photocurrent without any multiplication gain (M = 1). On the other hand, if the photodetector is a multiplying one, such as an avalanche photodiode (APD), additional shot noise is introduced. This is because, the internal gain mechanism of an APD is inherently random and this randomness in the multiplication process is manifested in the form of additional shot noise. The excess noise introduced by an APD is measured in terms of excess noise factor $P(M)$ which is a function of the multiplication gain, M. The excess noise factor is defined as the ratio of the mean-square gain divided by the square of the mean gain, given by

$$F(M) = \frac{\langle m^2 \rangle}{\langle m \rangle^2} = \frac{\langle m^2 \rangle}{M^2} \tag{8.1}$$

The excess-noise factor $F(M)$, depends on the ionization rate ratio and the type of carrier (electron or hole) initiating the ionization. For electron-initiated ionization, the excess noise factor can be expressed in terms of multiplication gain, M as

$$F(M) = kM + \left(2 - \frac{1}{M}\right)(1-k) \tag{8.2}$$

where, $k = \beta/\alpha$.

As discussed in the previous chapter the excess noise factor is often expressed empirically as

$$F(M) \approx M^x \tag{8.3}$$

where, x is a fraction ranging from 0 and 1 depending on the material.

The other noise components in a photodetector both *p-i-n* and APD arise from the dark current and additional leakage current that may arise out of absorption of background radiation or tunneling across the junction etc.

The analyses of quantum noise and the noise arising out of avalanche multiplication are quite complex and do not follow Gaussian statistics.

8.1.3 Quantum Noise

The mechanism of photodetection by a conventional photodetector is based on the theory of interaction of matter with light which is considered to be a stream of discrete particles called quanta or photons with energy $E = h\nu$, h being the Planck's constant and ν the frequency of the light. The absorption of a photon in the photodetector creates an electron–hole pair. The discrete nature of the arrival of photons and their subsequent creation of electron–hole pairs randomly in the photodetector give rise to a form of noise called quantum noise. This noise originates from the very mechanism of the photodetection process. The arrival of photons at the receiver is statistical. The primary photogenerated current arising from the random arrival of photons at the photodetector is a time-varying Poisson's process. If the photodetector on average detects $\bar{N}$ number of photons, then the probability $P(n)$ of detecting n number of photons in period τ is governed by Poisson's distribution given by (Russer, 1980)

$$P(n) = \frac{\bar{N}^n}{n!}\exp\left(-\bar{N}\right) \tag{8.4}$$

It should be noted that there is an uncertainty associated with the number of electrons and holes generated in the detector by a given optical power incident on the receiver for a fixed duration. This randomness in the total number of electron–hole pairs generated by the incident radiation is the source of this form of shot-noise called quantum noise.

In the above probability distribution function given by Eq. (8.4), $\bar{N}$ which is the mean value also corresponds to the variance of the distribution function. It is interesting to note that the special characteristic of Poisson's distribution is that the mean and the variance of the distribution function are the same. The rate of generation of electron–hole pairs by the incident photons can be expressed as

$$R = \eta\frac{P_{op}}{h\nu}\,\mathrm{s}^{-1} \tag{8.5}$$

where, η is the quantum efficiency of the photodetector.

If the detector receives an optical radiation $P(t)$ over a period τ, the average number of electron–hole pairs during the period can be expressed as

$$\bar{N} = \frac{\eta}{h\nu}\int_0^{\tau} P(t)dt = \frac{\eta E}{h\nu} \tag{8.6}$$

where, E is the total optical energy received by the photodetector over the duration τ.

8.1.4 Quantum Limit

It has been seen that the quantum noise arises from the fundamental mechanism of operation of a photodetector. If we ignore the presence of all other forms of noise in an optical detector system, we are left with the quantum noise which sets the fundamental lower limit of a photodetector system. To appreciate the actual meaning of quantum limit consider a digital optical communication system transmitting light in the form of optical pulses such that the presence of an optical pulse in a given bit period corresponds to the bit "1" and the absence of light during a given bit period corresponds to a bit "0". Consider an ideal situation where there is no dark current produced by the photodetector and there is no other form of noise. In such a situation no current will be produced by the photodetector in the absence of light and even a single electron–hole pair produced by the light in the detector can be detected. Under such a situation, an error can only occur when a light pulse is present and no electron–hole pair is produced by the photodetector to cause a flow of current in the external load. The probability that no electron–hole pair is produced in the presence of a light pulse can be obtained from Eq. (8.4) as

$$P(0/1) = P(0) = \exp(-\overline{N}) \tag{8.7}$$

where, $P(0/1)$ corresponds to the probability that the received bit is interpreted as a "0" given that the bit "1" is being transmitted. Note that this probability equals the probability that no electron–hole pair is produced in the presence of a light pulse.

Equation (8.7) is valid under the ideal condition that no electron–hole pair is produced in the dark and there is no other component of noise present in the receiver circuit. Thus, Eq. (8.7) allows one to compute the ultimate limit called the quantum limit in the digital optical communication system. The quantum limit is defined as the minimum energy of the light pulse required to maintain a given bit error rate (BER). The following example illustrates the significance of the quantum limit.

Example 8.1 A fiber optic link operating at 870 nm is required to maintain a bit-error-rate of 10^{-9}. Determine the theoretical quantum limit of the receiver system in terms of the quantum efficiency of the photodetector and the energy of the incident photon. Also, determine the minimum optical power required to maintain the above bit-error-rate assuming that the system uses a binary signaling operating at 500 Mbps and the quantum efficiency of the photodetector is 100%.

Solution The probability of error in this case is decided by the BER to be maintained by the system.

Thus,

$$P(0) = \exp(-\overline{N}) = \text{BER} = 10^{-9}$$

That is,

$$\overline{N} = 20.7$$

This means that around 20.7 number of photons are required to detect a binary "1" in digital binary signaling with a bit error rate of 10^{-9}.

Using Eq. (8.7), the minimum pulse energy $E = E_{\min}$ (say) corresponding to the quantum limit can be obtained as

$$20.7 = \frac{\eta E_{\min}}{h\nu}$$

That is,

$$E_{\min} = 20.7\left(\frac{hc}{\lambda\eta}\right) \tag{8.8}$$

For binary signaling with a bit period of τ and average power P_0, the energy can be expressed as

$$E_{\min} = P_0\tau \tag{8.9}$$

Using Eqs (8.8) and (8.9), the average power for binary signaling can be obtained as

$$P_0 = 20.7\left(\frac{hc}{\lambda\tau\eta}\right) \tag{8.10}$$

The bit period is related to the bit-rate, B can be obtained by assuming an equal number of 0 and 1 bits as

$$\frac{B}{2} = \frac{1}{\tau}$$

Using the above relationship, the average power can be obtained as

$$P_0 = 20.7\left(\frac{hcB}{2\lambda\eta}\right)$$

Given: λ = 870 nm, B = 500 MHz and $\eta = 1$
Therefore,

$$P_0 = 20.7\left(\frac{6.62\times10^{-34}\times3\times10^{8}\times500\times10^{6}}{2\times870\times10^{-9}\times1}\right) \approx 57 \text{ pW}$$

In terms of dBm, the average power can be estimated as

$$= 10\log_{10}\left(\frac{57\times10^{-12}}{10^{-3}}\right)$$

$$= 10\log_{10}(57\times10^{-9})$$

$$= 17.55 - 90 = -72.45 \text{ dBm}$$

It may be pointed out here that the minimum power estimated in the above example is the theoretical limit in the sense that no optical receiver can exhibit sensitivity better than this. It may be recalled here that all other noise sources have been neglected at the time of determining the quantum limit. As a result, a practical receiver generally exhibits sensitivity at least 10–12 dB less than what has been estimated in the above example.

8.2 RECEIVER CONFIGURATIONS

The block diagram of a typical receiver of a digital optical communication system is shown in Fig. 8.2. The first two blocks consisting of the photodetector and the low-noise preamplifier constitutes the front-end of the optical receiver. The front-end is followed by an equalizer. The function of the equalizer is to remove the signal distortion caused by the

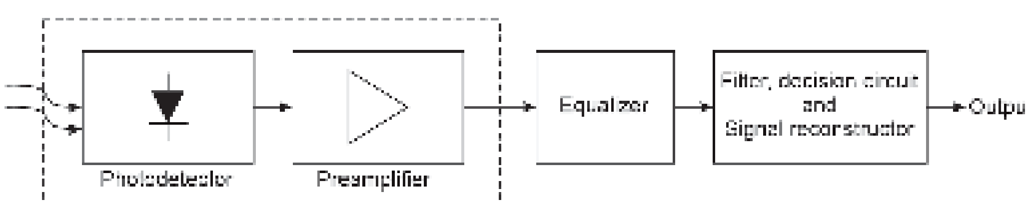

Fig. 8.2 Block diagram of an optical receiver of a digital optical communication system

non-linearity of the front-end and the dispersive effects in the fiber medium. The function of the filter is to maximize the signal-to-noise ratio while preserving the essential features of the signal. The decision circuit compares the signal at the input with a pre-decided threshold voltage to identify the received signal as "1" or "0". The bits ("1" or "0") are subsequently given the required pulse shape by the signal reconstructor circuit to deliver the final digital signal output. In the ideal situation in the absence of noise, the output digital electrical pulses would be a replica of the original optical pulses transmitted by the optical transmitter. However, in practice, the reproduced signal at the receiver output differs from the transmitted signal due to various noise components that try to mutilate the transmitted signal. In digital optical communication, the error is measured in terms of bit-error rate as discussed earlier. An important figure of merit of a receiver is the sensitivity which corresponds to the minimum optical power that must be received by the receiver to reproduce the signal with the given bit-error rate. The bit-error rate in a digital optical communication system is related to the signal-to-noise ratio. For an analog optical communication system in which the signal is in the analog form, the sensitivity is measured in terms of signal-to-noise ratio rather than bit-error rate. It may be pointed out here that the overall signal-to-noise ratio of the receiver is effectively decided by the signal-to-noise ratio at the output of the front-end stage (shown by the dotted line box). The desired signal-to-noise ratio must be maintained in the front-end stage when the signal is weak because there is no way to improve the signal-to-noise ratio after the front-end stage. In other words, the front-end of the receiver in a way decides the overall sensitivity of an optical receiver. Because of this fact, more emphasis shall be given to the study of the front-end of an optical receiver in the subsequent sections. The equivalent circuit of the receiver front-end is shown in Fig. 8.3. In the equivalent circuit shown in Fig. 8.3, i_p corresponds to the photogenerated signal current, C_d corresponds to the capacitance of the photodetector, R_b corresponds to the bias resistance of the photodetector usually considered as the load resistance and R_a and C_a correspond to the input resistance and input capacitance of the following amplifier stage, respectively. In the figure, $\langle i_s^2 \rangle$ and $\langle i_t^2 \rangle$ correspond to the mean square value of the shot-noise current and the mean square value of the thermal noise current in the circuit, respectively. The input amplifier noise arising out of the thermal noise associated with the amplifier resistance, R_a is represented as i_{amp} in the figure. The thermal noise associated with the amplifier channel noise is represented by $e_n(t)$. The noise introduced by the amplifier stage depends largely on the configuration of the amplifier and the detailed noise analysis of the front-end of a digital optical communication receiver can be found in the literature (Personick, 1973). As already pointed out, it is necessary to reduce the noise components shown in the equivalent diagram in Fig. 8.3 to enhance the receiver sensitivity. The front-end of an optical receiver comprising a photodetector and a preamplifier can be realized in three basic forms, e.g. low-impedance (LZ), high-impedance (HZ), and transimpedance (TZ) configurations.

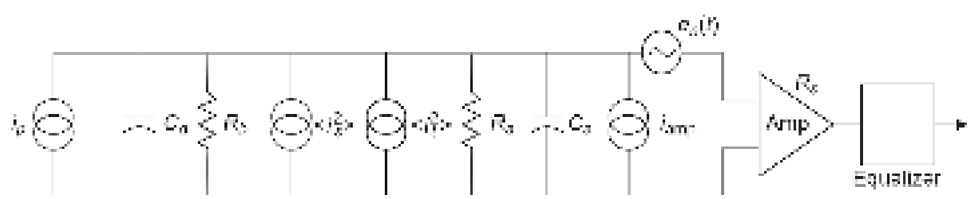

Fig. 8.3 The equivalent circuit of an optical receiver of a digital optical communication system showing various noise components

In a digital optical communication system, the rectangular optical pulses transmitted by the optical transmitter are received by the receiver in the form of distorted binary pulses. The received optical power can thus be represented as a train of binary pulses expressed as (Keiser, 2001)

$$P_{op}(t) = \sum_{n=-\infty}^{\infty} b_n h_p(t - nT_b) \tag{8.11}$$

where, $P_{op}(t)$ is the optical power received by the receiver, b_n corresponds to the amplitude of the nth digit of the received pulse, T_b is the bit period. Here, $h_p(t)$ corresponds to the shape of the received pulse. For a normalized pulse

$$\sum_{-\infty}^{\infty} h_p(t)dt = 1 \tag{8.12}$$

and in that case, b_n corresponds to the energy of the nth pulse.

The received optical power pulse is converted into output photocurrent by the photodetector. The mean output detector current can be obtained as

$$\langle i(t) \rangle = \frac{\eta q}{h\nu} MP_{op}(t) \tag{8.13}$$

where, M is the multiplication gain of the amplifier. For a p-i-n photodetector, $M = 1$. Using Eq. (8.11), Eq. (8.13) can be expressed as

$$\langle i(t) \rangle = \mathcal{R}_0 M \sum_{n=-\infty}^{\infty} b_n h_p(t - nT_b) \tag{8.14}$$

where, $\mathcal{R}_0(=\eta q/h\nu)$ is the responsivity of the photodetector. The output current of the photodetector is subsequently amplified by the preamplifier to produce a mean voltage at the output of the equalizer. The equalizer output voltage is subsequently compared with a pre-decided threshold to decide for "1" or "0".

8.2.1 Frequency Domain Representation

It is often convenient to calculate the mean voltage at the output of the equalizer by using frequency domain representation with the help of Fourier transforms. The mean voltage at the equalizer output can be obtained from the mean photodetector output current in the time domain with the help of convolution. It can be easily seen that the mean voltage at the equalizer output can be expressed as

$$\langle(v_{\text{out}}(t)\rangle = A\mathcal{R}_0 MP_{op}(t) * h_B(t) * h_{eq}(t) \tag{8.15}$$

where, $h_B(t)$ is the impulse response of the bias circuit comprising the photodetector bias resistance, R_b; the amplifier input resistance, R_a and the photodetector capacitance, C_d and the amplifier input capacitance, C_a. Here, $h_{eq}(t)$ is the impulse response of the equalizer circuit.

Assuming the form of the mean voltage at the equalizer output to be analogous to the received optical power pulse train given by Eq. (8.11), we may write

$$\langle v_{\text{out}}(t) \rangle = \sum_{n=-\infty}^{\infty} b_n h_{\text{out}}(t - nT_b) \tag{8.16}$$

where, $h_{\text{out}}(t)$ is the pulse shape produced at the output of the equalizer.

Comparing Eqs (8.16) and (8.15) in conjunction with Eq. (8.11), we obtain

$$h_{out}(t) = AR_0Mh_p(t) * h_B(t) * h_{eq}(t) \tag{8.17}$$

Taking the Fourier transform on both sides of Eq. (8.17), we may write

$$H_{out}(f) = AR_0MH_p(f)H_B(f)H_{eq}(f) \tag{8.18}$$

where, $H_{out}(f)$, $H_p(f)$, $H_B(f)$ and $H_{eq}(f)$ are the Fourier transforms of $h_{out}(t)$, $h_p(t)$, $h_B(t)$ and $h_{eq}(t)$, respectively. From Fig. 8.3, the transfer function $H_B(f)$ of the photodetector bias circuit which is essentially the Fourier transform of the impulse response of the circuit given by

$$H_B(f) = \frac{1}{\dfrac{1}{R_T} + \dfrac{1}{1/j2\pi fC_T}} = \frac{R_T}{1 + j2\pi fR_TC_T} \tag{8.19}$$

where,

$$\frac{1}{R_T} = \frac{1}{R_b} + \frac{1}{R_a} \tag{8.20}$$

and

$$C_T = C_a + C_d \tag{8.21}$$

8.2.2 Low-impedance (LZ) Front-end

The simplest form of receiver front-end is the so-called low-impedance configuration. As the name implies, this configuration consists of a preamplifier with a low value of input impedance (~50 Ω) which is matched with a low value of the photodetector bias resistance as shown in Fig. 8.4. It can be easily seen that the bias resistance of the photodetector is modified by the input resistance of the following stage amplifier. The effective resistance can be expressed as

$$R_T = \frac{R_bR_a}{R_b + R_a} \tag{8.22}$$

The most important features of an optical receiver front-end are the bandwidth and total noise introduced in this section. The bandwidth of the LZ front-end can be obtained from the effective RC time constant of the receiver front-end. If the overall capacitance of the front-end is C_T as given by Eq. (8.21), the bandwidth of the LZ front-end can be obtained as

$$B = \frac{1}{2\pi R_TC_T} \tag{8.23}$$

In the LZ front-end configuration, both the values of R_b and R_a are small and as a result the available bandwidth is very high.

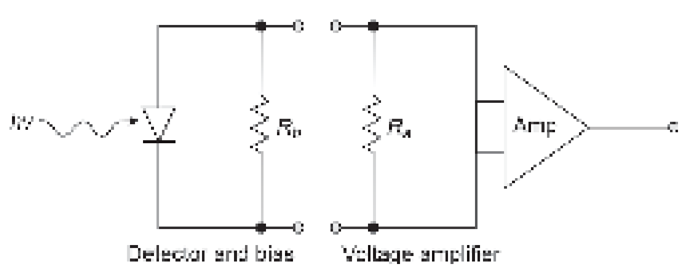

Fig. 8.4 Low-impedance (LZ) receiver front-end configuration

The mean-square value of the thermal noise generated by the front-end can be estimated as

$$\left\langle i_T^2 \right\rangle = 4kT\left(\frac{1}{R_T}\right)B \tag{8.24}$$

It can be easily seen from the Eq. (8.24) that the thermal noise produced by the front-end in the LZ configuration is very high because R_T is smaller than both R_b and R_a which have low values.

Therefore, a low-impedance (LZ) receiver front-end offers a high value of bandwidth but a low value of sensitivity due to increased thermal noise current.

Example 8.2 A low-impedance (LZ) optical receiver front-end uses a photodetector bias resistance of 50 Ω which is matched to the following stage amplifier with the same value of input resistance. The input capacitance of the photodetector is 5 pF while the input capacitance of the amplifier is only 3 pF. Estimate the values of the bandwidth and the mean-square value of the thermal noise component of the receiver front-end.

Solution The values bias resistance and the input resistance of the amplifier are given by

$$R_b = 50\ \Omega \text{ and } R_b = 50\ \Omega$$

The effective resistance is given by

$$R_T = \frac{50\times 50}{50+50} = 25\ \Omega$$

The overall capacitance of the front-end is

$$C_T = 5\text{ pF} + 3\text{ pF} = 8\text{ pF}$$

The bandwidth of the given LZ front-end can be estimated as

$$B = \frac{1}{2\times 3.14\times 25\times 8\times 10^{-12}} = 796.18\text{ MHz}$$

The mean square value of the thermal noise current generated by the LZ front-end can be obtained as

$$\left\langle i_T^2 \right\rangle = 4\times 1.38\times 10^{-23}\times 300\times \frac{1}{25}\times 796.18\times 10^{6}\text{ A}^2$$

$$= 5.27\times 10^{-13}\text{ A}^2$$

8.2.3 High-impedance (HZ) Front-end

Unlike LZ configuration, a high-impedance (HZ) receiver front-end configuration makes use of a high value of photodetector bias resistance which is matched with the high value of input impedance of a suitable preamplifier stage. A schematic of the HZ front-end configuration is shown in Fig. 8.5. The high value of the effective resistance eventually reduces the value of the mean square value of the thermal-noise component of the front-end. However, a high value of the effective resistance enhances the Resistor-Capacitor (RC) time constant and reduces the bandwidth of the HZ front-end configuration [see Eq. (8.23)]. As the photodetector output in this configuration is effectively integrated over a large time constant, the signal gets distorted and a heavy equalization is needed for restoration of the signal at the later stage as indicated in Fig. 8.5. This is a significant drawback of this configuration.

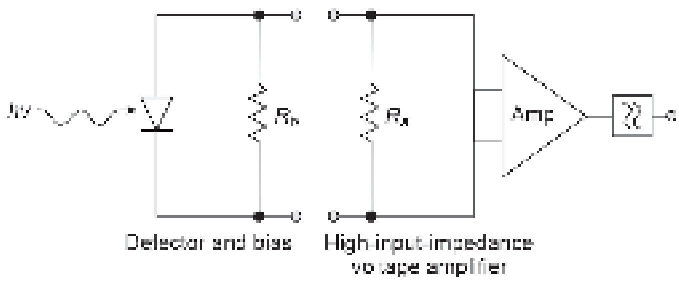

Fig. 8.5 High-impedance front-end configuration illustrating the requirement of post-amplification equalization

The high-impedance receiver front-end configuration exhibits a better sensitivity because of a reduced thermal noise current but a smaller bandwidth accompanied by the requirement of heavy equalization.

Example 8.3 A high-impedance (HZ) optical receiver front-end uses a photodetector bias resistance of 5 MΩ which is matched to the following stage amplifier with the same value of input resistance. The input capacitance of the photodetector is 4 pF while the input capacitance of the amplifier is only 6 pF. Estimate the values of the bandwidth and the mean-square value of the thermal noise component per unit bandwidth of the receiver front-end.

Solution The values bias resistance and the input resistance of the amplifier are given by

$$R_b = 5\ \text{M}\Omega \text{ and } R_b = 5\ \text{M}\Omega$$

The effective resistance is given by

$$R_T = \frac{5\times 5}{5+5} = 2.5\ \text{M}\Omega$$

The overall capacitance of the front-end is

$$C_T = 4\ \text{pF} + 6\ \text{pF} = 10\ \text{pF}$$

The bandwidth of the given HZ front-end without equalization can be estimated as

$$B = \frac{1}{2\times 3.14\times 2.5\times 10^{6}\times 8\times 10^{-12}} = 6.34\ \text{kHz}$$

The bandwidth of the HZ front-end configuration in practice can be improved by using an equalizer circuit following the preamplifier.

The mean square value of the thermal noise current generated by the HZ front-end can be obtained as

$$\left\langle i_T^2 \right\rangle = 4\times 1.38\times 10^{-23}\times 300\times \frac{1}{2.5}\times 10^{-6}\ \text{A}^2\text{Hz}^{-1}$$
$$= 6.624\times 10^{-27}\ \text{A}^2\text{Hz}^{-1}$$

8.2.4 Transimpedance (TZ) Front-end

From the foregoing discussion, it is apparent that the low-impedance (LZ) front-end offers a large bandwidth and a relatively low sensitivity while the high-impedance (HZ) front-end offers a smaller bandwidth without equalization and a relatively better sensitivity.

The main advantage of the HZ configuration is the improvement in the sensitivity over the simplistic LZ configuration. A compromise between the two extreme configurations leads to a transimpedance (TZ) configuration which is essentially similar to an HZ front-end configuration with negative feedback as shown in Fig. 8.6. The negative feedback reduces the high input impedance of the amplifier and the circuit, essentially works as a current mode amplifier in which the detected photocurrent is translated into a voltage at the output of the amplifier as shown in Fig. 8.6.

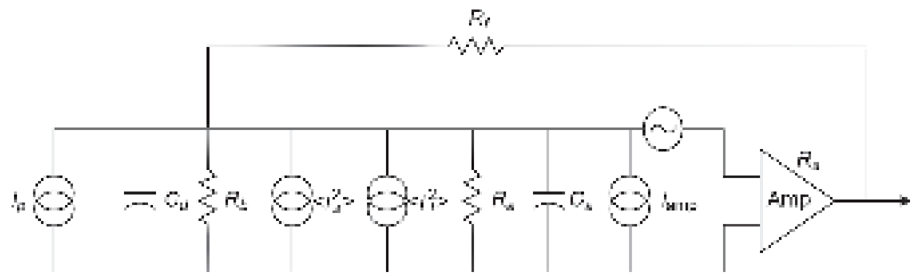

Fig. 8.6 A schematic of a transimpedance (TZ) front-end

The equivalent circuit of the transimpedance (TZ) front-end for computation of the current-to-voltage transfer function is shown in Fig. 8.7. The preamplifier in the equivalent circuit is considered to be a differential amplifier operating in the inverted mode with a gain $-G$.

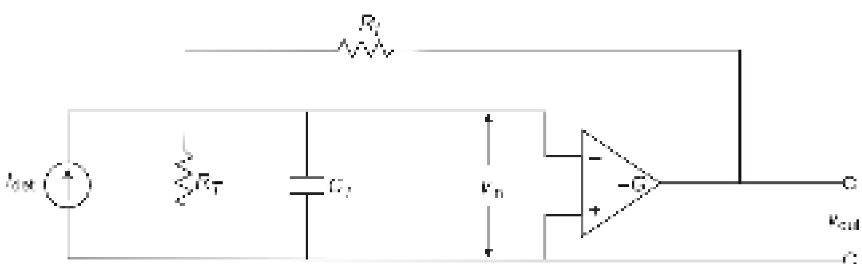

Fig. 8.7 Equivalent circuit for calculating the current-to-voltage transfer function

Applying KCL at the node connected to the inverting input of the differential amplifier, we may write

$$i_{\text{det}} + \frac{v_{\text{out}} - v_{\text{in}}}{R_f} = v_{\text{in}}\left(\frac{1}{R_T} + j2\pi fC_T\right) \tag{8.25}$$

where the effective values of resistance, R_T and capacitance C_T are given by

$$R_T = \frac{R_a R_b}{R_b + R_a}$$

and $$C_T = C_a + C_b$$

In the absence of feedback resistance, the amplifier is in an open-loop, and Eq. (8.25) becomes

$$i_{\text{det}} = v_{\text{in}}\left(\frac{1}{R_T} + j2\pi fC_T\right) \tag{8.26}$$

The open-loop gain of the differential amplifier can be expressed as

$$-G = \frac{v_{\text{out}}}{v_{\text{in}}} \tag{8.27}$$

Using Eqs (8.26) and (8.27), we get

$$i_{\text{det}} = -\frac{v_{\text{out}}}{G}\left(\frac{1}{R_T} + j2\pi fC_T\right) \tag{8.28}$$

where, f is the frequency of the input signal.

The open-loop transfer function of the front-end configuration without feedback can be expressed as

$$H_{OL}(f) = \frac{v_{\text{out}}}{i_{\text{det}}} = \frac{-GR_T}{1 + j2\pi fR_TC_T} VA^{-1} \tag{8.29}$$

In the absence of feedback (e.g. LZ, HZ cases), the bandwidth without equalization is determined by Eq. (8.23).

In the presence of feedback (closed loop), use of Eqs (8.25) and (8.27) yields

$$i_{\text{det}} = -v_{\text{out}}\left(\frac{1}{R_f} + \frac{1}{GR_f} + \frac{1}{GR_T} + \frac{j2\pi fC_T}{G}\right) \tag{8.30}$$

The closed-loop transfer function of the TZ front-end can be expressed as

$$H_{CL}(f) = \frac{v_{\text{out}}}{i_{\text{det}}} = \frac{-R_f}{1 + \frac{1}{G} + \frac{R_f}{GR_T} + \frac{j2\pi fR_TC_T}{G}}$$

That is,

$$H_{CL}(f) = \frac{-R_f \Big/ \left(1 + \frac{1}{G} + \frac{R_f}{GR_T}\right)}{1 + \dfrac{j2\pi fR_TC_T}{1 + G + \frac{R_f}{R_T}}} \tag{8.31}$$

The open-loop gain of the amplifier is generally high such that

$$G \gg 1 + \frac{R_f}{R_T} \tag{8.32}$$

Under the above condition, the closed-loop transfer function of the TZ front-end can be approximated as

$$H_{CL}(f) \approx \frac{-R_f}{1 + \frac{j2\pi fR_TC_T}{G}} \text{VA}^{-1} \tag{8.33}$$

The closed-loop transfer function of a TZ front-end configuration corresponds to gain which is measured in ohms.

The closed-loop transfer function of transimpedance front-end can be expressed as

$$H_{CL}(f) \approx \frac{-R_f}{1+j\left(\frac{f}{B}\right)} \mathrm{VA}^{-1} \tag{8.34}$$

where, B is the bandwidth of the TZ front-end given by

$$B = \frac{G}{2\pi R_T C_T} \tag{8.35}$$

It is interesting to note that the bandwidth of a TZ front-end without equalization can be made much larger as compared to that obtained in front-end configurations without feedback. The gain of the amplifier can be adjusted to increase the bandwidth.

A transimpedance configuration is essentially a HZ configuration that uses negative feedback to improve the bandwidth of the system without equalization. However, it may be stressed here that the increase in bandwidth is obtained at the expense of an increased thermal noise component arising out of the extra feedback resistance, R_f (Hullett *et al* 1977). It can be easily appreciated that when $R_f \ll R_T$ the thermal noise introduced by the feedback resistance dominates over the other thermal noise components and the former tends to dictate the overall sensitivity of the receiver front-end. On the other hand, when the feedback resistance R_f has a comparatively larger value, the thermal noise current introduced by the feedback resistance becomes low. When $R_f = R_T$, the sensitivity of a TZ front-end approaches that of a HZ front-end. However, a large value of feedback resistance may lead to instability problems.

Example 8.4 The high-impedance (HZ) front-end described in Example 8.3 is converted into a transimpedance (TZ) configuration by using a feedback resistance of 50 kΩ. If the open-loop gain of the amplifier is 500, estimate the bandwidth of the TZ front-end without equalization. Also, calculate the value of the mean square thermal noise current component introduced by the feedback resistance per unit bandwidth at 300 K.

Solution The value of the feedback resistance is 50 kΩ which is much less than the effective value of the resistance of 2 MΩ so that

$$1 + \frac{50 \times 10^3}{2.5 \times 10^6} \ll G(=400)$$

Therefore, the bandwidth of the TZ front-end without equalization can be obtained from Eq. (8.35) as

$$B = \frac{400}{2 \times 3.14 \times 2.5 \times 10^6 \times 8 \times 10^{-12}} = 3.18\ \mathrm{MHz}$$

The mean square value of the thermal noise current introduced by the feedback resistance per unit bandwidth can be estimated as

$$\begin{aligned}\left\langle i_f^2 \right\rangle &= 4 \times 1.38 \times 10^{-23} \times 300 \times \frac{1}{50} \times 10^{-3}\ \mathrm{A^2Hz^{-1}} \\ &= 33.12 \times 10^{-26}\ \mathrm{A^2Hz^{-1}}\end{aligned}$$

8.3 PREAMPLIFIER

The preamplifiers are based on active components such as transistors. In principle both the forms of transistors, e.g. BJT and FET can be used for designing the preamplifier stage following the photodetector stage. From the foregoing discussion, it is clearly understood that the preamplifier is desired to have low-noise characteristics because the photodetector converts the incoming optical signal to a weak electrical signal which is subjected to different noise components. From this view point, a FET which offers low noise as compared to its bipolar counterpart (BJT) turns out to be more attractive for designing the preamplifier of an optical receiver. Further, the gate of a FET is reverse biased the input impedance of a FET amplifier shown in Fig. 8.8 is extremely high. The low noise and low value of input capacitance of a Si-FET make it most attractive for application in optical receiver front-end. However, FET-based amplifiers usually offer much less gain as compared to that offered by their bipolar counterparts. The common source (grounded source) FET amplifier shown in Fig. 8.8 offers high input impedance and therefore can be used to realize a high impedance (HZ) receiver front-end by making use of a large bias resistance. In this form of configuration even though the thermal noise current is greatly reduced, its bandwidth is marginalized without equalization. The bandwidth can however be improved by making use of an equalization circuit at a later stage. The current gain of a Si-FET usually drops to unity at 25 MHz.

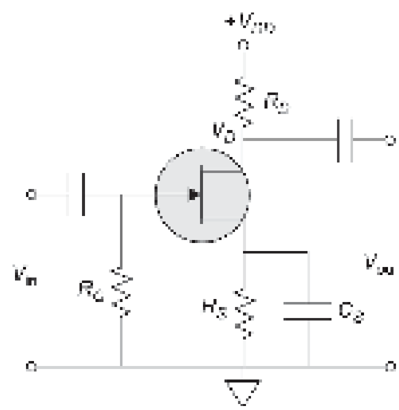

Fig. 8.8 Common source configuration of a FET-based preamplifer

8.4 FET-BASED RECEIVER FRONT-END

The high-frequency limitation of conventional Si-based junction field effect transistor (JFET) restricted its application in high-speed optical communication systems. On the other hand, high-performance microwave field-effect transistor based on Schottky gate configuration became very popular for use as a preamplifier in an optical receiver front-end. The Schottky gate field-effect transistors popularly known as metal semiconductor field effect transistors (MESFETs) in the mid-1970s and early 1980s were primarily based on GaAs material because of their high electron mobility as compared to Si. Moreover, the Schottky contact enables the device to be operated at a very high speed. The high gain at microwave frequency (~GHz) together with a relatively low-noise made GaAs-based MESFET as an attractive component for designing the preamplifier of high-speed optical receivers. MESFET-based receiver front-end usually contains a *p-i-n* photodetector or an APD as the preceding photodetector stage. The *p-i-n* photodetector and the MESFET preamplifier are generally realized in the form of a hybrid integrated circuit using thick film technology. The hybrid IC is called a PINFET. The thick film substrate reduces the stray capacitance value. Early PINFET structures were mostly focused in the 0.8–0.9 μm window (Hooper *et al* 1979; Smith *et al* 1978). Later on, PINFET-based receivers were reported for operation at longer wavelength.

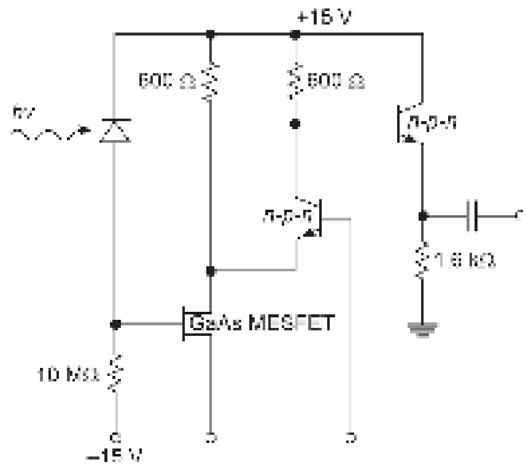

Fig. 8.9 Circuit diagram of a PINFET hybrid receiver (after Hooper *et al* 1980)

In FET-based photoreceivers, the preamplifier is made of a FET such as JFET, metal semiconductor field effect region around 1.3 μm (Ahmad *et al* 1978; Hooper *et al* 1980; Smith *et al* 1980). A typical circuit diagram of a PINFET hybrid receiver designed in the HZ configuration is shown in Fig. 8.9. The HZ stage integrates the signal and the integrated output after pulse shaping with the help of a suitable filter is passed through an equalizer (a differentiator in this case) before decision-making. The receiver is reported to be capable of handling a transmission rate of 140 Mbps. The PINFET receiver has been demonstrated to exhibit a sensitivity of –44.2. dBm at a BER 10^{-9}. A PINFET-based high-speed optical receiver front-end designed in the transimpedance configuration was reported by Ogawa *et al* (Ogawa *et al* 1979). The circuit schematic is shown in Fig. 8.10. The receiver front-end makes use of a GaAs MESFET followed by two complementary microwave BJTs. Negative feedback is provided from the output to the gate of the MESFET to convert it into a transimpedance amplifier. The receiver is capable of handling a transmission rate of

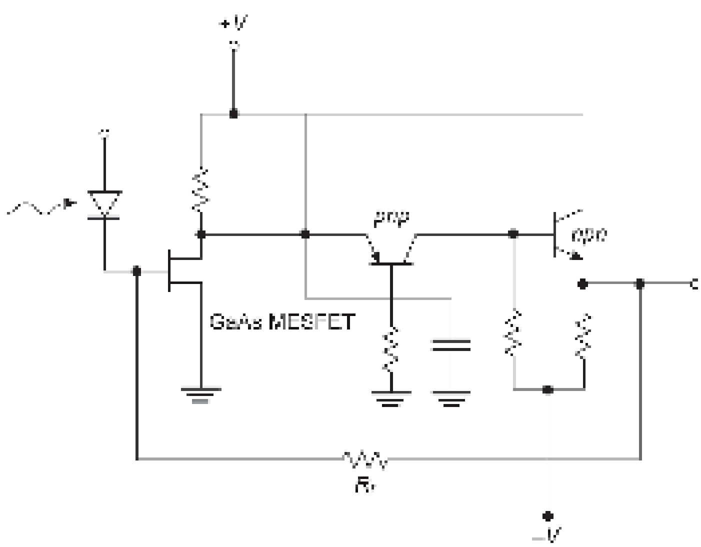

Fig. 8.10 Circuit schematic of a PINFET hybrid receiver front-end configured in TZ mode (after Ogawa *et al* 1979)

274 Mbps without any equalization. The receiver offers a sensitivity of –35 dBm at a BER of 10^{-9}. In an attempt to improve the sensitivity of the receiver a variety of structures has been tried. The basic pin detector has been replaced by avalanche photodetectors in some cases. In addition, a variety of transistors (FET and heterojunction bipolar transistor (HBT)) have been used in the preamplifier stage. Further, different III-V material combinations have been used for designing the photodetector to make the receiver suitable for operation in the longer wavelength region (1.3 μm and 1.55 μm). Moreover, it has been demonstrated that the use of monolithic integration in place of hybrid integration can greatly improve the performance of optical receivers in various wavelength regions of operation. The next section is a review of the state-of-the-art optical receivers.

8.5 STATE-OF-THE-ART OPTICAL RECEIVERS

In an attempt to improve the sensitivity and bit rate of operation of optical receivers a large number of combinations of photodetectors (photoconductor, pin detector, APD, heterojunction phototransistor, etc.) and transistors as active components (JFET, MESFET, MISFET, heterojunction field transistor (HFET), HBT) have been used by the researchers in the past. The integration of the detector and preamplifier along with other components leads to an optoelectronic integrated circuit (OEIC) receiver. It is understood that monolithic integration can greatly improve performance as compared to its hybrid counterpart. The photodetectors may be photoconductor, *p-i-n* photodiode, metal-semiconductor-metal (MSM) photodetector or APD. The early efforts in the direction of integrating the optical detector and preamplifier (Kolbas *et al* 1983) monolithically led to the development of first-generation planar OEIC receivers. The OEIC receiver chip comprises a Be-implanted GaAs *p-i-n* photodetector integrated with a transimpedance preamplifier containing six GaAs depletion mode MESFET and five Schottky diodes fabricated by selective implantation in a semi-insulating substrate (Kolbas *et al* 1983). A circuit schematic of the PINFET OEIC receiver front-end is reproduced in Fig. 8.11 (Kolbas *et al* 1983). An $In_{0.53}Ga_{0.43}As$ photoconductor-based receiver front-end was demonstrated for use in moderate-to-high bit rates (Forrest, 1985). This receiver sensitivity is reported to compete with *p-i-n*-based optical receivers in the bit rate region of 500 Mb/s–2 Gb/s. The sensitivity of the receiver has been reported to be –49 dBm and –45 dBm for operation bit rates of 500 Mb/s and 1 Gb/s, respectively, for a gain of 80 of the photoconductive detector. An optical receiver front-end consisting of InGaAs *p-i-n* photodiode integrated with an InP MISFET for the 1.3 μm wavelength region has been reported to exhibit a sensitivity of –34.5 dBm at a speed of 10 Mb/s for the BER of 10^{-9} (Kasahara *et al* 1984). A large number of studies involving *p-i-n* photodetector and FET based on III-V materials were subsequently reported. These include an InGaAs/InP *p-i-n* photodiode along with a FET preamplifier at the front-end of a lightwave optical receiver for operation in 1.3 μm wavelength region (Lee *et al* 1980) with sensitivity values of –46.7 dBm and –36 dBm at a speed of 44.7 Mb/s and 274 Mb/s, respectively; a monolithic photoreceiver based on a *p-i-n* photodetector and a GaAs preamplifier in TZ mode for operation at a speed of 140 Mb/s (Archambault *et al* 1987) was reported to be –38 dBm; an InGaAs/InP *p-i-n* photodiode and JFET preamplifier-based OEIC receiver with a sensitivity of –35 dBm at a bit rate of 200 Mbps reported to be grown by Metal-Organic-Vapor Phase Epitaxy (MOVPE) (Baur *et al* 1992) with a cut-off frequency of the photoreceiver of 7 GHz and a photresponsivity of 1.1 A/W; an InGaAs *p-i-n* photodiode monolithically integrated with and an InP MISFET

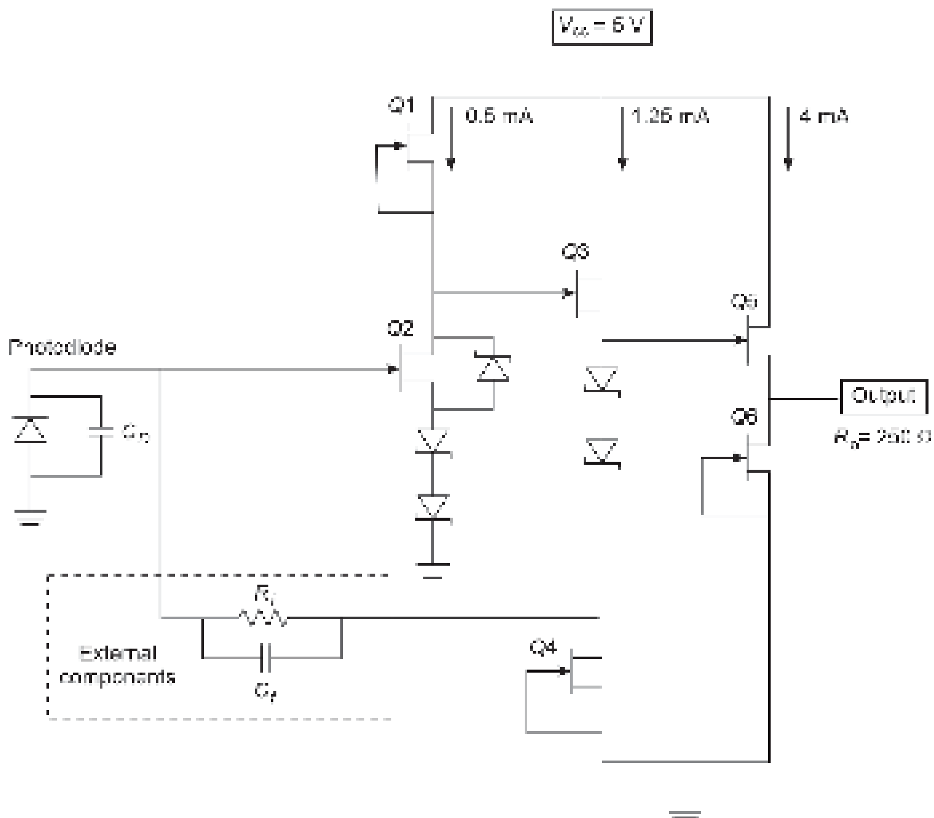

Fig. 8.11 Schematic of a planar *p-i-n* photodetector/GaAs preamplifier OEIC receiver front-end

using vertical integration scheme (Chen *et al* 1988) with a sensitivity of –27 dBm at the speed of 400 Mb/s; a monolithically integrated InGaAs *p-i-n* photodiode with 4 JFETs biased using a single 5 V supply to constitute the transimpedance amplifier stage exhibiting a transimpedance gain of 965 ohms at speed of 400 Mbps (Matsuda *et al* 1988); a photoreceiver consisting of an InGaAs *p-i-n* photodiode and InP-MISFET preamplifier reportedly grown by vapor phase epitaxy exhibiting a speed of 600 Mb/s with a sensitivity of –25.4 dBm (Antresyan *et al* 1989); a 622 Mb/s monolithic InGaAsP PINFET OEIC receiver with a sensitivity of –35.4 dBm using a cascode amplifier (Uchida *et al* 1991); a new *p-i-n*/FET lightwave receiver with active optical feedback for operation at 1 Gb/s speed with a sensitivity of –31 dBm (Williams *et al* 1986); an InGaAs *p-i-n* photodiode followed by an arch type JFET preamplifier based receiver offering a sensitivity of –37.3 dBm at a speed of 1 Gb/s (Lo *et al* 1989); a novel OEIC photoreceiver configuration consisting of an InGaAs *p-i-n* photodiode, InGaAs self-aligned JFET and a bias resistor fabricated by MOVPE technique with a bandwidth of 2.5 Gb/s (Park *et al* 1992); another photoreceiver consisting of back illuminated *p-i-n* photodiode with small junction area and a GaAs high impedance preamplifier with two-stage amplification for operation at 2 Gbps with a sensitivity of –26.9 dBm (Makiuchi *et al* 1988) having quantum efficiency and cut-off frequency of 81% and 19 GHz, respectively; a photoreceiver front-end consisting of *p-i-n* photodiode and InGaAs FET reportedly exhibiting a sensitivity of –18.5 dBm with a speed of 2 Gb/s (Miura *et al* 1988); a high-performance, high-reliability InP/GaInAs *p-i-n* photodiodes and flip-chip integrated receiver fabricated for lightwave communications at 2 Gbps, a sensitivity of –27.4 dBm at a speed of 2 Gb/s (Wada *et al* 1991); an ultra-broadband GaAs MESFET preamplifier IC with a transimpedance gain of 44 dBΩ for a bandwidth of 12 GHz fabricated for 10 Gb/s optical communication system (Miyashita *et al* 1992); a photoreceiver configuration consisting of *p-i-n* photodiode and distributed MESFET preamplifier having a sensitivity of –21 dBm for a speed of 10Gb/s with calculated input noise current density and bandwidth of 8 pA/Hz$^{1/2}$ and 8 GHz, respectively (Freundorfer *et al* 1998). The sensitivity limit of monolithically integrated pin/JFET photodetector was reported by Yoshida *et al* (Yoshida *et al* 1996). The speed of the receiver was reported to be limited to 622 Mb/s for the *p-i-n*/JFET configuration. This photoreceiver with a *p-i-n* photodiode and JFET preamplifier driven by a single and multi-power supply was tested. Three different types of photoreceivers using MOVPE grown crystals and Be-ion implantation technology were reported to be fabricated. The receiver using a cascade input stage had an extremely high sensitivity of –34.5 dBm for the speed of 622 Mb/s. The sensitivities of the other two receivers using the inverter input stage were –33.6 dBm for the multi-power supply and –31.4 dBm for the single supply input at this speed (622 Mb/s). A high-sensitivity and wide dynamic range *p-i-n* photodiode and MESFET preamplifier photoreceiver for the use in 1.55 μm wavelength region at a low bit rate employing an optically coupled feedback rather than conventional feedback resistor was also demonstrated (Kasper *et al* 1988). The sensitivity of the photoreceiver was reported to be –63.8 dBm at a speed of 1.5 Mb/s for a BER of 10^{-7}. The circuit schematic of an optical-feedback transimpedance amplifier is shown in Fig. 8.12 vis-à-vis the schematic of a conventional transimpedance receiver front-end (Kasper *et al* 1988). The use of optical feedback eliminates the thermal noise component associated with the feedback resistor of a conventional TZ configuration.

The design and fabrication of 1 Gb/s OEIC receiver based on GaAs-MSM photodetector and GaAs-MESFET preamplifier were reported for operation at 0.8 μm wavelength region exhibiting a high gain of 90 dB and sensitivity of –20.5 dBm at 1 Gb/s (Shih *et al* 1996). A fully-integrated photoreceiver comprising an ion-implanted GaAs-MESFET and an MSM photodetector was demonstrated to operate at a speed of 2.5 Gb/s (Chang, 1995). Forrest (Forrest 1985) reported a photoreceiver configuration consisting of $In_{0.53}Ga_{0.47}As/InP$ APD and GaAs FET preamplifier. The sensitivity of this photoreceiver was reported to be –53.2 dBm at a speed of 45 Mb/s for a BER of 10^{-9}. At this sensitivity, the APD exhibited a gain

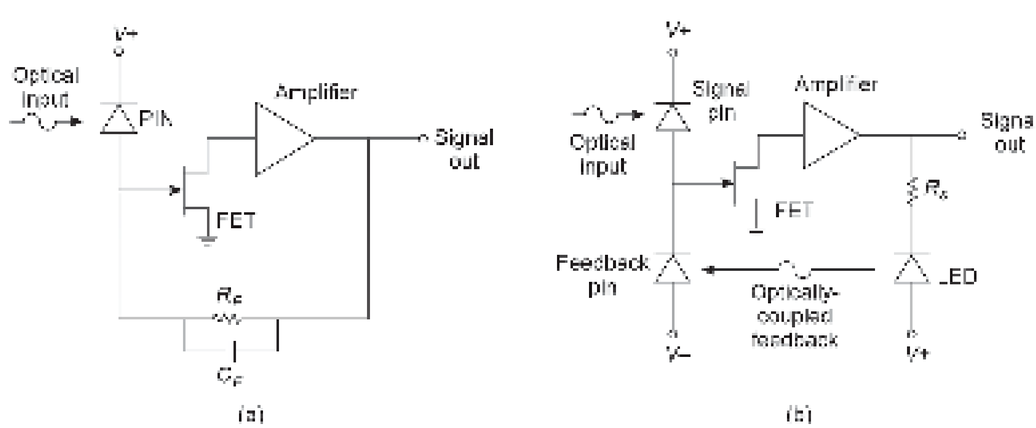

Fig. 8.12 Transimpedance optical receiver front-end: (a) Conventional circuit–schematic (b) Proposed optical-feedback transimpedance photoreceiver front-end

of 20 and unity quantum efficiency. Another avalanche photodiode-based photoreceiver for application in the 1.5–1.6 μm wavelength region was reported to exhibit a sensitivity of –35 dBm at a speed of 2 Gb/s (Matsushima *et al* 1983). A theoretical model of MESFET preamplifier-based photoreceiver consisting of an avalanche photodiode was reported by Fyath and O'Reilly (Fyath *et al* 1989). The speed of response of the photoreceiver was estimated to be 2 Gb/s with a sensitivity of –40 dBm. An APD/FET photoreceiver for operation at a bit rate of 8 Gb/s with a sensitivity of –25.8 dBm at a BER of 10^{-9} for operation in the 1.3–1.5 μm wavelength region was demonstrated by Kasper *et al* (Kasper *et al* 1987). The receiver exhibited a gain bandwidth product of 60 GHz for InGaAs/InGaAsP/InP heterojunction avalanche photodiode followed by a hybrid GaAs MESFET preamplifier. This photoreceiver consists of separate absorption, grading, and multiplication avalanched photodiode (SAGM-APD) with GaAs preamplifier. The gain bandwidth product of SAGM-APD was reported to be 140 GHz for a 2 μm wide multiplication layer. For this gain bandwidth product, the SAGM-APD photoreceiver exhibited a speed of 8 Gb/s with a sensitivity of –37 dBm for a BER of 10^{-9}. A delta-doped avalanche photodiode for high-bit rate lightwave receivers was later reported (Kuchibhotla *et al* 1991).

8.5.1 Heterojunction Field Transistors (HFET)-based Receivers

To improve the speed of photoreceivers, the preamplifier should use high-speed devices, such as HFET, e.g. modulation doped field effect transistors (MODFET) also known as high electron mobility transistors (HEMT). Based on this fact, several photoreceivers have been fabricated, tested, and studied by several researchers. For the HFET-based photoreceivers, the photodetector can be a *p-i-n* photodiode, MSM photodetector, or APD.

A monolithically integrated InP/GaInAs *p-i-n* photodiode and an AllnAs/GaInAs HEMT preamplifier-based photoreceiver for long-wavelength optical communication systems was reported for operation at 2 Gbps (Nobuhara *et al* 1988). The schematic cross-section of the *p-i-n*/HEMT OEIC front-end is shown in Fig. 8.13 (Nobuhara *et al* 1988). The transconductance of the HEMT preamplifier was reported to be 270 mS/mm for 1μm gate length. The receiver exhibited a sensitivity of –23.7 dBm with a speed of 2 Gb/s for the NRZ random signal. A novellong wavelength OEIC receiver for 1.3 μm wavelength application was reported in the same year by Spear *et al* (Spear *et al* 1988). This monolithic photoreceiver front-end consists of a *p-i-n* photodetector, load resistor, and inverter amplifier fabricated using a novel quasi-planar process. This quasi-planar process is based on a single-step MOVPE growth, allowing independent optimization of both optical and electronic components. The photoreceiver based on InGaAs/InP exhibited a sensitivity of –32.7 dBm with a speed of 560 Mb/s. Monolithically integrated optical receivers based on GalnAs *p-i-n* photodiode and an n-AllnAs/GalnAs HEMT preamplifier on an InP substrate grown by MOVPE were also reported by Sasaki *et al* (Sasaki *et al* 1988; Sasaki *et al* 1989) for long-wavelength fiber optic communication systems. The receivers exhibited a speed of 1.6 Gb/s with a sensitivity of –28 dBm. Later on, a high-impedance HEMT preamplifier-based photoreceiver with *p-i-n* photodiode as the front-end of the receiver was demonstrated (Yano *et al* 1990). The receiver exhibited a sensitivity of –30.4 dBm with a speed of 1.6 Gbps.

Three different types of photoreceiver structures, e.g. high impedance, transimpedance and straight forward type consisting of InGaAs *p-i-n* photodetector and InAlAs/InGaAs HEMT preamplifier were fabricated and tested (Hayashi *et al* 1991). The three different types of receivers exhibited sensitivities of –30.4 dBm, –27.1 dBm, and –25.5 dBm, respectively, for the high impedance, transimpedance, and straight forward cases. A speed of 10 Gb/s was reported to be achieved for these different photoreceiver configurations.

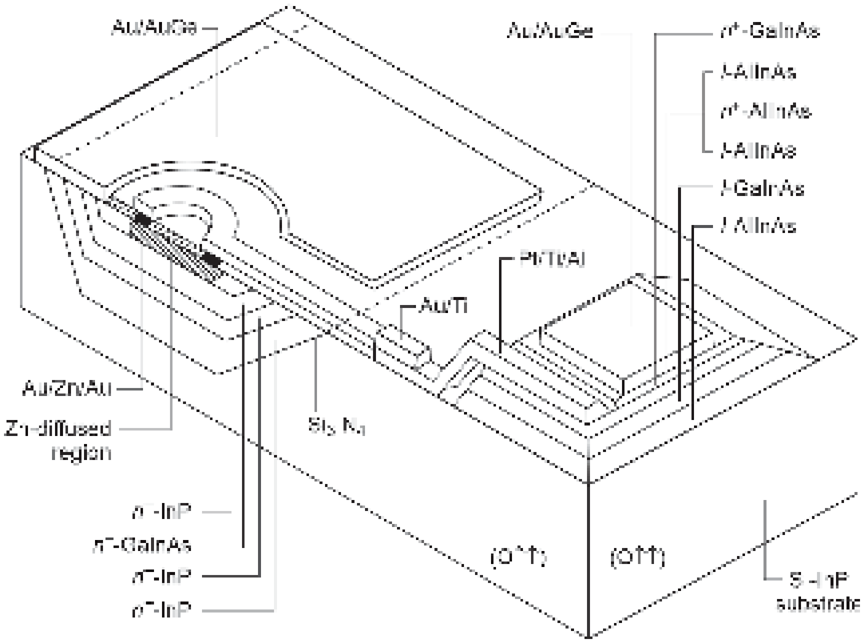

Fig. 8.13 Schematic of a InP-GaInAs*p-i-n*/AllnAs/GaInAs HEMT OEIC receiver front-end (cross-sectional view) (after Nobuhara *et al* 1988)

A photoreceiver consisting of $In_{0.53}Ga_{0.47}As$ *p-i-n* photodiode and pseudomorphic $In_{0.65}Ga_{0.35}$ AS/$In_{0.52}Al_{0.48}$ As MODFET using molecular beam epitaxy (MBE) was reported for operation in 1.55 µm wavelength region (Berger *et al* 1992). The cutoff frequency of the MODFET is reported to be 24 GHz. The transconductance of the MODFET was found to be 495 mS/mm. The 3 dB bandwidth of the photoreceiver was measured to be 1 GHz. The photoreceiver exhibited a sensitivity of –29.6 dBm at a speed of 1 Gb/s. The dynamic range of this photoreceiver was measured to be 25 dB. Later on, Berthier *et al* (Berthier *et al* 1994) studied and measured the noise performance of the high-impedance photoreceiver configuration consisting of InGaAsP channel HFETs on InP for OEIC applications. The front-end of the photoreceiver consisted of an InGaAs p-channel HFET preamplifier on InP and *p-i-n* as the photodetector. The photoreceiver exhibited f_t and f_{max} of 18 GHz, and 40 GHz, respectively. The measured values were in good agreement with the predicted values. Another photoreceiver based on $In_{0.65}Ga_{0.35}As$/$In_{0.52}Al_{0.48}As$ MODFET was reported for operation at a high bit rate (Berger *et al* 1992). The photoreceiver configuration consists of a *p-i-n* photodiode monolithically integrated with a MODFET to form the front-end. The photoreceiver exhibits a sensitivity of –29.6 dBm at a speed of 6 Gb/s. The dynamic range of the receiver was measured to be 25 dB. Yano *et al* (Yano *et al* 1990) reported the fabrication of an ultra-high speed optoelectronic integrated receiver consisting of GaInAs *p-i-n* photodiode and transimpedance AllnAs/InGaAs high electron mobility transistor amplifier on InP substrate. The 3 dB bandwidth and the transimpedance gain of the photoreceiver were reported to be 6 GHz and 50 dB, respectively. The sensitivity value of the OEIC receiver at this transimpedance gain was –21.2 dBm at a speed of 8 Gb/s for a BER of 10^{-9}. Akahori *et al* (Akahori *et al* 1994) reported a photoreceiver based on Be-ion implanted InGaAs *p-i-n* photodetector integrated with an InAlAs/InGaAs HEMT grown by MOVPE. The cross-sectional view of the *p-i-n*/HEMT

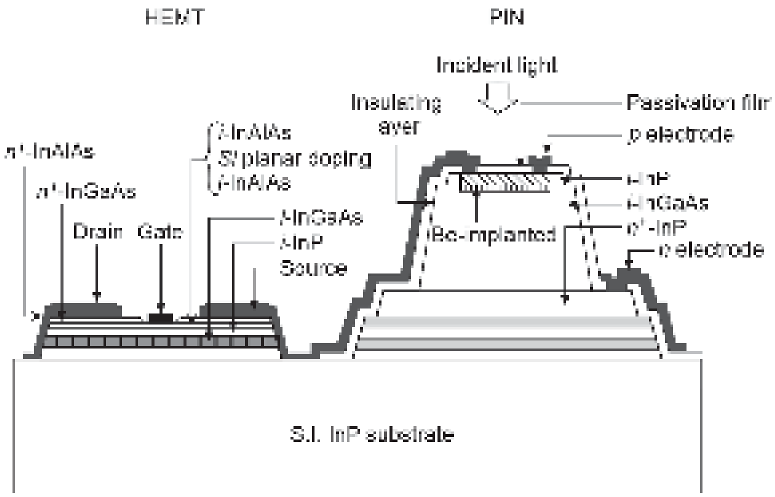

Fig. 8.14 Cross-sectional view of monolithic OEIC *p-i-n*/HEMT receiver front-end (after Akahori *et al* 1994)

receiver front-end is shown in Fig. 8.14 (Akahori *et al* 1992). The 3 dB bandwidth of the receiver was measured to be 8 GHz. The receiver also exhibited a good eye-pattern definition at 10 Gb/s non-return-to-zero (NRZ) optical signal. The reported sensitivity of this receiver is –16.5 dBm. In the following year, a photoreceiver configuration consisting of an InGaAs *p-i-n* photodetector and transimpedance MODFET preamplifier was reported to exhibit a bandwidth of 6 GHz for a speed of 10 Gb/s (Aitken *et al* 1993). Park and Minasian (Park *et al* 1994) reported a photoreceiver front-end based on an InGaAs *p-i-n* photodiode and HEMT preamplifier. The design of a photoreceiver was based on the optimum noise-matching network. The sensitivity measured at a speed of 10 Gb/s was reported to be –27 dBm. This photoreceiver demonstrated a bandwidth of 6.3 GHz with an input average noise current of 5 pA/Hz$^{1/2}$. A photoreceiver consisting of an InGaAs *p-i-n* photodetector and an InAlAs/InGaAs HEMT preamplifier with a speed of 15 Gb/s and a bandwidth of 11 GHz at this speed was reported to be achieved subsequently (Akahori *et al* 1994). A *p-i-n*/HFET OEIC receiver was designed and analyzed for high sensitivity and maximally flat frequency response characteristics (Das *et al* 1995). The HFET preamplifier worked in the transimpedance mode of operation. The photoreceiver exhibited a sensitivity of –28 dBm at a speed of 10 Gb/s. The receiver also exhibited a transimpedance gain of 75 dBΩ. The photoreceiver 3 dB frequency was reported to be 57 GHz. In the same year, a number of *p-i-n*/HEMT receiver front-end were reported. These include a side illuminated *p-i-n*/HEMT configuration with a speed of 10 Gb/s (Muramota *et al* 1995); an InP-based *p-i-n*/HEMT OEIC receiver for bit rates up to 10 Gb/s (Kuebart *et al* 1995) for operation in high impedance amplifier, transimpedance and cascade design modes with a responsivity of 12.9 dB A/W and a bandwidth of 5 GHz for the transimpedance case having a low average noise current of 11.5 pA/Hz$^{1/2}$ and a high sensitivity of –19.2 dBm; a monolithically fabricated and characterized integrated photoreceiver that exhibited a handwidth of 18 GHz in high-impedance (HZ) mode with the provision of input capacitance compensation through equalization (Kleper *et al* 1995); a 10 Gb/s hybrid *p-i-n* HEMT photoreceiver employing a loss-less tuned noise-matching technique between a *p-i-n* and HEMT amplifier stages for high sensitivity and a transimpedance feedback scheme for a wide dynamic range (Yun *et al* 1995) with a sensitivity of –23.5 dBm at a speed of 10 Gb/s. A photoreceiver based on an InGaAs photodetector followed by InAlAs/InGaAs HEMT preamplifier was reported by Akasharu *et al* for operation at 10 Gb/s (Akasharu *et al* 1992). Another OEIC photoreceiver based on GaInAs/AllnAs HEMT and InGaAs *p-i-n* photodetector was reported by Umbach *et al*. (Umbach *et al* 1996). This receiver exhibited a bandwidth of 27 GHz with a speed of 15 Gb/s.

The low-noise study of the performance of a monolithic OEIC optical receiver based on *p-i-n*/HEMT for long-wavelength application at 12 Gb/s was reported (Fay[1] *et al* 1998). The monolithic integration in this case was achieved by using a stacked layer structure and growing the *p-i-n* structure on a pregrown HEMT structure. The circuit schematic is shown in Fig. 8.15. The sensitivity of the receiver was measured to be –17.7 dBm at 10 Gb/s and –5.8 dBm at 12 Gb/s at a BER of 10^{-9}. In the same year, a comparative study of photoreceiver front-ends based on monolithically integrated MSM/HEMT and PIN/HEMT was reported by the same group (Fay[2] *et al* 1998; Fay *et al* 2002). The InGaAs MSM or InP/InGaAs *p-i-n* photodetector and InAlAs/InGaAs HEMT preamplifier were fabricated by using a stacked-layer structure design grown by MOVPE. The MSM/HEMT receiver front-end exhibited a sensitivity of –10.7 dBm at a speed of 10 Gb/s and –16.9 dBm at 5 Gb/s for a BER of 10^{-9} (Fay *et al* 1998) and –17.7 dBm at 10 Gb/s and –15.8 dBm at 12 Gb/s (Fay *et al* 2002). A low-noise monolithically integrated *p-i-n*/HEMT photoreceiver for long-wavelength transmission systems exhibiting a cutoff frequency of 150 GHz, a transconductance of the HEMT preamplifier of 80 mS/mm was also reported in the same year (Fay *et al* 1998). The photoreceiver exhibited an input-referred noise current spectral density of 8.82 pA/$Hz^{1/2}$. The receiver exhibited a sensitivity of –15.8 dBm at a speed of 12 Gb/s. A large bandwidth monolithically integrated photoreceiver consisting of a stacked structure of waveguide *p-i-n* photodiode and InAlAs–InGaAs high electron mobility transistor for 1.55 μm wavelength operation was reported (Takahata *et al* 1998). The photoreceiver exhibited a responsivity of 0.5 A/W. A bandwidth of 45.5 GHz was reported to be obtained. The speed of 40 Gb/s was measured for this receiver using an eye diagram. A novel optoelectronic integrated circuit based on InP consisting of a waveguide *p-i-n* photodiode and a distributed amplifier comprising four high-electron mobility transistors was also reported for operation with a bandwidth of 37 GHz (Mekonnen *et al* 1999).

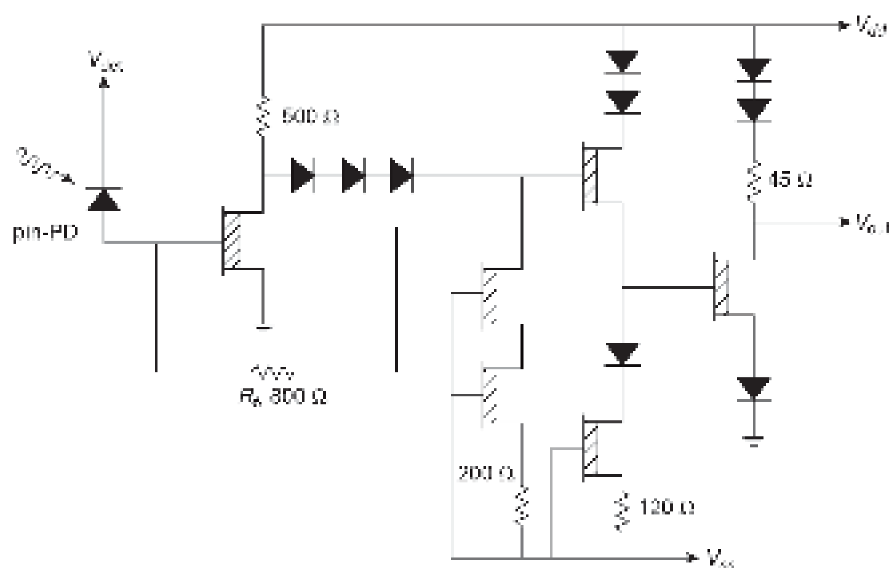

Fig. 8.15 The circuit schematic of a 12 Gbps monolithically integrated *p-i-n*/HEMT optical receiver (after Fay[2] *et al* 1998)

A photoreceiver consisting of AllnAs/GaInAs MSM photodetector and AllnAs/GaInAs HEMT preamplifier was reported to exhibit a speed of 1.7 Gb/s for NRZ signal (Hong *et al* 1989). Fujii *et al* fabricated a monolithically integrated InGaAs/InAlAs on InP MSM/HEMT photoreceiver for a speed of operation of 2 Gb/s (Fujii *et al* 1991). Design, Fabrication, and characterization of a monolithically integrated photoreceiver based on a pseudomorphic InGaAs on GaAs MODFET and MSM photodetector was reported to exhibit a sensitivity of –25 dBm for a speed of 10 Gb/s (Ketterson *et al* 1993). The transimpedance amplifier bandwidth was measured to be 14 GHz and the overall photoreceiver bandwidth was 11 GHz. A photoreceiver configuration consisting of MODFET and MSM photodetectors with bandwidths of 15 GHz and 18.5 GHz for the application in the 1.55 μm wavelength region was later reported (Fay *et al* 1996). A photoreceiver consisting of an MSM photodiode, a transimpedance amplifier, and three limiting amplifier stages for high-speed optical fiber link was proposed by Lao *et al* (Lao *et al* 1998). The schematic cross-section of the InGaAs MSM integrated with AlGaAs/GaAs HEMT is shown in Fig. 8.16 (Lao *et al* 1998). The IC was fabricated using a 0.2 μm gate length HEMT with a cutoff frequency of 60 GHz. The bandwidth of this photoreceiver was 16 GHz. A high transimpedance gain of 14 kΩ at 10 Gb/s bit-rate was achieved for this receiver. This photoreceiver was designed for use in the 1.3–1.55 μm wavelength applications. The sensitivity was measured to be –15 dBm for the speed of 10 Gb/s. An AlGaAs/GaAs MSM-HEMT OEIC receiver with 8.2 GHz bandwidth was reported initially for operation at 10 Gb/s (Hurm *et al* 1991). The receiver was operated at 0.85 μm wavelength region. Later on, the speed of the photoreceiver was improved to 20 Gb/s (Hurm *et al* 1996). This photoreceiver consisted of an MSM photodetector and AlGaAs/GaAs HEMT OEIC receiver operating at 0.85 μm wavelength applications. A novel monolithically integrated an InP-based photoreceiver operating in the narrow-band around 38 GHz at a wavelength of 1.55 μm was developed by Engel *et al* (Engel *et al* 1998). The optoelectronic integrated circuit incorporated two types of high-speed devices, a submicrometer metal-semiconductor-metal photodetector made of InGaAs-InP and a quarter micrometer high electron mobility transistor based on a lattice-matched InGaAs-InAIAs-InP layer stack. A responsivity of 3.5 A/W at 38 GHz bandwidth was obtained for this receiver. A HEMT preamplifer-based lightwave receiver using a Ge-APD as a photodetector was described by Walker *et al* (Walker *et al* 1989). The receiver was constructed on a standard Teflon printed circuit board with a packaged-tailed 30 μm diameter Ge-APD. The sensitivity of the photoreceiver was –25.5 dBm for the BER of 10^{-9}. A sensitivity of –25.5 dBm with a speed of 5 Gb/s for a BER of 10^{-9}. A sensitivity of –25.5 dBm with a speed of 5 Gb/s

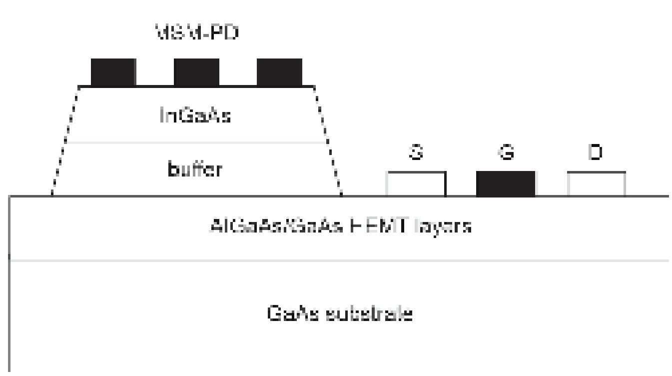

Fig. 8.16 Cross-sectional schematic of a MSM/HEMT optical receiver front-end (after Lao *et al* 1998)

for a BER of 10^{-9} was also demonstrated for this receiver. The receiver was operated at 1.31 μm wavelength region for the NRZ pseudorandom sequence. Zebda *et al* (Zebda *et al* 1988) reported a monolithic fabricated pseudomorphic InGaAs/AlGaAs MODFET-APD receiver. This photoreceiver exhibited a gain of 8 dB at 1 GHz bandwidth. A 10 Gb/s hybrid APD-HEMT photoreceiver was also reported for high-speed application (Yun *et al* 1996). This photoreceiver employed a loss-less tuned noise-matching technique between an APD and HEMT amplifier stages for high sensitivity and a transimpedance feedback scheme for a wide dynamic range. A sensitivity of –29.4 dBm at a speed of 10 Gb/s was obtained for this photoreceiver.

8.5.2 Heterojunction Bipolar Transistor (HBT)-based Receivers

HBTs have been demonstrated to exhibit a very large bandwidth exceeding 100 GHz and low-noise capabilities as discrete devices, particularly those based on InP substrates (John *et al* 1994). An InP/InGaAs *p-i-n* HBT monolithically integrated photoreceiver grown by MOVPE exhibited a transimpedance gain of 1375 ohm and a dynamic range of 25 dB was reported by Chandrasekhar *et al* (Chandrasekhar *et al* 1990). The cross-sectional view of the *p-i-n*/HBT structure is shown in Fig. 8.17. A sensitivity of –26.1 dBm at a speed of 1 Gb/s was reported to be achieved. The circuit schematic of the monolithic photoreceiver front-end is shown in Fig. 8.18. The HBT Q_1 and Q_2 are connected in the transimpedance mode while the third HBT Q_3 acts as a buffer stage for the output (Chandrasekhar *et al* 1990). A *p-i-n*/HBT photoreceiver was later reported for operation at a speed of 4 Gb/s (Chandrasekhar *et al* 1990). The speed of the *p-i-n*/HBT photoreceiver was further increased to 5 Gb/s by improving the fabrication technique (Chandrasekhar *et al* 1991). This photoreceiver was fabricated with InP/InGaAs heterostructures grown by chemical beam epitaxy (CBE). The OEIC photoreceiver exhibited a cut-off frequency of 32 GHz. The mid-band transimpedance gain was found to be 400 Ω. The photoreceiver was operated in the 1.5 μm wavelength range at a speed of 5 Gb/s. At this speed, the receiver exhibited a sensitivity of –18.8 dBm. A wide bandwidth monolithic OEIC receiver based on an HBT preamplifier and *p-i-n* photodetector was demonstrated to provide a low-noise and

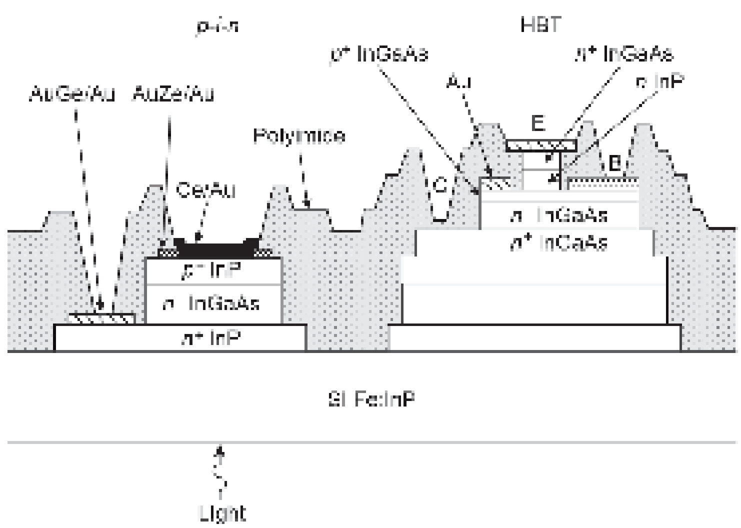

Fig. 8.17 Schematic cross-section of the post-fabricated *p-i-n*/HBT integrated photoreceiver front-end (after Chandrasekhar *et al* 1990)

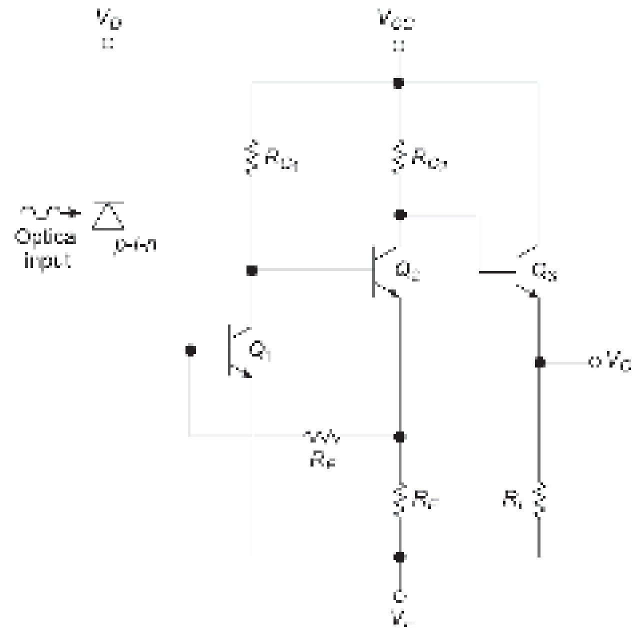

Fig. 8.18 Circuit schematic of a *p-i-n*/HBT photoreceiver front-end in transimpedance configuration (after Chadrasekhar *et al* 1990)

ultra-high performance in a preamplifier with an easily integrable optical detector (Pedrotti *et al* 1991). Later on, the same group reported a photoreceiver configuration consisting of the *p-i-n* photodetector and HBT preamplifier with a bandwidth of 3 GHz (Pedrotti *et al* 1991). Cowles *et al*. (Cowles *et al* 1994) fabricated and measured the performance characteristics of a monolithically integrated photoreceiver using an $In_{0.53}Ga_{0.47}As/In_{0.52}Al_{0.48}As$ *p-i-n*-HBT. The discrete transistor (HBT) demonstrated f_t and f_{max} of 54 GHz and 51 GHz, respectively. The photoreceiver amplifier exhibited a gain of 40 dB with a bandwidth of 10 GHz. The combined unit of the photoreceiver showed a bandwidth of 7.1 GHz. Chandrasekhar *et al*. further developed a photoreceiver module consisting of an InGa As/InP *p-i-n* photodetector and HBT preamplifier. The photoreceiver exhibited a speed of 5 Gb/s (Chandrasekhar *et al* 1993). A high performance *p-i-n*/HBT InAlAs/InGaAs photoreceiver was reported by Lunardi *et al* for operation at 12 Gb/s (Lunardi *et al* 1995). A 16 GHz bandwidth photoreceiver based on InAlAs/InGaAs *p-i-n*/HBT front-end was also reported in the same year by Aitken *et al* (Aitken *et al* 1995).

An OEIC receiver based on a *p-i-n* photodetector and an HBT-based preamplifier connected in the transimpedance configuration was reported for high sensitivity and maximally flat response (Das *et al* 1995). The HBT-based OEIC receiver exhibited a sensitivity of –28 dBm at a speed of 10 Gb/s. The estimated transimpedance gain of this receiver was reported to be 72 dBΩ for the cascaded HBT input stage. The HBT input stage OEIC receiver demonstrated a 3 dB frequency of 21.6 GHz. Later on, Lunardi *et al*. (Lunardi[1] *et al* 1995) fabricated a *p-i-n*/HBT photoreceiver with a sensitivity of –7.6 dBm at a speed of 12 Gb/s. Further, an InP/InGaAs *p-i-n*/HBT OEIC with a sensitivity of –17.3 dBm at 15 Gb/s was reported (Lunardi[2] *et al* 1995). The sensitivity of the *p-i-n* photodetector integrated with the InP/InGaAs-based HBT in the transimpedance mode was reported to improve further at a higher bit rate. The measured sensitivity was reported to be –17 dBm at a speed of 20 Gb/s (Lunardi[3] *et al* 1995).

Yang *et al* (Yang *et al* 1996) designed, fabricated, and characterized a monolithically integrated photoreceiver based on a *p-i-n* photodetector and HBT preamplifier. The photoreceiver had a three-stage transimpedance amplifier with a feedback resistance of 550 Ω. The schematic cross-section of the *p-i-n*/HBT after fabrication is shown along with the circuit schematic in Fig. 8.19. The photoreceiver demonstrated a transimpedance gain of 46 dBΩ and a 3 dB bandwidth of 20 GHz. The dynamic range was found to be 20 dB. A monolithic integrated optoelectronic receiver with a bandwidth of more than 20 GHz and a transimpedance gain of 300 Ω was reported (Westergren *et al* 1996). The possibility of producing a compact component with a single power supply voltage for the conversion of modulated light into electrical signals with a bandwidth that may be large enough for 40 Gb/s was also demonstrated. Sano *et al* (Sano *et al* 1996) fabricated an InP/InGaAs double heterostructure bipolar transistor-based *p-i-n*/HBT OEIC receiver operating at 20 Gb/s with a sensitivity of –9.5 dBm. The OEIC receiver exhibited a gain of 54.8 dBΩ. Despite the poor sensitivity, this receiver was envisaged to operate up to 40 Gb/s. The

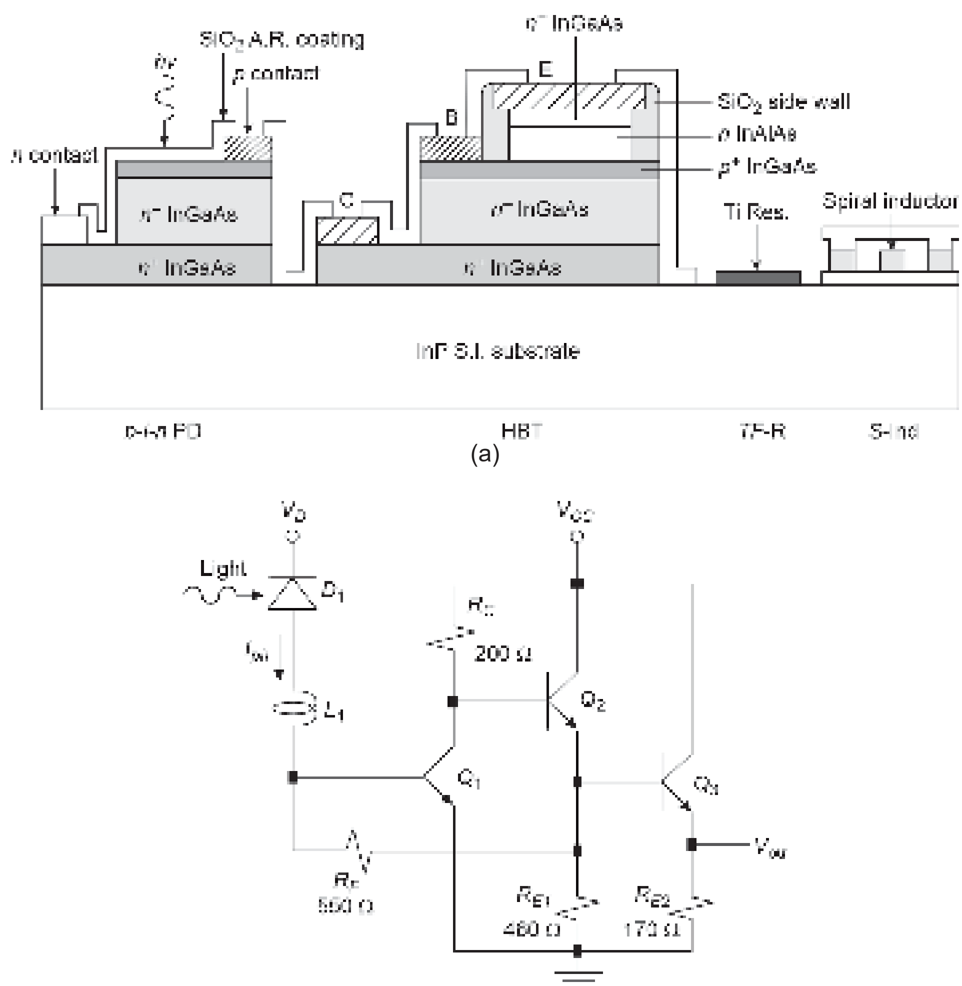

Fig. 8.19 *p-i-n*/HBT OEIC receiver front-end: (a) Cross-sectional schematic after fabrication (b) Circuit schematic in the transimpedance configuration

speed and sensitivity limit of *p-i-n*/HBT based photoreceivers were reported by Kim *et al* (Kim *et al* 1997).

8.5.3 Advanced Receivers

Photoreceivers based on preamplifiers other than those described above have also been reported. Silicon-based integrated NMOS photoreceiver involving *p-i-n* photodetector and MOSFET preamplifier has been reported (Garret *et al* 1996). This receiver is reported to exhibit a bandwidth of 30 MHz. The speed of the photoreceiver was measured with the help of an eye diagram. This photoreceiver showed an eye diagram at 400 Mb/s. A SiGe/Si-based monolithically integrated eight-channel photoreceiver array has been demonstrated and tested by Qasaimeh *et al* (Qasaimeh *et al* 2000). The photoreceiver consisting of a p^+-n-n^+ photodiode and a transimpedance preamplifier based on three stages SiGe/Si HBTs with a feedback resistance of 500 Ω and a 50 Ω matching resistance and other passive matching elements was reported to be grown using molecular beam epitaxy (MBE). The photoreceiver exhibited a transimpedance gain of 43 dBΩ and a bandwidth of 5.5 GHz.

Chandrasekhar *et al* (Chandrasekhar *et al* 1993) suggested and demonstrated HPT/HBT receivers. A heterojunction phototransistor (HPT) was used in the front-end of the photoreceiver and HBT as the preamplifier. Later on, Kamitsuna presented monolithically integrated receivers based on HPT. Two types of receivers were demonstrated. The first receiver consisted of only an HPT while the second receiver consisted of an HPT followed by an HBT preamplifier (Kamitsuna, 1995). The HPT-based receiver consisted of an inductor and series resistor. The schematic circuit diagram is shown in Fig. 8.20. This receiver exhibited an ultra-broadband operation with 3 dB bandwidth from 0.43–12.1 GHz and over 11 dB gain as compared to a photodiode with identical quantum efficiency. The HPT/HBT receiver had an ultra-wideband operation from 8.5–20.5 GHz with over 20 dB gain. This HPT/HBT photoreceiver exhibited a bandwidth of 12 GHz. A detailed review of the OEIC receiver front-end can be found in the literature (Rajamani, 1999).

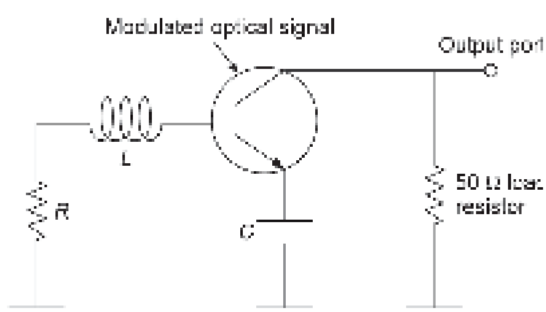

Fig. 8.20 Schematic circuit diagram of a single HPT-based photoreceiver

A novel OEIC receiver based on a single MESFET working as a photodetector-cum preamplifier was proposed and modeled by Chakrabarti *et al* (Chakrabarti *et al* 1999). The schematic circuit diagram of the proposed receiver is shown in Fig. 8.21. The sensitivity of the OEIC receiver designed in the transimpedance mode was estimated for an InGaAs-based MESFET for operation in a 1.6 μm wavelength region. The proposed OEIC receiver was reported to exhibit a wider bandwidth, higher sensitivity, and wider dynamic range at a speed of 20 Gb/s as compared to the OEIC receiver based on conventional photodetectors. The estimated values of sensitivity of –48 dBm and –30 dBm at operating bit rates of 10 Gb/s and 20 Gb/s,

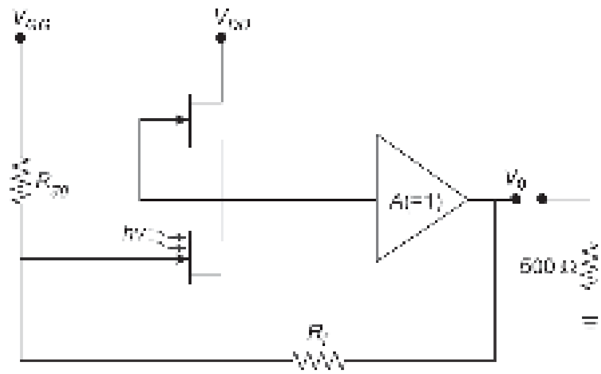

Fig. 8.21 Proposed OEIC receiver using a single MESFET to work as a photodetector-cum-preamplifier (after Chakrabarti *et al* 1999)

respectively, were reported for this receiver. A transimpedance gain over 50 dBΩ was reported to be estimated by adjusting the feedback resistance. A rigorous noise analysis of the OEIC photoreceiver was later reported by Chakrabarti *et al* (Chakrabarti *et al* 2004). An HPT/HBT optical receiver front-end based one InP/InGaAs was also proposed to design a novel OEIC photoreceiver front-end for operation at 1.55 μm (Chakrabarti *et al* 2005). The schematic circuit diagram of the OEIC receiver front-end is shown in Fig. 8.22. The results of the analysis revealed that the receiver has a high transimpedance gain over 60 dBΩ, a large bandwidth of nearly 30 GHz, and a reasonably high sensitivity of the order of –34.8 dBm at a BER of 10^{-9}.

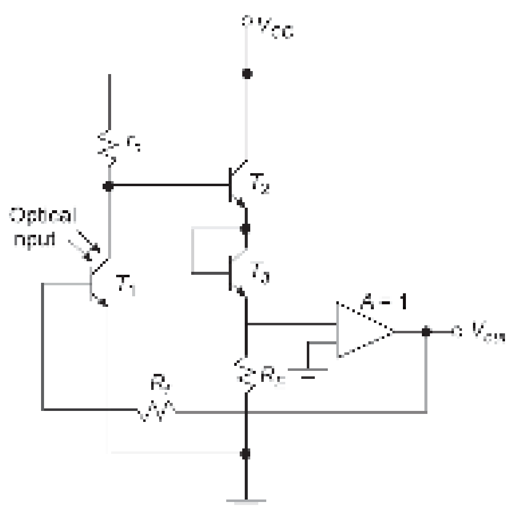

Fig. 8.22 Schematic circuit diagram of a HPT/HBT photoreceiver front-end (after Chakrabarti *et al* 2005)

The performance ratings of various photoreceiver configurations proposed over the past decades discussed in the foregoing section are required to be compared and contrasted to meet the specific requirements. Figure 8.23 depicts a quantitative comparison of the sensitivity values of different photoreceiver configurations with the variation of operating bit rate at a bit-error rate of 10^{-9} (Chakrabarti *et al* 2005). From the figure, it can be seen that a single HBT-based photoreceiver has a higher sensitivity as compared to other contemporary photoreceivers. It can be further seen that the APD/MESFET-based photoreceiver has a better sensitivity as compared to the single

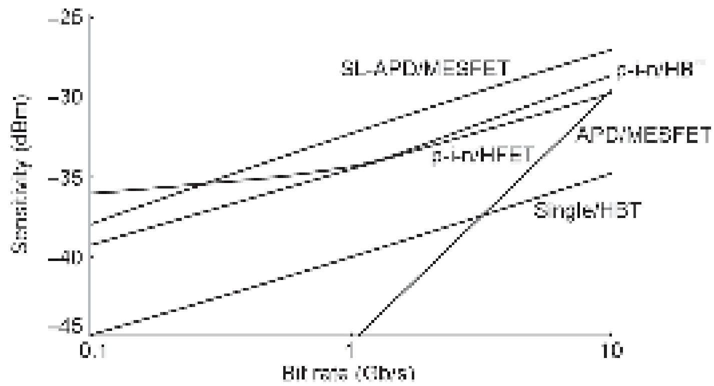

Fig. 8.23 Variation of sensitivity with bit rate for different receiver front-end configurations (after Chakrabarti *et al* 2005)

HBT-based receiver at 1 Gb/s. However, the sensitivity of the APD/MESFET-based receiver degrades very fast beyond 1 Gb/s. This is attributed to the inherent drawback of APD-based receivers in which the excess noise increases rapidly at a higher bit rate of operation (>1 Gb/s). A comparison of the sensitivity values at a BER of 10^{-9} for different photoreceiver configurations is listed in Table 8.1 for operating bit rates of 1 Gb/s and 20 Gb/s.

Table 8.1 Comparison of sensitivity values for different receiver configurations

Type of photoreceiver	*Sensitivity (dBm) at*		*References*
	1 Gb/s	*10 Gb/s*	
APD/MESFET	–45.41	–29.58	O'Reilly, 1988
SL-APD/MESFET	–32.25	–27.00	Rajamani *et al* 2003
p-i-n/HFET	–34.31	–29.76	Das *et al* 1995
p-i-n/HBT	–34.48	–28.62	Das *et al* 1995
Single MESFET	–	–44.00	Chakrabarti *et al* 1999
Single HBT	–40.05	–34.77	Chakrabarti *et al* 2005

Further improvement in the photoreceiver sensitivity at a given bit rate of operation can be achieved by using an in-built semiconductor laser amplifier (SLA)-based preamplifier (Marshall *et al* 1987). A schematic of such an optical receiver is illustrated in Fig. 8.24.

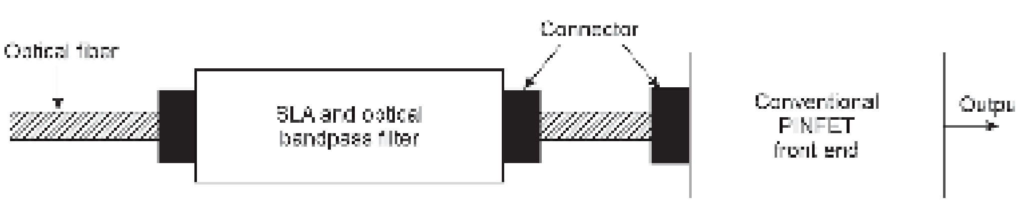

Fig. 8.24 Block diagram of an SLA-based optical receiver

It is also possible to use a fiber amplifier (FA), such as an erbium-doped fiber amplifier (EDFA) in place of an SLA. An improvement in the sensitivity of the photoreceiver of the order of 10 dB can be achieved through the use of an optical amplifier. Further improvement in the sensitivity by increasing the gain of the optical amplifier is not feasible because of the increased noise associatevd with the optical amplifier at a very high value of optical gain.

Summary

- The chapter discusses optical receivers for IM/DD systems based on photodetectors discussed in the previous section.
- For a digital optical communication receiver, the effect of noise is measured in terms of Bit Error Rate (BER) which is related to *S/N* ratio.
- The noise performance of the receiver is decided by the noise generated by the front-end of the receiver comprising the photodetector and the pre-amplifier.
- The front-end of the receiver is followed by an equalizer and decision-making circuit with filter and signal reconstructor.
- The major noise components of an optical receiver include quantum noise associated with the photodetector, thermal noise and shot-noise associated with the photodetector–preamplifier combination and the excess noise introduced by the APD in case the photodetector used is an APD.
- The excess noise factor of an APD is given by

  $$F(M) = kM + \left(2 - \frac{1}{M}\right)(1-k)$$

  where, $k = \beta/\alpha$.
- The front-end of an optical receiver comes in three standard configurations, e.g. low impedance (LZ), high impedance (HZ) and Transimpedance (TZ) forms.
- LZ configuration has a large bandwidth but poor sensitivity. On the other hand, HZ configuration offers poor bandwidth but high sensitivity. TZ configuration is a compromise between the two extremes and is most commonly used.
- The bandwidth of the transimpedance (TZ) front-end given by,

  $$B = \frac{G}{2\pi R_T C_T}$$

  where, G is the open-loop gain, R_T and C_T are the values of the effective resistance and capacitance of the receiver front-end.
- Receiver front-ends can be designed with various combinations of photodetectors and active components for amplifications. PINFET comprising a pin photodetector and an FET amplifier is a commonly used front-end. Others include pin/MESFET, pin/HEMT, pin/HBT, pin/MISFET, APD/MESFET, APD/HEMT, MSM/HEMT, MSM/MESFET, etc.
- Front-end of an optical receiver can be obtained in a monolithic form. The entire receiver can be obtained in the hybrid or monolithic form.
- The sensitivity of an optical receiver can be improved by making use of a semiconductor laser amplifier at the front-end prior to photodetection.

Problems

8.1 A fiber optic link operating at 900 nm is required to maintain a bit-error-rate of 10^{-10}. Determine the theoretical quantum limit of the receiver system in terms of the quantum efficiency of the photodetector and the energy of the incident photon. Also, determine the minimum optical power required to maintain the above bit-error-rate assuming that the system uses a binary signaling operating at 250 Mbps and the quantum efficiency of the photodetector is 100%.

8.2 A low-impedance (LZ) optical receiver front-end uses a photodetector bias resistance of 100 Ω which is matched to the following stage amplifier with the same value of input resistance. The net capacitance of the front-end including the capacitance of the photodetector is

10 pF. Estimate the values of the bandwidth and the mean-square value of the thermal noise component of the receiver front-end per unit bandwidth without equalization at 300 K.

8.3 A high-impedance (HZ) optical receiver front-end uses a photodetector bias resistance of 3 MΩ which is matched to the following stage amplifier with the same value of input resistance. The input capacitance of the photodetector is 5 pF which is the same as the input capacitance of the amplifier. Estimate the values of the bandwidth and the mean-square value of the thermal noise component per unit bandwidth of the receiver front-end at 300 K.

8.4 A transimpedance (TZ) optical receiver front-end uses a photodetector bias resistance of 5 MΩ which is matched to the following stage amplifier with the same value of input resistance. The net capacitance of the front-end including the photodetector is 5 pF. The feedback resistance has a value of 22 kΩ. The open-loop gain of the amplifier is 500. Estimate the values of the bandwidth and the mean-square value of the thermal noise component per unit bandwidth introduced by the feedback resistance at 300 K.

8.5 Estimate the bandwidth of the front-end described in Problem 8.4 in the absence of the feedback resistance. Compare and contrast the values in the two cases (with feedback and without feedback).

8.6 Estimate the magnitudes of the transimpedance gain of the TZ front-end described in Problem 8.4 at 50 kHz, 1 MHz and 100 MHz. Plot the variation of the magnitude of transimpedance gain of the TZ front-end with frequency.

8.7 Repeat Problem 8.4 assuming the feedback resistance to be R_f = 2.5 MΩ.

8.8 Derive the expression for the closed-loop transfer function of the TZ amplifier shown in Fig. P8.8 considering the effect of the stray feedback capacitance C_f-associated with the feedback resistance as indicated in the figure.

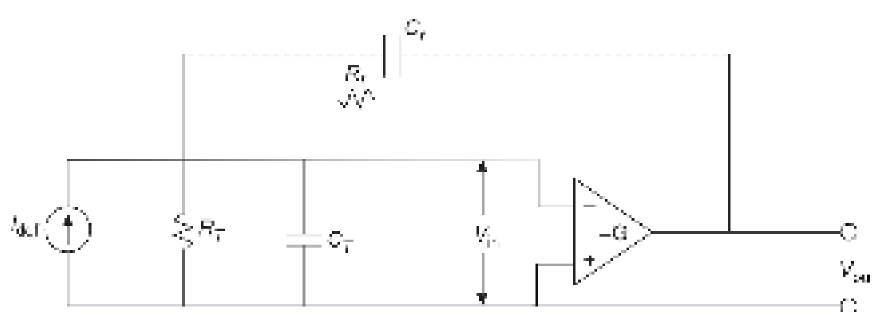

Fig. P8.8

8.9 What is the basic difference between monolithic and hybrid OEIC? Discuss their merits and demerits.

8.10 What is a PINFET? What are the major noise components arising out of the receiver front-end?

8.11 What are the relative merits and demerits of *p-i-n* and APD-based photoreceivers?

8.12 List the III-V materials that are suitable for making PINFET operating in the following wavelength regions:

(i) 0.85 μm (ii) 1.3 μm and (iii) 1.55 μm

8.13 Discuss the trade-off between the transimpedance gain and bandwidth of a *p-i-n*/HFET optical receiver.

8.14 How does an SLA improve the sensitivity of an optical receiver?

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Analog and Digital Optical Link Design

CHAPTER 9

A typical optical fiber link comprises an optical transmitter, an optical receiver, and optical fiber as the channel along with optical repeaters at intermediate points along the link. A modern optical link, however, may contain other optical components to perform the desired operations. A point-to-point optical link generally connects two points between which transmission and reception of information is made possible by using information-carrying optical signals propagating through the optical fibers. The purpose of a generalized optical link is to accomplish reliable transmission and reception of information by using information-bearing light (analog or digital) between any number of points connected by the link.

9.1 A TYPICAL OPTICAL FIBER LINK

A typical optical fiber link between two points is shown in Fig. 9.1 with the help of a block diagram. Like electrical communication, optical communication can either be in analog or digital form. Depending on the nature of optical communication, the actual components used in the optical link may vary. An optical transmitter consists of an encoder (for digital optical communication) or a signal-shaping circuit, such as an electrical filter (or analog optical communication) followed by a driver circuit which is generally used to modulate the optical source. The modulated light is launched into the optical fiber which acts as the transmission channel. The information-bearing light propagates along the fiber and reaches the receiver-end where it is first converted into an electrical signal.

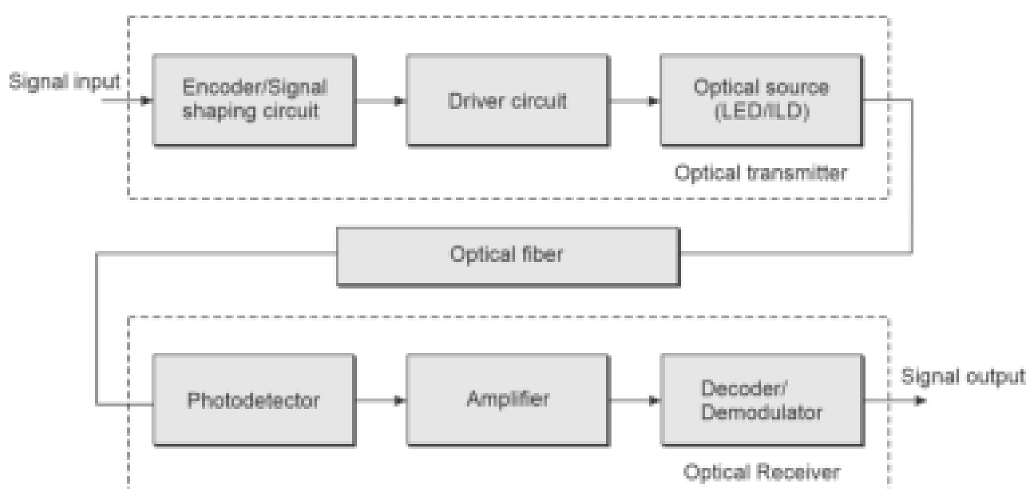

Fig. 9.1 Schematic block diagram of an optical communication system showing the transmitter, receiver and the channel. The arrows indicate electrical path and the dark solid lines indicate optical path

The transformed electrical signal is subsequently amplified using a low-noise amplifier. The amplified signal is finally demodulated or decoded to obtain the original message signal. It may be pointed out here that a majority of commercial optical fiber communication systems make use of some form of intensity modulation (IM) of light to carry the information from the source to the destination. At the receiver-end, a simple direct detection (DD) of the intensity-modulated light is carried out by the photodetector to convert the intensity variation into the corresponding electrical signal. This form of optical communication system is usually referred to as IM/DD system which finds extensive applications as compared to the more sophisticated system known as coherent optical communication system. It may be further stressed that in the IM/DD system, the low-frequency information signal is not used directly to vary the intensity of the optical source. In actual practice, the information signal is used to modulate a high-frequency microwave subcarrier using conventional electrical modulation techniques. This is generally referred to as subcarrier modulation. The resultant signal is subsequently used to modulate the intensity of the optical source. As a result, the output of the photodetector does not correspond to the information signal but the subcarrier-modulated information signal. A conventional electrical demodulation is required in such cases to obtain the information signal. The topics discussed in this chapter are relevant to IM/DD-based optical communication systems. The essence of coherent optical communication is discussed in Chapter 11. In coherent optical communication, the optical signal is used as a carrier in the true sense to the effect that the information signal is used in this case to vary the amplitude, frequency, phase, or polarization of the optical carrier.

On the other hand, in IM/DD systems, only the intensity or optical power output of the light source is made to vary following the information signal.

This chapter deals with the details of the techniques and systems which are involved in the optical transmitter and the receiver units. The details of the various components that are used in the optical transmitter and receiver have already been discussed in previous sections. The characteristics of the transmission channel (optical fiber) have also been discussed in previous chapters. In this chapter, we shall focus our attention on the design of the overall optical link and make use of our previous understanding of the functions of different components. We begin our discussion with the optical transmitter unit.

9.1.1 Optical Transmitters

The information signal to be communicated is processed by this unit. The information signal can be either in the analog or in the digital form. The signal is processed initially before using it to drive an optical source to make it suitable for transmission using an optical fiber cable. In actual practice, only special types of optical sources can be used for optical fiber communication systems. The requirements of such sources have already been studied in Chapter 5. It has been understood that both light-emitting diode (LED) and semiconductor laser diode, also known as injection laser diode (ILD) are suitable for use as optical sources at the transmitter-end. The major requirements of these sources include size compatibility with optical fibers, linearity of the light power output versus drive current characteristics, speed of response or modulation capability, spectral response, thermal stability, etc. It has already been pointed out that light-emitting diodes generally exhibit low output power, smaller bandwidth, and a relatively broad spectral width as compared to their laser counterpart. Nevertheless, light-emitting diodes do find applications in short-distance optical fiber communication systems for operating bit rates up to 200 Mbps. In this chapter, we discuss different methods of direct modulation of

LEDs and ILDs using different drive circuits suitable for analog and digital transmissions. It may be pointed out here that in optical communication systems, the major emphasis has been on digital transmission. However, analog optical transmission remains attractive for video transmission in wide-band distribution and CATV networks.

9.1.2 Analog LED Drive Circuits

The LED drive circuit design depends on the type of optical communication, e.g. analog or digital. For analog optical communication, the linearity of the LED power versus drive current is an important consideration. For practical diodes, the characteristics are far from an ideal linear variation. As a result, the LED introduces undesirable distortion in analog optical communication systems. LED drive circuits generally call for suitable compensation circuits for combating the non-linearity. A large number of LED drive circuits have been proposed and studied for application in analog optical communication systems (Shumate *et al* 1982). The simplest DC LED drive circuit consists of a current- limiting resistor connected in series with a voltage source and the LED in a manner shown in Fig. 9.2(a). For driving the LED by an AC source the circuit can be modified as shown in Fig. 9.2(b). The capacitor used in series with the resistance is the coupling capacitor while the diode connected in parallel with the LED holds the reverse voltages across the LED to a low value.

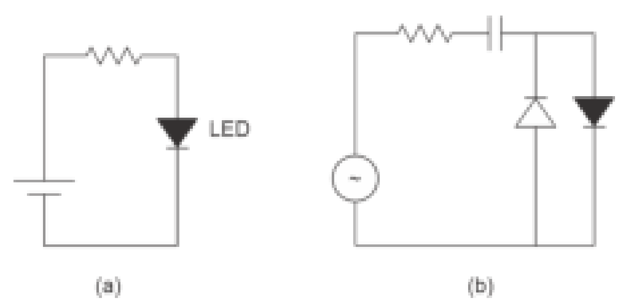

Fig. 9.2 LED drive circuits (a) DC source (b) AC source

For high-frequency analog optical communication applications the signal is generally impressed on the continuous wave (CW) light output from a light source. This can be achieved with the help of internal or direct modulation by varying the drive current through a light-emitting diode following the modulating signal. A simple circuit arrangement to achieve this is illustrated in Fig. 9.3. In this circuit, the modulating signal is applied to the base–emitter circuit of an *n-p-n* transistor with the base–emitter junction already forward-biased with the help of a DC source. The modulating signal causes a change in the base current to cause a variation in the quiescent collector current thereby resulting in a variation of the drive current placed on the collector path along with the current-limiting

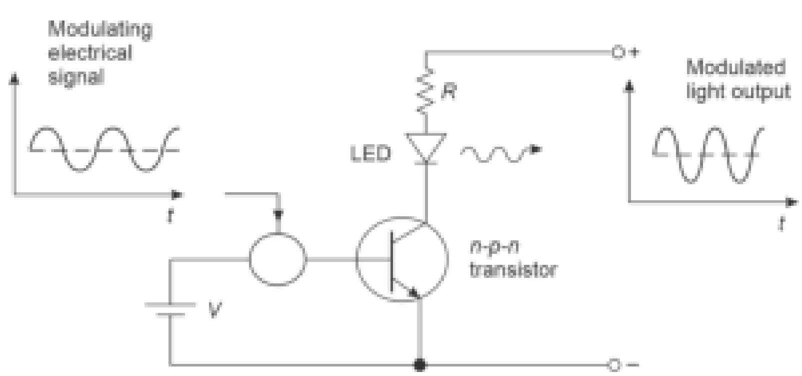

Fig. 9.3 Drive circuit for internal modulation of LED source by modulating signal (P Bhattacharya, Optoelectronic Devices)

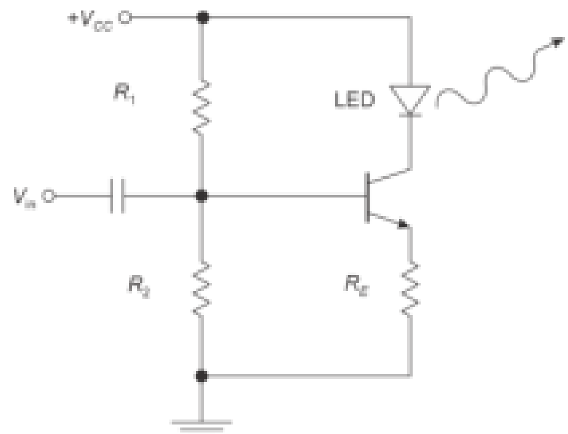

Fig. 9.4 BJT-based transconductance drive circuit for analog modulation of LED source

series resistance. As a result of variation in the forward drive current through the LED, the output light from the LED becomes intensity-modulated. For analog applications, LED as well as the transistor must operate in their linear regions only (Bhattacharya, 2002). Under this condition, the intensity of the light emitted by the LED varies exactly in the same fashion as the modulating signal. Another simple LED circuit based on direct modulation using self-biased bipolar junction transistor (BJT) is shown in Fig. 9.4. In the common-emitter transconductance amplifier based on an *n-p-n* bipolar transistor, the applied modulating voltage signal is converted into collector current which alters the driving forward current through the LED placed in the collector path. The amplifier is designed to work in Class A mode. The circuit shown in Fig. 9.4 exhibits a large impedance of the source. To reduce the large impedance of the source a Darlington transistor pair can be used to modify the above circuit. The modified circuit shown in Fig. 9.5 can be used to drive high-radiance LED up to a frequency of the order of 70 MHz.

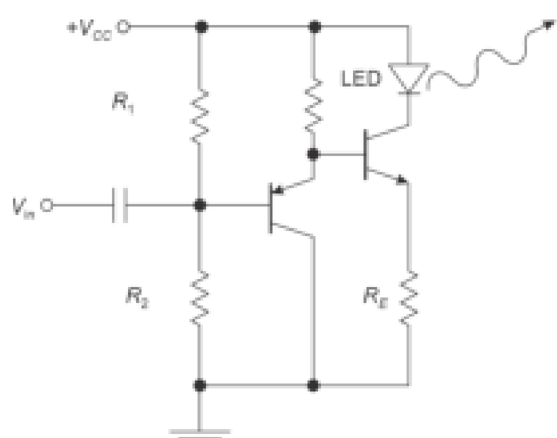

Fig. 9.5 LED drive circuit using common-emitter BJT in Darlington pair

LED driver circuits can also be achieved with the help of a differential amplifier operated in the linear region. A differential amplifier-based LED modulator is shown in Fig. 9.6. The differential amplifier consists of the bipolar transistors T_1 and T_2. The drive current operating point of the LED is set with the help of reference voltage applied to the base of the transistor T_2 while the modulating voltage is applied at the base terminal of the other

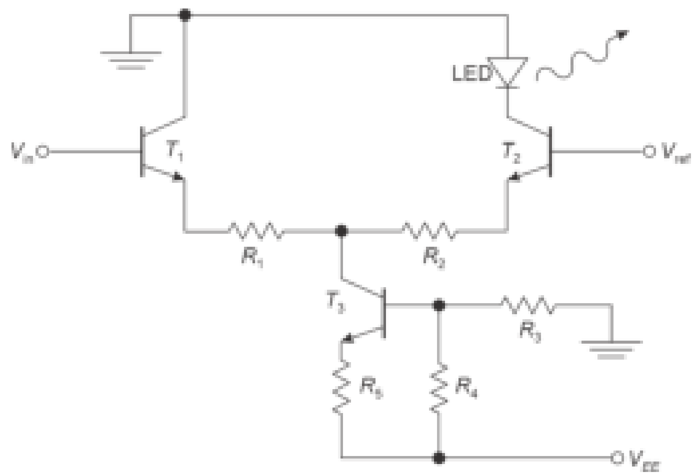

Fig. 9.6 Differential amplifier-based LED modulator

transistor T_1 of the differential amplifier. The transistor T_3 feeds current to the differential stage thereby controlling the maximum current through the LED (Senior, 2008).

A practical LED circuit based on a high-frequency operational amplifier from Linear Technology (LT1363) with a again-bandwidth product of 70 MHz is shown in Fig. 9.7. The circuit can be operated safely up to a bandwidth of 50 MHz. The circuit has a transconductance modulator configuration in which the input voltage is converted into output current. The DC biasing of the output is ensured by applying a fixed voltage at the non-inverting input of the operational amplifier while the inverting input of the operational amplifier is feedbacked with the help of the voltage drop across the resistance R_i at the output. The input modulating signal is capacitively coupled to the non-inverting put with the help of an RC network with a low-frequency modulation cutoff of a few Hz. Since the dynamic resistance of the LED is very small, the output resistance R_i can be adjusted to have a relatively high value so that the feedback fraction is nearly unity. Therefore, the expected modulation bandwidth will be on the order of 70 MHz (Gallant, 2008).

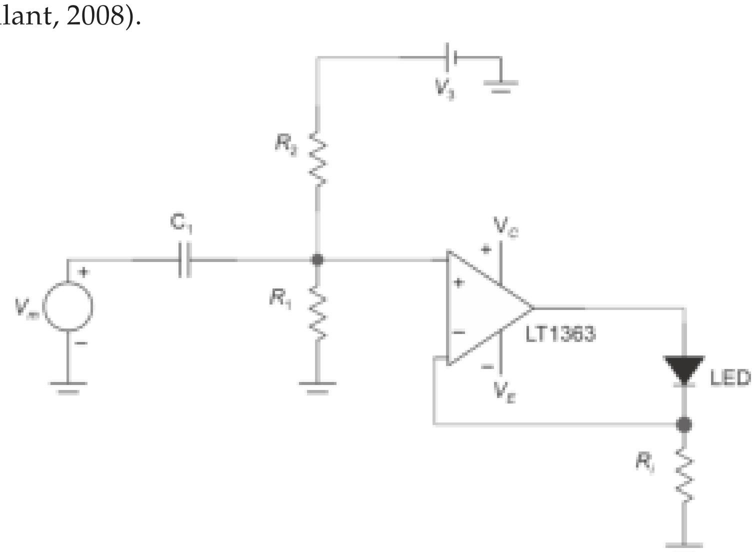

Fig. 9.7 An operational amplifier-based LED modulator (Gallant M, 2008)

Analog optical modulation requires a good linearity of the optical source with respect to light output power versus drive current characteristics. Any form of non-linearity of the source characteristics may cause a severe distortion of the transmitted signal. The distortion is manifested in the form of either amplitude distortion or phase distortion. For a single-channel transmission, the distortion may be acceptable up to a certain limit but for multiplexed signals in a multi-channel transmission, this type of non-linearity may lead to intermodulation distortion. Therefore, it is necessary to incorporate some form of linearization circuit in the modulators discussed earlier so that the distortion can be minimized. Several methods have been proposed in the past for improving the non-linearity of LEDs (Asatani[2] *et al* 1978). These methods include optical negative feedback (Ueno *et al* 1974), phase-shift modulation (Straus[1] *et al* 1977), feedforward (Straus[2] *et al* 1977), quasi-feedforward (Straus[2] *et al* 1977), and direct compensation by pre-distortion (Asatani[1] *et al* 1977). In the negative feedback approach, a photodetector is used to generate an electrical signal that monitors the optical source via a feedback circuit. A schematic of the approach is illustrated in Fig. 9.8 (Senior, 2008). The amount of reduction of dispersion in this scheme is decided by the linearities of the components and the closed loop gain. This technique is not very suitable for wide-band transmission applications because of unacceptable phase shifts caused by the feedback loop. It has been demonstrated that the improvement in third harmonic distortion arising out of intermodulation of the fundamental frequency and the second harmonic is extremely poor in this method. For a single-channel color television transmission the improvement in the second and third harmonic distortion is reported to be 12 dB and 4 dB, respectively (Ueno *et al* 1974).

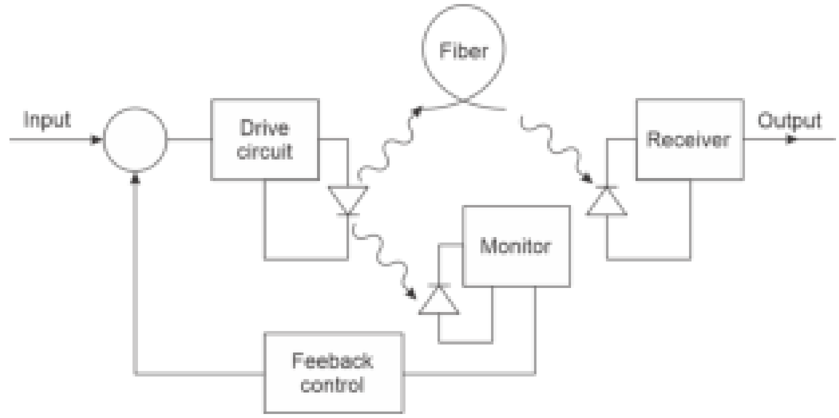

Fig. 9.8 Block diagram of a feedback control system for harmonic distortion compensation in LED transmitter

The second- and third-order harmonic distortion can be greatly reduced in the phase-shift modulation method. To compensate for second and third-order distortion, this technique involves the use of two LEDs with identical properties. The schematic arrangement of the phase-shift compensation for harmonic distortion is shown in Fig. 9.9. The input signal is divided into two components with a relative phase-shift between them. The two components are used to modulate two different LED sources of identical characteristics. The combined signal causes cancellation of the second and third harmonics. However, both second and third harmonic components cannot be reduced simultaneously by this simple arrangement. For simultaneous improvement of second and third harmonic distortion by phase-shift modulation one needs to use four LEDs with identical characteristics. In the feedforward method (Straus[2] *et al* 1977), the main optical signal is generated by an LED while the other generates only the error signal to linearize the whole optical output.

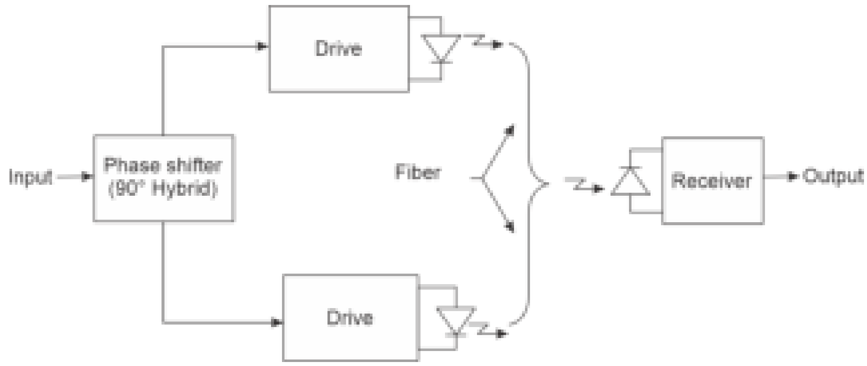

Fig. 9.9 Schematic arrangement of phase-shift method of harmonic distortion compensation (Straus[1] *et al* 1977)

This method makes use of a stable optical coupler. In quasi-feedforward, the incoming signal modulates two matched LEDs (1 and 2) to generate an equal amount of distortion. The distortion from LED-1 is used to create a compensating signal equal in amplitude and opposite in sign to the distortion generated by LED-2. By accurate error leveling and by proper control of the delays, it is possible to achieve distortion cancellation over a wide range of modulation levels. The schematic block diagram arrangements for the feedforward and quasi-feedforward methods of harmonic distortion compensation of LED modulators are shown in Fig. 9.10. An improvement of second and third harmonic

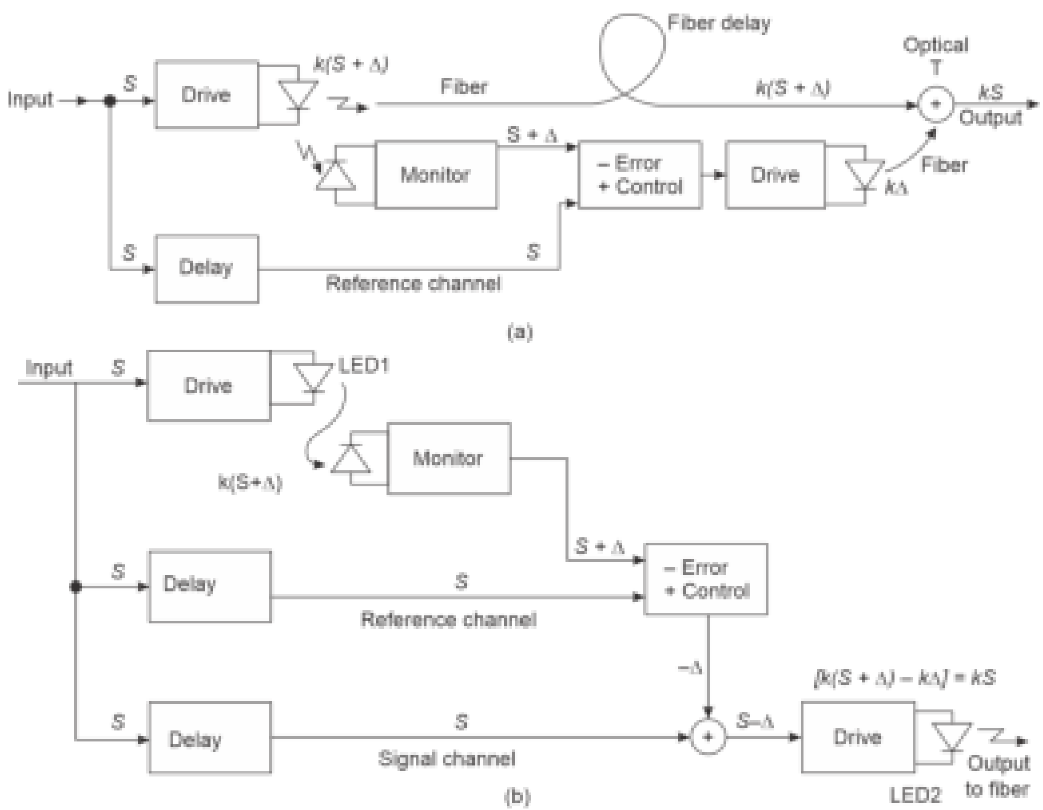

Fig. 9.10 Schematic diagram (a) Feedforward compensation scheme (b) Quasi-feedforward compensation scheme (Straus[2] *et al*)

distortion to the tune of 33 dB and 17 dB respectively are reported to be achieved by the quasi-feedforward compensation technique (Straus[2] *et al* 1977).

9.1.3 Digital LED Drive Circuits

Because of a relatively smaller bandwidth available with LEDs as compared to that of their ILD counterparts, the former devices find limited applications in practical optical communication systems. Nevertheless, LEDs do find applications for low-bit-rate operation because of the simple drive circuit requirement and low cost of these devices. In digital optical communication systems, LEDs used as optical sources need to switch "ON" and "OFF" at the desired speed depending on the driving pulse. LED digital drive circuits can be realized by using a *p-n-p* or an *n-p-n* transistor as a switch as shown in Fig. 9.11. The current through the LED can be controlled with the help of a current-limiting resistor.

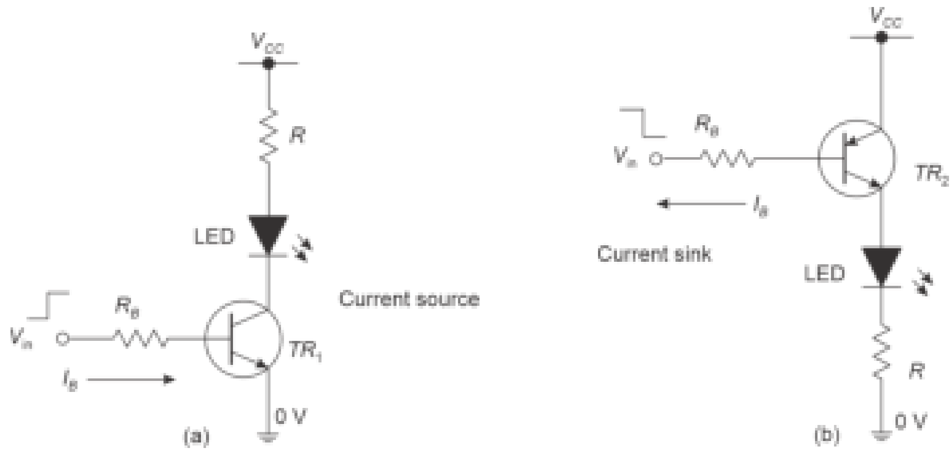

Fig. 9.11 LED driver circuits for digital switching (a) Current sourcing (b) Current sinking

In Fig. 9.11(a), when the input voltage at the base is "HIGH", both the base–emitter junction and the collector–base junction are forward-biased and the transistor is switched to saturation state resulting in a current gain in the collector circuit, and a low voltage drop across the collector–emitter junction. As a result, the drive current through the LED becomes large enough to switch it "ON". When the input voltage at the base is "LOW" the transistor of both the junctions are reversed-biased and the transistor is switched "OFF" and the LED is turned "OFF". Likewise in Fig. 9.11(b), the LED is turned "ON" or "OFF" depending on whether the base input is "LOW" or "HIGH". The simple common-emitter circuit shown in Fig. 9.11(a) has a limited switching speed due to space charge and diffusion capacitance. There exists a trade-off between the current gain of the transistor and the bandwidth. This can be compensated to some extent by using pre-emphasis which overdrives the base current during the switch "ON" period. The pre-emphasis can be achieved by using a speed-up capacitor C in parallel with the base resistor (Senior, 2008) as shown in Fig. 9.12. The switching speed of the digital drive circuit can be improved by making use of an emitter follower circuit shown in Fig. 9.12(b).

Switching of LEDs for digital transmission can also be achieved by interfacing the LED drive circuit with common logic families directly. The output stages of both transistor–transistor logic (TTL) and complementary metal-oxide semiconductor (CMOS) logic gates can be interfaced with the LED drive circuits in both current source and sink mode. The output drive current of the normal digital integrated circuit is sufficient to drive the LED

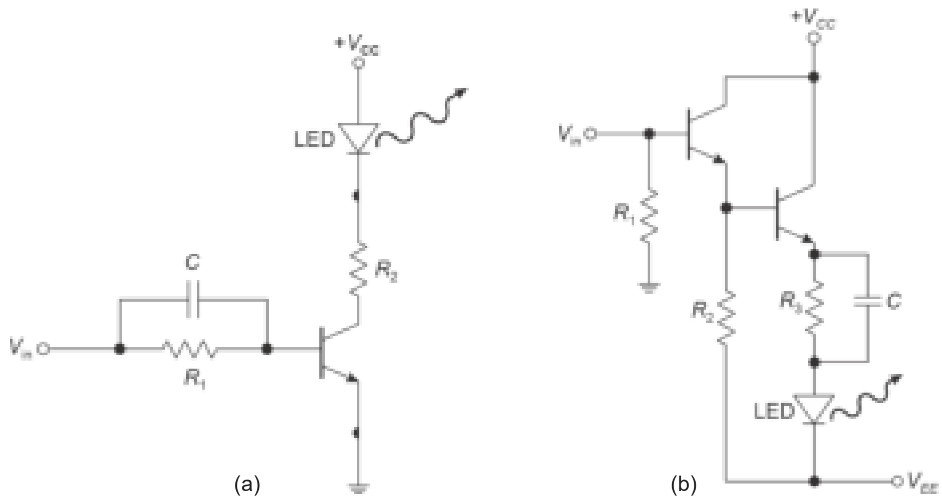

Fig. 9.12 (a) Pre-emphasized LED drive circuit for binary logic (b) Low impedance drive circuit using emitter follower

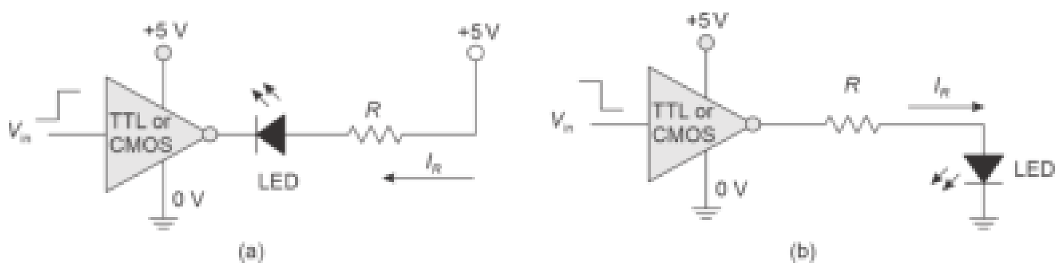

Fig. 9.13 Digital IC-based LED drivers for digital transmission (a) Current sink mode (b) Current source mode

connected at the output. This is illustrated in Fig. 9.13. In Fig. 9.13(a), the LED is "ON" when the output is low and the TTL/CMOS gate works in current sink mode. On the other hand in Fig. 9.13(b), the gate sources the current to turn the LED "ON" when the output voltage is high. Commercial TTL gates from Texas Instruments can also be used for driving LED for digital transmission. Fig. 9.14(a) shows a simple TTL-compatible LED drive circuit employing the 74S140 line driver series from Texas Instruments (Shumate Jr *et al* 1982). A current value of up to 60 mA can be achieved by using 50 Ω series resistance with the

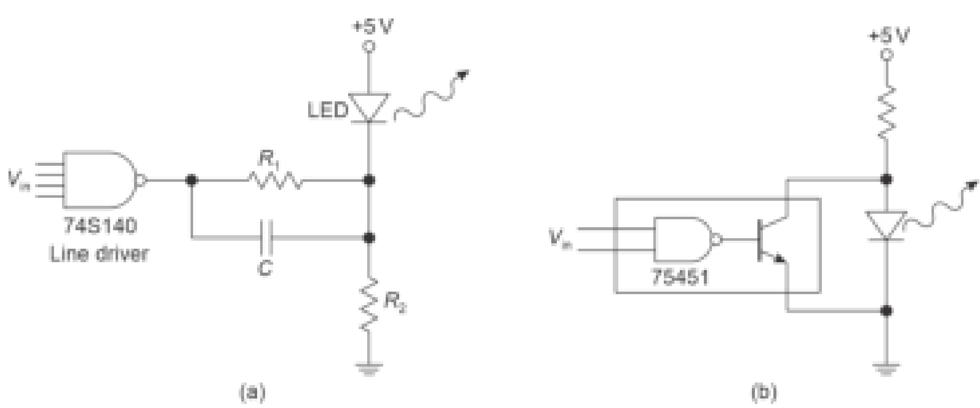

Fig. 9.14 TTL compatible LED drive circuits (a) TI 74S140 line driver (b) Shunt drive circuit

LED (Senior, 2008). The rise times of the order of a few nanoseconds can be achieved with the help of a speed-up capacitor. A TTL shunt drive circuit using a commercial logic circuit is shown in Fig. 9.14(b) (Shumate Jr *et al* 1982).

9.1.4 LASER Drive Circuits

Semiconductor laser diodes can be modulated by the information signal for analog as well as digital transmission systems. Semiconductor lasers are generally preferred over LEDs in long-haul high-speed, especially suited to optical fiber communication because of their small size, high efficiency, and large bandwidth in direct modulation. When a laser diode is switched from zero current to a large current level there is a delay caused by the time taken by the laser diode to reach the threshold for lasing after the application of the current pulse. This delay (of the order of a few nanoseconds) sets an upper limit to the modulation capability of a laser diode to 100 Mbps range. In Chapter 5, it has been demonstrated that the pulse rate can be greatly improved by pre-biasing the diode by a suitable applied voltage to bias the diode just below the threshold requirement for lasing. This approach can significantly improve the digital modulation rates in the range of a few Gbps (Abbott *et al* 1978; Albanese *et al* 1978).

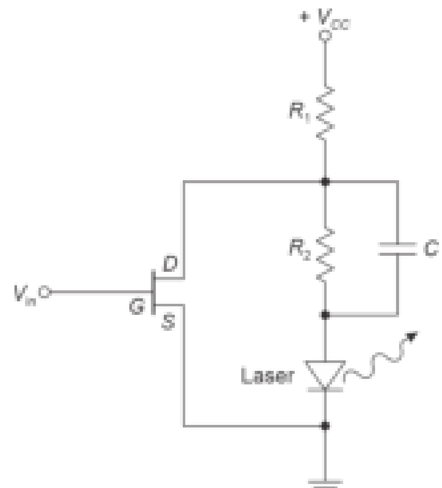

Fig. 9.15 FET-based laser diode drive circuit for digital transmission

The gain-bandwidth product and voltage and current levels of GaAs MESFET are suitable for direct modulation of injection lasers. A simple arrangement for driving a laser diode for a digital transmission system by utilizing a field-effect transistor is shown in Fig. 9.15 (Senior, 2008). The shunt-type drive circuit ensures sufficient voltage in series with the laser diode with the help of the resistance R_2 and the compensating capacitor C such that the field effect transistor (FET) is biased in the active or pinch-off region. When an input voltage is applied across the gate-to-source terminals of the FET, the total current flowing through R_1 is split into two components. A major portion of the current is diverted through the FET and a small amount of the remaining current flows through R_2 turning the laser diode off (Senior, 2008).

The schematic of a direct modulation of a double-heterostructure laser diode with a DC bias voltage and operation in the range of 1 Gbps is shown in Fig. 9.16 (Chown *et al* 1973). The circuit uses a single-stage emitter-coupled modulating circuit to drive the laser diode by allowing the bias as well as the signal currents to be varied. The drive circuit has been reported to be tested for operation around 1 Gbps. Direct modulation of injection laser diodes with high bit rate Pulse-code Modulation (PCM) signals has been reported to be developed using a hybrid integrated laser driver based on bipolar transistors (Gruber *et al* 1978). A GaAs-based DH-ILD has a threshold current of the order of a few hundred milliamperes. As discussed earlier, the laser diode must be biased just below the threshold to achieve a high bit rate. Figure 9.17 shows the circuit diagram of a high-speed laser driver using differential amplifiers. The circuit consists of an input stage and two differential amplifiers connected in parallel. To make the circuit ECL compatible with 50 Ω input impedance, a 50 Ω resistance is connected parallel to the input of the

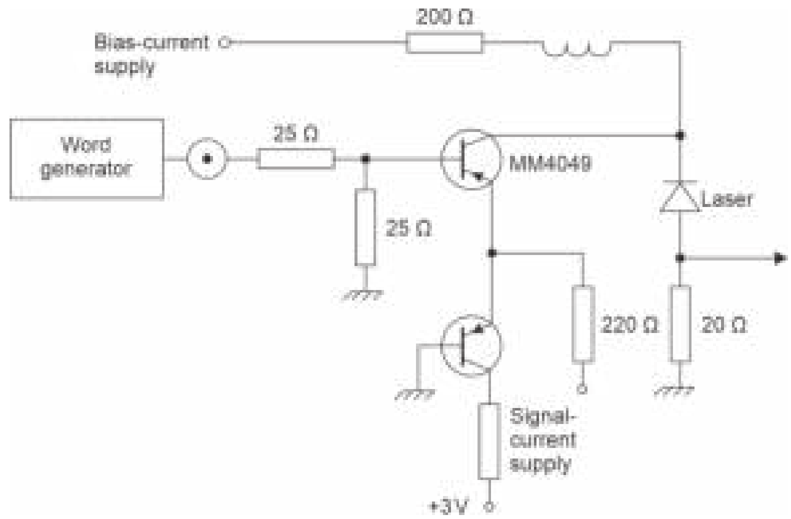

Fig. 9.16 Schematic of a laser drive circuit for operation in the range of 1 Gbps

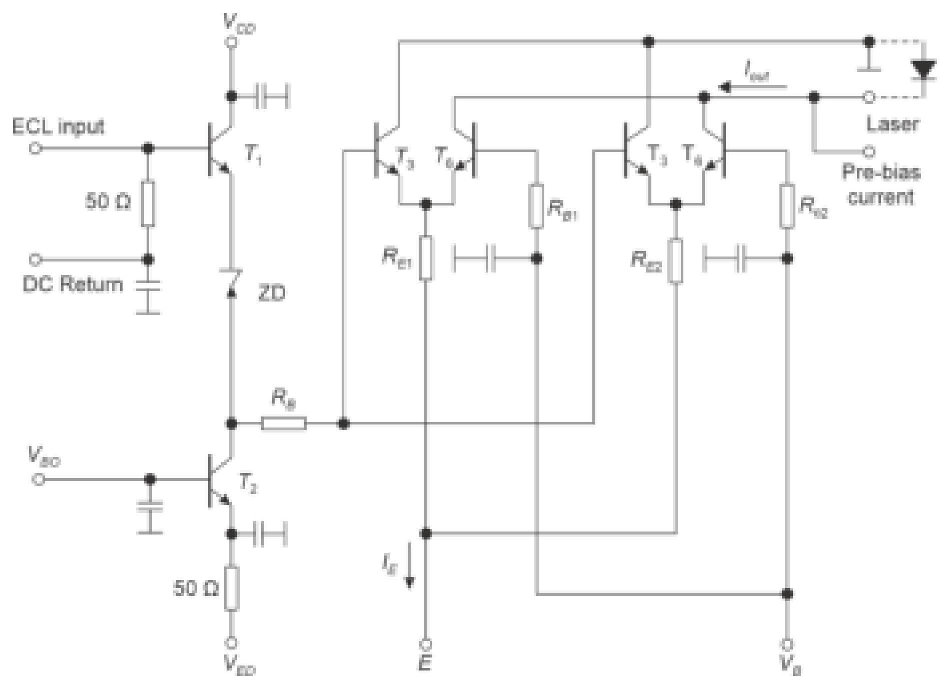

Fig. 9.17 Circuit diagram of an ECL compatible laser driver circuit for high speed digital transmission (Gruber *et al* 1978)

emitter follower T_1. In the circuit, T_2 acts as a current source and zener diode (ZD) is used to shift the signal baseline so that the input can be driven from standard ECL levels. (T_3–T_4) and (T_5–T_6) constitute two parallel differential amplifiers and are used to deliver a sufficient modulation current amplitude. Further, equal distribution of current between the two differential amplifiers is ensured by the emitter resistors R_{E1} and R_{E2}. The input current is controlled by the DC current I_E derived by an optical feedback control circuit necessary to stabilize the laser light output signal. The pre-bias current is applied to the laser source from a separate current source to reduce the delay and thereby improve the bit rate. The drive circuit is reported to be used up to a bit rate of 1 Gb/s by utilizing microwave transistors (Gruber *et al* 1978).

The slope of the output *versus* drive current characteristic of a laser diode tends to vary under the influences of temperature changes arising out of junction heating, long-duration operation, and aging. These parameters tend to change the threshold current of a laser diode. Thus, both the laser threshold current and the slope of the light output power versus drive current characteristic may change. Therefore, it is necessary to stabilize the output of the laser diode against the above changes by suitably adjusting the DC drive current. Further, to keep the modulation amplitude of the laser output signal constant to maintain a constant signal level at the input of the receiver. Moreover, the laser diode must be DC-biased slightly above the threshold. This is because if the bias is too low, there is a possibility of bit pattern-induced modulation distortions (Arnold *et al* 1977). On the other hand, an excessive bias level enhances the shot noise in the receiver and also increases the laser degradation rate (Gruber, 1978). The basic circuit diagram of the laser stabilization circuit that controls the DC bias current and the modulation current amplitude is shown in Fig. 9.18. The basic operation of the circuit involves monitoring the laser diode output by suitable feedback provided by the *p-i-n* photodiode that picks up a small power from the laser output. This can be achieved by coupling the photodiode to the rear facet of the laser diode. The electrical output signal of the photodiode is amplified with the help of a DC amplifier A_1 and subsequently by a broadband AC amplifier A_2. The amplifier A_1 produces output proportional to the temporal mean value of the laser light output power and the AC part of the monitoring signal is peak-detected after amplification by A_2. The positive and negative peaks are detected with the help of Schottky diode-capacitor combinations (SD_1, C_3) and (SD_2, C_4), respectively. The reference voltages provided by voltage divider networks compensate for the temperature dependence of SD_1, SD_2, and the subsequent amplifiers A_3 and A_4. The following stages of differential amplifiers (A_5–A_8) generate the control signals for controlling the modulation amplitude and bias current (Gruber *et al* 1978).

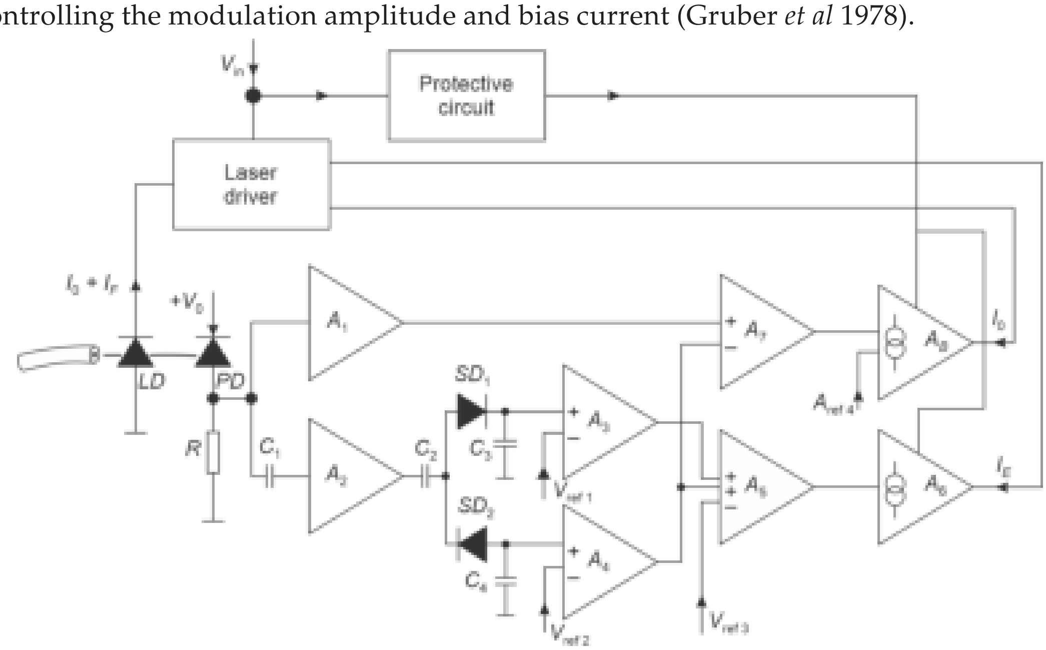

Fig. 9.18 Control circuit diagram for stabilizing the laser source of the drive circuit

9.2 SYSTEM DESIGN

Different optical receiver configurations along with circuit schematics have been extensively discussed in the previous chapter (Chapter 8). The following sections discuss various issues related to the design of a complete optical fiber communication link. The block diagram of a complete optical fiber link is shown in Fig. 9.19. In general, the distance between the terminal equipment is very large for a long-haul communication link. The distance that can be traversed by the optical signal without significant problems in terms of attenuation and dispersion depend on the characteristics of the connecting fibers. As a result, the entire distance cannot be covered in one go, for long-distance communication where the distance between the terminal equipment is longer than the maximum permissible distance. In such cases, it is necessary to insert regenerative repeaters at intermediate points in regular intervals to maintain the quality of the transmitted signal to an acceptable level. A regenerative repeater comprises a receiver at the input port through which the signal is received. The receiver in the repeater converts the optical signal into an electrical signal which is subsequently processed by an associated electronic circuit before finally delivering it to the in-built transmitter of the repeater. The regenerated signal modulates the transmitter and finally emerges from the output port of the repeater. In analog transmission, the function of the regenerative repeaters is primarily to provide amplification and equalization for removal of channel distortion before the signal is retransmitted. In the case of digital optical communication, the information is in the coded form which appears as binary pulses. The intersymbol interference (ISI), if not taken care of properly may lead to wrong interpretation of 1's and 0's at the receiving-end. The regenerative repeater in such cases has to perform amplification and equalization and additionally pulse regeneration.

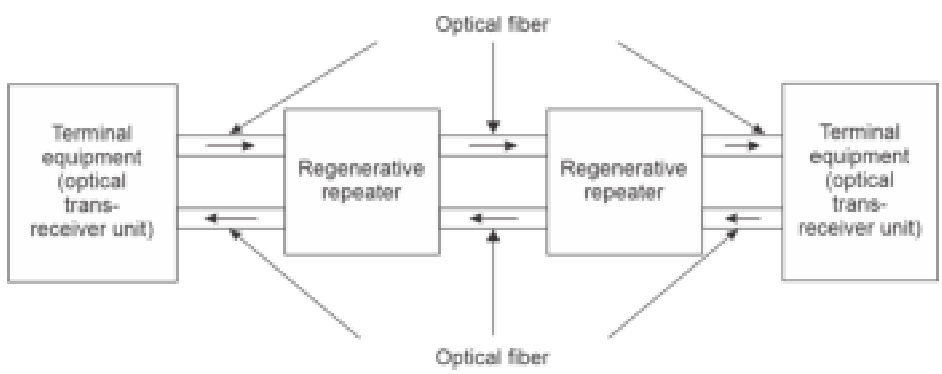

Fig. 9.19 Block diagram of a typical optical fiber communication link

For standard optical fiber link, the separation between the regenerative repeaters is usually 25–30 km. This means that for long-distance optical communication, a large number of intermediate regenerative repeaters are required for the implementation of a successful optical link. However, the repeaters increase the cost of the link substantially and increase the complexity of the design. Further, the repeaters also tend to create bottlenecks and tend to slow down the speed of the link because every time within a repeater the signal is taken to the electrical domain where the speed is limited by the delay caused by the time constants of the electrical circuits. This delay can be avoided if the entire processing in the repeater unit could be done in the optical domain. To cut down on the cost of the repeaters, it is necessary to make attempts to enhance the repeaterless distance between the end equipment by managing attenuation and dispersion properly. In the present generation of optical communication use of high-quality single mode fiber has significantly enhanced the repeater spacing. In a single-mode fiber-based optical link, it is often possible to boost the optical signal in the time domain itself to compensate for attenuation by making use of optical amplifiers (fiber amplifiers or semiconductor laser amplifiers). The block diagram of a typical regenerative repeater for a digital system is shown in Fig. 9.20. It can be seen that the digital data is sent over the fiber in the form of bits (1's and 0's) in the form of the rectangular pulse train of optical power. In this illustration, a bit '1' is represented by a positive voltage pulse over a given bit period while a bit '0' is represented by zero voltage over the bit period. The optical pulse train launched into the fiber gets progressively attenuated and dispersed. The impaired optical pulse train is received at the input port of the repeater. The weak dispersed optical signal is converted into the corresponding electrical signal by the photodetector. The electrical signal so produced is subsequently amplified with the help of a low-noise pre-amplifier. The amplified signal is further boosted in power level with the help of a post-amplifier equipped with an automatic gain control circuit before equalization necessary for compensating the distortion introduced by the non-ideal characteristics of the optical fiber channel. Before deciding the bits (1's and 0's) and reconstructing the pulses it is necessary to extract the accurate timing (clock) information for checking the equalizer output. The timing information is extracted with the help of a phase-locked loop. Extraction of the timing information enables the regenerator circuit to reconstruct the original pulse train under ideal conditions. This is accomplished by setting a threshold (depending on the voltage levels used to denote 1s and 0s and the rms value of the noise) above which a binary 1 is registered and below which a binary 0 is registered. The regeneration circuit makes this decision at the instants of time corresponding to the center of the bit intervals as extracted by the timing circuit. The decision times are generally set at the mid-points between the decision level crossing the pulse train. The reproduced pulse train is sampled at a rate equal to the bit-rate and depending on the sample value, a decision regarding the most probable bit (1 or 0) being transmitted is taken for each sample. The bit symbols are then given an appropriate pulse shape depending on the rule followed at the transmitter end. The regenerated pulse is subsequently used to drive the source driver circuit for retransmission. Under ideal conditions an exact regeneration of the original transmitted pulse train is possible in a digital optical communication link. This is, however, not possible in the case of analog optical transmission. The repeaters of analog optical communication do perform operations, such as filtering, equalization, and amplification but cannot remove the distortion completely to regenerate the exact

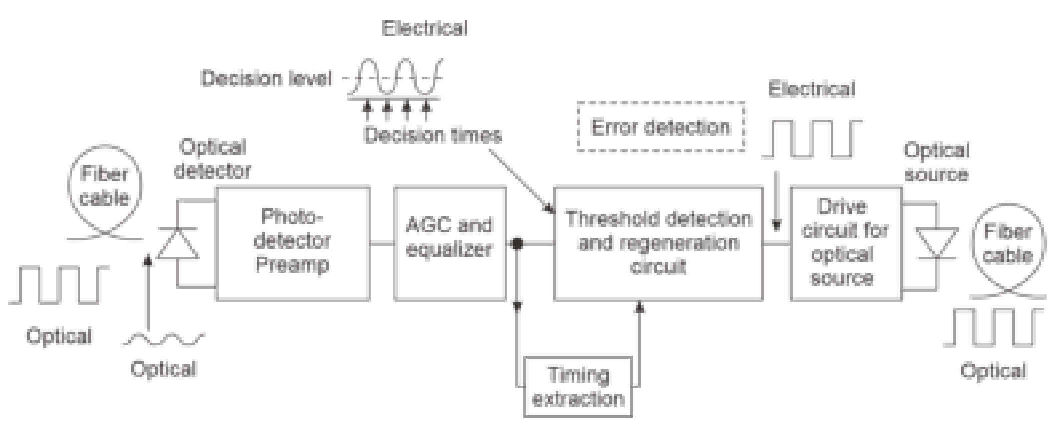

Fig. 9.20 Block diagram of a regenerative repeater exhibiting detection of impaired signal and its regeneration

original signal. In fact, in a long-haul analog optical communication system, the signal degradation in each stage of the repeater accumulates and the net degradation is a function of the number of repeater stages (Senior, 2008).

In a digital optical communication link, errors may occur under the following situations:

1. Large ISI arising out of dispersion introduced by the optical fiber waveguide. This can be cured to a large extent with the help of equalization.
2. The S/N ratio is unacceptably low to make an incorrect decision about the most probable bit being transmitted. Under this situation, the equalizer output voltage corresponding to a transmitted binary bit '0', may often exceed the predefined threshold value to register a binary 0 wrongly as a binary bit '1'.
3. Phase distortion (jitter) and a variation in the clock rate cause obscure zero crossing and decision time misalignment.

The performance of a regenerative repeater can be examined with the help of the so-called eye-pattern[1]. The eye pattern can be created on the oscilloscope screen by applying the received waveform across the vertical (Y) plates and using a sweep rate that is a fraction of the bit rate. A typical eye-pattern displays over a two-bit intervals' duration for a binary system with a little distortion and in the absence of additive noise is shown in Fig. 9.21(a). The pulse signal can be easily regenerated without error when the eye is completely open indicating the space for decision-making. The decision crosshair provided by the decision time and decision threshold lies in the open space within the eye. In the presence of intersymbol interference and additive noise, the opening area of the eye is reduced and may even close in extreme cases. This situation is illustrated in Fig. 9.21(b). Therefore, for reliable transmission, the eye must be open. The minimum distance between the decision crosshair and the edge of the eye is a measure of the margin against the occurrence of an error.

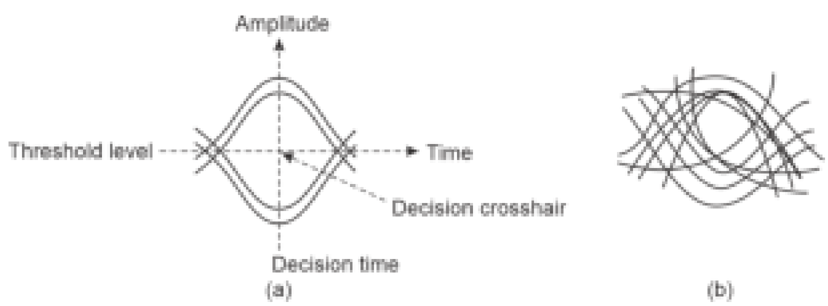

Fig. 9.21 Eye pattern in digital binary transmission (a) Pattern obtained with nominal distortion but without any additive noise (situation perfect for reliable regeneration) (b) Pattern in presence of distortion and additive noise

It is interesting to point out that in the presence of intersymbol interference and additive noise, there is a considerable chance of making the wrong decision about the received bit symbol (1 or 0) irrespective of the transmitted bit. The error in the process is measured in terms of Bit error-rate (BER) discussed in the previous chapter. For general-purpose reliable digital optical transmission, the BER should be at least 10^{-9} or less. However, for computer communication (data communication) any error is unacceptable. In such cases, it is necessary to make some arrangement for error detection and error correction by inserting a small amount of redundancy.

[1]The pattern looks very similar to human eye and hence the name.

The spacing between the repeaters is decided by several factors ranging from the characteristics of the optical source, optical detector, optical fiber, and even the couplers and joints. Specifically, the spacing between the regenerative repeaters is decided by the following parameters that take into consideration the deterioration of the components involved during the lifetime of the system (Senior, 2008).

1. Average power coupled by the source into the optical fiber.
2. For a given BER and at a given bit rate the minimum power required by the receiver to reproduce the signal.
3. The loss components over the entire link include average loss of the fiber, joint loss (both splice and demountable), and coupling loss at the source and detector-ends.
4. The temporal response of the system including the effect of pulse dispersion on the channel. This factor becomes significant for multimode fibers operating at a high bit rate.

9.2.1 Choice of Components

To design a practical optical communication link, it is necessary to choose various components judiciously from a variety of available components whose cost and characteristics vary over a wide range. The cost of the design is directly related to the performance of the link with respect to distance, speed, and quality. The fundamental requirement for designing an optical fiber link is to identify the wavelength range in which the system is supposed to operate. An optical fiber communication link is generally designed for application in the near-infrared region (NIR) (0.8–1.65 μm) of the optical fiber. This is because the loss in glass fiber is very low in this wavelength range. Optical fiber links are generally used in three principal wavelength regions around 0.85 μm, 1.3 μm and 1.55 μm. The range of choices of various components for the design of an optical fiber communication link is outlined below.

9.2.1.1 Fibers

The fibers are to be chosen properly based on the type, e.g. multimode or single mode to meet the requirements in terms of size, material (glass or plastic), index profile (in the case of MM fiber), numerical aperture, attenuation, dispersion, mode coupling, strength (depending on the type of installation such as aerial, underground or underwater), cabling, splicing, etc.

9.2.1.2 Sources

Optical sources are chosen from the two categories of sources, e.g. LED or laser diode (LD). The choice is dependent on the requirement with respect to optical power to be launched, type of fiber used, desired bit-rate, stability, and cost.

9.2.1.3 Photodetectors

A variety of photodetectors such as *p-n*, *p-i-n*, avalanche photodiode (APD), phototransistor, metal-semiconductor-metal (MSM), etc. are available. The selection of the right type of photodetector depends on the requirement with respect to responsivity, speed, detector gain, and sensitivity which is dependent on the dark current.

9.2.1.4 Transmitter

The configuration of optical transmitters depends on the type of optical communication, e.g. analog communication or digital communication, and also on the requirements in terms of desired power output level, bandwidth, input impedance, dynamic range, etc.

9.2.1.5 Receiver

The design of the receiver depends on the type of modulation used at the transmitter, available power at the input, BER ((or signal-to-noise ratio (SNR)) desired to be maintained, dynamic range, and bandwidth in addition to other requirements, such as automatic gain control (AGC).

9.2.1.6 Modulation and Coding

In optical communication, intensity modulation followed by direct detection is most commonly used. In this case, the intensity of the light source is made to vary following the message signal. Optical communication is possible by using a baseband signal directly or by using the baseband signal to modulate a microwave subcarrier frequency first and later the composite signal is used to modulate the intensity of the light source. For data communication when the signal is available in the form of a bit stream, the pulse-coded form of the data can be used directly to modulate the intensity of the light source. Different types of pulse formatting including PCM, delta modulation, adaptive delta modulation, etc. can be used together with return-to-zero (RZ), non-return-to-zero (NRZ) schemes including Manchester (biphase) or Miller (delay modulation) coding (Senior, 2008). In subcarrier modulation, the baseband signals can be impressed on the high-frequency microwave subcarrier frequency via conventional amplitude modulation (AM) or suppressed carrier AM (AM-SC) in the form of double-sideband SC (DSB-SC) or single-sideband SC (SSB-SC) or frequency modulation or phase modulation. The resultant electrically modulated subcarrier is finally used to vary the intensity of the light source analogously.

9.2.2 Multiplexing

The multiplexing of the baseband signal is done with the help of subcarrier modulation. However, multiplexing in the optical domain can be achieved to further enhance the information-carrying capacity of the optical fiber link. This can be achieved with the help of space-division multiplexing (SDM) or wavelength-division-multiplexing (WDM). In the former case, multiple fibers from the same fiber cable are used to transmit signals at different wavelengths. The multiplicity of the wavelength used in the process enables one to handle a larger volume of data. On the other hand, WDM involves the use of multiple wavelengths each of which carries different data to be transmitted through a single fiber. This is possible because the fiber supports different wavelengths to be transmitted through it simultaneously. As the optical signals are isolated in terms of wavelength, the data carried by each wavelength remain isolated from those carried by the other. A close examination of the two techniques reveals that SDM is not economical in terms of resources as it involves a larger number of fibers and consequently a larger number of additional optical components. However, optical isolation is much better in this case because each fiber carries one wavelength.

9.3 DIGITAL OPTICAL LINK

A vast section of the optical telecommunication network makes use of the transmission of information or data in the form of binary ("1" or "0") bit stream. For example, information in the form of binary PCM can be used to modulate the intensity of the optical source to produce PCM-IM. In this case, the output of the optical source will appear as optical pulses with "1" (light ON) or "0" (light "OFF"). In PCM transmission, bit-error rate in the range of $10^{-11} – 10^{-9}$ can be tolerated for all practical purposes. However, such errors are not acceptable in the case of data transmission. As a result, some form of error detection and correction techniques are generally incorporated in data transmission systems by introducing some form of redundancy in the bit stream. In optical communication systems, the errors may be caused because of the inter-symbol interference (ISI) arising out of the dispersion characteristics of the fiber and/or that caused by the electrical noise components that tend to mutilate the signal during the processing of the optical signal in the electrical domain after O/E conversion. Given this fact, in a long-haul optical communication system it is required to use many regenerative repeaters at intermediate points to maintain the integrity of the transmitted signal/data. The spacing between the regenerative repeaters is decided by many factors such as the average power launched into the fiber by the transmitter, minimum power required by the receiver (measured as sensitivity) to maintain the given bit-error rate, optical fiber characteristics such as average loss in the fiber and inter- and intra-modal dispersion of the fibers, other link losses, such as joint and splice losses. Optical sources can launch power into optical fibers which is much higher than the minimum power required by the receiver to reproduce the received signal reliably even after encountering loss of power during propagation. The repeaterless distance between the transmitter and the receiver or regenerative repeater is usually determined by the total loss in the fiber link in the absence of significant distortion of the signal. An LED source can launch power to the tune of tens of microwatt optical power whereas an ILD can launch power of the order of fractions of milliwatts of power in multimode fibers.

9.3.1 Digital Optical Receiver Design

The design of an optical receiver is much more complex as compared to the design of an optical transmitter. This is because at the receiver-end the received optical signal is usually low and distorted. Every receiver needs a minimum amount of optical power to be received depending on the receiver configuration and the transmission characteristics of the channel. In the IM/DD system, the optical signal is first converted into an electrical signal which gets affected by various noise components including the quantum noise arising out of the photodetection process and other components, such as thermal noise, shot noise arising out of the passive and active components of the receiver circuit. It is understood that under ideal conditions (in the absence of all other noise components) a photodetector needs at least twenty-one photons to interpret a binary "1" for a given BER of 10^{-9}. In practice, therefore every receiver needs a minimum amount of optical power to be received to maintain a certain BER. In this section, the sensitivity of a binary optical receiver system is estimated under certain simplified assumptions.

The SNR required to maintain a given BER in a digital optical communication receiver can be obtained by the method applied in the case of an electrical communication system by assuming the noise to be Gaussian in nature. It may however, be pointed out here that the quantum noise arising out of the photodetection process (O/E conversion) follows Poisson's distribution while other components, such as thermal noise follow Gaussian distribution. Accurate modeling of the receiver performance needs to take into account different noise components with different distributions. However, the modeling becomes very complex. On the other hand, a Gaussian approximation (Stillman, 1980) leads to a fairly accurate modeling with must less complexity. The receiver sensitivities calculated based on this simplified analysis are generally within 1 dB of those calculated by using other methods based on the exact probability distribution of the noise components (Webb *et al* 1974; Smith *et al* 1982).

To calculate the BER at the output of the receiver, we need to understand the mechanism of interpreting the bits in the presence of noise which tends to mutilate the signal. It is worth noting that the transmitted signal consists of two well-defined levels of optical power levels representing binary "1" and "0". On the other hand, the signal available at the receiver (generally considered the output of the equalizer) in the presence of noise does not exhibit well-defined levels. The available binary signal at the equalizer output looks something similar to that depicted in Fig. 9.22. To decide binary "1" or "0" at the output of the equalizer it is necessary to know the probability distribution of the signal plus the additive noise at this point. The probability density functions of the two transmitted binary states ("1" and "0") are depicted in Fig. 9.22. The probability density function (PDF) corresponds to the probability that the equalizer output has a value v within an incremental range dv. Assuming the additive noise to have a Gaussian distribution, the probability density functions for the two states can also be considered to be Gaussian. The Gaussian probability density function, in general, can be expressed as

$$f(y) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(y-m)^2}{2\sigma^2}\right] \tag{9.1}$$

where, m is the mean value and σ is the standard deviation (alternatively, σ^2 is the variance) of the distribution function.

If the decision threshold voltage is v_{th} between the two signal states, then a signal at the output of the equalizer is registered as binary "1" when the voltage exceeds the value v_{th}. Similarly, when the voltage at the output of the equalizer is less than the threshold value, v_{th} the received signal is registered as binary "0". However, in the presence of a large noise in optical transmission, the binary level voltages can be altered so much so that a binary "1" level could be reduced to a binary "0" level or a binary "0" level could be increased to a voltage level corresponding to a binary "1" level. These situations will lead to wrong decisions about the originally transmitted bits. The error probabilities can be calculated by integrating the probability densities outside the decision region as indicated in Fig. 9.22. For example, the probability, $p(0|1)$ that a signal transmitted as binary "1" is registered wrongly as a binary "0" due to noise is highlighted by the shaded area under the $p(y|1)$ distribution below v_{th} as indicated in Fig. 9.22. It is interesting to note that the shape of the probability distribution function for the binary "1" is not the same as that for a binary "0" of the transmitted signal. This indicates that the noise power associated with a binary "0" is not the same as that associated with a binary "1" in the transmitted data. This arises because various transmission impairments occur in the transmitted optical signal due to dispersion, amplifier noise, ISI effects, etc. before it is detected at the equalizer output. The probability, $p(1|0)$ that a signal transmitted originally as a binary "0" but registered as a binary "1" in the presence of large noise is indicated by the area under the $p(y|0)$ distribution curve above v_{th} as depicted in the figure. The two probabilities leading to errors can be expressed mathematically as

$$p_1(v_{\text{th}}) = \int_{-\infty}^{v_{\text{th}}} p(y|1)\,dy \tag{9.2}$$

corresponding to the probability that the equalizer output is less than v_{th} and

$$p_0(v_{\text{th}}) = \int_{v_{\text{th}}}^{\infty} p(y|0)\,dy \tag{9.3}$$

indicating the probability that the equalizer output voltage exceeds v_{th} when a binary "0" is originally transmitted.

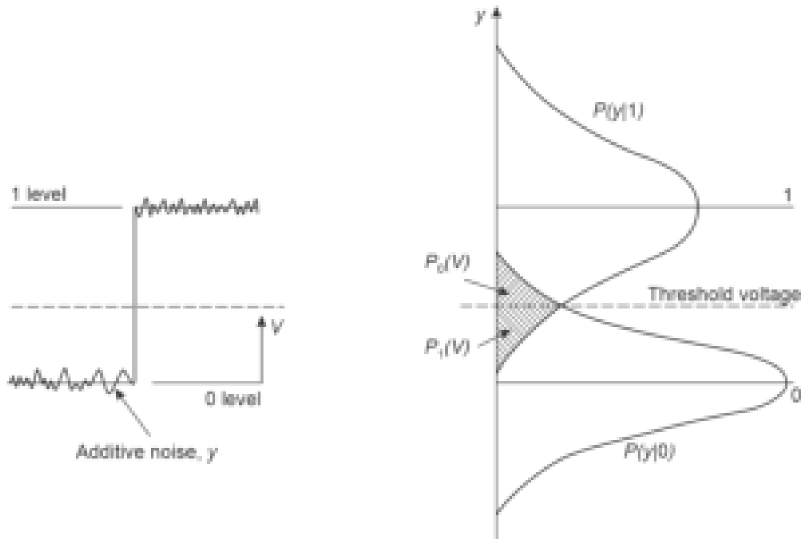

Fig. 9.22 Probability distributions for the received binary "0" and "1" signals in presence of noise indicating the situations under which error may occur in making the decision

The error probability can be expressed as

$$p_e = a\,p_1(v_{\text{th}}) + b\,p_0(v_{\text{th}}) \tag{9.4}$$

where, the factors a and b indicate the *a priori* distribution of the binary "1" and "0" data. That is, the factors a and b correspond, respectively, to the probabilities that either a binary "1" or a binary "0" occurs in the data. For unbiased data, the binary "1" and "0" occur with the same probability such that $a = b = 0.5$. To minimize the error given by Eq. (9.4), it is necessary to select the optimum value of v_{th}. For this purpose, it is essential to have quantitative knowledge about the mean square noise voltage which is superimposed on the signal voltage at the output of the equalizer. In the Gaussian approximation, it is assumed that the equalizer output voltage $v_{\text{out}}(t)$ is a Gaussian random variable with a probability density function described by Eq. (9.1). Therefore, to calculate the error probability all we need to know are the mean and the standard deviation of the equalizer output voltage.

Consider a binary signal transmission where a binary "1" is represented by a voltage level of amplitude V. The mean and the variance of the Gaussian output for a binary "1" are assumed to be b_{on} and σ_{on}^2 respectively, as illustrated in Fig. 9.23 and the corresponding parameters for a binary "0" are respectively b_{off} and σ_{off}^2. Consider the situation when a binary "0" pulse is being sent so that no signal pulse is present at the time of checking the equalizer output. The probability of an error in this case is the probability that the noise alone will exceed the threshold voltage, to get wrongly registered as "1′′. Thus, the probability $p_0(v_{\text{th}})$ is the probability that the equalizer output voltage will fall anywhere between v_{th} and ∞. Using Eqs (9.1) and (9.3) as

$$\begin{aligned} p_0(v_{\text{th}}) &= \int_{v_{\text{th}}}^{\infty} p(y\,|\,0)dy = \int_{v_{\text{th}}}^{\infty} f_0(y)dy \\ &= \frac{1}{\sqrt{2\pi}\sigma_{\text{off}}} \int_{v_{\text{th}}}^{\infty} \exp\left[-\frac{(v-b_{\text{off}})^2}{2\sigma_{\text{off}}^2}\right] dv \end{aligned} \tag{9.5}$$

where, $f_0(y)$ corresponds to the probability density function corresponding to the binary "0" bit.

Similarly, the probability of error that a transmitted binary "1" is misinterpreted as a "0" at the equalizer output can be obtained as

$$p_1(v_{\text{th}}) = \int_{-\infty}^{v_{\text{th}}} p(y|1)dy = \int_{-\infty}^{v_{\text{th}}} f_1(y)dy$$
$$= \frac{1}{\sqrt{2\pi}\sigma_{\text{on}}}\int_{-\infty}^{v_{\text{th}}} \exp\left[-\frac{(b_{\text{on}}-v)^2}{2\sigma_{\text{on}}^2}\right]dv \tag{9.6}$$

where, $f_1(y)$ corresponds to the probability density function corresponding to the binary "1" bit.

The integrals in Eqs (9.5) and (9.6) cannot be evaluated easily. These integrals can be expressed in terms of *error function* defined as

$$erf(u) = \frac{2}{\sqrt{\pi}}\int_0^u \exp(-z^2)dz \tag{9.7}$$

and the *complementary error function* is given by

$$erfc(u) = 1 - er\,f(u) = \frac{2}{\sqrt{\pi}}\int_u^{\infty} \exp(-z^2)dz \tag{9.8}$$

Equation (9.5) can be expressed in terms of *error function* as

$$p_0(v_{\text{th}}) = \frac{1}{\sqrt{2\pi}\sigma_{\text{off}}}\int_{v_{\text{th}}}^{\infty} \exp\left[-\frac{(v-b_{\text{off}})^2}{2\sigma_{\text{off}}^2}\right]dv$$
$$= \frac{1}{\sqrt{2\pi}\sigma_{\text{off}}}\left\{\int_{-\infty}^{\infty} \exp\left[-\frac{(v-b_{\text{off}})^2}{2\sigma_{\text{off}}^2}\right]dv - \int_0^{v_{\text{th}}} \exp\left[-\frac{(v-b_{\text{off}})^2}{2\sigma_{\text{off}}^2}\right]dv\right\}$$
$$= \frac{1}{2}\left\{1 - erf\left(\frac{v_{\text{th}}-b_{\text{off}}}{\sqrt{2}\sigma_{\text{off}}}\right)\right\}$$
$$= \frac{1}{2}erfc\left(\frac{v_{\text{th}}-b_{\text{off}}}{\sqrt{2}\sigma_{\text{off}}}\right) \tag{9.9}$$

Equation (9.6) can be similarly expressed as

$$p_1(v_{\text{th}}) = \frac{1}{\sqrt{2\pi}\sigma_{\text{on}}}\int_{-\infty}^{v_{\text{th}}} \exp\left[-\frac{(v-b_{\text{on}})^2}{2\sigma_{\text{on}}^2}\right]dv$$

$$= \frac{1}{\sqrt{2\pi}\sigma_{\text{on}}}\left\{\int_{-\infty}^{\infty} \exp\left[-\frac{(b_{\text{on}}-v)^2}{2\sigma_{\text{on}}^2}\right]dv - \int_{v_{\text{th}}}^{\infty} \exp\left[-\frac{(v-b_{\text{on}})^2}{2\sigma_{\text{on}}^2}\right]dv\right\}$$
$$= \frac{1}{2}erfc\left(\frac{b_{\text{on}}-v_{\text{th}}}{\sqrt{2}\sigma_{\text{on}}}\right) \tag{9.10}$$

If we assume that the apriori probabilities of binary 0s and 1s are the same, that is the number of transmitted 1s and 0s are equal, then

$$a = b = \frac{1}{2} \tag{9.11}$$

The error probability can be expressed using Eq. (9.4) as

$$P_e = \frac{1}{2}\left[\frac{1}{2}erfc\left(\frac{v_{\text{th}} - b_{\text{off}}}{\sqrt{2}\sigma_{\text{off}}}\right) + \frac{1}{2}erfc\left(\frac{b_{\text{on}} - v_{\text{th}}}{\sqrt{2}\sigma_{\text{on}}}\right)\right] \tag{9.12}$$

The error expressed by Eq. (9.12) can be viewed as the BER and can be subsequently expressed as

$$\begin{aligned} \text{BER} = P_e(Q) &= \frac{1}{2}erfc\left(\frac{Q}{\sqrt{2}}\right) \\ &= \frac{1}{2}\left[1 - erf\left(\frac{Q}{\sqrt{2}}\right)\right] \approx \frac{1}{\sqrt{2\pi}Q}\exp\left(-\frac{Q^2}{2}\right) \end{aligned} \tag{9.13}$$

The above approximation can be obtained by expanding the *error function* asymptotically. The parameter Q is given by

$$Q = \frac{v_{\text{th}} - b_{\text{off}}}{\sigma_{\text{off}}} = \frac{b_{\text{on}} - v_{\text{th}}}{\sigma_{\text{on}}} \tag{9.14}$$

The parameter Q is related to signal-to-noise ratio and determines the noise performance of the receiver in terms of BER for a given SNR. The variation of BER with Q as estimated by the actual expression given in Eq. (9.13) is shown in Fig. 9.23. The dotted line shows the variation of BER with Q under the asymptotic approximation of the error function by the exponential function given by Eq. (9.13). It can be easily seen that the approximate relation matches closely with the actual value for higher values of Q(>3). The standard BER for reliable digital transmission should be less than or at least equal to 10^{-9}. For example, the bit rate in a digital signal level 1 (DS1) telephone system is 1.544 Mbps, and a BER in the

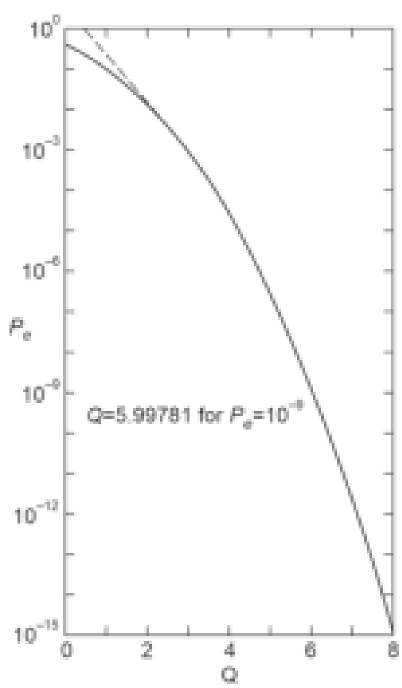

Fig. 9.23 Variation of bit-error-rate (BER) with Q (solid line). The dotted line shows the variation estimated on the basis of asymptotic approximation

tune of 10^{-9} amounts to a misinterpretation of bit every 650s or 11 min which is acceptable (Keiser, 2000). From the BER *vs* Q plot, it can be seen that

$$\text{BER} = 10^{-9} \text{ for } Q = 5.9978 \tag{9.15}$$

To maintain a BER of less than 10^{-9}, it is recommended that $Q \approx 6$.

It may be pointed out here that the variances in the noise powers are different for received binary "1" and "0" signals. That is, σ_{on} and σ_{off} have different values. Assuming the variances of the noise powers to be the same in both the cases (binary "1" and "0") and a binary "1" be represented by a voltage pulse of amplitude V over the bit period while a binary "0" as 0 V maintained for the same bit period, then we may write

$$\sigma_{\text{off}} = \sigma_{\text{on}} = \sigma$$

and $$b_{\text{on}} = V \text{ and } b_{\text{off}} = 0.$$

Under the above assumptions, Eq. (9.14) yields

$$V_{\text{th}} = \frac{V}{2}$$

Therefore,

$$Q = \frac{V}{2\sigma} \tag{9.16}$$

The variances of the noise powers given by σ_{off}^2 and σ_{on}^2 for binary "0" and "1" under the idealized condition attain the same value and as such the parameter σ may be viewed as the standard deviation or root-mean-square (rms) value of the noise power. That is, V/σ is the ratio of peak signal to rms value of noise. Under the above condition

$$\text{BER} = P_e(\sigma_{\text{on}} = \sigma_{\text{off}} = \sigma) = \frac{1}{2}\left[1 - erf\left(\frac{V}{2\sqrt{2}\sigma}\right)\right] \tag{9.17}$$

Example 9.1 A digital baseband binary optical communication link working at 100 Mbps offers a signal-to-noise ratio of 20. Estimate the bit-error-rate for the system assuming all associated noise components for the system are Gaussian in nature.

Solution Under the given condition

$$\text{BER} = \frac{1}{2}\left[1 - erf\left(\frac{V}{2\sqrt{2}\sigma}\right)\right]$$

$$= \frac{1}{2}\left[1 - erf\left(\frac{(S/N)^{1/2}}{2\sqrt{2}}\right)\right]$$

It should be noted here that the signal-to-noise ratio is generally measured in terms of power rather than current or voltage ratio unless stated otherwise.

Therefore,

$$\text{BER} = \frac{1}{2}\left[1 - erf\left(\frac{\sqrt{20}}{2\sqrt{2}}\right)\right] \approx \frac{1}{2}[1 - erf(1.6)]$$

$$= \frac{1}{2}(1 - 0.9763) = 11.83 \times 10^{-3}$$

Example 9.2 Calculate the voltage peak signal-to-noise ratio (V/σ) required to maintain a BER of 10^{-9} for a binary baseband digital optical communication link assuming that all noise components are Gaussian in nature.

Solution For the given situation

$$\text{BER} = 10^{-9} = \frac{1}{2}\left[1 - erf\left(\frac{V}{2\sqrt{2}\sigma}\right)\right]$$

Solving this (using the error function table), we find the peak signal-to-noise ratio as

$$\frac{V}{\sigma} \approx 12$$

It is interesting to examine how this signal-to-noise ratio translates into the number of photons required to be received by the photodetector at the receiver-end to ensure the desired bit-error-rate. Consider an avalanche photodiode-based optical receiver operating under quantum noise-limited conditions. Under quantum noise-limited condition, we ignore the shot noise components arising out of dark current and current due to background radiation and thermal noise components. The mean square value of the shot-noise current arising under the quantum limit can be expressed as

$$\langle i_s^2 \rangle = 2qI_p BM^2 F(M) \tag{9.18}$$

The signal-to-noise ratio can be expressed as

$$\frac{S}{N} = \frac{(I_p M)^2}{2qI_p BM^2 F(M)} = \frac{I_p}{2qBF(M)} \tag{9.19}$$

If the average number of photons incident on the APD over a bit duration period of τ is $\bar{n}$, then the photocurrent generated can be expressed as

$$I_p = \frac{q\eta\bar{n}}{\tau} \tag{9.20}$$

The signal-to-noise ratio can be expressed using Eqs (9.19) and (9.20) as

$$\frac{S}{N} = \frac{\bar{n}\eta}{2BF(M)} \tag{9.21}$$

Alternatively, the average number of photons required to be received by the photons over the bit period, τ to detect a binary "1" for a given signal-to-noise ratio can be expressed as

$$\bar{n} = \frac{2B\tau F(M)}{\eta}\left(\frac{S}{N}\right) \tag{9.22}$$

To avoid inter-symbol interference, the desired signal at the receiver should have a raised cosine spectrum. For a raised cosine pulse shaping the product of bandwidth and bit duration period for minimum ISI is (Senior, 2008)

$$B\tau \approx 0.6 \tag{9.23}$$

9.3.2 System Design Considerations

From the foregoing discussion, it is clearly understood that the BER for a given bit rate of transmission in a digital optical fiber link is related to the available signal-to-noise ratio at the receiver. In other words, the maximum possible transmission bit rate through a digital optical link for a given BER is dictated by the overall noise characteristics of the channel which includes the noise added in the channel as well as that added by the front-end of the receiver preceding the equalizer stage. The noise introduced by the channel in the form of ISI is dictated by the transmission characteristics of the fiber used as the channel as well as the characteristics of the source (output power, spectral width) while the noise introduced by the front-end of the receiver is decided by the characteristics of the photodetector and the following preamplifier. In the design of an optical fiber communication link the designer has a wide range of flexibility for selecting optical sources (LED or ILD), optical fiber (glass or plastic in terms of materials, multimode in the form of step-index or graded index, single mode, etc.) and photodetectors (pin, APD, MSM, etc.). As a result, overall characteristics of the optical fiber link largely depend on the chosen components from the available options. It is important to mention here that the cost of the optical link also depends on the choice of the components. A designer has to choose the components judiciously by keeping in mind the requirements specified by the user to keep the cost of the link appropriate for the specific application. For example, a bit rate of transmission of the order of 100 Mbps over a short distance can be easily managed with the help of an LED as the optical source, multimode glass fiber (may be good quality plastic fiber or plastic clad silica fiber), and a pin detector at the receiver instead of using a more expensive laser source, single mode fiber, and APD. The design of an optical link depends on the complexity of the architecture of the optical network. In order to have a general idea of the design considerations, we consider the simplest optical network architecture in the form of a point-to-point optical link.

9.3.3 Point-to-point Link Design

The block diagram of a point-to-point digital optical fiber link is shown in Fig. 9.24. For a regenerative repeaterless link, the transmitter and the receivers at the two ends are connected with the help of a fiber which may contain several fiber pieces spliced together to form a long-length fiber depending on the distance between the transmitter and the receiver. If the distance between the transmitter and the receiver is very large it may not be possible to transmit optical signal reliably through the link. This happens because the optical fiber attenuates the signal propagating through it and if the distance between the transmitter and the receiver is very large, the available power at the receiver-end may be lower than the minimum power required (depending on the sensitivity of the receiver) by the receiver to reproduce the signal reliably. For a long-distance link, it is often required to make use of regenerative repeaters which take care of attenuation and dispersion introduced by the optical fibers to regenerate them accordingly at intermediate points. The design of an optical fiber communication link involves analysis with respect to the following parameters:

1. Maximum possible length of the link.
2. The maximum rate at which data can be transmitted over the link (for a digital fiber optic link).
3. The BER of the overall link.

Fig. 9.24 A repeaterless point-to-point optical fiber link

For an analog optical communication link the bit rate of transmission shall be considered in terms of bandwidth of transmission while the BER shall translate in terms of SNR for reliable transmission.

It may be noted that the performance parameters of a digital optical communication link stated above largely depend on the characteristics of the components chosen for the particular design from the host of choices available to the designer. Moreover, all the three performance parameters of the link are interrelated in some form or the other. It is, therefore, necessary to keep in view how each of the parameters are affected at the time of optimizing any one of them. The characteristics of different components that are important for the link design are listed in Table 9.1.

Table 9.1 Important parameters for different components in optical link design

Components	*Key parameters*
Optical source	Operating wavelength, spectral response (spectral width), radiance and output optical power, radiation pattern (beam-width), emitting area, number of emitted modes, modulation capability (or bandwidth)
Optical fiber	Material (glass or plastic), core size, average loss at the given wavelength of transmission, index profile, index deviation, fiber geometry, numerical aperture for multimode fiber or mode-field diameter in the case of single-mode fiber, dispersion characteristics, inhomogeneity in the fiber, microbending/macrobending of the fiber
Optical detector	The operating wavelength, responsivity, gain (multiplying or non-multiplying), speed of response, noise-equivalent power (NEP), or more accurately sensitivity.

The system design for an optical link is generally carried out in the following two steps:

1. Link power budget
2. Rise-time budget

9.4 LINK POWER BUDGET

The link power budget analysis enables the designer to estimate the maximum permissible distance or separation between the transmitter and the receiver in a repeaterless digital link or the same between two consecutive regenerative repeaters in a link comprising regenerative repeaters. For a repeaterless optical link, the distance is determined by the power margin between the optical power transmitted by the transmitter and the minimum power required to be received by the receiver (or front-end of the next regenerative repeater) to maintain the given BER for the desired transmission speed. For an optical link involving multiple regenerative repeater, the permissible distance is calculated based on the power margin between the power emitted by the transmitter located at the rear-end of one repeater to the minimum power required by the receiver located at the front-end of the next regenerative repeater to maintain the specified BER at the desired bit rate. The minimum power required by the receiver depends on its sensitivity which is primarily decided by the photodetector and the following stage pre-amplifier. For a given set of power transmitted by the transmitter and the sensitivity of the receiver, the permissible length between the transmitter and the receiver is determined by the overall loss or attenuation provided by the optical fiber over the link. It may be recalled here that the permissible distance between the optical transmitter and the receiver or that between two consecutive regenerative repeaters also depends on the dispersion characteristics of the receiver. Assuming that there is no dispersion penalty on the link, the permissible distance can be calculated by estimating the total loss on the link. The total loss over the link consists of the following components:

1. Average loss of the optical fiber over the link as specified by the manufacturer at the desired wavelength of operation
2. Additional loss arising out of various fiber joints/splices

3. Coupling loss at the source end (transmitter side) and the detector end (receiver side)
4. A safety margin that accounts for any other extra loss arising out of component degradation and for fiber impairment caused by micro- or macro-bending or variation in the characteristics caused by temperature variations.

Out of the above four components, the average loss is generally specified by the manufacturers in terms of decibels per kilometer (dB/km). The joint or splice loss over the link depends on the actual number of joints or splices present over the link. The loss or attenuation caused by a joint/splice may vary from one joint to the other largely depending on the perfection of the joint. It is often convenient to express the fiber joint/splice loss in terms of an equivalent loss in decibels per kilometer by considering the combined joint/splice loss introduced by all the joints/splices present over the link and the total length of the link. The coupling loss at the transmitter and the receiver-end is usually determined by the loss introduced by the demountable connectors which connect the transmitter-end with the line fiber and the other which connects the same line fiber to the receiver. It may be recalled here that more often the user needs to connect the line fiber to a demountable connector available at the end of the fiber pigtail which is already connected to the optical source/detector in an optimum fashion by the supplier. In case, the user needs to couple power from the source to the line fiber or line fiber to the detector directly there may be additional coupling loss depending on the perfection of the coupling as well as other associated loss factors such as Fresnel loss. The permissible length between the transmitter and the detector can be adjusted for a given BER at the desired bit rate by changing the components accordingly and/or incorporating in-line optical amplifiers discussed in the next chapter. Once the link power budget is completed, the designer analyses the rise time budget for the link to ensure that the requirements specified by the user are met by the link.

The power transmitted by an optical transmitter is usually measured in terms of watts or dBm. Similarly, the sensitivity of an optical receiver is also measured in terms of watts or dBm. In the link power budget, it is customary to express both the transmitter power and the receiver sensitivity in dBm because all the loss components including the safety margin are expressed in dB. The optical power of 0.5 mW emitted by an optical transmitter can be expressed equivalently by –3 dBm which means that the power emitted by the source id 3 dB below 1 mW power. Similarly, an optical receiver sensitivity of –30 dBm suggests that the minimum power required by the receiver to reproduce the received signal reliably is 30 dB below 1 mW power. The power link budget thus involves the calculation of overall link loss and the power margin between the transmitted power and the receiver power both expressed in dBm.

A typical long-distance optical fiber link without a regenerative repeater is shown in Fig. 9.25.

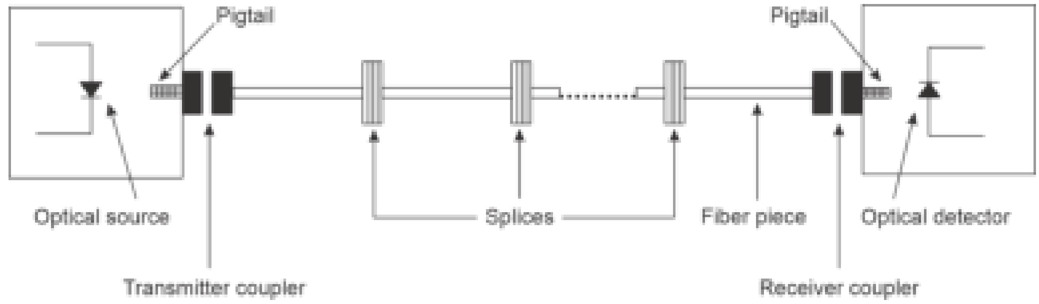

Fig. 9.25 Schematic of a point-to-point optical fiber link illustrating different loss components over the link

The total loss over the fiber link can be expressed as

$$L(dB) = \alpha_f L + S_l + C_t + C_r \tag{9.24}$$

where, α_f is the average fiber loss in dB per kilometer, S_l is the total splice loss (in dB) over the link length depending on the number of splices and the loss introduced by each splice, C_t and C_r are the coupling loss in dB at the transmitter-end and the receiver-end, respectively. The total splice loss can be calculated as

$$S_l = \sum_{i=1}^{n} S_i \tag{9.25}$$

where, S_i is the loss in dB introduced by the i^{th} splice on the link.

When the splice loss is expressed in terms of dB/km, the total splice loss can be calculated as

$$S_l(dB) = \alpha_s L \tag{9.26}$$

where, α_s is the effective attenuation in dB/km introduced by all the splices together and L is the length of the link.

Example 9.3 An optical fiber link of 5 km length is established by end-to-end splicing of ten pieces of fibers of length 500 m each. If the splice loss at each joint is 0.2 dB and the average loss of the optical fiber at the operating wavelength is 2 dB km^{-1}, calculate the total loss over the link. Assume that the connector losses at the transmitter and receiver ends are 2 dB each.

Solution The number of splices between the ten pieces of fiber is

$$n = 10 - 1 = 9$$

The total splice loss over the link is

$$S_l = 0.2 \times 9 = 1.8 \text{ dB}$$

The connector losses are

$$C_t = C_r = 2 \text{ dB}$$

The overall loss of the link is given by

$$\begin{aligned} L &= 2 \times 5 + 1.8 + 2 + 2 \\ &= 15.8 \text{ dB} \end{aligned}$$

Example 9.4 Repeat Problem 9.3 by assuming that the splice loss per kilometer for the above link is 1.0 dB/km.

Solution The total loss of the link in this case can be estimated as

$$\begin{aligned} L &= 2 \times 5 + 1 \times 5 + 2 + 2 \\ &= 24 \text{ dB/km} \end{aligned}$$

For given values of the transmitted optical power and receiver sensitivity, the maximum link length is calculated based on the available power margin using the following formula:

$$P_M = P_T - P_R = \alpha_f L + \alpha_s L + C_t + C_r + M_s \tag{9.27}$$

That is,

$$L\,(\text{km}) = \frac{P_T - P_R - C_t - C_r}{\alpha_f + \alpha_s} \tag{9.28}$$

where, P_T is the mean optical power (in dBm) launched into the optical fiber by the transmitter, P_R is the sensitivity of the receiver (in dBm) which corresponds to the mean power required by the receiver to ensure the given bit-error-rate for the desired bit rate and M_s is the desired safety power margin in dB. For general-purpose optical links the safety margin is kept at approximately 6 dB. For a laser diode-based optical fiber link, it is recommended to keep an additional safety margin of 2–3 dB to account for the variation of the laser output power because of temperature variation and/or aging.

Example 9.5 A long-haul optical communication system working at 56 Mbps uses various optical components with the following parameters.

Parameters	*Values*
Mean power launched by the transmitter into the optical fiber	0 dBm
Receiver sensitivity	–45 dBm
Average fiber loss	0.5 dBkm^{-1}
Splice loss	0.2 dBkm^{-1}
Transmitter connector loss	3.5 dB
Receiver connector loss	1.5 dB

Estimate the maximum possible link length without a repeater by incorporating a safety margin of 6 dB at the operating bit rate.

Solution The available margin of power between the mean launched power and the minimum power required by the receiver is

$$P_M = P_T - P_R = 0\text{ dBm} - (-45\text{ dBm}) = 45\text{ dBm}$$

Here, $\alpha_f = 0.5\text{ dBkm}^{-1}; \alpha_s = 0.2\text{ dBkm}^{-1}; C_t = 3.5\text{ dB}; C_r = 1.5\text{ dB}$ and $M_s = 6\text{ dB}$

The maximum link length can be estimated using Eq. (9.28) as

$$L = \frac{45 - 3.5 - 1.5 - 6}{0.5 + 0.2} = \frac{34}{0.7} = 48.6\text{ km}$$

Example 9.6 An average optical power of –6 dBm is launched by the transmitter of an optical fiber data link operating at 100 Mbps. The receiver uses an avalanche photodiode with a sensitivity of –50 dBm. If the average loss introduced by the fiber is 0.5 dBkm^{-1} and the splice loss is 0.2 dB per splice, calculate the repeaterless link length by assuming that the link contains 20 splices and the connection loss at the transmitter and the receiver-ends are 2.5 dB and the system requires a safety margin of 8 dB.

Solution The total fiber loss over the link is

$$L_f = 0.5 \times L\text{ dB}$$

The total splice loss is

$$S_l = 0.2 \times 20 = 4\text{ dB}$$

Therefore,

$$-6 - (-50) = 0.5 \times \text{L} + 4 + 2.5 + 2.5 + 8$$

That is,

$$L = \frac{44 - 17}{0.5} = 54\text{ km}$$

Example 9.7 The mean power available at the end of the transmitter pigtail of an optical fiber data link is 0.5 mW. The receiver sensitivity is –40 dBm at the operating bit rate. The average loss introduced by the fiber is 1.0 dBkm^{-1} and the average splice loss over the link is 0.5 dBkm^{-1}, calculate the repeaterless link length by assuming the connection losses at the transmitter and the receiver are 2.5 dB and 1.5 dB, respectively. The required safety margin for the system is desired to be 6 dB.

Solution The optical power output of the transmitter is

$$P_T = 0.5 \text{ mW} = -3 \text{ dBm}$$

The maximum link length without a repeater can be estimated as

$$L = \frac{-3-(-40)-2.5-1.5-6}{1.0+0.5} = \frac{27}{1.5} = 18 \text{ km}$$

Example 9.8 A 30 km long optical fiber data link operating at 100 Mbps uses an optical receiver with a sensitivity of –50 dBm. The mean power available at the end of the transmitter pigtail of an optical fiber data link is –6 dBm. The average loss introduced by the fiber is 0.8 dBm^{-1} and the average splice loss over the link is 0.5 dBkm^{-1}. The connector losses at the transmitter and the receiver are 3.5 dB and 1.5 dB, respectively. Estimate the value of the safety margin for the link.

Solution Using Eq. (9.27), we may write

$$-6 - (-50) = 0.8 \times 30 + 0.5 \times 30 + 3.5 + 1.5 + M_s$$

That is,

$$M_s = 0$$

This means that the link design is not viable because there is no safety margin. To have a suitable safety margin the designer needs to change the source to increase the transmitted power, reduce the link length or use a receiver with a better sensitivity.

It is often convenient to examine the viability of an optical link by writing down the optical power budget on the spreadsheet as illustrated below.

Example 9.9 A 10 km long optical fiber data link is to be designed to operate at 25 Mbps using an LED source operating at 1300 nm. The LED is capable of launching an optical power of 100 μW at the operating wavelength into the line fiber including the transmitter connector loss. The fiber used in the link offers an average attenuation of 1.5 dBkm^{-1}. The average splice loss on the link is 0.5 dBkm^{-1} and the receiver connector loss is 2.5 dB. The receiver uses a pin detector that requires a mean optical power of –40 dBm at this bit rate to maintain the desired bit-error rate. A safety margin of 6 dB is desired for the link. Write down the link power budget in the form of a spreadsheet to examine the viability of the link.

Solution The link power budget can be written in the form

Mean optical power launched into the line fiber at the transmitter end 100 μW =	–10 dBm
Sensitivity of the receiver at 25 Mbps	–40 dBm
Available power margin of the system to accommodate the total link loss	+ 30 dB
Fiber attenuation (1.5 × 10)	(–)15 dB
Splice loss (0.5 × 10)	(–)5 dB
Receiver connector loss	(–)2.5 dB
Desired safety margin	(–)6 dB
Total link loss	(–)28.5 dB
Excess power margin	1.5 dB

In the above example, the available power margin is shown as positive while all the loss components to be adjusted against the available power margin are shown as negative. Since there exists a positive excess power margin in the design after adjusting all possible loss components over the link including the safety margin, the data link is adjudged to be viable.

9.5 DISPERSION-EQUALIZATION PENALTY

In the above link power budget no additional power penalty required for combating the pulse dispersion is considered. In actual practice, both intramodal and intermodal dispersion cause a broadening of the pulse propagating through a multimode fiber. This pulse broadening gives rise to ISI resulting in a reduction of receiver sensitivity for maintaining the desired BER. If the ISI is not compensated by proper equalization within the receiver the desired BER cannot be ensured. On the other hand, equalization compensation within the receiver effectively calls for an increased optical power to be received by the receiver than prescribed by the sensitivity. This additional power requirement is generally viewed as an additional loss penalty to be accommodated along with our loss components of the link in the link power budget. This penalty is known as dispersion equalization or ISI penalty. For a single-mode fiber, there is no intermodal dispersion and the overall pulse broadening is usually very small and the dispersion power penalty is negligible. The dispersion power penalty is significant in the case of optical links involving multimode fibers operating at a high bit rate. The dispersion power penalty can be estimated by assuming a Gaussian pulse shape as (Midwinter, 1979)

$$D_L = \left(\frac{\tau_e}{\tau}\right)^4 \text{ dB} \tag{9.29}$$

where, τ_e is the $(1/e)$—full-width pulse broadening due to dispersion on the link and τ is the bit period.

For Gaussian pulse shape, τ_e is related to the rms pulse width σ as [see Eq. (4.29)]

$$\tau_e = 2\sqrt{2}\sigma \tag{9.30}$$

Using Eqs (9.29) and (9.30), we get

$$D_L = 2(2\sqrt{2}\sigma B_T)^4 \text{ dB} \tag{9.31}$$

where, $B_T = 1/\tau$ is the bit rate of transmission.

The dispersion-equalization penalty is expressed in dB and therefore can be easily incorporated in the link power budget equation the total loss of the link given by the equation can be modified to incorporate the dispersion-equalization power penalty as

$$\mathcal{L}(\text{dB}) = \alpha_f L + S_l + C_t + C_r + D_L \tag{9.32}$$

Similarly, the maximum link length without a regenerative repeater in the presence of a dispersion-equalization power penalty can be estimated as

$$L(\text{km}) = \frac{P_T - P_R - C_t - C_r - D_L}{\alpha_f + \alpha_s} \tag{9.33}$$

Example 9.10 A 10 km long repeaterless optical fiber data link operates at 50 Mbps. The rms pulse broadening caused by the fiber over the link is 0.5 nskm^{-1} due to intermodal dispersion. Assuming the intramodal dispersion is negligible, calculate the dispersion-equalization power penalty of the link. Compare and contrast the value when the bit rate is increased to 100 Mbps. Calculate the dispersion-equalization penalty in the two cases in the presence of mode coupling.

Solution The total rms pulse broadening is obtained as

$$\sigma = 0.5 \times 10 = 5 \text{ ns}$$

The dispersion-equalization penalty can be calculated as

$$D_L = 2(2\sqrt{2}\sigma B_T)^4 = 2(2\sqrt{2} \times 5 \times 10^{-9} \times 25 \times 10^6)^4$$
$$\approx 0.03 \text{ dB}$$

When the bit rate is increased to 100 Mbps, the dispersion-equalization power penalty becomes

$$D_L = 2(2\sqrt{2}\sigma B_T)^4 = 2(2\sqrt{2} \times 5 \times 10^{-9} \times 100 \times 10^6)^4$$
$$= 7.99 \text{ dB}$$

The result demonstrates that the dispersion-equalization penalty becomes significant at higher operating bit rates.

In the presence of mode coupling, the rms pulse broadening becomes

$$\sigma_{mc} = 0.5 \times \sqrt{10} = 1.58 \text{ ns}$$

The dispersion-equalization penalty can be calculated as

$$D_L = 2(2\sqrt{2}\sigma_{mc} B_T)^4$$
$$= 2(2\sqrt{2} \times 1.58 \times 10^{-9} \times 25 \times 10^6)^4$$
$$\approx 3 \times 10^{-4} \text{ dB}$$

When the bit-rate is increased to 100 Mbps, the dispersion-equalization power penalty becomes

$$D_L = 2(2\sqrt{2}\sigma_{mc} B_T)^4$$
$$= 2(2\sqrt{2} \times 1.58 \times 10^{-9} \times 100 \times 10^6)^4$$
$$= 0.079 \text{ dB}$$

9.6 RISE-TIME BUDGET

In the link power budget, the effect of dispersion characteristics of the fiber is generally ignored. The dispersion caused by the fiber indirectly affects the link power budget and the same can be accounted for by introducing the dispersion-equalization power penalty. However, the most rigorous method for determining the effect of dispersion is to carry out the rise-time budget for the link. In the rise-time budget analysis, for the entire system is done by considering the contribution of each component to the overall degradation of the pulse measured in terms of total rise-time of the system. The components contributing to the system rise-time include the transmitter, fiber cable, and receiver. The rise-time for each component is defined in terms of Gaussian response to rise from 10–90%. The contribution of the fiber to the system rise-time can be separated into two subcomponents arising from the intermodal dispersion and the intramodal dispersion. The overall system rise-time can be expressed as

$$T_{\text{sys}} = (T_T^2 + T_C^2 + T_M^2 + T_R^2)^{1/2} \tag{9.34}$$

where, T_T is the source rise-time, T_C and T_M are the rise-time due to intramodal or chromatic and intermodal dispersion, respectively, and T_R is the detector rise-time. All rise times are measured in nanoseconds. The values of the rise time for the transmitter and the receiver are generally available with the link designer.

The transmitter rise-time is attributed to the delay caused by E/O conversion by the source and the associated drive circuit. The receiver rise-time is contributed by the photodetector response and the 3 dB electrical bandwidth of the front end. The response of the receiver front-end can be approximated by an RC low-pass filter shown in Fig. 9.26.

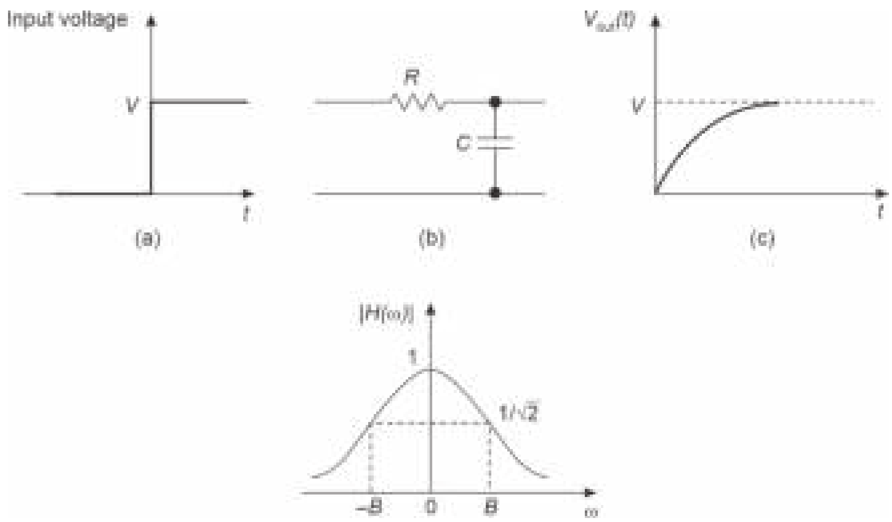

Fig. 9.26 An optical receiver approximated by a RC low-pass filter

The rise-time of the low-pass filter approximated as the receiver can be obtained from the response of the RC circuit to the turn-on transient shown in Fig. 9.26(a).

The response of the RC circuit [Fig. 9.26(b)] is given by

$$v_{\text{out}}(t) = V[1 - \exp(-t/RC)] \tag{9.35}$$

The 10–90% rise-time of the RC circuit can be obtained as

$$t_r = 2.2\ \text{RC} \tag{9.36}$$

The transfer function of the *RC* circuit can be expressed as

$$|H(f)| = \frac{1}{(1 + 4\pi^2 f^2 C^2 R^2)^{1/2}} \tag{9.37}$$

That is,

$$|H(f)| = \frac{1}{\left[1 + \left(\frac{f}{B}\right)^2\right]^{1/2}} \tag{9.38}$$

where, B is the 3 dB bandwidth of the *RC* circuit given by

$$B = \frac{1}{2\pi RC} \tag{9.39}$$

Substituting the value of RC from Eq. (9.39) into Eq. (9.36), the rise-time of the RC circuit can be obtained as

$$t_r = \frac{2.2}{2\pi B} = \frac{0.35}{B} \tag{9.40}$$

Expressing the rise-time in nanoseconds and the bandwidth in MHz and approximating the receiver by an RC low-pass circuit, the rise-time of the receiver can be expressed as

$$T_R(ns) = \frac{350}{B_R\ (\text{MHz})} \tag{9.41}$$

where, B_R is the 3dB bandwidth of the equivalent RC circuit of the receiver.

The intramodal dispersion comprises material dispersion and waveguide dispersion components. For a multimode fiber, the waveguide dispersion is generally negligible as compared to material dispersion whereas in a single-mode fiber waveguide dispersion is significant. The overall dispersion caused by the components is called chromatic dispersion or group velocity dispersion and is dependent on the spectral width of the source. The fiber rise-time rising out of chromatic dispersion can be approximated as

$$T_C \approx |D| L \sigma_\lambda \tag{9.42}$$

where, L is the length of the fiber, σ_λ is the half-power spectral width of the source, D is the resultant intramodal or chromatic of the fiber given by

$$D = \frac{d\tau_g}{d\lambda} \tag{9.43}$$

τ_g being the overall group delay caused by the final spectral width of the source in a given mode. The rise-time due to chromatic dispersion can be greatly reduced by making use of dispersion-shifted fiber.

The computation of the rise-time caused by the intermodal dispersion in a multimode fiber link is rather complex and tricky. This parameter largely depends on the order in which the fibers are joined to form the link. The overall bandwidth of the link limited by modal dispersion can be optimized by making careful choices in selecting the adjoining fibers to be spliced. The modal dispersion limited bandwidth over a link of length, L can be empirically expressed as (Keiser, 2000)

$$B_{\text{mod}} = \frac{B_0}{L^q} \tag{9.44}$$

where, B_0 is the bandwidth of 1 km long fiber in the link and q is an empirical constant.

The relation between the rise-time and 3 dB bandwidth of the fiber due to modal dispersion can be obtained by considering a Gaussian distribution of the optical power emerging from the fiber [see Eq. (4.27)]. The Gaussian pulse can be expressed as

$$p(t) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2\sigma^2}\right) \tag{9.45}$$

where, σ is the rms spectral width of the pulse caused by modal dispersion.

The time required to attain its half-maximum, $t_{1/2}$ of the pulse can be obtained as

$$p(t = t_{1/2}) = \frac{1}{2} p(0) \tag{9.46}$$

That is,

$$t_{1/2} = (2\ln 2)^{1/2}\sigma \tag{9.47}$$

The full-width half-maximum time can thus be written as

$$t_{\text{FWHM}} = 2(2\ln 2)^{1/2}\sigma \tag{9.48}$$

The 3dB bandwidth of the pulse can be obtained from the spectrum of the pulse. The Fourier transform of the Gaussian pulse defined by Eq. (9.45) can be obtained as

$$P(f) = \frac{1}{\sqrt{2\pi}}\exp(-2\pi^2 f^2\sigma^2) \tag{9.49}$$

The 3dB bandwidth can be obtained from

$$P(f = f_{3\,\text{dB}}) = \frac{1}{2}P(0) \tag{9.50}$$

That is,

$$\frac{1}{\sqrt{2\pi}}\exp(-2\pi^2 f_{3\,\text{dB}}^2\sigma^2) = \frac{1}{2}\left(\frac{1}{\sqrt{2\pi}}\right) \tag{9.51}$$

Therefore,

$$f_{3\,\text{dB}} = \frac{\sqrt{\ln(2)}}{\sqrt{2}\pi\sigma} \tag{9.52}$$

In terms of full-width half-maximum rise time, the 3-dB optical bandwidth limited by modal dispersion can be expressed as

$$B_{\text{mod}} = f_{3\,\text{dB}} = \frac{\sqrt{\ln(2)}}{\sqrt{2}\pi}\cdot\frac{2\sqrt{\ln(2)}}{t_{FWHM}} = \frac{\sqrt{2}}{\pi t_{FWHM}} \approx \frac{0.45}{t_{FWHM}} \tag{9.53}$$

Therefore, the rise-time arising from the modal dispersion can be expressed as

$$T_{\text{mod}} = \frac{0.45}{B_{\text{mod}}} = \frac{0.45L^q}{B_0} \tag{9.54}$$

Expressing the rise-time due to modal dispersion in nanoseconds and B_0 in MHz, we may write

$$T_{\text{mod}}(\text{ns}) = \frac{450L^q}{B_0(\text{MHz})} \tag{9.55}$$

The overall system rise time can be obtained using Eqs (9.34), (9.41), (9.42) and (9.55) as

$$T_{\text{sys}}(\text{ns}) = \left[\left(T_T^2 + \frac{450\,L^q}{B_0}\right)^2 + D^2\sigma_\lambda^2 L^2 + \left(\frac{350}{B_R}\right)^2\right]^{1/2} \tag{9.56}$$

It should be noted here that the transmitter rise-time must be expressed in nanoseconds, the dispersion coefficient D in terms of ns nm^{-1} km^{-1}.

Example 9.11 Consider a 10 km long digital optical fiber data link using an LED source with a spectral width of 40 nm. The receiver is based on a pin photodetector with a bandwidth of 40 MHz. The multimode fiber used in the link offers a chromatic dispersion of 10 ps nm^{-1} km^{-1}. The multimode fiber used in the link has a bandwidth-distance product of 300 MHz km and the modal dispersion can be expressed empirically with the help of Eq. (9.44) with q = 0.6. Estimate the system rise-time assuming that the rise-time of the optical transmitter end is 10 ns.

Solution The rise-time due to modal dispersion can be obtained as

$$T_{\text{mod}} = \frac{450 \times 10^{0.6}}{300} = 5.97 \text{ ns}$$

The overall system rise-time can be obtained as

$$T_{\text{sys}}(\text{ns}) = \left[(10 \text{ ns})^2 + (5.97)^2 + (10 \times 10^{-3} \times 40 \times 10 \text{ ns})^2 + \left(\frac{350}{40} \text{ns} \right)^2 \right]^{1/2}$$

That is,

$$T_{\text{sys}} = 15.10 \text{ ns}$$

Example 9.12 A 30 km long digital optical fiber data link operating at 1330 nm uses an injection laser diode with a spectral width of 0.2 nm. The transmitter offers a rise-time of 20 ps. The APD-based receiver operates at 2 Gbps. The multimode graded-index fiber used in the link offers a chromatic dispersion and modal dispersion of 5 ps and 7.0 ps, respectively, over the entire length of the link. Estimate the system rise-time.

Solution The receiver rise-time is given by (assuming NRZ format)

$$T_R = \frac{350}{2000} = 0.175 \text{ ns}$$

$$T_{\text{sys}} = [(0.02 \text{ ns})^2 + (0.007 \text{ ns})^2 + (0.005 \text{ ns})^2 + (0.175 \text{ ns})^2]^{1/2}$$

$$\approx 0.175 \text{ ns}$$

This means that the overall system rise-time in this case is decided by the rise-time of the receiver circuit.

9.7 ANALOG SYSTEMS

A major section of modern electrical telecommunication networks makes use of digital modes for transmission and reception of voice, video, and data. The popularity of digital communication lies in the fact that digital integrated circuits are cost-effective and reliable. However, there are many situations where transmission of the signal in the analog form becomes more convenient. In optical fiber communication, the mode may be either analog or digital. Even though digital forms of optical communication are more common because of obvious reasons, analog optical communication does find applications in microwave subcarrier multiplexing systems, video distribution in Community Antenna Television (CATV), direct cable television systems, etc. In such cases, the communication is generally over a short distance and the use of analog mode enables one to avoid the cost and complexity associated with digital systems involving A/D and D/A converters.

For successful implementation of an analog optical communication system, a few stringent conditions are to be met. These include the desired signal-to-noise ratio at the receiver output, bandwidth, and a high degree of linearity of the link to avoid cross-talk between the multiplexed signals. A typical analog communication link is shown with the help of a block diagram in Fig. 9.27. In a practical analog optical communication system the baseband signal is not used directly to modulate the intensity of the light source for transmission. More frequently, the baseband signal is first translated onto a high-frequency electrical subcarrier (usually at microwave frequency) before intensity modulation of the source. The subcarrier electrical modulation is done using standard modulation schemes such as AM, AM-SC, frequency modulation (FM), and phase modulation (PM). For a proper design of analog optical communication, one needs to take into consideration various issues including the linearity of the source, relative-intensity-noise (in the case of ILD) at the transmitter side, the attenuation of the fiber and its bandwidth limited by the dispersion characteristics and various noise components at the receiver-end those determine the signal-to-noise ratio. In addition, the noise associated with the amplified spontaneous emission of the in-line optical amplifier needs to be considered. In the subcarrier modulation case, the baseband signal is impressed upon the high-frequency carrier. In such cases, carrier-to-noise ratio (CNR) is used as the figure of merit in place of SNR.

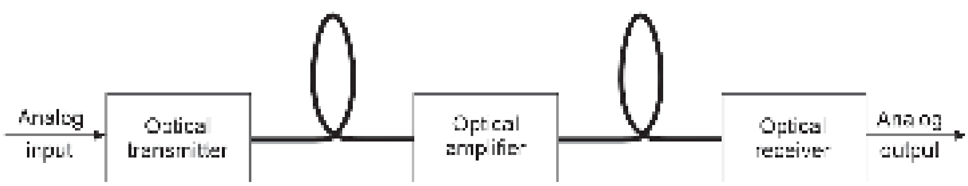

Fig. 9.27 Block diagram of a typical analog link

9.7.1 Direct Intensity Modulation (D-IM)

In this scheme, the analog baseband signal is used to modulate the intensity of the optical source directly as illustrated in Fig. 9.28.

When the optical source is modulated directly by the baseband signal, the power output from the transmitter is a function of time given by

$$P_{op}(t) = P_i[1 + k_a m(t)] \tag{9.57}$$

where, k_a is the sensitivity of the modulator, P_i is the average transmitted optical power in the absence of modulation and $m(t)$ is the baseband (message) signal.

For a single-tone baseband signal, we may write

$$m(t) = A_m \cos(2\pi f_m t) \tag{9.58}$$

where, A_m is the amplitude of the sinusoidal baseband signal and f_m is the frequency.

Therefore, the modulated signal from the transmitter can be expressed as

$$P_{op}(t) = P_i[1 + k_a A_m \cos(2\pi f_m t)] = P_i[1 + \mu \cos(2\pi f_m t)] \tag{9.59}$$

where, $\mu = k_a A_m$ is the index of modulation signifying the maximum excursion of the modulated signal on both sides from the average unmodulated power.

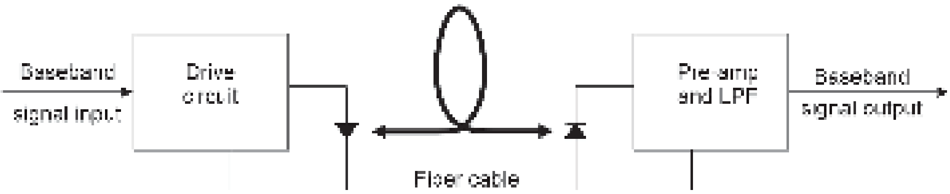

Fig. 9.28 Direct intensity modulation

The transmitted power propagates through the fiber and is received by the photodetector at the receiver end. Assuming the dispersion in the fiber is negligible, the photocurrent generated by the receiver will be proportional to the received power, P_0 which also varies in the similar form as does $P_{op}(t)$. The photocurrent generated by the photodetector can be expressed as

$$I_p(t) = \frac{q\eta}{h\nu} P_0[1 + \mu \cos(2\pi f_m t)] \tag{9.60}$$

The mean square signal current can be expressed as

$$\overline{i^2_{\text{signal}}} = \left(\frac{1}{\sqrt{2}} \frac{q\eta}{h\nu} \mu P_0 \right)^2 = \frac{1}{2}(\mu I_p)^2 \tag{9.61}$$

where, I_p is the average photocurrent produced at the receiver end by the unmodulated carrier.

The mean square value of the average shot-noise current at the receiver can be expressed as

$$\overline{i_s^2} = 2q(I_D + I_P)B \tag{9.62}$$

where, q is the electronic charge, I_D is the dark current and B is the post-detection bandwidth.

The mean square value of the thermal noise component can be expressed in terms of the amplifier noise figure F_n referred to the load resistance R_L as

$$\overline{i_T^2} = \frac{4kTBF_n}{R_L} \tag{9.63}$$

The signal-to-noise ratio in terms of mean-square rms signal current and mean-square total noise current can be expressed as

$$\frac{S}{N} = \frac{\frac{1}{2}(\mu I_p)^2}{2q(I_D + I_P)B + \dfrac{4kTBF_n}{R_L}} \tag{9.64}$$

For a receiver based on a multiplying photodetector such as an avalanche photodiode (APD) with a gain, M the signal-to-noise power can be obtained as

$$\left(\frac{S}{N}\right)_{APD} = \frac{\frac{1}{2}(M\mu I_p)^2}{2q(I_D + I_P)\mathrm{M}^2F(M)B + \dfrac{4kTBF_n}{R_L}} \tag{9.65}$$

where, $F(M)$ is the excess noise factor of the APD given by

$$F(M) = \frac{\overline{m^2}}{M^2} \tag{9.66}$$

$\overline{m^2}$ being the mean square value of the gain introduced by the APD.

For a non-multiplying photodetector (such as *p-i-n* photodetector) under the quantum shot-noise limit condition the signal-to-noise ratio can be calculated using Eq. (9.64) and assuming the dark current and thermal current components to be zero. Thus, the maximum rms signal-to-noise ratio available from the photodetector under quantum limit conditions can be expressed as

$$\frac{S}{N} = \frac{\mu^2 I_p}{4qB} \text{ (under quantum limit)} \tag{9.67}$$

The optical power received by the receiver can be alternatively expressed as

$$P_0 = \frac{4hv}{\mu^2\eta}\left(\frac{S}{N}\right)B \tag{9.68}$$

For a low value of signal-to-noise ratio, I_p is usually low and the thermal noise component dominates. Under this condition of thermal noise limit, the signal-to-noise ratio is obtained as

$$\frac{S}{N} = \frac{(\mu I_p)^2 R_L}{8kTBF_n} \text{ (under thermal noise limit)} \tag{9.69}$$

Consequently, the received power under thermal noise limit conditions can be obtained as

$$P_0 = \frac{hv}{q\eta\mu^2}\left(\frac{8kTF_n}{R_L}\right)^{1/2}\left(\frac{S}{N}\right)^{1/2} B^{1/2} \tag{9.70}$$

A comparison of Eqs (9.68) and (9.70) reveals that under quantum noise limit condition, the incident power on the receiver for a given signal-to-noise ratio is proportional to the post-detection bandwidth while at low power level under thermal noise limit condition, the power received by the detector is proportional to the square root of the post-detection bandwidth.

Example 9.13 Calculate the value of the average power required by an optical receiver of an analog optical link working with a D-IM scheme with a modulation index of 0.6 at 1550 nm under quantum noise limit condition to maintain an rms signal-to-noise ratio of 30 dB. The quantum efficiency of the photodetector is 65% at the operating wavelength. The post-detection bandwidth of the receiver is 1 MHz.

Solution The desired rms signal-to-noise ratio (dimensionless) can be obtained as

$$30 = 10\log_{10}\left(\frac{S}{N}\right)$$

That is,

$$\frac{S}{N} = 1000$$

The average power required by the receiver to maintain the above signal-to-noise ratio can be obtained under quantum noise limit condition by using Eq. (9.67) as

$$P_0 = \frac{4hc}{\lambda\mu^2\eta}\left(\frac{S}{N}\right)B$$

That is,

$$P_0 = \frac{4\times 6.626\times 10^{-34}\times 3\times 10^{8}}{1550\times 10^{-9}\times 0.36\times 0.65}\times 1000\times 10^{6}$$

$$= 2.19 \text{ nW} = -56.59 \text{ dBm}$$

Example 9.14 Consider an analog optical fiber link operating at 1.55 μm using D-IM with 60% modulation. The receiver of the link has a post-detection bandwidth of 1 MHz and is dominated by the thermal noise component. The photodetector used at the receiver has a quantum efficiency of 70% at the operating wavelength. Calculate the value of the average optical power required by the receiver to maintain an rms signal-to-noise ratio (in terms of mean square signal current and noise current) of 30 dB at room temperature. Assume that the effective load resistance of the photodetector is 0.5 MΩ and the noise figure of the pre-amplifier is 6 dB.

Solution The corresponding signal-to-noise ratio and the noise figure of the preamplifier specified in the problem can be expressed in terms of dimensionless ratio as

$$30 = 10\log_{10}\left(\frac{S}{N}\right)$$

That is,

$$\frac{S}{N} = 1000$$

Similarly, $F_n = 4$

The average power required to be received by the receiver to maintain the given signal-to-noise ratio under the thermal noise limit can be obtained from Eq. (9.70) as

$$P_0 = \frac{hc}{\lambda q\eta\mu^2}\left(\frac{8kTF_n}{R_L}\right)^{1/2}\left(\frac{S}{N}\right)^{1/2} B^{1/2}$$

That is,

$$P_0 = \frac{6.626\times10^{-34}\times3\times10^8}{1.55\times10^{-6}\times1.6\times10^{-19}\times0.7\times0.36}\times\left(\frac{8\times1.38\times10^{-23}\times300\times4}{0.5\times10^6}\right)^{1/2}\times(1000)^{1/2}\times(10^6)^{1/2}$$
$$= 3.18\times5.14\times10^{-13}\times31.63\times10^3 = 0.05\ \mu\text{W}$$

In the foregoing discussion, the baseband signal is assumed to be a single-tone signal. In the case of a generalized baseband signal $m(t)$ modulating the intensity of the light directly as given by Eq. (9.57), the time-dependent photocurrent generated by the photodetector at the receiver end can be estimated as

$$I(t) = l_p[1 + k_a m(t)] \tag{9.71}$$

It may be pointed out that the modulated intensity follows the baseband signal only when $|k_a m(t)| < 1$. The average photocurrent generated by the photodetector can be obtained as

$$I_p = R P_0 \tag{9.72}$$

where, P_0 is the average received optical and R is the responsivity of the photodetector.

The mean square value of the signal current can be expressed as

$$\overline{i_{\text{sig}}^2} = I_p^2 k_a^2 \overline{m^2(t)} \tag{9.73}$$

where, $\overline{m^2(t)}$ is the mean-square value or the average power of the baseband signal given by

$$\overline{m^2(t)} = \int_{-B_m}^{B_m} S_m(f)df \tag{9.74}$$

Here, B_m is the one-sided bandwidth of the baseband signal and $S_m(f)$ is the power spectral density of the baseband signal.

The signal-to-noise in terms of mean-square current for a generalized baseband signal can be consequently expressed as

$$\frac{S}{N} = \frac{k_a^2 I_p^2 \overline{m^2(t)}}{2q(I_D + I_P)B_m + \dfrac{4kTBF_n}{R_L}} \qquad (p\text{-}i\text{-}n \text{ based receiver}) \tag{9.75}$$

and
$$\frac{S}{N} = \frac{k_a^2 I_p^2 M^2 \overline{m^2(t)}}{2q(I_D + I_P)M^2 F(M)B_m + \dfrac{4kTBF_n}{R_L}} \quad \text{(APD-based receiver)} \tag{9.76}$$

9.7.2 Subcarrier Double-sideband Intensity Modulation (DSB-IM)

So far, we have considered direct modulation of the intensity by the baseband signal. The baseband signal can be used to modulate the high-frequency subcarrier by using any one of the techniques, such as double-sideband suppressed carrier (DSB-SC), frequency modulation (FM), or phase modulation (PM). The translated signal can subsequently be used to modulate the intensity of the source at the transmitter.

The baseband signal $m(t)$ can be used to modulate the subcarrier signal just by multiplying the signal by the high-frequency carrier. The result is a double-sideband suppressed carrier signal given by

$$s(t) = k_b m(t) \cos 2\pi f_c t \tag{9.77}$$

where k_b is a multiplying constant attributed to the sensitivity of the DSB modulator, f_c is the frequency of the subcarrier signal.

The optical power output of the transmitter modulated by this DSB-SC signal can be expressed as

$$P_{\text{opt}}(t) = P_i[1 + k_b m(t) \cos 2\pi f_c t] \tag{9.78}$$

The time-dependent photogenerated current for a DSB-SC subcarrier IM can be expressed as

$$I(t) = I_p[1 + k_b m(t) \cos 2\pi f_c t\,] \tag{9.79}$$

where I_p is the average photocurrent generated by the photodetector.

The signal-to-noise ratio can be expressed as

$$\frac{S}{N} = \frac{(RP_0)^2 \left[\frac{1}{2} k_b^2 \overline{m^2(t)}\right]}{4B_m q(I_D + I_P) + \dfrac{4kTBF_n}{R_L}} \text{(}p\text{-}i\text{-}n \text{ detector case)} \tag{9.80}$$

In the above expression, the bandwidth of the DSB-SC signal is used as

$$BW = 2B_m \tag{9.81}$$

9.7.3 Subcarrier Frequency Modulation Followed by Intensity Modulation (FM-IM)

The subcarrier frequency-modulated signal can be expressed as

$$s(t) = A_c \cos\left[2\pi f_c t + 2\pi k_f \int_0^t m(t)dt\right] \tag{9.82}$$

The peak frequency deviation of the FM signal can be estimated as

$$\Delta f = \max\left|k_f m(t)\right| \tag{9.83}$$

The bandwidth of the FM signal can be estimated by Carson's rule given by (Chakrabarti, 2011)

$$\begin{aligned} B_T &= 2\Delta f + 2B_m \\ &= 2B_m (D_f + 1) \end{aligned} \tag{9.84}$$

where, the frequency deviation ratio, D_f is given by

$$D_f = \frac{\Delta f}{B_m} \tag{9.85}$$

where, B_m is the bandwidth of the message signal.

The signal-to-noise ratio at the input of the subcarrier demodulator that is, at the output of the preamplifier of the receiver is

$$\left(\frac{S}{N}\right)_{\text{input}} = \frac{(RP_0)^2\left(\frac{A_c^2}{2}\right)}{2B_T q(I_D + I_P) + \frac{4kTBF_n}{R_L}} \tag{9.86}$$

9.7.4 Multichannel Transmission

In the foregoing discussion, we have considered intensity modulation of the optical source directly by the baseband signal or by a high-frequency subcarrier signal modulated by the baseband signal. In the latter case, the receiver needs to be equipped with a subcarrier demodulator to get back the intended baseband signal. However, in broad-band analog optical communication applications, such as in the case of cable television (CATV) distribution system, it is necessary to send multiple analog baseband signals over the same fiber. This can be achieved by superimposing these baseband signals over subcarrier signals operating at different frequencies f_{c1}, f_{c2}, f_{c3}, ... The modulated subcarriers constitute a multiplexed signal in the electrical domain which is subsequently used to modulate the intensity of the source at the optical signal. For cable TV distribution vestigial sideband suppressed carrier (VSB-SC) amplitude modulation for transmission of video signal and frequency modulation (FM) is used for audio transmission in the form of VSB-IM and FM-IM.

In a more advanced high-capacity lightwave system, a technique called microwave subcarrier multiplexing (SCM) is used to multiplex both multichannel analog and digital signals in the same system.

Summary

- The chapter discusses typical analog and digital optical fiber communication link consisting of the transmitter, fiber optic channel, and the receiver.
- An optical transmitter consists of an encoder (for digital optical communication) or a signal-shaping circuit such as an electrical filter (for analog optical communication) followed by a driver circuit which is generally used to modulate the optical source.
- The modulated light is launched into the optical fiber which acts as the transmission channel.
- The information bearing light propagates along the fiber and reaches the receiver-end where it is first converted into an electrical signal and the processed to recover the signal.
- Digital optical communication is used in general purpose optical communication system; analog optical communication finds application in the distribution of cable TV network.
- Optical transmitters use LED or ILD sources and transistor driver circuits. A variety of drive circuits are available for both analog and digital modulation.
- ILD drive circuits are much more complex than LED drive circuit because the former needs special attention for thermal stabilization.
- In a digital optical communication link errors may occur under the three situations, e.g. large intersymbol interference (ISI) arising out of dispersion introduced by the optical fiber waveguide, unacceptably high S/N ratio and phase distortion (jitter).

- For binary digital optical communication, the bit-error rate can be obtained as:

$$\text{BER} = P_e(Q) = \frac{1}{2} erfc\left(\frac{Q}{\sqrt{2}}\right)$$

where, $$Q = \frac{v_{\text{th}} - b_{\text{off}}}{\sigma_{\text{off}}} = \frac{b_{\text{on}} - v_{\text{th}}}{\sigma_{\text{on}}}$$

- In order to maintain a BER less than 10– 9 it is recommended that $Q \approx 6$.
- For, $\sigma_{\text{off}} = \sigma_{\text{on}} = \sigma$ and $b_{\text{on}} = V$ and $b_{\text{off}} = 0$ so that $v_{\text{th}} = V/2$

$$\text{BER} = P_e(\sigma_{\text{on}} = \sigma_{\text{off}} = \sigma) = \frac{1}{2}\left[1 - erf\left(\frac{V}{2\sqrt{2}\sigma}\right)\right]$$

- The design of an optical fiber communication link involves analysis in respect of maximum possible length of the link, maximum rate at which data can be transmitted over the link (for a digital fiber optic link) and the Bit Error Rate (BER) of the overall link.
- The system design for an optical link is generally carried out in the following two-steps e.g. Link power budget and Risetime budget.
- The total loss over the link consists of components such as average loss of the optical fiber over the link at the desired wavelength of operation, additional loss arising out of various fiber joints/splices, coupling loss at the source end (transmitter side) and at the detector end (receiver side) and a safety margin.
- The total loss over the fiber link can be expressed as

$$\mathcal{L}(dB) = \alpha_f L + S_l + C_t + C_r$$

- The additional power required for combating the pulse dispersion is called dispersion-equalization penalty.
- The maximum link length without regenerative repeater in the presence of dispersion-equalization power penalty can be estimated as

$$L(km) = \frac{P_T - P_R - C_t - C_r - D_L}{\alpha_f + \alpha_s}$$

- The overall system rise-time can be expressed as

$$T_{sys} = \left(T_T^2 + T_C^2 + T_M^2 + T_R^2\right)^{\frac{1}{2}}$$

where, T_T is the source rise-time, T_C and T_M are the rise-time due to intramodal orchromatic and intermodal dispersion respectively and TR is the detector rise-time.
- The analog optical communication does find applications in microwave subcarrier multiplexing system, video distribution in CATV (Community Antenna Television), direct cable television system, etc.
- Baseband and Subcarrier Double Side Band Intensity Modulation (DSB-IM) forms of analog optical communication are discussed.

Problems

9.1 Calculate the voltage peak signal-to-noise ratio (V/σ) in dB required to maintain a BER of 10^{-11} for a binary baseband digital optical communication link assuming that all noise components are Gaussian in nature.

9.2 A high-speed synchronous optical network operating at a bit rate of 622 Mbps has a peak signal-to-noise ratio (V/σ) of 10.8 dB. Estimate the time rate at which a bit is misinterpreted at the receiver-end.

9.3 Consider a digital binary baseband data link that uses an APD-based receiver operating at a multiplication gain of 100 with binary "1" designated by a pulse of amplitude V and a binary "0" as 0 V over the bit period. Assuming that the decision threshold is set at the middle of "0" and "1" signal levels compute the average number of photons required to be received by the APD to ensure a bit-error-rate of 10^{-9}. The excess noise factor of the APD can be expressed as

$$F(M) \approx M^{0.4}$$

Estimate the number of photons required to be received by the APD to ensure a bit-error-rate of 10^{-9}.

9.4 An optical fiber link of 10 km length is established by end-to-end splicing of twenty pieces of fibers of length 500 m each. If the splice loss is 1.5 dB km^{-1} and the average loss of the optical fiber at the operating wavelength is 2.5 dB km^{-1}, calculate the total loss over the link. Assume that the connector losses at the transmitter and receiver-ends are 3.5 dB and 2.5 dB, respectively.

9.5 Repeat Problem 9.4 by assuming the splice loss to be 0.3 dB each.

9.6 A long-haul optical communication system working at 48 Mbps uses various optical components with the following parameters.

Ratio	*Values*
Mean power launched by the transmitter into the optical fiber	–3 dBm
Receiver sensitivity	–42 dBm
Average fiber loss	0.5 dBkm^{-1}
Splice loss	0.2 dBkm^{-1}
Transmitter connector loss	3.5 dB
Receiver connector loss	1.5 dB

Estimate the maximum possible link length without a repeater by incorporating a safety margin of 5 dB at the operating bit rate.

9.7 An average optical power of –3 dBm is launched by the transmitter of an optical fiber data link operating at 25 Mbps. The receiver uses an avalanche photodiode with a sensitivity of –45 dBm. If the average loss introduced by the fiber is 0.8 dBkm^{-1} and the splice loss is 0.2 dB per splice, calculate the repeaterless link length by assuming that the link contains 20 splices, the connection loss at the transmitter and the receiver-ends are 2.5 dB and the system requires a safety margin of 8 dB.

9.8 A 25 km long optical fiber data link operating at 25 Mbps uses an optical receiver with a sensitivity of –48 dBm. The mean power available at the end of the transmitter pigtail of an optical fiber data link is –3 dBm. The average loss introduced by the fiber is 1.0 dBkm^{-1} and the average splice loss over the link is 0.5 dBkm^{-1}. The connector losses at the transmitter and the receiver-ends are 3.5 dB and 1.5 dB, respectively. Examine the viability of the link by considering the desired safety margin of 6 dB.

9.9 Repeat problem 9.8 by considering the dispersion-equalization penalty for the above link by assuming the rms pulse broadening resulting from intermodal dispersion to be 0.5 nskm^{-1}.

9.10 Repeat problem 9.9 in the presence of mode coupling.
[Hint: In the presence of mode coupling, the rms pulse broadening becomes

$$\sigma_{mc} = \sigma\sqrt{L} = 0.5\sqrt{25} = 2.5 \text{ ns}]$$

9.11 Repeat Problem 9.9 by assuming the link to operate at a bit rate of 100 Mbps. Compare and contrast the two results.

9.12 The spreadsheet of a link power budget is presented below. Fill in the minimum power in dBm required to be launched by the transmitter into the fiber.

Parameters	*Values*
Minimum optical power to be launched into the line fiber at the transmitter-end	
Sensitivity of the receiver at 25 Mbps	–45 dBm
Total fiber attenuation	20 dB
Total splice loss	10 dB
Receiver connector loss	3.5 dB
Desired safety margin	6 dB
Dispersion-equalization power penalty	2.5 dB

9.13 A long-haul digital optical communication link operating at 1550 nm using a single-mode optical fiber has the following parameters.

Parameters	*Values*
Mean power launched by the transmitter into the optical fiber	–2 dBm
Receiver sensitivity at 56 Mbps	–50 dBm
Receiver sensitivity at 200 Mbps	–45 dBm
Average fiber loss	0.3 dBkm^{-1}
Splice loss	0.2 dBkm^{-1}
Transmitter connector loss	2.5 dB
Receiver connector loss	1.5 dB
Safety margin	6 dB

Estimate the following:

i. Maximum possible repeaterless link length without dispersion-equalization penalty at 56 Mbps.

ii. Maximum possible link length without repeater at 200 Mbps assuming the dispersion equalization penalty to be 2.5 dB for the link at this operating bit rate.

9.14 A 12 km long digital optical fiber data link using an LED source with a spectral width of 50 nm. The receiver is based on a *p-i-n* photodetector with a bandwidth of 100 MHz. The multimode fiber used in the link offers a chromatic dispersion of 15 ps nm^{-1} km^{-1}. The multimode fiber used in the link has a bandwidth-distance product of 300 MHz.km. The modal dispersion can be expressed empirically as

$$T_{\text{mod}} = \frac{0.45\sqrt{L}}{B_0}$$

where, B_0 is the dispersion limited bandwidth per km of the link.

Prepare the system rise-time budget assuming the rise-time of the optical transmitter-end is 15 ns.

9.15 Repeat the rise-time budget for problem 9.14 assuming that the LED is replaced by an ILD with a spectral width of 2 nm to have a transmitter rise-time of 25 ps. Compare and contrast the results in the two cases.

9.16 A 10 km long digital optical communication link operating at 1330 nm using a single mode optical fiber has the following parameters.

Parameters	*Values*
Mean power launched by the transmitter into the optical fiber	–2 dBm
Transmitter rise-time	2 ns
Receiver sensitivity at 56 Mbps	–50 dBm
Average fiber loss	0.3 dBkm^{-1}
Splice loss	0.2 dBkm^{-1}

(*Contd.*)

(*Contd.*)

Chromatic dispersion	12 nsnm^{-1} km^{-1}
Transmitter connector loss	2.5 dB
Receiver connector loss	1.5 dB
Safety margin	6 dB

Prepare the link power budget and the system rise-time analysis to examine the viability of the link.

9.17 The receiver of a D-IM analog optical link working with an index of modulation of 0.5 at 1330 nm is required to maintain an rms signal-to-noise ratio of 40 dB. The quantum efficiency of the photodetector is 80% at the operating wavelength. The post-detection bandwidth of the receiver is 10 MHz. Estimate the power required to be received by the receiver in dBm under quantum noise-limit conditions.

9.18 An analog optical link operating at 850 nm using a D-IM scheme has the following parameters.

Parameters	*Values*
Index of modulation	50%
Quantum efficiency	70%
Effective load resistance of the photodetector	500 kΩ
Preamplifier noise figure	12 dB
Post detection bandwidth	500 kHz

Estimate the optical power required by the receiver to ensure an rms signal-to-noise ratio of 30 dB at 300 K when the performance of the receiver is dominated by thermal noise component. Compare and contrast this value with that obtained when the dark current of the photodetector is as high as 1 nA.

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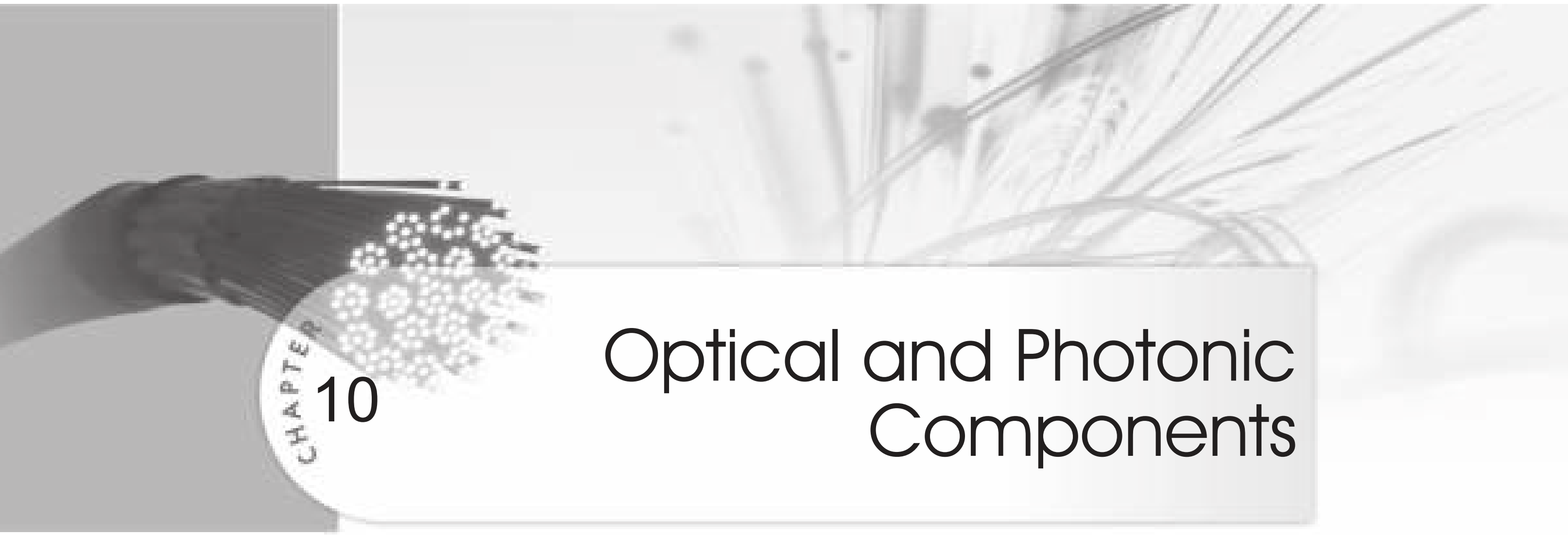

CHAPTER 10 Optical and Photonic Components

The previous chapters focused on the components and devices used in traditional optical fiber communication systems. Conventional optical fiber communication involves several optoelectronic components involving conversion of the signal from the optical to the electrical domain and vice-versa. This frequent change over the domain drastically affects the speed and the reliability of the system. Because of this fact, it has been envisaged that the performance of optical fiber communication and other allied systems can be greatly improved by confining all the processing of the signal in the optical domain only and avoiding the conversion in the electrical domain where the performance gets constrained by the limitations of electrical circuits. A large number of active optical devices and components have been developed to manipulate the signal in the optical domain itself. This chapter deals with major developments in the area of all-optical or photonic devices that do not call for the conversion of the signal in the electrical domain for processing. The possible applications of these devices and components in optical fiber communication systems are also discussed in this chapter. The major optical devices and components may be used either in the discrete or in the integrated form. Integrated optics (IO) deals with the integration of optical components and devices on a single substrate. The major optical devices and components include optical amplifiers, planar waveguides, optical splitters, modulators, switches, logic gates, optical filters, etc.

10.1 OPTICAL AMPLIFIERS

An optical amplifier is an in-line optical device that amplifies the optical signal directly in the optical domain without converting it into the electrical domain as usually done by a regenerative repeater in the traditional optical fiber communication system. The advent of optical amplifiers in the late 1980s paved the way for the development of the present generation fiber optic communication systems. The optical amplifiers are bidirectional and offer enough linearity for simultaneously amplifying signals in a multiplexed form without any cross-talk. The optical amplifiers can provide amplification to any optical signal irrespective of the modulation format and bit rate of transmission. These features of optical amplifiers make them the ideal choice in low-dispersion long-haul communication links for boosting optical signal levels without making use of regenerative repeaters. It may be recalled here that regenerative repeaters not only boost the signal by processing it in the electrical domain but also reshape the signal which is distorted during propagation due to the dispersion characteristics of the fiber. However, by making use of high-quality single mode fibers that do not suffer from intermodal dispersion in long-haul communication, it is possible to enhance the distance between the repeaters by making use of optical amplifiers to take care of the attenuation (Shimada *et al* 1984).

There are two major types of optical amplifiers available at present. These are:

1. Semiconductor optical amplifier (SOA), also known as semiconductor laser amplifier (SLA) exploits the injected carriers to produce stimulated emission.
2. Doped fiber amplifier (DFA) or simply fiber amplifier (FA) which provides gain through stimulated raman scattering (SRS) or stimulated brillouin scattering (SBS).

Both amplifiers can provide high gain over the spectral range, primarily used for present-generation optical fiber communication systems. The semiconductor amplifiers may be further subdivided into three classes depending on the configuration and mode of operation

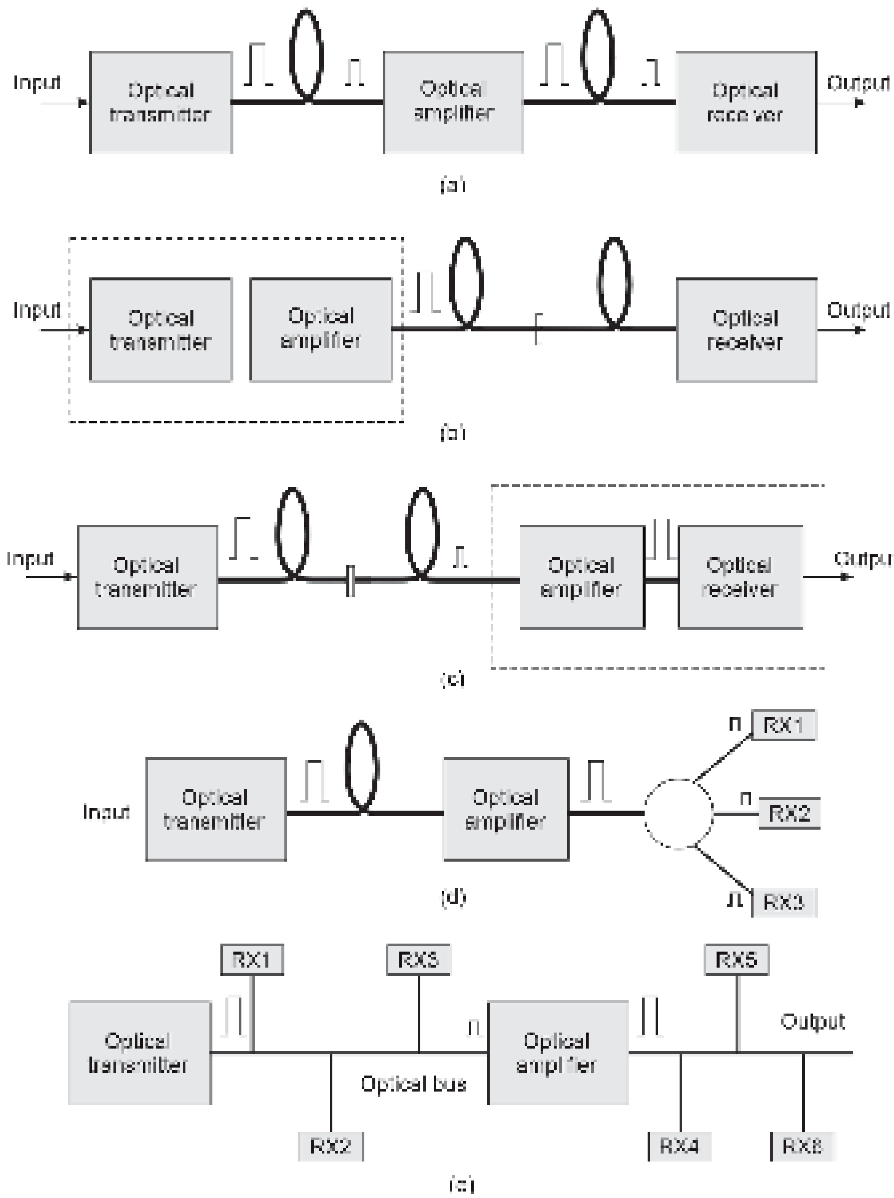

Fig.10.1 Schematic diagram illustrating the use of optical amplifiers for (a) In-line amplification for boosting optical signal at an intermediate point (b) Power boosting of optical transmitter (c) Increasing the sensitivity of the optical receiver (d) Boosting optical signal for distribution in local area network (e) Intermediate boosting for cable TV distribution using a single optical bus

e.g. resonant or Fébry-Perot amplifier which is essentially an oscillator-biased below threshold (Buus *et al* 1985), the traveling wave (TW), and the near traveling wave (NTW) amplifiers which are single-pass devices (O'Mahony, 1989; Eisenstein *et al* 1987). The optical amplifiers can be used for amplifying the optical signal at different points on the optical link as illustrated in Fig. 10.1. In a long-distance optical communication link, optical amplifiers can be used at intermediate points to boost the optical signal (Fig.10.1(a)). The incorporation of an optical amplifier in the transmitter module can significantly improve the optical power output of the transmitter (Fig. 10.1(b)). Similarly, the sensitivity of an optical receiver can be significantly improved by incorporating the optical amplifier before the photodetection (Chakrabarti *et al* 2003) as illustrated in Fig. 10.1(c). A semiconductor laser amplifier can be integrated with the transmitter or receiver module. Further optical amplifiers can be used as boosters in local area networks (Fig. 10.1(d)) or cable TV distribution systems (Fig. 10.1(e)).

10.1.1 Semiconductor LASER Amplifier (SLA)

The amplification of light in an optical amplifier (SLA or FA) is achieved with the help of stimulated emission. The mechanism of amplification in a semiconductor laser amplifier is similar to that of an Injection Laser Diode (ILD) except for the fact that the optical feedback is forbidden in the former case purposely to stop the device from lasing. The absence of optical feedback enables it to work as an amplifier rather than an oscillator which generates the coherent output without any input. Semiconductor laser amplifiers are generally available in two forms, e.g. Fabry-pérot amplifier (FPA) and travelling wave amplifier (TWA) (O'Mahony, 1988). The difference between the modes of operation of the two forms of SLA lies in the reflectivity of the front and rear facets. The schematic of an SLA is shown in Fig. 10.2. Structurally, the device resembles an ILD in the gain-guided or index-guided form discussed earlier. When the reflectivities of the facets of a typical injection laser diode are reduced to values in the range of 1–30% (O'Mahony, 1988), the structure behaves like a resonant amplifier rather than an oscillator. Reduction of the reflectivities of the facets enhances the value of the threshold current required for lasing. This is illustrated in Fig. 10.3. The device is biased by a current less than the threshold current required for lasing. The device is called the FPA. The amplifier is available in the form of a chip with fiber pigtails bonded at both ends for guiding light in and out of the amplifier.

By reducing the reflectivities of the two facets further down to the range of 10^{-3} or so, the structure can be used as a TWA. Low reflectivities of the facets can be achieved by

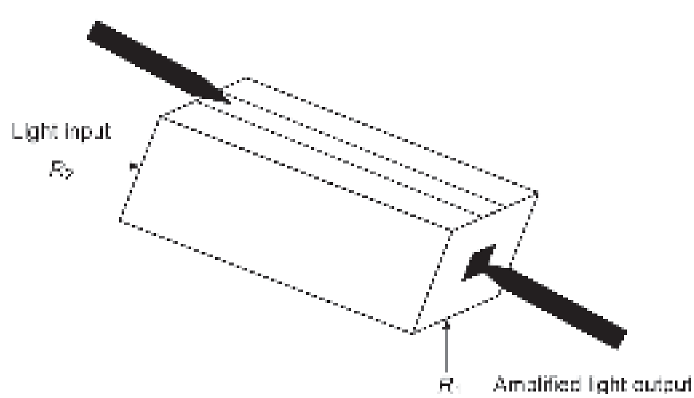

Fig. 10.2 Schematic of a semiconductor laser diode with tapered fibers at the two ends for signal in and out

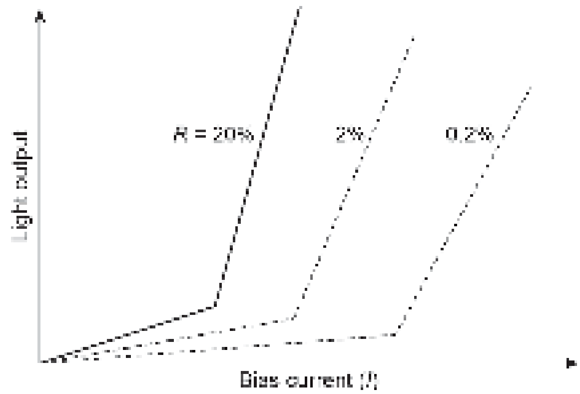

Fig. 10.3 Light output *vs* bias current characteristics of a FP cavity laser diode

making use of antireflection coatings of SiO_2 or Si_3N_4. Under this condition, the device works under single-pass amplifier mode. Reduction of the reflectivities suppresses the resonance in the FP cavity enabling the amplifier to operate over a broad spectral range and also enhances the threshold current for lasing. The dependence of threshold current on the reflectivities of the two facets ($R_1 = R_2 = R$) is shown in Fig. 10.3. For a true traveling wave amplifier, the facet reflectivities should be ideally zero. In practice, the reflectivities are generally low but have finite values. For this reason, TWA is often referred to as near travelling wave amplifier (NTWA). Semiconductor laser amplifiers generally make use of III-V semiconductor heterostructures and can be used in the wavelength ranges around 1.3 μm and 1.5 μm. In a semiconductor laser amplifier, the carriers may recombine spontaneously and radiatively to yield photons which are subsequently amplified and manifested in the form of amplified spontaneous emission (ASE).

10.1.1.1 Optical Gain

In a semiconductor laser amplifier population inversion (pumping) required for substantial stimulated emission is created by the injection of external current into the device like that done in the case of a laser diode. The effective gain of the amplifier, g (m^{-1}) can be defined in terms of material gain, g_m (m^{-1}), effective loss coefficient, $\overline{\alpha}$ (m^{-1}) and the optical confinement factor as (O' Mahony, 1988)

$$g = \Gamma g_m - \overline{\alpha} \tag{10.1}$$

The rate equation governing the carrier density $n(t)$ in the excited state in the presence of spontaneous as well as stimulated emission can be expressed as

$$\frac{dn(t)}{dt} = \frac{J(t)}{qd} - R(n) - R_{st}(t) \tag{10.2}$$

where, $n(t)$ is the number of carriers per unit volume in the active region at any instant of time, $J(t)$ is the current density, and q is the electronic charge. The second term on the right-hand side of Eq. (10.2) corresponds to the rate of recombination by spontaneous emission, τ_r being the radiative recombination lifetime of the carriers in the excited state and the last term corresponds to the recombination rate of carriers responsible for stimulated emission and is given by

$$R_{st} = \frac{\Gamma g_m}{E}(\beta I_{sp} + I) \tag{10.3}$$

where, I is the signal intensity, I_{sp} is the intensity of the total spontaneous emission, E is the photon energy, β is the spontaneous emission coefficient which is attributed to the fraction of the total spontaneous emission coupled to the travelling wave, given by (O'Mahony, 1988)

$$\beta = \frac{c}{N}\frac{a\tau}{V} \tag{10.4}$$

where, c is the velocity of light, N is the refractive index of the material, a is the gain cross-section and V is the volume of the active region which is the product of the lengths of the active region in the longitudinal, lateral, and transverse directions.

Further, the recombination rate can be approximated as

$$R(n) = \frac{n}{\tau} \tag{10.5}$$

Under steady-state condition

$$\frac{dn(t)}{dt} = 0 \tag{10.6}$$

Therefore, under steady-state condition using Eqs (10.3)–(10.4) and (10.5)–(10.6), we may write (O'Mahony, 1988)

$$\frac{J}{qd} = \frac{n}{\tau} + \frac{\Gamma g_m}{E} I \tag{10.7}$$

The material gain coefficient can be expressed as (Mukai *et al* 1983; Adams *et al* 1985)

$$g_m = \frac{g_0}{1 + \dfrac{I}{I_s}} \tag{10.8}$$

where g_0 is the unsaturated material gain coefficient in the absence of input signal and I_s is the saturation intensity given by (O' Mahony, 1988)

$$I_s = \frac{E}{\Gamma a \tau} \tag{10.9}$$

The single pass gain can be expressed as

$$G_s = \exp\left[\left(\frac{\Gamma g_0}{1 + \dfrac{I}{I_s}} - \bar{\alpha}\right)L\right] \tag{10.10}$$

10.1.1.2 Phase-shift

The total phase-shift associated with the single-pass amplifier can be expressed as (Adams *et al* 1985)

$$\Phi_s = \frac{2\pi n L}{\lambda} + \frac{g_0 b L}{2}\left(\frac{I}{I + I_s}\right) \tag{10.11}$$

where the first term on the right-hand side of Eq. (10.11) corresponds to the nominal phase-shift, the second term attributes to an additional phase-shift caused by the change in carrier density from the nominal value in the absence of the input signal, b corresponds to the line-width broadening factor, and n is the refractive index of the active region.

In a practical SLA, the reflectivity of the facets is generally kept very low by making use of anti-reflection coatings. However, in a practical FPA, the reflectivity of each facet has a finite value. The single-pass gain, G_s corresponds to the one-round trip gain of the cavity. Assuming that the time period corresponding to the highest signal frequency is much less than the round trip time of the cavity, the frequency-dependent gain of the cavity can be expressed as (O'Mahony 1988; Saitoh *et al* 1988)

$$G(\nu) = \frac{(1-R_1)(1-R_2)G_s}{(1-\sqrt{R_1 R_2}G_2)^2 + 4\sqrt{R_1 R_2}G_s \sin^2 \Phi} \tag{10.12}$$

where, R_1 and R_2 are the reflectivities of the front and the rear facets, respectively, and Φ is the single pass phase shift caused by the amplifier given by

$$\Phi = \frac{\pi(\nu - \nu_0)}{\delta\nu} \tag{10.13}$$

where, ν_0 is the resonant frequency of the FP cavity and $\delta\nu$ is the free spectral range of the SLA.

The 3dB bandwidth of the FPA is measured in terms of full width at half maximum (3dB) of single longitudinal mode bandwidth illustrated in Fig.10.4. The 3-dB bandwidth of the SLA can be expressed as (Saitoh *et al* 1988)

$$\begin{aligned} B = 2(\nu - \nu_0) &= \frac{2\delta\nu}{\pi}\sin^{-1}\left[\frac{1-\sqrt{R_1R_2}G_s}{2(\sqrt{R_1R_2}G_s)^{1/2}}\right] \\ &= \frac{c}{\pi n L}\sin^{-1}\left[\frac{1-\sqrt{R_1R_2}G_s}{2(\sqrt{R_1R_2}G_s)^{1/2}}\right] \end{aligned} \tag{10.14}$$

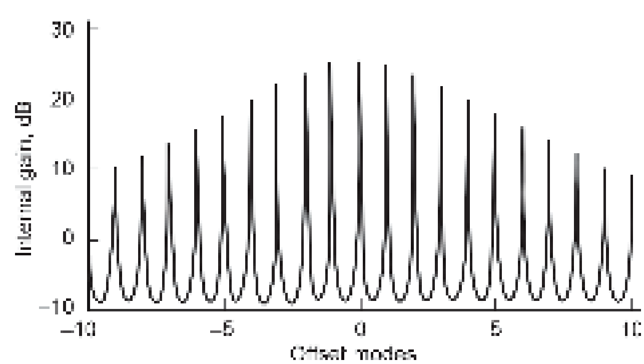

Fig. 10.4 An illustration of the passband of a Fabry-Perot amplifier wherein mode 0 corresponds to the peak gain wavelength (O' Mahony, 1988)

The 3dB optical bandwidth may alternatively expressed as a function of the FP cavity gain as (O'Mahony *et al* 1987; O'Mahony, 1988)

$$B = \frac{c}{\pi n L}\sin^{-1}\left[\frac{1}{2}\left(\frac{(1-R_1)(1-R_2)}{\sqrt{R_1R_2}G}\right)\right] \tag{10.15}$$

Example 10.1 A Fabry-Perot amplifier (FPA) has uncoated facets with reflectivities of 20% each. The single-pass gain of the cavity is 6.02 dB, the mode spacing is 1 nm and the peak-gain wavelength of the amplifier is 1550 nm. If the refractive index of the cavity is 3.7, estimate the length of the cavity and the 3dB bandwidth of the amplifier.

Solution The length of the cavity can be

$$L = \frac{\lambda^2}{2n\Delta\lambda} = \frac{(1550\times 10^{-9})^2}{2\times 3.6\times 10^{-9}}$$

$$= 325\ \mu\text{m}$$

Gain of the cavity (in terms of ratio) is

$$G_s = \text{antilog } (0.602) = 4$$

The 3 dB bandwidth of the FP laser amplifier can be obtained using Eq. (10.15) as

$$B = \frac{3\times 10^8}{3.14\times 3.7\times 325\times 10^{-6}} \sin^{-1}\left[\frac{1}{2}\times\frac{1-\sqrt{0.2\times 0.2}\times 4}{(\sqrt{0.2\times 0.2}\times 4)^{1/2}}\right]$$

$$= 7.94\times 10^{10}\sin^{-1}\left(\frac{0.2}{2\times(0.2\times 4)^{1/2}}\right)$$

$$= 7.94\times 10^{10}\times 0.112 = 8.89\text{ GHz}$$

It may be pointed out here that the under ideal condition $R_1 = R_2 = 0$ and the FPA behaves in single-pass mode and is known as a pure TWA. The TWA is viewed as a non-resonant counterpart of resonant FPA. The spectral bandwidth of the TWA corresponds to the full gain width of the amplifying medium. This is illustrated in Fig. 10.5. For NTWA the passband consists of ripples, alternate peaks and troughs. The gain undulation is measured in terms of the peak-trough ratio of the passband ripples. The relative amplitudes of the peaks and troughs are dependent on the facet reflectivities, bias current, and the input signal level. The peak–trough ratio of the passband ripple can be obtained as (O'Mahony, 1988)

$$\Delta G = \left(\frac{1+G_s\sqrt{R_1R_2}}{1-G_s\sqrt{R_1R_2}}\right)^2 \tag{10.16}$$

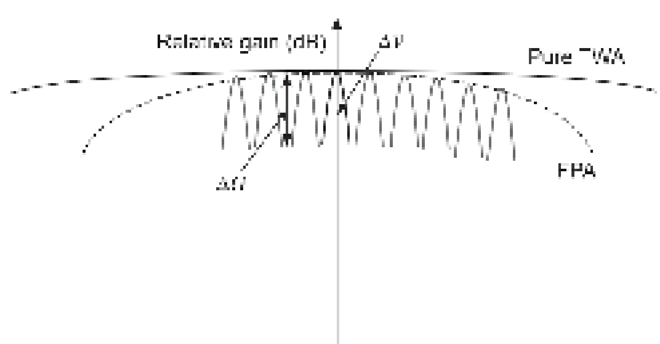

Fig. 10.5 Passband characteristics of pure TWA and FPA illustrating ripples

For a 3dB peak to trough ratio, we may write

$$\left(\frac{1+G_s\sqrt{R_1R_2}}{1-G_s\sqrt{R_1R_2}}\right)^2 = 0.5$$

Therefore, the single-pass cavity gain of the TWA becomes

$$G_s = \frac{0.172}{\sqrt{R_1 R_2}} \tag{10.17}$$

The reflectivities of the two facets can be significantly reduced by making use of antireflection coatings. Under this condition, the FPA behaves like a TWA with wide spectral bandwidth. TWA generally requires a higher bias current as compared to that required by an FPA. Moreover, a large spectral bandwidth of TWA is responsible for a higher noise level as compared to that of FPA which has a smaller bandwidth and consequently a lower noise level.

Further, the residual reflectivity of the facets introduces an additional problem when used in an optical communication system. The effect of residual reflectivity is manifested in the form of gain of the backward traveling signal. The gain, G_b of the backward traveling signal is defined as the ratio of the power in the backward traveling signal, P_b to the input signal power, P_{in} into the amplifier. The gain of the backward traveling signal can be obtained as (Henning *et al* 1985)

$$G_b = \frac{P_b}{P_{\text{in}}} = \frac{(\sqrt{R_1} - \sqrt{R_2}G_s)^2 + 4\sqrt{R_1 R_2}G_s \sin^2 \Phi}{(1 - \sqrt{R_1 R_2}G_s)^2 + 4\sqrt{R_1 R_2}G_s \sin^2 \Phi} \tag{10.18}$$

The effect of backward gain can be avoided by making use of an optical isolator.

10.1.2 Fiber Amplifiers

Optical fibers are generally viewed as passive components in the sense that optical power decreases as the light propagates down the fiber. This means that the power available at the output-end of the fiber is less than that of the input. The loss or attenuation of the fiber is expressed as

$$\alpha(\text{dB}) = 10\log_{10}\left(\frac{P_{\text{in}}}{P_{\text{out}}}\right) \tag{10.19}$$

A glass fiber can however be doped with a suitable rare-earth element, such as erbium (Er), neodymium (Nd), holmium (Ho), and yetterbium (Yb) to act as an active component. The rare earth element creates an energy state in the material in such a way as to provide gain due to stimulated emission (Koester *et al* 1964; Stone *et al* 1973; Cochrane, 1990; Urquhart, 1988; Aoki, 1988; Mears *et al* 1987). These fiber amplifiers can be used for optical amplification without O–E conversion in the same manner as in the case of semiconductor laser amplifiers. The most popular fiber amplifier makes use of silica fiber doped with erbium. Such a fiber amplifier is therefore known as an erbium-doped fiber amplifier (EDFA). EDFAs are very popular because of the fact these amplifiers work in the spectral region close to 1.55 μm which is the target wavelength for the present generation optical communication system. The schematic of a fiber amplifier is shown in Fig. 10.6. A typical fiber consists of a rare-earth-doped silica fiber of a short length (10–30 m).

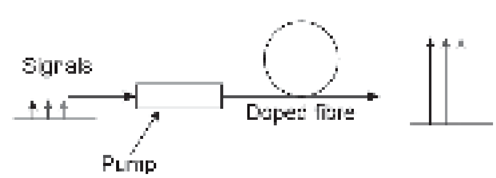

Fig. 10.6 Schematic of a fiber amplifier

To create population inversion in the doped fiber, it is necessary to use a pump source in the form of another laser diode operating at the pump wavelength, λ_p. In an EDFA, the pump photons excite Er^{3+} ions to produce population inversion. The input signal while propagating through the doped fiber in which population inversion is already created triggers stimulated emission at the wavelength λ_s, which is the same as that of the input signal and thereby causes amplification at the operating wavelength. The amplified signal is subsequently coupled to the desired section of the optical fiber link as required. A typical setup of an EDFA amplifier is illustrated in Fig. 10.7.

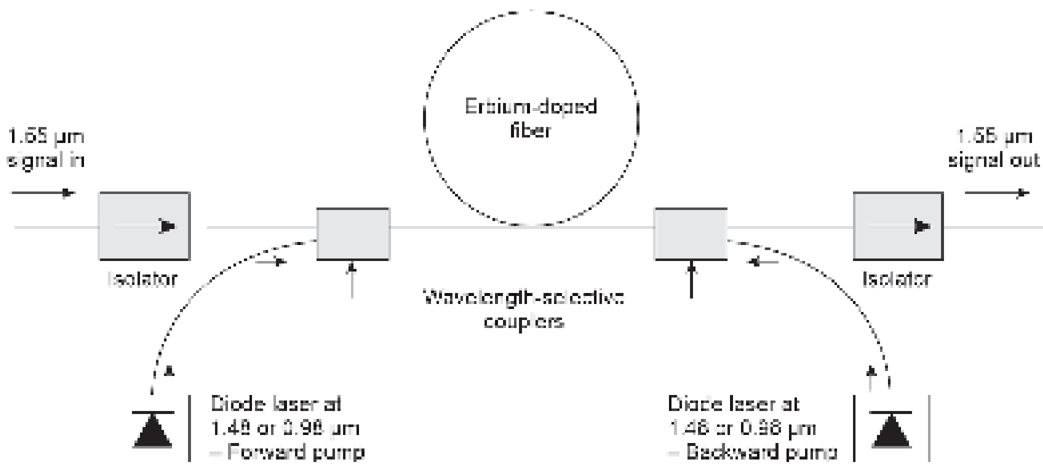

Fig. 10.7 A schematic illustrating the use of EDFA in an optical fiber link

10.1.2.1 Mechanism of operation of ebrium-doped fiber amplifier

An EDFA consists essentially of Er-doped silica fiber. To understand the basic mechanism of operation of an EDFA, it is necessary to know the energy level diagram of Er-doped silica. When Er^{3+} ions are embedded into amorphous silica, the individual energy levels are split into many sublevels which form energy bands as illustrated in Fig. 10.8. In EDFA, optical pumping is used to create population inversion. Three energy levels, e.g. ground level, higher excited state (pump level), and a metastable level are primarily involved in the stimulated emission process in EDFA. The pumping is usually achieved with the help of a laser diode operating at a suitable wavelength to irradiate the doped fiber. The energy of the pumping photon is decided by the difference in the energy between the ground-level ($^4I_{15/2}$) and the pumping level ($^4I_{13/2}$). For an Er-doped fiber, a laser diode operating at 980 nm is used as the pump source. The photons emitted by the pump source are absorbed by the doped fiber and the system is raised to a higher excited state ($^4I_{11/2}$). As the pumping level is an unstable state, the electrons excited to this state rapidly lose a part of their energy to decay non-radiatively

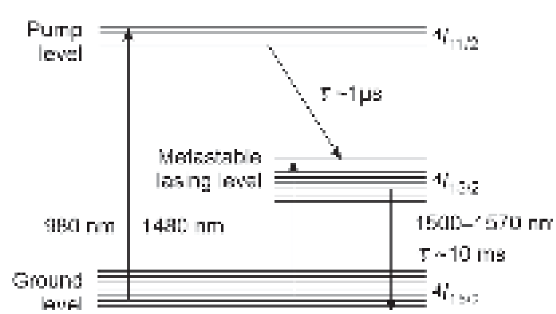

Fig. 10.8 Energy-band diagram of Er^{3+}-doped silica fiber

and fall to the metastable state (also called lasing level). When the pump power is sufficiently high, the population in the metastable state may exceed that in the ground state. Thus, a population inversion can be created through pumping. In EDFA, the energy difference between the ground state and the metastable state corresponds to the emission wavelength of 1550 nm. If an input signal consisting of photons of the same wavelength is allowed to pass through the doped fiber under population inversion conditions, the propagating photons can trigger a stimulated emission from the metastable state to the ground state to produce new photons that are identical to the input photons and thereby giving significant amplification of the input light signal.

It is also possible to use a pump source operating at 1480 nm to excite the electrons from the bottom of the ground state ($^4I_{11/2}$) directly to the lightly populated top metastable state ($^4I_{13/2}$). These excited electrons subsequently relax to drop down to the bottom-end of the metastable state which is relatively more densely populated. In the presence of incoming photons stimulated emission finally occurs between the lower-end states of the metastable band and the upper states of the ground state resulting in light amplification in the range of 1550 nm. Normally, stimulated emission may occur in the wavelength range between 1520 nm and 1570 nm. However, in the absence of incoming photons the electrons from the metastable states may drop down randomly to the ground state giving rise to spontaneous emission.

Practical EDFA makes use of pump laser sources operating at 980 nm to create population inversion as they are readily available in the market and can provide pump power in the range of 50–100 mW. Pump laser sources operating at 1480 nm can also be used for creating population inversion. However, in this case, higher pump power is required for creating population inversion. EDFAs can offer gain in the range of 30–40 dB with low noise (Mears *et al* 1987; Senior *et al* 1989).

The transitions within the energy bands in Er-doped silica fiber can be described in terms of absorption and emission cross-sections (Desurvire et al 1989; Desurvire, 1994). The total gain of an EDFA fiber of length L can be expressed as

$$G = \Gamma \exp\left[\int_0^L (\sigma_{se} N_2 - \sigma_{sa} N_1)dz\right] \tag{10.20}$$

where Γ is the confinement factor, σ_{se} and σ_{sa} are the emission and absorption cross-sections respectively, N_1 is the rate of absorption per unit volume from the ground level E_1 to the excited pump state E_3 and N_2 is the rate of emission per unit volume from the metastable state E_2 to the ground state E_1. The variation of the EDFA amplifier with pump power for different lengths of the fiber is shown in Fig. 10.10(a). Fig. 10.10b shows the variation of the gain of the EDFA amplifier with the length of the fiber for different pump power (Giles *et al* 1991).

10.1.2.2 Applications of eribum-doped fiber amplifier

Fiber amplifiers find several applications in optical fiber communication and data links. The in-line amplification has several advantages over conventional techniques of using regenerative repeaters in long-haul optical communication systems. A few possible applications of EDFA are illustrated in Fig. 10.9 (Urquhart, 1988). Figure 10.9(a) illustrates the use of an EDFA for boosting the output power of a normal optical transmitter. This power boosting enables one to increase the repeater-less distance between the transmitter and the receiver. Fig. 10.9(b) shows the use of an EDFA for boosting the optical signal level at any intermediate point on the link. Fig. 10.9(c) shows that the weak optical signal can be

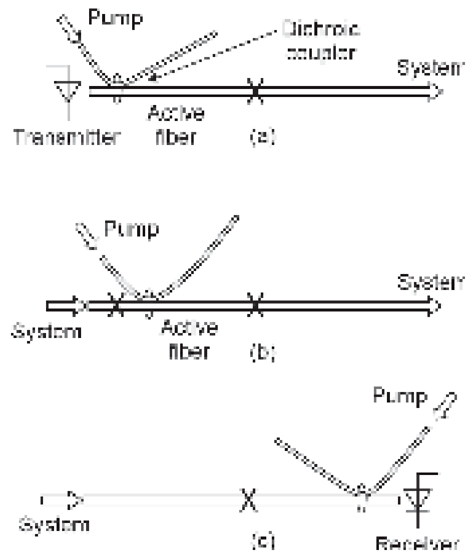

Fig. 10.9 Schematic diagram illustrating various uses of EDFA (after Urquhart, 1988)

boosted in the optical domain with the help of an EDFA just before the receiver. This can reduce the high sensitivity requirement of the receiver in long-haul optical communication systems. In certain receivers, semiconductor laser amplifiers (SLAs) or fiber amplifiers are built-in as the front-end to enhance the receiver sensitivity.

10.1.2.3 Raman and Brillouin fiber amplifiers

In Chapter 4, it is understood that non-linear scattering in optical fibers is an undesirable effect that can create problems for reliable optical fiber communication. Interestingly, non-linear scattering does not become dominant in the level of power range used in optical fiber communication. In any case, the undesirable non-linear scattering properties of an optical fiber can be exploited to achieve amplification. Such amplifications can be achieved by using SRS or SBS. Amplifiers based on SRS are often referred to as fiber Raman amplifiers (FRA). These devices are superior in performance compared to their SBS counterparts. FRAs provide larger gain bandwidth and high-speed response (Aoki, 1988). A typical Raman gain spectrum of a silica glass fiber is illustrated in Fig. 10.10. The Raman gain of a silica fiber has a bandwidth of nearly 40 THz. The high gain region spans between 9 THz and 16 THz. In SRS, the incident lightwave at a frequency v induces a gain in the silica glass fiber (scattering medium) at a different frequency v' given by

$$v' = v - v_r \tag{10.21}$$

where, v_r is the characteristic frequency of Raman-active vibration.

When the incident optical power is above the threshold value, the gain exceeds the loss and the scattered light gets amplified at a frequency, v'. For Raman scattering to occur the material is usually irradiated with the help of an intense monochromatic light provided by the pump source. The basic configuration of a fiber-based Raman amplifier is illustrated in Fig. 10.11. The Raman gain is inversely proportional to the wavelength of the pump source. The gain also depends on the polarization state of the wave. The Raman gain is much higher for the same state of polarization of the pump and Stokes waves as compared to that when they have an orthogonal polarization state. Raman scattering can be obtained both in the forward as well as in the backward direction as demonstrated in Fig. 10.12.

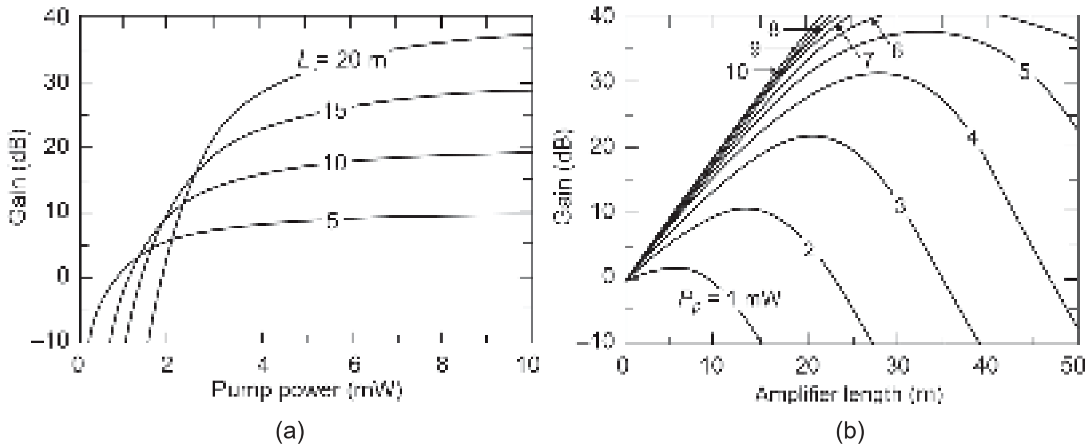

Fig. 10.10 Variation of the gain of an EDFA (a) With pump power for different length of the fiber (b) With amplifier length for different pump power (after Giles *et al* 1991)

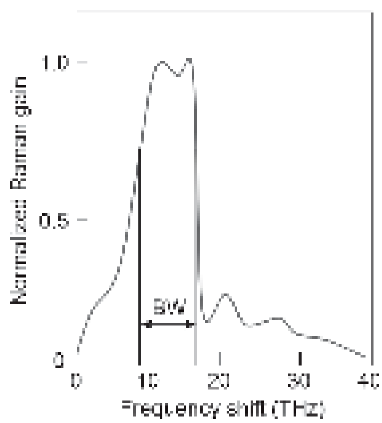

Fig. 10.11 Normalized Raman gain spectrum of a typical silica fiber

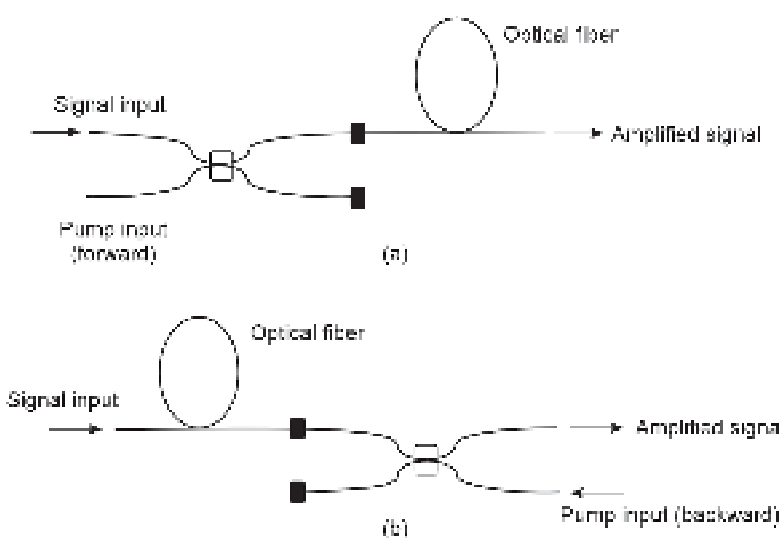

Fig. 10.12 Schematic of Raman fiber amplifier (a) Forward pumping (b) Backward pumping

10.2 ELEMENTS OF INTEGRATED OPTICS (IO)

Integrated optics also known as photonic integrated circuits (PICs) involve the integration of planar lightwave components on a single substrate to perform a variety of complex optical functions. Some of the integrable optical components include planar optical waveguides, optical fibers, optical filters, optical couplers, injection laser diodes, photodetectors, optical amplifiers, optical switches, etc. The term integrated optics was coined by Miller way back in 1969 (Miller, 1969 & Tien, 1971). The motivation behind integrated optics originated from the concept of the technology of microelectronic circuits in the form of integrated circuit chips. The integrated circuit technology of electronic components witnessed dramatic development resulting in the evolution of very large scale integration (VLSI) involving billions of transistors on a single chip. On the other hand, integrated optics could not grow that way as was envisaged in the early years of its inception. Major factors responsible for the limited growth of integrated optics include the following:

1. Integrated optics require optical interconnects to connect various optical components on the chip. The dimension of the optical waveguide cannot be reduced much below the operating wavelength.
2. The design of optical components is much more complex than compared of electronic components.
3. The integration of planar optical devices with vertical structures is technologically quite challenging. For example, integration of an optical amplifier with a directional coupler is much more complex than integrating electronic amplifiers based on transistors in electronic ICs.
4. Unlike electronic components, the size of the optical components cannot be scaled down below a certain limit.

Nevertheless, integrated optics can deliver fairly complex integrated optical devices on a single chip involving optical waveguides, couplers, filters, distributed feedback lasers, optical modulators, and photodetectors. Such integrated optical devices can find useful applications in processing light signals in the optical domain itself without requiring a conversion in the electrical domain. This enables the integrated optic devices to work faster than the traditional optoelectronic circuits where frequent E/O and O/E conversion turn out to be the bottleneck for achieving high-speed processing. These integrated circuits find potential application in the current generation of optical fiber communication based on single-mode fibers and wavelength division multiplexing (WDM). Many of the WDM devices and components are integrable with other optical devices. The optical components can be integrated on the surface of a crystalline material, such as silicon, silicon-on-insulator (SOI), silica, lithium niobate ($LiNbO_3$) (Baumann *et al* 1996), or even on an III-V material substrate such as InP. It may be pointed out that in electronic integrated circuits, Si is used as the principal substrate. Because of its versatility and low cost, Si has been tried extensively to build IO devices. The emergence of silicon photonics shows much promise for future-generation IO devices (Okamoto *et al* 1999; Almeida, 2004; Lipson, 2005; Jalali *et al* 2006).

10.2.1 Planar Waveguides

Planar waveguide is an important component that is extensively used for interconnecting various optical devices and components in an integrated optical device. The schematic of the simplest optical waveguide is shown in Fig. 10.13. It consists of a thin dielectric slab

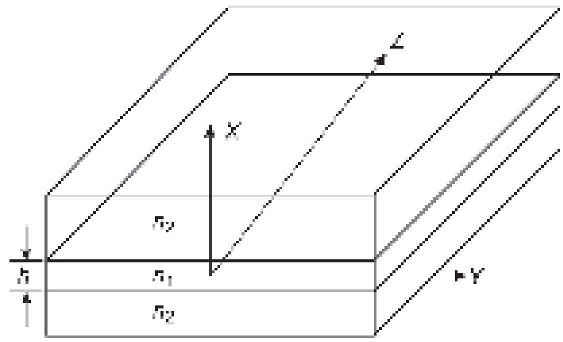

Fig. 10.13 Planar waveguide structure

of refractive index, n_1 sandwiched between two symmetrical dielectric slabs of refractive index, n_2 ($<n_1$). The thickness of the sandwiched layer, h is assumed to be very small as compared to the thicknesses of the confining slabs. The light is launched into the slab in such a way that it propagates in the z-direction. The thicknesses of the confining layers are assumed to be of infinite extent.

The number of guided modes can be obtained by ray analysis and applying the condition for constructive interference of the reflected waves associated with the rays to create the standing wave pattern inside the waveguide. The number of modes can be estimated as

$$M = \frac{2hn_1 \sin\theta_c}{\lambda} = \frac{2h}{\lambda}(n_1^2 - n_2^2)^{1/2} \tag{10.22}$$

where, θ_c corresponds to the critical angle subtended by the most oblique ray at the core-cladding interface.

Modal analysis of planar symmetrical planar waveguide with a step-index profile can be obtained by solving Maxwell's equation under appropriate boundary conditions. Both transverse electric (TE) and transverse magnetic (TM) modes are created in the planar waveguide structure. The modes are designated in terms of TE_m or TM_m modes (m = 0, 1, 2, 3, … … …). The field distribution of the first three order transverse electric modes, e.g. TE_0, TE_1, TE_2 is illustrated in Fig. 10.14.

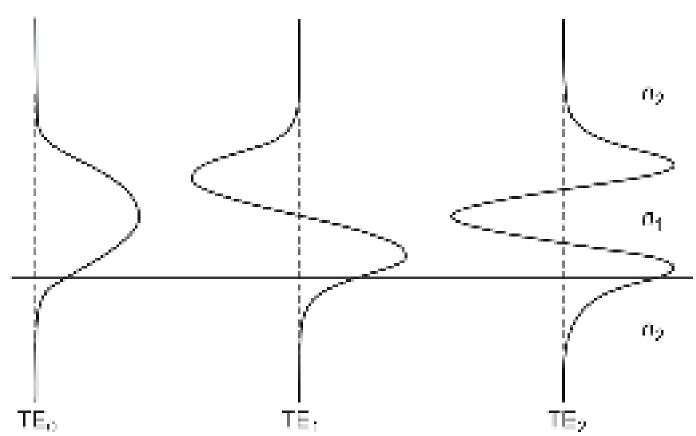

Fig. 10.14 The field distribution of the first three order TE modes (m = 0,1, 2)

The lowest order or fundamental modes are designated as TE_0 or TM_0. A symmetrical planar waveguide supports only one TE and TM mode when the V-number becomes

$$V = \frac{\pi h}{\lambda}(n_1^2 - n_2^2)^{1/2} < \frac{\pi}{2} \tag{10.23}$$

In other words, the thickness of the guiding layer for single-mode operation can be expressed as

$$h < \frac{\lambda}{2(n_1^2 - n_2^2)^{1/2}} \tag{10.24}$$

Example 10.2 Consider a planar waveguide with a step-index profile. The refractive index of the central waveguide region is 1.458 and that of the surrounding region is 1.405 at an operating wavelength of 850 nm. Estimate the maximum number of modes that would be guided by the planar waveguide. The thickness of the central wave-guiding layer is 10 μm. Also, find the range of propagation constant, β.

Solution The maximum number of modes can be obtained by using Eq. (10.22) as

$$M = \frac{2 \times 10 \times 10^{-6}}{850 \times 10^{-9}}[(1.458)^2 - (1.405)^2]^{1/2} = 9$$

The propagation constant lies in the range given by

$$\beta_2 < \beta < \beta_1$$

Here, $$\beta_1 = \frac{2\pi n_1}{\lambda} = \frac{2 \times 3.14 \times 1.458}{850 \times 10^{-9}} = 1.07 \times 10^7 \text{ m}^{-1}$$

and $$\beta_2 = \frac{2\pi n_2}{\lambda} = \frac{2 \times 3.14 \times 1.405}{850 \times 10^{-9}} = 1.03 \times 10^7 \text{ m}^{-1}$$

Therefore, $$1.03 \times 10^7 \text{ m}^{-1} < \beta < 1.07 \times 10^7 \text{ m}^{-1}$$

Example 10.3 Consider a planar waveguide with a step-index profile. The refractive index of the central waveguide region is 1.52 and that of the surrounding region is 1.48 at an operating wavelength of 900 nm. Calculate the maximum thickness of the guiding slab so that the waveguide supports only the fundamental TE mode.

Solution In order that the waveguide to support only the fundamental mode, it is required (see Eq. (10.24))

$$h < \frac{\lambda}{2(n_1^2 - n_2^2)^{1/2}}$$

That is, $$h < \frac{900 \times 10^{-9}}{2[(1.52)^2 - (1.48)^2]^{1/2}}$$

Therefore, $$h < 1.3 \text{ μm}$$

This means that the thickness of the guiding slab should not be more than 1.3 μm.

An asymmetrical waveguide structure consists of a planar film of refractive index n_1 sandwiched between a substrate with a refractive index of n_2 at the bottom and a cover layer of refractive index n_3 on the top such that $n_1 > n_2 \geq n_3$. The top layer often air with a refractive index $n_3 = 1$. When the refractive index of the top and the bottom layer is the same, i.e. $n_2 = n_3$ the waveguide structure becomes symmetrical. For a symmetrical waveguide structure, a finite number of TE or TM modes propagate through the waveguide of the given dimensions at a particular wavelength. As the dimensions of the waveguide are reduced without changing the wavelength the number of modes decreases. Finally, a stage is reached when only one mode survives and the waveguide supports only single-mode propagation. The fundamental mode cannot be eliminated. In the case of asymmetric structure, a situation may be so created by changing the dimensions that even the fundamental mode gets eliminated.

For an asymmetrical waveguide structure with a central guiding region of thickness, h the number of modes, m supported by the waveguide at a given length can be estimated using the following relation (Levi, 1980).

$$h \geq \frac{\left(m + \frac{1}{2}\right)\lambda}{2(n_1^2 - n_2^2)^{1/2}} \tag{10.25}$$

Example 10.4 An asymmetrical ZnS planar waveguide structure has

$$n_1 = 2.35 \text{ and } n_3 = 1$$

at an operating wavelength of 650 nm.

If the refractive index of the substrate is 1.5, estimate the range of values of the thickness of the ZnS film to support at least one guiding mode.

Solution For single mode operation, the thickness h of the film must lie between the values obtained from Eq. (10.25), when $m = 0$ and $m = 1$.

For $m = 1$, the value of the thickness is

$$h = \frac{3\lambda}{4(n_1^2 - n_2^2)^{1/2}}$$

$$= \frac{3 \times 650 \times 10^{-9}}{4[(2.35)^2 - (1.5)^2]^{1/2}} = 0.27\ \mu\text{m}$$

For $m = 0$, the thickness value becomes

$$h = \frac{\lambda}{4(n_1^2 - n_2^2)^{1/2}}$$

$$= \frac{650 \times 10^{-9}}{4[(2.35)^2 - (1.5)^2]^{1/2}} = 0.09\ \mu\text{m}$$

Therefore, for single-mode operation, the thickness of the guiding film must lie between

$$0.09\ \mu\text{m} \leq h \leq 0.27\ \mu\text{m}$$

A variety of materials are used for making planar waveguides for integrated circuit applications. The simplest material that is used for making optical waveguides is silica glass and a few transparent polymers. However, the optical properties of these materials cannot be changed by external energy sources. On the other hand, there is a host of other transparent dielectric materials whose properties can be changed by a variety of external energy sources, such as electrical, magnetic, or acoustic. These materials are called electro-optic, magneto-optic, or acousto-optic materials depending on the form of energy that controls the optical property of the material. Among these materials, electro-optic materials are most widely used for the development of IO devices and components, such as optical modulators, beam splitters, switches, etc.

The principal materials used for making electro-optic IO devices include lithium niobate ($LiNbO_3$); lithium tantalite ($LiTaO_3$); zinc oxide (ZnO); zinc sulphide (ZnS); titanium oxide

(TiO_2), etc. Among the III-V semiconductors GaAs, InP, GaSb, InAs, and their ternary and quaternary alloys are widely used for making optical waveguides (Payne *et al* 1987; Personick, 1987; Joyner *et al* 1987; Thylen, 1988). In addition, a large number of organic polymers are also used for IO applications. The biggest advantage of $LiNbO_3$ is that it is compatible with silicon and therefore can be used for easy integration with Si-based integrated circuits. IO devices can be fabricated by using technology that is very similar to that used in semiconductor device processing including thin-film deposition techniques. Standard processes such as diffusion, ion implantation, RF magnetron sputtering, vacuum deposition, e-beam deposition, pulsed laser deposition, chemical vapor deposition, and sol-gel techniques can be used to develop a variety of IO devices and components.

In the simple waveguide structures discussed above the light is confined in one direction only. However, it is often necessary to confine light in two dimensions to guide them along a channel as observed in the case of injection laser diodes. A variety of methods are used to achieve this. Fig. 10.15 shows some of these structures in which either selective doping, metal stripes, or complex geometries are used for two-dimensional confinement.

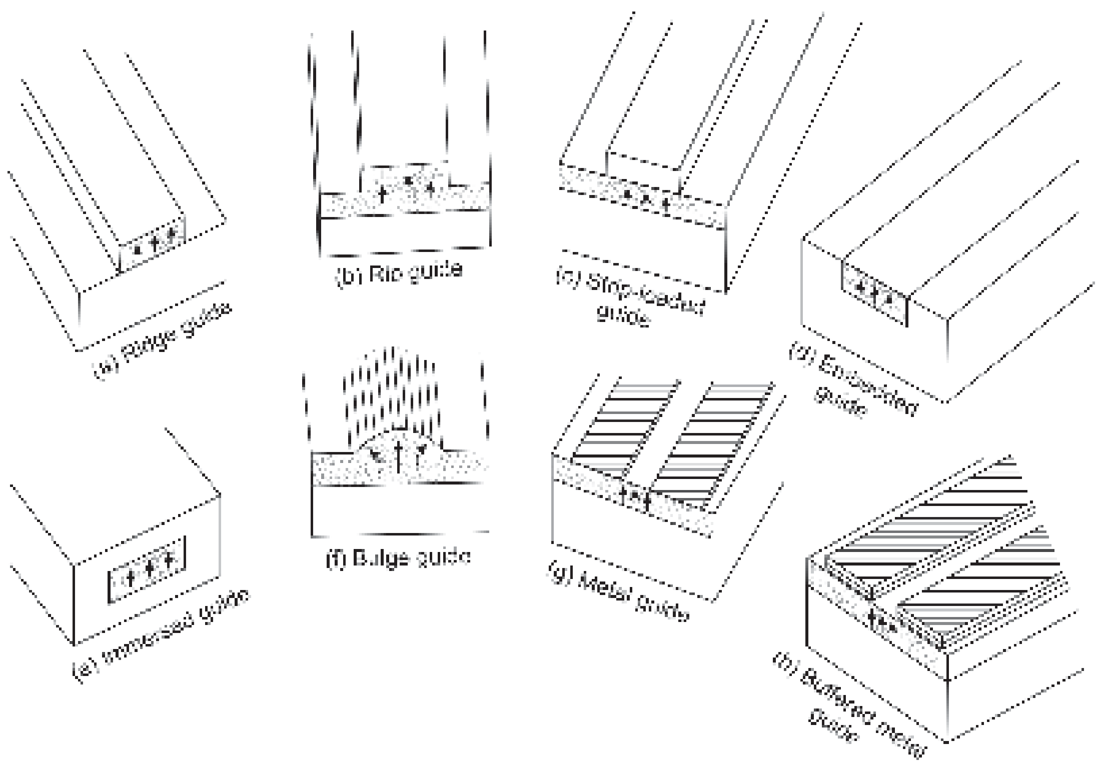

Fig. 10.15 Planar optical waveguide structures

10.2.2 Optical Modulators

Optical modulators are used for intensity modulation of light by applying external energy in the form of electrical, magnetic, or acoustic energy. It has already been seen that the intensity of a light source can be varied directly by controlling the flow of the bias current through the device. This mode of intensity modulation of an optical source is viewed as a direct modulation technique. There are a few materials whose optical properties (such as refractive index) can be changed by an external energy source in a suitable form (electrical, magnetic, or acoustic). When the light propagates through the material, the intensity of the light coming out of the material gets modulated following the variation of the external energy. This form of intensity modulation is called indirect modulation of light. Modulation at a very high speed is possible with the help of this IO device. In this section, different types of optical modulators are discussed.

10.2.3 Electro-optic Modulators

An electro-optic modulator (EOM) is an "IO" device in which the electro-optic effect of an electro-optic material is exploited to modulate a beam of light with respect to phase, frequency, amplitude, or polarization of the beam. Modulation bandwidths over gigahertz can be obtained by making use of a laser beam (Alferness, 1981; Karna *et al* 1996). A variety of non-linear optical materials including some organic polymers exhibit electro-optic effects by which the refractive index of the material can be changed by applying an external electric field. If the refractive index of the material medium varies linearly with the applied electric field, the effect is called the linear electro-optic effect known as the Pockels effect (after the name of the German Physicist, Friedrich Pockels). This effect is observed in non-centrosymmetric materials, such as lithium niobate ($LiNbO_3$), lithium tantalate ($LiTaO_3$), potassium dideuterium phosphate (KDP), ammonium dihydrogen phosphate (ADP), potassium titanium oxide phosphate (KTP), and III-V compound semiconductors such as GaAs) and InP and II-VI material such as CdTe. In certain centro-symmetric materials in the form of gases, liquids, or crystals, the refractive index is found to vary with the square of the applied electric field. This electro-optic effect is known as the quadratic electro-optic effect or the Kerr effect (after the name of John Kerr) (Saleh *et al* 1991; Kaminow, 1974). Some important electro-optic materials and their properties are listed in Table 10.1.

The refractive index of an electro-optic material can be expressed as the function of the applied electric field, E using Taylor's series about $E = 0$ as (Saleh *et al* 1991)

$$n(E) = n + a_1 E + \frac{1}{2} a_2 E^2 + \ldots \tag{10.26}$$

where the coefficients of the series are as under

$$n = n(0)$$

$$a_1 = \left.\frac{dn}{dE}\right|_{E=0}$$

$$a_2 = \left.\frac{d^2 n}{dE^2}\right|_{E=0}$$

Equation (10.26) can be expressed in terms of electro-optic coefficients as (Saleh *et al* 1991)

$$n(E) = n - \frac{1}{2} r n^3 E - \frac{1}{2} \zeta n^3 E^2 + \ldots \tag{10.27}$$

where, r and ζ are the electro-optic coefficients that depend on the direction of the electric field and polarization of the light (Saleh *et al* 1991). The higher-order terms are generally negligible.

For certain materials, the third term on the right-hand side is negligible and Eq. (10.27) can be approximated as

$$n(E) = n - \frac{1}{2} r n^3 E \quad \text{(Pockels effect)} \tag{10.28}$$

The corresponding material is called Pockels cell or Pockels medium.

Table 10.1 List of some important electro-optic materials and their properties

Material	*Chemical formula*	*Refractive index for*		*Bandwidth*	*Electro-optic coefficient, r (10^{-12} m/V)*
		o-ray	*e-ray*		
Ammonium Dihydrogen Phosphate (ADP)	$NH_4H_2PO_4$	1.52 @ λ = 550 nm	1.48 @ λ = 550 nm	~500 MHz	8.5
Potassium Dihydrogen Phosphate (KDP)	KH_2PO_4	1.51 @ λ = 550 nm	1.47 @ λ = 550 nm	~100 MHz	10.5
Potassium Dideuterium Phosphate (KD*P)	KD_2PO_4	1.49 @ λ = 1050 nm	1.46 @ λ = 1050 nm	~300 MHz	26.4
Lithium niobate	$LiNbO_3$	2.23 @ λ = 1060 nm	2.16 @ λ = 1060 nm	~8 GHz	30.8
Lithium tantalate	$LiTaO_3$	2.175 @ λ = 550 nm	2.18 @ λ = 550 nm	~1 GHz	30.3
Gallium arsenide	GaAs	3.6 @ λ = 550 nm	–	~1 GHz	1.6
Cadmium telluride	CdTe	2.6 @ λ = 10 μm	–	~1 GHz	6.8
Zinc oxide	ZnO	2.0 @ λ = 580 nm	–	~1 GHz	1.5

For some centrosymmetric material, the first derivative of the refractive index with respect to the electric field vanishes, and Eq. (10.27) becomes

$$n(E) = n - \frac{1}{2}\zeta n^3 E^2 \text{ (Kerr effect)} \tag{10.29}$$

The material exhibiting the Kerr effect is known as the Kerr medium or Kerr cell.

10.2.4 Phase Modulator

If a light beam is allowed to pass through a Pockels medium of a certain length L, in which an electric field is already applied the lightwave undergoes a phase change given by

$$\Phi = n(E)kL = \frac{2\pi n(E)}{\lambda} L \tag{10.30}$$

where, λ is the free space wavelength.

Substituting the value of field-dependent refractive index of the medium from Eq. (10.28) into Eq. (10.30), we get

$$\Phi = \Phi_0 - \frac{\pi r n^3 EL}{\lambda} \tag{10.31}$$

where, Φ_0 is the phase angle corresponding to $E = 0$, given by

$$\Phi_0 = \frac{2\pi nL}{\lambda} \tag{10.32}$$

Equation (10.31) shows that the phase of the lightwave can be modulated by the electric field and thereby the applied voltage.

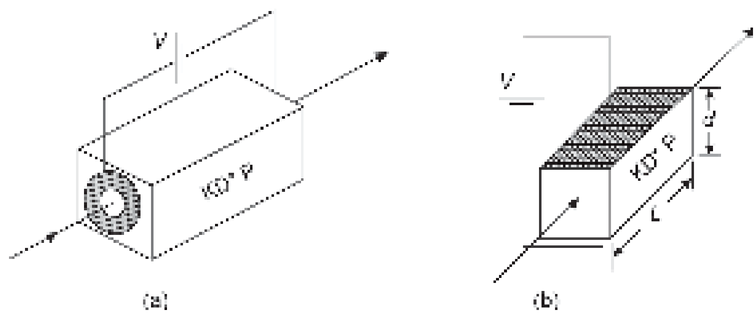

Fig. 10.16 Electro-optic phase modulator (a) Longitudinal modulator; (b) Transverse modulator

The electric field can be applied either in the direction of the propagation of light (longitudinal modulator) or a direction perpendicular to the direction of propagation (transverse modulator). If a voltage V is applied across the two facets of the Pockels cell in the transverse direction having a separation of d, the applied electric field can be expressed as

$$E = \frac{V}{d} \tag{10.33}$$

Therefore, the phase deviation can be expressed using Eqs (10.31) and (10.33) as

$$\Delta\Phi = \Phi_0 - \Phi = \frac{\pi r n^3 V L}{\lambda d} \tag{10.34}$$

The voltage required to create a phase difference of π can therefore be expressed as

$$V_\pi = \frac{\lambda d}{r n^3 L} \tag{10.35}$$

V_π is known as the half-wave voltage which is required for creating a phase-shift of π (Saleh *et al* 1991). The schematics of a longitudinal and a transverse electro-optic phase modulator are shown in Fig. 10.16.

When the electric field is applied in the longitudinal direction that is, in the direction parallel to the direction of propagation of the lightwave, then $d = L$. The phase difference created by the applied voltage thus depends on the free space propagation wavelength of the light, the ratio d/L, the electro-optic coefficient, r. The value of r depends on the direction of the electric field and the direction of propagation of light. For transverse modulators, the half-wave voltage is of the order of hundreds of volts whereas for longitudinal modulators the half-wave voltage can be as high as a few kilovolts (Saleh *et al* 1991). This type of electro-optic modulator is inconvenient for application in integrated optics. However, electro-optic modulators can be fabricated easily as an IO device in the form of strip waveguide in which the L/d ratio is very high. This type of electro-optic modulator can be fabricated by diffusing Ti in a lithium niobate ($LiNbO_3$) electro-optic substrate or diffusing Nb in a lithium tantalate ($LITaO_3$) substrate for enhancing the refractive index. The electric field can be applied with the help of deposited electrodes on the substrate. The schematic of an IO electro-optic modulator is shown in Fig. 10.17. Since the spacing between the electrodes is very small, a large value of the electric field can be easily generated by applying a small voltage. The half-wave voltage of such a strip waveguide IO electro-optic

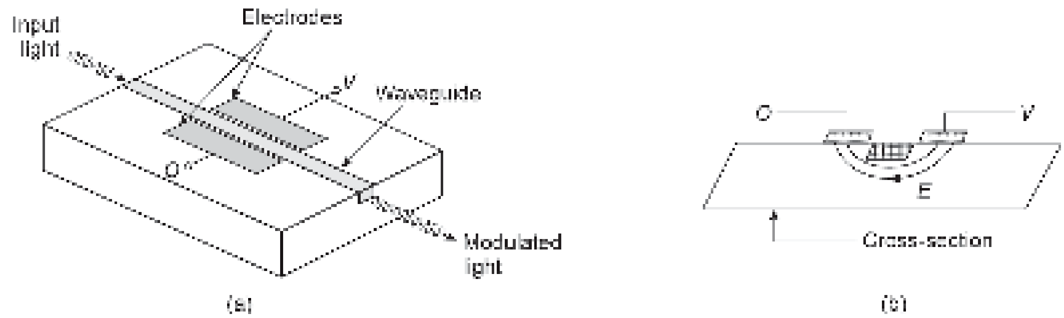

Fig. 10.17 Strip-waveguide transverse electro-optic modulator in IO form (a) Schematic (b) cross-sectional view showing the electric field.

modulator is only a few volts. The modulators are reported to operate at a high speed (~100 GHz) (Saleh *et al* 1991). The light input and output to such modulators can be managed by using optical fibers as shown in the figure.

Example 10.5 A transverse electro-optic modulator uses a KD*P Pockels cell of 5 cm length having a width of 0.5 cm. A light beam operating at a wavelength of 550 nm is allowed to pass through the material. Calculate the half-wave voltage of the modulator. Assume the electro-optic coefficient of the KD*P crystal to be 26.5 $\times 10^{-12}$ mV^{-1} at the operating wavelength. Given that the refractive index of KD*P cell is 1.51.

Solution The half-wave voltage in this case corresponds to the voltage required to be applied in the transverse direction to create a phase difference of π in the lightwave passing through the cell. The value of V_π can be estimated by using Eq. (10.35) as

$$V_\pi = \frac{\lambda d}{rn^3 L} = \frac{550\times 10^{-9}\times 0.5\times 10^{-2}}{26.5\times 10^{-12}\times (1.51)^3\times 5\times 10^{-2}}$$

$$= 602.8 \text{ V}$$

Example 10.6 A transverse electro-optic modulator is fabricated as an IO device by using a strip-waveguide configuration based on lithium niobate. The modulator is designed to operate at 1330 nm. If the length of the waveguide is 3 cm and the electrode spacing is 20 µm, determine the half-wave voltage required to be applied to the transverse EO modulator. The electro-optic coefficient and the refractive index of lithium niobate at the operating wavelength are 30.8 $\times$ 10^{-12} mV^{-1} and 2.17, respectively. Compare and contrast the result with that obtained in Example 10.5. Estimate the phase-shift caused by the waveguide when the applied voltage is 10 V.

Solution The half-wave voltage corresponding to a phase difference of π can be obtained by using Eq. (10.35) as

$$V_\pi = \frac{\lambda d}{rn^3 L} = \frac{1330\times 10^{-9}\times 20\times 10^{-6}}{30.8\times 10^{-12}\times (2.17)^3\times 3\times 10^{-2}} = 2.81 \text{ V}$$

By comparing this value of half-wave voltage with that obtained in Example 10.5 for a KD*P crystal, it is seen that phase modulation can be achieved at a much lower voltage in a strip-waveguide structure. This is because the phase difference depends on the transverse electric field. Since the spacing between the electrodes is very small, a high electric field can be obtained in an IO modulator by applying a small voltage.

The phase-shift for an applied voltage of 10 V can be obtained by using Eq. (10.34) as

$$\Delta\Phi = \frac{\pi r n^3 VL}{\lambda d} = \frac{\pi \times 30.8 \times 10^{-12} \times (2.17)^3 \times 10 \times 3 \times 10^{-2}}{1330 \times 10^{-9} \times 25 \times 10^{-6}}$$

$$= 2.83\pi$$

10.2.5 Acousto-optic Modulator

An altogether different approach based on the interaction of light and sound (acoustic) waves can be used to modulate the intensity, phase, and frequency of light. The change in refractive index of a dielectric medium in this case is caused by mechanical strain produced in the medium during the passage of an acoustic wave through the medium. This effect is known as an acousto-optic effect. This effect can be exploited to make an IO acousto-optic modulator. Acousto-optic modulators are based on the diffraction of light by a column of sound in a suitable interaction medium. When an acoustic wave travels through a transparent material, it causes periodic variations of the index of refraction. The sound wave can be viewed as a series of compressions and rarefactions moving through the material. In the compressed regions, the sound pressure is high as a result the material is compressed slightly. This compression thus leads to an increase in the refractive index of the material in the localized region. The series of compressions and rarefactions propagating through the transparent medium produces a periodic variation in the density caused by the mechanical strain. This variation in density causes a change in the refractive index in the medium. The propagating acoustic wave produces a moving optical phase-diffraction grating which diffracts any light beam passing through the medium and interacts with it. An acousto-optic device needs a material or medium with good acoustic and optical properties. Acousto-optic materials generally used in IO modulators include fused silica, GaAs, GaP, lead molybdate, tellurium oxide, etc.

An acousto-optic modulator (AOM) works in two regimes, e.g. the Bragg regime observed at high acoustic frequencies and the Raman-Nath regime operative at low acoustic frequencies. In the Bragg regime, the diffraction grating created by the acoustic wave is generally thick while in the Raman-Nath regime the grating is so thin that it behaves almost like a transmission grating. Therefore, maximum interaction of the light beam with the acoustic wave occurs in the Bragg regime. In this case, a zero-order mode is partially deflected into the first-order mode. IO-based electro-optic modulators generally work in the Bragg regime. A simple arrangement of an AOM is shown in Fig. 10.18. It consists of a piezoelectric transducer attached to an electro-acoustic material such as glass or quartz. An oscillating electric signal is used to drive the transducer to vibrate. The transducer creates acoustic waves in the glass. The propagating sound wave creates a thick Bragg diffraction grating. The incoming light gets scattered by the periodic variation in the refractive index created by the acoustic wave-induced grating. The diffracted beam emerges at an angle θ_B which depends on the wavelength of the incoming light and the wavelength of the sound wave. The angle θ_B can be expressed as (Laybourne *et al* 1981)

$$\sin\theta_B = \frac{\lambda}{2\Lambda} \text{ (Bragg regime)} \qquad (10.36)$$

where, λ is the wavelength of the light and Λ is the wavelength of the sound wave.

An IO acousto-optic Bragg modulator can be fabricated on a lithium niobate piezoelectric substrate by forming a strip waveguide by diffusing titanium. Acoustic waves in the form of surface acoustic waves (SAW) can be launched into the waveguide with the help of an interdigitated electrode system.

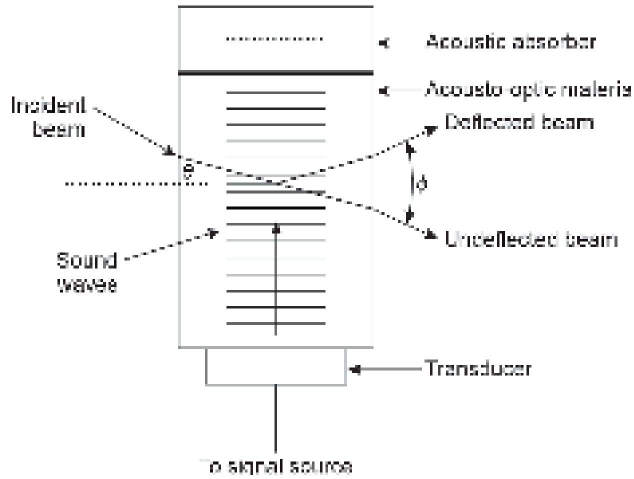

Fig. 10.18 Schematic of acousto-optic modulator

10.2.6 Optical Beam Splitters, Directional Couplers, and Switches

In optical fiber communication applications, it is often necessary to split a single beam into two or more branches or combine several beams into one channel. The simplest structure is a Y-splitter (coupler) which can divide the light signal from one channel to two different channels or combine light signals from two different channels and combine them to guide them through a single channel. The schematic of a Y-junction splitter is illustrated in Fig. 10.19. This device enables the light from the input port (stem) to be divided into two output ports (arms) as illustrated in the figure. It is rather difficult to build a Y-junction splitter (coupler) as an independent optical component. However, it is very convenient to fabricate a Y-junction splitter using planar waveguide technology. A passive Y-splitter can be fabricated using $LiNbO_3$. These splitters distribute almost equal power in the two arms. The major drawback of the Y-splitter is that the power coupled in the arms decreases drastically as the angle between the two arms increases. When the angle between the arms is very large, a significant power is radiated into the substrate. Y-couplers are generally used as one of the components of IO systems for coupling purposes. To have a large separation between the two arms it is necessary to have a very wide junction width as compared to the width of the guide. This limitation makes the size of the splitter unacceptably high.

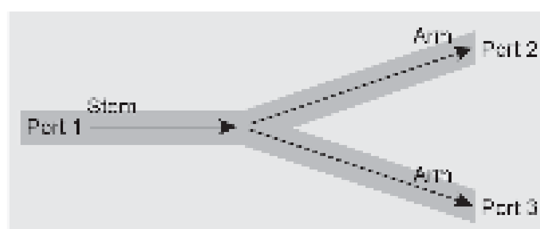

Fig. 10.19 Schematic of a passive Y-splitter (coupler)

However, a Y-junction splitter can be realized in the active form by making use of electro-optic materials making them more versatile. The schematic of a single-mode channel waveguide Y-junction using an electro-optic material is shown in Fig. 10.20. This splitter has three overlaid electrodes as shown in the figure (Sasaki *et al* 1976). In the absence of any electrical voltage across the electrodes, the optical power distribution is symmetrical and the splitter works as a symmetrical power divider (Sasaki *et al* 1976; Sasaki *et al* 1978). If the central electrode along the stem of the Y-coupler is grounded and

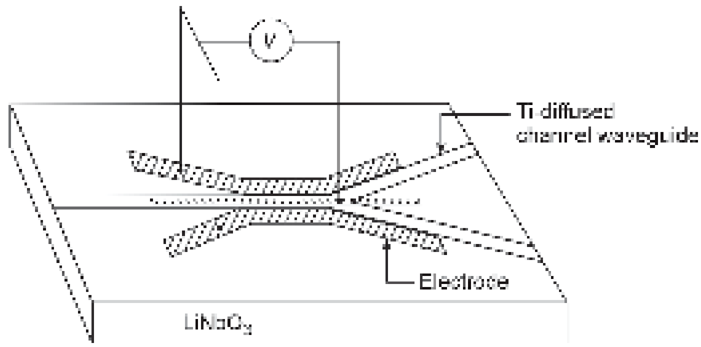

Fig. 10.20 An electro-optic Y-junction coupler/switch (after Sasaki *et al* 1978)

a bias voltage is applied between the electrodes along the arms of the Y-splitter then one side of the waveguide structure experiences an increase in the refractive index due to an electro-optic effect while the other side of the waveguide experiences a reduction in the refractive index. The change in the refractive index in the presence of the applied electric field can be expressed as

$$\delta n = \pm \frac{1}{2} n^3 r E \qquad (10.37)$$

Therefore, the light coming from the input port is deflected towards the region of higher refractive index thereby guiding the light in the corresponding output arm of the splitter. This type of active Y-splitter allows the light to be diverted to the output arms having larger junction angles as compared to its passive counterpart (Sasaki *et al* 1978). A large angle between the output arms enables one to achieve negligible coupling between the output waveguide arms. Y-splitters can be used for signal routing purposes.

10.2.7 Electro-optical Switches

Electro-optical switches can be fabricated in the IO form by placing two parallel strip waveguides made of electro-optic materials with electrodes overlaid on them for controlling purposes. Figure 10.21 shows one such arrangement in which the evanescent field generated outside the guiding region can be coupled transversally to an adjacent waveguide. The coupling can be facilitated by making use of electro-optic materials for fabricating the strip waveguide structures.

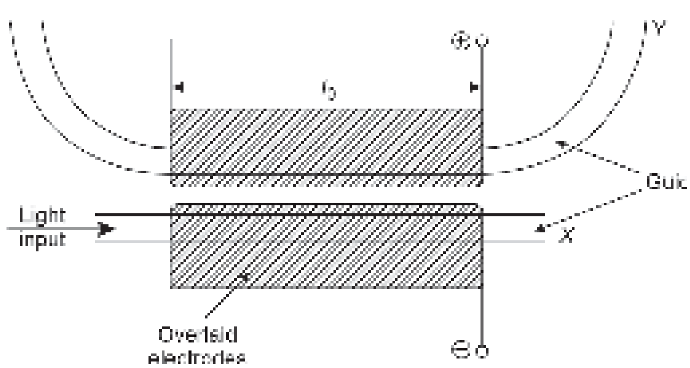

Fig. 10.21 Electro-optically switched directional coupler

10.2.8 Mach-Zehnder (MZ) Interferometer

An interferometer is an optical device that utilizes the effect of phenomenon of interference of light. In an interferometer, the input light beam is first split into two separate beams by using a beam splitter. The beams are allowed to undergo some phase changes before they are recombined by making use of a coupler. The two beams interfere with each other to perform the desired function. The electro-optic beam splitter (coupler) discussed earlier can be used to form a Mach-Zehnder-type interferometer in the IO form. A schematic of the electro-optic Mach-Zehnder electro-optic interferometer is shown in Fig. 10.22. It consists of two Y-junctions made of electro-optic materials in the form of a strip waveguide structure connected back-to-back. The electrodes are overlaid for applying an external bias voltage to control the device electro-optically. The input Y-junction acts as a splitter which divides the input light signal into the two arms of the Y-junction splitter. In the absence of any bias voltage applied to the electrodes, the input optical power is divided between the two arms without having any phase difference. The two components are subsequently combined with the help of the second Y-junction coupler to make the two beams interfere at the output junction in the same phase. The two beams interfering in the same phase give rise to interference maximum at the output. When an appropriate bias voltage is applied across the electrodes, a differential phase-shift is created between the signals in the two arms of the input Y-splitter. Depending on the phase of the light in the two arms, their subsequent combination at the output Y-junction causes either constructive (phase difference 0 or 2π) or destructive interference (phase difference of π). A constructive interference results in an intensity maximum while a destructive one results in an intensity minimum of the light signal. In a way, the phase modulation caused by the electro-optic effect is translated into intensity modulation of the signal in an MZ interferometer. Lithium niobate doped with titanium has been reported to be used for making high-speed (1.1 GHz) interferometric modulator (Thylen *et al* 2008; Ostrowsky, 1980).

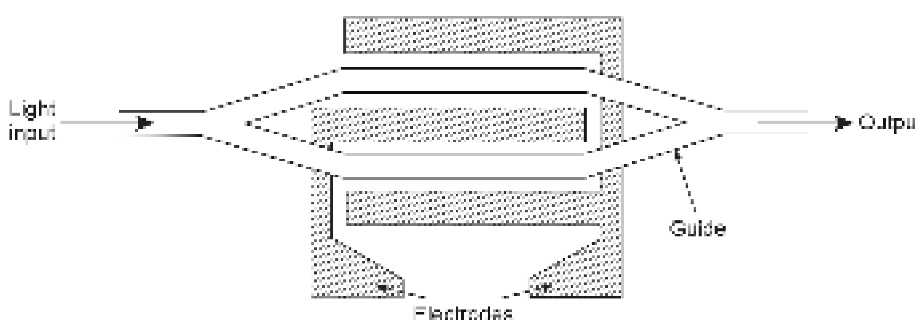

Fig. 10.22 Mach-Zehnder interferometric modulator

In addition, a variety of optical devices can be realized in the integrated optics form to perform a large number of functions ranging from optical resonators, filters, polarization controllers, and coherent optical receivers. Lithium niobate substrate is extensively used for developing IO-based IO devices on a single substrate.

Summary

- This chapter deals with major developments in the area of all optical/photonic devices that can manipulate light in the optical domain itself.
- The major optical devices and components include optical amplifiers, planar waveguides, optical splitters, modulators, switches, logic gates, optical filters, etc.

- Optical amplifiers amplify the optical signal in the optical domain without taking the signal to the electrical domain and using electronic amplifiers.
- Two major types of optical amplifiers are semiconductor optical amplifier (SOA), also known as semiconductor laser amplifier (SLA) and doped fiber amplifier (DFA) or simply fiber amplifier (FA).
- Both the amplifiers can provide high gain over the spectral range primarily used for present generation optical fiber communication system.
- The semiconductor amplifiers are of three types, e.g. resonant or Fabry-Perot amplifier, the travelling wave (TW) and the near travelling wave (NTW) amplifiers.
- The optical amplifiers find applications is boosting the transmitter power, increasing the receiver sensitivity or boosting the power level at intermediate points on the fiber link.
- The single pass gain of an FP SLA can be expressed as:

$$G_s = \exp\left[\left(\frac{\Gamma g_0}{1+\dfrac{I}{I_s}} - \bar{\alpha}\right)L\right]$$

- The 3 dB optical bandwidth is given by,

$$B = \frac{c}{\pi n L}\sin^{-1}\left[\frac{1}{2}\left(\frac{(1-R_1)(1-R_2)}{\sqrt{R_1 R_2 G}}\right)\right]$$

- Rare earth elements such as erbium and neodymium are used for doping silica fibers to provide optical gain.
- The commonly used fiber amplifier is obtained by doping silica fiber with erbium. The fiber amplifier is called erbium doped fiber amplifier (EDFA).
- Non-linear scattering such as stimulated Raman scattering (SRS) and stimulated Brillouin scattering (SBS) can be used to make fiber amplifiers.
- Integrated optics (IO) also known as photonic integrated circuits (PICs), involve integration of planar lightwave components on a single substrate to perform a variety of complex optical functions.
- Some of the integrable optical components include planar optical waveguide, optical fibers, optical filters, optical couplers, injection laser diodes, photodetectors, optical amplifiers, optical switches, etc. The dimension of the optical waveguide cannot be reduced much below the operating wavelength. Planar waveguide consists of a thin dielectric slab of refractive index n_1 sandwiched between two symmetrical dielectric slabs of refractive index n_2 ($< n_1$).
- The simplest material that is used for making optical waveguide is the silica glass and a few transparent polymers.
- Electro-optic and electro-acoustic materials are preferred for making optical waveguides because the characteristics can be controlled externally by electric fields.
- The principal materials used for making electro-optic IO devices include lithium niobate ($LiNbO_3$); lithium tantalite ($LiTaO_3$); zinc oxide (ZnO); zinc sulphide (ZnS); titanium oxide (TiO_2) including III-V semiconductors GaAs, InP, GaSb, InAs, and their ternary and quaternary alloys.
- Optical modulators are used for intensity modulation of light by applying an external energy in the form of electrical, magnetic or acoustic energy.
- An Electro-optic Modulator (EOM) is an IO device in which the electro-optic effect of an electro-optic material is exploited to modulate a beam of light in respect of phase, frequency, amplitude, or polarization of the beam.
- In an electro-acoustic modulator, the change in refractive index of a dielectric medium is caused by mechanical strain produced in the medium during the passage of an acoustic wave through the medium.

- The simplest beam splitter/coupler structure is a *Y*-splitter (coupler) which can divide the light signal from one channel to two different channels or combine light signals from two different channels and combine them to guide them through a single channel.
- Electro-optical switches can be fabricated in the IO form by placing two parallel strip waveguides made of electro-optic materials with electrodes overlaid on them for controlling purpose.
- Two electro-optic beam splitters/couplers can be combined to form an electro-optic interferometer popularly known as Mach-Zehnder type interferometer.

Problems

10.1 A 250 μm long Fabry-Perot cavity laser amplifier has uncoated facets with reflectivities of 30% each. The single-pass gain of the cavity is 2.8 dB, the mode spacing is 1 nm and the peak-gain wavelength of the amplifier is 850 nm. If the refractive index of the cavity is 3.6, estimate the 3 dB bandwidth of the amplifier.

10.2 A Fabry-Perot cavity laser amplifier has uncoated facets with reflectivities of 30% each. The single-pass gain of the cavity is 3.2 dB, the mode spacing is 1 nm and the peak-gain wavelength of the amplifier is 1550 nm. If the length of the cavity is 275 μm, estimate the refractive index of the cavity material and the 3 dB bandwidth of the amplifier.

10.3 Derive equation (10.16) for the peak–trough ratio of passband ripple.

10.4 For a TWA with R_1, R_2 << 1 shows that the cavity gain can be expressed as

$$G = \frac{0.25}{\sqrt{R_1 R_2}}$$

10.5 For a symmetrical SI planar waveguide structure, calculate the maximum thickness of the guiding slab to support the fundamental TE mode at 860 nm. The refractive index of the guiding slab and the surrounding cladding regions are 3.56 and 3.52, respectively.

10.6 In problem 10.5 if the thickness of the guiding slab is doubled, calculate the maximum wavelength at which the waveguide will support the fundamental mode only. Assume that the refractive index is independent of wavelength.

10.7 Calculate the minimum value of the thickness required for an asymmetrical planar slab waveguide to support the propagation of the lowest order TE mode with the following parameters:

i. Index of refraction of the guiding film = 3.5
ii. Substrate refractive index = 3.4
iii. Wavelength of operation = 900 nm

Assume that the waveguide is surrounded by air.

10.8 A transverse electro-optic modulator uses an ADP Pockels cell of 2-inch length having a width of 0.5 cm. A light beam operating at a wavelength of 550 nm is allowed to pass through the material. Calculate the half-wave voltage of the modulator. Assume the electro-optic coefficient and the refractive index of the ADP crystal to be 8.5×10^{-12} mV^{-1} and 1.52, respectively, at the operating wavelength. Estimate the phase-shift for an applied voltage of 500 V.

10.9 A transverse electro-optic IO modulator is fabricated by using a strip-waveguide configuration based on lithium tantalate. The modulator is designed to operate at 1300 nm. If the length of the waveguide is 2 cm and the electrode spacing is 25 μm, determine the voltage required to be applied to the transverse EO modulator to create a phase-shift of π radians. The electro-optic coefficient and the refractive index of lithium tantalate at the operating wavelength are 30.3×10^{-12} mV^{-1} and 2.2, respectively. Calculate the phase-shift for an applied voltage of 7 V.

10.10 A transverse electro-optic IO modulator is fabricated by using a strip-waveguide configuration based on lithium niobate. The half-wave voltage of the transverse modulator designed to operate at 1300 nm is 5 V. If the length of the waveguide is 1 inch, calculate the spacing between the electrodes. The electro-optic coefficient and the refractive index of lithium niobate at the operating wavelength are 30.8×10^{-12} mV^{-1} and 2.3, respectively.

10.11 Compare and contrast the speed for electro-optic and acousto-optic light-beam modulators.
10.12 List the important electro-optic materials. Name the materials that are most suitable for the fabrication of IO electro-optic modulators.
10.13 What is Pockels effect? List the materials that exhibit the Pockels effect.
10.14 What is the Kerr effect?
10.15 Distinguish between the Raman–Nath modulator and the Bragg deflection modulator.
10.16 Describe the mechanism of operation of an electro-optic IO device and mention some of its potential applications.
10.17 Distinguish between a longitudinal and a transverse electro-optic modulator.
10.18 Compare and contrast the half-wave voltage values for the phase modulators using different electro-optic materials listed in Table 10.1. The length of the strip waveguide may be assumed to be 2 cm and the separation between the electrodes to be 25 μm.

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Advanced Optical Communication Systems and Optical Networks

CHAPTER 11

This chapter deals with some advanced technology on the growth and widespread applications of optical fiber communication systems. It also outlines the elements of optical networking architectures.

11.1 EVOLUTION OF OPTICAL COMMUNICATION TECHNOLOGY

Among the various technologies that have contributed to the dramatic development and growth of optical fiber communication is certainly the concept of wavelength division multiplexing (WDM). This technology can significantly enhance the capacity of an existing optical fiber link by many folds. Using the WDM technique, it is possible to modulate numerous signals in the form of data, text, audio, and video signals from multiple channels simultaneously by using optical signals of different wavelengths (colors) of laser light on a single optical fiber. This technology can be used for bidirectional optical communication. WDM technology is often compared with frequency division multiplexing (FDM) in which a single channel is used for sending signals from different sources by using carrier signals operating at different frequencies.

So far in this book, we have discussed optical communication systems based on intensity modulation/direct detection scheme in which the intensity of light is varied following the information signal at the transmitter-end. At the receiver-end, the intensity-modulated optical signal is converted into a corresponding electrical signal with the help of a photodetector. The photodetector essentially acts as a photon counter which reproduces the electrical signal based on the photons received by the photodetector. The performance of an optical receiver in an IM/DD system is greatly constrained by the associated electrical noise. An alternative approach to improve the performance of optical fiber communication systems is to replace the conventional IM/DD system with the so-called coherent optical fiber communication system. The high sensitivity of the coherent receiver makes coherent optical communication more attractive by enhancing the repeaterless distance between the transmitter and the receiver. However, coherent optical communication could not develop much in the beginning because of several practical limitations. The widespread use of WDM and the advent of SLA/EDFA in the conventional IM/DD system made it even more popular for commercial applications. However, in the recent past coherent optical communication witnessed a renewed interest because of its ability to employ a variety of efficient modulation techniques (Nakazawa *et al* 2010; Harry, 1988). Another approach to a new generation of optical communication is expected to be based on soliton pulses. This chapter deals with all the above aspects of optical communication systems including some rudimentary discussion on optical networking.

11.2 WAVELENGTH DIVISION MULTIPLEXING

The wavelength division multiplexing involves the technology of combining multiple wavelengths carrying different signals onto a single fiber. WDM technique is similar to FDM used in radio frequency modulation systems. In WDM systems, the signals are isolated in the wavelength domain by properly spacing them in the wavelength range. In contrast to FDM, WDM system of each channel has access to the entire intensity modulated fiber bandwidth (~several Gbps). The WDM technique can thus improve the capacity of an existing network significantly without laying additional fibers. In the WDM system, each wavelength may carry transmission data in any format. For example, different wavelengths can be used to support synchronous and asynchronous digital data, analog signals, and digital data. Any combination of slow/fast, digital/analog signals can be sent independently and simultaneously using a single fiber with the help of the WDM technique. The principles of WDM and various components and devices required for the successful implementation of WDM are discussed in the following sections.

11.3 DENSE WAVELENGTH DIVISION MULTIPLEXING (DWDM)

A point-to-point optical fiber link involves one transmitter at the source-end, one receiver at the destination-end, and a fiber connecting the transmitter and the receiver. This simple system is called a simplex link. This link can provide one-way communication only. For two-way communication for the exchange of information, it is necessary to have one transmitter and receiver pair at each end. This type of system is called a duplex link. To transmit signals from multiple sources from one end to the other in a Simplex system or both ways in a duplex system, one would generally require independent fiber connections for each of these transmitter–receiver modules. This straightforward approach does the desired job but is not necessarily the only way to have simultaneous transmission and reception of signals from multiple sources. It is important to recall here that every optical source operates at a peak wavelength (wavelength corresponding to peak emitted optical power) with a finite spectral width. For LED sources, the spectral width is generally very large (~40 nm). On the other hand, injection laser diodes used for optical communication purposes have extremely narrow spectral ranges. The narrow spectral width of these sources makes them especially attractive for simultaneous transmission of signals from different sources by modulating the intensity of light emitted from multiple sources operating at different peak wavelengths. For example, consider N number of sources operating at different peak emission wavelength values of λ_1, λ_2, λ_3, ..., λ_N. If the wavelength values are so chosen that the fiber gives low values of attenuation for all these wavelength values, then it is possible to modulate each of these optical sources independently by the information signals coming through different channels and launch them simultaneously over the same single fiber which supports transmission of all these wavelengths. This approach enables one to overcome the requirement of additional fibers for each wavelength operation. Moreover, the the capacity of the existing link is significantly improved at a reduced cost. This is the essence of WDM. To probe further the immense potential of WDM, let us have a closer look at the attenuation characteristics of a high-quality silica fiber shown in Fig. 11.1 showing the variation of attenuation in dB/km against wavelength. From the characteristics, it is clear that there exist two low-loss regions in the single-mode fiber separated by the OH^- peak at around 1400 nm. The low-loss attenuation window on the left-hand side of the OH^- peak ranges from 1270–1350 nm and that on the right-hand side from 1480–1600 nm. The first window is often referred to as the 1310 nm window and the other is known as the 1550 nm window.

It is interesting to note that AllWave® fibers do not exhibit OH^- absorption peak and therefore can offer a wider window (Keiser, 2002).

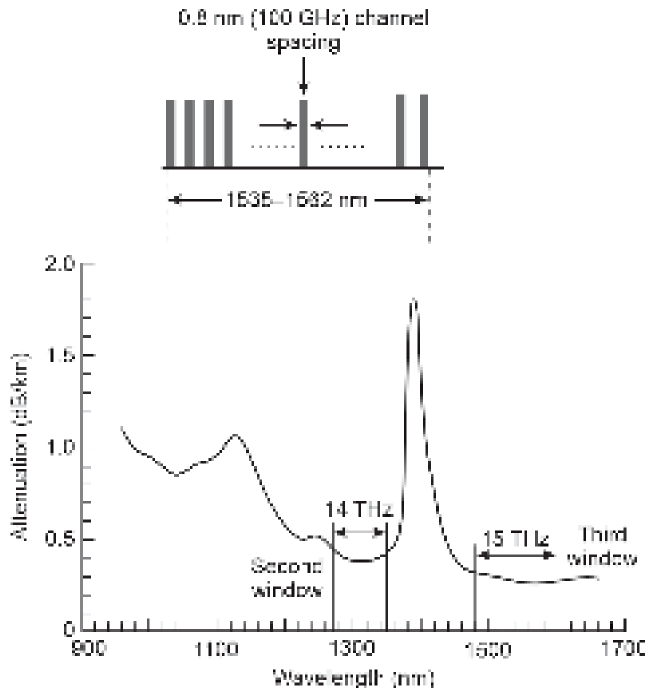

Fig. 11.1 Attenuation characteristics of a silica fiber illustrating the low attenuation windows corresponding to spectral bands centered around 1310 nm and 1550 nm. These spectral bands can be used for transmission of different signals simultaneously using multiple sources operating at different peak wavelengths via WDM technique

The two low-loss optical windows at 1310 nm and 1550 nm have spectral widths of 80 nm and 120 nm, respectively. These spectral widths can be utilised for sending intensity-modulated light signals from different sources operating at different peak wavelengths in this range with a certain amount of spectral width. The number of light signals that can be accommodated in the available spectral band is determined by the spectral width of individual light sources plus the guard bands in between them. It is often convenient to describe the available wavelength spectral band in terms of frequency bandwidth. The conversion from the wavelength spectral band to the frequency band can be obtained by noting that

$$\frac{\Delta \nu}{\nu} = \frac{\Delta \lambda}{\lambda} \tag{11.1}$$

Further,

$$c = \nu \lambda \tag{11.2}$$

Substituting the value ν from Eq. (11.2) into Eq. (11.1), we may write

$$|\Delta \nu| = \left(\frac{c}{\lambda^2}\right)|\Delta \lambda| \tag{11.3}$$

Equation (11.3) translates the wavelength deviation, $\Delta\lambda$ about a given wavelength, λ into the corresponding frequency deviation $\Delta\nu$. Let us now examine the frequency span of the two windows discussed earlier. It can be easily seen with the help of Eq. (11.3) that the spectral width of 80 nm around 1310 nm optical window translates into a frequency band of $\Delta\nu = 14$ THz (1 THz = 10^{12} Hz). Likewise, the available spectral band (in terms of wavelength) of 120 nm around the optical window at 1550 nm corresponds to a frequency band of 15 THz. Therefore, the total available spectral width of 200 nm (= 80 nm + 120 nm) amounts to a frequency band of 29 THz in the two windows. The WDM technology makes use of these optical windows to interleave optical channels to accommodate optical signals from independent optical sources operating at different peak wavelengths in these regions, each with a small spectral width. These operating wavelengths are so spaced by incorporating guard bands that they do not interfere with adjacent channels. The availability of laser sources with extremely small linewidth has made the implementation of WDM possible. Using this approach enables one to make a more efficient utilisation of the available spectral bands in the windows and thereby enhance the capacity of the existing optical fiber link.

A simple arrangement of increasing the capacity of a point-to-point optical fiber link using the WDM technique is illustrated in Fig. 11.2. The system consists of multiple transmitters operating at different wavelengths $\lambda_1, \lambda_2, \lambda_3, \lambda_4$. The intensity-modulated light signals are combined with the help of a multiplexer. The combined signals remain isolated in the wavelength domain. The combined signal is launched onto the common fiber. At the receiving-end, the signals are isolated based on their assigned wavelength with the help of a demultiplexer. The intensity-modulated light signals are then detected by the individual receivers operating at the corresponding wavelength. This simple arrangement can be extended to a duplex system as well as in a more complicated optical network architecture.

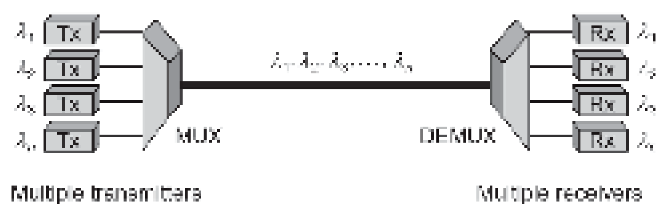

Fig. 11.2 Schematic block diagram of a WDM system

Example 11.1 An Allwave® fiber exhibits low attenuation in the entire range of transmission ranging from 850–1650 nm. It is intended to use the spectral range of 1250–1650 nm around 1450 nm for implementation of WDM using laser transmitters with narrow spectral width. If the required channel spacing is 500 GHz, calculate the number of channels that can be supported by the system.

Solution The available spectral range is

$$\Delta\lambda = 1650–1250 = 400 \text{ nm}$$

The corresponding separation can be estimated using Eq. (11.3) as

$$\Delta\nu = \frac{c}{\lambda^2}\Delta\lambda = \frac{3\times 10^8}{(1450\times 10^{-9})^2}\times 400\times 10^{-9} = 57 \text{ THz}$$

$$= 57 \text{ THz}$$

Since the required channel spacing is 500 GHz, the number of channels in the available spectral band is

$$N = \frac{57\times 10^{12}}{500\times 10^9} = 114$$

From the foregoing discussion, it is apparent that WDM is essentially a FDM of the optical carrier signals with the exception that the light carrier signals are specified in terms of wavelength rather than frequency. In the early days of the inception of WDM system, two primary attenuation windows in the near-infrared (NIR) region were used per fiber as separate channels. At the receiver-end, the near-infrared NIR channels used to be demultiplexed with the help of a dichroic filter. A dichroic filter is essentially a two-wavelength filter that has a cut-off wavelength located almost midway between the wavelengths of the two channels. In the early generation WDM, only a few selected wavelengths which are separated by several hundreds of nanometers in wavelength were used. It was soon realised that even more than two wavelengths can be used in WDM by using cascaded dichroic filters. This concept led to the development of coarse wavelength division multiplexing (CWDM) where the wavelengths used as channels are separated in the wavelength domain by several tens of nanometers. However, with the advent of high-quality laser sources with extremely narrow linewidth, it has become possible to pack the channels more densely within the attenuation windows of the silica fiber. These densely packed channels with wavelength separation to the tune of 1 nm or so have been successfully implemented in the recent past. The new generation wavelength division multiplexing is known as dense wavelength division multiplexing (DWDM). In CWDM, the maximum number of channels used to be in the range of 8-18 only. In DWDM, the number of channels is usually in multiples of 16. Further, each channel contains multiplexed information signal/ data through microwave subcarrier modulation with several gigabits per second. Therefore, using the WDM technique, it is possible to attain an effective transmission rate of several hundreds of Gbps.

Dense wavelength division multiplexing is thus characterised by narrower channel spacing as compared to that of CWDM. In contrast to CWDM, the DWDM application requires a stringent control requirement with respect to frequency stability to avoid any interference or overlapping of the channels. The International Telecommunication Union (ITU) is the regulatory body of the United Nations and is responsible for allocation of the channel spacing.

11.3.1 International Telecommunication Union–(ITU-T)

The Standardisation Sector (ITU-T) is a permanent wing of ITU. Even though in optical communication, it is customary to specify a channel in terms of wavelength rather than frequency, the ITU specifies the channel spacing in terms of frequency[1]. The CWDM specification ITU-T G.671 generally refers to transmission frequency allocated to coarse WDM and is uncontrolled in terms of frequency stability. The latest recommendation of ITU-T for DWDM allows fixed channel spacings ranging from 12.5 GHz to 100 GHz and wider (integer mutiples of 100 GHz) as well as flexible grid. It is also permissible to use uneven channel spacing. The channel spacing for the fixed grid has been obtained by subdividing the original 100 GHz grid successively by factors of 2. The nominal central frequency or channel frequency (CF) in THz for DWDM anchored at 193.1 THz is computed as follows (ITU-T G.694.1)

$$CF = 193.1 + n \times 0.0125 \quad \text{(for 12.5 GHz channel spacing)}$$
$$CF = 193.1 + n \times 0.025 \quad \text{(for 25 GHz channel spacing)}$$
$$CF = 193.1 + n \times 0.05 \quad \text{(for 50 GHz channel spacing)}$$
$$CF = 193.1 + n \times 0.01 \quad \text{(for 100 GHz channel spacing)}$$

[1] As a convention, the frequency of a laser source is fixed for locking the source to a particular mode of operation.

where,

$$n = 0, \pm 1, \pm 2, \pm 3, \cdots$$

Some nominal central frequencies in the C and L bands based on channel spacing in the range of 12.5 to 100 GHz anchored to 193.1 THz can be found in the latest records of ITU-T. For the computation of nominal wavelength, the velocity of light should be considered as $c = 2.99792458 \times 10^8$ m/s. As per ITU-T G.694.1 standards, the nominal central frequency referenced to 193.1000 THz corresponds to an approximate nominal central wavelength of 1552.5244 nm and spacing them at 12.5 GHz (0.1 nm) apart. Alternatively, the spacing could be adjusted to 25 GHz (0.2 nm); 50 GHz (0.4 nm) or 100 GHz (0.8 nm) apart.

Practical DWDM systems are much more complex than the one illustrated in Fig. 11.2. A practical DWDM optical fiber link based on NORTEL OpTera 640 system with 64 wavelengths and each carrying 10 Gbps data is shown in Fig. 11.3. Practical DWDM link

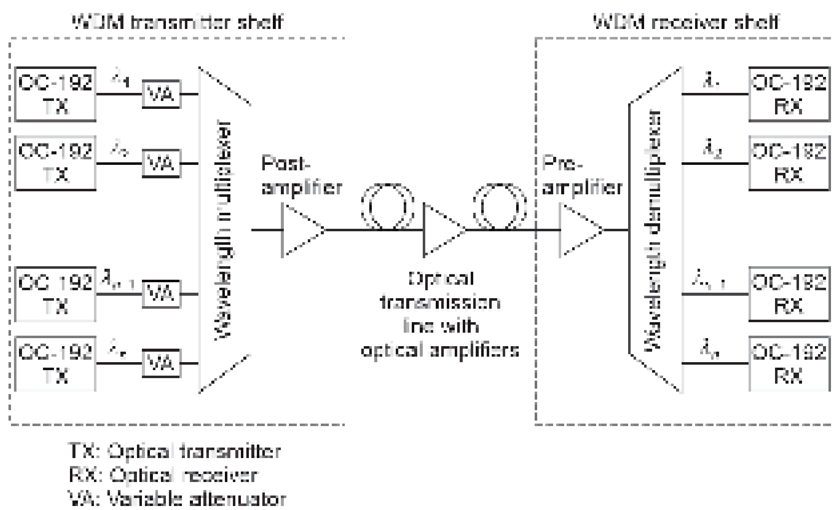

Fig. 11.3 Illustrating the application of NORTEL OpTera 640 DWDM system

contains optical amplifiers along the path. These in-line amplifiers can boost the optical power level irrespective of the optical wavelength without transforming the light signal into the electrical domain. On the transmitter side, the sources are primarily tunable laser sources with extremely narrow line widths. The power from the transmitter operating at different wavelengths can be attenuated with the help of variable attenuators as shown in Fig. 11.3. The multiplexer on the transmitter side combines the signals modulated at different wavelengths $\lambda_1, \lambda_2, \lambda_3, \ldots \ldots \lambda_n$ onto the fiber. The demultiplexer on the receiver side separates the signals based on the wavelength and sends them to the receiver units for reproduction of the signals in the electrical domain.

The DWDM is a very powerful technique for infusing high capacity and multi-functionality in an existing optical fiber network. Implementation of a WDM network needs different passive and active components for coupling, splitting, adding/dropping, isolating, distributing, and amplifying light signals. Some of these important components, such as the coupler, beam-splitter, and optical amplifiers have already been discussed in the previous chapter in connection with IO devices. The active WDM components include an optical filter, optical amplifier, optical source, optical detectors, optical add/drop multiplexers/demultiplexers, etc. The passive components include beam splitter, combiners, circulators, isolators, etc. The

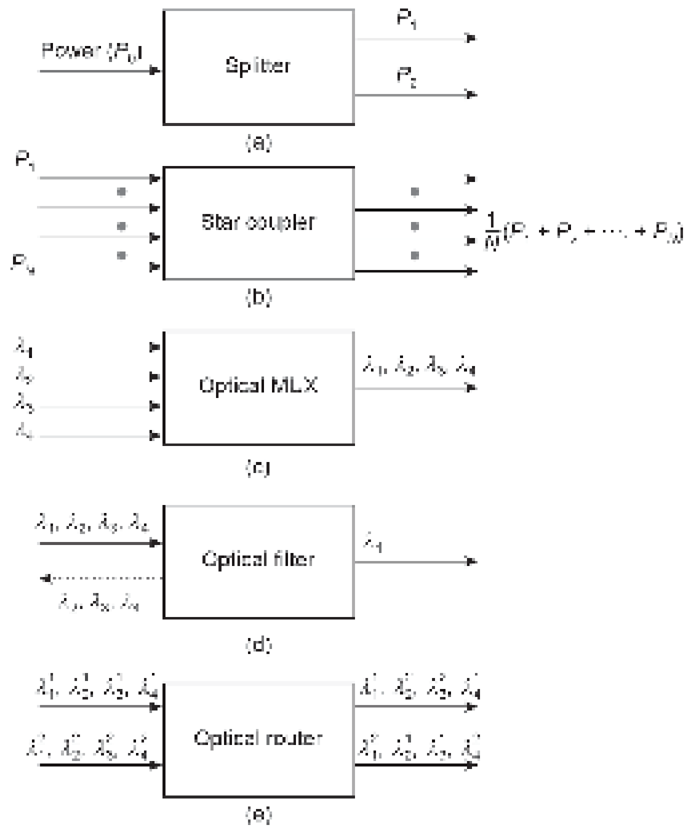

Fig. 11.4 Schematic diagram of different passive DWDM components (a) Splitter (b) Star coupler (c) Optical multiplexer (d) Optical filter (e) Optical router

passive WDM components work in the optical domain for combining, splitting, and filtering light signals in the electrical domain itself. The functions of some of the WDM components are illustrated in Fig. 11.4 with the help of a schematic diagram. The function of the beam splitter is to distribute the optical power from a single beam to two different paths as shown in Fig. 11.4(a). The star coupler (Fig. 11.4(b)) on the other hand couples or splits power available at the input ports to each of the output ports. The function of an optical multiplexer is to combine the signals at different wavelengths available at the input port to one output port to contain all the wavelength components (Fig. 11.4(c)). Fig. 11.4(d) shows the function of an optical filter that selects only one wavelength component from multiple wavelength components present at the input port and delivers the single selected component to the output port. Fig.11.4(e) shows the function of an optical router which diverts selected wavelengths present in the input ports to the output port.

11.4 PASSIVE DENSE WAVELENGTH DIVISION MULTIPLEXING COMPONENTS

The passive DWDM components do not require any external control for their operation. The passive components are designed either by making use of optical fibers or by integrated optics technology discussed in the previous section.

11.4.1 Beam Splitters/Couplers

In a DWDM optical network, it is often necessary to combine signals and/or to split them in a variety of ways. These couplers are described in terms of many inputs and outputs.

Couplers/splitters are available in the form of 1 × 2, 2 × 2, 2 × 1. A 2 × 2 coupler has two inputs and 2 outputs. In general, a $M \times N$ coupler has M inputs and N outputs. In WDM network design, it is often necessary to use a reflective star coupler which accepts many input signals, combines them, and splits the combined signal as many output ports as there are input ports. This type of coupler can be realised by cascading the basic 2 × 2 coupler.

A beam splitter can be realised by the lateral offset method illustrated in Fig. 11.5(a). In this method, the end face of the emitting fiber is placed with axial offset with reference to the end faces of two receiving fibers (van Dorn, 1985). Fractions of light power output from the emitting fiber are coupled to the two receiving fibers depending on the degree of overlapping of the cores of the fibers. The amount of lateral offset can be adjusted to ensure the desired amount of power to be coupled. This type of fiber coupler is not very efficient because of excessive coupling loss at the interface. The device is bidirectional. Only multimode step-index fibers can be used to design this type of coupler/splitter.

Alternatively, a fiber splitter/coupler can be designed by incorporating semi-transparent elements between the fibers as illustrated in Fig. 11.5(b). A partially reflecting surface can be directly incorporated in the fiber end-face cut at an angle of 45° with reference to the axis to form the beam splitter. The splitting of the power in the ports can be adjusted by controlling the reflection/transmission characteristics of the semi-transparent mirror (Agarwal, 1987).

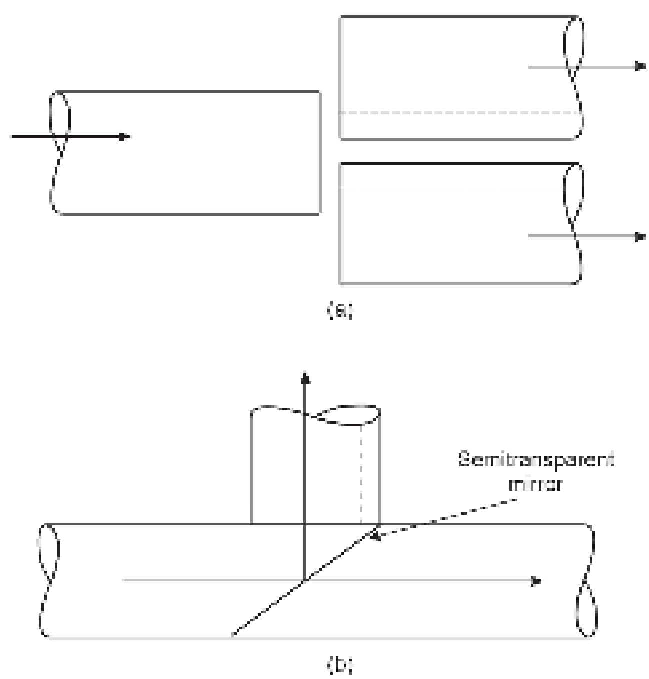

Fig. 11.5 Fiber coupler/splitter: (a) Lateral off-set method (b) Semitransparent mirror method

Another kind of optical fiber coupler is based on micro-optic fiber components in the form of a graded-index (GRIN) rod lens. Beam expansion and collimation properties of a GRIN rod lens can be utilised for the development of a wide range of couplers/splitters by incorporating retro-reflecting mirrors (Kobayashi *et al* 1979). Figure 11.6 shows a micro-optic fiber coupler based on GRIN rod microlens arrangement. The parallel surface type GRIN rod lens arrangement consists of two quarter-pitch lenses with a semi-transparent mirror incorporated within as illustrated in Fig. 11.5(a). In this arrangement, the light rays from the input fiber F_1 are collimated by the first GRIN rod and subsequently fall on the semi-transparent mirror fixed in the middle. A portion of the incident beam is reflected and gets focused on the fiber F_2. The transmitted beam gets focused to a third fiber F_3 placed suitably at the end of the second GRIN rod. The beam coupler can also be realised by the slant surface arrangement shown in Fig. 11.6(b).

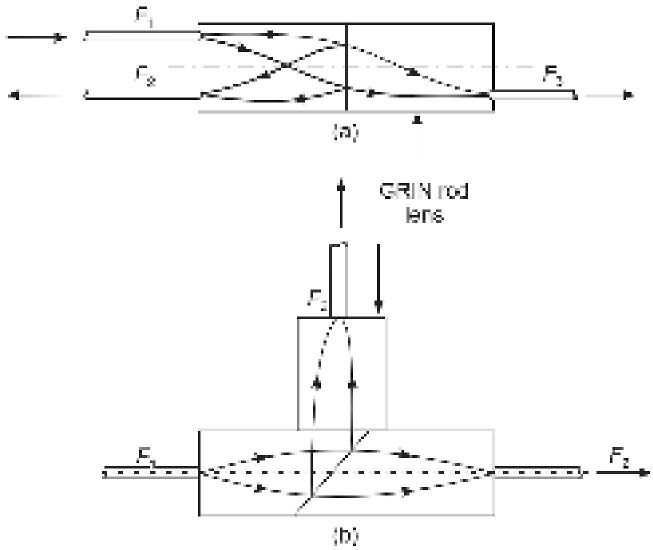

Fig. 11.6 Micro-optic couplers based on GRIN rod lenses: (a) Parallel surface style (b) Slant surface style

The semi-transparent mirror at the interface of the two GRIN rods is inclined at an angle of 45° with reference to the axis of the GRIN rod lenses. The third GRIN rod lens is attached perpendicular to the first two rod lenses. The light beam entering into the GRIN rod lens from fiber F_1 is collimated by the rod before it encounters the semi-transparent slant mirror. The transmitted beam gets focused with the help of an in-line second GRIN rod lens to the fiber F_2. The reflected beam is focused by the third GRIN rod lens fixed in the perpendicular direction as shown in Fig. 11.5(b). The focused beam is received by a third fiber F_3.

11.4.2 Fused Fiber Couplers

Fiber couplers can be designed by fusing fibers in such a way that the modes propagating through one fiber are coupled to the other. The degree of coupling is decided by the length of the coupling region and other factors (Hunsperger *et al* 1995; Yariv, 1991; Srivastava *et al* 1997).

11.4.3 2 × 2 Couplers

Figure 11.6 shows a simple 2 × 2 coupler with two input ports and two output ports realised with the help of two optical fibers. The fibers are fused in the middle and subsequently stretched to create a coupling region as illustrated. The optical power launched to fiber 1 through the input port gets coupled to output ports (2 and 3). The fraction of the input launched power available at each output port is called the coupling ratio. By suitably designing the coupler, it is possible to control the coupling ratio at the output ports.

A coupler with a 50 : 50 coupling ratio at the output ports is called a 3 dB coupler. In this case, half of the input power is available at each of the output ports, hence the name. Two identical single-mode fibers can be fused and stretched to form a coupling region of the desired length. The power flowing through the first single-mode fiber gets coupled to the second one because a significant amount of power leaks out through the fiber due to stretching which reduces the diameter of the fiber. The reduction of core radius to wavelength causes the V-number to decrease sharply. As a result, a significant amount of power spreads out of the core of the fiber. The evanescent field gets coupled into the second fiber. The remaining power continues to flow through the first fiber and becomes available at the output port 2 as illustrated in Fig. 11.7. Ideally, no power should be available at the other end towards the input side of the second fiber. However, because of the reflection

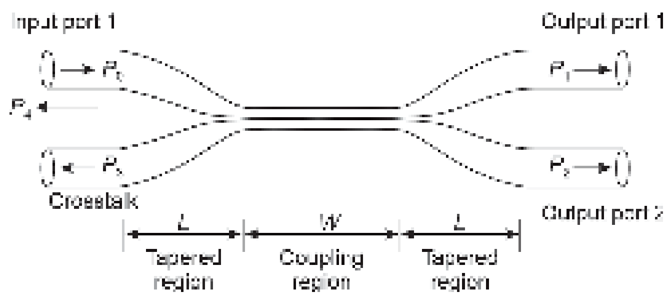

Fig. 11.7 Fused fiber coupler showing the power coupled to output ports

of the power may return through the input ports (Vance *et al* 1995). The reflected powers manifest in the form of cross-talk as illustrated in the figure. The reflected power at the input ports can be made negligibly small by making the tapers very gradual. The coupler is also referred to as a directional coupler because it diverts the power from the input port to the desired output ports. The coupling fractions depend on the length of the coupling region, and the ratio of core radius to wavelength due to stretching, length, and slope of the tapered region. This structure can be designed in a way to allow light to get coupled in either direction. This means that the input and output ports can be exchanged. Such a coupler becomes a bidirectional coupler. This type of coupler can also be realised in the form of an integrated optics waveguide using lithium niobate ($LiNbO_3$) on Si or InP substrate.

Consider the ideal situation in which the two fused fibers are identical single-mode fibers and are completely lossless. The power P_z coupled from the input port of the first fiber to the output port 2 of the second fiber over the axial length z can be expressed as (Ghatak *et al* 1999; Snyder *et al* 1983; Ankiewicz *et al* 1986)

$$P_2 = P_0 \sin^2(\text{kz}) \tag{11.4}$$

where, P_0 is the power launched at the input port 1 of the first fiber and k is the coupling coefficient determined by the interaction of the propagating field in the two fibers (Tewari *et al* 1986).

Assuming conservation of power and $P_3 = P_4 = 0$, we may write

$$\begin{aligned} P_1 &= P_0 - P_2 = P_0[1 - \sin^2(\text{kz})] \\ &= P_0 \cos^2(kz) \end{aligned} \tag{11.5}$$

Equations (11.4) and (11.5) reveal that there is a phase difference between the power carried by the driving fiber (fiber 1) and the driven fiber. The driven fiber lags behind the driving fiber by $\pi/2$. The variation of normalised power with the coupler draw length, z is shown in Fig. 11.8. It can be seen that

$$P_2 = P_0 \text{ for } kz = \left(m + \frac{1}{2}\right)\pi \tag{11.6}$$

$$P_1 = P_0 \text{ for } kz = m\pi \tag{11.7}$$

where, $m = 0, 1, 2, \ldots \ldots \ldots \ldots$

From Eqs (11.6) and (11.7), it can be easily seen that for $z = m\pi/k$, the entire power at the input port is available at the output port 1 and the power at the output port 2 is zero, that is $P_2 = 0$. Similarly, when $z = (2m + 1)\pi/2k$, the entire power is available at output port 2 and no power is available at output port 1, that is, $P_1 = 0$. In other words, as z increases from zero, the power available at the output port 1, of the first fiber decreases while that in the output port 2 of the second fiber increases. The power becomes zero at the output port 1 when $kz = \pi/2$ after which the power starts increasing again and becomes maximum when $kz = \pi$. This suggests that there is a continuous exchange of power between the two fibers. The fiber 1 in the beginning drives the fiber 2 until $kz = \pi/2$ following which fiber 2 starts driving fiber 1 until $kz = \pi$. The parameter k depends on the wavelength of operation (Eisenmann *et al* 1988).

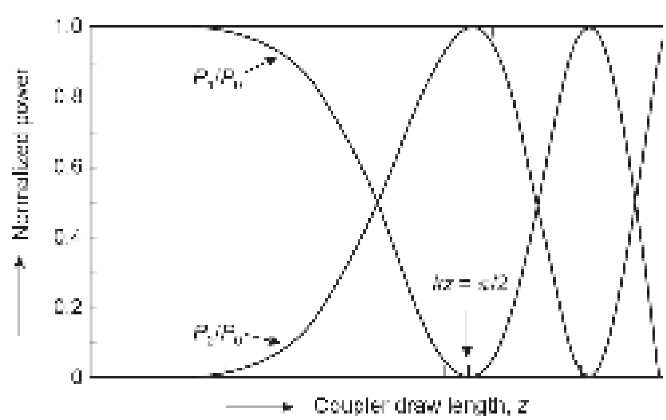

Fig. 11.8 Variation of the normalised power coupled to the output ports with the coupler draw length, z

The minimum interaction length over which the power is completely transferred from the first fiber to the second fiber is

$$z = z_c = \frac{\pi}{2k} \tag{11.8}$$

Therefore, for the coupling length $L = z_c = \pi/2k$ the entire power launched at the input port will be available at the output port 2 of the second fiber.

Example 11.2 It is intended to design a 3-dB coupler using a fused biconical tapered fiber coupler. Estimate the coupling length for interaction.

Solution For a 3-dB coupler, it is necessary that the power available at each of the output ports 1 and 2 of the coupler must be one-half of the input power.

This means that

$$P_1 = P_0 \cos^2(kz) = \frac{1}{2} P_0$$

and also

$$P_2 = P_0 \sin^2(kz) = \frac{1}{2} P_0$$

This situation will arise when the value of z is such that

$$\cos^2(kz) = \sin^2(kz) = \frac{1}{2}$$

That is,

$$kz = \frac{\pi}{4}$$

Therefore, the required length of interaction or coupling length should be

$$L = \frac{\pi}{4k}$$

The performance parameters of an optical coupler are coupling ratio or splitting ratio, excess loss, insertion loss, and cross-talk. The coupling or splitting ratio is defined as the fraction of the total power available at the output port 2 of the second fiber. Referring to Fig. 11.6, the coupling or splitting ratio can be expressed as

$$\text{Splitting or coupling ratio} = \frac{P_2}{P_1 + P_2} \tag{11.9a}$$

$$\text{Splitting or coupling ratio (\%)} = \frac{P_2}{P_1 + P_2} \times 100 \tag{11.9b}$$

$$\text{Splitting or coupling ratio (dB)} = -10\log_{10}\left(\frac{P_2}{P_1 + P_2}\right) \tag{11.9c}$$

In our foregoing discussion, we have assumed that the coupler is lossless. However, this is an ideal situation. In a practical coupler, there is always some loss associated with the transmission of light through the coupler due to attenuation, reflection, and scattering. The loss in the coupler is generally expressed in terms of excess loss and insertion loss. The excess loss is defined as the ratio of the input power to the total power available at the output ports. Referring to Fig. 11.6, we may express the excess loss as

$$\text{Excess loss (dB)} = 10\log_{10}\left(\frac{P_0}{P_1 + P_2}\right) \tag{11.10}$$

The insertion loss generally refers to a loss for a particular port-to-port path from the input to the output side. The insertion loss for the path from input port i to the output port j can be expressed as

$$\text{Insertion loss} = 10\log_{10}\left(\frac{P_i}{P_j}\right) \tag{11.11}$$

The insertion loss between the input port (0) and the output port 2 can be expressed as

Insertion loss (input port 0 and output port 2)

$$= 10\log_{10}\left(\frac{P_0}{P_2}\right)$$

$$= 10\ \log_{10}\left[\left(\frac{P_0}{P_1+P_2}\right)\left(\frac{P_1+P_2}{P_2}\right)\right]$$

$$= 10\ \log_{10}\left[\left(\frac{P_0}{P_1+P_2}\right)\right] - 10\log_{10}\left[\left(\frac{P_2}{P_1+P_2}\right)\right]$$

= Coupling ratio + excess loss

The cross-talk is a measure of the degree of isolation between the input at one port and the optical power reflected into the other input port. Referring to Fig. 11.6, the cross-talk can be expressed as

$$\text{Cross-talk} = 10\log_{10}\left(\frac{P_3}{P_0}\right) \tag{11.12}$$

Example 11.3 An optical power of 100 μW is launched into the input port of a fused biconical tapered fiber coupler. Estimate the values of coupling ratio, excess loss, insertion loss, and cross-talk of the coupler assuming that $P_1 = 40$ μW; $P_2 = 35$ μW and $P_3 = 7.5$ nW.

Solution The percentage coupling ratio can be estimated using Eq. (11.9b) as

$$\text{Coupling ratio (\%)} = \frac{P_2}{P_1+P_2}\times 100 = \frac{35}{40+35}\times 100 = 46.66\%$$

The excess loss can be estimated by using Eq. (11.10) as

$$\text{Excess loss (dB)} = 10\ \log_{10}\left(\frac{P_0}{P_1+P_2}\right)$$

$$= 10\log_{10}\left(\frac{100}{40+35}\right) = 1.25\text{ dB}$$

Using Eq. (11.11), the insertion loss can be calculated as

Insertion loss (input port to output port 1)

$$= 10\log_{10}\left(\frac{P_0}{P_1}\right)$$

$$= 10\log_{10}\left(\frac{100}{40}\right) = 3.98\text{ dB}$$

Insertion loss (input port to output port 2)

$$= 10\log_{10}\left(\frac{P_0}{P_2}\right)$$

$$= 10\log_{10}\left(\frac{100}{35}\right) = 4.56\text{ dB}$$

The cross-talk can be estimated by using Eq. (11.11) as

$$\text{Cross-talk} = 10\log_{10}\left(\frac{P_3}{P_0}\right) = 10\log_{10}\left(\frac{7.5\times10^{-3}}{100}\right) = -41.25\ \text{dB}$$

11.4.4 Star Coupler

A star coupler is a logical extension of the concept of 2 × 2 couplers. An $N \times N$ coupler has N inputs and N outputs. The main purpose of a generalised $M \times N$ coupler is to combine the powers from M inputs and distribute them equally among M output ports. A variety of fabrication techniques can be used to make star couplers. The simplest technique involves three major process steps, i.e. twisting fibers, fusing the fibers, and finally tapering the fibers. A (4 × 4) fused fiber star coupler is shown in Fig. 11.9. The major process steps involved in making the star coupler are illustrated in Fig. 11.10. These include fusion of fiber, use of grating, integrated optics approach using strip waveguide structure, micro-optic technology, etc. Several monolithic and wavelength-flattened (1 × 4), (1 × 7), (4 × 4), (1 × 19) fused fiber couplers have been reported (Arkwright *et al* 1990; Arkwright *et al* 1991; Mortimore, 1986; Mortimore, 1989; Mortimore[1] *et al* 1990; Mortimore[2] *et al* 1990; Mortimore *et al* 1991; Multimore 1986). In principle, N number of identical single-mode fibers can be fused and stretched to produce an ($N \times N$) coupler. However, for $N > 2$, it is practically difficult to control the coupling response between the fused fibers.

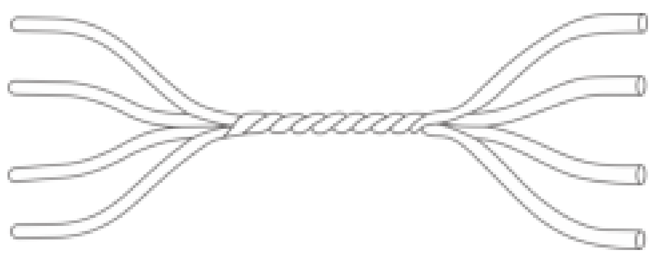

Fig. 11.9 A typical (4 × 4) fused fiber star coupler

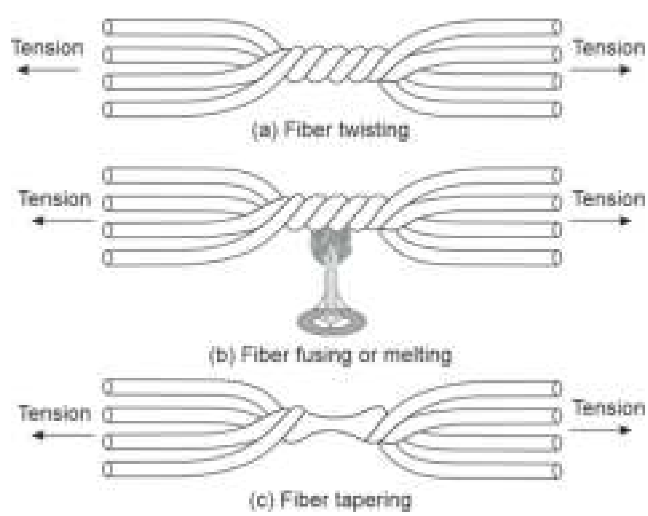

Fig. 11.10 The major steps involved in the process of fabrication of fused fiber star coupler (a) Twisting of fibers (b) Fusion of fibers (c) Tapering of fused fibers

In principle, the optical power from any input in an ($N \times N$) star coupler is evenly divided between the N number of output ports of the coupler. Extending the concept of a (2×2) coupler, the splitting loss can be calculated in the case of an ideal ($N \times N$) star coupler as

$$\text{Splitting loss (dB)} = -10\log_{10}\left(\frac{1}{N}\right) = 10\log_{10}(N) \tag{11.13}$$

The excess loss can be calculated by extending Eq. (11.10) in the case of ($N \times N$) star coupler. If a single input power P_{in} is distributed among N number of output ports, then the excess loss of the fiber star coupler can be obtained as

$$\text{Excess loss (fused fiber star coupler) (dB)} = 10\log_{10}\left(\frac{P_{in}}{\sum_{i=1}^{N} P_{out,\,i}}\right) \tag{11.14}$$

where, $P_{out,\,i}$ is the power available at the ith output port.

It may be pointed out that it is often more convenient to adopt other methods of fabrication of a star coupler including IO techniques and micro-optic techniques.

Example 11.4 An optical power of 100 μW is launched into a single input port of a 16×16 fused fiber star coupler. Estimate the values of splitting loss, excess loss, and the insertion loss of the coupler. The power measured at each output port is 5 μW.

Solution The splitting loss can be estimated by using Eq. (11.13) as

Splitting loss = $10 \log_{10}(N) = 10 \log_{10}(16) = 12.04$ dB

The excess loss can be estimated using Eq. (11.13) as

$$\text{Excess loss (fused fiber star coupler) (dB)} = 10\log_{10}\left(\frac{P_{in}}{\sum_{i=1}^{N} P_{out,\,i}}\right)$$

$$= 10\log_{10}\left(\frac{100}{16 \times 5}\right) = 0.97 \text{ dB}$$

The average insertion loss from the input port to any output port can be obtained as

$$\text{Insertion loss} = 10\log_{10}\left(\frac{100}{5}\right) = 13.01 \text{ dB}$$

It may be noted here that the total loss which is the sum of the splitting loss and the excess loss turns out to be the insertion loss as discussed earlier.

A star coupler can also be realised by cascading several fused biconical taper (FBT) couplers $N \times N$ coupler can be designed by cascading (2×2) 3-dB couplers (4-port) (Marhic *et al* 1984; Mortimore, 1986). This type of star couplers is known as ladder couplers. It is interesting to note here that three-port couplers cannot be cascaded to form a symmetrical ($N \times N$) star couplers. The three-port couplers can produce ($1 \times N$) star coupler. This type of coupler is known as tree coupler. A true symmetrical ($N \times N$) star coupler can only be realised with the help of four port-couplers. This type of cascaded star coupler also provides a low insertion loss and is very attractive for use as a WDM component. The only disadvantage of the cascaded star coupler is that an addition of a few extra nodes calls for the replacement of ($N \times N$) coupler by a ($2N \times 2N$) coupler leaving many of the ports unused. The schematic of an (8×8) star coupler realised by using four-port FBT is shown in Fig. 11.11. From Fig. 11.9, it can

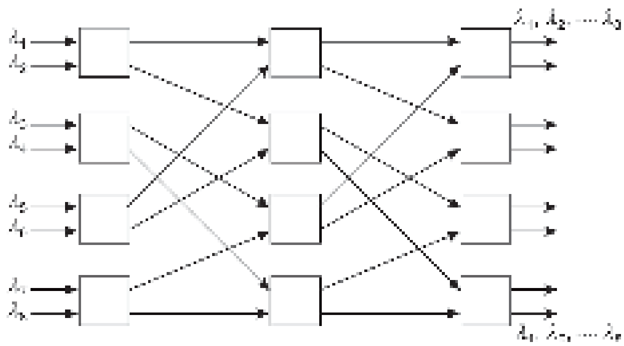

Fig. 11.11 (8 × 8) star coupler formed by cascading 12 four-port couplers (ladder couplers)

be deduced that the number of output ports N obtained with M number of stages of ladder coupler using four port star couplers is 2^M. Therefore,

$$N = 2^M \tag{11.15}$$

That is,

$$M = \log_2(N) \tag{11.16}$$

The number of four-port 3-dB couplers needed to construct the ($N \times N$) star coupler is

$$N_c = \frac{N}{2}\log_2(N) = \frac{N}{2}\frac{\log_{10}(N)}{\log_{10}2} \tag{11.17}$$

For an (8 × 8) cascaded star coupler the number of stages required is

$$M = \log_2(8) = 3$$

The number of four-port 3-dB coupler required is

$$N_c = 4\log_2(8) = 12$$

If the fractional power flowing through each 3 dB coupler element is F_T, then the excess loss can be estimated as (Keiser, 2000)

$$\text{Excess loss (dB)} = -10\log_{10}(F_T^M) = -10\log_{10}(F_T^{\log_2(N)}) \tag{11.18}$$

The total loss can be expressed as

$$\begin{aligned}\text{Total loss (dB)} &= \text{Splitting loss} + \text{excess loss}\\ &= 10\log_{10}(N) - 10\log_{10}(F_T^{\log_2(N)})\end{aligned}$$

That is,

$$\text{Total loss (dB)} = -10\log_{10}\left(\frac{F_T^{\log_2(N)}}{N}\right) \tag{11.19}$$

The Eq. (11.19) can be rearranged as

$$\text{Total loss (dB)} = 10\left(1 - \frac{\log_{10}(F_T)}{\log_{10}2}\right)\log_{10}(N) \tag{11.20}$$

The Eq. (11.20) shows that the total loss increases logarithmically with the number of ports.

Example 11.5 A 16 × 16 ladder type star coupler is designed by cascading four-port (2 × 2) 3 dB couplers. Calculate the number of stages and the total number of the (2 × 2) couplers required to realise the star coupler. If 90% of the power is transmitted through each element, calculate the excess loss and the total loss in dB.

Solution The number of stages required to realise the cascaded (16 × 16) star coupler using four-port (2 × 2) couplers are

$$M = \log_2(16) = 4$$

Total number of four-port (2 × 2) couplers needed is

$$N_c = \frac{16}{2} \times 4 = 32$$

The excess loss can be obtained as

$$\text{Excess loss (dB)} = -10\log_{10}(F_T^M)$$
$$= -10\log_{10}((0.9)^4) = 1.83 \text{ dB}$$

Splitting loss is

$$\text{Splitting loss (dB)} = 10\log_{10}(16) = 12.04 \text{ dB}$$

The total loss is

$$\text{Total loss} = 1.83 + 12.04 = 13.87 \text{ dB}$$

Alternatively, the total loss can be computed using Eq. (11.20).

Significantly low excess loss can be achieved by using a cascaded ladder-type star coupler using single mode (2 × 2) couplers. Excess loss in the range of 0.13 dB only has been reported for an (8 × 8) star coupler (Khoe *et al* 1986).

11.4.5 Multiplexers and Demultiplexers

For successful implementation of WDM, it is necessary to design multiplexers that can combine optical signals from different sources operating at different wavelengths into a single fiber. Further, a demultiplexer is also necessary at the receiver-end to separate the multiplexed signals based on the wavelength and divert them in appropriate channels. The optical sources used in DWDM systems generally have a stable peak wavelength output with extremely narrow linewidth. As a result, interchannel cross-talk is negligible at the transmitter-end. The photodetectors at the receiver-end on the other hand are sensitive to a wider range of wavelengths that may include even the entire DWDM channel. The design of a demultiplexer is therefore much more complex and the designer needs to take care of good channel isolation for different wavelengths used in the multiplexed signal. The DWDM multiplexers and demultiplexers can be designed by adopting any one of the two basic approaches, e.g. exploiting wavelength-dependent angular dispersion by fiber grating filters and surface interaction based on interference filters. A combination of these two techniques can also be used to form a hybrid kind of device. Multiplexers/demultiplexers are generally reversible devices. This means that a demultiplexer can be used as a multiplexer simply by interchanging the input and output ports.

11.4.6 Fiber Grating Filter (Demultiplexer)

In principle, a glass prism can be used as an angularly dispersive element to achieve wavelength multiplexing and demultiplexing of optical signals. The most important WDM component used for wavelength combination and separation is a diffraction grating. A grating consists of parallel equidistant slits of the same width cut on a substrate. A grating can be obtained by blazing lines on an epoxy layer pre-deposited on a glass substrate. The blazing can be done mechanically by scribing lines. Alternatively, grating can also be obtained by etching a single crystal silicon substrate anisotropically (Fujii *et al* 1980). This type of grating is called silicon grating and is superior to a conventional mechanically blazed grating.

A light beam incident on a diffraction grating gets reflected in a particular direction depending upon the grating constant, angle of incidence on the grating surface, and wavelength of the light. Grating-based WDM multiplexers/demultiplexers are generally based on a littrow grating or a plane grating with a lens for collimation/focusing or a concave grating without any lens since focusing is also done by the grating itself. GRIN rod lenses are also used for collimation/focusing purposes. A typical blazed grating, also called echelette (from the French word '*echelle' meaning 'ladder'*) grating is shown in Fig. 11.12. The incident ray makes an angle, α with reference to the grating normal and the diffracted ray makes an angle β. A littrow type grating is one in which the blaze angle of the grating is so adjusted that the incident ray and the reflected ray follow the same path, i.e. $\alpha = \beta$ as illustrated in Fig. 11.12(b).

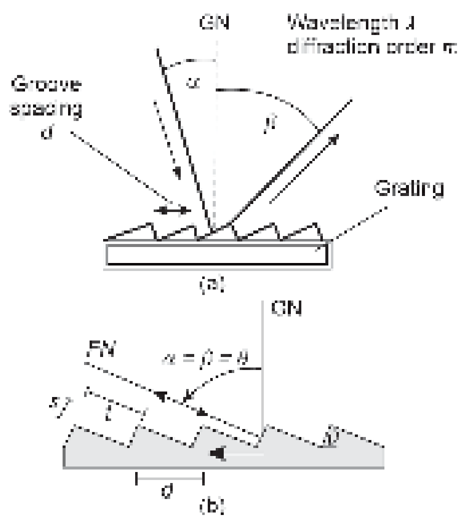

Fig. 11.12 Blazed grating (a) Typical ladder grating (b) Littrow grating (showing single step)

For a littrow grating (Fig. 11.10(b)), the blaze angle can be expressed in terms of the wavelength of the incident light, λ and the line spacing d on the grating as

$$2d \sin \theta_B = \lambda$$

That is,

$$\theta_B = \sin^{-1}\left(\frac{m\lambda}{2d}\right) \tag{11.21}$$

where, m is the diffracted order.

A demultiplexer based on littrow grating mount suitable for WDM application reported by Fujii *et al* (Fujii *et al* 1980) is shown in Fig. 11.13. It consists of a single transmission fiber and five output fibers arranged side-by-side in a row and placed on the focal plane of the lens arrangement. The transmission fiber emits a wavelength multiplexed beam which is diffracted by the littrow grating and subsequently diffracted at different angles depending on the wavelengths and focused to the appropriate output fiber as shown in the figure.

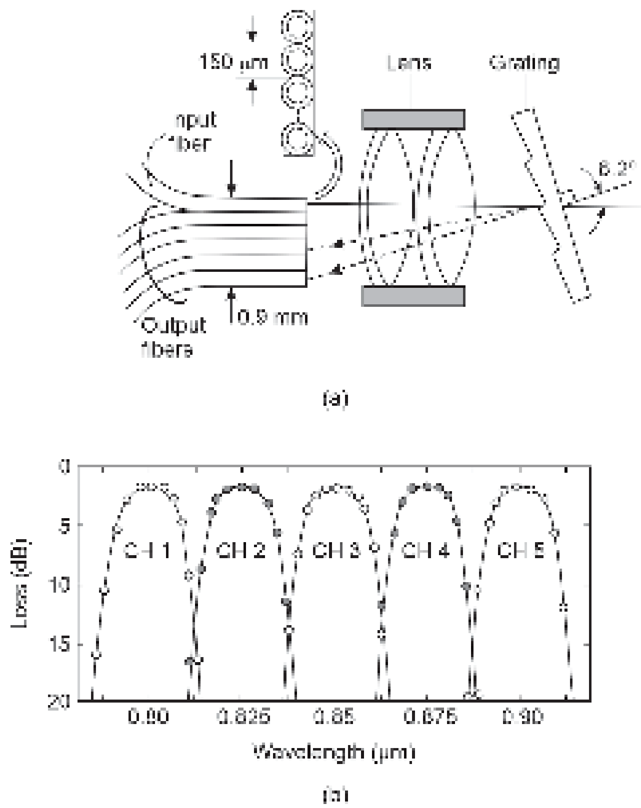

Fig. 11.13 (a) Littrow type grating demultiplexer (b) Spectral response of a typical output fiber of the five channel WDM demultiplexer

The spectral performance characteristics of the 5-channel WDM demultiplexer are shown in Fig. 11.11(b). The figure depicts the variation of the attenuation of light in a typical output fiber over a particular wavelength band (Minowa *et al* 1979). The WDM demultiplexer is to be so designed to exhibit low attenuation in the desired wavelength band and exhibit high interband isolation. The insertion losses of the 5 channels are reported to be of the order of 2dB each while the cross-talk level is < –20 dB (Minowa *et al* 1979).

Another form of WDM multiplexer/demultiplexer device based on an angularly dispersive element is a GRIN rod lens-based configuration shown in Fig. 11.13 (Erdmann, 1983). The assembly consists of a single input fiber carrying the WDM signal consisting of multiple wavelengths λ_1, λ_2, λ_3 and multiple output fibers to receive the light beams carrying the individual wavelengths. The end faces of the fibers are so arranged that they lie on the focal plane of the quarter-pitch GRIN rod lens as shown in the figure (Senior *et al* 1989). The incoming multiplexed beam is collimated by the lens to fall onto the diffraction grating. The grating is offset at the blaze angle so that the beam falls normally on the two groove faces. The desired offset can be achieved by incorporating a prism between the rear end face of the GRIN rod lens and the grating as shown in Fig. 11.13. Alternatively, the end face of the GRIN rod lens can be cut at a suitable angle with reference to the axis and polished before mounting the grating at the end face. The incident beam is subsequently diffracted by the grating to produce angularly dispersed separate beams according to the optical wavelength. The diffracted beams are finally focused by the lens on the end faces of the different output optical fibers as shown in the figure (Erdmann, 1983; Senior *et al* 1989). This type of demultiplexer generally exhibits low insertion loss and cross-talk.

11.4.7 Interference Filter-based WDM Devices

WDM devices can also be realized by using optical filter technology. Two basic forms of optical filter technology are available, e.g. interference filter technology and absorption filter technology. An interference filter can be constructed by depositing alternate layers of high-refractive index materials (such as zinc sulfide) and low-refractive index materials (such as magnesium fluoride). The thickness of each layer is equal to one-quarter of the wavelength of light (Fujii *et al* 1983; Senior *et al* 1989). Such a filter is known as a dielectric thin film (DTF) interference filter. The structure is shown schematically in Fig. 11.14. When a light beam is allowed to pass through such a structure, it undergoes multiple reflections from various interfaces. When the light is reflected within the high refractive index layers, it does not undergo any phase change. On the other hand, when the light beam is reflected within the low refractive index materials it undergoes a phase shift of 180º. The successive reflected beams interfere constructively at the front face of the structure to produce high reflectance over a limited wavelength range. Outside this band, the quarter-wavelength DTF stack gives very low reflectance. A quarter wavelength stack filter can be designed suitably to produce high reflectance in one wavelength region and high transmittance in the other wavelength region.

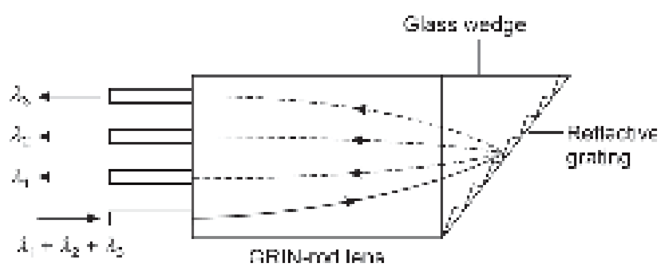

Fig. 11.14 WDM multiplexer/demultiplexer device based on an angularly dispersive element in a GRIN rod lens

Absorption filters on the other hand consist of a thin film of material (such as germanium) that exhibits high absorption at a specific wavelength. By fabricating an interference filter on an absorption layer substrate, it is possible to combine the sharp rejection of the absorption filter along with the flexibility of the interference filter. The combined structure can be used as high-performance edge filters or band-pass filters. An edge filter includes both the long wavelength pass filter (LWPF) and the short wavelength pass filter (SWPF) (Bandetti, 1983). Band-pass type filter has the advantage of having rejection bands on both sides of the pass band making the channel rejection very high (Ishio *et al* 1984).

Two typical WDM filter-based demultiplexer designs are illustrated in Fig. 11.15. Edge filters are generally used in WDM devices for separating two wavelength bands which are at least separated by 10% of the median wavelength. A simple edge filter configuration is illustrated in Fig. 11.15(a). It consists of a fiber cleaved at a specific angle and then an edge filter is incorporated between the end faces of the cleaved fiber (Winzer *et al* 1981). For demultiplexer operation, the multiplexed light is allowed to enter into the fiber from the front-end. After encountering the DTF-edge filter, one wavelength is transmitted through while the other wavelength is reflected by the filter and is collected by a suitably

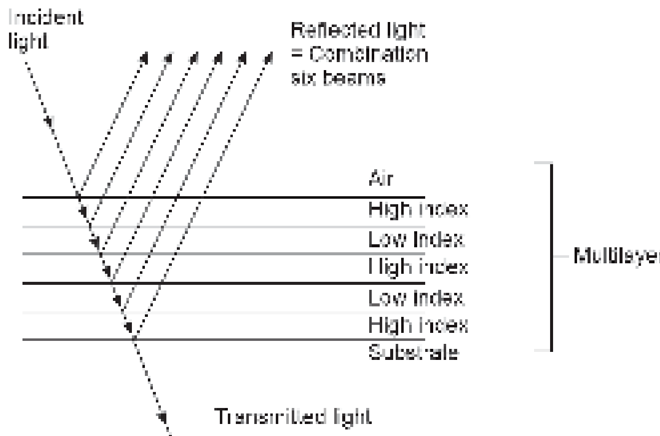

Fig. 11.15 Dielectric Thin Film (DFT) interference filter

positioned receiving fiber. The transmitted wavelength is collected by another fiber. This demultiplexer is reported to exhibit an insertion loss of 2–3 dB and a cross-talk level less than –60 dB (Winzer *et al* 1981) when tested with LED sources emitting at 755 nm and 825 nm, respectively.

A two-wavelength WDM demultiplexer device using a cascaded Bandpass Filter (BPF) sandwiched between two quarter-pitch GRIN rod lenses is illustrated in Fig. 11.15(b). The filter is deposited between the two GRIN rod lenses by DTF-deposition. The lenses are used for collimating/focusing as shown in the figure. When a multiplexed beam (containing two wavelengths (λ_1 and λ_2) is allowed to pass through the device, the beam is collimated onto the cascaded BPF in the middle. One of the wavelength components is reflected and focused by the GRIN rod lens onto the end-face of one of the receiving fibers suitably cemented to the front-end of the GRIN rod lens combination. The transmitted beam containing the other wavelength component is focused by the other GRIN rod lens to be collected by another suitably positioned receiving fiber at the rear-end of the GRIN rod lens combination. Multiple-wavelength multiplexer/demultiplexer devices can be designed by extending the concept of two-wavelength WDM devices. A 4-wavelength WDM multiplexer/demultiplexer device (Sano *et al* 1986) is shown in Fig. 11.16. The structure consists of a series of band-pass filters (BPF) with different pass-band regions cascaded in such a way that the filter transmits a particular wavelength but reflects all others. One of the major disadvantages of the device is that it exhibits a high value of insertion loss which increases linearly with the number of added channels (Sano *et al* 1986).

11.4.8 Mach-Zehnder (MZ) Interferometer-based Multiplexer

A WDM multiplexer device can be designed with the help of a Mach-Zehnder (MZ) interferometer (Syms *et al* 1992; Takato *et al* 1990). An MZ interferometer-based WDM multiplexer device is illustrated in Fig. 11.17. The device consists of two MZ interferometers connected back-to-back with the help of two 3-dB couplers via a central region consisting of two arms, one arm being longer than the other by a length ΔL. The signal traveling through the longer arm develops a phase shift with reference to the signal propagating through the other arm. The 3-dB coupler-1 splits the signal and directs them along the two arms of different lengths. The second coupler combines the signals propagating through the arm at the output port. It is possible to select the length

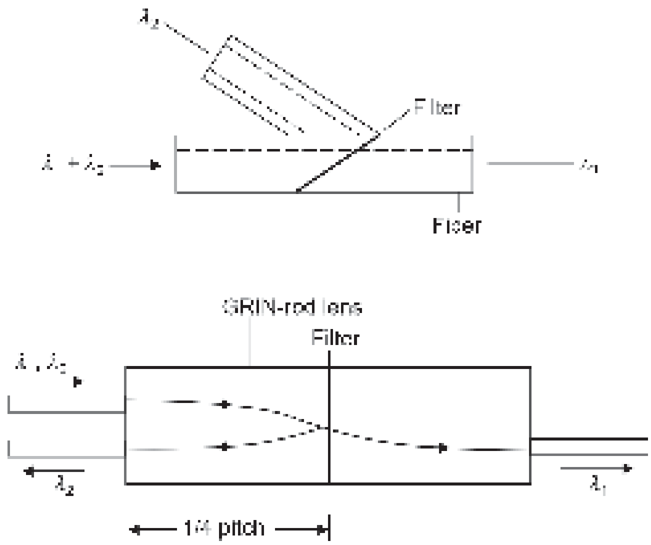

Fig. 11.16 WDM filter-based demultiplexer (a) Edge filter configuration (b) GRIN rod configuration

difference ΔL between the arms in such a way that the signals interfere constructively at one output port and destructively at the other output port as illustrated in Fig. 11.17. The multiplexed (combined) signal emerges from the port where the interference is constructive. This simple arrangement can be extended to achieve an $(N \times N)$ WDM multiplexer device by cascading the basic (2×2) MZ interferometers.

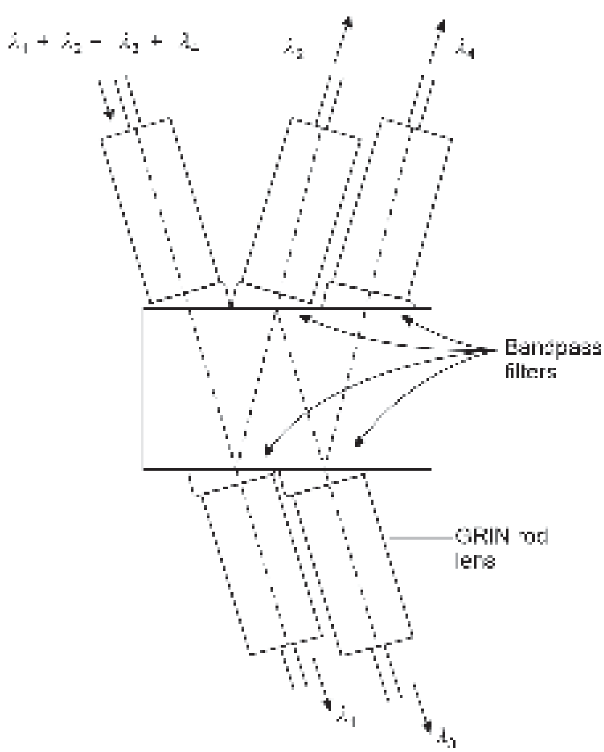

Fig. 11.17 A 4-wavelength WDM multiplexer/demultiplexer

The length difference between the arms of the interferometer can be expressed as

$$\Delta L = \left[2n_{\text{eff}}\left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2}\right)\right]^{-1} = \frac{c}{2n_{\text{eff}}\Delta\nu} \tag{11.22}$$

where n_{eff} is the effective refractive index in the waveguide and $\Delta\nu$ is the frequency separation of the two wavelengths.

11.4.9 Design and Application of Fiber Bragg Grating Demultiplexer

An all-fiber demultiplexer can be designed by making use of fiber Bragg grating (Kashyap, 1999). The photosensitivity of Ge-doped silica fiber can be exploited to make a fiber Bragg grating. When Ge-doped silica glass is exposed to intense UV light, the refractive index of the glass is changed. A periodic variation in the refractive index of the core of a Ge-doped silica fiber can be created by exposing the core to a holographic fringe pattern of two interfering UV beams as indicated in Fig. 11.18. The regions of high intensity shown by shaded portions cause a local increase in the refractive index of the photosensitive core while the local refractive index remains unaltered in the regions of zero intensity shown by the unshaded portions. In this way, a permanent Bragg grating can be written on the photosensitive core of a fiber. When a multiplexed signal comprising several wavelength components propagates through such a Bragg grating fiber, the wavelength that matches with the Bragg wavelength is not allowed to pass.

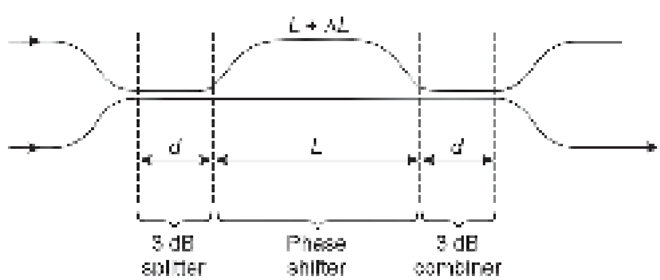

Fig. 11.18 A (2 × 2) MZ interferometer-based WDM Multiplexer

The refractive index along the core varies sinusoidally for the rag grating given by

$$n(z) = n_{\text{core}} + \delta n \cos\left(\frac{2\pi z}{\Lambda}\right) \tag{11.23}$$

where, n_{core} is the core refractive index in the unexposed region, δn is the change in refractive index caused by the exposure and Λ is the period of the interference pattern.

The maximum reflectivity of the grating occurs at the Bragg wavelength given by

$$\lambda_B = 2n_{\text{eff}}\Lambda \tag{11.24}$$

where, n_{eff} is the mode effective index of the core.

The above technique of using fiber Bragg grating for demultiplexing function can be extended to achieve multiple functions including add/drop wavelength (Ramaswami *et al* 1998; Giles *et al* 1999). An example of drop wavelength using a Bragg grating and a circulator is illustrated in Fig. 11.19. The arrangement consists of a three-port circulator and a fiber Bragg grating designed to reflect the desired wavelength. A signal comprising wavelength components λ_1, λ_2, λ_3, λ_4 enters the circulator through port 1 and emerges through port 2 as shown in the figure. The signal subsequently flows through a fiber

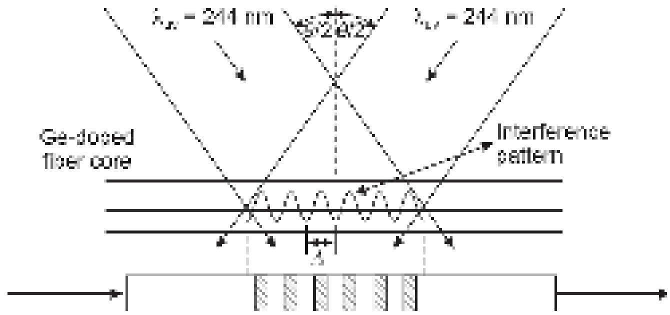

Fig. 11.19 Formation of Bragg grating in a fiber core by two intersecting UV beams

Bragg grating with a Bragg wavelength λ_2. All the wavelength components except λ_2 pass through the grating. The reflected wavelength component λ_2 enters through port 2 and finally exits through port 3 of the circulator. Using multiple circulators and fiber Bragg gratings, it is possible to design more complex multiplexers and demultiplexers (Fig. 11.20).

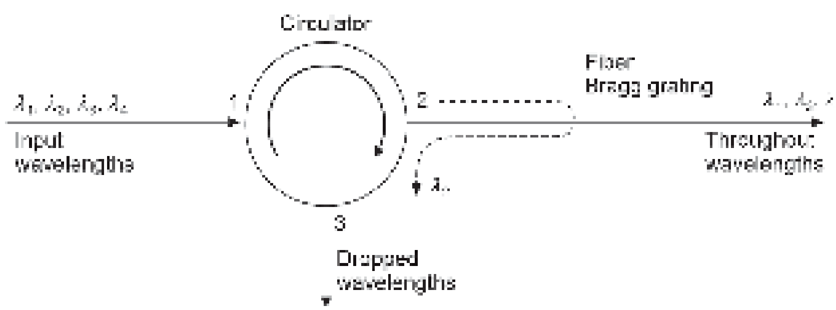

Fig. 11.20 Demultiplexer based on a fiber grating and an optical circulator

11.5 ACTIVE WDM COMPONENTS

Large sections of WDM components used in modern optical communication networks are active devices. These devices include tunable laser sources, optical amplifiers, and optical filters. Active WDM components are those whose characteristics can be controlled electronically. Active components provide greater flexibility in the design of optical networks.

11.5.1 Tunable Laser Sources

For WDM applications, it is necessary to have optical sources emitting different wavelengths for the desired applications. This can be accomplished by using discrete laser sources emitting different wavelengths with extremely narrow line widths for DWDM applications. This approach turns out to be expensive and less reliable. The drift of wavelength due to temperature variations and other factors may become quite troublesome for WDM applications. Tunable laser sources can largely resolve these issues by providing highly stable laser output at desired wavelengths. A tunable laser source is one in which the wavelength of the emitted light from a laser source can be controlled by external agencies. There are three basic approaches to design a tunable laser source. These devices are based on distributed feedback (DFB) or distributed bragg reflector (DBR) lasers discussed earlier in Chapter 5. A tunable laser source may be either a wavelength tunable or a frequency tunable device.

A frequency tunable laser source consists of a single source based on DFB or DBR configuration and incorporates one waveguide-type grating filter within the laser cavity itself (Murata *et al* 1990; Zirngibl, 1998). Frequency tuning can be done by varying the temperature of the device. This is because, the wavelength of emission of a laser source changes approximately by 1 nm/ºC. Alternatively, the emission wavelength of the laser can be changed by varying the injection current in the active region of the laser source. The change in the current into the active region alters the gain of the amplifier in the region and changes the wavelength of emission in the range of 0.8×10^{-2} to 4×10^{-2} nm/mA. This order of changes in the wavelength of emission provides a frequency tunability in the range of 1–5 GHz/mA. For tuning of wavelength of a Laser source, one needs to use two currents, one for lasing action and the other for tuning purposes. A variety of tunable laser sources have been reported (Todt *et al* 2004; Schmidt *et al* 1999). The primary objectives behind the design of tunable laser sources are to achieve stable and high power outputs at the desired wavelengths, a wide range of tuning, and a low-cost of production. There is a trade-off between the maximum power output of a tunable laser and the tuning range. Tunable laser sources can be designed for both the edge emitting and the vertical cavity surface emitting laser (VCSEL) configurations.

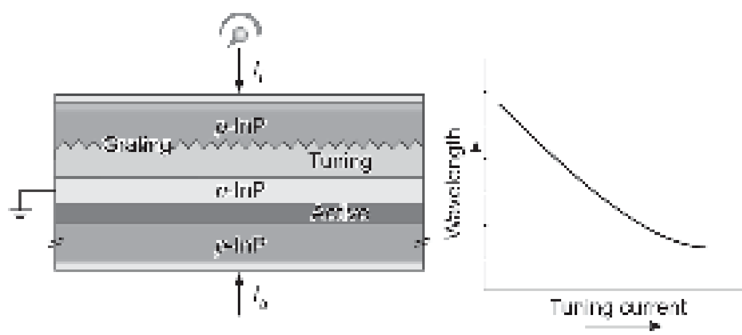

Fig. 11.21 Tunable twin guide laser

The tuning of an edge-emitting laser source is made possible because of the change in the refraction index of one layer caused by injected current. The injected current from the tuning control terminal causes injection of excess carrier in the region resulting in a change in the refractive index due to plasma effect. If the tuning current flows through the DFB grating part of the source, the optical thickness of the grating period length varies. This results in a shift of wavelength of emission of the laser source. As the plasma effect reduces the refractive index following the injection of the tuning current, the wavelength shifts towards the lower-end (blue sift). A simple tunable laser source in the form of a tunable twin guide laser (TTG-Laser) is shown in Fig. 11.21. This structure enables one to have a continuous wavelength variation over a range of about 8 nm by increasing the tuning current I_t.

The tuning range caused by the variation in the effective value of the refractive index resulting from the injected tuning current can be obtained as

$$\frac{\Delta\lambda_{\text{tune}}}{\lambda} = \frac{\Delta n_{\text{eff}}}{n_{\text{eff}}} \tag{11.25}$$

where, Δn_{eff} is the change in the effective refractive index.

To avoid cross-talk between the adjacent channels, a channel spacing of at least ten times the source spectral width $\Delta\lambda_s$ is necessary. This means that

$$\Delta\lambda_{\text{channel}} \approx 10\Delta\lambda_s \tag{11.26}$$

The maximum number of channels that can be accommodated within the tuning range can be obtained as (Keiser, 2002)

$$N \approx \frac{\Delta\lambda_{\text{tune}}}{\Delta\lambda_{\text{channel}}} \tag{11.27}$$

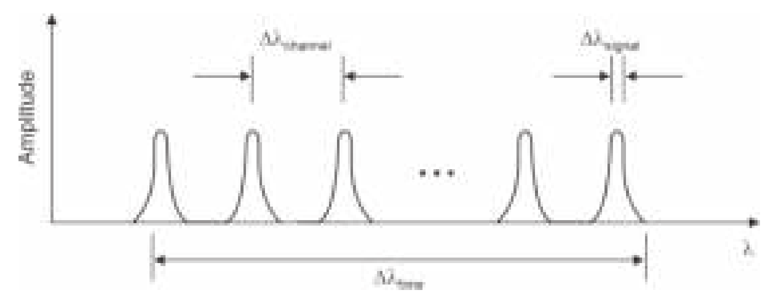

Fig. 11.22 Illustrating the relationship between tuning range, channel spacing and spectral width

The relationship between the tuning range, channel spacing, and the spectral width of the laser source is illustrated with the help of Fig. 11.22. For a large WDM network, it is often necessary to use an array of tunable laser sources. Such an array of practical WDM networks involves complex tunable laser structures based on complex buried heterostructure, MQW, and superlattice configurations (Todt *et al* 2004; Schmidt *et al* 1999).

Example 11.6 The tuning current in a DFB laser operating at 1550 nm can cause a maximum change in the effective refractive index of 0.5%. Calculate the tuning range and compute the number of channels that can be safely accommodated within this range for WDM applications. The spectral width of the source may be assumed to be 0.01 nm.

Solution The tuning range of the DFB laser operating at 1550 nm can be estimated using Eq. (11.25) as

$$\Delta\lambda_{\text{tune}} = \lambda \frac{\Delta n_{\text{eff}}}{n_{\text{eff}}} = 1550 \times 0.005 \approx 8 \text{ nm}$$

Using Eq. (11.26), the channel spacing can be approximated as

$$\Delta\lambda_{\text{channel}} = 10 \times 0.01 = 0.1 \text{ nm}$$

The number of channels that can be accommodated within the tuning range can be estimated using Eq. (11.26) as

$$N \approx \frac{\Delta\lambda_{\text{tune}}}{\Delta\lambda_{\text{channel}}} = \frac{8 \text{ nm}}{1.0 \text{ nm}} = 80$$

11.5.2 Tunable Filters

Tunable optical filters constitute a major component of complex optical networks. A variety of tunable optical filter configurations are available. A tunable optical filter is an active component of a WDM network that allows the flexibility to select, drop, and add the desired wavelengths. The major difference between a passive optical filter and a tunable optical filter lies in the fact that the wavelength selectivity in the latter case can be controlled by some external agencies (Kobrinski *et al* 1989).

The major parameters of interest of a tunable filter for the design of WDM networks include

(i) The tuning range of the filter accounts for the range of wavelength over which the filter can be tuned. This wavelength range may be as high as 200 nm for applications in the range of 1300–1500 nm.

(ii) The spectral bandwidth which accounts for the range of wavelength passed by the filter at 3 dB insertion loss

(iii) Tuning speed that accounts for the time needed to tune the filter at the desired wavelength.

A large number of techniques are available for designing tunable filters. These involve the use of tunable (2 × 2) directional couplers, tunable MZ interferometers, fiber Fabry-Pérot filters, tunable fiber Bragg gratings, tunable waveguide arrays, Acousto-optic Tunable Filters (AOTFs). Figure 11.23 illustrates the application of a directional coupler to work as a tunable filter. The structure consists of a multi-electrode directional coupler made of some electro-optic material such as lithium niobate ($LiNbO_3$) so that tuning can be done with the help of an external electric field. The asymmetric directional coupler has one arm thinner than the other as shown in the figure. There are N number of electrodes to enable one to apply a suitable voltage to select the desired wavelength. The multiplexed signal containing N wavelengths is applied to the waveguide through port 1. The application of a suitable voltage to the electrodes causes the refractive index of the waveguide to change. By suitably adjusting the voltage, it is possible to select a specific wavelength (say, λ_{N-1}) to be coupled to the other arm to exit through port 4 (Brooks *et al* 1995). Further, it is possible to add the dropped wavelength λ_{N-1} to those emerging from port 3. The dropped wavelength can be inserted through the input port 2 and subsequently coupling it to the upper waveguide by applying a suitable voltage across the electrodes. The added wavelength finally exits through the output port along with other wavelength components.

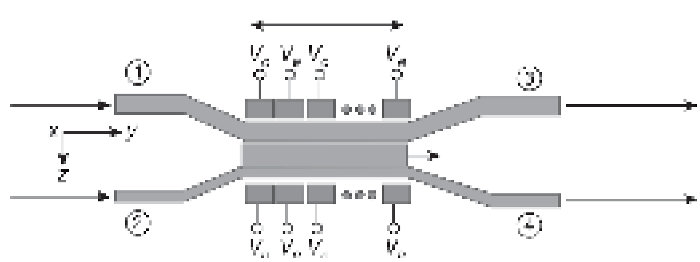

Fig. 11.23 Tunable filter-based directional coupler

A typical application of tunable filters in the design of a WDM network is illustrated in Fig. 11.24. It consists of two three-port circulators and a chain of tunable fiber-based reflection gratings on the path. Each grating can be tuned to the desired wavelength to be dropped. The demultiplexer separates the dropped wavelengths and the multiplexer combines the desired wavelengths for transmission.

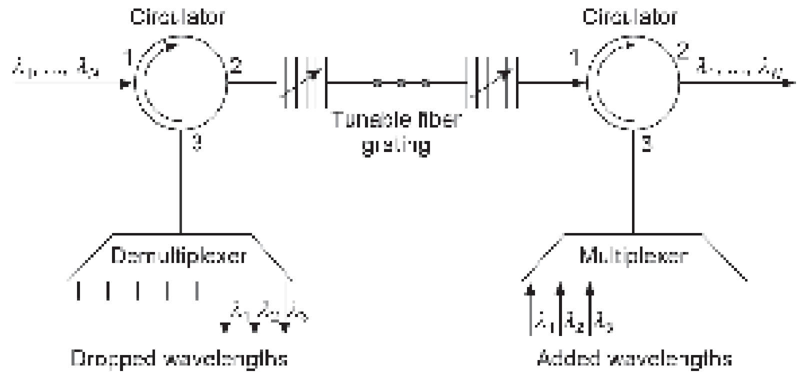

Fig. 11.24 Wavelength add/drop using multiple tunable fiber gratings and circulators

11.6 ELEMENTS OF OPTICAL NETWORKS

This section describes a rudimentary description of optical networks. So far, we have mostly concentrated on point-to-point optical fiber links. A practical optical fiber communication system is based on a much more complex network architecture. To have an overview of the complex optical network, we examine a few basic configurations used in the design of an optical network. It is understood that WDM technology plays a vital role in the design of complex optical fiber networks. With the advent of optical amplifiers in the form of a semiconductor laser amplifier (SLA) or a fiber amplifier (such as EDFA), it is possible to design optically amplified WDM networks. Advanced optical communication techniques, such as soliton transmission, and optical code-division multiple access (optical CDMA) are also becoming increasingly attractive for advanced applications. Some of these topics are beyond the scope of this book.

11.6.1 Basic Network Terminologies

An optical network may be viewed as a data communication network established with optical fiber technology. In optical networks, the optical fiber cables act as the primary communication medium for converting data and transporting the data as light pulses between sender and receiver nodes. The purpose of an optical network is to establish a link between a collection of devices through which the subscribers intend to communicate. These devices may be in different forms, such as telephone receivers, FAX, computer terminals, or any other device for communication. In network terminology, these devices are referred to as *stations*. The stations are also known as data terminal equipment (DTE) in network language. To establish an interconnection between the stations one needs to connect them with the help of a transmission path which may be an optical fiber or wireless (free space optics). The transmission paths interconnecting all these devices constitute an *optical network.* Within the network, a *node* refers to a joint where several stations are connected. A *node* also refers to a joint where several network paths terminate. The network *topology* refers to the physical or logical manner in which the nodes are connected to form a network. The connecting path between nodes or stations is referred to as a *link. Switching* involves the transfer of information from the source to the destination through a series of intermediate nodes. *Routing* on the other hand refers to the selection of a suitable path through the network for the desired purpose. *Protocol* refers to a set of rules and conventions that governs the generation, formatting, control, exchange, and interpretation of information sent through the network or that is stored in a database. A *switched network* consists of a portion interconnected collection of nodes through which the information data entering into the network from a station are routed to the destination after being switched from node-to-node. A *router* is used to interconnect two networks that follow different information exchange rules (protocols) (Dutton, 1998; Keiser, 2002).

The design of an optical network critically depends on its purpose as well as the environment in which it has to operate. The optical network can be classified under the following categories:

1. Point-to-point link refers to an optical network that is limited to direct interconnection between two stations. Such links may be either in the simplex or in duplex form as discussed earlier.
2. Local area network (LAN) generally interconnects stations in a localised area, such as within a building, department, section, factory area, residential complex, or even a University campus where the subscribers can transmit/receive data randomly from any of the stations connected to the network.
3. Metropolitan-area network (MAN) interconnects stations which are located in different buildings and premises of a city and the metropolitan area surrounding the city.
4. Wide-area networks (WAN) interconnects stations located over a large area covering the geographical boundaries of several cities within a country and even beyond.
5. Under-sea network is generally an advanced optical network that uses unmanned submarine repeaters/optical amplifiers making it expensive.

11.6.2 Network Topology

It is often convenient to describe any communication network including optical communication networks in terms of topology. A topology refers to a schematic description of a network arrangement comprising nodes and interconnecting lines (links). There are two forms of topology, e.g. the physical topology and the logical or signal topology.

A physical topology of an optical network refers to a geometric layout of the interconnected stations. There are various forms of physical topologies of an optical network. The following topologies are commonly used to describe an optical network.

1. **Bus topology:** In the bus network topology, each station of the optical network is connected to the main optical fiber cable called *bus.* The bus therefore connects each station to every other station in the network. The termination of stations to the optical bus is somewhat more complex than that in conventional electrical communication networks. Connection of stations to the optical bus is achieved with the help of a passive or an active coupler. An active coupler converts the optical signal present in the data bus into baseband data in the electrical domain before coupling to the station. A passive coupler on the other hand taps a fraction of the optical power flowing through the bus to the station without converting it to the electrical domain. Passive couplers include optical couplers discussed earlier. A typical bus topology is shown in Fig. 11.25(a).
2. **Ring topology:** In the ring network topology all the stations are connected in a closed-loop configuration. The adjacent stations are connected directly while all other pairs of stations are indirectly connected and therefore data pass through one or more nodes. In a ring topology, the interface at each node is an active device to interpret its address

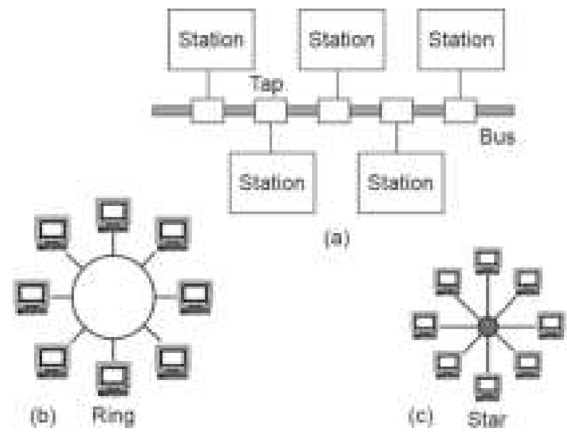

Fig. 11.25 Basic network topology (a) Bus topology (b) Ring topology (c) Star topology

in the data packet received from the other node. The message is accepted by the node if the address matches itself otherwise it is diverted to the next node. A typical optical ring topology is shown in Fig. 11.25(b).

3. **Star topology:** In a star topology, there is a central node to which all the stations are connected directly. The central node may be a passive (hub) or an active (switch) node. An active node at the ercenter helps in the routing of data in the network. This type of network is useful when the communication is primarily between the central node and the outlying stations. A passive node at the ercenter generally acts as a power splitter to distribute the optical signal among the outlying stations connected to the hub. The schematic of a star network topology is illustrated in Fig. 11.25(c).
4. **Mesh topology:** In a mesh topology, the stations are interconnected by point-to-point link. The mesh topology employs either of two schemes, called full mesh and partial mesh. In the full mesh topology, a station in the network is connected directly to every other station. In the partial mesh topology, some workstations are connected to all the others, and some are connected only to those other nodes with which they exchange the most data. This topology is illustrated in Fig. 11.26(a).
5. **Tree topology:** In tree network topology, two or more star networks are connected. The central station of the star network is connected to a main bus. Thus, a tree network is essentially a bus network of star networks. This is illustrated in Fig. 11.26(b).

Physical topology refers to the actual path by which the stations are interconnected (star, ring, bus, etc.) while logical topology refers to the actual path followed by the signal from one node to another node. For example, the logical topology of a network may be the same as the physical topology in many cases. However, it may so happen that some networks are physically laid out in a star configuration, but they operate logically as bus or ring networks.

11.6.3 Active Optical Coupler

An active optical coupler converts the optical signal from the optical bus into an electrical signal and transfers it to the terminal. After appropriate processing, the terminal sends the electrical signal to an optical transmitter which converts the electrical signal back into the optical domain. The optical signal is coupled to the bus for onward transmission to the next terminal (Keiser, 2002; Khare, 2004). The advantage of using an active optical filter

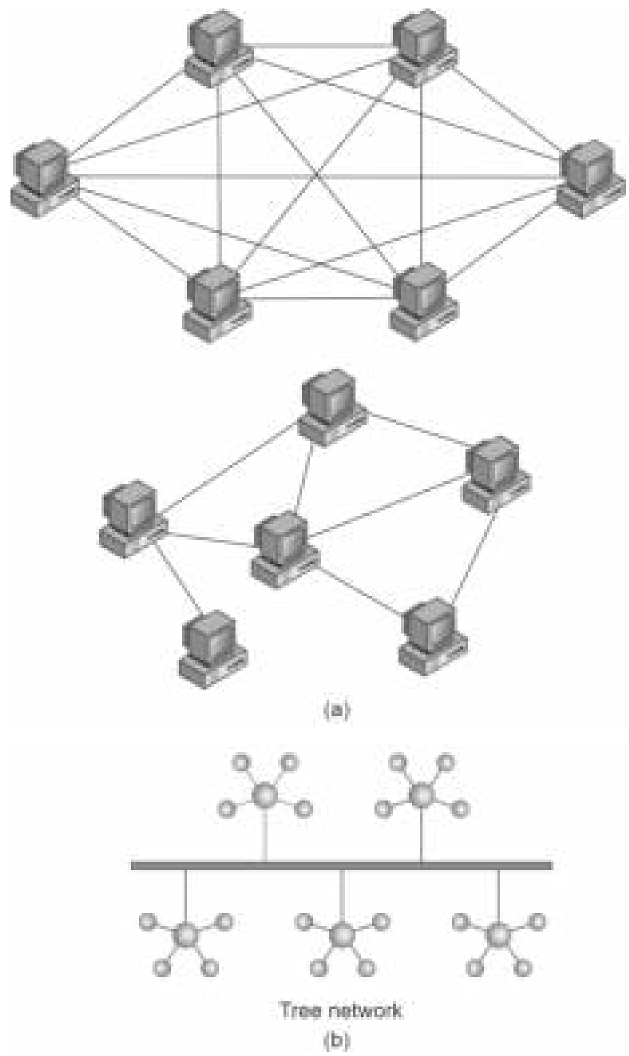

Fig. 11.26 (a) Mesh network topology Full mesh (upper one) and partial mesh (lower one) (b) Tree network topology

lies in the fact that each terminal effectively acts as an optical repeater. As the signal is regenerated at each terminal, an infinite number of terminals can be coupled to the bus without affecting the performance. A major disadvantage is that the bus network fails when any one of the terminals stops functioning. The block diagram of an active optical coupler is shown in Fig. 11.27.

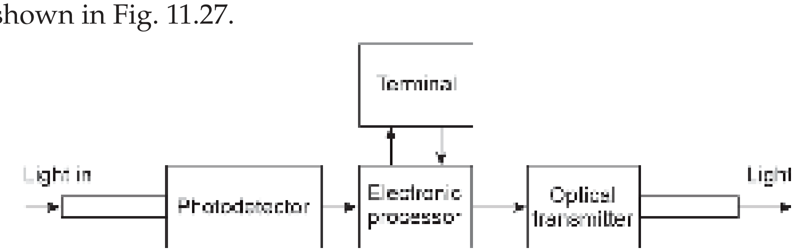

Fig. 11.27 Schematic of an active optical coupler

11.6.4 Passive Optical Coupler

A passive optical coupler couples the optical signal from the main optical bus to the terminal without converting it into the electrical domain. The optical power flowing through the bus decreases continuously due to the tapping of optical power at each terminal connected to the bus. There is no regeneration of the signal in the case of a passive coupler, unlike an active optical coupler. As a result, a finite number of terminals can be connected to this type of network in a few coupling losses at each terminal as well as average loss in the fiber. In a passive optical coupler, light signal can be inserted from the transmitter directly into the bus. A linear passive bus couple consists of four ports. The input and output ports are connected with the inline fiber bus as shown in Fig. 11.28. Out of the remaining two ports, one is used for coupling the light to the receiver of the terminal while the other is used for inserting the light after tap-off to keep the light out of the local terminal.

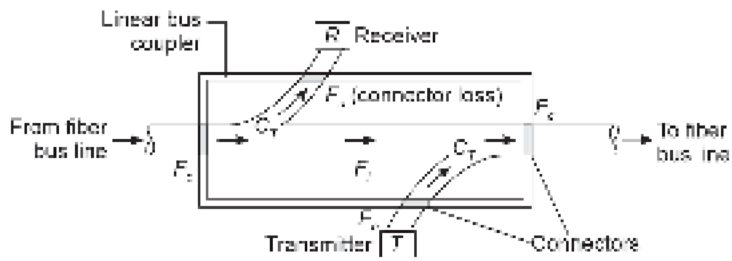

Fig. 11.28 Passive linear bus coupler (after Keiser)

It may be noted here that a fraction of the optical power is coupled to every port connected to the bus. At each connector, there is an additional connecting loss. If a fraction F_c of the optical power is lost at each coupling port, the connecting loss at each coupling can be expressed as

$$L_c = -10\log_{10}(1 - F_c) \tag{11.28}$$

Consider that a fraction C_T of the total power flowing through the bus is removed from it and delivered to the receiver port of the terminal. The tap loss which corresponds to the power extracted from the bus can be expressed as

$$L_T = -10\log_{10} C_T \tag{11.29}$$

For a symmetrical coupler, the same fraction C_T of the power is coupled from the transmitting input port to the bus. The power coupled to the bus is $C_T P_0$, P_0 being the power launched from the source flylead.

The throughput coupling loss can be expressed as (Keiser, 2002)

$$L_{\text{thru}} = -10\log_{10}(1 - C_T)^2 \tag{11.30}$$

In addition to the above loss components, each coupler introduces an intrinsic transmission loss L_i given by

$$L_i = -10\log_{10}(1 - F_i) \tag{11.31}$$

where, F_i is the fraction of the power lost in the coupler.

Let us assume that N number of stations which are uniformly separated by a distance L are connected to the bus as shown in Fig.11.29. The fiber attenuation between two adjacent nodes can be expressed as

$$L_f = -10\log_{10}\left(\frac{P(L)}{P_0}\right) = \alpha L \tag{11.32}$$

where, α is the fiber attenuation in dB/km.

11.7 POWER BUDGET

Let us carry out a power budgeting for adjacent stations (say stations 1 and 2) (Fig. 11.29). Let P_0 stands for the optical power launched from the source fly-lead of station 1. The power detected at station 2 can be expressed as

$$P_{1,2} = A_0 C_T^2 (1 - F_c)^4 (1 - F_i)^2 P_0 \tag{11.33}$$

where, A_0 is the attenuation measured as the fractional power loss over the fiber length L given by

$$A_0 = \frac{P(L)}{P(0)} \tag{11.34}$$

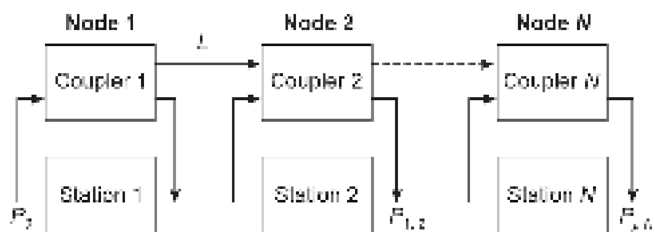

Fig. 11.29 Linear optical bus connecting *N* uniformly spaced stations

The overall loss between station 1 and station 2 can be expressed using Eqs (11.28) to (11.34) in decibels as

$$10 \log_{10}\left(\frac{P_0}{P_{1,2}}\right) = \alpha L + 2L_T + 4L_c + 2L_i \tag{11.35}$$

By extending the above calculation, the power received at the Nth station from station 1 can be obtained as (Keiser, 2002)

$$P_{1,N} = A_0^{N-1} (1 - F_c)^{2N} (1 - C_T)^{2(N-2)} C_T^2 (1 - F_i)^N P_0 \tag{11.36}$$

The power budget for the system can therefore be expressed as

$$\begin{aligned} 10 \log_{10}\left(\frac{P_0}{P_{1,N}}\right) &= (N-1)\alpha L + 2NL_c + (N-2)L_{\text{thru}} + 2L_T \\ &= N(\alpha L + 2L_c + L_{\text{thru}} + L_i) - \alpha L - 2L_{\text{thru}} + 2L_T \end{aligned} \tag{11.37}$$

It can be easily seen that the loss in dB is directly proportional to the number of stations.

Example 11.6 Ten uniformly spaced stations are connected to an optical bus having the following parameters:

C_T	L_T *(dB)*	L_{thru} *(dB)*	L_i *(dB)*	L_c *(dB)*	α *(dB)*
5%	7.5	1.0	0.6	1.0	0.5

Assuming that the separation between the adjacent stations is 500 m, prepare a power budget for the optical bus.

Solution For 10 uniformly spaced stations with a separation of 500 m, the power budget can be estimated as

$$10\log_{10}\left(\frac{P_0}{P_{1,N}}\right) = N(\alpha L + 2L_c + L_{\text{thru}} + L_i) - \alpha L - 2L_{\text{thru}} + 2L_T$$
$$= 10\times(0.5\times\frac{1}{2} + 2\times1 + 1 + 0.6) - 0.5\times\frac{1}{2} - 2\times1 + 2\times7.5$$
$$= 10\times3.85 - 2.25 + 15 = 51.25 \text{ dB}$$

11.8 SYNCHRONOUS OPTICAL NETWORK (SONET)/SYNCHRONOUS DIGITAL HIERARCHY (SDH)

Synchronous optical networking in North America and synchronous digital hierarchy in other parts of the world are standardized time division multiplexing (TDM) formats for transmission of multiple-bit streams over optical fiber using optical sources (LEDs/ILDs). Lower data rates can also be transferred using an electrical interface. SONET/SDH are essentially the same and were designed to transport circuit mode communication from different sources. The basic advantage of SONET/SDH is that it allows simultaneous transport of many different circuits of different origins within a single framing protocol. Because of transport-oriented features, SONET/SDH is the ideal choice for transporting asynchronous transfer mode (ATM) frames.

11.9 TRANSMISSION FORMAT

The basic unit of framing in SDH is the synchronous transport module, level 1 (STM-1). It operates at 155.52 Mbps. SONET refers to this basic unit as a synchronous transport signal 3, concatenated (STS-3c) when the signal is carried electrically and OC-3c when the signal is transmitted optically. SONET also offers an additional basic unit of transmission called Synchronous Transport Signal 1 (STS-1). The basic structure of the STS-1 SONET frame is shown in Fig. 11.30. It is essentially a two-dimensional structure comprising 90 columns and 9 rows of bytes. Each byte contains 8 bits. As per standard SONET terminologies, a section refers to the shortest connection between adjacent submodules of a SONET device, a line connects two SONET devices (a longer link) and a path is a complete end-to-end

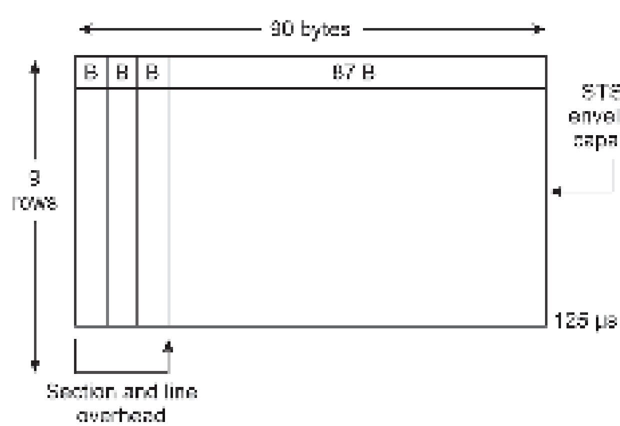

Fig. 11.30 Basic structure of STS-1 SONET frame (B: One byte consisting of 8-bits)

link. The duration of the fundamental frame of a SONET is 124 μs. The transmission bit rate of the basic SONET signal is given by

$$\text{STS-1 rate} = \frac{(90 \text{ bytes/row})\,(9 \text{ rows/frame})\,(8 \text{ bits/byte})}{125\ \mu\text{s/frame}}$$

$$= 51.84 \text{ Mbps} \tag{11.38}$$

This value is exactly one-third of an STM-1/STS-3c/OC-3c carrier which has a bit rate of 155.52 Mbps. This speed is decided by the bandwidth requirements for transmission of standard PCM-encoded telephonic voice signals. STS-1/OC-1 circuit at this rate supports the bandwidth requirement of a standard 672 voice channels of 64 Kbps of DS-3 standard. All other SONET signals are integer multiples of the basic STS-1. For example, in SONET, the STS-3c/OC-3c signal is composed of three multiplexed STS-1 signals. In general, an STS-N signal or in SONET terminology OC-N signal has a bit rate given by

$$\text{STS-}N \text{ rate} = 51.8 \times N \tag{11.39}$$

Some manufacturers also support the SDH equivalent of the STS-1/OC-1, known as synchronous transport module-level 0 (STM-0). More conventionally, the basic SDH rate is identified with STS-3 of SONET having a bit rate of 155.52 Mbps and is called STM-1 in SDH. Higher rates of SDH are designated by STM-M. The values of M supported by the International Telecommunication Union (ITU-T) are (M =) 1, 4, 16, 64 while in SONET OC-N signals $N = 3M$ that is, the values of N are N = 3, 12, 48, 192. In practice, the compatibility between SONET and SDH is maintained by choosing N as a multiple of 3. In SDH logical electrical signal, STS-N and physical optical signal OC-N are not distinguished and both are designated by STM-M. The commonly used SONET and SDH transmission rates are listed in Table 11.1.

In packet-oriented data transmission, such as Ethernet, a packet frame generally consists of a header carrying the network management information followed by the remaining field constituting the synchronous payload envelope (SPE) or simply payload. The header is first transmitted and then the payload. In SONET, the header is called overhead and is not transmitted before the payload. Generally, the overhead is interleaved with the payload during transmission in the sense that a part of overhead is transmitted followed by a part of the payload and then the next part of overhead is

Table 11.1 SONET and SDH standards

Optical carrier	*Data rate (Line rate) (Mbps)*	*Overhead rate (Mbps)*	*Payload-SONET (SPE) (Data rate-overhead)*	*User data rate (Mbps)*	*SONET STS (ANSI)*	*SDH STM (CCITT)*
OC-1	51.84	1.728	50.112	49.536	STS-1	–
OC-3	155.52	5.184	150.336	148.608	STS-3	STM-1
OC-9	466.56		451.044	445.824	STS-9	STM-3
OC-12	622.08	20.736	601.344	594.824	STS-12	STM-4
OC-18	933.12		902.088	891.648	STS-18	STM-6
OC-24	1244.16		1202.784	1188.864	STS-24	STM-8
OC-36	1866.24		1804.176	1783.296	STS-36	STM-12
OC-48	2488.32	82.944	2400	2377.728	STS-48	STM-16
OC-192	9953.28	331.776	9600	9510.912	STS-192	STM-64

followed by the next part of the payload, and so on until the entire frame is transmitted. For example, the frame of STS-1 is 810 octets in size. The frame is transmitted as 3 octets of overhead followed by 87 octets of payload. This is repeated 9 times until all 810 octets are transmitted. The entire process is completed within 125 μs. There is some minor difference in the internal structure of the overhead and payload in SONET and SDH. Further, different nomenclatures are used in the standards to describe these structures. Nevertheless, the implementation of their standards is very similar making it easy to interoperate between SDH and SONET at any given bandwidth. In practice, STS-1 designation refers to a signal in the electrical domain while OG refers to a signal in the optical domain. However, the designations STS-1 and OC-1 are sometimes used interchangeably.

SONET/SDH specifications provide details for optical source and detector characteristics and transmission distances of various types of fibers (GI fiber in the 1310 nm window, single mode fibers in the 1310 and 1550 window range, and dispersion-shifted SM fibers in the 1550 nm window). The details are specified in American National Standard Institute (ANSI) standards such as ANSI T1.105.06 as well as International Telecommunication Union (ITU) standards such as ITU-T G.957. Table 11.2 lists the specifications of different components and standards (ITU-T, 1995). These standards enable different manufacturers of optical components to ensure interconnection compatibility between equipment from different manufacturers.

Table 11.2 SONET/SDH standards and specifications for optical sources and detectors

Distance	*SONET terminology*	*SDH terminology*	*SONET rate*	*SDH rate*	*Source output power (dBm)*	*Receiver sensitivity (dBm)*
≤15 km	Intermediate-reach	Short-haul	OC-3	STM-1	–15 to –8	–23
			OC-12	STM-4	–15 to –8	–23
≤40 km			OC-48	STM-16	–10 to –3	–18
			OC-3	STM-1	–15 to –8	–28
			OC-12	STM-4	–15 to –8	–28
			OC-48	STM-16	–5 to 0	–18
≤80 km	Long-reach	Long-haul	OC-3	STM-1	–5 to 0	–34
			OC-12	STM-4	–3 to +2	–28
			OC-48	STM-16	–2 to +3	–27

11.10 SOLITON

In Chapter 4, it is discussed that dispersion in optical fiber causes the spreading of the optical pulses as they propagate through the fiber. The spreading or broadening of the pulses arises from group velocity dispersion (GVD) which is viewed as a linear effect in which the refractive index of the material depends on the wavelength of the optical signal. Further, when an optical signal propagates through a silica fiber, the non-linear effects in silica cause self-phase modulation (SPM) of the signal due to the Kerr non-linearity. The SPM is essentially caused by intensity-induced change in the refractive index of the material. While GVD is a linear effect, SPM is a non-linear effect. The SPM is considered to be positive if the refractive index increases with the increase in the intensity of the light. The change in the refractive index is caused by the time-varying intensity of the light.

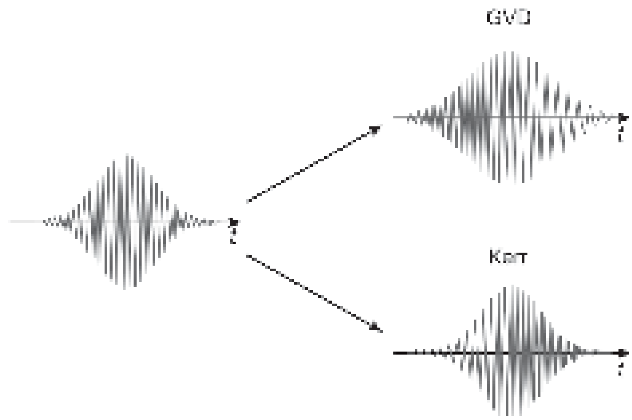

Fig. 11.31 Illustrating the effect of GVD and SPM on a Gaussian pulse propagating through the medium

This results in a time-dependent phase-shift. As the rate of change of phase is related to the frequency, the intensity-induced SPM is finally manifested in the form of a change in the pulse spectrum. The effect of GVD and SPM on a Gaussian pulse is illustrated in Fig. 11.31. When both GVD and SPM are positive then the light pulse propagating through the fiber undergoes temporal and spectral broadening. On the other hand, if a fiber offers a negative GVD (anomalous dispersion), then it may be used to compensate for the positive SPM of the fiber. By suitably adjusting the two effects it is possible to create a situation where an optical pulse can propagate through the fiber without any spreading. Under this circumstance, it is possible to transmit the signal over a large distance without any significant distortion of the pulses that may warrant the use of regenerative repeaters. The distance is, however, limited by the attenuation of the fiber. We have already seen that dispersion is much more a severe problem in optical fiber transmission than attenuation as the former tends to limit the speed of transmission (bit rate) by causing intersymbol interference (ISI). With the advent of single-mode fibers and optical amplifiers, the attenuation problem can be tackled by using in-line optical amplifiers to boost the optical power as and when required. Soliton is a special type of pulse that exploits the SPM caused by Kerr effect to compensate for the pulse-broadening effect induced by GVD (Mollenauer *et al* 1991; Haus, 1993; Haus *et al* 1996). The temporal changes in high-intensity narrow pulses subjected to the Kerr effect as it propagates through a fiber that has a positive GVD parameter and through a fiber that has a negative GVD parameter are shown in Fig. 11.32(a) and (b), respectively. It is seen that in the second case, the pulses neither change their shape nor their spectra as they propagate through the fiber. These pulses constitute fundamental soliton. It may appear to be a little baffling to appreciate the compensation of SPM by GVD because the former effect is manifested in the frequency domain while the GVD affects the pulse in the time domain. It is interesting to note here that the addition of a small time-dependent phase shift to a Fourier-transform-limited pulse does not change the spectrum to the first order. If the GVD of the medium can be so adjusted that the phase shift is cancelled then the pulse does not change its shape or spectrum as it propagates through the medium. Solitons are very narrow and high-intensity optical pulses that retain their shape by balancing pulse dispersion due to GVD with the non-linear properties of the fiber. The invention of the soliton pulse turned out to be a remarkable breakthrough in the field of optical fiber communications. The possibility of stationary transmission of bright or dark soliton in the anomalous and normal dispersion regimes, respectively,

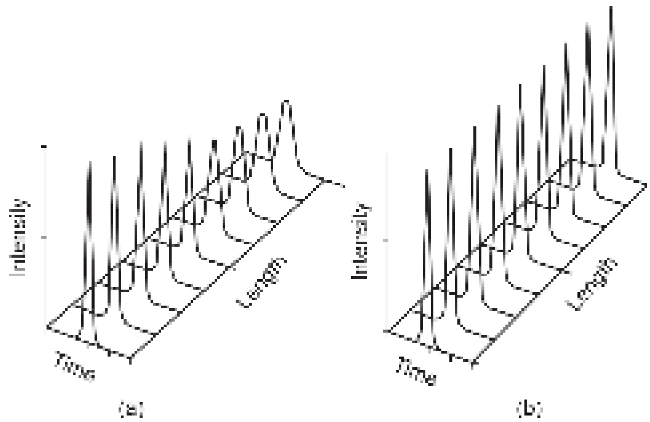

Fig. 11.32 Temporal changes in a narrow high-intensity pulse subjected to SPM caused by non-linear effect and linear GVD (a) For a positive GVD parameter (b) For a negative GVD parameter

in the single mode fiber, was predicted by Hasegawa *et al* (Hasegawa *et al* 1973). This prediction was then successfully demonstrated experimentally by Mollenauer *et al* (Mollenauer *et al* 1980). Solitons are localised solitary waves that exhibit following given properties:

(i) Propagate at a constant speed without changing their shape.
(ii) Extremely stable to perturbations, and in particular to collisions with small amplitude linear waves.
(iii) Also stable concerning collisions with other solitons.

The immunity of soliton pulses against distortion from nonlinearity effects and dispersion effects, which are inherent in fibers, motivated researchers to develop an all-optical transmission system without involving regenerative repeaters. The regenerative repeater in conventional optical fiber communication systems is not only expensive but also creates a major bottleneck for enhancement of the speed of transmission. This is because a regenerative repeater needs the light signal to be converted in the electrical domain for processing, boosting and reshaping. This frequent conversion from E/O and O/E slows down the speed of the transmission system. An all-optical link does not suffer from this problem. However, all-optical transmission needs optical amplifiers to compensate for the attenuation caused by the fiber. It may be pointed out here that the optical amplifiers tend to solve the loss problem but at the cost of increased dispersion problems. An optical amplifier cannot restore the amplified signal to its original state.

Dispersion-management (DM) schemes are used to compensate for this problem.

Summary

- Some advanced techniques which find applications in modern and futuristic optical fiber communication system, such as WDM, coherent optical communication and optical soliton propagation are discussed.
- The WDM involves the technology of combining multiple wavelengths carrying different signals onto a single fiber.
- The early generation WDM contains 8–18 channels on a single fiber. This type of WDM is called Coarse WDM (CWDM).

- With the advent of high quality laser source with extremely narrow line width it is possible to multiplex a very large number of channels (usually in the multiples of 16). Such WDM systems are called Dense WDM (DWDM) systems.
- Some important passive DWDM components include beam splitter, star coupler, optical multiplexer, optical filter, optical router, etc.
- Fiber coupler/splitter can be realized by using lateral off-set method of fibers or by using semitransparent mirror built within the fiber.
- A simple 2 × 2 coupler has two input ports and two output ports. The important parameters of the coupler are splitting/coupling ratio, excess loss, insertion loss, crosstalk, etc.
- A star coupler is a logical extension of the concept of 2 × 2 couplers. An $N \times N$ coupler has N inputs and N outputs.
- Multiplexers/demultiplexers are important components of WDM system and can be used interchangeably. In principle, a glass prism can be used as an angularly dispersive element to achieve wavelength multiplexing and demultiplexing of optical signals.
- WDM multiplexer/demultiplexer device can be designed by using an angularly dispersive element in a GRIN rod lens.
- WDM filter-based demultiplexer can be realized using edge filter configuration and GRIN rod configuration.
- WDM demultiplexer can be designed with the help of MZ interferometers and Bragg's grating structure.
- Active WDM components include tunable laser and tunable filter.
- The elements of optical communication network are discussed along with the basis of classifications of optical networks (LAN, MAN, and WAN) in terms of coverage.
- The network topologies include bus, ring, star, mesh, tree, etc.
- Passive and active couplers used in bus network are discussed.
- The essence of SONET/SDH is discussed with special reference to various standards.
- The principle of soliton pulse propagation is discussed. It is envisaged that soliton pulses can be used for long distance communication without using any repeater.
- The chapter concludes with discussion of an entirely different mode of optical communication known as coherent optical communication.
- Coherent optical receivers are more sensitive than their counterparts in IM/DD systems.
- They are more selective because the channel selection is done in the electrical domain by using sharp microwave filters (~ several GHz) in place of broad band optical filters (~ 10^5 GHz) used in direct-detection receivers.
- Unlike an IM/DD system where the light is directly modulated by using a drive circuit to cause an intensity modulator, coherent system makes use of a CW narrow linewidth laser source as an optical oscillator.
- Coherent detection can be done by homodyning (incoming signal frequency equals the LO frequency) or by heterodyning (LO frequency is different from incoming signal) technique.

Problems

11.1 A silica fiber exhibits low attenuation in the entire range of transmission ranging from 1260–1340 nm about 1300 nm. It is intended to use this spectral range for implementation of WDM using laser transmitters with narrow spectral widths. If the required channel spacing is 100 GHz, calculate the number of channels that can be supported by the system.

11.2 Estimate the wavelength corresponding to the nominal center frequency of 193.1125 THz anchored about 193.1000 THz for a channel spacing of 12.5 GHz as per ITU-T G.694.1 standard. Assume the velocity of light to be $c = 2.99792458 \times 10^8$ m/s.

11.3 Estimate the value of the required coupling length of a fused biconical fiber coupler to ensure that 75% of the launched power at the input port is available at the output port 2 of the coupler.

11.4 Refer to the biconical fused fiber coupler shown in Fig. P11.1. An optical power of 150 μW is launched into the input port 1 of the coupler. Estimate the values of coupling ratio, excess loss, insertion loss, and cross-talk of the coupler assuming that the power available at the output ports 3 and 4 is 75 μW and 55 μW, respectively. The reflected power at the input port 2 is 10 nW.

11.5. The coupling coefficient of a 3 dB biconical fused silica fiber coupler operating at 1550 nm is $k = 1.2\ \text{mm}^{-1}$. Calculate the interaction length of the coupler.

11.6 An optical power of 200 μW is launched into a single input port of a 32 × 32 fused fiber star coupler. Estimate the values of total loss comprising the splitting loss and the excess loss. Compare and contrast with value with the insertion loss of the coupler. The power measured at each output port is 5 μW.

11.7 Calculate the number of stages required to construct a (32 × 32) cascaded ladder star coupler using four-port 3 dB couplers. Also, calculate the required number of such 3 dB couplers.

11.8 A 64 × 64 ladder type star coupler is designed by cascading four-port (2 × 2) 3 dB couplers. Calculate the number of stages and the total number of the (2 × 2) couplers required to realise the star coupler. If 3% of the power is lost in each element, calculate the excess loss and the total loss in Decibels.

11.9 Distinguish between CWDM and DWDM. What is the special feature of an AllWave® fiber?

11.10 List the major passive components required for the implementation of DWDM.

11.11 What is an active WDM component? List the major active WDM components.

11.12 Design a (4 × 4) WDM multi/demultiplexer device using the basic (2 × 2) Mach-Zehnder (MZ) interferometers.

11.13 A 2-wavelength multiplexer is to be designed with the help of basic (2 × 2) MZ interferometer. If the wavelength separation between the two wavelength components is $\Delta\lambda = 0.085$ nm at 1550 nm, estimate the differential length ΔL between the two arms required for the multiplexer. Assume the effective refractive index of the silica waveguide to be 1.5.

11.14 Sketch and design a (4 × 4) WDM multiplexer device using the basic (2 × 2) MZ interferometers.

11.15 Estimate the value of the period of the interference pattern used for writing Bragg grating on a photosensitive fiber if the mode effective refractive index is 1.5 and the Bragg wavelength is 1550 nm.

11.16 List the major active components for WDM applications.

11.17 Explain the mechanism of optical tuning in a tunable laser source.

11.18 The tuning current in a DFB laser operating at 1330 nm can cause a maximum change in the effective refractive index of 0.6%. Calculate the tuning range and compute the number of channels that can be safely accommodated within this range for WDM applications. The spectral width of the source may be assumed to be 0.02 nm.

11.19 Ten uniformly spaced stations are connected to an optical bus having the following parameters:

C_T	L_T (dB)	L_{thru} (dB)	L_i (dB)	L_c (dB)	α (dB/km)
5%	8.0	1.5	0.5	1.5	0.6

Assuming that the separation between the adjacent stations is 500 m prepare a power budget for the optical bus.

11.20 Discuss with schematic the basic topologies of the optical fiber network.

11.21 What is a soliton? What is an all-optical link?

11.22 What is SPM? How is this effect used to balance the GVD of a fiber?

11.23 A coherent optical heterodyne receiver is operating at 1300 nm. The incoming signal power at its shot-noise limit of the receiver is –80.0 dBm. If the bandwidth of the IF amplifier is 2.4 GHz and the received SNR is 12 dB, determine the quantum efficiency of the photodetector.

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CHAPTER 12

Optical Fiber Measurement and Testing

Successful implementation of optical fiber communication systems needs careful design, measurement, and testing of various optoelectronic, and optical devices and components. The major components of a typical optical fiber communication system include

- Optical fibers/cables
- Optical sources
- Optical detectors
- Optical amplifiers
- Optical splitters/connectors/couplers

12.1 MEASUREMENT ASPECTS

Characterisation of the above components is required because this information is necessary for the designer of optical fiber communication system to ensure that each component satisfies the specific requirement of the network. The characterisation of these devices and components is done by the manufacturer. Measurement and testing of these devices and components are very important for the design engineers at the time of installation and commissioning. Different portable equipment and measuring instruments are available commercially for testing and finding faults in the optical fiber link or components. These equipment and measuring instruments can be deployed in the laboratory as well as in field situations.

In optical fiber communication systems, both multimode and single-mode fibers are used. The selection of the fiber is generally based on the desired application. For multimode fiber, the basic parameters that need to be measured include diameters of the core and the cladding, numerical aperture, refractive indices of the core and the cladding (refractive index profile for a graded-index fiber), average loss or attenuation per unit length of the fiber and the dispersion. For a single-mode fiber, the parameters of interest include the mode field diameter, cut-off wavelength, attenuation, and dispersion. These parameters are measured and tested by the manufacturer and provided to the user in the form of specifications. Some of the parameters (such as attenuation and dispersion) are dependent on the wavelength of operation. The manufacturer also provides the wavelength along with the specifications. Optical fibers are generally drawn by a fiber drawing machine which is equipped with a feedback arrangement to monitor the uniformity of the fiber diameter. The buffered fibers are subsequently incorporated in the form of a fiber cable as discussed earlier. A fiber cable consists of multiple optical fibers placed safely and securely within the cable to protect the fibers from external shocks and abusive environments during and after installation. Some of the basic parameters of the fiber are likely to be affected during fiber cabling and/or installation. These include attenuation and dispersion because of the microbends which may occur during cabling and installation. For a single-mode fiber, polarisation mode dispersion is an important factor that ultimately decides the maximum bit rate that can be transmitted over the fiber. The fiber joints, couplers, and splices can affect several performance parameters of the fiber. Therefore, the user needs to measure and test the parameters of these fibers at the time of installation.

The design engineers also need to know the characteristics of other active and passive components listed above. These components form an integral part of the communication network. The characteristics and performance parameters of the optical source are very important for designing the optical transmitter of the link. Some of the important characteristics of an optical source include peak wavelength of emission, optical power output at the peak wavelength, spectral response for determining the rms spectral width, modulation bandwidth, radiation pattern such as half-power beam-width (HPBW), etc. For the design of the optical receiver, the design engineer must know all the important parameters of the photodetector which is the key component of the receiver. These parameters include the wavelength of operation, responsivity, multiplication gain, noise equivalent power, speed, or bandwidth. For complex network architectures involving WDM, one needs to know the complete specification of passive (active) optical couplers/splitters, connectors, circulators, MUX/DMUX, tunable sources, etc. Most importantly once an optical fiber link is established it is necessary to test the performance of the overall link. The major operational parameters of interest include signal-to-noise ratio or bit-error rate in the case of digital optical communication and timing jitter. The standard procedure for this testing is done with the help of eye-pattern measurement. The fault in the optical fiber path can be determined with the help of optical time domain reflectometer measurements. A portable machine called optical time domain reflectometer (OTDR) is available for testing and fault detection purposes. Further, any discontinuity in the link can be fixed by splicing the fiber at the faulty joint. Portable splicing machines are commercially available for field applications. Testing/measurement is not only essential at the time of installation and commissioning but also essential for continuous monitoring and fixing faults in the link. This chapter deals with some standard procedures for the measurement and testing of optical devices and components. It also introduces some of the important testing equipment and instruments used for the above purpose. We begin with the discussion of various standards that are prevalent for the application and deployment of optical components and devices, further for the development of optical fiber links that need to meet certain standards decided by the international bodies.

12.2 MEASUREMENT STANDARDS

Optical fiber components and devices are manufactured by a large number of manufacturers with slightly different characteristics. There are certain issues related to the compatibility between the products from different manufacturers. However, all these components and devices need to maintain certain standards for their deployment in optical fiber networks. Different organisations at the national and international levels are responsible for the standardisation. For testing and characterisation of fundamental parameters of optical fibers such as fiber attenuation, dispersion, bandwidth, and mode-field diameter in the case of single-mode fiber various national-level standardisations are available in different countries. The National Institute of Standards and Technology in the United States, and the National Physical Laboratory in the United Kingdom, are organisations that are involved in the primary standardisation of optical devices and components. Similar other organisations and associations are involved in the process at the national level in other countries.

Several international bodies and organisations are involved in setting up standards for testing optical devices and components and also formulate procedures for calibration of optical measuring and testing equipment. The major organisations include the Telecommunication Industries Association (TIA) in association with the Electronic Industries Alliance (EIA) in the form of TIA/EIA (Table 12.1); American National Standard Institute (ANSI); International Telecommunication Association for the Telecommunication Sector (ITU-T); International Electrotechnical Association (IEC). In addition, the Institute of Electrical and Electronic Engineers (IEEE) also sets system standards for optical fiber links and networks. The TIA/EIA standards aim to serve the public interest to facilitate the interchangeability, compatibility, and improvement of optical fiber component products from different manufacturers. A large number of telecommunication industries, organisations including manufacturers, professionals, and users contribute to framing the standards. ANSI/TIA/EIA reviews the standards after every five years. The standards are published by TIA regularly with different reference numbers for standard measurement and testing procedures. A few of them are listed below from TIA publication, 2000 (TIA, 2000).

Table 12.1 TIA/EIA standards

Reference number	*Specification items*
ANSI/EIA/TIA-455-A-1994	Standard test procedures for optical fibers, cables, transducers, sensors, connectors, and other components and terminal devices
ANSI/TIA/EIA-526-7-1998	Optical power loss measurements of installed single-mode fiber cable
ANSI/TIA/EIA-526-14-A-1998	Optical power loss measurements of installed multimode fiber cable
ANSI/TIA/EIA-598-A-1995	Optical fiber cable color coding
ANSI/TIA/EIA-604-3-1997	Fiber optic connector intermateability standard

The optical fiber industries in different countries developed connectors with different dimensions and standards. Intermateability of these connectors becomes one major issue in interconnecting optical components from different manufacturers. As per recommendations of TIA/EIA, all connectors, adapters, and cable assemblies must comply with the dimensional requirements of the corresponding Fiber Optic Connector Intermateability Standard (FOCIS). TIA prescribes fiber optic test procedure (FOTP) under various clauses. All multimode connectors, adapters, and cable assemblies are required to meet the conditions of the prescribed clause at both 850 nm and 1300 nm ± 30 nm wavelengths. Likewise, all single-mode connectors, adapters, and cable assemblies have to meet the requirements of the specified clause at both 1310 nm and 1550 nm ± 30 nm wavelengths. Qualification testing must be conducted following the specified in the prescribed FOTP. For example, FOTP-13 refers to visual and mechanical inspection of mated connectors, FOTP-34 for attenuation measurements, and FOTP-A for return loss measurement.

The system standards for optical links and networks are prescribed by ANSI, ITU-T, and IEEE. Some of the recent recommendations of ITU-T relevant to various aspects of optical fiber links are listed in Table 12.2.

Table 12.2 Some ITU-T standard terms of reference

Recommendation reference	*Year*	*Relevant network aspect*
ITU-T G.652	2003	Characteristics of single-mode optical fiber
ITU-T G.982	1996	Optical access networks to support services up to the ISDN primary rate
ITU-T G.983.3	2001	Broadband optical access system with increased service capability by wavelength allocation
ITU-T G.984.2	2003	Gigabit-capable passive optical networks (GPON): Physical media dependent (PMD) layer specification
ITU-T G.984.3	2008	Gigabit-capable passive optical networks (GPON): Transmission convergence layer specification
ITU-T G.691	2006	Optical interfaces for single-channel STM-64 and other SDH systems with optical amplifiers
ITU-T G.694.1	2002	Spectral grids for WDM applications: DWDM frequency grid
ITU-T G.709/Y.1331	2012	Interfaces for the optical transport network
ITU-T G.783	2006	Characteristics of synchronous digital hierarchy (SDH) equipment functional blocks
ITU-T G.872	2001	Architecture of optical transport networks
ITU-T G.873.1	2006	Optical transport network (OTN): Linear protection
ITU-T G.874	2008	Management aspects of optical transport network elements
ITU-T G.957	2006	Optical interfaces for types of equipment and systems relating to the synchronous digital hierarchy
ITU-T G.959.1	2009	Optical transport network physical layer interfaces

12.3 TYPES OF TEST EQUIPMENT

Optical fiber communication systems make use of numerous passive and active optical components and devices including optical fiber as the main channel. For successful implementation and operation of the system, it is necessary to test the characteristics of these devices during installation as well as afterward for monitoring. At the time of any fault in the system these testing and measuring equipment help to diagnose the problem and fix the same. For field applications, these instruments should be portable and rugged. More sophisticated and advanced equipment is generally used in the laboratory environment for precise characterisation and testing. However, in recent times sophisticated advanced measuring and equipment with embedded systems are available commercially which are harsh enough to be used in adverse environments in the field situation. Some of the major equipment and measuring instruments include optical power meters, optical attenuators, splicing machines, tunable laser sources, optical spectrum analysers, optical time domain reflectometers (OTDRs), and multifunction optical test equipment.

12.3.1 Optical Power Meter

This is the basic instrument for all types of optical fiber communication applications. It is very common and a must just like multimeters for electronic circuits. An optical power meter measures the optical power level available at the tip of a fiber or pigtail end of an optical source. The power is coupled to the input port of the power meter through a connector/adaptor. Both single-mode and multimode fibers can be connected to the input port by using different connectors. The power meters are available either in the table-top form or as a handheld instrument. For field applications handheld meters are more suitable. The power meter can display the optical power directly in mw or mW or μW depending on the capability of the meter/selected range. The meter can also read the value directly in dBm or dBμ. A selector switch enables one to select the option in which the reading is required. The values are displayed on the LCD screen. The power meters can be operated either by the normal electrical supply by using an adaptor or using the built-in battery. Either option can be selected with the help of a changeover switch. The power meter uses a photodetector at the input port to convert the optical signal into an electrical signal which is subsequently amplified and processed to measure the corresponding electrical energy. The schematic of the internal block diagram of an optical power meter is shown in Fig. 12.1. The optical power output is a function of wavelength. The power meters are generally calibrated over different wavelength ranges depending on the type of photodetector used at the input port of the power meter. A Ge photodetector can be used at the input port for measuring optical power in the range of 780–1600 nm. Optical power in the wavelength range of 840–1650 nm can be measured using an InGaAs photodetector at the input port. A sophisticated power meter provides an option for selecting several calibrated wavelengths in the range.

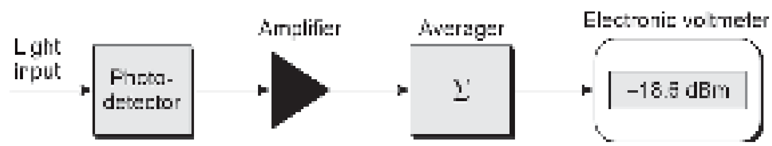

Fig. 12.1 Schematic block diagram of the internal units of an optical power meter

Various models of optical power meters are commercially available in the market from different manufacturers. The photo-image of a typical handheld power meter from UNIWAY is shown in Fig. 12.2. The optical power meter is best suited for the measurement of absolute and relative optic power levels of optical signal in CATV, CCTV, LAN, and Wide-Area-Network (WAN). The meter is equipped with a self-calibration function and universal adaptors. It is an ideal test instrument to measure, analyse, and maintain optical fiber cable TV (CATV) networks and other general field applications. The major features of the instrument include easy operation (handheld and battery operated), large measurement range –50 to + 26 dBm, absolute power measurement, wavelength selectivity (980, 1310, 1490 and 1550 nm), relative optical power measurement, very high accuracy (+ 0.15 dB). More advanced power meters such as the highly versatile FPM-600 power meter from ANSTEL are an ideal tool for link and system testing, and certification. It has a memory capacity of 1000 data items and converter software to facilitate data management and data transfer to a PC via a USB connection.

Fig. 12.2 Photo-image of a power meter from UNIWAY

More advanced tools in the form of handheld optical testers are also available commercially for multiple functions. The FOT-930 optical tester from ANSTEL delivers fully automated loss results in 10 seconds for up to three wavelengths, as well as automatic optical return loss (ORL) and fiber-length measurements. The tester combines a powerful light source, a power meter, a visual fault locator, a full-duplex digital talk set, and a video fiber inspection probe. It allows for the testing of passive optical networks (PONs) at the three main wavelengths, e.g. 1310, 1490, and 1550 nm used in fiber-to-the-home (FTTH) and fiber-to-the-premises (FTTP) networks. The tester complies with the ITU-T G.983 and G.984 recommendation series and the IEEE 802.3ah standard (ANSTEL Pvt Ltd.).

12.3.2 Optical Attenuators

Fiber optic attenuators are used in the fiber optic links to reduce the optical power to a desired level. Sometimes the power level from an optical source or an optical amplifier may be far above the measuring range of a normal power meter. Attenuators are used to bring down the power level to match the range. Various types of optical fiber attenuators, including LC, SC, and ST, FC are commercially available. Commonly used optical fiber attenuators come as female to male types called plug fiber attenuators. These attenuators come with ceramic ferrules and there are various types to fit different kinds of optical fiber connectors. Attenuators can be of fixed type or variable type. Fixed-value fiber optic attenuators can reduce the optical light power by a fixed factor. For example, a 3 dB attenuator reduces the power by a factor of 2. Attenuators are available for large values such as 60 dB which corresponds to the reduction of power by a factor of 10^6. Variable optical fiber attenuators are also available. The variable attenuator allows one to adjust the attenuation continuously. The fiber attenuators are required to meet TIA/EIA standards. Photo-images of fixed optical fiber attenuators from Fibertronics Inc. are shown in Fig. 12.3 (Fibertronics Inc., Melbourne, Florida).

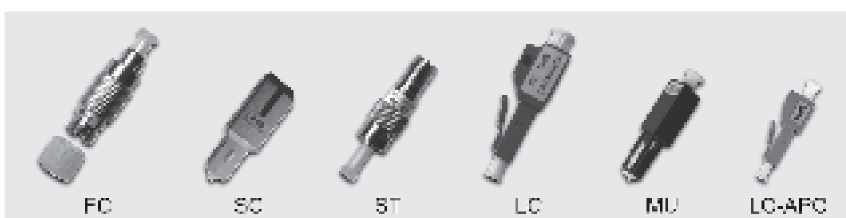

Fig.12.3 Different types of fixed attenuators

12.3.3 Tunable Laser Sources

A tunable laser source can be used as an optical source whose wavelength can be precisely set at any desired value in the specified wavelength band. Tunable laser sources have very narrow line widths. They are useful for studying wavelength-dependent characteristics of optical components, devices, and networks. Tunable laser sources of different makes and models are commercially available. A tunable laser source generally consists of a single-mode laser source with an external cavity. A tunable filter in the form of a movable diffraction grating is used to tune the laser source. The wavelength range over which the equipment can be tuned is decided by the laser source and the optical filter. Typically tunable laser sources for near-infrared wavelength applications can be tuned in the ranges 1250–1330 nm and 1450–1600 nm. In a single piece of equipment, the wavelength range can be changed by changing the source module. The equipment comes in the table-top form as well as in the portable that can be used in laboratory and field applications. Among various other available tunable laser sources Agilent/HP 8168F is a high-performance tunable laser source with a wavelength stability of $< \pm$ 100 MHz, wavelength accuracy of ±0.04 nm and a wavelength resolution of 0.001 nm. The equipment is very useful for the characterisation of optical devices over the wavelength range of 1450–1590 nm. The source can deliver a maximum optical power output of 7 dB and is also equipped with a variable optical attenuator to adjust the power level at the desired wavelength. Independent control of features of the instrument parameters ensures that the output power is kept stable over time and wavelength. The tunable laser source has an emission line width of 100 KHz and a side mode suppression ratio of < 50 dB. The equipment can be used as production line equipment in manufacturing industries and can be easily integrated into a fully automated production test environment for precise, fast, and repeatable testing. The equipment enables one to carry out several other tests and measurements with the help of a built-in Agilent 8153A lightwave multimeter and related application software. The equipment provides GPIB and RS-232 remote interfaces as well. The photo-image of the Agilent/HP 8168F tunable laser source is shown in Fig. 12.4.

Fig.12.4 Agilent/HP 8168F tunable laser source

12.3.4 Optical Spectrum Analyser

A general-purpose spectrum analyser (Fig. 12.5) is used to study the amplitude spectra (amplitude of various frequency components) of a signal. RF and microwave spectrum analysers are widely used for this purpose. Optical spectrum analysers carry out similar functions in the optical range. In the optical range, it is customary to describe the behavior of optical components and devices in terms of wavelength rather than frequency. In the measurement and testing of optical communication systems and networks, it is necessary to study the wavelength spectrum of light signals. For

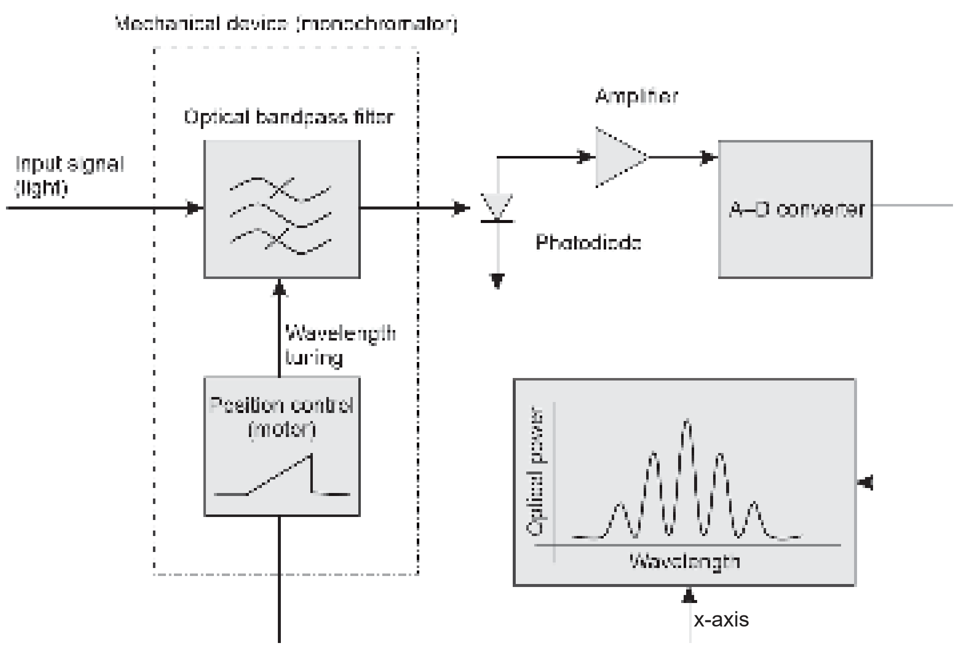

Fig.12.5 Internal functional building block diagram of a spectrum analyser

example, one may need to study the spectral response of an optical source or examine the wavelength spectrum of a WDM system. Such requirements are quite common for testing and fault detection purposes. A spectrum analyser displays the signal power at various wavelengths by scanning across a range of wavelengths. The basic principle of operation of a spectrum analyser is shown with the help of basic building blocks constituting the analyser. The input test signal passes through a monochromator. The optical bandpass filter restricts the light within a narrow wavelength slot. The output light is detected with the help of a photodetector to convert the signal into an electrical signal (O–E conversion). The electrical signal is subsequently amplified and converted to a digital signal with the help of a D–A converter as shown in the figure. The electronic signal is then fed to the y-axis control of an oscilloscope. The x-axis control of the oscilloscope is swept across in synchronism with the wavelength setting of the FP filter. The net result is a display on the CRO screen of the type.

This results in a display similar to that in the figure. The accuracy and precision cost of the spectrum analyser is determined by the sophistication of the bandpass filter. For a WDM system, the spectrum analyser can provide information such as the power levels of each channel, spectral width of the channel, and interference between adjacent channels. The key performance parameters of a spectrum analyser include wavelength accuracy, resolution, optical dynamic range, and wavelength sweeping speed. A variety of spectrum analysers of different makes and models are commercially available in the market. The photo-images of commercial spectrum analysers (from Agilent 8614xB series) in the form of portable and benchtop models are shown in Fig. 12.6. The 8614xx series allows fast, accurate, and comprehensive measurements for spectral analysis. The portable model is best suited for field application. The instrument is useful for testing and measurement on a variety of optical and network components including s for WDM systems, lasers, optical amplifiers, and other active and passive components/devices. The HP Agilent 86146B operates in the wavelength range of 600 nm –1700 nm. The spectrum analyser is compatible with a variety of connectors (Agilent 8614xB OSA Series datasheet).

(a) (b)

Fig.12.6 Photo-image of a spectrum analyser from Agilent 8614xB family (a) Portable (b) Benchtop

12.3.5 Optical Time Domain Reflectometer (OTDR)

The optical time-domain reflectometer is a versatile instrument that enables one to examine, test, and measure numerous parameters related to an optical fiber link such as the length of the fiber, attenuation of the whole fiber link in dB, connector and splice losses, locations of connector joints and faults in the fiber. An OTDR is essentially a fiber radar that examines the link from inside the fiber. It consists of an optical source, a receiver, a data acquisition system, a central processing unit, and an information storage system. It is a portable instrument with a CRT screen for displaying the results in the form of a trace. High-intensity pulses are sent from the specially designed laser source into the fiber. The pulse travels down the fiber and gets reflected from different discontinuity interfaces such as connectors, joints, splices, or any other serious imperfection such as a crinkle in the cable due to poor installation and or other physical damages or fiber rupture. The pulse returns through the same fiber which is then received by the receiver and processed in such a way as to be displayed in the form of a trace on the CRO screen of the instrument.

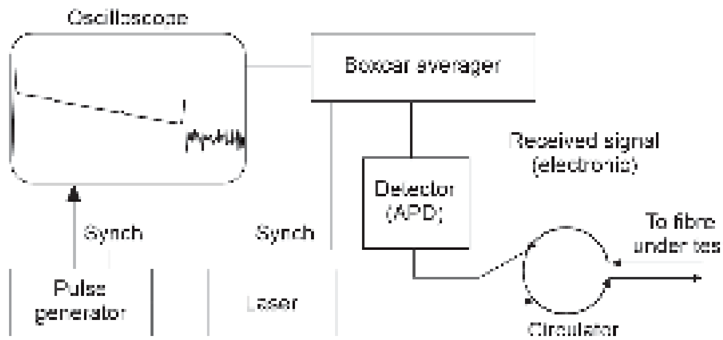

Fig. 12.7 Functional block diagram of an OTDR

The functional block diagram of a typical OTDR is illustrated in Fig. 12.7. It consists of a pulse generator which drives a high-power laser source. The light pulse is sent through one of the ports of a 3-port circulator into the optical fiber under test. The returned reflected signal enters through the same port which also acts as the exit port for the transmitted pulse. The reflected light is sent to a sensitive photodetector (usually an APD) through the third port of the circulator. One of the major problems with an OTDR is that the returning signal has a very low level especially when the length of the fiber is very large or the location of the fault is far away. The problem associated with low returned power can be tackled by using an ultra-sensitive photodetector such as an APD and a boxcar average circuit to average many thousands of returning pulses (Dutton). The average process removes a large amount of noise. A typical trace of an OTDR following the testing of a fiber link looks similar to one shown in Fig. 12.8. A basic advantage of OTDR is that for

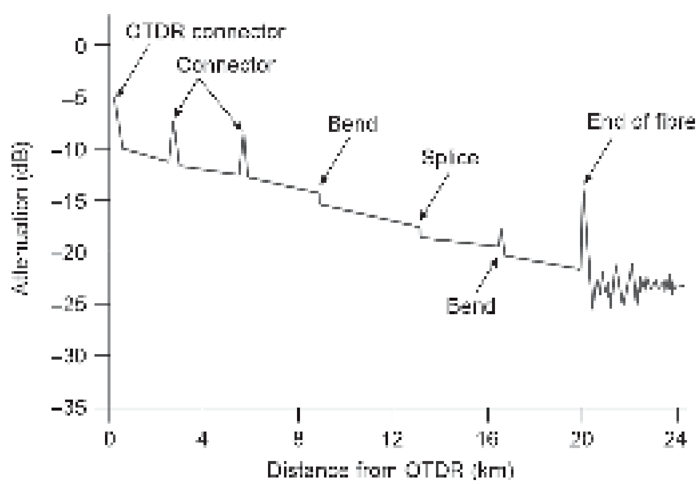

Fig. 12.8 Typical OTDR test result in the form of a trace showing the presence of various discontinuities due to connector, splices, bend, etc.

any fault in the fiber cable or measurement of attenuation, one needs to access the fiber from one end only. Modern OTDRs are extremely sophisticated and are available both in portable and bench-top forms. Portable OTDRs are especially attractive for field applications. OTDRs can be used for testing both single as well as multimode fibers. Some OTDRs are also equipped with additional laser sources and optical power meters. The photo-image of an OTDR from JDSU (Model T-BERD 6000 OTDR) is shown in Fig. 12.9. The T-BERD 6000 is a compact and lightweight portable test instrument for installation and maintenance of optical fiber communication networks (JDSU datasheet).

Fig. 12.9 T-BERD 6000 OTDR (photo-image)

12.4 MEASUREMENT AND TEST PROCEDURES

In this section measurement and testing of various optical components and fiber optic networks are discussed. The instruments discussed in the previous section are used for measurement and testing purposes. The measurement and test procedures are followed as per the recommendations of FOTP and ITU-T.

12.4.1 Attenuation

There are two basic approaches for measuring the attenuation of an optical fiber as prescribed by ITU-T. The simple and straightforward approach is the "cut-back" technique. In this approach, optical power is transmitted through a long length and a short length of the same fiber using the same coupling and making necessary measurements. This is, however, a destructive process. An alternative non-destructive approach is also available. This method is less accurate and is known as the "insertion-loss" method. There is a third method that is most convenient for field applications in installed fiber cables makes use of an OTDR.

12.4.1.1 The cut-back method

In this approach, one needs to have access to both ends of the fiber to make necessary measurements. A typical setup for measurement of attenuation/loss by this technique is shown in Fig. 12.10 (after ITU-T G.650). The optical source may be a lamp with a monochromator/laser or LED depending on the type of measurement. The spectral linewidth of the source should be narrow compared to fiber spectral attenuation. Some kind of modulation of the source is recommended to improve the S/N ratio at the receiver end. The launching condition must ensure the excitation of the fundamental mode in the

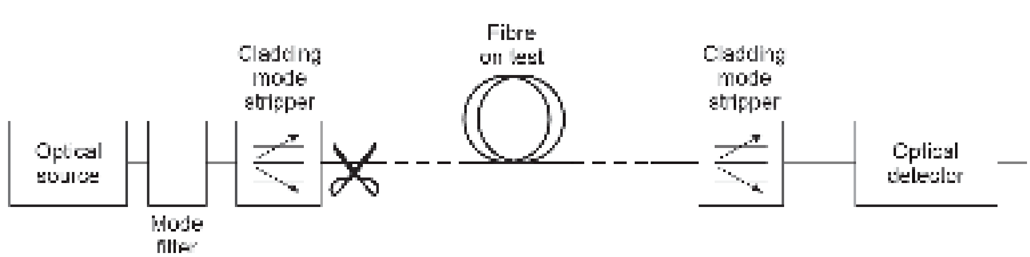

Fig.12.10 Experimental setup for attenuation measurement by cut-back method

fiber. The mode filter is used to ensure that the higher-order modes are eliminated. The cladding stripper is used to strip off the cladding modes. At the receiver end a sensitive photodetector is used for E–O conversion. The spectral response of the photodetector should match that of the source. The photodetector should have a linear sensitivity. The fiber under test must have smooth and clean endfaces perpendicular to the axis of the fiber. The fiber under test is placed in the setup as shown in Fig. 12.10. The optical power at the far end of the fiber is measured. Let this value be P_F. Keeping the launching condition the same, the fiber is cut at the cut-back length of a few meters (typically 2 m from the launching point) and the output power at the cut end is measured. Let the measured value of the power at this near end be P_N. The attenuation of the fiber can be estimated as

$$\alpha = \frac{10}{L}\log_{10}\left(\frac{P_N}{P_F}\right) \tag{12.1}$$

where L is the length of the fiber in kilometer between the two measurement points. The rationale behind the method is that it is very difficult to measure the exact value of launched power and therefore the power available at the cut-back end of the fiber may be considered as the power launched into the fiber. In any case, the method is destructive and may not be suitable for all kinds of applications. This method can also be used to study the dependence of attenuation on the wavelength of the light by making use of a tunable laser source.

12.4.1.2 The insertion-loss method

The cut-back technique cannot be applied in the case of installed cables connected with the help of connectors. Under this condition, the method of insertion loss can be used to find the attenuation of the fiber. This method is relatively less accurate but suitable for field applications. The experimental setup for this measurement is shown in Fig. 12.11. It can be seen from the figure that the coupling to the launching system and the detector is done with the help of connectors. The source is generally a tunable laser source or any other broad-band source equipped with a wavelength-selective device. The measurement procedure involves two steps. The first one is the reference measurement and the second one is carried out with the sample under test. The reference test involves coupling the source to a short length single-mode fiber having the equivalent nominal characteristics as the fiber under test and assembled with a mode filter and cladding mode stripper. The optical power measured at the end of the short length fiber is recorded as the launched power

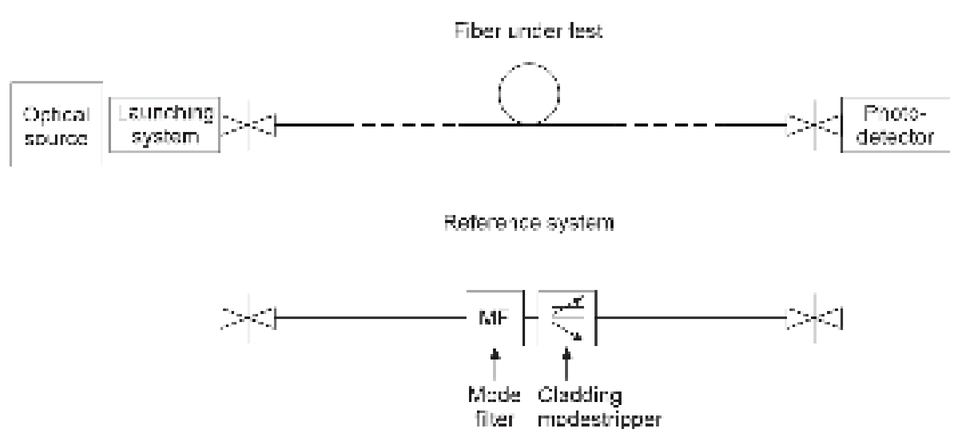

Fig. 12.11 Experimental setup for attenuation measurement using the "insertion-loss" technique

$P_1(\lambda)$. In the next step, the fiber cable assembly under test is connected between the launching and the detecting system. Let the optical power measured at the far end of the cable under test be $P_2(\lambda)$. The attenuation of the cable in dB can be expressed as

$$A = 10\log_{10}\left(\frac{P_1(\lambda)}{P_2(\lambda)}\right) \tag{12.2}$$

The third method using OTDR is discussed afterward.

12.4.2 Dispersion Measurement

The dispersion characteristics of an optical fiber are very important for determining the maximum rate at which information can be sent through the fiber. The dispersion measurement is done with respect to the following

Intermodal dispersion arising out of different modes in a multimode fiber traveling with different velocities to reach the destination at different instants of time giving rise to pulse spreading

Chromatic dispersion arises from the difference in velocity of individual wavelength components within a given mode and gives rise to the spreading of the pulse

Polarisation mode dispersion arising out of splitting of the fundamental mode in a single mode fiber into orthogonal polarisation modes.

12.4.2.1 Intermodal dispersion

To estimate the intermodal dispersion it is convenient to consider the optical fiber as a filter. Under this assumption, the filter can be characterised in terms of impulse response $h(t)$ in the time domain or terms of transfer function $H(f)$ in the frequency domain (Personick, 1973). As the impulse response of a system is the inverse Fourier transform of the transfer function we may write

$$h(t) = \int_{-\infty}^{\infty} H(f)\exp(j2\pi ft)df \tag{12.3}$$

or conversely,

$$H(f) = \int_{-\infty}^{\infty} h(t)\exp(j2\pi ft)dt \tag{12.4}$$

In the measurement (time domain and frequency domain) it is assumed that the fiber has a quasi-linear behavior with respect to power. This means that the total power received at the output of the fiber can be calculated as the linear sum of the contribution of individual overlapping output pulses from the fiber. The output power of the fiber in response to an input power $p_{\text{in}}(t)$ can be estimated with the help of the convolution integral given by (Personick, 1973)

$$p_{\text{out}}(t) = h(t) * p_{\text{in}}(t) = \int_{-\frac{T}{2}}^{\frac{T}{2}} p_{\text{in}}(t)h(t-\tau)d\tau \tag{12.5}$$

where T is the period between the input pulses.

Taking the Fourier transform on both sides of Eq. (12.5), we may write

$$P_{\text{out}}(f) = H(f)P_{\text{in}}(f) \tag{12.6}$$

where $P_{\text{in}}(f)$ and $P_{\text{out}}(f)$ are the Fourier transforms of the input and output pulse responses of the fiber considered as a filter. Therefore,

$$P_{\text{out}}(f) = \int_{-\infty}^{\infty} P_{\text{out}}(t)\exp(j2\pi ft)dt \tag{12.7}$$

$$P_{\text{in}}(f) = \int_{-\infty}^{\infty} P_{\text{in}}(t)\exp(j2\pi ft)dt \tag{12.8}$$

The transfer function of an optical fiber gives important information about the bandwidth.

12.4.2.2 Time domain measurement

An experimental setup for measuring the pulse dispersion in the time domain is shown in Fig. 12.12. The simplest way to estimate the pulse broadening caused by an optical fiber is to launch a narrow optical pulse at one end of the fiber and detect the broadened pulse at the other end with the help of the above setup (Hernday, 1998; Keiser, 2002). The standard procedure for measurement can be found in TIA/EIA literature (TIA/EIA FOTP-51). The light output from a laser source driven by a pulse drive circuit is launched into the fiber under test through a mode scrambler as shown in the Fig. 12.12. The output broadened pulse at the other end of the fiber is displayed on the screen of a sample oscilloscope having an in-built optical receiver or an external photodetector for O–E conversion. Similarly, the input pulse launched into the fiber is also displayed on the CRO screen by replacing the fiber under test with a short length reference fiber and adjusting the delay appropriately.

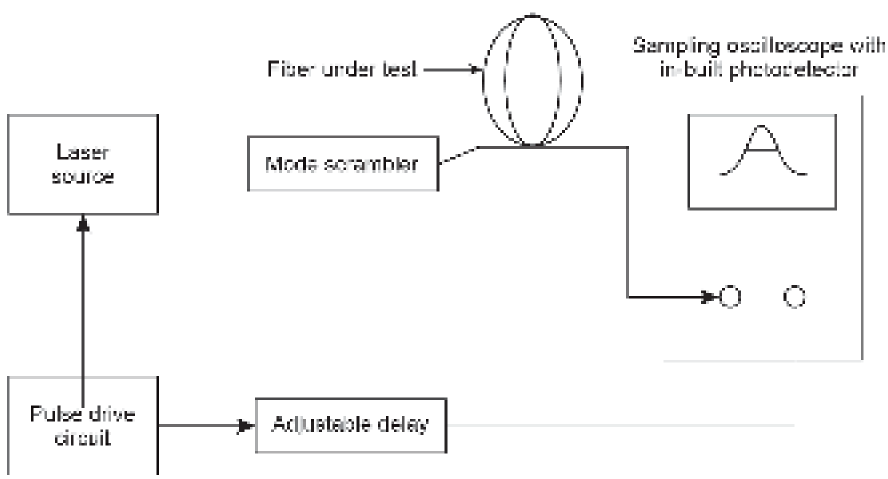

Fig. 12.12 Measurement setup for intermodal dispersion in the time domain

Care should be taken that the reference fiber has the same characteristics as the fiber under test. The synchronisation is maintained by deriving the trigger pulse of the CRO sweep circuit from the electrical drive unit of the source and using a delay circuit as shown in Fig. 12.13.

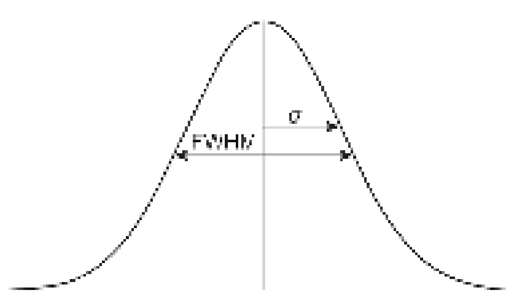

Fig.12.13 Displayed pulse shape for estimation of σ

From the displayed pulse shape the rms pulse width σ can be calculated as

$$\sigma = \left[\frac{\int_{-\infty}^{\infty} (t - \bar{t})^2 p_{\text{out}}(t) dt}{\int_{-\infty}^{\infty} p_{\text{out}}(t) dt} \right]^{\frac{1}{2}} \tag{12.9}$$

where the meantime $\bar{t}$ is determined as

$$\bar{t} = \frac{\int_{-\infty}^{\infty} p_{\text{out}}(t) dt}{\int_{-\infty}^{\infty} p_{\text{out}}(t) dt} \tag{12.10}$$

The above integration can be carried out numerically. As discussed earlier it is often convenient to assume the pulse shape at the output of the fiber to be Gaussian in nature (Fig. 12.13) expressed as

$$p_{\text{out}}(t) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{t^2}{2\sigma^2} \right) \tag{12.11}$$

where σ is the rms width of the pulse. The estimated value of σ by this method can be used to determine the bandwidth of the fiber using the following formula derived earlier in Chapter 4 and reproduced below.

$$B = \frac{0.186}{\sigma} \text{ Hz} \tag{12.12}$$

12.4.2.3 Frequency-domain measurement

The bandwidth of the test fiber can be easily estimated by measuring in the frequency domain (Hernday, 1998). The frequency domain intermodal dispersion directly yields the amplitude versus frequency as well as phase versus frequency response of the optical fiber. The experimental setup for frequency domain measurement of intermodal dispersion is shown in Fig. 12.14. The standard procedure for frequency domain measurement of the information transmission rate in a multimode fiber can be obtained from TIA/EIA recommendations (TIA/EIA FOTP-30). The light from a narrow linewidth CW laser source is sinusoidally modulated about a fixed level. The signal is launched into the fiber. The frequency response of the fiber is obtained from the ratio of the amplitudes of the

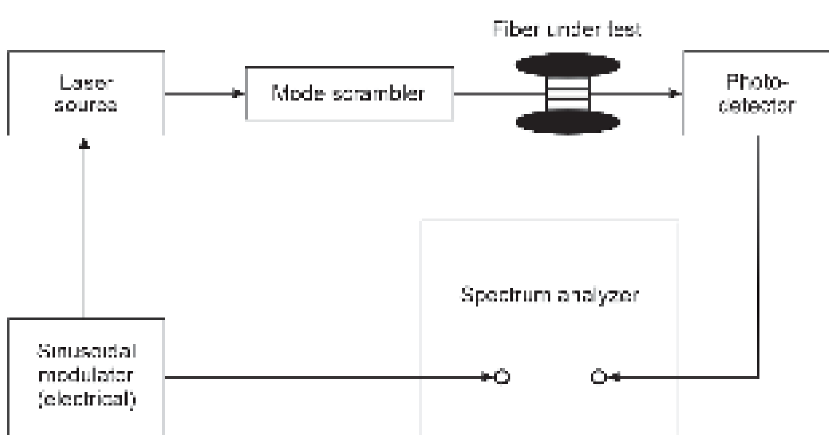

Fig. 12.14 Experimental setup for finding the frequency response of the fiber

sinusoidally modulated wave at the input end and the output end of the fiber. At the exit end of the fiber, a photodetector is used to convert the intensity-modulated optical signal to the corresponding electric power $P_{out}(f)$ as a function of frequency. The input signal power $P_{in}(f)$ is subsequently measured by replacing the test fiber with a short length reference fiber. The transfer function of the fiber under test can be obtained from the measurement as

$$H(f) = \frac{P_{out}(f)}{P_{in}(f)} \tag{12.13}$$

By increasing the frequency of the sinusoidal modulator it is possible to find the frequency at which $H(f)$ falls to one-half of its value at the baseband. This frequency corresponds to the fiber bandwidth.

12.4.2.4 Chromatic dispersion

Dependence of the propagation velocity of a mode on wavelength manifests in the form of chromatic dispersion. The chromatic dispersion has two components, e.g. material dispersion and waveguide dispersion. In a multimode fiber, the waveguide dispersion is usually very small. For a single mode, fiber is comparatively more. Nevertheless, material dispersion dominates over waveguide dispersion. Several methods for measurement of chromatic dispersion of optical fibers have been reported (Thevenaz *et al* 1989). The arrangement for measuring the chromatic dispersion of a single-mode fiber is shown in Fig. 12.15 is based on the modulation phase-shift method.

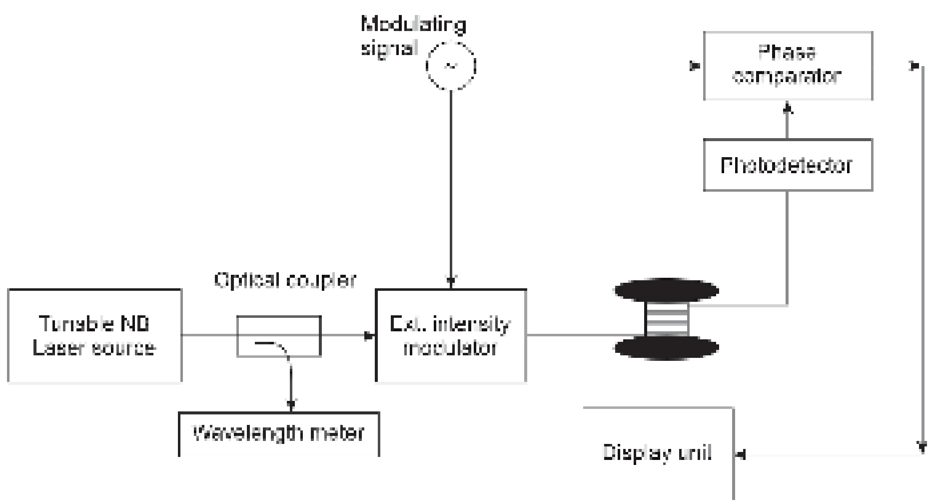

Fig. 12.15 Measurement setup for chromatic dispersion of a test fiber

The light from a tunable laser source is intensity-modulated externally with the help of a signal generator. The signal is launched into the fiber under test. At the receiving end of the fiber, the signal is detected with the help of a photodetector. A vector voltmeter is used to measure the phase of the modulation of the received signal with reference to the electrical modulation source. The phase measurement is repeated at a wavelength interval of $\Delta\lambda$ over the desired spectral band. The change of group delay can be estimated by using the results of the measurement at two adjacent wavelengths. The change in the group delay can be obtained as (Hernday *et al* 1998).

$$\Delta\tau_{\lambda} = \frac{\Phi_{\lambda+\Delta\lambda/2} - \Phi_{\lambda-\Delta\lambda/2}}{360 f_m} \times 10^6 \tag{12.14}$$

where λ is the central wavelength about which the measurements are made, f_m is the modulation frequency in MHz and Φ is the phase of the measured modulation. Several other methods have been reported in the literature for the measurement of chromatic dispersion in single-mode as well as multimode fibers (Thévenaz, 1988; Cohen, 1985; Hackert, 1992; TIA/EIA FOTP-175, 1992; TIA/EIA-169, 1992; TIA/EIA FOTP-168, 1992).

12.4.2.5 Polarisation Mode Dispersion

The polarisation mode dispersion arises from the birefringence property of the fiber material. The fundamental mode launched into a single-mode fiber decomposes into two components having orthogonal polarisation. This is caused by fiber inhomogeneity due to several factors discussed earlier. The difference in the propagation time of these two components leads to pulse spreading and is called polarisation mode dispersion. PMD limits the ultimate bandwidth of a single-mode fiber. There are different methods of measuring the polarisation mode dispersion of a fiber. The standardised methods are available in the published documents of ITU-T G 650.2. The EIA/TIA provides a recommendation for individual test solutions (TIA/EIA FOTP-113, 1997). In this section, we discuss the fixed analyser method. The measurement set up for polarisation mode dispersion of a single mode fiber is shown schematically in Fig. 12.16. It consists of a broadband polarised source and a polarised (variable) optical spectrum analyser. The mean period of the intensity modulation is measured from the power fluctuations spectrum. This is done by measuring the rate at which the state of polarisation changes as wavelength changes. This corresponds to the number of maxima and minima. Using this value the mean differential group delay is estimated as

$$\left\langle \Delta\tau_{\text{pol}} \right\rangle = \frac{kN_e \lambda_{\text{start}} \lambda_{\text{stop}}}{2(\lambda_{\text{start}} - \lambda_{\text{stop}})c} \tag{12.15}$$

where λ_{start} and λ_{stop} are the wavelengths corresponding to the starting and ending of the wavelength sweep used in the measurement, N_e is the number of extrema (maxima and minima) within the scanning range, c is the velocity of light and k is the mode coupling factor.

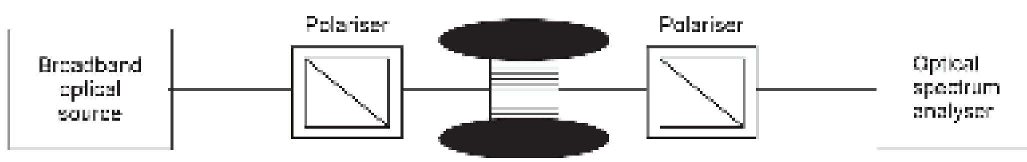

Fig. 12.16 Experimental setup for measurement of polarisation mode dispersion

12.5 MEASUREMENT WITH OTDR

The OTDR is a multipurpose instrument that can be used to make single-ended measurements of optical fiber links. The parameters such as attenuation, splice loss, connector loss, return loss, chromatic dispersion, etc of an optical network can be measured for installed optical fiber link. The measurement is completely non-destructive. One of the major applications of OTDR is to detect the location of any fault in the link arising out of fiber damage or rupture.

A typical trace on the screen of an OTDR for an installed optical fiber link looks like the one shown in Fig. 12.17. The OTDR acts like an optical RADAR which sends an intense optical pulse of narrow linewidth through the fiber. The ordinate of the display screen represents the back-reflected optical power in dB while the abscissa corresponds to the distance between the instrument and the measurement point in the fiber link. For a normal optical link without any major fault/discontinuity exhibit the following distinctive features in the trace obtained with the help of the back-reflected optical power

i. A large initial peak at the starting point. This peak is attributed to the Fresnel reflection loss at the input end. If the light enters from the air to the fiber, the Fresnel reflection coefficient can be expressed as

$$R = \left(\frac{n_1 - 1}{n_1 + 1}\right)^2 \tag{12.16}$$

where n_1 is the refractive index of the fiber core at the operating wavelength. The back-reflected power can be estimated as

$$P_{\text{ref}} = P_0 \times R \tag{12.17}$$

ii. A long decaying tail till the end of the trace. This is due to the Rayleigh scattering of the back reflected wave in the reverse direction as the wave advances through the fiber.

iii. Small peaks are occurring in between. These are attributed to strong reflections of the light resulting in larger power of the back-reflected light arising from connectors, joints, or other minor discontinuities in the fiber link.

iv. A positive spike at the end occurs because of Fresnel reflection from the back end of the fiber.

v. In addition there are abrupt shifts of the curve caused by additional optical loss at various joints and discontinuities.

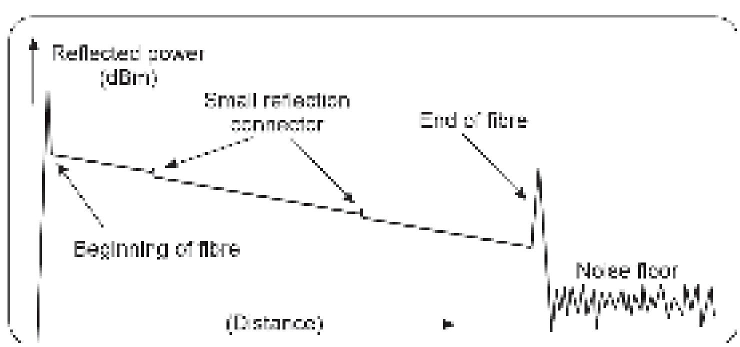

Fig.12.17 Typical display on an OTDR screen showing back-scattered optical power

As the trace plots the returned power against distance the instrument helps one to pinpoint the position of joints, connectors, splices, any fiber discontinuity, etc.

In addition to measurement of various parameters such as attenuation, dispersion, component loss, splice loss, etc. an OTDR can be used to detect the location of any fault in the optical fiber link arising out of any damage/break in the fiber. Any major break or damage of the fiber will give rise to an additional spike in the trace and the distance can be read directly.

12.6 EYE DIAGRAMS

The data handling capability of an optical fiber can be best adjudged with the help of an eye diagram for a digital transmission system. This technique is extensively used in digital electrical communication systems. The same may be used in the case of digital optical transmission case by converting the optical signal to the corresponding electrical signal.

A typical setup for generating the eye diagram on the screen of a CRO is shown in Fig. 12.18. The output of a pseudorandom data pattern generator is applied to the optical transmitter to drive the laser source. The bit pattern is allowed to travel through the test fiber. The bits get distorted while traveling through the fiber. The distorted optical pulses are converted to an electrical bit pattern which is subsequently applied to the vertical input of the oscilloscope. The data rate of the pseudorandom bit generator is used to trigger the time base circuit of the oscilloscope. This results in the formation of the eye pattern or eye diagram. A typical eye diagram is shown in Fig. 12.19. The details concerning the formation of eye patterns can be found in any standard textbook on digital communication (Chakrabarti, 1999).

The eye diagram provides the following important information on the system's performance.

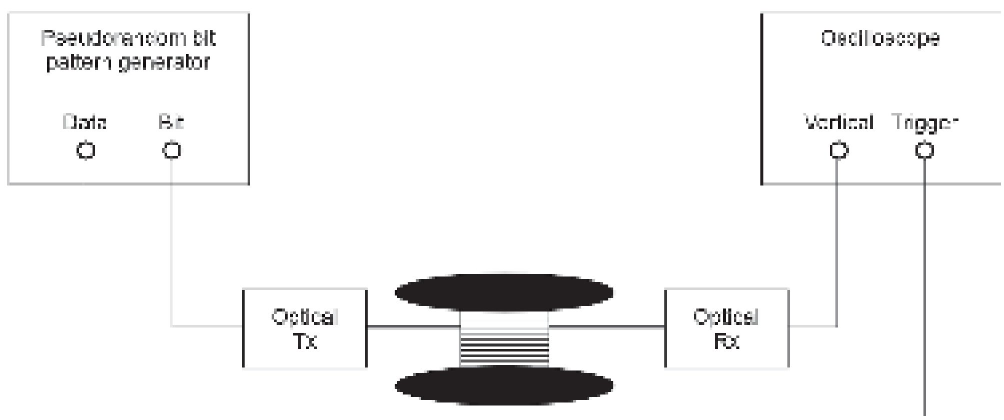

Fig. 12.18 Experimental setup for generating eye diagram

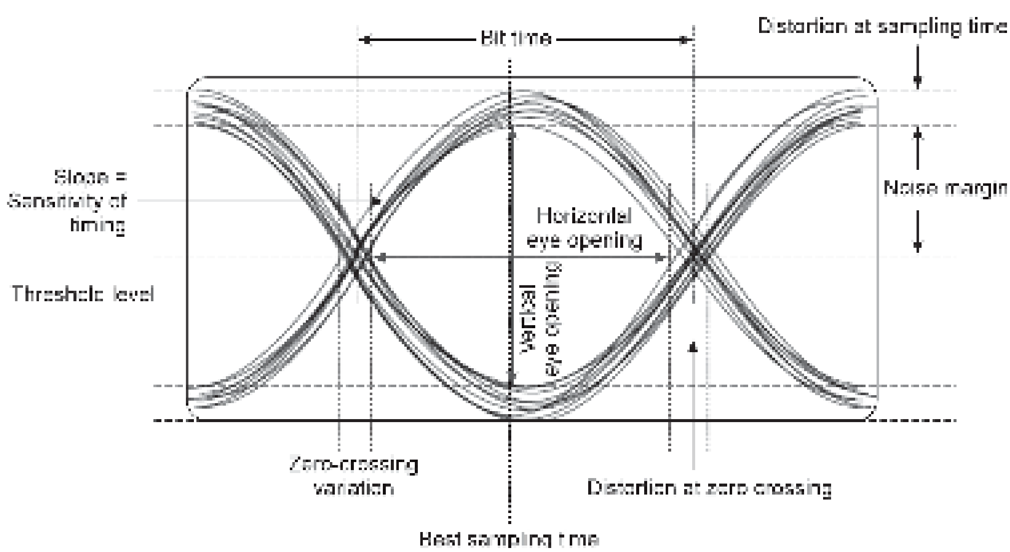

Fig. 12.19 Simplified eye diagram showing the key performance pattern

i. The width of the eye-opening corresponds to the time interval over which the received signal can be sampled.
ii. The best sampling time corresponds to the time when the height of the eye-opening is maximum. The height of the eye-opening is reduced because of amplitude distortion. The separation between the top of the eye-opening and the maximum amplitude corresponds to maximum distortion.
iii. The height of the eye-opening corresponds to the noise margin at the specified sampling time.
iv. The sensitivity of the system to the timing error is determined by the rate at which the eye closes, i.e. by the slope of the eye pattern side.
v. Time jitter arises out of the noise introduced by the optical receiver and pulse distortion introduced by the fiber. If the signal is sampled at the middle of the bit period then the amount of distortion at the threshold level indicates the amount of jitter.

$$\%\text{ timing jitter} = \frac{\Delta T}{T_b} \times 100 \tag{12.18}$$

where T_b is the bit period.

12.7 MEASUREMENT WITH OPTICAL SPECTRUM ANALYSER

Optical Spectrum Analyser is a versatile instrument that can be used for quick measurement and characterisation of a large number of optical components and devices. This instrument is particularly useful for the characterisation of optical sources used in optical fiber communication. It can be used to study the following characteristics of the following components

- Light emitting diode
- FP laser source
- DFB laser
- Semiconductor laser amplifier
- EDFA

The instrument can determine the spectral response of the above devices by measuring the emitted power output at a different wavelengths and thereby helps to calculate the full-width half maximum (FWHM), rms spectral width, the peak wavelength of emission, mean wavelength of emission, and peak value of the emitted power. An optical spectrum analyser is also very important in studying and testing DWDM networks. The optical spectrum analyser can display the spectral response of an optical source on the screen directly. It can also be used for measuring the gain and noise figure of optical amplifiers. A high-end optical spectrum analyser can have a resolution bandwidth as small as 0.1 nm making this instrument especially attractive for characterizing a wide variety of optical sources.

Summary

- The major components of a typical optical fiber communication system include optical fibers/cables, optical sources, optical detectors, optical amplifiers, optical splitters/connectors, and couplers.
- For installation and maintenance of optical fiber communication system, it is necessary to follow some standard test and measurement procedures.

- Several international bodies and organizations are involved in setting up standards for testing optical devices and components and also formulate procedures for calibration of optical measuring and testing equipment.
- The major organizations include Telecommunication Industries Association (TIA) in association with Electronic Industries Alliance (EIA) in the form of TIA/EIA; American National Standard Institute (ANSI); International Telecommunication Association for the Telecommunication sector (ITU-T); International Electrotechnical Association (IEC).
- In addition to these, Institute of Electrical and electronic engineers (IEEE) also sets system standards for optical fiber links and networks.
- Optical fiber communication systems make use of numerous passive and active optical components and devices including optical fiber as the main channel.
- Some of the major equipment and measuring instruments include optical power meter, optical attenuators, splicing machines, tunable laser sources, optical spectrum analyzer, optical time domain reflectometer (OTDR), and multifunction optical test equipment.
- Optical power meter is very common and a must just like multimeters for electronic circuits. An optical power meter measures the optical power level available at the tip of a fiber or pigtail end of an optical source directly in dBm.
- Fiber optic attenuators are used in the fiber optic links to reduce the optical power to a desired level.
- A tunable laser source can be used as an optical source whose wavelength can be precisely set at any desired value in the specified wavelength band. Tunable laser sources have very narrow line width. They are useful for studying wavelength dependent characteristics of optical components, devices and networks.
- An optical spectrum analyzer displays the amplitude of different wavelength components present in the light. It is most convenient to study Light emitting diode, FP laser source, DFB laser, Semiconductor laser amplifier and EDFA.
- An optical time-domain reflectometer is a versatile instrument that enables one to examine, test and measure numerous parameters related to an optical fiber link such as the length of the fiber, attenuation of the whole fiber link in dB, connector and splice losses, locations of connector joints and faults in the fiber.
- An OTDR is essentially a fiber radar which examines the link from inside the fiber.
- The data handling capability of an optical fiber can be best adjudged with the help of eye diagram for digital transmission system.
- The eye diagram provides information about the time interval over which the received signal can be sampled, the best sampling time, noise margin, the sensitivity of the system to the timing error, and the time jitter arising out of the noise introduced by the optical receiver.

Problems

12.1 List the different international standards for optical devices and components.

12.2 Discuss the standard procedures for the measurement of the following parameters of an optical fiber
(i) Attenuation (ii) Intermodal dispersion.

12.3 What is the major disadvantage of the cut-back method for measuring attenuation?

12.4 Why is the “insertion loss” method of measuring attenuation less accurate as compared to the cut-back method?

12.5 List the major types of equipment used for testing and measurement in optical communication systems.

12.6 What is an OTDR? Explain how it can be used for optical cable faults.

12.7 If the time difference between the application of pulse from an OTDR to an input fiber and receiving the back-reflected pulse from the end of the fiber is T, calculate the length of the fiber in terms of the refractive index of the fiber core and other parameters.

12.8 Explain the importance of an eye diagram for assessing the data-handling capacity of a fiber cable.

12.9 Explain how does a tunable laser source work?

12.10 What are the major uses of an optical spectrum analyser?

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Appendices

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Appendix A

STANDARD INTERNATIONAL (SI) UNITS

Quantity	*Unit*	*Unit symbol*	*Dimension*
Mass	kilogram	kg	
Length	meter	m	
Time	second	s	
Temperature	Kelvin	K	
Current	Ampere	A	C.s
Electric charge	Coulomb	C	A.s
Frequency	Hertz	Hz	s^{-1}
Force	Newton	N	$kg.m/s^2$
Pressure	Pascal	Pa	N/m^2
Energy	Joules	J	N.m
Potential	Volt	V	J/C
Resistance	Ohm	W	V/A
Capacitance	Farad	F	C/V
Inductance	Henry	H	Wb/A
Conductance	Siemens	S	A/V
Magnetic flux	Weber	Wb	V.s
Magnetic induction	Tesla	T	Wb/m^2

Appendix B

PHYSICAL CONSTANTS

Quantity	*Symbol*	*Value*
Electronic charge	q	$1.60217733 \times 10^{-19}$ C
Rest mass of electron	m_0	$9.10938975 \times 10^{-31}$ kg
Rest mass of proton	m_P	$1.67262311 \times 10^{-27}$ kg
Speed of light in vacuum	c	2.99792458×10^{8} m/s
Avogadro's number	N	$6.02213673 \times 10^{23}$/mole
Permeability of vacuum	μ_0	$4\pi \times 10^{-7}$ H/m
Permittivity of vacuum	ϵ_0	$8.854187817 \times 10^{-12}$ F/m
Planck constant	h	$6.62607554 \times 10^{-34}$ J.s
Reduced Planck constant ($h/2\pi$)	$\hbar$	1.05458×10^{-34} J.s
Boltzmann constant	k	$1.38065812 \times 10^{-23}$ J/K
Electron volt	eV	$1.60217733 \times 10^{-19}$ J
Thermal voltage (300K)	V_T	0.02585 eV
Gas constant	R	1.98719 cal/mol-K

Appendix C

USEFUL MATHEMATICAL RELATIONS

This supplement lists some important mathematical relationships which are relevant to the subject matter dealt in this text. More exhaustive lists are available in mathematical handbooks and other sources (Kurtz, 1991 and Gradshteyn *et al* 1994).

C.1 Trigonometric Identities

$$\exp(\pm j\theta) = \cos\theta \pm j\sin\theta$$

$$\sin^2\theta + \cos^2\theta = 1$$

$$\cos\theta = \frac{1}{2}[\exp(j\theta) + \exp(-j\theta)] = \sin\left(\theta + \frac{\pi}{2}\right)$$

$$\sin\theta = \frac{1}{2}[\exp(j\theta) - \exp(-j\theta)] = \cos\left(\theta - \frac{\pi}{2}\right)$$

$$\cos^2\theta - \sin^2\theta = \cos 2\theta$$

$$\cos^2\theta = \frac{1}{2}[1 + \cos 2\theta]$$

$$\sin^2\theta = \frac{1}{2}[1 - \cos 2\theta]$$

$$2\sin\theta\cos\theta = \sin 2\theta$$

$$\cos^3\theta = \frac{1}{4}[3\cos\theta + \cos 3\theta]$$

$$\sin^3\theta = \frac{1}{4}[3\sin 3\theta - \sin 3\theta]$$

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$$

$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$$

$$\tan(\alpha \pm \beta) = \frac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}$$

$$2\sin\alpha\sin\beta = \cos(\alpha - \beta) - \cos(\alpha + \beta)$$

$$2\cos\alpha\cos\beta = \cos(\alpha - \beta) + \cos(\alpha + \beta)$$

$$2\sin\alpha\sin\beta = \sin(\alpha - \beta) + \sin(\alpha + \beta)$$

C.2 Series Expansions

Taylor Series

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \cdots$$

where,

$$f^{(n)}(a) = \left.\frac{d^n f(x)}{dx^n}\right|_{x=a}$$

MacLaurin Series

$$f(x) = f(0)\frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(a)}{n!}x^n + \cdots$$

where,

$$f^{(n)}(0) = \left.\frac{d^n f(x)}{dx^n}\right|_{x=0}$$

Binomial Series

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \cdots \quad |nx| < 1$$

Exponential Series

$$\exp(x) = 1 + x + \frac{x^2}{2!} + \cdots$$

Logarithmic Series

$$\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \ldots$$

Trigonometric Series

$$\sin x = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \cdots$$

$$\cos x = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \cdots$$

$$\tan x = x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + \cdots$$

$$\sin^{-1} x = x + \frac{1}{6}x^3 + \frac{3}{40}x^5 + \cdots$$

$$\tan^{-1} x = x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \ldots \quad |x| < 1$$

$$= \frac{\pi}{2} - \frac{1}{x} + \frac{1}{3x^3} - \ldots \quad |x| > 1$$

$$\sin x = 1 - \frac{1}{3!}(\pi x)^2 + \frac{1}{5!}(\pi x)^4 - \cdots$$

Summation

$$\sum_{m=1}^{M} m = \frac{M(M+1)}{2}$$

$$\sum_{m=1}^{M} m^2 = \frac{M(M+1)(2M+1)}{6}$$

$$\sum_{m=1}^{M} m^3 = \frac{M^2(M+1)^2}{4}$$

$$\sum_{m=0}^{M} x^m = \frac{x^m - 1}{x - 1}$$

C.3 Integrals

$\int udv$	$= uv - \int vdu$
$\int \sin xdx$	$= -\cos x$
$\int \cos xdx$	$= \sin x$
$\int \sqrt{a^2 - x^2}dx$	$= \frac{1}{2}\int\left[x\sqrt{a^2 - x^2} + a^2 \sin^{-1}\left(\frac{x}{a}\right)\right]$
$\int x\sqrt{a^2 - x^2}dx$	$= -\frac{1}{3}(a^2 - x^2)^{\frac{3}{2}}$
$\int x\sin(ax)dx$	$= \frac{1}{a^2}[\sin(ax) - ax\cos(ax)]$
$\int x\cos(ax)dx$	$= \frac{1}{a^2}[\cos(ax) + ax\sin(ax)]$
$\int \sin^2 xdx$	$= \frac{x}{2} - \frac{1}{4}\sin 2x$
$\int \cos^2 xdx$	$= \frac{x}{2} + \frac{1}{4}\sin 2x$
$\int \sin^n xdx$	$= \frac{\sin^{n-1} x\cos x}{n} + \frac{n-1}{n}\int \sin^{n-2} xdx$
$\int \exp(ax)dx$	$= \frac{1}{a}\exp(ax)$
$\int x\exp(ax)dx$	$= \frac{1}{a^2}(ax - 1)\exp(ax)$

(*Contd.*)

$\int x \exp(ax^2) dx$	$= \frac{1}{2a} \exp(ax^2)$
$\int \exp(ax) \sin(bx) dx$	$= \frac{1}{a^2 + b^2} \exp(ax)[a \sin(bx) - b \cos(bx)]$
$\int \exp(ax) \cos(bx) dx$	$= \frac{1}{a^2 + b^2} \exp(ax)[a \cos(bx) + b \sin(bx)]$
$\int \frac{dx}{a^2 + b^2 x^2}$	$= \frac{1}{ab} \tan^{-1}\left(\frac{bx}{a}\right)$
$\int \frac{x^2 dx}{a^2 + b^2} dx$	$= \frac{x}{b^2} - \frac{a}{b^3} \tan^{-1}\left(\frac{bx}{a}\right)$

Appendix D

BESSEL FUNCTIONS

D.1 Bessel Function of the First Kind

The Bessel function of the first kind of order n and argument x, denoted by $J_n(x)$ is defined as

$$J_n(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \exp(jx\sin\theta - jn\theta)d\theta \qquad (D1)$$

$$= \frac{1}{\pi}\int_{-0}^{\pi} \cos(x\sin\theta - n\theta)d\theta \qquad (D2)$$

Bessel function defined by Eqs ($D1$) and ($D2$) can be expanded in power series as

$$J_n(x) = \sum_{m=0}^{\infty} \frac{(-1)^m \left(\frac{1}{2}x\right)^{n+2m}}{m!(n+m)!} \qquad (D3)$$

The various orders of Bessel function of the first kind can be written using ($D3$) as

$$J_0(x) = 1 - \frac{x^2}{2^2} + \frac{x^4}{2^2 \cdot 4^2} - \frac{x^6}{2^2 \cdot 4^2 \cdot 6^2} + \cdots \qquad (D4)$$

$$J_1(x) = \frac{x}{2} - \frac{x^3}{2^2 \cdot 4} + \frac{x^5}{2^2 \cdot 4^2 \cdot 6} \qquad (D5)$$

and

$$J_2(x) = \frac{x^2}{2 \cdot 4} - \frac{x^4}{2^2 \cdot 4 \cdot 6} + \frac{x^6}{2^2 \cdot 4^2 \cdot 6 \cdot 8} - \ldots \qquad (D6)$$

Bessel function has the following properties:

(i) $J_n(x) = (-1)^n J_{-n}(x)$ $\qquad (D7)$

(ii) $J_n(x) = (-1)^n J_n(-x)$ $\qquad (D8)$

(iii) $J_{n-1}(x) - J_{n+1}(x) = \dfrac{2n}{x} J_n(x)$ $\qquad (D9)$

(iv) $J_{n-1}(x) - J_{n+1}(x) = 2J'_n(x)$ $\qquad (D10)$

(v) $J'_n(x) = J_{n-1}(x) - \dfrac{n}{x} J_n(x)$ $\qquad (D11)$

(vi) $J'_n(x) = -J_{n-1}(x) + \dfrac{n}{x} J_n(x)$ $\qquad (D12)$

(vii) For small values of x,

$$J_n(x) = \frac{x^n}{2^n n!} \qquad (D13)$$

$$J_0(x) \approx 1 \qquad (D14a)$$

$$J_1(x) \approx \frac{x}{2} \tag{D14b}$$

$$J_n(x) \approx 0 \quad \text{for } n > 1 \tag{D14c}$$

(viii) For large values of x,

$$J_n(x) = \sqrt{\frac{2}{\pi x}} \cos\left(x - \frac{\pi}{4} - \frac{n\pi}{2}\right) \tag{D15}$$

(ix) $\sum_{n=-\infty}^{\infty} J_n^2(x) = 1 \quad \text{for all } x$ (D16)

where, $J_n'(x)$ is the first derivative of $J_n(x)$ with respect to x.

D.2 Bessel Function of the Second Kind

The Bessel function of the second kind can be expressed in the integral form as

$$K_0(x) = -\frac{1}{\pi}\int_0^{\pi} \exp(\pm x \cos\theta)\,[\gamma + \ln(2x \sin^2\theta)]d\theta \tag{D17}$$

where, $\gamma = 0.57722$ is the Euler's constant.

$$K_n(x) = \frac{\pi^{\frac{1}{2}}\left(\frac{x}{2}\right)^m}{\Gamma\left(n + \frac{1}{2}\right)} \int_0^{\infty} \cos^{-x \cos ht} \sin h^{2n} t dt \tag{D18}$$

where, $\Gamma(z)$ is the Gamma function given by

$$\Gamma(z) = \int_0^{\infty} t^{z-1} \exp(-t) dt$$

For integer n,

$$\Gamma(n+1) = n!$$

For fractional values,

$$\Gamma\left(\frac{1}{2}\right) = \pi^{\frac{1}{2}} = \left(-\frac{1}{2}\right)! \approx 1.77245$$

$$K_0(x) = \int_0^{\infty} \cos(x \sin h\, t) dt = \int_0^{\infty} \frac{\cos(xt)}{\sqrt{t^2+1}} dt \quad (x > 0) \tag{D19}$$

$$K_n(x) = \sec\left(\frac{n\pi}{2}\right)\int_0^{\infty} \cos(x \sin h\, t)\cos h(nt) dt \quad (x > 0) \tag{D20}$$

D.2.1 Recurrence Relation

Let $L_n = \exp(j\pi n) K_n$

$$L_{n-1}(x) - L_{n+1}(x) = \frac{2n}{x} L_n(x) \tag{D21}$$

$$L_n'(x) = L_{n-1}(x) - \frac{n}{x} L_n(x) \tag{D22}$$

$$L_{n-1}(x) + L_{n+1}(x) = 2L_n'(x) \tag{D23}$$

$$L_n'(x) = L_{n+1}(x) + \frac{n}{x} L_n(x) \tag{D24}$$

D.3 Values of Bessel Function of the First Kind

↓x →n	0	1	2	3	4	5	6	7	8	9	10	11
0.00	1.00											
0.10	0.997	0.050	0.001									
0.20	0.990	0.099	0.005									
0.30	0.978	0.148	0.011	0.001								
0.40	0.960	0.196	0.020	0.000								
0.50	0.938	0.242	0.031	0.002								
0.60	0.912	0.287	0.044	0.004								
0.70	0.881	0.329	0.059	0.007								
0.8	0.846	0.369	0.076	0.010	0.001							
0.9	0.807	0.406	0.095	0.014	0.002							
1.0	0.765	0.440	0.115	0.019	0.002							
1.25	0.646	0.511	0.171	0.037	0.006	0.001						
1.50	0.512	0.558	0.232	0.061	0.012	0.002						
1.75	0.370	0.580	0.294	0.092	0.021	0.004						
2.00	0.224	0.577	0.353	0.129	0.034	0.007	0.001					
2.50	–0.048	0.497	0.446	0.217	0.074	0.019	0.004	0.001				
3.00	–0.260	0.339	0.486	0.309	0.132	0.043	0.011	0.003				
3.50	–0.380	0.137	0.459	0.387	0.204	0.080	0.025	0.007	0.002			
4.00	–0.397	–0.006	0.364	0.430	0.281	0.132	0.049	0.015	0.004	0.001		
4.50	–0.321	–0.231	0.218	0.425	0.348	0.195	0.084	0.030	0.009	0.002		
5.00	–0.178	–0.328	0.047	0.365	0.391	0.261	0.131	0.053	0.018	0.005	0.001	
5.50	–0.007	–0.341	–0.117	0.256	0.397	0.362	0.187	0.087	0.034	0.011	0.003	0.001
6.00	0.151	–0.277	0.243	0.115	0.358	0.374	0.246	0.130	0.056	0.021	0.007	0.002
6.50	0.260	–0.154	–0.307	–0.035	0.275	0.348	0.300	0.180	0.090	0.037	0.013	0.004
7.00	0.300	–0.005	–0.301	–0.168	0.158	0.283	0.339	0.234	0.128	0.059	0.024	0.008
7.50	0.266	0.135	–0.230	–0.258	0.024	0.186	0.354	0.283	0.174	0.089	0.039	0.015
8.00	0.171	0.234	–0.113	–0.291	–0.105	0.067	0.338	0.321	0.223	0.126	0.061	0.026
8.50	0.042	0.273	0.022	–0.263	–0.208	0.055	0.287	0.338	0.269	0.169	0.089	0.041
9.00	–0.091	0.245	0.145	–0.181	–0.65	–0.055	0.204	0.327	0.305	0.215	0.125	0.062
9.50	–0.194	0.161	0.227	–0.06	–0.269	–0.161	0.009	0.287	0.323	0.258	0.165	0.090
10.0	–0.245	0.044	0.255	–0.058	–0.220	–0.234	–0.014	0.217	0.318	0.0290	0.207	0.213

Appendix E

PARAMETERS OF IMPORTANT SEMICONDUCTORS

Semiconductor	*Crystal structure*	*Band*	*Lattice constant (Å)*	*Bandgap (300K) (eV)*	*Mobility (m^2V^{-1} s^{-1})*		*Effective mass*		*Relative permittivity*
					μ_n	μ_p	(m^*_n/m_0)	(m^*_p/m_0)	ϵ_s/ϵ_0
				Elemental					
Si	Diamond	Indirect	5.43102	1.12	0.145	0.05	1.18	0.16^{lh}	11.9
Ge	Diamond	Indirect	5.64613	0.66	0.39	0.19	0.082	0.49^{hh}	16.0
				III–V Compound					
GaAs	Zinc blende	Direct	5.6533	1.42	0.8	0.04	0.063	0.076^{lh} 0.05^{hh}	12.9
AlAs	Zinc blende	Indirect	5.6605	2.36	0.018		0.11	0.22	10.1
InP	Zinc blende	Direct	5.8686	1.35	0.46	0.015	0.077	0.64	12.6
GaP	Zinc blende	Indirect	5.4512	2.26	0.011	0.0075	0.82	0.60	11.1
AlP	Zinc blende	Indirect	5.4635	2.42	0.0060	0.045	0.212	0.145	9.8
GaN	Wurtzite	Direct	a = 3.189 c = 5.182	3.44	0.0400	0.0010	0.27	0.8	10.4
InSb	Zinc blende	Direct	6.4794	0.17	8.0	0.125	0.0145	0.40	16.8
InAs	Zinc blende	Direct	6.0584	0.36	3.3	0.046	0.023	0.40	15.1
AlSb	Zinc blende	Indirect	6.1355	1.58	0.020	0.0420	0.12	0.98	14.4
BN	Zinc blende	Indirect	6.3157	6.4	0.020	0.050	0.26	0.36	7.1
				II–VI					
CdS	Zinc blende	Direct	5.825	2.5			0.14	0.51	5.4
	Wurtzite	Direct	a = 4.136 c = 6.714	2.49	0.035	0.0040	0.20	0.70	9.1
ZnS	Zinc blende	Direct	5.410	3.66	0.060		0.039	0.23	8.4
	Wurtzite	Direct	a = 3.822 c = 6.26	3.78	0.028	0.080	0.29	0.49	9.6
ZnO	Zinc blende	Direct	4.580	3.35	0.020	0.018	0.27		9.0
	Wurtzite	Direct	a = 3.25 c = 5.2	3.437	0.022	0.005	0.24	0.59	
CdTe	Zinc blende	Direct	6.482	1.56	0.105	0.010			10.2
				IV–IV					
SiC	Wurtzite	Indirect	a = 3.086 c = 15.117	2.996	0.040	0.005	0.60	1.0	9.66
				IV–VI					
PbS	Rock salt	Indirect	5.9363	0.41	0.060	0.070	0.25	0.25	17.0
PbTe	Rock salt	Indirect	6.4620	0.31	0.60	0.40	0.17	0.20	30.0

***lh*: Light hole; *hh*: Heavy hole**

Index

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T

U

V

Revised New Edition of Optical Fiber Communication
Previously Published by McGraw Hill

Optoelectronic Devices and Optical Fiber Communication

deals with semiconductor optoelectronic devices and components that find a major application in optical fiber communication including those which find other non-telecommunication applications. The primary focus is on the basic understanding of optical communication systems and their application in modern low-loss and high-bandwidth transmission systems as an alternative to conventional electrical communication systems.

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